On Langlands Conjectures

Probabilistic Methods

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Source, code and data: https://github.com/VadimSokolov/langlands

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To A.L.S. Corner

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PREFACE

In 1967 Robert Langlands wrote a letter to André Weil proposing a web of conjectures that would connect automorphic forms, the representation theory of reductive groups, and the arithmetic of Galois representations. Six decades later the Langlands programme is a central organising project of number theory. Many of its cases are theorems, among them the modularity of elliptic curves over the rationals, the local Langlands correspondence for GL_n, the fundamental lemma, and symmetric power functoriality for holomorphic modular forms. The general conjectures remain open.

This book asks what probability can contribute. Our answer has two sides.

The first is a limitation. No probabilistic argument known to us proves reciprocity or functoriality, and in Chapter 1 we prove a proposition explaining why no argument that sees only the limiting distribution of Frobenius data could. Equidistribution is a consequence of functoriality, not a route to it. We do not claim to prove the Langlands conjectures or any Riemann hypothesis.

The second is what probability does supply: consequences of the conjectures, calibration against data, and probabilistic reformulations. The Langlands conjectures predict that the Frobenius conjugacy classes attached to an automorphic representation behave like Haar random elements of a compact group. That prediction is the probabilistic shadow of the programme.

Part II is about equidistribution: Weyl's criterion and the Sato Tate law, Frobenius statistics at scale, random matrix models of the zeros of L functions, Katz's vertical law for Kloosterman sums and the random Fourier series of Kowalski and Sawin, and the Cohen Lenstra and Poonen Rains heuristics, in which class groups and Selmer groups behave like random groups.

Part III treats Hecke operators as random walks on Bruhat Tits trees, Ihara zeta functions, Ramanujan graphs, and hitting times on supersingular isogeny graphs.

Part IV starts from the infinitely divisible laws attached to Euler products and follows Selberg's central limit theorem to the log correlated picture of the critical line, with its branching random walk, multiplicative chaos and the conjecture of Fyodorov, Hiary and Keating. At its centre is the Thorin framework, which extends earlier work of the first author [442]. For every cuspidal automorphic representation of GL_n over $\mathbb{Q}$, the generalized Riemann hypothesis is equivalent to the statement that the reciprocal of an even completion of its L function, read in the variable $s = z^2$, is the Laplace transform of a generalized gamma convolution; equivalently, that a Lévy function built from the zeros is completely monotone, that a logarithmic derivative is a Stieltjes function, or that a sequence computable from

central derivatives alone is a Stieltjes moment sequence. These are reformulations, not proofs. The same structure appears locally, where the Ramanujan conjecture at a prime is exactly a local Thorin property; for zeta functions of curves over finite fields, where Weil's theorem makes it unconditional; on supersingular isogeny graphs, where the weight two bound of Eichler, Shimura and Weil does; and for Hecke random walks on Bruhat Tits trees, whose passage times are generalized gamma convolutions with Thorin measure filling the Kesten band. The framework also connects to van Dantzig pairs and Wald couples, to the sine squared transport that carries the primes, through the explicit formula, to the Thorin measure, and to the work of Konstantopoulos, Patie and Sarkar on the van Dantzig problem, where we give a necessary condition for their class and test it on fourteen L functions. Chapter 17 reads the error term of the squarefree counting function through the Pólya law of ξ and its Thorin atoms; Chapter 18 carries the framework to Dirichlet's β function, whose Pólya kernel is built from the cube of Dedekind's eta function, and proves, up to the heights to which the Riemann hypotheses for ζ and β are verified, an inequality on the lines $\frac{1}{2} < \operatorname{Re} s \leq 3.15$ that Lagarias proved for ξ on every line to the right of the centre under the Riemann hypothesis; Chapter 19 computes signed Thorin densities by Stieltjes inversion and counts zeros numerically by the argument principle for ζ, $L(s, \chi_{12})$ and $L(s, \Delta)$. None of the three proves a Riemann hypothesis. Chapter 20 turns these devices on Langlands reciprocity: it derives quadratic reciprocity from the normal law, treats Frobenius traces as random variables and as the labels of a Gamma process, and recasts Hecke's converse theorem at level one as a symmetry of a Pólya kernel, and the holomorphy and the generalized Riemann hypothesis of a self dual Artin L function as the complete monotonicity of a heat trace computed from Frobenius traces, the conductor and the gamma factor. Again these are reformulations, and no new case of reciprocity is proved. Chapter 21 reads reciprocity laws as identities between hitting times: on supersingular isogeny graphs in characteristic N, Kemeny constants and, for $N \equiv 1 \bmod 12$, numbers of spanning trees are computed from the reduction of $J_0(N)$ modulo the degree of the isogenies, Chebotarev densities are hitting rates for the largest prime factors of a random integer, the Thorin converse theorem concerns explosion times of pure birth processes, and the functional equation of the L function of an odd primitive Dirichlet character is an identity between two expansions of a signed exit law of Brownian motion. Here too no new case of reciprocity is proved. Chapter 22 follows the proof of Fermat's last theorem through these devices: weight one forms, through which the theorem of Langlands and Tunnell gives the residual modularity that Wiles's argument starts from, are the newforms whose label laws are supported on finitely many points; the modularity of an elliptic curve is, by Weil's converse theorem, a family of reflection symmetries of twisted Pólya kernels built from point counts and the conductor; Mazur's torsion theorem, through an

argument of Serre, makes positive, for a Frey curve and $p \geq 5$, the Gamma shape of the primes ℓ at which p does not divide $|E(\mathbb{F}_\ell)|$, the local density of Koblitz's conjecture; and the genus zero of $X_0(2)$ empties a Thorin measure built from Hecke eigenvalues at level 2. These restate the proof, reprove no step of it and prove no new case of modularity; its content is the theorems of Langlands and Tunnell, Mazur, Ribet, Wiles, and Taylor and Wiles, on the curve of Hellegouarch and Frey and the reduction of Frey and Serre. Chapter 23 collects the inverse Laplace transforms behind these devices: Bohr means read Frobenius traces off the transported density of $\eta(z)\eta(23z)$; Bromwich inversion of the Artin L function of $x^3 - x - 1$ on the critical line returns that weight one form, given the holomorphy that Hecke's identification of the two L functions supplies; moving the basepoint adds Cauchy noise to the zeros; the Thorin variable of an Euler product with nonnegative coefficients has a Pareto tail of index $\frac{3}{2}$, as Roynette and Yor found; atomic Thorin laws invert to B splines; and the Sato Tate law minimises the roughness penalty of Good and Gaskins among densities that vanish on the walls of the alcove. Most of these results are known, and none of them proves a case of reciprocity or a Riemann hypothesis. Chapter 24 relates the Gamma process of Chapter 20 to classical laws: the normalised sizes of the prime factors of a random integer follow the Poisson Dirichlet law, as the normalised cycle lengths of a random permutation do; normalised, the Gamma process with Frobenius labels is Ferguson's Dirichlet process, with the Haar measure of Chebotarev's theorem as base measure; and the random integers weighted by the coefficients of the powers of ζ give the whole Ewens family. For uniform random integers below x, the class of a size biased prime factor approaches Haar measure at rate $1/\log x$, the class of the largest one faster still, but the classes of the distinct prime factors only at rate $1/\log\log x$, with Mertens constants per Chebotarev class computed from $L(1,\chi)$ and $\frac{L'}{L}(1,\chi)$. These are theorems about random integers, most of them known, and none proves a case of reciprocity. Chapter 25 treats degree one, where reciprocity is a theorem: Dirichlet's theorem on primes in progressions and Hecke's theorem on the angles of Gaussian primes become Dirichlet processes, Landau's theorem on the integers with no prime factor in a given class becomes an atom at zero, and for an elliptic curve with complex multiplication, whose L function is that of a Hecke character by Deuring's theorem, the Gamma shape test of Chapter 20 detects the complex multiplication. Chapter 26 turns to the Maass forms, the other half of the automorphic spectrum of $\mathrm{SL}_2(\mathbb{Z})$: it reads the Selberg trace formula as a heat trace, whose terms are return probabilities of Brownian motion on a hyperbolic surface, restates Selberg's eigenvalue conjecture as the complete monotonicity of that heat trace, treats the closed geodesics as primes to which the sine squared transport and the Poisson Dirichlet limits apply, and reads the Eichler Selberg trace formula as the moment identity behind Serre's vertical Sato Tate law. These too are reformulations, and no new case of Selberg's

conjecture is proved. Part IV ends with an extreme value representation of the L function of a multiplicative eta product as the expectation of a regularised incomplete gamma function at a maximum of independent exponential variables.

Part V turns to inference: right Haar measure, transformation models and Barndorff Nielsen's p^* formula, reference priors for orbits after Chang and Eaves, equidistribution of Heegner points, and Bayesian identification of Sato Tate groups. Chapter 30 states twelve open problems.

Results new to this book include the limit $\ell/(\ell-1)$ for Kemeny constants of isogeny graphs, the Thorin criteria for the generalized Riemann hypothesis of automorphic L functions and for the Ramanujan conjecture, the Thorin structure of Hecke passage times, the representation of the squarefree error under Gamma(ν) smoothing as an average of the Mertens function, with positive weights exactly when $\nu \le 2$, the signs of the Pólya kernels of imaginary quadratic fields, the minimum on the real axis of the logarithmic derivative of the completed Dirichlet β function on the lines $\frac{1}{2} < \operatorname{Re} s \le 3.15$, up to the height to which its Riemann hypothesis is verified, the Gamma process limit of the Frobenius labelled factorisation of a random integer, the expansion in $1/\log x$ of the probability that a size biased prime factor of a uniform random integer below x has Frobenius in a given conjugacy class, the extreme value representation, and the consistency theorem for Bayesian identification.

Computation runs through the book as the experimental side of the subject. We computed Frobenius traces for all primes below 10^7, Hecke spectra on supersingular isogeny graphs at 207 prime levels, zeros of $L(s, \Delta)$, and posterior distributions over Sato Tate groups of genus two curves. Every chapter before Chapter 31 ends with open computational problems, stated with their scale and expected deliverables but deliberately left unsolved. Chapter 31 collects them into a programme, the agenda for the second author's computational work and, we hope, for some of our readers. Appendix B prints the core routines.

The book is written for probabilists and statisticians who want to see where their tools meet the Langlands programme, and for number theorists curious about what probabilistic structure lies inside their objects. We assume graduate probability and some familiarity with modular forms; Chapter 2 and Appendix A fill the gaps on each side.

Nick Polson, Chicago
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October 2026

PART I

FOUNDATIONS

CHAPTER 1

WHAT PROBABILITY CAN AND CANNOT DO

In January 1967 Robert Langlands, then a young mathematician at Princeton, wrote a letter of seventeen handwritten pages to André Weil [349]. In a covering note he asked Weil to read it as speculation; the letter set out conjectures that would organise much of number theory for the next six decades. Weil had it typed, the typed copy circulated widely, and its ideas were developed in Langlands' lectures *Problems in the theory of automorphic forms* [350]. This book asks a narrow question about that programme: what can probability theory say about it? The answer has two parts. No probabilistic method now known proves the central conjectures, and in Section 1.4 we prove that no argument that sees only the limiting distribution of Frobenius data can. But the conjectures make sharp probabilistic predictions, many of those predictions are theorems that can be measured in data, and they connect to classical objects of probability: random walks on trees, Haar measure on compact groups, infinitely divisible laws, extreme values and Bayes factors.

1.1 RECIPROCITY BEFORE LANGLANDS

The prehistory of the programme is the theory of reciprocity laws. Gauss' law of quadratic reciprocity says that whether a prime q is a square modulo an odd prime $p \neq q$ depends only on p modulo $4q$. Equivalently, whether p splits in the quadratic field $\mathbb{Q}(\sqrt{q})$ is decided by a congruence condition on p. Artin's reciprocity law, proved in 1927 [28], extends this to every abelian extension of number fields. Over $\mathbb{Q}$ it says that if $F/\mathbb{Q}$ is abelian and p is unramified in F, then the Frobenius element $\mathrm{Frob}_p \in \mathrm{Gal}(F/\mathbb{Q})$, which records how p splits in F, is determined by p modulo the conductor of F. By the Kronecker Weber theorem F lies in a cyclotomic field $\mathbb{Q}(\zeta_N)$, and Frob_p is the image of p mod N under $(\mathbb{Z}/N\mathbb{Z})^\times \cong \mathrm{Gal}(\mathbb{Q}(\zeta_N)/\mathbb{Q}) \to \mathrm{Gal}(F/\mathbb{Q})$.

The analytic side begins with Dirichlet and Hecke. A character of $\mathrm{Gal}(F/\mathbb{Q})$ defines an Artin L function, an Euler product over primes; Artin reciprocity identifies it with a Dirichlet L function $L(s,\chi) = \sum \chi(n) n^{-s}$, which has analytic continuation and a functional equation. Hecke proved the same properties for the L functions of Grössencharacters of number fields, and in the 1930s built the theory of Hecke operators and Euler products for modular forms that is the subject of Chapter 2 [251]. Tate's thesis of 1950 recast Hecke's work on Grössencharacters in the language of harmonic analysis on adeles [539], the language Langlands

would use.

What is the nonabelian analogue? If $\mathrm{Gal}(F/\mathbb{Q})$ is not abelian, Frob_p is only defined up to conjugacy, and no congruence condition on p determines it. Langlands' proposal was that the correct replacement for a congruence condition is an automorphic form: the conjugacy class of Frob_p under an n dimensional representation of the Galois group should match the Hecke eigenvalues at p of an automorphic representation of GL_n. For $n = 1$ this is Artin reciprocity. For $n = 2$ the best known instance is the modularity of elliptic curves, the Taniyama Shimura Weil conjecture: for every elliptic curve E over $\mathbb{Q}$ there is a weight two newform f with $L(s, E) = L(s, f)$. Taniyama raised a version of this question in 1955, Shimura sharpened it, and Weil, in the 1967 paper that proved his converse theorem [571], gave it a precise and testable form by predicting that the level of f equals the conductor of E. Wiles, with Taylor, proved it for semistable curves in 1995 [581, 541], which was enough for Fermat's last theorem, and Breuil, Conrad, Diamond and Taylor completed the proof for all elliptic curves over $\mathbb{Q}$ [41].

The local Langlands correspondence for GL_n over p adic fields was proved by Harris and Taylor and by Henniart [241, 257]. The fundamental lemma, a family of identities between orbital integrals on a group and on its endoscopic groups that is needed to stabilise the Arthur Selberg trace formula, was formulated by Langlands and Shelstad and proved by Ngô [421]. Newton and Thorne proved symmetric power functoriality for holomorphic modular forms [419, 420]. In 2024 Gaitsgory, Raskin and their collaborators announced a proof of the unramified global geometric Langlands conjecture in a series of long preprints [206]; it is a categorical statement about sheaves on the moduli space of bundles on a curve over a field of characteristic zero, and it does not by itself settle the conjectures over number fields. The general conjectures of reciprocity and functoriality over number fields remain open, and nothing in this book changes that.

1.2 The conjectures for a probabilist

We state the conjectures only to the precision needed later; Chapters 2 and 3 develop the objects and record which cases are known. Readers who want a survey from the arithmetic side should consult Gelbart [211] and Arthur [24].

An *automorphic representation* π of GL_n over $\mathbb{Q}$ is, roughly, an irreducible representation of the adelic group $\mathrm{GL}_n(\mathbb{A})$ occurring in a space of functions on $\mathrm{GL}_n(\mathbb{Q})\backslash\mathrm{GL}_n(\mathbb{A})$. It factors as a restricted tensor product $\pi = \bigotimes_v \pi_v$ over places v. At all but finitely many primes p the component π_p is *unramified*, and by the Satake isomorphism an unramified representation of $\mathrm{GL}_n(\mathbb{Q}_p)$ is determined by a semisimple conjugacy class

$$A_p(\pi) \in \mathrm{GL}_n(\mathbb{C}),$$

its *Satake parameter*. We normalise $A_p(\pi)$ unitarily, so that for cuspidal π the Ramanujan conjecture says that its eigenvalues have absolute value one. The standard L function is the Euler product $L(s,\pi) = \prod_p \det(I - A_p(\pi)p^{-s})^{-1}$, with suitable factors at the ramified primes.

On the Galois side, let $\rho : \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \to \mathrm{GL}_n(\overline{\mathbb{Q}}_\ell)$ be a continuous representation that is unramified outside a finite set and comes from geometry, for instance the action on the ℓ adic cohomology of a variety. For unramified $p \neq \ell$ the matrix $\rho(\mathrm{Frob}_p)$ is defined up to conjugacy. Fix an isomorphism $\iota : \overline{\mathbb{Q}}_\ell \cong \mathbb{C}$.

CONJECTURE 1.1 (Reciprocity). *For every irreducible geometric ρ of dimension n there is a cuspidal automorphic representation π of $\mathrm{GL}_n(\mathbb{A})$ such that, for all but finitely many p, the characteristic polynomial of $\rho(\mathrm{Frob}_p)$, transported to $\mathbb{C}$ by ι, equals that of $A_p(\pi)$, after a fixed normalisation of weights.*

Functoriality is the second pillar. To a reductive group G over $\mathbb{Q}$ Langlands attached a complex dual group $\widehat{G}$ and an L group ${}^L G$, a semidirect product of $\widehat{G}$ with a Galois or Weil group that may be replaced by $\widehat{G}$ when G is split, and to the unramified components of automorphic representations of G Satake parameters, semisimple conjugacy classes in ${}^L G$.

CONJECTURE 1.2 (Functoriality). *Let G, H be reductive groups over $\mathbb{Q}$ with G quasisplit, and $r : {}^L H \to {}^L G$ an admissible homomorphism of L groups. For each automorphic representation π of H there is an automorphic representation Π of G with $A_p(\Pi) = r(A_p(\pi))$ for almost all p.*

The case most used in this book is $H = \mathrm{GL}_2$, $G = \mathrm{GL}_{n+1}$, $r = \mathrm{Sym}^n$, the nth symmetric power. Newton and Thorne proved it for every n when π comes from a holomorphic newform of weight at least two without complex multiplication [419, 420]; for forms with complex multiplication it reduces to automorphic induction from Hecke characters. For general cuspidal π, Maass forms included, it is known for $n \leq 4$ [212, 316, 315] and open for $n \geq 5$.

1.2.1 The probabilistic shadow

Take a holomorphic newform f of weight $k \geq 2$ with trivial character and without complex multiplication, for instance Ramanujan's Δ or the weight two form attached to an elliptic curve over $\mathbb{Q}$ without complex multiplication. For p not dividing the level, normalise its Hecke eigenvalue at p to $x_p = a_p / p^{(k-1)/2}$. By Deligne's theorem [140] $x_p \in [-2, 2]$, so $x_p = 2\cos\theta_p$ for an angle $\theta_p \in [0, \pi]$, and the unitarily normalised Satake parameter is the conjugacy class of $\mathrm{diag}(e^{i\theta_p}, e^{-i\theta_p})$ in the compact group $\mathrm{SU}(2)$.

DEFINITION 1.3. Let K be a compact group with normalised Haar measure h, and let $\pi(x)$ be the number of primes $p \leq x$ (context distinguishes this func-

tion from an automorphic representation π). A sequence (g_p) of conjugacy classes in K indexed by primes is *equidistributed* with respect to Haar measure if $\pi(x)^{-1}\sum_{p\le x}\phi(g_p)\to\int_K\phi\,dh$ as $x\to\infty$ for every continuous class function ϕ on K.

For $K=\mathrm{SU}(2)$ a class function is a function of the trace $x=2\cos\theta$, and the image of Haar measure under the trace is the semicircle law

$$\mu_{\mathrm{ST}}(dx)=\frac{1}{2\pi}\sqrt{4-x^2}\,dx\qquad\text{on }[-2,2],$$

whose even moments $\int x^{2m}\,d\mu_{\mathrm{ST}}$, $m=1,2,3,\dots$, are the Catalan numbers $1,2,5,14,42,\dots$, as for Wigner's semicircle law for large random matrices.

THEOREM 1.4 (Sato Tate; Barnet Lamb, Geraghty, Harris, Taylor). *For every holomorphic newform of weight $k\ge 2$ with trivial character and without complex multiplication, the normalised eigenvalues x_p are equidistributed with respect to μ_{ST}.*

The conjecture was formulated by Sato and Tate in the 1960s [538]. For elliptic curves over totally real fields with nonintegral j invariant it was proved through potential automorphy by Clozel, Harris, Shepherd Barron and Taylor [103, 540, 266], and for all newforms as in the theorem, which include the form of every elliptic curve over $\mathbb{Q}$ without complex multiplication, by Barnet Lamb, Geraghty, Harris and Taylor [67]. The argument, due in outline to Serre [499], is short once the hard input is available. The traces of the irreducible representations of $\mathrm{SU}(2)$ are $\operatorname{tr}\mathrm{Sym}^n=U_n(x/2)$, with U_n the Chebyshev polynomials of the second kind, and by the Peter Weyl theorem equidistribution is equivalent to $\pi(x)^{-1}\sum_{p\le x}U_n(x_p/2)\to 0$ for every $n\ge 1$. That vanishing follows by a Tauberian theorem from holomorphy and nonvanishing of $L(s,\mathrm{Sym}^n f)$ on the line $\Re s=1$, which is what (potential) automorphy of $\mathrm{Sym}^n f$ provides. Chapter 4 gives the argument in full.

Functoriality is the *input*; equidistribution is the *output*. In general the Langlands conjectures predict the following.

> *Normalised Frobenius conjugacy classes are equidistributed with respect to Haar measure on a compact group, the Sato Tate group, attached to the motive or automorphic representation.*

This prediction is the probabilistic shadow of the Langlands conjectures, and it is concrete and testable. Figure 1.1 shows it for two elliptic curves. For $E_1: y^2=x^3+x+1$, which has discriminant -496, conductor 496 and no complex multiplication, we computed $a_p=p+1-\#E_1(\mathbb{F}_p)$ for all 664,577 good primes below 10^7 with PARI/GP. The histogram of $x_p=a_p/\sqrt{p}$ sits on the semicircle and its even moments approach $1,2,5,14,42,132$. For $E_2: y^2=x^3-x$, which

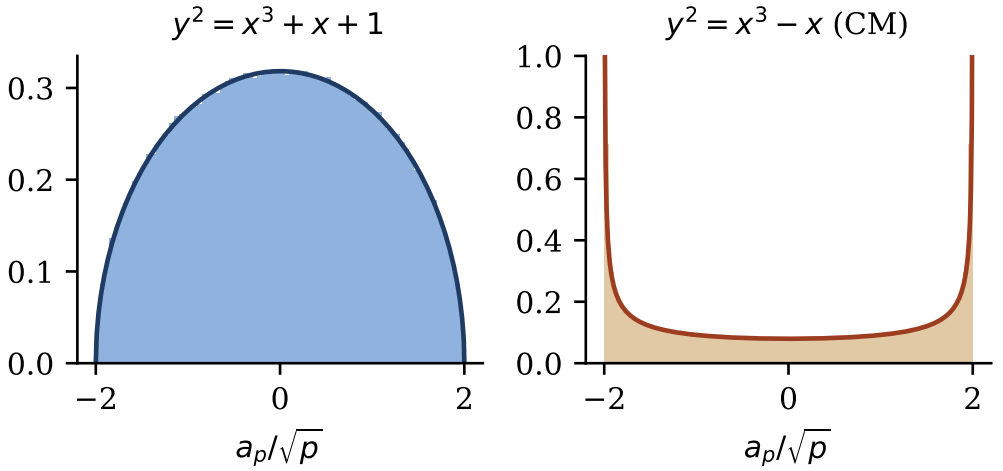


Figure 1.1: Normalised Frobenius traces $a_p/\sqrt{p}$ for all good primes below 10^7. Left: $E_1 : y^2 = x^3 + x + 1$, histogram of all traces against the Sato Tate density $\sqrt{4 - x^2}/(2\pi)$. Right: $E_2 : y^2 = x^3 - x$, which has complex multiplication; histogram of the nonzero traces (the primes $p \equiv 1 \bmod 4$), scaled to their share of all good primes (about $\frac{1}{2}$), against the continuous part $1/(2\pi\sqrt{4 - x^2})$ of μ_{CM}, which has mass $\frac{1}{2}$. The atom of mass $\frac{1}{2}$ at zero, from the primes $p \equiv 3 \bmod 4$, is not drawn.

has complex multiplication by $\mathbb{Z}[i]$ and conductor 32, the primes $p \equiv 3 \bmod 4$ are supersingular, with $a_p = 0$, and for $p \equiv 1 \bmod 4$ the angles θ_p are uniformly distributed on $[0, \pi]$ by Hecke's theorem [248]; the law is

$$\mu_{\mathrm{CM}} = \tfrac{1}{2}\delta_0 + \tfrac{1}{2}\,\frac{dx}{\pi\sqrt{4 - x^2}},$$

the image of Haar measure on the normaliser of a maximal torus in $\mathrm{SU}(2)$, with even moments $\frac{1}{2}\binom{2m}{m} = 1, 3, 10, 35, 126, 462$ for $m = 1, \dots, 6$. Among the good primes below 4×10^4 the fraction with $a_p = 0$ is 0.5038.

The two histograms distinguish two compact groups: the Sato Tate group of E_1 is $\mathrm{SU}(2)$, and that of E_2 is the normaliser $N(\mathrm{U}(1))$ of the diagonal torus. Telling them apart from data is a statistical problem, the subject of Chapter 29.

1.3 A map of methods

Table 1.1 lists the main families of probabilistic and computational methods that touch the programme, the target in each case, and what each delivers. "Proves" means the method is the engine of a theorem, usually with arithmetic input

supplied from elsewhere. "Reformulates" means it recasts a conjecture as an equivalent probabilistic statement. "Tests" means it produces evidence, calibration or conjectural constants. "None known" means that no contribution is known.

The last line of the table is the subject of the next section.

1.4 THE KEY LIMITATION

Could one prove reciprocity probabilistically, by showing that some sequence of Galois data has the statistics that automorphy forces? The following proposition shows that the limiting empirical law of the data cannot do it.

Let $\mathcal{P}$ be the set of primes. A set $S \subset \mathcal{P}$ has *density zero* if $\#\{p \le x : p \in S\}/\pi(x) \to 0$. Say that a sequence $(x_p)_{p\in\mathcal{P}}$ of real numbers *comes from* a cuspidal automorphic representation π of $\mathrm{GL}_2(\mathbb{A})$ with trivial central character if $x_p = \operatorname{tr} A_p(\pi)$ for all but finitely many primes p, and call it *automorphic* if it comes from some such π. Both notions are unchanged when finitely many terms are changed. A sequence $(x_p) \in [-2,2]^{\mathcal{P}}$ is equidistributed with respect to μ_{ST} if $\pi(x)^{-1}\sum_{p\le x}\phi(x_p) \to \int \phi\, d\mu_{\mathrm{ST}}$ for every continuous ϕ on $[-2,2]$.

PROPOSITION 1.5 (Equidistribution does not determine automorphy; this book). *Let $\mathcal{E}$ be the set of sequences $(x_p) \in [-2,2]^{\mathcal{P}}$ that are equidistributed with respect to μ_{ST}.*

1. *$\mathcal{E}$ has the cardinality of the continuum and contains continuum many sequences no two of which agree at all but finitely many primes. The automorphic sequences fall into countably many classes under the relation of agreeing at all but finitely many primes, so all but countably many of those sequences are not automorphic.*
2. *Let $(x_p) \in [-2,2]^{\mathcal{P}}$ come from π, and let $(x'_p) \in [-2,2]^{\mathcal{P}}$ differ from (x_p) exactly on an infinite set S of density zero. Then (x'_p) is not automorphic, and $(x'_p) \in \mathcal{E}$ if and only if $(x_p) \in \mathcal{E}$.*

PROOF. We first show that density zero alterations do not change limits of empirical averages. Let $(x_p), (x'_p) \in [-2,2]^{\mathcal{P}}$ agree outside a density zero set S, and let ϕ be continuous on $[-2,2]$. Then

$$\Big|\frac{1}{\pi(x)}\sum_{p\le x}\phi(x'_p) - \frac{1}{\pi(x)}\sum_{p\le x}\phi(x_p)\Big| \le 2\|\phi\|_\infty \frac{\#\{p\le x : p\in S\}}{\pi(x)} \longrightarrow 0,$$

so either both empirical averages converge, to the same limit, or neither does. In particular $(x'_p) \in \mathcal{E}$ if and only if $(x_p) \in \mathcal{E}$.

(1) By Theorem 1.4 the normalised eigenvalues $x_p = \tau(p)/p^{11/2}$ of Δ form an element of $\mathcal{E}$. Let S be an infinite set of density zero, for instance $S = \{p_{k^2} : k \ge 1\}$, where p_j is the jth prime; it has $\lfloor \pi(x)^{1/2} \rfloor$ elements up to x. For $c \in [-2,2]$ let $(x_p^{(c)})$ equal c on S and agree with (x_p) off S. By the first paragraph each $(x_p^{(c)})$ lies

in $\mathcal{E}$, and for $c \neq c'$ the sequences $(x_p^{(c)})$ and $(x_p^{(c')})$ differ at every prime of the infinite set S. These are continuum many elements of $\mathcal{E}$, no two agreeing at all but finitely many primes, and $|\mathcal{E}| \leq |[-2,2]^{\mathcal{P}}| = 2^{\aleph_0}$.

On the other side, every cuspidal automorphic representation π of $\mathrm{GL}_2(\mathbb{A})$ has nonzero vectors fixed by some open compact subgroup K of $\mathrm{GL}_2(\mathbb{A}_f)$. There are countably many such K, since each contains some principal congruence subgroup $K(N)$, is a finite union of cosets of $K(N)$, and $\mathrm{GL}_2(\mathbb{A}_f)/K(N)$ is countable. For fixed K, the cuspidal functions on $\mathrm{GL}_2(\mathbb{Q})Z(\mathbb{A})\backslash\mathrm{GL}_2(\mathbb{A})$ that are right invariant under K form a separable Hilbert space on which $\mathrm{GL}_2(\mathbb{R})$ and the Hecke operators of level K act with discrete spectrum and finite multiplicities, and each π with trivial central character and nonzero K fixed vectors occurs in it; so there are countably many such π. Hence there are countably many π in all, and the sequences that come from a given π form a single class under agreement at all but finitely many primes. A class contains at most one of the sequences $(x_p^{(c)})$, so all but countably many of them are not automorphic.

(2) The second assertion was proved in the first paragraph. Suppose (x'_p) came from some π'. Then $\operatorname{tr} A_p(\pi') = x'_p = x_p = \operatorname{tr} A_p(\pi)$ for all primes outside $S \cup F$, with F finite, and $S \cup F$ has natural density zero, hence Dirichlet density zero. Ramakrishnan's refinement of strong multiplicity one [451] states that two cuspidal automorphic representations of $\mathrm{GL}_2(\mathbb{A})$ with trivial central character are isomorphic if their Hecke eigenvalues $\operatorname{tr} A_p$ agree at all primes outside a set of density less than $1/8$. Therefore $\pi' \cong \pi$, and hence $x'_p = \operatorname{tr} A_p(\pi) = x_p$ for all but finitely many p. This contradicts the hypothesis that the sequences differ at every prime of the infinite set S. □

COROLLARY 1.6 (this book). *Let Φ be any function on $[-2,2]^{\mathcal{P}}$ that is unchanged when a sequence is altered on a set of density zero; for instance any function of the set of weak limit points of the empirical measures $\pi(x)^{-1}\sum_{p\leq x}\delta_{x_p}$ as $x \to \infty$, which by the first paragraph of the proof above is unchanged by such alterations. Then for every automorphic $(x_p) \in [-2,2]^{\mathcal{P}}$ there are continuum many sequences in $[-2,2]^{\mathcal{P}}$ that are not automorphic and on which Φ takes the value it takes on (x_p); if $(x_p) \in \mathcal{E}$, they all lie in $\mathcal{E}$. In particular Φ cannot characterise automorphy.*

PROOF. Fix an infinite set S of density zero and apply part (2) of Proposition 1.5 to every $(x'_p) \in [-2,2]^{\mathcal{P}}$ that agrees with (x_p) off S and differs from it at every prime of S. There are continuum many such sequences, since each x'_p with $p \in S$ ranges over $[-2,2] \setminus \{x_p\}$; none is automorphic, and Φ takes the same value on all of them and on (x_p). □

Any argument that uses only the limiting distribution of Frobenius traces therefore sees the same object for an automorphic representation and for continuum many nonautomorphic impostors. Rates of convergence do not help either.

If $\epsilon(x)\pi(x) \to \infty$, there is an infinite set S with $\#\{p \le x : p \in S\} \le \epsilon(x)\pi(x)$ for all x, and an alteration on S changes every empirical average $\pi(x)^{-1}\sum_{p\le x}\phi(x_p)$ by at most $2\|\phi\|_\infty\epsilon(x)$. With $\epsilon(x) = \pi(x)^{-3/4}$, for instance, which satisfies both $\epsilon(x)\pi(x) \to \infty$ and $\epsilon(x) = o(\pi(x)^{-1/2})$, this gives impostors that the discrepancy statistics of Chapter 5, which live at the scale $\pi(x)^{-1/2}$, cannot see. What distinguishes the genuine sequence is structure that couples different primes: the analytic continuation and functional equation of the Dirichlet series built from it and of all its twists, and the Galois representation attached to it, which for an elliptic curve makes each $a_p = \sqrt{p}\,x_p$ an integer whose residue modulo a prime $\ell \neq p$ is determined by the Frobenius class of p in the field generated by the ℓ torsion points. Converse theorems, from Weil [571] to Cogdell and Piatetski Shapiro [108], characterise automorphy through the first kind of coupling; modularity lifting theorems, from Wiles onward [581, 541], establish it through the second, by Galois deformation theory and congruences between automorphic forms. No probabilistic method currently exploits either. Chapter 20 restates part of the first kind of coupling as a symmetry of functions computed from Frobenius traces, the conductor and the gamma factor, and that part together with the generalized Riemann hypothesis as a positivity property of such functions; this is a reformulation, not a method of proof. Probability therefore enters the programme through consequences of the conjectures, as a calibration tool and as a source of reformulations, not as a proof technique for reciprocity itself.

REMARK 1.7. The same argument applies to any statistic that is a limit of averages over primes in which each prime carries weight $O(1/\pi(x))$, for instance the empirical joint law of consecutive traces $(x_{p_n}, x_{p_{n+1}})$. It says nothing about statistics that weight individual primes heavily, or that use the analytic properties of the full Dirichlet series and its twists. A sparser alteration, on an infinite set S with $\sum_{p\in S,\, p\le x}\log p = o(\sqrt{x})$, moves every twisted sum $\sum_{p\le x} x_p\chi(p)\log p$ by $o(\sqrt{x})$, so properties of these sums at the square root scale cannot characterise automorphy either; whether, uniformly over characters χ of growing conductor, they can decide closeness to a newform is Open Problem 30.4 of Chapter 30.

1.5 WHAT THE METHODS DELIVER

Within these limits there is a great deal to prove. The book is organised in five parts.

1.5.1 Part I: Foundations

Chapter 2 develops modular forms, Hecke operators, Euler products and functional equations for readers trained in probability, and isolates two facts used throughout: normalised Hecke eigenvalues are traces of elements of $\mathrm{SU}(2)$, and

the completed L function of Δ is the Mellin transform of a positive function. Chapter 3 treats Galois representations and the Chebotarev density theorem, reciprocity, the dual group and the Satake isomorphism, functoriality and the Sato Tate group, and records which cases are known. Appendix A collects the probability used later.

1.5.2 Part II: Equidistribution

The Weyl criterion on compact groups turns equidistribution into the vanishing of averages of characters. For $\mathrm{SU}(2)$ these are Chebyshev polynomials, and for the CM law μ_{CM} the averages $\int U_n(x/2)\, d\mu_{\mathrm{CM}}$ equal 1 when $4 \mid n$ and 0 otherwise, a fingerprint of the Sato Tate group. Chapter 4 gives Serre's derivation of Sato Tate from symmetric power functoriality. Chapter 5 studies the rate of convergence (the discrepancy D_n of the first n traces, which for our curves sits at the scale $n^{-1/2}$ of independent sampling), the bias of the weighted sums $\sum_{p \le x} a_p \log p/p$, which through the explicit formula and under the Riemann hypothesis for $L(s, E)$ encodes the analytic rank of an elliptic curve in logarithmic mean, and the Lang Trotter conjecture on how often a_p takes a fixed value.

Chapter 6 turns to the other random matrix limit, in which the dimension of the group tends to infinity and eigenvalues model zeros of L functions: the pair correlation of Montgomery and Dyson, the symmetry types of Katz and Sarnak, low lying zeros of families of modular forms and the moment conjectures of Keating and Snaith. Chapter 7 finds the same semicircle in Kloosterman sums, through Katz's vertical Sato Tate theorem [296] and the random Fourier series of Kowalski and Sawin [331]. Chapter 8 replaces Haar random elements of a compact group by random finite abelian groups: the Cohen Lenstra heuristics for class groups and the Poonen Rains model for Selmer groups, with what they predict for ranks, compared with class groups and Selmer groups computed for this book.

1.5.3 Part III: Random walks and hitting times

Hecke operators, suitably normalised, are Markov operators. The Bruhat Tits tree of the group $\mathrm{PGL}_2(\mathbb{Q}_q)$ is the $(q+1)$ regular tree, T_q acts as its adjacency operator, and the spectral measure of simple random walk on the tree is the Kesten McKay law [309, 387], which is also the q adic Plancherel measure. Chapter 9 shows that its Chebyshev moments are $q^{-n/2}$ for even n and 0 for odd n, and that it converges to the semicircle as $q \to \infty$, which explains heuristically, but does not prove, why the horizontal Sato Tate law (one form, the prime varying) and the vertical one (one prime, the form varying) are compatible. Chapter 10 turns to finite quotients, whose Ihara zeta functions make the Ramanujan property a graph Riemann hypothesis, and to the supersingular isogeny graphs of Pizer [438], on

which the Ramanujan bound for Hecke eigenvalues of weight two forms becomes an optimal spectral gap. We prove that the Kemeny constant of the supersingular ℓ isogeny graph, the expected time to hit a target drawn from the stationary distribution, is asymptotic to $\ell/(\ell-1)$ times the number of vertices.

1.5.4 Part IV: Lévy and Thorin measures

Chapter 11 starts from Khinchine's observation that for $\sigma > 1$ the function $t \mapsto \zeta(\sigma+it)/\zeta(\sigma)$ is the characteristic function of an infinitely divisible law. For an automorphic L function the corresponding Lévy Khintchine exponent has weights of both signs, and positivity returns for Rankin Selberg L functions: for $\sigma > 1$, $t \mapsto L(\sigma+it, \pi \times \tilde{\pi})/L(\sigma, \pi \times \tilde{\pi})$ is again the characteristic function of an infinitely divisible law. Chapter 12 follows $\log|L|$ on the critical line, through Selberg's central limit theorem, the log correlated field of Fyodorov, Hiary and Keating and its maximum, and multiplicative chaos, tested on $L(s, \Delta)$.

For Ramanujan's Δ the completed L function $\Lambda(s, \Delta)$ is the Mellin transform of the positive function $y \mapsto \Delta(iy)$, so $\Lambda(6+it)/\Lambda(6)$ is the characteristic function of a symmetric random variable, and the Riemann hypothesis for Δ says that this characteristic function has only real zeros. In Chapter 13 we prove that the Riemann hypothesis for Δ is equivalent to $s \mapsto \Lambda(6)/\Lambda(6+\sqrt{s})$ being the Laplace transform of a generalized gamma convolution in the sense of Thorin and Bondesson [546, 73], with Thorin measure carried by the squared ordinates of the zeros. This is a reformulation, not a proof. Chapters 14 and 15, extending the first author's earlier work [442], turn it into a framework. For every cuspidal automorphic representation of GL_n over $\mathbb{Q}$ the generalized Riemann hypothesis is equivalent to the Thorin property, to complete monotonicity of a Lévy function built from the zeros, and to positive semidefiniteness of Hankel matrices of power sums computable from the central Taylor coefficients alone; the order of vanishing at the centre, which for an elliptic curve is the Mordell Weil rank under the conjecture of Birch and Swinnerton Dyer, appears as a degenerate Thorin atom at the origin of mass half the order. Locally the Ramanujan conjecture at a prime is exactly a Thorin property of the Euler factor, and the framework becomes unconditional for curves over finite fields, for Ramanujan graphs, including the supersingular isogeny graphs, and for Hecke random walks on Bruhat Tits trees, whose passage times are generalized gamma convolutions with explicit large deviations. Chapter 16 recasts all of this, as in [442], as van Dantzig pairs and Wald couples, proves the sine squared transport from Euler products to Thorin densities, and tests the class of Konstantopoulos, Patie and Sarkar [330] against L functions.

Chapters 17 to 19 test the framework on three cases. Chapter 17 follows the squarefree integers: their counting error is governed by the zeros of ζ, the constant

of the squarefree problem is a combination of two tilted means of the Pólya law of ξ, a Gamma smoothing turns the error into an average of the Mertens function, and the Thorin picture extends into the critical strip and to imaginary quadratic fields, whose Pólya kernels are positive when $|d| < \pi^2$ and take negative values whenever 2 and 3 are inert. Chapter 18 does the same for Dirichlet's β function, whose kernel is built from the cube of Dedekind's eta function, and proves for β and for ξ, on the lines $\operatorname{Re} s = \alpha$ with $\frac{1}{2} < \alpha \leq 3.15$ and up to the heights to which their Riemann hypotheses are verified, that the real part of the logarithmic derivative is smallest on the real axis, which for ξ Lagarias proved on every line to the right of the centre under the Riemann hypothesis, and Garunkštis unconditionally for $\alpha \geq 1$. Chapter 19 computes signed Thorin densities by Stieltjes inversion and counts zeros numerically by the argument principle for ζ, $L(s, \chi_{12})$ and $L(s, \Delta)$; its first section says where Scholze's work enters the theory of L functions, and that the computations do not use it. None of the three chapters proves a Riemann hypothesis.

Chapter 20 turns these devices on reciprocity. It derives quadratic reciprocity and Gauss's sign from the characteristic function of the normal law, puts Frobenius traces and Hecke eigenvalues on one probability space, where irreducible Artin representations and cuspidal automorphic representations are both orthonormal systems of random variables, reads the Frobenius labelled prime factorisation of a random integer as a Gamma process, and restates Hecke's converse theorem at level one as a symmetry of a Pólya kernel, and the holomorphy and the generalized Riemann hypothesis of a self dual Artin L function as the complete monotonicity of a heat trace computed from Frobenius traces, the conductor and the gamma factor. These are reformulations. Finitely many evaluations of the heat trace can disprove holomorphy or the generalized Riemann hypothesis but are not known to verify them, even to a finite height, and no new case of reciprocity is proved.

Chapter 21 reads reciprocity through hitting times. For the walk on supersingular elliptic curves in characteristic N along isogenies of degree ℓ, the Kemeny constant is a harmonic sum of the factors $\ell + 1 - a_\ell(f)$ of $|J_0(N)(\mathbb{F}_\ell)|$, and for $N \equiv 1 \bmod 12$ the number of spanning trees is $|J_0(N)(\mathbb{F}_\ell)|$ divided by the number of vertices, a formula of Hashimoto [246, 364] that Munier and Shnidman [408] read as a reciprocity law exchanging the two primes. Chebotarev densities become hitting rates for the largest prime factors of a random integer, the Thorin converse theorem a statement about explosion times of pure birth processes, and the functional equation of the L function of an odd primitive Dirichlet character an identity between two expansions of a signed exit law of Brownian motion. These too are reformulations, and no new case of reciprocity is proved.

Chapter 22 follows the proof of Fermat's last theorem through the same language. Weight one newforms, through which the theorem of Langlands and Tunnell gives the residual modularity that Wiles's argument starts from, are

exactly the newforms whose label laws are supported on finitely many points; modularity becomes, through Weil's converse theorem, a family of reflection symmetries of twisted Pólya kernels built from point counts and the conductor; the density of the primes ℓ at which p does not divide $|E(\mathbb{F}_\ell)|$, the local density of Koblitz's conjecture, becomes the shape of a Gamma law, positive for a Frey curve and $p \geq 5$ by Serre's argument from Mazur's torsion theorem; and the genus zero of $X_0(2)$, which ends the proof, becomes the emptiness of a Thorin measure built from Hecke eigenvalues, or a Kemeny constant equal to zero. These are restatements: no step of the proof is reproved and no new case of modularity is proved, and the content of the proof is in the theorems of Langlands and Tunnell, Mazur, Ribet, Wiles, and Taylor and Wiles, on the curve of Hellegouarch and Frey and the reduction of Frey and Serre.

Chapter 23 collects the inverse Laplace transforms behind the earlier chapters of this part and computes with them. Bohr means invert the sine squared transport, so that the transported density of an L function on any half line determines its Frobenius traces, which we recover at the primes below 64 from the transported density of $\eta(z)\eta(23z)$. Bromwich inversion of the Artin L function of $x^3 - x - 1$ on the critical line, computed as ζ_K/ζ, returns the weight one form; it uses the holomorphy of this L function, which Hecke's identification of it with the L function of the form supplies, and in general the modularity of the result is the content of Weil's converse theorem. As in [441], the shifted Thorin densities of Chapter 16 form a Cauchy semigroup in the basepoint, in which each zero is seen through Cauchy noise, and by a theorem of Roynette and Yor [467] the Thorin variable of an Euler product with nonnegative coefficients has a Pareto tail of index $\frac{3}{2}$. A last section relates atomic Thorin laws to the B splines of Curry and Schoenberg, and reads the Sato Tate law as the density that minimises the roughness penalty of Good and Gaskins among those vanishing on the walls of the alcove. Most of these results are known, and no case of reciprocity is proved.

Chapter 24 relates the Gamma process of Chapter 20 to classical laws. The normalised sizes of the prime factors of a uniform random integer follow the Poisson Dirichlet law by Billingsley's theorem, as the normalised cycle lengths of a random permutation do; normalised, the Gamma process with Frobenius labels is Ferguson's Dirichlet process, whose base measure is the Haar measure that Chebotarev's theorem gives, so that the limit of every Frobenius functional is a standard exponential variable times a Dirichlet mean; and the random integers weighted by the coefficients of the powers of ζ give the whole Ewens family, with the exponent of the singularity at $s = 1$ as Ewens parameter. For a uniform random integer below x, the class of a size biased prime factor approaches Haar measure at rate $1/\log x$, the class of the largest one faster still, but the classes of the distinct prime factors only at rate $1/\log\log x$, with Mertens constants per class computed from $L(1,\chi)$ and $\frac{L'}{L}(1,\chi)$: among the distinct prime factors of random

integers below 10^7, the primes that split completely in the field of $x^3 - x - 1$ have less than half their Haar share. A last section asks which of Bondesson's theorems on generalized gamma convolutions do arithmetic work. These are theorems about random integers, most of them known, and no case of reciprocity is proved.

Chapter 25 treats degree one, where reciprocity is a theorem: by Kronecker and Weber over $\mathbb{Q}$, by Artin's reciprocity law over a number field, and for Hecke's characters of infinite order. Dirichlet's theorem on primes in progressions becomes a Dirichlet process on the classes modulo q, with Mertens constants computed from $L(1, \chi)$ and with Landau's theorem on the integers that have no prime factor in a given class as the atom at zero of Chapter 24; Hecke's theorem on the angles of Gaussian primes becomes a Dirichlet process on the circle; and the Thorin measure of a degree one L function is checked against its zeros. By Deuring's theorem the L function of an elliptic curve with complex multiplication is that of a Hecke character of the imaginary quadratic field, and seen from that field the Rankin Selberg subordinator of Chapter 20 is the sum of two independent copies of itself seen from $\mathbb{Q}$, so that the Gamma shape test of Chapter 20 detects complex multiplication, here as a theorem.

Chapter 26 turns to the Maass forms, the eigenfunctions of the hyperbolic Laplacian on the modular surface and the other half of the automorphic spectrum of $\mathrm{SL}_2(\mathbb{Z})$. With the normalised hyperbolic area the modular surface is a probability space, and Brownian motion on it is ergodic with that law. Read as a heat trace, the Selberg trace formula of a compact hyperbolic surface says that the total return probability of Brownian motion is the free return probability plus a sum over closed geodesics, which is McKean's reading [389]. Selberg's eigenvalue conjecture becomes the complete monotonicity of the heat trace over the cusp forms, as the holomorphy and the generalized Riemann hypothesis of a self dual Artin L function became that of a heat trace in Chapter 20. Weyl's law keeps the eigenvalues out of the Thorin class, while the closed geodesics, ordered by length, form a system of primes to which the sine squared transport of Chapter 16 and the Poisson Dirichlet limits of Chapter 24 apply unconditionally. A last section states the Eichler Selberg trace formula and reads it as the moment identity behind Serre's vertical Sato Tate law. These are reformulations, and no new case of Selberg's conjecture is proved.

Finally, in Chapter 27, the product $\prod_n (1 - e^{-2\pi n y})$ is the distribution function of the maximum of independent exponentials $E_n/(2\pi n)$, which gives an extreme value representation of the L functions of multiplicative eta products. For example, $L(E_{11a}, 1) = 0.2538419\ldots$ is the probability that an independent standard exponential exceeds $2\pi M$, where M is the maximum of independent variables $E/(2\pi n)$ and $E/(22\pi n)$, $n \geq 1$, two of each, read off from the eta product $\eta(\tau)^2\eta(11\tau)^2$ of E_{11a}.

1.5.5 Part V: Inference and open problems

Chapter 28 treats Haar measure on locally compact groups. On a compact Sato Tate group left and right Haar measure coincide, but on the Borel subgroup that acts on the upper half plane they differ: hyperbolic area is left Haar measure, while right Haar measure is the prior whose invariant credible sets are exact confidence sets in a transformation model. We connect this to Barndorff Nielsen's p^* formula, which is exact in such models, to the reference priors of Chang and Eaves for orbits, which bear on inference about Satake parameters as conjugacy classes, and to Duke's equidistribution of Heegner points for the invariant measure, which we check on 28,251 points. Chapter 29 asks which Sato Tate group a curve has, given its traces at the first n primes. We prove that Bayes factors between candidate groups are consistent when the traces are equidistributed for the true law, atoms included, with exponential rates given by differences of Kullback Leibler divergences. In genus two we compare three candidate groups and study a curve whose group lies outside the candidate set. Chapters 30 and 31 collect the open problems and the computational programme.

1.6 How to read this book

A reader comfortable with measure theoretic probability, characteristic functions and Markov chains, and with complex analysis at the level of Mellin transforms and the Gamma function, has the prerequisites for this book. Algebraic number theory and representation theory are used, but each chapter states the results it needs and gives references.

Conjectures are labelled as conjectures. Known theorems are attributed, and those with long proofs are stated with a sketch and a reference. Results that are new in this book, including Proposition 1.5, the Kemeny constant limit, the Bayesian consistency theorem, the Thorin criteria for the generalized Riemann hypothesis of automorphic L functions and for the Ramanujan conjecture, the Thorin structure of Hecke passage times, and the extreme value representation, are identified as ours and are given complete proofs. Numerical values are from computations carried out for this book, mainly in PARI/GP and Python, and are reported to the precision we verified.

Every chapter except the last two ends with exercises and a short list of open computational problems. These are posed and not solved. Each specifies an objective, the objects and the data scale, a suggested software route, the expected deliverable, and the reason it matters. Together they form a programme of computational experiments, collected and prioritised in Chapter 31. Their purpose is to measure the probabilistic shadow at scales where the predictions become sharp, and to look for rates, biases or constants that depart from what the theory predicts.

1.7 Notes

Langlands' letter to Weil [349] and his 1970 lectures [350] remain the best primary sources for the original vision. Gelbart's survey [211] is the standard introduction, Arthur [24] explains functoriality, and Frenkel [201] gives the geometric side. The naming of the modularity conjecture for elliptic curves has been disputed; Lang [348] collects the documents. Serre's book on ℓ adic representations [499] contains the derivation of Sato Tate from analytic properties of symmetric power L functions, and Tate's paper [538] records the conjecture from the side of algebraic cycles. Strong multiplicity one for GL_n is due to Jacquet and Shalika [279]. The refinement for GL_2 that we use is Ramakrishnan's [451], and an example of Serre built from dihedral forms shows that its threshold $1/8$ for the density of the exceptional set cannot be raised. Katz and Sarnak [298, 299] developed the random matrix side of the shadow, the subject of Chapter 6. The point of view that probability sees the consequences of functoriality rather than functoriality itself is implicit in much of this literature; Proposition 1.5 makes one version of it precise, and given Ramakrishnan's theorem its proof is elementary.

1.8 Exercises

1. Show that the even moments of μ_{ST} are the Catalan numbers $\frac{1}{m+1}\binom{2m}{m}$ and that its odd moments vanish.

2. Let θ have density $\frac{2}{\pi}\sin^2\theta$ on $[0,\pi]$. Show that $2\cos\theta$ has law μ_{ST}, and that $\mathbb{E}[U_n(\cos\theta)] = 0$ for every $n \geq 1$, where $U_n(\cos\theta) = \sin((n+1)\theta)/\sin\theta$.

3. Show that the set of primes $\{p_{k^2} : k \geq 1\}$ has density zero, where p_j is the jth prime. Show that density zero cannot be weakened to positive upper density in Proposition 1.5: for every set S of primes of positive upper density, construct an alteration on S of an μ_{ST} equidistributed sequence that is not μ_{ST} equidistributed.

4. Let (x_p) come from a newform f of weight $k \geq 2$ with trivial character and without complex multiplication, and let (x'_p) be obtained by replacing x_p by $-x_p$ for p in a set S of density δ. Show that if $\delta < 1/8$ and $x_p \neq 0$ for infinitely many $p \in S$, then (x'_p) is not automorphic. Now let S be the set of primes p with $\left(\frac{p}{5}\right) = -1$, of density $\frac{1}{2}$. Show that (x'_p) comes from the twist of f by the quadratic character modulo 5, and is therefore automorphic and μ_{ST} equidistributed. Why does this not contradict Proposition 1.5?

5. Let $\nu(dx) = dx/(\pi\sqrt{4-x^2})$ be the arcsine law on $[-2,2]$, the continuous part of μ_{CM} normalised to mass one. Compute the Kullback Leibler divergence $\int \log \frac{d\nu}{d\mu_{\mathrm{ST}}}\, d\nu$, using $\int_0^\pi \log\sin\theta\, d\theta = -\pi\log 2$. Interpret the answer

as the rate at which a likelihood ratio test tells a CM curve from a generic one using only the traces at the split primes.

6. Let χ be a nontrivial Dirichlet character modulo N. Using the prime number theorem for arithmetic progressions, show that, as p runs over the primes not dividing N, the values $\chi(p)$ are equidistributed with respect to the uniform measure on the image $\chi((\mathbb{Z}/N\mathbb{Z})^\times)$, a finite cyclic group of roots of unity, and explain why this is the case $n = 1$ of the probabilistic shadow.

1.9 Open computational problems

Computational Problem 1.1 (A Sato Tate atlas for elliptic curves). Objective: compute and publish the empirical Sato Tate law, the Weyl averages $\pi(x)^{-1}\sum_{p\le x} U_k(x_p/2)$ for $k \le 30$, and the normalised Kolmogorov discrepancy $\sqrt{n}\,D_n$ of the first n traces, for every isogeny class of elliptic curves over $\mathbb{Q}$ of conductor up to 10^5 in the LMFDB [371], using all good primes below 10^9. Route: Sutherland's `smalljac` library or PARI/GP `ellap`, distributed over a cluster; the traces depend only on the isogeny class, so store results by class. Deliverable: a public table, and a plot of the Weyl averages against k for each of the two Sato Tate groups that occur, $\mathrm{SU}(2)$ and $N(\mathrm{U}(1))$. Why it matters: it gives the empirical baseline against which every rate in Part II is calibrated, and it would detect any curve whose Weyl averages decay anomalously slowly.

Computational Problem 1.2 (Detecting impostors). Objective: to quantify Proposition 1.5 in data. Take the traces of E_1 for primes below 10^8 and construct impostor sequences by replacing them, on density zero sets of increasing size (the primes p_{k^2}, the primes $p_{\lfloor k^{3/2}\rfloor}$, and random sets of x^β primes up to x for $\beta < 1$), by independent draws from μ_{ST}. For each impostor compute (a) distributional statistics, such as the scaled discrepancy $\sqrt{n}\,D_n$ of the first n traces against μ_{ST} and the Weyl averages, whose limits cannot separate impostor from genuine sequence; and (b) coupling statistics, such as the defect of the functional equation of the Dirichlet series obtained by propagating the altered traces multiplicatively, evaluated on the critical line at heights up to about $2\pi x/\sqrt{496}$, so that coefficients up to x carry weight, and Hecke consistency checks at prime squares when the traces at p^2 are included. Route: PARI/GP `lfuncreate` with user supplied coefficients and `lfuncheckfeq` at low height, and a smoothed approximate functional equation in C or Python at large height. Deliverable: power curves for each test as a function of the alteration size and the number of primes used. Why it matters: it measures how quickly coupling statistics see what distributional statistics cannot, and it is a first experiment towards the probabilistic converse theorem of Open Problem 30.4.

Computational Problem 1.3 (Near coincidences of newforms). Objective: for all pairs of weight two newforms with rational coefficients and level up to 10^4,

compute the proportion of primes below 10^6 at which their Hecke eigenvalues agree, and find the pairs with the largest such proportion. Route: LMFDB downloads or Sage modular symbols, with locality sensitive hashing on random sets of primes (two forms that agree at a proportion α of primes collide on k sampled primes with probability about α^k) to avoid comparing all pairs. Deliverable: a table of the most nearly coincident pairs, each explained where possible (quadratic twists, for instance, agree at the primes where the twisting character equals one and at the supersingular primes), and a histogram of agreement proportions, compared with the threshold $7/8$ of Ramakrishnan. Why it matters: it shows how far the threshold of strong multiplicity one is from the agreement actually seen between distinct forms, and so how many primes a certificate of equality based on agreement of traces needs in practice.

Method	Langlands target	Status
Weyl criterion, moments	Sato Tate and general Sato Tate laws	proves, given functoriality
Tauberian theorems	Prime number theorems for $L(s, \mathrm{Sym}^n \pi)$	proves, given automorphy
Random walks on trees	p adic Plancherel measure, Hecke operators	proves
Trace formula as expectation	Vertical Sato Tate [508, 95]	proves
Random walks on isogeny graphs	Kemeny constants, spanning trees, mixing	proves (Part III and Chapter 21)
Random matrix theory	Low lying zeros [298, 271]	proves for restricted support (often under GRH), tests beyond
Interlacing, random graphs	Bipartite Ramanujan graphs [406]	proves
Lévy and Thorin measures	Positivity, Riemann hypothesis and Artin conjecture reformulations; squarefree integers (Chapter 17); the steps of the proof of Fermat's last theorem (Chapter 22); inversions between Frobenius traces, Pólya kernels and zeros (Chapter 23)	reformulates, tests; proves identities (Chapters 17 and 23)
Poisson Dirichlet law, Dirichlet processes	Chebotarev densities in the prime factors of random integers, Mertens constants per class (Chapter 24)	proves
Extreme values	L values of eta products	proves identities
Bayesian model choice	Sato Tate group identification	proves consistency, given equidistribution
Point counting, modular symbols	Data for all of the above	tests
Any of the above	Reciprocity, functoriality	no proof known

Table 1.1: Probabilistic and computational methods in the Langlands programme, and what each delivers.

CHAPTER 2

MODULAR FORMS AND L FUNCTIONS FOR PROBABILISTS

In 1916 Srinivasa Ramanujan published a paper on certain arithmetical functions [453] in which he studied the coefficients $\tau(n)$ of the power series $q\prod_{n\geq 1}(1-q^n)^{24}$. From a table of thirty values he guessed three properties: τ is multiplicative, it satisfies a second order recursion at prime powers, and $|\tau(p)| \leq 2p^{11/2}$. Mordell proved the first two a year later [403]; the third waited until Deligne's proof of the Weil conjectures in 1974 [140]. Hecke's theory of operators on modular forms explains the first two observations, and the third is the prototype of the Ramanujan conjecture for automorphic representations. This chapter develops the objects of that story for readers whose training is in probability: modular forms, Hecke operators, Euler products, Mellin transforms and functional equations. Along the way we isolate two facts that drive the later parts of the book: by Deligne's theorem the normalised Hecke eigenvalues are traces of elements of $\mathrm{SU}(2)$, and the completed L function of Δ is the Mellin transform of a *positive* function.

2.1 THE MODULAR GROUP AND MODULAR FORMS

The group $\mathrm{SL}_2(\mathbb{Z})$ of integer matrices of determinant one acts on the upper half plane $\mathbb{H} = \{\tau \in \mathbb{C} : \Im\tau > 0\}$ by Möbius transformations

$$\gamma\tau = \frac{a\tau + b}{c\tau + d}, \qquad \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}),$$

and $\Im(\gamma\tau) = \Im\tau/|c\tau+d|^2$. The group is generated by the matrices T and S that act by $\tau \mapsto \tau + 1$ and $\tau \mapsto -1/\tau$, and every orbit meets the closed fundamental domain $\mathcal{F} = \{\tau \in \mathbb{H} : |\Re\tau| \leq \frac{1}{2},\ |\tau| \geq 1\}$. The quotient $\mathrm{SL}_2(\mathbb{Z})\backslash\mathbb{H}$ has finite hyperbolic area $\pi/3$ for the measure $dx\,dy/y^2$, where $\tau = x + iy$, so $\frac{3}{\pi}\,dx\,dy/y^2$ is a probability measure on it; it is the limit law of Heegner points in Chapter 28.

DEFINITION 2.1. Let k be an even integer. A *modular form* of weight k for $\mathrm{SL}_2(\mathbb{Z})$ is a holomorphic function f on $\mathbb{H}$ with

$$f(\gamma\tau) = (c\tau + d)^k f(\tau) \qquad \text{for all } \gamma \in \mathrm{SL}_2(\mathbb{Z}),$$

which is holomorphic at infinity in the following sense. Invariance under T gives a Fourier expansion $f(\tau) = \sum_{n\in\mathbb{Z}} a(n)q^n$ in $q = e^{2\pi i\tau}$, and we require $a(n) = 0$ for $n < 0$. If also $a(0) = 0$, f is a *cusp form*. The spaces are M_k and S_k.

Invariance under S is the nontrivial condition: $f(-1/\tau) = \tau^k f(\tau)$. The spaces are finite dimensional: for even $k \geq 0$, $\dim M_k = \lfloor k/12 \rfloor + 1$ unless $k \equiv 2 \bmod 12$, in which case $\dim M_k = \lfloor k/12 \rfloor$, and $\dim S_k = \dim M_k - 1$ for $k \geq 4$ [503]. In particular $S_k = 0$ for $k < 12$ and for $k = 14$, and $\dim S_{12} = 1$.

2.1.1 Eisenstein series

For even $k \geq 4$ the series $\sum_{(c,d)\neq(0,0)} (c\tau + d)^{-k}$ converges absolutely and is visibly of weight k. Normalised to have constant term one it becomes

$$E_k(\tau) = 1 - \frac{2k}{B_k} \sum_{n\geq 1} \sigma_{k-1}(n) q^n, \qquad \sigma_{k-1}(n) = \sum_{d|n} d^{k-1},$$

with B_k the Bernoulli numbers. Thus $E_4 = 1 + 240 \sum \sigma_3(n) q^n$ and $E_6 = 1 - 504 \sum \sigma_5(n) q^n$. The coefficients $\sigma_{k-1}(n)$ are multiplicative, and at a prime $\sigma_{k-1}(p) = 1 + p^{k-1}$. This is the degenerate case of everything that follows. The form E_k is an eigenvector of the Hecke operator T_p of the next section with eigenvalue $\sigma_{k-1}(p)$; the roots of $X^2 - \sigma_{k-1}(p) X + p^{k-1}$ are 1 and p^{k-1}, and after division by $p^{(k-1)/2}$ they are $p^{\pm(k-1)/2}$, far from the unit circle. The normalised eigenvalues $p^{(k-1)/2} + p^{-(k-1)/2}$ are unbounded, so they are not traces of elements of a compact subgroup of $\mathrm{GL}_2(\mathbb{C})$: equidistribution in a compact group, the probabilistic shadow of Chapter 1, is a phenomenon of cusp forms.

2.1.2 Ramanujan's Delta

The normalised cusp form of weight 12 is

$$\Delta(\tau) = \frac{E_4(\tau)^3 - E_6(\tau)^2}{1728} = q \prod_{n\geq 1} (1 - q^n)^{24} = \sum_{n\geq 1} \tau(n) q^n = \eta(\tau)^{24},$$

where $\eta(\tau) = q^{1/24} \prod_n (1 - q^n)$ is the Dedekind eta function. The equality of the Eisenstein expression with the product is Jacobi's; it can be proved by checking that the product is a cusp form of weight 12, using the transformation law $\eta(-1/\tau) = \sqrt{\tau/i}\, \eta(\tau)$, and comparing the first coefficient.

EXAMPLE 2.2. Expanding $q(1-q)^{24}(1-q^2)^{24}(1-q^3)^{24} \cdots$ to order q^4: the coefficient of q^2 comes only from $(1-q)^{24}$ and is $\tau(2) = -24$. The coefficient of q^3 receives $\binom{24}{2} = 276$ from $(1-q)^{24}$ and -24 from $(1-q^2)^{24}$, so $\tau(3) = 252$. For q^4 we collect $-\binom{24}{3} = -2024$ from q^3 in $(1-q)^{24}$, then $(-24)\cdot(-24) = 576$ from $q \cdot q^2$, and -24 from q^3 in $(1-q^3)^{24}$, so $\tau(4) = -2024 + 576 - 24 = -1472$. Continuing, $\tau(5) = 4830$ and $\tau(6) = -6048 = \tau(2)\tau(3)$.

THEOREM 2.3 (Ramanujan's observations). *The coefficients $\tau(n)$ satisfy:*

1. $\tau(mn) = \tau(m)\tau(n)$ *when* $\gcd(m,n) = 1$ *(Mordell, 1917);*

2. $\tau(p^{r+1}) = \tau(p)\tau(p^r) - p^{11}\tau(p^{r-1})$ *for primes* p *and* $r \geq 1$ *(Mordell, 1917);*
3. $|\tau(p)| \leq 2p^{11/2}$ *for all primes* p *(Deligne [140]).*

The first two follow from the theory of Hecke operators in the next section, since Δ spans the one dimensional space S_{12}. The third follows from the Weil conjectures, which Deligne proved, applied to a Kuga Sato variety: a smooth compactification of the tenth fibre power of the universal elliptic curve (with auxiliary level structure), a variety of dimension 11 in whose cohomology $\tau(p)$ is the trace of Frobenius on a two dimensional piece of H^{11} [139]. Example 2.2 checks (2) at $p = 2$, $r = 1$: $\tau(2)^2 - 2^{11} = 576 - 2048 = -1472 = \tau(4)$.

Whether $\tau(n)$ is ever zero is a question Lehmer raised in 1947 [360], still open. Lehmer showed that the least n with $\tau(n) = 0$, if any, is prime; computations of Derickx, van Hoeij and Zeng [145] based on congruences for τ have shown that $\tau(n) \neq 0$ for all n below 8×10^{23}. Serre proved that the primes with $\tau(p) = 0$ have density zero [506]. Lehmer's question is the weight 12 analogue of the Lang Trotter question for the trace $t = 0$ (Chapter 5). If $\tau(p)/p^{11/2}$ behaved like an independent draw from the Sato Tate law of Chapter 1, rounded to the lattice $p^{-11/2}\mathbb{Z}$, the event $\tau(p) = 0$ would have probability of order $p^{-11/2}$, a summable sequence. The same heuristic gives probability of order $p^{-1/2}$ for $a_p = 0$ on an elliptic curve and so predicts infinitely many supersingular primes; for Δ it predicts only finitely many zeros of $\tau(p)$, and probably none.

2.2 Hecke operators

For a prime p the Hecke operator T_p on M_k sums f, viewed as a function of lattices, over the $p + 1$ sublattices of index p. In coordinates,

$$(T_p f)(\tau) = p^{k-1} f(p\tau) + \frac{1}{p} \sum_{j=0}^{p-1} f\Big(\frac{\tau + j}{p}\Big),$$

and on q expansions

$$a(T_p f, n) = a(np) + p^{k-1} a(n/p),$$

where $a(n/p) = 0$ if $p \nmid n$. The operators T_n for all $n \geq 1$ are defined similarly by summing over sublattices of index n, and satisfy:

Proposition 2.4 (Hecke relations). *The operators* T_n *commute,* $T_{mn} = T_m T_n$ *for coprime* m, n*, and for each prime* p *and* $r \geq 1$*,*

$$T_p T_{p^r} = T_{p^{r+1}} + p^{k-1} T_{p^{r-1}}.$$

Proof sketch. Identify f with the function $F(\mathbb{Z}\omega_1 + \mathbb{Z}\omega_2) = \omega_2^{-k} f(\omega_1/\omega_2)$ of lattices, where $\Im(\omega_1/\omega_2) > 0$; it satisfies $F(\lambda L) = \lambda^{-k} F(L)$, and T_n becomes

$T_n F(L) = n^{k-1} \sum_{[L:L']=n} F(L')$. Coprime indices factor uniquely through intermediate lattices, which gives multiplicativity. For the prime power relation, count pairs $L'' \subset L' \subset L$ with $[L : L'] = p$ and $[L' : L''] = p^r$: each sublattice L'' of index p^{r+1} arises once unless $L'' \subset pL$, in which case it arises $p + 1$ times. The excess p copies of each sublattice of pL of index p^{r-1} in pL give the term $p^{k-1}T_{p^{r-1}}$ after the weight normalisation, since $F(pL) = p^{-k}F(L)$. Commutativity follows, since each T_n is then a polynomial in the T_p with $p \mid n$, and $T_p T_q = T_{pq} = T_q T_p$ for primes $p \neq q$. Details are in [503]. □

The relation at a prime is a three term recursion, the same one satisfied by Chebyshev polynomials. If f is a simultaneous eigenvector of all T_n with $a(1) = 1$, an *eigenform*, then $a(n)$ is the eigenvalue of T_n (compare the coefficients of q on both sides of $T_n f = \lambda_n f$, using $a(T_n f, 1) = a(n)$), and the Hecke relations become

$$a(mn) = a(m)a(n) \quad (\gcd(m, n) = 1),$$

$$a(p^{r+1}) = a(p)a(p^r) - p^{k-1}a(p^{r-1}).$$

Since S_{12} is one dimensional, Δ is an eigenform and Theorem 2.3(1),(2) follow.

2.2.1 Euler products and Satake parameters

Let f be an eigenform, put $x_p = a(p)/p^{(k-1)/2}$, and let α_p, β_p be the roots of $X^2 - x_p X + 1$, so that $a(p) = p^{(k-1)/2}(\alpha_p + \beta_p)$ and $\alpha_p \beta_p = 1$. The recursion becomes

$$\frac{a(p^r)}{p^{r(k-1)/2}} = \frac{\alpha_p^{r+1} - \beta_p^{r+1}}{\alpha_p - \beta_p} = U_r(x_p/2),$$

by induction, since the Chebyshev polynomials U_r of the second kind satisfy $U_{r+1}(x/2) = xU_r(x/2) - U_{r-1}(x/2)$; the middle term is read as $(r+1)\alpha_p^r$ when $\alpha_p = \beta_p$. Multiplicativity and the generating function $\sum_{r \geq 0} U_r(x/2)t^r = (1 - xt + t^2)^{-1}$ give, where the series converges absolutely (for $\Re s > k/2 + 1$ by Hecke's bound in Section 2.3),

$$\begin{aligned} L(s, f) = \sum_{n \geq 1} \frac{a(n)}{n^s} &= \prod_p \left(1 - a(p)p^{-s} + p^{k-1-2s}\right)^{-1} \\ &= \prod_p \left(1 - \alpha_p p^{\frac{k-1}{2} - s}\right)^{-1} \left(1 - \beta_p p^{\frac{k-1}{2} - s}\right)^{-1}, \end{aligned}$$

an Euler product of degree two. The pair $\{\alpha_p, \beta_p\}$ is the Satake parameter, the conjugacy class of $\mathrm{diag}(\alpha_p, \beta_p)$ in $\mathrm{SL}_2(\mathbb{C})$. Since $a(p)$ is real (T_p is self adjoint for the Petersson inner product of Section 2.4), Deligne's bound $|a(p)| \leq 2p^{(k-1)/2}$ is

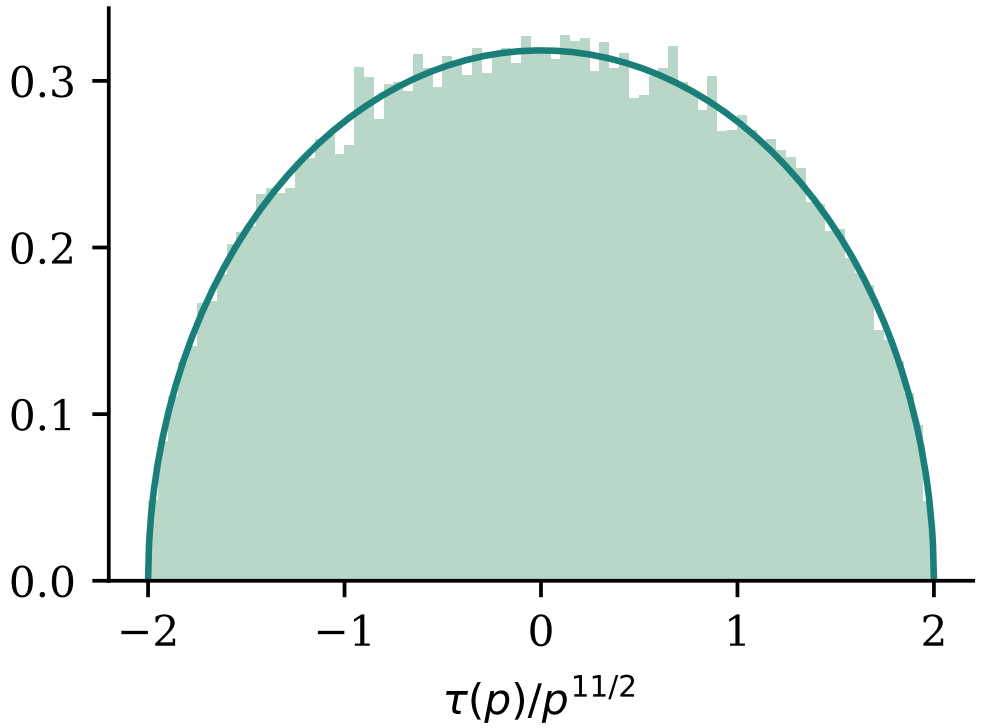


Figure 2.1: Histogram of the normalised Ramanujan coefficients $x_p = \tau(p)/p^{11/2}$ for the 78,498 primes $p < 10^6$, against the Sato Tate density $\frac{1}{2\pi}\sqrt{4-x^2}$ on $[-2,2]$.

equivalent to $|\alpha_p| = |\beta_p| = 1$, that is, to $\alpha_p = e^{i\theta_p}$, $\beta_p = e^{-i\theta_p}$ with $\theta_p \in [0,\pi]$, and then

$$x_p = \frac{a(p)}{p^{(k-1)/2}} = 2\cos\theta_p \in [-2,2].$$

In probabilistic language: the Satake parameter is a conjugacy class in $\mathrm{SU}(2)$, the normalised eigenvalue x_p is its trace, and the normalised coefficient $a(p^r)/p^{r(k-1)/2} = U_r(x_p/2)$ is the character of Sym^r evaluated on it. The Sato Tate law, a theorem for Δ (Theorem 1.4), says that the θ_p are equidistributed with respect to $\frac{2}{\pi}\sin^2\theta\, d\theta$ on $[0,\pi]$, the law of the angle of a Haar random element of $\mathrm{SU}(2)$; equivalently, the x_p are equidistributed with respect to μ_{ST}.

Figure 2.1 shows this for Δ and the primes below 10^6; Chapter 5 measures the fit by moments (Table 5.1), extreme values and discrepancy.

2.3 Mellin transforms and the functional equation

The Mellin transform passes from modular forms to L functions through the Gamma integral

$$\int_0^\infty e^{-2\pi n y} y^{s-1}\,dy = (2\pi n)^{-s}\Gamma(s) \qquad (\Re s > 0,\ n \geq 1).$$

Let $f = \sum a(n)q^n$ be a cusp form of weight k. The function $y^{k/2}|f(x+iy)|$ is invariant under $\mathrm{SL}_2(\mathbb{Z})$ and bounded, since f decays at the cusp, and $a(n) = e^{2\pi n y}\int_0^1 f(x+iy)e^{-2\pi i n x}\,dx$ for every $y > 0$; the choice $y = 1/n$ gives Hecke's bound $a(n) = O(n^{k/2})$. Hence $L(s,f)$ converges absolutely for $\Re s > k/2 + 1$, and on that half plane termwise integration gives

$$\Lambda(s,f) := (2\pi)^{-s}\Gamma(s)L(s,f) = \int_0^\infty f(iy)\, y^{s-1}\,dy.$$

The modular transformation under S now gives the analytic continuation.

THEOREM 2.5 (Hecke [250]). *The function $\Lambda(s,\Delta) = (2\pi)^{-s}\Gamma(s)L(s,\Delta)$ extends to an entire function of s, bounded in vertical strips, and satisfies*

$$\Lambda(s,\Delta) = \Lambda(12-s,\Delta).$$

PROOF. Take $\tau = iy$ in $\Delta(-1/\tau) = \tau^{12}\Delta(\tau)$. Since $-1/(iy) = i/y$ and $(iy)^{12} = y^{12}$,

$$\Delta(i/y) = y^{12}\Delta(iy) \qquad (y > 0).$$

As $y \to \infty$ the q expansion gives $\Delta(iy) = e^{-2\pi y}\bigl(1 + O(e^{-2\pi y})\bigr)$, and then the identity gives $\Delta(iy) = y^{-12}e^{-2\pi/y}\bigl(1+o(1)\bigr)$ as $y \to 0$. Thus $\Delta(iy)$ decays faster than any power of y at both ends. For $\Re s > 7$ split the integral at $y = 1$ and substitute $y = 1/u$ in the piece over $(0,1)$:

$$\int_0^1 \Delta(iy)y^{s-1}\,dy = \int_1^\infty \Delta(i/u)u^{-s-1}\,du = \int_1^\infty \Delta(iu)u^{11-s}\,du.$$

Hence

$$\Lambda(s,\Delta) = \int_1^\infty \Delta(iy)\bigl(y^{s-1} + y^{11-s}\bigr)\,dy.$$

The right side converges absolutely and locally uniformly for every $s \in \mathbb{C}$, by the exponential decay of $\Delta(iy)$ on $[1,\infty)$, so it defines an entire function, bounded on vertical strips since $|y^{s-1}| = y^{\Re s - 1}$. It is visibly invariant under $s \mapsto 12 - s$. □

The same argument for a cusp form f of weight k and level one gives $\Lambda(s,f) = i^k\Lambda(k-s,f)$ (Exercise 6). For a newform f of level N (Section 2.4) the completed function is

$$\Lambda(s,f) = N^{s/2}(2\pi)^{-s}\Gamma(s)L(s,f) = \int_0^\infty f\bigl(iy/\sqrt{N}\bigr)\, y^{s-1}\,dy,$$

and the Fricke involution $\tau \mapsto -1/(N\tau)$ takes the place of S. A newform satisfies $f(-1/(N\tau)) = \pm N^{k/2}\tau^k f(\tau)$ for one of the two signs, and the same argument gives $\Lambda(s,f) = \varepsilon_f \Lambda(k-s,f)$ with root number $\varepsilon_f = \pm i^k$. In the same paper Hecke proved a converse: if $\sum a(n)n^{-s}$ has polynomially bounded coefficients and its completion $\Lambda(s) = (2\pi)^{-s}\Gamma(s)\sum a(n)n^{-s}$ is entire, bounded in vertical strips and satisfies $\Lambda(s) = i^k\Lambda(k-s)$, then $\sum a(n)q^n$ is a cusp form of weight k and level one. Weil [571] extended the converse theorem to level N, at the price of assuming functional equations for the twists $\sum a(n)\chi(n)n^{-s}$ by Dirichlet characters χ. Chapter 20 restates Hecke's converse at level one as a symmetry of a Pólya kernel built from the coefficients (Proposition 20.27).

Since $\Gamma(s)$ has no zeros and $\Lambda(s,\Delta)$ is entire, $L(s,\Delta)$ vanishes at the poles $s = 0, -1, -2, \dots$ of Γ; these are its trivial zeros, and every other zero of $L(s,\Delta)$ is a zero of $\Lambda(s,\Delta)$. The Euler product converges absolutely for $\Re s > 13/2$ by Deligne's bound, so $\Lambda(s,\Delta) \neq 0$ there, and by the functional equation also for $\Re s < 11/2$. Since $L(s,\Delta)$ does not vanish on the line $\Re s = 13/2$ either (Jacquet and Shalika [278]; Chapter 11), the zeros of $\Lambda(s,\Delta)$ lie in the open critical strip $11/2 < \Re s < 13/2$, symmetric about the line $\Re s = 6$. The Riemann hypothesis for Δ is the conjecture that they all lie on $\Re s = 6$. It is open.

2.3.1 Positivity: a probabilistic observation

The later parts of the book build on one fact. From the product,

$$\Delta(iy) = e^{-2\pi y}\prod_{n\geq 1}\bigl(1-e^{-2\pi n y}\bigr)^{24} > 0 \qquad (y>0).$$

So $\Lambda(s,\Delta)$ is the Mellin transform of a positive function. Put $y = e^u$. Then

$$\Lambda(6+z,\Delta) = \int_{-\infty}^{\infty} \Delta(ie^u)e^{6u}\, e^{zu}\, du,$$

and the symmetry $\Delta(i/y) = y^{12}\Delta(iy)$ says exactly that $u \mapsto \Delta(ie^u)e^{6u}$ is an even function of u. Normalising,

$$\Phi_\Delta(u) = \frac{\Delta(ie^u)e^{6u}}{\Lambda(6,\Delta)}$$

is the density of a symmetric random variable W, and

$$\frac{\Lambda(6+it,\Delta)}{\Lambda(6,\Delta)} = \mathbb{E}\bigl[e^{itW}\bigr]$$

is its characteristic function, real and even. The Riemann hypothesis for Δ becomes the statement that this characteristic function has only real zeros. This is the same structure that Pólya [444] and de Bruijn [133] exploited for the Riemann

zeta function, where the role of Δ is played by a Jacobi theta function. Our computations give $L(6,\Delta) = 0.792123$, and the first zeros on the critical line are at $6 \pm i\gamma$ with $\gamma = 9.22238,\ 13.90755,\ 17.44278, \ldots$. Chapter 13 develops the Lévy and Thorin measure theory of W, and Chapter 27 reads the product $\prod_n (1 - e^{-2\pi n y})$ as the distribution function of a maximum of independent exponential variables.

2.4 Level, newforms and elliptic curves

For $N \geq 1$ let

$$\Gamma_0(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}) : c \equiv 0 \bmod N \right\}.$$

Modular and cusp forms of weight k and level N, $M_k(\Gamma_0(N))$ and $S_k(\Gamma_0(N))$, are defined as before, with the transformation law for $\gamma \in \Gamma_0(N)$ and holomorphy (or vanishing) at every cusp. The Hecke operators T_p for $p \nmid N$ act and commute, and are self adjoint for the Petersson inner product

$$\langle f, g \rangle = \int_{\Gamma_0(N)\backslash \mathbb{H}} f(\tau)\overline{g(\tau)}\, y^k \, \frac{dx\, dy}{y^2},$$

so $S_k(\Gamma_0(N))$ has a basis of simultaneous eigenvectors for them, with real eigenvalues. Forms of lower level $M \mid N$ embed in level N via $f(\tau) \mapsto f(d\tau)$ for $d \mid N/M$; the orthogonal complement of these *oldforms* is spanned by *newforms*. Atkin and Lehner [29] proved that a newform is determined by its eigenvalues at almost all primes (multiplicity one), that it is an eigenvector of all T_n and of the Fricke involution, and that its L function has an Euler product with degree two factors at $p \nmid N$ and degree at most one at $p \mid N$.

2.4.1 Elliptic curves and modularity

Let E be an elliptic curve over $\mathbb{Q}$, given by a minimal Weierstrass equation, with conductor N_E. For a prime p of good reduction set

$$a_p(E) = p + 1 - \#E(\mathbb{F}_p).$$

At a prime of bad reduction $a_p(E)$ is 1, -1 or 0 according as the reduction is split multiplicative, nonsplit multiplicative or additive. For a curve $y^2 = h(x)$ with h a cubic and p an odd prime of good reduction, $\#E(\mathbb{F}_p) = p + 1 + \sum_{x \bmod p} \left(\frac{h(x)}{p}\right)$, counting the point at infinity, so $a_p = -\sum_x \left(\frac{h(x)}{p}\right)$: a sum of p Legendre symbols, each equal to ± 1 except at the at most three roots of h modulo p.

Theorem 2.6 (Hasse). *$|a_p(E)| \leq 2\sqrt{p}$ for every prime of good reduction.*

It is the Riemann hypothesis for the zeta function of the curve $E/\mathbb{F}_p$, whose numerator is $1 - a_pT + pT^2$ with roots of absolute value $p^{-1/2}$. Compare a sum of p independent fair signs: it is typically of order $\sqrt{p}$, but its absolute value exceeds $2\sqrt{p}$ with probability tending to $\mathbb{P}(|Z| > 2) \approx 0.0455$, Z standard normal (Exercise 4). Hasse's bound holds for every p, and the constant 2 is sharp. The L function of E is

$$L(s,E) = \prod_{p\nmid N_E} \left(1 - a_p p^{-s} + p^{1-2s}\right)^{-1} \prod_{p\mid N_E} \left(1 - a_p p^{-s}\right)^{-1},$$

in the classical normalisation, with centre of symmetry $s = 1$.

THEOREM 2.7 (Modularity: Wiles, Taylor and Wiles, Breuil, Conrad, Diamond and Taylor [581, 541, 41]). *For every elliptic curve $E/\mathbb{Q}$ there is a newform $f \in S_2(\Gamma_0(N_E))$ with rational coefficients such that $L(s,E) = L(s,f)$.*

It is a reciprocity law in the sense of Chapter 1, and Chapter 3 returns to it: the two dimensional ℓ adic Galois representation on the Tate module of E corresponds to an automorphic representation of GL_2. As a consequence $\Lambda(s,E) = N_E^{s/2}(2\pi)^{-s}\Gamma(s)L(s,E)$ continues to an entire function with $\Lambda(s,E) = w(E)\Lambda(2-s,E)$, where the root number $w(E) = \pm 1$ equals ε_f. For the curve $11a$ of conductor 11, $y^2 + y = x^3 - x^2 - 10x - 20$, the newform is the eta product $\eta(\tau)^2\eta(11\tau)^2 = q\prod_{n\geq 1}(1-q^n)^2(1-q^{11n})^2$, with root number $+1$; Chapter 27 studies such products.

EXAMPLE 2.8. Take the curve $E_1 : y^2 = x^3 + x + 1$ of Chapter 1, with discriminant $-16(4+27) = -496$ and conductor $496 = 2^4 \cdot 31$. At $p = 5$ the values of $x^3 + x + 1$ for $x = 0, 1, 2, 3, 4$ are $1, 3, 1, 1, 4$ modulo 5; the squares modulo 5 are 1 and 4, so four values are nonzero squares and one is a nonsquare. Hence $\#E_1(\mathbb{F}_5) = 1 + 2 \cdot 4 = 9$ and $a_5 = 5 + 1 - 9 = -3$. The same count for the odd primes below 20 gives

p	3	5	7	11	13	17	19
a_p	0	-3	3	-2	-4	0	-1

each within the Hasse bound (for instance $|a_{13}| = 4 \leq 2\sqrt{13} \approx 7.2$). The primes 3 and 17 are supersingular. By modularity these are the coefficients at primes of a newform of weight two and level 496. The curve has no complex multiplication, and the $x_p = a_p/\sqrt{p}$ follow the semicircle of Figure 1.1.

2.4.2 Dirichlet characters: degree one

A Dirichlet character χ modulo N is a homomorphism $(\mathbb{Z}/N\mathbb{Z})^\times \to \mathbb{C}^\times$, extended by zero. Its L function $L(s,\chi) = \prod_p (1 - \chi(p)p^{-s})^{-1}$ has degree one, and when χ is primitive its completion is

$$\Lambda(s,\chi) = \Big(\frac{N}{\pi}\Big)^{(s+\delta)/2}\Gamma\Big(\frac{s+\delta}{2}\Big)L(s,\chi), \qquad \chi(-1) = (-1)^\delta,$$

which satisfies $\Lambda(s,\chi) = \varepsilon(\chi)\Lambda(1-s,\bar{\chi})$, where the root number is $\varepsilon(\chi) = g(\chi)/(i^\delta\sqrt{N})$. Here $g(\chi)$ is the Gauss sum, the sum of $\chi(a)e^{2\pi ia/N}$ over a mod N, and $|g(\chi)| = \sqrt{N}$, so $|\varepsilon(\chi)| = 1$. The Satake parameter at $p \nmid N$ is the number $\chi(p)$ on the unit circle, an element of the compact group $\mathrm{U}(1)$. Dirichlet's theorem on primes in progressions, in its density form, says that the $\chi(p)$ are equidistributed with respect to Haar measure on the finite group of values of χ: the simplest instance of the probabilistic shadow.

2.5 Automorphic representations in outline

The objects above are the cases $n = 1$ and $n = 2$ of a general theory, whose shape Chapter 1 described. A cuspidal automorphic representation π of $\mathrm{GL}_n(\mathbb{A}_\mathbb{Q})$ with unitary central character has, at each prime p not dividing its conductor N_π, a Satake parameter $A_p(\pi)$, a semisimple conjugacy class in $\mathrm{GL}_n(\mathbb{C})$ with eigenvalues $\alpha_{1,p},\dots,\alpha_{n,p}$, and a standard L function

$$L(s,\pi) = \prod_{p\nmid N_\pi}\prod_{j=1}^{n}\bigl(1-\alpha_{j,p}p^{-s}\bigr)^{-1}\prod_{p\mid N_\pi}L_p(s,\pi),$$

where each factor $L_p(s,\pi)$ at a ramified prime is the reciprocal of a polynomial in p^{-s} of degree at most n. Godement and Jacquet [217] proved that the completed function $\Lambda(s,\pi) = N_\pi^{s/2}L_\infty(s,\pi)L(s,\pi)$, where $L_\infty(s,\pi)$ is a product of n Gamma factors, is entire (unless $n = 1$ and π is a character $|\cdot|^{it}$) and satisfies $\Lambda(s,\pi) = \varepsilon(\pi)\Lambda(1-s,\tilde{\pi})$, where $\tilde{\pi}$ is the contragredient and the root number $\varepsilon(\pi)$ has absolute value one. A holomorphic newform f of weight k gives such a π_f for $n = 2$, with $A_p(\pi_f) = \{\alpha_p,\beta_p\}$ and $L(s,\pi_f) = L(s+\frac{k-1}{2},f)$. This *unitary*, or *analytic*, normalisation moves the centre of symmetry from $s = k/2$ to $s = \frac{1}{2}$; the completed function is $\Lambda(s+\frac{k-1}{2},f)$ up to a constant factor, and $\Lambda(s,\pi_f) = \varepsilon_f\Lambda(1-s,\pi_f)$. For Δ it gives $L(s,\pi_\Delta) = \sum_n \tau(n)n^{-11/2-s}$, and the zeros $6 \pm i\gamma$ of $\Lambda(s,\Delta)$ become $\frac{1}{2} \pm i\gamma$. Later chapters say which normalisation they use, the classical one with centre $k/2$ or the unitary one with centre $\frac{1}{2}$.

For cuspidal π on GL_n and π' on GL_m, both with unitary central character, the Rankin Selberg L function has Satake parameters $A_p(\pi)\otimes A_p(\pi')$:

$$L(s,\pi\times\pi') = \prod_{p\nmid N_\pi N_{\pi'}}\prod_{i,j}\bigl(1-\alpha_{i,p}\alpha'_{j,p}p^{-s}\bigr)^{-1},$$

up to finitely many factors at the ramified primes. Jacquet, Piatetski Shapiro and Shalika [286], Shahidi [510] and Moeglin and Waldspurger [397] established its continuation and functional equation; it has a pole at $s = 1$ if and only if $\pi' \cong \tilde{\pi}$ (Jacquet and Shalika [279]), and it does not vanish on the line $\Re s = 1$ (Shahidi [510]). For $\pi' = \tilde{\pi}$ the coefficient of p^{-s} at an unramified prime is $|\operatorname{tr} A_p(\pi)|^2$,

because the Satake parameters of $\tilde{\pi}$ are the complex conjugates of those of π, and $\log L(s, \pi \times \tilde{\pi})$ has nonnegative coefficients (Chapter 11). The pole, the nonvanishing and the Wiener Ikehara theorem [579, 580] give $\sum |\operatorname{tr}(A_p(\pi)^m)|^2 \log p \sim x$, the sum over unramified primes p and integers $m \geq 1$ with $p^m \leq x$. When the Satake parameters are bounded, as for holomorphic newforms by Deligne's theorem, the terms with $m \geq 2$ contribute $O(\sqrt{x})$, and

$$\sum_{p \leq x} |\operatorname{tr} A_p(\pi)|^2 \log p \sim x :$$

the second moment of the traces is one, an orthogonality relation that in the compact group picture says that the standard representation is irreducible.

CONJECTURE 2.9 (Generalised Ramanujan conjecture). *If π is a cuspidal automorphic representation of $\mathrm{GL}_n(\mathbb{A}_{\mathbb{Q}})$ with unitary central character, then $|\alpha_{j,p}| = 1$ for every unramified p and every j; equivalently $A_p(\pi)$ is conjugate into the unitary group $\mathrm{U}(n)$.*

For π attached to holomorphic newforms this is Deligne's theorem [140] in weight $k \geq 2$ and a theorem of Deligne and Serre [142] in weight one. For Maass forms on GL_2 it is open; the best bound $|\alpha_{j,p}| \leq p^{7/64}$ is due to Kim and Sarnak [315]. The conjecture is what makes the probabilistic picture possible: it places the Satake parameters in a compact group, where Haar measure exists and the language of equidistribution applies. Functoriality then predicts which compact subgroup, the Sato Tate group, and that the parameters are Haar distributed in it.

2.6 NOTES

Serre's *A Course in Arithmetic* [503] gives a short complete account of modular forms of level one, Hecke operators and the formula for Δ; Diamond and Shurman [155] cover levels, newforms and modularity. Mordell's 1917 proof of multiplicativity [403] anticipated Hecke's general theory [251] by twenty years. Building on the work of Eichler, Shimura, Kuga and Ihara, who related Hecke eigenvalues to the cohomology of modular curves and of fibre varieties over them, Deligne reduced the Ramanujan conjecture to the Weil conjectures in 1969 [139] and deduced it from his proof of them [140]. Atkin and Lehner developed newforms [29]. Hasse's theorem dates from 1933 to 1936. The automorphic theory is surveyed by Gelbart [211]; Godement and Jacquet [217] and Jacquet and Shalika [279] are the basic analytic papers. The positivity of $\Delta(iy)$ and its consequences for W are developed in Part IV; the analogous construction for the Riemann ξ function goes back to Pólya [444], and its probabilistic interpretation is surveyed by Biane, Pitman and Yor [82].

2.7 Exercises

1. Use the Hecke recursion and Example 2.2 to compute $\tau(8)$ and $\tau(9)$, and check them against the expansion of $q\prod(1-q^n)^{24}$. Verify $\tau(10)=\tau(2)\tau(5)$.

2. Prove by induction that $a(p^r)/p^{r(k-1)/2} = U_r(x_p/2)$ for an eigenform, and deduce the identity of power series in p^{-s}

$$\left(1-a(p)p^{-s}+p^{k-1-2s}\right)^{-1} = \sum_{r\ge 0} U_r(x_p/2)\, p^{-r(s-(k-1)/2)}.$$

   Interpret $\sum_r U_r(\cos\theta)t^r = (1-2t\cos\theta+t^2)^{-1}$ as the generating function of the characters of $\mathrm{SU}(2)$.

3. Let W have density Φ_Δ. Show that $\mathbb{E}[W^2] = -\frac{d^2}{dt^2}\log\Lambda(6+it,\Delta)\big|_{t=0}$ (numerically 0.112352) and that $\mathbb{E}[e^{zW}]$ is finite for every $z\in\mathbb{C}$. Show that $\Phi_\Delta(u)\sim e^{-2\pi e^{|u|}+6|u|}/\Lambda(6,\Delta)$ as $|u|\to\infty$, and that

$$\mathbb{P}(|W|>u) \sim \frac{e^{-2\pi e^{u}+5u}}{\pi\Lambda(6,\Delta)} \qquad (u\to\infty),$$

   so the tails of W decay doubly exponentially, far faster than any Gaussian tail.

4. Let ε_x, for x mod p, be independent uniform signs. Show that, as $p\to\infty$, $\mathbb{P}\big(|\sum_x \varepsilon_x| > 2\sqrt{p}\big)$ tends to $\mathbb{P}(|Z|>2)\approx 0.0455$, where Z is standard normal. Contrast this with the Hasse bound, by which the Legendre symbols $\left(\frac{h(x)}{p}\right)$ of a curve $y^2=h(x)$ with good reduction at p never produce such a sum.

5. For $E_2: y^2=x^3-x$ show directly that $a_p=0$ for every prime $p\equiv 3 \bmod 4$, using the substitution $x\mapsto -x$. Deduce from Dirichlet's theorem that the empirical law of the $x_p=a_p/\sqrt{p}$, $3\le p\le x$, gives mass at least $\frac12+o(1)$ to the point 0 as $x\to\infty$.

6. Prove that $\Lambda(s,f)=i^k\Lambda(k-s,f)$ for any cusp form f of level one and weight k, and conclude that $L(k/2,f)=0$ when $k\equiv 2 \bmod 4$. Using the dimension formula of Section 2.1, find the smallest weight for which this forces a nonzero cusp form to vanish at the centre, and identify the form.

2.8 Open computational problems

Computational Problem 2.1 (Ramanujan tau at scale). Objective: compute $\tau(p)$ for all primes $p<10^{11}$ and record (a) the empirical law of $x_p=\tau(p)/p^{11/2}$ and its discrepancy from μ_{ST}, (b) the extreme values of $|x_p|$ near ± 2, (c) the residues

of $\tau(p)$ modulo the classical congruence primes $2, 3, 5, 7, 23, 691$ (for 23, Wilton's congruence of Chapter 22), and (d) the smallest values of $|\tau(p)|/p^{11/2}$ as a probe of Lehmer's question. Route: compute all coefficients of $q\prod(1-q^n)^{24}$ up to 10^{11} by fast power series arithmetic (Jacobi's identity $\prod(1-q^n)^3 = \sum_{m\geq 0}(-1)^m(2m+1)q^{m(m+1)/2}$ reduces $\prod(1-q^n)^{24}$ to three FFT squarings), working modulo four primes of about 2^{63} (seven if 32 bit words are used), which suffice since $|\tau(n)| \leq d(n)n^{11/2} < 2^{213}$ for $n \leq 10^{11}$, and reconstructing by the Chinese remainder theorem, blockwise on GPUs; spot check individual values with the Eichler Selberg trace formula. Deliverable: a public table, the distribution of $\min_{p\leq x}|x_p|$ against the prediction from μ_{ST} for independent draws, and the discrepancy curve. Why it matters: Lehmer's question is the weight 12 analogue of the Lang Trotter problem, and the scaling of $\min|x_p|$ is a direct test of the independence heuristic behind the Lang Trotter conjecture (Chapter 5).

COMPUTATIONAL PROBLEM 2.2 (Hecke eigenvalue tables). Objective: compute the Hecke eigenvalues a_p for $p < 10^5$ of all newforms in $S_k(\Gamma_0(N))$ with trivial character, $N \leq 10^4$ and even $k \leq 12$, together with their coefficient fields and whether they have complex multiplication. Route: modular symbols in Magma or Sage, or the trace formula implementation in PARI/GP (`mfinit`, `mfeigenbasis`), parallelised by level; cross check against LMFDB [371] where it overlaps. Deliverable: a database of eigenvalue vectors and, for each newform and each embedding of its (totally real) coefficient field, the first ten Weyl averages $\pi(x)^{-1}\sum_{p\leq x}U_n(x_p/2)$ and the normalised discrepancy from μ_{ST}, or from μ_{CM} for forms with complex multiplication. Why it matters: these are the data for the vertical and horizontal equidistribution theorems of Part II and for the Bayesian identification of Part V, and a uniform table across weights lets one test whether the rate of convergence depends on the weight or the level.

COMPUTATIONAL PROBLEM 2.3 (Functional equations as a test statistic). Objective: turn the functional equation into a statistic that certifies a candidate set of coefficients. For a newform of weight k, level N and root number ε, the theta series $\theta(t) = \sum_n a(n)e^{-2\pi nt/\sqrt{N}} = f(it/\sqrt{N})$ satisfies $\theta(1/t) = \varepsilon t^k\theta(t)$, which is the functional equation of Section 2.3 after Mellin inversion. Given $a(n)$ for $n \leq X$, compute the defect $\delta_X = \max_t|\theta_X(1/t) - \varepsilon t^k\theta_X(t)|$ of the truncated series θ_X over 50 points $t \in [1,2]$, for genuine newforms of level up to 10^3 taken from LMFDB [371] and for impostor sequences obtained by altering the coefficients on density zero sets of primes (and propagating the alteration multiplicatively). Route: PARI/GP `lfuncreate` with user supplied coefficients and `lfuncheckfeq`, which compares the theta function at t and $1/t$ and returns the base two logarithm of the resulting defect. Deliverable: the distribution of the defect for genuine and altered sequences as a function of X and of the size of the alteration, and the smallest X that separates them. Why it matters: Proposition 1.5 shows that

distributional statistics cannot detect automorphy; the functional equation couples all primes, and this experiment, which extends the impostor test of Chapter 1 from one curve to all newforms of small level, measures how much coupling a finite sample reveals.

CHAPTER 3

GALOIS REPRESENTATIONS, RECIPROCITY AND FUNCTORIALITY

In 1880 Frobenius observed that the way a prime p factors in a Galois extension is governed by a single conjugacy class in the Galois group, and he conjectured that every class occurs with a frequency proportional to its size. Chebotarev proved the conjecture in 1922; his proof appeared in Russian in 1923 and in German in 1926. Read with a probabilist's eye, his theorem says that the Frobenius class of a random prime is distributed as the conjugacy class of a uniformly random element of a finite group. Much of this book elaborates that sentence, replacing the finite group by a compact Lie group. This chapter assembles the arithmetic objects involved: Galois representations, reciprocity, the dual group and the Satake isomorphism, functoriality, and the Sato Tate group. We prove nothing new about reciprocity or functoriality; our purpose is to state the conjectures precisely, to say which cases are proved, and to isolate the probabilistic statements they imply.

3.1 FROBENIUS ELEMENTS AND THE CHEBOTAREV DENSITY THEOREM

The absolute Galois group $G_{\mathbb{Q}} = \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ is the inverse limit of $\mathrm{Gal}(L/\mathbb{Q})$ over finite Galois extensions L. Let $L/\mathbb{Q}$ be finite Galois with group G and absolute discriminant d_L, and let p be unramified in L, that is, $p \nmid d_L$. Choose a prime $\mathfrak{P}$ of L above p, with residue field $\mathbb{F}_{\mathfrak{P}}$. Since the inertia group of $\mathfrak{P}$ is trivial, the decomposition group $\{\sigma \in G : \sigma\mathfrak{P} = \mathfrak{P}\}$ maps isomorphically onto $\mathrm{Gal}(\mathbb{F}_{\mathfrak{P}}/\mathbb{F}_p)$, which is generated by $t \mapsto t^p$, and the preimage of this generator is the Frobenius element $\mathrm{Frob}_{\mathfrak{P}} \in G$, characterised by

$$\mathrm{Frob}_{\mathfrak{P}}(\alpha) \equiv \alpha^p \pmod{\mathfrak{P}} \qquad \text{for all algebraic integers } \alpha \in L.$$

Changing $\mathfrak{P}$ to $\sigma\mathfrak{P}$ conjugates $\mathrm{Frob}_{\mathfrak{P}}$ by σ, so p determines a conjugacy class $\mathrm{Frob}_p \subset G$. For a representation ρ of $G_{\mathbb{Q}}$ unramified at p, that is, trivial on the inertia groups of the primes above p, the matrix $\rho(\mathrm{Frob}_p)$ is therefore defined up to conjugacy, and its trace and characteristic polynomial are well defined. If L is the splitting field of a monic irreducible $f \in \mathbb{Z}[x]$ and $p \nmid \mathrm{disc}(f)$, the cycle type of Frob_p on the roots is the list of degrees of the irreducible factors of f modulo p (Dedekind).

THEOREM 3.1 (Chebotarev density theorem). *Let $L/\mathbb{Q}$ be a finite Galois extension with group G and let $C \subset G$ be a conjugacy class. Then*

$$\lim_{x\to\infty} \frac{\#\{p \le x :\ p \text{ unramified in } L,\ \mathrm{Frob}_p = C\}}{\pi(x)} = \frac{|C|}{|G|}.$$

The theorem is due to Chebotarev [102]. Effective forms are due to Lagarias and Odlyzko [340]: under the generalized Riemann hypothesis for the Dedekind zeta function of L, the count in the numerator is $(|C|/|G|)\,\mathrm{Li}(x)$ up to an error $O\big((|C|/|G|)\sqrt{x}\log(d_L x^{[L:\mathbb{Q}]}) + \log d_L\big)$, with an absolute implied constant, where $\mathrm{Li}(x) = \int_2^x dt/\log t$. Serre [506] develops applications.

COROLLARY 3.2 (Chebotarev as equidistribution). *Let h be the uniform probability measure on G. For every class function $\phi : G \to \mathbb{C}$,*

$$\frac{1}{\pi(x)} \sum_{p\le x,\ p\nmid d_L} \phi(\mathrm{Frob}_p) \longrightarrow \int_G \phi\, dh.$$

Equivalently, if P is chosen uniformly from the unramified primes up to x, then the class Frob_P converges in law, as $x \to \infty$, to the conjugacy class of a uniformly random element of G.

PROOF. Indicators of conjugacy classes span the class functions, and $h(C) = |C|/|G|$. Ramified primes are finite in number. □

Since irreducible characters span the class functions, Corollary 3.2 is equivalent to $\pi(x)^{-1}\sum_{p\le x,\,p\nmid d_L}\chi(\mathrm{Frob}_p) \to 0$ for every nontrivial irreducible character χ of G. This is the finite group case of the Weyl criterion (Theorem 4.3), and it gives a modern proof: the limit follows by a Tauberian argument from the holomorphy and nonvanishing on $\Re s \ge 1$ of the Artin L functions of Section 3.2. Chebotarev's own proof reduced the theorem to abelian extensions by adjoining roots of unity.

EXAMPLE 3.3 (The cubic $x^3 - x - 1$). Let $f = x^3 - x - 1$, with discriminant -23. Its splitting field L has group S_3 and is the Hilbert class field of $\mathbb{Q}(\sqrt{-23})$, which has class number 3. For $p \ne 23$ the identity class has size 1 (f splits into linear factors), the transpositions have size 3 (f has exactly one root modulo p), and the 3 cycles have size 2 (f is irreducible modulo p). Chebotarev predicts frequencies $1/6, 1/2, 1/3$. We computed the class of Frob_p for all 664,578 primes $p < 10^7$ with $p \ne 23$, using the Kronecker symbol $(-23/p)$ and a test of $x^p \equiv x$ in $\mathbb{F}_p[x]/(f)$. The observed frequencies are

$$0.16637, \qquad 0.50027, \qquad 0.33336,$$

against 0.16667, 0.5, 0.33333; Figure 3.1 shows the convergence. The two dimensional irreducible representation ρ of S_3 has character values $2, 0, -1$ on the three

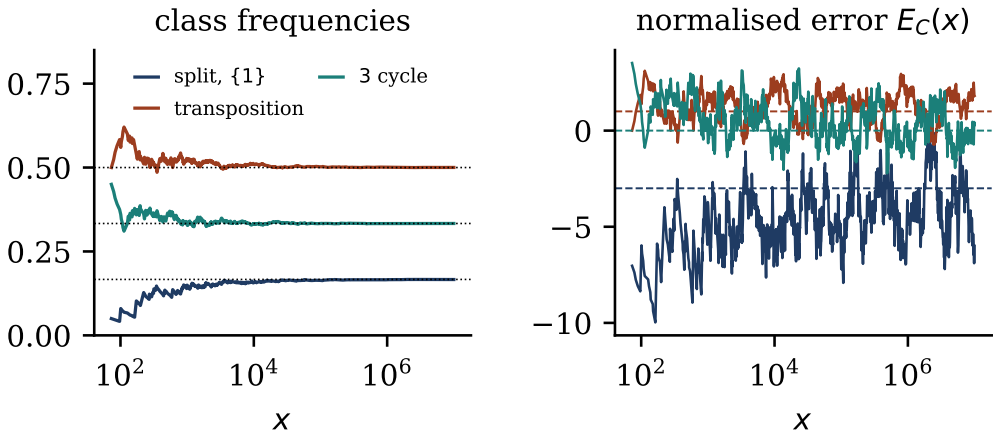


Figure 3.1: Chebotarev for $x^3 - x - 1$, with Galois group $G = S_3$, primes below 10^7. Left: frequencies of the three Frobenius classes (split completely, transposition, 3 cycle) among the unramified primes up to x, against $1/6, 1/2, 1/3$. Right: the normalised error $E_C(x) = (\log x/\sqrt{x})\,(|G|\pi_C(x)/|C| - \pi^{\mathrm{ur}}(x))$, where $\pi_C(x)$ and $\pi^{\mathrm{ur}}(x)$ count the unramified primes $p \le x$ with $\mathrm{Frob}_p = C$ and in total; dashed lines mark the limiting means $1 - r_C/|C| = -3, 1, 0$ produced by the prime squares, where $r_C = \#\{g \in G : g^2 \in C\}$.

classes. It is induced from a character of order 3 of the class group of $\mathbb{Q}(\sqrt{-23})$, so Hecke's theory of theta series shows that the numbers $\operatorname{tr}\rho(\mathrm{Frob}_p)$, $p \neq 23$, are the Fourier coefficients at primes of a weight one cusp form of level 23, namely $\eta(z)\eta(23z)$. This classical nonabelian instance of reciprocity reappears in Chapter 20 as the main test case of its reciprocity criteria, in Chapter 27 as an extreme value law, and in Chapter 22, through Wilton's congruence for $\tau(p)$ modulo 23, as the prototype of the congruences between forms of weight one and of higher weight used in the proof of Fermat's last theorem. In Chapter 23 the Frobenius traces are read off the transported density of $\eta(z)\eta(23z)$, and the form is rebuilt from the Artin L function by Bromwich inversion on the critical line, which uses the holomorphy that Hecke's theorem supplies.

The right panel shows a second order effect. Write $\pi_C(x)$ for the number of unramified $p \le x$ with $\mathrm{Frob}_p = C$, $\pi^{\mathrm{ur}}(x) = \sum_C \pi_C(x)$ for the number of unramified $p \le x$, and $E_C(x) = (\log x/\sqrt{x})(|G|\pi_C(x)/|C| - \pi^{\mathrm{ur}}(x))$ for the normalised error. The prime squares enter the explicit formula through Frob_p^2, distributed as g^2 for g uniform in G. Under the Riemann hypothesis for the Artin L functions of L, and since neither $L(s, \rho)$ nor the L function of the sign character, which is the Dirichlet L function of $(-23/\cdot)$, vanishes at $s = \frac{1}{2}$, $E_C(x)$ is the

constant $1 - r_C/|C|$, up to $o(1)$, plus an oscillation driven by the zeros whose logarithmic mean is zero (Exercise 5). For S_3 the limiting means are -3, 1 and 0; our logarithmic averages over $10^3 < x < 10^7$ are -4.3, 1.2 and 0.3.

The gap is a lower order effect: the limits are approached only at the rate $1/\log x$. The explicit formula counts the prime powers p^k with weight $1/k$, and $\pi_C(x)$ is what remains when the terms with $k \geq 2$ are removed. Replacing the number of unramified primes $p \leq y$ with $\mathrm{Frob}_p^k = C$ in those terms by its expected value $r_C^{(k)} \operatorname{li}(y)/|G|$ and the number of all unramified primes $p \leq y$ by $\operatorname{li}(y)$, with $r_C^{(k)} = \#\{g \in G : g^k \in C\}$ (so $r_C = r_C^{(2)}$) and $\operatorname{li}(y) = \operatorname{Li}(y) + 1.045\ldots$ the principal value logarithmic integral, adds $(\log x/\sqrt{x}) \sum_{k\geq 2} k^{-1}(1 - r_C^{(k)}/|C|) \operatorname{li}(x^{1/k})$ to $E_C(x)$, and the term $k = 2$ equals $(1 - r_C/|C|)(1 + 2/\log x + O(\log^{-2} x))$. The logarithmic means of this sum over $10^3 < x < 10^7$ are -4.4, 1.3 and 0.2 if only $k = 2, 3$ are kept, which is close to the observed values, but the truncation is arbitrary: keeping all k with $2^k \leq x$ gives -5.0, 1.5 and 0.3. The prime powers therefore explain the sign and the order of magnitude of the gap, not its exact value. The smallest primes are not spread over the classes as expected counts assume (the first prime that splits completely is 59), and the remaining terms of the explicit formula oscillate with amplitude of order one and need not average out over a window of this length. Rubinstein and Sarnak [470] put Chebyshev's bias for primes in arithmetic progressions on a probabilistic footing, and their argument extends to Chebotarev classes: under the generalized Riemann hypothesis the law of $E_C(e^y)$, for y uniform in $[0, Y]$, converges as $Y \to \infty$, and under a linear independence hypothesis on the zeros the limit is the law of the constant plus a weighted sum of independent uniformly distributed phases. In our data, primes that split completely are rarer than Chebotarev predicts, $6\pi_{\{1\}}(x) < \pi^{\mathrm{ur}}(x)$, for every x with $2 \leq x < 10^7$ outside $[2.02 \times 10^6, 2.51 \times 10^6]$, and inside that interval except on short stretches of total length below 10^4.

3.2 Artin L functions and ℓ adic representations

Let $\rho : G_{\mathbb{Q}} \to \mathrm{GL}(V)$ be a continuous representation on a finite dimensional complex vector space, necessarily with finite image. Artin [27] defined

$$L(s, \rho) = \prod_p \det\bigl(1 - \rho(\mathrm{Frob}_p) p^{-s} \mid V^{I_p}\bigr)^{-1},$$

where I_p is an inertia group at p, V^{I_p} is the subspace it fixes, and Frob_p acts on V^{I_p} through any of its lifts; the product converges for $\Re s > 1$. Brauer's induction theorem [84] writes $L(s, \rho)$ as a product of integral powers of L functions of one dimensional characters over intermediate fields, which class field theory identifies with Hecke L functions. Hence $L(s, \rho)$ is meromorphic on $\mathbb{C}$ with a functional

equation, and holomorphic and nonzero on $\Re s = 1$ except for a pole at $s = 1$ whose order is the multiplicity of the trivial representation in ρ. That is all that Chebotarev requires.

CONJECTURE 3.4 (Artin). *If ρ is irreducible and nontrivial, then $L(s, \rho)$ is entire.*

The Artin conjecture follows from reciprocity: if ρ corresponds to a cuspidal automorphic representation of GL_n, then $L(s, \rho)$ is entire by Godement and Jacquet [217]. For two dimensional ρ, odd or even, such a correspondence is known when the projective image is dihedral (Hecke), tetrahedral (Langlands [351]) or octahedral (Langlands and Tunnell [553]). Every odd irreducible two dimensional ρ comes from a weight one newform, by the proof of Serre's modularity conjecture by Khare and Wintenberger [311, 312], which also uses work of Kisin [320]. The even icosahedral case is open, and in dimension three and higher the conjecture is open in general.

Geometry supplies representations with infinite image. For an elliptic curve $E/\mathbb{Q}$ and a prime ℓ, the ℓ adic Tate module $T_\ell E = \varprojlim E[\ell^n]$, with $E[\ell^n]$ the ℓ^n torsion of $E(\overline{\mathbb{Q}})$, is free of rank 2 over $\mathbb{Z}_\ell$ with a continuous action of $G_\mathbb{Q}$, giving $\rho_{E,\ell} : G_\mathbb{Q} \to \mathrm{GL}_2(\mathbb{Z}_\ell)$. It is unramified at every $p \nmid \ell N_E$, where N_E is the conductor, and there

$$\det\bigl(1 - \rho_{E,\ell}(\mathrm{Frob}_p)T\bigr) = 1 - a_p T + pT^2, \qquad a_p = p + 1 - \#E(\mathbb{F}_p),$$

independently of ℓ; by Hasse the reciprocal roots have absolute value $\sqrt{p}$.

THEOREM 3.5 (Serre's open image theorem [502]). *Let $E/\mathbb{Q}$ be an elliptic curve without complex multiplication. Then $\rho_{E,\ell}(G_\mathbb{Q})$ is open in $\mathrm{GL}_2(\mathbb{Z}_\ell)$ for every ℓ, and equals $\mathrm{GL}_2(\mathbb{Z}_\ell)$ for all but finitely many ℓ. The image of the adelic representation $G_\mathbb{Q} \to \mathrm{GL}_2(\widehat{\mathbb{Z}})$ is open.*

Chebotarev, applied to the field cut out by $E[\ell^n]$, shows that the classes of $\rho_{E,\ell}(\mathrm{Frob}_p)$ mod ℓ^n are equidistributed for the uniform measure on the image of $G_\mathbb{Q}$ in $\mathrm{GL}_2(\mathbb{Z}/\ell^n\mathbb{Z})$. Hence, when E has no complex multiplication, for all but finitely many ℓ and every $d \in \mathbb{F}_\ell^\times$, the residues a_p mod ℓ over primes $p \equiv d$ mod ℓ are distributed as the trace of a uniform element of $\mathrm{GL}_2(\mathbb{F}_\ell)$ with determinant d. These congruence laws are the correction factors in the Lang and Trotter heuristic of Chapter 5. They say nothing about the archimedean size of $a_p/\sqrt{p}$; that is the content of Sato Tate, and its only known proof goes through automorphy.

The family $(\rho_{E,\ell})_\ell$ is the model of a compatible system: representations $\rho_\ell : G_\mathbb{Q} \to \mathrm{GL}_n(\overline{\mathbb{Q}}_\ell)$, one for each prime ℓ, unramified outside $S \cup \{\ell\}$ for a fixed finite set S, such that for each $p \notin S$ the characteristic polynomial of $\rho_\ell(\mathrm{Frob}_p)$, $\ell \neq p$, has coefficients in a fixed number field embedded in every $\overline{\mathbb{Q}}_\ell$ and does not depend on ℓ. The cohomology $H^i_{\mathrm{et}}(X_{\overline{\mathbb{Q}}}, \mathbb{Q}_\ell)$ of a smooth projective variety $X/\mathbb{Q}$ gives one, with S the primes of bad reduction, and by Deligne [140] for $p \notin S$ the

eigenvalues of the geometric Frobenius $\rho_\ell(\mathrm{Frob}_p)^{-1}$ have absolute value $p^{i/2}$.

3.3 Reciprocity

The reciprocity conjecture of Langlands [350] predicts that every irreducible compatible system arising from geometry corresponds to a cuspidal automorphic representation π of $\mathrm{GL}_n(\mathbb{A}_\mathbb{Q})$: for almost all p the characteristic polynomial of $\rho_\ell(\mathrm{Frob}_p)$ is the Satake polynomial of π_p, suitably normalised. We record what is known for $n \leq 2$ over $\mathbb{Q}$ in the cases this book uses.

3.3.1 The case GL_1*: class field theory*

A continuous character $G_\mathbb{Q} \to \mathbb{C}^\times$ has finite image, so by the Kronecker and Weber theorem it factors through $\mathrm{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q}) \cong (\mathbb{Z}/m\mathbb{Z})^\times$ for some m, and is a Dirichlet character. Over a number field F, global class field theory identifies the abelianisation of G_F with the group of connected components of the idele class group $F^\times\backslash\mathbb{A}_F^\times$, so that characters of finite order of G_F correspond to Hecke characters of finite order. Reciprocity for GL_1 is a theorem.

3.3.2 The case GL_2 *over* $\mathbb{Q}$*: modularity*

Let $f = \sum a_n q^n$ be a normalised cuspidal newform of weight $k \geq 1$, level N and nebentypus ε.

Theorem 3.6 (Eichler, Shimura, Deligne, Deligne and Serre). *There is a number field K_f containing the a_n, and for each prime λ of K_f, of residue characteristic ℓ, a continuous irreducible representation $\rho_{f,\lambda} : G_\mathbb{Q} \to \mathrm{GL}_2(K_{f,\lambda})$ unramified outside $N\ell$ with*

$$\operatorname{tr}\rho_{f,\lambda}(\mathrm{Frob}_p) = a_p, \qquad \det\rho_{f,\lambda}(\mathrm{Frob}_p) = \varepsilon(p)p^{k-1} \qquad (p \nmid N\ell).$$

For $k = 2$ this is due to Eichler [171] and Shimura [514]; for $k > 2$ to Deligne [139], whose proof of the Weil conjectures [140] then gives $|a_p| \leq 2p^{(k-1)/2}$; for $k = 1$ to Deligne and Serre [142], who obtain an odd irreducible representation with finite image in $\mathrm{GL}_2(\mathbb{C})$, unramified outside N. Irreducibility for $k \geq 2$ is due to Ribet [458].

Conversely, by the modularity theorem (Theorem 2.7) every elliptic curve $E/\mathbb{Q}$ arises from a newform f of weight 2 and level N_E, so that $L(s, E) = L(s, f)$. The proof is by modularity lifting, deformation theory matched against Hecke algebras; no probabilistic argument enters.

For the probabilist the Galois side gives congruences, with $\rho(\mathrm{Frob}_p)$ equidistributed in every finite quotient of the image; the automorphic side gives analysis, namely continuation and functional equations for symmetric powers and tensor products. Equidistribution of $a_p/p^{(k-1)/2}$ in $[-2, 2]$ is proved through the second,

which is why probability enters as a consequence of reciprocity and functoriality (Chapter 1).

3.4 The dual group and the Satake isomorphism

Let G be a split connected reductive group over $\mathbb{Q}_p$, with split maximal torus T and root datum $(X^*, \Phi, X_*, \Phi^\vee)$, where $X^* = X^*(T)$ and $X_* = X_*(T)$. The Langlands dual group $\widehat{G}$ is the complex reductive group with the dual root datum $(X_*, \Phi^\vee, X^*, \Phi)$. Thus GL_n is self dual, SL_2 and PGL_2 are dual to each other, and SO_{2n+1} is dual to Sp_{2n}. For a connected reductive group G over a number field F, not necessarily split, the L group is ${}^L G = \widehat{G} \rtimes W_F$, the Weil group W_F acting through a finite quotient. Regard the split group G as defined over $\mathbb{Z}_p$, let $K = G(\mathbb{Z}_p)$, and let $\mathcal{H}(G(\mathbb{Q}_p), K)$ be the spherical Hecke algebra of compactly supported K biinvariant functions on $G(\mathbb{Q}_p)$ under convolution.

Theorem 3.7 (Satake isomorphism [483]). *There is a canonical isomorphism of* $\mathbb{C}$ *algebras*

$$\mathcal{H}(G(\mathbb{Q}_p), K) \;\cong\; \mathbb{C}[X_*(T)]^W \;\cong\; \mathbb{C}[\widehat{G}]^{\widehat{G}},$$

where W is the Weyl group and the last algebra is the algebra of regular class functions on $\widehat{G}$, spanned by the characters of finite dimensional representations of $\widehat{G}$.

An irreducible smooth representation π_p of $G(\mathbb{Q}_p)$ is unramified if it has a nonzero K fixed vector. The space π_p^K is then one dimensional, and $\mathcal{H}(G(\mathbb{Q}_p), K)$ acts on it by a character, which by Satake is evaluation at a semisimple conjugacy class $A(\pi_p) \subset \widehat{G}$, the Satake parameter of π_p. The map $\pi_p \mapsto A(\pi_p)$ is a bijection from isomorphism classes of unramified irreducible representations onto semisimple conjugacy classes in $\widehat{G}$; this is the unramified local Langlands correspondence. For GL_2 and a newform f of weight k with $p \nmid N$, the unitarily normalised Satake parameter is the class of $\mathrm{diag}(\alpha_p, \beta_p)$ with $\alpha_p + \beta_p = a_p / p^{(k-1)/2}$ and $\alpha_p \beta_p = \varepsilon(p)$.

If π_p is tempered, as Conjecture 2.9 predicts for the components of cuspidal representations of GL_n and as is known for holomorphic newforms, then $A(\pi_p)$ meets a maximal compact subgroup of $\widehat{G}$. The Hecke operator corresponding to the character χ_V of a representation V of $\widehat{G}$ acts on π_p^K by $\operatorname{tr} V(A(\pi_p))$. For GL_2, a newform with trivial nebentypus and $V = \mathrm{Sym}^n$ this is $U_n(x_p/2)$ with $x_p = a_p / p^{(k-1)/2}$, the Chebyshev polynomial of the Weyl criterion for $\mathrm{SU}(2)$ in Chapter 4; with nontrivial nebentypus the identity fails (Exercise 6). After Satake, the spherical Hecke algebra is the complexified representation ring of the dual group. For PGL_2, Chapter 9 realises T_p as the adjacency operator of the Bruhat Tits tree, $p+1$ times the transition operator of simple random walk.

3.5 THE PRINCIPLE OF FUNCTORIALITY

CONJECTURE 3.8 (Functoriality; Langlands [350]). *Let G and H be connected reductive groups over a number field F, with G quasisplit, and let $r : {}^LH \to {}^LG$ be an L homomorphism, that is, a continuous homomorphism compatible with the projections to W_F and holomorphic on $\widehat{H}$. For every automorphic representation π of $H(\mathbb{A}_F)$ there is an automorphic representation Π of $G(\mathbb{A}_F)$ such that for almost all places v the Satake parameter of Π_v is $r(A(\pi_v))$.*

Here, at a finite place v where H and π_v are unramified, $A(\pi_v)$ is the Satake parameter of π_v: a semisimple $\widehat{H}$ conjugacy class in the coset of ${}^LH$ above a Frobenius element of W_{F_v} [78], which is the class in $\widehat{H}$ of Section 3.4 when H is split; the same convention applies to G and Π_v. The quasisplit hypothesis falls on the target: Jacquet and Langlands transfer from the multiplicative group of a quaternion algebra to GL_2, but not every representation of GL_2 transfers back. Taking $G = \mathrm{GL}_m$ and r an m dimensional representation of ${}^LH$, functoriality predicts that the automorphic L function $L(s, \pi, r)$ is, up to finitely many Euler factors, a standard L function of GL_m, with continuation and a functional equation; see Arthur [24] and Gelbart [211]. Here are the cases we use.

1. Symmetric powers. For π on GL_2 and $r = \mathrm{Sym}^n$, functoriality was proved for $n = 2$ by Gelbart and Jacquet [212], for $n = 3$ by Kim and Shahidi [316], and for $n = 4$ by Kim [315]. Newton and Thorne [419, 420] proved it for every n when π comes from a holomorphic newform of weight $k \geq 2$ without complex multiplication. For Maass forms the cases $n \geq 5$ are open.

2. Base change for GL_n along cyclic extensions of prime degree, and so along solvable Galois extensions by iteration, by Langlands [351] for $n = 2$ and Arthur and Clozel [26] for general n.

3. Jacquet Langlands. If D is a quaternion algebra over F, Jacquet and Langlands [277] transferred automorphic representations of $D^\times(\mathbb{A}_F)$ of dimension greater than one to cuspidal representations of $\mathrm{GL}_2(\mathbb{A}_F)$, discrete series where D ramifies. This identifies Hecke eigenvalues on the supersingular isogeny graphs of Chapter 10 with Fourier coefficients of modular forms.

4. Tensor products. $\mathrm{GL}_2 \times \mathrm{GL}_2 \to \mathrm{GL}_4$ is functorial by Ramakrishnan [452], and $\mathrm{GL}_2 \times \mathrm{GL}_3 \to \mathrm{GL}_6$ by Kim and Shahidi [316]. The general case is open.

The local Langlands correspondence for GL_n over a finite extension F_v of $\mathbb{Q}_p$ is a bijection from isomorphism classes of irreducible smooth representations of $\mathrm{GL}_n(F_v)$ to isomorphism classes of n dimensional Frobenius semisimple Weil Deligne representations of W_{F_v}, characterised by local class field theory for $n = 1$ and by compatibility with twists, central characters, duals and the L and ε factors of pairs. On unramified representations it is the bijection $\pi_p \mapsto A(\pi_p)$ of Section 3.4.

It was proved by Harris and Taylor [241] and by Henniart [257], with a later proof by Scholze [489].

3.5.1 Potential automorphy and the Sato Tate theorem

The Sato Tate theorem was proved before Newton and Thorne, from a weaker statement. A compatible system over a number field F is potentially automorphic if its restriction to $G_{F'}$ is automorphic for some finite Galois extension F'/F; in the results quoted next, F' can be chosen totally real when F is. Clozel, Harris and Taylor [103], Taylor [540], and Harris, Shepherd Barron and Taylor [266] proved potential automorphy of the odd symmetric powers $\mathrm{Sym}^{2m-1}\rho_{E,\ell}$ for elliptic curves over totally real fields with nonintegral j invariant, which together with Rankin Selberg theory controls every symmetric power L function. Barnet Lamb, Geraghty, Harris and Taylor [67] proved potential automorphy of all symmetric powers of holomorphic newforms of weight $k \geq 2$ without complex multiplication, and noted that the hypothesis on the j invariant can now be removed for elliptic curves without complex multiplication over totally real fields.

Potential automorphy suffices because Brauer's induction theorem, applied to $\mathrm{Gal}(F'/F)$, expresses $L(s, \mathrm{Sym}^n \rho)$ as a product of integral powers of L functions over intermediate fields. The subgroups in Brauer's theorem are elementary, hence solvable, so solvable descent [26] makes these standard or Rankin Selberg L functions of automorphic representations. Each is nonvanishing on $\Re s \geq 1$ by the results of Jacquet and Shalika [278] and Shahidi [510], and holomorphic there except for the simple pole at $s = 1$ of the Rankin Selberg factors $L(s, \pi \times \tilde{\pi})$, which cancels in $L(s, \mathrm{Sym}^n \rho)$ for $n \geq 1$. So $L(s, \mathrm{Sym}^n \rho)$ is meromorphic on $\mathbb{C}$, and holomorphic and nonzero on $\Re s \geq 1$ for $n \geq 1$, though not known from this argument to be entire. This is exactly the hypothesis of Serre's theorem (Theorem 4.8).

THEOREM 3.9 (Sato Tate; Clozel, Harris, Shepherd Barron, Taylor, Barnet Lamb, Geraghty). *Let $E/\mathbb{Q}$ be an elliptic curve without complex multiplication and, for $p \nmid N_E$, write $a_p = 2\sqrt{p}\cos\theta_p$ with $\theta_p \in [0, \pi]$. Then (θ_p) is equidistributed with respect to $\frac{2}{\pi}\sin^2\theta\, d\theta$.*

The proof is the archetype of the slogan that functoriality implies probability: potential automorphy gives the analysis of $L(s, \mathrm{Sym}^n E)$ for every irreducible representation Sym^n of $\mathrm{SU}(2)$, and the Weyl criterion converts it into a law of large numbers; Chapter 4 gives the details. Harris [240] proved the joint version, assuming results on Galois representations that were established soon afterwards: for two elliptic curves over a totally real field, without complex multiplication and not isogenous over an algebraic closure, the pairs (θ_p, θ'_p) are equidistributed for the product of the two Sato Tate measures (over $\mathbb{Q}$ this is Theorem 4.11, where the hypotheses are discussed).

3.6 The Sato Tate group

Which compact group replaces $\mathrm{SU}(2)$ for a general motive? Serre gave the definition [509], and Fité, Kedlaya, Rotger and Sutherland [192] showed that for abelian surfaces it is determined by the Galois module of endomorphisms. Let A be an abelian variety of dimension g over a number field F.

Definition 3.10 (Serre). Fix a prime ℓ and let $\rho_\ell : G_F \to \mathrm{GSp}_{2g}(\mathbb{Q}_\ell)$ be the action on $V_\ell A = T_\ell A \otimes \mathbb{Q}_\ell$, symplectic for the Weil pairing of a polarisation. Let G_ℓ be the Zariski closure of $\rho_\ell(G_F)$ in GSp_{2g}, and $G_\ell^1 = G_\ell \cap \mathrm{Sp}_{2g}$. Choose a field embedding $\iota : \mathbb{Q}_\ell \to \mathbb{C}$. The Sato Tate group $\mathrm{ST}(A)$ is a maximal compact subgroup of $G_\ell^1 \otimes_\iota \mathbb{C}$, a compact subgroup of $\mathrm{USp}(2g)$ well defined up to conjugacy.

For a prime $\mathfrak{p} \nmid \ell$ of F at which A has good reduction, with norm $N\mathfrak{p}$, the normalised Frobenius $\iota\rho_\ell(\mathrm{Frob}_\mathfrak{p})/\sqrt{N\mathfrak{p}}$ lies in $G_\ell^1(\mathbb{C})$ and is semisimple with eigenvalues of absolute value one, so it defines a conjugacy class $s_\mathfrak{p}$ in $\mathrm{ST}(A)$. Independence of ℓ and ι is known for $g \le 3$. Let K be the smallest extension of F over which all endomorphisms of A are defined. Then $\mathrm{Gal}(K/F)$ is a quotient of the component group $\mathrm{ST}(A)/\mathrm{ST}(A)^0$, and the two are isomorphic when $g \le 3$ [34, 192, 532], so that Chebotarev for K/F governs which component contains $s_\mathfrak{p}$; for general g the same role is played by the extension of F cut out by the component group, which contains K.

Conjecture 3.11 (Generalised Sato Tate). *The classes $s_\mathfrak{p}$, over the primes $\mathfrak{p}$ of good reduction ordered by norm, are equidistributed with respect to the pushforward of Haar measure on $\mathrm{ST}(A)$ to its space of conjugacy classes.*

For an elliptic curve E/F without complex multiplication over $\overline{F}$ the group is $\mathrm{ST}(E) = \mathrm{SU}(2) = \mathrm{USp}(2)$. For E with complex multiplication by an imaginary quadratic field M, the group is $\mathrm{U}(1)$ if $M \subseteq F$ and its normaliser $N(\mathrm{U}(1))$ in $\mathrm{SU}(2)$ otherwise; the pushforward of Haar measure on $N(\mathrm{U}(1))$ to the trace is the law $\mu_{\mathrm{CM}} = \frac{1}{2}\delta_0 + \frac{1}{2}$ arcsine of Section 4.6, with Weyl averages in Proposition 4.10.

Theorem 3.12 (Fité, Kedlaya, Rotger and Sutherland [192]). *Up to conjugacy in* $\mathrm{USp}(4)$, *there are exactly* 52 *compact groups that occur as Sato Tate groups of abelian surfaces over number fields, and* 34 *of them occur over* $\mathbb{Q}$. *Their identity components are* $\mathrm{U}(1)$, $\mathrm{SU}(2)$, $\mathrm{U}(1) \times \mathrm{U}(1)$, $\mathrm{U}(1) \times \mathrm{SU}(2)$, $\mathrm{SU}(2) \times \mathrm{SU}(2)$ *and* $\mathrm{USp}(4)$.

For abelian surfaces Conjecture 3.11 is open in the generic case $\mathrm{ST}(A) = \mathrm{USp}(4)$, which occurs exactly when $\mathrm{End}(A_{\overline{F}}) = \mathbb{Z}$, as for the Jacobian of a typical genus 2 curve; Johansson [284] proved many of the nongeneric cases from the potential automorphy theorems. Kedlaya and Sutherland [302] compare moments of normalised L polynomial coefficients over large ranges of primes with those of candidate groups, and for dimension three Fité, Kedlaya and Sutherland [193] announced a classification with 410 groups. Chebotarev governs the components;

automorphy governs the identity component. Chapter 29 turns the moment comparison into a consistent Bayesian model selection.

3.7 Notes

Frobenius's density theorem (1896) gives the density of the primes whose Frobenius generates a given cyclic subgroup up to conjugacy, hence of each factorisation type, and is a weaker form of Chebotarev's; Chebotarev's device of adjoining roots of unity was then used by Artin to prove his reciprocity law [28]. For the elliptic curve analogue of Chebyshev's bias see Sarnak's letter to Mazur [480] and Mazur's survey [386]. Borel's Corvallis article [78] remains the standard account of L groups. Beyond the cases listed in Section 3.5, the stabilisation of the trace formula, which rests on Ngô's proof of the fundamental lemma [421], underlies Arthur's classification of the automorphic representations of quasisplit symplectic and orthogonal groups, which includes their transfer to general linear groups [25]. Serre's book [509] treats Sato Tate groups, with Sutherland's survey [532] as a computational companion. Rouse, Sutherland and Zureick Brown [468] classified the ℓ adic images of Galois for elliptic curves over $\mathbb{Q}$ without complex multiplication, up to the possible existence of unexpected rational points on the modular curves $X_{\mathrm{ns}}^{+}(\ell)$ of nonsplit Cartan type for primes $\ell \geq 19$ and on six further modular curves, of levels 25, 27, 49 and 121.

3.8 Exercises

1. Let $f = x^3 - x - 1$ and let N_p be the number of roots of f modulo p. Show that N_p converges in law, along primes, to the number of fixed points of a uniform permutation in S_3. Deduce that the mean of N_p tends to 1 and its second moment to 2, and check these against the frequencies 0.16637, 0.50027, 0.33336 observed below 10^7.

2. (Burnside) Let $f \in \mathbb{Z}[x]$ be monic and irreducible with Galois group G, and let N_p be the number of roots of f modulo p. Prove that the average of N_p over primes $p \leq x$ tends to 1, and that the limiting kth factorial moment of N_p equals the number of orbits of G on ordered k tuples of distinct roots.

3. Let A be Haar distributed in $\mathrm{SU}(2)$ and $t = \operatorname{tr} A$. Show that the characters $\operatorname{tr}\mathrm{Sym}^m(A)$ are orthonormal in L^2 of Haar measure and that $\operatorname{tr}\mathrm{Sym}^2(A) = t^2 - 1$. Deduce from Theorem 3.9 that if $E/\mathbb{Q}$ has no complex multiplication then $\pi(x)^{-1} \sum_{p \leq x,\, p \nmid N_E} (a_p^2/p - 1) \to 0$. Then prove the same limit for every $E/\mathbb{Q}$, with or without complex multiplication, using only the Rankin Selberg L function $L(s, f \times f)$ of the newform f attached to E. (Hint: $\log L(s, f \times f)$ has nonnegative Dirichlet coefficients, and $L(s, f \times f)$ continues meromor-

phically with a simple pole at $s = 1$ and no other pole or zero on $\Re s = 1$; apply the Wiener Ikehara theorem of Chapter 4.)

4. Let Γ be one of $\mathrm{SU}(2)$, $N(\mathrm{U}(1))$ or $\mathrm{USp}(4)$, and let $t = \operatorname{tr}\gamma$ for a Haar random $\gamma \in \Gamma$. Show that $(\mathbb{E}t^2, \mathbb{E}t^4, \mathbb{E}t^6)$ equals $(1,2,5)$, $(1,3,10)$ and $(1,3,14)$ respectively, using the Weyl density (6.1) of Chapter 6 for $\mathrm{USp}(4)$. Conclude that $N(\mathrm{U}(1))$ and $\mathrm{USp}(4)$ are not separated by second and fourth moments, and explain how the proportion of primes with Frobenius trace 0 separates them.

5. Let G be a finite group, C a conjugacy class, $r_C = \#\{g \in G : g^2 \in C\}$, and $\varepsilon_\chi = |G|^{-1}\sum_{g\in G}\chi(g^2)$ the Frobenius Schur indicator of an irreducible character χ. Show that $\sum_\chi \varepsilon_\chi\overline{\chi(C)} = r_C/|C|$, the sum running over all irreducible characters. With $\pi_\chi(x) = \sum_{p\le x,\, p\nmid d_L}\chi(\mathrm{Frob}_p)$, show that $|G|\pi_C(x)/|C| - \pi^{\mathrm{ur}}(x) = \sum_{\chi\neq 1}\overline{\chi(C)}\,\pi_\chi(x)$, and deduce that if each $(\log x/\sqrt{x})\pi_\chi(x)$, $\chi \neq 1$, has logarithmic mean $-\varepsilon_\chi$, then E_C has logarithmic mean $1 - r_C/|C|$. Check the values $-3, 1, 0$ for S_3, and show that $\sum_C |C|(1 - r_C/|C|) = 0$ for every G.

6. Let $A = \mathrm{diag}(\alpha, \beta)$ with $\alpha\beta = 1$. Show that $\operatorname{tr}\mathrm{Sym}^n A = U_n(\operatorname{tr}A/2)$ for every $n \ge 0$. If instead $\alpha\beta = \varepsilon$ with $|\varepsilon| = 1$, show that $\operatorname{tr}\mathrm{Sym}^n A = \eta^n U_n(\operatorname{tr}A/(2\eta))$ for either square root η of ε; in particular $\operatorname{tr}\mathrm{Sym}^2 A = (\operatorname{tr}A)^2 - \varepsilon$, so for a newform with nontrivial nebentypus the Weyl averages of $U_n(x_p/2)$ are the wrong test of equidistribution.

3.9 Open computational problems

Computational Problem 3.1 (Chebotarev statistics and bias races). Objective: measure the convergence rate and the bias in Chebotarev's theorem for nonabelian Galois groups at scale. Objects: the splitting fields of $x^3 - x - 1$ (S_3), of $x^4 - x - 1$ (S_4, discriminant -283) and of $x^5 + 20x + 16$ (A_5, discriminant $2^{16}5^6$); all primes $p \le 10^{10}$. Route: factor f modulo p by computing x^p modulo f on GPUs, separating the two classes of 5 cycles in A_5, which share a cycle type, by a resolvent or a p adic Frobenius algorithm in PARI/GP; compute the first 10^4 zeros of each Artin L function with `lfun`, using weight one forms in LMFDB where they exist. Deliverable: tables of $\pi_C(x)$ at $x = 10^k$, plots of $E_C(e^y)$, the logarithmic densities of $\{E_C > E_{C'}\}$, and a comparison with the Rubinstein and Sarnak model built from the computed zeros. Why it matters: the bias measures Galois structure (r_C) and the low zeros of Artin L functions, and the comparison tests consequences of the linear independence hypothesis in a nonabelian setting.

Computational Problem 3.2 (Empirical Sato Tate groups in genus 2 and 3). Objective: assign a Sato Tate group to every curve in a large database from Frobe-

nius data alone, with calibrated confidence. Objects: all genus 2 curves in LMFDB (about 66,000) and a comparable sample of genus 3 curves, with L polynomials at all good primes $p \leq 2^{30}$. Route: point counting in the style of Kedlaya and Sutherland, or the average polynomial time computation of Hasse Witt matrices of Harvey and Sutherland [244] for hyperelliptic curves, which gives the L polynomial modulo p and, in genus at most 3, is completed by a generic group algorithm; moment sequences of the coefficients a_1, a_2 of the normalised L polynomial and the proportion of primes with $a_1 = 0$; the sequential Bayesian test of Chapter 29 with the 52 candidate measures of [192] and the 410 of [193]. Deliverable: a table of posterior probabilities, agreement rates with the endomorphism data in LMFDB, and the number of primes needed for a posterior of $1 - 10^{-6}$ as a function of the group. Why it matters: it turns Conjecture 3.11 into a calibrated statistical procedure and flags curves with atypical Frobenius data.

COMPUTATIONAL PROBLEM 3.3 (Joint Sato Tate independence). Objective: test the asymptotic independence of Frobenius angles at the same prime for unrelated motives, and measure the rate. Objects: pairs of elliptic curves over $\mathbb{Q}$ of conductor below 10^4, without complex multiplication and not isogenous over $\overline{\mathbb{Q}}$, with all primes below 10^9; a random sample of 10^4 pairs (E, f) with E as before and f a newform of weight 4, trivial nebentypus and level at most 100, without complex multiplication, with all primes below 10^7; and, as a control, pairs $(E, E^{(D)})$ with $E^{(D)}$ a quadratic twist, for which θ'_p equals θ_p or $\pi - \theta_p$ according as $(D/p) = 1$ or -1, so that the pairs (θ_p, θ'_p) lie on the two diagonals of $[0, \pi]^2$ with frequency $\frac{1}{2}$ each. Route: PARI/GP `ellap` in batch for the curves and `mfcoefs` (after `mfinit` and `mfeigenbasis`) for f, in a fixed real embedding of its coefficient field, or average polynomial time algorithms for the curves; joint Weyl averages $\pi(x)^{-1} \sum_{p \leq x} U_m(x_p/2) U_n(x'_p/2)$ for $0 \leq m, n \leq 20$, $(m, n) \neq (0, 0)$, a two dimensional discrepancy, and a permutation test that shuffles one sequence across primes. Deliverable: heat maps of the Weyl averages, the growth of $\sqrt{\pi(x)}$ times the discrepancy, and test statistics with null distributions. Why it matters: joint equidistribution holds in the first two cases, by symmetric power functoriality and Rankin Selberg theory (Theorem 4.11 for two curves over $\mathbb{Q}$), and its rate is governed by the zeros of $L(s, \mathrm{Sym}^m E \times \mathrm{Sym}^n E')$, where E' is the second curve or, for the pairs (E, f), the form f, so the data probe the Riemann hypothesis for these L functions.

Part II

Equidistribution

Chapter 4

Haar Measure, the Weyl Criterion and Sato Tate

In 1916 Hermann Weyl showed that a sequence in $[0,1)$ is equidistributed if and only if the averages of the characters $e^{2\pi i h x}$ along it tend to zero for every integer $h \neq 0$ [576]. He used the criterion to reprove the equidistribution of the fractional parts of $n\alpha$ for irrational α, found around 1910 by Bohl, Sierpiński and Weyl himself, and to extend it to polynomial sequences; the criterion has outlived the problem. The Peter Weyl theorem [436], proved in 1927 for compact Lie groups, makes the matrix coefficients of irreducible representations dense among continuous functions and so gives the same criterion on compact groups, with irreducible characters in place of exponentials. In the early 1960s Sato and Tate, working independently and from quite different directions, predicted that the angles of Frobenius of an elliptic curve over $\mathbb{Q}$ without complex multiplication are distributed according to the image of Haar measure on $\mathrm{SU}(2)$ [538]. Serre saw that Weyl's criterion converts this prediction into a statement about L functions [499]: the conjecture follows if every symmetric power L function is holomorphic and nonvanishing on the line $\Re s = 1$. That statement is a consequence of functoriality. For elliptic curves over $\mathbb{Q}$ and holomorphic newforms it was proved through potential automorphy in papers published between 2008 and 2011, and in 2021 Newton and Thorne proved that every symmetric power is automorphic; this is one of the clearest successes of the Langlands programme. We prove the probabilistic side completely and state the arithmetic input precisely. The logical direction matters throughout: automorphy is the input, equidistribution is the output.

4.1 Equidistribution on compact metric spaces

Let X be a compact metric space and μ a Borel probability measure on X. A sequence (x_j) in X is *equidistributed* with respect to μ if the empirical measures $\nu_n = n^{-1}\sum_{j \le n} \delta_{x_j}$ converge weakly to μ, that is $\int f\,d\nu_n \to \int f\,d\mu$ for every $f \in C(X)$. For sequences indexed by primes we use the ordering by size, so the natural normalisation is $\pi(x)^{-1}\sum_{p \le x}$.

Weak convergence can be tested on any family of functions whose linear span is dense in $C(X)$ for the supremum norm. This observation is the whole content

of the Weyl criterion; what makes it useful is that representation theory supplies a canonical dense family. For the circle $\mathbb{R}/\mathbb{Z}$ the family is $x \mapsto e^{2\pi i h x}$, $h \in \mathbb{Z}$, whose span is dense by Fejér's theorem; this recovers Weyl's original criterion. We shall also use the portmanteau theorem: if $\nu_n \to \mu$ weakly, then $\limsup \nu_n(F) \le \mu(F)$ for every closed $F \subset X$.

4.2 Peter Weyl and Schur orthogonality

Let K be a compact metrisable group, for instance a closed subgroup of $\mathrm{U}(n)$, with normalised Haar measure h. Representations of K are continuous and finite dimensional. A *class function* is a function constant on conjugacy classes. Write $K^\natural$ for the space of conjugacy classes with the quotient topology, a compact metrisable space, and $h^\natural$ for the pushforward of h. Continuous class functions on K are exactly the continuous functions on $K^\natural$. For a representation ρ of K we write $\chi_\rho = \operatorname{tr} \rho$ for its character.

Theorem 4.1 (Schur orthogonality). *If ρ and σ are irreducible representations of K, then*

$$\int_K \chi_\rho(g)\overline{\chi_\sigma(g)}\, dh(g) = \begin{cases} 1 & \rho \cong \sigma, \\ 0 & \text{otherwise.} \end{cases}$$

In particular $\int \chi_\rho \, dh = 0$ for every nontrivial irreducible ρ.

Theorem 4.2 (Peter and Weyl [436]). *Matrix coefficients of finite dimensional irreducible representations span a dense subspace of $C(K)$. Consequently the characters of irreducible representations span a dense subspace of the continuous class functions on K.*

The second statement follows from the first. The averaging operator $P\phi(g) = \int_K \phi(kgk^{-1})\, dh(k)$ fixes class functions, does not increase the supremum norm, and, by Schur's lemma, maps each matrix coefficient of an irreducible ρ to a multiple of χ_ρ. So if a class function f satisfies $\|f - \phi\|_\infty < \delta$ for a combination ϕ of matrix coefficients, then $\|f - P\phi\|_\infty < \delta$, and $P\phi$ is a combination of irreducible characters.

4.3 The Weyl criterion on compact groups

We prove the form of the criterion used throughout the book.

Theorem 4.3 (Weyl criterion [576, 499]). *Let K be a compact metrisable group and (θ_p) a sequence in $K^\natural$ indexed by primes. Then (θ_p) is equidistributed with respect to $h^\natural$ if and only if, for every nontrivial irreducible representation ρ of K,*

$$\lim_{x \to \infty} \frac{1}{\pi(x)} \sum_{p \le x} \chi_\rho(\theta_p) = 0.$$

More generally, (θ_p) is equidistributed with respect to a probability measure μ on $K^\natural$ if and only if $\pi(x)^{-1}\sum_{p\le x}\chi_\rho(\theta_p)\to\int\chi_\rho\,d\mu$ for every irreducible ρ.

PROOF. We prove the general statement; the first follows from Schur orthogonality, which gives $\int\chi_\rho\,dh^\natural=0$ for nontrivial ρ and 1 for the trivial representation. Write $\nu_x=\pi(x)^{-1}\sum_{p\le x}\delta_{\theta_p}$.

Necessity is immediate: each χ_ρ is continuous on $K^\natural$.

For sufficiency, let $f\in C(K^\natural)$ and $\delta>0$. By the Peter Weyl theorem there is a finite linear combination $g=\sum_\rho c_\rho\chi_\rho$ with $\|f-g\|_\infty<\delta$. By hypothesis and linearity, $\int g\,d\nu_x\to\int g\,d\mu$. Then

$$\left|\int f\,d\nu_x-\int f\,d\mu\right|\le\left|\int(f-g)\,d\nu_x\right|+\left|\int g\,d\nu_x-\int g\,d\mu\right|+\left|\int(g-f)\,d\mu\right|,$$

and the first and third terms are each at most δ because ν_x and μ are probability measures. Hence $\limsup_x|\int f\,d\nu_x-\int f\,d\mu|\le 2\delta$ for every $\delta>0$, and the limit is 0. □

The proof uses only that each ν_x is a probability measure on $K^\natural$. The criterion therefore holds for any sequence of probability measures on $K^\natural$, for instance the empirical measures $n^{-1}\sum_{j\le n}\delta_{\theta_j}$ of a sequence indexed by the integers, or those of the finite families of conjugacy classes that arise in vertical problems (Chapter 7).

The limits $\int\chi_\rho\,d\mu$ for a measure other than Haar are themselves informative: when μ is the image of Haar measure on a closed subgroup $H\subset K$, the integral is the multiplicity of the trivial representation in $\rho|_H$, a nonnegative integer. We call the function $\rho\mapsto\int\chi_\rho\,d\mu$ on the irreducible representations of K the *Weyl fingerprint* of μ; for $\mathrm{SU}(2)$ it is a sequence indexed by $n\ge 0$ (next section).

4.4 THE GROUP SU(2)

Every element of $\mathrm{SU}(2)$ is conjugate to $\mathrm{diag}(e^{i\theta},e^{-i\theta})$ for a unique $\theta\in[0,\pi]$, so $\mathrm{SU}(2)^\natural$ is identified with $[0,\pi]$, and equally with the interval $[-2,2]$ through the trace $x=2\cos\theta$. The irreducible representations are the symmetric powers Sym^n of the standard representation, of dimension $n+1$, with eigenvalues $e^{i(n-2j)\theta}$ for $0\le j\le n$. Summing the geometric progression,

$$\chi_n(\theta)=\operatorname{tr}\mathrm{Sym}^n(\theta)=\sum_{j=0}^{n}e^{i(n-2j)\theta}=\frac{\sin((n+1)\theta)}{\sin\theta}=U_n(\cos\theta)=U_n(x/2),$$

where U_n is the Chebyshev polynomial of the second kind. The recurrence $U_1(t)U_n(t)=U_{n+1}(t)+U_{n-1}(t)$ is the Clebsch Gordan rule $\mathrm{std}\otimes\mathrm{Sym}^n\cong\mathrm{Sym}^{n+1}\oplus\mathrm{Sym}^{n-1}$.

PROPOSITION 4.4 (Weyl integration formula for SU(2)). *The pushforward of Haar measure to* $[0,\pi]$ *is* $\frac{2}{\pi}\sin^2\theta\,d\theta$. *Equivalently, the trace* $x = 2\cos\theta$ *of a Haar random element has the semicircle law*

$$\mu_{\mathrm{ST}}(dx) = \frac{1}{2\pi}\sqrt{4-x^2}\,dx, \qquad x \in [-2,2].$$

PROOF. Identify SU(2) with the unit quaternions S^3; Haar measure is normalised surface measure, since multiplications by unit quaternions are isometries. The trace is $2a$ for $g = a + bi + cj + dk$, and the slice $\{a = \cos\theta\}$ is a two sphere of radius $\sin\theta$, so surface measure on S^3 projects to a density proportional to $\sin^2\theta$ on $[0,\pi]$. Since $\int_0^\pi \sin^2\theta\,d\theta = \pi/2$, the normalised density is $\frac{2}{\pi}\sin^2\theta$. Changing variables $x = 2\cos\theta$, $dx = -2\sin\theta\,d\theta$ and $\sin\theta = \frac{1}{2}\sqrt{4-x^2}$ gives the semicircle. □

PROPOSITION 4.5 (Orthonormality and Catalan moments). *The polynomials* $U_n(x/2)$ *are orthonormal in* $L^2(\mu_{\mathrm{ST}})$, *and*

$$\int_{-2}^{2} x^k\,\mu_{\mathrm{ST}}(dx) = \begin{cases} C_m = \dfrac{1}{m+1}\dbinom{2m}{m} & k = 2m, \\ 0 & k \text{ odd}. \end{cases}$$

PROOF. In the variable θ, the factors $\sin\theta$ cancel against the density $\frac{2}{\pi}\sin^2\theta$, and

$$\int U_m(x/2)\,U_n(x/2)\,\mu_{\mathrm{ST}}(dx) = \frac{2}{\pi}\int_0^\pi \sin((m+1)\theta)\sin((n+1)\theta)\,d\theta = \delta_{mn}$$

by orthogonality of sines on $[0,\pi]$.

For the moments, iterate the Clebsch Gordan rule. Since $x = U_1(x/2)$, we have $x\,U_n(x/2) = U_{n+1}(x/2) + U_{n-1}(x/2)$ for $n \geq 1$ and $x\,U_0(x/2) = U_1(x/2)$. By induction $x^k = \sum_{n\geq 0} N_k(n)\,U_n(x/2)$, where $N_k(n)$ is the number of paths of k steps ± 1 on the nonnegative integers from 0 to n that never go below 0. By orthonormality, $\int x^k\,d\mu_{\mathrm{ST}} = N_k(0)$, the number of Dyck paths of length k. This vanishes for k odd, and for $k = 2m$ the reflection principle gives $N_{2m}(0) = \binom{2m}{m} - \binom{2m}{m+1} = C_m$. □

The moments of μ_{ST} thus count returning paths of the simple walk on the nonnegative integers. Spectral measures whose moments count returning paths recur throughout the book; in Chapter 9 the same computation on a regular tree produces the Kesten McKay laws.

4.5 THE SATO TATE CONJECTURE AND SERRE'S THEOREM

Let $E/\mathbb{Q}$ be an elliptic curve of conductor N. For $p \nmid N$ put $a_p = p + 1 - \#E(\mathbb{F}_p)$. Hasse's bound $|a_p| \leq 2\sqrt{p}$ says that $x_p = a_p/\sqrt{p}$ lies in $[-2,2]$, so we may write

$x_p = 2\cos\theta_p$ with $\theta_p \in [0,\pi]$, and regard θ_p as the conjugacy class in $\mathrm{SU}(2)$ of the normalised Frobenius. More generally, for a holomorphic newform $f = \sum a_n q^n$ of weight $k \geq 2$, level N and trivial character, Deligne's theorem [140] gives $|a_p| \leq 2p^{(k-1)/2}$ for $p \nmid N$, and we set $x_p = a_p / p^{(k-1)/2}$.

CONJECTURE 4.6 (Sato and Tate). *If $E/\mathbb{Q}$ has no complex multiplication, the angles θ_p are equidistributed in $[0,\pi]$ with respect to $\frac{2}{\pi}\sin^2\theta\, d\theta$; equivalently (x_p) is equidistributed with respect to μ_{ST}.*

The conjecture is now a theorem, and we explain how. The analytic objects are the symmetric power L functions. Write the normalised Satake parameters at a good prime as $\{\alpha_p, \beta_p\} = \{e^{i\theta_p}, e^{-i\theta_p}\}$ and define, for $n \geq 1$,

$$L(s, \mathrm{Sym}^n f) = \prod_{p \nmid N} \prod_{j=0}^{n} \left(1 - \alpha_p^j \beta_p^{n-j} p^{-s}\right)^{-1},$$

which converges absolutely for $\Re s > 1$. (Including suitable factors at the finitely many bad primes changes nothing below, because those factors are holomorphic and nonvanishing on $\Re s > 0$.) The tool that converts analytic information on $\Re s = 1$ into asymptotics is the Tauberian theorem of Wiener and Ikehara [579, 580].

THEOREM 4.7 (Wiener and Ikehara). *Let $F(s) = \sum_{n \geq 1} b_n n^{-s}$ with $b_n \geq 0$, convergent for $\Re s > 1$. Suppose there is a constant c such that $F(s) - c/(s-1)$ extends to a continuous function on the closed half plane $\Re s \geq 1$. Then $\sum_{n \leq x} b_n \sim c\, x$ as $x \to \infty$.*

The prime number theorem is the case $F = -\zeta'/\zeta$, $b_n = \Lambda(n)$, $c = 1$, the hypothesis being that ζ has no zeros on $\Re s = 1$.

THEOREM 4.8 (Serre [499]). *Let f be a holomorphic newform of weight $k \geq 2$ with trivial character. Suppose that for every $n \geq 1$ the function $L(s, \mathrm{Sym}^n f)$ extends holomorphically to an open set containing $\Re s \geq 1$ and has no zeros on $\Re s = 1$. Then (x_p) is equidistributed with respect to μ_{ST}.*

PROOF. Sums over primes in this proof run over $p \nmid N$; omitting finitely many primes changes no limit below. Fix $n \geq 1$. Taking the logarithmic derivative of the Euler product,

$$-\frac{L'}{L}(s, \mathrm{Sym}^n f) = \sum_{p \nmid N} \sum_{m \geq 1} \Big(\sum_{j=0}^{n} (\alpha_p^j \beta_p^{n-j})^m \Big) \log p\, p^{-ms}.$$

The term $m = 1$ has coefficient $\mathrm{tr}\,\mathrm{Sym}^n(\theta_p) \log p = U_n(x_p/2) \log p$. Each inner sum has absolute value at most $n+1$ because $|\alpha_p| = |\beta_p| = 1$, so the terms with $m \geq 2$ form a series dominated by $(n+1)\sum_p \log p \sum_{m \geq 2} p^{-m\sigma}$, which converges uniformly on $\Re s \geq \sigma_0$ for any $\sigma_0 > 1/2$. Hence

$$D_n(s) := \sum_p U_n(x_p/2) \log p\, p^{-s} = -\frac{L'}{L}(s, \mathrm{Sym}^n f) - H_n(s),$$

with H_n holomorphic on $\Re s > 1/2$. By hypothesis $L(s, \mathrm{Sym}^n f)$ is holomorphic and zero free on a neighbourhood of $\Re s \geq 1$ (on $\Re s > 1$ the Euler product does not vanish), so D_n extends holomorphically to a neighbourhood of $\Re s \geq 1$.

The coefficients of D_n are not of one sign, so we add a multiple of the prime number theorem. Let $\vartheta(s) = \sum_p \log p \, p^{-s}$; by the same estimate, and because the finitely many terms with $p \mid N$ are entire, $\vartheta(s) = -\zeta'(s)/\zeta(s) - H_0(s)$ with H_0 holomorphic on $\Re s > 1/2$, so $\vartheta(s) - 1/(s-1)$ extends continuously to $\Re s \geq 1$. Put

$$F(s) = (n+1)\vartheta(s) + D_n(s) = \sum_p \bigl(n + 1 + U_n(x_p/2)\bigr) \log p \, p^{-s}.$$

Since $|U_n(t)| \leq n+1$ on $[-1,1]$, the coefficients are nonnegative, and $F(s) - (n+1)/(s-1)$ extends continuously to $\Re s \geq 1$. The Wiener Ikehara theorem, applied to F and to ϑ, gives

$$\sum_{p \leq x} \bigl(n + 1 + U_n(x_p/2)\bigr) \log p \sim (n+1)x, \qquad \sum_{p \leq x} \log p \sim x,$$

and subtracting, $A_n(x) := \sum_{p \leq x} U_n(x_p/2) \log p = o(x)$.

Now remove the weight $\log p$ by partial summation:

$$\sum_{p \leq x} U_n(x_p/2) = \frac{A_n(x)}{\log x} + \int_2^x \frac{A_n(t)}{t \log^2 t} \, dt.$$

Given $\varepsilon > 0$ choose T with $|A_n(t)| \leq \varepsilon t$ for $t \geq T$. The first term is at most $\varepsilon x / \log x$ for $x \geq T$. The integral over $[2, T]$ is a constant, and the integral over $[T, x]$ is at most $\varepsilon \int_2^x dt/\log^2 t = O(\varepsilon x/\log^2 x)$. Hence $\sum_{p \leq x} U_n(x_p/2) = o(x/\log x) = o(\pi(x))$. Theorem 4.3 for $K = \mathrm{SU}(2)$ completes the proof. □

Functoriality would supply the hypothesis of Theorem 4.8. If $\mathrm{Sym}^n f$ corresponds to a cuspidal automorphic representation Π_n of $\mathrm{GL}_{n+1}(\mathbb{A}_{\mathbb{Q}})$, then $L(s, \mathrm{Sym}^n f)$ agrees with the standard L function of Π_n up to finitely many Euler factors, which are holomorphic and nonvanishing on $\Re s \geq 1$; Godement and Jacquet [217] proved that this L function is entire for $n \geq 1$, and Jacquet and Shalika [278] that it does not vanish on $\Re s = 1$. Potential automorphy also suffices: by Brauer's induction theorem and solvable descent, as explained before Theorem 3.9, it makes $L(s, \mathrm{Sym}^n f)$ holomorphic and nonzero on $\Re s \geq 1$.

THEOREM 4.9 (Sato Tate for elliptic curves and newforms [67]). *The Sato Tate conjecture holds for every elliptic curve over* $\mathbb{Q}$ *without complex multiplication. More generally, if* f *is a holomorphic newform of weight* $k \geq 2$ *with trivial character and without complex multiplication, then* (x_p) *is equidistributed with respect to* μ_{ST}.

The proofs are long and we only describe them. Clozel, Harris and Taylor [103] and Taylor [540] proved automorphy lifting theorems for ℓ adic representations

of unitary type. Harris, Shepherd Barron and Taylor [266] proved potential automorphy theorems by realising the relevant Galois representations modulo ℓ in the cohomology of a family of Calabi Yau hypersurfaces of Dwork type. What these papers and Barnet Lamb, Geraghty, Harris and Taylor [67] proved is recorded before Theorem 3.9: potential automorphy of the odd symmetric powers of elliptic curves with nonintegral j invariant over totally real fields, then of all symmetric powers of newforms without complex multiplication. The latter gives Theorem 4.9 for such newforms and, by modularity (Theorem 2.7), for elliptic curves over $\mathbb{Q}$ without complex multiplication. Newton and Thorne [419, 420] then proved full automorphy, that $\mathrm{Sym}^n f$ is automorphic for every n and every such newform f, which gives the hypothesis of Theorem 4.8 directly.

Figure 1.1 of Chapter 1 shows the theorem at work for $E_1 : y^2 = x^3 + x + 1$ (conductor 496), using all 664,577 good primes below 10^7. The even moments of x_p up to order 12 are within 0.06% of the Catalan numbers of Proposition 4.5 (Table 5.1). The Weyl averages $\pi(x)^{-1} \sum_{p \le x} U_n(x_p/2)$ at $x = 10^7$ are all below 0.0016 in absolute value for $1 \le n \le 12$, the size of the sampling error $\pi(x)^{-1/2} \approx 0.0012$ of an average of independent draws from μ_{ST}.

Figure 4.1 shows the Frobenius angles for the primes below 4×10^5 on the circle, each $\theta_p \in [0, \pi]$ together with its mirror image $-\theta_p$. The Sato Tate density becomes $\sin^2\theta/\pi$ on $[-\pi, \pi]$, a figure of eight. For the curve E_2 with complex multiplication (Section 4.6) the angles at split primes are uniform on the circle; the inert primes, which form an atom at $\pm\pi/2$ carrying half the mass, are omitted.

4.6 Complex multiplication and the Weyl fingerprint

Suppose now that E has complex multiplication by an order in an imaginary quadratic field M. Deuring [149] showed that $L(s, E) = L(s, \psi)$ for a Hecke Grössencharacter ψ of M of infinity type $(1, 0)$. For a prime $p \nmid N$ inert in M the curve is supersingular and $a_p = 0$. For p split, $p = \mathfrak{p}\bar{\mathfrak{p}}$, we have $a_p = \psi(\mathfrak{p}) + \overline{\psi(\mathfrak{p})}$, and the argument of $\psi(\mathfrak{p})/\sqrt{p}$ is $\pm\theta_p$. Hecke proved [248] that these angles are equidistributed on the circle. In the language of this chapter, the Weyl criterion on $\mathrm{U}(1)$ applies because the L functions of the unitary characters $(\psi/|\psi|)^n$, $n \neq 0$, are entire and nonvanishing on $\Re s = 1$. By the Chebotarev density theorem half the primes split. Combining these, (x_p) is equidistributed with respect to

$$\mu_{\mathrm{CM}}(dx) = \frac{1}{2}\delta_0(dx) + \frac{1}{2}\,\frac{dx}{\pi\sqrt{4 - x^2}}, \qquad x \in [-2, 2],$$

a point mass at 0 plus an arcsine law. Group theoretically, μ_{CM} is the image of Haar measure on the normaliser $N(\mathrm{U}(1))$ of the diagonal torus in $\mathrm{SU}(2)$: the identity component contributes the arcsine law, since a uniform angle θ gives

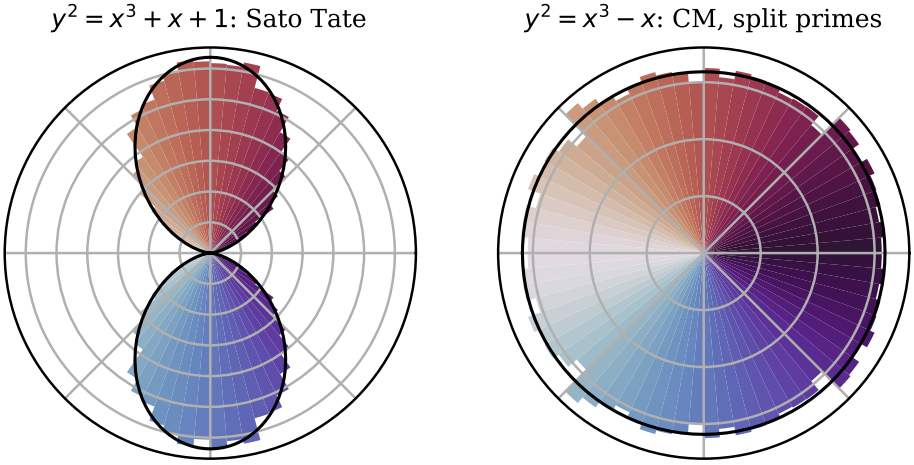


Figure 4.1: Polar histograms of the Frobenius angles $\pm\theta_p$ for the primes below 4×10^5; the radius is the density, the radial labels are omitted, and each panel has its own radial scale. Left: E_1, with the Sato Tate density $\sin^2\theta/\pi$ (black). Right: E_2, split primes only, with the uniform density $1/(2\pi)$ (black); the inert primes, where $\theta_p = \pi/2$, are omitted.

$2\cos\theta$ arcsine distributed, and the other component consists of elements of trace 0. The moments are $\int x^{2m}\,d\mu_{\mathrm{CM}} = \frac{1}{2}\binom{2m}{m}$ for $m \geq 1$.

PROPOSITION 4.10 (Weyl fingerprint of the CM law). *For $n \geq 1$,*

$$\int_{-2}^{2} U_n(x/2)\,\mu_{\mathrm{CM}}(dx) = \begin{cases} 1 & 4 \mid n, \\ 0 & \textit{otherwise.} \end{cases}$$

PROOF. Under the arcsine law the angle θ is uniform on $[0,\pi]$, so the continuous part contributes

$$\frac{1}{\pi}\int_0^{\pi} \frac{\sin((n+1)\theta)}{\sin\theta}\,d\theta = \frac{1}{\pi}\int_0^{\pi} \sum_{j=0}^{n} e^{i(n-2j)\theta}\,d\theta.$$

Pairing the terms j and $n-j$ writes the sum as 1 (present only for n even, from $j = n/2$) plus terms $2\cos((n-2j)\theta)$ with $n - 2j \neq 0$, each of which integrates to 0 over $[0,\pi]$. So the arcsine average of $U_n(x/2)$ is 1 for n even and 0 for n odd. At the atom, $U_n(0) = \sin((n+1)\pi/2) = \cos(n\pi/2)$, which is 0 for n odd and $(-1)^{n/2}$ for n even. Averaging the two halves gives 0 for n odd and $\frac{1}{2}(1+(-1)^{n/2})$ for n even, which is 1 when $4 \mid n$ and 0 when $n \equiv 2 \pmod 4$. □

As predicted by the remark after Theorem 4.3, this is the dimension of the

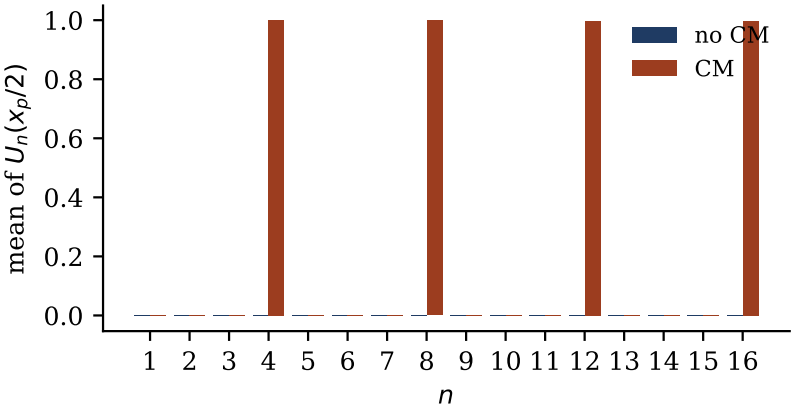


Figure 4.2: Weyl averages $\pi(x)^{-1}\sum_{p\le x} U_n(x_p/2)$ for $1 \le n \le 16$ over the good primes below 10^7. For E_1 (no CM, left bar of each pair) they are close to 0, the limit that Theorem 4.3 requires of a sequence equidistributed for μ_{ST}; for the CM curve E_2 (right bar) they are close to 1 when $4 \mid n$ and to 0 otherwise, the limits of Proposition 4.10.

$N(\mathrm{U}(1))$ invariants in Sym^n: the torus fixes only the weight zero vector (for n even), and the other component acts on it by $(-1)^{n/2}$.

Figure 4.2 shows the fingerprint in our data. For $E_2 : y^2 = x^3 - x$ (conductor 32) and the good primes below 10^7, the averages at $n = 4, 8, 12$ are 0.9995, 0.9988, 0.9979, and the other averages with $n \le 12$ lie within 0.0018 of zero. The proportion of these primes with $a_p = 0$ is 0.5002. The even moments of x_p for E_2 are within 0.14% of $\frac{1}{2}\binom{2m}{m}$ (Table 5.1).

Chapter 29 turns this separation into a sequential Bayesian test with explicit error exponents.

4.7 Joint Sato Tate and coincidences

For two curves, the pairs $(\theta_p(E), \theta_p(E'))$ are classes in $\mathrm{SU}(2) \times \mathrm{SU}(2)$, whose irreducible representations are $\mathrm{Sym}^m \boxtimes \mathrm{Sym}^n$. By Theorem 4.3, joint equidistribution with respect to $\mu_{\mathrm{ST}} \otimes \mu_{\mathrm{ST}}$ is equivalent to

$$\frac{1}{\pi(x)} \sum_{p \le x} U_m(x_p/2)\, U_n(x'_p/2) \to 0 \qquad \text{for all } (m,n) \neq (0,0).$$

By the argument of Theorem 4.8, with $(m+1)(n+1)$ in place of $n+1$, this follows if for each $(m,n) \neq (0,0)$ the Rankin Selberg L function $L(s, \mathrm{Sym}^m E \times \mathrm{Sym}^n E')$,

with Euler factors

$$\prod_{i=0}^{m}\prod_{j=0}^{n}\left(1-\alpha_p^i\beta_p^{m-i}\alpha_p'^j\beta_p'^{n-j}p^{-s}\right)^{-1}$$

at the primes p of good reduction for both curves (the factors at the finitely many other primes are holomorphic and nonvanishing on $\Re s>0$, as for a single curve), extends holomorphically to a neighbourhood of $\Re s\geq 1$ and has no zeros on $\Re s=1$. Isogenous curves have equal traces, and a quadratic twist by χ has $a_p(E')=\chi(p)a_p(E)$, which puts the pairs on the lines $x'=\pm x$; so the right hypothesis is that E and E' are not isogenous over $\bar{\mathbb{Q}}$.

THEOREM 4.11 (Joint Sato Tate; Harris [240]). *Let E and E' be elliptic curves over $\mathbb{Q}$ without complex multiplication that are not isogenous over $\bar{\mathbb{Q}}$. Then the pairs $(x_p(E),x_p(E'))$ are equidistributed in $[-2,2]^2$ with respect to $\mu_{\mathrm{ST}}\otimes\mu_{\mathrm{ST}}$.*

Harris [240] reduced the theorem to the potential automorphy of the symmetric powers of E and E' over a common totally real field, and proved it assuming results on the Galois representations attached to automorphic representations of unitary groups that were then expected; those results were established soon afterwards (see the introduction of [67]). Given the automorphy of all symmetric powers [419, 420], the theorem also follows directly from Rankin Selberg theory. For unitary cuspidal automorphic representations Π and Π', the L function $L(s,\Pi\times\Pi')$ has no zeros on $\Re s=1$ [510], and it is holomorphic on $\Re s\geq 1$ unless $\Pi'\cong\tilde{\Pi}\otimes|\det|^{it}$ for some real t, in which case its only pole there is a simple pole at $s=1-it$ [286, 279, 397].

For $(m,n)\neq(0,0)$ the hypothesis excludes this exceptional isomorphism when Π and Π' are the cuspidal automorphic representations attached to $\mathrm{Sym}^m E$ and $\mathrm{Sym}^n E'$, by an argument of Harris that rests on Faltings' isogeny theorem [182]. Both have trivial central character (symmetric powers of $\mathrm{SU}(2)$ have determinant 1), while $\tilde{\Pi}\otimes|\det|^{it}$ has central character $|\cdot|^{it(m+1)}$, so $t=0$, and the dimensions give $m=n$. Then $\mathrm{Sym}^n\rho_{E,\ell}\cong\mathrm{Sym}^n\rho_{E',\ell}$ for the ℓ adic representations on the Tate modules (Chapter 3), which, as their images are open in $\mathrm{GL}_2(\mathbb{Z}_\ell)$ (Theorem 3.5), makes the adjoint representations agree and so forces $\rho_{E',\ell}\cong\rho_{E,\ell}\otimes\eta$ for a character η, quadratic because both determinants are the cyclotomic character. By Faltings' theorem E' is then isogenous to a quadratic twist of E, hence to E over an extension of $\mathbb{Q}$ of degree at most 2. For curves over $\mathbb{Q}$ without complex multiplication an isogeny over $\bar{\mathbb{Q}}$ is already defined over such an extension, so our hypothesis is equivalent to Harris's, that E and E' do not become isogenous over an abelian extension of $\mathbb{Q}$.

COROLLARY 4.12. *Under the hypotheses of Theorem 4.11, the set of primes p with $a_p(E)=a_p(E')$ has natural density zero.*

PROOF. If $a_p(E)=a_p(E')$ then $x_p=x_p'$, so (x_p,x_p') lies on the diagonal

$\Delta = \{(u,u) : u \in [-2,2]\}$. The set Δ is closed, and $(\mu_{\mathrm{ST}} \otimes \mu_{\mathrm{ST}})(\Delta) = \int \mu_{\mathrm{ST}}(\{u\})\, \mu_{\mathrm{ST}}(du) = 0$ because μ_{ST} has no atoms. The portmanteau theorem and Theorem 4.11 give

$$\limsup_{x\to\infty} \frac{\#\{p \le x : a_p(E) = a_p(E')\}}{\pi(x)} \le (\mu_{\mathrm{ST}} \otimes \mu_{\mathrm{ST}})(\Delta) = 0. \qquad \square$$

The heuristic behind the Lang Trotter conjecture (Chapter 5) predicts the size of this set. If $a_p(E)$ and $a_p(E')$ were independent integers distributed like $\sqrt{p}$ times Sato Tate variables, they would agree with probability about $p^{-1/2} \int_{-2}^{2} \big(\sqrt{4-u^2}/(2\pi)\big)^2\, du = \frac{8}{3\pi^2}\, p^{-1/2}$, so the number of coincidences up to x should be of order $\sqrt{x}/\log x$, with a constant that also reflects congruences between the two curves.

4.8 The general Sato Tate conjecture

By Theorem 4.3 and Serre's argument, the general Sato Tate conjecture for an abelian variety $A/\mathbb{Q}$ (Conjecture 3.11, with $\mathrm{ST}(A)$ as in Definition 3.10) follows if the L functions attached to the nontrivial irreducible representations of $\mathrm{ST}(A)$ extend holomorphically to a neighbourhood of $\Re s \ge 1$ without zeros on $\Re s = 1$, a consequence of the Langlands conjectures. For the 52 groups of abelian surfaces (Theorem 3.12), the moment statistics of the curves that Fité, Kedlaya, Rotger and Sutherland [192] exhibit for each group, integer combinations of values of the Weyl fingerprint, match the predicted ones numerically; the conjecture is open for $\mathrm{ST}(A) = \mathrm{USp}(4)$ and known in many other cases, as recorded after that theorem.

4.9 Notes

Weyl's paper [576] contains the criterion and the equidistribution of polynomial sequences by differencing. The appendix to Chapter I of Serre's book [499] is the standard reference for the compact group version and the source of Theorem 4.8. Sato found the $\sin^2$ law numerically; Tate derived it from the expectation that algebraic cycles on powers of E account for the poles of symmetric power zeta functions [538]. Birch [64] proved a vertical version: averaged over all elliptic curves over $\mathbb{F}_p$, the even moments of $a_p/\sqrt{p}$ tend to the Catalan numbers as $p \to \infty$. Chapter 6 surveys such vertical laws, and Chapter 9 treats the one for Hecke eigenvalues.

The route through potential automorphy is surveyed in the introductions of [266, 67]; the tensor product device of Harris [240] is one of the ingredients of [67]. Newton and Thorne's proof [419, 420] combines automorphy lifting with an analytic continuation argument along the Coleman Mazur eigencurve, and Thorner [548] used their theorem to prove effective forms of Sato Tate. Serre's

lecture notes [509] give the general formalism, Sutherland's survey [532] describes Sato Tate groups and their moment sequences, and the database [371] records Sato Tate groups of many curves and abelian surfaces. Katz and Sarnak [298] develop the analogous theory over finite fields, where Deligne's equidistribution theorem [141] proves the analogue of the conjecture for families.

Nothing in this chapter proves functoriality: Theorem 4.8 derives a probabilistic consequence, and the arithmetic input of Theorem 4.9 was supplied by others. Proposition 1.5 records the converse limitation: no statistic of the empirical law of (x_p) alone can detect automorphy.

4.10 Exercises

1. Show that the irreducible characters of $\mathrm{U}(1)$ are $\theta \mapsto e^{ih\theta}$, $h \in \mathbb{Z}$, and deduce from Theorem 4.3, in the form for sequences indexed by the integers noted after its proof, Weyl's theorem that $(n\alpha \bmod 1)$ is equidistributed for irrational α. Show that the same sequence is not equidistributed for any rational α by exhibiting a Weyl average that does not tend to zero.

2. Prove that the arcsine law $dx/(\pi\sqrt{4-x^2})$ on $[-2,2]$ has even moments $\binom{2m}{m}$, by counting all (not only nonnegative) paths of ± 1 steps that return to 0. Deduce the moments of μ_{CM}, and compare with the Catalan numbers by the ballot theorem.

3. Let $E/\mathbb{Q}$ be an elliptic curve without complex multiplication, and let g be a Riemann integrable function on $[-2,2]$. Show that $\pi(x)^{-1}\sum_{p\le x} g(x_p)$ tends to $\int g\, d\mu_{\mathrm{ST}}$ (approximate g from above and below by continuous functions), and deduce that the primes with $|a_p| \le \sqrt{p}$ have natural density $\frac{1}{3} + \frac{\sqrt{3}}{2\pi}$.

4. Let $H \subset \mathrm{SU}(2)$ be a finite subgroup and μ the image on $[-2,2]$, under the trace, of the uniform probability measure on H. Show that $\int U_n(x/2)\,\mu(dx)$ is a nonnegative integer for every n, and that it is at least 1 whenever n is a multiple of $2|H|$. Conclude that no such μ can be the limiting law of (x_p) for a newform without complex multiplication.

5. (Probabilistic.) Let $Y_1, Y_2, \dots$ be independent with law μ_{ST}, and $W_n(T) = T^{-1}\sum_{j\le T} U_n(Y_j/2)$. Show that for every finite set S of positive integers the vector $(\sqrt{T}\,W_n(T))_{n\in S}$ converges in law as $T\to\infty$ to a vector of independent standard normal variables. What changes when the Y_j have law μ_{CM}? Compute the means and covariances of the $U_n(Y_j/2)$ in that case.

4.11 Open computational problems

COMPUTATIONAL PROBLEM 4.1 (Weyl averages at scale). Objective: measure how fast the Weyl averages $W_n(x) = \pi(x)^{-1} \sum_{p \le x} U_n(x_p/2)$ tend to zero, and how the rate depends on n, for E_1 and for the 200 random curves of Chapter 5, for $1 \le n \le 50$ and all primes up to 10^{10} (about 4.6×10^8 primes per curve). Route: PARI/GP `ellap` over blocks of primes in parallel, GPU baby step giant step, or Harvey's average polynomial time algorithm [243], accumulating $U_n(x_p/2)$ for all $n \le 50$ by the three term recurrence. Deliverable: plots on logarithmic axes of $\sqrt{\pi(x)}\,|W_n(x)|$ against x and n, with a fitted law for its growth in n. Why it matters: for independent draws from μ_{ST} the quantity $\sqrt{\pi(x)}\,W_n(x)$ is asymptotically standard normal for every n (Exercise 5), while the generalized Riemann hypothesis for $L(s, \mathrm{Sym}^n E)$ gives, through the explicit formula, only $W_n(x) = O(n\, x^{-1/2} \log x \, \log(nNx))$ with N the conductor of E (Chapter 5 of [276]); the data show which of the two describes the dependence on n.

COMPUTATIONAL PROBLEM 4.2 (Fingerprints of genus two Jacobians). Objective: compute the exact Weyl fingerprints of the 52 Sato Tate groups G of abelian surfaces [192], that is the integers $\int \chi_\rho \, d\mu_G = \dim \rho^G$ for all irreducible representations ρ of $\mathrm{USp}(4)$ with highest weight (a, b), $a \ge b \ge 0$, $a + b \le 12$; find a smallest set of such ρ that separates the 52 groups, or the pairs that this range cannot separate; and compare the fingerprints with the empirical Weyl averages of the 66,158 genus two curves in the LMFDB (absolute discriminant at most 10^6), using all primes up to 10^8. Route: branching from $\mathrm{USp}(4)$ to the identity component of G, followed by the action of the component group, in Sage or Magma; L polynomials by the methods of [302] or by the average polynomial time algorithm of [244]. Deliverable: the table of fingerprints, a minimal separating set of representations, and for each curve the empirical averages on that set with the group they single out. Why it matters: Weyl averages are sufficient in the limit for the Bayesian identification of Chapter 29, so a minimal separating set measures how few averages identify a Sato Tate group; agreement for each group that occurs among these curves tests the general Sato Tate conjecture group by group, including groups for which it is not known.

COMPUTATIONAL PROBLEM 4.3 (Joint laws of pairs and twists). Objective: count the coincidences of Corollary 4.12, and contrast the joint laws of unrelated pairs with those of twists. Problem 3.3 already tests the independence of unrelated pairs, with twists as a control, by joint Weyl averages and discrepancy; here the focus is the density zero set of coincidences. Objects: 10^3 pairs of curves of conductor below 10^4 without complex multiplication that are not isogenous over $\bar{\mathbb{Q}}$, and E_1 with its quadratic twists by the fundamental discriminants $d \ne 1$ with $|d| \le 10^3$; all primes up to 10^9. Route: LMFDB curve lists and PARI/GP `ellap` in batch; a

twist has traces $\chi_d(p)\,a_p(E_1)$ and needs no new point counts. Deliverable: for the unrelated pairs, the counts $\#\{p \le x : a_p(E) = a_p(E')\}$ at $x = 10^6, \dots, 10^9$ fitted to $c\sqrt{x}/\log x$ and compared with $c_0 = 16/(3\pi^2) \approx 0.54$, the value for independent Sato Tate traces, corrected for congruences from the mod ℓ images as Lang and Trotter do for $a_p = 0$; and heat maps of the joint Weyl averages $\pi(x)^{-1}\sum_{p \le x} U_m(x_p/2)U_n(x'_p/2)$, $m, n \le 10$, whose limits are 0 for unrelated pairs when $(m, n) \ne (0, 0)$, and for twists δ_{mn} for even n and 0 for odd n. Why it matters: Corollary 4.12 gives only density zero; the constants c are the analogue for pairs of the Lang Trotter constants of Chapter 5, and quadratic twists, up to isogeny over $\mathbb{Q}$, are exactly the pairs excluded from Theorem 4.11.

Chapter 5

Frobenius Data: Statistics at Scale

Sato did not derive the semicircle; he saw it, in point counts of specific curves modulo many primes. Birch and Swinnerton Dyer, at about the same time, found their conjecture in plots of $\log \prod_{p \le x} N_p/p$, with $N_p = \#E(\mathbb{F}_p)$, against $\log \log x$, computed on the EDSAC 2 computer in Cambridge [89]. Both were statistical observations before they were theorems or programmes. This chapter treats Frobenius data in the same spirit and asks three questions of it. How fast does the empirical law approach the Sato Tate law? What correction at the scale of $\log x$ encodes the rank? How often does a trace take a prescribed value? The first leads to discrepancy, the second to the explicit formula, the third to the Lang Trotter conjecture. For each we state what is proved, what is conjectured, and what our computations show, keeping the distance between a stable statistic and a theorem in view.

5.1 Computing traces of Frobenius

Let $E : y^2 = x^3 + ax + b$ with $a, b \in \mathbb{Z}$ and discriminant $\Delta = -16(4a^3 + 27b^2) \neq 0$. For a prime $p \nmid 2\Delta$ we have

$$a_p = p + 1 - \#E(\mathbb{F}_p) = -\sum_{x \in \mathbb{F}_p} \Big(\frac{x^3 + ax + b}{p} \Big),$$

since each x contributes $1 + \big(\frac{x^3+ax+b}{p}\big)$ affine points and there is one point at infinity. This gives an $O(p)$ algorithm, adequate for p up to about 10^5 and convenient for checking.

Baby step giant step methods reduce the cost to $O(p^{1/4})$ group operations: by Hasse, $\#E(\mathbb{F}_p)$ lies in an interval of length $4\sqrt{p}$, and one searches it for a multiple m of the order of a random point; Mestre's trick with the quadratic twist makes the answer unique for $p > 229$. Schoof's algorithm [491] computes a_p mod ℓ from the action of Frobenius on ℓ torsion and recovers a_p by the Chinese remainder theorem in time polynomial in $\log p$; the Elkies and Atkin improvements (SEA) are the method of choice at cryptographic sizes. PARI/GP [432] chooses among these in `ellap`.

For all primes up to a bound the cost per prime is tiny: the 664,577 traces of E_1 below 10^7 took about 20 seconds on one core. For much larger ranges Harvey's

$2m$	E_1	τ	C_m	E_2	$\frac{1}{2}\binom{2m}{m}$
2	1.0000	0.9987	1	0.9995	1
4	1.9994	1.9939	2	2.9980	3
6	4.9976	4.9772	5	9.9916	10
8	13.9930	13.9204	14	34.9642	35
10	41.9808	41.7250	42	125.8514	126
12	131.9488	131.0436	132	461.3908	462

Table 5.1: Even moments of normalised traces: E_1 and E_2 over good primes below 10^7, $\tau(p)/p^{11/2}$ over primes below 10^6, against the Sato Tate (Catalan) and CM predictions.

average polynomial time method [243] computes a_p for all $p \le X$ at once, at a cost per prime polynomial in $\log X$.

5.2 The data sets of this book

All computations were carried out for this book with PARI/GP and Python, on these data.

1. $E_1 : y^2 = x^3 + x + 1$, discriminant -496, conductor $496 = 2^4 \cdot 31$ (LMFDB label 496.a1 [371]), without complex multiplication; all 664,577 good primes below 10^7.
2. $E_2 : y^2 = x^3 - x$, complex multiplication by $\mathbb{Z}[i]$, conductor 32; all good primes below 10^7.
3. Ramanujan's $\tau(p)/p^{11/2}$ for the 78,498 primes below 10^6.
4. Two hundred curves $y^2 = x^3 + ax + b$ with integers a, b drawn uniformly from $[-50, 50]$ (four curves occur twice), singular curves and curves with complex multiplication being discarded; all good primes p with $5 \le p < 10^6$, about 7.8×10^4 per curve.
5. The curves $11a1$, $37a1$, $389a1$ and $5077a1$ of the Cremona tables, of rank 0, 1, 2 and 3 (the smallest conductors of these ranks); all odd primes below 10^7.

Table 5.1 shows the even moments of the normalised traces. They agree with the Catalan numbers C_m (for E_1 and τ) and with the moments $\frac{1}{2}\binom{2m}{m}$ of the CM law μ_{CM} of Section 4.6 (for E_2) to relative errors at most 0.06%, 0.73% and 0.14% respectively, in every case within one standard error of the corresponding average of independent draws. For τ, Deligne's bound is visible directly: the largest value of $|\tau(p)|/p^{11/2}$ below 10^6 is 1.99920.

Write W_j for the Weyl average of $U_j(x_p/2)$ over the good primes $p \le x$ (Chapter 4). Since $x^{2m} = \sum_{k=0}^{m} b_{m,k} U_{2k}(x/2)$ with $b_{m,k} = \binom{2m}{m-k} - \binom{2m}{m-k-1} \ge 0$ and $b_{m,0} = C_m$, the $2m$th moment equals $C_m + \sum_{k=1}^{m} b_{m,k} W_{2k}$. The coefficients sum to $\binom{2m}{m} \sim 4^m/\sqrt{\pi m}$, so high moments amplify small errors; for questions of rate the right object is the distribution function.

5.3 Discrepancy

Let $F_{\mathrm{ST}}(t) = \mu_{\mathrm{ST}}([-2,t])$ and let F_n be the empirical distribution function of the first n normalised traces $x_{p_1}, \dots, x_{p_n}$, where $p_1 < p_2 < \cdots$ are the good primes. The *Kolmogorov discrepancy* is

$$D_n = \sup_{t \in [-2,2]} \bigl| F_n(t) - F_{\mathrm{ST}}(t) \bigr|.$$

By Theorem 4.9 and the continuity of F_{ST}, $D_n \to 0$ for every curve without complex multiplication. The question is the rate.

The classical link between D_n and Weyl sums is the inequality of Erdős and Turán [178].

Theorem 5.1 (Erdős and Turán). *There is an absolute constant C such that for any points $u_1, \dots, u_n$ in $\mathbb{R}/\mathbb{Z}$ and any integer $M \ge 1$,*

$$\sup_{I} \Bigl| \frac{\#\{j \le n : u_j \in I\}}{n} - |I| \Bigr| \le C\Bigl(\frac{1}{M} + \sum_{h=1}^{M} \frac{1}{h} \Bigl| \frac{1}{n} \sum_{j=1}^{n} e^{2\pi i h u_j} \Bigr| \Bigr),$$

the supremum being over intervals I.

An analogue for the Sato Tate measure, with Weyl averages of Chebyshev polynomials U_m in place of exponential sums, was proved by Niederreiter [422]. Combined with bounds for Weyl averages from the analytic properties of $L(s, \mathrm{Sym}^m E)$, it converts zero free regions into rates for D_n. In the opposite direction, sharp rates for D_n constrain the zeros.

Theorem 5.2 (Akiyama and Tanigawa [11]; Mazur [386]). *Let $E/\mathbb{Q}$ be an elliptic curve without complex multiplication. If $D_n = O(n^{-1/2+\varepsilon})$ for every $\varepsilon > 0$, then the Riemann hypothesis holds for $L(s, \mathrm{Sym}^m E)$ for every $m \ge 1$.*

Akiyama and Tanigawa conjectured the rate in the hypothesis and proved the case $m = 1$, the Riemann hypothesis for $L(s, E)$. The extension to all m is stated by Mazur under analytic hypotheses on $L(s, \mathrm{Sym}^m E)$ that the theorem of Newton and Thorne [419, 420] now supplies (Chapter 4). The argument is short. The function $U_m(x/2)$ has bounded variation on $[-2,2]$ and Sato Tate mean zero, so integration by parts gives $\bigl|\sum_{j \le n} U_m(x_{p_j}/2)\bigr| \le V_m \, n D_n$, with V_m its total variation, and the hypothesis gives $\sum_{p \le x} U_m(x_p/2) \ll x^{1/2+\varepsilon}$. In the unitary normalisation,

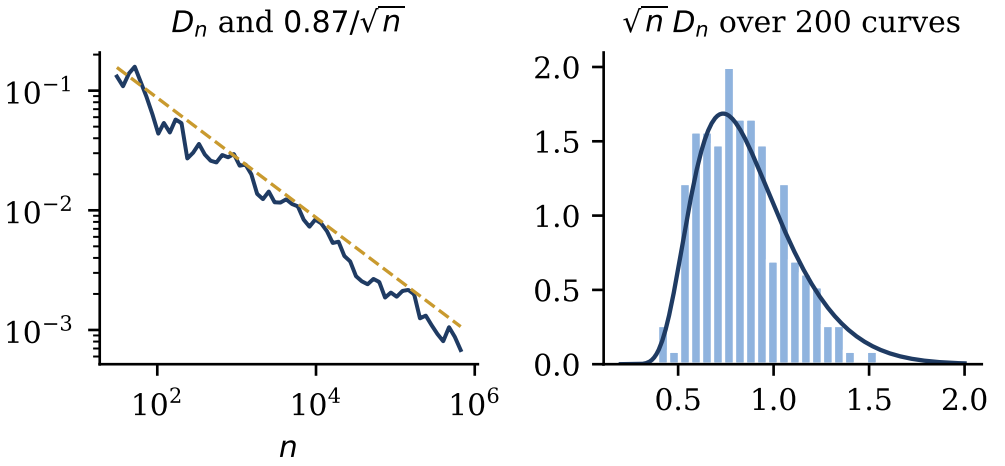


Figure 5.1: Left: Kolmogorov discrepancy D_n for E_1 against $0.87/\sqrt{n}$, the asymptotic mean of D_n for n independent draws from μ_{ST}. Right: histogram of $\sqrt{n}\,D_n$ over 200 random curves without complex multiplication (primes below 10^6) against the density of the Kolmogorov distribution.

with centre $s = \frac{1}{2}$, partial summation shows that $\sum_p U_m(x_p/2)p^{-s}$ converges on $\Re s > \frac{1}{2} + \varepsilon$, while the prime powers and the bad primes contribute a function holomorphic on $\Re s > \frac{1}{2}$. Hence $\log L(s, \mathrm{Sym}^m E)$ continues holomorphically to $\Re s > \frac{1}{2} + \varepsilon$, and $L(s, \mathrm{Sym}^m E)$ has no zeros there (Exercise 3); since ε is arbitrary, the functional equation does the rest. Results in the other direction are much weaker. Using the functoriality of Newton and Thorne, Thorner [548] proved unconditionally that $D_n \ll \log(N \log p_n)/\sqrt{\log p_n}$, with N the conductor of E and an absolute implied constant. For squarefree N and assuming the Riemann hypothesis for every $L(s, \mathrm{Sym}^m E)$, Rouse and Thorner [466] obtained the explicit form $D_n \ll p_n^{-1/4} \log(N p_n)$ of V. K. Murty's bound [409] of order $n^{-1/4}$ up to logarithms, still far from $n^{-1/2}$.

5.3.1 Our evidence

For E_1 the scaled discrepancy $\sqrt{n}\,D_n$ equals 0.60, 0.79, 0.80, 0.66, 0.55 at $n = 10^2, 10^3, 10^4, 10^5$ and at $n = 664{,}577$ (all good primes below 10^7). The values stay of order one over almost four orders of magnitude (Figure 5.1), as for independent draws; five values of a statistic whose spread under independence is about 0.26 cannot reveal a slow trend. For $\tau(p)$ over primes below 10^6 the value is 0.646.

If the traces were independent draws from μ_{ST}, Kolmogorov's theorem [328] would give $\sqrt{n}\,D_n \Rightarrow K = \sup_{0 \le u \le 1} |B(u)|$, with B a Brownian bridge (Exercise 2);

K has mean $\sqrt{\pi/2}\,\log 2 = 0.8687$, standard deviation 0.2603, and 10th, 50th and 90th percentiles 0.571, 0.828, 1.224. Over our 200 random curves, $\sqrt{n}\,D_n$ has mean 0.837, standard deviation 0.219, and percentiles 0.576, 0.828, 1.147. A Kolmogorov Smirnov test of the 200 values against the law of K gives $p = 0.40$, so it detects no departure from the Kolmogorov law. It has limited power, however, against a contraction of the spread that moves the mean far less than the standard deviation, as here. For 200 independent copies of K the standard errors of the sample mean and standard deviation are 0.018 and 0.016; the observed mean is 1.7 standard errors below 0.8687, the standard deviation 2.7 standard errors below 0.2603, and the upper tail is visibly lighter (Figure 5.1, right). Across curves, the discrepancy at about 7.8×10^4 primes has the size expected for independent draws and is, if anything, less dispersed.

5.3.2 Interpretation

Three qualifications apply.

First, the sequence (x_p) is deterministic. A p value of 0.40 says that a summary of a deterministic object is unremarkable under a reference law; it does not say the object is random. What is random in our experiment is the choice of curve, so the law of $\sqrt{n}\,D_n$ across curves is a vertical statistic, closer to the family averages of Chapter 6 than to a statement about one curve.

Second, the explicit formula predicts structure that independent sampling lacks. In the unitary normalisation let r_m be the order of $L(s, \mathrm{Sym}^m E)$ at $s = \frac{1}{2}$. The prime squares enter through the identity $\operatorname{tr} \mathrm{Sym}^m(g^2) = \sum_{j=0}^{m} (-1)^j \operatorname{tr} \mathrm{Sym}^{2m-2j}(g)$ (Exercise 4), whose right side has Sato Tate mean $(-1)^m$, and the Riemann hypothesis for $L(s, \mathrm{Sym}^m E)$ gives

$$\sum_{p \le x} U_m(x_p/2) \log p = -\big((-1)^m + 2r_m\big)\sqrt{x} - \sum_{\gamma \neq 0} \frac{x^{1/2+i\gamma}}{\frac{1}{2} + i\gamma} + o(\sqrt{x}),$$

with γ running over the ordinates of the nontrivial zeros $\frac{1}{2} + i\gamma$ of $L(s, \mathrm{Sym}^m E)$, counted with multiplicity. By the method of Rubinstein and Sarnak [470], the left side divided by $\sqrt{x}$ has a limiting distribution in logarithmic scale, with mean $-((-1)^m + 2r_m)$. For $m = 1$, where $r_1 = r$ is the analytic rank of E, the mean is $1 - 2r$: this is the rank bias of Section 5.4. By partial summation $\sum_{p \le x} U_m(x_p/2)$ is then of order $\sqrt{x}/\log x$, a factor $\sqrt{\log x}$ below the $\sqrt{\pi(x)}$ of independent sampling. Thus, for each fixed m, $\sqrt{n}\,W_m \to 0$ in logarithmic density, n being the number of good primes up to x, whereas for independent draws $\sqrt{n}\,W_m$ is asymptotically standard normal. Since $\sqrt{n}\,W_m = -\frac{1}{2}\int_{-2}^{2} \sqrt{n}\,(F_n - F_{\mathrm{ST}})(t)\,U_m'(t/2)\,dt$, it follows that under the Riemann hypothesis for every $L(s, \mathrm{Sym}^m E)$ the process $\sqrt{n}\,(F_n - F_{\mathrm{ST}})$ has no nondegenerate limit in law in logarithmic scale, Brownian bridge or other: every Chebyshev coefficient of a limit would vanish.

The decay is slow, however. The variance of the limiting distribution grows with the logarithm of the analytic conductor of $L(s, \mathrm{Sym}^m E)$, which is at least linear in m, and heuristically frequency m is damped only once $\log x$ is large compared with it. Below 10^7 this can affect only the first few frequencies, consistent with a spread close to that of the Kolmogorov law; whether it accounts for the smaller spread across curves is open. For E_1 at $x = 10^7$ the twelve values of $\sqrt{n}\, W_m$, $m \le 12$, have mean square 0.49 against 1 for independent draws; a sum of twelve squares of independent standard normal variables falls this low with probability 0.08, so the data neither confirm nor exclude damping at this scale. The first of them, -0.36, may be compared with the leading order prediction $(1 - 2r)/\sqrt{\log x} = -0.25$ for its mean, E_1 having analytic rank $r = 1$.

Third, statements at the scale $n^{-1/2}$ are beyond present methods. For independent draws $\sqrt{n}\, D_n = O(\sqrt{\log\log n})$ almost surely, by the law of the iterated logarithm; for the traces, even $\sqrt{n}\, D_n = O(n^{\varepsilon})$ for every $\varepsilon > 0$ would imply the Riemann hypothesis for every $L(s, \mathrm{Sym}^m E)$, by the theorem above. We state the questions as follows.

CONJECTURE 5.3 (Size of the Sato Tate discrepancy; open). *Let $E/\mathbb{Q}$ be without complex multiplication, and let n be the number of good primes up to x.*

(a) (Akiyama and Tanigawa.) $D_n = O(n^{-1/2+\varepsilon})$ for every $\varepsilon > 0$.

(b) $\sqrt{n}\, D_n \to 0$ in logarithmic density: for every $\delta > 0$ the set of x with $\sqrt{n}\, D_n > \delta$ has logarithmic density zero.

Part (b) is our reading of the explicit formula. For each fixed frequency the corresponding statement, $\sqrt{n}\, W_m \to 0$, follows from the Riemann hypothesis for $L(s, \mathrm{Sym}^m E)$, as shown above; the open point is uniformity in m, since D_n sees frequencies up to about $\sqrt{n}$. Part (b) strengthens the conditional statement above that no nondegenerate limit exists along the primes of one curve. Open Problem 30.3 of Chapter 30 asks whether a Brownian bridge arises under another randomisation, for instance over curves of growing conductor, and Computational Problem 5.1 proposes a test.

5.4 THE EXPLICIT FORMULA AND RANK BIAS

Let $E/\mathbb{Q}$ have conductor N and analytic rank $r = \mathrm{ord}_{s=1} L(s, E)$. Write

$$S(x) = \sum_{p \le x,\ p \nmid N} \frac{a_p \log p}{p} = \sum_{p \le x,\ p \nmid N} \frac{x_p \log p}{\sqrt{p}}.$$

For $p \nmid N$ we have $\log(N_p/p) = -a_p/p + (1 - x_p^2/2)/p + O(p^{-3/2})$, and $\sum_{p \nmid N} (x_p^2 - 1)/p$ converges by the prime number theorem for $L(s, \mathrm{Sym}^2 E)$; so the asymptotic $\prod_{p \le x} N_p/p \sim A(\log x)^r$ that Birch and Swinnerton Dyer read from

their plots [89] amounts to $\sum_{p \le x,\, p \nmid N} a_p/p = (\tfrac{1}{2} - r)\log\log x + O(1)$. The same prediction with weights $\log p$ is

$$S(x) \approx \Big(\frac{1}{2} - r\Big)\log x.$$

We explain where the two terms come from. Normalise $L(s,E)$ so that the centre is at $s = 1$ and write $a_p = 2\sqrt{p}\cos\theta_p$. Then

$$-\frac{L'}{L}(s,E) = \sum_n \frac{\Lambda_E(n)}{n^s}, \qquad \Lambda_E(p^k) = p^{k/2}\,2\cos(k\theta_p)\log p \quad (p \nmid N).$$

Assuming the Riemann hypothesis for $L(s,E)$, the explicit formula, applied with a smooth cutoff at the centre, expresses $\sum_{n \le x} \Lambda_E(n)/n$ as $-r\log x$ (the contribution of the zero of order r at $s = 1$) minus a sum over the other zeros $\rho = 1 + i\gamma$, oscillating like $\sum_{\gamma \ne 0} x^{i\gamma}/(i\gamma)$, plus a bounded term. The prime terms give $S(x)$. The prime square terms give

$$\sum_{p \le \sqrt{x}} \frac{2\cos(2\theta_p)\log p}{p} = \sum_{p \le \sqrt{x}} \frac{(U_2(x_p/2) - 1)\log p}{p} = -\frac{1}{2}\log x + o(\log x),$$

where we used $2\cos 2\theta = U_2(\cos\theta) - 1$, Mertens' theorem $\sum_{p \le y} \log p/p = \log y + O(1)$, and the fact that $\sum_{p \le y} U_2(x_p/2)\log p/p = o(\log y)$, which follows from the holomorphy and nonvanishing of $L(s, \mathrm{Sym}^2 E)$ on $\Re s = 1$ in the unitary normalisation (Chapter 4). Cubes and higher powers contribute $O(1)$. Hence

$$S(x) - \frac{1}{2}\log x = -r\log x - \sum_{\gamma \ne 0} \frac{x^{i\gamma}}{i\gamma} + O(1),$$

heuristically. The $\tfrac{1}{2}$ is the shadow of the symmetric square: the prime square coefficient $U_2(x_p/2) - 1$ has average -1 over primes because $L(s, \mathrm{Sym}^2 E)$ has no pole at $s = 1$. Under the Riemann hypothesis for $L(s,E)$ the oscillatory sum has bounded mean square in $\log x$, so $S(x)/\log x \to \tfrac{1}{2} - r$ in logarithmic mean; a zero off the line would instead produce oscillations of exponential amplitude in $\log x$. The pointwise statement is not known to follow from the Riemann hypothesis, but Kim and Murty [314] showed that if $S(x)/\log x$ has a limit, the limit is $\tfrac{1}{2} - r$. For the product itself, Goldfeld [219] showed that if $\prod_{p \le x} N_p/p \sim A(\log x)^r$ for some $A > 0$ and real r, where N_p counts the nonsingular points when $p \mid N$, then $L(s,E)$ satisfies the Riemann hypothesis, $r = \mathrm{ord}_{s=1} L(s,E)$, and $A = \sqrt{2}\, e^{r\gamma_{\mathrm{E}}}\, r!/L^{(r)}(E,1)$, with γ_{E} Euler's constant. The factor $\sqrt{2}$ comes from the same prime squares, which the product cuts off at $p \le x$ rather than at $p^2 \le x$; Conrad [113] relates it to second moments for general L functions. Kuo and Murty

Curve	r	$\frac{1}{2}-r$	slope	computed $S(10^7)$	$(\frac{1}{2}-r)\log 10^7$
$11a1$	0	0.5	0.46	6.46	8.06
$37a1$	1	-0.5	-0.59	-9.76	-8.06
$389a1$	2	-1.5	-1.43	-26.73	-24.18
$5077a1$	3	-2.5	-2.51	-43.54	-40.30

Table 5.2: Rank bias. The slope of $S(x) = \sum_{p \le x} a_p \log p/p$ against $\log x$ is the chord slope $(S(10^7) - S(10^3))/\log 10^4$, compared with $\frac{1}{2} - r$. The computed S runs over the odd primes, bad prime included; adding $p = 2$, where $a_2 = -2$ for all four curves, would lower every value of S by $\log 2$ and leave the slopes unchanged.

[336] and, independently, Conrad showed that the asymptotic for the product is equivalent to $\sum_{n \le x} \Lambda_E(n) = o(x \log x)$, a bound stronger than the $O(x \log^2 x)$ that the Riemann hypothesis gives.

Table 5.2 and Figure 5.2 show the four curves. The slopes match $\frac{1}{2} - r$ to within 0.1 in every case, and the computed $S(10^7)$ lies below $(\frac{1}{2} - r)\log 10^7$ by between 1.6 and 3.2, offsets that the explicit formula attributes to the $O(1)$ term and the low lying zeros. The rank is visible with a ruler, and so is the extra $\frac{1}{2}$.

Nothing here bears on the Birch and Swinnerton Dyer conjecture itself: r is the analytic rank, and our data test the explicit formula, not the equality of ranks (known when the analytic rank is at most 1, by Gross and Zagier [228] and Kolyvagin [329]).

5.5 The Lang Trotter conjecture

For a fixed integer t let $\pi_{E,t}(x) = \#\{p \le x : p \nmid N,\ a_p = t\}$. When $t = 0$ and $p \ge 5$, $a_p = 0$ means that p is a supersingular prime of E.

Conjecture 5.4 (Lang and Trotter [353]). *Let $E/\mathbb{Q}$ be without complex multiplication and $t \in \mathbb{Z}$. There is a constant $C_{E,t} \ge 0$ such that*

$$\pi_{E,t}(x) \sim C_{E,t} \frac{\sqrt{x}}{\log x},$$

where for $C_{E,t} = 0$ this means that $\pi_{E,t}(x)$ is bounded.

5.5.1 The probabilistic heuristic

The conjecture concerns a set of primes of density zero, about which Sato Tate says nothing; the heuristic is a local limit version of it. The integer a_p lies in an interval of length $4\sqrt{p}$, so the lattice spacing is 1 for a_p and $1/\sqrt{p}$ for x_p. If x_p followed

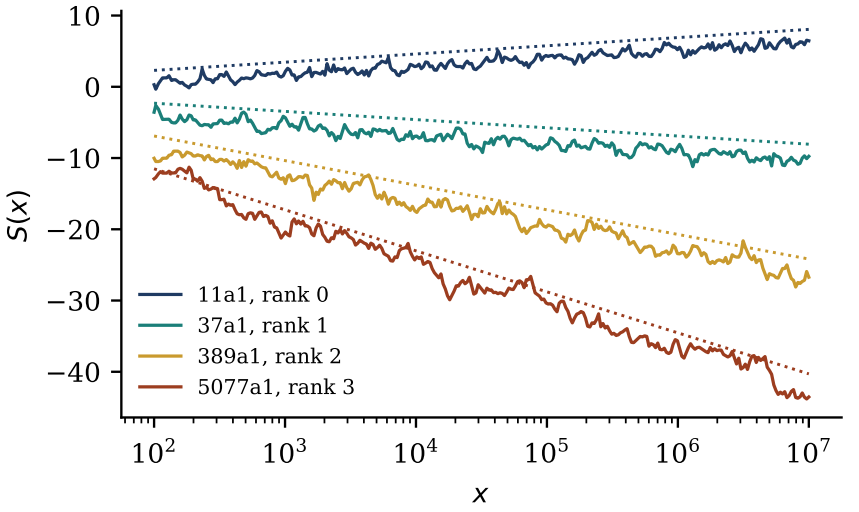


Figure 5.2: $S(x) = \sum_{p \le x} a_p \log p/p$ for $11a1$, $37a1$, $389a1$, $5077a1$ (ranks 0 to 3), $10^2 \le x \le 10^7$, summed over odd primes as in Table 5.2, against the lines $(\frac{1}{2} - r)\log x$.

μ_{ST} with integrality the only constraint, then

$$\mathbb{P}(a_p = t) \approx \mathbb{P}\Big(|x_p - t/\sqrt{p}| \le \frac{1}{2\sqrt{p}}\Big) \approx \frac{1}{\sqrt{p}} \cdot \frac{\sqrt{4 - t^2/p}}{2\pi} \approx \frac{1}{\pi\sqrt{p}},$$

since the semicircle density at 0 is $\frac{1}{2\pi} \cdot 2 = \frac{1}{\pi}$. Summing over p, the expected count is

$$\sum_{p \le x} \frac{1}{\pi\sqrt{p}} \approx \frac{1}{\pi} \int_2^x \frac{du}{\sqrt{u}\log u} = \frac{2}{\pi}\,\pi_{1/2}(x),$$

where $\pi_{1/2}(x) = \int_2^x du/(2\sqrt{u}\log u) \sim \sqrt{x}/\log x$, using the prime number theorem in the form that primes near u have density $1/\log u$. This is the shape $c\sqrt{x}/\log x$ with $c = 2/\pi$.

Congruence conditions correct this. Modulo a prime ℓ, a_p is the trace of Frob_p in the mod ℓ Galois image $G(\ell) \subset \mathrm{GL}_2(\mathbb{Z}/\ell\mathbb{Z})$, and by Chebotarev [102] the probability that $a_p \equiv t \pmod{\ell}$ is $\#\{g \in G(\ell) : \operatorname{tr} g = t\}/\#G(\ell)$ rather than $1/\ell$. In the same way the probability that ℓ divides $|E(\mathbb{F}_p)| = p + 1 - a_p$, the value at 1 of the characteristic polynomial of Frob_p, is the proportion of elements of $G(\ell)$ with eigenvalue 1; Koblitz [326] built from these proportions his conjecture on the primes p for which $|E(\mathbb{F}_p)|$ is prime, and Chapter 22 reads their complements as the shapes of Gamma laws. By Serre's open image theorem [502] there is an integer m such that the image of Galois in $\mathrm{GL}_2(\hat{\mathbb{Z}})$ is the full inverse image of its

reduction $G(m) \subset \mathrm{GL}_2(\mathbb{Z}/m\mathbb{Z})$, and the constant of Lang and Trotter is

$$C_{E,t} = \frac{2}{\pi} \cdot \frac{m\,\#\{g \in G(m) : \operatorname{tr} g = t\}}{\#G(m)} \cdot \prod_{\ell \nmid m} \frac{\ell\,\#\{g \in \mathrm{GL}_2(\mathbb{F}_\ell) : \operatorname{tr} g = t\}}{\#\mathrm{GL}_2(\mathbb{F}_\ell)}.$$

The factor at m does not in general split over the primes dividing m, because the images modulo different primes can be entangled.

For $t = 0$ and full image $\mathrm{GL}_2(\mathbb{F}_\ell)$ the number of trace zero elements is $\ell^3 - \ell^2$ (Exercise 5), so the factor at ℓ is $\ell \cdot \ell^2(\ell-1)/\big((\ell^2-1)(\ell^2-\ell)\big) = \ell^2/(\ell^2-1)$; for $\ell = 2$ it is $4/3$, as four of the six elements of $\mathrm{GL}_2(\mathbb{F}_2) \cong S_3$ have trace 0. The mod 2 image of E_1 is all of $\mathrm{GL}_2(\mathbb{F}_2)$, since $x^3 + x + 1$ is irreducible over $\mathbb{Q}$ with nonsquare discriminant -31. If the image of Galois were all of $\mathrm{GL}_2(\hat{\mathbb{Z}})$, the constant would be

$$\frac{2}{\pi} \prod_{\ell} \frac{\ell^2}{\ell^2 - 1} = \frac{2}{\pi}\,\zeta(2) = \frac{\pi}{3} \approx 1.047.$$

Over $\mathbb{Q}$ this never happens [502]. For E_1 the field $\mathbb{Q}(\sqrt{-31})$ lies in $\mathbb{Q}(E_1[2])$, which contains the square root of the discriminant, and in $\mathbb{Q}(\zeta_{31}) \subset \mathbb{Q}(E_1[31])$, so the images modulo 2 and 31 are entangled. According to the LMFDB [371], E_1 is a Serre curve: its image in $\mathrm{GL}_2(\hat{\mathbb{Z}})$ has index 2, the least possible, and is cut out by this single condition. The constant is then a finite computation in $\mathrm{GL}_2(\mathbb{Z}/62\mathbb{Z})$ (Exercise 5):

$$C_{E_1,0} = \frac{\pi}{3}\Big(1 + \frac{1}{62}\Big) \approx 1.064.$$

5.5.2 What is known

The conjecture is open for every curve without complex multiplication and every t. Elkies [175] proved that every elliptic curve over $\mathbb{Q}$ has infinitely many supersingular primes; the lower bounds for $\pi_{E,0}(x)$ that are known are far below $\sqrt{x}/\log x$. Upper bounds $\pi_{E,t}(x) = O(x^\theta)$ with $\theta < 1$ follow from effective Chebotarev under the generalized Riemann hypothesis (Serre [506]; Murty, Murty and Saradha [411]), and averages over families of curves confirm the order $\sqrt{x}/\log x$ (Fouvry and Murty [199] for $t = 0$, David and Pappalardi [128] for $t \neq 0$). For a curve with complex multiplication half the primes are supersingular, which is why the conjecture excludes such curves.

5.5.3 Our counts

For E_1 the supersingular primes begin $3, 17, 179, 227, 523, 1031$. The counts below $x = 10^4, 10^5, 10^6, 10^7$ are 10, 23, 79, 210, and the ratios $\pi_{E_1,0}(x)/\pi_{1/2}(x)$ are 0.66, 0.65, 0.89, 0.91 (Figure 5.3). At $x = 10^7$, where $\pi_{1/2}(x) = 231.5$, the Poisson standard error of the ratio is $\sqrt{210}/231.5 \approx 0.06$. The data indicate a constant of

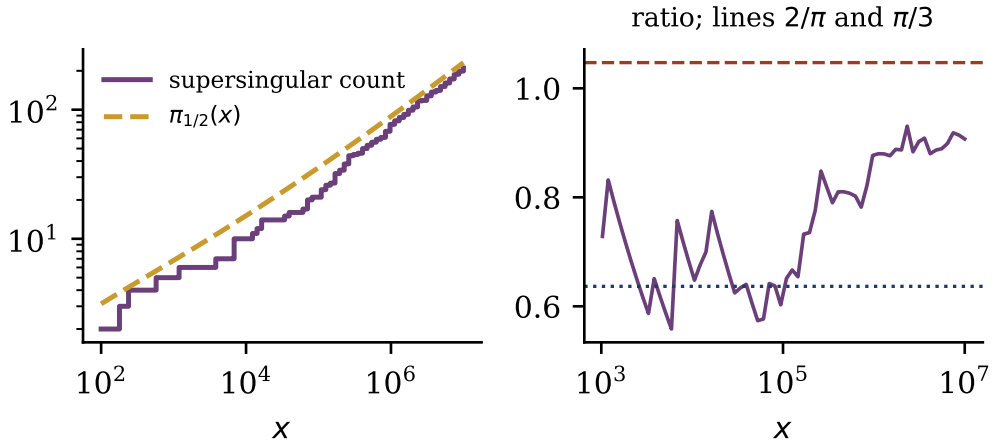


Figure 5.3: Supersingular primes of E_1 below 10^7: the count $\pi_{E_1,0}(x)$ against $\pi_{1/2}(x) = \int_2^x du/(2\sqrt{u}\log u)$ (left), and their ratio (right), with the bare heuristic constant $2/\pi$ and the value $\pi/3$ for a surjective adelic image. The Lang Trotter constant of E_1 is $(\pi/3)(1+1/62) \approx 1.064$.

order one, above the bare $2/\pi \approx 0.64$ and more than two standard errors below $C_{E_1,0} \approx 1.064$; the ratio has stayed near 0.9 since 10^6. Counts of a few hundred are a consistency check only; Computational Problem 5.2 asks for counts up to 10^{12}.

5.6 Notes

Schoof's algorithm [491] was the first polynomial time point counting method. Kedlaya and Sutherland [302] computed the distributions of normalised L polynomials of hyperelliptic curves of genus at most 3 at scale, the numerical side of the classification of Sato Tate groups of abelian surfaces [192]. Harvey's average polynomial time method [243], made practical by Harvey and Sutherland [244], extends such computations much further; Sutherland's survey [532] covers the methods, and the LMFDB [371] holds much of the resulting data. Akiyama and Tanigawa [11] supported their conjecture with numerical computations of the discrepancy.

The rank bias was implicit in the computations of Birch and Swinnerton Dyer [89] and made precise by Goldfeld [219]. Sarnak's letter [480] treats the biases in the signs of $\tau(p)$ and a_p through all the symmetric powers, and Mazur [386] discusses the meaning of such error terms. The mechanism is the one behind Chebyshev's bias, where the prime squares produce the bias [470]. Chapter 29 uses the Sato Tate law, with a small atom at 0 for the supersingular primes, as a likelihood for Bayesian inference.

5.7 Exercises

1. Prove that $a_p = -\sum_{x \bmod p} \left(\frac{x^3+ax+b}{p}\right)$ for $p \nmid \Delta$, and deduce that a_p is even if and only if $x^3 + ax + b$ has a root modulo p. Using the Chebotarev density theorem, compute the proportion of primes with a_p even for E_1.

2. Let $Y_1, \dots, Y_n$ be independent with law μ_{ST}, and let F_n be their empirical distribution function. Show that $\sqrt{n}\,(F_n - F_{\mathrm{ST}})$ converges in law to $B \circ F_{\mathrm{ST}}$ with B a Brownian bridge, and that $\sqrt{n}\,D_n \Rightarrow \sup_u |B(u)|$. Verify that $\mathbb{E}\sup_u |B(u)| = \sqrt{\pi/2}\,\log 2$.

3. Suppose that $|\sum_{p \le x} U_m(x_p/2)| \le A_m x^{1/2+\varepsilon}$ for all $x \ge 2$. Show that the series $\sum_p U_m(x_p/2)\,p^{-s}$ converges for $\Re s > \frac{1}{2} + \varepsilon$, and deduce that $L(s, \mathrm{Sym}^m E)$ has no zeros with $\Re s > \frac{1}{2} + \varepsilon$ in the unitary normalisation. (Show that the prime powers and the bad primes contribute to $\log L(s, \mathrm{Sym}^m E)$ a function holomorphic on $\Re s > \frac{1}{2}$.)

4. (Prime squares and bias.) (a) Show that for $g \in \mathrm{SU}(2)$

$$\operatorname{tr}\mathrm{Sym}^m(g^2) = \sum_{j=0}^{m} (-1)^j \operatorname{tr}\mathrm{Sym}^{2m-2j}(g),$$

   so that $\operatorname{tr}\mathrm{Sym}^m(g^2)$ has Sato Tate mean $(-1)^m$. (b) Let x_p be independent with law μ_{ST}. Show that $\sum_{p \le x} x_p \log p/\sqrt{p}$ has mean 0 and variance $\sim \frac{1}{2}\log^2 x$, so that a random model without the explicit formula would not produce a drift of size $\frac{1}{2}\log x$ distinguishable from noise. Explain why the arithmetic sum has much smaller fluctuations.

5. (a) Show that the number of $g \in \mathrm{GL}_2(\mathbb{F}_\ell)$ with trace t is $\ell^3 - \ell^2$ for $t = 0$, and compute it for $t \neq 0$. Deduce the local Lang Trotter factor at ℓ for $t \neq 0$ when the image is full. (b) Since E_1 is a Serre curve, its image in $\mathrm{GL}_2(\hat{\mathbb{Z}})$ is the subgroup of index 2 on which $\mathrm{sgn}(g \bmod 2) = \left(\frac{\det g}{31}\right)$, with sgn the sign character of $\mathrm{GL}_2(\mathbb{F}_2) \cong S_3$. Show that $C_{E_1,0} = (\pi/3)(1 + e_2 e_{31})$, where e_2 is the average of sgn over the trace zero elements of $\mathrm{GL}_2(\mathbb{F}_2)$ and e_{31} that of $\left(\frac{\det g}{31}\right)$ over the trace zero elements of $\mathrm{GL}_2(\mathbb{F}_{31})$. Show that $e_2 = -\frac{1}{2}$ and, by counting the solutions of $a^2 + bc = -d$, that $e_{31} = -\frac{1}{31}$, so that $C_{E_1,0} = (\pi/3)(1 + \frac{1}{62})$.

6. Under the heuristic that the events $\{a_p = 0\}$ are independent with probabilities $C/(2\sqrt{p})$, show that $\pi_{E,0}(x)$ has mean and variance both asymptotic to $C\,\pi_{1/2}(x)$ and is asymptotically normal. Using $\pi_{1/2}(10^7) = 231.5$, construct an approximate 95% confidence interval for C from the count 210 of supersingular primes of E_1 below 10^7, and compare it with $2/\pi$ and with $C_{E_1,0}$.

5.8 Open computational problems

Computational Problem 5.1 (Brownian bridge test at scale). Objective: test Conjecture 5.3(b) and the predictions for single frequencies in Section 5.3, for E_1 with all primes up to 10^{10} (about 4.6×10^8 primes) and for 10^3 random curves with all primes up to 10^9. Route: GPU or multicore baby step giant step, or Harvey's average polynomial time algorithm [243]; streaming accumulation of F_n on a fixed grid and of the Weyl averages W_m for $m \le 100$. Deliverable: plots of $\sqrt{n}\, D_n$ and of $\sqrt{n}\, W_m$ against $\log x$, the latter with the means $-((-1)^m + 2r_m)/\sqrt{\log x}$ that the explicit formula predicts; the empirical covariance function of the process at $n = 10^8, 10^9$, compared with $\min(s,t) - st$ after the transform F_{ST}; Kolmogorov Smirnov, Anderson Darling and variance tests of the family values of $\sqrt{n}\, D_n$ against the law of K, with the atom at 0 created by the supersingular primes treated separately. Why it matters: a decline of $\sqrt{n}\, D_n$, or of the low frequency Weyl averages, would show the zeros of symmetric power L functions acting on the empirical process and would separate the traces from independent draws; the family test decides whether the smaller spread seen at 10^6 persists.

Computational Problem 5.2 (Lang Trotter constants). Objective: test the Lang Trotter constants $C_{E,t}$, $t = 0, 1, 2$, of E_1 and of two further Serre curves against counts up to 10^{12}. Route: adelic images from the LMFDB [371] and the algorithms of Zywina [598] and of Rouse, Sutherland and Zureick Brown [468]; exact evaluation of the Lang Trotter formula by counting in $G(m)$, as in Exercise 5 for $C_{E_1,0}$; counts of $a_p = t$ by testing $(p + 1 - t)P = O$ for a few random points P, much cheaper than computing a_p, with hits confirmed by `ellap`. Deliverable: the constants to six digits; the ratios $\pi_{E,t}(x)/(C_{E,t}\, \pi_{1/2}(x))$ for $x = 10^8, \dots, 10^{12}$ with Poisson error bars; a fit of the rate at which they approach 1. Why it matters: it is the first serious test of the conjecture's constants for these curves, it decides whether the deficit of E_1 at 10^7 is a slow transient, and it measures whether the independence heuristic survives the move from density one statements to density zero ones.

Computational Problem 5.3 (Bias at high rank). Objective: check the slope $\frac{1}{2} - r$ for curves of rank 4 to 8 and beyond, including Elkies's curves of large rank, using all primes below 10^9. Route: curves of rank 4 to 8 of small conductor and Elkies's curves, from the LMFDB [371] and the literature, with r the analytic rank, which for $r \ge 4$ is determined only numerically; PARI/GP `ellap` for the traces; regression of $S(x)$ on $\log x$ with a model for the oscillatory term from the low lying zeros computed by `lfunzeros` where the conductor permits. Deliverable: a table of fitted slopes against $\frac{1}{2} - r$ with confidence bands, and residual plots showing the contribution of the first zeros. Why it matters: for large conductor the oscillatory term is larger and the logarithmic convergence slower, and the data

show at what scale the rank becomes legible from Frobenius data alone, a question of practical importance for rank estimation.

COMPUTATIONAL PROBLEM 5.4 (A GPU point counting pipeline). Objective: build an open pipeline that computes a_p for all primes $p \leq 10^{10}$ and all curves of conductor up to 10^5 in the Cremona database, with verification. Route: batched baby step giant step on GPUs with Montgomery arithmetic, cross checked against PARI/GP on random samples and against the Hasse bound and known L values via the approximate functional equation. Deliverable: compressed trace tables, a reproducible build, and timing benchmarks per prime. Why it matters: every statistical question in this chapter and in Chapters 6 and 29 is limited by data scale, and a shared resource at 10^{10} would move discrepancy, bias and Lang Trotter questions into the regime where finite sample effects are small.

CHAPTER 6

RANDOM MATRICES AND FAMILIES OF L FUNCTIONS

In spring 1972 Hugh Montgomery visited the Institute for Advanced Study and, over tea, described to Freeman Dyson the function $1-(\sin \pi u/\pi u)^2$ that he had found governing the pair correlation of zeros of the Riemann zeta function. Dyson recognised it at once as the pair correlation of eigenvalues of a random Hermitian matrix from the Gaussian unitary ensemble. That conversation set the agenda for half a century: the conjecture, supported by theorems for restricted test functions and by extensive computation, that zeros of L functions have the local statistics of eigenvalues of random matrices from compact classical groups. This chapter develops the theory, from the Weyl integration formula to the Katz and Sarnak symmetry types, low lying zeros and the Keating and Snaith moment conjectures. Throughout we keep two random matrix limits apart: the Sato Tate limit, in which a fixed compact group of small rank governs Frobenius classes at varying primes, and the Katz and Sarnak limit, in which groups of growing dimension govern the zeros near a fixed point.

6.1 HAAR MEASURE AND THE WEYL INTEGRATION FORMULA

Every compact group K carries a unique left and right invariant probability measure, the Haar measure (Chapter 4). When K is a connected Lie group, the integral $\int_K f\,dh$ of a class function f (one constant on conjugacy classes) reduces to an integral over a maximal torus T, because every element of K is conjugate into T and two elements of T are conjugate in K precisely when they lie in one orbit of the Weyl group $W = N(T)/T$. The density of the pushforward to T/W is given by Weyl's formula [577].

THEOREM 6.1 (Weyl integration formula). *Let K be a compact connected Lie group with maximal torus T, Weyl group W and set of roots Φ. For every continuous class function f,*

$$\int_K f\,dh = \frac{1}{|W|}\int_T f(t)\prod_{\alpha\in\Phi}\bigl(1-\alpha(t)\bigr)\,dt,$$

where dt is Haar probability measure on T.

For the unitary group $\mathrm{U}(N)$, with eigenvalues $e^{i\theta_1},\dots,e^{i\theta_N}$, the formula gives

the joint density

$$\frac{1}{(2\pi)^N N!}\prod_{j<k}\left|e^{i\theta_j}-e^{i\theta_k}\right|^2$$

on $[0,2\pi)^N$. This is the circular unitary ensemble (CUE). For the compact symplectic group $\mathrm{USp}(2g)$, eigenvalues come in pairs $e^{\pm i\theta_j}$ with $\theta_j \in [0,\pi]$, and the joint density of $(\theta_1,\dots,\theta_g)$ is

$$\frac{2^{g^2}}{g!\,\pi^g}\prod_{j<k}(\cos\theta_j-\cos\theta_k)^2\prod_{j=1}^{g}\sin^2\theta_j. \tag{6.1}$$

For the even and odd orthogonal groups the factor $\prod\sin^2\theta_j$ is replaced by 1 and by $\prod\sin^2(\theta_j/2)$, with different constants [298]. The factor $\prod_{j<k}(\cos\theta_j-\cos\theta_k)^2$, common to the three groups, is eigenvalue repulsion; the single product, which depends on the group, is the interaction with the symmetry point $\theta=0$, and it is this factor that the low lying zeros of Section 6.4 detect. For $\mathrm{SU}(2)=\mathrm{USp}(2)$, Chapter 4 derives the density from the geometry of the three sphere; here is a proof through Gaussian vectors.

PROPOSITION 6.2. *Let A be Haar distributed in $\mathrm{SU}(2)$, with eigenvalues $e^{\pm i\theta}$, $\theta\in[0,\pi]$. Then θ has density $\frac{2}{\pi}\sin^2\theta$, and $x=\operatorname{tr}A=2\cos\theta$ has the semicircle density $\frac{1}{2\pi}\sqrt{4-x^2}$ on $[-2,2]$.*

PROOF. As in Chapter 4, Haar measure on $\mathrm{SU}(2)$ is uniform measure on the unit quaternions $S^3\subset\mathbb{R}^4$ and $\operatorname{tr}A=2u$ with u the first coordinate, so u has the law of $Z_1/(Z_1^2+Z_2^2+Z_3^2+Z_4^2)^{1/2}$ for independent standard Gaussian Z_j, and u^2 has the $\mathrm{Beta}(\frac12,\frac32)$ law. Since u is symmetric, its density is $\frac{2}{\pi}\sqrt{1-u^2}$ on $[-1,1]$, and the substitutions $u=\cos\theta$ and $x=2u$ give both claims. □

This agrees with (6.1) for $g=1$. The semicircle is both the Sato Tate law and the Wigner law of large random matrices, but here it is the exact law of the trace of a 2×2 matrix. For large N, the traces of powers of a Haar unitary matrix are nearly independent Gaussians.

THEOREM 6.3 (Diaconis and Shahshahani [154]). *Let U_N be Haar distributed in $\mathrm{U}(N)$ and let $Z_1,Z_2,\dots$ be independent standard complex Gaussian variables, $\mathbb{E}|Z_j|^2=1$. For each fixed k,*

$$\left(\operatorname{tr}U_N,\operatorname{tr}U_N^2,\dots,\operatorname{tr}U_N^k\right)\Rightarrow\left(Z_1,\sqrt{2}\,Z_2,\dots,\sqrt{k}\,Z_k\right)\qquad(N\to\infty).$$

Moreover, for nonnegative integers m_j,n_j with $N\ge\max(\sum jm_j,\sum jn_j)$, the mixed moments $\mathbb{E}\prod_j(\operatorname{tr}U_N^j)^{m_j}\overline{(\operatorname{tr}U_N^j)}^{n_j}$ equal those of the Gaussian limit exactly, namely $\delta_{m,n}\prod_j j^{m_j}m_j!$.

The proof uses power sum symmetric functions and Schur orthogonality; Diaconis and Evans [151] extended the central limit theorem to linear statistics $\sum_j f(\theta_j)$ with f in the Sobolev space of order one half. Diaconis and Shahshahani also treat the orthogonal and symplectic groups, whose traces are real and whose Gaussian limits acquire a mean shift: for $\mathrm{USp}(2N)$, $\mathbb{E}\,\mathrm{tr}\,U^j = -1$ for even $j \le 2N$ and $\mathbb{E}\,\mathrm{tr}\,U^j = 0$ otherwise. As a check we drew 4000 Haar unitary matrices with $N = 50$; the sample means of $|\mathrm{tr}\,U^j|^2/j$ were 1.001, 0.994, 1.013 for $j = 1, 2, 3$, each with standard error about 0.016, and the correlation between $|\mathrm{tr}\,U|^2$ and $|\mathrm{tr}\,U^2|^2$ was -0.012. Heuristically, through the explicit formula, the Gaussian behaviour of $\mathrm{tr}\,U^j$ mirrors the near independence of the contributions $\mathrm{tr}\,A_p^j \log p / p^{j/2}$ of distinct prime powers.

6.2 Montgomery, Dyson and Odlyzko

Assume the Riemann hypothesis and write the nontrivial zeros of ζ as $\frac{1}{2} + i\gamma$. Near height T the zeros have density $\frac{1}{2\pi}\log\frac{T}{2\pi}$. We list the ordinates in the upper half plane with multiplicity, $0 < \gamma_1 \le \gamma_2 \le \cdots$, and unfold them by $\tilde\gamma_j = \frac{\gamma_j}{2\pi}\log\frac{\gamma_j}{2\pi e}$, the main term of the number of zeros up to height γ_j, so that the mean spacing of the $\tilde\gamma_j$ tends to one.

Theorem 6.4 (Montgomery [399]). *Assume the Riemann hypothesis. Let f be a Schwartz function whose Fourier transform $\hat f(\xi) = \int f(x) e^{-2\pi i x\xi} dx$ is supported in $(-1, 1)$. Then, as $T \to \infty$,*

$$\frac{1}{N(T)} \sum_{\substack{j \ne l \\ \gamma_j, \gamma_l \le T}} f(\tilde\gamma_j - \tilde\gamma_l) \longrightarrow \int_{-\infty}^{\infty} f(u)\Big(1 - \Big(\frac{\sin \pi u}{\pi u}\Big)^2\Big) du,$$

where $N(T)$ is the number of zeros with $0 < \gamma \le T$, counted with multiplicity.

The pair correlation conjecture, that the limit holds for all Schwartz f, remains open. The function $1 - (\sin \pi u / \pi u)^2$ is the limiting pair correlation of the CUE. Rudnick and Sarnak [471] extended Montgomery's theorem to all n level correlations and to cuspidal automorphic L functions of GL_m, again for restricted support, assuming the Riemann hypothesis for the L function and, for larger m, an auxiliary hypothesis on the coefficients at prime powers (Theorem 12.15 in Chapter 12 gives the precise statement). In that range the bulk statistics, those of zeros at large height, do not depend on the L function and agree with those of the CUE.

Odlyzko [425] computed large blocks of zeros near the 10^{12}th zero and later near the 10^{20}th, with an algorithm developed with Schönhage [426]. The nearest neighbour spacings agree closely with Gaudin's spacing law for the unitary ensemble [209]. The small systematic deviations that remain are lower order

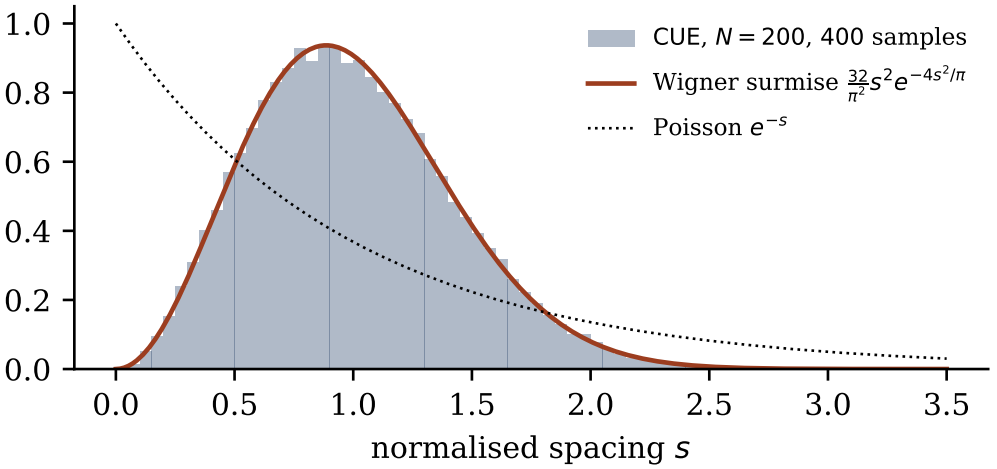


Figure 6.1: Nearest neighbour spacings of the eigenvalue angles of 400 Haar unitary matrices of size $N = 200$, scaled by $N/2\pi$ to have mean one (80,000 spacings), with the Wigner surmise $\frac{32}{\pi^2} s^2 e^{-4s^2/\pi}$ for $\beta = 2$ and the Poisson density e^{-s}. The quadratic vanishing at $s = 0$ is eigenvalue repulsion.

effects: Bogomolny, Bohigas, Leboeuf and Monastra [40] matched them with Haar unitary matrices of an effective size proportional to $\log \frac{T}{2\pi}$. Figure 6.1 shows the random matrix experiment: spacings of 400 independent CUE matrices of size 200, against the Wigner surmise $\frac{32}{\pi^2} s^2 e^{-4s^2/\pi}$, the exact spacing law for 2×2 matrices of the Gaussian unitary ensemble and a close approximation to Gaudin's law. The sample variance of the spacings is 0.180, the Gaudin value, against $\frac{3\pi}{8} - 1 \approx 0.178$ for the surmise and 1 for Poisson spacings; repetitions of the experiment fluctuate by about 0.001, so a single run of this size separates the Gaudin law from the surmise only marginally.

6.3 The Katz and Sarnak philosophy

Katz and Sarnak [298, 299] found that, unlike the bulk statistics, the statistics of zeros near the central point $s = \frac{1}{2}$, averaged over a family, distinguish families, and that the distinction is by the symmetry type of a classical group.

In function fields the analogous statements are theorems. Let C be a smooth projective curve of genus g over $\mathbb{F}_q$. Its zeta function is

$$Z(C, u) = \frac{P_C(u)}{(1-u)(1-qu)}, \qquad P_C(u) = \det\bigl(1 - u\,F \mid H^1\bigr),$$

a polynomial of degree $2g$. By Weil's theorem its roots have absolute value

$q^{-1/2}$, so $P_C(q^{-1/2}u) = \det(1 - u\,\vartheta_C)$ for a conjugacy class ϑ_C in $\mathrm{USp}(2g)$, the unitarised Frobenius; an eigenvalue $e^{i\theta}$ of ϑ_C corresponds to the zeros $s = \frac{1}{2} + i(\theta + 2\pi m)/\log q$, $m \in \mathbb{Z}$, of $Z(C, q^{-s})$. Let C vary in an algebraic family, for instance the hyperelliptic curves $y^2 = f(x)$ with q odd and f squarefree of degree $2g+1$. Deligne's equidistribution theorem [141] says that as $q \to \infty$ the classes ϑ_C become equidistributed for the Haar measure of a maximal compact subgroup of the geometric monodromy group of the family, here all of $\mathrm{USp}(2g)$. Letting first $q \to \infty$ and then $g \to \infty$, Katz and Sarnak proved for many families over function fields that the low lying zeros follow one of a few limiting laws, labelled by the symmetry types

$$\mathrm{U}, \quad \mathrm{USp}, \quad \mathrm{SO(even)}, \quad \mathrm{SO(odd)}, \quad \mathrm{O}.$$

Over number fields the symmetry type of a family is conjectural: Dirichlet L functions $L(s, \chi)$ with χ ranging over primitive characters of a large modulus are unitary; quadratic Dirichlet characters χ_d are symplectic; holomorphic newforms of weight k and level N with $N \to \infty$ are orthogonal, splitting into $\mathrm{SO(even)}$ and $\mathrm{SO(odd)}$ according to the root number; quadratic twists of a fixed elliptic curve are orthogonal, again split by root number. For several of these families the prediction has been proved, for test functions of restricted support only and assuming the generalized Riemann hypothesis where needed; Section 6.4 treats the newforms.

To make this precise, fix an even Schwartz function ϕ whose Fourier transform $\hat{\phi}$ has compact support, so that ϕ extends to an entire function. For an L function $L(s, f)$ in the unitary normalisation of Chapter 2 (centre $\frac{1}{2}$), with analytic conductor c_f and nontrivial zeros $\frac{1}{2} + i\gamma_f$, counted with multiplicity and not assumed to be real, the one level density is $D(f;\phi) = \sum_{\gamma_f} \phi\big(\frac{\gamma_f}{2\pi}\log c_f\big)$, scaled so that the mean spacing near the central point is one. For a classical group $G(N)$ of $N \times N$ matrices, the analogous statistic $\sum_j \phi(N\theta_j/2\pi)$ of a Haar random matrix, summed over its N eigenvalue angles $\theta_j \in (-\pi, \pi]$, has expectation converging to $\int \phi(x) W(G)(x)\,dx$ as $N \to \infty$, where, with $\mathrm{si}(x) = \frac{\sin 2\pi x}{2\pi x}$,

$$\begin{aligned} W(\mathrm{U}) &= 1, & W(\mathrm{USp}) &= 1 - \mathrm{si}, \\ W(\mathrm{SO(even)}) &= 1 + \mathrm{si}, & W(\mathrm{SO(odd)}) &= 1 - \mathrm{si} + \delta_0, \\ W(\mathrm{O}) &= 1 + \tfrac{1}{2}\delta_0. \end{aligned}$$

In Fourier terms, with $\hat{W}$ in the sense of distributions and $\mathbb{I} = \mathbb{I}_{[-1,1]}$ the indicator, $\hat{W}(\mathrm{U}) = \delta_0$ and

$$\begin{aligned} \hat{W}(\mathrm{USp}) &= \delta_0 - \tfrac{1}{2}\mathbb{I}, & \hat{W}(\mathrm{SO(even)}) &= \delta_0 + \tfrac{1}{2}\mathbb{I}, \\ \hat{W}(\mathrm{O}) &= \delta_0 + \tfrac{1}{2}, & \hat{W}(\mathrm{SO(odd)}) &= \delta_0 + 1 - \tfrac{1}{2}\mathbb{I}. \end{aligned}$$

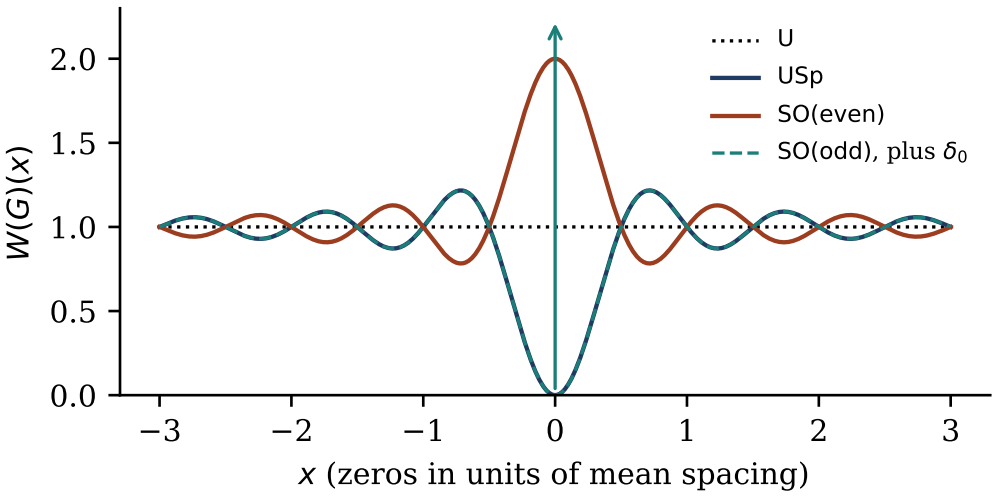


Figure 6.2: One level densities $W(G)(x)$ of the symmetry types U, USp, SO(even) and SO(odd), with x in units of the mean spacing. The symplectic density vanishes quadratically at the origin; the even orthogonal density equals 2 there, twice the unitary value; the odd orthogonal density is the symplectic one plus a unit point mass at the origin (arrow), from the eigenvalue forced at 1. The full orthogonal density $1 + \frac{1}{2}\delta_0$, not drawn, is the average of the even and odd orthogonal densities.

Figure 6.2 plots the densities of the first four types. The three orthogonal densities have the same Fourier transform on $(-1,1)$, so test functions with $\operatorname{supp}\hat{\phi} \subset (-1,1)$ cannot tell them apart; only support beyond $[-1,1]$ separates them.

6.4 Low lying zeros of modular forms

The first theorem over $\mathbb{Q}$ that separates the orthogonal types is due to Iwaniec, Luo and Sarnak [271], and its method is the template for later work.

Theorem 6.5 (Iwaniec, Luo and Sarnak [271]). *Assume the generalized Riemann hypothesis for the L functions of holomorphic newforms and for Dirichlet L functions. Fix an even weight k, let N run through squarefree integers, and let $H_k^*(N)$ be the set of holomorphic newforms of weight k and level N, with analytic conductor $c_f = k^2N$, and $H_k^{\pm}(N)$ the subsets of root number ± 1. Let ϕ be an even Schwartz function with $\operatorname{supp}\hat{\phi} \subset (-2,2)$. Then*

$$\lim_{N\to\infty} \frac{1}{|H_k^{\pm}(N)|} \sum_{f\in H_k^{\pm}(N)} D(f;\phi) = \int \phi(x) W(\mathrm{SO}(\pm))(x)\,dx,$$

where $\mathrm{SO}(+) = \mathrm{SO}(\mathrm{even})$ *and* $\mathrm{SO}(-) = \mathrm{SO}(\mathrm{odd})$*; for the full family* $H_k^*(N)$ *the limit is* $\int \phi\, W(\mathrm{O})$*. For* $\operatorname{supp}\hat{\phi} \subset (-1,1)$ *the limits for* $H_k^{\pm}(N)$ *hold as* $kN \to \infty$*, without the hypothesis on Dirichlet* L *functions.*

Averaging also over the weight brings in level one, where the root number is i^k: under the same hypotheses Iwaniec, Luo and Sarnak prove these limits for the forms of weight $k \le K$ and fixed level as $K \to \infty$, with support in $(-2,2)$. For the level one forms of weight $k \le K$, not split by root number and weighted by the harmonic weights of the Petersson formula, proportional to $L(1, \mathrm{Sym}^2 f)^{-1}$, they prove the orthogonal limit $\int \phi\, W(\mathrm{O})$ for $\operatorname{supp}\hat{\phi} \subset (-2,2)$ unconditionally.

The proof starts from the explicit formula, which for $f \in H_k^*(N)$, assuming also the Riemann hypothesis for $L(s, \mathrm{Sym}^2 f)$, gives

$$D(f;\phi) = \hat{\phi}(0) + \tfrac{1}{2}\phi(0) - \frac{2}{\log c_f} \sum_{p \nmid N} \lambda_f(p) \hat{\phi}\Big(\frac{\log p}{\log c_f}\Big) \frac{\log p}{\sqrt{p}} + O\Big(\frac{\log\log 3kN}{\log c_f}\Big),$$

with $\lambda_f(p) = x_p \in [-2,2]$ the normalised Hecke eigenvalues. The term $\frac{1}{2}\phi(0)$ comes from the prime squares. The coefficient of p^{-2s} in $-L'/L(s,f)$ is $(\lambda_f(p)^2 - 2)\log p = (\lambda_f(p^2) - 1)\log p$; the part with $\lambda_f(p^2)$ is negligible (this is where the Riemann hypothesis for the symmetric square enters; Iwaniec, Luo and Sarnak note that on average over the family Petersson's formula replaces it, for the same support, so the theorem does not assume it), and the constant -1, summed by the prime number theorem, gives $\frac{1}{2}\phi(0)$. For a symplectic family such as $L(s,\chi_d)$ the coefficient is $\chi_d(p)^2 \log p = \log p$ and the same sum gives $-\frac{1}{2}\phi(0)$. This is the arithmetic shadow of the mean shift $\mathbb{E}\,\mathrm{tr}\, U^2 = \pm 1$ in orthogonal and symplectic groups. Petersson's formula then averages the sum over primes. For $\operatorname{supp}\hat{\phi} \subset (-1,1)$ its Kloosterman terms are negligible, so the prime sum averages to $o(1)$ and the average of $D(f;\phi)$ is $\hat{\phi}(0) + \frac{1}{2}\phi(0)$ for both root numbers, in keeping with the agreement of the three orthogonal densities there. Beyond $(-1,1)$ these terms, evaluated with the generalized Riemann hypothesis for Dirichlet L functions, contribute main terms of opposite signs to the two root numbers, through $\varepsilon_f = i^k \mu(N)\sqrt{N}\lambda_f(N)$; they cancel in the full family, and they are what separates $\mathrm{SO}(\mathrm{even})$ from $\mathrm{SO}(\mathrm{odd})$.

Under the generalized Riemann hypothesis the zeros are real, so if $\phi \ge 0$ and $\phi(0) = 1$ the order of vanishing of $L(s,f)$ at $\frac{1}{2}$ is at most $D(f;\phi)$. Averaging over the family and choosing ϕ well, Iwaniec, Luo and Sarnak deduce that for all large squarefree N more than $\frac{9}{16}$ of the forms in $H_k^+(N)$ have $L(\frac{1}{2},f) \ne 0$ and more than $\frac{15}{16}$ of those in $H_k^-(N)$ have $L'(\frac{1}{2},f) \ne 0$; the extension of the support beyond $(-1,1)$ is essential for these constants. For test functions of unrestricted support the Katz and Sarnak prediction is not known for any family over a number field.

6.4.1 Why the central point sees the group

For a self dual L function the zeros are symmetric about $s = \frac{1}{2}$, just as the eigenvalues of orthogonal and symplectic matrices come in pairs $e^{\pm i\theta}$; a matrix in $\mathrm{SO}(2N+1)$ always has the eigenvalue 1, as an L function of root number -1 vanishes at $\frac{1}{2}$. Away from the symmetry point all groups look unitary; near it, the factor $\prod \sin^2 \theta_j$ or $\prod \sin^2(\theta_j/2)$ in the Weyl density decides whether eigenvalues are repelled from 1. Low lying zeros thus measure the Weyl density, near the identity, of the symmetry group attached to the family.

6.4.2 Ranks of elliptic curves

If the two root numbers occur equally often and, as the Katz and Sarnak model predicts, the analytic rank is 0 or 1 according to the root number outside a set of density zero (the minimalist conjecture of Section 8.5), then the average analytic rank should be $\frac{1}{2}$, provided that exceptional set contributes nothing to the average. Brumer [88], Heath Brown [247] and Young [591] used one level density with restricted support to prove, under the generalized Riemann hypothesis, upper bounds for the average analytic rank of all elliptic curves ordered by height, with Young obtaining $\frac{25}{14}$. These are conditional theorems; the value $\frac{1}{2}$ is a conjecture. Unconditional bounds for the average algebraic rank come from Selmer groups (Theorem 8.11).

In computable ranges, families show an excess of rank two curves; Bektemirov, Mazur, Stein and Watkins [68] examine this tension between data and conjecture and argue that slow convergence can reconcile the two. Miller [395, 396] found that for small conductors the first zero above the central point is repelled more strongly than the limiting density predicts. Dueñez, Huynh, Keating, Miller and Snaith [150] reproduce this repulsion with an excised orthogonal ensemble, which discards the matrices in $\mathrm{SO}(2N)$ whose characteristic polynomial at 1 is too small, mimicking the discreteness of central values imposed by the formulas of Waldspurger and of Kohnen and Zagier; this is a model, not a theorem.

6.5 Moments and the Keating and Snaith conjectures

Random matrices also predict the size of L functions at the central point. Let $Z_N(\theta) = \det(I - U_N e^{-i\theta})$ be the characteristic polynomial of a Haar unitary matrix. Keating and Snaith [300] computed, using Selberg's integral,

$$\mathbb{E}|Z_N(0)|^{2k} = \prod_{j=1}^{N} \frac{\Gamma(j)\Gamma(j+2k)}{\Gamma(j+k)^2} \sim \frac{G(k+1)^2}{G(2k+1)} N^{k^2} \qquad (N \to \infty),$$

for $\Re k > -\frac{1}{2}$, where G is the Barnes G function, $G(n+1) = \prod_{j=0}^{n-1} j!$ for integers $n \geq 0$. Equating the mean density of zeros, $N \leftrightarrow \log \frac{T}{2\pi}$, and inserting an

arithmetic factor that random matrices cannot see, they proposed the following.

CONJECTURE 6.6 (Keating and Snaith). *For each $k > -\frac{1}{2}$, as $T \to \infty$,*

$$\frac{1}{T}\int_0^T \left|\zeta(\tfrac{1}{2}+it)\right|^{2k} dt \sim a_k \,\frac{G(k+1)^2}{G(2k+1)}\,(\log T)^{k^2},$$

with $a_k = \prod_p (1-p^{-1})^{k^2} \sum_{m\ge 0} \left(\frac{\Gamma(m+k)}{m!\,\Gamma(k)}\right)^2 p^{-m}$.

For $k = 1$ the right side is $\log T$ (Hardy and Littlewood), and for $k = 2$ it is $\frac{1}{2\pi^2}(\log T)^4$ (Ingham), since $G(3)^2/G(5) = \frac{1}{12}$ and $a_2 = 6/\pi^2$. For $k = 3$ and $k = 4$ the leading coefficients are $42a_3/9!$ and $24024a_4/16!$, in agreement with the conjectures of Conrey and Ghosh [116] and of Conrey and Gonek [117]. No case other than $k = 0, 1, 2$ is proved. Lower order terms are predicted by the recipe of Conrey, Farmer, Keating, Rubinstein and Snaith [96].

For families with symplectic or orthogonal symmetry, Keating and Snaith [301] computed the moments of $\det(I-U)$ for Haar U in $\mathrm{USp}(2N)$ and in $\mathrm{SO}(2N)$; for instance, for each positive integer k, as $N \to \infty$,

$$\mathbb{E}_{\mathrm{USp}(2N)} \det(I-U)^k \sim g_k N^{k(k+1)/2}, \qquad g_k = 2^{k(k+1)/2} \prod_{j=1}^{k} \frac{j!}{(2j)!}.$$

A matrix in $\mathrm{USp}(2N)$ has $2N$ eigenvalues, so equating densities of zeros matches $2N$, not N, with the logarithm of the conductor, and they conjectured, for example, that

$$\frac{1}{X^*} \sum_{0<d\le X}^{\flat} L(\tfrac{1}{2},\chi_d)^k \sim b_k\, g_k \Big(\frac{\log X}{2}\Big)^{k(k+1)/2},$$

where the sum runs over the $X^* \sim 3X/\pi^2$ fundamental discriminants $0 < d \le X$ and b_k is an explicit Euler product, the analogue of a_k. The exponent $k(k+1)/2$ identifies the family as symplectic; for orthogonal families it is $k(k-1)/2$. Jutila [288] proved the cases $k = 1, 2$, and Soundararajan [522] the case $k = 3$ for the closely related family of characters χ_{8d}; Shen [511] proved the case $k = 4$ for that family under the generalized Riemann hypothesis, and Shen and Stucky [512] remove the hypothesis in a preprint. The ratios conjecture of Conrey, Farmer and Zirnbauer [97], applied by Conrey and Snaith [118], predicts the one level density with all lower order terms, for test functions of arbitrary support.

6.6 TWO RANDOM MATRIX LIMITS

The Sato Tate limit of Chapters 3 and 4 fixes π and varies p: the Satake parameters $A_p(\pi)$ are conjectured to become equidistributed for the Haar measure of a fixed compact group $\mathrm{ST}(\pi)$ (Conjecture 3.11 for abelian varieties). This is a theorem

for elliptic curves over $\mathbb{Q}$ without complex multiplication, where the group is $\mathrm{SU}(2)$ (Theorem 4.9), and open for a generic abelian surface, where it is $\mathrm{USp}(4)$. Its vertical versions fix p and vary the object. For the eigenvalues of T_p on spaces of modular forms of growing level or weight, the limit is the Kesten McKay law μ_p (Serre [508]; Conrey, Duke and Farmer [95]; Theorem 9.4), which depends on p and tends to the Sato Tate law as $p \to \infty$. For all elliptic curves $y^2 = x^3 + ax + b$ over $\mathbb{F}_p$, Birch [64] computed the moments of the traces of Frobenius and showed that their normalised values become semicircle distributed as $p \to \infty$; Deligne's equidistribution theorem of Section 6.3 recovers this as a case of genus one.

The Katz and Sarnak limit of this chapter models the zeros of $L(s, f)$ near the central point by the eigenvalues of a random matrix whose size grows like $\log c_f$, so the groups have dimension tending to infinity. For curves over $\mathbb{F}_q$ both limits concern the same classes $\vartheta_C \in \mathrm{USp}(2g)$: at fixed g, Deligne's theorem as $q \to \infty$ is a vertical Sato Tate law for ϑ_C, whose trace is $-(\#C(\mathbb{F}_q) - q - 1)/\sqrt{q}$, and Katz and Sarnak then let $g \to \infty$ and study the eigenvalues near 1. Over number fields the link between the two limits is the explicit formula, which expresses sums over zeros through the Satake parameters $A_p(\pi)$.

6.7 Notes

Montgomery's paper [399] states the pair correlation conjecture explicitly, and Odlyzko's computations [425] remain the principal numerical evidence for the unitary model. For eigenvalue statistics of compact classical groups, Mehta's book [391], the monograph of Katz and Sarnak [298] and Meckes's book [390] are standard, and the survey of Katz and Sarnak [299] is the shortest route to the symmetry types. Beyond Theorem 6.3, suitably normalised powers $|Z_N|^{2\alpha}$ converge to Gaussian multiplicative chaos for α below a critical value (Webb [565]; Nikula, Saksman and Webb [424]), and the maximum of $\log|Z_N|$ on the circle is known to order one (Chhaibi, Madaule and Najnudel [106]), as conjectured by Fyodorov, Hiary and Keating [187]; Chapter 12 takes up these log correlated questions.

6.8 Exercises

1. Using Proposition 6.2, show that $\operatorname{tr} A^2 = x^2 - 2 = 2\cos 2\theta$ for Haar $A \in \mathrm{SU}(2)$, that $\mathbb{E}\operatorname{tr} A^2 = -1$, and that $\mathbb{E}\operatorname{tr} A^j = 0$ for every other $j \geq 1$. Compare with the symplectic mean shift for $\mathrm{USp}(2N)$ stated after Theorem 6.3.

2. Let U be Haar in $\mathrm{U}(N)$. Show directly from the invariance $U \mapsto e^{i\alpha}U$ that $\mathbb{E}(\operatorname{tr} U^j) = 0$ for $j \geq 1$. Using the Weyl density for $\mathrm{U}(N)$, show that $\mathbb{E}|\operatorname{tr} U|^2 = 1$ for $N \geq 1$ and more generally $\mathbb{E}|\operatorname{tr} U^j|^2 = \min(j, N)$. Hint: the

density is $\frac{1}{N!}\det[K_N(\theta_j,\theta_l)]_{j,l\le N}$, where

$$K_N(\theta,\theta') = \frac{1}{2\pi}\sum_{m=0}^{N-1} e^{im(\theta-\theta')}.$$

3. Let $\theta_1,\dots,\theta_g$ have the density (6.1) and put $t = \sum_j 2\cos\theta_j$, the trace of a Haar element of $\mathrm{USp}(2g)$. Show that $\mathbb{E}t^2 = 1$ for every g, using that t is the character of an irreducible representation. Deduce from Deligne's equidistribution theorem, as stated in Section 6.3, that for the hyperelliptic curves of genus g over $\mathbb{F}_q$ the mean of $\#C(\mathbb{F}_q)$ is $q+1+o(\sqrt{q})$ and its variance is $(1+o(1))q$ as $q\to\infty$.

4. Compute the Fourier transforms of $1 \pm \frac{\sin 2\pi x}{2\pi x}$ and verify the formulas for $\hat{W}$; note that the point mass δ_0 in $W(\mathrm{SO(odd)})$ becomes the constant 1 in $\hat{W}(\mathrm{SO(odd)})$. Deduce that if $\operatorname{supp}\hat{\phi} \subset (-1,1)$ then $\int \phi\, W(G) = \hat{\phi}(0) + \frac{1}{2}\phi(0)$ for all three orthogonal types and $\hat{\phi}(0) - \frac{1}{2}\phi(0)$ for USp, and compare with the explicit formula of Section 6.4. Check that $W(\mathrm{O})$ is the average of $W(\mathrm{SO(even)})$ and $W(\mathrm{SO(odd)})$.

5. Verify the identities

$$\frac{G(3)^2}{G(5)} = \frac{1}{12}, \qquad \frac{G(4)^2}{G(7)} = \frac{42}{9!}, \qquad \frac{G(5)^2}{G(9)} = \frac{24024}{16!}.$$

Show that $\log\bigl(G(k+1)^2/G(2k+1)\bigr) \sim -k^2\log k$ as $k\to\infty$. Since $\log|Z_N(0)|$ is approximately Gaussian with variance $\frac{1}{2}\log N$, a Gaussian model predicts $\mathbb{E}|Z_N(0)|^{2k} \approx N^{k^2}$ with no constant depending on k; interpret the decay of $G(k+1)^2/G(2k+1)$ in terms of the upper tail of $\log|Z_N(0)|$ at the scale $k\log N$.

6.9 Open computational problems

Computational Problem 6.1 (One level density for quadratic twists of 11a). Objective: measure the one level density of the family of quadratic twists of the curve $11a$, separated by root number, and compare it with $W(\mathrm{SO(even)})$ and $W(\mathrm{SO(odd)})$ including lower order terms. Objects: all twists E_d of $y^2+y = x^3-x^2-10x-20$ by fundamental discriminants d coprime to 11 with conductor $11d^2 \le 10^8$, so that $|d|$ runs up to about 3000 and the root number of E_d is $\chi_d(-11)$; the lowest ten zeros of each $L(s,E_d)$. Route: PARI/GP `lfunzeros` or Rubinstein's `lcalc`, with $a_p(E_d) = \chi_d(p)a_p(E)$ computed once; parallel over d. Deliverable: histograms of scaled zeros near the central point for each sign, the empirical one level density for several test functions with Fourier support up to 2 and beyond, the distribution of the first zero, and a fit to the ratios conjecture prediction

with finite conductor corrections. Why it matters: it tests the Katz and Sarnak prediction in a family where theorems of the type of Theorem 6.5 cover only restricted support, quantifies the repulsion of the first zero observed by Miller and modelled by the excised orthogonal ensemble, and connects with the excess rank question.

COMPUTATIONAL PROBLEM 6.2 (Keating and Snaith versus numerical moments of quadratic L functions). Objective: compare the conjectured asymptotics for $\sum_{|d|\le X} L(\frac{1}{2},\chi_d)^k$ with numerical data, for $k = 1,\dots,6$, and estimate the lower order terms. Objects: all fundamental discriminants $|d| \le 10^9$, or a random subset of comparable size at larger X. Route: compute $L(\frac{1}{2},\chi_d)$ by the approximate functional equation with smoothed sums, on GPUs, or with PARI/GP `lfun` for smaller ranges; evaluate the full CFKRS polynomial in $\log X$ for each k and the ratios conjecture predictions. Deliverable: tables of empirical moments against the leading term and against the full polynomial, plots of the relative error against X, and an empirical estimate of the power savings in the error term. Why it matters: the symplectic exponent $k(k+1)/2$ is the random matrix signature of the family; the numerical residuals measure how accurately the CFKRS recipe captures arithmetic beyond the leading constant, and for $k \ge 5$ no asymptotic is known, even conditionally.

COMPUTATIONAL PROBLEM 6.3 (Nearest neighbour spacings for the zeros of $L(s,\Delta)$ at large height). Objective: test unitary statistics for the zeros of the L function of the Ramanujan Δ function, a degree two L function, at heights comparable to those of Odlyzko's zeta computations, continuing Problem 12.3, whose highest block of zeros lies near height 10^7. Objects: blocks of 10^6 to 10^7 consecutive zeros of $L(s,\Delta)$ near height 10^8, where the zero density is about 5 per unit height, and blocks of up to 10^9 zeros near 10^{10} and, as a long term target, near 10^{20}. Route: at moderate heights use the approximate functional equation in `lcalc` or ARB; at large heights an analogue of the Odlyzko and Schönhage algorithm for degree two (multipoint evaluation of $\sum \tau(n) n^{-11/2-s}$ with fast summation, or Hiary's methods adapted to modular forms) must be developed, and the feasibility at 10^{20} is itself part of the problem. Deliverable: empirical spacing distributions against the Gaudin law, pair correlation, number variance, and the arithmetic lower order corrections predicted by Bogomolny and Keating type formulas. Why it matters: Rudnick and Sarnak prove GUE correlations for $L(s,\Delta)$ only for restricted test functions, and between heights 500 and 2000 the spacings of its zeros are markedly more rigid than the Gaudin law (variance 0.138 against 0.180, Chapter 12); large height data would show how fast a degree two L function approaches the unitary limit, where the arithmetic corrections involve $\mathrm{Sym}^2\,\Delta$.

CHAPTER 7

KLOOSTERMAN SUMS: VERTICAL SATO TATE AND RANDOM FOURIER SERIES

In 1926 Kloosterman, counting representations of integers by quaternary quadratic forms with the circle method, met the exponential sums that now bear his name and proved that they are smaller than trivial by a power of the modulus [323]. Twenty two years later Weil deduced the optimal bound from the Riemann hypothesis for curves over finite fields [568], and forty years after that Katz determined the statistical law of the normalised sums: as the prime grows, Kloosterman angles are distributed like the angles of Haar random matrices in $\mathrm{SU}(2)$ [296]. The same semicircle that governs Frobenius traces of elliptic curves (Chapter 4) and the eigenvalues of random matrices (Chapter 6) thus governs a family of exponential sums with no elliptic curve in sight. Kloosterman sums are also tied to the spectral theory of automorphic forms: the Bruggeman Kuznetsov formula equates sums of them with spectral averages of Fourier coefficients of Maass forms, and Selberg's bound $\lambda_1 \geq 3/16$ is Weil's bound read spectrally. In the last part of the chapter we follow Kowalski and Sawin [331], who joined partial Kloosterman sums into polygonal paths and found in the limit a random Fourier series with independent Sato Tate coefficients. We prove the analytic reduction behind their theorem, compute the covariance of the limit in closed form, and show exactly how a change of parametrisation removes one Sato Tate coefficient from the limit, the effect behind a variance deficit of $1/\pi^2$ that our computations ran into.

7.1 KLOOSTERMAN SUMS AND WEIL'S BOUND

Let p be an odd prime and write $e(z) = e^{2\pi i z}$. For $x \in \mathbb{F}_p^\times$ let $\bar{x}$ denote its inverse. For $a, b \in \mathbb{F}_p$ the *Kloosterman sum* is

$$\mathrm{Kl}(a, b; p) = \sum_{x \in \mathbb{F}_p^\times} e\Big(\frac{ax + b\bar{x}}{p}\Big),$$

and we abbreviate $\mathrm{Kl}(c; p) = \mathrm{Kl}(c, 1; p)$. Three elementary properties are used constantly (Exercise 1). First, $\mathrm{Kl}(a, b; p)$ is real, because $x \mapsto -x$ maps the sum to its complex conjugate. Second, if $a \neq 0$ the substitution $y = ax$ gives $\mathrm{Kl}(a, b; p) = \mathrm{Kl}(1, ab; p)$, so the sum depends only on the product ab when $a \neq 0$; by symmetry the same holds when $b \neq 0$, so $\mathrm{Kl}(a, b; p) = \mathrm{Kl}(ab; p)$ unless $a = b = 0$. Third, if

exactly one of a, b vanishes the sum is a Ramanujan sum: $\sum_{x\neq 0} e(cx/p) = -1$ for $c \neq 0$, so $\mathrm{Kl}(0;p) = -1$.

The trivial bound is $|\mathrm{Kl}(a;p)| \leq p-1$. The moments show that the typical size is $\sqrt{p}$.

PROPOSITION 7.1 (Second and fourth moments). *For every odd prime p,*

$$\sum_{a\in\mathbb{F}_p^\times} \mathrm{Kl}(a;p) = 1, \qquad \sum_{a\in\mathbb{F}_p^\times} \mathrm{Kl}(a;p)^2 = p^2-p-1,$$
$$\sum_{a\in\mathbb{F}_p^\times} \mathrm{Kl}(a;p)^4 = 2p^3-3p^2-3p-1.$$

Consequently the normalised values $x_{p,a} = \mathrm{Kl}(a;p)/\sqrt{p}$ have empirical second moment $1-1/(p(p-1))$ and empirical fourth moment $2-1/p+O(p^{-2})$ over $a\in\mathbb{F}_p^\times$.

These identities are classical; we include the proof because the fourth moment is the first place where the semicircle appears.

PROOF. We sum over all $a\in\mathbb{F}_p$ and subtract the term $a=0$, where $\mathrm{Kl}(0;p) = -1$. By orthogonality of additive characters, $\sum_a e(au/p) = p\,\mathbb{1}_{u=0}$, we have $\sum_a \mathrm{Kl}(a;p) = \sum_{x\neq 0} e(\bar{x}/p)\sum_a e(ax/p) = 0$, and, since $\mathrm{Kl}(a;p)$ is real,

$$\sum_a \mathrm{Kl}(a;p)^2 = \sum_{x,y\neq 0} e\Big(\frac{\bar{x}-\bar{y}}{p}\Big)\sum_a e\Big(\frac{a(x-y)}{p}\Big) = p(p-1).$$

For the fourth moment write $\mathrm{Kl}(a;p)^2 = \sum_u e(au/p)T(u)$ with

$$T(u) = \sum_{\substack{x,y\neq 0\\ x+y=u}} e\Big(\frac{\bar{x}+\bar{y}}{p}\Big).$$

Since Kl^2 is real, Parseval gives $\sum_a \mathrm{Kl}(a;p)^4 = \sum_a |\mathrm{Kl}(a;p)^2|^2 = p\sum_u |T(u)|^2$. For $u=0$ we have $y=-x$, hence $\bar{y}=-\bar{x}$, and $T(0)=p-1$. For $u\neq 0$ we have $\bar{x}+\bar{y} = u\,\overline{x(u-x)}$, and the substitution $x=us$ gives $T(u) = \sum_{s\neq 0,1} e(\bar{u}\,\overline{s(1-s)}/p)$. The equation $s(1-s)=v$ has $1+\left(\frac{1-4v}{p}\right)$ solutions, with $\left(\frac{\cdot}{p}\right)$ the Legendre symbol χ. Writing $w=\bar{v}$ and using $\chi(1-4\bar{w}) = \chi(w(w-4))$,

$$T(u) = -1 + G(\bar{u}), \qquad G(c) = \sum_{w\in\mathbb{F}_p} \chi\big(w(w-4)\big)e(cw/p).$$

Now $G(0) = \sum_w \chi(w^2-4w) = -1$ (a quadratic with nonzero discriminant), $\sum_c G(c) = p\,\chi(0) = 0$, and by Parseval $\sum_c |G(c)|^2 = p\,\#\{w : w(w-4)\neq 0\} = p(p-2)$. Hence $\sum_{c\neq 0}|G(c)|^2 = p^2-2p-1$ and $\sum_{c\neq 0} G(c) = 1$, so

$$\sum_{u\neq 0}|T(u)|^2 = \sum_{c\neq 0}|G(c)-1|^2 = (p^2-2p-1)-2+(p-1) = p^2-p-4.$$

With $T(0)^2 = (p-1)^2$ added, $\sum_u |T(u)|^2 = 2p^2 - 3p - 3$, so $\sum_{a \in \mathbb{F}_p} \mathrm{Kl}(a;p)^4 = 2p^3 - 3p^2 - 3p$, and removing $a = 0$ proves the claim. □

The limits 1 and 2 are the first two even moments of the Sato Tate law μ_{ST}, the Catalan numbers C_1, C_2 of Chapter 4. The second moment shows that $\max_a |\mathrm{Kl}(a;p)| \geq \sqrt{p}\,(1 + o(1))$. The fourth moment gives $\max_a |\mathrm{Kl}(a;p)|^4 \leq 2p^3$, that is $|\mathrm{Kl}(a;p)| \leq 2^{1/4} p^{3/4}$, which is essentially Kloosterman's bound. A bound of the right order p^{k+1} for the moment of order $2k$ would give the exponent $\frac{1}{2} + \frac{1}{2k}$, but proving it already requires the geometry behind the optimal bound.

THEOREM 7.2 (Weil [568]). *For $p \nmid ab$, $|\mathrm{Kl}(a,b;p)| \leq 2\sqrt{p}$.*

Idea of proof. By the second property above we may take $b = 1$ and $a \neq 0$. For $n \geq 1$ let $\mathrm{Kl}_n(a) = \sum_{x \in \mathbb{F}_{p^n}^\times} \psi\big(\mathrm{Tr}_{\mathbb{F}_{p^n}/\mathbb{F}_p}(ax + \bar{x})\big)$ with $\psi(z) = e(z/p)$, so that $\mathrm{Kl}_1(a) = \mathrm{Kl}(a;p)$. To a monic $f \in \mathbb{F}_p[X]$ with $f(0) \neq 0$ attach $\lambda(f) = \psi\big(a\sigma(f) + \sigma'(f)\big)$, where $\sigma(f)$ and $\sigma'(f)$ are the sums of the roots of f and of their inverses. Then λ is completely multiplicative, and grouping the $x \in \mathbb{F}_{p^n}^\times$ by their minimal polynomials over $\mathbb{F}_p$ turns the Euler product over irreducible f into

$$L_a(T) = \exp\Big(\sum_{n \geq 1} \mathrm{Kl}_n(a) \frac{T^n}{n}\Big) = \sum_f \lambda(f)\, T^{\deg f}.$$

For $f = X^d + c_{d-1}X^{d-1} + \cdots + c_0$ we have $\sigma(f) = -c_{d-1}$ and $\sigma'(f) = -c_1/c_0$. When $d \geq 3$ the coefficient c_{d-1} is free and the terms of degree d cancel because $a \neq 0$; counting the terms of degrees 1 and 2 gives $L_a(T) = 1 + \mathrm{Kl}(a;p)\,T + pT^2$. This is Weil's argument [568]. Factor $L_a(T) = (1 - \alpha T)(1 - \beta T)$; then $\mathrm{Kl}_n(a) = -(\alpha^n + \beta^n)$ and $\alpha\beta = p$. Counting the points of the Artin Schreier curve $y^p - y = ax + 1/x$, which has genus $p - 1$, over $\mathbb{F}_{p^n}$ shows that α and β are among the reciprocal roots of the numerator of its zeta function, so the Riemann hypothesis for curves gives $|\alpha| = |\beta| = \sqrt{p}$, hence the bound. Conversely, a bound $|\mathrm{Kl}_n(a)| \leq C p^{n/2}$ with C independent of n already forces $|\alpha| = |\beta| = \sqrt{p}$ (Exercise 4). The elementary method of Stepanov [526], in Bombieri's form [71], proves point count bounds of exactly this kind over every extension $\mathbb{F}_{p^n}$, and so reaches the same conclusion.

Since $\alpha\beta = p$, $|\alpha| = |\beta| = \sqrt{p}$ and $\alpha + \beta = -\mathrm{Kl}(a;p)$ is real, we have $\beta = \bar{\alpha}$, and we may write $\{\alpha, \beta\} = \{-\sqrt{p}\,e^{i\theta}, -\sqrt{p}\,e^{-i\theta}\}$ with a unique *Kloosterman angle* $\theta_{p,a} \in [0, \pi]$, so that

$$x_{p,a} = \frac{\mathrm{Kl}(a;p)}{\sqrt{p}} = 2\cos\theta_{p,a} \in [-2, 2].$$

In the language of ℓ adic sheaves, Deligne and Katz construct a rank two lisse sheaf $\mathcal{K}l$ on $\mathbb{G}_m$ over $\mathbb{F}_p$, pure of weight one, whose Frobenius at $a \in \mathbb{F}_p^\times$ has

eigenvalues α, β and trace $-\mathrm{Kl}(a;p)$ [296]. After the half Tate twist the Frobenius at a determines a conjugacy class in $\mathrm{SU}(2)$, exactly like the Satake parameter $\mathrm{diag}(e^{i\theta_p}, e^{-i\theta_p})$ of a modular form at a prime p. Kloosterman angles and Satake angles live on the same circle; the sign in the trace is immaterial for us because μ_{ST} is invariant under $\theta \mapsto \pi - \theta$.

7.2 Katz's theorem: vertical Sato Tate

There are two ways to let the Kloosterman angles vary. *Vertically*, fix p and let a run over $\mathbb{F}_p^\times$, then let $p \to \infty$. *Horizontally*, fix a and let p run over the primes. The vertical problem is the analogue of the vertical Sato Tate theorem for Hecke eigenvalues in Chapter 9; the horizontal problem is the analogue of Sato Tate for a fixed elliptic curve.

THEOREM 7.3 (Katz [296]). *As $p \to \infty$ the multisets $\{\theta_{p,a} : a \in \mathbb{F}_p^\times\}$ become equidistributed in $[0,\pi]$ with respect to the Sato Tate measure $\frac{2}{\pi}\sin^2\theta\, d\theta$. Equivalently, for every continuous f on $[-2,2]$,*

$$\frac{1}{p-1}\sum_{a\in\mathbb{F}_p^\times} f\Big(\frac{\mathrm{Kl}(a;p)}{\sqrt{p}}\Big) \longrightarrow \int_{-2}^{2} f\, d\mu_{\mathrm{ST}}.$$

Since $\mathrm{Kl}(a,b;p) = \mathrm{Kl}(ab;p)$ and $a \mapsto ab$ permutes $\mathbb{F}_p^\times$ when $p \nmid b$, the same holds for $\mathrm{Kl}(a,b;p)$ with b fixed.

Proof sketch. By the Weyl criterion (Theorem 4.3, whose proof applies verbatim to any sequence of probability measures on $[0,\pi]$) it suffices to show that the averages of $U_n(\cos\theta_{p,a})$ tend to zero for each $n \geq 1$. Up to the sign $(-1)^n$, $U_n(\cos\theta_{p,a})$ is the trace of Frobenius at a on Sym^n of the normalised sheaf. By the Grothendieck Lefschetz trace formula,

$$\sum_{a\in\mathbb{F}_p^\times} \mathrm{tr}\big(\mathrm{Frob}_a \mid \mathrm{Sym}^n \mathcal{K}l\big) = \sum_{i=0}^{2} (-1)^i \,\mathrm{tr}\big(\mathrm{Frob} \mid H_c^i(\mathbb{G}_{m,\bar{\mathbb{F}}_p}, \mathrm{Sym}^n \mathcal{K}l)\big).$$

The group H_c^0 vanishes because $\mathbb{G}_m$ is affine. The group H_c^2 is the space of coinvariants of the geometric monodromy group acting on Sym^n, and Katz proved that this group is SL_2 [296]. The inputs are local: the sheaf has geometrically trivial determinant, it is tamely ramified at 0 with unipotent local monodromy consisting of a single Jordan block, and it is wildly ramified at ∞ with both breaks equal to $\frac{1}{2}$ (Swan conductor 1); Katz deduces from them that no proper subgroup of SL_2 can be the geometric monodromy group. Since Sym^n is an irreducible nontrivial representation of SL_2 for $n \geq 1$, the coinvariants vanish. Finally, by Deligne's Weil II [141] the eigenvalues of Frobenius on H_c^1 have absolute value

at most $\sqrt{p}$ after normalisation, and the Grothendieck Ogg Shafarevich formula gives $\dim H^1_c = \mathrm{Swan}_\infty(\mathrm{Sym}^n \mathcal{K}l) \le \frac{n+1}{2}$, because $\mathrm{Sym}^n \mathcal{K}l$ is tame at 0 and its breaks at ∞ are at most $\frac{1}{2}$. Therefore

$$\Big|\frac{1}{p-1}\sum_{a\in\mathbb{F}_p^\times} U_n(\cos\theta_{p,a})\Big| \le \frac{(n+1)\sqrt{p}}{2(p-1)} \le \frac{n+1}{\sqrt{p}}.$$

This is Deligne's equidistribution theorem [141] applied to the Kloosterman sheaf, and the monodromy computation is the heart of the matter; Katz and Sarnak use exactly this mechanism to produce the random matrix statistics of Chapter 6 [298]. The bound $(n+1)/\sqrt{p}$ for the Weyl sums gives an explicit rate in the discrepancy, worked out by Niederreiter [422]. Katz also computed the monodromy of the hyper Kloosterman sheaves of rank n: it is the symplectic group Sp_n for even n, and SL_n for odd n when p is odd [296]. The case $n = 2$ of this chapter is the first member of the symplectic series, since $\mathrm{SL}_2 = \mathrm{Sp}_2$ and $\mathrm{SU}(2) = \mathrm{USp}(2)$.

For the paths of Section 7.4 we need a joint statement for additive shifts.

THEOREM 7.4 (Joint equidistribution of shifts). *Let $h_1, \dots, h_k$ be distinct integers. As $p \to \infty$, with a uniform among the elements of $\mathbb{F}_p$ for which $a + h_j \neq 0$ for all j, the random vector $\big(\mathrm{Kl}(a+h_1;p), \dots, \mathrm{Kl}(a+h_k;p)\big)/\sqrt{p}$ converges in law to $(\mathrm{ST}_1, \dots, \mathrm{ST}_k)$, independent random variables with law μ_{ST}.*

This is due to Kowalski and Sawin [331], who deduce it from Katz's computation of the monodromy. The relevant monodromy group is now that of the direct sum of the translated sheaves $[x \mapsto x + h_j]^*\mathcal{K}l$, a subgroup of SL_2^k projecting onto each factor. By the Goursat Kolchin Ribet criterion [297] it is all of SL_2^k unless two of the translates become geometrically isomorphic after tensoring with a rank one sheaf. The ramification at 0 rules this out. Pulling back by $x \mapsto x - h_i$, we may compare $\mathcal{K}l$ itself with its translate $[x \mapsto x + h]^*\mathcal{K}l$, where $h = h_j - h_i \neq 0$, and an isomorphism $[x \mapsto x + h]^*\mathcal{K}l \simeq \mathcal{K}l \otimes \mathcal{L}$ is impossible: the translate is lisse at 0, while $\mathcal{K}l \otimes \mathcal{L}$ is ramified at 0 for every rank one sheaf $\mathcal{L}$, because the local monodromy of $\mathcal{K}l$ at 0 is a nontrivial unipotent block. Joint equidistribution then follows from Deligne's theorem with $\mathrm{SU}(2)^k$ in place of $\mathrm{SU}(2)$: the Kloosterman sums at distinct additive shifts behave like independent traces of Haar random matrices.

7.2.1 Data

For this book we computed $\mathrm{Kl}(a;p) = \sum_{x=1}^{p-1} e((x + a\bar{x})/p)$ for all a at $p = 1009$ and $p = 10007$, and for 3000 random a at $p = 100003$. Table 7.1 compares the even moments of $\mathrm{Kl}/\sqrt{p}$ with the Catalan numbers. The second and fourth moments at $p = 1009$ and $p = 10007$ agree with the exact values of Proposition 7.1 to all printed

	$m=1$	$m=2$	$m=3$	$m=4$
$p=1009$, all a	0.999999	1.99900	4.98271	13.90784
$p=10007$, all a	1.0000000	1.99990	5.00198	14.03501
$p=100003$, 3000 values	0.9918	1.9794	4.9547	13.9179
Catalan C_m	1	2	5	14

Table 7.1: Empirical moments $\mathbb{E}(\mathrm{Kl}(a;p)/\sqrt{p})^{2m}$, with a uniform over the values listed, against the Sato Tate moments C_m. For the random sample at $p=100003$ the standard errors under μ_{ST}, $\big((C_{2m}-C_m^2)/3000\big)^{1/2}$, are about 0.02, 0.06, 0.19 and 0.64 in the four columns.

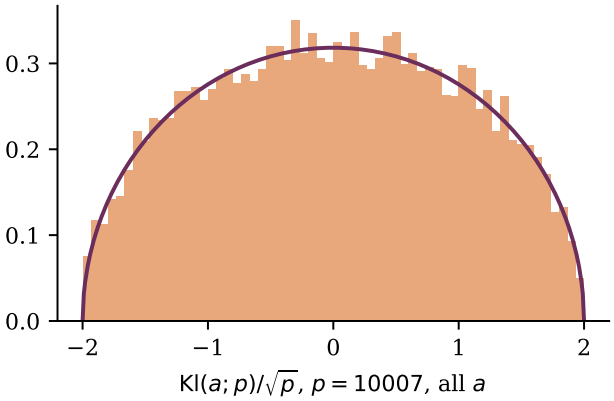


Figure 7.1: Histogram of the normalised Kloosterman sums $\mathrm{Kl}(a;p)/\sqrt{p}$ for all $a\in\mathbb{F}_p^\times$, $p=10007$, against the semicircle density $\frac{1}{2\pi}\sqrt{4-x^2}$ of μ_{ST} (Theorem 7.3).

digits: for $p=1009$ the identity predicts $1-1/(1009\cdot 1008)=0.999999$ and $(2p^3-3p^2-3p-1)/((p-1)p^2)=1.999005$. The largest values of $|\mathrm{Kl}|/\sqrt{p}$ are 1.98077 and 1.99876, inside Weil's interval and close to its edge, and Kolmogorov Smirnov tests against μ_{ST} give p values 0.918 and 0.993 (descriptive only, since the values $\mathrm{Kl}(a;p)$ are not independent samples). Figure 7.1 shows the histogram at $p=10007$.

7.2.2 The horizontal question

For a fixed holomorphic newform of weight at least 2 with trivial character and without complex multiplication, Sato Tate in the horizontal direction is a theorem (Theorem 4.9), because the Satake parameters at varying p are the local components of a single global automorphic representation, and the potential automorphy [67] and then automorphy [419, 420] of its symmetric powers supply L functions holomorphic and nonvanishing on $\Re s\geq 1$ (Chapter 4). For Kloosterman sums

there is no such global object: as p varies, the sheaves $\mathcal{K}l$ live over different finite fields and are not known to be the reductions of a compatible system attached to a motive or automorphic representation over $\mathbb{Q}$. Katz conjectured the horizontal law nevertheless.

CONJECTURE 7.5 (Katz's horizontal conjecture). *Fix a nonzero integer a. As p runs over the odd primes not dividing a, ordered by size, the angles $\theta_{p,a}$ are equidistributed in $[0,\pi]$ with respect to $\frac{2}{\pi}\sin^2\theta\,d\theta$. Equivalently,*

$$\frac{1}{\pi(x)}\sum_{p\le x,\ p\nmid a} U_n\Big(\frac{\mathrm{Kl}(a;p)}{2\sqrt{p}}\Big)\longrightarrow 0 \qquad (n\ge 1).$$

Very little is known. Even the statement that $\mathrm{Kl}(1;p)$ is positive for infinitely many primes and negative for infinitely many primes is, to our knowledge, open; Fouvry and Michel proved sign changes of $\mathrm{Kl}(1;c)$ along moduli c with a bounded number of prime factors [198], using sieve methods together with Katz's equidistribution results. The vertical theorem, by contrast with the horizontal conjecture, is geometry over a single finite field.

7.3 KLOOSTERMAN SUMS AND AUTOMORPHIC FORMS

For an integer $c\ge 1$ let $S(m,n;c)=\sum_{x\in(\mathbb{Z}/c\mathbb{Z})^\times} e((mx+n\bar{x})/c)$. For prime $c=p$ this is $\mathrm{Kl}(m,n;p)$. The Chinese remainder theorem gives a twisted multiplicativity, and Weil's bound together with an elementary evaluation at prime powers yields

$$|S(m,n;c)|\le \tau(c)\,(m,n,c)^{1/2}\,c^{1/2},$$

with τ the divisor function [276].

7.3.1 The Bruggeman Kuznetsov formula

Let $\{u_j\}$ be an orthonormal basis of Maass cusp forms for $\mathrm{SL}_2(\mathbb{Z})$, eigenfunctions of the hyperbolic Laplacian with eigenvalues $\lambda_j=\frac14+t_j^2$, with Fourier expansions $u_j(z)=\sum_{n\neq0}\rho_j(n)W_{s_j}(nz)$, where $s_j=\frac12+it_j$ and $W_s(z)=2y^{1/2}K_{s-1/2}(2\pi y)e(x)$ is the Whittaker function, as in [275]. Put $\nu_j(n)=(4\pi|n|/\cosh\pi t_j)^{1/2}\rho_j(n)$.

THEOREM 7.6 (Bruggeman [87], Kuznetsov [337]). *Let $\delta>0$ and let h be even and holomorphic in $|\Im t|\le\frac12+\delta$ with $h(t)\ll(1+|t|)^{-2-\delta}$. For integers $m,n\ge1$,*

$$\sum_j h(t_j)\overline{\nu_j(m)}\nu_j(n)+\frac{1}{4\pi}\int_{-\infty}^{\infty}h(t)\,\overline{\eta_m(t)}\eta_n(t)\,dt$$
$$=\frac{\delta_{m,n}}{\pi}\int_{-\infty}^{\infty}t\tanh(\pi t)h(t)\,dt+\sum_{c\ge1}\frac{S(m,n;c)}{c}\,h^+\Big(\frac{4\pi\sqrt{mn}}{c}\Big),$$

where $\eta_n(t) = (4\pi|n|/\cosh \pi t)^{1/2}\rho_t(n)$, *with* $\rho_t(n)$ *the coefficient of* $W_{1/2+it}(nz)$ *in the Eisenstein series* $E(z, \frac{1}{2}+it)$, *and*

$$h^+(x) = 2i\int_{-\infty}^{\infty} J_{2it}(x)\frac{h(t)\,t}{\cosh \pi t}\,dt.$$

The normalisation of W_s, ρ_j, ν_j and of $E(z,s) = \sum_{\gamma\in\Gamma_\infty\backslash \mathrm{SL}_2(\mathbb{Z})} \mathrm{Im}(\gamma z)^s$ is that of Iwaniec [275]. Constants differ between sources: dividing by π gives the equivalent form with spectral weights $\overline{a_j(m)}a_j(n)/\cosh \pi t_j$, where $a_j(n) = 2|n|^{1/2}\rho_j(n)$ is the coefficient of $y^{1/2}K_{it_j}(2\pi|n|y)e(nx)$, diagonal constant π^{-2} and prefactor $2i/\pi$ in h^+. The formula holds for congruence subgroups $\Gamma_0(N)$ with c restricted to multiples of N. Its structure parallels the trace formula picture of Chapter 9. The left side is a spectral average of products of Fourier coefficients, that is of Hecke eigenvalues, since $\nu_j(n) = \nu_j(1)\lambda_j(n)$ for a basis of Hecke eigenforms, with $\lambda_j(n)$ the eigenvalues normalised so that the Ramanujan conjecture reads $|\lambda_j(p)| \le 2$. The diagonal term $\delta_{m,n}$ carries the archimedean Plancherel density $t\tanh(\pi t)$, the analogue of the Kesten McKay density μ_q, and the Kloosterman terms are the analogue of the cycle counts that measure the deviation of a finite quotient from its universal cover. The formula can be read in either direction: as a tool for averaging Fourier coefficients (the Kloosterman side is small by Weil), or as a tool for bounding sums of Kloosterman sums (the spectral side is controlled). The second reading is Kuznetsov's.

7.3.2 Selberg's 3/16 theorem

Let $\Gamma = \Gamma_0(N)$ and let $\lambda_1(\Gamma)$ be the smallest nonzero eigenvalue of the Laplacian on $L^2(\Gamma\backslash\mathbb{H})$. The continuous spectrum begins at $\frac{1}{4}$, the bottom of the spectrum of the hyperbolic plane itself; an eigenvalue $\lambda = s(1-s) < \frac{1}{4}$ with $\frac{1}{2} < s < 1$ is called exceptional.

THEOREM 7.7 (Selberg [497]). *For every congruence subgroup* $\Gamma_0(N)$, $\lambda_1 \ge \frac{3}{16}$.

Sketch of Selberg's argument. Consider the Kloosterman zeta function

$$Z_{m,n}(s) = \sum_{c\equiv 0\ (N)} \frac{S(m,n;c)}{c^{2s}}.$$

Selberg showed, by unfolding the inner product of two Poincaré series and inserting the spectral decomposition, that $Z_{m,n}$ continues meromorphically, and that in $\Re s > \frac{1}{2}$ its only possible poles are at the points $s_j \in (\frac{1}{2}, 1)$ with $s_j(1-s_j)$ an exceptional eigenvalue, with residue a nonzero multiple of $\sum \overline{\rho(m)}\rho(n)$ summed over an orthonormal basis of that eigenspace. Taking $m = n$ with $\rho(n) \ne 0$ for some form in the eigenspace, the residue is a positive multiple of $\sum|\rho(n)|^2$, so the

pole is genuine. On the other hand Weil's bound gives $|S(n,n;c)| \ll_{n,\varepsilon} c^{1/2+\varepsilon}$, so the series converges absolutely for $\Re s > \frac{3}{4}$ and $Z_{n,n}$ is holomorphic there. Hence every exceptional s_j satisfies $s_j \le \frac{3}{4}$, and since $s \mapsto s(1-s)$ decreases on $[\frac{1}{2}, 1]$, $\lambda_j \ge \frac{3}{4} \cdot \frac{1}{4} = \frac{3}{16}$.

Selberg conjectured that $\lambda_1 \ge \frac{1}{4}$ for all congruence subgroups. In representation theoretic language this is the Ramanujan conjecture at the archimedean place for Maass forms: the archimedean component of the automorphic representation is tempered exactly when $\lambda \ge \frac{1}{4}$. The best bound known is due to Kim and Sarnak in the appendix to Kim's paper [315]: using the automorphy of Sym^4 for GL_2, they show that the archimedean Satake parameter of a Maass cusp form on a congruence subgroup has real part at most $\frac{7}{64}$, hence $\lambda_1 \ge \frac{1}{4} - \left(\frac{7}{64}\right)^2 = \frac{975}{4096} \approx 0.238$. Earlier improvements on $\frac{3}{16}$ were obtained by Luo, Rudnick and Sarnak [374] and by Kim and Shahidi [316].

7.3.3 Linnik and Selberg on sums of Kloosterman sums

Weil's bound alone gives $\sum_{c \le x} |S(m,n;c)|/c \ll x^{1/2+\varepsilon}$. Linnik [367] and Selberg [497] conjectured that the signs of the Kloosterman sums cancel as c varies.

CONJECTURE 7.8 (Linnik, Selberg). *For fixed $m, n \ge 1$ and every $\varepsilon > 0$,*

$$\sum_{c \le x} \frac{S(m,n;c)}{c} \ll_{m,n,\varepsilon} x^{\varepsilon}.$$

Kuznetsov proved the bound $x^{1/6+\varepsilon}$ [337], and to our knowledge the exponent $\frac{1}{6}$ has not been improved for fixed m, n. He obtained it from the version of Theorem 7.6 read in the opposite direction, in which a test function on the Kloosterman side is prescribed (here one that approximates the indicator of $c \le x$) and the spectral side is bounded [276]. The connection with Theorem 7.7 is direct. If the analogue of Conjecture 7.8 held for $\Gamma_0(N)$, with c restricted to multiples of N, then partial summation would make $Z_{n,n}(s)$ holomorphic in $\Re s > \frac{1}{2}$, and Selberg's argument would exclude exceptional eigenvalues altogether. Conversely, an exceptional eigenvalue $s_j(1-s_j)$ produces a term of order x^{2s_j-1} in the smoothed sums.

The probabilistic reading. Think of $c \mapsto S(m,n;c)/\sqrt{c}$ as a sequence of random variables. For prime c these are $2\cos\theta_{c,mn}$, and if the angles behaved like independent Sato Tate variables (the horizontal model of Conjecture 7.5) the sum $\sum_{p \le x} 2\cos\theta_{p,mn}/\sqrt{p}$ would have mean zero and variance $\sum_{p \le x} 1/p \sim \log\log x$, comfortably inside x^{ε}. Square root cancellation in sums of Kloosterman sums is therefore the statement that this sequence has no drift, and Theorem 7.6 identifies the possible drifts: they are the exceptional eigenvalues, each contributing a

deterministic term x^{2s_j-1}. The analogy with Chapters 9 and 10 is close. There, a connected finite quotient of the tree is Ramanujan exactly when the return probabilities of simple random walk on it approach equilibrium at an exponential rate no slower than the Kesten rate $2\sqrt{q}/(q+1)$ (Proposition 10.3); here, $\lambda_1 \geq \frac{1}{4}$ means that Brownian motion on $\Gamma\backslash\mathbb{H}$, with generator the Laplacian, mixes at least at the rate $e^{-t/4}$ dictated by the spectrum of the hyperbolic plane. In both settings the Ramanujan property is a spectral gap, and a trace formula converts it into cancellation in an arithmetic sum: counts of closed nonbacktracking walks in the first case, through the Ihara Bass formula, and Kloosterman sums over moduli in the second, through Kuznetsov's.

7.4 KLOOSTERMAN PATHS

Kowalski and Sawin [331] asked what the partial sums of a Kloosterman sum look like. For $a \in \mathbb{F}_p^\times$ and $0 \leq t \leq 1$ let

$$S_p(a,t) = \frac{1}{\sqrt{p}} \sum_{1 \leq x \leq t(p-1)} e\Big(\frac{ax+\bar{x}}{p}\Big),$$

and let $K_p(a,\cdot) \in C([0,1])$ be the polygonal path that interpolates linearly between the values at $t = j/(p-1)$, $0 \leq j \leq p-1$. Then $K_p(a,0) = 0$, $K_p(a,1) = \mathrm{Kl}(a;p)/\sqrt{p}$, and $|K_p(a,t) - S_p(a,t)| \leq p^{-1/2}$, so the two have the same limiting finite dimensional distributions. With a uniform in $\mathbb{F}_p^\times$, K_p is a random continuous function.

DEFINITION 7.9. Let $(\mathrm{ST}_h)_{h\in\mathbb{Z}}$ be independent random variables with law μ_{ST} and put $c_0(t) = t$ and $c_h(t) = (e(ht)-1)/(2\pi i h)$ for $h \neq 0$. The *Kloosterman random Fourier series* is

$$K(t) = \sum_{h\in\mathbb{Z}} c_h(t)\,\mathrm{ST}_h = t\,\mathrm{ST}_0 + \sum_{h\neq 0} \frac{e(ht)-1}{2\pi i h}\mathrm{ST}_h,$$

the series being the limit of its symmetric partial sums over $|h| \leq H$ as $H \to \infty$.

Since $\mathbb{E}\,\mathrm{ST}_h = 0$, $\mathbb{E}\,\mathrm{ST}_h^2 = 1$ and $\sum_h |c_h(t)|^2 < \infty$, the series converges in L^2 and almost surely for each t. Kowalski and Sawin show, with Kahane's theory of random Fourier series, that K is almost surely continuous, indeed Hölder continuous of every order below $\frac{1}{2}$, and almost surely nowhere differentiable [331]; so K is a random element of $C([0,1])$. Since $c_h(t) = \int_0^t e(hu)\,du$, the process K is formally the integral $\int_0^t W(u)\,du$ of the random Fourier series $W(u) = \sum_h \mathrm{ST}_h e(hu)$, whose Fourier coefficients are independent Sato Tate variables. Equivalently $c_h(t) = \widehat{\mathbb{1}_{[0,t]}}(h)$, where $\hat{f}(h) = \int_0^1 f(u)e(-hu)\,du$, and in this form the coefficients arise from the discrete Fourier analysis of an interval, as we now show.

7.4.1 The completion formula

LEMMA 7.10 (Completion). *Let $I = \{1, \dots, m\}$ with $m = \lfloor t(p-1) \rfloor$ and, for $h \in \mathbb{Z}$,*

$$\alpha_p(h,t) = \frac{1}{p} \sum_{y=1}^{m} e\Big(\frac{hy}{p}\Big).$$

Then, for every $a \in \mathbb{F}_p$,

$$S_p(a,t) = \sum_{|h|<p/2} \alpha_p(h,t)\, \frac{\mathrm{Kl}(a-h;p)}{\sqrt{p}}. \tag{7.1}$$

Further $\alpha_p(0,t) = m/p$, $|\alpha_p(h,t)| \le 1/(2|h|)$ for $1 \le |h| < p/2$, and $\alpha_p(h,t) \to c_h(t)$ as $p \to \infty$ for each fixed h and t.

PROOF. By finite Fourier inversion on $\mathbb{Z}/p\mathbb{Z}$, for every residue x,

$$\mathbb{I}_I(x) = \sum_{h \bmod p} \alpha_p(h,t)\, e\Big(-\frac{hx}{p}\Big),$$

since $\sum_h \alpha_p(h,t) e(-hx/p) = p^{-1} \sum_{y \in I} \sum_h e(h(y-x)/p)$ and the inner sum is p when $y = x$ and 0 otherwise. Insert this into the definition of S_p and interchange the sums:

$$\begin{aligned} \sqrt{p}\, S_p(a,t) &= \sum_{h \bmod p} \alpha_p(h,t) \sum_{x \in \mathbb{F}_p^\times} e\Big(\frac{(a-h)x + \bar{x}}{p}\Big) \\ &= \sum_{h \bmod p} \alpha_p(h,t)\, \mathrm{Kl}(a-h;p); \end{aligned}$$

then represent the residues h by $|h| < p/2$. For $h \neq 0$ the geometric sum gives $|\sum_{y=1}^{m} e(hy/p)| = |\sin(\pi h m/p)/\sin(\pi h/p)| \le 1/|\sin(\pi h/p)|$, and $\sin(\pi u) \ge 2u$ for $0 \le u \le \tfrac{1}{2}$ gives $|\alpha_p(h,t)| \le 1/(2|h|)$. Finally $\alpha_p(h,t)$ is a Riemann sum for $\int_0^{m/p} e(hu)\,du$ with mesh $1/p$, and $m/p \to t$, so $\alpha_p(h,t) \to \int_0^t e(hu)\,du = c_h(t)$. □

Formula (7.1) is the starting point of Kowalski and Sawin [331]. Its term $h \equiv a$ involves $\mathrm{Kl}(0;p) = -1$; it is the only one that is not a genuine Kloosterman sum, and for $1 \le a \le p-1$ its coefficient is at most $1/(2\min(a, p-a))$ in absolute value. The completion formula exhibits the path as a linear functional of the shifted sums $\mathrm{Kl}(a-h;p)$, $|h| < p/2$, which by Theorem 7.4 are asymptotically independent Sato Tate variables. Renaming $h \mapsto -h$ writes the same identity with the shifts $\mathrm{Kl}(a+h;p)$ and the limiting coefficients $c_{-h}(t)$; the law of K is unchanged because the ST_h are independent and identically distributed.

The next lemma controls the tail of (7.1) uniformly in p; it uses only orthogonality.

LEMMA 7.11 (Uniform tail bound; this book). *For every $H \geq 1$, every $p > 2H+1$ and every $t \in [0,1]$,*

$$\frac{1}{p-1}\sum_{a\in\mathbb{F}_p^\times}\Big|\sum_{H<|h|<p/2}\alpha_p(h,t)\frac{\mathrm{Kl}(a-h;p)}{\sqrt{p}}\Big|^2 \leq \frac{1}{H}.$$

PROOF. For $k \in \mathbb{F}_p$, orthogonality in c gives

$$\begin{aligned}\sum_{c\in\mathbb{F}_p}\mathrm{Kl}(c;p)\mathrm{Kl}(c+k;p) &= \sum_{x,y\neq 0} e\Big(\frac{\bar{x}+\bar{y}+ky}{p}\Big)\sum_{c} e\Big(\frac{c(x+y)}{p}\Big)\\ &= p\sum_{x\neq 0} e\Big(-\frac{kx}{p}\Big),\end{aligned}$$

which is $p(p-1)$ for $k=0$ and $-p$ for $k\neq 0$. Hence, since the $\mathrm{Kl}(c;p)$ are real, for any coefficients β_h indexed by a set $\mathcal{H}$ of distinct residues,

$$\frac{1}{p}\sum_{c\in\mathbb{F}_p}\Big|\sum_{h\in\mathcal{H}}\beta_h\frac{\mathrm{Kl}(c-h;p)}{\sqrt{p}}\Big|^2 = \sum_h|\beta_h|^2 - \frac{1}{p}\Big|\sum_h\beta_h\Big|^2,$$

which is at most $\sum_h|\beta_h|^2$. Take $\mathcal{H} = \{H<|h|<p/2\}$ and $\beta_h = \alpha_p(h,t)$. By Lemma 7.10, $\sum_{|h|>H}|\alpha_p(h,t)|^2 \leq 2\sum_{h>H}\frac{1}{4h^2} \leq \frac{1}{2H}$. Passing from $\mathbb{F}_p$ to $\mathbb{F}_p^\times$ multiplies the average of a nonnegative quantity by at most $p/(p-1)\leq 2$. □

THEOREM 7.12 (Kowalski and Sawin [331]). *With a uniform in $\mathbb{F}_p^\times$, the finite dimensional distributions of the Kloosterman paths $K_p(a,\cdot)$ converge to those of K: for all $0\leq t_1<\cdots<t_k\leq 1$, $(K_p(a,t_i))_{i\leq k} \Rightarrow (K(t_i))_{i\leq k}$ in $\mathbb{C}^k$.*

Proof, given Theorem 7.4. It suffices to treat $S_p(a,t_i)$. Fix H and let $S_p^{(H)}(a,t)$ be the part of (7.1) with $|h|\leq H$, and $K^{(H)}(t)=\sum_{|h|\leq H}c_h(t)\mathrm{ST}_h$. Excluding the at most $2H+1$ values $a\equiv h$, Theorem 7.4 gives $(\mathrm{Kl}(a-h;p)/\sqrt{p})_{|h|\leq H}\Rightarrow(\mathrm{ST}_h)_{|h|\leq H}$, and these vectors lie in the compact cube $[-2,2]^{2H+1}$ by Weil's bound. The vector $(S_p^{(H)}(a,t_i))_i$ is the image of this vector under a linear map whose coefficients $\alpha_p(h,t_i)$ converge to $c_h(t_i)$ (Lemma 7.10), so $(S_p^{(H)}(a,t_i))_i \Rightarrow (K^{(H)}(t_i))_i$ as $p\to\infty$. Next, $(K^{(H)}(t_i))_i\to(K(t_i))_i$ in L^2 as $H\to\infty$, because $\sum_{|h|>H}|c_h(t)|^2 \leq \sum_{|h|>H}\pi^{-2}h^{-2}\leq 2/(\pi^2H)$. Finally, by Lemma 7.11, Chebyshev's inequality and a union bound, $\mathbb{P}(\max_i|S_p(a,t_i)-S_p^{(H)}(a,t_i)|>\varepsilon)\leq k/(H\varepsilon^2)$ for all $p>2H+1$. The approximation theorem for weak convergence [60, Theorem 4.2] now gives the claim. □

Thus Katz's joint equidistribution, which is the deep input, enters only through finitely many shifts at a time, and the infinite series is assembled by an L^2 argument that needs nothing beyond orthogonality of additive characters.

Tightness. Convergence in law in $C([0,1])$ requires in addition tightness, for instance a moment bound of the form $\mathbb{E}|K_p(a,t)-K_p(a,s)|^4 \ll |t-s|^{1+\delta}$ uniformly in p, which amounts to square root cancellation in *short* Kloosterman sums averaged over a. Kowalski and Sawin prove convergence in law in $C([0,1])$ in two cases where such an estimate is available [331]: the two parameter family $\sum_x e((ax+b\bar{x})/p)$ with (a,b) uniform in $(\mathbb{F}_p^\times)^2$ (they phrase it as an extra average over the additive character), where the extra average supplies the fourth moment bound, and the Birch sums $\sum_x e((ax+x^3)/p)$. For the one parameter family above the required estimate for short sums averaged over a alone is not known, and they explain that convergence in $C([0,1])$ would follow from natural conjectures on short sums. In later work they describe the support of the limiting law in $C([0,1])$ [332].

7.4.2 Exact second moments and the convention lesson

The second moments of the paths can be computed exactly at every finite p, without Katz's theorem. Let $I=\{1,\dots,m\}$ be the interval of Lemma 7.10 and $-I=\{p-x : x\in I\}$.

PROPOSITION 7.13 (Second moments of Kloosterman paths; this book). *Let $R_I=\sum_{x\in I}e(\bar{x}/p)$ and $G_I=\sum_{x\in I}e(x/p)$. With a uniform in $\mathbb{F}_p^\times$:*

1. *for $S_p(a,t)=p^{-1/2}\sum_{x\in I}e((ax+\bar{x})/p)$,*

$$\mathbb{E}|S_p|^2=\frac{p|I|-|R_I|^2}{p(p-1)},\qquad \mathbb{E}S_p^2=\frac{p\,|I\cap(-I)|-R_I^2}{p(p-1)};$$

2. *for $S'_p(a,t)=p^{-1/2}\sum_{x\in I}e((x+a\bar{x})/p)$,*

$$\mathbb{E}|S'_p|^2=\frac{p|I|-|G_I|^2}{p(p-1)},\qquad \mathbb{E}S_p'^2=\frac{p\,|I\cap(-I)|-G_I^2}{p(p-1)}.$$

In both cases $\mathbb{E}S=-p^{-1/2}(p-1)^{-1}R_I$, respectively $-p^{-1/2}(p-1)^{-1}G_I$, which tends to 0.

PROOF. For the first family, summing over all $a\in\mathbb{F}_p$,

$$\sum_a\Big|\sum_{x\in I}e\Big(\frac{ax+\bar{x}}{p}\Big)\Big|^2=\sum_{x,y\in I}e\Big(\frac{\bar{x}-\bar{y}}{p}\Big)\sum_a e\Big(\frac{a(x-y)}{p}\Big)=p|I|,$$

and similarly $\sum_a(\sum_{x\in I}e((ax+\bar{x})/p))^2=p\sum_{x,y\in I,\,x+y\equiv 0}e((\bar{x}+\bar{y})/p)=p|I\cap(-I)|$, because $\bar{y}=-\bar{x}$ when $y=-x$. Removing the term $a=0$, which equals R_I, and dividing by $p(p-1)$ gives (1). The second family is identical with the roles of x and $\bar{x}$ exchanged in the orthogonality step; now the term $a=0$ is the geometric

sum G_I. The means follow from $\sum_{a\in\mathbb{F}_p} e(a\bar{x}/p) = \sum_a e(ax/p) = 0$ for $x \neq 0$, and $|R_I| \le p$, $|G_I| \le p$. □

The two families have the same complete sums, since $\mathrm{Kl}(a,1) = \mathrm{Kl}(1,a)$, and the same variance over all $a \in \mathbb{F}_p$. They differ only through the single excluded value $a = 0$, and that value matters in exactly one of them. In the first family R_I is an incomplete Kloosterman sum, and the completion method gives $|R_I| \ll \sqrt{p}\log p$, so $|R_I|^2/p^2 \to 0$. In the second family G_I is a geometric sum of size comparable to p: $G_I/p \to \int_0^t e(u)\,du = c_1(t)$. Therefore, as $p \to \infty$,

$$\mathbb{E}|S_p|^2 \to t, \qquad \mathbb{E}S_p^2 \to (2t-1)_+,$$
$$\mathbb{E}|S_p'|^2 \to t - |c_1(t)|^2, \qquad \mathbb{E}S_p'^2 \to (2t-1)_+ - c_1(t)^2.$$

The same phenomenon is visible in the completion formula. Expanding $\mathbb{I}_I$ as in Lemma 7.10 in the second family gives

$$\begin{aligned}\sqrt{p}\,S_p'(a,t) &= \sum_{h \bmod p} \alpha_p(h,t) \sum_{x\neq 0} e\Big(\frac{(1-h)x + a\bar{x}}{p}\Big)\\ &= \sum_{h\neq 1} \alpha_p(h,t)\,\mathrm{Kl}\big(a(1-h);p\big) - \alpha_p(1,t),\end{aligned}$$

because for $h = 1$ the inner sum is the Ramanujan sum $\sum_{x\neq 0} e(a\bar{x}/p) = -1$. After division by $\sqrt{p}$ the term $h = 1$ is $O(p^{-1/2})$: the Sato Tate variable that should carry the coefficient $c_1(t)$ is simply absent, and the remaining coefficients multiply Kloosterman sums at the multiplicative translates $a(1-h)$. Kowalski and Sawin point this out in a remark of [331], where the shifts are written $a(1+h)$ and the missing index is $h = -1$, and they show that in this parametrisation the paths do converge in law in $C([0,1])$, because here the short sum estimate needed for tightness follows from a counting result of Bourgain and Garaev; the limit is $K - c_1\mathrm{ST}_1$ in our notation. Their displayed formula has $c_{-1}\mathrm{ST}_{-1}$ in place of $c_1\mathrm{ST}_1$, a process with the law of the complex conjugate of $K - c_1\mathrm{ST}_1$; with the conventions of Lemma 7.10 the missing coefficient is c_1, as the limit of $\mathbb{E}S_p'^2$ above shows, because $c_1(t)^2$ is not real for $0 < t < 1$, $t \neq \frac{1}{2}$. Proposition 7.13 confirms, unconditionally and at every finite p, that the second order structure is that of $K - c_1\mathrm{ST}_1$, and not that of K.

COROLLARY 7.14 (Variance deficit; this book). *In the parametrisation $e((x+a\bar{x})/p)$ the limiting variances are*

$$\mathrm{Var}\,\Re S' = \tfrac{1}{2}\big(t + (2t-1)_+\big) - \frac{\sin^2(2\pi t)}{4\pi^2},$$
$$\mathrm{Var}\,\Im S' = \tfrac{1}{2}\big(t - (2t-1)_+\big) - \frac{\sin^4(\pi t)}{\pi^2}.$$

At $t = \frac{1}{2}$ the real part is unaffected and the imaginary part loses exactly $1/\pi^2$: $\operatorname{Var} \Im S' \to \frac{1}{4} - \frac{1}{\pi^2} = 0.1487$.

PROOF. Write $c_1(t) = \frac{\sin 2\pi t}{2\pi} + i\frac{1-\cos 2\pi t}{2\pi}$, so $(\Re c_1)^2 = \sin^2(2\pi t)/(4\pi^2)$ and $(\Im c_1)^2 = \sin^4(\pi t)/\pi^2$. Since the means tend to zero, $\operatorname{Var} \Re S' = \frac{1}{2}(\mathbb{E}|S'|^2 + \Re \mathbb{E} S'^2)$ and $\operatorname{Var} \Im S' = \frac{1}{2}(\mathbb{E}|S'|^2 - \Re \mathbb{E} S'^2)$; insert the limits above and use $|c|^2 \pm \Re c^2 = 2(\Re c)^2$ or $2(\Im c)^2$. □

Our first run of the path computations, at $p = 10007$, used the parametrisation $e((x + a\bar{x})/p)$, natural for the vertical statistics of Section 7.2. It gave $\operatorname{Var} \Im S'(\frac{1}{2}) = 0.149$, where the random Fourier series sample of that run gave 0.258. Corollary 7.14 explains the gap: the limiting value is $\frac{1}{4} - \frac{1}{\pi^2} = 0.1487$, a deficit of $1/\pi^2 = 0.101$ against the value $\frac{1}{4}$ for K. At $p = 1009$ and $t = \frac{1}{2}$, the exact formulas of Proposition 7.13, averaged over all a, give 0.14848, against $\frac{1}{4} - \frac{1}{\pi^2} = 0.14868$. The lesson is general: vertical statistics of a complete sum are blind to the parametrisation, but partial sums are not, and a single Ramanujan sum can remove one independent coordinate of the limit.

7.4.3 Numerical check

For $p = 10007$ and 3000 random a we compared the paths $K_p(a, \cdot)$, in the parametrisation of Kowalski and Sawin, with 4000 samples of the random Fourier series truncated at $|h| \le 3000$. Two sample Kolmogorov Smirnov tests give the following p values for the real and imaginary parts: at $t = \frac{1}{4}$, 0.79 and 0.80; at $t = \frac{1}{2}$, 0.82 and 0.77; at $t = \frac{3}{4}$, 0.70 and 0.80. At $t = \frac{1}{2}$ the variances are 0.2502 (paths) and 0.2529 (series) for the real part and 0.2459 and 0.2463 for the imaginary part, against the value $\frac{1}{4}$ for both from Proposition 7.15 below. For the sup norm $\|K_p\|_\infty$ the means are 1.159 and 1.138, with a two sample Kolmogorov Smirnov p value of 0.16. The sup norm is not a finite dimensional functional, so this agreement is consistent with convergence in law in $C([0,1])$ for the one parameter family, the part of the theory that is not proved. Figure 7.2 shows typical paths. At $p = 1009$, averaging over all a, the exact formulas of Proposition 7.13 give $\operatorname{Var} \Re S_p(\frac{1}{2}) = 0.25000$ and $\operatorname{Var} \Im S_p(\frac{1}{2}) = 0.24992$.

7.4.4 Properties of the limit

PROPOSITION 7.15 (Covariance and symmetry of K; this book). *For $s, t \in [0,1]$,*

$$\mathbb{E}\, K(s)\overline{K(t)} = \min(s,t), \qquad \mathbb{E}\, K(s)K(t) = (s + t - 1)_+.$$

Consequently $\operatorname{Var} \Re K(t) = \frac{1}{2}\big(t + (2t-1)_+\big)$, $\operatorname{Var} \Im K(t) = \frac{1}{2}\big(t - (2t-1)_+\big)$, *and* $\Re K(t)$, $\Im K(t)$ *are uncorrelated. Moreover, almost surely,*

$$K(1-t) = K(1) - \overline{K(t)} \qquad (0 \le t \le 1),$$

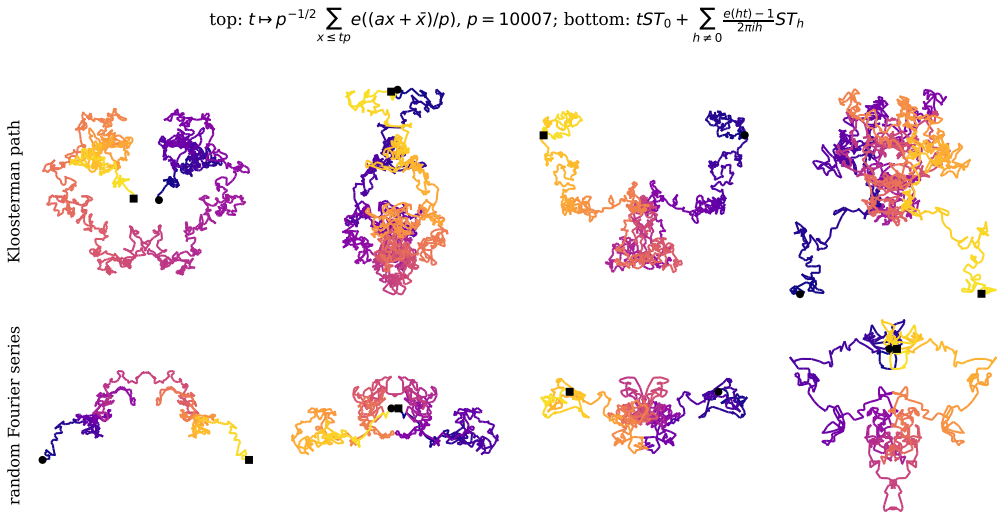


Figure 7.2: Top: Kloosterman paths $t \mapsto p^{-1/2} \sum_{x \le t(p-1)} e((ax+\bar{x})/p)$ in the complex plane for four values of a, $p = 10007$, coloured by t. Bottom: four samples of the random Fourier series $K(t)$ of Definition 7.9. Every path starts at 0 and ends on the real axis at a point of $[-2, 2]$; its imaginary part is symmetric about $t = \frac{1}{2}$, and the path is symmetric under reflection in the vertical line through the midpoint of its endpoints (Proposition 7.15).

so $\Im K(1-t) = \Im K(t)$, $K(1) = 2\Re K(\frac{1}{2}) = \mathrm{ST}_0$, *and* $\Im K(0) = \Im K(1) = 0$.

PROOF. Since the ST_h are independent, real, centred and of variance one, $\mathbb{E}K(s)\overline{K(t)} = \sum_h c_h(s)\overline{c_h(t)}$ and $\mathbb{E}K(s)K(t) = \sum_h c_h(s)c_h(t)$. With $f = \mathbb{I}_{[0,s]}$ and $g = \mathbb{I}_{[0,t]}$ we have $c_h(s) = \overline{\hat{f}(h)}$, so Parseval gives $\sum_h \overline{\hat{f}(h)}\hat{g}(h) = \int_0^1 fg = \min(s,t)$. For the second identity, $\hat{g}(h) = \overline{\check{\tilde{g}}(h)}$ with $\tilde{g}(u) = g(-u)$ on $\mathbb{R}/\mathbb{Z}$, so $\sum_h \overline{\hat{f}(h)}\,\overline{\hat{g}(h)} = \sum_h \overline{\hat{f}(h)}\check{\tilde{g}}(h) = \int_0^1 f(u)g(-u)\,du$, the length of $[0,s] \cap [1-t,1]$, which is $(s+t-1)_+$. Both sums are real, so the variances follow from $\mathrm{Var}\,\Re K = \frac{1}{2}(\mathbb{E}|K|^2 + \mathbb{E}K^2)$ and $\mathrm{Var}\,\Im K = \frac{1}{2}(\mathbb{E}|K|^2 - \mathbb{E}K^2)$, and the covariance of the real and imaginary parts is $\frac{1}{2}\Im\mathbb{E}K^2 = 0$. For the symmetry, $e(h(1-t)) = e(-ht)$ gives $c_h(1-t) = -\overline{c_h(t)}$ for $h \neq 0$ and $c_h(1) = 0$; summing term by term, $K(1-t) = (1-t)\mathrm{ST}_0 - \overline{(K(t) - t\,\mathrm{ST}_0)} = \mathrm{ST}_0 - \overline{K(t)}$, and $K(1) = \mathrm{ST}_0$. □

At $t = \frac{1}{4}$ the formulas of Proposition 7.15 give $\frac{1}{8}$ and $\frac{1}{8}$, against the observed path variances 0.1234 and 0.1249, and at $t = \frac{3}{4}$ they give $\frac{5}{8}$ and $\frac{1}{8}$, against 0.6308 and 0.1249; the two observed imaginary variances coincide, as the symmetry below forces already at finite p. The limits of the exact second moments in

Proposition 7.13, $\mathbb{E}|S_p|^2 \to t$ and $\mathbb{E}S_p^2 \to (2t-1)_+$, are exactly these covariances at $s = t$, and the same computation with two intervals gives $\min(s,t)$ and $(s + t - 1)_+$ for $s \neq t$: the second order structure of the paths agrees with that of K unconditionally, without Katz's theorem.

The arithmetic meaning of the symmetry is simple: $x \mapsto p - x$ maps $e((ax + \bar{x})/p)$ to its conjugate, so $K_p(a, 1-t) = \overline{K_p(a,t)}$ for all t. The second half of a Kloosterman path is therefore the mirror image of the first half in the vertical line $\Re z = \mathrm{Kl}(a;p)/(2\sqrt{p})$, which is the symmetry visible in every path of Figure 7.2.

PROPOSITION 7.16 (Unboundedness and a Gaussian tail bound; this book). *The random variable $\Im K(\frac{1}{2}) = \frac{1}{\pi}\sum_{h \text{ odd}} \mathrm{ST}_h/h$ is unbounded above and below, and for every $M > 0$,*

$$\mathbb{P}\big(\Im K(\tfrac{1}{2}) > M\big) \le e^{-2M^2},$$

the Chernoff bound of a centred Gaussian variable with the same variance $\frac{1}{4}$. More generally, every real random variable $X = \sum_h \gamma_h \mathrm{ST}_h$ with $\sum_h \gamma_h^2 < \infty$, in particular every real linear combination of the real and imaginary parts of finitely many values of K, satisfies $\mathbb{E}e^{\lambda X} \le e^{\lambda^2 \mathrm{Var}(X)/2}$ for all real λ.

PROOF. At $t = \frac{1}{2}$, $c_h = ((-1)^h - 1)/(2\pi i h)$ vanishes for even $h \neq 0$ and equals $i/(\pi h)$ for odd h, which gives the formula; pairing h with $-h$, $\Im K(\frac{1}{2}) = \frac{1}{\pi}\sum_{h>0 \text{ odd}}(\mathrm{ST}_h - \mathrm{ST}_{-h})/h$. Given M, choose N with $\frac{3}{\pi}\sum_{h \le N,\, h \text{ odd}} \frac{1}{h} > M + 1$, possible since the harmonic series over odd integers diverges. With positive probability $\mathrm{ST}_h > 1.5$ and $\mathrm{ST}_{-h} < -1.5$ for all odd $h \le N$, so the partial sum exceeds $M + 1$. The remainder is independent of this event and symmetric, so it is nonnegative with probability at least $\frac{1}{2}$. Hence $\mathbb{P}(\Im K(\frac{1}{2}) > M + 1) > 0$, and the law is symmetric. For the tail, the law μ_{ST} is subgaussian with variance proxy equal to its variance: its odd moments vanish and its even moments are the Catalan numbers $C_k = (2k)!/(k!\,(k+1)!)$, so

$$\mathbb{E}e^{\lambda \mathrm{ST}_h} = \sum_{k \ge 0} \frac{\lambda^{2k}}{k!\,(k+1)!} \le \sum_{k \ge 0} \frac{\lambda^{2k}}{2^k k!} = e^{\lambda^2/2},$$

because $2^k \le (k+1)!$. For $X = \sum_h \gamma_h \mathrm{ST}_h$, independence gives $\mathbb{E}e^{\lambda X_N} \le e^{\lambda^2 \sum_{|h| \le N} \gamma_h^2/2}$ for the partial sums X_N, which converge to X almost surely, and Fatou's lemma gives $\mathbb{E}e^{\lambda X} \le e^{\lambda^2 \mathrm{Var}(X)/2}$, since $\mathrm{Var}(X) = \sum_h \gamma_h^2$. For $\Im K(\frac{1}{2})$ we have $\gamma_h = 1/(\pi h)$ for odd h and $\sum_{h \text{ odd}} \gamma_h^2 = \frac{2}{\pi^2} \cdot \frac{\pi^2}{8} = \frac{1}{4}$, and Chernoff's bound with $\lambda = 4M$ gives $\mathbb{P}(\Im K(\frac{1}{2}) > M) \le e^{-4M^2 + 2M^2} = e^{-2M^2}$. □

So although each normalised Kloosterman sum lies in $[-2, 2]$, the limit of the partial sums is unbounded, consistent with the fact that partial sums of such exponential sums can be as large as a constant times $\sqrt{p}\log\log p$; for Birch sums

Lamzouri showed that some partial sum reaches $(\frac{2}{\pi}+o(1))\sqrt{p}\log\log p$ [343]. The true tails are much thinner than Gaussian. Kowalski and Sawin prove that for $0<t<1$ there is $c=c(t)>0$ with

$$c^{-1}\exp\bigl(-\exp(cM)\bigr)\le \mathbb{P}\bigl(|\Im K(t)|\ge M\bigr)\le c\exp\bigl(-\exp(M/c)\bigr)\quad (M>0),$$

the same bounds for $|\Re K(t)|$ when $t\neq\frac{1}{2}$, and bounds of the same shape for $\|K\|_\infty$ [331]. For the maximum of the partial sums at finite p, Lamzouri obtained nearly matching bounds of this type for Birch sums [343], and Autissier, Bonolis and Lamzouri showed, for Birch sums and for the two parameter Kloosterman family, that the proportion of parameters for which the maximum exceeds $V\sqrt{p}$ is $\exp(-\exp(\pi V/2+O(1)))$, uniformly for V up to about $\frac{2}{\pi}\log\log p$; for the further families of trace functions that they treat, whose normalised Fourier transforms take values in $[-N,N]$, they prove the same with $\frac{\pi}{2}$ replaced by $\frac{\pi}{N}$ and the range by $\frac{N}{\pi}\log\log p$ [5].

7.5 Probabilistic interpretation

A Sato Tate Brownian motion. Proposition 7.15 says that on $[0,\frac{1}{2}]$ the process K has exactly the second order structure of a standard complex Brownian motion Z: $\mathbb{E}K(s)\overline{K(t)}=\min(s,t)$ and $\mathbb{E}K(s)K(t)=0$ for $s,t\le\frac{1}{2}$. On $[\frac{1}{2},1]$ it is determined by its first half through the reflection $K(t)=2\Re K(\frac{1}{2})-\overline{K(1-t)}$. The Gaussian process with the same covariance is therefore complex Brownian motion on $[0,\frac{1}{2}]$ folded back on itself: the imaginary part is a Brownian path on $[0,\frac{1}{2}]$ followed by its time reversal, so it is pinned at 0 at both ends like a bridge, and the real part is a Brownian path followed by its reflection through the midpoint value, ending at $2\Re Z(\frac{1}{2})$, a centred Gaussian of variance one. The Kloosterman process differs from this Gaussian model in two visible ways. Its endpoint $K(1)=\mathrm{ST}_0$ is semicircular, hence bounded, and its marginals are not Gaussian although their tails are subgaussian (Proposition 7.16), indeed doubly exponential. The three processes are one series with three choices of coefficients. With independent standard complex Gaussian coefficients ($\mathbb{E}|\zeta_h|^2=1$, $\mathbb{E}\zeta_h^2=0$) the series $\sum_h c_h(t)\zeta_h$ is a centred complex Gaussian process with covariances $\min(s,t)$ and 0, by the proof of Proposition 7.15, that is, a standard complex Brownian motion on $[0,1]$; with independent real standard Gaussian coefficients it is the folded process just described (Exercise 6); and with independent real Sato Tate coefficients, the traces of independent Haar elements of $\mathrm{SU}(2)$, it is K. In this sense K is a Sato Tate Brownian motion.

Relation to random matrices. Writing $\mathrm{ST}_h=\operatorname{tr}g_h$ with g_h independent Haar elements of $\mathrm{SU}(2)$, the limit is a linear statistic of an infinite sequence of independent random matrices. Other one variable sums lead to other groups. For

the additive twists $x \mapsto \mathrm{Kl}^{(r)}(\overline{ax}; p)\, e(bx/p)$ of the hyper Kloosterman sums $\mathrm{Kl}^{(r)}(c; p) = \sum_{y_1 \cdots y_r = c} e((y_1 + \cdots + y_r)/p)$, the discrete Fourier transforms are, after normalisation, the hyper Kloosterman sums $\mathrm{Kl}^{(r+1)}(\bar{a}(b+h); p)$ for $h \neq -b$, so Katz's monodromy computation makes the coefficients of the limit traces of Haar elements of $\mathrm{USp}(r+1)$ when r is odd; for odd $r \geq 3$ these are among the families of Autissier, Bonolis and Lamzouri recalled in Section 7.4, with $N = r+1$. For prime power moduli Ricotta and Royer found a limit of the same shape whose coefficients have an atom of mass $\frac{1}{2}$ at 0 and an arcsine part [462]. This law is the measure μ_{CM} of elliptic curves with complex multiplication (Chapter 4), because Kloosterman sums to prime power moduli have an elementary evaluation: for odd p, modulus p^k with $k \geq 2$ and parameters prime to p, they vanish unless the product of the parameters is a square modulo p, and are otherwise, up to sign, $p^{k/2}$ times twice the cosine of an explicit angle.

Open questions. We record four.

1. The law of $\|K\|_\infty$ is not known in closed form. Autissier, Bonolis and Lamzouri determine its tail up to the constant in the exponent, $\mathbb{P}(\|K\|_\infty > V) = \exp(-\exp(\pi V/2 + O(1)))$ for large V [5], but the distribution function, and even its mean (1.138 in our simulation), have no formula.

2. Extreme values. In the two parameter family the normalised maximum of the partial sums is $O(\log p)$ for every (a, b) by the completion method, and exceeds $(\frac{2}{\pi} + o(1)) \log\log p$ for many (a, b) [5]. Whether it is $O(\log\log p)$ for all (a, b), which is what the estimate of Autissier, Bonolis and Lamzouri would give if it held for all V, is open; they show that the corresponding bound fails for some families of periodic functions to which their theorem applies, although these families, unlike Kloosterman sums, are not uniformly bounded.

3. For the one parameter family, convergence in law in $C([0,1])$ depends on estimates for short Kloosterman sums averaged over the parameter that are not known (see the discussion of tightness above).

4. Horizontal paths. Fix an integer $a \neq 0$ and let p be uniform among the primes up to x. Do the paths $K_p(a, \cdot)$ converge in finite dimensional distributions, as $x \to \infty$, to $K - c_a \mathrm{ST}_a$? For fixed a the term $h = a$ of (7.1) is $-\alpha_p(a,t)/\sqrt{p}$, so one Sato Tate coefficient disappears, as in the second parametrisation. The endpoint $K_p(a, 1) = 2\cos\theta_{p,a}$ alone is the subject of Conjecture 7.5, and the finite dimensional distributions would also need the angles $\theta_{p,a-h}$ at distinct shifts $h \neq a$ to become independent as p varies.

7.6 NOTES

Kloosterman introduced his sums in 1926 to treat quaternary quadratic forms by the circle method, obtaining a bound of order $p^{3/4}$ [323]. Weil's paper [568] deduced $2\sqrt{p}$ from his proof of the Riemann hypothesis for curves over finite fields [567]; Stepanov's method [526], in Bombieri's formulation [71], gives an elementary proof. The theory of Kloosterman sheaves, their monodromy, and the equidistribution theorem are in Katz's monograph [296]; Deligne's equidistribution theorem is in Weil II [141], and the Goursat Kolchin Ribet criterion is in [297]. Katz and Sarnak [298] develop the general passage from monodromy to random matrices used in Chapter 6. Iwaniec and Kowalski [276] treat exponential sums over finite fields, Kloosterman sums included, in Chapter 11 and sums of Kloosterman sums with the Kuznetsov formula in Chapter 16; Iwaniec's book [275] is the standard reference for the spectral theory. The Kuznetsov formula is due independently to Bruggeman [87] and Kuznetsov [337]; Deshouillers and Iwaniec [146] combined it with large sieve inequalities for Fourier coefficients and turned it into a standard tool of analytic number theory. Selberg's $3/16$ theorem and the Kloosterman zeta function are in [497], and the record $\frac{975}{4096}$ is in the appendix of Kim and Sarnak to [315]. The paths, the completion formula of Lemma 7.10 and the observation that the parametrisation $e((x + a\bar{x})/p)$ loses one coefficient of the limit are due to Kowalski and Sawin [331], who describe the support of the limit in [332]. Lemma 7.11, Proposition 7.13, Corollary 7.14, Proposition 7.15 and Proposition 7.16 are our own formulations, with complete proofs, of facts that are implicit in or easily derived from their work; the tail bound of Proposition 7.16 is far weaker than their doubly exponential estimates. Lamzouri [343] and Autissier, Bonolis and Lamzouri [5] study the maximum of the partial sums at finite p, and Ricotta and Royer [462] the paths for prime power moduli.

7.7 EXERCISES

1. Show that $\mathrm{Kl}(a, b; p)$ is real, that $\mathrm{Kl}(a, b; p) = \mathrm{Kl}(1, ab; p)$ for $a \neq 0$, and that $\mathrm{Kl}(a, b; p) = \mathrm{Kl}(b, a; p)$. Deduce that for $ab \neq 0$ the sum depends only on ab.

2. Show that $\sum_{x \in \mathbb{F}_p^\times} e(c\bar{x}/p) = -1$ for $c \neq 0$ (a Ramanujan sum), and deduce $\mathrm{Kl}(0; p) = -1$ and $\sum_{a \in \mathbb{F}_p^\times} \mathrm{Kl}(a; p) = 1$. Let m_2 and m_4 be the empirical second and fourth moments of $x_{p,a}$ over $a \in \mathbb{F}_p^\times$ (Proposition 7.1). Using the Paley Zygmund inequality $\mathbb{P}(Z > \theta\,\mathbb{E}Z) \geq (1-\theta)^2(\mathbb{E}Z)^2/\mathbb{E}Z^2$ for $Z \geq 0$ and $0 \leq \theta \leq 1$, show that the proportion of a with $|\mathrm{Kl}(a; p)| > \frac{1}{2}\sqrt{p}$ is at least $(m_2 - \frac{1}{4})^2/m_4$, which tends to $\frac{9}{32}$. Compare with the Sato Tate value $\mu_{\mathrm{ST}}(\{x : |x| > \frac{1}{2}\}) = 0.685$.

3. For $p = 5$, compute by hand $\mathrm{Kl}(1; 5) = \frac{3-\sqrt{5}}{2}$, $\mathrm{Kl}(2; 5) = -1 - \sqrt{5}$, $\mathrm{Kl}(3; 5) =$

$\sqrt{5}-1$, $\mathrm{Kl}(4;5)=\frac{3+\sqrt{5}}{2}$. Check Weil's bound $2\sqrt{5}$ and the three identities of Proposition 7.1 (the fourth moment is 159), and find the Kloosterman angles.

4. Suppose $L_a(T)=1+\mathrm{Kl}(a;p)T+pT^2=(1-\alpha T)(1-\beta T)$ and $\mathrm{Kl}_n(a)=-(\alpha^n+\beta^n)$. Show that $\mathrm{Kl}_2(a)=2p-\mathrm{Kl}(a;p)^2$. Show that if $|\mathrm{Kl}_n(a)|\le Cp^{n/2}$ for all n, with C independent of n, then $|\alpha|=|\beta|=\sqrt{p}$. (Hint: consider the radius of convergence of $\sum_n \mathrm{Kl}_n(a)T^n$.)

5. (Probabilistic.) Compute $\mathrm{Var}\,\Re K(t)$ and $\mathrm{Var}\,\Im K(t)$ directly from the series, using $\sum_{h\neq 0}\frac{\sin^2(\pi ht)}{\pi^2h^2}=t(1-t)$ for $0\le t\le 1$, which follows from the Fourier expansion of the Bernoulli polynomial $B_2(t)=t^2-t+\frac{1}{6}$. Check the answer against Proposition 7.15, and show that $\mathrm{Var}\,\Im K(t)=\frac{1}{2}\min(t,1-t)$, maximal at $t=\frac{1}{2}$.

6. (Probabilistic.) Let ξ_h be independent real standard Gaussians and $B(t)=\sum_h c_h(t)\xi_h$. Show that B is a complex Gaussian process with the covariance of Proposition 7.15, identify it in terms of a standard complex Brownian motion on $[0,\frac{1}{2}]$ as in Section 7.5, and compute $\mathbb{P}(\Im B(\frac{1}{2})>M)$. Compare with the bound e^{-2M^2} of Proposition 7.16. Show that if the ξ_h are replaced by independent standard complex Gaussians, the same series is a standard complex Brownian motion on $[0,1]$.

7.8 Open computational problems

Computational Problem 7.1 (Horizontal statistics of Kloosterman angles). Objective: test Conjecture 7.5 for $a=1$ and a few other small a. Objects: $\mathrm{Kl}(a;p)/\sqrt{p}$ for all primes $p\le 10^8$ and for a random sample of 10^5 primes in $[10^8,10^9]$. Route: each sum costs $O(p)$ operations with modular inversion by a batched Montgomery trick; GPU kernels accumulating $\cos(2\pi(x+a\bar{x})/p)$ in double precision, with compensated summation; cross check against PARI/GP at small p. Deliverable: Weyl averages $\pi(x)^{-1}\sum_{p\le x}U_n(x_p/2)$ for $n\le 20$ as functions of x, $\sqrt{\pi(x)}$ times the Kolmogorov distance to μ_{ST}, and the counts of positive and negative values. Why it matters: unlike Satake angles of a fixed modular form, Kloosterman angles at varying primes come from no known global object, so no automorphic proof of the horizontal law is in sight, and the data measure whether they are nevertheless as random as Satake angles.

Computational Problem 7.2 (Law of the maximum of Kloosterman paths). Objective: estimate the distribution of $\|K_p(a,\cdot)\|_\infty$ and compare it with that of $\|K\|_\infty$. Objects: all $a\in\mathbb{F}_p^\times$ for a sequence of primes p up to 10^6 (about 10^{12} terms for the largest), in both parametrisations of Proposition 7.13 (the second should be compared with $K-c_1\mathrm{ST}_1$), and 10^7 samples of the random Fourier series truncated at $|h|\le 10^4$. Route: cumulative sums of $e((ax+\bar{x})/p)$ from a

table of inverses, vectorised over a on GPUs, keeping only the running maximum of the modulus; the series evaluated on a fine grid by fast Fourier transforms. Deliverable: empirical distribution functions and upper tails of the maxima, a fit of $\log(-\log \mathbb{P}(\|K\|_\infty > V))$ against V to measure the doubly exponential rate, and two sample tests by p. Why it matters: agreement at large p is evidence for convergence in $C([0,1])$ for the one parameter family, which is not proved, and the fitted slope can be compared with the value $\frac{\pi}{2}$ of Autissier, Bonolis and Lamzouri [5].

COMPUTATIONAL PROBLEM 7.3 (Sums of Kloosterman sums and the spectral gap). Objective: measure $\Sigma_N(x) = \sum_{c \le x,\ N \mid c} S(m,n;c)/c$ against Conjecture 7.8. Objects: $m, n \le 10$, levels $N \le 100$, and x up to 10^7. Route: compute $S(m,n;p)$ for primes $p \le x$ on GPUs, obtain prime power moduli from the explicit evaluations, and assemble composite moduli by twisted multiplicativity; smooth the sums with a fixed weight. Deliverable: plots of $\Sigma_N(x)$ against $\log x$, fitted growth exponents with confidence bands, and a comparison of $|\Sigma_N(x)|^2$ with $V_N(x) = \sum_{c \le x,\ N \mid c} S(m,n;c)^2/c^2$, the variance of the model in which the terms carry independent random signs. Why it matters: Conjecture 7.8 is open even for $N = 1$, where Kuznetsov's exponent $\frac{1}{6}$ stands, and an exceptional eigenvalue $s(1-s)$ of $\Gamma_0(N)$ would produce a drift of order x^{2s-1}; the experiment turns Selberg's eigenvalue conjecture into a test for the absence of drift in an arithmetic random walk.

COMPUTATIONAL PROBLEM 7.4 (Birch paths). Objective: study the paths $t \mapsto p^{-1/2} \sum_{0 \le x \le tp} e((ax + bx^3)/p)$ for the one parameter family ($b = 1$) and the two parameter family (a, b). Objects: all a (and 10^6 random pairs (a,b)) for a sequence of primes up to 10^6. Route: as in the previous problems, with the cubic phase computed exactly modulo p. Deliverable: Kolmogorov Smirnov comparisons with the random Fourier series of Kowalski and Sawin at fixed t, the variance functions $\mathrm{Var}\,\Re$ and $\mathrm{Var}\,\Im$ against the formulas of Proposition 7.15 (for $b = 1$ the limit is again K [331]), the law of the maximum, and a check of the vertical Sato Tate law for the complete sums. Why it matters: Birch sums are the case in which convergence in $C([0,1])$ is a theorem, so they calibrate the numerical methods used for Kloosterman paths.

CHAPTER 8

RANDOM GROUPS: CLASS GROUPS, SELMER GROUPS AND RANKS

In 1801 Gauss tabulated class numbers of binary quadratic forms and noticed regularities that he could not explain, among them that the determinants he called irregular, those whose principal genus (the subgroup of squares in the form class group) is not cyclic, are comparatively rare. Gauss had no notion of a random group, and the probabilistic reading of his tables had to wait until 1983, when Henri Cohen and Hendrik Lenstra proposed that class groups of quadratic fields behave like finite abelian groups drawn at random, each group G weighted in proportion to $1/|\operatorname{Aut} G|$ [110]. The same principle now organises a large part of arithmetic statistics. Selmer groups and Shafarevich Tate groups of elliptic curves, and with them the ranks of Mordell Weil groups, are modelled by intersections of random isotropic subspaces and by cokernels of random alternating matrices, and several of their predictions are now theorems. This chapter develops the random group models, proves their basic properties, records what is known, and compares the predictions with data computed for this book.

8.1 WHY RANDOM GROUPS

The previous chapters studied measures on compact groups: Frobenius conjugacy classes in $\mathrm{SU}(2)$ or $\mathrm{USp}(4)$, and eigenvalues of Haar random matrices in large classical groups. Those measures govern the *analytic* side of the Langlands programme, the Satake parameters and the zeros of L functions. The *arithmetic* side, the groups that L functions are conjectured to measure, has its own random models, and the measures that appear are again Haar measures, now on profinite objects such as $M_n(\mathbb{Z}_p)$.

Two instances anchor the connection with the Langlands programme.

The first is class field theory, the case GL_1 of the programme. For a number field K the Artin map identifies the class group $\mathrm{Cl}(K)$ with the Galois group of the Hilbert class field, the maximal abelian extension unramified at all places. Characters of $\mathrm{Cl}(K)$ are exactly the finite order automorphic representations of $\mathrm{GL}_1(\mathbb{A}_K)$ that are unramified at every finite place and trivial at the infinite places, and their Hecke L functions factor the Dedekind zeta function of the Hilbert class field. A random model for class groups is therefore a random model for

the unramified automorphic spectrum of GL_1 over K, and questions about the p part of $\mathrm{Cl}(K)$ are questions about cyclic unramified extensions of degree p, that is, about number fields of a prescribed type. The theorem of Davenport and Heilbronn in Section 8.3 is proved by counting cubic fields.

The second is the conjecture of Birch and Swinnerton Dyer [89]. For an elliptic curve $E/\mathbb{Q}$, modularity [581, 541, 41] attaches to E a holomorphic newform of weight 2, so $L(s,E)$ is an automorphic L function of GL_2, entire, with a functional equation and a root number $w(E)=\pm 1$. The conjecture predicts that the order of vanishing at the centre equals the rank $r(E)$ of $E(\mathbb{Q})$, and that the leading coefficient involves the order of the Shafarevich Tate group $\mathrm{Sha}(E)$. Selmer groups sit between the two: they are finite, computable, and bound the rank from above. A random model for Selmer groups and Sha is therefore a random model for the arithmetic data that the central behaviour of an automorphic L function is conjectured to encode.

The guiding heuristic, in both settings, is that an arithmetic group should be distributed like a random object chosen with probability inversely proportional to its number of symmetries, after the constraints imposed by the arithmetic (units, duality pairings, roots of unity) have been accounted for. When the constraints are correctly identified the models agree with the theorems and with the data; when they are missed the models fail in instructive ways, as Malle discovered for class groups in the presence of roots of unity [380].

8.2 The Cohen Lenstra measure

Throughout, p is a prime and G runs through isomorphism classes of finite abelian p groups. Write

$$\eta_n(p)=\prod_{k=1}^{n}(1-p^{-k}), \qquad \eta_\infty(p)=\prod_{k=1}^{\infty}(1-p^{-k}),$$

so that $\eta_\infty(2)=0.288788$, $\eta_\infty(3)=0.560126$, $\eta_\infty(5)=0.760333$ and $\eta_\infty(7)=0.836795$. The p rank of G is $\mathrm{rk}_p\,G=\dim_{\mathbb{F}_p}G/pG=\dim_{\mathbb{F}_p}G[p]$.

Definition 8.1. The Cohen Lenstra measure on finite abelian p groups is

$$\mu_{\mathrm{CL}}(G)=\frac{\eta_\infty(p)}{|\operatorname{Aut}G|}.$$

The cyclic group $\mathbb{Z}/p^2$ has $p(p-1)$ automorphisms while $(\mathbb{Z}/p)^2$ has $(p^2-1)(p^2-p)$, so among groups of order p^2 the measure prefers the cyclic one by the factor p^2-1, in agreement with Gauss's observation that irregular determinants, those with noncyclic principal genus, are rare. That μ_{CL} is a probability measure

is equivalent to the identity

$$\sum_{G} \frac{1}{|\operatorname{Aut} G|} = \prod_{k=1}^{\infty} (1 - p^{-k})^{-1}, \tag{8.1}$$

which goes back to P. Hall [234]. We prove it, together with the random matrix model of Friedman and Washington [204], in a single argument.

8.2.1 Cokernels of random p adic matrices

Let A_n be a random matrix in $M_n(\mathbb{Z}_p)$ drawn from Haar probability measure, so that its entries are independent and uniform on $\mathbb{Z}_p$. The set $\{\det A = 0\}$ is a proper analytic subset and has Haar measure zero, so almost surely $\det A_n \neq 0$ and $\operatorname{cok} A_n = \mathbb{Z}_p^n / A_n \mathbb{Z}_p^n$ is a finite abelian p group of order $p^{v_p(\det A_n)}$. For a finite abelian p group B of p rank r, a homomorphism $\mathbb{Z}_p^n \to B$ is surjective if and only if its reduction $\mathbb{F}_p^n \to B/pB \cong \mathbb{F}_p^r$ is, so

$$|\operatorname{Sur}(\mathbb{Z}_p^n, B)| = |B|^n \prod_{j=0}^{r-1} (1 - p^{j-n}) = |B|^n \prod_{k=n-r+1}^{n} (1 - p^{-k}), \tag{8.2}$$

an empty product being 1 (and the count is 0 when $r > n$).

THEOREM 8.2 (Hall; Friedman and Washington). *For every finite abelian p group G of p rank $r \leq n$,*

$$\mathbb{P}(\operatorname{cok} A_n \cong G) = \frac{1}{|\operatorname{Aut} G|}\, \eta_n(p) \prod_{k=n-r+1}^{n} (1 - p^{-k}),$$

and as $n \to \infty$ this tends to $\mu_{\mathrm{CL}}(G)$. Moreover μ_{CL} is a probability measure, that is, Hall's identity (8.1) *holds, and* $\operatorname{cok} A_n$ *converges in total variation to μ_{CL}.*

PROOF. A finite module C is isomorphic to G if and only if there is a surjection $\varphi : \mathbb{Z}_p^n \to G$ with kernel $A_n \mathbb{Z}_p^n$, and when $C = \operatorname{cok} A_n \cong G$ the number of such φ is exactly $|\operatorname{Aut} G|$, since any two differ by an automorphism of G. Hence

$$\mathbb{P}(\operatorname{cok} A_n \cong G) = \frac{1}{|\operatorname{Aut} G|} \sum_{\varphi \in \operatorname{Sur}(\mathbb{Z}_p^n, G)} \mathbb{P}\big(A_n \mathbb{Z}_p^n = \ker \varphi\big).$$

Fix φ and put $L = \ker \varphi$, a sublattice of index $|G|$. The columns of A_n are independent and uniform on $\mathbb{Z}_p^n$, and each lies in L with probability $|G|^{-1}$. Conditionally on all columns lying in L, they are independent and uniform on L; choosing a basis of L we may write $A_n = MC$ with M a fixed matrix whose columns are that basis and C Haar distributed in $M_n(\mathbb{Z}_p)$. Then $A_n \mathbb{Z}_p^n = L$ if and only if $C\mathbb{Z}_p^n = \mathbb{Z}_p^n$, that is $C \in \mathrm{GL}_n(\mathbb{Z}_p)$, which happens if and only if $C \bmod p \in \mathrm{GL}_n(\mathbb{F}_p)$, with

probability $|\mathrm{GL}_n(\mathbb{F}_p)|/p^{n^2} = \eta_n(p)$. So each φ contributes $|G|^{-n}\eta_n(p)$, and (8.2) gives the formula. As $n \to \infty$ the two products tend to $\eta_\infty(p)$ and 1.

For the remaining statements let $P_n(G)$ denote this probability. As $\det A_n \neq 0$ almost surely, $\sum_G P_n(G) = 1$ for every n. By Fatou's lemma $\sum_G \mu_{\mathrm{CL}}(G) \leq \liminf_n \sum_G P_n(G) = 1$, so that

$$\sum_G |\operatorname{Aut} G|^{-1} \leq \eta_\infty(p)^{-1} < \infty.$$

Now $P_n(G) \leq |\operatorname{Aut} G|^{-1}$ for all n and G, because the two products are at most 1; this domination is summable, and dominated convergence gives $\sum_G \mu_{\mathrm{CL}}(G) = \lim_n \sum_G P_n(G) = 1$, which is (8.1). Pointwise convergence of probability mass functions to a probability mass function implies convergence in total variation (Scheffé's lemma). □

The case $G = 0$ is worth isolating, since it is the simplest instance and the one most often used.

COROLLARY 8.3. *For Haar random $A_n \in M_n(\mathbb{Z}_p)$, $\mathbb{P}(\operatorname{cok} A_n = 0) = \mathbb{P}(A_n \in \mathrm{GL}_n(\mathbb{Z}_p)) = \eta_n(p) \to \eta_\infty(p)$.*

PROOF. The cokernel vanishes if and only if A_n is invertible over $\mathbb{Z}_p$, if and only if $\det A_n$ is a unit, if and only if $A_n \bmod p$ is invertible. The reduction is uniform on $M_n(\mathbb{F}_p)$, and $|\mathrm{GL}_n(\mathbb{F}_p)| = \prod_{i=0}^{n-1}(p^n - p^i) = p^{n^2}\eta_n(p)$, counting bases column by column. □

The proof shows what the weight $1/|\operatorname{Aut} G|$ means probabilistically: the natural Haar measure is on presentations (surjections from a free module), and a group with many automorphisms has proportionally fewer distinct presentations. This is the same mechanism as the orbit counting behind the weight $1/|\operatorname{Aut}|$ in every mass formula of Siegel type.

8.2.2 The p rank law and the probability that p divides the class number

PROPOSITION 8.4 (Cohen and Lenstra). *Under μ_{CL},*

$$\mu_{\mathrm{CL}}(\mathrm{rk}_p\, G = r) = \frac{p^{-r^2}\,\eta_\infty(p)}{\prod_{k=1}^{r}(1-p^{-k})^2}, \qquad r \geq 0,$$

and $\mu_{\mathrm{CL}}(G \neq 0) = 1 - \eta_\infty(p)$.

PROOF. For each n, $\mathrm{rk}_p \operatorname{cok} A_n = \dim \operatorname{cok}(A_n \bmod p) = n - \mathrm{rank}_{\mathbb{F}_p}(A_n \bmod p)$, and $A_n \bmod p$ is uniform on $M_n(\mathbb{F}_p)$. The number of $n \times n$ matrices of rank k over $\mathbb{F}_p$ is $\prod_{i=0}^{k-1}(p^n - p^i)^2/(p^k - p^i)$: choose the image, a k dimensional subspace,

in $\prod_{i<k}(p^n-p^i)/(p^k-p^i)$ ways, then a surjection onto it in $\prod_{i<k}(p^n-p^i)$ ways. With $k=n-r$,

$$\frac{1}{p^{n^2}}\prod_{i=0}^{k-1}\frac{(p^n-p^i)^2}{p^k-p^i}=p^{2nk-k^2-n^2}\,\frac{\prod_{j=r+1}^{n}(1-p^{-j})^2}{\prod_{j=1}^{k}(1-p^{-j})},$$

and $2nk-k^2-n^2=-r^2$. Letting $n\to\infty$ gives $p^{-r^2}\eta_\infty^2/(\eta_r^2\eta_\infty)$. By Theorem 8.2 the laws of $\operatorname{cok}A_n$ converge in total variation to μ_{CL}, so the laws of the p rank converge to its law under μ_{CL}. The last claim is the case $G=0$ of the definition. $\square$

For $p=3$ the rank law is 0.5601, 0.4201, 0.0197 for $r=0,1,2$, and the Cohen Lenstra prediction for the proportion of imaginary quadratic fields with $3\mid h(D)$ is $1-\eta_\infty(3)=0.4399$, well above the naive $1/3$.

8.2.3 Moments

For a fixed finite abelian p group B, the number $|\operatorname{Sur}(G,B)|$ is a natural test function on groups: $|\operatorname{Sur}(G,\mathbb{Z}/p)|=|G[p]|-1$, since every nonzero homomorphism to $\mathbb{Z}/p$ is surjective and $|\operatorname{Hom}(G,\mathbb{Z}/p)|=|G[p]|$.

PROPOSITION 8.5 (Moments of the Cohen Lenstra measure). *For every finite abelian p group B, $\mathbb{E}_{\mu_{\mathrm{CL}}}|\operatorname{Sur}(G,B)|=1$. In particular $\mathbb{E}_{\mu_{\mathrm{CL}}}|G[p]|=2$.*

PROOF. Let r be the p rank of B. Surjections $\operatorname{cok}A_n\to B$ correspond to surjections $\psi:\mathbb{Z}_p^n\to B$ with $\psi\circ A_n=0$, and for fixed ψ this event has probability $|B|^{-n}$, each column independently lying in $\ker\psi$. Hence, by (8.2), $\mathbb{E}|\operatorname{Sur}(\operatorname{cok}A_n,B)|=\prod_{k=n-r+1}^{n}(1-p^{-k})$, which is at most 1 and tends to 1. To pass the limit inside the expectation we use uniform integrability: $|\operatorname{Sur}(G,B)|^2\le|\operatorname{Hom}(G,B\times B)|=\sum_{B'\le B\times B}|\operatorname{Sur}(G,B')|$, so applying the bound just proved to each of the finitely many subgroups B' of $B\times B$ shows that $\mathbb{E}|\operatorname{Sur}(\operatorname{cok}A_n,B)|^2$ is bounded in n. Since $\operatorname{cok}A_n$ converges in law to μ_{CL} (Theorem 8.2), the expectations converge to $\mathbb{E}_{\mu_{\mathrm{CL}}}|\operatorname{Sur}(G,B)|$, which is therefore 1. The second statement is the case $B=\mathbb{Z}/p$. $\square$

The moments characterise the measure. Ellenberg, Venkatesh and Westerland [180] proved that μ_{CL} is the only probability measure on finite abelian p groups all of whose B moments equal 1, in a robust form: random groups whose B moments converge to 1 for every B converge in distribution to μ_{CL}. Wood [588] showed more generally that moments $M_B=O(|\wedge^2 B|)$ determine the distribution, again robustly under limits. This moment method is the route by which most theorems in the subject are now proved: one counts surjections, which is a counting problem for number fields or for points on moduli spaces, and deduces the distribution.

8.2.4 The real quadratic variant

For real quadratic fields the unit group has rank one, and Cohen and Lenstra proposed the weights

$$\mu^{+}_{\mathrm{CL}}(G) = \frac{c_p^{+}}{|G|\,|\operatorname{Aut} G|},$$

the constant being fixed by normalisation: $\sum_G |G|^{-1}|\operatorname{Aut} G|^{-1} = \prod_{k\ge 2}(1-p^{-k})^{-1}$ (the matrix model below proves this as Theorem 8.2 proves (8.1); it is also the value at $x = 1/p$ of the generating function in the first exercise), so $c_p^{+} = \eta_\infty(p)/(1-p^{-1})$. The p rank law becomes

$$\mu^{+}_{\mathrm{CL}}(\mathrm{rk}_p\, G = r) = \frac{p^{-r(r+1)}\,\eta_\infty(p)}{\prod_{k=1}^{r}(1-p^{-k})\prod_{k=1}^{r+1}(1-p^{-k})},$$

the moments become $\mathbb{E}|\operatorname{Sur}(G,B)| = |B|^{-1}$, and in particular $\mathbb{E}|G[p]| = 1+p^{-1}$, equal to $\frac{4}{3}$ for $p=3$. The matrix model is the cokernel of a Haar random $n\times(n+1)$ matrix over $\mathbb{Z}_p$, the extra column playing the role of the unit (Wood [589] proves universality for such $n\times(n+u)$ models). The proof of Theorem 8.2 adapts line by line: each φ now contributes $|G|^{-n-1}\prod_{k=2}^{n+1}(1-p^{-k})$, which produces the extra factor $|G|^{-1}$. The convergence in n is fast: for a uniformly random 8×9 matrix over $\mathbb{F}_3$ the exact probabilities of cokernel dimension 0, 1, 2 already agree with the limits 0.8402, 0.1575, 0.0023 to four decimal places.

8.3 Theorems and conjectures for class groups

Conjecture 8.6 (Cohen and Lenstra [110]). *Let p be an odd prime and G a finite abelian p group. As $X\to\infty$, the proportion of negative fundamental discriminants D with $|D|<X$ and $\mathrm{Cl}(D)[p^\infty]\cong G$ tends to $\mu_{\mathrm{CL}}(G)$, and the proportion of positive fundamental discriminants $D<X$ with $\mathrm{Cl}(D)[p^\infty]\cong G$ tends to $\mu^{+}_{\mathrm{CL}}(G)$.*

The prime 2 is excluded because genus theory determines $\mathrm{Cl}(D)[2]$ from the number of prime factors of D. Gerth's modification treats $(2\mathrm{Cl}(D))[2^\infty]$; Fouvry and Klüners [197] proved the predicted distribution of the 4 rank, and Smith [521] proved, in a preprint, the predicted distribution of $(2\mathrm{Cl}(D))[2^\infty]$ for imaginary quadratic fields. For odd p the only proved case concerns $p=3$, and only the first moment.

Theorem 8.7 (Davenport and Heilbronn [127]). *As $X\to\infty$, the average of $|\mathrm{Cl}(D)[3]|$ over fundamental discriminants $-X<D<0$ tends to 2, and over fundamental discriminants $0<D<X$ it tends to $\frac{4}{3}$.*

The proof is a count of cubic fields. A cubic field K of discriminant D fundamental has Galois closure containing $\mathbb{Q}(\sqrt{D})$, and by class field theory the

nontrivial 3 torsion characters of $\mathrm{Cl}(D)$, up to inversion, correspond to cubic fields of discriminant D. Cubic rings correspond to $\mathrm{GL}_2(\mathbb{Z})$ orbits of integral binary cubic forms (Delone and Faddeev), so counting cubic fields by discriminant becomes counting lattice points in a fundamental domain for $\mathrm{GL}_2(\mathbb{Z})$ acting on a four dimensional real space, followed by a sieve to maximal rings. Bhargava's parametrisations of rings of rank up to 5 extended the method, and in particular gave the average size of the 2 torsion in class groups of cubic fields, $\frac{5}{4}$ for totally real and $\frac{3}{2}$ for complex cubic fields [53], in agreement with the Cohen and Martinet extension of the heuristic.

THEOREM 8.8 (Bhargava, Shankar and Tsimerman; Taniguchi and Thorne). *Let $N_3^{\pm}(X)$ be the number of cubic fields K with $0 < \pm \operatorname{disc} K < X$, and put $C^+ = K^+ = 1$, $C^- = 3$, $K^- = \sqrt{3}$. There is $\delta > 0$ such that*

$$N_3^{\pm}(X) = \frac{C^{\pm}}{12\zeta(3)} X + K^{\pm} \frac{4\zeta(\frac{1}{3})}{5\Gamma(\frac{2}{3})^3 \zeta(\frac{5}{3})} X^{5/6} + O(X^{5/6-\delta}),$$

and, with D running over fundamental discriminants,

$$\sum_{0<\pm D<X} |\mathrm{Cl}(D)[3]| = \frac{3 + C^{\pm}}{\pi^2} X + K^{\pm} \frac{8\zeta(\frac{1}{3})}{5\Gamma(\frac{2}{3})^3} \prod_p \Big(1 - \frac{p^{1/3}+1}{p(p+1)}\Big) X^{5/6} + O(X^{5/6-\delta}). \quad (8.3)$$

The secondary term was conjectured by Roberts on the basis of numerical data and proved independently in [90] and [536]; the formulas above are those of Taniguchi and Thorne, whose error terms are $O(X^{7/9+\epsilon})$ and $O(X^{18/23+\epsilon})$, while Bhargava, Shankar and Tsimerman obtain $O(X^{13/16+\epsilon})$ for the cubic fields. Since $\zeta(\frac{1}{3}) = -0.9734$, both secondary terms are negative. The coefficient of $X^{5/6}$ in (8.3) is -0.41478 for $D < 0$ and -0.23947 for $D > 0$, and there are $(3/\pi^2)X + O(X^{1/2})$ fundamental discriminants of each sign up to X, so the mean of $|\mathrm{Cl}(D)[3]|$ over $0 < -D < X$ is $2 - 1.365X^{-1/6} + O(X^{-1/6-\delta})$, and over $0 < D < X$ it is $\frac{4}{3} - 0.788X^{-1/6} + O(X^{-1/6-\delta})$. In the imaginary case the correction is still 0.14 at $X = 10^6$ and 0.014 at $X = 10^{12}$: averages of 3 torsion converge so slowly that no computation in the accessible range sees the limit directly.

For $p \geq 5$, and for the full distribution at $p = 3$, Conjecture 8.6 is open: not even the average of $|\mathrm{Cl}(D)[5]|$ is known. The strongest evidence beyond number fields comes from function fields. For an odd prime ℓ and a finite abelian ℓ group G, Ellenberg, Venkatesh and Westerland [180] used homological stability for Hurwitz spaces to show that the upper and lower densities of imaginary quadratic extensions of $\mathbb{F}_q(t)$ whose class group has ℓ part isomorphic to G both tend to $\mu_{\mathrm{CL}}(G)$ as $q \to \infty$ through odd prime powers prime to ℓ with $q \not\equiv 1$

$(\mathrm{mod}\ \ell)$. The theorem is a limit in q and does not by itself give the distribution for any fixed q; we refer to the paper and its sequels for the exact statements.

When the base field contains pth roots of unity the heuristics require modification. Malle [380] found in extensive computations that the p parts of class groups of extensions of a base field containing μ_p do not follow the Cohen and Martinet predictions, and proposed corrected distributions; the correction reflects an additional pairing coming from the roots of unity, and the analogous phenomenon in function fields is explained by the Weil pairing on the Jacobian.

8.3.1 Our data

We computed with PARI/GP (`quadclassunit`, whose output is guaranteed under the generalized Riemann hypothesis) the class groups of all 303963 fundamental discriminants $D \in [-2 \cdot 10^6, -10^6]$ and of all 91185 fundamental discriminants $D \in [10^5, 4 \cdot 10^5]$. Table 8.1 and Figure 8.1 compare the p rank frequencies with Proposition 8.4 and its real analogue.

		$r = 0$	$r = 1$	$r = 2$	$\mathbb{P}(p \mid h)$
imaginary, $p = 3$	observed	0.5901	0.3976	0.0123	0.4099
	Cohen Lenstra	0.5601	0.4201	0.0197	0.4399
imaginary, $p = 5$	observed	0.7662	0.2321	0.0017	0.2338
	Cohen Lenstra	0.7603	0.2376	0.0021	0.2397
imaginary, $p = 7$	observed	0.8406	0.1590	0.0003	0.1594
	Cohen Lenstra	0.8368	0.1627	0.0005	0.1632
real, $p = 3$	observed	0.8768	0.1227	0.0004	0.1232
	Cohen Lenstra	0.8402	0.1575	0.0023	0.1598
real, $p = 5$	observed	0.9603	0.0397	0.0000	0.0397
	Cohen Lenstra	0.9504	0.0495	0.0001	0.0496
real, $p = 7$	observed	0.9814	0.0186	0.0000	0.0186
	Cohen Lenstra	0.9763	0.0237	0.0000	0.0237

Table 8.1: Distribution of the p rank of class groups: 303963 imaginary fundamental discriminants in $[-2 \cdot 10^6, -10^6]$ and 91185 real ones in $[10^5, 4 \cdot 10^5]$, against the Cohen Lenstra predictions.

Three features stand out. First, the agreement improves with p. In the imaginary range every frequency for $p = 7$ is within 0.004 of the prediction and every frequency for $p = 5$ within 0.006, while for $p = 3$ the deficit in $\mathbb{P}(3 \mid h)$ is 0.030; in the real range the deficits in $\mathbb{P}(p \mid h)$ are 0.037, 0.010 and 0.005 for $p = 3, 5, 7$. Even the small deficits are significant: in the imaginary range the binomial standard error of $\mathbb{P}(p \mid h)$ is below 0.001, so the deficits for $p = 5$ and $p = 7$ are about 8 and 6 standard errors. Second, the deviations have a consistent sign: the data show too few groups with nontrivial p part. Third, the mean of $|\mathrm{Cl}(D)[3]|$ is 1.8933 in the

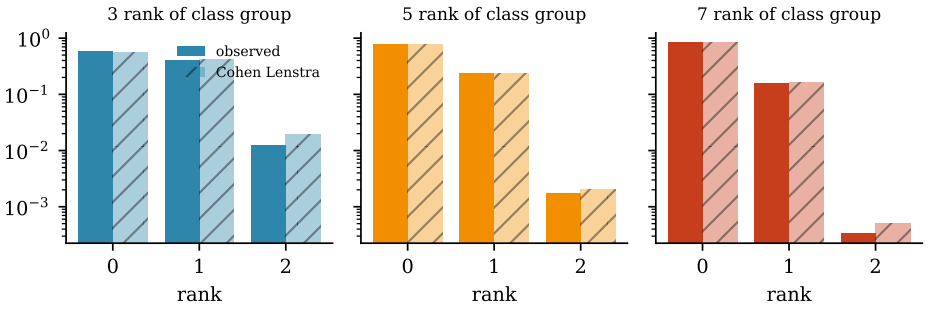


Figure 8.1: Frequencies of 3, 5 and 7 ranks of class groups of imaginary quadratic fields with $10^6 \le |D| \le 2 \cdot 10^6$ (logarithmic scale), observed against the Cohen Lenstra law $p^{-r^2}\eta_\infty(p)/\prod_{k\le r}(1-p^{-k})^2$.

imaginary range against the limit 2, with window means in Figure 8.2 between 1.877 and 1.907 and no visible trend, and 1.2489 in the real range against $\frac{4}{3}$.

The secondary term of Theorem 8.8 accounts for the 3 torsion deficit quantitatively. The nontrivial part of the mean, $\mathbb{E}|\mathrm{Cl}(D)[3]| - 1$, is 0.893 against 1 in the imaginary range and 0.249 against $\frac{1}{3}$ in the real range, relative deficits of 0.107 and 0.253. Taking differences of (8.3) at the ends of our two ranges and dividing by the number of fundamental discriminants in them predicts means of 1.8933 for $-2\cdot 10^6 \le D \le -10^6$ and 1.2496 for $10^5 \le D \le 4\cdot 10^5$, against the observed 1.8933 and 1.2489; with the variance of $|G[3]|$ under the Cohen Lenstra measures, the standard error of each observed mean is about 0.003. The larger relative deficit of the real range has two sources: its discriminants are about six times smaller, which multiplies $X^{-1/6}$ by $6^{1/6} = 1.35$, and its secondary coefficient, measured against the main term of the nontrivial part ($3/\pi^2$ for $D<0$, $1/\pi^2$ for $D>0$), is larger by the factor $\sqrt{3}$. Across the imaginary range (8.3) predicts window means rising only from 1.887 to 1.898, a change comparable to the standard error 0.008 of a single window mean, so the flat window means of Figure 8.2 are expected; across the real range it predicts a rise from 1.239 to 1.256, and the data go from 1.238 to 1.259. For $p = 5$ and $p = 7$ no secondary term is known; the smaller deficits suggest one that is smaller or decays faster. The lesson for the computational programme is that data should be compared with Cohen and Lenstra only after the secondary term is accounted for, from (8.3) where it is known and by fitting a correction $c'X^{-\theta}$ with a free exponent where it is not (Computational Problem 8.1).

8.4 Selmer groups and the Poonen Rains model

Let $E/\mathbb{Q}$ be an elliptic curve and $n \ge 2$. The Kummer sequence $0 \to E[n] \to E(\bar{\mathbb{Q}}) \xrightarrow{n} E(\bar{\mathbb{Q}}) \to 0$ gives an injection $E(\mathbb{Q})/nE(\mathbb{Q}) \hookrightarrow H^1(\mathbb{Q}, E[n])$, and similarly

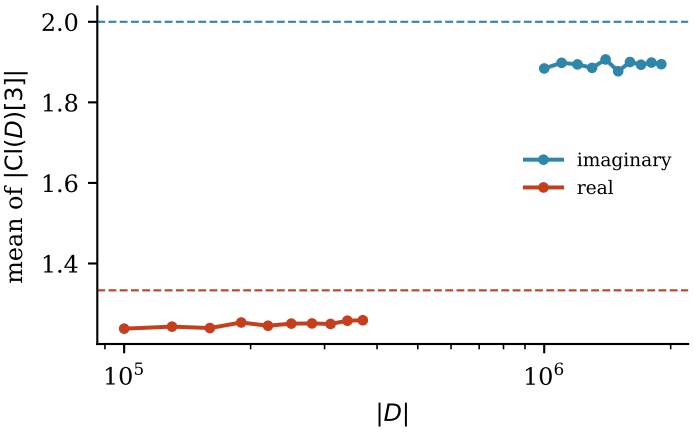


Figure 8.2: Window means of $|\mathrm{Cl}(D)[3]|$ for imaginary ($10^6 \le |D| \le 2 \cdot 10^6$) and real ($10^5 \le D \le 4 \cdot 10^5$) quadratic fields: each range is split into ten windows with equal numbers of discriminants, and each mean is plotted at the smallest $|D|$ of its window. Dashed lines: the Davenport Heilbronn limits 2 and $\frac{4}{3}$.

over every completion $\mathbb{Q}_v$. The n Selmer group $\mathrm{Sel}_n(E)$ is the subgroup of classes in $H^1(\mathbb{Q}, E[n])$ whose restriction to every $\mathbb{Q}_v$ lies in the image of $E(\mathbb{Q}_v)/nE(\mathbb{Q}_v)$; it is finite and effectively computable. The Shafarevich Tate group $\mathrm{Sha}(E)$ is the group of isomorphism classes of torsors under E that have a point over every completion of $\mathbb{Q}$, and the descent sequence reads

$$0 \to E(\mathbb{Q})/nE(\mathbb{Q}) \to \mathrm{Sel}_n(E) \to \mathrm{Sha}(E)[n] \to 0. \tag{8.4}$$

When $E(\mathbb{Q})[2] = 0$ we have $E(\mathbb{Q})/2E(\mathbb{Q}) \cong (\mathbb{Z}/2)^r$, so $r(E) \le \dim_{\mathbb{F}_2} \mathrm{Sel}_2(E)$, with equality if and only if $\mathrm{Sha}(E)[2] = 0$. If $\mathrm{Sha}(E)[2^\infty]$ is finite, the Cassels Tate pairing on it is alternating and nondegenerate, so $|\mathrm{Sha}(E)[2]|$ is a square and $\dim \mathrm{Sel}_2 \equiv r \pmod 2$.

8.4.1 Intersections of random maximal isotropic subspaces

Poonen and Rains [447] proved that $\mathrm{Sel}_p(E)$ is naturally an intersection of two maximal isotropic subspaces. The restricted product $V = \prod_v' H^1(\mathbb{Q}_v, E[p])$, taken with respect to the unramified classes, carries a quadratic form, built from Mumford's Heisenberg group following Zarhin, whose associated bilinear form is the sum of the local Tate pairings (cup product followed by the Weil pairing). The product Z of the local Kummer images, a compact open subspace, and the image W of $H^1(\mathbb{Q}, E[p])$, a discrete one, are both maximal isotropic, and arithmetic duality gives $\mathrm{Sel}_p(E) \cong W \cap Z$. The model replaces V by a finite hyperbolic quadratic space and W, Z by independent uniformly random maximal isotropic subspaces.

Let Q be the hyperbolic quadratic form on $V_n = \mathbb{F}_p^{2n}$ (orthogonal sum of n hyperbolic planes; for odd p any nondegenerate form of this discriminant). Its maximal isotropic subspaces have dimension n, and the following counts are classical:

(a) the number of maximal isotropic subspaces of V_n is $\prod_{i=0}^{n-1}(1+p^i)$;

(b) for a fixed maximal isotropic W, the number of maximal isotropic Z with $Z \cap W = 0$ is $p^{n(n-1)/2}$.

For (b), choose a maximal isotropic W' with $V_n = W \oplus W'$, so that the bilinear form identifies W with the dual of W'; a complement of W is the graph of a linear map $T : W' \to W$, and it is isotropic if and only if the bilinear form $(x,y) \mapsto \langle Tx, y\rangle$ on W' is alternating (with zero diagonal, which is the quadratic condition in characteristic 2). The alternating forms form a space of dimension $n(n-1)/2$. Statement (a) follows from (b), the decomposition by $W \cap Z$ in the proof below, and the q binomial theorem.

THEOREM 8.9 (Poonen and Rains [447]). *Let W, Z be independent uniformly random maximal isotropic subspaces of V_n. Then*

$$\mathbb{P}(\dim W \cap Z = d) = \binom{n}{d}_p \frac{p^{(n-d)(n-d-1)/2}}{\prod_{i=0}^{n-1}(1+p^i)},$$

with $\binom{n}{d}_p$ the Gaussian binomial coefficient, and as $n \to \infty$

$$\mathbb{P}(\dim W \cap Z = d) \to \prod_{j\geq 0}(1+p^{-j})^{-1} \prod_{j=1}^{d} \frac{p}{p^j - 1}. \tag{8.5}$$

PROOF. Fix W. A maximal isotropic Z with $Z \cap W = U$, $\dim U = d$, is contained in $U^\perp$ and its image in $U^\perp/U$, a hyperbolic space of dimension $2(n-d)$, is maximal isotropic and meets $(W \cap U^\perp)/U = W/U$ trivially; conversely each such image lifts uniquely. By (b) there are $p^{(n-d)(n-d-1)/2}$ choices for each of the $\binom{n}{d}_p$ subspaces $U \subset W$, and dividing by (a) gives the finite formula. For the limit, write $\binom{n}{d}_p = p^{nd-d(d-1)/2} \prod_{i<d}(1-p^{i-n}) / \prod_{i=1}^{d}(p^i-1)$ and $\prod_{i<n}(1+p^i) = p^{n(n-1)/2}\prod_{j<n}(1+p^{-j})$. The power of p is $nd - \frac{d(d-1)}{2} + \frac{(n-d)(n-d-1)}{2} - \frac{n(n-1)}{2} = d$, and letting $n \to \infty$ gives $p^d \prod_{j\geq 0}(1+p^{-j})^{-1} / \prod_{i=1}^{d}(p^i-1)$, which is (8.5). □

CONJECTURE 8.10 (Poonen and Rains [447]). *For each prime p, as E runs through elliptic curves over $\mathbb{Q}$ ordered by height, $\dim_{\mathbb{F}_p} \mathrm{Sel}_p(E)$ has the limiting distribution* (8.5).

For $p = 2$ the law is 0.2097, 0.4194, 0.2796, 0.0799, 0.0107 for $d = 0, \dots, 4$, with constant $c_2 = \prod_{j\geq 0}(1+2^{-j})^{-1} = 0.20971$. The model also predicts that $\dim \mathrm{Sel}_p$ is even with probability exactly $\frac{1}{2}$, a point we return to in Proposition 8.13.

8.4.2 The theorems of Bhargava and Shankar

The Poonen Rains model predicts $\mathbb{E}|\operatorname{Sel}_p| = p+1$ (Proposition 8.13). For $p \leq 5$ the corresponding average over all curves has been proved, the case $p = 5$ in a preprint. Every elliptic curve over $\mathbb{Q}$ has a unique model $y^2 = x^3 + Ax + B$ with $A, B \in \mathbb{Z}$ and no prime ℓ such that $\ell^4 \mid A$ and $\ell^6 \mid B$; we order curves by the naive height $H = \max(4|A|^3, 27B^2)$ of this model.

THEOREM 8.11 (Bhargava and Shankar). *Over all elliptic curves $E/\mathbb{Q}$ ordered by height, the average size of $\operatorname{Sel}_2(E)$ is 3 [55], and the average size of $\operatorname{Sel}_3(E)$ is 4 [56]. In later work, still a preprint, Bhargava and Shankar showed that the average size of $\operatorname{Sel}_5(E)$ is 6 [54].*

The method is again geometry of numbers: 2 Selmer elements correspond to locally soluble integral binary quartic forms up to $\mathrm{PGL}_2(\mathbb{Z})$, and 3 Selmer elements to ternary cubic forms; the average is a ratio of volumes, and the answer $p+1$ comes out as 1 (the identity element) plus the Tamagawa number p of PGL_p. Since $r \leq \dim \operatorname{Sel}_p$ and $p^r \leq |\operatorname{Sel}_p|$, these averages bound the average rank. Bhargava and Shankar obtain average rank at most 1.5 from Sel_2 and less than 1.17 from Sel_3; combining Sel_5 with the p parity theorem and with their proof that root numbers are equidistributed in a family of density at least 55%, they obtain less than 0.885, which shows that the average rank is less than 1.

Combining the 3 Selmer theorem with the p parity theorem and with the Iwasawa main conjecture of Skinner and Urban [520], Bhargava and Shankar showed that a positive proportion of elliptic curves over $\mathbb{Q}$ have rank 0 and analytic rank 0. For rank 1 the analogous statement uses in addition the theorems of Gross and Zagier [228] and of Kolyvagin [329] and the converse theorems of Skinner and of W. Zhang, and Bhargava, Skinner and Zhang [91] combined these ideas, in a preprint, to show that more than 66% of elliptic curves over $\mathbb{Q}$ ordered by height satisfy the conjecture of Birch and Swinnerton Dyer in its rank part (analytic rank equals algebraic rank, both 0 or 1, with Sha finite). These results are unconditional, and the statements about analytic rank rest on modularity, through which $L(s, E)$ is known to be entire.

8.4.3 The model of Bhargava, Kane, Lenstra, Poonen and Rains

The Poonen Rains model concerns one prime at a time and only Sel_p. Bhargava, Kane, Lenstra, Poonen and Rains [65] lifted it to $\mathbb{Z}_p$, modelling the whole sequence $0 \to E(\mathbb{Q}) \otimes \mathbb{Q}_p/\mathbb{Z}_p \to \operatorname{Sel}_{p^\infty}(E) \to \operatorname{Sha}(E)[p^\infty] \to 0$ by intersections of random maximal isotropic $\mathbb{Z}_p$ submodules, and showed that for curves of rank r the predicted law of $\operatorname{Sha}(E)[p^\infty]$ is that of the torsion of the cokernel of a random alternating matrix over $\mathbb{Z}_p$ with kernel of rank r. In the compact form used by Park, Poonen, Voight and Wood, choose a large n of uniformly random parity and

a Haar random alternating matrix $A \in M_n(\mathbb{Z}_p)$, and put

$$\operatorname{rank} \leftrightarrow \operatorname{rk}_{\mathbb{Z}_p} \ker A, \qquad \text{Sha}[p^\infty] \leftrightarrow (\operatorname{cok} A)_{\text{tors}}.$$

An alternating matrix of odd size is singular, so $\ker A$ has rank 1 almost surely for odd n and 0 for even n: the model predicts rank 0 and rank 1 each with probability $\frac{1}{2}$, rank at least 2 with probability 0, and $\text{Sha}[p^\infty]$ finite with probability 1. The torsion of the cokernel of an alternating matrix carries a nondegenerate alternating pairing, matching the Cassels Tate pairing, and for curves of rank r the model gives $\text{Sha}[p^\infty] \cong G$ with probability proportional to $|G|^{1-r}/|\operatorname{Sp}(G)|$, where G runs over finite abelian p groups with a nondegenerate alternating pairing and $\operatorname{Sp}(G)$ is the group of automorphisms preserving it. This symplectic analogue of Cohen and Lenstra is the distribution proposed by Delaunay [136], in the form he later refined. The model reproduces Conjecture 8.10 for Sel_p, and with it the averages $p+1$ of Theorem 8.11, and it also predicts the laws of the groups Sel_{p^e}.

Park, Poonen, Voight and Wood [448] replaced the p adic matrices by alternating integer matrices whose size n and entry bound X both grow with the height H, calibrated by $X^n = H^{1/12+o(1)}$ so that the cokernel, which models Sha, has the size that the conjecture of Birch and Swinnerton Dyer suggests. In their model a curve of height H has rank at least $r \geq 1$ with probability $H^{-(r-1)/24+o(1)}$, and summing over all curves predicts that only finitely many elliptic curves over $\mathbb{Q}$ have rank greater than 21, so that ranks are bounded. This is a heuristic and not a theorem; it is in tension with the long held view that ranks are unbounded, and the record curves (Elkies found one of rank at least 28 in 2006, and Elkies and Klagsbrun one of rank at least 29 in 2024) are compatible with it, since the heuristic allows finitely many exceptions.

8.4.4 Our data

We took the 6122 curves $y^2 = x^3 + ax + b$ with $|a|, |b| \leq 40$, nonzero discriminant and $x^3 + ax + b$ irreducible over $\mathbb{Q}$, so that $E(\mathbb{Q})[2] = 0$. For each we computed the root number and ran PARI/GP `ellrank`, which performs a 2 descent, computes the 2 part of the Cassels Tate pairing on Sel_2 and searches for points; its output gives $\dim \text{Sel}_2$ and lower and upper bounds for the rank, which coincided for all but one curve. Table 8.2 and Figure 8.3 summarise the 6121 curves whose rank was determined.

Every frequency departs from the limiting law in the same direction: small Selmer groups are too common, and large ones too rare. The mean of $|\text{Sel}_2|$ is 2.119, not 3. Yet the mean rank, 0.853, is far above the conjectural limit $\frac{1}{2}$, and one curve in six has rank 2. The two effects are compatible. The mean of $\dim \text{Sel}_2$ is 0.894, only 0.041 above the mean rank. For a curve whose rank the descent determines, $\dim \text{Sel}_2 - r = \dim \text{Sha}(E)[2]$ is even (Section 8.5.2), and

	0	1	2	3	4
$\dim \mathrm{Sel}_2$, observed	0.3093	0.4976	0.1826	0.0105	0
$\dim \mathrm{Sel}_2$, Poonen Rains	0.2097	0.4194	0.2796	0.0799	0.0107
rank, observed	0.3277	0.4996	0.1642	0.0085	

Table 8.2: The 6121 curves $y^2 = x^3 + ax + b$, $|a|, |b| \le 40$, with no rational 2 torsion and rank determined by `ellrank`: distribution of $\dim \mathrm{Sel}_2$ against the Poonen Rains law, and of the rank. Mean $|\mathrm{Sel}_2| = 2.119$ (Bhargava and Shankar: 3 in the limit); mean rank 0.853; root number $+1$ for 49.2%.

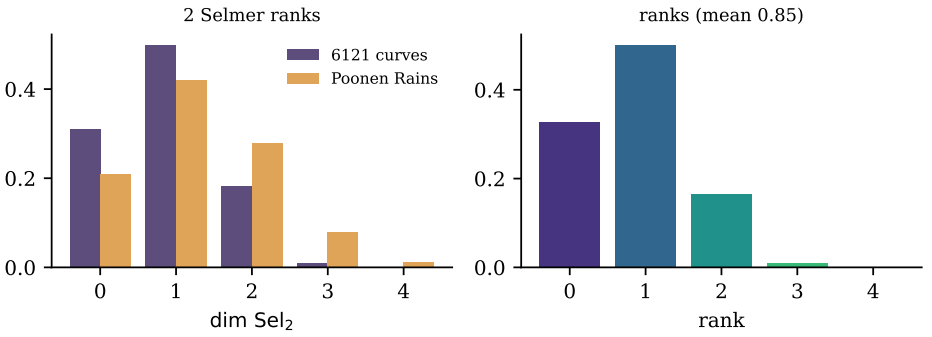


Figure 8.3: Left: $\dim \mathrm{Sel}_2$ for the 6121 curves of small height of Table 8.2 against the Poonen Rains law. Right: their ranks (mean 0.85).

here $\dim \mathrm{Sel}_2 \le 3$, so $\mathrm{Sha}(E)[2] \cong (\mathbb{Z}/2)^2$ for exactly 125 curves (2.0%) and $\mathrm{Sha}(E)[2] = 0$ for the rest: at small height, Sel_2 is almost entirely accounted for by rational points. This is what the conjecture of Birch and Swinnerton Dyer leads one to expect, since together with standard conjectures on $L(E,1)$ it gives $|\mathrm{Sha}|$ an average size $H^{1/12+o(1)}$ over curves of rank 0 [448], which is small at our heights ($H \le 256000$). In the model of Park, Poonen, Voight and Wood both features appear at small height: the cokernel that models Sha is small, and the probability of rank at least 2 decays only like $H^{-1/24+o(1)}$. The excess of rank 2 in small height families was documented systematically by Bektemirov, Mazur, Stein and Watkins [68], who argued that the convergence to the limiting average is slow (a heuristic of Watkins discussed there makes the proportion of analytic rank at least 2 among curves of root number $+1$ and discriminant near X decay like $(\log X)^{3/8} X^{-1/24}$) and that the data do not contradict the minimalist conjecture. We read our table in the same cautious way: a box with $|a|, |b| \le 40$ is far from the asymptotic regime. Section 8.5.2 examines the statistic that should already be close to its limit, the proportion of curves with root number $+1$.

8.5 Ranks and L functions

8.5.1 Goldfeld's conjecture and the minimalist conjecture

For a fixed $E/\mathbb{Q}$ of conductor N_E and fundamental discriminants d, the quadratic twists E_d have $L(s, E_d) = \sum \chi_d(n) a_n n^{-s}$ (away from the conductor), and for d prime to N_E the root number is $w(E_d) = \chi_d(-N_E) w(E)$, so these twists split evenly between root numbers $+1$ and -1.

Conjecture 8.12 (Goldfeld [218]). *In the family of quadratic twists of a fixed elliptic curve over $\mathbb{Q}$, ordered by $|d|$, the average analytic rank is $\frac{1}{2}$: asymptotically half the twists have analytic rank 0 and half have analytic rank 1.*

The minimalist conjecture is the analogous statement for all elliptic curves ordered by height, or for any reasonable family: the rank is as small as the root number allows, so rank 0 when $w = +1$ and rank 1 when $w = -1$, outside a set of density zero. Both are predictions of the Katz and Sarnak philosophy (Chapter 6): for a family with orthogonal symmetry split by root number, the zeros near the centre behave like eigenvalues near 1 of Haar random matrices in $\mathrm{SO}(2N)$ or $\mathrm{SO}(2N+1)$, and a random matrix in $\mathrm{SO}(2N)$ has eigenvalue 1 with probability 0, while one in $\mathrm{SO}(2N+1)$ has it with multiplicity exactly 1 almost surely. The random alternating matrices of the BKLPR model make the same prediction on the arithmetic side: kernel rank 0 or 1 according to the parity of n. That two random matrix models, one for zeros (orthogonal) and one for groups (alternating p adic), agree on the 50/50 split is an instance of the dictionary between the analytic and arithmetic sides of the conjecture of Birch and Swinnerton Dyer.

The rigorous results are bounds. Under the generalized Riemann hypothesis, the one level density methods of Brumer, Heath Brown and Young described in Section 6.4 bound the average analytic rank; the Bhargava and Shankar theorems give unconditional bounds for the average algebraic rank. In quadratic twist families Smith [521] proved, in a preprint, that for $E/\mathbb{Q}$ with full rational 2 torsion and no rational cyclic subgroup of order 4 the 2^∞ Selmer groups of the twists E_d have the distribution predicted by Delaunay's heuristic, and deduced that the twists of rank at least 2 have density zero, the algebraic heart of Goldfeld's conjecture for these families.

8.5.2 Root numbers and parity

Among the 6121 curves of Table 8.2 the root number is $+1$ for 49.2%, and for every one of them the rank is even exactly when $w(E) = +1$. This is what the parity conjecture $(-1)^{r(E)} = w(E)$ predicts; it follows from the conjecture of Birch and Swinnerton Dyer and is open in general. What is proved is the p parity conjecture: Monsky [398] showed that $(-1)^{\dim \mathrm{Sel}_2(E) - \dim E(\mathbb{Q})[2]} = w(E)$ for every

$E/\mathbb{Q}$, and T. and V. Dokchitser [160] proved that the $\mathbb{Z}_p$ corank of the p^∞ Selmer group has the parity of the root number for every $E/\mathbb{Q}$ and every prime p. Rank parity follows for every curve with $Ш(E)[p^\infty]$ finite for one p, since that corank is then the rank. Our check is therefore not independent evidence for the parity conjecture. The upper bound that `ellrank` computes exceeds the rank by the dimension of $2\,Ш(E)[4]$, so when the bounds coincide $Ш(E)$ has no element of order 4; then $Ш(E)[2^\infty] = Ш(E)[2]$ is finite, of even dimension, and Monsky's theorem forces the agreement. For the same reason the even parts of the two distributions, $0.3093 + 0.1826$ and $0.3277 + 0.1642$, both equal the root number proportion 0.4919. What the data test is the equidistribution of root numbers, and 0.4919 is within 1.3 standard errors of $\frac{1}{2}$.

8.5.3 The rank as a Thorin atom and as a bias

The rank enters the analytic theory of this book in two further guises. In Chapter 5 we saw that the partial sums $\sum_{p \le x} a_p \log p / p$ drift like $(\frac{1}{2} - r)\log x$ in logarithmic mean under the Riemann hypothesis for $L(s, E)$ (Section 5.4, Table 5.2): the central zero of order r appears, through the explicit formula, as a bias in Frobenius traces. In Chapter 14 (Proposition 14.14) the same central zero appears as a degenerate Thorin atom: $\log|\Lambda(\frac{1}{2} + \sqrt{s}, E)|$ has slope $r/2$ in $\log s$ near $s = 0$, a gamma factor of shape $r/2$ whose Thorin mass has escaped to the origin. In the language of the present chapter the minimalist conjecture, in its analytic form, says that in a family this atom has mass 0 or $\frac{1}{2}$ according to the root number, each with probability $\frac{1}{2}$, and that the larger masses $1, \frac{3}{2}, \dots$ occur with density zero. Our small height data, with mean rank 0.853, put visible weight on the larger masses: by the theorems of Gross and Zagier [228] and Kolyvagin [329] a curve of rank at least 2 has analytic rank at least 2, and 17.3% of our curves have rank at least 2. The rank bias and the Thorin atom read the order of vanishing at the centre directly; the Selmer statistics read it through the conjecture of Birch and Swinnerton Dyer.

8.5.4 The Poonen Rains law as a sum of independent coins

We close the mathematical part of the chapter with a structural description of the Poonen Rains law that makes its moments transparent.

PROPOSITION 8.13 (Poonen and Rains [447]). *Let $\xi_0, \xi_1, \dots$ be independent with $\mathbb{P}(\xi_i = 1) = 1/(1 + p^i) = 1 - \mathbb{P}(\xi_i = 0)$. Then:*

(a) for each n, $\dim W \cap Z$ in Theorem 8.9 has the law of $\xi_0 + \cdots + \xi_{n-1}$, and the Poonen Rains law (8.5) *is the law of $S = \sum_{i \ge 0} \xi_i$, which is almost surely finite;*

(b) $\mathbb{E}t^S = \prod_{i \ge 0}(t + p^i)/(1 + p^i)$ for all complex t;

(c) $\mathbb{P}(S \text{ even}) = \frac{1}{2}$, and S is even with probability $\frac{1}{2}$ conditionally on $\xi_1, \xi_2, \dots$;

(d) $\mathbb{E}p^S = p+1$, *and more generally* $\mathbb{E}\prod_{j=0}^{k-1}(p^S - p^j) = p^{k(k+1)/2}$ *for* $k \geq 1$;

(e) $\mathbb{E}S = \sum_{i\geq 0}(1+p^i)^{-1}$ *and* $\operatorname{Var} S = \sum_{i\geq 0} p^i(1+p^i)^{-2}$; *for* $p=2$ *these are* 1.2645 *and* 0.8463;

(f) $\mathbb{P}(S=0) = c_p := \prod_{j\geq 0}(1+p^{-j})^{-1}$ *and* $\mathbb{P}(S\geq 2) = 1 - \frac{2p-1}{p-1}\,c_p$, *equal to* $1-3c_2 = 0.3709$ *for* $p=2$.

PROOF. (a) By Theorem 8.9 the generating function of $\dim W\cap Z$ is

$$\begin{aligned}\mathbb{E}t^{\dim W\cap Z} &= \frac{1}{\prod_{i<n}(1+p^i)}\sum_{d=0}^{n}\binom{n}{d}_p p^{(n-d)(n-d-1)/2}t^d \\ &= \frac{t^n}{\prod_{i<n}(1+p^i)}\sum_{e=0}^{n}\binom{n}{e}_p p^{e(e-1)/2}t^{-e},\end{aligned}$$

substituting $e = n-d$ and using $\binom{n}{d}_p = \binom{n}{n-d}_p$. The q binomial theorem $\sum_e \binom{n}{e}_p p^{e(e-1)/2}u^e = \prod_{i<n}(1+p^iu)$ with $u = 1/t$ gives $\prod_{i<n}(t+p^i)/(1+p^i)$, the generating function of $\xi_0+\cdots+\xi_{n-1}$, since $(t+p^i)/(1+p^i) = \mathbb{P}(\xi_i = 0) + t\mathbb{P}(\xi_i = 1)$. Since $\sum_i \mathbb{P}(\xi_i=1) < \infty$, Borel and Cantelli show that only finitely many ξ_i equal 1, so the partial sums converge almost surely to $S<\infty$; hence the laws converge pointwise, and the limit is (8.5) by Theorem 8.9. (b) The partial products converge for every t because $\sum_i |t-1|/(1+p^i) < \infty$, and $\mathbb{E}t^{S_n} \to \mathbb{E}t^S$ by dominated convergence for $|t|\leq 1$ and, for larger $|t|$, by the bound $|t|^{S_n} \leq |t|^S$ and the finiteness of $\mathbb{E}|t|^S = \prod_i(|t|+p^i)/(1+p^i)$ (monotone convergence). (c) ξ_0 is a fair coin independent of $\xi_1,\xi_2,\ldots$, so the parity of S is a fair coin given the rest. (d) Fix W in V_n and $k\leq n$. A k tuple of linearly independent vectors in $W\cap Z$ spans a k dimensional subspace $U\subset W$, and, as in the proof of Theorem 8.9, the maximal isotropic Z containing a given U correspond to the maximal isotropic subspaces of $U^\perp/U \cong V_{n-k}$, of which there are $\prod_{i<n-k}(1+p^i)$. Counting pairs,

$$\mathbb{E}\prod_{j<k}(p^{\dim W\cap Z} - p^j) = \prod_{j<k}(p^n-p^j)\,\frac{\prod_{i<n-k}(1+p^i)}{\prod_{i<n}(1+p^i)} = \prod_{j<k}\frac{p^n-p^j}{1+p^{n-k+j}},$$

which tends to $\prod_{j<k}p^{k-j} = p^{k(k+1)/2}$ as $n\to\infty$. By (a) the left side is $\mathbb{E}f(S_n)$ with $S_n = \xi_0+\cdots+\xi_{n-1}$ and $f(m) = \prod_{j<k}(p^m-p^j)$, which is nonnegative and nondecreasing on the integers; since S_n increases to S, monotone convergence gives $\mathbb{E}f(S) = p^{k(k+1)/2}$. The case $k=1$ is $\mathbb{E}p^S = p+1$. (e) Sum the Bernoulli means and variances. (f) $\mathbb{P}(S=0) = \prod_i \mathbb{P}(\xi_i=0) = \prod_i p^i/(1+p^i)$, and $\mathbb{P}(S=1) = \mathbb{P}(S=0)\sum_i \mathbb{P}(\xi_i=1)/\mathbb{P}(\xi_i=0) = \mathbb{P}(S=0)\sum_i p^{-i} = \mathbb{P}(S=0)\,p/(p-1)$. □

Parts (a) and (b) are Proposition 2.6 of Poonen and Rains, (c) is their Proposition 2.22(b), and (d) is equivalent to their Proposition 2.22(a); in the form stated, (d) is the prime case of a theorem of Bhargava, Kane, Lenstra, Poonen and Rains [65], proved there by the same count. The independent coin form separates parity from size. For a curve with $E(\mathbb{Q})[p] = 0$ the theorem of Dokchitser and Dokchitser gives $(-1)^{\dim \mathrm{Sel}_p(E)} = w(E)$, because $\mathrm{Sel}_p(E) = \mathrm{Sel}_{p^\infty}(E)[p]$ and the quotient of $\mathrm{Sel}_{p^\infty}(E)$ by its divisible part, which carries the nondegenerate alternating Cassels Tate pairing, has p torsion of even dimension. In the model the parity of S plays the role of $w(E)$, and by (c) it is a fair coin even given $\xi_1, \xi_2, \dots$, as equidistributed root numbers require. In the BKLPR model the rank r is 0 or 1 almost surely and $\dim \mathrm{Sel}_p = r + \dim \mathrm{Sha}[p]$ with $\dim \mathrm{Sha}[p]$ even, so r is the parity of S and $\dim \mathrm{Sha}[p] = 2\lfloor S/2 \rfloor$; for $p = 2$ the coins $\xi_1, \xi_2, \dots$, with probabilities $\frac{1}{3}, \frac{1}{5}, \frac{1}{9}, \dots$, govern its size.

COROLLARY 8.14 (Rank bounds from the model). *Consider a family of elliptic curves over $\mathbb{Q}$ with $E(\mathbb{Q})[2] = 0$, ordered by a height, in which $\dim \mathrm{Sel}_2(E)$ converges in distribution to the Poonen Rains law* (8.5) *for $p = 2$ and the average of $|\mathrm{Sel}_2(E)|$ over the curves of height at most X stays bounded as $X \to \infty$. Then the average rank $\bar{r}(X)$ of the curves of height at most X satisfies*

$$\limsup_{X \to \infty} \bar{r}(X) \le \mathbb{E}S = \sum_{i \ge 0} \frac{1}{1 + 2^i} = 1.2645 < \log_2 \mathbb{E}2^S = \log_2 3 = 1.585,$$

where S has the law (8.5) *for $p = 2$.*

PROOF. Since $E(\mathbb{Q})[2] = 0$, the Kummer map embeds $E(\mathbb{Q})/2E(\mathbb{Q}) \cong (\mathbb{Z}/2)^r$ in $\mathrm{Sel}_2(E)$, so $r \le \dim \mathrm{Sel}_2(E)$. For every $\varepsilon > 0$ there is C_ε with $m \le C_\varepsilon + \varepsilon 2^m$ for all $m \ge 0$, so bounded averages of $2^{\dim \mathrm{Sel}_2}$ make $\dim \mathrm{Sel}_2$ uniformly integrable along the family, and its averages converge to $\mathbb{E}S$, which Proposition 8.13(e) evaluates. The last comparison is Jensen's inequality for the concave logarithm, strict because S is not constant, with $\mathbb{E}2^S = 3$ from Proposition 8.13(d). □

The comparison clarifies what an average of $|\mathrm{Sel}_2|$ alone can give. Knowing only that the average of $|\mathrm{Sel}_2|$ is 3, as Bhargava and Shankar proved, Jensen yields average rank at most $\log_2 3 \approx 1.585$; the integer inequality $2r \le 2^r$ does better and yields their bound 1.5. Knowing in addition the full Poonen Rains law would give 1.2645. The same argument with the integer inequalities $r \le \frac{1}{2} + \frac{1}{6}3^r$ and $r \le \frac{3}{4} + \frac{1}{20}5^r$, both equalities at $r = 1, 2$, turns the 3 Selmer average 4 into their bound $\frac{7}{6} \approx 1.17$ and the 5 Selmer average 6 into 1.05, before the parity input described after Theorem 8.11. The value $\frac{1}{2}$ is out of reach of the Selmer group at any single prime, even with its full law: Selmer data alone cannot exclude $r = \dim \mathrm{Sel}_p$, so the best bound they give is $\mathbb{E} \dim \mathrm{Sel}_p = \sum_{i \ge 0}(1 + p^i)^{-1}$ in the model, which exceeds $\frac{1}{2}$ for every p. It tends to $\frac{1}{2}$ as $p \to \infty$, and Rubin observed

[447] that the law (8.5) itself tends to the fair coin on $\{0,1\}$. Hence the average rank would be at most $\frac{1}{2}$ if Conjecture 8.10 and the average $p+1$ of $|\,\mathrm{Sel}_p\,|$, which Poonen and Rains also conjecture, held for infinitely many p. The law alone for infinitely many p would give the density statements that almost all curves have rank at most 1 and at least half have rank 0, but not the bound on the average, since a set of curves of density zero and large rank could raise it.

8.6 Notes

Cohen and Lenstra [110] introduced the heuristics in the Noordwijkerhout proceedings, with extensive tables; Cohen and Martinet [111] extended them to general number fields. The random matrix interpretation is due to Friedman and Washington [204], in the function field setting, where the ℓ part of the class group is the cokernel of $1-\mathrm{Frob}$ acting on the ℓ adic Tate module of the Jacobian. Hall's identity [234] predates both. Wood developed the moment method [588] and showed [589] that cokernels of random integral matrices with quite general independent entries have Cohen Lenstra limits, a universality phenomenon analogous to the universality of eigenvalue statistics in Chapter 6. The sandpile groups of supersingular isogeny graphs, which are modules over a Hecke algebra $\mathbb{T}$, behave differently: for a fixed characteristic $N \equiv 1 \bmod 12$, growing isogeny degree and all but finitely many primes r, their r parts follow the law of $\mathbb{T}_r/\det(\gamma-1)\mathbb{T}_r$ for γ Haar random in the subgroup of $\mathrm{GL}_2(\mathbb{T}_r)$ with determinant in $\mathbb{Z}_r^\times$, where $\mathbb{T}_r = \mathbb{T}\otimes\mathbb{Z}_r$, and this law is not the Cohen Lenstra law (Munier and Shnidman [408]; Chapter 21).

The theorem of Davenport and Heilbronn [127] is still the only determination of the mean of $|\mathrm{Cl}(D)[p]|$ over quadratic fields for an odd prime p. Bhargava's higher composition laws and his count of quartic rings and fields [53] carried such averages beyond the quadratic case; the secondary term [90, 536] confirmed Roberts's conjecture. The function field results [180] brought topology (homological stability of Hurwitz spaces) into arithmetic statistics. The corrections at roots of unity are in Malle [380].

The Poonen Rains model [447] was motivated in part by the earlier theorems of Heath Brown on 2 Selmer groups in the congruent number family and of Swinnerton Dyer and Kane on quadratic twists of curves with full rational 2 torsion and no rational cyclic subgroup of order 4, which give the law (8.5) for $p=2$, shifted by 2 to account for $E[2]$, in these twist families. The theorems of Bhargava and Shankar [55, 56, 54] and the synthesis of [65] and [448] now frame the subject, and the database of Balakrishnan, Ho, Kaplan, Spicer, Stein and Weigandt [57] tests these models on hundreds of millions of curves ordered by height. Bektemirov, Mazur, Stein and Watkins [68] discuss the tension between data and the minimalist conjecture. The proposal that the Langlands programme

be read alongside random group models is not ours; it is implicit in the way the conjecture of Birch and Swinnerton Dyer connects automorphic L functions to Selmer and Shafarevich Tate groups.

8.7 Exercises

1. (Hall's identity in small order.) List the abelian groups of order p^2 and p^3 with their automorphism groups, and verify that $\sum_{|G|=p^n} |\operatorname{Aut} G|^{-1}$ equals the coefficient of x^n in $\prod_{k\geq 1}(1-xp^{-k})^{-1}$ for $n = 1, 2, 3$. Deduce (8.1) by setting $x = 1$, assuming the generating function identity.

2. (Rank 0 over a finite field.) Show that the probability that a uniformly random $n \times n$ matrix over $\mathbb{F}_p$ is invertible is $\eta_n(p)$, that it decreases in n, and that $0 \leq \eta_n(p) - \eta_\infty(p) \leq p^{-n}/(p-1)$. Compute $\eta_n(2)$ for $n = 1, \dots, 5$ and compare with $\eta_\infty(2) = 0.2888$.

3. (Poonen Rains at $d = 0$.) Using Theorem 8.9, show that $\mathbb{P}(\mathrm{Sel}_2 = 0)$ is predicted to be $\prod_{j\geq 0}(1+2^{-j})^{-1} = 0.2097$, and that the finite n value is $2^{n(n-1)/2}/\prod_{i<n}(1+2^i)$. Compute it for $n = 1, \dots, 6$.

4. (The constant at 3.) Compute $\eta_\infty(3)$ to six digits using Euler's pentagonal number theorem, $\eta_\infty(p) = \sum_{k\in\mathbb{Z}}(-1)^k p^{-k(3k-1)/2}$, and determine how many terms are needed for six digits. Deduce the Cohen Lenstra prediction $\mathbb{P}(3 \mid h(D)) = 0.4399$.

5. (Mean of $|G[p]|$.) Let $\bar{A}$ be a uniformly random $n \times n$ or $n \times (n+1)$ matrix over $\mathbb{F}_p$. By summing $\mathbb{P}(\bar{A}^{\mathsf{T}} v = 0)$ over $v \in \mathbb{F}_p^n$, show that $\mathbb{E}|\ker \bar{A}^{\mathsf{T}}|$ equals $2 - p^{-n}$ in the square case and $1 + p^{-1} - p^{-n-1}$ in the rectangular case. Since $|G[p]| = p^{\mathrm{rk}_p G}$ and $p^{\mathrm{rk}_p \operatorname{cok} A} = |\ker \bar{A}^{\mathsf{T}}|$ when $\bar{A}$ is the reduction of A modulo p, deduce $\mathbb{E}_{\mu_{\mathrm{CL}}}|G[p]| = 2$ and $\mathbb{E}_{\mu^+_{\mathrm{CL}}}|G[p]| = 1 + p^{-1}$, justifying the passage to the limit as in the proof of Proposition 8.5. Compare with Theorem 8.7.

6. (Parity.) In the setting of Proposition 8.13, show that $\mathbb{E}(-1)^S = 0$, by evaluating the generating function at $t = -1$, and that $\mathbb{P}(\dim W \cap Z \text{ even}) = \tfrac{1}{2}$ for every finite $n \geq 1$. Show that if root numbers in a family of curves without rational 2 torsion are $+1$ and -1 with limiting proportions $\tfrac{1}{2}(1+\beta)$ and $\tfrac{1}{2}(1-\beta)$, then by Monsky's theorem $\dim \mathrm{Sel}_2$ cannot follow the Poonen Rains law unless $\beta = 0$. Check that our data, with $\beta = -0.016$, are consistent with $\beta = 0$ at the usual significance levels.

8.8 Open computational problems

Computational Problem 8.1 (Cohen Lenstra at scale with the secondary term). Objective: measure the convergence of 3, 5 and 7 parts of class groups of quadratic fields to the Cohen Lenstra laws, separating the secondary term. Objects: fundamental discriminants D, imaginary and real, all those with $|D| \leq 10^9$ and all those in the windows $X - 10^8 < |D| \leq X$ for $X = 10^{10}$ and 10^{11}. Route: PARI/GP `quadclassunit` (conditional on the generalized Riemann hypothesis, as for our data) or the subexponential class group algorithms of Jacobson and collaborators, distributed over discriminant windows on a cluster; for 3 torsion alone, count cubic fields with Belabas's program as an unconditional check. Deliverable: tables of p rank frequencies and of the means of $|\mathrm{Cl}(D)[p]|$ per window; a weighted least squares fit of $c + c' X^{-1/6} + c'' X^{-1/3}$ to the 3 torsion means, with the fitted c' compared with the constants of Bhargava, Shankar and Tsimerman and of Taniguchi and Thorne; analogous fits for $p = 5, 7$ with a free exponent. Why it matters: Table 8.1 shows a 3 deficit of the size of the secondary term; a free exponent fit for $p \geq 5$ would give an empirical rate of convergence where no theorem is known.

Computational Problem 8.2 (Selmer groups and ranks by height). Objective: test the Poonen Rains law and the boundedness heuristic of Park, Poonen, Voight and Wood on curves ordered by height. Objects: all curves $y^2 = x^3 + Ax + B$ with naive height $H \leq 10^{10}$ (about 10^8 curves), with the database of Balakrishnan, Ho, Kaplan, Spicer, Stein and Weigandt [57] as a starting point. Route: 2 descent by PARI/GP `ellrank` or Magma, 3 Selmer groups by Magma where feasible, root numbers from local data, and analytic ranks by `lfun` for a random subsample, distributed over curves on a cluster. Deliverable: the distribution of $\dim \mathrm{Sel}_2$ and $\dim \mathrm{Sel}_3$ in height windows, the mean of $|\mathrm{Sel}_2|$ against 3, the proportion of rank ≥ 2 as a function of H with a fitted power law, and the tail counts of rank $\geq r$ against the exponents predicted by [448]. Why it matters: the excess of curves of rank at least 2 in Table 8.2 must decay if the minimalist conjecture holds, and its rate of decay is what separates slow convergence from a genuine bias.

Computational Problem 8.3 (Shafarevich Tate groups against BKLPR). Objective: compare the analytic orders of Sha with the alternating cokernel model of [65] and with Delaunay's distribution. Objects: all curves of conductor up to $5 \cdot 10^5$ in Cremona's tables and the LMFDB [371], plus quadratic twist families of $11a1$ and $37a1$ with $|d| \leq 10^7$. Route: the analytic order of Sha from the leading term of $L(s, E)$ via the conjecture of Birch and Swinnerton Dyer (PARI/GP `ellbsd` and `lfun`), checked against 2, 3 and 4 descents in Magma. Deliverable: the empirical law of $\mathrm{Sha}[p^\infty]$ for $p = 2, 3, 5$ in conductor windows, the proportion of nontrivial Sha as a function of conductor, and a fit of the effective matrix size in the model of Park, Poonen, Voight and Wood as a function of the conductor. Why it matters:

our data show Sha[2] nontrivial for only 2% of small curves; tracking how this grows tests the calibration on which the boundedness heuristic rests.

COMPUTATIONAL PROBLEM 8.4 (Class groups of cubic and quartic fields). Objective: test the Cohen Martinet heuristics, with Malle's corrections at roots of unity, for class groups of cubic and quartic fields. Objects: all cubic fields with $|D| \leq 10^9$ (enumerated by Belabas's algorithm) and all quartic fields with $|D| \leq 10^8$ with Galois groups D_4, A_4, S_4 and C_4; class groups for primes $p \leq 7$, with separate treatment of fields containing μ_p or with Galois closure containing it. Route: PARI/GP `bnfinit` under GRH, with spot certification by `bnfcertify`; Bhargava's parametrisations to enumerate quartic fields. Deliverable: p rank frequencies against Cohen Martinet, including the average of $|\mathrm{Cl}[2]|$ for cubic fields against Bhargava's $\frac{5}{4}$ and $\frac{3}{2}$, and a table of the discrepancies at roots of unity against Malle's proposed corrections. Why it matters: these are the class groups attached to GL_1 over fields of degree 3 and 4, and the cases where the heuristics are known to need correction are precisely where the arithmetic constraints are subtle.

PART III

RANDOM WALKS AND HITTING TIMES

CHAPTER 9

HECKE OPERATORS AS RANDOM WALKS ON TREES

In 1959 Harry Kesten computed the probability that a symmetric random walk on a free group returns to its starting point, and found that it decays exponentially, like ρ^n with $\rho < 1$ up to a polynomial factor [309]. The work of Ihara on discrete subgroups of $\mathrm{PGL}_2(\mathbb{Q}_q)$ and Serre's lectures on trees [269, 505] identified the regular tree on which such a walk lives with the Bruhat Tits tree of $\mathrm{PGL}_2(\mathbb{Q}_q)$, on which the Hecke operator T_q of the theory of modular forms acts as the adjacency operator. In 1981 McKay found Kesten's spectral law again as the limiting eigenvalue distribution of large random regular graphs [387], and in 1997 Serre found it once more as the limiting distribution of the eigenvalues of T_q on spaces of modular forms of growing level or weight [508]. This chapter explains why these are one law. We prove the tree theorem (Theorem 9.1), state the vertical Sato Tate theorem with a probabilistic proof idea, and test it on supersingular isogeny graphs at 207 prime levels.

9.1 THE BRUHAT TITS TREE

Fix a prime q. A lattice in $\mathbb{Q}_q^2$ is a $\mathbb{Z}_q$ submodule $\mathbb{Z}_q u \oplus \mathbb{Z}_q v$ spanned by a basis u, v of $\mathbb{Q}_q^2$, and two lattices L, L' are homothetic if $L' = cL$ for some $c \in \mathbb{Q}_q^\times$. The vertices of $\mathcal{T}_q$ are homothety classes $[L]$. Two classes are joined by an edge if they have representatives with $qL \subsetneq L' \subsetneq L$. Since $L/qL \cong \mathbb{F}_q^2$ has exactly $q+1$ lines, every vertex has $q+1$ neighbours, and one shows that the resulting graph is connected and has no cycles, so $\mathcal{T}_q$ is the $(q+1)$ regular tree [505]. The group $\mathrm{GL}_2(\mathbb{Q}_q)$ acts on lattices, the centre acts trivially on classes, and $\mathrm{PGL}_2(\mathbb{Q}_q)$ acts on $\mathcal{T}_q$ by automorphisms, transitively on vertices, with the stabiliser of $[\mathbb{Z}_q^2]$ equal to $\mathrm{PGL}_2(\mathbb{Z}_q) = \mathrm{GL}_2(\mathbb{Z}_q)/\mathbb{Z}_q^\times$, a maximal compact subgroup. Thus the vertex set is $\mathrm{PGL}_2(\mathbb{Q}_q)/\mathrm{PGL}_2(\mathbb{Z}_q)$, and functions on vertices are right $\mathrm{PGL}_2(\mathbb{Z}_q)$ invariant functions on the group.

If $L' \subseteq L$ and $L/L' \cong \mathbb{Z}/q^a \oplus \mathbb{Z}/q^b$ with $a \geq b$, then $d([L],[L']) = a - b$. For $n \geq 1$ the sphere $\{w : d(v,w) = n\}$ around a vertex v has $(q+1)q^{n-1}$ elements.

9.1.1 Hecke operators

The Hecke operator T_q acts on functions on lattice classes by $T_q f([L]) = \sum f([L'])$, summed over the $q+1$ sublattices of index q. This is the adjacency operator

A of $\mathcal{T}_q$. More generally let A_n be the operator $A_n f(v) = \sum_{d(v,w)=n} f(w)$, with $A_0 = I$; equivalently $A_n(v,w)$ counts nonbacktracking walks of length n from v to w, since on a tree such a walk is a geodesic and there is exactly one between vertices at distance n. The classical Hecke operator T_{q^n} sums over all sublattices of index q^n; each of them is $q^b L''$ with L/L'' cyclic of order q^{n-2b}, so on classes $T_{q^n} = A_n + A_{n-2} + A_{n-4} + \cdots$ (Exercise 1). All these operators commute: they lie in the spherical Hecke algebra, which is the polynomial algebra $\mathbb{C}[T_q]$.

9.2 THE TREE THEOREM

Let o be a fixed vertex and let μ_q be the spectral measure of the bounded self adjoint operator $X = q^{-1/2}A$ on $\ell^2(\mathcal{T}_q)$ at δ_o, so that $\langle \delta_o, \phi(X)\delta_o \rangle = \int \phi \, d\mu_q$ for continuous ϕ. Write U_n for the Chebyshev polynomials of the second kind, $U_n(\cos\theta) = \sin((n+1)\theta)/\sin\theta$, and recall from Chapter 4 that the $U_n(x/2)$ are orthonormal for the Sato Tate measure $\mu_{\mathrm{ST}}(dx) = \sqrt{4-x^2}\,dx/(2\pi)$ on $[-2,2]$.

THEOREM 9.1 (Tree theorem, this book). *Let $q \geq 2$ be an integer, let $\mathcal{T}_q$ be the $(q+1)$ regular tree, and put $B_n = q^{-n/2}A_n$.*

1. *$B_1 = U_1(X/2)$ and $B_n = U_n(X/2) - q^{-1}U_{n-2}(X/2)$ for $n \geq 2$.*
2. *$c_n = \int U_n(x/2)\, d\mu_q(x)$ equals $q^{-n/2}$ for n even and 0 for n odd.*
3. *μ_q is supported on $[-2,2]$ with the Kesten McKay density*

$$\mu_q(dx) = \frac{q+1}{(q^{1/2}+q^{-1/2})^2 - x^2} \cdot \frac{\sqrt{4-x^2}}{2\pi}\, dx.$$

4. *If S is simple random walk on $\mathcal{T}_q$ started at o, then for $m \geq 0$*

$$\int x^{2m}\, d\mu_q = (q+1)^{2m} q^{-m} \mathbb{P}(S_{2m} = o).$$

5. *The walk returns to o with probability $1/q$, and the Green function $G(1) = \sum_{n\geq 0} \mathbb{P}(S_n = o)$, the expected number of visits to o, equals $q/(q-1)$.*
6. *$\mu_q \Rightarrow \mu_{\mathrm{ST}}$ weakly as $q \to \infty$.*

PROOF. (1) A vertex at distance $n \geq 1$ from w has one neighbour at distance $n-1$ and q neighbours at distance $n+1$, while w itself has $q+1$ neighbours at distance 1. For $n \geq 1$ the entry $(AA_n)(v,w)$ counts the neighbours of v at distance n from w, so it is 1 if $d(v,w) = n+1$, q if $d(v,w) = n-1 \geq 1$, $q+1$ if $n = 1$ and $v = w$, and 0 otherwise. Hence

$$AA_1 = A_2 + (q+1)I, \qquad AA_n = A_{n+1} + qA_{n-1} \quad (n \geq 2).$$

Dividing by $q^{(n+1)/2}$, $XB_1 = B_2 + (1+q^{-1})I$ and $XB_n = B_{n+1} + B_{n-1}$ for $n \geq 2$. The Chebyshev recursion $xU_n(x/2) = U_{n+1}(x/2) + U_{n-1}(x/2)$, with $U_0 = 1$ and $U_1(x/2) = x$, gives $B_1 = X = U_1(X/2)$ and

$$B_2 = XU_1(X/2) - (1+q^{-1})I = U_2(X/2) - q^{-1}U_0(X/2).$$

For $n \geq 2$ the recursion $B_{n+1} = XB_n - B_{n-1}$ is linear and is satisfied separately by the sequences $U_n(X/2)$ and $q^{-1}U_{n-2}(X/2)$; since the formula holds for $n = 2$ and, with the convention $U_{-1} = 0$, also for $n = 1$, induction proves (1).

(2) The tree has no cycles, so a nonbacktracking walk of positive length never returns: $\langle \delta_o, B_n \delta_o \rangle = 0$ for $n \geq 1$. Taking the (o,o) entry of (1) gives $c_1 = 0$ and $c_n = q^{-1}c_{n-2}$ for $n \geq 2$, and $c_0 = 1$. Hence $c_{2m} = q^{-m}$ and $c_{2m+1} = 0$.

(3) Put $R = (q+1)/\sqrt{q}$, so that μ_q lives on $[-R, R]$ because $\|X\| \leq R$. Since $\sum_n c_n^2 < \infty$, the series $f = \sum_n c_n U_n(x/2)$ converges in $L^2(\mu_{\mathrm{ST}})$. With $x = 2\cos\theta$ and $U_{2m}(\cos\theta) = \sum_{|j| \leq m} e^{2ij\theta}$, summing over $m \geq |j|$ first (the double series converges absolutely),

$$\begin{aligned} f &= \sum_{m \geq 0} q^{-m} \sum_{|j| \leq m} e^{2ij\theta} = \frac{q}{q-1} \sum_{j \in \mathbb{Z}} q^{-|j|} e^{2ij\theta} \\ &= \frac{q+1}{q} \cdot \frac{1}{1 - 2q^{-1}\cos 2\theta + q^{-2}}, \end{aligned}$$

a Poisson kernel. Since $2\cos 2\theta = x^2 - 2$, the denominator equals $q^{-1}\big((q^{1/2} + q^{-1/2})^2 - x^2\big)$, which gives the stated density, positive and bounded on $[-2,2]$. The measure $f\,d\mu_{\mathrm{ST}}$ has Chebyshev moments c_n by orthonormality, hence the same moments $\int x^k$ as μ_q, because the $U_n(x/2)$ span the polynomials. Both laws live on $[-R, R]$, so they coincide by Theorem A.1. In particular μ_q is supported on $[-2,2]$.

(4) By definition $\int x^{2m}\,d\mu_q = q^{-m}A^{2m}(o,o)$, and $A^{2m}(o,o)$ is the number of closed walks of length $2m$ at o, each of which simple random walk follows with probability $(q+1)^{-2m}$.

(5) Let $D_n = d(o, S_n)$. By the neighbour count in (1), D is a Markov chain on $\{0, 1, 2, \dots\}$: from 0 it moves to 1, and from $k \geq 1$ it moves up with probability $q/(q+1)$ and down with probability $1/(q+1)$. By the gambler's ruin formula for a walk with upward drift, the probability of ever reaching 0 from 1 is $(1/(q+1))/(q/(q+1)) = 1/q$. Since the first step from o goes to distance one, the return probability is $1/q$, the number of visits to o (counting time 0) is geometric with mean $1/(1-1/q)$, and $G(1) = q/(q-1)$.

(6) For each $n \geq 1$, $c_n \to 0 = \int U_n(x/2)\,d\mu_{\mathrm{ST}}$ as $q \to \infty$. All measures live on $[-2,2]$, where polynomials are dense in the continuous functions, so $\int \phi\,d\mu_q \to \int \phi\,d\mu_{\mathrm{ST}}$ for every continuous ϕ. □

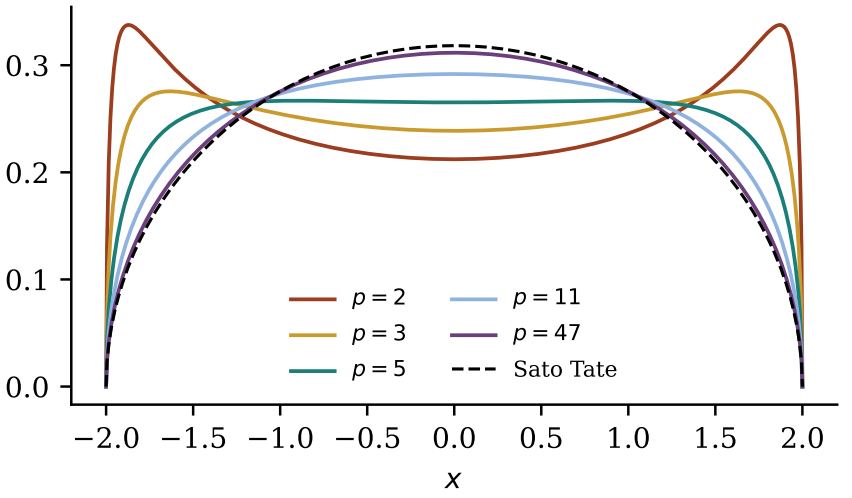


Figure 9.1: Densities of the Kesten McKay laws μ_q for the primes $q = 2, 3, 5, 11, 47$ (labelled p in the legend) against the Sato Tate semicircle (dashed). As q grows the mass moves from the edges towards the centre (Theorem 9.1(6)).

The density of μ_q relative to μ_{ST} is $q/(q-1)$ times the Poisson kernel with parameter q^{-1} at the angle 2θ. It increases with $|x|$, so μ_q puts more mass near ± 2 than the semicircle does, and it tends to 1 uniformly on $[-2, 2]$ as $q \to \infty$ (Figure 9.1). Since $\mathbb{P}(S_n = o) = \langle \delta_o, P^n \delta_o \rangle$ with $P = \sqrt{q}\, X/(q+1)$, and $\sqrt{q}\,|x|/(q+1) \le 2\sqrt{q}/(q+1) < 1$ on $[-2, 2]$, summing the geometric series gives a second, spectral, expression $G(1) = \int (1 - \sqrt{q}\, x/(q+1))^{-1}\, d\mu_q(x)$, which drives the Kemeny constant theorem of Chapter 10. We verified numerically that quadrature of the density reproduces $c_{2m} = q^{-m}$ for $q = 2, 3, 5, 11$ to 10^{-12}, that exact dynamic programming for the distance chain reproduces the moments in (4) through $2m = 12$ (for $q = 2$, $\int x^{12}\, d\mu_2 = 372.234375$), and that the spectral integral for $G(1)$ equals 2 and $3/2$ for $q = 2, 3$ to 10^{-13}.

9.3 Kesten and McKay

Statement (4) of Theorem 9.1 can be rewritten as

$$\mathbb{P}(S_{2m} = o) = \rho^{2m} \int \left(\frac{x}{2}\right)^{2m} d\mu_q(x), \qquad \rho = \frac{2\sqrt{q}}{q+1} < 1,$$

and since μ_q lives on $[-2, 2]$ with a density vanishing like a square root at ± 2, the integral decays like $m^{-3/2}$. So return probabilities decay like ρ^n up to a polynomial factor. This is Kesten's computation.

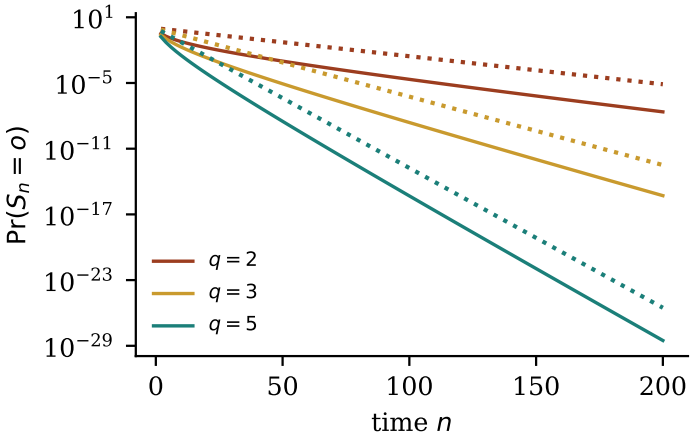


Figure 9.2: Return probabilities $\mathbb{P}(S_n = o)$ of simple random walk on $\mathcal{T}_q$, computed by dynamic programming on the distance chain at even times $n \le 200$, for $q = 2, 3, 5$ (solid) against the Kesten rate $(2\sqrt{q}/(q+1))^n$ (dotted), on a logarithmic scale. Each solid curve lies below its dotted curve by a factor of order $n^{-3/2}$.

THEOREM 9.2 (Kesten [309, 310]). *The spectral radius* $\limsup_n \mathbb{P}(S_n = o)^{1/n}$ *of simple random walk on the* $(q+1)$ *regular tree is* $2\sqrt{q}/(q+1)$. *For a symmetric random walk on a finitely generated group whose step law has finite support generating the group, the spectral radius equals* 1 *if and only if the group is amenable.*

The tree is a Cayley graph of the free product $(\mathbb{Z}/2)^{*(q+1)}$, which is not amenable. Figure 9.2 compares exact return probabilities with ρ^n.

THEOREM 9.3 (McKay [387]). *Let* G_n *be* $(q+1)$ *regular graphs on* n *vertices such that for every fixed* k *the number of cycles of length* k *is* $o(n)$. *Then the empirical distribution of the eigenvalues of* $q^{-1/2}A(G_n)$ *converges weakly to* μ_q. *The hypothesis holds with probability tending to one for uniformly random* $(q+1)$ *regular graphs.*

McKay's proof counts closed walks: when short cycles are rare, almost every vertex sees a tree out to any fixed radius, so the empirical moments converge to those of μ_q. In later language the graphs converge locally to the tree in the sense of Benjamini and Schramm [43], and local convergence implies weak convergence of the empirical spectral measures, which are the averages over vertices of the spectral measures at the δ_v. We use the same argument for the vertical theorem below.

9.4 THE p ADIC PLANCHEREL MEASURE

The spectral theory of A on $\ell^2(\mathcal{T}_q)$ is the spherical harmonic analysis of $\mathrm{PGL}_2(\mathbb{Q}_q)$. An irreducible unramified representation π of $\mathrm{PGL}_2(\mathbb{Q}_q)$ has a one dimensional space of $\mathrm{PGL}_2(\mathbb{Z}_q)$ fixed vectors on which T_q acts by a scalar, and by the Satake isomorphism (Theorem 3.7) this scalar is $q^{1/2}(\alpha+\alpha^{-1})$, where $\{\alpha,\alpha^{-1}\}$ is the Satake parameter, a semisimple conjugacy class in the dual group $\mathrm{SL}_2(\mathbb{C})$. The representation is tempered exactly when $|\alpha|=1$, that is when $x=\alpha+\alpha^{-1}=2\cos\theta$ lies in $[-2,2]$. Thus

$$\text{unramified tempered dual of } \mathrm{PGL}_2(\mathbb{Q}_q) \;\longleftrightarrow\; [0,\pi] \;\longleftrightarrow\; [-2,2],$$

and, with Haar measure normalised so that $\mathrm{PGL}_2(\mathbb{Z}_q)$ has volume one, the Plancherel measure restricted to this set is μ_q: for $\mathrm{PGL}_2(\mathbb{Q}_q)$, Macdonald's Plancherel formula for spherical functions [379] is the identity $\langle \delta_o, \phi(X)\delta_o\rangle = \int \phi\, d\mu_q$ with the density of Theorem 9.1(3). The nontempered parameters, including the trivial representation at $x=q^{1/2}+q^{-1/2}$ (the pole of the density), carry no Plancherel mass.

The Ramanujan conjecture for GL_2 says that the local components of cuspidal automorphic representations are tempered, so their Satake parameters live in the support of μ_q. For holomorphic forms of weight $k\geq 2$ it is Deligne's theorem [140], in weight one it is due to Deligne and Serre [142], and for Maass forms it is open. In the next section the same measure appears as the limiting distribution of these parameters.

9.5 THE VERTICAL SATO TATE THEOREM

Fix a prime q. For $N\geq 1$ with $q\nmid N$ and even $k\geq 2$, let $x_1,\dots,x_d$ be the eigenvalues, with multiplicity, of $T_q/q^{(k-1)/2}$ on $S_k(\Gamma_0(N))$, where $d=\dim S_k(\Gamma_0(N))$; they lie in $[-2,2]$ by Deligne.

THEOREM 9.4 (Serre [508]; Conrey, Duke and Farmer [95]). *As $k+N\to\infty$ with k even and $q\nmid N$, the empirical measure $d^{-1}\sum_j \delta_{x_j}$ converges weakly to μ_q.*

Conrey, Duke and Farmer treated level one; Serre treated the general case, including joint equidistribution at finitely many primes.

Proof idea. For $q\nmid N$ the Hecke relation $T_qT_{q^n}=T_{q^{n+1}}+q^{k-1}T_{q^{n-1}}$ gives $q^{-n(k-1)/2}T_{q^n}=U_n(x/2)$ on an eigenform whose normalised T_q eigenvalue is x, the analogue of Exercise 1. The Eichler Selberg trace formula expresses $\mathrm{tr}\,T_{q^n}$ on $S_k(\Gamma_0(N))$ as an identity term, present only for even n, plus elliptic terms given by class numbers of imaginary quadratic orders of discriminant t^2-4q^n, plus smaller hyperbolic and parabolic terms. After normalisation the identity term divided

by the dimension tends to $q^{-n/2}$ for even n, and for fixed n the other terms are of lower order than the dimension. Hence $d^{-1}\sum_j U_n(x_j/2) \to \int U_n(x/2)\,d\mu_q$ for every n, which gives the vertical law by the method of moments (Theorem A.1).

The probabilistic reading is cleanest in weight two and prime level $N \equiv 1 \bmod 12$, where the space of forms can be realised on a finite graph. The supersingular q isogeny graph $G_N(q)$ of Chapter 10 has $h = (N-1)/12$ vertices, is $(q+1)$ regular, and is a finite quotient $\Gamma\backslash\mathcal{T}_q$ of the tree by a discrete cocompact group Γ coming from a definite quaternion algebra. Its adjacency eigenvalues are $q+1$ together with those of T_q on $S_2(\Gamma_0(N))$ (Theorem 10.4). Identity (1) of Theorem 9.1 holds on any $(q+1)$ regular graph, with A_n counting nonbacktracking walks: a step appended to a nonbacktracking walk of length n either extends it to one of length $n+1$ or undoes its last step, and the walks of the second kind number q for each nonbacktracking walk of length $n-1$ ($q+1$ when $n=1$). Taking normalised traces,

$$\frac{1}{h}\sum_{\lambda}\Big(U_n\big(\tfrac{\lambda}{2\sqrt{q}}\big) - q^{-1}U_{n-2}\big(\tfrac{\lambda}{2\sqrt{q}}\big)\Big) = \frac{q^{-n/2}}{h}\ \operatorname{tr} A_n,$$

which is $q^{-n/2}$ times the expected number of closed nonbacktracking walks of length n from a uniformly random vertex. A closed nonbacktracking walk of length n at a curve E determines, up to sign, an endomorphism of E of degree q^n with cyclic kernel, and by Deuring and Eichler [148, 171, 227] the total number of these over all E is bounded by a sum of class numbers $H(4q^n - t^2)$ that does not depend on N. So the right side tends to zero as $N \to \infty$: with probability tending to one, a uniformly random vertex of $G_N(q)$ sees no cycle of bounded length, that is, $G_N(q)$ converges locally to $\mathcal{T}_q$ in the sense of Benjamini and Schramm. The left side is an empirical version of $c_n = q^{-1}c_{n-2}$, and its limit forces the moments of μ_q. In this setting the trace formula is an expectation over a uniformly random vertex of a finite quotient of the tree, its elliptic terms count short cycles, and the vertical theorem is local convergence.

REMARK 9.5 (Finite level correction). The same identity gives the exact finite level bias. The constant eigenvector, with eigenvalue $\lambda = q+1$, contributes $q^{-n/2}(q+1)q^{n-1}/h$ to the left side, where $(q+1)q^{n-1}$ is the size of a sphere of radius n in $\mathcal{T}_q$. Removing it, and writing $x_j = \lambda_j/\sqrt{q}$ for the $h-1$ normalised nontrivial eigenvalues, the average over the cusp forms is

$$\frac{1}{h-1}\sum_{j}\Big(U_n(x_j/2) - q^{-1}U_{n-2}(x_j/2)\Big) = \frac{q^{-n/2}}{h-1}\Big(\operatorname{tr} A_n - (q+1)q^{n-1}\Big).$$

For fixed n both terms on the right are $O(1/h)$: the negative one is the Eisenstein contribution, the positive one counts cycles and is class number data. Call the right side e_n. If $\hat{c}_n$ is the average of $U_n(x_j/2)$ over j, with $\hat{c}_{-1} = 0$, the left side is $\hat{c}_n - q^{-1}\hat{c}_{n-2}$; since $\hat{c}_0 = c_0 = 1$ and $c_n = q^{-1}c_{n-2}$, induction gives $\hat{c}_n - c_n =$

	$q=2$				$q=3$			
n	2	4	6	8	2	4	6	8
data	0.4960	0.2443	0.1194	0.0573	0.3299	0.1071	0.0360	0.0145
$q^{-n/2}$	0.5	0.25	0.125	0.0625	0.3333	0.1111	0.0370	0.0123

Table 9.1: Pooled averages of $U_n(x/2)$ over the nontrivial spectra of the supersingular q isogeny graphs at 207 prime levels, against the tree moments of Theorem 9.1.

$\sum_{0\le k<n/2} q^{-k} e_{n-2k}$, so the balance of the two terms in $e_n, e_{n-2}, \dots$ decides the sign of the correction.

9.5.1 Data

We use the data set of Chapter 10, the supersingular 2 and 3 isogeny graphs at the 207 primes $N \equiv 1 \bmod 12$ between 1000 and 8000. For $N = 1009$ and $q = 3$ the nontrivial spectrum agrees with that of T_3 on $S_2(\Gamma_0(1009))$ computed by PARI to 10^{-13}. Pooling the normalised nontrivial eigenvalues $x = \lambda/\sqrt{q}$ gives 75,497 values for each q. Table 9.1 compares their Chebyshev averages with $c_n = q^{-n/2}$; the odd averages ($n = 1, 3$) lie between -0.0041 and -0.0028 for both q. The small deviations are the finite level correction of Remark 9.5: for $n = 2$ and $q = 2$ the deviation is e_2, whose pooled Eisenstein term is $-3 \cdot 207/75497 = -0.0082$; the observed deviation is -0.0040, and the difference is the positive cycle term. At $n = 8$ and $q = 3$ the cycle terms win and the average exceeds its limit. Figure 9.3 shows the histograms.

9.6 Horizontal versus vertical

There are two limits and two laws. Horizontally, a single newform f without complex multiplication is fixed and p varies; the Satake parameters equidistribute for μ_{ST} (Theorem 4.9). Vertically, q is fixed and the form varies over a growing family; the parameters equidistribute for μ_q. Theorem 9.1(6) reconciles them. The horizontal statement over primes $p \le x$ samples the local laws μ_p at ever larger p, which by (6) look more and more like the semicircle: the semicircle is the large degree limit of the tree, the free semicircular law of Voiculescu's theory. This is a heuristic for why the two laws are compatible, not a proof of either: the horizontal theorem needs the analytic behaviour of all symmetric power L functions on $\Re s = 1$, which comes from their potential automorphy [67], and the vertical theorem needs the trace formula.

The vertical picture generalises to reductive groups.

Theorem 9.6 (Shin [515]). *Let G be a connected reductive group over a totally real field*

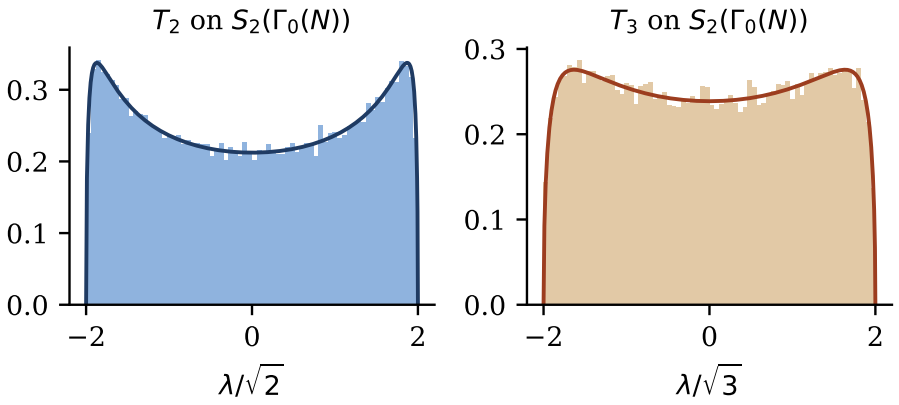


Figure 9.3: Pooled normalised eigenvalues $x = \lambda/\sqrt{q}$ of T_2 (left) and T_3 (right) on $S_2(\Gamma_0(N))$ for the 207 primes $N \equiv 1 \bmod 12$ between 1000 and 8000 (75,497 values for each q), computed as nontrivial spectra of supersingular isogeny graphs, against the Kesten McKay densities μ_2 and μ_3.

F such that $G(F \otimes_{\mathbb{Q}} \mathbb{R})$ has discrete series, under conditions on the centre. For families of cuspidal automorphic representations with level shrinking to the identity at fixed weight, or with weight tending to infinity at fixed level, the normalised counting measures of the local components at a fixed finite place v converge to the Plancherel measure of $G(F_v)$ (tested against the Hecke algebra).

For PGL_2 and spherical test functions the weight aspect recovers Theorem 9.4 at fixed level; the level aspect concerns levels such as those of $\Gamma(N)$, which shrink to the identity, rather than $\Gamma_0(N)$. In higher rank the tree becomes a building, simple random walk becomes several commuting Hecke walks, and μ_q becomes Macdonald's measure on a torus modulo the Weyl group.

9.7 Notes

The tree of $\mathrm{PGL}_2(\mathbb{Q}_q)$ is constructed in Serre's book *Trees* [505], which also explains Ihara's theorem that torsion free discrete subgroups of $\mathrm{SL}_2(\mathbb{Q}_q)$ are free [269]. Kesten's two papers of 1959 [309, 310] gave the return probabilities on free groups and the amenability criterion. McKay [387] found the same law for random regular graphs; Macdonald [379] is the reference for spherical functions. The vertical law is due to Serre [508] and, for level one, Conrey, Duke and Farmer [95]; Sarnak [476] had earlier proved the same law for the Hecke eigenvalues of Maass forms on $\mathrm{SL}_2(\mathbb{Z})$ ordered by Laplace eigenvalue, and Murty and Sinha [412] gave effective versions. Shin's theorem [515] and its quantitative refinement by Shin and Templier [516] put the vertical law in the general setting of the Langlands

programme. The individual statements of Theorem 9.1 are classical, and the recursion for the sphere operators A_n is standard in the theory of Ramanujan graphs [373]; what is ours is the formulation and the single proof organised around the Chebyshev identity (1).

9.8 Exercises

1. Let $T_{q^n} f([L]) = \sum f([L'])$, summed over the sublattices $L' \subseteq L$ of index q^n, with $T_1 = I$. Show that these sublattices are the $q^b L''$ with $0 \le b \le n/2$ and L/L'' cyclic of order q^{n-2b}, so that $T_{q^n} = \sum_{0 \le k \le n/2} A_{n-2k}$ on $\mathcal{T}_q$. Show that $T_q T_{q^n} = T_{q^{n+1}} + q T_{q^{n-1}}$ for $n \ge 1$, and deduce $q^{-n/2} T_{q^n} = U_n(X/2)$, either from this recursion or by telescoping part (1) of Theorem 9.1.

2. Use the density in Theorem 9.1(3) and vertex transitivity to show that $\|A\| = 2\sqrt{q}$ on $\ell^2(\mathcal{T}_q)$, and that $\mathbb{P}(S_{2m} = o) \sim C_q\, m^{-3/2} \rho^{2m}$; identify C_q.

3. Compute the generating function $G(u) = \sum_n \mathbb{P}(S_n = o) u^n$ by first passage decomposition of the distance chain and check $G(1) = q/(q-1)$. Show that the radius of convergence of G is $1/\rho$, with branch points at $u = \pm 1/\rho$, and relate this to Kesten's theorem.

4. Let $d = d(o, v)$, $U_{-1} = 0$ and $U_{-2} = -1$. Show that the function

$$\varphi_\theta(v) = \tfrac{q}{q+1} q^{-d/2} \big(U_d(\cos\theta) - q^{-1} U_{d-2}(\cos\theta) \big)$$

   satisfies $\varphi_\theta(o) = 1$ and $A\varphi_\theta = 2\sqrt{q}\cos\theta\, \varphi_\theta$. For $\theta \in \mathbb{C}$, show that it is bounded exactly when $|\Im\theta| \le \tfrac{1}{2}\log q$.

5. Let μ_q be as in Theorem 9.1. Show that $\int \log(4 - x^2)\, d\mu_q$ is finite and compute it, using $\log(4\sin^2\theta) = -2\sum_{k \ge 1} \cos(2k\theta)/k$ and the Poisson kernel in the proof of (3); compare with $\int \log(4 - x^2)\, d\mu_{\mathrm{ST}} = 1$.

9.9 Open computational problems

Computational Problem 9.1 (Rate of vertical convergence). Objective: measure the rate at which the eigenvalues of T_p on $S_2(\Gamma_0(N))$ approach μ_p, and separate the exact finite level correction from the rest. Data: all primes $N \equiv 1 \bmod 12$ with $37 \le N < 10^5$ and all primes $p \le 50$ with $p \ne N$, so that Remark 9.5 applies as stated, using PARI/GP `mfheckemat` for $N \le 10^4$ and, beyond, sparse Brandt matrices (Sage or Magma) or supersingular isogeny graphs built as in Chapter 10. Deliverable: the Chebyshev averages $\hat{c}_n(N, p)$ for $n \le 20$, split into the Eisenstein and class number terms of Remark 9.5, and the Kolmogorov distance to μ_p as a function of N, compared with the effective bounds of Murty and Sinha [412]. Why it matters: each Chebyshev average is given exactly by the trace formula, but the

Kolmogorov distance involves all n at once and has no closed form; its decay, and its fluctuation from level to level, can be compared with random matrix heuristics for families (Chapter 6).

COMPUTATIONAL PROBLEM 9.2 (Noncongruence quotients). Objective: compare quotients of the tree by congruence and by noncongruence subgroups of one arithmetic lattice. For a prime $q \equiv 1 \bmod 4$ the lattice $\Gamma \subset \mathrm{PGL}_2(\mathbb{Q}_q)$ of Lubotzky, Phillips and Sarnak [373], built from integral Hamilton quaternions of norm a power of q, is free on $(q+1)/2$ generators and acts simply transitively on the vertices of $\mathcal{T}_q$. A subgroup of index n is therefore given by a transitive action of these generators on n points, and its quotient of $\mathcal{T}_q$ is the $(q+1)$ regular Schreier graph of that action. Data: $q = 5$ and $q = 13$; congruence subgroups of index up to 10^3, such as the stabilisers of points of $\mathbb{P}^1(\mathbb{F}_M)$ under reduction of Γ modulo odd primes $M \neq q$; for each such index, 100 transitive actions by independent uniform random permutations, which define noncongruence subgroups with probability tending to one (GAP or Magma). Algorithm: sparse eigensolvers and 10^6 simulated walks per graph up to time 200. Deliverable: histograms against μ_q, the largest nontrivial eigenvalue against $2\sqrt{q}$, and return probability curves against Figure 9.2. Why it matters: congruence quotients are Ramanujan by the Ramanujan bound for weight two forms, while random quotients are only nearly Ramanujan (Friedman [203]); local convergence to the tree is a property of the graph, the Ramanujan property is arithmetic, and the experiment measures the gap between the two.

COMPUTATIONAL PROBLEM 9.3 (Plancherel statistics for GL_3). Objective: test Shin's theorem in rank two. Data: quotients of the building of $\mathrm{PGL}_3(\mathbb{Q}_2)$ by cocompact arithmetic lattices, up to 10^5 vertices, with the two commuting Hecke operators on vertices. Such lattices come from unitary groups in three variables over $\mathbb{Q}$ that are compact at infinity and attached to an imaginary quadratic field in which 2 splits, so that at 2 the projective unitary group becomes $\mathrm{PGL}_3(\mathbb{Q}_2)$. Algorithm: for the unitary group of a definite hermitian form, enumerate the genus of a hermitian lattice and compute the Hecke operators by neighbour methods (Magma), diagonalise both, and map joint eigenvalues to Satake parameters in the maximal torus of $\mathrm{SU}(3)$. Deliverable: histograms on the Weyl alcove against Macdonald's Plancherel density, moments of $\mathrm{SU}(3)$ characters, and the proportion of nontempered parameters. Why it matters: it is the first case in which the tree becomes a building, and the Ramanujan property of a quotient is no longer automatic: in Rogawski's classification of the automorphic representations of unitary groups in three variables, the endoscopic lifts of one dimensional representations have nontempered unramified local components and do occur, so the naive Ramanujan conjecture fails for general lattices of this kind, although the bound is known for all other infinite dimensional representations; the proportion

of nontempered parameters detects these lifts and measures how far the quotients are from Ramanujan complexes.

CHAPTER 10

IHARA ZETA FUNCTIONS, RAMANUJAN GRAPHS AND HITTING TIMES

In 1966 Ihara attached to a discrete, torsion free, cocompact subgroup Γ of PGL_2 over a p adic field with residue field of order q a zeta function, an Euler product over the primitive closed geodesics of the finite quotient graph $\Gamma\backslash\mathcal{T}_q$, and proved that it is a rational function whose poles are governed by the eigenvalues of the Hecke operator [269]. Sunada, Hashimoto and Bass later recognised that the construction makes sense for any finite graph [531, 246, 39]. For a regular graph the analogue of the Riemann hypothesis is a statement about eigenvalues that Lubotzky, Phillips and Sarnak, motivated by the Ramanujan conjecture, called the Ramanujan property [373]. We read the graph Riemann hypothesis three ways: as a statement about poles, about eigenvalues and about how fast a random walk forgets its start. We then turn to supersingular isogeny graphs, whose spectra are Hecke eigenvalues, and prove that their Kemeny constant, the expected time to hit a uniformly random target, is asymptotic to $\ell/(\ell-1)$ times the number of vertices.

10.1 THE IHARA ZETA FUNCTION

Let G be a finite connected graph, possibly with loops and multiple edges, with vertex set V, $n = |V|$, and edge set E. Orient each edge both ways, and let $A(x,y)$ count the oriented edges from x to y, so that a loop contributes 2 to the diagonal of the adjacency matrix A. A closed path $C = (e_1, \dots, e_m)$ of oriented edges is *reduced* if $e_{i+1} \neq \bar{e}_i$ for all i (indices mod m, so there is no backtracking and no tail), and *primitive* if it is not a power D^k with $k \geq 2$. A *prime* $\mathfrak{p}$ is an equivalence class of primitive reduced closed paths under cyclic rotation; its length is $\nu(\mathfrak{p})$. The Ihara zeta function is

$$Z_G(u) = \prod_{\mathfrak{p}} \left(1 - u^{\nu(\mathfrak{p})}\right)^{-1},$$

convergent for small $|u|$. Then $u\,Z_G'(u)/Z_G(u) = \sum_{m\geq 1} N_m u^m$, where N_m counts reduced closed paths of length m: Z_G counts closed nonbacktracking walks as ζ counts prime powers.

THEOREM 10.1 (Ihara [269]; Bass [39]; Hashimoto [246]). *If G is $(q+1)$ regular with*

adjacency matrix A, then

$$Z_G(u)^{-1} = (1-u^2)^{r-1} \det\bigl(I - Au + qu^2 I\bigr),$$

where $r = |E| - n + 1$ is the rank of the fundamental group of G, so that $r - 1 = (q-1)n/2$.

The proof writes $N_m = \operatorname{tr} H^m$, where H is the nonbacktracking (Hashimoto) matrix indexed by the $2|E|$ oriented edges, so that $Z_G(u)^{-1} = \det(I - uH)$, and evaluates this determinant by a block matrix identity equivalent to the recursions for A_m in Theorem 9.1. For each eigenvalue λ of A the factor $1 - \lambda u + qu^2$ contributes two poles $u_\pm$ with $u_+ u_- = 1/q$ and $u_+ + u_- = \lambda/q$. For $q \geq 2$ the factor $(1-u^2)^{r-1}$ contributes poles at $u = \pm 1$; the eigenvalue $\lambda = q+1$ gives poles at $u = 1$ and $u = 1/q$, and if G is bipartite the eigenvalue $-(q+1)$ gives poles at $u = -1$ and $u = -1/q$. The poles in $\{\pm 1, \pm 1/q\}$ are the *trivial* poles. In the variable $u = q^{-s}$ the pole at $u = 1/q$ is $s = 1$, and the circle $|u| = q^{-1/2}$ is the line $\Re s = \frac{1}{2}$.

DEFINITION 10.2. A $(q+1)$ regular graph satisfies the *graph Riemann hypothesis* if every nontrivial pole of Z_G lies on $|u| = q^{-1/2}$. It is *Ramanujan* if every eigenvalue λ of A with $|\lambda| \neq q+1$ satisfies $|\lambda| \leq 2\sqrt{q}$.

The name reflects the fact that for the graphs of Lubotzky, Phillips and Sarnak and for the isogeny graphs below the bound is the Ramanujan conjecture for weight two modular forms, which is a theorem: Eichler [171] deduced it for prime level from Weil's Riemann hypothesis for curves over finite fields through the Eichler Shimura relation, and Deligne [140] proved it in all weights. The bound $2\sqrt{q}$ is the edge of the support of the Kesten McKay law μ_q of Chapter 9 after rescaling, and by the Alon Boppana theorem it cannot be improved for infinite families [423] (Exercise 4 proves a weak form).

10.2 THREE FORMS OF THE RIEMANN HYPOTHESIS FOR GRAPHS

Let $P = A/(q+1)$ be the transition matrix of simple random walk on G; its stationary law is uniform. Order the eigenvalues of A as $q+1 = \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$, let $\mu_j = \lambda_j/(q+1)$, and define

$$\lambda_* = \max\{|\lambda_j| : |\lambda_j| \neq q+1\}, \qquad \delta_t = \max_{x \in V} \Bigl| P^t(x,x) - \frac{1 + \beta(-1)^t}{n} \Bigr|,$$

where $\beta = 1$ if G is bipartite and $\beta = 0$ otherwise, and $\lambda_* = 0$ if no eigenvalue has $|\lambda_j| \neq q+1$. Thus δ_t measures how far return probabilities are from their stationary (or, in the bipartite case, periodic stationary) values. Write $\rho_q = 2\sqrt{q}/(q+1)$ for the Kesten rate of the tree (Chapter 9).

PROPOSITION 10.3 (this book). *For a connected $(q+1)$ regular graph G the following are equivalent.*

1. *Z_G satisfies the graph Riemann hypothesis.*
2. *G is Ramanujan.*
3. *$\limsup_{t\to\infty} \delta_t^{1/t} \le \rho_q$.*

Moreover $\limsup_t \delta_t^{1/t} = \lambda_*/(q+1)$ exactly, so the return probabilities of a Ramanujan graph approach equilibrium at an exponential rate no slower than the rate ρ_q at which the return probabilities of the tree tend to zero.

PROOF. (1)⇔(2). By the Ihara Bass formula every pole of Z_G is ± 1 or a root of $1 - \lambda_j u + q u^2$ for some j, and every such root is a pole. If $|\lambda_j| \le 2\sqrt{q}$ the discriminant $\lambda_j^2 - 4q$ is nonpositive, the roots are complex conjugates (or a double real root) and $|u_\pm|^2 = u_+ u_- = 1/q$. If $2\sqrt{q} < |\lambda_j| < q+1$, the roots are real and distinct with product $1/q$, so one of them has modulus strictly less than $q^{-1/2}$; neither is trivial, since ± 1 or $\pm 1/q$ is a root only when $\lambda_j = \pm(q+1)$. The eigenvalues $\pm(q+1)$ give only trivial poles. Hence all nontrivial poles lie on the circle if and only if every λ_j with $|\lambda_j| \ne q+1$ satisfies $|\lambda_j| \le 2\sqrt{q}$.

(2)⇔(3). Let $\phi_1, \dots, \phi_n$ be an orthonormal eigenbasis of A in $\ell^2(V)$ with $\phi_1 \equiv n^{-1/2}$. Then $P^t(x,x) = \sum_j \mu_j^t \phi_j(x)^2$ and $\sum_j \phi_j(x)^2 = 1$. Since G is connected, $\lambda_1 = q+1$ is simple, and $-(q+1)$ is an eigenvalue (then simple, with eigenvector $n^{-1/2}(\pm 1)$ on the two sides) if and only if G is bipartite. Removing these terms,

$$P^t(x,x) - \frac{1+\beta(-1)^t}{n} = \sum_{j\in J} \mu_j^t \phi_j(x)^2,$$

where J indexes the eigenvalues with $|\lambda_j| \ne q+1$. The right side is bounded in absolute value by $(\lambda_*/(q+1))^t$, so $\limsup \delta_t^{1/t} \le \lambda_*/(q+1)$. Conversely, for even t all terms are nonnegative, and averaging over x,

$$\delta_t \ge \frac{1}{n} \sum_x \sum_{j\in J} \mu_j^t \phi_j(x)^2 = \frac{1}{n} \sum_{j\in J} \mu_j^t \ge \frac{1}{n} \Big(\frac{\lambda_*}{q+1} \Big)^t,$$

so $\limsup \delta_t^{1/t} \ge \lambda_*/(q+1)$. Hence the limsup equals $\lambda_*/(q+1)$, and it is at most $\rho_q = 2\sqrt{q}/(q+1)$ if and only if $\lambda_* \le 2\sqrt{q}$. □

The Riemann hypothesis for Z_G is therefore a statement about mixing. Applied to off diagonal entries, the same argument gives the uniform bound $|P^t(x,y) - 1/n| \le (\lambda_*/(q+1))^t$ in the nonbipartite case (Appendix A), so $\max_{x,y} |nP^t(x,y) - 1| \le \varepsilon$ once $t \ge \log(n/\varepsilon)/\log((q+1)/\lambda_*)$.

10.3 Supersingular isogeny graphs

Let N be a prime and $\ell \neq N$ another prime. Every supersingular elliptic curve over $\overline{\mathbb{F}}_N$ has a model over $\mathbb{F}_{N^2}$, and there are finitely many up to isomorphism; by Deuring [148] and Eichler's mass formula, when $N \equiv 1 \bmod 12$ there are exactly $h = (N-1)/12$ of them and none has automorphisms other than ± 1. The supersingular ℓ isogeny graph $G_N(\ell)$ has these h curves (or their j invariants) as vertices, with an edge for each of the $\ell+1$ subgroups of order ℓ of a curve, joining it to the quotient curve. Dual isogenies pair edges, so for $N \equiv 1 \bmod 12$ the graph is undirected and $(\ell+1)$ regular, possibly with loops and multiple edges. The neighbours of j are the roots in $\mathbb{F}_{N^2}$ of the modular polynomial $\Phi_\ell(j, Y)$, counted with multiplicity, as in Mestre's *méthode des graphes* [394].

Through Deuring's correspondence the vertices are also the left ideal classes of a maximal order $\mathcal{O}$ in the definite quaternion algebra B ramified at N and ∞, and the adjacency matrix of $G_N(\ell)$ is the Brandt matrix $B(\ell)$ [171, 227]. The graph is a quotient $\Gamma \backslash \mathcal{T}_\ell$ of the Bruhat Tits tree of Chapter 9, with Γ the image of $(\mathcal{O}[1/\ell])^\times$ in $\mathrm{PGL}_2(\mathbb{Q}_\ell)$.

A loop at a curve E is an endomorphism of degree ℓ, up to sign. When its dual has the same kernel, the loop is a *half loop*: one oriented edge equal to its own reverse, entered once on the diagonal of $B(\ell)$, and it comes from an element of Γ that inverts an edge of the tree. For $\ell = 2$ and $N \equiv 1 \bmod 12$ this happens only at the curve with $j = 8000$, which has complex multiplication by $\mathbb{Z}[\sqrt{-2}]$ and is supersingular exactly when $N \equiv 13 \bmod 24$; then h is odd, and a 3 regular graph on an odd number of vertices must have a half loop. (For $\ell = 3$ and $N \equiv 1 \bmod 12$ there are none.) For a $(q+1)$ regular graph with s half loops the Ihara Bass formula becomes

$$Z_G(u)^{-1} = (1-u)^{((q-1)n-s)/2}(1+u)^{((q-1)n+s)/2} \det\bigl(I - Au + qu^2 I\bigr).$$

To see this, let Σ be the matrix with rows indexed by the vertices and columns by the $(q+1)n$ oriented edges, with $\Sigma(v,e) = 1$ if e starts at v, and let R be the permutation matrix of the reversal $e \mapsto \bar{e}$, which has s fixed points. Then $\Sigma\Sigma^\top = (q+1)I$ and $\Sigma R \Sigma^\top = A$, and $H = R\Sigma^\top\Sigma - R$, because f follows e exactly when f starts at the origin of $\bar{e}$ and $f \neq \bar{e}$. Since $(I+uR)^{-1}R = (R - uI)/(1-u^2)$, the identity $\det(I - XY) = \det(I - YX)$ gives

$$\begin{aligned} \det(I - uH) &= \det(I + uR)\det\bigl(I - u\Sigma(I+uR)^{-1}R\Sigma^\top\bigr) \\ &= \det(I+uR)\,\frac{\det(I - Au + qu^2 I)}{(1-u^2)^n}, \end{aligned}$$

and $\det(I+uR) = (1-u^2)^{((q+1)n-s)/2}(1+u)^s$, because $I+uR$ is block diagonal with a block $\left(\begin{smallmatrix} 1 & u \\ u & 1 \end{smallmatrix}\right)$ for each pair $\{e, \bar{e}\}$ and the entry $1+u$ for each half loop. Hence only the trivial poles change, and Proposition 10.3 applies to every $G_N(\ell)$.

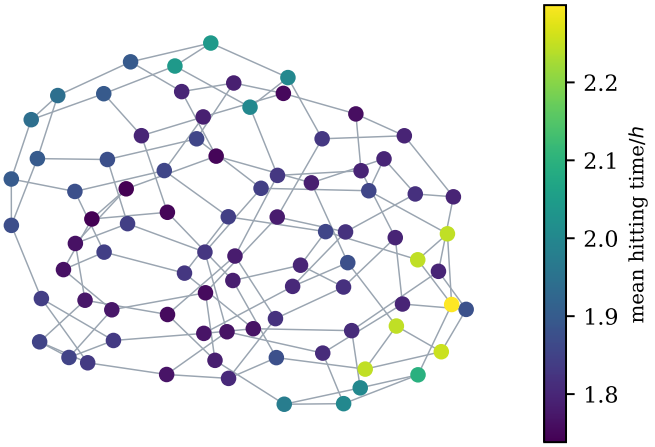


Figure 10.1: The supersingular 2 isogeny graph for $N = 1009$: 84 vertices, 3 regular, without loops or multiple edges. Each vertex y is coloured by $\mathbb{E}_\pi \tau_y / h$, its mean hitting time from a uniformly random start divided by h; the average over all y is the Kemeny constant divided by h, 1.866.

THEOREM 10.4 (Eichler; Jacquet Langlands; Pizer [171, 277, 438]). *For $N \equiv 1$ mod 12 the spectrum of the adjacency matrix of $G_N(\ell)$ is $\{\ell+1\}$ together with the eigenvalues, with multiplicity, of T_ℓ on $S_2(\Gamma_0(N))$. In particular $G_N(\ell)$ is Ramanujan.*

The eigenvalue $\ell + 1$ belongs to the Eisenstein series, and the dimension $h - 1$ of $S_2(\Gamma_0(N))$ is the genus of $X_0(N)$. The cuspidal part is Eichler's basis problem, in modern terms the Jacquet Langlands correspondence between $B^\times$ and GL_2 (Chapter 3). The Ramanujan property is the bound $|a_\ell(f)| \le 2\sqrt{\ell}$ for weight two newforms, and connectivity follows because $\ell + 1$ is then simple. Lubotzky, Phillips and Sarnak [373] and independently Margulis [381] built Ramanujan Cayley graphs of PGL_2 and PSL_2 over finite fields from Hamilton quaternions with the same input, and Morgenstern [404] extended the construction to every prime power q.

10.3.1 Data

For this book we built $G_N(2)$ and $G_N(3)$ explicitly with PARI/GP [432] for all 207 primes $N \equiv 1$ mod 12 between 1000 and 8000, with h from 84 to 666 (code in Appendix B). For every level and both ℓ, every nontrivial pole of Z_G satisfies $\left||u| - \ell^{-1/2}\right| < 5 \times 10^{-16}$, and $\lambda_* / (2\sqrt{\ell})$ is at most 0.99978 for $\ell = 2$ and 0.99983 for $\ell = 3$. Theorem 10.4 guarantees both statements, so these numbers check the computation rather than the theory. Figure 10.1 draws $G_{1009}(2)$.

10.4 Random regular graphs

Alon conjectured that random regular graphs are almost Ramanujan, and Friedman proved it.

Theorem 10.5 (Friedman [203]). *Fix $d \geq 3$ and $\varepsilon > 0$. With probability tending to one as $n \to \infty$ (n even if d is odd), a uniformly random d regular graph on n vertices has all eigenvalues other than d in $[-2\sqrt{d-1} - \varepsilon,\ 2\sqrt{d-1} + \varepsilon]$.*

Bordenave [77] gave a shorter proof through the nonbacktracking matrix H that defines the Ihara zeta function, and extended the theorem to random lifts of a fixed graph. Marcus, Spielman and Srivastava [406, 407] proved by interlacing polynomials that bipartite Ramanujan multigraphs exist for every degree and every even number of vertices. Huang, McKenzie and Yau [263] showed, in a 2024 preprint, that the extreme nontrivial eigenvalues of a random d regular graph fluctuate around $\pm 2\sqrt{d-1}$ on the scale $n^{-2/3}$ with the Tracy Widom law of the Gaussian orthogonal ensemble; a consequence is that the probability that a random d regular graph is Ramanujan tends to a constant strictly between 0 and 1 (their analysis puts it near 69 percent). In a computation for this book a random 4 regular graph on 3000 vertices had $\lambda_* = 3.46376$, just below $2\sqrt{3} = 3.46410$, so that sample is Ramanujan.

10.5 Hitting times and Kemeny's constant

For simple random walk on G let $h(x,y) = \mathbb{E}_x \tau_y$ with $\tau_y = \min\{t \geq 0 : X_t = y\}$, as in Appendix A; the letter h without arguments keeps its meaning as the number of vertices of an isogeny graph. Kemeny's constant $K = \sum_y \pi(y) h(x,y)$ does not depend on x, and for a connected $(q+1)$ regular graph (Theorem A.4)

$$K = \sum_{j \geq 2} \frac{1}{1 - \lambda_j/(q+1)} = \sum_{t \geq 0} \Big(\sum_x P^t(x,x) - 1 \Big),$$

where the second form, obtained by expanding each term as a geometric series, requires G not to be bipartite. Since π is uniform, K is also the average of $h(x,y)$ over all ordered pairs. The mean hitting time of a fixed target from a uniform start, $\mathbb{E}_\pi \tau_y = n^{-1} \sum_x h(x,y)$, does depend on y (Figure 10.1); its average over y is K.

By the second form, K/n is the expected number of visits to the starting point in excess of equilibrium, averaged over starting points. On the tree the expected number of visits to the start is the Green function $G(1) = q/(q-1)$ of Theorem 9.1(5). A graph that looks like a tree from almost every vertex should therefore have $K \approx G(1)\, n$. For isogeny graphs this heuristic is a theorem.

Theorem 10.6 (Kemeny constant of isogeny graphs, this book). *Let ℓ be prime and let K_N be the Kemeny constant of simple random walk on $G_N(\ell)$, with $h = (N-1)/12$*

vertices. As $N \to \infty$ through primes $N \equiv 1 \bmod 12$,

$$\frac{K_N}{h} \longrightarrow \frac{\ell}{\ell - 1}.$$

Equivalently, the mean hitting time of a uniformly random target, from any fixed start or from a uniformly random one, is asymptotic to $\ell h/(\ell-1)$; for $\ell = 2$ it is about $2h$.

PROOF. By Theorem 10.4 the eigenvalues of the adjacency matrix are $\ell + 1$ and $\lambda_j = \sqrt{\ell}\, x_j$ for $2 \le j \le h$, where $x_2, \dots, x_h$ are the normalised eigenvalues of T_ℓ on $S_2(\Gamma_0(N))$, counted with multiplicity; by the Ramanujan bound they lie in $[-2, 2]$. Since $|\lambda_j| \le 2\sqrt{\ell} < \ell + 1$ the graph is connected and not bipartite, and Theorem A.4 gives

$$\frac{K_N}{h} = \frac{h-1}{h} \cdot \frac{1}{h-1} \sum_{j=2}^{h} g(x_j), \qquad g(x) = \Big(1 - \frac{\sqrt{\ell}\, x}{\ell + 1}\Big)^{-1}.$$

On $[-2, 2]$ we have $\sqrt{\ell}|x|/(\ell+1) \le \rho_\ell < 1$, so g is continuous there, with values in $[1/(1+\rho_\ell), 1/(1-\rho_\ell)]$. By the vertical Sato Tate theorem (Theorem 9.4, weight 2, prime level $N \to \infty$, $\ell \nmid N$) the empirical measures $\nu_N = (h-1)^{-1} \sum_{j=2}^{h} \delta_{x_j}$ converge weakly to the Kesten McKay law μ_ℓ. All these measures live on the compact interval $[-2, 2]$, where g is continuous, so $\int g\, d\nu_N \to \int g\, d\mu_\ell$, that is,

$$\frac{1}{h-1} \sum_{j=2}^{h} g(x_j) \longrightarrow \int_{-2}^{2} g\, d\mu_\ell.$$

It remains to evaluate the integral. With $q = \ell$, g is the integrand of the spectral expression for the Green function given after the proof of Theorem 9.1, so $\int_{-2}^{2} g\, d\mu_\ell = G(1) = \ell/(\ell-1)$ by Theorem 9.1(5). Since $(h-1)/h \to 1$, the theorem follows. The statement about mean hitting times is the identity $K = n^{-2} \sum_{x,y} h(x,y)$ for uniform π. □

Without the Ramanujan bound the proof would fail: the trivial bound $|x_j| \le (\ell+1)/\sqrt{\ell}$ does not keep the x_j away from the pole of g at $(\ell+1)/\sqrt{\ell}$. Chapter 21 reads the terms arithmetically. By the Eichler Shimura congruence relation $\prod_{j \ge 2} (\ell + 1 - \lambda_j) = |J_0(N)(\mathbb{F}_\ell)|$, and $\ell + 1 - \lambda_j$ is the number of points modulo ℓ of an elliptic curve of conductor N when the eigenform has rational coefficients; so $K_N = (\ell+1) \sum_{j \ge 2} (\ell + 1 - \lambda_j)^{-1}$ is a harmonic sum of these factors of $|J_0(N)(\mathbb{F}_\ell)|$ (Theorem 21.1), and $G_N(\ell)$ has $|J_0(N)(\mathbb{F}_\ell)|/h$ spanning trees, a formula of Hashimoto (Theorem 21.2).

REMARK 10.7 (An exact series). The generating function $\sum_{m \ge 0} U_m(x/2) z^m = (1 - xz + z^2)^{-1}$ at $z = \ell^{-1/2}$ gives $g(x) = \frac{\ell+1}{\ell} \sum_{m \ge 0} \ell^{-m/2} U_m(x/2)$, uniformly

on $[-2,2]$ because $|U_m(x/2)| \le m+1$ there. Since $T_\ell T_{\ell^m} = T_{\ell^{m+1}} + \ell T_{\ell^{m-1}}$ on $S_2(\Gamma_0(N))$ for $\ell \nmid N$, as on the tree (Exercise 1 of Chapter 9), the eigenvalue of T_{ℓ^m} on the jth eigenform is $\ell^{m/2}U_m(x_j/2)$, and therefore

$$K_N = \frac{\ell+1}{\ell} \sum_{m\ge 0} \ell^{-m} \operatorname{tr}\big(T_{\ell^m} \mid S_2(\Gamma_0(N))\big).$$

Dividing by $h-1$ and letting $N \to \infty$ term by term (Theorems 9.4 and 9.1(2), with $|U_m(x/2)| \le m+1$) gives, by dominated convergence with the summable bound $(m+1)\ell^{-m/2}$, the limit $\frac{\ell+1}{\ell}\sum_{k\ge 0}\ell^{-2k} = \frac{\ell}{\ell-1}$ again. The Eichler Selberg trace formula writes each trace through class numbers, so this series is the natural route to the second order term (Computational Problem 10.1).

10.5.1 Spectral bounds

The Ramanujan property also gives uniform bounds valid at every level. Write $\mu_* = \lambda_*/(\ell+1) \le \rho_\ell$, as in Appendix A. Since $1/(1+\mu_*) \le 1/(1-\mu_j) \le 1/(1-\mu_*)$ for $j \ge 2$,

$$\frac{h-1}{1+\rho_\ell} \le K_N \le \frac{h-1}{1-\rho_\ell}.$$

For individual pairs, with $n = h$, the representation $h(x,y) = n\big(Z(y,y) - Z(x,y)\big)$ of Appendix A, where Π is the matrix whose rows all equal π and $Z = \sum_t (P^t - \Pi)$, and the bound $|P^t(x,y) - 1/n| \le \mu_*^t$ give

$$\max_{x,y} h(x,y) \le n \sum_{t\ge 0} \big(|P^t(y,y) - \tfrac{1}{n}| + |P^t(x,y) - \tfrac{1}{n}|\big) \le \frac{2n}{1-\rho_\ell}.$$

For $\ell = 2$ these read $0.51(h-1) \le K_N \le 17.5(h-1)$ and $\max h(x,y) \le 35h$, while Theorem 10.6 gives $K_N \sim 2h$.

10.5.2 Data

For each of the 207 levels we computed K_N from the spectrum. For $\ell = 2$, K_N/h rises from 1.866 at $h = 84$ and lies between 1.971 and 2.012 for all 59 levels with $h \ge 500$ (mean 1.988). For $\ell = 3$, K_N/h lies between 1.435 and 1.515 throughout, approaching $3/2$. A least squares fit of $a + c/h$ over all levels gives $a = 2.004$, $c = -8.2$ for $\ell = 2$ and $a = 1.498$, $c = -3.0$ for $\ell = 3$, consistent with the limits. The scatter about the fit is of the same order as the correction, and it is arithmetic: for $\ell = 2$ the mean of $K_N - 2h$ is -3.3 over the 108 levels inert in $\mathbb{Q}(\sqrt{-7})$ and -11.0 over the other 99. At the inert levels the curves with complex multiplication by orders of $\mathbb{Q}(\sqrt{-7})$ are supersingular, and the largest single contribution to the difference comes from the one with $j = -3375$ described below. Whether

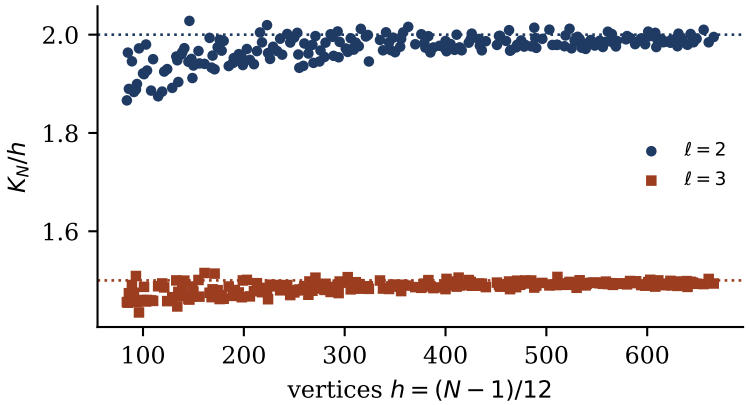


Figure 10.2: Kemeny constants K_N of the supersingular ℓ isogeny graphs for $\ell = 2, 3$ at the 207 primes $N \equiv 1 \bmod 12$ between 1000 and 8000, divided by the number of vertices $h = (N-1)/12$, against the limits 2 ($\ell = 2$) and $3/2$ ($\ell = 3$) of Theorem 10.6 (dotted lines).

$K_N - 2h$ stays bounded is Open Problem 30.10 of Chapter 30. Figure 10.2 shows the convergence.

As an independent check we solved the hitting time equations directly for every target on $G_N(2)$ for $N = 1009, 2017, 4129$ ($h = 84, 168, 344$). The Kemeny constants are 156.766, 324.736 and 682.586, so that $K_N/h = 1.866, 1.933, 1.984$, and they agree with the spectral formula to a relative error below 10^{-13}. The maximal hitting times divided by h are 2.58, 3.35 and 5.98, far below the bound 35; they reflect the neighbourhood of the hardest target, not growth in h. At $N = 4129$ that target is the curve with $j = -3375$, which has complex multiplication by the maximal order of $\mathbb{Q}(\sqrt{-7})$ and is supersingular because 4129 is inert in that field. Its endomorphisms $(1 \pm \sqrt{-7})/2$ of degree 2 form a loop counted twice, so the walk stays put with probability $2/3$. If the rest of its neighbourhood is a tree, a walk started there returns with probability $\frac{2}{3} + \frac{1}{3} \cdot \frac{1}{2} = \frac{5}{6}$ and makes 6 visits on average, against $G(1) = 2$ at a vertex whose neighbourhood is a tree, so its mean hitting time is about $6h$ (Exercise 5). At $N = 2017$ the hardest targets are the two curves with complex multiplication by the maximal order of $\mathbb{Q}(\sqrt{-15})$, which are joined by a double edge.

10.6 Isogeny graphs in cryptography

Charles, Goren and Lauter [98] proposed a hash function built from walks on $G_N(2)$: the bits of the message choose, at each step, one of the two edges that do not backtrack, starting from a fixed curve, and the hash value is the endpoint.

Because the graph is Ramanujan, a walk of $O(\log N)$ steps ends at a nearly uniform vertex. A collision gives two distinct paths between the same curves, hence a cycle and an endomorphism of the starting curve that is not multiplication by an integer, so security rests on the presumed difficulty of finding paths in the graph. Under the generalized Riemann hypothesis this problem is equivalent to computing endomorphism rings of supersingular curves [574]. All known classical algorithms for it take time exponential in $\log N$; the method of Delfs and Galbraith [138] takes about $\sqrt{N}$ steps up to logarithmic factors. We make no claim about particular schemes and note only that the key exchange SIDH of Jao and De Feo [281], which also publishes the images of torsion points under the secret isogeny, was broken in 2022 by an attack that exploits this extra information and does not solve the general path problem [94]. Theorem 10.6 quantifies the cost of naive search: from any fixed start, a random walk looking for a target curve chosen uniformly at random needs $K_N \approx 2h \approx N/6$ steps on average, far more than $\sqrt{N}$.

10.7 Notes

Ihara’s zeta function appeared in his work on discrete subgroups of p adic groups [269]; Serre’s book [505] made the link with graphs explicit, Sunada [531] suggested general graphs, and Hashimoto [246] and Bass [39] proved the determinant formula in general. Terras’s book [545] is an introduction. The equivalence of the graph Riemann hypothesis and the Ramanujan property is standard; the third form in Proposition 10.3 is a reformulation we find useful because it ties the Riemann hypothesis to the Kesten rate. Supersingular graphs go back to Deuring, Eichler and Mestre [394]; Pizer [438] identified them as Ramanujan graphs, and Gross [227] used Brandt matrices for special values of L functions. Kemeny and Snell [304] introduced the constant that bears Kemeny’s name. Theorem 10.6 is new as far as we know; it is a direct consequence of Serre’s theorem once the Kemeny constant is recognised as a spectral statistic whose tree value is the gambler’s ruin constant. The same constant appears for random regular graphs: Cooper and Frieze [119] proved that the cover time of a random r regular graph is asymptotic to $\frac{r-1}{r-2}\, n \log n$, through hitting times of order $\frac{r-1}{r-2}\, n$ at vertices whose neighbourhoods are trees, and $\frac{r-1}{r-2} = \frac{\ell}{\ell-1}$ when $r = \ell + 1$.

10.8 Exercises

1. Prove that $u\, Z_G'(u)/Z_G(u) = \sum_m N_m u^m$. For a $(q+1)$ regular graph deduce from the Ihara Bass formula that $N_m = \sum_j (\alpha_j^m + \beta_j^m) + (|E| - n)(1 + (-1)^m)$, where α_j, β_j are the roots of $t^2 - \lambda_j t + q$.

2. For the complete graph on $q+2$ vertices compute Z_G, verify the graph Riemann hypothesis, and check $K=(n-1)^2/n$.
3. Show that a connected $(q+1)$ regular graph with $q \geq 2$ is bipartite if and only if $-(q+1)$ is an eigenvalue, and that $u=-1/q$ is a pole of Z_G exactly in that case. Find the order of the pole of Z_G at $u=-1$ in both cases.
4. Prove the Alon Boppana bound in the weak form $\liminf \lambda_* \geq 2\sqrt{q}$ for a sequence of $(q+1)$ regular graphs with $n \to \infty$. (Hint: by the recursions of Theorem 9.1(1), which hold on any $(q+1)$ regular graph, A^{2k} is a combination with nonnegative coefficients of the nonbacktracking matrices A_m, whose traces are nonnegative, and the coefficient of $A_0=I$ is the number $q^k \int x^{2k}\, d\mu_q$ of closed walks of length $2k$ at a vertex of the tree. Compare with $\operatorname{tr} A^{2k} \leq 2(q+1)^{2k} + n\lambda_*^{2k}$, which allows for a bipartite graph, and let $k \to \infty$ slowly with n.)
5. Show that for every target y, $\mathbb{E}_\pi \tau_y = \frac{1}{n}\sum_x h(x,y) = n\, Z(y,y)$, and interpret $Z(y,y)$ as an excess number of visits. For $q=2$, compute the expected number of visits to y of simple random walk started at y on an infinite 3 regular graph that is a tree except that y carries a loop counted twice, and compare with $G(1)=2$ and with the hardest target at $N=4129$. Do the same when y carries a half loop instead, and when y is joined to a neighbour by a double edge (answers 6, 3 and $10/3$).

10.9 Open computational problems

Computational Problem 10.1 (Second order term of the Kemeny constant). Objective: determine the correction $K_N - \ell h/(\ell-1)$ as a function of N. Data: all primes $N \equiv 1 \bmod 12$ up to 10^6, $\ell = 2,3$. Algorithm: the series of Remark 10.7, with $\operatorname{tr} T_{\ell^m}$ from the Eichler Selberg trace formula, whose elliptic term is a sum of Hurwitz class numbers $H(4\ell^m - s^2)$ weighted by Legendre symbols at N and whose identity, hyperbolic and weight two divisor terms are explicit (the identity term, $(N+1)/12$ for even m, contributes $\ell/(6(\ell-1))$ to the correction); the class numbers do not depend on N and can be tabulated once, and the series is truncated at m of order $\log N$. Check against PARI/GP spectra for $N \leq 10^5$. Deliverable: the empirical law of the correction, its regression on the symbols (D/N) for small discriminants D, and a conjectural formula in terms of class numbers. Why it matters: the correction is where the elliptic terms, invisible in the limit, become measurable; it explains the scatter in Figure 10.2, which for $\ell=2$ is led by $D=-7$, and it is the computational side of Open Problem 30.10 of Chapter 30.

Computational Problem 10.2 (Hitting times on large supersingular graphs). Objective: the distribution over targets y of the mean hitting time $\mathbb{E}_\pi \tau_y / h =$

$Z(y, y)$ on $G_N(2)$, and of $\max_{x,y} h(x, y)/h$, which the spectral bound keeps below 35. Data: $G_N(2)$ for N up to 10^7 (h near 8×10^5), walks generated on the fly from Φ_2 over $\mathbb{F}_{N^2}$. Algorithm: 10^5 Monte Carlo walks per level on GPU, run in continuous time with exponential holding times of mean one, which leaves the means unchanged, with exact sparse solves for $N \leq 10^5$ as ground truth. Deliverable: histograms of $\mathbb{E}_\pi \tau_y / h$ over targets, with the atoms predicted by Exercise 5, and the law of $\tau_y / \mathbb{E}_\pi \tau_y$ against the exponential law that Corollary 15.19 of Chapter 15 guarantees up to $O(1/h)$. Why it matters: the exponential law predicts the cost of naive search once the mean is known, and the means carry the arithmetic, through curves with complex multiplication by small discriminants.

COMPUTATIONAL PROBLEM 10.3 (Other degrees and ordinary curves). Objective: repeat the analysis for $\ell = 5, 7$ and for ℓ isogeny graphs of ordinary curves. Data: supersingular $G_N(\ell)$ for $N \leq 10^5$; ordinary isogeny volcanoes over $\mathbb{F}_N$ for $N \leq 10^6$ (Sutherland's algorithms or Sage). Deliverable: K_N/h against $\ell/(\ell - 1)$, and for volcanoes the Kemeny constant as a function of the class group and the depth. Why it matters: volcanoes are governed by class groups of imaginary quadratic orders rather than by the Ramanujan bound, so the comparison isolates what Theorem 10.6 owes to the Ramanujan property.

COMPUTATIONAL PROBLEM 10.4 (Ihara zeta as a test for random lifts). Objective: the probability that a random k fold lift of $G_N(2)$ satisfies the graph Riemann hypothesis for its new poles. Data: lifts of index $k \leq 100$ of graphs with $h \leq 500$, 10^4 lifts each. Algorithm: largest new eigenvalue of the nonbacktracking matrix H by sparse Arnoldi iteration. Deliverable: the empirical probability as a function of k and h, with Tracy Widom fits at the edge. Why it matters: Bordenave proved that random lifts are almost Ramanujan; whether lifts of arithmetic graphs behave like random graphs probes how much arithmetic survives a random perturbation.

Part IV

Lévy and Thorin Measures

CHAPTER 11

INFINITE DIVISIBILITY OF EULER PRODUCTS

In his 1938 monograph on limit laws for sums of independent random variables, Khinchine observed that the Riemann zeta function, read along a vertical line to the right of 1 and divided by its value on the real axis, is the characteristic function of an infinitely divisible law. The proof is one line long: the logarithm of an Euler product is a sum over prime powers, and a sum of exponentials with positive weights is the exponent of a compound Poisson law. This chapter develops two consequences. First, positivity of the weights, which for automorphic L functions is the positivity of Rankin Selberg convolutions, underlies the usual proofs of nonvanishing on the line $\operatorname{Re} s = 1$ and of the classical zero free regions; for the standard L function of a cusp form the weights change sign and the characteristic function property fails. Second, once p^{-it} is replaced by an independent uniform phase, the logarithm of an Euler product becomes a sum of independent random variables, and the theory of Bohr and Jessen describes the values of ζ to the right of the critical line by the law of that sum. We carry the first picture to GL_n and the second to GL_2, where the natural random model draws Satake parameters from the Sato Tate measure, and we compute what Sato Tate implies for the law of that model.

11.1 INFINITELY DIVISIBLE LAWS

A probability law μ on $\mathbb{R}$ is *infinitely divisible* if for every n it is the n fold convolution power of some probability law μ_n. The class is closed under convolution and weak limits, and it is described completely by the following theorem.

THEOREM 11.1 (Lévy Khintchine). *A function $\varphi : \mathbb{R} \to \mathbb{C}$ is the characteristic function of an infinitely divisible law if and only if $\varphi = e^{\eta}$ with*

$$\eta(t) = iat - \tfrac{1}{2}\beta^2 t^2 + \int \left(e^{itx} - 1 - itx\, 1_{\{|x|<1\}}\right) \nu(dx)$$

for some $a \in \mathbb{R}$, some $\beta \geq 0$ and some positive measure ν on $\mathbb{R} \setminus \{0\}$ satisfying $\int \min(1, x^2)\, \nu(dx) < \infty$. The triple (a, β^2, ν) is unique.

The measure ν is the *Lévy measure* and η the Lévy Khintchine exponent. A finite measure ν of total mass $M > 0$ defines the *compound Poisson law*, the law of $\sum_{j=1}^{\mathcal{N}} X_j$ with $\mathcal{N}$ Poisson of mean M and $X_1, X_2, \ldots$ independent of $\mathcal{N}$ and of each

other, each with law ν/M; its characteristic function is

$$\varphi(t) = \exp\Big(\int \big(e^{itx} - 1\big)\, \nu(dx) \Big),$$

the case $\beta = 0$, $a = \int_{|x|<1} x\, \nu(dx)$ of the theorem. The uniqueness clause is what makes positivity a usable test. Its proof recovers ν from η by Fourier inversion and applies verbatim to finite signed measures, so if ν is a finite signed measure with a nonzero negative part, then $\exp \int (e^{itx} - 1)\, \nu(dx)$ is not the characteristic function of an infinitely divisible law. It may still be the characteristic function of some other law: for $0 < r < 1$ the two point law with $\mathbb{P}(X = 1) = r/(1+r) = 1 - \mathbb{P}(X = 0)$ has

$$\mathbb{E}e^{itX} = \exp\Big(\sum_{k \geq 1} \frac{(-1)^{k+1} r^k}{k} \big(e^{ikt} - 1\big) \Big),$$

a signed exponent with alternating weights. Or it may be no characteristic function at all, as for $L(s, \Delta)$ in Proposition 11.3 below. We refer to Sato's monograph [485] for proofs.

11.2 Khinchine's observation and the zeta distribution

THEOREM 11.2 (Khinchine [313]). *For every $\sigma > 1$ the function $t \mapsto \zeta(\sigma + it)/\zeta(\sigma)$ is the characteristic function of an infinitely divisible law. It is compound Poisson with Lévy measure*

$$\nu_\sigma = \sum_p \sum_{k \geq 1} \frac{1}{k\, p^{k\sigma}}\, \delta_{-k \log p},$$

whose total mass is $\log \zeta(\sigma)$.

PROOF. For $\operatorname{Re} s > 1$ the Euler product $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$ converges absolutely, and expanding $-\log(1 - p^{-s})$ gives

$$\log \zeta(\sigma + it) = \sum_p \sum_{k \geq 1} \frac{p^{-k\sigma}}{k}\, e^{-itk \log p},$$

with $\sum_{p,k} p^{-k\sigma}/k = \log \zeta(\sigma) < \infty$. Subtracting the value at $t = 0$,

$$\frac{\zeta(\sigma + it)}{\zeta(\sigma)} = \exp\Big(\sum_{p,k} \frac{1}{k p^{k\sigma}} \big(e^{-itk \log p} - 1\big) \Big) = \exp\Big(\int \big(e^{itx} - 1\big)\, \nu_\sigma(dx) \Big),$$

which is the compound Poisson exponent of the finite positive measure ν_σ. □

The range $\sigma > 1$ is sharp. As $\sigma \to 1^+$ the mass $\log \zeta(\sigma)$ of ν_σ tends to infinity, and for $\frac{1}{2} < \sigma \leq 1$ the function $t \mapsto \zeta(\sigma + it)$ is unbounded: since $\sum_p p^{-\sigma}$ diverges, the limit law of Theorem 11.9 below gives positive mass to every half

plane $\{\operatorname{Re} z > R\}$, so $\operatorname{Re}\log\zeta(\sigma+it)$ exceeds R for a positive proportion of t. A characteristic function is bounded by 1.

There is a second, more concrete description. The *zeta distribution* with parameter $\sigma > 1$ is the law of a positive integer N with $\mathbb{P}(N=n) = n^{-\sigma}/\zeta(\sigma)$. Then $\mathbb{E}[N^{-it}] = \zeta(\sigma+it)/\zeta(\sigma)$, so the law of Theorem 11.2 is the law of $-\log N$. The law of $\log N$ itself has characteristic function $\zeta(\sigma-it)/\zeta(\sigma)$ and Laplace transform $\mathbb{E}N^{-\lambda} = \zeta(\sigma+\lambda)/\zeta(\sigma)$, and its Lévy measure, the reflection of ν_σ, has its atoms at the points $k\log p$; it is the law at time 1 of the zeta subordinator of Proposition 16.10. Unique factorisation turns the Euler product into independence: $N = \prod_p p^{G_p}$ where the exponents G_p are independent and geometric, $\mathbb{P}(G_p = k) = (1-p^{-\sigma})p^{-k\sigma}$. A geometric law is itself compound Poisson (Exercise 1), so $-\log N = -\sum_p G_p \log p$, a sum with almost surely finitely many nonzero terms, is a sum of independent compound Poisson variables. This is the probabilistic content of the theorem.

11.3 Euler products as signed Lévy Khintchine exponents

Let π be a cuspidal automorphic representation of $\mathrm{GL}_n(\mathbb{A}_{\mathbb{Q}})$ with unitary central character and let S be a finite set of primes containing the ramified ones. For $p \notin S$ the Satake parameter is a semisimple class $A_p \in \mathrm{GL}_n(\mathbb{C})$ and the partial L function is $L^S(s,\pi) = \prod_{p\notin S} \det(I - A_p p^{-s})^{-1}$, absolutely convergent for $\operatorname{Re} s > 1$ by Jacquet and Shalika [279]. Exactly as for ζ,

$$\frac{L^S(\sigma+it,\pi)}{L^S(\sigma,\pi)} = \exp\Big(\sum_{p\notin S}\sum_{k\geq 1} \frac{\operatorname{tr}(A_p^k)}{k\,p^{k\sigma}}\big(e^{-itk\log p} - 1\big)\Big), \qquad \sigma > 1. \tag{11.1}$$

Formally this is a compound Poisson exponent with atoms at $-k\log p$ and weights $\operatorname{tr}(A_p^k)/(kp^{k\sigma})$. The weights are complex in general and real when π is self dual, but even then they need not be positive. We call (11.1) a *signed Lévy Khintchine exponent*. A simpler example, the law of a random squarefree integer, whose Dirichlet series is $\zeta(s)/\zeta(2s)$, is in Section 17.9.

For the discriminant form Δ we use in this chapter the unitary normalisation $L(s,\Delta) = \sum_n \tau(n) n^{-11/2-s}$, with centre $s = \frac12$; Chapter 13 uses the classical normalisation with centre 6. Since Δ has level 1, every prime is unramified and we may take $S = \emptyset$. Write $\tau(p)/p^{11/2} = 2\cos\theta_p$ with $\theta_p \in [0,\pi]$, by Deligne's theorem [140], so that A_p has eigenvalues $e^{\pm i\theta_p}$ and the weights are $2\cos(k\theta_p)/(kp^{k\sigma})$. Already at $p=2$ we have $\tau(2) = -24$ and $2\cos\theta_2 = -0.53033$, a negative first weight.

Proposition 11.3. *For every $\sigma > 1$ we have $\sup_{t\in\mathbb{R}} |L(\sigma+it,\Delta)| > L(\sigma,\Delta) > 0$. In particular $t \mapsto L(\sigma+it,\Delta)/L(\sigma,\Delta)$ is not a characteristic function.*

PROOF. Put $q = p^{-\sigma}$ and $x_p = 2\cos\theta_p$. Writing $p^{-it} = e^{i\phi}$, the modulus of the local factor of $L(\sigma+it,\Delta)$ at p is $f_p(\phi) = |1 - x_p q e^{i\phi} + q^2 e^{2i\phi}|^{-1}$, and $f_p(0) = (1 - x_p q + q^2)^{-1} = |1 - e^{i\theta_p} q|^{-2} > 0$, so $L(\sigma,\Delta) = \prod_p f_p(0) > 0$. Since $1 + x_2 q + q^2 < 1 - x_2 q + q^2$ when $x_2 < 0$, we have $f_2(\pi) > f_2(0)$. Fix $\varepsilon > 0$ and choose P so large that the modulus of the tail product over $p > P$ lies in $[1-\varepsilon, 1+\varepsilon]$ uniformly in t, which is possible by absolute convergence. The numbers $\log p$, $p \le P$, are linearly independent over $\mathbb{Q}$, so by Kronecker's theorem the vector $(p^{-it})_{p\le P}$ comes arbitrarily close to $(-1, 1, \dots, 1)$, the entry -1 in the coordinate $p = 2$. For such t, $|L(\sigma+it,\Delta)| \ge (1-\varepsilon)^2 f_2(\pi) \prod_{2<p\le P} f_p(0)$, while $L(\sigma,\Delta) \le (1+\varepsilon) f_2(0) \prod_{2<p\le P} f_p(0)$; the first bound exceeds the second once $(1-\varepsilon)^2 f_2(\pi) > (1+\varepsilon) f_2(0)$, which holds for small ε. A characteristic function has modulus at most its value 1 at the origin. □

The proof uses a single prime p_0 with $x_{p_0} < 0$, and we now prove that every holomorphic newform f with real Hecke eigenvalues has infinitely many such primes. Write $x_p \in [-2,2]$ for its normalised eigenvalues. If $x_p \ge 0$ for all p outside a finite set, then $x_p^2 \le 2x_p$ there, so $\sum_p x_p^2 p^{-\sigma} \le 2\sum_p x_p p^{-\sigma} + O(1) = 2\log L(\sigma, f) + O(1)$ would stay bounded above as $\sigma \to 1^+$, since $L(s,f)$ is entire; but $\sum_p x_p^2 p^{-\sigma} = \log L(\sigma, f\times f) + O(1)$ tends to infinity, because the Rankin Selberg function has a pole at $s = 1$ (next section). Hence for no such f and no $\sigma > 1$ is $t \mapsto L(\sigma+it,f)/L(\sigma,f)$ a characteristic function. The obstruction is the sign of the Hecke eigenvalues; for Δ they are negative on a set of primes of density $\frac12$, by the Sato Tate theorem [67]. Positivity has to be manufactured, and the Rankin Selberg convolution manufactures it.

11.4 RANKIN SELBERG POSITIVITY

For π as above the Rankin Selberg L function $L(s, \pi\times\tilde\pi)$ has, at $p \notin S$, local factor $\det(I - A_p \otimes \tilde A_p\, p^{-s})^{-1}$. Unitarity of π_p implies that the multiset of eigenvalues of $\tilde A_p$, which is $\{\alpha^{-1}\}$ with α running over the eigenvalues of A_p, coincides with $\{\bar\alpha\}$. Hence $\operatorname{tr}(\tilde A_p^k) = \overline{\operatorname{tr}(A_p^k)}$ and $\operatorname{tr}\big((A_p\otimes\tilde A_p)^k\big) = |\operatorname{tr}(A_p^k)|^2$. By Jacquet and Shalika [279], $L(s,\pi\times\tilde\pi)$ converges absolutely for $\operatorname{Re} s > 1$ and has a simple pole at $s = 1$; its meromorphic continuation and functional equation are due to Jacquet, Piatetski Shapiro and Shalika [286], Shahidi [510] and Moeglin and Waldspurger [397].

THEOREM 11.4 (Rankin Selberg positivity). *For $\sigma > 1$ the function $t \mapsto L^S(\sigma+it, \pi\times\tilde\pi)/L^S(\sigma,\pi\times\tilde\pi)$ is the characteristic function of a compound Poisson law with Lévy measure*

$$\nu_{\pi,\sigma} = \sum_{p\notin S}\sum_{k\ge1} \frac{|\operatorname{tr}(A_p^k)|^2}{k\,p^{k\sigma}}\,\delta_{-k\log p}.$$

PROOF. By the computation above,

$$\log L^S(\sigma+it,\pi\times\tilde\pi)=\sum_{p\notin S}\sum_{k\ge1}\frac{|\operatorname{tr}(A_p^k)|^2}{k}\,p^{-k(\sigma+it)}$$

has nonnegative coefficients, and the series converges absolutely for $\sigma>1$, so $\nu_{\pi,\sigma}$ is a finite positive measure of mass $\log L^S(\sigma,\pi\times\tilde\pi)$. Subtracting the value at $t=0$ and exponentiating gives the compound Poisson characteristic function, as in the proof of Theorem 11.2. As there, the reflected law, with characteristic function $L^S(\sigma-it,\pi\times\tilde\pi)/L^S(\sigma,\pi\times\tilde\pi)$, has the same weights at the points $k\log p$. □

The restriction to $p\notin S$ is a convenience. At a ramified prime the local factor $L(s,\pi_p\times\tilde\pi_p)$ of Jacquet, Piatetski Shapiro and Shalika is still the reciprocal of a polynomial in p^{-s} with constant term 1, but its roots are not in general the products of the roots of $L(s,\pi_p)$ and $L(s,\tilde\pi_p)$, so the formula $|\operatorname{tr}(A_p^k)|^2$ is not available there. Rudnick and Sarnak computed these factors explicitly [471] (Appendix A.2), and their formula exhibits each coefficient of $\log L(s,\pi_p\times\tilde\pi_p)$ as a sum of nonnegative terms. Hence the complete finite L function $L(s,\pi\times\tilde\pi)=\prod_p L(s,\pi_p\times\tilde\pi_p)$ also has nonnegative logarithmic coefficients, and Theorem 11.4 holds for it, the primes of S adding further nonnegative atoms at the points $-k\log p$. By the simple pole, the mass $\log L^S(\sigma,\pi\times\tilde\pi)=\log\frac{1}{\sigma-1}+O(1)$ of $\nu_{\pi,\sigma}$ tends to infinity as $\sigma\to1^+$; since every atom lies at or below $-\log 2$, the laws of Theorem 11.4 escape to $-\infty$.

For Δ the weights are $(2\cos k\theta_p)^2/(kp^{k\sigma})$: the trace $2\cos k\theta_p$, which enters the signed exponent of $L(s,\Delta)$ with its sign, enters that of $L(s,\Delta\times\Delta)$ squared. The computation uses nothing but the Satake parameters, so it applies to any isobaric sum $\Pi=\pi_1\boxplus\cdots\boxplus\pi_r$ of unitary cuspidal representations. At p outside a finite set the Satake parameter of Π is the block diagonal matrix $A_p(\Pi)$ with blocks $A_p(\pi_1),\dots,A_p(\pi_r)$, the partial function $L^S(s,\Pi\times\tilde\Pi)=\prod_{i,j}L^S(s,\pi_i\times\tilde\pi_j)$ has local factor $\det(I-A_p(\Pi)\otimes A_p(\tilde\Pi)\,p^{-s})^{-1}$, and the coefficient of p^{-ks} in its logarithm is $|\operatorname{tr}A_p(\Pi)^k|^2/k\ge0$. The next section exploits this freedom.

11.5 LANDAU'S LEMMA AND NONVANISHING ON THE LINE

LEMMA 11.5 (Landau). *Let $F(s)=\sum_n a_n n^{-s}$ with $a_n\ge0$ have abscissa of convergence σ_c finite. Then F has no analytic continuation to any neighbourhood of the real point $s=\sigma_c$.*

In probabilistic language $F(\sigma)$ is the Laplace transform of the positive measure $\sum_n a_n\delta_{\log n}$, and the lemma says that such a transform is singular at the real point of its abscissa. For the proof, suppose F continues analytically to a disc about σ_c. Then the Taylor series of F about a real point $\sigma_0>\sigma_c$ close enough to σ_c converges at some real $\sigma_1<\sigma_c$. The derivatives $F^{(j)}(\sigma_0)=\sum_n a_n(-\log n)^j n^{-\sigma_0}$

have sign $(-1)^j$, so every term $F^{(j)}(\sigma_0)(\sigma_1-\sigma_0)^j/j!$ is a series of nonnegative terms, and summing the double series in the other order gives $\sum_n a_n n^{-\sigma_1} < \infty$, which contradicts the definition of σ_c.

EXAMPLE 11.6. Let χ be a real nonprincipal Dirichlet character. The coefficients of $\zeta(s)L(s,\chi) = \sum_n (\sum_{d\mid n}\chi(d))n^{-s}$ are nonnegative and at least 1 at squares. If $L(1,\chi) = 0$ then $\zeta(s)L(s,\chi)$ would be entire, so by Landau's lemma the series would converge for all real s, which is false at $s = 1/2$ because the terms at squares contribute $\sum_m m^{-1}$. Hence $L(1,\chi) \neq 0$.

For the line $\operatorname{Re} s = 1$ the classical tool is Mertens' inequality $3 + 4\cos\theta + \cos 2\theta = 2(1+\cos\theta)^2 \geq 0$. In the language of Theorem 11.2 it becomes an inequality for the Lévy measure. Write $\psi_\sigma(t) = \int (1-\cos tx)\,\nu_\sigma(dx)$, so that $\log|\zeta(\sigma+it)| = \log\zeta(\sigma) - \psi_\sigma(t)$. Integrating $3+4\cos(tx)+\cos(2tx) \geq 0$ against ν_σ gives

$$4\psi_\sigma(t) + \psi_\sigma(2t) \leq 8\log\zeta(\sigma). \tag{11.2}$$

THEOREM 11.7 (Hadamard, de la Vallée Poussin). *$\zeta(1+it) \neq 0$ for real $t \neq 0$.*

PROOF. Suppose $\zeta(1+it) = 0$ with $t \neq 0$. Then $|\zeta(\sigma+it)| \leq C(\sigma-1)$ and $|\zeta(\sigma+2it)| \leq C$ for $1 < \sigma < 2$, and $\log\zeta(\sigma) = \log\frac{1}{\sigma-1} + O(\sigma-1)$. So $\psi_\sigma(t) \geq 2\log\frac{1}{\sigma-1} - C'$ and $\psi_\sigma(2t) \geq \log\frac{1}{\sigma-1} - C'$, whence the left side of (11.2) is at least $9\log\frac{1}{\sigma-1} - 5C'$ while the right side is $8\log\frac{1}{\sigma-1} + O(1)$. Let $\sigma \to 1^+$. □

The Rankin Selberg version replaces the trigonometric inequality by the positivity of $|\operatorname{tr}|^2$ for a well chosen isobaric sum. Its shape is the identity $(1+2\cos\theta)^2 = 3 + 4\cos\theta + 2\cos 2\theta$: the coefficients 3, 4 and 2 count the three groups of factors in the auxiliary function D below.

PROPOSITION 11.8. *Let π be a unitary cuspidal automorphic representation of $\mathrm{GL}_n(\mathbb{A}_{\mathbb{Q}})$, $n \geq 2$, and let $t \in \mathbb{R}$ with $\pi \otimes |\det|^{2it} \not\cong \tilde{\pi}$. Then $L(1+it,\pi) \neq 0$.*

PROOF. Put $\pi_t = \pi \otimes |\det|^{it}$, so $L(s,\pi_t) = L(s+it,\pi)$, and consider the isobaric sum $\Pi = 1 \boxplus \pi_t \boxplus \tilde{\pi}_{-t}$, where $\tilde{\pi}_{-t} = \tilde{\pi} \otimes |\det|^{-it}$ is the contragredient of π_t. Then $\tilde{\Pi} \cong \Pi$, and expanding $\Pi \times \tilde{\Pi}$ into its nine constituents gives

$$\begin{aligned} D(s) &:= L^S(s, \Pi \times \tilde{\Pi}) \\ &= \zeta^S(s)\, L^S(s, \pi\times\tilde{\pi})^2 \\ &\quad \times L^S(s+it,\pi)^2\, L^S(s-it,\tilde{\pi})^2 \\ &\quad \times L^S(s+2it, \pi\times\pi)\, L^S(s-2it, \tilde{\pi}\times\tilde{\pi}). \end{aligned}$$

At $p \notin S$ the kth power of the Satake parameter of Π has trace $c_{p,k} = 1 + 2\operatorname{Re}\bigl(p^{-ikt}\operatorname{tr} A_p^k\bigr)$, a real number, and the coefficient of p^{-ks} in $\log D$ is $c_{p,k}^2/k \geq 0$. Hence $D(\sigma) \geq 1$ for real $\sigma > 1$.

Now count orders at $s = 1$. Each partial L function is the complete one times finitely many polynomials in the p^{-s}, $p \in S$, so its poles are among those of the complete function and its zeros include those of the complete function. The standard L functions $L(s + it, \pi)$ and $L(s - it, \tilde{\pi})$ are entire, since $n \geq 2$ [217]. $L(s, \pi \times \tilde{\pi})$ has a simple pole at $s = 1$, and $L(s, \pi \times \pi')$ for cuspidal π' is holomorphic at $s = 1$ unless $\pi' \cong \tilde{\pi}$ [279]; by hypothesis $L(s + 2it, \pi \times \pi) = L(s, \pi \times \pi_{2t})$ and its conjugate partner are holomorphic at $s = 1$. So D has a pole of order at most 3 at $s = 1$. If $L(1 + it, \pi) = 0$ then $L(1 - it, \tilde{\pi}) = \overline{L(1 + it, \pi)} = 0$ as well, and the squared factors contribute a zero of order at least 4. Thus D is holomorphic at $s = 1$ with $D(1) = 0$, contradicting $D(\sigma) \geq 1$ for $\sigma > 1$ and continuity. □

The hypothesis says exactly that π_t is not self dual, and the excluded case is $L(1, \pi_t) \neq 0$ for self dual π_t. There the last two factors of D acquire poles, so D has a pole of order 5 at $s = 1$, while a simple zero of $L(s, \pi_t)$ at $s = 1$ contributes a zero of order only 4: the auxiliary function has no zero to spare, as for a real character in the classical theory.

The full theorem, $L(1 + it, \pi) \neq 0$ for all t and all unitary cuspidal π, is due to Jacquet and Shalika [278], and Shahidi [510] proved $L(1 + it, \pi \times \pi') \neq 0$ for Rankin Selberg L functions, both through the theory of Eisenstein series. The positivity proof above follows Section 1 of Sarnak's note [479], which also makes the Eisenstein series proof effective for ζ alone: $|\zeta(1 + it)| \gg (\log |t|)^{-3}$ for $|t| \geq 2$, hence a zero free region $\sigma > 1 - c/(\log |t|)^5$. Quantifying the positivity argument with the Hadamard factorisation of D gives zero free regions of the classical shape $\sigma > 1 - c/\log(\mathfrak{q}(\pi)(|t| + 3))$, with $\mathfrak{q}(\pi)$ the analytic conductor and $c > 0$ depending only on n, apart from the possibility of one exceptional real zero when π is self dual; see Chapter 5 of Iwaniec and Kowalski [276] for the precise statements. In every positivity proof the input is the same: a Dirichlet series F with nonnegative logarithmic coefficients, for which, as in the proofs of Theorems 11.2 and 11.4, $t \mapsto F(\sigma + it)/F(\sigma)$ is the characteristic function of a compound Poisson law at every σ where the series for $\log F$ converges absolutely.

11.6 Value distribution in the strip

To the left of $\operatorname{Re} s = 1$ the Euler product diverges and Theorem 11.2 fails, but its probabilistic model survives. Let (X_p) be independent random variables uniform on the unit circle and define the random Euler product $\zeta(s, X) = \prod_p (1 - X_p p^{-s})^{-1}$ through its logarithm

$$\log \zeta(\sigma, X) = -\sum_p \log\bigl(1 - X_p p^{-\sigma}\bigr).$$

The summands are independent, the terms with $k \geq 2$ in the expansion converge absolutely for $\sigma > 1/2$, and the $k = 1$ part $\sum_p X_p p^{-\sigma}$ has mean zero and summable variances $p^{-2\sigma}$, so by Kolmogorov's two series theorem (Appendix A) the sum converges almost surely for every $\sigma > 1/2$. On the side of ζ, for t outside a countable set the horizontal half line from $\sigma + it$ to $+\infty + it$ meets no zero and not the pole, and $\log \zeta(\sigma + it)$ is defined by continuous variation along it.

THEOREM 11.9 (Bohr and Jessen). *Fix $\sigma > 1/2$. As $T \to \infty$ the law of $\log \zeta(\sigma + it)$, with t uniform on $[0, T]$, converges weakly to the law of $\log \zeta(\sigma, X)$. The limit law has a continuous density on $\mathbb{C}$.*

Sketch. For $\sigma > 1$ the map $t \mapsto (p^{-it})_{p \leq P}$ equidistributes on the torus $\prod_{p \leq P} S^1$ as t ranges over $[0, T]$, by Weyl's criterion and the linear independence of the $\log p$; truncation at P costs $O(\sum_{p > P} p^{-\sigma})$ uniformly. For $1/2 < \sigma \leq 1$ one replaces uniform control by a mean square approximation of $\log \zeta(\sigma + it)$ by short Euler products for most t, which follows from Carlson's mean value theorem and zero density estimates. The continuity of the density is part of Bohr and Jessen's theorem; Jessen and Wintner [283] reproved the theorem with their theory of infinite convolutions and showed that the density has continuous partial derivatives of all orders. □

The real part of the $k = 1$ term has variance $\frac{1}{2} \sum_p p^{-2\sigma}$, which grows like $\frac{1}{2} \log \frac{1}{2\sigma - 1}$ as $\sigma \to \frac{1}{2}^+$. This is the random model counterpart of Selberg's central limit theorem [494]: for t uniform on $[T, 2T]$, $\log |\zeta(\frac{1}{2} + it)| / \sqrt{\frac{1}{2} \log \log T}$ converges in law to the standard normal, and the two variances match when $\sigma - \frac{1}{2}$ is of order $1/\log T$. Chapter 12 takes up the critical line.

Bagchi [31] lifted Theorem 11.9 from values to functions: on the strip $\frac{1}{2} < \operatorname{Re} s < 1$, with t uniform on $[0, T]$, the random holomorphic function $s \mapsto \zeta(s + it)$ converges in law as $T \to \infty$ to $\zeta(s, X)$, for the topology of uniform convergence on compact sets, and the support of the limit is the set of nonvanishing holomorphic functions together with 0. The support statement is universality.

THEOREM 11.10 (Voronin, Bagchi). *Let K be a compact subset of $\frac{1}{2} < \operatorname{Re} s < 1$ with connected complement, and f continuous and nonvanishing on K and holomorphic in its interior. Then for every $\varepsilon > 0$*

$$\liminf_{T \to \infty} \frac{1}{T} \operatorname{meas}\Big\{ t \in [0, T] : \max_{s \in K} |\zeta(s + it) - f(s)| < \varepsilon \Big\} > 0,$$

where meas *is Lebesgue measure.*

Voronin [558] proved this for discs; the general form is Bagchi's. Universality is known for many L functions with Euler products, including those of holomorphic

cusp forms, through the same probabilistic route: a limit theorem plus a support computation for a random Euler product.

Large values are a large deviation question for the random model. For fixed $\sigma \in (\frac{1}{2}, 1)$ and $\tau \to \infty$,

$$\mathbb{P}\big(\operatorname{Re}\log\zeta(\sigma, X) > \tau\big) = \exp\Big(-c(\sigma)\,\tau^{1/(1-\sigma)}(\log\tau)^{\sigma/(1-\sigma)}\big(1+o(1)\big)\Big)$$

for a constant $c(\sigma) > 0$: to reach τ the phases X_p must align for all primes up to a point of size about $(\tau\log\tau)^{1/(1-\sigma)}$. Lamzouri [342] proved that the distribution of $\log|\zeta(\sigma+it)|$ obeys the same asymptotics uniformly in a range of τ that grows with T, with analogues for Dirichlet L functions in the conductor aspect and for L functions of weight 2 cusp forms in the level aspect. We state the shape only; constants and ranges are in the cited work.

11.7 Random multiplicative functions and random Euler products for GL_2

Expanding the random Euler product gives a random multiplicative function. Steinhaus' model takes $f(n)$ completely multiplicative with $f(p) = X_p$ uniform on the circle, so that $\sum_n f(n)n^{-s} = \prod_p (1 - X_p p^{-s})^{-1}$; Rademacher's model, introduced by Wintner [586], takes f supported on squarefree integers with $f(p) = \pm 1$ by fair coin tosses. Wintner showed that the partial sums of the Rademacher model are $O(x^{1/2+\varepsilon})$ almost surely for every $\varepsilon > 0$, the random analogue of the Riemann hypothesis. Harper [238] showed that in both models $\mathbb{E}|\sum_{n\le x} f(n)| \asymp \sqrt{x}/(\log\log x)^{1/4}$, better than square root cancellation, and traced the factor $(\log\log x)^{-1/4}$ to critical multiplicative chaos.

For a cusp form the natural model keeps the shape of the Euler factor and randomises the Satake parameter.

Definition 11.11. Let (θ_p) be independent with the Sato Tate law $\frac{2}{\pi}\sin^2\theta\,d\theta$ on $[0,\pi]$. The *Sato Tate random Euler product* is

$$L_{\mathrm{rand}}(s) = \prod_p \big(1 - e^{i\theta_p}p^{-s}\big)^{-1}\big(1 - e^{-i\theta_p}p^{-s}\big)^{-1}.$$

Write $x_p = 2\cos\theta_p$, so x_p has the semicircle law μ_{ST} on $[-2,2]$. For $x = 2\cos\theta$ and $k \ge 1$ we have $2\cos k\theta = U_k(x/2) - U_{k-2}(x/2)$, with U_k the Chebyshev polynomials of the second kind and $U_{-1} = 0$; the $U_k(x/2)$ are the characters of Sym^k and are orthonormal in $L^2(\mu_{\mathrm{ST}})$ (Chapter 4). Since $\int U_k(x/2)\,\mu_{\mathrm{ST}}(dx) = \delta_{k0}$ we get

$$\mathbb{E}[2\cos k\theta_p] = 0 \ (k = 1 \text{ or } k \ge 3), \qquad \mathbb{E}[2\cos 2\theta_p] = -1. \tag{11.3}$$

PROPOSITION 11.12 (this book). *For real $\sigma > 1/2$ and real t the product $L_{\mathrm{rand}}(\sigma+it)$ converges almost surely and in L^2, and*

$$\mathbb{E}\big[L_{\mathrm{rand}}(\sigma+it)\big] = 1, \qquad \mathbb{E}\big|L_{\mathrm{rand}}(\sigma+it)\big|^2 = \zeta(2\sigma).$$

PROOF. Put $q = p^{-\sigma-it}$. The generating function of the U_k gives

$$F_p = \big(1-e^{i\theta_p}q\big)^{-1}\big(1-e^{-i\theta_p}q\big)^{-1} = \sum_{k\ge 0} U_k(x_p/2)\,q^k,$$

a series that converges absolutely since $|U_k(x/2)| \le k+1$ and $|q|<1$. The factors F_p are independent, and by orthonormality $\mathbb{E}[F_p]=1$ and $\mathbb{E}|F_p|^2 = \sum_k |q|^{2k} = (1-p^{-2\sigma})^{-1}$. The partial products $M_P = \prod_{p\le P} F_p$ therefore form a complex martingale with $\mathbb{E}|M_P|^2 = \prod_{p\le P}(1-p^{-2\sigma})^{-1} \le \zeta(2\sigma)$. A martingale bounded in L^2 converges almost surely and in L^2, and the limit has mean 1 and second moment $\zeta(2\sigma)$. $\square$

The mean one property is a multiplicative chaos normalisation: L_{rand} is a product of independent mean one factors, and its size is governed by the logarithm. The next theorem identifies the law of the logarithm at the real point as $\sigma \to \frac12^+$.

THEOREM 11.13 (this book). *Let $V(\sigma) = \sum_p p^{-2\sigma}$, so $V(\sigma) = \log\frac{1}{2\sigma-1} + O(1)$ as $\sigma \to \frac12^+$. Then*

$$\frac{\log L_{\mathrm{rand}}(\sigma) + \frac12 V(\sigma)}{\sqrt{V(\sigma)}} \Longrightarrow N(0,1), \qquad \sigma \to \tfrac12^+.$$

PROOF. For real σ, $\log L_{\mathrm{rand}}(\sigma) = \sum_p \sum_k 2\cos(k\theta_p)/(kp^{k\sigma}) = A + B + C$ with $A = \sum_p x_p p^{-\sigma}$, $B = \frac12\sum_p (x_p^2-2)p^{-2\sigma}$ and C the terms with $k \ge 3$. By (11.3), $\mathbb{E}C = 0$, and $\mathbb{E}|C| \le \sum_p \sum_{k\ge3} 2p^{-k\sigma}/k$ is bounded uniformly for $\sigma \ge \frac12$. Next $\mathbb{E}B = -\frac12 V(\sigma)$ and, since $\mathbb{E}x^4 = 2$ under μ_{ST}, $\mathrm{Var}(x^2) = 1$ and $\mathrm{Var}\,B = \frac14\sum_p p^{-4\sigma}$ is bounded. So $B + \frac12 V(\sigma)$ and C are $O_{\mathbb{P}}(1)$. Finally A is a sum of independent centred variables with $\mathrm{Var}\,A = V(\sigma) \to \infty$ and summands bounded by $2p^{-\sigma} \le 2\cdot 2^{-1/2}$, so the Lindeberg condition holds and $A/\sqrt{V(\sigma)} \Rightarrow N(0,1)$. Slutsky's lemma completes the proof. $\square$

Two remarks place this in context. First, the negative drift $-\frac12 V$ is forced by the single moment $\mathbb{E}[x_p^2 - 2] = -1$, which is $\int (U_2(x/2) - U_0(x/2))\,\mu_{\mathrm{ST}}(dx)$; in the Sato Tate average the terms with $k=2$ behave like those of $\log\zeta(2s)^{-1/2}$. Drift and variance match Proposition 11.12: if the logarithm were exactly $N(-\frac12 V, V)$ then $\mathbb{E}L = 1$ and $\mathbb{E}L^2 = e^V$, and $\zeta(2\sigma) = e^{V(\sigma)+O(1)}$. Second, the Gaussian limit sees μ_{ST} only through $\mathbb{E}x_p = 0$, $\mathbb{E}x_p^2 = 1$ and the boundedness of x_p: the proof needs no more. The law $\frac12\delta_0 + \frac12$ (arcsine) of curves with complex multiplication

has the same first two moments, so the same limit, although its fourth moment is 3 and its higher Weyl averages differ. The distinction between Sato Tate groups lives in those higher averages and at finite σ, not in the central limit.

Two random models for an individual form must be kept apart. The t aspect model for $L(\sigma+it,\Delta)$ keeps the true θ_p and replaces the phases p^{-it} by independent uniform X_p, as Bohr and Jessen did for ζ. For $\sigma>1$ the equidistribution argument in the proof of Theorem 11.9 shows that the law of $\log L(\sigma+it,\Delta)$ converges to that of the model. For $\frac12<\sigma<1$ the analogous limit theorem for $L(s+it,\Delta)$ itself, in Bagchi's form, is part of the proof of universality for Hecke eigenforms of level 1 by Laurinčikas and Matsumoto [355]. In the t aspect model the first order term $\sum_p x_p X_p p^{-\sigma}$ has uncorrelated real and imaginary parts, each of variance $\frac12\sum_p x_p^2p^{-2\sigma}$, and the second order term has mean zero because $\mathbb{E}X_p^2=0$. The Sato Tate model randomises θ_p and keeps the phases. At a point $\sigma+it$ with $t\neq0$ fixed, its first order term $\sum_p x_p p^{-\sigma-it}$ has real and imaginary parts of variance $\frac12V(\sigma)+O(1)$ and bounded covariance, because $\sum_p p^{-2\sigma-2it}=\log\zeta(2\sigma+2it)+O(1)$ stays bounded as $\sigma\to\frac12{}^+$, and its drift $-\frac12\sum_p p^{-2\sigma-2it}$ is bounded as well. So the two models agree to first order near the critical line precisely when $\sum_p x_p^2p^{-2\sigma}\sim V(\sigma)$, and this holds unconditionally: $\sum_p x_p^2p^{-s}=\log L(s,\Delta\times\Delta)+O(1)=\log\frac{1}{s-1}+O(1)$ as $s\to1^+$, by the simple pole of the Rankin Selberg function. The Sato Tate theorem for Δ [67] supplies more: every moment of the empirical law of the x_p, $p\le y$, tends to the corresponding moment of μ_{ST} as $y\to\infty$.

At the real point $t=0$ the Sato Tate model and the true function part company. The Sato Tate drift $-\frac12V(\sigma)$ of Theorem 11.13 is unbounded, and the true second order terms carry the same drift,

$$\begin{aligned}\frac12\sum_p(x_p^2-2)p^{-2\sigma}&=\frac12\log L(2\sigma,\operatorname{Sym}^2\Delta)-\frac12\log\zeta(2\sigma)+O(1)\\&=-\frac12V(\sigma)+O(1),\end{aligned}$$

because $x_p^2-2=U_2(x_p/2)-1$ and $L(s,\operatorname{Sym}^2\Delta)$ is holomorphic and nonzero at $s=1$ [212, 278]. But $L(\frac12,\Delta)$, which is $L(6,\Delta)=0.792123$ in the classical normalisation, is not zero. So on the Riemann hypothesis for $L(s,\Delta)$, which makes $\sum_p x_p p^{-\sigma}$ converge for $\sigma>\frac12$, the first order sum compensates: $\sum_p x_p p^{-\sigma}=\frac12V(\sigma)+O(1)$. This bias towards positive x_p is the Abelian form of the bias $(\frac12-r)\log x$ of Section 5.4, with $r=0$, and no model with independent centred x_p reproduces it.

11.8 Notes

The zeta distribution, its infinite divisibility and its history are surveyed by Lin and Hu [366], and Sato [485] is our reference for infinitely divisible laws. The prime number theorem was proved independently by Hadamard [232] and de la Vallée Poussin [137] in 1896, and the trigonometric inequality is due to Mertens. Landau's lemma is from [344]. The value distribution theory begins with Bohr and Jessen [70]. Proposition 11.12 and Theorem 11.13 are elementary, but we have not found them stated in this form; we record them because they isolate exactly which features of μ_{ST} the random model uses. Our extension of Proposition 11.3 to all holomorphic newforms with real eigenvalues is elementary too; its point is that the Rankin Selberg pole, without the Sato Tate theorem, forces infinitely many negative x_p.

11.9 Exercises

1. Show that the geometric law $\mathbb{P}(G = k) = (1-q)q^k$ is infinitely divisible, identify its Lévy measure, and deduce Theorem 11.2 from the factorisation $N = \prod_p p^{G_p}$ of a zeta distributed integer.

2. Prove $3 + 4\cos\theta + \cos 2\theta = 2(1+\cos\theta)^2$ and $(1+2\cos\theta)^2 = 3 + 4\cos\theta + 2\cos 2\theta$. Use the second identity and the function $\zeta(s)^3 L(s+it,\chi)^2 L(s-it,\bar{\chi})^2 L(s+2it,\chi^2)L(s-2it,\bar{\chi}^2)$ to show $L(1+it,\chi) \neq 0$ for a Dirichlet character χ and real t, unless χ^2 is principal and $t = 0$. (That case, for nonprincipal χ, is the example after Lemma 11.5.)

3. Prove Landau's lemma in detail, and show that the abscissa of convergence of $\sum_n |\tau(n)|^2 n^{-s}$ is 12, the location of the pole of $L(s, \Delta \times \Delta)$ in the classical normalisation.

4. Let ν be a finite measure on $\mathbb{R}$ and put $\psi(t) = \int (1-\cos tx)\,\nu(dx)$. Show that $4\psi(t) + \psi(2t) \leq 8\nu(\mathbb{R})$ and that equality forces ν to be carried by $\{x : tx \in \pi + 2\pi\mathbb{Z}\}$.

5. In the setting of Proposition 11.12, compute $\mathbb{E}|L_{\mathrm{rand}}(\sigma)|^2$ when θ_p is drawn instead from the law of curves with complex multiplication, $\frac{1}{2}\delta_{\pi/2} + \frac{1}{2}\,d\theta/\pi$, and show that it is finite for $\sigma > 1/2$ but differs from $\zeta(2\sigma)$. (Hint: for real σ and $q = p^{-\sigma}$ the factor $F_p = (1 - 2q\cos\theta_p + q^2)^{-1}$ is positive. The atom at $\pi/2$ contributes $\frac{1}{2}(1+q^2)^{-2}$ to $\mathbb{E}F_p^2$, and the uniform part contributes half of $\frac{1}{\pi}\int_0^\pi (1-2q\cos\theta+q^2)^{-2}\,d\theta = (1+q^2)/(1-q^2)^3$. The factors now have mean $(1-q^4)^{-1}$, so run the martingale argument on the normalised factors $(1-q^4)F_p$.)

11.10 Open computational problems

Computational Problem 11.1 (Value distribution of $\log L(\sigma + it, \Delta)$). Objective: compare the empirical law of $\log L(\sigma + it, \Delta)$ in the strip with its two random models. Objects and scale: the five abscissae $\sigma = 0.55, 0.6, 0.7, 0.8, 0.9$ and t on a grid of spacing 0.1 in $[10^3, 10^6]$, about 10^7 evaluations per σ, using $\tau(n)$ for n up to the approximate functional equation length. Route: PARI/GP `lfun` with precomputed $\tau(n)$ via the product formula, or a GPU implementation of the approximate functional equation; Monte Carlo of the t aspect model (true θ_p, uniform phases) and of the Sato Tate model, with primes up to 10^8 in the random products. Deliverable: two dimensional histograms of $\log L$, Kolmogorov and Wasserstein distances to both models as functions of σ and T, and a plot of the variance of $\operatorname{Re} \log L$ against $\frac{1}{2}\sum_p x_p^2 p^{-2\sigma}$. Why it matters: it tests the Bohr Jessen theorem for GL_2 quantitatively and measures how far the Sato Tate model, which forgets the actual θ_p, is from the truth.

Computational Problem 11.2 (Large deviation rate functions). Objective: estimate $-\log \mathbb{P}(\operatorname{Re} \log L > \tau)$ for ζ, for $L(s, \Delta)$, and for L_{rand}, and compare with the predicted shape $c(\sigma)\tau^{1/(1-\sigma)}(\log \tau)^{\sigma/(1-\sigma)}$. Objects and scale: $\sigma \in \{0.6, 0.7, 0.8\}$, $t \leq 10^7$ for the L functions, and 10^9 samples of the random models using importance sampling by exponential tilting of each prime factor. Route: GPU Monte Carlo with tilted phases; saddle point evaluation of the cumulant generating function $\lambda \mapsto \sum_p \log \mathbb{E}|1 - X_p p^{-\sigma}|^{-\lambda}$ of $\operatorname{Re} \log \zeta(\sigma, X)$ and of its analogues for the other two models; for the L functions, Odlyzko Schönhage multi evaluation of the Riemann Siegel formula for ζ and of the approximate functional equation for $L(s, \Delta)$, checked against PARI/GP `lfun`. Deliverable: rate function plots and fitted constants $c(\sigma)$ with confidence intervals, and the largest τ at which the empirical L function tail still follows the model. Why it matters: it locates the range in Lamzouri's theorem numerically and shows where arithmetic correlations between primes begin to matter.

Computational Problem 11.3 (Testing universality). Objective: we ask for the density in Theorem 11.10 for concrete targets. Objects and scale: K a disc of radius 0.1 centred at 0.75, targets $f(s) = e^{a(s-0.75)}$ with $a \in \{0.5, 1, 2\}$ and $f(s) = L(s, \chi_{-4})$, tolerances $\varepsilon \in \{0.5, 0.2, 0.1\}$, shifts $t \leq 10^7$, for ζ and for $L(s, \Delta)$ in the unitary normalisation. Route: Odlyzko Schönhage style multi evaluation on a fine grid of K and t, or on GPU the Riemann Siegel formula for ζ and the approximate functional equation for $L(s, \Delta)$; for the model, Monte Carlo of random Euler products with primes to 10^7. Deliverable: table of empirical densities against the probability that the random Euler product lies within ε of f, and the scaling of the density with ε. Why it matters: the universality density is a small ball probability for a random Euler product, and its size as $\varepsilon \to 0$ is poorly understood;

numerics would suggest its true order.

CHAPTER 12

LOG CORRELATED FIELDS: SELBERG'S CENTRAL LIMIT THEOREM, MAXIMA AND MULTIPLICATIVE CHAOS

In 1946 Selberg proved that the logarithm of the Riemann zeta function on the critical line is asymptotically Gaussian: if t is chosen at random between T and $2T$, then $\log|\zeta(\frac{1}{2}+it)|$ is approximately normal with mean zero and variance $\frac{1}{2}\log\log T$ [494]. The variance grows, but so slowly that at every height a computer will ever reach it is a single digit number, and for six decades the theorem was read as a statement about the value at one random point, a part of the value distribution theory of Chapter 11. Its modern meaning emerged when Fyodorov, Hiary and Keating [187] looked at $t \mapsto \log|\zeta(\frac{1}{2}+it)|$ on a whole window of unit length. Two points at distance h share the primes p with $\log p < 1/h$ and see independent phases for the others, so for $1/\log T \le h \le 1$ the covariance of the field is about $\frac{1}{2}\log(1/h)$: the field is *log correlated*, the same structure that governs branching random walks, the two dimensional Gaussian free field, and the characteristic polynomials of random unitary matrices. From that structure they predicted the maximum over a unit window up to a bounded random error, and Arguin, Bourgade and Radziwiłł have since proved the prediction to that accuracy; the law of the error is still open. The exponential of a log correlated field is a multiplicative chaos, and Saksman and Webb showed that the zeta function on a random window converges in law, as a random generalised function, to a complex multiplicative chaos times a smooth random function. Harper found the same chaos, at its critical point, inside the partial sums of random multiplicative functions. This chapter develops these ideas for ζ and for the discriminant form Δ, tests them on our own computations of $L(s,\Delta)$, and connects them with the Thorin and extreme value representations of the chapters that follow.

Conventions. We write $L(s,\Delta)=\sum_n \tau(n)n^{-s}$ in the classical normalisation of Chapter 2, so that the centre is $s=6$ and the critical line is $6+it$. Put $\lambda(n)=\tau(n)/n^{11/2}$ and $\lambda(p)=x_p=2\cos\theta_p$ with $\theta_p\in[0,\pi]$ (Deligne). Then

$$L(6+it,\Delta)=\prod_p\Big[\big(1-e^{i\theta_p}p^{-\frac{1}{2}-it}\big)\big(1-e^{-i\theta_p}p^{-\frac{1}{2}-it}\big)\Big]^{-1}$$

formally, and $\log|L(6+it,\Delta)|$ is $\log|L(\frac12+it)|$ in the unitary normalisation used in Chapter 11. The completed function is $\Lambda(s,\Delta)=(2\pi)^{-s}\Gamma(s)L(s,\Delta)$, and the Hardy function

$$Z_\Delta(t)=e^{i\vartheta(t)}L(6+it,\Delta),\qquad \vartheta(t)=\arg\big((2\pi)^{-it}\Gamma(6+it)\big),$$

with ϑ continuous and $\vartheta(0)=0$, is real for real t, and $|Z_\Delta(t)|=|L(6+it,\Delta)|$. The zeros of $L(s,\Delta)$ near height t have density $\frac{1}{\pi}\log\frac{t}{2\pi}$, twice that of ζ, because the analytic conductor is of size t^2.

12.1 Selberg's central limit theorem

THEOREM 12.1 (Selberg [494]). *Let t be uniformly distributed on $[T,2T]$. As $T\to\infty$,*

$$\frac{\log\zeta(\frac12+it)}{\sqrt{\frac12\log\log T}}\Longrightarrow X+iY,$$

with X and Y independent standard normal variables. In particular $\log|\zeta(\frac12+it)|/\sqrt{\frac12\log\log T}$ is asymptotically standard normal.

Selberg's original account is brief, and a detailed proof was written out by Tsang [551]. The heuristic behind it is the random model of Chapter 11. Formally $\log\zeta(\frac12+it)=\sum_p p^{-1/2-it}+O(1)$, and for t uniform on $[T,2T]$ the phases p^{-it}, $p\le T^\varepsilon$, behave like independent uniform variables on the circle, because a monomial $\prod p^{-ik_pt}$ averages to almost zero over $[T,2T]$ unless it is trivial. The real part of $p^{-1/2-it}$ then has variance $\frac{1}{2p}$, and the variance of the prime sum over $p\le x$ is $\frac12\sum_{p\le x}\frac1p=\frac12\log\log x+O(1)$ by Mertens' theorem. The difficulty, which occupies the whole proof, is that the Dirichlet series diverges on the line and that zeros of ζ make $\log|\zeta|$ singular.

12.1.1 The variance for the discriminant form

For Δ the prime sum is $\operatorname{Re}\sum_p\lambda(p)p^{-1/2-it}$, whose variance in the random phase model is $\frac12\sum_{p\le x}\lambda(p)^2/p$. Its growth is controlled by the Rankin Selberg convolution, exactly as the positivity of Chapter 11 was.

LEMMA 12.2. *There is a constant B_Δ such that $\sum_{p\le x}\lambda(p)^2/p=\log\log x+B_\Delta+o(1)$.*

PROOF. The Hecke relation $\tau(p)^2=\tau(p^2)+p^{11}$ gives $\lambda(p)^2=1+\lambda(p^2)$, and $\lambda(p^2)=e^{2i\theta_p}+1+e^{-2i\theta_p}$ is the coefficient at p of the symmetric square $L(s,\operatorname{Sym}^2\Delta)$ in the unitary normalisation. Since Δ has no complex multiplication, $\operatorname{Sym}^2\Delta$ is a cuspidal automorphic representation of GL_3 by Gelbart and Jacquet [212], so $L(s,\operatorname{Sym}^2\Delta)$ is entire and does not vanish on $\operatorname{Re}s=1$ [278].

A zero free region of de la Vallée Poussin type (Chapter 5 of [276]; a possible exceptional real zero $\beta < 1$ of this single function only contributes $O(x^{\beta})$) gives $\sum_{p \le x} \lambda(p^2) \log p = O(x/\log^2 x)$, and partial summation shows that $\sum_p \lambda(p^2)/p$ converges. Mertens' theorem $\sum_{p \le x} 1/p = \log\log x + M + o(1)$ finishes the proof, with $B_\Delta = M + \sum_p \lambda(p^2)/p$. □

So the variance of the prime sum for Δ is again $\frac{1}{2}\log\log x$ to leading order; the degree of the L function does not enter, because the Rankin Selberg function $L(s, \Delta \times \Delta) = \zeta(s) L(s, \mathrm{Sym}^2 \Delta)$ has a simple pole at $s = 1$. The constant, however, is far from zero. From $\tau(p)$ for all $p \le 10^5$, computed with PARI/GP for this book, we find $B_\Delta \approx -0.688$: the partial sums $\sum_{p \le x} \lambda(p)^2/p - \log\log x$ are -0.6854 and -0.6874 at $x = 10^4$ and 10^5, and the limit, computed from $L(1, \mathrm{Sym}^2 \Delta)$ by the method of Exercise 2, is -0.6879. The value is driven by the small primes: $\lambda(2)^2 = 0.281$ and $\lambda(3)^2 = 0.358$, against the Sato Tate average 1.

THEOREM 12.3 (Selberg; Bombieri and Hejhal). *Assume the Riemann hypothesis for $L(s, \Delta)$. For t uniform on $[T, 2T]$,*

$$\frac{\log |L(6 + it, \Delta)|}{\sqrt{\frac{1}{2}\log\log T}} \Longrightarrow N(0, 1),$$

and the same holds for the argument, jointly, with independent limits.

Selberg announced, under a zero density hypothesis, the central limit theorem for the class of Dirichlet series that now bears his name [498], and Bombieri and Hejhal [72] proved joint versions for several L functions under hypotheses on their zeros, which they used to study zeros of linear combinations of Euler products. We state the theorem under the Riemann hypothesis for $L(s, \Delta)$, under which the zero density hypotheses of both works hold trivially. The short proof described next needs no zero density estimate at all. Its one input specific to ζ is the twisted second moment of Step 2, and Radziwiłł and Soundararajan remark that the analogous estimate in degree two can be obtained from bounds for shifted convolution sums; that would give the statement for $\log |L(6 + it, \Delta)|$ without the Riemann hypothesis, but we know of no published account for Δ, so we keep the hypothesis.

12.1.2 The proof of Radziwiłł and Soundararajan

Radziwiłł and Soundararajan [450] gave a proof of Theorem 12.1 that fits on a few pages. We sketch it for ζ; for Δ one replaces p^{-s} by $\lambda(p)p^{-s}$, uses Lemma 12.2 for the variance, and needs the degree two analogue of Step 2 discussed above.

Sketch. Put $W = (\log\log\log T)^4$, $\sigma_0 = \frac{1}{2} + W/\log T$, $X = T^{1/(\log\log\log T)^2}$, and $\mathcal{P}(s) = \sum_{2 \le p \le X} p^{-s}$.

Step 1 (moments of the prime sum). For a Dirichlet polynomial $A(s) = \sum_{n\le N} a_n n^{-s}$ with $N \le T$ one has $\frac{1}{T}\int_T^{2T} |A(\sigma+it)|^2 dt = \sum_n |a_n|^2 n^{-2\sigma}(1 + O(N/T))$. Expanding $\operatorname{Re}\mathcal{P}(\sigma_0+it)^k$ for k up to a slowly growing bound gives Dirichlet polynomials of length $X^k \le T^{o(1)}$, and the diagonal terms reproduce the moments of a sum of independent random variables $\operatorname{Re}(U_p p^{-\sigma_0})$ with U_p uniform on the circle. Hence $\operatorname{Re}\mathcal{P}(\sigma_0+it)$ has asymptotically the moments of $N(0, \frac{1}{2}\sum_{p\le X} p^{-2\sigma_0})$, and the variance is $\frac{1}{2}\log\log T + O(\log\log\log T)$.

Step 2 (off the line, a mollifier). Let $\mathcal{M}(s)$ be the Dirichlet polynomial obtained by truncating the Taylor series of $\exp(-\mathcal{P}(s))$ suitably. A twisted second moment computation shows that $\frac{1}{T}\int_T^{2T} |1-\zeta(\sigma_0+it)\mathcal{M}(\sigma_0+it)|^2 dt$ tends to zero. On the set where $|\mathcal{P}(\sigma_0+it)|$ is not too large, $\mathcal{M} = \exp(-\mathcal{P})(1+o(1))$, so $\log|\zeta(\sigma_0+it)| = \operatorname{Re}\mathcal{P}(\sigma_0+it) + o(1)$ outside a set of measure $o(T)$.

Step 3 (back to the line). Using the Hadamard factorisation, $\log|\zeta(\sigma_0+it)| - \log|\zeta(\frac{1}{2}+it)|$ is a sum over zeros ρ of $\log|(\sigma_0+it-\rho)/(\frac{1}{2}+it-\rho)|$ plus a smooth term, and integrating over t gives

$$\int_T^{2T} \bigl|\log|\zeta(\tfrac{1}{2}+it)| - \log|\zeta(\sigma_0+it)|\bigr|\, dt \ll (\sigma_0-\tfrac{1}{2})\, T\log T = WT.$$

Hence the difference is $O(W)$ on average, which is $o(\sqrt{\log\log T})$.

Step 4. Combining the three steps, $\log|\zeta(\frac{1}{2}+it)|$ equals $\operatorname{Re}\mathcal{P}(\sigma_0+it) + o(\sqrt{\log\log T})$ outside a set of measure $o(T)$, and Step 1 gives the Gaussian limit. This is the real part of the theorem, and it is all that Radziwiłł and Soundararajan treat. The imaginary part does not follow verbatim, since knowing that $\zeta\mathcal{M}$ is close to 1 fixes $\operatorname{Im}\log\zeta(\sigma_0+it)$ only modulo 2π; its Gaussian limit goes back to Selberg's computation of the moments of $S(t) = \pi^{-1}\arg\zeta(\frac{1}{2}+it)$ [494]. □

The arithmetic enters through mean values: of Dirichlet polynomials in Step 1, the precise sense in which the p^{-it} are independent, and of ζ twisted by a short Dirichlet polynomial in Step 2. The analytic content is the passage from σ_0 to the line across the zeros.

12.1.3 Finite height: a heavy left tail

The central limit theorem is a statement about $\log\log T \to \infty$, and at any accessible height the distribution of $\log|L|$ is visibly not normal. The most visible departure comes from the zeros. Near a simple zero γ of Z_Δ, $\log|Z_\Delta(t)| = \log|t-\gamma| + \log|Z_\Delta'(\gamma)| + O(|t-\gamma|)$, which tends to $-\infty$, while large positive values require many primes to conspire.

PROPOSITION 12.4 (Exponential left tail; this book). *Let f be real analytic on a neighbourhood of $[a,b]$, with finitely many zeros $\gamma_1,\dots,\gamma_r$ in (a,b), all simple, and none*

at a or b. Then, as $x \to \infty$,

$$\Big|\big\{t \in [a,b] : \log|f(t)| < -x\big\}\Big| = 2e^{-x} \sum_{j=1}^{r} \frac{1}{|f'(\gamma_j)|} + O(e^{-2x}).$$

PROOF. Away from the zeros $|f|$ is bounded below by a positive constant, so for large x the set lies in small neighbourhoods of the γ_j. Near γ_j write $f(t) = f'(\gamma_j)(t-\gamma_j)(1+g_j(t))$ with g_j analytic and $g_j(\gamma_j) = 0$. The set $\{|f(t)| < e^{-x}\}$ near γ_j is an interval with endpoints $\gamma_j \pm e^{-x}/|f'(\gamma_j)|\,(1+O(e^{-x}))$, whose length is $2e^{-x}/|f'(\gamma_j)| + O(e^{-2x})$. □

Applied to $f = Z_\Delta$ on a window $[T, 2T]$ whose zeros are simple, the proposition says that the empirical law of $\log|L(6+it,\Delta)|$ has a left tail $\mathbb{P}(\log|L| < -x) \approx Ce^{-x}$, exponential with rate one, while a normal law of variance V has tail $\exp(-x^2/(2V))$. In standard units the exponential tail is $\exp(-\sqrt{V}\,y)$, heavier than Gaussian for every finite V; it disappears from the limit only because $\sqrt{V} \to \infty$. The right tail is different: it is governed by the large deviations of the prime sum, and it is close to Gaussian in the range of the central limit theorem.

Data. For this book we evaluated $\log|L(6+it,\Delta)|$ with PARI/GP at 50,000 points $t \in [500, 2500)$ with step 0.04. The sample mean is 0.0020 and the sample variance is 1.5265, against $\frac{1}{2}\log\log t = 0.9948$ at the midpoint $t = 1500$. The skewness is -1.135 and the excess kurtosis is 2.493, and a Kolmogorov Smirnov test of the standardised values against $N(0,1)$ rejects decisively: the distance is 0.069. Neighbouring values are strongly dependent, so we quote no p value for all 50,000 of them; each of the 15 subsamples of every 15th value, spaced by about one mean zero spacing, has a nominal $p < 10^{-10}$. Figure 12.1 shows the shape of the departure: a long left tail, and a bulk shifted to the right with a right tail lighter than the normal one. This does not conflict with Theorem 12.3, which is a limit statement: at height 10^3 the normal law describes the centre of the distribution but not its tails, and the variance is about one and a half times its leading asymptotic. Section 12.1.4 accounts for the variance.

12.1.4 Keating and Snaith, and the hybrid product

Keating and Snaith [300] modelled $\zeta(\frac{1}{2}+it)$ near height T by the characteristic polynomial of a Haar random matrix in $\mathrm{U}(N)$ with $N = \log\frac{T}{2\pi}$, the number of zeros in a window of length 2π, and computed its moments exactly.

PROPOSITION 12.5 (Keating and Snaith). *Let U be Haar distributed in* $\mathrm{U}(N)$ *and*

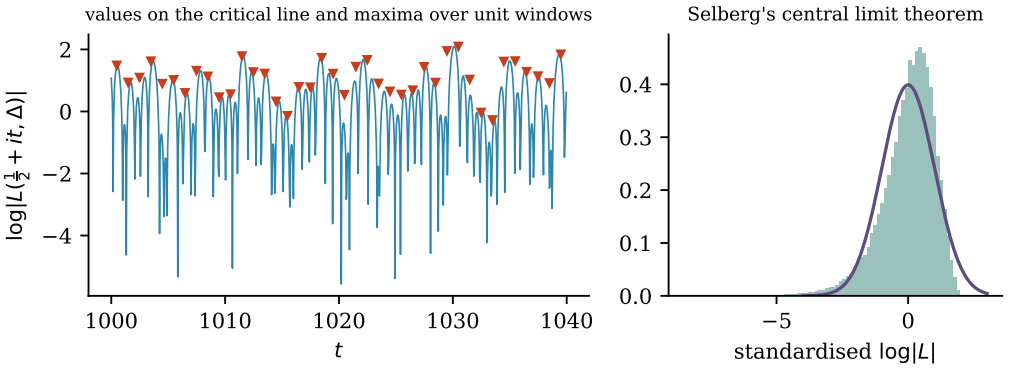


Figure 12.1: Left: $\log|L(6+it,\Delta)|$ for $t \in [1000, 1040]$ (the axis label uses the unitary normalisation, with centre $\frac{1}{2}$), with the maximum over each unit window marked; the downward spikes are zeros. Right: histogram of the standardised values of $\log|L(6+it,\Delta)|$ at 50,000 points of $[500, 2500)$ against the standard normal density. The excess on the left is the exponential tail of Proposition 12.4.

$\Lambda_N = \det(I-U) = Z_N(0)$ *(Section 6.5). Then*

$$\operatorname{Var}\log|\Lambda_N| = \frac{1}{2}\sum_{j=1}^{N}\psi'(j) = \frac{1}{2}\Big(H_N + N\sum_{k>N}\frac{1}{k^2}\Big)$$
$$= \frac{1}{2}\big(\log N + \gamma_{\mathrm{E}} + 1\big) + O\big(N^{-1}\big),$$

where $\psi = \Gamma'/\Gamma$, H_N is the harmonic number and γ_{E} is Euler's constant. For $j \geq 3$ the cumulants $\kappa_j(N)$ of $\log|\Lambda_N|$ converge as $N \to \infty$ to

$$\kappa_j(\infty) = (1-2^{1-j})(-1)^j (j-1)!\,\zeta(j-1);$$

in particular $\kappa_3(\infty) = -\pi^2/4$ and $\kappa_4(\infty) = \frac{21}{4}\zeta(3)$.

PROOF. Keating and Snaith evaluate $\mathbb{E}|\Lambda_N|^s = \prod_{j=1}^{N}\Gamma(j)\Gamma(j+s)/\Gamma(j+\frac{s}{2})^2$ for $\operatorname{Re} s > -1$ by Selberg's integral [300]. The cumulant generating function of $\log|\Lambda_N|$ is therefore

$$K(s) = \sum_{j\leq N}\big[\log\Gamma(j) + \log\Gamma(j+s) - 2\log\Gamma(j+\tfrac{s}{2})\big],$$

and $\kappa_m = K^{(m)}(0) = (1-2^{1-m})\sum_{j\leq N}\psi^{(m-1)}(j)$. For $m=2$ use $\psi'(j) = \sum_{k\geq j}k^{-2}$ and exchange the sums: $\sum_{j\leq N}\psi'(j) = \sum_k \min(k,N)k^{-2} = H_N + N\sum_{k>N}k^{-2}$. Since $H_N = \log N + \gamma_{\mathrm{E}} + O(N^{-1})$ and $N\sum_{k>N}k^{-2} = 1 + O(N^{-1})$, the variance follows. For $m \geq 3$ use $\psi^{(m-1)}(j) = (-1)^m(m-1)!\sum_{k\geq j}k^{-m}$; then $\sum_{j\geq 1}\sum_{k\geq j}k^{-m} = \sum_k k^{1-m} = \zeta(m-1)$ converges. $\square$

The random matrix model thus predicts a variance of the right growth, $\frac{1}{2}\log\log T$, and also a constant, $\frac{1}{2}(\gamma_{\mathrm{E}}+1)$. It also predicts that the higher cumulants stay bounded: the non Gaussian features of $\log|\zeta|$ at finite height should be of fixed absolute size, decaying in standard units only like powers of $\log\log T$. For our sample, with variance 1.5265, the limits $\kappa_3(\infty)$ and $\kappa_4(\infty)$ correspond to skewness -1.308 and excess kurtosis 2.708, against the observed -1.135 and 2.493. The signs and sizes agree; the model is not expected to be exact here, since it ignores the arithmetic.

The arithmetic is restored by the hybrid Euler Hadamard product of Gonek, Hughes and Keating [214]. For a parameter $X \geq 2$ they write $\zeta(s) = P_X(s)Z_X(s)$ up to a small error, where P_X is a smoothed Euler product over the primes $p \leq X$ and Z_X is a smoothed Hadamard product over the zeros within about $1/\log X$ of s. They prove that $\mathbb{E}|P_X|^{2k} \sim a_k(e^{\gamma_{\mathrm{E}}}\log X)^{k^2}$ as $X, T \to \infty$ with $X = O((\log T)^{2-\varepsilon})$, with the arithmetic factor

$$a_k = \prod_p \Big(1-\frac{1}{p}\Big)^{k^2} \sum_{m\geq 0} \frac{d_k(p^m)^2}{p^m},$$

conjecture that $\mathbb{E}|Z_X|^{2k} \sim \frac{G(k+1)^2}{G(2k+1)}\big(\log T/(e^{\gamma_{\mathrm{E}}}\log X)\big)^{k^2}$, the Keating Snaith moment of a matrix of size $\log T/(e^{\gamma_{\mathrm{E}}}\log X)$, and conjecture that the two factors are asymptotically independent, so that their moments multiply; they establish both conjectures when $k=1$ and $k=2$. The product recovers the moment conjecture $\mathbb{E}|\zeta|^{2k} \sim a_k \frac{G(k+1)^2}{G(2k+1)}(\log T)^{k^2}$ of Keating and Snaith and of Conrey, Farmer, Keating, Rubinstein and Snaith [96], and the dependence on X cancels. Here G is Barnes' function and d_k is the k fold divisor function. Reading off the coefficient of k^2 in the logarithm of each factor splits the variance into a prime part and a zero part. Since $\log\frac{G(k+1)^2}{G(2k+1)} = (1+\gamma_{\mathrm{E}})k^2 + O(k^3)$ and $d_k(p^m) = k/m + O(k^2)$, the variance implied for ζ is

$$V_\zeta(T) = \frac{1}{2}\Big(\log\log\frac{T}{2\pi} + \gamma_{\mathrm{E}} + 1 + A_\zeta\Big), \quad A_\zeta = \sum_p \Big(\log\Big(1-\frac{1}{p}\Big) + \mathrm{Li}_2\Big(\frac{1}{p}\Big)\Big), \tag{12.1}$$

and we computed $A_\zeta = -0.17625$, so that the constant is $\frac{1}{2}(\gamma_{\mathrm{E}}+1+A_\zeta) = 0.70048$.

For a form of degree two the same recipe gives a conjecture we can test. Write A_p for the Satake parameter of Δ at p, so $\mathrm{tr}\, A_p^m = 2\cos m\theta_p$.

CONJECTURE 12.6 (Variance at finite height; our formulation for Δ). *For t uniform on $[T,2T]$, $\mathrm{Var}\log|L(6+it,\Delta)| = V_\Delta(T) + o(1)$, where*

$$V_\Delta(T) = \frac{1}{2}\Big(\log\Big(2\log\frac{T}{2\pi}\Big) + \gamma_{\mathrm{E}} + 1 + A_\Delta\Big)$$

and

$$A_\Delta = \sum_p \Big(\log\Big(1 - \frac{1}{p}\Big) + \sum_{m \ge 1} \frac{(2\cos m\theta_p)^2}{m^2 p^m} \Big).$$

The term $\log(2\log\frac{T}{2\pi})$ is $\log N$ for the matrix size $N = 2\log\frac{T}{2\pi}$ that matches the doubled density of zeros, and A_Δ is the coefficient of k^2 in the CFKRS arithmetic factor of Δ, obtained exactly as for ζ from $d_k(\Delta; p^m) = k \operatorname{tr} A_p^m / m + O(k^2)$. The series converges by Lemma 12.2, and Exercise 2 expresses A_Δ through $L(1, \operatorname{Sym}^2 \Delta)$.

The arithmetic. From $\tau(p)$ for $p \le 10^5$ we find $A_\Delta = -0.898$: the partial sums are -0.8946, -0.8966, -0.8976 at $p \le 2500$, 10^4, 10^5, and the identity of Exercise 2 gives the limit -0.8979. Below we use the value -0.8976. At $T = 1500$ the five pieces of V_Δ are

$$\underbrace{\tfrac{1}{2}\log\log T}_{0.9948} + \underbrace{\tfrac{1}{2}\log 2}_{0.3466} + \underbrace{\tfrac{1}{2}\log\frac{\log(T/2\pi)}{\log T}}_{-0.1447} + \underbrace{\tfrac{1}{2}(\gamma_{\mathrm{E}} + 1)}_{0.7886} + \underbrace{\tfrac{1}{2}A_\Delta}_{-0.4488} = 1.5365.$$

Our sample is spread over $t \in [500, 2500]$, and the average of $V_\Delta(t)$ over that range is 1.5268, against the observed variance 1.5265. The leading term $\frac{1}{2}\log\log t$ accounts for only 65 per cent of the observed variance. The random matrix constant without the arithmetic factor, $\frac{1}{2}(\log(2\log\frac{t}{2\pi}) + \gamma_{\mathrm{E}} + 1)$, averages 1.9756 and overshoots by 0.45. The prime model alone, the variance of the random Euler product with the true Satake angles at the primes $p \le t$, gives 0.76, 0.81, 0.84, 0.87 at $t = 500$, 1000, 1500, 2500, and undershoots by about 0.7; its deficit against $\frac{1}{2}\log\log t$ comes from the negative constant B_Δ of Lemma 12.2, partly offset by the prime powers. Only the combination of Conjecture 12.6 accounts for the data, and it does so within the sampling error. The 50,000 values are strongly dependent, but batch means, a moving block bootstrap and a Bartlett long run variance, with blocks or bandwidths of 1 to 100 units of t, put the standard error of the sample variance between 0.012 and 0.018, far below the distance of 0.45 or more from each of the three alternatives above. Corrections of relative order $1/\log t$ are also expected. The agreement to the third decimal is therefore luck; the agreement within the sampling error, over a single range of heights, is evidence for the conjecture, not a test of its error term. The split of V_Δ into a prime part and a zero part is not unique, and the moment theorem of Gonek, Hughes and Keating concerns $X = O((\log T)^{2-\varepsilon})$, not $X = T$; what is tested is the combination V_Δ, in which X does not appear.

12.2 A LOG CORRELATED FIELD

12.2.1 Covariance across scales

The variance $\frac{1}{2}\log\log T$ is the sum of contributions $\frac{1}{2p}$ from the primes, and the dependence between nearby points is decided prime by prime. The following proposition makes this exact for the random model.

PROPOSITION 12.7 (Covariance of the random model; this book). *Let (U_p) be independent and uniform on the unit circle, and for $P \geq 3$ and real h put $X_P(h) = \operatorname{Re}\sum_{p\leq P} U_p p^{-1/2-ih}$. Then*

$$\begin{aligned}\operatorname{Cov}\big(X_P(h), X_P(h')\big) &= \frac{1}{2}\sum_{p\leq P}\frac{\cos\big((h-h')\log p\big)}{p}\\ &= \frac{1}{2}\log\min\Big(\log P, \frac{1}{|h-h'|}\Big) + O(1),\end{aligned}$$

uniformly in P and in $0 < |h-h'| \leq 1$. The same holds for the field $X_P^{\Delta}(h) = \operatorname{Re}\sum_{p\leq P}\lambda(p)U_p p^{-1/2-ih}$, with $\lambda(p)^2/p$ in place of $1/p$ in the sum.

PROOF. For independent uniform U, $\mathbb{E}\operatorname{Re}(Ua)\operatorname{Re}(Ub) = \frac{1}{2}\operatorname{Re}(a\bar{b})$, which gives the first identity. Put $u = |h-h'|$ and $Q = \min(P, e^{1/u})$. For $p \leq Q$ we have $\cos(u\log p) = 1 + O(u^2\log^2 p)$, and $\sum_{p\leq Q} u^2\log^2 p/p \ll u^2\log^2 Q \leq 1$ by Mertens' estimate $\sum_{p\leq y}\log p/p = \log y + O(1)$ and partial summation. So the primes $p \leq Q$ contribute $\frac{1}{2}\sum_{p\leq Q} 1/p + O(1) = \frac{1}{2}\log\log Q + O(1)$. If $P > e^{1/u}$ the remaining primes contribute $\frac{1}{2}\sum_{e^{1/u}<p\leq P}\cos(u\log p)/p$. By the prime number theorem with remainder $O(y/\log^2 y)$ and partial summation, this equals $\frac{1}{2}\int_{1/u}^{\log P}\cos(uv)\,\frac{dv}{v} + O(u)$, and the integral is $\int_1^{u\log P}\cos w\,\frac{dw}{w}$, which is bounded uniformly. Since $\log\log Q = \log\min(\log P, 1/u)$, the formula follows. For X_P^{Δ} the same argument applies with Lemma 12.2 and the prime number theorem for $L(s, \Delta\times\Delta) = \zeta(s)L(s, \operatorname{Sym}^2\Delta)$ in the form $\sum_{p\leq y}\lambda(p)^2\log p = y + O(y/\log^2 y)$, which follows from the zero free regions used in the proof of the lemma, in place of Mertens and the prime number theorem. □

At $h = h'$ the variance is $\frac{1}{2}\log\log P + O(1)$. Two points at distance u are almost perfectly correlated in the primes up to $e^{1/u}$ and almost uncorrelated beyond, so for $1/\log P \leq u \leq 1$ the correlation coefficient is $\log(1/u)/\log\log P + O(1/\log\log P)$. This is the defining property of a log correlated field. For ζ itself the analogous statement is a theorem of Bourgade [80]: for shifts $h_1, \dots, h_k$ with $|h_i - h_j| = (\log T)^{-c_{ij}+o(1)}$, $0 \leq c_{ij} \leq 1$, the vector $(\log\zeta(\frac{1}{2} + i(t+h_j)))_j/\sqrt{\frac{1}{2}\log\log T}$ is asymptotically complex Gaussian, with correlation c_{ij} between the real parts. Thus, to leading order, the covariance of

$\log|\zeta|$ at distance u is $\frac{1}{2}\log(1/u)$ for $1/\log T \le u \le 1$, saturating at $\frac{1}{2}\log\log T$ below the scale $1/\log T$ of the zero spacings.

12.2.2 The branching picture

Group the primes into blocks $e^{e^{k-1}} < p \le e^{e^k}$, $k = 1, \dots, n$ with $n = \log\log T$, and write the field of Proposition 12.7 with $P = T$ as $X_T(h) = \sum_k Y_k(h)$, with Y_k the contribution of block k. Each block has variance about $\frac{1}{2}$, since $\sum 1/p$ over a block is about 1. Inside a window $|h| \le 1$, Y_k is essentially constant on intervals of length e^{-k} and essentially independent across such intervals, because its frequencies $\log p$ are about e^k. So X_T restricted to the window is approximately a branching random walk: a tree in which every vertex at level k has about e children, each level adds an independent Gaussian increment of variance $\frac{1}{2}$, and the leaves at level n are the effectively distinct values of the field, about $e^n = \log T$ per unit length, one for each interval of length $1/\log T$, the scale of the zero spacings. Figure 12.2 shows the picture for the field

$$R_X(t) = \sum_{p \le X} \frac{2\cos\theta_p \cos(t\log p)}{\sqrt{p}} = \operatorname{Re} \sum_{p \le X} 2\cos\theta_p \, p^{-1/2-it},$$

the prime part of $\log|L_{\text{rand}}(\frac{1}{2}+it)|$ for the Sato Tate random Euler product of Chapter 11, with θ_p independent with the Sato Tate law, built up through the levels $X = e^{e^{k/2}}$, $k = 1, \dots, 5$ (that is 5.2, 15.2, 88.4, 1618, 195339), and then $X = 10^6$.

The maximum of a branching random walk is understood in detail, and it is the source of the constant $\frac{3}{4}$. We isolate the computation in a continuous model where it can be done exactly. Let $b \ge 2$ be an integer, $\sigma > 0$, and let $\{B_v(s) : 0 \le s \le n\}$, indexed by the b^n leaves v of the b ary tree of depth n, be Brownian motion indexed by the tree: each edge carries an independent Brownian path of duration one and variance σ^2 per unit time, and B_v is the concatenation along the path from the root to v. At integer times this is a branching random walk with $N(0,\sigma^2)$ increments. Put

$$c = \sigma\sqrt{2\log b}, \qquad \theta = \frac{c}{\sigma^2} = \frac{\sqrt{2\log b}}{\sigma}, \qquad x_n = cn - a\log n$$

for a constant $a > 0$.

PROPOSITION 12.8 (First moments with and without the ballot constraint; this book). *Fix $y \ge 0$. Let $N_n(y)$ be the number of leaves with $B_v(n) \ge x_n + y$, and $Z_n(y)$ the number of leaves with $B_v(n) \in [x_n + y, x_n + y + 1]$ and $B_v(s) \le \frac{s}{n}x_n + y + 1$ for all $s \in [0, n]$. As $n \to \infty$,*

$$\mathbb{E}N_n(y) \sim \frac{\sigma}{c\sqrt{2\pi}}\, n^{\theta a - \frac{1}{2}} e^{-\theta y}, \qquad \mathbb{E}Z_n(y) \sim \kappa\,(1+y)\, n^{\theta a - \frac{3}{2}} e^{-\theta y},$$

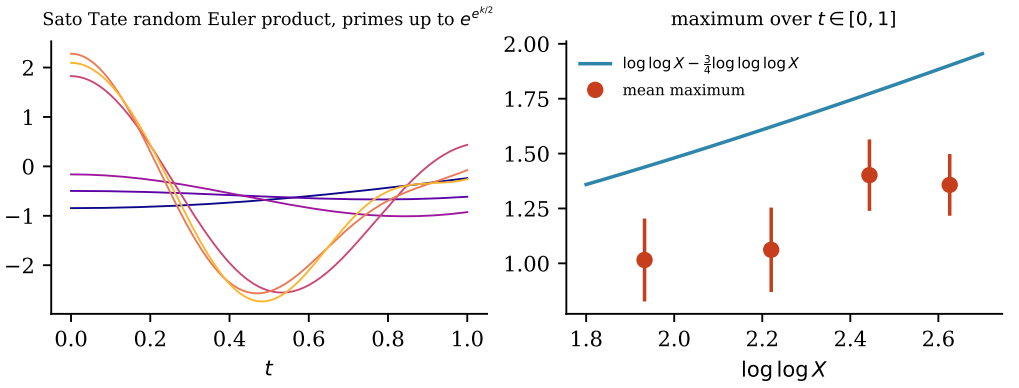


Figure 12.2: Left: one realisation of the field $R_X(t) = \sum_{p \le X} 2\cos\theta_p \cos(t\log p)/\sqrt{p}$ of the Sato Tate random Euler product on $t \in [0,1]$, built up by adding the primes in successive ranges $p \le e^{e^{k/2}}$, $k = 1, \dots, 5$, and then $p \le 10^6$; each level refines the previous one on a finer scale, which is the branching structure. Right: mean of $\max_{t\in[0,1]} R_X(t)$ for $X = 10^3, 10^4, 10^5, 10^6$ (300, 200, 80 and 30 draws) against the curve $\log\log X - \frac{3}{4}\log\log\log X$. The error bars are one standard deviation divided by $\sqrt{30}$ at every point, which overstates the standard error of the first three means. The offset is an additive constant that the theory does not determine.

with $\kappa = 2(\theta - 1 + e^{-\theta})/(\sigma^3\theta^2\sqrt{2\pi})$. *In particular the choice* $a = \frac{1}{2\theta}$ *makes* $\mathbb{E}N_n$ *of order one, while* $a = \frac{3}{2\theta}$ *makes* $\mathbb{E}Z_n(y) \to \kappa(1+y)e^{-\theta y}$, *and for that choice* $\mathbb{E}N_n(y)$ *grows like* n.

PROOF. Each $B_v(n)$ is $N(0, \sigma^2 n)$ with density φ_n. For $z = x_n + y + u$ with u bounded,

$$\begin{aligned} \frac{z^2}{2\sigma^2 n} &= \frac{c^2 n}{2\sigma^2} - \frac{c}{\sigma^2}(a\log n - y - u) + O\Big(\frac{\log^2 n}{n}\Big) \\ &= n\log b - \theta a \log n + \theta(y+u) + o(1), \end{aligned}$$

so $b^n \varphi_n(z) = (\sigma\sqrt{2\pi n})^{-1} n^{\theta a} e^{-\theta(y+u)}(1 + o(1))$. The first claim follows from $\mathbb{E}N_n(y) = b^n \mathbb{P}(B(n) \ge x_n + y)$ and the Mills ratio $\mathbb{P}(B(n) \ge z) \sim \sigma^2 n\, \varphi_n(z)/z$ with $z \sim cn$. For the second, condition on $B_v(n) = x_n + y + u$, $u \in [0,1]$. The process $R(s) = B_v(s) - \frac{s}{n}x_n$ is then a Brownian bridge of variance σ^2 per unit time from 0 to $y + u$ on $[0, n]$, and the constraint is $R \le y + 1$. By the reflection principle a bridge from 0 to e of duration n stays below a level $h > \max(0, e)$ with probability $1 - \exp(-2h(h-e)/(\sigma^2 n))$. With $h = y + 1$ and $e = y + u$ this is

$\frac{2(1+y)(1-u)}{\sigma^2 n}(1+o(1))$. Integrating against $b^n \varphi_n(x_n+y+u)\,du$ over $[0,1]$ gives

$$\mathbb{E}Z_n(y) \sim \frac{2(1+y)}{\sigma^3\sqrt{2\pi}}\, n^{\theta a - \frac{3}{2}} e^{-\theta y} \int_0^1 (1-u) e^{-\theta u}\, du,$$

and $\int_0^1 (1-u)e^{-\theta u}du = (\theta - 1 + e^{-\theta})/\theta^2$. □

The proposition explains the two corrections. The naive first moment places the maximum at $cn - \frac{1}{2\theta}\log n$. But the leaves that reach that height are overwhelmingly descendants of ancestors that were already too high at intermediate levels, and such ancestors are rare; they occur with probability tending to zero, while each of them produces a large number of high leaves, which is how the expectation can be large while the typical count is zero. Requiring the path to stay below the straight line to its endpoint, a ballot constraint, costs a factor $1/n$, and this moves the correct centring to $m_n = cn - \frac{3}{2\theta}\log n$. The factor $1+y$ is the ballot probability for a path that starts at distance $y+1$ below the barrier, and it produces a right tail of the form $ye^{-\theta y}$ for the maximum. That the constrained first moment gives the right answer is the content of the following theorem.

THEOREM 12.9 (Bramson; Addario Berry and Reed; Aïdékon). *For the branching random walk above, with $M_n = \max_v B_v(n)$ and $m_n = cn - \frac{3}{2\theta}\log n$, the sequence $M_n - m_n$ is tight, and it converges in law: there are a constant $C^* > 0$ and a positive random variable D_∞, the limit of the derivative martingale, such that*

$$\mathbb{P}(M_n \le m_n + y) \longrightarrow \mathbb{E}\big[\exp\big(-C^* D_\infty e^{-\theta y}\big)\big] \qquad (y \in \mathbb{R}).$$

The limit is a Gumbel law randomly shifted by $\theta^{-1}\log(C^* D_\infty)$. Bramson [83] proved the $\frac{3}{2\theta}\log n$ correction for branching Brownian motion, where Lalley and Sellke [341] identified the derivative martingale; Addario Berry and Reed [8] proved tightness for branching random walks, and Aïdékon [9] the convergence in law. The book of Bovier [81] and the lecture notes of Kistler [321] explain the method, whose heart is a refinement of Proposition 12.8.

For the zeta field the parameters are fixed by the variance and the number of leaves. Take $b = 2$ and $n_2 = \lfloor \log\log T / \log 2 \rfloor$ levels of variance $\sigma^2 = \frac{1}{2}\log 2$, so that a leaf has variance $\frac{1}{2}\log\log T$ and there are $2^{n_2} \approx \log T$ leaves. Then $c = \log 2$ and $\theta = 2$, so $cn_2 = \log\log T + O(1)$, $\frac{3}{2\theta}\log n_2 = \frac{3}{4}\log\log\log T + O(1)$, and the right tail is ye^{-2y}. The value $\theta = 2$ does not depend on the choice of tree: per unit of $\log\log T$ the field gains variance $\frac{1}{2}$ while the number of leaves is multiplied by e, so the speed is $c = 1$ per unit and $\theta = c/\sigma^2 = 2$.

12.2.3 The Fyodorov Hiary Keating conjecture

CONJECTURE 12.10 (Fyodorov, Hiary and Keating [187, 205]). *Let τ be uniform on $[T, 2T]$. Then*

$$\max_{|h|\le 1} \log\bigl|\zeta\bigl(\tfrac{1}{2}+i(\tau+h)\bigr)\bigr| = \log\log T - \tfrac{3}{4}\log\log\log T + \mathcal{M}_T,$$

where $\mathcal{M}_T$ converges in law to a random variable $\mathcal{M}$ with $\mathbb{P}(\mathcal{M} > y) \sim Cye^{-2y}$ as $y \to \infty$.

Fyodorov, Hiary and Keating reached the conjecture through random matrix theory, predicting by a freezing transition that $\max_\phi \log|\det(I - Ue^{-i\phi})| = \log N - \frac{3}{4}\log\log N + O(1)$ for Haar unitary U, and matching $N = \log T$. For the matrices they also predicted the law of the bounded term, and they compared their predictions with Hiary's computations of ζ at large height. The tail ye^{-2y} is expected to carry over to ζ, but the law itself is not expected to coincide with the random matrix one [205, 7]. The branching random walk gives the same centring.

What is proved is the following.

THEOREM 12.11. *Let τ be uniform on $[T, 2T]$, and write*

$$M_T = \max_{|h|\le 1} \log\bigl|\zeta\bigl(\tfrac{1}{2}+i(\tau+h)\bigr)\bigr|, \qquad m_T = \log\log T - \tfrac{3}{4}\log\log\log T.$$

1. *(First order.) For every $\varepsilon > 0$, with probability tending to one, $|M_T - \log\log T| \le \varepsilon \log\log T$. This is due to Najnudel [414], assuming the Riemann hypothesis, and to Arguin, Belius, Bourgade, Radziwiłł and Soundararajan [3] unconditionally.*
2. *(Upper tail.) Arguin, Bourgade and Radziwiłł [6] prove that $\mathbb{P}(M_T > m_T + y) \ll ye^{-2y}$ for all large T, uniformly in $y \ge 1$; in particular $M_T - m_T$ is bounded above in probability.*
3. *(Lower bounds.) In a second paper [7] the same authors prove that $\mathbb{P}(M_T < m_T - y) \to 0$ as $y \to \infty$, uniformly in large T, and that $\mathbb{P}(M_T > m_T + y) \gg ye^{-2y}$ for $10 \le y \le \sqrt{\log\log T}$. Together the two papers show that $M_T - m_T$ is tight, with right tail of exact order ye^{-2y} in that range.*

Before the work of Arguin, Bourgade and Radziwiłł, Harper [236] proved that with probability tending to one

$$\max_{|h|\le 1/2} \log\bigl|\zeta\bigl(\tfrac{1}{2}+i(\tau+h)\bigr)\bigr| \le \log\log T - \bigl(\tfrac{3}{4}+o(1)\bigr)\log\log\log T,$$

the first upper bound with the correct second term, by bounding the critical partition function $\int_{|h|\le 1/2} |\zeta(\frac{1}{2}+i(\tau+h))|^2 dh$. What remains open is the convergence in law of $\mathcal{M}_T$ (tightness gives only subsequential limits), the tail constant C, and the form of the limit, which by analogy with Theorem 12.9 and with the random

matrix case one expects to be a randomly shifted Gumbel law. For random unitary matrices more is known: the leading order is due to Arguin, Belius and Bourgade [2], the second order term $-\frac{3}{4}\log\log N$ to Paquette and Zeitouni [430], tightness at the scale $\log N - \frac{3}{4}\log\log N$ to Chhaibi, Madaule and Najnudel [106], and Paquette and Zeitouni [431] proved, for the circular β ensembles, convergence in law to the sum of a Gumbel variable and an independent variable defined through a derivative martingale.

For a degree two form the Rankin Selberg variance and the doubled density of zeros change only $O(1)$ terms in the heuristic: the leaves are the $2\log T$ effectively distinct values per unit window, and $\log\log(T^2) = \log\log T + \log 2$. We therefore expect $\max_{|h|\le 1}\log|L(6+i(\tau+h),\Delta)| = \log\log T - \frac{3}{4}\log\log\log T + O_{\mathbb{P}}(1)$, with an additive constant that differs from that of ζ. We are not aware of a proof for Δ, even to first order, though we expect the methods of [3] to extend.

Data. We split $[500, 2500]$ into 2000 unit windows and recorded the maximum of $\log|L(6+it,\Delta)|$ in each (Figure 12.1, left). The mean of the window maxima is 1.1010 with standard deviation 0.5674, and the mean of $\log\log t - \frac{3}{4}\log\log\log t$ at the window centres is 1.4655. The difference has mean -0.3645 and standard deviation 0.5657, so the standard error of the mean difference is about 0.013 (the windows are adjacent, so slightly more in truth). This is consistent with a bounded additive correction, which neither the conjecture nor the theorem determines. It is not evidence for the $\frac{3}{4}$: over $[500, 2500]$ the term $\frac{3}{4}\log\log\log t$ varies only between 0.45 and 0.54, so it cannot be separated from an unknown constant. For the field R_X of Figure 12.2 the mean maxima over $t \in [0,1]$ are 1.015, 1.062, 1.402, 1.358 for $X = 10^3, 10^4, 10^5, 10^6$, from 300, 200, 80 and 30 draws with standard deviations 1.03, 1.05, 0.89, 0.77, against $\log\log X - \frac{3}{4}\log\log\log X = 1.438, 1.622, 1.773, 1.902$. The differences, -0.42, -0.56, -0.37, -0.54, with standard errors 0.06, 0.07, 0.10, 0.14, are consistent with a constant offset (weighted mean -0.47, $\chi^2 = 3.3$ on 3 degrees of freedom), and the decrease of the mean maximum from 10^5 to 10^6 is within noise; the range of X is too short to separate the $\frac{3}{4}\log\log\log X$ term from the constant. The model is also not stationary: $\mathbb{E}R_X(t)^2 = \frac{1}{2}\sum_{p\le X}(1+\cos(2t\log p))/p$ is close to $\log\log X$ rather than $\frac{1}{2}\log\log X$ near $t = 0$, so we use it only to show the branching mechanism.

12.3 Multiplicative chaos

Exponentiating a log correlated field produces a random measure. Let Y be a centred Gaussian field on an interval with $\mathbb{E}Y(x)Y(x') = \log\frac{1}{|x-x'|} + g(x,x')$, g bounded and continuous. The field is not defined pointwise, but its regularisations Y_ε at scale ε are, and Kahane [289] showed that, for suitable regularisations and

$0 < \gamma < \sqrt{2}$, the measures

$$M_{\gamma,\varepsilon}(dx) = e^{\gamma Y_\varepsilon(x) - \frac{\gamma^2}{2}\mathbb{E}Y_\varepsilon(x)^2}\,dx$$

converge, as $\varepsilon \to 0$, to a nontrivial random measure M_γ, the *Gaussian multiplicative chaos* of parameter γ; later work showed that the limit does not depend on the regularisation [457], and it is carried by a set of points where Y is atypically large, of dimension $1 - \gamma^2/2$. For $\gamma \geq \sqrt{2}$ the normalised measures tend to zero. At the critical value $\gamma = \sqrt{2}$ a nontrivial limit is obtained either from the derivative martingale or by the Seneta Heyde normalisation $\sqrt{\log(1/\varepsilon)}\, M_{\sqrt{2},\varepsilon}$, and the two constructions agree up to a constant factor, as shown by Duplantier, Rhodes, Sheffield and Vargas [165, 166]. The survey of Rhodes and Vargas [457] gives the theory.

The dictionary with ζ follows from Proposition 12.7. The field $\log|\zeta(\frac{1}{2} + i(\tau + x))|$ has covariance $\frac{1}{2}\log\frac{1}{|x-x'|}$ down to the scale $\varepsilon = 1/\log T$, so $Y = \sqrt{2}\log|\zeta|$ is normalised as above, and

$$|\zeta|^{2\beta} = e^{2\beta\log|\zeta|} = e^{\sqrt{2}\beta Y}: \qquad \gamma = \sqrt{2}\,\beta.$$

The measures $|\zeta(\frac{1}{2} + i(\tau + x))|^{2\beta}dx$, normalised by their mean $(\log T)^{\beta^2}$ up to constants, are therefore expected to converge to subcritical chaos for $\beta < 1$, and $\beta = 1$, the measure $|\zeta|^2 dx$ whose mean is the classical Hardy Littlewood second moment, is exactly critical. The critical measure carries the information about the maximum. Its Seneta Heyde factor $\sqrt{\log\log T}$ is the inverse of the probability, of order $1/\sqrt{n}$ with $n = \log\log T$ levels, that a random walk stays below a barrier; Proposition 12.8 computes the same constraint for a bridge, whose cost is $1/n$.

Saksman and Webb proved a statement of this kind for ζ itself rather than for powers of its modulus.

THEOREM 12.12 (Saksman and Webb [473]). *Let ω be uniform on $[0,1]$. As $T \to \infty$, the random generalised function $x \mapsto \zeta(\frac{1}{2} + ix + i\omega T)$ on a fixed bounded interval converges in law, in the Sobolev space $W^{-\alpha,2}$ for any $\alpha > \frac{1}{2}$, to the random generalised function*

$$\zeta_{\mathrm{rand}}(\tfrac{1}{2} + ix) = \lim_{N\to\infty} \prod_{p \leq N} \left(1 - U_p p^{-1/2-ix}\right)^{-1},$$

with (U_p) independent and uniform on the unit circle, the limit taken in the same space. Moreover $\zeta_{\mathrm{rand}}(\frac{1}{2} + ix)$ is the product of a random smooth function and a complex Gaussian multiplicative chaos, the exponential of a complex Gaussian field whose real and imaginary parts are log correlated.

No normalisation is needed: the shifted zeta function itself has a limit, but only as a generalised function. The chaos is complex: the exponent $\sum_p U_p p^{-1/2-ix}$ has real and imaginary parts, each with covariance $\frac{1}{2}\log\frac{1}{|x-x'|} + O(1)$. Saksman

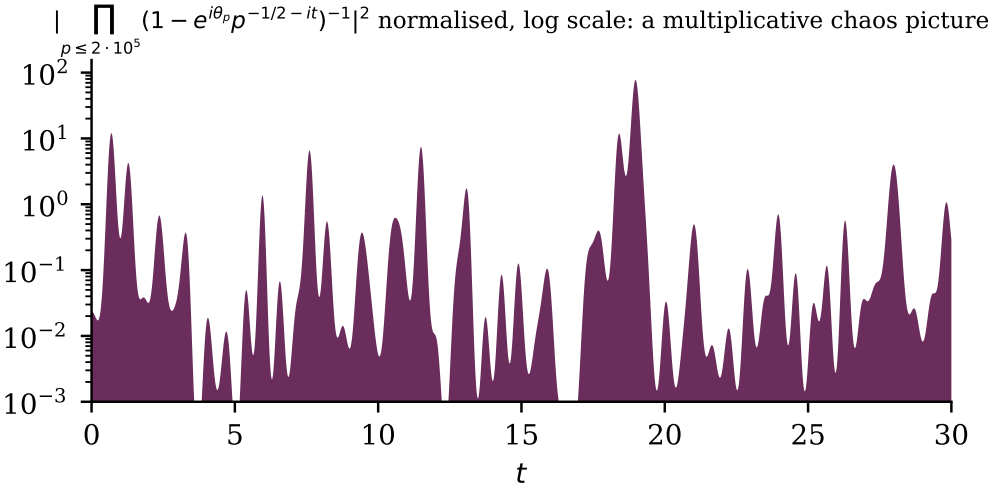


Figure 12.3: One realisation of $|\prod_{p\le 2\cdot 10^5}(1-U_p p^{-1/2-it})^{-1}|^2$ for $t \in [0,30]$, with U_p independent and uniform on the unit circle (the Steinhaus random Euler product), divided by its mean over the window, on a logarithmic scale. The mass sits on a sparse set of spikes: a picture of critical multiplicative chaos ($\beta = 1$, $\gamma = \sqrt{2}$), cut off at the scale $1/\log(2\cdot 10^5)$.

and Webb formulate the convergence of $|\zeta|^{2\beta}dx$ to subcritical chaos for $\beta < 1$ as a conjecture, together with the critical case: that

$$\sqrt{\log\log T}\,(\log T)^{-1}\left|\zeta\left(\tfrac{1}{2}+ix+i\omega T\right)\right|^2 dx$$

converges in law to a constant multiple of a critical chaos measure. Both conjectures are open in print as we write; Gorodetsky and Wong [224] report that the critical one will be established in forthcoming work of Harper, Saksman and Webb. For the truncated random Euler product the analogous critical statement is a theorem of Saksman and Webb.

Figure 12.3 shows, for a single realisation, the squared modulus of the random Euler product $\prod_{p\le N}(1-U_p p^{-1/2-it})^{-1}$ truncated at $N = 2\times 10^5$, on a logarithmic scale. The measure is concentrated in thin spikes separated by long stretches where it is small. This is the behaviour expected near criticality: for $\gamma < \sqrt{2}$ the chaos is carried by a set of dimension $1-\gamma^2/2$, which shrinks to dimension zero at the critical value.

Moments of moments. The freezing transition behind Conjecture 12.10 is visible in the moments of the chaos. For U Haar in $\mathrm{U}(N)$ and $Z_N(\phi) = \det(I - Ue^{-i\phi})$,

Fyodorov and Keating [205] studied

$$\mathrm{MoM}_N(k,\beta) = \mathbb{E}\Big[\Big(\frac{1}{2\pi}\int_0^{2\pi} |Z_N(\phi)|^{2\beta} d\phi\Big)^k\Big]$$

and conjectured that it grows like $N^{k\beta^2}$ when $k\beta^2 < 1$ and like $N^{k^2\beta^2-k+1}$ when $k\beta^2 > 1$. Below the transition the kth moment of the total mass of the limiting chaos is finite, and MoM_N is of the order of the kth power of the mean; above it, the moment is dominated by the neighbourhood of the maximum. Bailey and Keating [32] proved that for positive integers k and β, $\mathrm{MoM}_N(k,\beta)$ is a polynomial in N of degree $k^2\beta^2 - k + 1$, confirming the exponent in the second regime for integer parameters. Relatedly, Arguin, Belius and Bourgade [2] proved that the free energy $\frac{1}{\log N}\log\frac{1}{2\pi}\int_0^{2\pi}|Z_N(\phi)|^{2\beta}d\phi$ converges in probability to β^2 for $\beta \le 1$ and to $2\beta - 1$ for $\beta \ge 1$, the slope 2 above the transition reflecting a maximum of $\log|Z_N|$ equal to $\log N$ to first order.

12.4 Random multiplicative functions

The random Euler product has a Dirichlet series, and its coefficients are a random multiplicative function. In the *Steinhaus model* $f(p) = U_p$ are independent and uniform on the unit circle and f is completely multiplicative; in the *Rademacher model* of Wintner [586] $f(p) = \pm 1$ with independent fair signs and f is supported on squarefree integers. Write $S(x) = \sum_{n\le x} f(n)$.

For the Steinhaus model $\mathbb{E} f(n)\overline{f(m)} = \mathbb{I}_{n=m}$ by unique factorisation, so $\mathbb{E}|S(x)|^2 = \lfloor x\rfloor$. If $S(x)/\sqrt{x}$ were approximately a standard complex Gaussian, we would have $\mathbb{E}|S(x)| \sim \frac{\sqrt{\pi}}{2}\sqrt{x} = 0.886\sqrt{x}$. Helson [256] conjectured instead that $\mathbb{E}|S(x)| = o(\sqrt{x})$, and Harper proved it with the exact rate.

Theorem 12.13 (Harper [238]). *Let f be a Steinhaus random multiplicative function. Uniformly for all large x and $0 \le q \le 1$,*

$$\mathbb{E}|S(x)|^{2q} \asymp \Big(\frac{x}{1+(1-q)\sqrt{\log\log x}}\Big)^q.$$

In particular $\mathbb{E}|S(x)|$ is of exact order $\sqrt{x}\,(\log\log x)^{-1/4}$. The same bounds hold for the Rademacher model.

Harper calls this better than square root cancellation. The mechanism is critical multiplicative chaos, and it can be seen through an exact identity.

Lemma 12.14 (Plancherel identity). *Let f be any function on the positive integers with $|f| \le 1$, $S(u) = \sum_{n\le u} f(n)$ and $F(s) = \sum_n f(n) n^{-s}$. For $\sigma > 1$,*

$$\int_1^\infty \frac{|S(u)|^2}{u^{1+2\sigma}}\,du = \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{|F(\sigma+it)|^2}{|\sigma+it|^2}\,dt.$$

PROOF. By partial summation $F(s) = s\int_1^\infty S(u)u^{-s-1}du$ for $\operatorname{Re} s > 1$. Put $g(v) = S(e^v)e^{-\sigma v}$ for $v \geq 0$ and $g = 0$ for $v < 0$; then $g \in L^1 \cap L^2$ since $|S(u)| \leq u$ and $\sigma > 1$, and its Fourier transform is $\int_0^\infty S(e^v)e^{-(\sigma+it)v}dv = F(\sigma+it)/(\sigma+it)$. Plancherel's theorem and the substitution $u = e^v$ give the identity. □

For the Steinhaus model the identity continues to hold almost surely for $\sigma > \frac{1}{2}$, by the same argument once one knows that $S(u) \ll u^{1/2+\varepsilon}$ almost surely, which follows from the method Wintner used for the Rademacher model. Harper works at $\sigma = \frac{1}{2} + 1/\log x$, where the left side is a weighted average of $|S(u)|^2/u$ over u up to about x, on a logarithmic scale, and the right side is, up to the factor $\frac{1}{2\pi}$, the total mass of the measure $|F(\frac{1}{2} + \frac{1}{\log x} + it)|^2 dt/|\frac{1}{2} + it|^2$. Since $\log|F(\frac{1}{2} + \frac{1}{\log x} + it)|$ is approximately the log correlated field of Proposition 12.7 with $P = x$, this is a critical chaos measure, cut off at scale $1/\log x$. Its expectation is of order $\log x$, but its typical size is of order $\log x/\sqrt{\log\log x}$, the Seneta Heyde factor again. Conditionally on the values $f(p)$ for the small primes, $S(x)$ is roughly Gaussian with variance of the order of $\frac{x}{\log x}$ times the total mass, so that

$$\mathbb{E}|S(x)| \approx \sqrt{\frac{x}{\log x}} \cdot \sqrt{\frac{\log x}{\sqrt{\log\log x}}} = \frac{\sqrt{x}}{(\log\log x)^{1/4}}.$$

Harper's proof turns each step of this heuristic into an inequality, and the hard step is the upper bound for low moments of the critical chaos mass, which he proves with a barrier argument of ballot type for the random walk formed by the logarithms of the partial Euler products.

Data. We drew 400 Steinhaus functions and computed $S(x)$ for $x \leq 10^6$. The estimates of $\mathbb{E}|S(x)|/\sqrt{x}$ at $x = 10^3, 3\cdot10^3, 10^4, 3\cdot10^4, 10^5, 3\cdot10^5, 10^6$ are 0.773, 0.765, 0.807, 0.798, 0.778, 0.777, 0.781, and their ratios to $(\log\log x)^{-1/4}$ are 0.91, 0.92, 0.98, 0.99, 0.97, 0.98, 0.99 (Figure 12.4). Since $\mathbb{E}|S(x)|^2 = \lfloor x \rfloor$, the standard deviation of $|S(x)|/\sqrt{x}$ is about $\sqrt{1-0.78^2} \approx 0.63$, so each estimate has a standard error of about 0.03; the estimates at different x come from the same 400 functions and are correlated. Every estimate lies between two and a half and four standard errors below the Gaussian value 0.886, which is the qualitative content of the theorem. The rate is another matter: over this range $(\log\log x)^{-1/4}$ falls only from 0.848 to 0.786, about two standard errors, and the estimates do not decrease monotonically, so the data are consistent with the theorem but cannot confirm the exponent $\frac{1}{4}$. Gorodetsky and Wong [224] have since proved that $(\log\log x)^{1/4}S(x)/\sqrt{x}$ converges in distribution, answering a question of Harper; since Theorem 12.13 with $q = \frac{3}{4}$ makes these variables uniformly integrable, $c(x) = \mathbb{E}|S(x)|(\log\log x)^{1/4}/\sqrt{x}$ converges too. We know of no closed form for the limit, and our ratios are only rough estimates of it (Computational Problem 12.4).

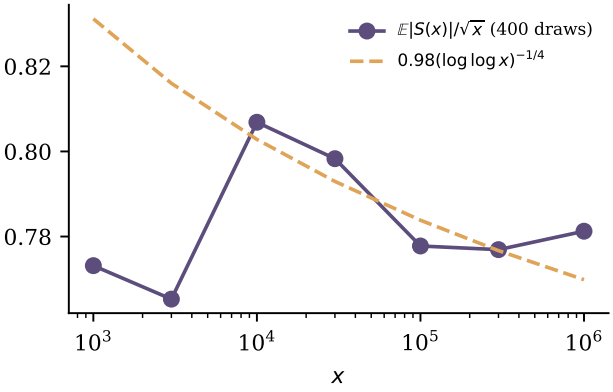


Figure 12.4: Estimates of $\mathbb{E}|S(x)|/\sqrt{x}$ for the Steinhaus random multiplicative function from 400 draws (standard error about 0.03 per point), against $0.98(\log\log x)^{-1/4}$: the rate of Theorem 12.13, with the constant 0.98 chosen to match the data, since the theorem gives none. A complex Gaussian with the same second moment would give 0.886.

Deterministic consequences. Harper transferred the mechanism to deterministic families. For a large prime r and Dirichlet characters χ modulo r, the sums $\sum_{n\le x}\chi(n)$ have mean square x over χ when $x < r$, by orthogonality; Harper [239] showed that their first absolute moment is $O(\sqrt{x}/(\log\log(10L))^{1/4})$ with $L = \min(x, r/x)$, hence $o(\sqrt{x})$ whenever $x \to \infty$ and $r/x \to \infty$, and proved the analogous statement for zeta sums $\sum_{n\le x} n^{it}$ with t uniform on $[0, T]$ and $L = \min(x, T/x)$. Averaging over characters makes the $\chi(p)$ behave like the Steinhaus $f(p)$ to the extent needed. Sharp bounds for higher moments of random multiplicative functions, where Gaussian heuristics fail in the opposite direction, are in the companion paper of Harper [237].

12.5 Pair correlation of the zeros of $L(s, \Delta)$

Up to a smooth term, $\log|L|$ is the logarithmic potential of the zeros, so its log correlated structure and its constants, such as the $\frac{1}{2}(\gamma_{\mathrm{E}} + 1)$ of Proposition 12.5, record how rigidly the zeros are spaced. We now look at the zeros directly.

Montgomery's theorem was stated in Chapter 6 (Theorem 6.4): assuming the Riemann hypothesis, the pair correlation of the normalised zeros of ζ agrees with $1 - (\sin \pi u/\pi u)^2$ for test functions whose Fourier transform is supported in $(-1, 1)$ [399]. The pair correlation conjecture asserts the same for all test functions; the stronger GUE conjecture asserts that all n level correlations, and with them all local statistics such as the nearest neighbour spacing law, are those of the Gaussian

unitary ensemble.

THEOREM 12.15 (Rudnick and Sarnak [471]). *Let π be a cuspidal automorphic representation of $\mathrm{GL}_m(\mathbb{A}_{\mathbb{Q}})$, assume the Riemann hypothesis for $L(s,\pi)$, and for $m \geq 4$ assume their Hypothesis H on the coefficients at prime powers (it has since been proved for $m = 4$, so it is needed only for $m \geq 5$). Let the ordinates of the zeros be normalised to have mean spacing one. Then for each $n \geq 2$ the n level correlations of the normalised ordinates agree with those of the Gaussian unitary ensemble, for test functions whose Fourier transform is supported in $\sum_j |\xi_j| < 2/m$.*

For $n = 2$ and a translation invariant test function the support condition reads $|\alpha| < 1/m$, so for ζ ($m = 1$) it is Montgomery's range and for $L(s,\Delta)$ ($m = 2$) it is $|\alpha| < \frac{1}{2}$. Within its range, the theorem shows that the local statistics of zeros at large height are universal: they do not see the degree, the level or the Sato Tate group, in contrast with the low lying zeros of families of Chapter 6. For Δ we normalise by the smooth counting function: with ϑ the phase of the Hardy function Z_Δ of the Conventions,

$$\vartheta(t) = \operatorname{Im}\log\Gamma(6+it) - t\log 2\pi = t\log\frac{t}{2\pi e} + \frac{11\pi}{4} + O\Big(\frac{1}{t}\Big),$$

the number of zeros with ordinates in $(T_1, T_2]$ is $(\vartheta(T_2) - \vartheta(T_1))/\pi + O(\log T_2)$, and the unfolded ordinates are $\tilde{\gamma} = \vartheta(\gamma)/\pi$.

The nearest neighbour spacing law of the Gaussian unitary ensemble is Gaudin's [209]: $p_2(s) = \frac{d^2}{ds^2}\det(I - K_s)$, where K_s is the sine kernel $\frac{\sin\pi(x-y)}{\pi(x-y)}$ acting on $L^2[0,s]$; its variance is about 0.180 [391]. The Wigner surmise $p_W(s) = \frac{32}{\pi^2}s^2e^{-4s^2/\pi}$, the exact law for 2×2 matrices, is a close approximation with variance $\frac{3\pi}{8} - 1 = 0.178$.

Data. We located the zeros of $L(s,\Delta)$ with ordinates in $[500, 2000]$ as sign changes of Z_Δ on a grid of step 0.04, computed with PARI/GP. There are 2495 of them, against $(\vartheta(2000) - \vartheta(500))/\pi = 2494.8$. Sign changes miss any pair of zeros that falls in one grid cell, so we repeated the search on a grid of step 0.01: it finds the same 2495 zeros, with smallest spacing 0.122 in normalised units. Under the Gaudin law, with density about $\frac{\pi^2}{3}s^2$ near 0, the expected number of the 2494 spacings below the grid step is 0.83 for the step 0.04 and 0.013 for the step 0.01, so it is very unlikely that any zero was missed. Figure 12.5 shows the results. Over 30 bins of width 0.1 on $[0,3]$ the empirical pair correlation differs from Montgomery's function by at most 0.122 (in the bin at $u = 1.95$); each bin contains about 250 pairs, so the relative sampling error per bin is about 0.06, and the largest of 30 deviations is expected to be of about twice that. Repulsion is clear, and stronger than in the limit law: the eight bins with $u < 0.8$ all lie below the curve and together hold 783 pairs where Montgomery's function predicts 878, and in the

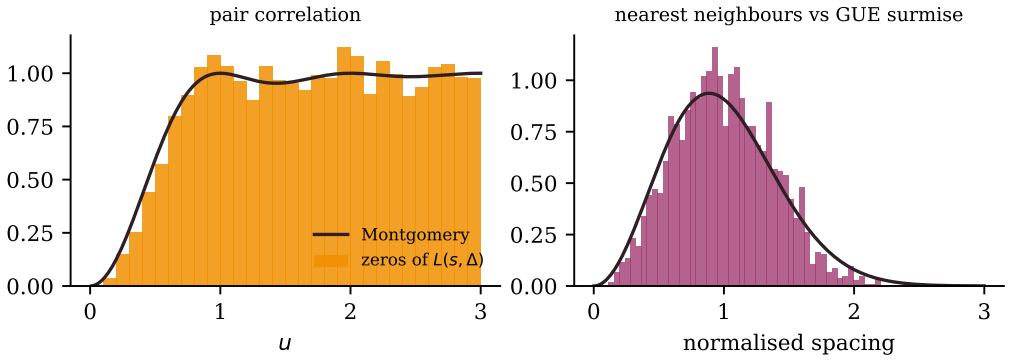


Figure 12.5: Zeros of $L(s,\Delta)$ with ordinates in $[500, 2000]$ (2495 zeros, located as sign changes of Z_Δ on a grid of step 0.04). Left: empirical pair correlation of the unfolded ordinates in bins of width 0.1, against Montgomery's function $1-(\sin \pi u/\pi u)^2$. Right: histogram of the 2494 normalised nearest neighbour spacings against the Wigner surmise $\frac{32}{\pi^2}s^2e^{-4s^2/\pi}$.

first bin, where $1-(\sin \pi u/\pi u)^2 \approx \frac{\pi^2}{3}u^2$ predicts about 2.7 pairs, there is none. The recount shows that missed close pairs do not account for this; it is consistent with the finite height rigidity that the spacings also show.

The nearest neighbour spacings are more informative. Their variance is 0.138, against 0.178 for the surmise and 0.180 for the Gaudin law, and the Kolmogorov distance to the surmise is 0.037, which for 2494 spacings corresponds to a nominal p value of about 0.002 (nominal, because spacings are not independent). The zeros of Δ at this height are thus more rigid than the asymptotic law predicts. This is the expected direction. Bogomolny, Bohigas, Leboeuf and Monastra [40] argued, with numerical support, that at finite height the spacings of the zeros of ζ are better described by the spacings of a unitary matrix of an effective size proportional to $\log \frac{T}{2\pi}$, with an explicit arithmetic constant of proportionality, and that this effective size is small at low height; the spacing law of a small unitary matrix is narrower than Gaudin's law. Behind this is the pair correlation formula of Bogomolny and Keating [69], derived from the Hardy Littlewood conjectures, which gives lower order arithmetic corrections to $1-(\sin \pi u/\pi u)^2$. We have not computed the analogous effective size for Δ, whose zeros are twice as dense, and we do not claim that it explains the value 0.138 quantitatively; this is part of Computational Problem 12.3.

12.6 The field through the Thorin lens

The chapters that follow reformulate $L(s,\Delta)$ in terms of a probability law, the Pólya variable W of Chapter 13 with density $\Phi_\Delta(u) \propto \Delta(ie^u)e^{6u}$, its Thorin measure (Chapter 14), and the extreme value variable M^* of Chapter 27. Here we record what the results of this chapter say in that language. Everything in this section is an exact consequence of the cited theorems, and none of it is evidence for or against the Riemann hypothesis.

Proposition 12.16 (Selberg's theorem as a statement about a characteristic function; this book). *Let $\varphi_W(t) = \mathbb{E}e^{itW}$. For real t,*

$$\log|L(6+it,\Delta)| = \log|\varphi_W(t)| - \log|\Gamma(6+it)| + \log\big(\Gamma(6)L(6,\Delta)\big),$$

and, as $|t| \to \infty$,

$$\log|L(6+it,\Delta)| = \log|\varphi_W(t)| + \frac{\pi}{2}|t| - \frac{11}{2}\log|t| + C_\Delta + O\Big(\frac{1}{|t|}\Big),$$

with $C_\Delta = \log(120\,L(6,\Delta)) - \frac{1}{2}\log 2\pi = 3.6355$. Consequently, under the hypothesis of Theorem 12.3, for t uniform on $[T,2T]$,

$$\frac{\log|\varphi_W(t)| + \frac{\pi}{2}t - \frac{11}{2}\log t + C_\Delta}{\sqrt{\frac{1}{2}\log\log T}} \Longrightarrow N(0,1).$$

Proof. By Theorem 13.6, $\varphi_W(t) = \Lambda(6+it,\Delta)/\Lambda(6,\Delta)$, and $|\Lambda(6+it,\Delta)| = (2\pi)^{-6}|\Gamma(6+it)||L(6+it,\Delta)|$, $\Lambda(6,\Delta) = (2\pi)^{-6}\Gamma(6)L(6,\Delta)$. This gives the identity. Stirling's formula $\log|\Gamma(\sigma+it)| = (\sigma-\frac{1}{2})\log|t| - \frac{\pi}{2}|t| + \frac{1}{2}\log 2\pi + O(1/|t|)$ at $\sigma = 6$ gives the expansion, with $L(6,\Delta) = 0.792123$. The last statement is Theorem 12.3 rewritten. □

The density Φ_Δ extends analytically to the strip $|\operatorname{Im} u| < \frac{\pi}{2}$, since ie^u stays in the upper half plane there, and it is this strip that forces $|\varphi_W(t)|$ to decay like $e^{-\pi|t|/2}$ up to powers of t. Under the Riemann hypothesis for Δ, Proposition 12.16 says that on top of this deterministic decay the characteristic function of an explicit, smooth, symmetric probability law carries lognormal fluctuations of variance $\frac{1}{2}\log\log t$. By Theorem 12.15 its real zeros, which are the zeros of $L(s,\Delta)$ on the critical line, have GUE correlations for test functions in the restricted range, and the heuristics of Section 12.2 predict that the maxima of $\log|L(6+it,\Delta)|$ over unit windows follow the Fyodorov Hiary Keating asymptotics. The same holds for the Pólya kernel of the Riemann Ξ function in Chapter 13, with the lognormal fluctuations unconditional by Theorem 12.1 and, under the Riemann hypothesis for ζ, Montgomery's theorem in place of Theorem 12.15.

Under the Riemann hypothesis for Δ the same function is a logarithmic potential. By Chapter 14 the function $s \mapsto \Lambda(6)/\Lambda(6+\sqrt{s})$ is then the Laplace transform of the generalized gamma convolution $Y = \sum_k E_k/\gamma_k^2$, with E_k independent standard exponentials and Thorin measure $\tau_\Delta = \sum_k \delta_{\gamma_k^2}$.

PROPOSITION 12.17 (The field as the potential of the Thorin measure; this book). *Assume the Riemann hypothesis for $L(s,\Delta)$, with zeros $6 \pm i\gamma_k$, $\gamma_k > 0$. For real $t \neq \pm\gamma_k$,*

$$\log|\varphi_W(t)| = \sum_k \log\Big|1 - \frac{t^2}{\gamma_k^2}\Big| = \int_{(0,\infty)} \log\Big|1 - \frac{t^2}{x}\Big|\,\tau_\Delta(dx) = -\log\big|\mathcal{L}_Y(t^2)\big|,$$

where $\mathcal{L}_Y(u) = \prod_k (1-u/\gamma_k^2)^{-1}$ is the meromorphic continuation of the moment generating function $\mathbb{E}e^{uY}$, $u < \gamma_1^2$, to $\mathbb{C}$. Moreover, on each interval between consecutive zeros, $t \mapsto \log|\varphi_W(t)|$ is strictly concave.

PROOF. Theorem 13.7 gives $\Lambda(6+z)/\Lambda(6) = \prod_k (1+z^2/\gamma_k^2)$ for all z, with $\sum_k \gamma_k^{-2} < \infty$; put $z = it$ and take logarithms of absolute values, which is legitimate because the product converges absolutely. The integral is the same sum written against τ_Δ. For $u < \gamma_1^2$, independence gives $\mathbb{E}e^{uY} = \prod_k \mathbb{E}e^{uE_k/\gamma_k^2} = \prod_k (1-u/\gamma_k^2)^{-1}$, and the product converges to a meromorphic function of u with simple poles at the γ_k^2. Finally $\frac{d^2}{dt^2}\log|1-t^2/\gamma^2| = -2(\gamma^2+t^2)/(\gamma^2-t^2)^2 < 0$, and a convergent sum of strictly concave functions is strictly concave. $\square$

So, under the Riemann hypothesis, the field of this chapter is, after the archimedean drift, the logarithmic potential of the Thorin measure, evaluated at the point t^2 of the half line that carries the measure. The log correlated picture predicts that this potential, for atoms with sine kernel statistics, fluctuates like a log correlated Gaussian field; for random unitary matrices Hughes, Keating and O'Connell [262] proved that the logarithm of the characteristic polynomial along the circle converges, as a random generalised function, to such a field. The hybrid product also has a reading here, which we offer only as a heuristic: Z_X involves the zeros within about $1/\log X$ of t, that is, the Thorin atoms within about $2t/\log X$ of t^2, while the potential of the far atoms is smooth in t and is carried by the primes $p \le X$ through the explicit formula used in Chapter 16. The prime and zero parts of V_Δ are then the variances of the far and near potentials.

The extreme value representation of Chapter 27 is a third view of the same object. There $\Delta(iy) = e^{-2\pi y}F(y)$ with F the distribution function of M^*, the maximum of the independent variables $E_{j,n}/(2\pi n)$, and $L(s,\Delta) = \mathbb{E}[Q(s, 2\pi M^*)]$ (Theorem 27.2). Proposition 27.6 there identifies Φ_Δ as a conditional law; on the critical line it reads as follows.

PROPOSITION 12.18 (The critical line through M^*; this book). *Let G have the gamma law of shape 6 and unit rate, independent of M^*. For real t,*

$$\frac{\Gamma(6+it)}{\Gamma(6)}\,L(6+it,\Delta) = \mathbb{E}\big[G^{it}\,;\,G \geq 2\pi M^*\big].$$

The finite measure $\nu(B) = \mathbb{P}(\log G \in B,\ G \geq 2\pi M^*)$ has total mass $L(6,\Delta)$ and equals $L(6,\Delta)$ times the law of $W + \log 2\pi$.

PROOF. Proposition 27.6, with $G_6 = G$, says that conditional on $\{G \geq 2\pi M^*\}$, an event of probability $L(6,\Delta)$ by Theorem 27.2, the variable $\log(G/2\pi)$ has density Φ_Δ, which gives the second statement. At $z = it$ the proposition yields $\mathbb{E}[(G/2\pi)^{it}; G \geq 2\pi M^*] = L(6,\Delta)\Lambda(6+it,\Delta)/\Lambda(6,\Delta)$, and $\Lambda(s,\Delta) = (2\pi)^{-s}\Gamma(s)L(s,\Delta)$ turns this into the first. □

So the Pólya variable is the logarithm of a gamma variable conditioned to beat $2\pi M^*$, recentred. Two kinds of extremes should not be confused here. The variable M^* is a maximum of independent exponentials on the Mellin side, and it determines $L(s,\Delta)$ for every s; the maxima of $\log|L(6+it,\Delta)|$ over unit windows, the analogue for Δ of Conjecture 12.10, are maxima of a log correlated field on the spectral side. Proposition 12.18 writes $L(6+it,\Delta)$ as the Fourier transform of a measure built from G and M^*, but we see no direct route from the law of M^* to those maxima beyond the decay rate of Proposition 12.16.

12.7 NOTES

Arguin, Belius and Harper [4] first proved the two leading terms of Conjecture 12.10 for a randomised model of ζ, and the proofs for ζ itself collected in Theorem 12.11 adapt the branching random walk methods behind Theorem 12.9.

Our own contributions are modest and labelled: Proposition 12.4 (an elementary fact we use to explain the left tail), Proposition 12.7 (the covariance of the random model, a standard computation written out), Proposition 12.8 (the ballot computation in a model where it is exact), Conjecture 12.6 (the CFKRS and hybrid heuristics written out for Δ, with the constant A_Δ computed), and the three propositions of Section 12.6. Proposition 12.5 is Keating and Snaith's, in our presentation. All data in this chapter come from our PARI/GP and Python computations: the 50,000 values of $\log|L(6+it,\Delta)|$, the 2000 window maxima, the 2495 zeros, the random field and the Steinhaus simulations, and the constants A_ζ, A_Δ, B_Δ.

12.8 EXERCISES

1. Using Proposition 12.5, compute $\kappa_5(\infty)$ and $\kappa_6(\infty)$ for $\log|\Lambda_N|$, and show that $(-1)^j\kappa_j(\infty) > 0$ for all $j \geq 3$. Deduce that, in standard units, the

skewness of $\log|\Lambda_N|$ is asymptotic to $-\frac{\pi^2}{4}\big(\frac{1}{2}\log N\big)^{-3/2}$. Compare with the data of Section 12.1.3.

2. Using $L(s,\Delta\times\Delta)=\zeta(s)L(s,\mathrm{Sym}^2\Delta)$ and the prime number theorem for $\mathrm{Sym}^2\Delta$, prove that

$$A_\Delta=\log L(1,\mathrm{Sym}^2\Delta)-\sum_p\sum_{m\ge 2}\Big(\frac{1}{m}-\frac{1}{m^2}\Big)\frac{(2\cos m\theta_p)^2}{p^m},$$

and show that the analogous identity for ζ reads $A_\zeta=-\sum_p\sum_{m\ge2}(\frac{1}{m}-\frac{1}{m^2})p^{-m}$. Check numerically that $A_\zeta=-0.1762$ using the primes below 10^4.

3. Prove that a Brownian bridge of variance σ^2 per unit time from 0 to e on $[0,n]$ stays below $h>\max(0,e)$ with probability $1-\exp(-2h(h-e)/(\sigma^2 n))$. Using Proposition 12.8, find m_n for the b ary tree with $\sigma^2=\frac{1}{2}\log b$ and check that cn and θ do not depend on b when n is measured in units of $\log b$.

4. (Probabilistic.) For the Steinhaus model show that $\mathbb{E}|S(x)|^4$ equals the number of quadruples (n_1,n_2,n_3,n_4) in $[1,x]^4$ with $n_1n_2=n_3n_4$, and that this is $\asymp x^2\log x$. Explain why this shows that $S(x)/\sqrt{x}$ is not asymptotically Gaussian, and why it is consistent with Theorem 12.13 by Hölder's inequality.

5. Prove Lemma 12.14 in detail, and use it with $f(n)=1$ to show that for $\sigma>1$

$$\int_1^\infty\frac{\lfloor u\rfloor^2}{u^{1+2\sigma}}\,du=\frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{|\zeta(\sigma+it)|^2}{|\sigma+it|^2}\,dt.$$

Evaluate both sides asymptotically as $\sigma\to1^+$.

6. Assume the Riemann hypothesis for Δ. Using Propositions 12.16 and 12.17, show that between two consecutive zeros $\gamma_k<\gamma_{k+1}$ the function $\log|L(6+it,\Delta)|$ has exactly one critical point once γ_k is large enough. (Hint: compare the second derivative of the archimedean term with that of the potential, and use that the gaps $\gamma_{k+1}-\gamma_k$ are bounded, by the counting formula of Section 12.5.)

12.9 Open computational problems

Computational Problem 12.1 (Selberg's theorem and the hybrid constant at height 10^8). Objective: test Conjecture 12.6 and the cumulant predictions of Proposition 12.5 for $L(s,\Delta)$ at heights where $\frac{1}{2}\log\log T$ has moved appreciably. Objects and scale: $\log|L(6+it,\Delta)|$ on grids of spacing 0.01 in windows of length 10^4 centred at $T=10^4,10^5,10^6,10^7,10^8$ (about 10^6 values per window), and A_Δ to five digits, from $\tau(p)$ for all $p\le10^9$ or, more cheaply, from the identity of Exercise 2

with $L(1, \mathrm{Sym}^2 \Delta) = \frac{\pi}{2} \frac{(4\pi)^{12}}{11!} \langle \Delta, \Delta \rangle$, where $\langle \Delta, \Delta \rangle = \int_{\mathrm{SL}_2(\mathbb{Z}) \backslash \mathbb{H}} |\Delta(\tau)|^2 y^{12} \, dx \, dy / y^2$ is computed by PARI/GP `mfpetersson`. Route: an Odlyzko Schönhage type multi evaluation [426] of the approximate functional equation for Δ, or interval arithmetic in the style of Platt for certified values, checked against PARI/GP `lfun`. Deliverable: a table of sample variance, κ_3, κ_4 and their standard errors (by block bootstrap over zero spacings) against $V_\Delta(T)$ and $\kappa_j(\infty)$; and a fit of the residual $\mathrm{Var} - V_\Delta(T)$ against $1/\log T$. Why it matters: it would show whether the agreement found at $T \approx 1500$ persists, and measure the lower order terms that the hybrid heuristic leaves out.

COMPUTATIONAL PROBLEM 12.2 (The law of the maximum). Objective: estimate the distribution of $M_T - m_T$ (Theorem 12.11) for ζ and for $L(s, \Delta)$ and compare it with a randomly shifted Gumbel law. Objects and scale: 10^5 unit windows at each of the heights 10^{10}, 10^{16}, 10^{22} for ζ and 10^5 windows at heights up to 10^8 for Δ; in parallel, 10^6 realisations of the Steinhaus and Sato Tate random Euler products and of a matched branching random walk, recording its derivative martingale. Route: for ζ and Δ, Odlyzko Schönhage multi evaluation [426] (for ζ, also Hiary's fast algorithms) on fine grids with local refinement of maxima; for the models, GPU evaluation of truncated Euler products, and a fit of $\mathbb{P}(M \le y)$ to $\mathbb{E} \exp(-C^* D_\infty e^{-2y})$. Deliverable: empirical distribution functions of $M_T - m_T$, estimates of the additive constants for ζ, Δ and the models, tail plots of $\log \mathbb{P}(M > y) + 2y - \log y$, and a test of the shape $y e^{-2y}$. Why it matters: the limit law is the open part of Conjecture 12.10.

COMPUTATIONAL PROBLEM 12.3 (Pair correlation and spacings for Δ to height 10^7). Objective: measure the finite height corrections to the GUE statistics of the zeros of $L(s, \Delta)$. Objects and scale: all zeros with $0 < \gamma \le 10^5$, about 2.8×10^5, certified by Turing's method, and blocks of 10^5 consecutive zeros near $\gamma = 10^6$ and 10^7. Route: PARI/GP `lfunzeros` or Rubinstein's `lcalc`, with a dense search near small gaps to recover close pairs; the Bogomolny Keating formula [69] adapted to Δ, and spacing laws of small unitary matrices from Fredholm determinants. Deliverable: pair correlation residuals against $1 - (\sin \pi u / \pi u)^2$ with error bands, the spacing variance as a function of height against 0.180, and the effective matrix size for Δ that best fits the spacing law at each height, compared with the analogue of the prediction of [40]. Why it matters: it would show whether finite height effects account for the narrow spacing law (variance 0.138) observed near $t \approx 10^3$, and test the arithmetic corrections in degree two.

COMPUTATIONAL PROBLEM 12.4 (Harper's constant and critical chaos). Objective: estimate the limit of $c(x) = \mathbb{E}|S(x)| (\log \log x)^{1/4} / \sqrt{x}$ for the Steinhaus model, which exists by the theorem of Gorodetsky and Wong (Section 12.4), and the speed of convergence; decide numerically whether $c(x)$ converges for the Rademacher model; and relate the limits to critical chaos. Objects and scale: 10^4 realisations of

each model with x up to 10^{10}, and for each realisation the total mass $\int_{-1/2}^{1/2} |F_x(\frac{1}{2} + \frac{1}{\log x} + it)|^2 dt$ of the truncated Euler product over $p \le x$. Route: segmented sieves on GPUs generating $f(n)$ from random values at primes; variance reduction by conditioning on the small primes. Deliverable: plots of $c(x)$ with confidence bands; the joint law of $S(x)/\sqrt{x}$ and the chaos mass, with a test of the conditional Gaussian picture of Section 12.4; and the same for the character sums of [239] with r prime near 10^{12}. Why it matters: Harper's theorem gives the order of magnitude but not a constant, and we know of no closed form for the limit of $c(x)$; a numerical value of the limit, its rate of approach, and its relation to the critical measure would indicate what the next theorem should be.

CHAPTER 13

PÓLYA KERNELS, GENERALIZED GAMMA CONVOLUTIONS AND THORIN MEASURES

In 1926 Pólya studied Riemann's Ξ function as the Fourier transform of an explicit positive even kernel and asked which properties of a kernel force its transform to have only real zeros. For Riemann's kernel the question is the Riemann hypothesis, still open; it led to de Bruijn's heat flow and the de Bruijn Newman constant. Two later strands from probability, Thorin's generalized gamma convolutions and the Brownian identities surveyed by Biane, Pitman and Yor, supply the language of this chapter. We transport Pólya's framework from ζ to an automorphic L function of GL_2. For the discriminant form Δ the kernel is Δ itself on the imaginary axis, read in logarithmic coordinates; the corresponding random variable has a characteristic function that vanishes at t exactly when $L(6+it,\Delta)=0$; and the Riemann hypothesis for Δ becomes the statement that a certain reciprocal is the Laplace transform of a generalized gamma convolution with an explicit Thorin measure. We prove these equivalences in full. They reformulate the Riemann hypothesis for Δ without proving it, and we say exactly what is missing.

13.1 THE PÓLYA KERNEL OF THE RIEMANN ZETA FUNCTION

Let $\xi(s)=\frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ and $\Xi(t)=\xi(\frac{1}{2}+it)$. Riemann [463] observed that Ξ is entire, even and real on $\mathbb{R}$, and the Riemann hypothesis is the statement that all zeros of Ξ are real. From the theta transformation formula one obtains

$$\Xi(t)=\int_{-\infty}^{\infty}\Phi(u)\,e^{itu}\,du, \tag{13.1}$$

$$\Phi(u)=2\sum_{n\geq 1}\bigl(2\pi^2n^4e^{9u/2}-3\pi n^2e^{5u/2}\bigr)\exp\bigl(-\pi n^2e^{2u}\bigr).$$

The kernel Φ is even, strictly positive, and decays like $\exp(-\pi e^{2|u|}+\frac{9}{2}|u|)$. After normalisation by $\Xi(0)$ it is the density of a symmetric random variable whose characteristic function is $\Xi(t)/\Xi(0)$.

Pólya [444] proved that an explicit approximation of Φ built from its leading term has a transform with only real zeros, and in a sequel [445] he identified multipliers of a kernel, among them $e^{\lambda u^2}$ with $\lambda>0$, that preserve the reality of

the zeros of its transform. De Bruijn [133] made the second idea quantitative. Put

$$H_\lambda(z) = \int_{-\infty}^{\infty} e^{\lambda u^2} \Phi(u)\, e^{izu}\, du,$$

so that $H_0 = \Xi$. If all zeros of H_0 lie in the strip $|\operatorname{Im} z| \le b_0$, then for $\lambda > 0$ all zeros of H_λ lie in $|\operatorname{Im} z| \le \sqrt{\max(b_0^2 - 2\lambda, 0)}$. For Ξ one may take $b_0 = \frac{1}{2}$, so H_λ has only real zeros for $\lambda \ge \frac{1}{8}$. Newman [417] showed that there is a real constant λ_0 such that H_λ has only real zeros exactly when $\lambda \ge \lambda_0$, and conjectured $\lambda_0 \ge 0$; the Riemann hypothesis is the statement $\lambda_0 \le 0$. The literature normalises the heat flow by $H_0(z) = \frac{1}{8}\Xi(z/2)$, in which the threshold is the de Bruijn Newman constant $\Lambda = 4\lambda_0$ and de Bruijn's bound reads $\Lambda \le \frac{1}{2}$. Rodgers and Tao [465] proved Newman's conjecture $\Lambda \ge 0$, and the Polymath project [446] proved $\Lambda \le 0.22$.

13.1.1 The Laguerre Pólya class and Pólya frequency functions

An entire function belongs to the *Laguerre Pólya class* if it is a locally uniform limit of real polynomials with only real zeros; by the theorem of Laguerre and Pólya these are the real entire functions of genus at most one with only real zeros, times a factor $e^{-\gamma z^2}$ with $\gamma \ge 0$ [361]. The Riemann hypothesis is the statement that Ξ is in the Laguerre Pólya class. A function f on $\mathbb{R}$ is a *Pólya frequency function* of infinite order if every determinant $\det[f(x_i - y_j)]$ with $x_1 < \cdots < x_n$ and $y_1 < \cdots < y_n$ is nonnegative.

THEOREM 13.1 (Schoenberg). *An integrable function f on $\mathbb{R}$, not almost everywhere zero, is a Pólya frequency function of infinite order if and only if its bilateral Laplace transform is $\int e^{-sx} f(x)\, dx = 1/\Psi(s)$ in a vertical strip containing the imaginary axis, with*

$$\Psi(s) = Ce^{-\gamma s^2 + \delta s} \prod_k (1 + \delta_k s) e^{-\delta_k s},$$

where $C > 0$, $\gamma \ge 0$, $\delta, \delta_k \in \mathbb{R}$ and $0 < \gamma + \sum_k \delta_k^2 < \infty$.

The functions Ψ in Schoenberg's theorem [488, 293] are exactly the members of the Laguerre Pólya class that are positive at the origin and not of the form $Ce^{\delta s}$. Thus a real entire function Ψ with $\Psi(0) > 0$, equal to $e^{-\gamma s^2}$ with $\gamma \ge 0$ times a function of genus at most one and not of the form $Ce^{\delta s}$, has only real zeros exactly when $1/\Psi$ is, near the imaginary axis, the bilateral Laplace transform of a Pólya frequency function.

13.2 GAMMA CONVOLUTIONS AND THORIN MEASURES

DEFINITION 13.2. A probability law on $[0, \infty)$ is a *generalized gamma convolution* (GGC) if it is a weak limit of finite convolutions of gamma laws. Equivalently

[546, 73], its Laplace transform is

$$\varphi(s) = \exp\Big(-as - \int_{(0,\infty)} \log\Big(1 + \frac{s}{t}\Big)\,\tau(dt)\Big), \qquad s \geq 0,$$

with $a \geq 0$ and a positive measure τ on $(0,\infty)$, finite on compact sets, the *Thorin measure*, satisfying $\int_{(0,1]} |\log t|\,\tau(dt) < \infty$ and $\int_{(1,\infty)} t^{-1}\,\tau(dt) < \infty$. These conditions say exactly that the integral in the exponent is finite for $s > 0$.

By Frullani's integral $\log(1 + s/t) = \int_0^\infty (1 - e^{-sx}) e^{-tx} x^{-1}\,dx$, so a GGC is infinitely divisible with drift a and Lévy measure $k(x)x^{-1}dx$, where

$$k(x) = \int e^{-tx}\,\tau(dt)$$

is completely monotone. Conversely an infinitely divisible law on $[0,\infty)$ whose Lévy measure has density $k(x)/x$ with k completely monotone is a GGC, and τ is the Bernstein measure of k. A gamma law with shape α and rate t has Thorin measure $\alpha\delta_t$. GGCs are self decomposable, hence unimodal, and the class is closed under convolution and weak limits [73].

LEMMA 13.3. *The Laplace transform of a GGC extends to a holomorphic function on* $\mathbb{C} \setminus (-\infty, 0]$ *with no zeros there.*

PROOF. For s in a compact $K \subset \mathbb{C} \setminus (-\infty, 0]$ and $t > 0$, the number $1 + s/t$ avoids $(-\infty, 0]$, so the principal logarithm $\log(1 + s/t) = \log(t + s) - \log t$ is holomorphic in s. It is bounded by $2|s|/t$ for $t \geq 2\max_K |s|$, by $|\log t| + C_K$ for $t \leq 1$, and by a constant on the compact range between. By the conditions on τ the integral converges absolutely and uniformly on K, so the exponent is holomorphic on $\mathbb{C} \setminus (-\infty, 0]$, it agrees with $\log\varphi$ for $s > 0$, and its exponential has no zeros. □

13.2.1 Brownian motion and the Riemann ξ function

Biane, Pitman and Yor [82] survey a probabilistic reading of ξ through Brownian motion. Let T be the first hitting time of 1 by a three dimensional Bessel process started at 0. Then $\mathbb{E}e^{-\lambda T} = \sqrt{2\lambda}/\sinh\sqrt{2\lambda}$, and the product formula for sinh gives

$$\mathbb{E}e^{-\lambda(T+\hat{T})} = \Big(\frac{\sqrt{2\lambda}}{\sinh\sqrt{2\lambda}}\Big)^2 = \prod_{n\geq 1}\Big(1 + \frac{2\lambda}{\pi^2 n^2}\Big)^{-2}$$

for an independent copy $\hat{T}$. Hence $T + \hat{T}$ is a GGC with Thorin measure $2\sum_n \delta_{\pi^2 n^2/2}$: it has the law of $\sum_n 2G_n/(\pi^2 n^2)$ with G_n independent gamma variables of shape 2. The variable $Y = \sqrt{\pi(T + \hat{T})/2}$, which has the law of $\sqrt{2/\pi}$ times the maximum of a standard Brownian excursion, satisfies

$$\mathbb{E}[Y^s] = 2\xi(s) \qquad \text{for all } s \in \mathbb{C}.$$

The functional equation $\xi(s) = \xi(1-s)$ becomes a distributional identity for Y, and the Euler product plays no role. The identity $\xi(\frac{1}{2}+z)/\xi(\frac{1}{2}) = \mathbb{E}[Y^{1/2+z}]/\mathbb{E}[Y^{1/2}]$ shows that Pólya's density $\Phi/\Xi(0)$ is the density of $\log Y$ when the law of Y is size biased by $Y^{1/2}$. Theorem 13.6 below gives the analogue for Δ: $\mathbb{E}[(e^W)^{s-6}] = \Lambda(s,\Delta)/\Lambda(6,\Delta)$, and the functional equation is the symmetry of W.

13.2.2 The reciprocal Xi function

The first author [441] studied the reciprocal of Ξ in the variable $\sqrt{s}$. Since Ξ is even and of order one, $t \mapsto \Xi(\sqrt{t})$ is entire of order one half, and if all zeros of Ξ are real, say $\pm\gamma_k$ with $\gamma_k > 0$, Hadamard's theorem gives

$$\frac{\xi(\frac{1}{2})}{\xi(\frac{1}{2}+\sqrt{s})} = \prod_k \Big(1 + \frac{s}{\gamma_k^2}\Big)^{-1},$$

which is the Laplace transform of the GGC $\sum_k E_k/\gamma_k^2$, with E_k independent standard exponential variables and Thorin measure $\sum_k \delta_{\gamma_k^2}$; conversely the GGC property forces the zeros onto the line, by the argument of Theorem 13.8 below. That paper develops this Thorin representation, its Lévy measure and a transform taking the Lévy measure of the primal function to the Thorin measure of its reciprocal, as a route to Hilbert's eighth problem. It does not prove the Riemann hypothesis, and neither does this book; what it supplies is a dictionary in which the Riemann hypothesis is a positivity statement. We now build the same dictionary for the discriminant form, where each identity can be checked numerically.

13.3 The Pólya kernel of the discriminant form

Let $\Delta(\tau) = q\prod_{n\geq 1}(1-q^n)^{24} = \sum_n \tau(n) q^n$ with $q = e^{2\pi i\tau}$, the cusp form of weight 12 for $\mathrm{SL}_2(\mathbb{Z})$. We use the classical normalisation of Chapter 2: $L(s,\Delta) = \sum_n \tau(n) n^{-s}$ and $\Lambda(s,\Delta) = (2\pi)^{-s}\Gamma(s)L(s,\Delta)$, which is entire with $\Lambda(s,\Delta) = \Lambda(12-s,\Delta)$ by Hecke's Theorem 2.5, so the centre is $s = 6$. All zeros of $\Lambda(s,\Delta)$ lie in $\frac{11}{2} < \mathrm{Re}\, s < \frac{13}{2}$ (Section 2.3; on the line $\mathrm{Re}\, s = \frac{13}{2}$ this is Proposition 11.8 away from $s = \frac{13}{2}$ and Proposition 13.4 below at $s = \frac{13}{2}$), and the Riemann hypothesis for Δ is the conjecture that they all lie on $\mathrm{Re}\, s = 6$.

Proposition 13.4. *For all $s \in \mathbb{C}$,*

$$\Lambda(s,\Delta) = \int_0^\infty \Delta(iy)\, y^{s-1}\, dy,$$

with $\Delta(iy) = e^{-2\pi y}\prod_n(1-e^{-2\pi n y})^{24} > 0$ for $y > 0$ and $\Delta(i/y) = y^{12}\Delta(iy)$. In particular $\Lambda(\sigma,\Delta) > 0$ for real σ.

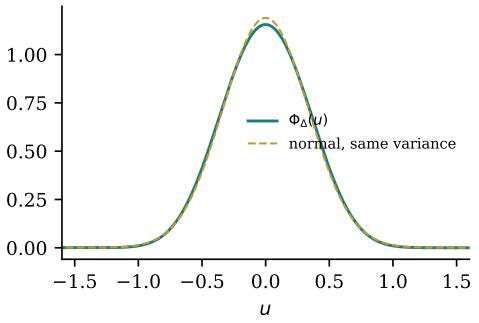


Figure 13.1: The kernel Φ_Δ against the normal density with the same variance 0.112352.

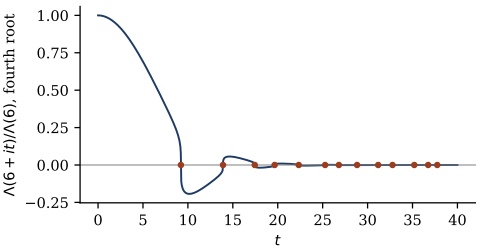


Figure 13.2: The characteristic function $\mathbb{E}e^{itW} = \Lambda(6+it,\Delta)/\Lambda(6,\Delta)$ for $0 \le t \le 40$, plotted as a signed fourth root to make the decaying oscillations visible, with its zeros γ_k marked.

PROOF. Section 2.3 gives the identity for $\operatorname{Re} s > 7$. The proof of Theorem 2.5 gives $\Delta(i/y) = y^{12}\Delta(iy)$ and shows that $\Delta(iy)$ decays faster than any power of y at both ends, so the integral converges absolutely for every s and defines an entire function, which equals Λ by analytic continuation. Positivity of $\Delta(iy)$ is clear from the product, and for real σ the integrand is positive. □

Substitute $y = e^u$. The function $g(u) = \Delta(ie^u)e^{6u}$ satisfies $g(-u) = \Delta(ie^{-u})e^{-6u} = e^{12u}\Delta(ie^u)e^{-6u} = g(u)$, and $\int g = \Lambda(6,\Delta)$.

DEFINITION 13.5. The *Pólya kernel of* Δ is the probability density

$$\Phi_\Delta(u) = \frac{\Delta(ie^u)\,e^{6u}}{\Lambda(6,\Delta)}, \qquad u \in \mathbb{R},$$

and W, the *Pólya variable* of Δ, is a random variable with density Φ_Δ.

The numbers are $L(6,\Delta) = 0.792123$ and

$$\Lambda(6,\Delta) = 120\,L(6,\Delta)/(2\pi)^6 = 0.0015449.$$

The kernel is even, strictly positive, and decays doubly exponentially, $\Phi_\Delta(u) \asymp \exp(-2\pi e^{|u|} + 6|u|)$. Figure 13.1 compares it with the normal density of the same variance. The two are close: the peak $\Phi_\Delta(0) = 1.1557$ lies below the normal peak 1.1902, and under the Riemann hypothesis for Δ the kernel is platykurtic (Exercise 1). Its tails, too small to see at this scale, are far lighter than Gaussian ones.

THEOREM 13.6 (this book). *The random variable W is symmetric, has moments of all orders and an entire moment generating function, and for all $z \in \mathbb{C}$*

$$\mathbb{E}\left[e^{zW}\right] = \frac{\Lambda(6+z,\Delta)}{\Lambda(6,\Delta)}.$$

In particular $\mathbb{E}[e^{itW}] = \Lambda(6+it,\Delta)/\Lambda(6,\Delta)$ is real and even, and the Riemann hypothesis for Δ is equivalent to the statement that the characteristic function of W has only real zeros.

PROOF. By Proposition 13.4 for $s = 6+z$ and $y = e^u$, $\Lambda(6+z) = \int g(u)e^{zu}\,du$, which converges absolutely for all z by the doubly exponential decay of g; dividing by $\Lambda(6)$ gives the identity. Symmetry of Φ_Δ makes the characteristic function real and even. Zeros of $\Lambda(s,\Delta)$ are the points $6+it$ with $\mathbb{E}e^{itW} = 0$, and they lie on $\operatorname{Re} s = 6$ exactly when the corresponding t is real. □

Our computations for this book located 404 zeros of $\Lambda(6+it,\Delta)$ on the critical line with $0 < t \le 400$, the first six being

$$9.22238, \quad 13.90755, \quad 17.44278, \quad 19.65651, \quad 22.33610, \quad 25.27464.$$

For the number $N(T)$ of zeros $6+i\gamma$ with $0 < \gamma \le T$, the Riemann von Mangoldt formula for Δ reads $N(T) = \frac{T}{\pi}\log\frac{T}{2\pi e} + \frac{11}{4} + O(\log T)$, where the error term is $\frac{1}{\pi}\arg L(6+iT,\Delta)$ up to $O(1/T)$; the main terms give 404.3 at $T = 400$. The characteristic function computed by quadrature against Φ_Δ agrees with $\Lambda(6+it)/\Lambda(6)$ to at least twelve significant digits at $t = 2, 5, 12$, and at $t = \gamma_1$ its computed value is 1.2×10^{-17}. Figure 13.2 shows it.

13.3.1 The Hadamard product and the variance identity

The zeros of $\Lambda(6+z,\Delta)$ are symmetric under $z \mapsto -z$ and $z \mapsto \bar{z}$, since $\Lambda(6+z)$ is even and real for real z, and none is real, since $\Lambda(6+x) > 0$. Writing a zero as $z = i\gamma$, we list one representative of each pair $\{\gamma, -\gamma\}$ with $\operatorname{Re}\gamma > 0$ as $\gamma_1, \gamma_2, \dots$, with multiplicity and in order of increasing real part. The Riemann hypothesis for Δ says all γ_k are real.

THEOREM 13.7 (this book). *Unconditionally, $\sum_k |\gamma_k|^{-2} < \infty$ and for all $z \in \mathbb{C}$*

$$\frac{\Lambda(6+z,\Delta)}{\Lambda(6,\Delta)} = \prod_k \Big(1 + \frac{z^2}{\gamma_k^2}\Big).$$

Consequently $\mathbb{E}W = 0$ and

$$\operatorname{Var} W = 2\sum_k \frac{1}{\gamma_k^2},$$

where the right side is real because nonreal γ_k occur in conjugate pairs.

PROOF. Since $\Delta(iy) > 0$, Proposition 13.4 gives $|\Lambda(s)| \le \Lambda(\operatorname{Re} s)$ for every s. For real $\sigma \ge 7$ we have $L(\sigma, \Delta) \le \sum_n |\tau(n)| n^{-7} < \infty$ and $\Gamma(\sigma) \le \sigma^\sigma$, and $\Lambda(\sigma) = \Lambda(12 - \sigma)$; hence $|\Lambda(6 + z)| \le \Lambda(6 + |\operatorname{Re} z|) \le C \exp\big((|z| + 7)\log(|z| + 7)\big)$, and $\Lambda(6 + z)$ is entire of order at most one. Define $G(w) = \Lambda(6 + \sqrt{w})/\Lambda(6)$. It is well defined because $\Lambda(6 + z)$ is even, it is entire because an even power series in z is a power series in $w = z^2$, and $|G(w)| \le C' \exp\big((|w|^{1/2} + 7)\log(|w|^{1/2} + 7)\big)$, so G has order at most one half (exactly one half, by Stirling's formula on the real axis). Its zeros are the points $w_k = -\gamma_k^2$, with multiplicity, and $G(0) = 1$. An entire function of order less than one has genus zero, so Hadamard's theorem gives $G(w) = \prod_k (1 - w/w_k)$ with $\sum_k |w_k|^{-1} < \infty$, with no exponential factor and with constant $G(0) = 1$. This is the convergence statement and, with $w = z^2$, the product formula. Finally $G(w) = \mathbb{E}[e^{\sqrt{w}W}] = \sum_n \mathbb{E}[W^{2n}] w^n/(2n)!$ since odd moments vanish, so $G'(0) = \frac{1}{2}\mathbb{E}W^2$, while the absolutely convergent product gives $G'(0) = \sum_k \gamma_k^{-2}$. □

The variance identity equates a moment of the kernel with a sum over zeros, a trace formula in miniature. Quadrature against Φ_Δ gives $\operatorname{Var} W = 0.112352$. The 404 computed zeros give $2\sum_{\gamma_k \le 400} \gamma_k^{-2} = 0.104146$. The density of zeros at height t is $\frac{1}{\pi}\log\frac{t}{2\pi}$, so the tail $\sum_{\gamma_k > T} \gamma_k^{-2}$ is approximately $\int_T^\infty \frac{\log(t/2\pi)}{\pi t^2}\,dt = \frac{\log(T/2\pi)+1}{\pi T}$, which at $T = 400$ is 0.004101; doubling and adding gives 0.112348, a relative difference of 3×10^{-5}.

13.4 THE RIEMANN HYPOTHESIS FOR Δ AS A GGC PROPERTY

THEOREM 13.8 (this book). *The following are equivalent.*

(i) All zeros of $\Lambda(s, \Delta)$ lie on the line $\operatorname{Re} s = 6$.

(ii) The function $s \mapsto \Lambda(6, \Delta)/\Lambda(6 + \sqrt{s}, \Delta)$, $s > 0$, is the Laplace transform of a generalized gamma convolution.

When they hold, the GGC has drift 0 and Thorin measure $\sum_k \delta_{\gamma_k^2}$, it is the law of

$$X = \sum_k \frac{E_k}{\gamma_k^2}$$

with E_k independent standard exponential variables, and its Lévy measure has density

$$\frac{k(x)}{x}, \qquad k(x) = \sum_k e^{-\gamma_k^2 x},$$

which is completely monotone.

PROOF. Let G be as in the proof of Theorem 13.7, so the function in (ii) is $1/G(s)$ for $s > 0$.

(ii) implies (i). Let φ be the Laplace transform of the GGC. By Lemma 13.3, φ is holomorphic and zero free on $\mathbb{C} \setminus (-\infty, 0]$. The function φG is holomorphic on this connected domain and equals 1 on $(0, \infty)$, hence everywhere by the identity theorem, so G has no zeros off $(-\infty, 0]$. The zeros of G are the points $-\gamma_k^2$, and $G(0) = 1$, so every γ_k^2 is a positive real number. As $\operatorname{Re} \gamma_k > 0$, every γ_k is real, and the zeros $6 \pm i\gamma_k$ lie on $\operatorname{Re} s = 6$.

(i) implies (ii). If all γ_k are real, hence positive, Theorem 13.7 gives $1/G(s) = \prod_k (1 + s/\gamma_k^2)^{-1}$ for $s > 0$. Each factor is the Laplace transform of E_k/γ_k^2. Since $\sum_k \mathbb{E}[E_k/\gamma_k^2] = \sum_k \gamma_k^{-2} < \infty$, the series X converges almost surely and in L^1, and by independence and dominated convergence its Laplace transform is the infinite product. Writing the product as $\exp(-\sum_k \log(1 + s/\gamma_k^2))$ exhibits it in the form of Definition 13.2 with $a = 0$ and $\tau = \sum_k \delta_{\gamma_k^2}$. The conditions on τ hold: $\sum_k \gamma_k^{-2} < \infty$ leaves only finitely many atoms γ_k^2 below any bound, so τ is finite on compact sets and $\int_{(0,1]} |\log t|\, \tau(dt)$ is a finite sum, while $\int t^{-1}\, \tau(dt) = \sum_k \gamma_k^{-2} < \infty$. Frullani's integral gives the Lévy density. The series $k(x)$ converges for $x > 0$ because $e^{-\gamma_k^2 x} \le (\gamma_k^2 x)^{-1}$, and k is completely monotone as the Laplace transform of the positive measure τ. □

The first Thorin atoms are

$$\gamma_k^2 = 85.05, \quad 193.42, \quad 304.25, \quad 386.38, \quad 498.90, \quad 638.81.$$

Figure 13.3 compares the product over the computed zeros, with the tail correction of the previous section in the exponent, to the exact value $\Lambda(6)/\Lambda(6 + \sqrt{s})$. The pairs (exact, corrected product over the 404 zeros) are $(0.945467, 0.945468)$ at $s = 1$, $(0.575602, 0.575612)$ at $s = 10$, $(0.0069732, 0.0069741)$ at $s = 100$ and $(5.6354, 5.6359) \times 10^{-8}$ at $s = 400$: a relative error below 2×10^{-4} over more than seven orders of magnitude. Figure 13.4 shows k; under the Riemann hypothesis for Δ, $k(x)/x$ is the Lévy density. As $x \to 0$ the zero density $\frac{1}{\pi} \log \frac{t}{2\pi}$ gives

$$k(x) = \frac{\log(1/x) - c_0 + o(1)}{4\sqrt{\pi x}}, \qquad c_0 = \log(16\pi^2) + \gamma_{\mathrm{E}} = 5.639,$$

with γ_{E} Euler's constant. The constant matters at the scale of Figure 13.4: at $x = 10^{-4}$ the numerator is about 3.57, against 9.21 for $\log(1/x)$ alone. The Lévy density therefore behaves like $x^{-3/2} \log(1/x)/(4\sqrt{\pi})$ at the origin: the Lévy measure is infinite, while its first moment near the origin, $\int_0^1 k(x)\, dx$, is finite, as for every subordinator.

REMARK 13.9 (A dual pair). Under the Riemann hypothesis for Δ let $V = \sqrt{2X}\, N$ with N standard normal independent of X. Then $\mathbb{E}e^{itV} = \mathbb{E}e^{-t^2 X} = \Lambda(6)/\Lambda(6 + t)$ for real t: on the real axis, the reciprocal of the moment generating function of W is the characteristic function of V. The bilateral Laplace transform of V is

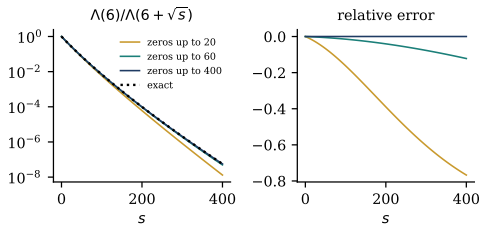


Figure 13.3: Left: the product $\prod_k (1 + s/\gamma_k^2)^{-1}$ over the zeros up to height 20, 60 and 400, each with its tail correction, against $\Lambda(6,\Delta)/\Lambda(6+\sqrt{s},\Delta)$ (log scale). Right: the relative errors.

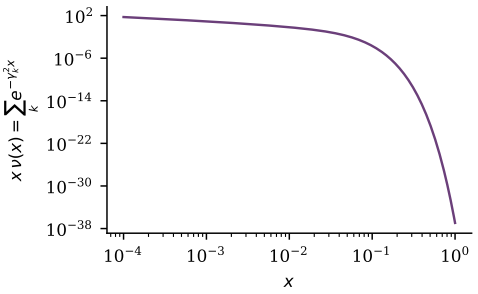


Figure 13.4: The function $k(x) = \sum_k e^{-\gamma_k^2 x}$ over the 404 zeros up to height 400, for $10^{-4} \le x \le 1$ on log scales; under the Riemann hypothesis for Δ it is $x\nu(x)$, with ν the Lévy density.

$1/\prod_k (1 - s^2/\gamma_k^2)$ in the strip $|\operatorname{Re} s| < \gamma_1$, so by Theorem 13.1, with $\delta_k = \pm\gamma_k^{-1}$, the density of V is a Pólya frequency function of infinite order. Conversely, if $\Lambda(6)/\Lambda(6+t)$ were the characteristic function of a Pólya frequency density, Schoenberg's theorem would put $\prod_k (1 - s^2/\gamma_k^2)$ in the Laguerre Pólya class and all γ_k would be real.

13.4.1 General self dual L functions and the rank

Apart from the variance identity, the proofs of Theorems 13.7 and 13.8 used only that $\Lambda(6+z)$ is entire of order at most one, even, real on the real axis and nonzero at $z = 0$; positivity of the kernel served only to bound the order quickly and to give the probabilistic reading of Theorem 13.6. Let π be a self dual cuspidal automorphic representation of $\mathrm{GL}_n(\mathbb{A}_{\mathbb{Q}})$, $n \ge 2$, with completed L function $\Lambda(s,\pi)$ in the unitary normalisation, conductor included, so that $\Lambda(s,\pi)$ is entire of order one [217] and $\Lambda(s,\pi) = \varepsilon\Lambda(1-s,\pi)$ with $|\varepsilon| = 1$. Self duality makes $\Lambda(s,\pi)$ real on the real axis, and then $\varepsilon = \pm 1$.

THEOREM 13.10 (this book). *Suppose $\Lambda(\frac{1}{2},\pi) \neq 0$. Then the generalized Riemann hypothesis for π holds if and only if $s \mapsto \Lambda(\frac{1}{2},\pi)/\Lambda(\frac{1}{2}+\sqrt{s},\pi)$ is the Laplace transform of a GGC, and then its Thorin measure is $\sum_k \delta_{\gamma_k^2}$ over the ordinates $\gamma_k > 0$ of the zeros $\frac{1}{2} \pm i\gamma_k$.*

PROOF. Since $\Lambda(\frac{1}{2}) \neq 0$ the functional equation at $s = \frac{1}{2}$ forces $\varepsilon = 1$, so $\Lambda(\frac{1}{2}+z)$ is even, and by the paragraph before the theorem the proofs of Theorems 13.7 and 13.8 apply to $G(w) = \Lambda(\frac{1}{2}+\sqrt{w})/\Lambda(\frac{1}{2})$. The one new case is a real zero $\frac{1}{2}+x$ of Λ, which would give a zero $x^2 > 0$ of G; the GGC property excludes it

too. Theorem 14.7 gives the general argument and the other equivalent forms. □

When $\Lambda(\frac{1}{2}+z,\pi)$ vanishes to order $r \geq 1$ at $z=0$, the functional equation forces $\varepsilon = (-1)^r$, the argument applies to the even function $\Lambda(\frac{1}{2}+z,\pi)/z^r$, and under the generalized Riemann hypothesis $1/\Lambda(\frac{1}{2}+\sqrt{s},\pi)$ is a constant times $s^{-r/2}$ times a GGC transform. The factor $s^{-r/2}$ is a degenerate Thorin atom of mass $r/2$ at the origin, half the analytic rank for an elliptic curve; Proposition 14.14 proves this, and Chapter 14 extends Theorem 13.10 to every cuspidal π, self dual or not.

13.5 What is known unconditionally

The product of Theorem 13.7 splits as

$$\frac{\Lambda(6,\Delta)}{\Lambda(6+\sqrt{s},\Delta)} = \prod_{\gamma_k\in\mathbb{R}}\Big(1+\frac{s}{\gamma_k^2}\Big)^{-1} \times \prod_{\gamma_k\notin\mathbb{R}}\Big(1+\frac{s}{\gamma_k^2}\Big)^{-1}.$$

The first factor is always a GGC transform, the law of $\sum E_k/\gamma_k^2$ over real γ_k. The second collects off line zeros in conjugate pairs. For one pair write $\gamma^2 = a+ib$ with $b\neq 0$; then for $s > -a$

$$\begin{aligned}\Big(1+\frac{s}{\gamma^2}\Big)^{-1}\Big(1+\frac{s}{\bar\gamma^2}\Big)^{-1} &= \frac{|\gamma|^4}{(s+a)^2+b^2}\\ &= \int_0^\infty e^{-sx}\,\frac{|\gamma|^4}{b}\,e^{-ax}\sin(bx)\,dx,\end{aligned}$$

the Laplace transform of a function that changes sign. The Riemann hypothesis for Δ is the statement that the signed part is empty.

How much of the Thorin measure is known to be positive? Hafner [233] proved, following the method of Selberg for ζ [493], that a positive proportion of the zeros of the L function of a holomorphic Hecke eigenform for $\mathrm{SL}_2(\mathbb{Z})$, counted by height, lie on the critical line. In our language, a positive proportion of the atoms γ_k^2 lie on $(0,\infty)$. But no proportion short of all of them yields the GGC property, because a single conjugate pair of nonreal γ already destroys it, by Theorem 13.8. For ζ the proportion is above two fifths [114], with the same caveat.

The heat flow transfers as well. Put $H_\lambda^\Delta(z) = \int e^{\lambda u^2}\Phi_\Delta(u)e^{izu}\,du$, so that $H_0^\Delta(z) = \Lambda(6+iz)/\Lambda(6)$. All zeros of H_0^Δ lie in $|\operatorname{Im} z| < \frac{1}{2}$, so by de Bruijn's theorem H_λ^Δ has only real zeros for $\lambda \geq \frac{1}{8}$. Dobner [158], by a method different from that of Rodgers and Tao, proved that every member of the extended Selberg class has a de Bruijn Newman constant and that it is nonnegative. The function $L(s,\Delta)$, in the unitary normalisation, belongs to that class, and Dobner's heat parameter is 4λ in our normalisation. Hence there is a constant λ_Δ with $0 \leq$

$\lambda_\Delta \le \frac{1}{8}$ such that H_λ^Δ has only real zeros exactly when $\lambda \ge \lambda_\Delta$, and the Riemann hypothesis for Δ is the statement $\lambda_\Delta = 0$; Dobner's constant for Δ is $4\lambda_\Delta$, as $\Lambda = 4\lambda_0$ for ζ.

REMARK 13.11. Why is this a reformulation and not a proof? Every equivalence above passes through the Hadamard product, that is, through the zeros themselves. A proof of the Riemann hypothesis along these lines would need an independent construction of a GGC with Laplace transform $\Lambda(6)/\Lambda(6+\sqrt{s})$, for example a stochastic process whose hitting times or occupation laws have that transform, as the Bessel hitting time T of Section 13.2.1 has the transform $\sqrt{2\lambda}/\sinh\sqrt{2\lambda}$, a GGC whose Thorin atoms $\pi^2 n^2/2$ come from the zeros $\pm i\pi n$ of $\sinh z/z$ (Exercise 2). The Euler product gives positivity of a different kind, the infinite divisibility of Chapter 11, which holds only to the right of the critical strip and, for Δ, holds for the Rankin Selberg function $L(s, \Delta \times \Delta)$ but not for $L(s,\Delta)$ itself: the logarithmic coefficients of $L(s,\Delta)$ change sign, and for $\sigma > \frac{13}{2}$ the function $t \mapsto L(\sigma+it,\Delta)/L(\sigma,\Delta)$ is not even a characteristic function (Proposition 11.3). The GGC property is a positivity statement about the reciprocal inside the strip, and no bridge between the two is known.

OPEN PROBLEM 13.1 (Hilbert's eighth problem for GL_2). Construct, without reference to the zeros, a random variable whose Laplace transform is $\Lambda(6,\Delta)/\Lambda(6+\sqrt{s},\Delta)$ and prove that it is a GGC; more generally do this for every self dual cuspidal π with $\Lambda(\frac{1}{2},\pi) \neq 0$. A positive solution for one π would prove the generalized Riemann hypothesis for that π.

13.6 NOTES

Pólya's kernel and real zero theorems are in [444, 445], the heat flow in [133, 417, 465, 446], its extension to the extended Selberg class in [158], and total positivity in [488, 293]. Thorin [546, 547] introduced generalized gamma convolutions to prove that the Pareto and lognormal laws are infinitely divisible, and Bondesson's monograph [73] is the standard reference. The Brownian interpretation of ξ and its relatives is the subject of the survey of Biane, Pitman and Yor [82]. The Thorin representation of the reciprocal Ξ function is from the first author [441], and his later paper [442] proves the equivalence of real zeros and the Thorin property for real entire functions of order at most one, with applications to Dirichlet and modular L functions. Theorems 13.6, 13.7, 13.8 and 13.10 are elementary consequences of Hecke's theory and Hadamard's factorisation, and we present them as the book's formulation of Pólya's programme for GL_2. The numerical values were computed for this book with PARI/GP; the first zeros and $L(6,\Delta)$ agree with those listed in the LMFDB [371].

13.7 Exercises

1. Show that $\mathbb{E}W^4 - 3(\mathbb{E}W^2)^2 = -12\sum_k \gamma_k^{-4}$, so that under the Riemann hypothesis for Δ the kernel Φ_Δ is platykurtic. Compare with Figure 13.1.

2. Let T be the hitting time of 1 by a three dimensional Bessel process from 0. Show that T is a GGC with Thorin measure $\sum_n \delta_{\pi^2 n^2/2}$. Writing $\mathbb{E}e^{-\lambda T} = 1/F(\sqrt{2\lambda})$ with $F(z) = \sinh z/z$, explain why this is the exact analogue of Theorem 13.8 for an entire function whose zeros $\pm i\pi n$ are known to lie on a line.

3. Show that if φ is the Laplace transform of a GGC with drift a and Thorin measure τ then $-\varphi'/\varphi = a + \int (t+s)^{-1}\tau(dt)$ is a Stieltjes function, that is, a function of the form $c/s + b + \int_{(0,\infty)} (s+t)^{-1}\sigma(dt)$ with $b, c \geq 0$ and σ a positive measure satisfying $\int (1+t)^{-1}\sigma(dt) < \infty$, hence completely monotone. Deduce that under the Riemann hypothesis for Δ the function $\frac{d}{ds}\log G(s) = \sum_k (s+\gamma_k^2)^{-1}$ is a Stieltjes function.

4. Show conversely that if $\frac{d}{ds}\log G$ agrees on a half line (s_0, ∞) with a Stieltjes function then the Riemann hypothesis for Δ holds. (A Stieltjes function extends holomorphically to $\mathbb{C} \setminus (-\infty, 0]$.) Show also that $\frac{d}{ds}\log G(s)$ is the Laplace transform of k for $s > \frac{1}{4}$, where $k(x) = \sum_k e^{-\gamma_k^2 x}$ is defined unconditionally with complex γ_k taken in conjugate pairs (note $|\operatorname{Im}\gamma_k| < \frac{1}{2}$, so $\operatorname{Re}\gamma_k^2 > -\frac{1}{4}$), and conclude by Bernstein's theorem that the Riemann hypothesis for Δ is equivalent to complete monotonicity of k on $(0, \infty)$.

5. Verify Remark 13.9: if X is a GGC with Thorin measure $\sum \delta_{\gamma_k^2}$ and $V = \sqrt{2X}N$, then V has the law of $\sum_k L_k/\gamma_k$ with independent standard Laplace L_k.

13.8 Open computational problems

Computational Problem 13.1 (The Thorin measure of Δ to height 10^5). Objective: compute the zeros of $\Lambda(6+it, \Delta)$ for $0 < t \leq 10^5$ and estimate the Thorin measure and the Lévy density of Theorem 13.8. Objects and scale: about 2.8×10^5 zeros. Route: PARI/GP `lfunzeros` or Rubinstein's `lcalc` with $\tau(n)$ precomputed to the needed length, with Turing's method to certify the count; the Mellin integral by which Chapter 19 evaluates $\Lambda(s, \Delta)$ needs a working precision that grows with t and is no route at this height. Deliverable: the zero list, the empirical measure $\sum \delta_{\gamma_k^2}$ against its smooth prediction, $k(x)$ on a logarithmic grid with tail corrections, and the convergence of $2\sum \gamma_k^{-2}$ and $-12\sum \gamma_k^{-4}$ to the cumulants κ_2 and κ_4 of W computed from the kernel. Why it matters: it verifies the Riemann hypothesis for

Δ to height 10^5 and shows, in the GGC language, how fast the truncated Thorin measures reproduce the kernel.

COMPUTATIONAL PROBLEM 13.2 (The de Bruijn Newman constant of Δ). Objective: bound λ_Δ from above by running the heat flow H_λ^Δ numerically. Objects and scale: $\lambda \in [0, \frac{1}{8}]$, zeros of H_λ^Δ to height 10^4, with a barrier argument in the style of the Polymath project for larger heights. Route: an effective approximation of H_λ^Δ by sums over $\tau(n)$, derived as the Polymath project did for ζ and evaluated in Arb with GPU parallelism over z; direct quadrature of $e^{\lambda u^2}\Phi_\Delta(u)\cos(zu)$ as a check at small z, where cancellation is mild. Deliverable: a certified upper bound $\lambda_\Delta \le \bar{\lambda}$ with $\bar{\lambda}$ well below $\frac{1}{8}$, and plots of zero trajectories. Why it matters: since $\lambda_\Delta \ge 0$ is known, the Riemann hypothesis for Δ is the statement $\lambda_\Delta = 0$, and $\bar{\lambda}$ measures how close unconditional knowledge comes to it, as the bound $\Lambda \le 0.22$, that is $\lambda_0 \le 0.055$ in our normalisation, does for ζ.

COMPUTATIONAL PROBLEM 13.3 (Monotonicity across weights and levels). Objective: test, for many forms, that $k(x) = \sum_k e^{-\gamma_k^2 x}$ is completely monotone and that $\frac{d}{ds}\log G(s)$ is a Stieltjes function, using data that do not presuppose the zeros: k computed from the primes through the explicit formula, and $\frac{d}{ds}\log G$ from values of Λ. (A k assembled from zeros located on the critical line is completely monotone by construction, so that test would be empty.) For Δ, Table 16.1 and Figure 16.3 of Chapter 16 compute k, there written $\mathcal{W}$, and its first three derivatives from the primes; the problem concerns the other forms. Objects and scale: all other newforms with rational coefficients and trivial character, of weight at most 24 and level $N \le 500$, with nonzero central value, from LMFDB. Route: compute Λ at the centre plus $\sqrt{s}$ with PARI/GP `lfun` on a grid $s \in [0, 10^4]$, and k and its derivatives from the explicit formula with Gaussian test functions; test the first twenty sign conditions $(-1)^n k^{(n)} \ge 0$ in multiprecision, and fit a positive measure σ in $\frac{d}{ds}\log G(s) = \int (s+t)^{-1}\sigma(dt)$ by nonnegative least squares. Deliverable: a table of forms with the smallest margin found in each inequality and the fitted Thorin measures compared with the squared zero ordinates. Why it matters: complete monotonicity of k and the Stieltjes property are each equivalent to the generalized Riemann hypothesis for the form (Exercises 3 and 4 for Δ, with the same proofs in general), and the margins measure the distance to the boundary of the GGC cone.

COMPUTATIONAL PROBLEM 13.4 (Rank two curves and the degenerate atom). Objective: exhibit the degenerate Thorin atom of mass $r/2$ at the origin for elliptic curves of analytic rank $r = 2$. Objects and scale: the first fifty curves of analytic rank two in LMFDB by conductor, starting with 389a1, with zeros to height 10^3. Route: PARI/GP `lfun` and `lfunzeros` in the classical normalisation with centre 1; evaluate $\frac{d}{ds}\log\Lambda(1+\sqrt{s}, E)$ on $s \in [10^{-4}, 10^3]$ and fit the representation $m/s + \int (s+t)^{-1}\tau(dt)$ with $m \ge 0$ and $\tau \ge 0$ by regularised nonnegative least

squares. Deliverable: the fitted mass near $t = 0$ against the predicted value $r/2 = 1$ (that is, the coefficient m of $1/s$), and the remainder of the measure against $\sum \delta_{\gamma_k^2}$. Why it matters: it checks the dictionary between order of vanishing and Thorin mass at zero by fitting the whole measure, not only the slope near $s = 0$ that Chapter 14 reports for 389a1.

CHAPTER 14

THE THORIN FRAMEWORK FOR AUTOMORPHIC L FUNCTIONS

Thorin introduced generalized gamma convolutions in 1977 to prove that the Pareto and lognormal laws are infinitely divisible [546, 547]. Much later they were proposed as a language for the reciprocal of Riemann's Ξ function [441]. In [442] the first author proved, for real entire functions of order at most one in the Laguerre Pólya setting, that all zeros lie on the critical axis if and only if Thorin's condition holds, identified the Thorin measure as $\sum_k \delta_{\gamma_k^2}$, and recast the Hadamard factorisation as a van Dantzig pair and a Wald couple (Chapter 16), with applications to ξ, Dirichlet and modular L functions and the Dedekind η function. Chapter 13 built that dictionary for the discriminant form Δ; this chapter shows that nothing in it is special to Δ. For every cuspidal automorphic representation π of GL_n over $\mathbb{Q}$ we build an even completion Ξ_π of $\Lambda(s,\pi)$, real and nonnegative on the real axis, and prove that the generalized Riemann hypothesis for π is equivalent to each of six positivity statements: a Thorin representation, complete monotonicity of a Lévy function, a Stieltjes property, a Herglotz inequality, a Stieltjes moment property of power sums computable from central Taylor coefficients alone, and, when the Pólya kernel is positive, a moment property of the even cumulants of the Pólya variable. We then relax the Thorin property into parabolic conditions, recover de Bruijn's bound as the unconditional case, read the order of central vanishing as a degenerate Thorin atom, test the criteria numerically on eleven L functions, and locate the boundary of the framework at the Selberg zeta function. All of it is reformulation; none of it proves a Riemann hypothesis.

14.1 THORIN, STIELTJES AND BERNSTEIN

We use the definition of a generalized gamma convolution (GGC) and its Thorin measure from Definition 13.2: a law on $[0,\infty)$ whose Laplace transform is

$$\varphi(s) = \exp\Big(-as - \int_{(0,\infty)} \log\Big(1 + \frac{s}{t}\Big)\tau(dt)\Big), \qquad s \geq 0, \tag{14.1}$$

with drift $a \geq 0$ and Thorin measure $\tau \geq 0$ subject to the integrability conditions stated there [546, 73]. The analytic fact behind every equivalence below is

Lemma 13.3: φ extends to a holomorphic function on $\mathbb{C} \setminus (-\infty, 0]$ *without zeros*. Three classical theorems complete the toolkit.

THEOREM 14.1 (Bernstein). *A function f on $(0,\infty)$ is completely monotone, that is $(-1)^m f^{(m)} \geq 0$ for all m, if and only if $f(x) = \int e^{-tx}\mu(dt)$ for a unique positive measure μ on $[0,\infty)$.*

DEFINITION 14.2. A *Stieltjes function* is a function $g : (0,\infty) \to [0,\infty)$ of the form

$$g(s) = \frac{a}{s} + b + \int_{(0,\infty)} \frac{\sigma(dt)}{s+t}, \qquad a, b \geq 0, \quad \int_{(0,\infty)} \frac{\sigma(dt)}{1+t} < \infty.$$

Differentiating (14.1) gives $-\varphi'/\varphi = a + \int \tau(dt)/(s+t)$, so minus the logarithmic derivative of a GGC transform is a Stieltjes function with no pole at the origin; conversely a Stieltjes function g with $a = 0$ and $\int_0^1 g(u)\,du < \infty$ integrates to a GGC transform $\exp(-\int_0^s g)$, with drift b and Thorin measure σ. The integral part is the Laplace transform of a completely monotone function, $\int \sigma(dt)/(s+t) = \int_0^\infty e^{-sx} \int e^{-tx}\sigma(dt)\,dx$, and Stieltjes functions have a classical complex analytic description.

THEOREM 14.3 (Nevanlinna Pick characterisation). *A function $g : (0,\infty) \to [0,\infty)$ is a Stieltjes function if and only if it extends holomorphically to $\mathbb{C} \setminus (-\infty, 0]$ and satisfies $\operatorname{Im} g(s) \leq 0$ whenever $\operatorname{Im} s > 0$.*

The direction from the representation to the extension is elementary: for s in a compact $K \subset \mathbb{C} \setminus (-\infty, 0]$ the continuous positive function $(s,t) \mapsto |s+t|/(1+t)$ on $K \times [0,\infty]$ is bounded below by some $c_K > 0$, so the integral converges locally uniformly, and $\operatorname{Im}(s+t)^{-1} = -\operatorname{Im} s/|s+t|^2 \leq 0$. The converse rests on the Herglotz representation [487]; we need only the elementary direction.

THEOREM 14.4 (Stieltjes). *A real sequence $(m_j)_{j\geq 0}$ is a Stieltjes moment sequence, $m_j = \int_{[0,\infty)} u^j \nu(du)$ for a positive measure ν on $[0,\infty)$, if and only if the Hankel matrices $(m_{i+j})_{i,j=0}^N$ and $(m_{i+j+1})_{i,j=0}^N$ are positive semidefinite for every $N \geq 0$.*

Necessity is the identity $\sum c_i c_j m_{i+j+\ell} = \int u^\ell (\sum c_i u^i)^2 \nu(du)$ for $\ell = 0, 1$; sufficiency is Stieltjes' theorem [528, 10].

14.2 THE SELF DUAL COMPLETION

Let π be a cuspidal automorphic representation of $\mathrm{GL}_n(\mathbb{A}_\mathbb{Q})$, $n \geq 1$, with unitary central character, and assume that π is not a unitary twist $|\cdot|^{iy}$ of the trivial representation of GL_1. We use the unitary normalisation: $L(s,\pi) = \sum_m a_\pi(m) m^{-s}$ converges absolutely for $\operatorname{Re} s > 1$ [279], the archimedean factor is $\gamma(s,\pi) = N_\pi^{s/2} \prod_{j=1}^n \Gamma_\mathbb{R}(s+\mu_j)$ with $\Gamma_\mathbb{R}(s) = \pi^{-s/2}\Gamma(s/2)$ and $\operatorname{Re}\mu_j > -\frac{1}{2}$, and $\Lambda(s,\pi) =$

$\gamma(s,\pi)L(s,\pi)$. By Godement and Jacquet [217] (Tate and Hecke for $n=1$) $\Lambda(s,\pi)$ is entire of order one and

$$\Lambda(s,\pi) = \varepsilon(\pi)\Lambda(1-s,\tilde\pi), \qquad |\varepsilon(\pi)| = 1,$$

where the contragredient $\tilde\pi$ is the complex conjugate of π, so that $a_{\tilde\pi}(m) = \overline{a_\pi(m)}$, $\mu_j(\tilde\pi) = \overline{\mu_j(\pi)}$, and hence

$$\Lambda(\bar s,\tilde\pi) = \overline{\Lambda(s,\pi)}. \tag{14.2}$$

The number of zeros ρ of $\Lambda(s,\pi)$ with $|\operatorname{Im}\rho| \le T$ is $O(T\log(N_\pi(T+3)^n))$ [276]. Applying the functional equation twice gives $\varepsilon(\pi)\varepsilon(\tilde\pi) = 1$. The generalized Riemann hypothesis (GRH) for π is the conjecture that all zeros of $\Lambda(s,\pi)$ lie on $\operatorname{Re} s = \frac12$; by (14.2) it holds for π if and only if for $\tilde\pi$.

DEFINITION 14.5. Let $r = \operatorname{ord}_{s=1/2}\Lambda(s,\pi)$, which equals $\operatorname{ord}_{s=1/2}\Lambda(s,\tilde\pi)$ by the functional equation. The *self dual completion* of π is

$$\Xi_\pi(z) = \frac{\Lambda(\frac12+z,\pi)\,\Lambda(\frac12+z,\tilde\pi)}{z^{2r}}, \qquad F_\pi(s) = \frac{\Xi_\pi(\sqrt{s})}{\Xi_\pi(0)}.$$

When $\pi \cong \tilde\pi$ we also use $\Xi^{\mathrm{sd}}_\pi(z) = \Lambda(\frac12+z,\pi)/z^r$, so that $\Xi_\pi = (\Xi^{\mathrm{sd}}_\pi)^2$. For Δ we keep the classical normalisation of Chapter 13, centre 6, for $\Lambda(s,\Delta)$, so that $\Xi^{\mathrm{sd}}_\Delta(z) = 2\Lambda(6+z,\Delta)$. For the Riemann zeta function, which is excluded above because of its pole, we put $\Xi_\zeta(z) = \xi(\frac12+z)$ with $\xi(s) = \frac12 s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s)$; the same recipe applies to a Dedekind zeta function after multiplying by $s(s-1)$.

LEMMA 14.6 (this book). *Let π be as above.*

(i) *Ξ_π is entire, even, and $\Xi_\pi(\bar z) = \overline{\Xi_\pi(z)}$; in particular Ξ_π is real on $\mathbb{R}$.*

(ii) *For real $x \neq 0$, $\Xi_\pi(x) = |\Lambda(\frac12+x,\pi)|^2/x^{2r} \ge 0$, $\Xi_\pi(0) > 0$, and $\Xi_\pi(x) > 0$ for $|x| \ge \frac12$.*

(iii) *Ξ_π has order at most one, and every zero z of Ξ_π satisfies $|\operatorname{Re} z| < \frac12$.*

(iv) *F_π is entire of order at most $\frac12$ with $F_\pi(0) = 1$. If the zeros of Ξ_π are listed in pairs $\pm z_k$ with multiplicity, then with $s_k = z_k^2$,*

$$F_\pi(s) = \prod_k \Big(1 - \frac{s}{s_k}\Big), \qquad \sum_k |s_k|^{-1} < \infty.$$

(v) *If $\pi \cong \tilde\pi$ then $\varepsilon(\pi) = (-1)^r$, and Ξ^{sd}_π is entire, even and real on $\mathbb{R}$ with $\Xi^{\mathrm{sd}}_\pi(0) \neq 0$.*

The same statements hold for Ξ_ζ, which is moreover strictly positive on $\mathbb{R}$.

PROOF. (i) The two functional equations and $\varepsilon(\pi)\varepsilon(\tilde\pi) = 1$ show that the numerator of Ξ_π is even; it vanishes to order exactly $2r$ at 0, so Ξ_π is entire and even. By (14.2), conjugation exchanges $\Lambda(\tfrac12 + z, \pi)$ and $\Lambda(\tfrac12 + \bar z, \tilde\pi)$, whence $\overline{\Xi_\pi(z)} = \Xi_\pi(\bar z)$.

(ii) For real x, (14.2) gives $\Lambda(\tfrac12 + x, \tilde\pi) = \overline{\Lambda(\tfrac12 + x, \pi)}$, so the numerator is $|\Lambda(\tfrac12 + x, \pi)|^2$. If $\Lambda(\tfrac12 + z, \pi) = cz^r + O(z^{r+1})$ with $c \neq 0$ then $\Xi_\pi(0) = |c|^2 > 0$. For $|x| \geq \tfrac12$ the point $\tfrac12 + x$ lies outside the open critical strip, where Λ does not vanish by (iii).

(iii) Products of entire functions of order at most one, divided by a polynomial, have order at most one. For $\operatorname{Re} s \geq 1$, $L(s, \pi) \neq 0$ (absolute convergence of the Euler product for $\operatorname{Re} s > 1$, Jacquet and Shalika [278] on $\operatorname{Re} s = 1$) and $\gamma(s, \pi)$ is finite and nonzero; the functional equation transfers this to $\operatorname{Re} s \leq 0$.

(iv) Since $\Xi_\pi(z) = \sum_m c_m z^{2m}$ is even, $G(w) = \sum_m c_m w^m$ is entire with $G(z^2) = \Xi_\pi(z)$ and $\max_{|w|=R} |G| = \max_{|z|=R^{1/2}} |\Xi_\pi|$, so G has order at most $\tfrac12$; zeros correspond with multiplicity because $z \mapsto z^2$ is locally biholomorphic away from 0. Hadamard's theorem for order less than one [361] gives the genus zero product with $\sum |s_k|^{-1} < \infty$.

(v) Write $\Lambda(\tfrac12 + z, \pi) = z^r H(z)$ with $H(0) \neq 0$. Self duality and the functional equation give $z^r H(z) = \varepsilon(\pi)(-z)^r H(-z)$, so $H(z) = \varepsilon(\pi)(-1)^r H(-z)$; evaluating at 0 gives $\varepsilon(\pi) = (-1)^r$ and H is even. Reality on $\mathbb{R}$ follows from (14.2) with $\tilde\pi = \pi$.

For ζ, ξ is entire of order one, $\xi(s) = \xi(1 - s)$, ξ is real on $\mathbb{R}$, $\xi(\tfrac12) > 0$, and $\xi(\sigma) > 0$ for real σ because $\zeta(\sigma) < 0$ on $(0, 1)$; nonvanishing on $\operatorname{Re} s = 1$ is the theorem of Hadamard and de la Vallée Poussin [232, 137]. □

Strict positivity of Ξ_π on $(-\tfrac12, \tfrac12)$ would exclude Siegel zeros; we never assume it. From now on we list the zeros of Ξ_π as pairs $\pm z_k$, $k = 1, 2, \ldots$, with multiplicity, and write

$$z_k = i\gamma_k, \qquad t_k = -z_k^2 = \gamma_k^2, \qquad F_\pi(s) = \prod_k \Big(1 + \frac{s}{t_k}\Big). \tag{14.3}$$

The zeros of Ξ_π are the points $\rho - \tfrac12$ and $\bar\rho - \tfrac12$ for the zeros $\rho \neq \tfrac12$ of $\Lambda(s, \pi)$, so GRH for π holds if and only if every γ_k is real, if and only if every t_k lies in $(0, \infty)$. The multiset $\{t_k\}$ is closed under complex conjugation. We call the t_k the *Thorin atoms* of π and introduce

$$k_\pi(x) = \sum_k e^{-t_k x} = \sum_k e^{x z_k^2}, \qquad S_n = \sum_k t_k^{-n} = \sum_k (-z_k^2)^{-n}, \quad n \geq 1. \tag{14.4}$$

14.3 THE THORIN EQUIVALENCE THEOREM

THEOREM 14.7 (Thorin equivalence, abstract form; after [442]). *Let E be an entire function with $E(-z) = E(z)$, $E(\bar z) = \overline{E(z)}$, $E(0) = 1$, of order less than two, whose*

zeros lie in a strip $|\operatorname{Re} z| \le \sigma_0$. *List its zeros as pairs* $\pm z_k$ *with multiplicity, put* $t_k = -z_k^2$ *and* $F(s) = E(\sqrt{s}) = \prod_k (1 + s/t_k)$. *Then* $k(x) = \sum_k e^{-t_k x}$ *converges absolutely and locally uniformly on* $(0,\infty)$, *the sums* $S_n = \sum_k t_k^{-n}$, $n \ge 1$, *converge absolutely and are real, and the following are equivalent.*

(a) Every zero of E lies on $i\mathbb{R}$; *equivalently* $t_k \in (0,\infty)$ *for all k.*

(b) $1/F$ *restricted to* $(0,\infty)$ *is the Laplace transform of a GGC.*

(c) $1/F(s) = \mathbb{E}e^{-sX}$ *for* $s \ge 0$, *where* $X = \sum_k E_k/t_k$ *with* E_k *independent standard exponential variables; that is,* $1/F$ *is a GGC transform with drift* 0 *and Thorin measure* $\sum_k \delta_{t_k}$.

(d) k is completely monotone on $(0,\infty)$.

(e) F has no zero in $(0,\infty)$ *and* F'/F *is a Stieltjes function on* $(0,\infty)$.

(e′) E has no real zero, and $\operatorname{Im}(F'/F)(s) \le 0$ *at every s with* $\operatorname{Im} s > 0$ *and* $F(s) \ne 0$.

(f) There is a positive measure ν *on* $[0,\infty)$ *with* $S_{j+1} = \int u^j \nu(du)$ *for all* $j \ge 0$; *equivalently, the Hankel matrices* $(S_{i+j+1})_{i,j=0}^N$ *and* $(S_{i+j+2})_{i,j=0}^N$ *are positive semidefinite for every N.*

When these hold, $\nu = \sum_k t_k^{-1} \delta_{1/t_k}$, *the Stieltjes function in (e) is* $\sum_k (s + t_k)^{-1}$, *and the Bernstein measure of k is* $\sum_k \delta_{t_k}$.

PROOF. *Preliminaries.* By the argument of Lemma 14.6(iv), F is entire of order less than one, $F(s) = \prod_k (1 + s/t_k)$ and $\sum_k |t_k|^{-1} < \infty$. This gives absolute convergence of S_n; reality follows because $\{t_k\}$ is closed under conjugation, as $E(\bar{s}) = \overline{E(z)}$. Write $z_k = a_k + ib_k$ with $|a_k| \le \sigma_0$, so that

$$\operatorname{Re} t_k = b_k^2 - a_k^2 \ge b_k^2 - \sigma_0^2, \qquad |t_k| = a_k^2 + b_k^2 \le b_k^2 + \sigma_0^2. \tag{14.5}$$

By (14.5), $\operatorname{Re} t_k \ge |t_k| - 2\sigma_0^2$, so for $x \ge x_0 > 0$ we get $|e^{-t_k x}| \le e^{2\sigma_0^2 x} e^{-x_0 |z_k|^2} \le e^{2\sigma_0^2 x}/(x_0 |z_k|^2)$, which is summable; the series for k therefore converges absolutely and locally uniformly on $(0,\infty)$, and the same bound with an extra factor $|t_k|^m$, absorbed by $e^{-x_0|z_k|^2/2}$, justifies termwise differentiation. The logarithmic derivative of the product is

$$\frac{F'}{F}(s) = \sum_k \frac{1}{s + t_k}, \tag{14.6}$$

locally uniformly away from the points $-t_k$, a meromorphic function on $\mathbb{C}$ with a simple pole of positive integer residue at each zero of F.

(a) implies (c) implies (b). This is the proof of Theorem 13.8, (i) implies (ii), with t_k in place of γ_k^2: X converges almost surely because $\mathbb{E}X = S_1 < \infty$, $\mathbb{E}e^{-sX} = \prod_k (1 + s/t_k)^{-1} = 1/F(s)$, and $\tau = \sum_k \delta_{t_k}$ satisfies the conditions of (14.1) because it has no mass below $\min_k t_k > 0$ and $\int t^{-1} \tau(dt) = S_1$.

(b) implies (a). This is the proof of Theorem 13.8, (ii) implies (i), with F in place of G: by Lemma 13.3 and the identity theorem, F has no zeros off $(-\infty, 0]$, and $F(0) = 1$, so every $t_k > 0$.

A rigidity step. Suppose there are $c \geq 0$ and a Stieltjes function g such that F has no zero in (c, ∞) and $F'/F = g$ on (c, ∞). By the elementary half of Theorem 14.3, g is holomorphic on $\mathbb{C} \setminus (-\infty, 0]$. The set $\Omega = \mathbb{C} \setminus ((-\infty, 0] \cup \{-t_k\})$ is open and connected, since the $-t_k$ form a discrete set, and $F'/F = g$ on Ω by the identity theorem. If some $-t_k$ were not in $(-\infty, 0]$, then g would be bounded near $-t_k$ while F'/F has a pole there, a contradiction. Hence every t_k is in $[0, \infty)$, and $t_k \neq 0$ because $F(0) = 1$. So (a) holds.

(a) implies (d), (e), (e′), (f). If all $t_k > 0$, then k is a convergent sum of decreasing exponentials and is completely monotone; (14.6) is a Stieltjes function with $a = b = 0$ and $\sigma = \sum \delta_{t_k}$, and $\int \sigma(dt)/(1+t) \leq S_1$; E has no real zero since a real zero $x \neq 0$ would give $t = -x^2 < 0$; and $\operatorname{Im}(s + t_k)^{-1} = -\operatorname{Im} s/|s + t_k|^2 \leq 0$. For (f) put $u_k = 1/t_k \in (0, 1/t_1']$, where $t_1' = \min t_k$, and $\nu = \sum_k u_k \delta_{u_k}$, a finite measure of mass S_1; then $\int u^j \nu(du) = \sum_k u_k^{j+1} = S_{j+1}$.

(d) implies (a). Let $s > 2\sigma_0^2$. Since $\operatorname{Re}(s + t_k) \geq s - 2\sigma_0^2 + |t_k|$, we have $\sum_k \int_0^\infty |e^{-(s+t_k)x}|\, dx = \sum_k (\operatorname{Re}(s + t_k))^{-1} < \infty$, and Fubini's theorem with (14.6) gives

$$\int_0^\infty e^{-sx} k(x)\, dx = \sum_k \frac{1}{s + t_k} = \frac{F'}{F}(s), \qquad s > 2\sigma_0^2.$$

If k is completely monotone, Theorem 14.1 gives $k(x) = \int_{[0,\infty)} e^{-tx} \mu(dt)$ with $\mu \geq 0$, and Tonelli's theorem gives $\int_0^\infty e^{-sx} k(x)\, dx = \int \mu(dt)/(s + t)$, finite for $s > 2\sigma_0^2$. Thus F'/F coincides on $(2\sigma_0^2, \infty)$ with the Stieltjes function $\mu(\{0\})/s + \int_{(0,\infty)} \mu(dt)/(s + t)$, and F has no zero there because $\operatorname{Re}(-t_k) \leq \sigma_0^2$. The rigidity step gives (a).

(e) implies (a). This is the rigidity step with $c = 0$.

(e′) implies (a). Suppose $F(s_0) = 0$ with $\operatorname{Im} s_0 > 0$, of multiplicity $m \geq 1$. Near s_0, $F'/F(s) = m/(s - s_0) + h(s)$ with h holomorphic. Take $s = s_0 - i\epsilon$ with $0 < \epsilon < \operatorname{Im} s_0$, so that $\operatorname{Im} s > 0$; then $m/(s - s_0) = im/\epsilon$ and $\operatorname{Im}(F'/F)(s) = m/\epsilon + O(1) > 0$ for small ϵ, contradicting (e′). So F has no zero in the open upper half plane, nor, by conjugate symmetry, in the lower one. A zero $s_0 > 0$ of F gives real zeros $\pm\sqrt{s_0}$ of E, excluded by hypothesis, and $F(0) = 1$. Hence every zero $-t_k$ of F is negative.

(f) implies (a). Let $u_k = 1/t_k$, $R = \max_k |u_k|$ (finite, as $|t_k| \to \infty$ and no t_k vanishes) and $A = \sum_k |u_k|$. Then $\int u^j \nu(du) = S_{j+1} \leq AR^j$ for all j. If $\nu((R + \eta, \infty)) > 0$ for some $\eta > 0$ then $\int u^j \nu(du) \geq (R + \eta)^j \nu((R + \eta, \infty))$, which eventually exceeds AR^j; hence ν is carried by $[0, R]$. For $|w| > R$ expand both

sides in powers of $1/w$:

$$\sum_k \frac{u_k}{w-u_k} = \sum_{j\ge 0} \frac{S_{j+1}}{w^{j+1}} = \int_{[0,R]} \frac{\nu(du)}{w-u},$$

both double series converging absolutely. The left side is meromorphic on $\mathbb{C}\setminus\{0\}$, with a pole at each distinct u_k of residue equal to u_k times its multiplicity, which is nonzero; the u_k accumulate only at 0. The right side is holomorphic off $[0,R]$. The compact set $K=[0,R]\cup\{u_k\}$ has connected complement: from a point of $\mathbb{C}\setminus K$ off the real axis, the rays pointing away from $\mathbb{R}$ miss $[0,R]$ and all but countably many of them miss $\{u_k\}$, so they reach infinity inside $\mathbb{C}\setminus K$; and a point of $\mathbb{C}\setminus K$ on $\mathbb{R}$ is joined to such a point by a short vertical segment in the open set $\mathbb{C}\setminus K$. By the identity theorem the two sides agree on $\mathbb{C}\setminus K$. If some $u_k\notin[0,R]$, it is an isolated point of K, the right side is holomorphic near it and the left side has a pole there, a contradiction. So every u_k lies in $(0,R]$ and every $t_k>0$. The equivalence with Hankel positivity is Theorem 14.4 applied to $m_j=S_{j+1}$.

The final assertions follow from the explicit formulas above, uniqueness in Bernstein's theorem, and uniqueness of ν: by the proof of (f) implies (a) every solution is carried by $[0,R]$, and a measure on a compact interval is determined by its moments. □

The strip hypothesis was used only for (d) and for the convergence of k: the equivalence of (a), (b), (c), (e), (e′) and (f) holds for every even entire E of order less than two, real on $\mathbb{R}$, with $E(0)=1$.

THEOREM 14.8 (Thorin equivalence for automorphic L functions; extending [442]). *Let π be a cuspidal automorphic representation of $\mathrm{GL}_n(\mathbb{A}_{\mathbb{Q}})$ as in Section 14.2, with Thorin atoms t_k, Lévy function k_π and power sums S_n as in* (14.3) *and* (14.4). *The following are equivalent.*

(a) The generalized Riemann hypothesis holds for π (equivalently, for $\tilde\pi$).

(b) $s\mapsto 1/F_\pi(s)=\Xi_\pi(0)/\Xi_\pi(\sqrt{s})$ is the Laplace transform of a GGC.

(c) $1/F_\pi$ is the Laplace transform of $\sum_k E_k/\gamma_k^2$, E_k independent standard exponential: a GGC with drift zero and Thorin measure $\sum_k \delta_{\gamma_k^2}$.

(d) The Lévy function $k_\pi(x)=\sum_k e^{xz_k^2}$, which converges for every $x>0$, is completely monotone on $(0,\infty)$.

(e) $\Lambda(s,\pi)$ has no real zero other than $\frac12$, and F_π'/F_π is a Stieltjes function; equivalently (Herglotz criterion) $\Lambda(s,\pi)$ has no real zero other than $\frac12$ and $\mathrm{Im}(F_\pi'/F_\pi)(s)\le 0$ whenever $\mathrm{Im}\, s>0$ and $F_\pi(s)\neq 0$.

(f) $(S_{j+1})_{j\ge0}$ is a Stieltjes moment sequence; equivalently (Thorin Hankel criterion) all Hankel matrices $(S_{i+j+1})_{i,j=0}^N$ and $(S_{i+j+2})_{i,j=0}^N$ are positive semidefinite.

The same holds for ζ with $\Xi_\zeta(z) = \xi(\frac{1}{2}+z)$, and, when $\pi \cong \tilde{\pi}$, with Ξ_π^{sd} in place of Ξ_π, in which case every Thorin atom has half the multiplicity.

PROOF. By Lemma 14.6, $E = \Xi_\pi / \Xi_\pi(0)$ satisfies the hypotheses of Theorem 14.7 with $\sigma_0 = \frac{1}{2}$; in particular its order is at most one. Real zeros of E are the points $x \neq 0$ with $\Lambda(\frac{1}{2}+x, \pi) = 0$, and positive zeros of F_π correspond to them. The statement follows from Theorem 14.7, with the remark before (14.4) identifying (a). □

Each of (b) to (f) is rigid enough to forbid a single pair of atoms off the half line: a GGC transform has no zeros off $(-\infty, 0]$, a Stieltjes or Herglotz function has no poles off $(-\infty, 0]$, and the Cauchy transform of a positive measure on $[0, R]$ has no poles off $[0, R]$. What is missing for a proof is an independent source of the positivity (Chapter 13).

COROLLARY 14.9 (Newton's identities). *Let $\Xi_\pi(z)/\Xi_\pi(0) = \sum_{m \geq 0} e_m z^{2m}$, so that $e_m = \Xi_\pi^{(2m)}(0)/((2m)!\,\Xi_\pi(0))$ and $F_\pi(s) = \sum e_m s^m$. Then for $|s| < \min_k |t_k|$*

$$\log F_\pi(s) = \sum_{n \geq 1} (-1)^{n+1} \frac{S_n}{n} s^n,$$

and $S_1 = e_1$, $S_n = (-1)^{n+1}\big(n e_n - \sum_{j=1}^{n-1} (-1)^{j+1} S_j e_{n-j}\big)$ for $n \geq 2$. In particular the Hankel matrices in Theorem 14.8(f) are computable from finitely many central derivatives of $\Lambda(s, \pi)$ and $\Lambda(s, \tilde{\pi})$, without knowledge of any zero.

PROOF. For $|s| \leq \rho < \min |t_k|$, $\sum_k \sum_n |s/t_k|^n / n = -\sum_k \log(1 - |s/t_k|) \leq C_\rho \sum_k |t_k|^{-1}$, so the double series $\sum_k \log(1 + s/t_k) = \sum_k \sum_n (-1)^{n+1} (s/t_k)^n / n$ may be rearranged. Comparing coefficients in $F_\pi' = F_\pi \cdot (\log F_\pi)'$ gives $n e_n = \sum_{j=1}^{n} (-1)^{j+1} S_j e_{n-j}$, which is the stated recursion. □

REMARK 14.10 (Comparison with Li's criterion). Li's criterion [363] states that the Riemann hypothesis is equivalent to $\lambda_n = \sum_\rho [1 - (1 - 1/\rho)^n] \geq 0$ for all $n \geq 1$; it uses $\rho \mapsto 1 - 1/\rho$, which sends the critical line to the unit circle. The Thorin Hankel criterion uses $\rho \mapsto -(\rho - \frac{1}{2})^{-2}$, which sends it to $(0, \infty)$, and replaces positivity of numbers by positivity of Hankel forms. Hankel criteria for real zeros go back to Hermite for polynomials and to Grommer [226] for entire functions; (f) is a half line version in the variable $s = z^2$.

REMARK 14.11 (What a finite computation sees). For large n, S_n is dominated by the atoms of smallest modulus; for Δ, $t_1/t_2 = 85.05/193.42 = 0.44$, so S_8 equals the first atom's contribution t_1^{-8} to within a relative $1.5 \cdot 10^{-3}$. Under GRH the measure ν of Theorem 14.7(f) has infinitely many support points, because $\Lambda(s, \pi)$ has infinitely many zeros, so every determinant $\det(S_{i+j+1})_{m \times m}$ and $\det(S_{i+j+2})_{m \times m}$ is strictly positive: one determinant that is zero or negative would disprove GRH

for π. The converse needs all of them. An off line pair at height T changes S_n by $O(T^{-2n})$, so a positive definite Hankel matrix of fixed size stays so under such a perturbation once T is large: finitely many Hankel inequalities probe only the lowest atoms, and GRH is the conjunction of all of them, as for Li's coefficients. Under GRH all $S_n > 0$, so the Taylor coefficients of $\log F_\pi$ at 0 alternate in sign.

14.4 PÓLYA VARIABLES AND THEIR CUMULANTS

Chapter 13 wrote $\Lambda(6+z,\Delta)/\Lambda(6,\Delta)$ as the moment generating function of a symmetric random variable. The kernel exists for every π; only its positivity is special.

LEMMA 14.12 (Pólya kernel; this book). *For every π as in Section 14.2 there is a real, even, continuous function Φ_π on $\mathbb{R}$ with $\int \Phi_\pi = 1$ and constants $c, C > 0$ such that $|\Phi_\pi(u)| \le C\exp(-ce^{c|u|})$ and*

$$\frac{\Xi_\pi(z)}{\Xi_\pi(0)} = \int_{-\infty}^{\infty} \Phi_\pi(u) e^{zu}\, du \qquad \text{for all } z \in \mathbb{C}.$$

PROOF. Let $E = \Xi_\pi/\Xi_\pi(0)$. By Stirling's formula and polynomial growth of $L(s,\pi)$ in vertical strips [276], $E(x+it)$ decays exponentially in $|t|$, uniformly for x in compacts, so $\Phi(u) = \frac{1}{2\pi}\int E(x+it)e^{-(x+it)u}\,dt$ is independent of x. At $x = 0$ it is the inverse Fourier transform of the integrable function $t \mapsto E(it)$, which is real and even since $\overline{E(it)} = E(-it) = E(it)$; so Φ is real and even. For $x \ge 1$, $|L(\frac12 + x + it, \pi)|$ is bounded by absolute convergence, and $|\Gamma(\sigma + i\tau)| \le \Gamma(\sigma)(1+\tau^2/\sigma^2)^{-1/2}$ for $\sigma > 0$. Using this bound for two of the $2n$ gamma factors in $\Lambda(\frac12 + x + it, \pi)\Lambda(\frac12 + x + it, \tilde\pi)$ and $|\Gamma(\sigma+i\tau)| \le \Gamma(\sigma)$ for the others, Stirling's formula gives $\int |E(x+it)|\,dt \le \exp(Ax\log x + Bx)$ for $x \ge 1$. Therefore $|\Phi(u)| \le \exp(-xu + Ax\log x + Bx)$ for all $x \ge 1$, and the choice $x = \exp((u - A - B)/A)$, admissible for $u \ge 2A + B$, gives $|\Phi(u)| \le \exp(-Ax)$, a bound of the stated doubly exponential form; evenness and continuity give it for all u. Then $\int \Phi(u)e^{zu}du$ is entire and equals $E(z)$ on $i\mathbb{R}$ by Fourier inversion, hence everywhere, and $z = 0$ gives $\int \Phi = 1$. □

When $\pi \cong \tilde\pi$ the same construction applied to Ξ_π^{sd} gives a kernel Φ_π^{sd} with $\Phi_\pi = \Phi_\pi^{\mathrm{sd}} * \Phi_\pi^{\mathrm{sd}}$, and everything below applies to either. When $\Phi_\pi \ge 0$ we call a random variable W_π with density Φ_π the *Pólya variable* of π; then $\mathbb{E}e^{zW_\pi} = \Xi_\pi(z)/\Xi_\pi(0)$ and $\mathbb{E}e^{itW_\pi} = \Xi_\pi(it)/\Xi_\pi(0)$. Positivity holds for ζ (Pólya's kernel (13.1)), for Δ with Ξ^{sd} (kernel $\Delta(ie^u)e^{6u}/\Lambda(6,\Delta)$, Chapter 13), and for every multiplicative eta product $f = \prod_m \eta(m\tau)^{r_m}$ of level N and weight k [157]: $f(iy) > 0$ (Chapter 27), so the kernel proportional to $f(ie^u/\sqrt{N})e^{ku/2}$ is positive, and the Fricke involution makes it even or odd according to the sign of the functional equation, which positivity forces to be $+1$. For $\Xi_\pi = (\Xi_\pi^{\mathrm{sd}})^2$ the Pólya variable is the

sum of two independent copies. Positivity also holds for $L(s,\chi_{-4})$ and $L(s,\chi_{12})$, whose kernels are built from η^3 and η (Chapter 18), and for the Dedekind zeta functions of the four imaginary quadratic fields with $|d|<\pi^2$ (Theorem 17.17); it fails for infinitely many imaginary quadratic fields (Theorem 17.18), and theta functions of real characters can change sign (Chapters 19 and 21).

PROPOSITION 14.13 (this book). *Suppose $\Phi_\pi \geq 0$. The odd cumulants of W_π vanish and the even ones are*

$$\frac{\kappa_{2n}(W_\pi)}{(2n)!} = (-1)^{n+1}\frac{S_n}{n}, \qquad n \geq 1.$$

Consequently GRH for π holds if and only if the sequence

$$m_j = (-1)^j (j+1)\,\frac{\kappa_{2j+2}(W_\pi)}{(2j+2)!}, \qquad j \geq 0,$$

is a Stieltjes moment sequence. In particular GRH implies $(-1)^{n+1}\kappa_{2n}(W_\pi) > 0$ for all n.

PROOF. By Corollary 14.9, for $|z|^2 < \min|t_k|$ the cumulant generating function is $\log \mathbb{E}e^{zW} = \log F_\pi(z^2) = \sum_n (-1)^{n+1} S_n z^{2n}/n$; compare with $\sum_m \kappa_m z^m/m!$. Then $m_j = S_{j+1}$ and Theorem 14.8(f) applies. □

For $n=1$ this is the variance identity $\operatorname{Var} W = 2S_1$ of Theorem 13.7. Our computations for Δ, with W of density proportional to $\Delta(ie^u)e^{6u}$, compare cumulants obtained by quadrature against the kernel with S_n obtained from central Taylor coefficients:

$$\begin{aligned} \kappa_2 &= 0.11235178504232986, & 2S_1 &= 0.11235178504232987,\\ \kappa_4 &= -0.0023969664979173883, & -12S_2 &= -0.002396966497917392,\\ \kappa_6 &= 0.00044094223922184345, & 240S_3 &= 0.000440942239221848, \end{aligned}$$

agreement to fourteen or more significant digits, with the signs $+,-,+$ predicted by GRH. By Proposition 14.13, the Riemann hypothesis for Δ is equivalent to the statement that the rescaled even cumulants of this explicit symmetric law form a Stieltjes moment sequence.

14.5 RANK AS A DEGENERATE THORIN ATOM

PROPOSITION 14.14 (this book). *Let $\pi \cong \tilde\pi$ with $r = \operatorname{ord}_{s=1/2}\Lambda(s,\pi) \geq 1$ and $H(z) = \Lambda(\tfrac12+z,\pi)/z^r$. Unconditionally,*

$$\lim_{s\to 0^+} \frac{d\log|\Lambda(\tfrac12+\sqrt{s},\pi)|}{d\log s} = \frac{r}{2}.$$

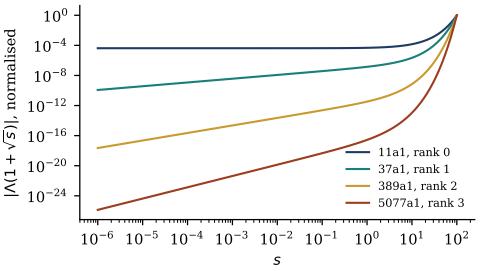


Figure 14.1: $|\Lambda(1+\sqrt{s}, E)|$ on log log axes, normalised to 1 at $s = 100$, for the elliptic curves 11a1, 37a1, 389a1, 5077a1 of ranks $r = 0$ to 3. Near $s = 0$ the slopes are 0, 0.500, 1.000, 1.500: a degenerate Thorin atom of mass $r/2$ at the origin.

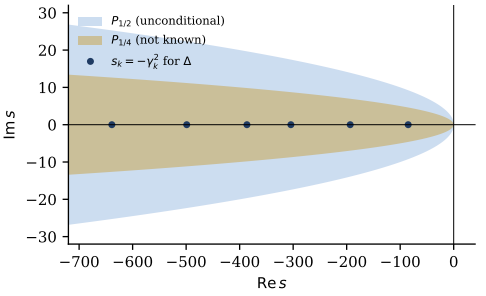


Figure 14.2: The parabolic regions $P_{1/2}$ and $P_{1/4}$ in the plane of $s = z^2$, with the points $s_k = -\gamma_k^2$ for the first six zeros of Δ. For every π all s_k lie in $P_{1/2}$ (condition $\Theta_{1/2}$, unconditional); $\Theta_{1/4}$ is known for no π. GRH puts every s_k on $P_0 = (-\infty, 0]$.

Under GRH for π, for $s > 0$,

$$\frac{H(0)}{\Lambda(\frac{1}{2}+\sqrt{s}, \pi)} = s^{-r/2}\,\mathbb{E}e^{-sX} = \int_0^\infty e^{-sx}\,(m_r * \mu_X)(dx),$$

with $m_r(dx) = x^{r/2-1}dx/\Gamma(r/2)$, X the GGC of Theorem 14.8(c) for Ξ_π^{sd} and μ_X its law. The infinite measure m_r is the increasing limit as $\epsilon \downarrow 0$ of $\epsilon^{-r/2}$ times the gamma law of shape $r/2$ and rate ϵ, whose Thorin measure is $\frac{r}{2}\delta_\epsilon$: a Thorin atom of mass $r/2$ escaping to the origin.

PROOF. $\log|\Lambda(\frac{1}{2}+\sqrt{s})| = \frac{r}{2}\log s + \log|H(\sqrt{s})|$ and $H(\sqrt{s}) \to H(0) \neq 0$ with derivative in $\log s$ tending to zero, which gives the slope. Under GRH, Theorem 14.8 gives $H(0)/H(\sqrt{s}) = \mathbb{E}e^{-sX}$. Also $\int_0^\infty e^{-sx}x^{r/2-1}dx/\Gamma(r/2) = s^{-r/2}$, and the Laplace transform of a convolution of positive measures is the product, by Tonelli. Finally the gamma density of shape $r/2$ and rate ϵ times $\epsilon^{-r/2}$ is $x^{r/2-1}e^{-\epsilon x}/\Gamma(r/2)$, which increases to the density of m_r, and its transform $\epsilon^{-r/2}(1+s/\epsilon)^{-r/2} = (\epsilon+s)^{-r/2}$ tends to $s^{-r/2}$. □

For $r = 2$ the density of $m_2 * \mu_X$ is $\mathbb{P}(X \leq x)$. In the product completion Ξ_π the central zero has order $2r$ and the degenerate mass is r. For an elliptic curve $E/\mathbb{Q}$ the conjecture of Birch and Swinnerton Dyer [89] identifies the analytic rank with the Mordell Weil rank, so under that conjecture *twice the Thorin mass at the origin of Ξ^{sd} is the Mordell Weil rank*; for analytic rank at most one this equality is a theorem of Gross and Zagier [228] and Kolyvagin [329].

Figure 14.1 shows the effect for four curves in the classical normalisation with centre 1: the slopes of $\log|\Lambda(1+\sqrt{s})|$ against $\log s$ between $s = 10^{-6}$ and 10^{-4} are 0.500, 1.000, 1.500 for ranks 1, 2, 3 and 0 for rank 0, until the atoms γ_k^2 take over near $s \approx \gamma_1^2$ (8.27 for 389a1, 4.21 for 5077a1).

14.6 Parabolic Thorin conditions and the heat flow

GRH asks that every Thorin atom lie on the half line. A quantitative relaxation asks that they lie in a region around it.

Definition 14.15. For $\sigma \geq 0$, condition Θ_σ for π is: every zero z of Ξ_π satisfies $|\operatorname{Re} z| \leq \sigma$; equivalently every zero of $\Lambda(s,\pi)$ lies in $|\operatorname{Re} s - \frac{1}{2}| \leq \sigma$. Let $P_\sigma = \{x+iy : x \leq \sigma^2 - y^2/(4\sigma^2)\}$ for $\sigma > 0$ and $P_0 = (-\infty, 0]$.

Lemma 14.16. *For $\sigma \geq 0$, $\{z^2 : |\operatorname{Re} z| \leq \sigma\} = P_\sigma$. Hence Θ_σ holds if and only if every $s_k = z_k^2$ lies in P_σ, if and only if every Thorin atom t_k lies in $-P_\sigma = \{x+iy : x \geq -\sigma^2 + y^2/(4\sigma^2)\}$, the region bounded by the parabola with vertex $-\sigma^2$ that opens along the positive axis. The regions increase with σ and $\bigcap_{\sigma>0}(-P_\sigma) = [0,\infty)$.*

Proof. The case $\sigma = 0$ is $\{(ib)^2\} = (-\infty, 0]$. Let $\sigma > 0$. The two square roots $\pm w$ of $s = x+iy$ have $|\operatorname{Re} w| = ((|s|+x)/2)^{1/2}$, so s is a square of some z with $|\operatorname{Re} z| \leq \sigma$ if and only if $|s| \leq 2\sigma^2 - x$. This requires $x \leq 2\sigma^2$, and then squaring gives $x^2 + y^2 \leq 4\sigma^4 - 4\sigma^2 x + x^2$, that is $x \leq \sigma^2 - y^2/(4\sigma^2)$; conversely this inequality implies $x \leq \sigma^2 \leq 2\sigma^2$. The remaining statements are immediate. □

By Lemma 14.6(iii), $\Theta_{1/2}$ holds unconditionally for every π, with strict inequality: all atoms lie in the interior of $-P_{1/2}$. The image under $z \mapsto z^2$ of the boundary line $\operatorname{Re} z = \frac{1}{2}$ is the parabola $x = \frac{1}{4} - y^2$, which is how the nonvanishing theorem of Jacquet and Shalika looks in the s plane. Zero free regions of de la Vallée Poussin type [276] narrow the strip in the z plane by a logarithmically small amount, which in the s plane is a band whose horizontal width grows like $y^2/\log|y|$ (Exercise 3), but they give Θ_σ for no $\sigma < \frac{1}{2}$, and for no automorphic L function is Θ_σ known with $\sigma < \frac{1}{2}$. Condition Θ_0 is GRH (Figure 14.2).

14.6.1 Gaussian tilting

The Pólya kernel of Lemma 14.12 can be tilted by a Gaussian factor. For $\lambda \in \mathbb{R}$ put

$$H_{\pi,\lambda}(t) = \int_{-\infty}^{\infty} e^{\lambda u^2} \Phi_\pi(u) e^{itu}\, du,$$

an entire function of t, with $H_{\pi,0}(t) = \Xi_\pi(it)/\Xi_\pi(0)$, so Θ_σ says exactly that the zeros of $H_{\pi,0}$ lie in $|\operatorname{Im} t| \leq \sigma$. We use de Bruijn's theorem [133] in the following form: if Φ is real, even and integrable with $\Phi(u) = O(\exp(-|u|^{2+\delta}))$ for some

$\delta > 0$, and all zeros of $H_0(t) = \int \Phi(u) e^{itu} du$ lie in $|\operatorname{Im} t| \le b_0$, then for $\lambda > 0$ all zeros of $H_\lambda(t) = \int e^{\lambda u^2} \Phi(u) e^{itu} du$ lie in $|\operatorname{Im} t| \le (\max(b_0^2 - 2\lambda, 0))^{1/2}$. Positivity of Φ is not needed; see [133, 417] for the precise hypotheses. By Lemma 14.12 they hold for every Φ_π, in particular for Δ and every multiplicative eta product, and Pólya's kernel for ζ satisfies them as well (Chapter 13).

THEOREM 14.17 (Gaussian tilting; (i) de Bruijn, (ii) this book). *Suppose Θ_σ holds for π, and let $\lambda \ge 0$.*

(i) The zeros of $H_{\pi,\lambda}$ lie in $|\operatorname{Im} t| \le \sigma_\lambda$, where $\sigma_\lambda = (\max(\sigma^2 - 2\lambda, 0))^{1/2}$; Gaussian tilting moves the Thorin atoms from $-P_\sigma$ into $-P_{\sigma_\lambda}$, and for $\lambda \ge \sigma^2/2$ all zeros of $H_{\pi,\lambda}$ are real.

(ii) If $\Phi_\pi \ge 0$, let $W_{\pi,\lambda}$ have density proportional to $e^{\lambda u^2}\Phi_\pi(u)$. For every $\lambda \ge \sigma^2/2$ the function $s \mapsto H_{\pi,\lambda}(0)/H_{\pi,\lambda}(i\sqrt{s}) = 1/\mathbb{E}e^{\sqrt{s}W_{\pi,\lambda}}$ is the Laplace transform of a GGC with drift zero and Thorin measure $\sum_k \delta_{x_k(\lambda)^2}$, where $\pm x_k(\lambda)$ are the real zeros of $H_{\pi,\lambda}$.

Unconditionally ($\sigma = \frac{1}{2}$), $H_{\pi,\lambda}$ has only real zeros for every $\lambda \ge \frac{1}{8}$, and (ii) holds for these λ.

PROOF. (i) is de Bruijn's theorem with $b_0 = \sigma$, translated by Lemma 14.16. For (ii), $E_\lambda(z) = H_{\pi,\lambda}(-iz)/H_{\pi,\lambda}(0) = \mathbb{E}e^{zW_{\pi,\lambda}}$ is well defined because $H_{\pi,\lambda}(0) = \int e^{\lambda u^2}\Phi_\pi > 0$, even because Φ_π is even, real on $\mathbb{R}$, and $E_\lambda(0) = 1$. Its order is at most one: with $|\Phi_\pi(u)| \le C\exp(-ce^{c|u|})$ one has $\sup_u(\lambda u^2 + |z||u| - \frac{c}{2}e^{c|u|}) = O(|z|\log(|z|+2))$. By (i) its zeros $z = ix_k(\lambda)$ lie on $i\mathbb{R}$, so Theorem 14.7, (a) implies (c), applies with $\sigma_0 = 0$. The unconditional case is $\sigma = \frac{1}{2}$, from Lemma 14.6(iii). $\square$

REMARK 14.18 (The factor four). The literature normalisation for ζ (Chapter 13) is $H_0^{\mathrm{cl}}(x) = \frac{1}{8}\Xi(x/2)$, $\Xi(t) = \xi(\frac{1}{2} + it)$, with kernel tilt $e^{\lambda_{\mathrm{cl}} v^2}$. Since $t = x/2$, the kernel variables satisfy $u = 2v$, so $e^{\lambda_{\mathrm{cl}} v^2} = e^{(\lambda_{\mathrm{cl}}/4)u^2}$ and $\lambda = \lambda_{\mathrm{cl}}/4$. De Bruijn's classical bound $\lambda_{\mathrm{cl}} \ge \frac{1}{2}$, from zeros of H_0^{cl} in $|\operatorname{Im} x| < 1$, is our $\lambda \ge \frac{1}{8}$, from zeros of H_0 in $|\operatorname{Im} t| < \frac{1}{2}$; the Polymath bound $\Lambda \le 0.22$ [446] is 0.055 in our units.

Define the de Bruijn Newman constant $\lambda_\pi \in [-\infty, \frac{1}{8}]$ of π, in our units, as the infimum of the λ for which $H_{\pi,\lambda}$ has only real zeros. If H_{π,λ_0} has only real zeros, de Bruijn's theorem applied to the kernel $e^{\lambda_0 u^2}\Phi_\pi$ with $b_0 = 0$ shows that $H_{\pi,\lambda}$ has only real zeros for every $\lambda > \lambda_0$. The set of such λ is also closed, by Hurwitz's theorem, so it is $[\lambda_\pi, \infty)$ when λ_π is finite, and GRH for π is equivalent to $\lambda_\pi \le 0$. When $\Phi_\pi \ge 0$, Theorem 14.7 (whose equivalence of (a) and (b) needs no strip hypothesis; see the paragraph after its proof) identifies this set with the set of λ for which the tilted reciprocal is a GGC transform: *the Thorin property holds exactly on* $[\lambda_\pi, \infty)$. Rodgers and Tao [465] proved Newman's conjecture $\lambda_\zeta \ge 0$, and Dobner's extension [158] to the extended Selberg class gives $\lambda_\Delta \ge 0$, as Chapter 13

explains; we expect, but do not verify here, that it gives $\lambda_\pi \geq 0$ for the other kernels Φ_π of this chapter. For ζ, and for every π with $\Phi_\pi \geq 0$ and $\lambda_\pi \geq 0$, GRH would therefore mean $\lambda_\pi = 0$: the tilted reciprocal is a GGC transform for every $\lambda \geq 0$ and for no $\lambda < 0$. *For such π, if GRH holds, $1/F_\pi$ lies exactly on the edge of the class of GGC transforms along the heat flow*, in line with Newman's remark that the Riemann hypothesis, if true, is only barely so.

PROPOSITION 14.19 (Motion of Thorin atoms). *Let I be an open interval of λ on which $H_{\pi,\lambda}(0) \neq 0$ and all zeros of $H_{\pi,\lambda}$ are real and simple, written $\pm x_k(\lambda)$, $x_k > 0$, and let $t_k(\lambda) = x_k(\lambda)^2$. Then each t_k is real analytic on I and*

$$\frac{dt_k}{d\lambda} = 2 + 8 \sum_{j \neq k} \frac{t_k}{t_k - t_j},$$

the series converging absolutely.

PROOF. $H(t, \lambda) = H_{\pi,\lambda}(t)$ is jointly analytic, and differentiating under the integral gives $\partial_\lambda H = -\partial_t^2 H$. At a simple zero the implicit function theorem gives analytic $x_k(\lambda)$ with $x_k' = -\partial_\lambda H / \partial_t H = H''(x_k)/H'(x_k)$, primes on H denoting t derivatives. As in the proof of Theorem 14.17, $H_{\pi,\lambda}(t) = H_{\pi,\lambda}(0)(1 - t^2/t_k)g(t)$ with $g(t) = \prod_{j \neq k}(1 - t^2/t_j)$, locally uniformly convergent. Writing $H = (t - x_k)q(t)$ with $q(t) = -H_{\pi,\lambda}(0) t_k^{-1}(t + x_k)g(t)$ and $q(x_k) \neq 0$, we get $H'(x_k) = q(x_k)$ and $H''(x_k) = 2q'(x_k)$, and $g'/g(t) = \sum_{j \neq k} 2t/(t^2 - t_j)$, so

$$x_k' = 2\Big(\frac{1}{2x_k} + \sum_{j \neq k} \frac{2x_k}{x_k^2 - x_j^2}\Big).$$

Multiplying by $2x_k$ gives the formula. Absolute convergence follows from $|t_k/(t_k - t_j)| \leq 2t_k/t_j$ for large j and $\sum_j t_j^{-1} < \infty$. □

The constant 2 is the repulsion of each zero from its mirror image. In the variable x the flow is the classical zero dynamics of the backward heat equation, $x_k' = 2\sum (x_k - x)^{-1}$ over all other zeros x, summed symmetrically, and formally $x_k'' = -8\sum (x_k - x)^{-3}$, an inverse cube interaction of Calogero Moser type; the proposition only rewrites it for the Thorin atoms $t_k = x_k^2$. As λ decreases the atoms attract; heuristically, a collision is the moment two real zeros meet and leave the axis as a conjugate pair, and λ_π is the supremum of the collision times. Csordas, Smith and Varga [120] obtained lower bounds for λ_ζ from close pairs of zeros along these lines. We use the dynamics only heuristically; continuation through collisions is not addressed.

14.7 A CATALOGUE OF APPLICATIONS

The proof of Theorem 14.8 uses only that Λ is entire of order one, satisfies a functional equation relating it to its conjugate, and is zero free off the strip $0 < \operatorname{Re} s < 1$. This is known in the following cases.

Dirichlet L functions. For primitive $\chi \neq 1$, $\Lambda(s,\chi)$ is entire; real χ are self dual, and for complex χ, $\Xi_\chi(x) = |\Lambda(\frac{1}{2}+x,\chi)|^2/x^{2r}$ on $\mathbb{R}$.

Dedekind and Hecke L functions. For a number field K, $s(s-1)\Lambda_K(s)$ is entire and symmetric under $s \mapsto 1-s$; Hecke L functions of nontrivial Größencharaktere are entire [248, 539].

Newforms and elliptic curves. For a holomorphic newform f, $\Lambda(s,f)$ is entire [251, 29]; by modularity [581, 541, 41] this covers every elliptic curve over $\mathbb{Q}$.

Symmetric powers. Newton and Thorne [419, 420] proved that for every holomorphic newform f of weight at least two without complex multiplication, $\operatorname{Sym}^m \pi_f$ is a cuspidal automorphic representation of $\mathrm{GL}_{m+1}(\mathbb{A}_\mathbb{Q})$ for every $m \geq 1$. Hence Theorem 14.8 applies to every symmetric power of every such form. (With complex multiplication $\operatorname{Sym}^m f$ splits into Hecke characters, and for $4 \mid m$ a trivial constituent gives a pole, removed as for ζ.)

Rankin Selberg L functions. For cuspidal π, π' with $\pi' \not\cong \tilde{\pi}$, $\Lambda(s, \pi \times \pi')$ is entire with a functional equation [286, 397], bounded in vertical strips away from the real axis by Gelbart and Shahidi [213], which gives order one, and nonzero on $\operatorname{Re} s = 1$ by Shahidi [510]. For $\pi' = \tilde{\pi}$ there are simple poles at $s = 0, 1$, removed by the factor $s(s-1)$. Theorem 14.7 applies with the same proof.

Artin L functions. The theorem applies to every irreducible ρ for which Artin's conjecture is known. By Khare and Wintenberger [311] every odd irreducible two dimensional ρ comes from a weight one newform, so this includes all of them. Our example is the representation of S_3 cut out by $x^3 - x - 1$, of conductor 23, with L function $\zeta_K(s)/\zeta(s)$ for the cubic field K, the L function of $\eta(\tau)\eta(23\tau)$.

Table 14.1 records computations carried out for this book on eleven self dual L functions, using Ξ^{sd}. We computed the zeros to height $T = 300$ (138 to 680 of them) and compared $\Xi^{\mathrm{sd}}(\sqrt{s})/\Xi^{\mathrm{sd}}(0)$ with $\prod(1 + s/\gamma_k^2)\exp(s\sum_{\gamma>T}\gamma^{-2})$, the tail estimated by $(\log N + d\log(T/2\pi) + d)/(2\pi T)$, which is the integral of t^{-2} against the zero density $(\log N + d\log(t/2\pi))/(2\pi)$ over $t > T$. The relative error is at most $2.8 \cdot 10^{-5}$ at $s = 4$, $1.8 \cdot 10^{-4}$ at $s = 25$ and $8 \cdot 10^{-4}$ at $s = 100$ in every case. The same tail estimate is added to S_1 from zeros in the table. For S_3 and S_4 the values from Taylor coefficients and from zeros agree to all printed digits, for instance $1.8373 \cdot 10^{-6}$ and $2.0017 \cdot 10^{-8}$ for Δ, $1.4417 \cdot 10^{-7}$ and $6.6303 \cdot 10^{-10}$ for ζ. For S_2 the zero sums carry no tail correction and fall short of the Taylor values ($1.9975 \cdot 10^{-4}$ for Δ, $3.7173 \cdot 10^{-5}$ for ζ) by the omitted tail $\sum_{\gamma>T}\gamma^{-4}$, a relative amount of at most $2.2 \cdot 10^{-4}$. The coefficients of $\log F(s)$ alternate in sign through $n = 8$ for all eleven functions, as GRH requires (Remark 14.11). The Hankel

L function	d	N	r	γ_1	S_1 Taylor	S_1 zeros	err(25)	Hankel
ζ (via ξ)	1	1	0	14.1347	0.023105	0.023108	$8.3 \cdot 10^{-5}$	yes
$L(s, \chi_{-4})$	1	4	0	6.0209	0.078015	0.078013	$3.9 \cdot 10^{-5}$	yes
$L(s, \chi_{-3})$	1	3	0	8.0397	0.056701	0.056696	$1.1 \cdot 10^{-4}$	yes
$L(s, \chi_5)$	1	5	0	6.6485	0.078457	0.078462	$1.3 \cdot 10^{-4}$	yes
Δ	2	1	0	9.2224	0.0561759	0.0561824	$1.7 \cdot 10^{-4}$	yes
11a1	2	11	0	6.3626	0.127589	0.127584	$1.1 \cdot 10^{-4}$	yes
37a1	2	37	1	5.0032	0.189609	0.189611	$5.4 \cdot 10^{-5}$	yes
389a1	2	389	2	2.8761	0.407048	0.407051	$1.0 \cdot 10^{-4}$	yes
5077a1	2	5077	3	2.0525	0.725919	0.725921	$6.0 \cdot 10^{-5}$	yes
Artin, $x^3 - x - 1$	2	23	0	5.1157	0.170287	0.170294	$1.8 \cdot 10^{-4}$	yes
Sym^2 11a1	3	121	0	3.8993	0.321206	0.321205	$2.1 \cdot 10^{-5}$	yes

Table 14.1: The global Thorin catalogue, computed with Ξ^{sd}: degree d, conductor N, central order r, first ordinate γ_1 beyond the centre, $S_1 = \sum \gamma_k^{-2}$ from central Taylor coefficients and from the zeros to height 300 plus the tail estimate, absolute relative error of the product formula at $s = 25$, and positivity of $\det(S_{i+j+1})_{m\times m}$, $m \le 4$, and $\det(S_{i+j+2})_{m\times m}$, $m \le 3$, computed from Taylor coefficients alone.

determinants $\det(S_{i+j+1})_{m\times m}$, $m \le 4$, and $\det(S_{i+j+2})_{m\times m}$, $m \le 3$, computed from central Taylor coefficients at 60 digit precision without using any zero, are positive in every case; for Δ the first family is $5.618 \cdot 10^{-2}$, $6.331 \cdot 10^{-8}$, $4.749 \cdot 10^{-19}$, $9.579 \cdot 10^{-36}$, and for ζ it is $2.310 \cdot 10^{-2}$, $1.949 \cdot 10^{-9}$, $2.173 \cdot 10^{-22}$, $1.178 \cdot 10^{-41}$. These are necessary conditions for GRH, so the computation is consistent with it; by Remark 14.11 it constrains only the lowest atoms.

14.8 The Thorin threshold

Everything above requires $\sum_k |t_k|^{-1} < \infty$, which makes F a genus zero product and $\sum E_k/t_k$ convergent; for automorphic L functions $N(T) \asymp T \log T$ gives it with room to spare. The Selberg zeta function crosses this threshold.

Let $\Gamma\backslash\mathbb{H}$ be a compact hyperbolic surface with Laplace eigenvalues $0 = \lambda_0 < \lambda_1 \le \lambda_2 \le \dots$ and write $\lambda_j = \frac{1}{4} + r_j^2$. Selberg's zeta function $Z_\Gamma(s)$ [496, 252] has zeros at $s = \frac{1}{2} \pm i r_j$, and by Sarnak [477] it equals, up to an explicit factor built from the Barnes double gamma function, the regularised determinant $\det(\Delta_\Gamma - s(1-s))$. Removing the zeros at $s = 0, 1$ from $\lambda_0 = 0$ and assuming for simplicity that $\frac{1}{4}$ is not an eigenvalue, we obtain an even entire $E_\Gamma(z)$, real on $\mathbb{R}$, $E_\Gamma(0) = 1$, with zeros $\pm i r_j$, $j \ge 1$, and put $F_\Gamma(s) = E_\Gamma(\sqrt{s})$. Its Thorin atoms are

$$t_j = r_j^2 = \lambda_j - \tfrac{1}{4},$$

the spectrum of $\Delta_\Gamma - \frac{1}{4}$. But Weyl's law $\#\{j : \lambda_j \le x\} \sim \frac{\mathrm{area}}{4\pi} x$ [575, 252] gives $\#\{j : |r_j| \le T\} \sim \frac{\mathrm{area}}{4\pi} T^2$, so E_Γ has order two, $\sum_j |t_j|^{-1}$ diverges logarithmically

and $\sum_j |t_j|^{-2} < \infty$. The reciprocal is no longer a GGC transform even when all $t_j > 0$, but a compensated version survives.

PROPOSITION 14.20 (Regularised Thorin equivalence; this book). *Let E be entire, even, real on $\mathbb{R}$, $E(0) = 1$, of order at most two with zeros $\pm z_k$ satisfying $\sum |z_k|^{-4} < \infty$, and let $t_k = -z_k^2$, $F(s) = E(\sqrt{s})$. Then $F(s) = e^{bs} \prod_k (1 + s/t_k) e^{-s/t_k}$ for some real b, and the following are equivalent: (a) all $t_k > 0$; (b) for independent standard exponential variables E_k the series $Y = b + \sum_k (E_k - 1)/t_k$, which converges in L^2, is real valued and satisfies $1/F(s) = \mathbb{E}e^{-sY}$ for $s \geq 0$; (c) $1/F$ agrees on $(0, \infty)$ with a function holomorphic and zero free on $\mathbb{C} \setminus (-\infty, 0]$. The law of Y in (b) is infinitely divisible with Lévy measure $x^{-1} \sum_k e^{-t_k x} dx$ on $(0, \infty)$; when $\sum_k t_k^{-1} = \infty$ its first moment $\int_0^1 \sum_k e^{-t_k x} dx$ near the origin is infinite, which is why the centring is needed.*

PROOF. F has order at most one with $\sum |t_k|^{-2} < \infty$, so Hadamard's theorem gives the genus one product with a factor e^{a+bs}, $a = 0$ since $F(0) = 1$; b is real since $\{t_k\}$ is closed under conjugation and F is real on $(0, \infty)$. The series for Y converges in L^2 because its summands are independent with mean zero and $\sum_k |t_k|^{-2} < \infty$. (a) implies (b): if all $t_k > 0$ the series also converges almost surely, and $\mathbb{E}e^{-s(E_k - 1)/t_k} = e^{s/t_k}(1 + s/t_k)^{-1}$. For fixed $s \geq 0$ the normalised partial products $e^{-sY_N}/\mathbb{E}e^{-sY_N}$, with $Y_N = \sum_{k \leq N} (E_k - 1)/t_k$, form a positive martingale whose second moments are $\prod_{k \leq N} (1 + s/t_k)^2 (1 + 2s/t_k)^{-1} \leq \prod_k (1 + s^2/t_k^2) < \infty$; it is therefore uniformly integrable, and passing to the limit gives (b). (b) implies (a): the imaginary part of Y is $\sum_k \operatorname{Im}(t_k^{-1})(E_k - 1)$, a sum of independent variables that has a continuous law as soon as one coefficient is nonzero, so Y real forces every t_k real; and a real $t_k < 0$ would make $\mathbb{E}e^{-s(E_k - 1)/t_k}$, hence $\mathbb{E}e^{-sY}$, infinite for every $s \geq |t_k|$, while $1/F$ is finite on $(0, \infty)$ outside a discrete set. (a) implies (c): each factor $e^{s/t_k}(1 + s/t_k)^{-1}$ is holomorphic and zero free on $\mathbb{C} \setminus (-\infty, 0]$ and the product converges locally uniformly there. (c) implies (a): the identity theorem shows that F has no zeros off $(-\infty, 0]$, and $F(0) = 1$, so all $t_k > 0$. The Lévy measure follows from Frullani's integral applied to each factor. □

For the surface, (a) says $\lambda_1 \geq \frac{1}{4}$ (strictly, under our simplifying assumption): the Thorin property of the regularised reciprocal of the Selberg zeta function is the absence of exceptional eigenvalues. On a general surface small eigenvalues give negative atoms $t_j \in (-\frac{1}{4}, 0)$. For arithmetic surfaces this is Selberg's eigenvalue conjecture [497], for congruence subgroups of $\mathrm{SL}_2(\mathbb{Z})$ and, through the Jacquet Langlands correspondence [277], for the compact quaternion surfaces over $\mathbb{Q}$. There the bound $\lambda_1 \geq \frac{975}{4096} = \frac{1}{4} - (\frac{7}{64})^2$ of Kim and Sarnak [315] says precisely that every atom satisfies $t_j \geq -(\frac{7}{64})^2$: condition $\Theta_{7/64}$ for the spectral zeros. (For noncompact congruence surfaces the zeros from the scattering determinant must first be separated.) In field theoretic terms, if all $t_j > 0$ then $\sum_j E_j/t_j$ is the formal squared norm of the complex Gaussian field with covariance $(\Delta_\Gamma - \frac{1}{4})^{-1}$, and

the centring in Y is its Wick renormalisation, needed because that norm diverges logarithmically in two dimensions.

OPEN PROBLEM 14.1 (Selberg's conjecture as a regularised Thorin problem). For the family of compact arithmetic surfaces attached to maximal orders in indefinite quaternion algebras over $\mathbb{Q}$, or for congruence surfaces after removal of the scattering zeros, construct directly, without spectral input and uniformly in the level, real random variables Y_Γ with $\mathbb{E}e^{-sY_\Gamma} = 1/F_\Gamma(s)$ for every $s \geq 0$; under (a) of Proposition 14.20 the variable of its part (b) is one. Since the atoms $t_j = \lambda_j - \frac{1}{4}$ are real, a negative atom would make $1/F_\Gamma$ infinite at the point $s = -t_j > 0$ but finite at 0 and at all large s, which is impossible for a Laplace transform, whose domain of finiteness is an interval. Such a construction would therefore prove Selberg's eigenvalue conjecture for the family. More modestly, develop the Thorin theory of order two: the analogues of the Lévy function criterion and of the Hankel criterion, Theorem 14.7(d) and (f), for compensated sums.

14.9 PICTURES OF THE THORIN PROPERTY

Figures 14.3 and 14.4 show the two sides of Theorem 14.8 for Δ. The phase portrait of $\Lambda(6+z,\Delta)$ (classical normalisation, centre 6) colours each point by the argument of the function, with brightness bands for its modulus; a zero is a point where every colour meets. In the window every zero lies on the axis $\operatorname{Re} z = 0$, as the Riemann hypothesis for Δ predicts; this is a numerical observation, not a proof. Since $\Lambda(6+z,\Delta)$ is even in z, the reciprocal $\Lambda(6)/\Lambda(6+\sqrt{s})$ is meromorphic in the whole plane of $s = z^2$, and in the window its only poles are the points $-\gamma_k^2$ on the negative axis: this is the analytic shape of a generalized gamma convolution whose Thorin measure is $\sum_k \delta_{\gamma_k^2}$.

14.10 NOTES

For generalized gamma convolutions, the reciprocal Ξ function, the Brownian reading of ξ and the heat flow see the Notes to Chapter 13. Bernstein's theorem is [49]; Stieltjes functions and their Nevanlinna Pick characterisation are treated in [487], and the moment problem on the half line in [528, 10]. The equivalence of (a), (b) and (c) of Theorem 14.7 for Laguerre Pólya functions, with its van Dantzig and Wald forms, is from [442]. As far as we know, the self dual completion for general π, the forms (d), (e) and (f) in this generality, the cumulant criterion, the rank atom, the parabolic and tilted versions and the computations have not appeared in this form, although each rests on classical ingredients: Hankel criteria for real zeros go back to Hermite and Grommer [226], and Newton's identities link the power sums S_n to the central Taylor coefficients. The results labelled as ours are elementary consequences of Hadamard's factorisation, the identity theorem and the classical

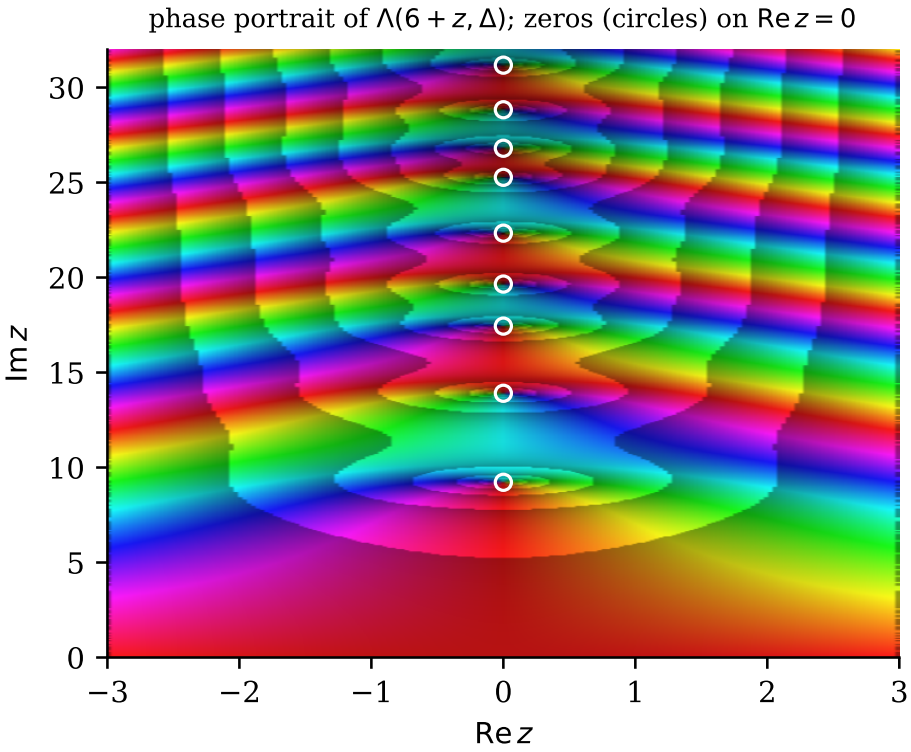


Figure 14.3: Phase portrait of $\Lambda(6+z,\Delta)$ for $|\mathrm{Re}\, z| \le 3$, $0 \le \mathrm{Im}\, z \le 32$, with the modulus normalised along each horizontal line. The white circles mark the zeros found on $\mathrm{Re}\, z = 0$ by PARI/GP; the colours show no other zero in the window.

facts cited; none proves a Riemann hypothesis. The zero dynamics under the heat flow, which Proposition 14.19 rewrites for the Thorin atoms, were used by Csordas, Smith and Varga [120] and by the Polymath project [446]. Numerical values were computed for this book with PARI/GP [432]; the first zeros agree with those listed in the LMFDB [371].

14.11 Exercises

1. Let $g(s) = \sum_{j=1}^{J} c_j/(s+t_j)$ with $c_j > 0$ and distinct real t_j. Show directly that $\mathrm{Im}\, g(s) \le 0$ on the upper half plane, and that g is a Stieltjes function if and only if every $t_j \ge 0$. Show that if the t_j may be complex, closed under conjugation with equal weights on conjugate pairs, then $\mathrm{Im}\, g \le 0$ on the upper half plane forces all t_j real.

2. For $\nu > -1$ let $E_\nu(z) = \Gamma(\nu+1)(2/z)^\nu I_\nu(z)$, whose zeros are $\pm i j_{\nu,k}$ with

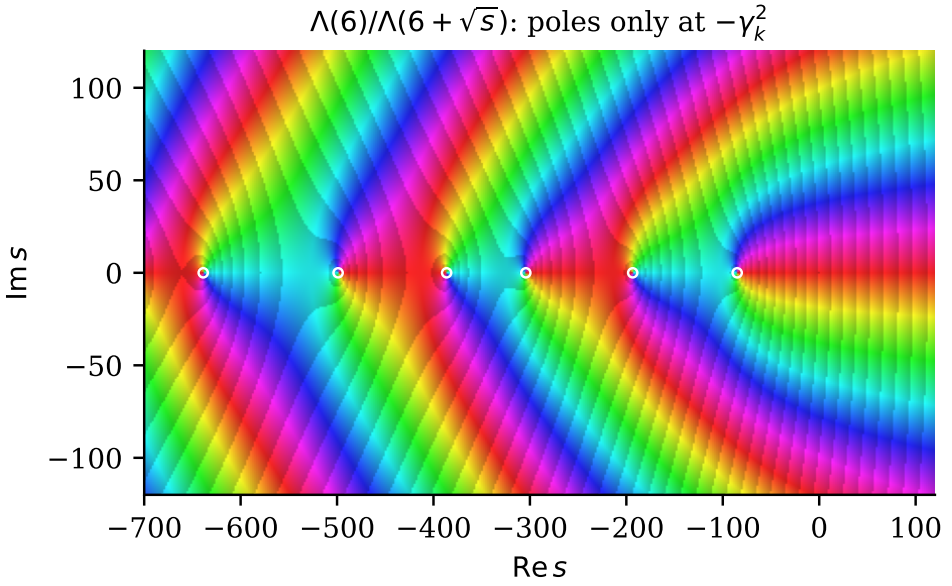


Figure 14.4: Domain colouring of $\Lambda(6,\Delta)/\Lambda(6+\sqrt{s},\Delta)$ in the s plane. The circles mark $-\gamma_k^2$ for the first six zeros of Δ; the colour wheel winds around each of them with the orientation of a pole, and around no other point of the window.

$j_{\nu,k}>0$ the zeros of J_ν. Show that E_ν satisfies the hypotheses of Theorem 14.7, that $S_1=\frac{1}{4(\nu+1)}$ and $S_2=\frac{1}{16(\nu+1)^2(\nu+2)}$, and check the case $\nu=\frac12$, $E(z)=\sinh z/z$, against $\sum(\pi k)^{-2}=\frac16$ and $\sum(\pi k)^{-4}=\frac{1}{90}$. Compute S_3 and S_4 by Corollary 14.9 and verify that the two 2×2 Hankel determinants $S_1S_3-S_2^2$ and $S_2S_4-S_3^2$ are positive.

3. Verify Lemma 14.16 by showing that $z\mapsto z^2$ maps the line $\operatorname{Re} z=a>0$ onto the parabola $x=a^2-y^2/(4a^2)$. Show that the image of the curve $\operatorname{Re} z=\frac12-c/\log(|\operatorname{Im} z|+3)$ lies at horizontal distance asymptotic to $4cy^2/\log|y|$ from the parabola $x=\frac14-y^2$ as $|y|\to\infty$: a zero free region that is logarithmically thin in the z plane is a wide region in the s plane.

4. Let $E/\mathbb{Q}$ have analytic rank 2, assume GRH for E, and put $H(z)=\Lambda(1+z,E)/z^2$. Show that $H(0)/\Lambda(1+\sqrt{s},E)=\int_0^\infty e^{-sx}\mathbb{P}(X\le x)\,dx$ with X the GGC of Proposition 14.14, and deduce that $s\mapsto H(0)/\Lambda(1+\sqrt{s},E)$ is completely monotone on $(0,\infty)$. For rank 3 find the corresponding fractional integral.

5. Let W be the Pólya variable of ζ. Using Proposition 14.13 and $S_1=0.023105$, $S_2=3.7173\cdot10^{-5}$, compute the variance and the excess kurtosis $\kappa_4/\kappa_2^2=$

$-3S_2/S_1^2$ of W. Prove that $-3S_2/S_1^2 \in (-3, 0)$ whenever all $t_k > 0$ and there are at least two atoms, and show by an example with one conjugate pair of nonreal atoms that $-3S_2/S_1^2$ can then be positive.

14.12 Open computational problems

Computational Problem 14.1 (Certified Thorin Hankel determinants). Objective: certify the positivity of the Hankel determinants $\det(S_{i+j+1})_{m\times m}$ and $\det(S_{i+j+2})_{m\times m}$ for $m \le 20$, computed from central derivatives only. Objects and scale: all L functions in LMFDB of degree at most 4 and analytic conductor below 10^3. Route: central derivatives $\Xi^{(2j)}(0)$, $2j \le 80$, in rigorous ball arithmetic (Arb, cross checked with PARI/GP `lfun`), Corollary 14.9 in interval arithmetic, and interval Cholesky factorisation; the precision needed grows roughly like $m^2 \log t_m$ digits. Deliverable: a table of certified lower bounds for the smallest eigenvalue of each Hankel matrix, and the smallest m at which certification fails for lack of precision. Why it matters: these are the finite sections of the Thorin Hankel criterion, Theorem 14.8(f), and the margins quantify how the criterion sees the lowest atoms (Remark 14.11).

Computational Problem 14.2 (The Herglotz test in the upper half plane). Objective: map $\operatorname{Im}(F'/F)(s)$ on a grid in $\{\operatorname{Im} s > 0\}$ and measure its distance from violating Theorem 14.8(e). Objects and scale: the eleven functions of Table 14.1 and all elliptic curves of conductor below 10^4, on a logarithmic grid with $\operatorname{Re} s \in [-10^4, 10^2]$, $\operatorname{Im} s \in [10^{-3}, 10^3]$, about 10^6 points each; on the lines $\operatorname{Im} s = \varepsilon$ the map gives, up to a constant factor and the change of variable $z = -\operatorname{Re} s$, the signed densities u_ε of Chapter 19. Route: Λ and Λ' at $\frac{1}{2} + \sqrt{s}$ with PARI/GP `lfunlambda`, parallelised over grid points. Deliverable: heat maps of $-\operatorname{Im}(F'/F) \cdot |s|$, the minimum over the grid, and an empirical law for how the minimum decays as the grid approaches the negative axis. Why it matters: an off line zero z_0 of Ξ would give a pole of F'/F at $s_0 = z_0^2$, with $\operatorname{Im} s_0 = 2|\operatorname{Re} z_0 \operatorname{Im} z_0|$, surrounded by a region where $\operatorname{Im}(F'/F) > 0$; the computation calibrates how such a region would be seen.

Computational Problem 14.3 (Collision times of the Thorin atoms of Δ). Objective: measure how close the low zeros of Δ are to leaving the critical line under the backward heat flow. Let $\lambda_c(K) < 0$ be the first collision time of the truncated system in which the lowest K atoms γ_k^2 move from $\lambda = 0$ downwards by the equation of Proposition 14.19, the atoms beyond them being replaced in the interaction sum, from γ_{K+1} on, by their fixed smooth density $\frac{1}{\pi} \log \frac{\gamma}{2\pi}$ in the ordinate γ; the atoms are those of the self dual kernel $\Phi_\Delta^{\mathrm{sd}}$, since the proposition needs simple zeros and for the product kernel $\Phi_\Delta^{\mathrm{sd}} * \Phi_\Delta^{\mathrm{sd}}$ every atom is double. Objects and scale: K from 10 to 10^5, hence the first 10^5 zeros of $L(s, \Delta)$, and $\lambda \in [-1, 0]$. Route: zeros

by PARI/GP `lfunzeros`, an adaptive implicit integrator with fast multipole summation, validated for small $|\lambda|$ against the zeros of $t \mapsto \int e^{\lambda u^2} \Phi_\Delta^{\mathrm{sd}}(u) \cos(tu)\, du$ located by quadrature. Deliverable: $\lambda_c(K)$ on a grid of K, the pair responsible (a Lehmer pair for Δ), and the dependence of $\lambda_c(K)$ on K. Why it matters: since $\lambda_\Delta \geq 0$ (Chapter 13), the Riemann hypothesis for Δ would make the Thorin property hold exactly for $\lambda \geq 0$ (Theorem 14.17 and the discussion after it) and, heuristically, 0 the supremum of the collision times; $\lambda_c(K)$ measures how narrowly the hypothesis would hold among the lowest K atoms.

COMPUTATIONAL PROBLEM 14.4 (A regularised Thorin computation for a Selberg zeta function). Objective: test Proposition 14.20 numerically for an arithmetic surface. Objects and scale: the first 10^4 Laplace eigenvalues of the compact quotient $\Gamma\backslash\mathbb{H}$, with Γ the norm one units of a maximal order in the quaternion algebra of discriminant 6 over $\mathbb{Q}$ (an orbifold of genus 0), and of the modular surface (Maass cusp forms, scattering part separated). Route: eigenvalues by Hejhal's method or finite elements, checked against the trace formula and Weyl's law; the law of $Y = b + \sum_j (E_j - 1)/t_j$ by Monte Carlo with 10^6 draws and the Weyl tail added as a Gaussian correction. Deliverable: the atoms $t_j = \lambda_j - \frac{1}{4}$, a plot of the Lévy function $\sum_j e^{-t_j x}$ against its Weyl asymptote $\frac{\mathrm{area}}{4\pi x}$ as $x \to 0$, and the density of Y. Why it matters: it tests the Thorin framework beyond the threshold, where the reformulated conjecture is Selberg's eigenvalue conjecture rather than a Riemann hypothesis.

CHAPTER 15

LOCAL THORIN STRUCTURE: RAMANUJAN BOUNDS, FUNCTION FIELDS AND HITTING TIMES

Ramanujan conjectured in 1916 that $|\tau(p)| \leq 2p^{11/2}$ [453]. The bound concerns a single Euler factor: it says exactly that the two roots of $1 - \tau(p)X + p^{11}X^2$ have the same absolute value. The Riemann hypothesis for $L(s, \Delta)$ is a statement about all Euler factors at once, and nothing in the classical theory makes the two look alike. Chapter 14 recast the generalized Riemann hypothesis for an automorphic L function as a positivity property: the reciprocal of the completed function, evaluated at $\sqrt{s}$, is the Laplace transform of a generalized gamma convolution, with Thorin measure carried by the squared ordinates of the zeros. In this chapter we apply the same test to a single Euler factor and find that it is passed exactly when the Ramanujan bound holds. The slogan is that *Ramanujan is local Thorin and Riemann is global Thorin*. We then follow the local Thorin measures in four directions: into the explicit formula, where they appear as damped periodic combs; into random Euler products with Sato Tate angles, where they become point processes; into function fields and regular graphs, where global Thorin is a theorem; and into hitting times of Hecke walks on trees and isogeny graphs, where Thorin measures are spectra and Keilson's theorem turns the Ramanujan bound into statements about passage times. The last section returns to the extreme value laws of Chapter 27 and identifies their exponent measures.

Throughout, a GGC is as in Definition 13.2, we write $L = \log p$ (or $\log q$) when the prime is fixed, and $E, E_1, E_2, \ldots$ are independent standard exponential variables.

15.1 THE LOCAL THORIN CRITERION

Let π be a cuspidal automorphic representation of $\mathrm{GL}_2(\mathbb{A}_{\mathbb{Q}})$ with trivial central character, in the unitary normalisation, and let p be a prime at which π is unramified. Its Euler factor is

$$L_p(s) = (1 - \alpha p^{-s})^{-1}(1 - \beta p^{-s})^{-1}, \qquad \alpha\beta = 1.$$

Unitarity of π_p makes the Satake set $\{\alpha, \beta\}$ stable under $z \mapsto \bar{z}^{-1}$, so either $|\alpha| = 1$ or α is real; in both cases $x_p = \alpha + \beta$ is real, and $|x_p| \leq 2$ exactly when $|\alpha| = 1$. The Ramanujan Petersson conjecture at p is the statement $|\alpha| = |\beta| = 1$. We complete

the local factor by the symmetrising factor p^{w}:

$$E_p(w) = p^{w} L_p(w)^{-1} = p^{w} - x_p + p^{-w} = 2\cosh(wL) - x_p, \qquad L = \log p.$$

It is entire, even and real on the real axis, the exact local analogue of Ξ_π. Its zeros are the local zeros of $L_p(w)^{-1}$, and the Ramanujan bound says they lie on $\Re w = 0$. So the local centre is $\Re w = 0$, while the global centre in the same normalisation is $\Re s = \frac{1}{2}$; Section 15.2 measures the difference.

LEMMA 15.1. *For $0 < \theta \le \pi$ and all $w \in \mathbb{C}$,*

$$\cosh w - \cos\theta = (1 - \cos\theta) \prod_{k \in \mathbb{Z}} \Big(1 + \frac{w^2}{(\theta + 2\pi k)^2}\Big),$$

the product converging absolutely and locally uniformly.

PROOF. The left side is even and entire, so it equals $G(w^2)$ with $G(u) = \sum_{n \ge 0} u^n/(2n)! - \cos\theta$ entire. Since $|G(u)| \le \cosh|u|^{1/2} + 1$, G has order at most $\frac{1}{2}$. Its zeros are the values $u = w^2$ with $\cosh w = \cos\theta$, that is $w = \pm i(\theta + 2\pi k)$, so $u_k = -(\theta + 2\pi k)^2$, $k \in \mathbb{Z}$. If $0 < \theta < \pi$ the u_k are distinct simple zeros, because $G'(u_k) = \sinh(w)/(2w)$ with $\sinh w = \pm i \sin\theta \ne 0$. If $\theta = \pi$ then $u_k = u_{-k-1}$, and both sides acquire double zeros, since $\cosh w + 1 = 2\cosh^2(w/2)$; listing u_k over $k \in \mathbb{Z}$ counts this multiplicity correctly. As $\sum_k |u_k|^{-1} < \infty$ and G has order less than one, Hadamard's theorem gives $G(u) = G(0)\prod_k (1 - u/u_k)$ with no exponential factor, and $G(0) = 1 - \cos\theta \ne 0$. □

The main result of this section needs a classical fact about Laplace transforms of positive measures, due in the setting of Dirichlet series to Landau [344] and going back to Pringsheim's theorem on power series; see Widder [578].

LEMMA 15.2 (Pringsheim and Landau). *Let μ be a positive measure on $[0, \infty)$ whose Laplace transform $\varphi(s) = \int e^{-sx}\mu(dx)$ converges for $s > \sigma_c$ and diverges for $s < \sigma_c$, where $-\infty < \sigma_c < \infty$. Then φ has no analytic continuation to a neighbourhood of σ_c.*

PROOF. The proof of Lemma 11.5 applies verbatim to μ, since it uses only the positivity of the measure $\sum_n a_n \delta_{\log n}$. □

THEOREM 15.3 (Local Thorin criterion, this book). *Let $L > 0$ and $x \in \mathbb{R}$, and put*

$$f_x(s) = \frac{2 - x}{2\cosh(L\sqrt{s}) - x}, \qquad s > 0.$$

Then f_x is the Laplace transform of a probability law on $[0, \infty)$ if and only if $-2 \le x < 2$. In that case, writing $x = 2\cos\theta$ with $0 < \theta \le \pi$, the law is a GGC with drift 0 and Thorin measure

$$\tau_{\theta, L} = \sum_{k \in \mathbb{Z}} \delta_{t_k}, \qquad t_k = \Big(\frac{\theta + 2\pi k}{L}\Big)^2,$$

namely the law of $\sum_{k\in\mathbb{Z}} E_k/t_k$, *with mean* $L^2/(2-x)$.

PROOF. Suppose $-2 \le x < 2$. By Lemma 15.1 with $w = L\sqrt{s}$, $f_x(s) = \prod_k (1 + s/t_k)^{-1}$ with all $t_k > 0$, so Theorem 14.7, (a) implies (c), applied to $z \mapsto (2\cosh(Lz) - x)/(2-x)$, shows that f_x is the Laplace transform of the GGC $\sum_k E_k/t_k$ with Thorin measure $\sum_k \delta_{t_k}$. Comparing the coefficients of w^2 in the lemma gives its mean $\sum_k t_k^{-1} = L^2/(4\sin^2(\theta/2)) = L^2/(2-x)$.

Suppose $x > 2$ and write $x = 2\cosh a$ with $a > 0$. Then $s \mapsto 2\cosh(L\sqrt{s}) - x$ increases on $[0,\infty)$ from $2 - x < 0$ and vanishes at $s_0 = (a/L)^2$, so $f_x(s) > 1$ on $(0, s_0)$ and $f_x(s) \to +\infty$ as $s \uparrow s_0$. A Laplace transform of a probability law is bounded by 1 on $(0,\infty)$, so f_x is not one.

Suppose $x < -2$ and write $x = -2\cosh a$ with $a > 0$. Since $\cosh(L\sqrt{s})$ is an entire function of s, f_x is meromorphic on $\mathbb{C}$, with poles where $\cosh(L\sqrt{s}) = -\cosh a$, that is at $s = \big((a + i(2k+1)\pi)/L\big)^2$, $k \in \mathbb{Z}$. These poles are not real, since $a \neq 0$ and $(2k+1)\pi \neq 0$. On the real axis f_x is finite: for $s \ge 0$ the denominator is at least $2 + 2\cosh a$, and for $s = -u^2 < 0$ it equals $2\cos(Lu) + 2\cosh a > 0$. Suppose $f_x(s) = \int e^{-sy}\mu(dy)$ for $s > 0$ with μ a probability. Let $\sigma_c \le 0$ be the abscissa of convergence. The Laplace integral is holomorphic on $\Re s > \sigma_c$ and agrees with f_x on $(0,\infty)$, so by uniqueness of continuation it agrees with f_x on the real interval (σ_c,∞). If $\sigma_c > -\infty$ then f_x continues it analytically to a neighbourhood of σ_c, contradicting Lemma 15.2. So $\sigma_c = -\infty$, the integral converges absolutely on $\mathbb{C}$ and defines an entire function equal to f_x, which is impossible because f_x has poles. Finally $x = 2$ makes the numerator vanish. □

The proof shows that for a single Euler factor the GGC property is automatic once f_x is a Laplace transform at all: positivity alone forces the Thorin structure. The atoms t_k are the squares of the ordinates $(\theta + 2\pi k)/L$ of the local zeros $w = \pm i(\theta + 2\pi k)/L$ of E_p, exactly as the global Thorin atoms of Chapter 14 are the squares γ_k^2. We call $\tau_{\theta,L}$ the *local Thorin comb*; on the ordinate axis it consists of the two arithmetic progressions $(\pm\theta + 2\pi\mathbb{Z})/L$.

COROLLARY 15.4 (this book). *Let* π *be as above and* p *unramified with* $x_p \neq 2$. *The Ramanujan Petersson conjecture holds at* p *if and only if* $s \mapsto E_p(0)/E_p(\sqrt{s})$ *is the Laplace transform of a probability law on* $[0,\infty)$, *and then this law is a GGC with Thorin measure* $\tau_{\theta_p,\log p}$, $x_p = 2\cos\theta_p$.

PROOF. $E_p(0)/E_p(\sqrt{s}) = f_{x_p}(s)$ with $L = \log p$, and $|\alpha| = 1$ if and only if $-2 \le x_p \le 2$. □

The excluded value $x_p = 2$ ($\alpha = \beta = 1$) is tempered, and there E_p has a double zero at $w = 0$: then $1/E_p(\sqrt{s}) = (L^2 s)^{-1}\prod_{k\neq 0}(1 + sL^2/(2\pi k)^2)^{-1}$, and, as in the rank discussion of Chapter 13, the factor $(L^2 s)^{-1}$ is a degenerate Thorin atom of mass 1 at the origin (Exercise 2).

EXAMPLE 15.5 (The discriminant form). For π attached to Δ, $x_p = \tau(p)p^{-11/2}$. Deligne's theorem [140], the Ramanujan conjecture for Δ, gives $|x_p| \le 2$, and $x_p \neq \pm 2$ since $p^{11/2}$ is irrational. So local Thorin holds at every prime, and by Corollary 15.4 this statement is equivalent to the Ramanujan conjecture for Δ. Global Thorin for Δ, by Theorem 13.8 and Chapter 14, is the Riemann hypothesis for $L(s,\Delta)$, which is open. At $p = 2$, $x_2 = -0.53033$, $\theta_2 = 1.83917$, and the first atoms of $\tau_{\theta_2,\log 2}$ are 7.0403 and 41.105; the local law has mean $(\log 2)^2/(2 - x_2) = 0.18988$.

The local criterion also grades failures of Ramanujan. Write $\alpha = e^{i(\varphi + i\eta)}$ with φ, η real, so that $|\alpha| = e^{-\eta}$ and $x_p = 2\cos(\varphi + i\eta)$. The zeros of E_p are $w = i(\pm(\varphi + i\eta) + 2\pi k)/L$, with real parts $\mp\eta/L$. For Maass forms on GL_2 over $\mathbb{Q}$, Kim and Sarnak proved $|\alpha_p| \le p^{7/64}$ [315], that is $|\eta| \le \frac{7}{64}\log p$, so every local zero lies in the strip $|\Re w| \le \frac{7}{64}$. Squaring, the poles $s = w^2$ of $s \mapsto 1/E_p(\sqrt{s})$ lie in the parabolic region $P_{7/64} = \{w^2 : |\Re w| \le \frac{7}{64}\}$ of Chapter 14. Thus, writing $\Theta_\sigma^{(p)}$ for the local parabolic condition that all poles of $1/E_p(\sqrt{s})$ lie in P_σ, the Kim Sarnak theorem is $\Theta_{7/64}^{(p)}$ at every unramified p, and the Ramanujan conjecture is $\Theta_0^{(p)}$, the degenerate parabola $P_0 = (-\infty, 0]$, which by Theorem 15.3 is local Thorin. Globally, $\Theta_{1/2}$ holds for every π (Chapter 14), but Θ_σ with $\sigma < \frac{1}{2}$, a zero free strip of positive width, is known for no automorphic L function over $\mathbb{Q}$, while the local exponent $\frac{7}{64}$ is a theorem.

15.2 LOCAL COMBS AND THE EXPLICIT FORMULA

For an even test function h we define the *comb sum* of p as the sum of h over the ordinates of the local zeros:

$$c_p(h) = \sum_{k\in\mathbb{Z}} \Big[h\Big(\frac{\theta_p + 2\pi k}{L}\Big) + h\Big(\frac{-\theta_p + 2\pi k}{L}\Big)\Big], \qquad L = \log p.$$

LEMMA 15.6. *Let h be an even Schwartz function and $H(x) = \int h(t)e^{-ixt}\,dt$. Then for every $\theta \in \mathbb{R}$ and $L > 0$,*

$$\sum_{k\in\mathbb{Z}} \Big[h\Big(\frac{\theta + 2\pi k}{L}\Big) + h\Big(\frac{-\theta + 2\pi k}{L}\Big)\Big] = \frac{L}{\pi}H(0) + \frac{2L}{\pi}\sum_{m\ge 1}\cos(m\theta)\,H(mL).$$

PROOF. Let $g(y) = h((\theta + 2\pi y)/L)$, a Schwartz function. Poisson summation gives $\sum_k g(k) = \sum_m \hat g(m)$ with $\hat g(\xi) = \int g(y)e^{-2\pi i\xi y}dy$. Substituting $t = (\theta + 2\pi y)/L$, so $2\pi y = Lt - \theta$,

$$\hat g(m) = \frac{L}{2\pi}\int h(t)e^{-im(Lt-\theta)}dt = \frac{L}{2\pi}e^{im\theta}H(mL).$$

The progression with $-\theta$ gives $\frac{L}{2\pi}\sum_m e^{-im\theta}H(mL)$. Adding, and using that H is even because h is, the total is $\frac{L}{2\pi}\sum_{m\in\mathbb{Z}} 2\cos(m\theta)H(mL)$, which is the stated formula after separating $m=0$ and pairing $\pm m$. $\square$

So the comb of p has mean density L/π and oscillatory Fourier coefficients $\frac{2L}{\pi}\cos(m\theta_p)$ at the frequencies $m\log p$. Compare Weil's explicit formula [569], which we state in the form of Iwaniec and Kowalski [276], Chapter 5, for Δ in the unitary normalisation: $L(s,\Delta)=\sum\tau(n)n^{-11/2-s}$, $\Lambda(s,\Delta)=(2\pi)^{-s-11/2}\Gamma(s+\frac{11}{2})L(s,\Delta)$, nontrivial zeros $\frac12+i\gamma$ counted with multiplicity (the zeros $6+i\gamma$ of the classical normalisation of Chapter 13).

THEOREM 15.7 (Weil; see [276]). *Let h be even and holomorphic in $|\Im t|<\frac12+\varepsilon$, with $h(t)\ll(1+|t|)^{-1-\delta}$ there, and $H(x)=\int h(t)e^{-ixt}dt$. Then*

$$\sum_\gamma h(\gamma)=\frac{1}{2\pi}\int_{\mathbb{R}} h(t)\,2\Re\Big[-\log 2\pi+\frac{\Gamma'}{\Gamma}(6+it)\Big]dt \\ -\frac{1}{\pi}\sum_p\sum_{m\ge1}2\cos(m\theta_p)\,\log p\;p^{-m/2}H(m\log p).$$

Here the number $(\alpha_p^m+\beta_p^m)\log p=2\cos(m\theta_p)\log p$ is the coefficient of p^{-ms} in $-L'/L(s,\Delta)$. Lemma 15.6 now gives a term by term dictionary.

PROPOSITION 15.8 (this book). *For every prime p and every h as in Theorem 15.7 that is also Schwartz on $\mathbb{R}$, the contribution of p to the prime side of the explicit formula is obtained from the oscillatory part $c_p(h)-\frac{L}{\pi}H(0)=\frac{2L}{\pi}\sum_{m\ge1}\cos(m\theta_p)H(mL)$ of the local comb sum by changing the sign and multiplying the mth Fourier coefficient by $p^{-m/2}$.*

The proof is the comparison of the two displays. The factor $p^{-m/2}=|p^{-ms}|$ at $\Re s=\frac12$ is the trace of the half unit shift from the local centre $\Re w=0$, where the comb lives, to the global centre $\Re s=\frac12$, where the zeros live; the sign reflects that $-L'/L$ counts zeros with a minus sign. The mean density L/π of the comb has no counterpart on the prime side: the global zero density comes from the archimedean term.

We checked Theorem 15.7 numerically for this book with the Gaussian $h(t)=e^{-t^2/40.5}$ (standard deviation 4.5), using the zeros of $\Lambda(s,\Delta)$ and $\tau(p)$ from PARI/GP. The zero side is $\sum_\gamma h(\gamma)=0.2630073607042847$. The archimedean term and the prime side are

$$0.25581384838722604 \quad\text{and}\quad 0.007193512317058718,$$

with total 0.26300736070428476, an agreement to sixteen digits. The prime 2 contributes 0.0072002, while the oscillatory comb term of 2 is -0.010183; the ratio is -0.707, which is $-2^{-1/2}$ to the digits shown, because $H(m\log 2)$ decays so fast

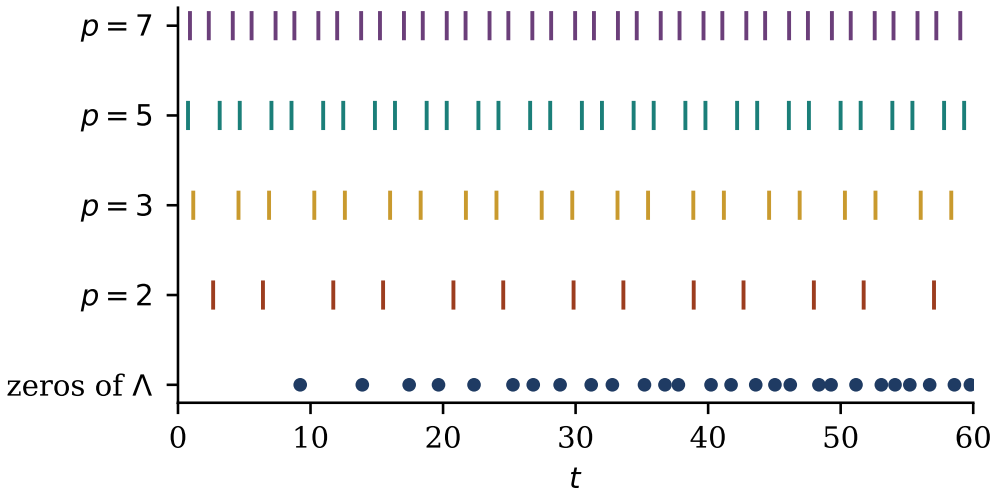


Figure 15.1: Ordinates of the local zeros (local Thorin combs) of Δ at $p = 2, 3, 5, 7$, the progressions $(\pm\theta_p + 2\pi\mathbb{Z})/\log p$, against the ordinates γ of the zeros of $\Lambda(s, \Delta)$ for $0 < t < 60$. Each comb is periodic with density $\log p/\pi$; the global zeros are not periodic.

that $m = 1$ dominates both, and the $m = 1$ coefficients differ exactly by the factor $-2^{-1/2}$. The contributions of $p = 3$ and $p = 5$ are -6.7×10^{-6} and -7.3×10^{-12}. Figure 15.1 shows the local combs of Δ at $p = 2, 3, 5, 7$ against the ordinates of the zeros.

REMARK 15.9 (Heuristic). The combs of the primes $p \le P$ have total density $\sum_{p \le P} \log p/\pi \approx P/\pi$ by the prime number theorem, while the zeros of $L(s, \Delta)$ near height t have density $\frac{1}{\pi} \log \frac{t}{2\pi}$. The two agree when $P \approx \log \frac{t}{2\pi}$: near height t, a truncated product of local GGCs over $p \lesssim \log t$ carries as many Thorin atoms per unit length as the global Thorin measure. This echoes the hybrid Euler Hadamard product $P_X Z_X$ of Gonek, Hughes and Keating [214] (Section 12.1.4), whose splitting conjecture is posed for $X = O((\log T)^{2-\varepsilon})$, a range that contains the matching scale $X \approx \log T$.

The Euler product itself does not converge to a global GGC: the local means $(\log p)^2/(2 - x_p)$ are not summable, and the global Thorin measure is not the sum of the local combs. The explicit formula, read through Proposition 15.8, is the precise relation between them.

15.3 Sato Tate random Euler products

Model the angles θ_p as independent random variables with the Sato Tate law $\frac{2}{\pi}\sin^2\theta\,d\theta$ on $[0,\pi]$. By Theorem 15.3 every local reciprocal factor f_{x_p}, $x_p = 2\cos\theta_p$, is the Laplace transform of a GGC almost surely (the only excluded value, $\theta_p = 0$, has probability zero), and for every P the truncated random Euler product

$$\Phi_P(s) = \prod_{p\le P} \frac{2 - x_p}{2\cosh(\sqrt{s}\log p) - x_p}$$

is the Laplace transform of a GGC whose Thorin measure $\tau_P = \sum_{p\le P} \tau_{\theta_p,\log p}$ is random. We call τ_P, or its preimage on the ordinate axis, a *Thorin point process*.

Proposition 15.10 (this book). *Let θ have the Sato Tate law and let $\epsilon = \pm 1$ be an independent fair sign.*

1. *The configuration $\{\theta, -\theta\}$ on the circle $\mathbb{R}/2\pi\mathbb{Z}$ has intensity $\frac{2}{\pi}\sin^2\psi\,d\psi$, and $\epsilon\theta$ has density $\frac{1}{\pi}\sin^2\psi$ on $(-\pi,\pi]$.*

2. *The ordinates $(\pm\theta + 2\pi\mathbb{Z})/\log p$ of the comb of p form a symmetric point process on $\mathbb{R}$ with intensity*

$$\frac{\log p}{\pi}\big(1 - \cos(2t\log p)\big)\,dt.$$

   Consequently the Thorin point process of Φ_P has intensity $I_P(t)\,dt$ on the ordinate axis, $I_P(t) = \frac{1}{\pi}\sum_{p\le P}\log p\,(1-\cos(2t\log p))$.

3. *$\mathbb{E}[(2-x_p)^{-1}] = 1$, so the local law has expected mean $(\log p)^2$.*

Proof. (1) For a test function ϕ on the circle, $\mathbb{E}[\phi(\theta)+\phi(-\theta)] = \int_0^\pi(\phi(\psi)+\phi(-\psi))\frac{2}{\pi}\sin^2\psi\,d\psi = \int_{-\pi}^{\pi}\phi(\psi)\frac{2}{\pi}\sin^2\psi\,d\psi$, and half of this is $\mathbb{E}\phi(\epsilon\theta)$.

(2) Let $L = \log p$ and ϕ be compactly supported on $\mathbb{R}$. By (1) and unfolding the sum over k,

$$\begin{aligned}\mathbb{E}\sum_{k\in\mathbb{Z}}\Big[\phi\Big(\frac{\theta+2\pi k}{L}\Big)+\phi\Big(\frac{-\theta+2\pi k}{L}\Big)\Big] &= \sum_k\int_{-\pi}^{\pi}\phi\Big(\frac{\psi+2\pi k}{L}\Big)\frac{2}{\pi}\sin^2\psi\,d\psi\\ &= \int_{\mathbb{R}}\phi\Big(\frac{u}{L}\Big)\frac{2}{\pi}\sin^2 u\,du,\end{aligned}$$

using periodicity of $\sin^2$. Substituting $u = Lt$ and $2\sin^2(Lt) = 1-\cos(2Lt)$ gives the intensity. Intensities of independent superpositions add.

(3) With $x = 2\cos\theta$, $2-x = 4\sin^2(\theta/2)$ and $\sin^2\theta = 4\sin^2(\theta/2)\cos^2(\theta/2)$, so $\mathbb{E}(2-x)^{-1} = \frac{2}{\pi}\int_0^\pi\cos^2(\theta/2)\,d\theta = 1$. The mean of the local law is $L^2/(2-x)$ by Theorem 15.3. □

Compare two other models. If θ_p is uniform on $[0,\pi]$ (the arcsine law for x_p), the comb of p has flat intensity $\log p/\pi$. Under the law $\mu_{\mathrm{CM}} = \frac{1}{2}\delta_0 + \frac{1}{2}\,\text{arcsine}$ of Chapter 4, half the primes have $\theta_p = \pi/2$ and contribute the deterministic comb $(\frac{1}{2}+\mathbb{Z})\pi/\log p$, the other half the flat intensity, so the intensity is $\frac{\log p}{2\pi} + \frac{1}{2}\sum_{k\in\mathbb{Z}}\delta_{(k+1/2)\pi/\log p}$. Only the Sato Tate model depletes the origin: $I_P(0) = 0$ and $I_P(t) \le \frac{2t^2}{\pi}\sum_{p\le P}(\log p)^3$, a quadratic vanishing at $t = 0$ (and at no other point once two primes are present). Part (3) shows the same effect in the mean: under the arcsine model $\mathbb{E}(2-x_p)^{-1} = \infty$, because angles near 0 put Thorin atoms near the origin, while under Sato Tate the depletion makes the expected mean finite.

We read this cautiously. Of the one level densities of Section 6.3, $W(\mathrm{USp})$ vanishes quadratically at the central point, and so does $W(\mathrm{SO(odd)})$ once the forced central zero is removed, while $W(\mathrm{SO(even)})$ does not [271, 298]. The Sato Tate depletion near $t = 0$ is a local mechanism with the sign and the order of vanishing of the symplectic case, built prime by prime from the factor $\sin^2$ in the Haar measure of $\mathrm{SU}(2) = \mathrm{USp}(2)$. At the scale $t \asymp 1/\log P$ the prime number theorem gives $I_P(t) = \frac{1}{\pi}\big(\vartheta(P) - \Re\frac{P^{1+2it}}{1+2it}\big) + o(P)$, with ϑ Chebyshev's function, so the depletion has width of order $1/\log P$. We do not claim that the global one level density is a consequence of this local picture; the heuristic of the previous section suggests matching $\log P$ with $\log\log t$, and making any such statement precise is open.

15.4 Function fields and graphs: unconditional global Thorin

Let C be a smooth projective geometrically connected curve of genus $g \ge 1$ over $\mathbb{F}_q$. Its zeta function is

$$Z(C,u) = \frac{P(u)}{(1-u)(1-qu)}, \qquad P(u) = \prod_{j=1}^{2g}(1-\alpha_j u) \in \mathbb{Z}[u], \qquad u = q^{-s},$$

and the functional equation makes the multiset $\{\alpha_j\}$ stable under $\alpha \mapsto q/\alpha$. Pair the roots accordingly, and for each pair put $c_j = (\alpha_j + q/\alpha_j)/\sqrt{q}$, $1 \le j \le g$. With $z = s - \frac{1}{2}$ and $u = q^{-1/2-z}$, each pair gives $(1-\alpha_j u)(1-(q/\alpha_j)u) = q^{-z}\big(2\cosh(z\log q) - c_j\big)$, so

$$\Xi_C(z) := q^{gz}P(q^{-1/2-z}) = \prod_{j=1}^{g}\big(2\cosh(z\log q) - c_j\big),$$

an even entire function, real on the real axis. The Riemann hypothesis for C is $|\alpha_j| = \sqrt{q}$ for all j, equivalently $c_j = 2\cos\theta_j \in [-2,2]$ with $\alpha_j = \sqrt{q}\,e^{i\theta_j}$. It was proved by Weil [567], and in all dimensions by Deligne [140]. Put $F_C(s) = \Xi_C(\sqrt{s})/\Xi_C(0)$.

THEOREM 15.11 (this book). *Assume $P(q^{-1/2}) \neq 0$, that is, no α_j equals $\sqrt{q}$. Then $s \mapsto 1/F_C(s)$ is the Laplace transform of a GGC if and only if the Riemann hypothesis holds for C. In that case the Thorin measure is the finite superposition of periodic combs*

$$\tau_C = \sum_{j=1}^{g} \sum_{k \in \mathbb{Z}} \delta_{((\theta_j + 2\pi k)/\log q)^2},$$

and the law is $\sum_j \sum_k E_{j,k}\,(\log q)^2/(\theta_j + 2\pi k)^2$, with mean $(\log q)^2 \sum_j (2 - c_j)^{-1}$. By Weil's theorem all of this holds unconditionally.

PROOF. If the Riemann hypothesis holds then every $c_j \in [-2, 2)$, the value 2 being excluded by $P(q^{-1/2}) \neq 0$, and $1/F_C = \prod_j f_{c_j}$ with $L = \log q$. Each factor is a GGC transform with Thorin measure $\tau_{\theta_j, \log q}$ by Theorem 15.3, and GGCs are closed under convolution, Thorin measures adding. Conversely, suppose $1/F_C$ agrees on $(0, \infty)$ with the Laplace transform φ of a GGC. By Lemma 13.3, φF_C is holomorphic on $\mathbb{C} \setminus (-\infty, 0]$, and it equals 1 on $(0, \infty)$, hence everywhere; so the entire function $s \mapsto \Xi_C(\sqrt{s})$ has no zeros off $(-\infty, 0]$, and since $\Xi_C(0) \neq 0$ every zero z of Ξ_C is purely imaginary. A zero of $2\cosh(z \log q) - c_j$ with $z = iy$ satisfies $\cos(y \log q) = c_j/2$, which forces $c_j \in [-2, 2]$. Hence every c_j is real in $[-2, 2]$ and $|\alpha_j| = \sqrt{q}$. □

EXAMPLE 15.12. For $C : y^2 = x^5 - x + 1$ over $\mathbb{F}_{101}$ (genus 2) the characteristic polynomial of Frobenius, computed for this book, is $T^4 + 8T^3 - 16T^2 + 808T + 10201$, so $\#C(\mathbb{F}_{101}) = 110$. All roots have absolute value $\sqrt{101}$, with angles $\theta_1 = 0.973934$, $\theta_2 = 2.858029$, so $c_1 = 1.124099$, $c_2 = -1.920129$ and $\Xi_C(0) = P(101^{-1/2}) = 3.433644$. The smallest Thorin atoms are 0.044534, 0.383502, 0.550801, 1.323428, and the GGC has mean 29.7504. The product identity of Lemma 15.1 at these angles, truncated at $|k| \leq 20000$, was checked to relative error 1.3×10^{-5}.

The same holds for regular graphs. Let G be a finite connected $(q+1)$ regular graph with adjacency eigenvalues λ_j. By the Ihara Bass formula of Chapter 10, each eigenvalue contributes $1 - \lambda_j u + q u^2 = q^{-z}(2\cosh(z \log q) - \lambda_j/\sqrt{q})$ at $u = q^{-1/2-z}$. The trivial eigenvalue $q + 1$ gives $x = \sqrt{q} + 1/\sqrt{q} > 2$, which in Theorem 15.3 is a positive real pole: it plays the role of the pole of ζ at $s = 1$, and the Thorin criterion rejects it, as the global criterion must reject the pole. Put

$$\Xi_G(z) = \prod_{|\lambda_j| < q+1} \Big(2\cosh(z \log q) - \frac{\lambda_j}{\sqrt{q}}\Big), \qquad F_G(s) = \frac{\Xi_G(\sqrt{s})}{\Xi_G(0)}.$$

THEOREM 15.13 (this book). *Suppose no eigenvalue of G equals $2\sqrt{q}$. Then G is Ramanujan if and only if $1/F_G$ is the Laplace transform of a GGC, and then its Thorin measure is $\sum_j \sum_{k \in \mathbb{Z}} \delta_{((\theta_j + 2\pi k)/\log q)^2}$ over the nontrivial eigenvalues $\lambda_j = 2\sqrt{q}\cos\theta_j$. For the supersingular ℓ isogeny graphs $G_N(\ell)$ of Chapter 10, with $N \equiv 1 \bmod 12$ and $\ell \neq N$ primes, both hold unconditionally.*

PROOF. The first statement is the proof of Theorem 15.11 with $c_j = \lambda_j/\sqrt{q}$. For $G_N(\ell)$, Theorem 10.4 identifies the nontrivial spectrum with the eigenvalues of T_ℓ on $S_2(\Gamma_0(N))$, which satisfy $|\lambda| \le 2\sqrt{\ell}$. The eigenvalue $2\sqrt{\ell}$ does not occur: $\lambda = \pm 2\sqrt{\ell}$ would mean equal Satake parameters, excluded in weight two by Coleman and Edixhoven [112]. □

The shape of the Thorin measures separates function fields from $\mathbb{Q}$. Over a function field the global Thorin measure is a finite superposition of g periodic combs with common period $2\pi/\log q$ on the ordinate axis, because Ξ_C is a polynomial in $\cosh(z \log q)$. Over $\mathbb{Q}$ the ordinates that carry the global Thorin measure of Chapter 14 are aperiodic, with density growing like $\log t$, and the explicit formula, read through Proposition 15.8, shows them as the interference pattern of infinitely many local combs, each damped by its half shift, whose base frequencies $\log p$ are linearly independent over $\mathbb{Q}$ by unique factorisation. Ramanujan is local Thorin in both worlds; global Thorin is known only where there is one comb period.

15.5 HECKE WALKS AND THORIN MEASURES OF PASSAGE TIMES

In this section the Thorin measures are spectra. Let Y be a birth death process on $\{0, 1, \dots, n\}$ with birth rates $b_k > 0$ $(0 \le k < n)$ and death rates $d_k > 0$ $(1 \le k \le n)$, reflecting at 0, started at 0, and let T be its first passage time to n. Let Q_n be the generator restricted to $\{0, \dots, n-1\}$ (killed at n).

THEOREM 15.14 (Keilson [303]). *The eigenvalues $\lambda_1 < \cdots < \lambda_n$ of $-Q_n$ are real, simple and positive, and T has the law of $\sum_{j=1}^{n} E_j/\lambda_j$. In particular T is a GGC with Thorin measure $\sum_j \delta_{\lambda_j}$, and $\mathbb{E}e^{-sT} = \det(-Q_n)/\det(sI - Q_n)$.*

PROOF. For $s > 0$ let $u_k = \mathbb{E}_k e^{-sT}$, $0 \le k < n$. First step analysis gives $(sI - Q_n)u = b_{n-1}e_{n-1}$. By Cramer's rule u_0 is b_{n-1} times the $(n-1, 0)$ cofactor of $sI - Q_n$, divided by $\det(sI - Q_n)$. Deleting row $n-1$ and column 0 of the tridiagonal matrix $sI - Q_n$ leaves a triangular matrix with diagonal $-b_0, \dots, -b_{n-2}$, so the cofactor equals $b_0 \cdots b_{n-2}$, independent of s. Hence $u_0 = b_0 \cdots b_{n-1}/\det(sI - Q_n)$, and letting $s \to 0$, where $u_0 \to \mathbb{P}_0(T < \infty) = 1$, gives $\det(-Q_n) = b_0 \cdots b_{n-1} \ne 0$. The matrix $-Q_n$ is similar, by the diagonal matrix of square roots of the reversible weights, to a symmetric Jacobi matrix with nonzero off diagonal entries $-\sqrt{b_k d_{k+1}}$; so its eigenvalues are real and simple. Each row of $-Q_n$ has diagonal entry at least the sum of the moduli of its off diagonal entries, so by Gershgorin's theorem the eigenvalues are nonnegative, and since $\det(-Q_n) \ne 0$ they are positive. Therefore $\mathbb{E}e^{-sT} = \prod_j \lambda_j/(s + \lambda_j)$, the Laplace transform of $\sum_j E_j/\lambda_j$. □

Kent proved the analogue for one dimensional diffusions, with infinitely many rates [308], and Fill gave a probabilistic proof by a stochastic construction of the

exponential stages [188]. An exercise of Chapter 13 uses the hitting time of a Bessel process in the same way.

Let $\mathcal{T}_q$ be the $(q+1)$ regular tree, the Bruhat Tits tree of $\mathrm{PGL}_2(\mathbb{Q}_q)$ for q prime, and run continuous time simple random walk on it, jumping at rate 1 to a uniformly chosen neighbour; this is the Hecke walk with generator $(q+1)^{-1}A - I$, where $A = T_q$ is the adjacency operator of Chapter 9. Let T_n be the first time at distance n from the start. The distance is a birth death process with $b_0 = 1$, and $b_k = \frac{q}{q+1}$, $d_k = \frac{1}{q+1}$ for $k \geq 1$. Symmetrising,

$$-Q_n \sim J_n = I - S_n, \qquad S_n = \begin{pmatrix} 0 & a & & & \\ a & 0 & c & & \\ & c & 0 & \ddots & \\ & & \ddots & \ddots & c \\ & & & c & 0 \end{pmatrix},$$

with $a = 1/\sqrt{q+1}$ and $c = \sqrt{q}/(q+1) = \rho/2$, where $\rho = 2\sqrt{q}/(q+1)$ is the Kesten spectral radius.

THEOREM 15.15 (Passage times on the tree, this book). *Let $q \geq 2$, $\rho = 2\sqrt{q}/(q+1)$, and let $\lambda_1 < \cdots < \lambda_n$ be the Thorin atoms of T_n.*

1. *$\lambda_j = 1 - \frac{\sqrt{q}}{q+1} x_j$, where $x_1 > \cdots > x_n$ are the zeros of the polynomial $U_n(x/2) - q^{-1}U_{n-2}(x/2)$ that expresses $B_n = q^{-n/2}A_n$ in terms of $X = q^{-1/2}A$ (Theorem 9.1). All x_j lie in $(-2,2)$, so all rates lie in the Kesten band $(1-\rho, 1+\rho)$.*
2. *$\frac{1}{n}\sum_j \delta_{\lambda_j}$ converges weakly to the law of $1 - \rho\cos(\pi U)$, U uniform on $(0,1)$: the arcsine law on the Kesten band.*
3. *$\mathbb{E}T_n = \frac{n(q+1)}{q-1} - \frac{2q(1-q^{-n})}{(q-1)^2}$, and $\mathbb{E}T_n/n \to \frac{q+1}{q-1} = \frac{1}{\sqrt{1-\rho^2}} = \int_0^1 \frac{du}{1-\rho\cos\pi u}$. The escape speed of the walk is $\frac{q-1}{q+1} = \sqrt{1-\rho^2}$.*
4. *$\mathrm{Var}\, T_n/n \to \left(\frac{q+1}{q-1}\right)^3 = (1-\rho^2)^{-3/2}$, and $(T_n - \mathbb{E}T_n)/\sqrt{\mathrm{Var}\, T_n}$ converges in law to a standard normal.*

PROOF. (1) Let D_k be the characteristic polynomial of the leading $k \times k$ block of $c^{-1}S_n$, whose first off diagonal entry is $a/c = \sqrt{1+1/q}$ and the others 1. Expanding along the last row, $D_1 = x$, $D_2 = x^2 - 1 - q^{-1}$ and $D_{k+1} = xD_k - D_{k-1}$ for $k \geq 2$. The polynomials $U_k(x/2) - q^{-1}U_{k-2}(x/2)$, with $U_{-1} = 0$, satisfy the same recursion for $k \geq 2$ and agree with D_1, D_2 and $D_3 = x^3 - (2+q^{-1})x$ at $k = 1,2,3$, so they equal D_k; by Theorem 9.1(1) this is the polynomial of B_k. The eigenvalues of S_n are cx_j, so $\lambda_j = 1 - cx_j$. To locate the zeros put $x = 2\cos\phi$, $0 < \phi < \pi$; then $D_n = g(\phi)/\sin\phi$ with

$$g(\phi) = \sin((n+1)\phi) - q^{-1}\sin((n-1)\phi).$$

At $\phi_k^* = (k+\frac12)\pi/(n+1)$, $0 \le k \le n$, $|\sin((n+1)\phi)| = 1 > q^{-1}$, so $g(\phi_k^*)$ has the sign $(-1)^k$. Hence g has a zero ϕ_j in each interval (ϕ_{j-1}^*, ϕ_j^*), $1 \le j \le n$. These are n distinct zeros in $(0,\pi)$, and D_n has degree n, so they are all the zeros and $x_j = 2\cos\phi_j \in (-2,2)$. Since $c \cdot 2 = \rho$, $\lambda_j = 1 - \rho\cos\phi_j \in (1-\rho, 1+\rho)$.

Conceptually, S_n is a compression of the radial part of $P = (q+1)^{-1}A$, whose spectrum lies in $[-\rho,\rho]$ by Kesten's theorem [309], so the rates cannot leave the closed Kesten band.

(2) The bracketing $\phi_j \in (\phi_{j-1}^*, \phi_j^*)$ gives $|\phi_j - j\pi/(n+1)| < \pi/(2(n+1))$, so for continuous ψ on $[1-\rho, 1+\rho]$, $\frac1n \sum_j \psi(1-\rho\cos\phi_j)$ is a Riemann sum converging to $\int_0^1 \psi(1-\rho\cos\pi u)\,du$. It also follows from Szegő's theorem for Toeplitz matrices [225], since S_n differs from the tridiagonal Toeplitz matrix with symbol $\rho\cos\phi$ by a perturbation of rank two, and by interlacing such a perturbation moves the empirical spectral measure by at most $2/n$ in Kolmogorov distance.

(3) Let m_k be the expected time to go from distance k to $k+1$. Then $m_0 = 1$, and for $k \ge 1$ the walk waits a mean unit time and then moves down with probability $\frac{1}{q+1}$, after which it needs $m_{k-1} + m_k$ more on average: $m_k = 1 + \frac{1}{q+1}(m_{k-1} + m_k)$, that is $m_k = \frac{q+1}{q} + \frac{m_{k-1}}{q}$. With $m^* = \frac{q+1}{q-1}$ this is $m_k - m^* = (m_{k-1} - m^*)/q$, so $m_k = m^* - \frac{2}{q-1}q^{-k}$, and summing $k = 0, \dots, n-1$ gives the formula. The limit of $\mathbb{E}T_n/n = \frac1n \sum_j \lambda_j^{-1}$ also follows from (2), since $1/\lambda$ is continuous on the band, together with $\int_0^\pi \frac{d\phi}{\pi(1-\rho\cos\phi)} = (1-\rho^2)^{-1/2}$ and $1-\rho^2 = \left(\frac{q-1}{q+1}\right)^2$. For the speed, away from 0 the distance jumps at rate 1, up with probability $\frac{q}{q+1}$ and down with probability $\frac{1}{q+1}$, and it visits 0 only finitely often, so by the law of large numbers the distance at time t divided by t tends almost surely to $\frac{q-1}{q+1} = 1/m^*$.

(4) $\operatorname{Var} T_n = \sum_j \lambda_j^{-2}$, and (2) with $\int_0^\pi \frac{d\phi}{\pi(1-\rho\cos\phi)^2} = (1-\rho^2)^{-3/2}$ gives the limit. The summands E_j/λ_j are independent with $\mathbb{E}|E_j - 1|^3 \lambda_j^{-3} \le C(1-\rho)^{-3}$, so the Lyapunov ratio is $O(n(1-\rho)^{-3}/n^{3/2}) \to 0$. □

Thus the Thorin rates of the passage time are $1 - \frac{\sqrt{q}}{q+1}x$ with x in the tempered spectrum $[-2,2]$: the statement *Thorin support in the Kesten band* is the random walk form of the Ramanujan bound. Kesten's spectral radius determines the escape speed through the arcsine average of the reciprocal rates; in Chapter 9 the same number is the drift $\frac{q-1}{q+1}$ of the distance chain, whose gambler's ruin gives the return probability $1/q$.

The theorem also gives exact large deviations. Since all rates exceed $1-\rho$, $\mathbb{E}e^{\theta T_n} = \prod_j (1-\theta/\lambda_j)^{-1}$ is finite for $\theta \le 1-\rho$, and we can compute the limit through the Thorin measure.

THEOREM 15.16 (Large deviations of passage times, this book). *For $\theta \le 1-\rho$,*

$$\lim_{n\to\infty}\frac{1}{n}\log \mathbb{E}e^{\theta T_n} = \Lambda(\theta) = -\log\frac{(q+1)\bigl(1-\theta+\sqrt{(1-\theta)^2-\rho^2}\bigr)}{2q},$$

and the limit is $+\infty$ for $\theta > 1-\rho$. In particular $\Lambda(1-\rho) = \frac{1}{2}\log q$. The function Λ is steep, $\Lambda'(\theta) = ((1-\theta)^2-\rho^2)^{-1/2} \to \infty$ as $\theta \uparrow 1-\rho$, so T_n/n satisfies a large deviation principle with good rate function

$$\Lambda^*(y) = y - \sqrt{1+\rho^2y^2} + \log\frac{(q+1)\bigl(1+\sqrt{1+\rho^2y^2}\bigr)}{2qy}, \qquad y>0,$$

which vanishes only at $y = \frac{q+1}{q-1}$.

PROOF. For $\theta < 1-\rho$, $\frac{1}{n}\log\mathbb{E}e^{\theta T_n} = \frac{1}{n}\sum_j\bigl[\log\lambda_j - \log(\lambda_j-\theta)\bigr]$, and the integrand is continuous on the band, so by Theorem 15.15(2) the limit is $\int_0^\pi[\log(1-\rho\cos\phi) - \log(1-\theta-\rho\cos\phi)]\frac{d\phi}{\pi}$. The classical integral $\int_0^\pi \log(a-b\cos\phi)\frac{d\phi}{\pi} = \log\frac{a+\sqrt{a^2-b^2}}{2}$ for $a \ge |b|$, together with $1+\sqrt{1-\rho^2} = \frac{2q}{q+1}$, gives Λ. At $\theta = 1-\rho$ the second term becomes $\log(\rho(1-\cos\phi))$, which is singular at $\phi=0$, but it is increasing in ϕ and integrable, and the bracketing of the ϕ_j in intervals of length $\pi/(n+1)$ shows that its Riemann sums converge to the integral (the term $j=1$ is $O(n^{-1}\log n)$). The value is $-\log\frac{(q+1)\rho}{2q} = \frac{1}{2}\log q$. For $\theta > 1-\rho$, $\lambda_1 \to 1-\rho$ by (2), so $\mathbb{E}e^{\theta T_n} = \infty$ for large n. Differentiating, $\Lambda'(\theta) = ((1-\theta)^2-\rho^2)^{-1/2}$, which tends to infinity at the edge, so Λ is essentially smooth with 0 in the interior of its domain and the Gärtner Ellis theorem [143] applies. Since Λ' maps $(-\infty, 1-\rho)$ onto $(0,\infty)$, the supremum in $\Lambda^*(y) = \sup_\theta[\theta y - \Lambda(\theta)]$ is attained at $1-\theta = \sqrt{\rho^2+y^{-2}}$, and substituting gives the formula. At $y = (1-\rho^2)^{-1/2}$ one finds $\sqrt{1+\rho^2y^2} = y$ and $\Lambda^*(y) = 0$. □

The value $\Lambda(1-\rho) = \frac{1}{2}\log q$ is half the logarithm of the branching number: at the edge of the Kesten band the exponential moment grows like $q^{n/2}$, and indeed the closed form $\mathbb{E}e^{(1-\rho)T_n} = \frac{q+1}{q}\,q^{n/2}\big/\bigl(n+1-\frac{n-1}{q}\bigr)$, which follows from Theorem 15.14 and the values of D_n at $x=2$ and $x = 1/c$, shows the polynomial correction. For $q=2$ and $n=6400$ it gives $\frac{1}{n}\log\mathbb{E}e^{(1-\rho)T_n} = 0.345376$, the value computed from the rates, against the limit $\frac{1}{2}\log 2 = 0.346574$.

The numbers, computed for this book from the Jacobi matrices J_n for $q = 2,3,5$ and $n = 100, 400, 1600, 6400$, confirm every statement. All rates lie in $(1-\rho, 1+\rho)$; at $n=6400$ the smallest rates are 0.05719, 0.13397, 0.25464 against gaps $1-\rho =$ 0.057191, 0.133975, 0.254644. The ratios $\mathbb{E}T_n/n$ are 2.99938, 1.99977, 1.49990 against 3, 2, 1.5, and $\operatorname{Var}T_n/n = 26.9856, 7.9975, 3.3744$ against 27, 8, 3.375. The trace $\operatorname{tr}J_n^{-1} = 19196$ for $q=2$, $n=6400$ equals the formula of (3). The Kolmogorov

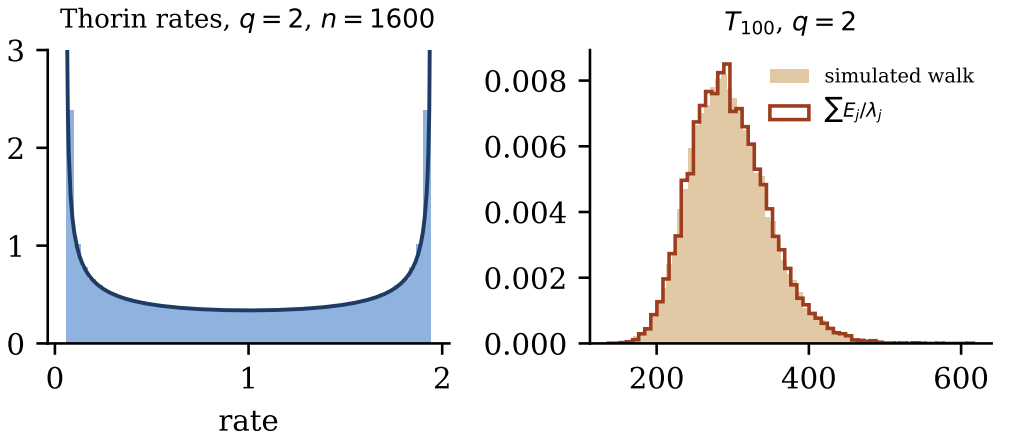


Figure 15.2: Left: the Thorin atoms of T_{1600} for $q = 2$ against the arcsine density on the Kesten band $(1-\rho, 1+\rho)$. Right: simulated passage times T_{100} ($q = 2$, 20000 walks) against the GGC $\sum_j E_j/\lambda_j$ of Theorem 15.14.

distance of the rescaled rates to the arcsine law is 8×10^{-5} at $n = 6400$. For $q = 2$ and $n = 6400$, $\frac{1}{n} \log \mathbb{E} e^{\theta T_n}$ is -0.693095 at $\theta = -0.5$ against $\Lambda = -0.693147$, and 0.100872 at $\theta = (1-\rho)/2 = 0.0286$ against 0.100899. A simulation of 20000 walks with $q = 2$, $n = 100$ gave mean 296.07, against 296.24 for an equal sample from $\sum E_j/\lambda_j$ and the exact value 296; the Kolmogorov Smirnov test comparing the two samples has $p = 0.57$ (Figure 15.2).

15.6 Exponential hitting times on Ramanujan graphs

On a finite graph a passage time is no longer a pure GGC, but the spectral structure survives. Let G be a connected $(q+1)$ regular graph with vertex set V of size h that is not bipartite. Loops and multiple edges are allowed, but the adjacency matrix A is symmetric, as for the Brandt matrices of Chapter 10 when $N \equiv 1 \bmod 12$. Let $P = (q+1)^{-1} A$, with eigenvalues $1 = p_1 > p_2 \geq \cdots \geq p_h > -1$ (written μ_j in Appendix A; below, μ_j is a rate), and run the continuous time walk with generator $P - I$. Fix a target v and let τ_v be its hitting time. Let P_v be the principal submatrix of P on $V \setminus \{v\}$, with eigenvalues $r_1 \geq r_2 \geq \cdots \geq r_{h-1}$ and orthonormal eigenvectors ψ_j, and put $\mu_j = 1 - r_j$, the *Dirichlet rates*.

Proposition 15.17. *Started from the uniform law π,*

$$\mathbb{P}_\pi(\tau_v > t) = \sum_{j=1}^{h-1} w_j e^{-\mu_j t}, \qquad w_j = \frac{1}{h} \langle \vec{1}, \psi_j \rangle^2 \geq 0, \qquad \sum_j w_j = \frac{h-1}{h},$$

a completely monotone mixture of exponentials, and $\mathbb{E}_\pi \tau_v = \sum_j w_j/\mu_j \geq (h-1)^2/h$.

PROOF. Killing at v, $\mathbb{P}_x(\tau_v > t) = (e^{-t(I-P_v)}\vec{1})(x)$ for $x \neq v$, and averaging over x with weight $1/h$ gives the mixture by the spectral theorem for the symmetric matrix P_v. Connectedness makes $I - P_v$ positive definite. For the mean, $\mathbb{E}_\pi \tau_v = \frac{1}{h}\langle \vec{1}, (I-P_v)^{-1}\vec{1}\rangle$, and the Cauchy Schwarz inequality for the positive definite form gives $\langle \vec{1}, M^{-1}\vec{1}\rangle\langle \vec{1}, M\vec{1}\rangle \geq \langle \vec{1}, \vec{1}\rangle^2$ with $M = I - P_v$. Here $\langle \vec{1}, \vec{1}\rangle = h-1$, and since P is symmetric and stochastic, $\langle \vec{1}, M\vec{1}\rangle = (h-1) - \sum_{x,y\neq v} P(x,y) = \sum_{x\neq v} P(x,v) = 1 - P(v,v) \leq 1$. □

The mixture form is classical (Keilson); see Aldous and Fill [14]. By Cauchy interlacing the Dirichlet rates interlace the graph rates $1 - p_j$, and these are what the Ramanujan bound controls.

THEOREM 15.18 (this book). *If G is Ramanujan then $\mu_2, \ldots, \mu_{h-1} \in [1-\rho, 1+\rho]$, $\rho = 2\sqrt{q}/(q+1)$: only the slowest Dirichlet rate $\mu_1 \leq 1 - p_2$ can leave the Kesten band. Moreover, for every target v,*

$$\sup_{t\geq 0} \left| \mathbb{P}_\pi(\tau_v > t) - e^{-t/\mathbb{E}_\pi \tau_v} \right| \leq \frac{\tau_2}{\mathbb{E}_\pi \tau_v} \leq \frac{(q+1)\,h}{(\sqrt{q}-1)^2(h-1)^2} \leq \frac{2(q+1)}{(\sqrt{q}-1)^2(h-1)},$$

where $\tau_2 = 1/(1-p_2)$ is the relaxation time.

PROOF. By Cauchy interlacing for the principal submatrix P_v of the symmetric matrix P, $p_j \geq r_j \geq p_{j+1}$ for $1 \leq j \leq h-1$. The Ramanujan property says $p_2, \ldots, p_h \in [-\rho, \rho]$. For $j \geq 2$, $r_j \leq p_j \leq p_2 \leq \rho$, and for all $j \leq h-1$, $r_j \geq p_{j+1} \geq p_h \geq -\rho$. Hence $\mu_j = 1 - r_j \in [1-\rho, 1+\rho]$ for $2 \leq j \leq h-1$, and $\mu_1 = 1 - r_1 \leq 1 - p_2$. The first inequality in the display is the Aldous Brown inequality for reversible chains started from stationarity [13, 14]. Ramanujan gives $1 - p_2 \geq 1 - \rho = (\sqrt{q}-1)^2/(q+1)$, so $\tau_2 \leq (q+1)/(\sqrt{q}-1)^2$, and Proposition 15.17 gives $\mathbb{E}_\pi \tau_v \geq (h-1)^2/h \geq (h-1)/2$. □

COROLLARY 15.19 (Unconditional exponential law, this book). *Let $N > 13$ be a prime with $N \equiv 1 \bmod 12$, so that $h = (N-1)/12 \geq 2$, and let $\ell \neq N$ be prime. For simple random walk in continuous time on the supersingular ℓ isogeny graph $G_N(\ell)$, every vertex v satisfies*

$$\sup_{t\geq 0} \left| \mathbb{P}_\pi(\tau_v > t) - e^{-t/\mathbb{E}_\pi \tau_v} \right| \leq \frac{2(\ell+1)}{(\sqrt{\ell}-1)^2(h-1)},$$

and all Dirichlet rates but the slowest lie in the Kesten band $[1-\rho_\ell, 1+\rho_\ell]$, $\rho_\ell = 2\sqrt{\ell}/(\ell+1)$.

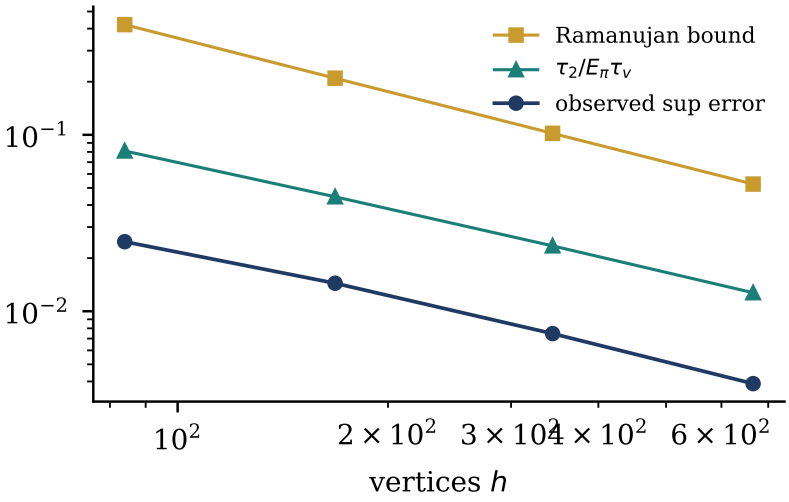


Figure 15.3: Supersingular 2 isogeny graphs for $N = 1009, 2017, 4129, 7993$ ($h = 84$ to 666 vertices), six random targets v each, logarithmic axes: the largest observed sup distance between $\mathbb{P}_\pi(\tau_v > t)$ and the exponential law with the same mean, against the largest Aldous Brown bound $\tau_2/\mathbb{E}_\pi\tau_v$ and the unconditional Ramanujan bound of Corollary 15.19.

PROOF. By Theorem 10.4 every nontrivial eigenvalue satisfies $|\lambda| \le 2\sqrt{\ell} < \ell + 1$. Hence $\ell + 1$ is a simple eigenvalue and $-(\ell + 1)$ is not an eigenvalue, so $G_N(\ell)$ is connected, not bipartite and Ramanujan, and Theorem 15.18 applies. □

Theorem 10.6 supplies the scale. In continuous time at rate 1 the expected hitting times are the discrete ones, so $\sum_v \pi(v)\mathbb{E}_\pi\tau_v$ is Kemeny's constant K_N, and $K_N/h \to \ell/(\ell-1)$. So on large isogeny graphs the time to find a given curve from a random start is, uniformly in time and up to $O(1/h)$, exponential, with a mean that averages over targets to about $\frac{\ell}{\ell-1}h$, that is about $2h$ for $\ell = 2$.

The data, computed for this book on the supersingular 2 isogeny graphs at $N = 1009, 2017, 4129, 7993$ ($h = 84, 168, 344, 666$) with six random targets each, are as follows. The largest observed $\sup_t |\mathbb{P}_\pi(\tau_v > t) - e^{-t/\mathbb{E}_\pi\tau_v}|$ is 0.0248, 0.0144, 0.0075, 0.0039; the Aldous Brown bound $\tau_2/\mathbb{E}_\pi\tau_v$ is 0.081, 0.045, 0.023, 0.0127; the Ramanujan bound of Corollary 15.19 is 0.421, 0.209, 0.102, 0.0526. The spectral gaps $1 - p_2$ are 0.0845, 0.0708, 0.0642, 0.0601, above the Ramanujan lower bound $1 - \rho_2 = 0.0572$; the smallest second Dirichlet rates are $\mu_2 = 0.0862, 0.0710, 0.0642, 0.0601 \ge 0.0572$, and the largest Dirichlet rate is at most $1.9406 \le 1 + \rho_2 = 1.9428$. All three quantities decay like $1/h$ (Figure 15.3); the observed error is about a third of the Aldous Brown bound.

15.7 Extreme values and exponent measures

Chapter 27 represents $\prod_n (1 - e^{-2\pi n y})$ as the distribution function of $M = \max_n E_n/(2\pi n)$. Here we identify the structure of M in the language of this chapter: just as a GGC has a completely monotone Lévy density, M has a completely monotone exponent function. Write $\sigma_{-1}(m) = \sum_{d|m} d^{-1}$ and $P(y) = \mathbb{P}(M \le y)$ as in Chapter 27.

Proposition 15.20 (this book). *1. Let $G = -\log P$. Then for $y > 0$*

$$G(y) = \sum_{m \ge 1} \sigma_{-1}(m)\, e^{-2\pi m y}.$$

Hence G is completely monotone, $G(y) = \int e^{-yu} \nu(du)$ with

$$\nu = \sum_{m \ge 1} \sigma_{-1}(m)\, \delta_{2\pi m},$$

and M is max infinitely divisible.

2. *For each n, $E_n/(2\pi n)$ has the law of $\max_{k \ge 1} \max_{i \le N_k} E_{k,i}/(2\pi n k)$, with N_k Poisson of mean $1/k$, all variables independent. Consequently M has the law of $\max_{m \ge 1} \max_{i \le N'_m} E'_{m,i}/(2\pi m)$ with independent N'_m Poisson of mean $\sigma_{-1}(m)$.*

3. *For $\Re s > 1$, $\int_0^\infty G(y) y^{s-1} dy = \Gamma(s)(2\pi)^{-s}\zeta(s)\zeta(s+1)$. The right side continues meromorphically to $\mathbb{C}$ with poles only at $s = 0, \pm 1$, and is invariant under $s \mapsto -s$.*

4. *For $f = \prod_m \eta(m\tau)^{r_m}$ with $r_m \ge 0$, $\sum_m m r_m = 24$, and M_f as in Theorem 27.7, $-\log \mathbb{P}(M_f \le y) = \sum_m r_m G(my)$ is completely monotone with exponent measure $\sum_n \big(\sum_{m|n} r_m \sigma_{-1}(n/m)\big)\delta_{2\pi n}$, and its Mellin transform is $\Gamma(s)(2\pi)^{-s}\zeta(s)\zeta(s+1)\sum_m r_m m^{-s}$.*

Proof. (1) $-\log(1 - e^{-2\pi n y}) = \sum_{k \ge 1} e^{-2\pi n k y}/k$; summing over n and collecting $m = nk$ gives $\sum_m e^{-2\pi m y} \sum_{k|m} k^{-1}$, and $\sum_{k|m} k^{-1} = \sigma_{-1}(m)$. A positive combination of decreasing exponentials is completely monotone. For every $t > 0$, $P^t = \exp(-tG)$ is the distribution function of the maximum of a Poisson process with intensity $t \sum_m \sigma_{-1}(m)\, \mathrm{Law}(E/(2\pi m))$ by the computation in (2), so M is max infinitely divisible [456].

(2) If N is Poisson of mean λ, independent of the Z_i, independent copies of Z, then $\mathbb{P}(\max_{i \le N} Z_i \le y) = \mathbb{E}\mathbb{P}(Z \le y)^N = \exp(-\lambda \mathbb{P}(Z > y))$, with an empty maximum read as 0. For $Z = E/(2\pi n k)$ and $\lambda = 1/k$ this is $\exp(-e^{-2\pi n k y}/k)$, and the product over k is $\exp(\log(1 - e^{-2\pi n y})) = 1 - e^{-2\pi n y} = \mathbb{P}(E_n/(2\pi n) \le y)$, by the identity $1 - e^{-x} = \exp(-\sum_k e^{-kx}/k)$ for $x > 0$. The maximum is almost surely over a nonempty set, since $\sum_k 1/k = \infty$. Taking independent copies for all

n and merging the Poisson families with the same $m = nk$ (superposition adds the means $1/k$ over $k \mid m$) gives the second statement.

(3) By Tonelli's theorem the Mellin integral equals $\sum_m \sigma_{-1}(m)\Gamma(s)(2\pi m)^{-s}$, and $\sum_m \sigma_{-1}(m)m^{-s} = \sum_{d,k} d^{-s}k^{-1-s} = \zeta(s)\zeta(s+1)$ for real $s > 1$, hence for $\Re s > 1$. For the continuation use the modular relation (27.1), which in terms of $G = -\log P$ reads $G(1/y) = G(y) - \frac{1}{2}\log y - \frac{\pi}{12y} + \frac{\pi y}{12}$. Splitting the integral at 1 and substituting $y \mapsto 1/y$ on $(0,1)$,

$$\int_0^\infty G(y)y^{s-1}dy = \int_1^\infty G(y)\big(y^{s-1} + y^{-s-1}\big)dy - \frac{1}{2s^2} - \frac{\pi}{12(s+1)} + \frac{\pi}{12(s-1)},$$

using $\int_1^\infty y^{-s-1}\log y\,dy = s^{-2}$, $\int_1^\infty y^{-s-2}dy = (s+1)^{-1}$ and $\int_1^\infty y^{-s}dy = (s-1)^{-1}$. The integral is entire because G decays exponentially, and it is invariant under $s \mapsto -s$, as are $-\frac{1}{2s^2}$ and the sum of the last two terms. Equivalently, the duplication formula gives $\Gamma(s)(2\pi)^{-s}\zeta(s)\zeta(s+1) = \frac{1}{2}\hat\xi(s)\hat\xi(s+1)$ with $\hat\xi(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$, and the symmetry is the functional equation $\hat\xi(s) = \hat\xi(1-s)$.

(4) $\mathbb{P}(M_f \le y) = \prod_m P(my)^{r_m}$ by definition, and (1) applied to each factor, with the Mellin substitution $y \mapsto my$, gives the rest. □

By (3), the exponent of the extreme value law of η has Mellin transform $\Gamma(s)(2\pi)^{-s}\zeta(s)\zeta(s+1)$, a product of two shifted zeta functions of Eisenstein type, symmetric under $s \mapsto -s$, with a double pole at $s = 0$ matching the term $\frac{1}{2}\log y$ in the expansion of G at 0. Read backwards, Mellin inversion turns this symmetry into the modular relation, which is a classical proof of the transformation law of η; the probabilistic reading adds the exponent measure ν and the Poisson structure of (2). For every multiplicative eta product the exponent is Eisenstein, while the cusp form L function enters only through the incomplete gamma expectation of Theorem 27.7. The Poisson splitting identity in (2) is a max analogue of Frullani's integral: one exponential clock is a maximum over infinitely many Poisson families of faster clocks, just as one gamma variable is an integral of Poisson jumps.

15.8 Notes

The Thorin criterion for single Euler factors and its corollaries are ours; the ingredients are classical. Thorin [546] and Bondesson [73] developed generalized gamma convolutions, and the Pringsheim Landau principle is in Widder's book [578]. Deligne [140] proved the Ramanujan Petersson conjecture for holomorphic forms; for Maass forms the bound $\frac{7}{64}$ is due to Kim and Sarnak [315], building on Kim and Shahidi [316]. Weil's explicit formula is in [569], and our normalisation follows Iwaniec and Kowalski [276]. The hybrid product is from Gonek, Hughes and Keating [214], and one level densities from Iwaniec, Luo and Sarnak [271] and Katz and Sarnak [298]. Weil proved the Riemann hypothesis for curves in

[567]; for graphs see Ihara [269], Lubotzky, Phillips and Sarnak [373] and Pizer [438]. Keilson's theorem on passage times of birth death processes is [303], Kent's diffusion version [308], and Fill's probabilistic proof [188]; the Aldous Brown inequality is [13], and Aldous and Fill [14] is the general reference for hitting times of reversible chains. Toeplitz spectra are in Grenander and Szegő [225], large deviations in Dembo and Zeitouni [143], and max infinite divisibility in Resnick [456]. The identification of passage time rates with zeros of the sphere polynomials B_n, the large deviation rate of Theorem 15.16, the exponential law on isogeny graphs and the max infinitely divisible reading of Proposition 15.20 are, as far as we know, new; the Mellin transform in Proposition 15.20(3) is classical.

15.9 Exercises

1. Comparing the coefficients of w^4 in Lemma 15.1, prove that
$$\sum_{k\in\mathbb{Z}} \frac{1}{(\theta+2\pi k)^4} = \frac{2+\cos\theta}{48\sin^4(\theta/2)}.$$
Deduce the variance of the local GGC of Theorem 15.3, and show that under the Sato Tate law the expected variance is infinite.

2. Show that for $x_p = 2$, $\frac{(\log p)^2 s}{2\cosh(\sqrt{s}\log p)-2} = \prod_{k\neq 0}\bigl(1+\frac{s(\log p)^2}{4\pi^2k^2}\bigr)^{-1}$, and interpret $1/E_p(\sqrt{s})$ as a GGC with an extra degenerate Thorin atom of mass 1 at the origin, as for a central zero of order two in Chapter 13.

3. Let E be an elliptic curve over $\mathbb{F}_q$ with $a = q+1-\#E(\mathbb{F}_q) \neq 2\sqrt{q}$. Show that the GGC of Theorem 15.11 has mean $(\log q)^2\sqrt{q}/(\#E(\mathbb{F}_q) - (\sqrt{q}-1)^2)$, and that this mean is positive if and only if $\#E(\mathbb{F}_q) > (\sqrt{q}-1)^2$, one half of Hasse's bound $|a| \leq 2\sqrt{q}$. Show that the GGC property itself is equivalent to the full bound, and identify which half of it comes from the rejection of a real pole and which from Lemma 15.2.

4. In Theorem 15.15 take $q = 1$ (the reflecting walk on $\{0,1,2,\dots\}$, which is the line folded at the origin). Show that the rates are $1-\cos\frac{(2j-1)\pi}{2n}$, $1 \leq j \leq n$, that $\mathbb{E}T_n = n^2$, and explain why the absence of a gap ($\rho = 1$) turns the ballistic law of large numbers into diffusive scaling.

5. Derive the formula for Λ^* in Theorem 15.16, and show that $\Lambda^*(y)/y \to 1-\rho$ as $y \to \infty$. Interpret: the probability of an atypically long passage time decays at the rate of the bottom of the Kesten band.

6. Using Proposition 15.20(4), show that for $f = \eta(\tau)^2\eta(11\tau)^2$ the exponent function $-\log\mathbb{P}(M_f \leq y)$ has Mellin transform
$$2(1+11^{-s})\,\Gamma(s)(2\pi)^{-s}\zeta(s)\zeta(s+1),$$

and compute the expected number of points of the Poisson representation of Proposition 15.20(2), applied to each factor of M_f, that exceed y.

15.10 Open computational problems

Computational Problem 15.1 (Comb statistics against the zeros of Δ). Objective: compare the Thorin point process intensity $I_P(t)$ of Proposition 15.10, with the actual angles θ_p of $\tau(p)$, against the zeros of $L(s,\Delta)$. Data: all zeros with $0 < \gamma \le 10^5$ (about 2.8×10^5), and $\tau(p)$ for $p \le 10^7$. Route: PARI/GP `lfunzeros` or `lcalc` for the zeros, `ramanujantau` for the coefficients, with the explicit formula of Theorem 15.7 for Gaussian test functions of widths from 0.1 to 10 as a certificate. Deliverable: the pair correlation of local comb points against that of the zeros as a function of P and t, a test of the heuristic matching $P \approx \log(t/2\pi)$, and a table of the per prime ratios of explicit formula terms to comb terms against $-p^{-1/2}$. Why it matters: it measures quantitatively how the aperiodic global Thorin measure is assembled from periodic local combs.

Computational Problem 15.2 (Large deviations on finite isogeny graphs). Objective: compare the cumulant generating function of distance passage times on supersingular isogeny graphs with the tree formula $\Lambda(\theta)$ of Theorem 15.16. Data: $G_N(\ell)$ for $\ell = 2, 3$ and primes $N \equiv 1 \bmod 12$ up to 10^5 (h up to about 8000), with passage times to distance n up to the injectivity radius. Route: build the graphs from modular polynomials in PARI/GP as in Chapter 10; compute $\mathbb{E}e^{\theta T_n}$ exactly by solving the killed linear systems for θ on a grid. Deliverable: plots of $n^{-1}\log \mathbb{E}e^{\theta T_n}$ against $\Lambda(\theta)$, the value at $\theta = 1-\rho$ against $\frac{1}{2}\log \ell$, and the distance $n(N)$ at which the finite graph departs from the tree. Why it matters: the departure measures how far the Ramanujan graph is from its universal cover at the scale of large deviations, a finer statistic than the spectrum.

Computational Problem 15.3 (Kim Sarnak versus Ramanujan through local parabolas). Objective: test the local parabolic conditions $\Theta_\sigma^{(p)}$ for Maass forms. Data: Hecke eigenvalues of Maass cusp forms for $\mathrm{SL}_2(\mathbb{Z})$ and $\Gamma_0(N)$ with small N from the LMFDB [371], for all available forms and primes. Route: compute x_p and the largest $|\Re w|$ among the zeros of E_p; record any $|x_p| > 2$. Deliverable: the empirical distribution of $\max_p |\Re w|$ per form against the proven bound $\frac{7}{64}$, the Sato Tate fit of the x_p, and a statistical test of the hypothesis that all local combs are on the axis. Why it matters: Ramanujan for Maass forms is open, and the local Thorin form makes every unramified prime a separate test with a quantitative margin.

Computational Problem 15.4 (Thorin combs of curves of genus three and four). Objective: measure how the periodic Thorin combs of Theorem 15.11 fill the line in families. Data: all curves of genus 3 and 4 over $\mathbb{F}_q$ for $q \le 13$ (hyperelliptic and

plane quartic models in genus 3), with their L polynomials. Route: point counting in Sage or Magma, or the tables of Sutherland and collaborators where available. Deliverable: histograms of the angles θ_j against the $\mathrm{USp}(2g)$ Weyl density (Katz Sarnak [298]), the distribution of the smallest Thorin atom $(\min_j \theta_j / \log q)^2$, and the low lying angle density near 0 compared with the Sato Tate depletion of Proposition 15.10. Why it matters: function fields are, with regular graphs, the setting where global Thorin is a theorem, and the statistics of the smallest atoms are the finite field model of low lying zeros.

CHAPTER 16

VAN DANTZIG PAIRS, WALD COUPLES AND THE SINE SQUARED TRANSPORT

In 1958 David van Dantzig posed a prize question in the Nieuw Archief voor Wiskunde [555]: describe the characteristic functions F for which $t \mapsto 1/F(it)$ is again a characteristic function. The cosine and the hyperbolic secant form the simplest example, and Lukacs [375, 376] turned the question into a theory. Roynette and Yor [467] rewrote the Riemann hypothesis in terms of Wald couples and Pólya frequency functions; the first author [442] observed that the Hadamard factorisation of a Laguerre Pólya function is the same thing as a van Dantzig pair, a Wald couple and a Thorin representation, three probabilistic readings of one product; and more than sixty years after the prize question Konstantopoulos, Patie and Sarkar [330] showed that the Riemann hypothesis is a statement about the Lukacs subclass of van Dantzig's class. This chapter assembles these ideas for automorphic L functions. We describe the van Dantzig problem and its Lukacs subclass, Wald couples and the duality theorem, a gallery of classical pairs, and then the analytic engine behind the Thorin densities of Chapter 14: the sine squared transport, which carries the Lévy measure of $\log f$ at a basepoint to the Thorin measure of $f(\alpha)/f(\alpha+\sqrt{s})$. At the centre it carries, formally and through Weil's explicit formula, the primes to the squared ordinates of the zeros. We then build subordinators from Euler products and compute their first passage laws, and we test numerically whether completed L functions belong to the Patie class of van Dantzig solutions. Nothing here proves a Riemann hypothesis; the point is to see the same positivity from several sides, and to make the positivity computable.

Throughout, Ξ_π, F_π, the Thorin atoms $t_k = \gamma_k^2$ and the Pólya kernel Φ_π are as in Chapter 14, GGC is as in Definition 13.2, $E, E_1, E_2, \dots$ are independent standard exponential variables and $L_1, L_2, \dots$ independent standard Laplace variables, with density $\frac{1}{2}e^{-|x|}$ and characteristic function $(1+t^2)^{-1}$. Since $\mathbb{E}e^{itN\sqrt{2E}} = \mathbb{E}e^{-t^2E}$, a standard Laplace variable is $\sqrt{2E}\,N$ with N standard normal and independent of E.

16.1 THE VAN DANTZIG PROBLEM

Let $\mathcal{P}$ be the set of characteristic functions of real random variables. For $F \in \mathcal{P}$ analytic in a neighbourhood of 0, put $VF(t) = 1/F(it)$ for small real t. A

characteristic function that is analytic near 0 is analytic in a strip around $\mathbb{R}$ [376], so it is determined by its values near 0; we write $VF \in \mathcal{P}$ when VF agrees near 0 with a characteristic function, and then use VF for that function.

DEFINITION 16.1. The *van Dantzig class* is $\mathcal{D} = \{F \in \mathcal{P} : F \text{ analytic near } 0,\ VF \in \mathcal{P}\}$, and a pair $[F, VF]$ with both members in $\mathcal{P}$ is a *van Dantzig pair*. The *Lukacs class* $\mathcal{D}_L$ is the set of even functions $F \in \mathcal{P}$ in the Laguerre Pólya class: even real entire functions with $F(0) = 1$ that are locally uniform limits of real polynomials with only real zeros.

If $F \in \mathcal{D}$ is the characteristic function of X, then $VF(t) = 1/\mathbb{E}e^{-tX}$ is real for small t, hence real on $\mathbb{R}$, hence even, because a real characteristic function is even; so F is even near 0, hence on $\mathbb{R}$, and $V(VF)(t) = 1/VF(it) = F(-t) = F(t)$. Thus every member of $\mathcal{D}$ is the characteristic function of a symmetric law, V is an involution of $\mathcal{D}$ [330], and $FG \in \mathcal{D}$ whenever $F, G \in \mathcal{D}$, because $V(FG) = VF \cdot VG$ and products of characteristic functions are characteristic functions. The first example is $F(t) = \cos t$, the characteristic function of a fair sign ± 1, with $VF(t) = 1/\cosh t$, the characteristic function of the density $1/(2\cosh(\pi x/2))$. The Lukacs class is contained in $\mathcal{D}$: if $F \in \mathcal{D}_L$, then by the Laguerre Pólya theorem $F(t) = e^{-ct^2}\prod_k (1 - t^2/c_k^2)$ with $c \geq 0$ and $\sum c_k^{-2} < \infty$, so $VF(t) = e^{-ct^2}\prod_k (1 + t^2/c_k^2)^{-1}$, the characteristic function of $\sqrt{2c}\,N + \sum_k L_k/c_k$.

THEOREM 16.2 (Konstantopoulos, Patie and Sarkar [330]). *(i) $\mathcal{D}$ and $\mathcal{D}_L$ are both closed under multiplication and under locally uniform limits; they are monoids.*

(ii) (Lukacs [375]) The Lukacs operators $L^{(p)}f(t) = t^{p-2}f^{(p)}(t)/f^{(2)}(0)$, $p = 1, 2$, map $\mathcal{D}_L$ into itself.

(iii) With $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s)$, the function $t \mapsto \xi(\frac{1}{2} + it)/\xi(\frac{1}{2})$ belongs to $\mathcal{D}_L$ if and only if the Riemann hypothesis holds; and $t \mapsto \xi(\frac{1}{2})/\xi(\frac{1}{2} + t)$ is the characteristic function of a symmetric Pólya frequency function if and only if the Riemann hypothesis holds.

(iv) A partition function P, an even entire function arising as the moment generating function of a symmetric law, has the Lee Yang property (all zeros purely imaginary) if and only if $t \mapsto P(it)$ belongs to $\mathcal{D}_L$.

In [330] these are Proposition 3.4 with Theorem 3.14, Theorem 3.5, Theorem 2.3 with Corollary 2.4, and Theorem 2.5; item (ii) is Theorem 4 of [375], and the second half of (iii) is, in substance, Théorème V.3.2 of Roynette and Yor [467]. Item (iii) uses Pólya's positive kernel (13.1): positivity makes $\xi(\frac{1}{2} + it)/\xi(\frac{1}{2})$ a characteristic function, and the Riemann hypothesis is then exactly the Laguerre Pólya property. Item (iv) places the Riemann hypothesis beside the circle theorem of Lee and Yang [359]: for ferromagnetic Ising systems the partition function, as

a function of the external field, has only purely imaginary zeros. Newman [416] characterised the single spin distributions for which the Lee Yang property persists under ferromagnetic coupling, and in [417] he read the Riemann hypothesis as the statement that the Pólya variable of ζ has the Lee Yang property, with the de Bruijn Newman constant measuring how far a Gaussian deformation can push it; Newman and Wu [418] survey this parallel. In the language of this book, *GRH for π is the Lee Yang property of the Pólya variable W_π*, whenever $\Phi_\pi \geq 0$.

16.2 Wald couples and the duality theorem

If B is a Brownian motion and T a bounded stopping time, Wald's identity gives $\mathbb{E}\exp(\theta B_T - \theta^2 T/2) = 1$: along the path, the exponential of the position is compensated exactly by a Gaussian transform of the time. A Wald couple asks for the same compensation between two separate laws.

Definition 16.3. A *Wald couple* is a pair (X, H) of random variables with $H \geq 0$ such that $\mathbb{E}e^{sX} < \infty$ for s near 0 and

$$\mathbb{E}e^{sX}\,\mathbb{E}e^{-s^2 H} = 1 \qquad \text{for all } s \in [0, s_0),\ \text{some } s_0 > 0.$$

Roynette and Yor [467] introduced Wald couples with both members infinitely divisible, normalised as $\mathbb{E}e^{\lambda X}\mathbb{E}e^{-\lambda^2 H/2} = 1$, and the first author [442] keeps infinite divisibility of both members, in the normalisation used here. We use the identity alone, for the following reason.

Remark 16.4. The characteristic function of an infinitely divisible law never vanishes on $\mathbb{R}$ (Exercise 5). The characteristic function of the Pólya variable W_π is $\Xi_\pi(it)/\Xi_\pi(0)$, which vanishes at every real ordinate γ_k; for ζ there are infinitely many by Hardy's theorem [235], and for Δ by Hafner's theorem [233]. No real ordinate is needed: W_π has exponential moments of all orders, so if it were infinitely divisible its Lévy measure would have them too ([485], Section 25), and $z \mapsto \mathbb{E}e^{zW_\pi} = \Xi_\pi(z)/\Xi_\pi(0)$ would be the exponential of an entire function, whereas it vanishes at every z_k. This is the argument of Roynette and Yor [467], Remarque V.3.1, for ξ. So W_π is never infinitely divisible, and the infinitely divisible member of the couple is always the second one. This takes nothing away from the duality theorem below, which is about H.

Theorem 16.5 (after Polson [442], Theorem 12). *Let f be a real entire function of order at most one, symmetric about α in the sense $f(\alpha + s) = f(\alpha - s)$, with $f(\alpha) > 0$. The zeros of $s \mapsto f(\alpha + s)$ are purely imaginary, $\pm i\rho_k$, if and only if Thorin's condition holds:*

$$\frac{f(\alpha)}{f(\alpha + \sqrt{s})} = \mathbb{E}e^{-sH}, \qquad s > 0,$$

for a GGC H. In that case the Thorin measure of H is $\sum_k \delta_{\rho_k^2}$; if moreover $f(\alpha+\cdot)/f(\alpha)$ is the moment generating function of a symmetric X, then $[f_\alpha, g_\alpha]$ with $f_\alpha(t) = f(\alpha+it)/f(\alpha)$, $g_\alpha(t) = f(\alpha)/f(\alpha+t)$ is a van Dantzig pair and (X,H) is a Wald couple.

The proof is the argument of Theorem 14.7, (a) if and only if (b), applied to $E(z) = f(\alpha+z)/f(\alpha)$. In [442] the variable X is also asserted to be infinitely divisible; that cannot hold when f has zeros, since f_α then vanishes at the real points $\pm\rho_k$ (Exercise 5), so we state the van Dantzig and Wald conclusions without it. For automorphic L functions we obtain the following, using Chapter 14.

THEOREM 16.6 (this book; for ζ, [467, 442, 330]). *Let π be as in Theorem 14.8, or ζ, and suppose its Pólya kernel is nonnegative, so that W_π exists (for instance ξ, Δ and every multiplicative eta product, using Ξ^{sd}). Let $Z_\pi = \sum_k E_k/\gamma_k^2$ and $V_\pi = \sum_k L_k/\gamma_k$ whenever all γ_k are real. The following are equivalent.*

(a) GRH holds for π.

(b) $t \mapsto \Xi_\pi(0)/\Xi_\pi(t)$ is the characteristic function of a symmetric Pólya frequency density; equivalently, of a series $\sum_k L_k/c_k$ with real c_k and $\sum c_k^{-2} < \infty$. Then $c_k = \gamma_k$, $[\,\Xi_\pi(it)/\Xi_\pi(0),\ \Xi_\pi(0)/\Xi_\pi(t)\,]$ is the van Dantzig pair of (W_π, V_π), and

$$V_\pi = \sum_k \frac{L_k}{\gamma_k} \ \stackrel{d}{=}\ \sqrt{2Z_\pi}\,N.$$

(c) There is a GGC H such that (W_π, H) is a Wald couple; then $H \stackrel{d}{=} Z_\pi$.

(d) $t \mapsto \Xi_\pi(it)/\Xi_\pi(0)$ belongs to the Lukacs class $\mathcal{D}_L$.

PROOF. (a) implies all. Under GRH, $\Xi_\pi(z)/\Xi_\pi(0) = \prod_k (1+z^2/\gamma_k^2)$ with real γ_k and $\sum \gamma_k^{-2} < \infty$ (Lemma 14.6). For real t, $\Xi_\pi(0)/\Xi_\pi(t) = \prod(1+t^2/\gamma_k^2)^{-1} = \mathbb{E}e^{itV_\pi}$ by independence, the series V_π converging in L^2. Writing $L_k = \sqrt{2E_k}N_k$, conditionally on the E_k the series is centred Gaussian with variance $2\sum E_k/\gamma_k^2 = 2Z_\pi$, which gives the mixture form. Schoenberg's theorem (Theorem 13.1) with $\Psi(s) = \prod_k (1+s/\gamma_k)(1-s/\gamma_k)$, a function of the stated form with $\delta_k = \pm\gamma_k^{-1}$, shows that the density of V_π is a Pólya frequency function, symmetric because Ψ is even. This gives (b). For (c), $\mathbb{E}e^{sW_\pi} = \Xi_\pi(s)/\Xi_\pi(0)$ by Lemma 14.12, and $\mathbb{E}e^{-s^2 Z_\pi} = \prod(1+s^2/\gamma_k^2)^{-1}$, a GGC transform by Theorem 14.8(c). For (d), $\Xi_\pi(it)/\Xi_\pi(0) = \prod(1-t^2/\gamma_k^2)$ is an even Laguerre Pólya function, and it is the characteristic function of W_π.

(b) implies (a). Suppose $\Xi_\pi(0)/\Xi_\pi(t) = \int e^{itx} g(x)\,dx$ for a Pólya frequency density g. By Schoenberg's theorem $\int e^{-sx} g(x)\,dx = 1/\Psi(s)$ on a strip containing $i\mathbb{R}$, with Ψ of Laguerre Pólya type. On $i\mathbb{R}$, $\Psi(-it) = \Xi_\pi(t)/\Xi_\pi(0)$; both sides are entire in t, so $\Psi(w) = \Xi_\pi(iw)/\Xi_\pi(0)$ for all w. Since Ψ has only real zeros, every zero $z = iw$ of Ξ_π is purely imaginary. In the second form, $\Xi_\pi(t)/\Xi_\pi(0) = \prod(1+t^2/c_k^2)$ on $\mathbb{R}$, hence on $\mathbb{C}$, so the zeros are $\pm ic_k$.

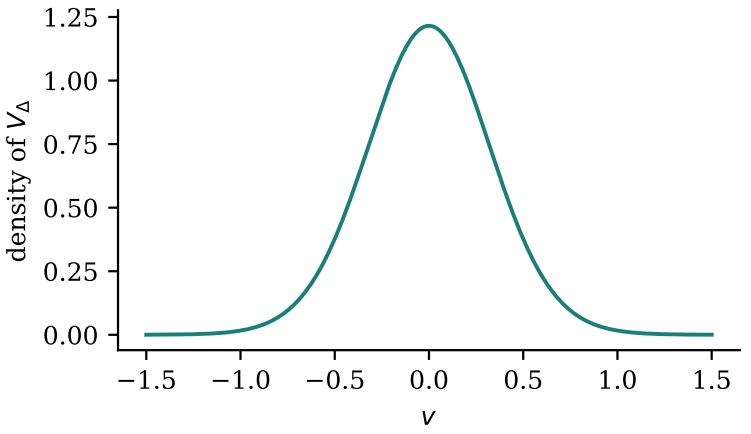


Figure 16.1: The density of V_Δ, the inverse Fourier transform of $t \mapsto \Lambda(6,\Delta)/\Lambda(6+t,\Delta)$, computed from L values for $|t| \le 60$ only. It is positive, of mass 0.99995, with value 1.21546 at the origin, and log concave on the computed range.

(c) implies (a). The Wald identity with $u = s^2$ gives $\varphi(u)F_\pi(u) = 1$ for $u \in (0, s_0^2)$, where $\varphi(u) = \mathbb{E}e^{-uH}$ is a GGC transform. By Lemma 13.3, φ is holomorphic on $\mathbb{C} \setminus (-\infty, 0]$, and F_π is entire, so the identity holds on that domain and F_π has no zero off $(-\infty, 0]$. This is Theorem 14.8(b), so GRH holds, and (c) of that theorem identifies H.

(d) implies (a). A Laguerre Pólya function has only real zeros, and the zeros of $t \mapsto \Xi_\pi(it)$ are the γ_k. □

Without a positive kernel, (b) and (c) remain equivalent to GRH with the Wald identity read for the transform $\Xi_\pi(s)/\Xi_\pi(0)$ in place of $\mathbb{E}e^{sW_\pi}$; only (d) needs the kernel. For ζ, the equivalence of (a) and (b) is Théorème V.3.2 of [467] and Corollary 2.4 of [330], that of (a) and (d) is Theorem 2.3 of [330], and that of (a) and (c) follows from Theorem 16.5; the general case rests on Chapter 14.

For Δ the partner can be computed without knowing a single zero. We evaluated $g(t) = \Lambda(6,\Delta)/\Lambda(6+t,\Delta)$ on $0 \le t \le 60$ with PARI/GP and inverted the Fourier transform by quadrature. The result (Figure 16.1) is a positive density (minimum $2.4 \cdot 10^{-4}$ on $[0, 1.5]$) of total mass 0.99995, with density 1.21546 at 0, and log concave on the computed range. Log concavity is the Pólya frequency property of order two, so the computation is numerical evidence for the two weakest consequences of Theorem 16.6(b).

16.3 A GALLERY OF VAN DANTZIG PAIRS AND WALD COUPLES

Each example below is a characteristic function f with a partner $1/f(i\cdot)$ and, where available, a Wald couple. We checked every displayed identity numerically to at least nine digits.

Cosine and sine. The function $\cos s = \prod_{k\ge1}(1 - 4s^2/((2k-1)\pi)^2)$ is the characteristic function of ±1, with partner $1/\cosh s$, the characteristic function of $\sum_k 2L_k/((2k-1)\pi)$. Similarly $\sin s/s$, the characteristic function of the uniform law on $(-1,1)$, has partner $s/\sinh s$, the characteristic function of $\sum_k L_k/(k\pi)$, and of the density $\frac{\pi}{4}\cosh^{-2}(\pi x/2)$. In the notation of Biane, Pitman and Yor [82], $\mathbb{E}e^{-\lambda C} = 1/\cosh\sqrt{2\lambda}$ and $\mathbb{E}e^{-\lambda S} = \sqrt{2\lambda}/\sinh\sqrt{2\lambda}$, so $(\pm1, C/2)$ and $(U, S/2)$, with U uniform on $(-1,1)$, are Wald couples; C and S are hitting times of 1 by the norms of one and three dimensional Brownian motion started at 0.

Bessel functions and the symmetric Beta laws. For $\nu > -\frac12$ let X_ν have the symmetric Beta (power semicircle) density $(1-x^2)^{\nu-1/2}/B(\nu+\frac12,\frac12)$ on $(-1,1)$: the arcsine law for $\nu = 0$, the uniform law for $\nu = \frac12$, Wigner's semicircle for $\nu = 1$, and the sign ±1 as the limit $\nu \downarrow -\frac12$. Then

$$f_\nu(s) = \mathbb{E}e^{isX_\nu} = \Gamma(\nu+1)\Big(\frac{s}{2}\Big)^{-\nu} J_\nu(s) = \prod_{n\ge1}\Big(1 - \frac{s^2}{j_{\nu,n}^2}\Big), \tag{16.1}$$

with $j_{\nu,n}$ the positive zeros of J_ν (Exercise 2), and the partner $1/f_\nu(is) = \Gamma(\nu+1)^{-1}(s/2)^\nu/I_\nu(s)$ is the characteristic function of $H_\nu = \sum_n L_n/j_{\nu,n}$ [260, 287, 18, 245]. The case $\nu = \frac12$ is the sine pair above and $\nu = -\frac12$ the cosine pair. The Wald couple is $(X_\nu, T_\nu/2)$ with T_ν the hitting time of 1 by a Bessel process of dimension $2\nu+2$ from 0, whose Laplace transform $(\sqrt{2\lambda}/2)^\nu/(\Gamma(\nu+1)I_\nu(\sqrt{2\lambda}))$ is Kent's [307].

Macdonald functions. Pólya [444] proved that for $a > 0$ the function $t \mapsto K_{it}(a) = \int_0^\infty e^{-a\cosh u}\cos(tu)\,du$ has only real zeros. It is the Fourier transform of the positive even density proportional to $e^{-a\cosh u}$, so $t \mapsto K_{it}(a)/K_0(a)$ lies in $\mathcal{D}_L$ and $t \mapsto K_0(a)/K_t(a)$ is a characteristic function. Biane [58] gave a probabilistic account of Pólya's theorem through matrix valued Brownian motion. The kernel $e^{-a\cosh u}$ is the simplest model of the Pólya kernel of ξ, which behaves like $\exp(-\pi e^{2|u|})$.

Symmetric z laws. For $\delta > 0$ let G, G' be independent with the gamma law of shape δ. Then $X = \log(G/G')$ has

$$\mathbb{E}e^{isX} = \frac{|\Gamma(\delta+is)|^2}{\Gamma(\delta)^2} = \prod_{k\ge0}\Big(1 + \frac{s^2}{(\delta+k)^2}\Big)^{-1},$$

and for $\delta > \frac12$ its van Dantzig partner $\Gamma(\delta)^2/(\Gamma(\delta+t)\Gamma(\delta-t)) = \prod_{k\ge0}(1 - t^2/(\delta+k)^2)$ is the characteristic function of 2Θ, where Θ has density proportional

to $\cos^{2\delta-2}\theta$ on $(-\frac{\pi}{2},\frac{\pi}{2})$, by Cauchy's beta integral (we checked this at $\delta = 1.3$). So $X = \sqrt{Y}\,N$ is a normal variance mixture with mixing law $Y = \sum_{k\ge 0} E_k/\lambda_k$, $\lambda_k = \frac{1}{2}(\delta+k)^2$, a GGC with Thorin measure $\sum \delta_{\lambda_k}$. Partial fractions give the mixing density as an alternating series,

$$p_Y(u) = \sum_{k\ge 0} \binom{-2\delta}{k} \frac{\delta+k}{B(\delta,\delta)}\, e^{-(\delta+k)^2 u/2}, \qquad u>0,$$

whose coefficients alternate in sign although $p_Y \ge 0$. For $\delta = 0.7$ we checked $\prod_k (1+s/\lambda_k)^{-1} = |\Gamma(\delta + i\sqrt{2s})|^2/\Gamma(\delta)^2$ to all printed digits at $s = 0.1,\ 1,\ 5$ (values 0.6076958543678661, 0.04998906407585931, $2.860474564093928\cdot 10^{-4}$), and the Laplace transform of the series above against the product at $s = 1.3$ to fifteen digits.

Gumbel and Hinds. If $X = -\log E$, the Gumbel law with distribution function $\exp(-e^{-x})$, then

$$\mathbb{E}e^{-sX} = \Gamma(1+s) = \exp\Big(-\gamma_{\mathrm{E}} s + \int_0^\infty (e^{-st} - 1 + st)\frac{dt}{t(e^t-1)}\Big), \qquad s>-1,$$

with γ_{E} Euler's constant. The Lévy density is $k(t)/t$ with $k(t) = 1/(e^t-1)$ completely monotone, so X is an extended GGC in Bondesson's sense [73]. Hinds [260] considered X_1 with density $\frac{2}{\sqrt{\pi}}\exp(x - e^{2x})$, that is $X_1 = \frac{1}{2}\log G$ with G of gamma law of shape $\frac{1}{2}$, and $X_2 = -X_1$. Their characteristic functions are

$$f_1(s) = \frac{1}{\sqrt{\pi}}\Gamma\Big(\frac{1+is}{2}\Big), \qquad f_2(s) = \frac{1}{\sqrt{\pi}}\Gamma\Big(\frac{1-is}{2}\Big),$$

with normalisation $1/\sqrt{\pi}$, as $f_1(0) = 1$ requires. Neither lies in $\mathcal{D}$: $Vf_1(t) = \sqrt{\pi}/\Gamma(\frac{1-t}{2})$ is real but not even, while a real characteristic function is even. Their product, the characteristic function of $X_1 - X_1'$ with X_1' an independent copy of X_1, is $|\Gamma(\frac{1+is}{2})|^2/\pi = 1/\cosh(\pi s/2)$ by the reflection formula, and $V(1/\cosh(\pi\cdot/2))(t) = \cos(\pi t/2)$: the cosine pair again.

The reciprocal Gamma function and a Gamma Wald couple. Hartman [242] proved that for real c the function $e^{c\sqrt{s}}/\Gamma(1+\sqrt{s})$ is the Laplace transform of a positive infinitely divisible law if and only if $c \le -\gamma_{\mathrm{E}}$ (see also [467]). Applying the sine squared lemma of the next section (Lemma 16.7) to $f = \Gamma$ at $\alpha = 1$, whose Lévy representation is the Gumbel formula above, gives the extreme case $c = -\gamma_{\mathrm{E}}$ with an explicit Thorin density (this book):

$$\frac{e^{-\gamma_{\mathrm{E}}\sqrt{s}}}{\Gamma(1+\sqrt{s})} = \mathbb{E}e^{-sH_\Gamma}, \qquad u_\Gamma(z) = \frac{\operatorname{Re}\psi(1+i\sqrt{z}) + \gamma_{\mathrm{E}}}{2\pi\sqrt{z}} \ge 0, \tag{16.2}$$

with H_Γ a GGC of Thorin density u_Γ, where $\psi = \Gamma'/\Gamma$; positivity is the identity $\operatorname{Re}\psi(1+iy) + \gamma_{\mathrm{E}} = \sum_{k\ge 1} y^2/(k(k^2+y^2))$. We checked (16.2) at $s = 2$ to nine

digits. The GGC property is stated for the family $\Gamma(\alpha)e^{\sqrt{s}\,\psi(\alpha)}/\Gamma(\alpha+\sqrt{s})$ in [442], Section 4; the density u_Γ is computed here. Since $\mathbb{E}e^{s(\log E+\gamma_{\mathrm{E}})} = e^{\gamma_{\mathrm{E}}s}\Gamma(1+s)$, the pair $(\log E+\gamma_{\mathrm{E}}, H_\Gamma)$ is a Wald couple with both members infinitely divisible: in our normalisation it is the couple that Roynette and Yor [467] build on Hartman's theorem.

Ostrovskii's hyperbolic class. Ostrovskii [429] studied characteristic functions built from hyperbolic functions. In the form presented in the 2018 to 2021 versions of [442], one takes $f_\delta(z) = z^{-2} - (Cz)^{-1} + \delta\sum_k a_k/(z+h_k)$ with $\frac{\pi}{2} < C < B$, $h_k \geq B$, $a_k \geq 0$, $\sum a_k < \infty$; then for δ small $\phi_\delta(t) = f_\delta(\cosh t)/f_\delta(1)$ and $1/\phi_\delta(it) = f_\delta(1)/f_\delta(\cos t)$ are both characteristic functions. The positivity of the Fourier transform of ϕ_δ rests on the elementary integrals

$$\int_{\mathbb{R}} \frac{e^{itx}\,dt}{\cosh t + b} = \frac{2\pi\sinh((\pi-a)x)}{\sin a\,\sinh \pi x}, \qquad |b|<1,\ \cos a = -b,$$
$$\int_{\mathbb{R}} \frac{e^{itx}\,dt}{\cosh t + b} = \frac{2\pi\sin(ax)}{\sinh a\,\sinh \pi x}, \qquad b>1,\ \cosh a = b,$$

and $\int e^{itx}\cosh^{-2} t\,dt = \pi x/\sinh(\pi x/2)$, which we verified (those versions misprint the second condition as $\cosh a = -b$). We refer to [429] for the precise class and do not reproduce the estimates.

16.4 The sine squared transport

The Thorin densities of this book have so far been read off from zeros. The following lemma, the $2\sin^2$ lemma of [442] (Lemma 14 there; see also [441], Section 5), produces them from the Lévy measure of $\log f$ at a basepoint, which for an L function is the arithmetic. Its first step, which writes $e^{-x\sqrt{s}} - 1 + x\sqrt{s}$ through the positive $\frac{1}{2}$ stable law, is Théorème I.1.1 of Roynette and Yor [467], which also shows that the resulting H is self decomposable. We state the lemma in our normalisation and prove it.

Lemma 16.7 (sine squared lemma). *Let f be holomorphic and zero free on $\operatorname{Re} w > \alpha - \eta$ for some $\eta > 0$, positive on (α,∞), and suppose that for $s \geq 0$*

$$\log f(\alpha+s) - \log f(\alpha) - bs = \int_0^\infty (e^{-sx} - 1 + sx)\,\frac{\nu(dx)}{x}, \qquad b = \frac{f'(\alpha)}{f(\alpha)}, \tag{16.3}$$

with ν a positive measure on $(0,\infty)$ such that $\int_0^\infty \min(1,x)\,\nu(dx) < \infty$. Then for $s > 0$

$$\frac{f(\alpha)}{f(\alpha+\sqrt{s})} = e^{-b\sqrt{s}}\,\mathbb{E}e^{-sH},$$

where H is a GGC with drift zero and Thorin density

$$U(z) = \frac{1}{2\pi\sqrt{z}}\int_0^\infty 2\sin^2\Big(\frac{x\sqrt{z}}{2}\Big)\nu(dx) = \frac{\operatorname{Re}(f'/f)(\alpha+i\sqrt{z}) - b}{2\pi\sqrt{z}}. \tag{16.4}$$

If $b \geq 0$, $e^{-b\sqrt{s}}$ is the Laplace transform of a positive $\frac{1}{2}$ stable law, a GGC with Thorin density $b/(2\pi\sqrt{z})$, and then $f(\alpha)/f(\alpha+\sqrt{s})$ is itself a GGC transform with Thorin density $\mathrm{Re}(f'/f)(\alpha+i\sqrt{z})/(2\pi\sqrt{z})$. The hypothesis (16.3) *with $\nu \geq 0$ holds if and only if $s \mapsto (\log f)''(\alpha+s)$ is completely monotone on $(0,\infty)$ and tends to 0 at infinity.*

PROOF. *Step 1: subordination.* For $x > 0$ the positive $\frac{1}{2}$ stable density $q_x(y) = \frac{x}{2\sqrt{\pi}} y^{-3/2} e^{-x^2/(4y)}$ has $\int_0^\infty e^{-sy} q_x(y)\, dy = e^{-x\sqrt{s}}$, and $\int_0^\infty (1-e^{-sy}) y^{-3/2} dy = 2\sqrt{\pi}\sqrt{s}$ (substitute $u = sy$ and integrate by parts). Hence

$$\begin{aligned} e^{-x\sqrt{s}} - 1 + x\sqrt{s} &= \int_0^\infty (1-e^{-sy}) \Big[\frac{x}{2\sqrt{\pi}} y^{-3/2} - q_x(y) \Big] dy \\ &= \int_0^\infty (1-e^{-sy}) K(x,y)\, dy, \end{aligned}$$

with $K(x,y) = \frac{x}{2\sqrt{\pi}\, y^{3/2}} (1 - e^{-x^2/(4y)}) \geq 0$.

Step 2: a Laplace pair. For $a \geq 0$ and $y > 0$,

$$\frac{1-e^{-a/y}}{\sqrt{y}} = \int_0^\infty e^{-yz} \frac{2\sin^2 \sqrt{az}}{\sqrt{\pi z}}\, dz. \tag{16.5}$$

Indeed $\int_0^\infty e^{-yz} (\pi z)^{-1/2} dz = y^{-1/2}$, and expanding the cosine,

$$\int_0^\infty \frac{e^{-yz} \cos(2\sqrt{az})}{\sqrt{\pi z}}\, dz = \sum_{n \geq 0} (-1)^n \frac{4^n a^n\, \Gamma(n+\frac{1}{2})}{(2n)!\sqrt{\pi}\, y^{n+1/2}} = \frac{e^{-a/y}}{\sqrt{y}},$$

using $\Gamma(n+\frac{1}{2}) = (2n)!\sqrt{\pi}/(4^n n!)$; the interchange is justified by the bound $\cosh(2\sqrt{az})$ for the absolute series, which is integrable against $e^{-yz} z^{-1/2}$. Subtract and use $1 - \cos 2\theta = 2\sin^2\theta$.

Step 3: the Lévy density. Replace s by $\sqrt{s}$ in (16.3) and insert Step 1. All integrands are nonnegative, so Tonelli's theorem gives

$$\log \frac{f(\alpha+\sqrt{s})}{f(\alpha)} - b\sqrt{s} = \int_0^\infty (1-e^{-sy}) \frac{k(y)}{y}\, dy,$$

with $k(y) = (4\pi y)^{-1/2} \int_0^\infty (1 - e^{-x^2/(4y)})\, \nu(dx)$. By (16.5) with $a = x^2/4$ and Tonelli again, $k(y) = \int_0^\infty e^{-yz} U(z)\, dz$ with U the first expression in (16.4). Frullani's integral $\int_0^\infty (1-e^{-sy}) e^{-zy} y^{-1} dy = \log(1 + s/z)$ then gives $\log(f(\alpha+\sqrt{s})/f(\alpha)) - b\sqrt{s} = \int_0^\infty \log(1+s/z) U(z)\, dz$, finite for each $s > 0$. Finiteness for $s = 1$ forces $\int_0^1 |\log z| U(z) dz < \infty$ and $\int_1^\infty U(z) z^{-1} dz < \infty$, the integrability conditions of Definition 13.2, so $\exp(-\int \log(1+s/z) U(z) dz)$ is the transform of a GGC H.

Step 4: the closed form. Differentiating (16.3), $(f'/f)(\alpha+s) - b = \int_0^\infty (1 - e^{-sx}) \nu(dx)$ for $s > 0$. The right side converges absolutely and uniformly for $\mathrm{Re}\, s \geq$

0 because $|1-e^{-sx}| \le \min(2,|s|x)$, so it is continuous there and holomorphic inside; the left side is holomorphic on $\operatorname{Re} s > -\eta$. They agree for $\operatorname{Re} s \ge 0$, and at $s = ir$ the real part of the right side is $\int (1-\cos rx)\nu(dx) = \int 2\sin^2(rx/2)\nu(dx)$, which gives the second expression in (16.4).

Step 5. $\int_0^\infty \log(1+s/z) z^{-1/2} dz = 2\pi\sqrt{s}$ (substitute $z = su$), so $b\sqrt{s} = \int \log(1+s/z)\, b(2\pi\sqrt{z})^{-1} dz$, which is the stated GGC form of $e^{-b\sqrt{s}}$. Finally, (16.3) gives $(\log f)''(\alpha+s) = \int_0^\infty x e^{-sx}\nu(dx)$, completely monotone and tending to 0. Conversely Bernstein's theorem writes such a function as $\int_{(0,\infty)} e^{-sx} m(dx)$ with $m \ge 0$ and no atom at 0; integrating twice from 0 gives (16.3) with $\nu(dx) = m(dx)/x$. □

The second form of (16.4) is Stieltjes inversion in disguise: on the cut $s = -z$ the principal root is $\sqrt{s} = i\sqrt{z}$. It shows that *the transported Thorin density can be computed from f alone on one vertical line*. The hypothesis $\nu \ge 0$ is sufficient, not necessary, and fails for completed L functions. The next theorem replaces it by a Hadamard argument.

THEOREM 16.8 (Shifted Thorin representation; this book, extending [441, 442]). *Let π be as in Theorem 14.8, or ζ, and $a > 0$. Put*

$$V_a(z) = \frac{\operatorname{Re}(\Xi_\pi'/\Xi_\pi)(a+i\sqrt{z})}{2\pi\sqrt{z}}, \qquad z > 0.$$

(i) If every zero z_k of Ξ_π satisfies $|\operatorname{Re} z_k| < a$, then $V_a \ge 0$ and $\Xi_\pi(a)/\Xi_\pi(a+\sqrt{s}) = \mathbb{E}e^{-sH_a}$ for $s > 0$, with H_a a GGC of drift zero and Thorin density V_a.

(ii) If $\Xi_\pi(a)/\Xi_\pi(a+\sqrt{s})$ is a GGC transform on $(0,\infty)$, every zero satisfies $|\operatorname{Re} z_k| \le a$.

(iii) Unconditionally, (i) holds for every $a \ge \frac12$, that is for every basepoint $\alpha = \frac12 + a \ge 1$ in the variable s of $\Lambda(s,\pi)$ in the unitary normalisation.

(iv) GRH for π holds if and only if $V_a \ge 0$ for every $a > 0$, if and only if the conclusion of (i) holds for every $a > 0$. Under GRH, $V_a(z)\,dz \to \sum_k \delta_{\gamma_k^2}$ vaguely and $H_a \to Z_\pi$ in law as $a \downarrow 0$.

PROOF. (i) For $c \in \mathbb{C}$ with $\operatorname{Re} c > 0$ and $s > 0$, closing the contour in the lower half plane, where the only pole is $r = -i\sqrt{s}$,

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{dr}{(c+ir)(s+r^2)} = \frac{1}{2\sqrt{s}\,(c+\sqrt{s})}.$$

Taking real parts, and adding the same identity for $\bar c$, the integrand $\operatorname{Re}\frac{1}{c+ir} + \operatorname{Re}\frac{1}{\bar c+ir}$ is even in r and positive, and the substitution $z = r^2$ gives

$$\frac{d}{ds}\log\Big|1+\frac{\sqrt{s}}{c}\Big|^2 = \int_0^\infty \frac{v_c(z)\,dz}{s+z},$$

with $v_c(z) = [\mathrm{Re}(c + i\sqrt{z})^{-1} + \mathrm{Re}(\bar{c} + i\sqrt{z})^{-1}]/(2\pi\sqrt{z}) > 0$; for real $c > 0$ use c alone and halve. Integrating from 0 (Tonelli) gives $\log|1 + \sqrt{s}/c|^2 = \int_0^\infty \log(1 + s/z) v_c(z)\, dz$. Now $\Xi_\pi(w) = \Xi_\pi(0) \prod_k (1 - w^2/z_k^2)$, so

$$\frac{\Xi_\pi(a+w)}{\Xi_\pi(a)} = \prod_k \Big(1 + \frac{w}{a - z_k}\Big)\Big(1 + \frac{w}{a + z_k}\Big),$$

the product over pairs converging absolutely because each pair factor is $1 + O((|w| + |w|^2)|z_k|^{-2})$. The numbers $c = a \mp z_k$ have positive real part by hypothesis and form a set closed under conjugation. Summing the single factor identities over this set (Tonelli), and halving because c and $\bar{c}$ both occur, gives $\log(\Xi_\pi(a + \sqrt{s})/\Xi_\pi(a)) = \int \log(1 + s/z) V_a(z)\, dz$, because $\sum_c \mathrm{Re}(c + i\sqrt{z})^{-1} = \mathrm{Re} \sum_k [(a + i\sqrt{z} - z_k)^{-1} + (a + i\sqrt{z} + z_k)^{-1}] = \mathrm{Re}(\Xi_\pi'/\Xi_\pi)(a + i\sqrt{z})$. The integrability conditions follow as in Step 3 of Lemma 16.7.

(ii) By Lemma 13.3 the GGC transform φ is holomorphic and zero free on $\mathbb{C} \setminus (-\infty, 0]$, and $s \mapsto \Xi_\pi(a + \sqrt{s})$ is holomorphic there; the identity $\varphi(s)\Xi_\pi(a + \sqrt{s}) = \Xi_\pi(a)$ on $(0, \infty)$ extends, so $\Xi_\pi(a + w) \neq 0$ for $\mathrm{Re}\, w > 0$. By evenness there is no zero with $\mathrm{Re}\, z < -a$.

(iii) By Lemma 14.6(iii) all zeros satisfy $|\mathrm{Re}\, z_k| < \frac{1}{2}$.

(iv) Under GRH, $\mathrm{Re}\, z_k = 0 < a$ for all k, so (i) applies for every $a > 0$. Conversely suppose $V_a \geq 0$ for all $a > 0$, and let z_0 be a zero of multiplicity m with $\beta = \mathrm{Re}\, z_0 > 0$. For small $\eta \in (0, \beta)$, $(\Xi_\pi'/\Xi_\pi)(z_0 - \eta) = -m/\eta + O(1)$, whose real part is negative; but $z_0 - \eta = a + iy$ with $a = \beta - \eta > 0$, and $V_a(y^2) \geq 0$ says $\mathrm{Re}(\Xi_\pi'/\Xi_\pi)(a + iy) \geq 0$ (for $y \neq 0$; if $y = 0$, then z_0 is real and the same argument applies to $(\Xi_\pi'/\Xi_\pi)(a)$, which is the limit of $\mathrm{Re}(\Xi_\pi'/\Xi_\pi)(a + iy)$ as $y \to 0$). This contradiction and evenness give GRH; the second equivalence follows from (i) and (ii). For the limit, under GRH $\mathrm{Re}(\Xi_\pi'/\Xi_\pi)(a + ir) = \sum_{\gamma_k > 0} [P_a(r - \gamma_k) + P_a(r + \gamma_k)]$ with the Poisson kernel $P_a(u) = a/(a^2 + u^2)$, and $\frac{1}{2\pi r} P_a(r - \gamma)\, 2r\, dr \to \delta_\gamma$ in the variable r, that is δ_{γ^2} in z; the terms $P_a(r + \gamma_k)$ tend to zero, uniformly summable. Convergence of the transforms, $\Xi_\pi(a)/\Xi_\pi(a + \sqrt{s}) \to \Xi_\pi(0)/\Xi_\pi(\sqrt{s})$, gives convergence in law. □

For ξ, parts (i) to (iii) and the limit in (iv) are Theorem 18 and Remark 21 of [442], and [441] relates the family $\xi(\alpha)/\xi(\alpha + \sqrt{s})$ to the criterion of Hinkkanen [261] and Lagarias [339]: the Riemann hypothesis holds if and only if $\mathrm{Re}\, \xi'/\xi(s) > 0$ for $\mathrm{Re}\, s > \frac{1}{2}$. Part (iv) is a Thorin form of that criterion. Parts (i) and (ii) are the parabolic picture of Section 14.6 seen from a shifted basepoint: the Thorin property at a is sandwiched between the conditions Θ_σ with $\sigma < a$ and $\sigma = a$. A zero $z_0 = \beta + i\gamma$ of Ξ_π with $\beta > a$ would contribute *locally* a negative dip to V_a near $z = \gamma^2$, of depth about $(\beta - a)^{-1}/(2\pi\gamma)$. Chapter 17 splits these densities for ξ into the contributions of single pairs of zeros, with a Brownian subordination (Corollary 17.15), Chapter 18 studies V_a for $L(s, \chi_{-4})$, and Chapter 19 computes

the Thorin measure itself, as a signed density smoothed at width ε, by Stieltjes inversion. For self dual π, Chapter 23 writes V_a, halved by passing to Ξ_π^{sd}, in the variable $r = \sqrt{z}$, as a superposition over the zeros of folded Cauchy laws of scales $a - \beta$, so that moving the basepoint adds Cauchy noise (Theorem 23.9).

16.4.1 At the centre: from the cosine measure to the zeros

Formally, for real $\gamma > 0$ and $\operatorname{Re} z > 0$, Frullani's integral with an Abel damping gives

$$\begin{aligned}\log\Big(1 + \frac{z^2}{\gamma^2}\Big) &= \lim_{\epsilon\downarrow 0} 2\int_0^\infty \big(e^{-\epsilon x} - e^{-(z+\epsilon)x}\big)\cos(\gamma x)\frac{dx}{x}\\ &= \int_0^\infty (1 - e^{-zx})\, 2\cos(\gamma x)\frac{dx}{x},\end{aligned}$$

so that under GRH, $\log(\Xi_\pi(z)/\Xi_\pi(0)) = -\int_0^\infty (e^{-zx} - 1 + zx)\mu(dx)/x$ formally, with the *cosine Lévy measure* $\mu(dx) = 2\sum_k \cos(\gamma_k x)\, dx$ (the linear term vanishes because $\Xi_\pi'(0) = 0$). Insert $-\mu$ for ν in the first form of (16.4) at $\alpha = 0$. As distributions in $r = \sqrt{z} > 0$, we have $\int_0^\infty \cos(\gamma x)dx = 0$ and $\int_0^\infty \cos(rx)\cos(\gamma x)dx = \frac{\pi}{2}\delta(r - \gamma)$, so

$$\begin{aligned}U(z) &= -\frac{1}{2\pi r}\sum_k 2\int_0^\infty (1 - \cos rx)\cos(\gamma_k x)\, dx\\ &= \frac{1}{2r}\sum_k \delta(r - \gamma_k) = \sum_k \delta(z - \gamma_k^2),\end{aligned}$$

using $\delta(\sqrt{z} - \gamma) = 2\gamma\,\delta(z - \gamma^2)$. *The sine squared transport carries the cosine measure of the zeros exactly to the Thorin measure* $\sum_k \delta_{\gamma_k^2}$. The damping $e^{-\epsilon x}$ makes this rigorous: it is the shift $a = \epsilon$, and the computation becomes Theorem 16.8.

PROPOSITION 16.9 (this book). *Assume GRH for π and let $\epsilon > 0$. With $P_\epsilon(u) = \epsilon/(\epsilon^2 + u^2)$ and $b_\epsilon = (\Xi_\pi'/\Xi_\pi)(\epsilon) = \sum_{\gamma_k > 0} 2P_\epsilon(\gamma_k)$,*

$$\frac{\Xi_\pi(\epsilon)}{\Xi_\pi(\epsilon + \sqrt{s})} = e^{-b_\epsilon\sqrt{s}}\exp\Big(-\int_0^\infty \log\Big(1 + \frac{s}{z}\Big)U_\epsilon(z)\, dz\Big),$$

$$U_\epsilon(z) = \frac{1}{2\pi\sqrt{z}}\sum_{\gamma_k > 0}\big[P_\epsilon(\sqrt{z} - \gamma_k) + P_\epsilon(\sqrt{z} + \gamma_k) - 2P_\epsilon(\gamma_k)\big],$$

and $U_\epsilon(z) = (2\pi\sqrt{z})^{-1}\int_0^\infty 2\sin^2(x\sqrt{z}/2)e^{-\epsilon x}(-\mu)(dx)$ is the sine squared transport of the damped cosine measure.

PROOF. $U_\epsilon = V_\epsilon - b_\epsilon/(2\pi\sqrt{z})$ in the notation of Theorem 16.8, and the identity is (i) of that theorem combined with Step 5 of Lemma 16.7. The transport formula is the elementary integral $\int_0^\infty (1 - \cos rx)\cos(\gamma x)e^{-\epsilon x}dx = P_\epsilon(\gamma) - \frac{1}{2}[P_\epsilon(r - \gamma) + P_\epsilon(r + \gamma)]$, summed over k. □

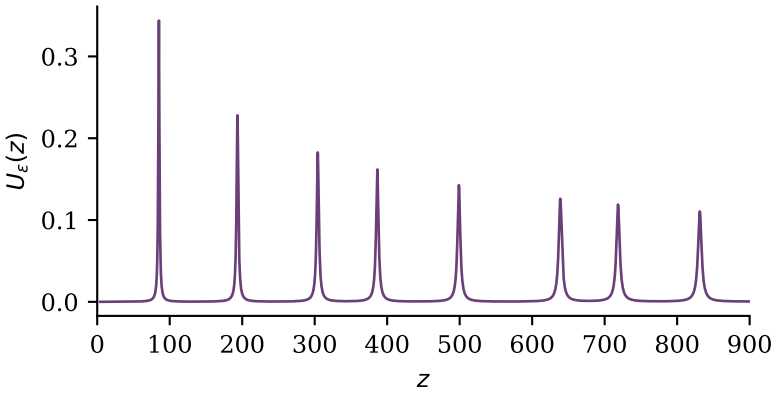


Figure 16.2: The sine squared transport $U_\epsilon(z)$ of the damped cosine Lévy measure $2\sum_k \cos(\gamma_k x)e^{-0.05x}$ over the first 60 zeros of Δ: a smoothed Thorin density with peaks at γ_k^2, mass 0.983 within 30 of $\gamma_1^2 = 85.052$.

For Δ, with $\epsilon = 0.05$ and the first 60 zeros, U_ϵ is positive on $4 \le z \le 900$ (minimum $2.0 \cdot 10^{-5}$) with peaks at the γ_k^2 and mass 0.983 within 30 of $\gamma_1^2 = 85.052$, unit mass up to the smoothing loss (Figure 16.2). Since U_ϵ contains the negative term $-b_\epsilon/(2\pi\sqrt{z})$, its positivity is a property of the zeros and not of the construction.

16.4.2 The explicit formula: from the primes to the Thorin measure

The cosine measure is the Fourier transform of the zero counting measure, and Weil's explicit formula (Theorem 15.7 for Δ; [569, 276] in general) computes that transform. Let π be self dual with $\Lambda(\frac{1}{2}, \pi) \neq 0$, and write $-L'/L(s,\pi) = \sum_n \Lambda_\pi(n)n^{-s}$ in the unitary normalisation. Taking for h the cosine kernel, the formula reads, as an identity of distributions on $(0,\infty)$,

$$\mu(dx) = 2\sum_{\gamma_k>0} \cos(\gamma_k x)\,dx = -\sum_{n\ge 2} \frac{\Lambda_\pi(n)}{\sqrt{n}}\,\delta_{\log n}(dx) + A_\pi(x)\,dx, \qquad (16.6)$$

where A_π is the archimedean density, the Fourier transform of $\frac{1}{2\pi}2\,\mathrm{Re}[\frac{\gamma'}{\gamma}(\frac{1}{2} + it, \pi)]$; for ζ the right side also contains the polar term $2\cosh(x/2)\,dx$. Combining (16.6) formally with the transport: *the Thorin density of $1/F_\pi$ is the sine squared transform of the explicit formula distribution, and GRH is its positivity.* Since the transport acts on $\nu = -\mu$, the atom at $\log n$ contributes $+\Lambda_\pi(n)n^{-1/2}(2\pi\sqrt{z})^{-1}2\sin^2(\frac{1}{2}\sqrt{z}\log n)$, of the sign of $\Lambda_\pi(n)$, and for Δ the atoms of (16.6) at $\log p^m$, $m \ge 1$, are, up to sign and the factor $p^{-m/2}$, the corresponding atoms of the Fourier transform of the local Thorin comb of p

x	quantity	from the zeros	from the primes
0.02	$\mathcal{W}$	0.2061515276408161	0.20615152764081612
	$-\mathcal{W}'$	20.450633094275233	20.450633094275236
	$\mathcal{W}''$	2391.2851645202795	2391.2851645202795
	$-\mathcal{W}'''$	359738.32402760646	359738.3240276065
0.05	$\mathcal{W}$	0.014290332953786858	0.01429033295378686
	$-\mathcal{W}'''$	9216.990197017574	9216.99019701757
0.1	$\mathcal{W}$	$2.0241135580126968 \cdot 10^{-4}$	$2.0241135580125608 \cdot 10^{-4}$
	$-\mathcal{W}'''$	124.56174145637718	124.56174145637709

Table 16.1: The heat trace $\mathcal{W}(x) = \sum_{\gamma>0} e^{-x\gamma^2}$ of Δ and its derivatives, from the zeros and from the primes through Weil's explicit formula.

(Proposition 15.8); only the sum over all n together with A_π is concentrated on the γ_k^2.

The rigorous form of this statement uses test functions. The heat trace

$$\mathcal{W}_\pi(x) = k_\pi(x) = \sum_k e^{-x\gamma_k^2},$$

summed over the pairs $\pm z_k = \pm i\gamma_k$ of (14.3), formed from Ξ_π^{sd} when π is self dual, so that each zero of $\Lambda(s,\pi)$ counts once, as in Table 16.1, is the Lévy function of Theorem 14.8(d), so *GRH holds if and only if $\mathcal{W}_\pi$ is completely monotone*. For ξ the first author states in [442] (Theorems 22 and 26) that the heat trace is positive and strictly decreasing unconditionally, so that the law with Laplace transform $\xi(\frac{1}{2})/\xi(\frac{1}{2}+\sqrt{s})$ is self decomposable, and that its complete monotonicity is equivalent to the Riemann hypothesis; the second statement is the case $\pi = \zeta$ of Theorem 14.8, and we do not use the first. With $h(t) = t^{2n}e^{-xt^2}$ in Weil's formula, $(-1)^n\mathcal{W}^{(n)}(x) = \sum \gamma_k^{2n} e^{-x\gamma_k^2}$ is a sum over primes plus an archimedean integral. Since the transform $H(y)$ of h in Theorem 15.7 then decays like $e^{-y^2/(4x)}$, only primes with $\log p \le \sqrt{240x}+1$ matter at double precision: $p \le 24$ at $x = 0.02$ and $p \le 365$ at $x = 0.1$. For Δ we obtained, from the zeros and independently from the primes, the values in Table 16.1: agreement to thirteen to sixteen significant digits, with the complete monotonicity signs through $n = 3$. Small x needs only a few primes and is dominated by the archimedean term; large x and high derivative orders need many primes, and that is where individual ordinates are resolved.

Figure 16.3 plots the same quantities as functions of x.

The same mechanism recovers the zeros themselves: in Figure 16.4 the smoothed zero count of Δ computed from the primes agrees with that computed from the zeros to $4.0 \cdot 10^{-14}$, against a peak value of 3.84.

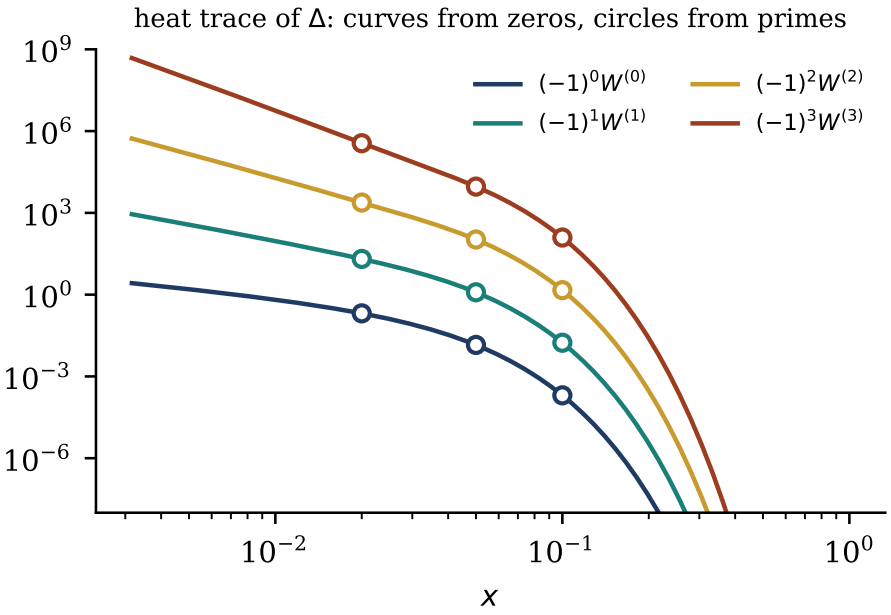


Figure 16.3: Heat trace $\mathcal{W}$ of Δ and its first three derivatives with alternating signs. Curves: sums over the zeros up to height 300. Circles: Weil's explicit formula, using only the primes and the archimedean term. Complete monotonicity is the statement that every such curve is positive for every n.

16.4.3 Away from the centre

For $\alpha > 1$ the Euler product gives the Lévy measure of $\log L(\alpha + s, \pi)$ explicitly:

$$\log \frac{L(\alpha+s,\pi)}{L(\alpha,\pi)} - s\frac{L'}{L}(\alpha,\pi) = \sum_{n\geq 2} \frac{\Lambda_\pi(n)}{n^\alpha \log n}\big(e^{-s\log n} - 1 + s\log n\big),$$

so $\nu_{\mathrm{ar}} = \sum_n \Lambda_\pi(n) n^{-\alpha}\delta_{\log n}$, whose transport is

$$U_{\mathrm{ar}}(z) = \frac{1}{2\pi\sqrt{z}} \sum_{n\geq 2} \frac{\Lambda_\pi(n)}{n^\alpha}\, 2\sin^2\Big(\frac{\sqrt{z}\log n}{2}\Big).$$

Chapter 23 inverts this transport, signed or not, by Gaussian means and recovers the $\Lambda_\pi(n)$ (Corollary 23.6). For ζ, for Dedekind zeta functions and for the Rankin Selberg function $L(s, \pi \times \tilde{\pi})$, either partial, as L^S with S a finite set of primes containing the ramified ones, or complete (Theorem 11.4 and the paragraph after it), the coefficients $\Lambda_\pi(n)$ are nonnegative, so $U_{\mathrm{ar}} \geq 0$. The completed function adds two archimedean pieces of opposite sign. Binet's formula $(\log\Gamma)''(w) = \int_0^\infty te^{-wt}(1-e^{-t})^{-1}dt$ shows that $\log\Gamma((\alpha+s)/2)$ contributes the positive density

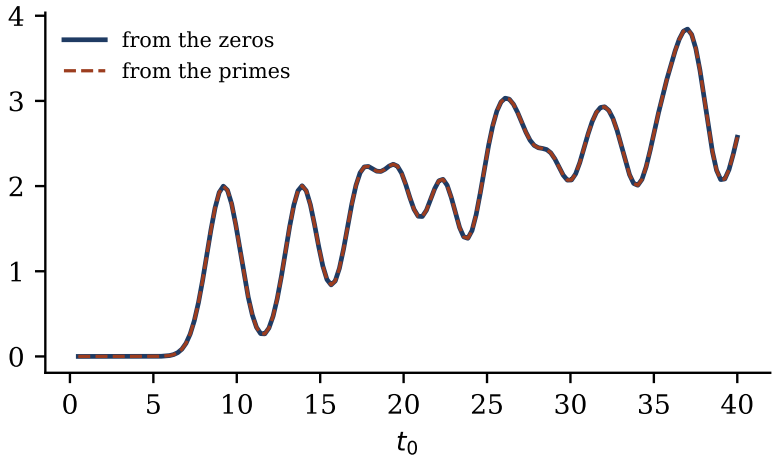


Figure 16.4: Zeros from primes for Δ: the smoothed zero count $\sum_\gamma [e^{-(\gamma-t_0)^2/2} + e^{-(\gamma+t_0)^2/2}]$ from the zeros and from the primes $p \le 1.6 \cdot 10^5$ with the archimedean term. Maximal difference $4.0 \cdot 10^{-14}$ for $t_0 \in [0.5, 40]$.

$e^{-\alpha x}(1-e^{-2x})^{-1}$ to ν, while Frullani's integral shows that the polynomial factors $s(s-1)$ of ξ contribute $-(e^{-\alpha x} + e^{-(\alpha-1)x})$, a negative density. Hence ν is a signed measure for ξ and Lemma 16.7 does not apply as stated; what must be checked is the positivity of the *total* transported density. The first author discusses this decomposition in [442] (Remark 19) and obtains the Thorin representation of $\xi(\alpha)/\xi(\alpha+\sqrt{s})$ for $\alpha \ge 1$ from the Hadamard product instead, as Theorem 16.8(iii) does for every Ξ_π. For signed coefficients, as for $L(s,\chi)$ with χ real taking the value -1, or $L(s,\Delta)$, the arithmetic part ν_{ar} is itself signed, and for complex χ it is complex; one must then work with the completion Ξ_χ of Chapter 14. Inside the critical strip the question is of Riemann hypothesis type: by the argument of Theorem 16.8(iv), nonnegativity of $V_{a'}$ for every $a' \ge a$ excludes zeros of $\Lambda(s,\pi)$ with real part above $\frac{1}{2}+a$.

We computed the density U of (16.4), in its second form, at the 15000 points $\sqrt{z} = 0.01, 0.02, \dots, 150$, for ξ at $\alpha = 0.55, 0.75, 0.99, 1.25, 2$ and for Δ at the base-points $6+\alpha-\frac{1}{2}$ of $\Lambda(s,\Delta)$, the same offsets $\alpha-\frac{1}{2}$ from the centre; it is positive at every point in every case, with its minimum at $\sqrt{z}=0.01$, where U vanishes like a multiple of $\sqrt{z}$ (the value there is $1.8 \cdot 10^{-8}$ for ξ at $\alpha = 0.55$ and $9.5 \cdot 10^{-8}$ for Δ at the same α). These are numerical observations at every α, and the grid sees only zeros of height below 150. For $\alpha \ge 1$, Theorem 16.8(iii) proves only $V_{\alpha-1/2} \ge 0$; since $U = V_{\alpha-1/2} - b/(2\pi\sqrt{z})$ with $b > 0$, positivity of U is a stronger statement: for ξ it says that $\mathrm{Re}\,\xi'/\xi$ is smallest on the real axis along the line $\mathrm{Re}\,s = \alpha$. Lagarias [339] proved this under the Riemann hypothesis for every

$\alpha > \frac{1}{2}$, and unconditionally for $\alpha \geq 10$; Garunkštis lowered the unconditional range to $\alpha > 4.5$ [207] and then to $\alpha \geq 1$ [208]. Unconditionally it is open only for $\frac{1}{2} < \alpha < 1$, where Chapter 18 proves it up to the height to which the Riemann hypothesis is verified, with the analogue for $L(s, \chi_{-4})$ (Theorem 18.21). Chapter 20 builds the corresponding family for the Rankin Selberg square of an Artin representation from its Frobenius traces, keeping the poles of the completed L function, so that no polynomial factor contributes a negative density; there the densities are nonnegative for $\alpha > 1$ (Proposition 20.34), and below $\alpha = 1$ the same question is Open Problem 20.2.

16.5 Subordinators of L functions and hitting times

Fix $\sigma > 1$. Since $\log\zeta(\sigma+\lambda) - \log\zeta(\sigma) = -\sum_{n\geq 2} \frac{\Lambda(n)}{n^{\sigma}\log n}(1 - n^{-\lambda})$, the function $\lambda \mapsto \log(\zeta(\sigma)/\zeta(\sigma+\lambda))$ is a Bernstein function with Lévy measure $\Pi_\sigma = \sum_n \frac{\Lambda(n)}{n^{\sigma}\log n}\delta_{\log n}$ of total mass $\log\zeta(\sigma)$. Let X_t be the compound Poisson subordinator with this Lévy measure:

$$\mathbb{E}e^{-\lambda X_t} = \Big(\frac{\zeta(\sigma+\lambda)}{\zeta(\sigma)}\Big)^t, \qquad \lambda \geq 0,\ t \geq 0.$$

Let $d_t(m)$ be the Dirichlet coefficients of ζ^t, defined as $\exp(t\log\zeta)$: d_t is multiplicative with $d_t(p^k) = \binom{t+k-1}{k}$.

Proposition 16.10 (this book). *For $t > 0$, X_t takes values in $\{\log m : m \geq 1\}$ and $\mathbb{P}(X_t = \log m) = d_t(m)m^{-\sigma}/\zeta(\sigma)^t$.*

Proof. The jumps of X lie in $\{\log n\}$, so X_t is the logarithm of a product of integers. For $\operatorname{Re} s > 1$ the Euler product gives

$$\zeta(s)^t = \prod_p (1-p^{-s})^{-t} = \prod_p \sum_k \binom{t+k-1}{k} p^{-ks} = \sum_m d_t(m) m^{-s},$$

with $d_t(m) \geq 0$. Hence $\mathbb{E}e^{-\lambda X_t} = \sum_m d_t(m)m^{-\sigma}\zeta(\sigma)^{-t}e^{-\lambda\log m}$ for $\lambda \geq 0$, and the coefficients sum to 1 at $\lambda = 0$. A law on $[0,\infty)$ is determined by its Laplace transform. □

As $\sigma \downarrow 1$ the ranked normalised sizes of the prime factors of the integer e^{X_t} converge in law to the Poisson Dirichlet law with parameter t (Theorem 24.4), so that the zeta subordinator carries the whole Ewens family.

Let $c > 0$ and $Z_t = t/c - X_t$, a spectrally negative Lévy process of bounded variation with drift $1/c$. Its Laplace exponent $\psi(\lambda) = \log\mathbb{E}e^{\lambda Z_1} = \lambda/c + \log(\zeta(\sigma+\lambda)/\zeta(\sigma))$ is convex on $[0,\infty)$ with $\psi(0) = 0$ and $\psi(\lambda) \to \infty$; let Φ be its right

inverse. Let $\tau_x = \inf\{t : Z_t > x\}$ for $x > 0$. Since Z moves up only continuously, $Z_{\tau_x} = x$ on $\{\tau_x < \infty\}$, so

$$\tau_x = c\big(x + \log M_x\big), \qquad M_x = e^{X_{\tau_x}} \in \{1, 2, 3, \dots\}.$$

We use two classical facts: $\mathbb{E}e^{-q\tau_x} = e^{-x\Phi(q)}$ for $q \geq 0$ [50, 338], and Kendall's identity [305, 50, 338]: for a spectrally negative Lévy process Z that is not the negative of a subordinator, as measures on $(0,\infty)^2$,

$$t\,\mathbb{P}(\tau_x \in dt)\,dx = x\,\mathbb{P}(Z_t \in dx)\,dt. \tag{16.7}$$

In outline, for compound Poisson X, (16.7) is a consequence of Takács' ballot theorem [534]: if S is a nondecreasing step process with cyclically exchangeable increments on $[0,t]$ and $S_0 = 0$, then $\mathbb{P}(S_u < u/c \text{ for all } u \in (0,t] \mid S_t = y) = (1 - cy/t)^+$. Applying it to the time reversal $S_u = X_t - X_{(t-u)-}$, which has the same law as X on $[0,t]$, the event that Z first exceeds level $x = t/c - y$ at time t is the event that the reversed path stays strictly below the line u/c, and its conditional probability $x/(t/c)$ given $X_t = y$ is (16.7) after the change of variables $x = t/c - y$. We refer to [50], Chapter VII, for the general case.

THEOREM 16.11 (First passage law; this book). *Let $\sigma > 1$, $c > 0$, $x > 0$. For every integer $m \geq 1$,*

$$\mathbb{P}(M_x = m) = \frac{x}{x + \log m} \cdot \frac{d_t(m)\, m^{-\sigma}}{\zeta(\sigma)^t}, \qquad t = c(x + \log m),$$

and $\sum_m \mathbb{P}(M_x = m)\, e^{-qc(x+\log m)} = e^{-x\Phi(q)}$. If $1/c \geq -\zeta'/\zeta(\sigma)$, then $\tau_x < \infty$ almost surely.

PROOF. By Proposition 16.10, $\mathbb{P}(Z_t \in dx)\,dt = \sum_m p_t(m)\,\delta_{t/c - \log m}(dx)\,dt$ with $p_t(m) = d_t(m) m^{-\sigma} \zeta(\sigma)^{-t}$. On the line $\{t = c(x + \log m)\}$, parametrised by x, the measure $\delta_{t/c-\log m}(dx)\,dt$ is $c\,dx$. On the other side, $\mathbb{P}(\tau_x \in dt)\,dx = \sum_m \mathbb{P}(M_x = m)\,\delta_{c(x+\log m)}(dt)\,dx$. Distinct m give disjoint lines, so (16.7) gives $t\,\mathbb{P}(M_x = m) = x\,c\,p_t(m)$ with $t = c(x + \log m)$, which is the formula. The Laplace transform identity is $\mathbb{E}e^{-q\tau_x} = e^{-x\Phi(q)}$ written out. Finally $\psi'(0+) = 1/c + \zeta'/\zeta(\sigma)$; if it is nonnegative, $\Phi(0) = 0$ and $\mathbb{P}(\tau_x < \infty) = e^{-x\Phi(0)} = 1$. □

The theorem is a Lagrange inversion for Dirichlet series: summing over m at $q = 0$ gives $\sum_m \frac{x}{x+\log m} d_{c(x+\log m)}(m) m^{-\sigma} \zeta(\sigma)^{-c(x+\log m)} = 1$ under the drift condition. We simulated 200000 paths with $\sigma = 2$, $c = 0.5$, $x = 1$ (so $1/c = 2 > -\zeta'/\zeta(2) = 0.5700$) and compared with the formula (Figure 16.5):

m	1	2	3	4	6	12
simulation	0.78075	0.08160	0.03273	0.01873	0.00970	0.00324
Theorem 16.11	0.77970	0.08202	0.03296	0.01892	0.00968	0.00349

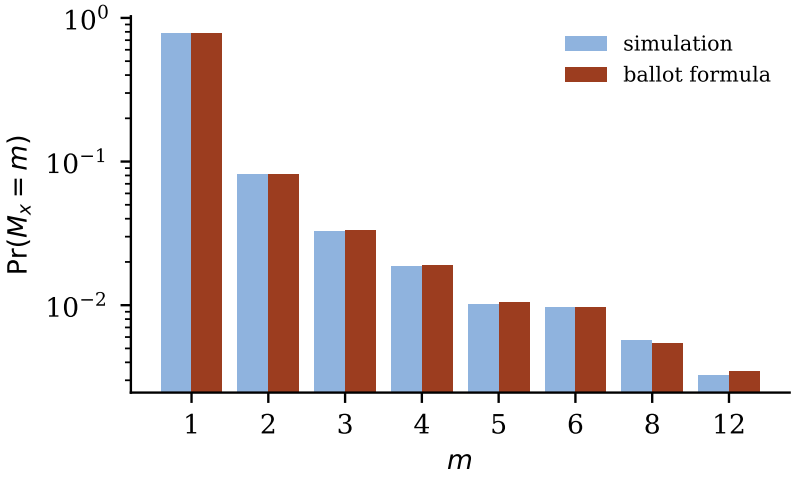


Figure 16.5: First passage law of the zeta subordinator ($\sigma = 2$, $c = 1/2$, $x = 1$): simulation of 200000 paths against the ballot formula of Theorem 16.11.

All differences are within two standard errors.

The construction needs only that $\log L$ has nonnegative Dirichlet coefficients, so it applies verbatim to a Dedekind zeta function ζ_K, with values in $\{\log \mathrm{N}\mathfrak{a}\}$, and to $L(s, \pi \times \tilde{\pi})$, partial or complete (Theorem 11.4 and the paragraph after it), whose subordinator is the probabilistic object attached to π to the right of 1. It fails for $L(s, \chi)$ when χ takes the value -1 and for $L(s, \Delta)$, whose Lévy measures are signed as in (11.1): at $t = 1$ the candidate weights $\chi(m)m^{-\sigma}/L(\sigma, \chi)$ take negative values, and even $L(s, \chi)L(s, \bar{\chi})$, whose coefficient at p^k is $2\,\mathrm{Re}\,\chi(p)^k/k$, does not suffice.

16.6 The Patie class

Patie and Savov [433] introduced *Bernstein gamma functions*: for a Bernstein function ϕ, W_ϕ is the unique solution, in a suitable class, of

$$W_\phi(z+1) = \phi(z)W_\phi(z), \qquad W_\phi(1) = 1,$$

so that $W_\phi(n+1) = \phi(1)\cdots\phi(n)$ and $W_\phi = \Gamma$ for $\phi(z) = z$. They govern exponential functionals of Lévy processes: for a subordinator η with Laplace exponent ϕ, the exponential functional $I_\phi = \int_0^\infty e^{-\eta_t}dt$ has $\mathbb{E}I_\phi^n = n!/W_\phi(n+1)$, and Patie and Savov extend this to the Mellin transform, $\mathbb{E}I_\phi^{z-1} = \Gamma(z)/W_\phi(z)$. Konstantopoulos, Patie and Sarkar use them to build a new class of van Dantzig solutions. Let Ψ be the Laplace exponent of a possibly killed spectrally negative

Lévy process,

$$\Psi(u) = -\kappa + au + \tfrac{1}{2}\sigma^2 u^2 - \int_0^\infty \bigl(1 - e^{-ur} - ur1_{\{r<1\}}\bigr)\Pi(dr),$$

with $\kappa \geq 0$, $\sigma^2 \geq 0$, $\int_0^\infty \min(1, r^2)\,\Pi(dr) < \infty$, excluding the degenerate case in which the process is the negative of a subordinator, and with $\Psi(\tfrac{1}{2}) \geq 0$ (equivalently, the largest zero of Ψ in $[0, \infty)$ lies in $[0, \tfrac{1}{2}]$). Put

$$J_\Psi(t) = \sum_{n\geq 0} \frac{(-1)^n t^{2n}}{W_\Psi(n+1)}, \qquad W_\Psi(n+1) = \Psi(1)\cdots\Psi(n), \qquad \mathcal{D}_P = \{J_\Psi\}.$$

THEOREM 16.12 (Konstantopoulos, Patie and Sarkar [330], Theorem 4.1 and Lemma 5.7). *$\mathcal{D}_P \subset \mathcal{D}$. The partner $1/J_\Psi(i\cdot)$ is the characteristic function of $D_\Psi = \sqrt{2}\,B_{T_\Psi}$, where B is a Brownian motion and T_Ψ an independent positive self decomposable variable. The Bessel functions of* (16.1) *are the case $\Psi(u) = u(u+\nu)$, up to the scaling $t \mapsto 2t$.*

They leave open whether $t \mapsto \xi(\frac{1}{2} + it)/\xi(\frac{1}{2})$ lies in $\mathcal{D}_P$, noting that the question does not seem straightforward. We give a necessary condition that can be tested from Taylor coefficients.

PROPOSITION 16.13 (this book). *Let $\Xi_\pi(z) = \sum_{n\geq 0} a_n z^{2n}$ with all $a_n > 0$, and suppose $\Xi_\pi(it)/\Xi_\pi(0) = J_\Psi(t)$ for some Ψ as above. Then $\Psi(n) = a_{n-1}/a_n$ for $n \geq 1$, and the sequence $\delta_n = \Psi(n+2) - 2\Psi(n+1) + \Psi(n)$ is completely monotone in the sense of Hausdorff:*

$$(-1)^k \Delta^k \delta_n \geq 0 \qquad \textit{for all } k \geq 0,\ n \geq 1,$$

where $\Delta\delta_n = \delta_{n+1} - \delta_n$.

PROOF. Comparing coefficients of t^{2n} in $\sum(-1)^n a_n t^{2n}/a_0$ and $J_\Psi(t)$ gives $W_\Psi(n+1) = a_0/a_n$, so $\Psi(n) = W_\Psi(n+1)/W_\Psi(n) = a_{n-1}/a_n$. For $u > 0$ differentiation under the integral, justified by $\int \min(1, r^2)\Pi(dr) < \infty$ and the bound $r^2 e^{-ur} \leq C_u \min(1, r^2)$, gives $\Psi''(u) = \sigma^2 + \int_0^\infty r^2 e^{-ur}\Pi(dr)$. Hence

$$\delta_n = \int_0^1\int_0^1 \Psi''(n+u+v)\,du\,dv = \sigma^2 + \int_0^\infty e^{-nr}(1-e^{-r})^2\,\Pi(dr),$$

which is $\int_{[0,1]} y^n\,\rho(dy)$, where ρ is the image of $(1-e^{-r})^2\Pi(dr)$ under $y = e^{-r}$ plus the atom $\sigma^2\delta_1$, a finite positive measure because $(1-e^{-r})^2 \leq \min(1, r^2)$. Then $(-1)^k\Delta^k\delta_n = \int y^n(1-y)^k\rho(dy) \geq 0$. □

We ran the test on the self dual completions of Table 16.2, using central Taylor coefficients computed for this book with PARI/GP at 150 to 500 digits. For ξ,

completed L function	d	N	$\Psi(1)$	$\Psi(2)$	$\Psi(3)$	Hausdorff test
ξ	1	1	43.2807	93.0400	148.3165	no violation, $k \le 30$
$L(s,\chi_{-4})$	1	4	12.8181	30.2532	51.2721	no violation, $k \le 30$
Δ	2	1	17.8012	38.0083	60.2084	no violation, $k \le 30$
$L(s,\chi_{-3})$	1	3	17.6363	39.5190	65.1347	fails: $k=5$, $n=1$, $-1.24\cdot10^{-3}$
$L(s,\chi_5)$	1	5	12.7458	28.8566	47.9351	fails: $k=7$, $n=3$, $-2.57\cdot10^{-6}$
11a1	2	11	7.8377	16.7656	26.6952	fails: $k=10$, $n=11$, $-1.04\cdot10^{-9}$
$L(\chi_{-4})L(\chi_5)$	2	20	6.3909	13.7035	21.8976	fails: $k=3$, $n=1$, $-2.92\cdot10^{-3}$
Sym^2 11a1	3	121	3.1133	6.7475	10.8728	fails: $k=3$, $n=1$, $-1.75\cdot10^{-3}$
Sym^2 37a1	3	1369	1.4173	3.2542	5.4717	fails: $k=3$, $n=2$, $-1.02\cdot10^{-4}$
Artin, x^3-x-1	2	23	5.8724	12.7516	20.5318	no violation, $k \le 8$
ξ_K, $K=\mathbb{Q}(x^3-x-1)$	3	23	5.1708	11.0277	17.4859	no violation, $k \le 8$
$L(\chi_5)L(\chi_{13})$	2	65	3.5773	8.4558	14.1903	no violation, $k \le 8$
Sym^3 11a1	4	1331	1.5923	3.6011	5.9504	no violation, $k \le 8$
Sym^4 11a1	5	14641	0.8871	2.1710	3.7238	no violation, $k \le 8$

Table 16.2: The Patie test of Proposition 16.13: implied $\Psi(n) = a_{n-1}/a_n$ and the first violation of $(-1)^k\Delta^k\delta_n \ge 0$, with order k, index n and value. Computed from central Taylor coefficients with 150 to 500 digit arithmetic.

Δ and $L(s,\chi_{-4})$ we used 40 coefficients, which reach order $k = 30$; results are identical at 350 and 500 digits, and there is no violation. The minima over n for ξ are 2.01, $1.79\cdot10^{-2}$, $6.75\cdot10^{-4}$, $4.47\cdot10^{-5}$ for $k = 0,1,2,3$, $1.90\cdot10^{-12}$ at $k = 20$ and $2.33\cdot10^{-10}$ at $k = 30$. This is evidence, not proof, on the question of Konstantopoulos, Patie and Sarkar. In six other cases the test fails decisively, with the violation stable under an increase of precision (for 11a1 identical at 350 and 500 digits, for Sym^2 11a1 at 150 and 300): *these completed L functions are not in $\mathcal{D}_P$*, up to the reliability of uncertified high precision arithmetic. The implied exponents of ξ are $\Psi(1),\dots,\Psi(5) = 43.2807, 93.0400, 148.3165, 208.4940, 273.1404$, and of Δ are 17.8012, 38.0083, 60.2084, 84.1425, 109.6331. No simple pattern by Gamma factor emerges: $L(s,\chi_{-4})$ passes while $L(s,\chi_{-3})$, with the same Gamma factor, fails. Figure 16.6 shows the minima against k.

Figure 16.7 shows the whole table rather than its minima: for ξ every entry of $(-1)^k\Delta^k\delta_n$ is positive, while for 11a1 a negative region opens from order $k = 10$.

By Theorem 16.12, if $\Xi_\pi(it)/\Xi_\pi(0) = J_\Psi$, the partner would be $\sqrt{2}B_T$ with T self decomposable and $\mathbb{E}e^{-sT} = \Xi_\pi(0)/\Xi_\pi(\sqrt{s})$. Under GRH this is consistent with Theorem 16.6, where $V_\pi = \sqrt{2Z_\pi}N$ and Z_π, a GGC, is self decomposable; for ξ, self decomposability of T is the unconditional statement of [442] quoted above. Membership would add a structure on the Taylor coefficients, the factorisation $a_0/a_n = \Psi(1)\cdots\Psi(n)$ through a Lévy exponent. A representing Ψ would have no Brownian component. Indeed $\Psi(u)/u^2 \to \sigma^2/2$ for any such exponent, and if $\Psi(n) \ge \epsilon n^2$ for all large n then $a_n \le Ca_0\epsilon^{-n}(n!)^{-2}$, so $|\Xi_\pi(z)| \le C'e^{2|z|/\sqrt{\epsilon}}$, contradicting the growth $\exp(cx\log x)$ of $\Xi_\pi(x)$ on the real axis forced by the

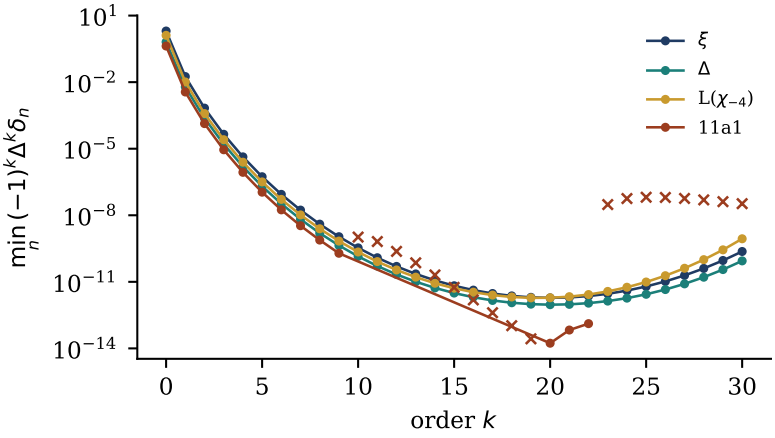


Figure 16.6: $\min_n (-1)^k \Delta^k \delta_n$ against the order k for ξ, Δ, $L(s, \chi_{-4})$ (no violation through $k = 30$) and 11a1 (crosses: absolute values of negative minima, from $k = 10$).

Gamma factors; hence $\liminf \Psi(n)/n^2 = 0$ and $\sigma = 0$. For ξ the ratios $\Psi(n)/n^2$ are 43.3, 23.3, 16.5, 13.0, 10.9 for $n = 1, \ldots, 5$. Interpolating the $\Psi(n)$ by a Bernstein gamma type function is a well posed computational problem (Computational Problem 16.1).

16.7 Notes

The van Dantzig problem was posed in 1958 [555]; Lukacs [375, 376] introduced the class we call $\mathcal{D}_L$ and the operators $L^{(p)}$, and Ostrovskii [429] constructed hyperbolic examples. Hinds [260] studied the Gamma and symmetric Beta examples; for the latter see also [18, 287, 245], and for Bessel hitting times Kent [307]. Thorin [546] introduced generalized gamma convolutions and Bondesson [73] developed them, including the extended class. Biane, Pitman and Yor [82] gave the Brownian reading of ξ, and Biane [58] the matrix Brownian reading of Pólya's Macdonald function theorem [444]. Wald couples with infinitely divisible members are due to Roynette and Yor [467], building on Hartman's theorem on the reciprocal Gamma function [242]; they also rewrote the Riemann hypothesis through Wald couples and Pólya frequency functions (their Théorème V.3.2) and observed that the Pólya variable of ξ is not infinitely divisible. The equivalence between Hadamard factorisation, van Dantzig pairs, Wald couples and Thorin's condition, the $2\sin^2$ lemma, the transport of Euler product Lévy measures, and the shifted Thorin representations of ξ are from the first author [441, 442]. Konstantopoulos, Patie and Sarkar [330] proved the Lukacs class characterisation of the Riemann hypothesis and of

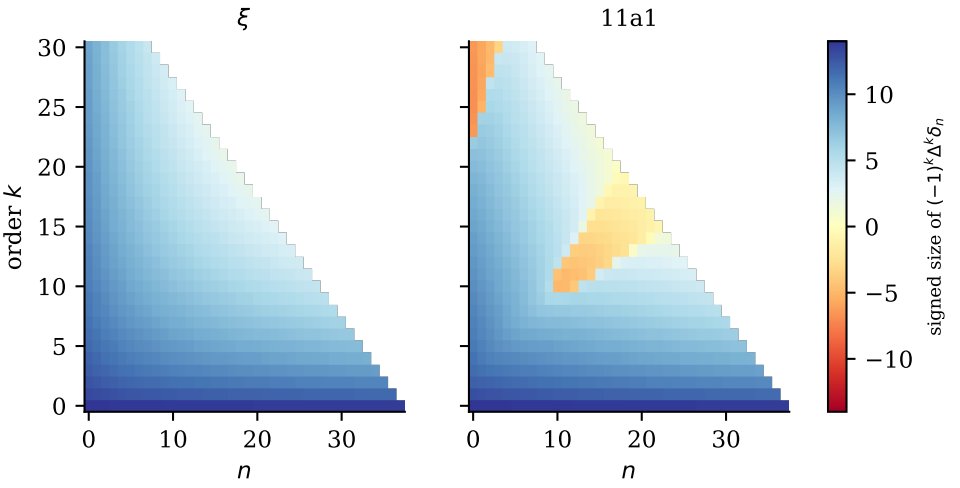


Figure 16.7: Signed size of $(-1)^k\Delta^k\delta_n$ (blue positive, red negative, logarithmic scale) for ξ and for the elliptic curve 11a1, computed from 40 central Taylor coefficients at 350 digits.

the Lee Yang property [359, 416, 418], and introduced $\mathcal{D}_P$ through the Bernstein gamma functions of Patie and Savov [433]. The positivity criterion for ξ is due to Hinkkanen [261] and Lagarias [339]; Weil's explicit formula [569] is used in the form of [276]. Kendall's identity is from [305], the ballot theorem from Takács [534], and fluctuation theory of spectrally negative processes from [50, 338]. Ours are Theorem 16.6 and Theorem 16.8 in the generality of Chapter 14 (for ζ they are due to the authors cited above), Proposition 16.9, the Thorin density u_Γ in (16.2), Proposition 16.10, Theorem 16.11, Proposition 16.13 and all computations; each is an elementary consequence of the cited results, and none proves a Riemann hypothesis.

16.8 Exercises

1. Verify that $\frac{2}{\sqrt{\pi}}e^{x-e^{2x}}$ is a probability density with characteristic function $\frac{1}{\sqrt{\pi}}\Gamma(\frac{1+is}{2})$, and prove $|\Gamma(\frac{1+is}{2})|^2 = \pi/\cosh(\pi s/2)$. Show that $t \mapsto \sqrt{\pi}/\Gamma(\frac{1-t}{2})$ takes the values $-\frac{1}{2}$ at $t = 2$ and 2 at $t = -2$, so that it is neither even nor bounded by 1; deduce that the Hinds law is not in $\mathcal{D}$, while the product is.

2. For $\nu > -\frac{1}{2}$ expand $\cos(sx)$ and use Beta integrals to show that the symmetric Beta law of density $(1-x^2)^{\nu-1/2}/B(\nu+\frac{1}{2},\frac{1}{2})$ has characteristic function $\Gamma(\nu+1)(s/2)^{-\nu}J_\nu(s)$. Identify the cases $\nu = 0, \frac{1}{2}, 1$, and the limit $\nu \downarrow -\frac{1}{2}$.

3. Prove the Laplace pair $\int_0^\infty e^{-yz}\cos(2\sqrt{az})(\pi z)^{-1/2}dz = y^{-1/2}e^{-a/y}$ and deduce (16.5). Then, for $f(w) = w^2 + \gamma^2$ at $\alpha = \epsilon$, where ν is a signed measure, check directly that the two expressions in (16.4) agree, recovering the single zero case of Proposition 16.9.

4. Show that $d_1 = 1$ and that d_2 is the divisor function. For $\sigma > 1$ and $t > 0$ compute $\mathbb{E}X_t$ and $\operatorname{Var} X_t$ in terms of ζ'/ζ and its derivative, and check the first entry $\mathbb{P}(M_1 = 1) = \zeta(2)^{-1/2} \cdot 1 = 0.77970$ of the table for $\sigma = 2$, $c = \frac{1}{2}$.

5. Show that the characteristic function of an infinitely divisible law has no real zero. (If $\varphi = \varphi_n^n$ for every n, then $|\varphi_n|^2 = |\varphi|^{2/n}$ converges to the indicator of $\{\varphi \neq 0\}$, which is therefore continuous at 0 and a characteristic function, hence identically 1.) Deduce that W_ζ and W_Δ are not infinitely divisible (Remark 16.4).

6. For $\cos t$ show that the Patie test of Proposition 16.13 gives $\Psi(n) = 2n(2n-1)$, the exponent $\Psi(u) = 4u^2 - 2u$ of Brownian motion with variance 8 and drift -2, with largest zero $\frac{1}{2}$, and that $\delta_n = 8$. Do the same for $\sin t/t$ and for f_ν of (16.1), and compare with Theorem 16.12.

16.9 Open computational problems

Computational Problem 16.1 (Certified Patie test and Bernstein gamma interpolation). Objective: certify the Hausdorff inequalities of Proposition 16.13 for ξ, Δ and $L(s,\chi_{-4})$ through order $k = 100$, and attempt to reconstruct Ψ. Objects and scale: the first 120 central Taylor coefficients of each completed function. Route: coefficients in rigorous ball arithmetic (Arb, cross checked with PARI/GP `lfunlambda` at 2000 digits), interval finite differences, then a least squares fit of $\Psi(u) = au + \int (e^{-ur} - 1 + ur1_{\{r<1\}})\Pi(dr)$ with Π a positive combination of $r^{-1-\beta}e^{-\lambda r}$, via the Bernstein gamma recursion of [433]. Deliverable: certified lower bounds for $\min_n (-1)^k \Delta^k \delta_n$, the first certified violation if any, and the fitted Lévy measure with residuals. Why it matters: a certified violation answers the question of Konstantopoulos, Patie and Sarkar negatively for ξ; a stable fit suggests a candidate Brownian subordination structure for the partner of ξ.

Computational Problem 16.2 (Transported Thorin densities in the strip). Objective: map the sign of the total transported density V_a of Theorem 16.8, and of its arithmetic and archimedean parts, for basepoints $\alpha = \frac{1}{2} + a \in (\frac{1}{2}, 1]$. Objects and scale: ξ, Δ and $L(s,\chi_{-4})$, $a \in \{0.01, 0.02, \dots, 0.5\}$, $\sqrt{z} \le 10^4$; the arithmetic part from all primes below 10^9 with smoothing. Route: PARI/GP `lfunlambda` derivatives on GPU grids; interval arithmetic near the minima. Deliverable: the region of the (a,z) plane where positivity is certified, the minimum of V_a for each a, the location of the minima relative to the γ_k^2, and, for $\Lambda(s,\chi_{-4})$, the margin

$\min_{t\ge 1}\big(\operatorname{Re}\Lambda'/\Lambda(\alpha+it)-\Lambda'/\Lambda(\alpha)\big)$ in Theorem 18.21. Why it matters: by the argument of Theorem 16.8(iv), certified positivity of $V_{a'}$ on $\sqrt{z}\le T$ for every $a'\ge a$ (interval arithmetic in a' as well as in z) excludes zeros with real part above $\frac{1}{2}+a$ and height at most T, so the map is a Thorin form of zero free region computations.

COMPUTATIONAL PROBLEM 16.3 (First passage tables and large deviations for Euler subordinators). Objective: tabulate the first passage laws of Theorem 16.11 and their large deviations. Objects and scale: ζ, ζ_K for the 100 quadratic and cubic fields of smallest discriminant, and the Rankin Selberg functions $L(s,\pi\times\tilde{\pi})$, partial or complete, for the eleven functions of Table 14.1; $\sigma\in\{1.1,1.5,2\}$, $x\le 50$, $m\le 10^8$. Route: generalized divisor functions by sieving, Φ by Newton iteration on ψ, Monte Carlo with 10^7 paths. Deliverable: tables of $\mathbb{P}(M_x=m)$, verification of $\sum_m \mathbb{P}(M_x=m)=1$ to 10^{-10}, and the rate function $\lim x^{-1}\log\mathbb{P}(\tau_x>\theta x)$ compared with the Legendre transform of ψ. Why it matters: these are the probabilistic objects attached to π to the right of 1, and their behaviour as $\sigma\downarrow 1$ is governed by the pole and the zero free region of $L(s,\pi\times\tilde{\pi})$.

COMPUTATIONAL PROBLEM 16.4 (Van Dantzig and Pólya frequency tests across LMFDB). Objective: test numerically, without zeros, whether the partner $\Xi_\pi(0)/\Xi_\pi(t)$ is the characteristic function of a Pólya frequency density. Objects and scale: all self dual L functions in LMFDB of degree at most 4 and conductor below 10^4, whether or not the Pólya kernel is positive. Route: evaluate $\Xi_\pi(t)$ on $|t|\le 200$ with PARI/GP, invert the Fourier transform by quadrature with error control, and compute Toeplitz determinants $\det[g(x_i-y_j)]$ of orders 2 to 6 on random grids. Deliverable: for each function the minimum density, the smallest Toeplitz minor found, and a table of failures with diagnosis (quadrature error or genuine). Why it matters: by Theorem 16.6 the Pólya frequency property of V_π is equivalent to GRH, and log concavity, observed here for Δ, is its first rung.

CHAPTER 17

SQUAREFREE INTEGERS AND THE THORIN ATOMS OF Ξ

The squarefree integers are a simple arithmetic sequence whose fine structure is controlled by the zeros of ζ, through the doubled argument $2s$. Their generating function is

$$D(s) = \sum_{n\geq 1} \frac{\mu(n)^2}{n^s} = \frac{\zeta(s)}{\zeta(2s)},$$

so every nontrivial zero ρ of ζ becomes a pole of D at $\rho/2$ (unless $\zeta(\rho/2) = 0$, which RH excludes), and the critical line $\operatorname{Re} s = \frac{1}{2}$ is carried to $\operatorname{Re} s = \frac{1}{4}$. This chapter reads that picture through two objects of Chapters 13 and 14: the Pólya density of Ξ and the Thorin measure of $1/\Xi$. Three things come out of it. Under RH the poles of D in the strip $0 < \operatorname{Re} s < \frac{1}{2}$ correspond to the Thorin atoms γ_k^2 through $\gamma_k^2 = -(2s - \frac{1}{2})^2$; the constant κ of the squarefree problem is a combination of two tilted means of the Pólya law; and smoothing the squarefree count by a Gamma(2) law turns its error into a positive average of the Mertens function, with Gamma(2) the largest shape for which that holds. Section 17.8 records how the numbers quoted are computed.

The starting point is a Math StackExchange question [383] on the asymptotics of $\sum_{n\leq x} \mu(n)^2 n^{-\alpha}$, especially at $\alpha = 1$. Section 17.1 answers it.

17.1 THE CLASSICAL PROBLEM

Let $Q(x) = \sum_{n\leq x} \mu(n)^2$ and $\Delta(x) = Q(x) - 6x/\pi^2$. From $\mu(n)^2 = \sum_{d^2\mid n} \mu(d)$,

$$Q(x) = \sum_{d\leq\sqrt{x}} \mu(d) \Big\lfloor \frac{x}{d^2} \Big\rfloor = \frac{6}{\pi^2} x - \sum_{d\leq\sqrt{x}} \mu(d) \Big\{ \frac{x}{d^2} \Big\} + O(\sqrt{x}), \tag{17.1}$$

so $\Delta(x) = O(\sqrt{x})$. The terms $\mu(d)\{x/d^2\}$ will return in Section 17.5.

17.1.1 Perron's formula over a number field

For a number field K with Dedekind zeta function ζ_K, the ideal analogue of D has Euler factor $1 + N(\mathfrak{p})^{-s} = (1 - N(\mathfrak{p})^{-2s})/(1 - N(\mathfrak{p})^{-s})$, hence equals

$\zeta_K(s)/\zeta_K(2s)$, and Perron's formula gives

$$Q_\alpha(x) = \sum_{N(\mathfrak{a})\le x} \frac{\mu^2(\mathfrak{a})}{N(\mathfrak{a})^\alpha} = \frac{1}{2\pi i}\int_{(c)} \frac{\zeta_K(s+\alpha)}{\zeta_K(2s+2\alpha)}\, x^s\, \frac{ds}{s}, \qquad c > \max(0, 1-\alpha), \tag{17.2}$$

for x not the norm of an ideal. The integrand has a simple pole at $s = 1-\alpha$, a simple pole at $s = 0$, and poles at $s = \rho/2 - \alpha$ for the zeros ρ of ζ_K with $\zeta_K(\rho/2) \ne 0$. Let $c_K = \operatorname{Res}_{s=1}\zeta_K(s)$, and define the Euler Kronecker constant γ_K [270] by $\zeta_K(s) = c_K/(s-1) + c_K\gamma_K + O(s-1)$.

PROPOSITION 17.1. *For $\alpha = 1$ the poles at $s = 1-\alpha$ and $s = 0$ merge into a double pole at $s = 0$, and*

$$\sum_{N(\mathfrak{a})\le x} \frac{\mu^2(\mathfrak{a})}{N(\mathfrak{a})} = \frac{c_K}{\zeta_K(2)}\big(\log x + \kappa_K\big) + o(1), \qquad \kappa_K = \gamma_K - \frac{2\zeta_K'(2)}{\zeta_K(2)}. \tag{17.3}$$

Over $\mathbb{Q}$, with A the Glaisher Kinkelin constant,

$$\sum_{n\le x} \frac{\mu(n)^2}{n} = \frac{6}{\pi^2}\big(\log x + \kappa\big) + O\big(x^{-1/2}\big), \tag{17.4}$$

where $\kappa = 24\log A - \gamma_{\mathrm{E}} - 2\log 2\pi = 1.7171376511\ldots$, and for real $\alpha > \frac12$ with $\alpha \ne 1$ the residues at $s = 1-\alpha$ and $s = 0$ give

$$\sum_{n\le x} \frac{\mu(n)^2}{n^\alpha} = \frac{6}{\pi^2}\,\frac{x^{1-\alpha}}{1-\alpha} + \frac{\zeta(\alpha)}{\zeta(2\alpha)} + O\big(x^{\frac12-\alpha}\big).$$

PROOF. Let $Q_1(x)$, as in (17.2), be the left side of (17.3). Summation by parts from the count of squarefree ideals, $\frac{c_K}{\zeta_K(2)}x + O(x^\theta)$ with some $\theta < 1$ (from $\mu^2(\mathfrak{a}) = \sum_{\mathfrak{d}^2\mid\mathfrak{a}}\mu(\mathfrak{d})$ and the ideal count $c_K y + O(y^{1-1/[K:\mathbb{Q}]})$), gives $Q_1(x) = \frac{c_K}{\zeta_K(2)}\log x + C + O(x^{\theta-1})$. As $s \downarrow 0$, $s\int_1^\infty Q_1(t)t^{-s-1}dt = \zeta_K(1+s)/\zeta_K(2+2s)$ is therefore $\frac{c_K}{\zeta_K(2)}\frac{1}{s} + C + o(1)$, and it is also $\frac{c_K}{\zeta_K(2)}(\frac1s + \kappa_K) + O(s)$, because κ_K is the logarithmic derivative of $(s-1)\zeta_K(s)/\zeta_K(2s)$ at $s = 1$. So $C = \frac{c_K}{\zeta_K(2)}\kappa_K$, and the main term is the residue at the double pole. Over $\mathbb{Q}$ use $\gamma_{\mathbb{Q}} = \gamma_{\mathrm{E}}$ and $\zeta'(2)/\zeta(2) = \gamma_{\mathrm{E}} + \log 2\pi - 12\log A$. For the error term, write $B(x) = \sum_{n\le x}(\mu(n)^2 - 6/\pi^2) = \Delta(x) + O(1) = O(\sqrt{x})$ and sum by parts: the tail $\int_x^\infty B(t)t^{-2}dt$ and the boundary term $B(x)/x$ are both $O(x^{-1/2})$. For $\alpha \ne 1$ the same summation by parts gives the error $O(x^{\frac12-\alpha})$ and a constant which, as a function of α, is meromorphic in $\operatorname{Re}\alpha > \frac12$ and equals $\zeta(\alpha)/\zeta(2\alpha)$ for $\alpha > 1$, hence for all $\alpha > \frac12$. □

The answers in [383] give the $\log x$ term and, for $\alpha \ne 1$, the first main term; the second answer reaches (17.4) with error $o(1)$ and the constant left as $\sum_n(\mu(n)^2 - 6/\pi^2)/n$, which equals $-2\zeta'(2)/\zeta(2)^2$. For $\frac12 < \alpha < 1$ the constant $\zeta(\alpha)/\zeta(2\alpha)$ is negative, for example $\zeta(\frac34)/\zeta(\frac32) = -1.3173\ldots$; Corollary 17.3 reads the sign through the Pólya density.

17.2 The Pólya density

Let ξ, Ξ and Pólya's kernel Φ be as in (13.1), so that $\xi(\frac12+z)=\int_{\mathbb R}\Phi(u)e^{zu}\,du$ for all complex z, and let W_ζ, with density $\Phi/\Xi(0)$, be the Pólya variable of ζ (Lemma 14.12 and the paragraph after it). Put $T=W_\zeta/2$, whose density is $p(t)=2\Phi(2t)/\Xi(0)$, and $M_T(\theta)=\mathbb E[e^{\theta T}]$. Then

$$\xi\big(\tfrac12+w\big)=\xi\big(\tfrac12\big)\,M_T(2w),\qquad \xi\big(\tfrac12\big)=0.4971207781\ldots \tag{17.5}$$

Applying the definition of ξ at s and $2s$, and the duplication formula for $\Gamma(s)$, gives the following.

Proposition 17.2. *For all s,*

$$\frac{\zeta(s)}{\zeta(2s)}=\omega(s)\,\frac{M_T(2s-1)}{M_T(4s-1)},\qquad \omega(s)=\frac{2s-1}{s-1}\,2^s\,\pi^{-(s+1)/2}\,\Gamma\Big(\frac{s+1}{2}\Big).$$

The factor $1/(s-1)$ carries the pole at $s=1$, and $2s-1$ gives the zero of D at $s=\frac12$, where $\zeta(2s)$ has its pole. In $\operatorname{Re}s>-1$ every other singularity of D is a zero of $M_T(4s-1)$; the poles of $\Gamma(\frac{s+1}{2})$ give the poles of D at $s=-1,-3,\ldots$.

Corollary 17.3 (sign of the constant). *For real $\alpha>-1$ with $\alpha\neq\frac12,1$, the sign of $\zeta(\alpha)/\zeta(2\alpha)$ is the sign of $(2\alpha-1)/(\alpha-1)$. In particular it is negative exactly on $\frac12<\alpha<1$.*

Proof. $M_T>0$ on the real line, and the gamma factor in ω is positive for $\alpha>-1$. Directly, $\zeta<0$ on $(-2,1)$ and $\zeta>0$ on $(1,\infty)$. □

At $\alpha=\frac34$, for instance,

$$\omega\big(\tfrac34\big)=-1.3461\ldots,\qquad \frac{M_T(\frac12)}{M_T(2)}=\frac{\xi(\frac34)}{\xi(\frac32)}=0.9786\ldots,$$

and their product is $\zeta(\frac34)/\zeta(\frac32)=-1.3173\ldots$.

17.2.1 The constant as tilted Pólya means

Let $\mathbb E_\theta$ denote expectation under the tilted law $e^{\theta t}p(t)/M_T(\theta)$, so that $\mathbb E_\theta[T]=(\log M_T)'(\theta)$. Index the zeros of ξ as $\rho=\frac12+i\gamma_\rho$ and let $\sum_\rho$ run over those with $\operatorname{Re}\gamma_\rho>0$, with multiplicity. By Hadamard's product and (17.5),

$$\log M_T(\theta)=\sum_\rho\log\Big(1+\frac{\theta^2}{4\gamma_\rho^2}\Big),\qquad \mathbb E_\theta[T]=\sum_\rho\frac{2\theta}{\theta^2+4\gamma_\rho^2}. \tag{17.6}$$

THEOREM 17.4. *The constant in* (17.4) *is*

$$\begin{aligned}\kappa &= 2 + \log 2 - \tfrac{1}{2}\log\pi - \tfrac{1}{2}\gamma_{\mathrm{E}} + 2\,\mathbb{E}_1[T] - 4\,\mathbb{E}_3[T] \\ &= 2 + \log 2 - \tfrac{1}{2}\log\pi - \tfrac{1}{2}\gamma_{\mathrm{E}} + \sum_{\rho}\Big(\frac{4}{1+4\gamma_\rho^2} - \frac{24}{9+4\gamma_\rho^2}\Big).\end{aligned}$$

PROOF. κ is the logarithmic derivative of $(s-1)D(s)$ at $s=1$. Take logarithms in Proposition 17.2, differentiate at $s=1$, use $\psi(1) = -\gamma_{\mathrm{E}}$, and apply (17.6). □

Numerically $\mathbb{E}_1[T] = 0.0115478545$ and $\mathbb{E}_3[T] = 0.0345331158$, and the theorem reproduces κ to twenty digits. The first sixty zeros contribute -0.0944 of the zero sum -0.1150, the tail being about $-5\sum_{k>60}\gamma_k^{-2}$. Thus the arithmetic content of the constant is two tilted means of the Pólya law, at the tilts $1 = 2s-1$ and $3 = 4s-1$ evaluated at $s=1$.

Comparing coefficients of θ^{2m} in (17.6) gives the cumulants of $W_\zeta = 2T$, as in Proposition 14.13:

$$\kappa_{2m}(2T) = (-1)^{m+1}\frac{(2m)!}{m}S_m, \qquad S_m = \sum_{\rho}\gamma_\rho^{-2m}, \tag{17.7}$$

and in particular $S_1 = 2\,\mathrm{Var}(T) = 0.0231049931\ldots$ These identities are unconditional. Under RH the even cumulants of T alternate in sign.

17.3 THORIN ATOMS ARE THE POLES IN THE STRIP

Generalized gamma convolutions (GGC) and their Thorin measures are as in Definition 13.2; their Laplace transforms continue analytically and without zeros to $\mathbb{C}\setminus(-\infty,0]$ (Lemma 13.3).

PROPOSITION 17.5 (Theorem 14.8 for ζ; after [441]). *RH holds if and only if the function $u \mapsto \xi(\frac12)/\xi(\frac12+\sqrt{u})$ is a GGC Laplace transform. Then*

$$\frac{\xi(\frac12)}{\xi(\frac12+\sqrt{u})} = \prod_{k\ge1}\Big(1+\frac{u}{\gamma_k^2}\Big)^{-1} = \mathbb{E}\big[e^{-uY}\big], \qquad Y = \sum_{k\ge1}\frac{E_k}{\gamma_k^2},$$

with E_k independent standard exponential variables, drift 0 and Thorin measure $\tau = \sum_k \delta_{\gamma_k^2}$. Moreover $\mathbb{E}[Y] = 2\,\mathrm{Var}(T)$.

PROOF. This is Theorem 14.7 for $E(z) = \xi(\frac12+z)/\xi(\frac12)$, the function Ξ_ζ of Definition 14.5 normalised at 0; the mean is S_1, by (17.7). □

Theorem 14.7 applies equally to Dedekind zeta functions (Definition 14.5). Let $\xi_K(s) = s(s-1)|d_K|^{s/2}\Gamma_{\mathbb{R}}(s)^{r_1}\Gamma_{\mathbb{C}}(s)^{r_2}\zeta_K(s)$, with $\Gamma_{\mathbb{R}}(s) = \pi^{-s/2}\Gamma(\frac{s}{2})$ and $\Gamma_{\mathbb{C}}(s) = 2(2\pi)^{-s}\Gamma(s)$.

PROPOSITION 17.6 (Theorem 14.7 for ζ_K). *If $\zeta_K(\tfrac12)\neq 0$, then $\xi_K(\tfrac12)/\xi_K(\tfrac12+\sqrt{u})$ is a GGC Laplace transform if and only if the Riemann hypothesis holds for ζ_K, and its Thorin measure is then $\sum_k \delta_{\gamma_k^2}$ over the positive ordinates of the zeros of ζ_K.*

A real zero $\tfrac12+x$ with $x>0$, a Siegel zero in particular, gives a pole at $u=x^2>0$, as in the proof of Theorem 13.10. The hypothesis $\zeta_K(\tfrac12)\neq 0$ fixes the normalisation: Armitage [21] exhibited a number field, from a construction of Serre, where it fails, and there one divides by u^m, where $2m$ is the order of the zero at $\tfrac12$, as in Definition 14.5.

17.3.1 The recentred series

Put $s=\tfrac14+\tfrac12\sqrt{u}$, so that $2s=\tfrac12+\sqrt{u}$. By Proposition 17.2 and (17.5),

$$D\big(\tfrac14+\tfrac12\sqrt{u}\big)=\omega\big(\tfrac14+\tfrac12\sqrt{u}\big)\,\xi\big(\tfrac14+\tfrac12\sqrt{u}\big)\,\frac{1}{\xi(\frac12+\sqrt{u})},$$

and under RH the last factor is $\xi(\tfrac12)^{-1}\mathbb{E}[e^{-uY}]$. The atom γ_k^2 of the Thorin measure gives the pole at $u=-\gamma_k^2$, that is at $s=\tfrac14\pm\tfrac{i}{2}\gamma_k=\rho_k/2$. Under RH the poles of the squarefree series in $0<\operatorname{Re}s<\tfrac12$ are therefore exactly the Thorin atoms of $1/\Xi$, and the same holds over K under the Riemann hypothesis for ζ_K by Proposition 17.6, with an extra pole at $s=\tfrac14$ when $\zeta_K(\tfrac12)=0$.

17.3.2 What each atom contributes

At a simple zero, the Perron residue of $D(s)x^s/s$ at $\rho/2$ is $R_\rho\,x^{\rho/2}$ with

$$R_\rho=\frac{\zeta(\rho/2)}{\rho\,\zeta'(\rho)},\qquad R_{\rho_1}=0.02880-0.09618\,i,\quad |R_{\rho_1}|=0.10040\ldots \tag{17.8}$$

Formally $\Delta(x)\approx 1+2\operatorname{Re}\sum_k R_{\rho_k}x^{\rho_k/2}$, where 1 is the residue at $s=0$. Under RH every term has size $x^{1/4}$. Figure 17.1 compares sixty atoms with the exact error.

The atom series is not known to converge, even on RH: the Perron weight $1/s$ decays only like $1/|\gamma|$, and nothing controls the sizes of $\zeta(\rho/2)$ and $1/\zeta'(\rho)$ well enough. On RH, Montgomery and Vaughan [400] proved $\Delta(x)\ll x^{1/3+\varepsilon}$ and Liu [369] $\Delta(x)\ll x^{11/35+\varepsilon}$; the conjectured bound $\Delta(x)\ll x^{1/4+\varepsilon}$ remains open on RH [405, 215] (Open Problem 17.1).

17.4 SMOOTHED, AVERAGED AND EXTREME VALUES

17.4.1 Exponential smoothing

Let $S(x)=\sum_n \mu(n)^2e^{-n/x}=\frac{1}{2\pi i}\int_{(2)}\Gamma(s)D(s)x^s\,ds$.

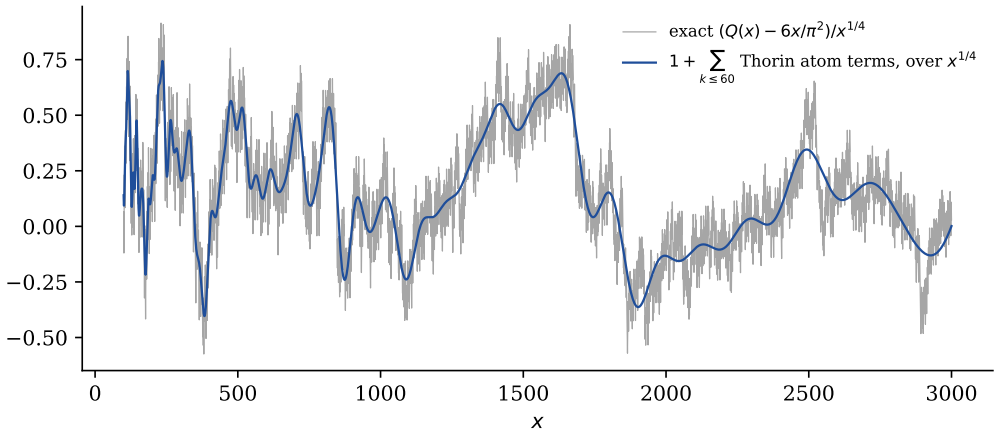


Figure 17.1: The normalised error $\Delta(x)/x^{1/4}$ on $[100, 3000]$ (grey) and the contribution of the first sixty Thorin atoms, $\big(1 + 2\,\mathrm{Re}\sum_{k\le 60} R_{\rho_k} x^{\rho_k/2}\big)/x^{1/4}$ (blue). Correlation about 0.92.

THEOREM 17.7. *RH holds if and only if $S(x) = 6x/\pi^2 + O_\varepsilon(x^{1/4+\varepsilon})$ for every $\varepsilon > 0$. Under RH with simple zeros, and with the zeros summed over $\gamma_k < T_\nu$ for a suitable sequence $T_\nu \to \infty$ [550, Ch. 14],*

$$S(x) = \frac{6}{\pi^2}x + 1 + 2\,\mathrm{Re}\sum_{k} \frac{\Gamma(\rho_k/2)\,\zeta(\rho_k/2)}{2\,\zeta'(\rho_k)}\, x^{\rho_k/2} - \frac{\zeta(-1)}{2\zeta'(-2)}\,\frac{\log x}{x} + O\Big(\frac{1}{x}\Big).$$

PROOF. Assume RH. Then $1/\zeta(\sigma+it) \ll |t|^{\varepsilon}$ for $\sigma \ge \frac12 + \varepsilon$ [550, Thm. 14.2], ζ grows polynomially on vertical lines and Γ decays exponentially, so the line of integration moves to $\mathrm{Re}\,s = \frac14 + \varepsilon$, crossing only the pole at $s = 1$. Moving further left collects the atoms at $\rho/2$, the residue 1 at $s = 0$, and the double pole at $s = -1$, where Γ has a pole and $\zeta(2s)$ a trivial zero.

Conversely, the Mellin transform of $S(1/y) - 6/(\pi^2 y)$ on $(0,1)$ plus that of $S(1/y)$ on $(1,\infty)$ equals $\Gamma(s)D(s) - 6/(\pi^2(s-1))$, and the hypothesis makes it analytic in $\mathrm{Re}\,s > \frac14$. Since Γ has no zeros, D is analytic there except at $s = 1$. Suppose $\zeta(\rho) = 0$ with $\beta = \mathrm{Re}\,\rho > \frac12$. Then $\rho/2$ lies in $\mathrm{Re}\,s > \frac14$, so its pole must cancel: $\zeta(\rho/2) = 0$. By the functional equation, $\rho' = 1 - \overline{\rho}/2$ is a zero with $\mathrm{Re}\,\rho' = 1 - \beta/2 > \frac12$ and $\mathrm{Im}\,\rho' = \mathrm{Im}\,\rho/2$. Iterating produces distinct zeros accumulating at $\frac23$, the fixed point of $\beta \mapsto 1 - \beta/2$; they are distinct because ζ has no real zeros in $(0,1)$. This is impossible for a function analytic and not identically zero at $\frac23$. This descent is due to Keith Conrad; Mossinghoff, Oliveira e Silva and Trudgian [405] give it for the sharp count. $\square$

Table 17.1 checks the expansion at four values of x.

x	$S(x) - 6x/\pi^2 - 1$	forty atoms	$-\frac{\zeta(-1)}{2\zeta'(-2)}\frac{\log x}{x}$
10^3	-9.711×10^{-3}	8.2×10^{-5}	-9.45×10^{-3}
10^4	-1.577×10^{-3}	-2.83×10^{-4}	-1.26×10^{-3}
$3 \cdot 10^4$	-7.41×10^{-4}	-2.59×10^{-4}	-4.70×10^{-4}
10^5	4.23×10^{-4}	5.83×10^{-4}	-1.58×10^{-4}

Table 17.1: Theorem 17.7. The left column equals the sum of the other two to within 4×10^{-5} for $x \geq 10^4$; at $x = 10^3$ the difference, -3.4×10^{-4}, is the $O(1/x)$ term $-0.339/x$ of the theorem. The atoms are damped by $\Gamma(\rho/2)$, of size $e^{-\pi\gamma/4}$.

17.4.2 Mean square and extreme values

THEOREM 17.8. *RH holds if and only if* $\int_X^{2X} \Delta(x)^2\,dx \ll_\varepsilon X^{3/2+\varepsilon}$ *for every* $\varepsilon > 0$.

PROOF. Let $F(s) = \int_1^\infty \Delta(x)x^{-s-1}dx = D(s)/s - 6/(\pi^2(s-1))$. Under RH, F is analytic in $\operatorname{Re} s > \frac14$, and the Lindelöf bounds $\zeta(\sigma+it) \ll |t|^{1/4+\varepsilon}$ and $1/\zeta(2\sigma+2it) \ll |t|^\varepsilon$ for $\sigma \geq \frac14 + \varepsilon$ give $|F(\sigma+it)| \ll |t|^{-3/4+2\varepsilon}$. The bound is uniform in $\sigma \geq c = \frac14 + \varepsilon$, and F is bounded near $s = 1$, so $\sup_{\sigma>c}\int_{\mathbb{R}} |F(\sigma+it)|^2\,dt < \infty$. Since $F(s) = \int_0^\infty \Delta(e^v)e^{-sv}\,dv$ for $\operatorname{Re} s > \frac12$, the Paley Wiener theorem for the half plane $\operatorname{Re} s > c$ gives $\int_0^\infty \Delta(e^v)^2e^{-2cv}\,dv < \infty$, that is $\int_1^\infty \Delta(x)^2x^{-2c-1}dx < \infty$, and hence $\int_X^{2X}\Delta^2 \ll X^{3/2+2\varepsilon}$. Conversely, Cauchy Schwarz turns the hypothesis into absolute convergence of the integral for F in $\operatorname{Re} s > \frac14$, and the descent in the proof of Theorem 17.7 gives RH. □

COROLLARY 17.9 (part (ii) after Evelyn and Linfoot [179] and Stark [524]). *(i) Under RH,* $|\Delta(x)| \leq X^{1/4+\varepsilon}$ *for all* $x \in [X, 2X]$ *outside a set of measure* $O(X^{1-\varepsilon})$.

(ii) Unconditionally, with $|R_{\rho_1}| = 0.10040\ldots$ *from* (17.8),

$$\limsup_{x\to\infty} \frac{\Delta(x)}{x^{1/4}} \geq |R_{\rho_1}|, \qquad \liminf_{x\to\infty} \frac{\Delta(x)}{x^{1/4}} \leq -|R_{\rho_1}|.$$

PROOF. (i) is Chebyshev's inequality. For (ii) suppose $\Delta(x) \leq c\,x^{1/4}$ for $x \geq x_0$, with $c < |R_{\rho_1}|$. The function $c\,x^{1/4} - \Delta(x)$ is eventually nonnegative and its Mellin transform $c/(s-\frac14) - F(s)$ has no real singularity in $(\frac14, \infty)$, since $\zeta(2s)$ has no real zeros there. By Landau's theorem (Lemma 15.2, [344]) the transform converges in $\operatorname{Re} s > \frac14$, which already forces RH by the descent in the proof of Theorem 17.7, and at $s = \sigma + \frac{i}{2}\gamma_1$

$$\Big|\frac{c}{s-\frac14} - F(s)\Big| \leq \frac{c}{\sigma-\frac14} - F(\sigma) + O(1) = \frac{c}{\sigma-\frac14} + O(1).$$

As $\sigma \downarrow \frac{1}{4}$ the left side is at least $|R_{\rho_1}|/(\sigma - \frac{1}{4}) + O(1)$, since F has a pole at $\rho_1/2$ with residue R_{ρ_1} if ρ_1 is simple and a pole of higher order otherwise. Hence $|R_{\rho_1}| \le c$, a contradiction. The lower bound is symmetric. $\square$

Part (ii) is Landau's oscillation argument applied to the first zero. Evelyn and Linfoot [179] proved $\Delta(x) = \Omega(x^{1/4})$, and Stark [524] gave the first effective constant, $2(1 - \gamma_1/\gamma_2)|R_{\rho_1}| = 0.0657\ldots$; Landau's argument gives the larger constant of (ii), but not effectively. Using many zeros, Mossinghoff, Oliveira e Silva and Trudgian [405] prove $\limsup_{x\to\infty} \Delta(x)/x^{1/4} > 3$ and $\liminf_{x\to\infty} \Delta(x)/x^{1/4} < -3$, and computing $Q(x)$ for all $x \le 10^{18}$ they find $|\Delta(x)| < 1.12543\, x^{1/4}$ there. Sieving to 2×10^7 and sampling at the integers gives root mean squares of $\Delta/X^{1/4}$ on $[X, 2X]$ of 0.280, 0.255, 0.281 and 0.353 for $X = 10^4, \ldots, 10^7$, and maxima of $|\Delta|/X^{1/4}$ between 0.68 and 0.98.

17.4.3 How much smoothing the sharp count needs

Let $G_{n,x}$ have the Gamma(n) law with mean x. It concentrates on a window of width $x/\sqrt{n}$, and $\mathbb{E}\, Q(G_{n,x}) \to Q(x)$ as $n \to \infty$ when x is not a squarefree integer.

PROPOSITION 17.10. *Under RH, uniformly in $n \ge 1$,*

$$\mathbb{E}\, Q(G_{n,x}) = \frac{6}{\pi^2} x + O_\varepsilon\big(x^{1/4+\varepsilon} n^{1/8+\varepsilon}\big).$$

PROOF. With rate $\lambda = n/x$, $\mathbb{E}\, Q(G) = \frac{1}{2\pi i}\int D(s)\frac{\Gamma(s+n)}{s\,\Gamma(n)}(x/n)^s\, ds$. Shift to $\operatorname{Re} s = c = \frac{1}{4} + \varepsilon$. From $|\Gamma(n + c + it)/\Gamma(n + c)|^{-2} = \prod_{k\ge 0}(1 + t^2/(n + c + k)^2)$, on that line

$$\frac{|\Gamma(n + c + it)|}{\Gamma(n)} \ll n^c \exp\big(-c_0 \min(t^2/n, |t|)\big)$$

for an absolute $c_0 > 0$. With $|D(c + it)| \ll |t|^{1/4+\varepsilon}$ the integral is therefore

$$\ll x^c \int (1 + |t|)^{-3/4+\varepsilon} e^{-c_0 \min(t^2/n, |t|)}\, dt \ll x^c n^{1/8+\varepsilon}.$$

$\square$

At $n \asymp x^2$ the window has width $O(1)$ and the bound is the trivial $x^{1/2}$. The loss $n^{1/8}$ is the growth $|t|^{1/4}$ of ζ on $\operatorname{Re} s = \frac{1}{4}$ integrated against $dt/|t|$ up to $|t| \asymp \sqrt{n}$. Removing it needs cancellation in $\int D(\frac{1}{4} + it) x^{it} dt/t$, not a bound on its size.

17.5 Gamma(2) smoothing and the Mertens function

Return to the terms $\mu(d)\{x/d^2\}$ of (17.1). Steutel and Thiemann [527] studied when the integer and fractional parts of a random variable are independent, the exponential law being the basic case. We randomise x by a Gamma(2) law and compute every term in closed form.

17.5.1 Expected fractional parts

Let G have the Gamma(2) density $\lambda^2 t e^{-\lambda t}$ and $\theta = \lambda a$. Since $P(G \ge ma) = (1+m\theta)e^{-m\theta}$,

$$\begin{aligned} \mathbb{E}\lfloor G/a \rfloor &= \frac{1}{e^\theta - 1} + \frac{\theta}{4\sinh^2(\theta/2)}, \\ \mathbb{E}\{G/a\} = g(\theta) &= \frac{2}{\theta} - \frac{1}{e^\theta - 1} - \frac{\theta}{4\sinh^2(\theta/2)}. \end{aligned} \tag{17.9}$$

With $\theta = 2y$ the term $\theta/(4\sinh^2(\theta/2))$ equals $\theta^{-1}(y/\sinh y)^2$, and $(y/\sinh y)^2$ is the Laplace transform at $y^2/2$ of the sum of two independent hitting times of 1 by a three dimensional Bessel process, through which Williams represents ξ (Section 13.2.1; [583, Section 5] and [82, (6) to (8) and Table 2]). Expanding $1/(e^\theta - 1) = \sum_m e^{-m\theta}$ gives the Mellin transform $\int_0^\infty g(\theta)\theta^{s-1}d\theta = -(1+s)\Gamma(s)\zeta(s)$ for $0 < \operatorname{Re} s < 1$. Take $\lambda = 2/X$, so that $\mathbb{E}G = X$, and write G_X. Summing (17.9) over $a = d^2$, now over all d since $0 \le g(\theta) \le \min(1, 2/\theta)$,

$$\mathbb{E}\, Q(G_X) = \sum_{n\ge1} \mu(n)^2 \Big(1 + \frac{2n}{X}\Big) e^{-2n/X} = \frac{6}{\pi^2} X - \sum_{d\ge1} \mu(d)\, g\Big(\frac{2d^2}{X}\Big). \tag{17.10}$$

Table 17.2 checks this identity numerically.

17.5.2 The Mertens representation

Theorem 17.11. *The function*

$$f_V(v) = -2g'(v) = \frac{4}{v^2} - \frac{v\cosh(v/2)}{2\sinh^3(v/2)}, \qquad v > 0,$$

is a probability density with $P(V \ge v) = 2g(v)$ and $\mathbb{E}[V^s] = -2\Gamma(s+2)\zeta(s)$ for $0 < \operatorname{Re} s < 1$. With $M(t) = \sum_{d\le t} \mu(d)$ the Mertens function,

$$\mathbb{E}\, Q(G_X) = \frac{6}{\pi^2} X - \frac{1}{2}\, \mathbb{E}\Big[M\Big(\sqrt{XV/2}\Big)\Big].$$

Moreover f_V is twice the difference between the continuous and the integer rate mixtures of Gamma(2) densities $\gamma_{2,\lambda}(v) = \lambda^2 v e^{-\lambda v}$:

$$f_V(v) = 2\Big(\int_0^\infty \gamma_{2,\lambda}(v)\, d\lambda - \sum_{m\ge1} \gamma_{2,m}(v)\Big).$$

X	E_X	$-\sum_{d\le 2\cdot 10^5}\mu(d)g(2d^2/X)$	$(E_X-1)/X^{1/4}$
10^3	0.9971612	0.9971610	-5.0×10^{-4}
10^4	1.0009452	1.0009431	9.5×10^{-5}
10^5	0.9966202	0.9965983	-1.9×10^{-4}
10^6	1.0061029	1.0058842	1.9×10^{-4}

Table 17.2: Identity (17.10), with $E_X=\mathbb{E}Q(G_X)-6X/\pi^2$ from a sieve to 4×10^7. The discrepancies are the truncation of the d sum. The last column divides the part of Corollary 17.12(iii) beyond the constant 1 by $X^{1/4}$; at $X=10^3$ it is mostly the term $-\pi^2/(3\zeta(3)X)$.

PROOF. With $x=v/2$, $f_V\ge0$ reads $(\sinh x/x)^3\ge\cosh x$, which is Lazarević's inequality [357]. The mass is $2g(0+)=1$. The moments follow by parts from the Mellin transform of g. Since $P(\sqrt{XV/2}\ge d)=2g(2d^2/X)$, the expectation of M is $2\sum_d\mu(d)g(2d^2/X)$, which is (17.10). Finally $\int_0^\infty\gamma_{2,\lambda}(v)d\lambda=2/v^2$ and $\sum_m\gamma_{2,m}(v)=v\cosh(v/2)/(4\sinh^3(v/2))$. □

After Gamma(2) smoothing, the squarefree error is exactly minus half the mean of the Mertens function at a random point, and the randomising law is itself the gap between a continuous and a lattice mixture of Gamma(2) laws. With exponential smoothing the same holds with $f_{V_1}(v)=2/v^2-1/(2\sinh^2(v/2))$, positive since $\sinh x\ge x$, and $S(x)=6x/\pi^2-\frac12\mathbb{E}M(\sqrt{xV_1})$.

COROLLARY 17.12. *(i) Unconditionally* $\mathbb{E}\,Q(G_X)=6X/\pi^2+o(X^{1/2})$. *(ii) RH holds if and only if* $\mathbb{E}\,Q(G_X)=6X/\pi^2+O_\varepsilon(X^{1/4+\varepsilon})$ *for every* $\varepsilon>0$. *(iii) Under RH with simple zeros, and with the zeros summed over* $\gamma_k<T_\nu$ *for a suitable sequence* $T_\nu\to\infty$ *[550, Ch. 14],*

$$\mathbb{E}\,Q(G_X)=\frac{6}{\pi^2}X+1+2\,\mathrm{Re}\sum_k\Big(1+\frac{\rho_k}{2}\Big)\frac{\Gamma(\rho_k/2)\,\zeta(\rho_k/2)}{2\,\zeta'(\rho_k)}\Big(\frac{X}{2}\Big)^{\rho_k/2}-\frac{\pi^2}{3\zeta(3)\,X}+O\Big(\frac{1}{X^2}\Big).$$

PROOF. By Littlewood's criterion [550, Ch. 14], RH is equivalent to $M(t)\ll t^{1/2+\varepsilon}$, which gives (ii) forwards since $\mathbb{E}V^\beta<\infty$ for $\beta<1$; $M(t)=o(t)$ and dominated convergence give (i). The converse of (ii) is the descent argument of Theorem 17.7 with kernel $(1+s)\Gamma(s)2^{-s}$, which has no zeros in $\mathrm{Re}\,s>0$. For (iii), shift the contour in $\frac{1}{2\pi i}\int(1+s)\Gamma(s)D(s)(X/2)^s\,ds$ to the left along the heights T_ν. The pole at $s=-1$ is simple, since $(1+s)\Gamma(s)$ is regular there, and its residue is $-\zeta(-1)/(\zeta'(-2)X)=-\pi^2/(3\zeta(3)X)$. □

17.5.3 Which Gamma shapes work

For Gamma(ν) smoothing the expected fractional part is

$$g_\nu(\theta) = \frac{\nu}{\theta} - \sum_{m\ge 1} \frac{\Gamma(\nu, m\theta)}{\Gamma(\nu)},$$

and the argument above gives

$$\mathbb{E}\, Q(G) = \frac{6}{\pi^2}\mathbb{E}G - \frac{1}{2}\int_0^\infty M\big(\sqrt{v/\lambda}\big) f_{V_\nu}(v)\, dv, \qquad f_{V_\nu} = -2g_\nu',$$

with moments

$$\int_0^\infty v^s f_{V_\nu}(v)\, dv = -\frac{2\,\Gamma(s+\nu)\,\zeta(s)}{\Gamma(\nu)}.$$

The poles of the Mellin transform at $s = -\nu - k$ give

$$g_\nu(\theta) = \frac{1}{2} + \frac{\zeta(-\nu)}{\Gamma(\nu+1)}\,\theta^\nu + O(\theta^{\nu+1}),$$

so $f_{V_\nu}(v) = O(v^{\nu-1})$ at 0 and the moment formula holds on $-\nu < \operatorname{Re} s < 1$ by continuation. Is f_{V_ν} a probability density, so that the error is a positive average of M? The answer is a sharp threshold.

LEMMA 17.13. *$f_{V_\nu}(\theta) = 2\nu\big(1 - R_\nu(\theta)\big)/\theta^2$, where $R_\nu(h) = h\sum_{m\ge 1}\gamma_{\nu+1}(mh)$ is the right Riemann sum with step h of the Gamma($\nu+1$,1) density $\gamma_{\nu+1}$. Thus $f_{V_\nu} \ge 0$ if and only if $R_\nu \le 1$.*

PROOF. Differentiate g_ν term by term, using $y\,\gamma_\nu(y) = \nu\,\gamma_{\nu+1}(y)$. □

For example

$$R_1(h) = \frac{(h/2)^2}{\sinh^2(h/2)}, \qquad R_2(h) = \frac{(h/2)^3\cosh(h/2)}{\sinh^3(h/2)}.$$

THEOREM 17.14. *$f_{V_\nu} \ge 0$ if and only if $0 < \nu \le 2$.*

PROOF. Necessity. For $\nu > 2$ the moment formula holds on $(-\nu, 1)$, where $\Gamma(s+\nu) > 0$. But $\zeta > 0$ on $(-4,-2)$, which meets $(-\nu,1)$, so $\int v^s f_{V_\nu} dv < 0$ for some real s and f_{V_ν} takes negative values.

Sufficiency. The cases $\nu = 1, 2$ are $\sinh x \ge x$ and Theorem 17.11. Let $\nu = k + \eta$ with $k \in \{0,1\}$ and $0 < \eta < 1$, and set $R_0(x) = x/(e^x - 1)$. The identity $y^\eta = \frac{\eta}{\Gamma(1-\eta)}\int_0^\infty (1 - e^{-sy}) s^{-\eta-1} ds$, the Lévy Khintchine form of the Bernstein function y^η, applied to the sum and the integral defining $R_\nu(h) - 1$, gives with $c = 1 + s$

$$\Gamma(\nu+1)\big(R_\nu(h) - 1\big) = \frac{\eta\, k!}{\Gamma(1-\eta)}\int_0^\infty s^{-\eta-1}\Big[\frac{1 - R_k(hc)}{c^{k+1}} - \big(1 - R_k(h)\big)\Big] ds.$$

The bracket is nonpositive for every $c > 1$ as soon as $x \mapsto (1 - R_k(x))/x^{k+1}$ is nonincreasing. For $k = 0$ this function is g_1, decreasing by the case $\nu = 1$. For $k = 1$ it is $\frac{1}{2} f_{V_1}(x) = x^{-2} - \frac{1}{4}\sinh^{-2}(x/2)$. Its derivative, with $u = x/2$, is $\frac{1}{4}(\cosh u/\sinh^3 u - u^{-3})$, nonpositive by Lazarević's inequality. □

The proof is a ladder: positivity on $(k, k+1)$ follows from monotonicity of $(1 - R_k(x))/x^{k+1}$, which is g_1 for $k = 0$ and $x^{1-k} f_{V_k}(x)/(2k)$ for $k \geq 1$. It stops after two rungs because $x^{-1} f_{V_2}(x) \sim x/60$ increases near 0, which is the statement $\zeta(-3) > 0$ with $\zeta(-2) = 0$. The substitution $x = v/2$ of the proof of Theorem 17.11 turns $f_{V_2}(v) = v^2/60 + O(v^4)$ into $(\sinh x/x)^3 - \cosh x = x^4/15 + O(x^6)$, and 3 is the smallest exponent for which Lazarević's inequality holds near 0.

17.6 Off the critical line

For $\sigma \geq \frac{1}{2}$ put

$$\varphi_\sigma(u) = \frac{\xi(\sigma)}{\xi(\sigma + \sqrt{u})}, \qquad u \in \mathbb{C} \setminus (-\infty, 0],$$

with the principal root, so that $\sigma + \sqrt{u}$ ranges over $\operatorname{Re} s > \sigma$. For $\sigma > \frac{1}{2}$ the function $\xi(\sigma + w)$ is not even in w, and the factors of its Hadamard product are no longer of the form $1 + u/r$ with $r > 0$. The Thorin structure nevertheless survives: atoms are smeared into Cauchy profiles.

By Theorem 16.8 with $a = \sigma - \frac{1}{2}$, which for ξ is Theorem 18 with Remark 21 of [442], φ_σ is a GGC Laplace transform if and only if $\sigma \geq \beta^*$, where $\beta^* = \sup\{\operatorname{Re}\rho : \zeta(\rho) = 0,\ 0 < \operatorname{Re}\rho < 1\}$; the case $\sigma = \beta^*$ follows by closure under limits. For $\sigma > \beta^*$ the Thorin measure τ_σ of φ_σ has the density $V_{\sigma-1/2}(r)$ of that theorem, which in the variable $t = \sqrt{r}$ reads

$$\frac{1}{\pi} \operatorname{Re} \frac{\xi'}{\xi}(\sigma + it)\, dt, \qquad \frac{\xi'}{\xi}(s) = \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2}\log\pi + \frac{1}{2}\psi\Big(\frac{s}{2}\Big) + \frac{\zeta'}{\zeta}(s), \quad (17.11)$$

so it can be computed from ζ'/ζ alone. A pair of zeros $\beta \pm it_\rho$ with $\delta = \sigma - \beta > 0$ and $c = \delta^2 + t_\rho^2$ contributes the factor $c/(c + B(u))$, with $B(u) = u + 2\delta\sqrt{u}$, to φ_σ, and the density

$$h_c^{(\delta)}(r) = \frac{\delta}{\pi} \frac{r + c}{\sqrt{r}\,\big((c - r)^2 + 4\delta^2 r\big)}$$

to τ_σ. This is the density of one pair of zeros in the proof of Theorem 16.8, since $2t\, h_c^{(\delta)}(t^2)$ is the sum of the Poisson kernels $\frac{1}{\pi}\delta/(\delta^2 + (t \mp t_\rho)^2)$. Each $h_c^{(\delta)}$ is a probability density, close to a Cauchy density at c of half width $2\delta\sqrt{c}$, and it tends to the unit mass at c, an atom of the Thorin measure of Proposition 17.5, as $\delta \downarrow 0$.

COROLLARY 17.15 (subordination). *For $\sigma \geq \beta^*$, $\varphi_\sigma(u) = \mathbb{E}[e^{-uY_\sigma}]$ with*

$$Y_\sigma = \sum_{\rho=\beta+it_\rho,\ t_\rho>0} \Big(\frac{E_\rho}{c_\rho} + T_\rho\Big), \qquad c_\rho = (\sigma-\beta)^2 + t_\rho^2,$$

where the zeros are counted with multiplicity, the E_ρ are independent standard exponentials and, given E_ρ, T_ρ is the first passage time of an independent standard Brownian motion to the level $\sqrt{2}(\sigma-\beta)E_\rho/c_\rho$.

PROOF. With $\delta = \sigma - \beta$, $c_\rho/(c_\rho + B(u)) = \mathbb{E}\exp(-B(u)E_\rho/c_\rho)$, and $e^{-yB(u)} = e^{-yu}e^{-a\sqrt{2u}}$ with $a = \sqrt{2}\delta y$ is the transform of y plus the passage time to the level a. □

The factor $e^{-a\sqrt{2u}}$ is the transform of the positive $\frac{1}{2}$ stable law, as in the first step of Lemma 16.7 [467]. Moving off the line therefore runs each exponential of Proposition 17.5 through its own Brownian clock. Since passage times have infinite mean, $\mathbb{E}Y_\sigma = \infty$ for $\sigma > \frac{1}{2}$, which matches the $\sqrt{u}$ branch point of φ_σ at 0.

Positivity of $\operatorname{Re}\xi'/\xi$ on the half plane $\operatorname{Re} s > \frac{1}{2}$ is equivalent to RH by the criterion of Hinkkanen [261] and Lagarias [339], and Theorem 16.8(iv) is its Thorin form; more generally, positivity on $\operatorname{Re} s > \sigma$ is equivalent to $\sigma \geq \beta^*$ ([339], Theorem 1.1), that is to the GGC property at σ. At $\sigma = 0.65$, $t = 20$ the closed form (17.11) gives $\operatorname{Re}\xi'/\xi = 0.15849$, a density of 0.05045, and the first 300 zero pairs give 0.15801, the difference being the tail. A negative value of $\operatorname{Re}\xi'/\xi$ on the line $\operatorname{Re} s = \sigma$, computed with a rigorous error bound, proves a zero with real part above σ; a positive value on one line proves nothing.

17.7 NUMBER FIELDS

17.7.1 The constant over K

Theorem 17.4 extends to every number field, which completes (17.3).

THEOREM 17.16. *With the zeros of ζ_K in the critical strip written $\frac{1}{2} + i\gamma_\rho$ and summed over all of them with multiplicity,*

$$\kappa_K = A_K + \sum_\rho \Big(\frac{2}{1+4\gamma_\rho^2} - \frac{12}{9+4\gamma_\rho^2}\Big),$$

$$A_K = 2 + \tfrac{1}{2}\log|d_K| + r_1\big(\log 2 - \tfrac{1}{2}\log\pi - \tfrac{1}{2}\gamma_{\mathrm{E}}\big) + r_2\big(2 - \gamma_{\mathrm{E}} - \log 2\pi\big).$$

PROOF. From the definition of ξ_K,

$$(s-1)\frac{\zeta_K(s)}{\zeta_K(2s)} = 2(2s-1)\,|d_K|^{s/2}\,\frac{\Gamma_{\mathbb{R}}(2s)^{r_1}\Gamma_{\mathbb{C}}(2s)^{r_2}}{\Gamma_{\mathbb{R}}(s)^{r_1}\Gamma_{\mathbb{C}}(s)^{r_2}}\cdot\frac{\xi_K(s)}{\xi_K(2s)}.$$

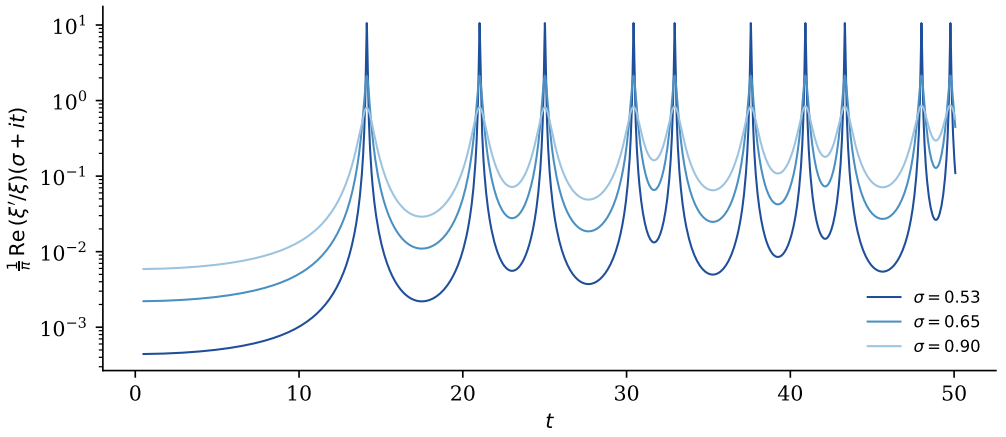


Figure 17.2: The function $\frac{1}{\pi}\,\mathrm{Re}(\xi'/\xi)(\sigma+it)$ of (17.11), computed from ζ'/ζ alone, for $\sigma = 0.53, 0.65, 0.90$ and $0.5 \le t \le 50$, on a logarithmic vertical scale. Under RH it is the density of τ_σ in the variable t, and each zero contributes a Cauchy profile of half width $\sigma - \frac{1}{2}$ centred at its ordinate, which sharpens into an atom as $\sigma \downarrow \frac{1}{2}$. The minima on $[0.5, 50]$, all at $t = 0.5$, are 4.4×10^{-4}, 2.2×10^{-3} and 5.9×10^{-3}.

The logarithmic derivative of the elementary factor at $s = 1$ is A_K, using $\psi(1) = -\gamma_{\mathrm{E}}$, $\psi(\frac{1}{2}) = -\gamma_{\mathrm{E}} - 2\log 2$ and $\psi(2) = 1 - \gamma_{\mathrm{E}}$. For the rest, $\frac{\xi_K'}{\xi_K}(\frac{1}{2}+w) = \sum_\rho \frac{w}{w^2+\gamma_\rho^2}$ over all zeros, taken at $w = \frac{1}{2}$ minus twice its value at $w = \frac{3}{2}$. □

For $K = \mathbb{Q}$ this is Theorem 17.4, whose sum runs over $\mathrm{Re}\,\gamma_\rho > 0$ with doubled weights. For $K = \mathbb{Q}(i)$, where $\zeta_K = \zeta \cdot L(s, \chi_{-4})$, both sides of the theorem give $\kappa_{\mathbb{Q}(i)} = 1.7846166671\ldots$ to twenty digits when the zero sum is evaluated as $\frac{\xi_K'}{\xi_K}(1) - 2\frac{\xi_K'}{\xi_K}(2)$; summed over the zeros below height 1000, with a tail estimate, it agrees to six digits. The zeros of ζ and of $L(s, \chi_{-4})$ simply pool their Thorin atoms.

17.7.2 A Pólya kernel for imaginary quadratic fields

Let K be imaginary quadratic of discriminant d, let $a_n = \sum_{m|n} \chi_d(m)$ count ideals of norm n, and put $\lambda_n = 2\pi n/\sqrt{|d|}$ and $F(k) = k(k-2)e^{-k}$. Write $\Lambda_K(s) = (\sqrt{|d|}/2\pi)^s \Gamma(s)\zeta_K(s)$, so that $\xi_K(s) = 2s(s-1)\Lambda_K(s)$.

THEOREM 17.17. *Let $H_K(x) = \sum_{n \ge 1} a_n F(\lambda_n x)$ and $\Phi_K(v) = e^{v/2} H_K(e^v)$. Then Φ_K is even, decays doubly exponentially, and $\xi_K(\frac{1}{2}+iz) = 2\int_{\mathbb{R}} \Phi_K(v) e^{izv}\,dv$. If $|d| < \pi^2$, that is $d \in \{-3, -4, -7, -8\}$, then $\Phi_K > 0$; consequently $\xi_K(\frac{1}{2}+w)/\xi_K(\frac{1}{2})$ is the moment generating function of a probability density, and $\zeta_K(\frac{1}{2}) < 0$.*

PROOF. With $\psi_K(x) = \sum_n a_n e^{-\lambda_n x}$ one has $\Lambda_K(s) = \int_0^\infty \psi_K(x) x^{s-1} dx$ and $H_K = (x^2\psi_K')'$, so two integrations by parts give $\int_0^\infty H_K(x) x^{s-1} dx = s(s-1)\Lambda_K(s) = \frac{1}{2}\xi_K(s)$. Substitute $x = e^v$; the functional equation $\Lambda_K(s) = \Lambda_K(1-s)$ makes Φ_K even. Every term of H_K is positive for $x > 2/\lambda_1 = \sqrt{|d|}/\pi$, which covers $x \geq 1$ when $|d| < \pi^2$, and evenness covers the rest. Finally $\xi_K(\frac{1}{2}) = -\frac{1}{2}\Lambda_K(\frac{1}{2}) = 2\int \Phi_K > 0$. □

The sign of $\zeta_K(\frac{1}{2}) = \zeta(\frac{1}{2})L(\frac{1}{2}, \chi_d)$ is known far beyond these four fields: $L(s, \chi_d)$ has no positive real zero for $-3 \cdot 10^8 \leq d < 0$ [564]. For $d = -4$ the Fourier identity holds to sixteen digits at $z = 0$ and $z = 2.5$. Beyond $|d| < \pi^2$ positivity is arithmetic. Replacing a_n by its mean $L(1, \chi_d)$ makes $H_K(1)$ small, about $L(1, \chi_d)\lambda_1/6$, since $\int_0^\infty F = 0$; the sign is decided by how the smallest ideal norms deviate from their average.

THEOREM 17.18. *If* 2 *and* 3 *are both inert in* K*, then* Φ_K *takes negative values. This holds for infinitely many fields, among them* $d = -19, -43, -67, -163$*.*

PROOF. Write $x = y\sqrt{|d|}/2\pi$, so $H_K = \sum_n a_n F(ny)$. The ideal counts are multiplicative with $a_{p^e} \leq e + 1$, and when 2, 3 are inert a_{2^e} and a_{3^e} vanish for odd e and equal 1 for even e. Hence $a_n \leq a_n^*$, where $a_n^* = 0$ unless $v_2(n)$ and $v_3(n)$ are even, in which case a_n^* is the number of divisors of $n 2^{-v_2(n)} 3^{-v_3(n)}$; in particular $a_2 = a_3 = 0$. At $y = 1.42$, $F(ny) > 0$ for $n \geq 2$, and

$$H_K \leq F(1.42) + \sum_{n \geq 4} a_n^* F(1.42\, n) = -0.19908 + 0.13918 < 0,$$

the sum computed to $n = 4000$ with negligible tail. Finally 2 and 3 are both inert exactly when $d \equiv 5 \bmod 24$, which holds for infinitely many fundamental discriminants. □

Since ξ_K is a constant multiple of $\xi(s)\Lambda(s, \chi_d)$ (proof of Theorem 18.25), Φ_K is a positive multiple of the convolution of Pólya's kernel with the theta kernel $v \mapsto e^{3v/2}\vartheta_{\chi_d}(e^{2v})$ of Section 21.4. So for $d = -4$ positivity also follows from Proposition 18.2, and when 2 and 3 are inert the theta function of χ_d takes negative values (compare Section 21.4).

All 6079 fundamental discriminants with $|d| \leq 20000$ were scanned on a grid of 200 points (Section 17.8); 2738 are positive at every grid point, and a grid ten times finer changes none. Table 17.3 sorts them by the behaviour of 2 and 3: the more small primes split, the more ideals of small norm, and the likelier positivity. Positivity is not decided at the centre: only 1904 of the 3341 failures have $\Phi_K(0) < 0$. The number of positive fields per block of 2000 discriminants falls from 337 to between 253 and 272 in the last five blocks.

	$\chi_d(3)=1$	$\chi_d(3)=0$	$\chi_d(3)=-1$
$\chi_d(2)=1$	676/767	334/507	412/757
$\chi_d(2)=0$	489/753	178/514	131/757
$\chi_d(2)=-1$	397/761	121/500	0/763

Table 17.3: Fields with $\Phi_K > 0$ at every point of the grid of Section 17.8, among fundamental discriminants $-20000 \le d < 0$. The empty cell is Theorem 17.18.

17.8 Computing the quantities in this chapter

The sieve and the ideal counts are exact integer computations, done with `numpy`. The analytic quantities are floating point, computed with `mpmath` at twenty digits or more unless stated otherwise, and carry no rigorous error bounds.

- Squarefree indicators to N: start from an array of ones and zero out multiples of d^2 for $d \le \sqrt{N}$; cumulative sums give Q exactly, and Δ in floating point. A sieve to 4×10^7 runs in seconds.
- Tilted Pólya means: $\mathbb{E}_\theta[T] = \frac{1}{2}(\xi'/\xi)(\frac{1}{2}+\frac{\theta}{2})$ by numerical differentiation of $\log \xi$; no quadrature of Φ is needed. The Pólya integral itself converges with forty terms of Φ on $[0,4]$.
- Atom coefficients R_ρ: `zetazero(k)` and `diff(zeta,rho)`. Sixty zeros take under a minute.
- Kernel signs Φ_K: Kronecker symbols by quadratic reciprocity and a_n by a divisor sieve to $12\sqrt{|d|}+50$, both exact; H_K in floating point on 200 points of $[1, \max(1, \sqrt{|d|}/\pi)]$. This covers $v \ge 0$, evenness covers $v < 0$, and beyond $\sqrt{|d|}/\pi$ every term is positive.
- The Thorin density of Figure 17.2: the closed form (17.11), on a grid refined near the first ten ordinates.

17.9 The Euler product as a law

For $\sigma > 1$, $P(N = n) = \mu(n)^2 n^{-\sigma}/D(\sigma)$ is a law on squarefree integers with $\log N = \sum_p B_p \log p$ for independent Bernoulli variables of success probability $p^{-\sigma}/(1+p^{-\sigma})$. As for the Euler products of Chapter 11, $\log \mathbb{E} N^{-w} = \log D(\sigma + w) - \log D(\sigma)$ has a Lévy Khintchine form, here with the signed Lévy measure $\sum_p \sum_k (-1)^{k+1} k^{-1} p^{-k\sigma} \delta_{k \log p}$ (compare (11.1)), so by uniqueness the law of $\log N$ is not infinitely divisible; it is quasi infinitely divisible in the sense of [365]. The

negative part sits at even k and contributes

$$-\sum_{p}\sum_{j\geq 1}\frac{p^{-2js}}{2j}=-\frac{1}{2}\log\zeta(2s)$$

to $\log D(s)$, the factor $\zeta(2s)^{-1/2}$: in $D(s)=\zeta(s)/\zeta(2s)$ the even k carry $+1/k$ from $\zeta(s)$ and $-2/k$ from $1/\zeta(2s)$. The factor $1/\zeta(2s)$, which makes the arithmetic side signed, becomes after the change of variable $u=(2s-\frac{1}{2})^2$ the positive Thorin measure of Proposition 17.5 on the analytic side, provided RH holds.

17.9.1 Open problems

OPEN PROBLEM 17.1 (The exponent one quarter). Prove that $\Delta(x)\ll x^{1/4+\varepsilon}$ on RH; the best exponent known on RH is $\frac{11}{35}$ (end of Section 17.3). By Proposition 17.10 it would follow from removing the factor $n^{1/8}$ at $n\asymp x^2$, and by Corollary 17.9 it amounts to emptying the exceptional set.

OPEN PROBLEM 17.2 (How often is Φ_K positive?). Does the set of negative fundamental discriminants d with $\Phi_K>0$ have positive lower density? Is there a finite set of splitting conditions on small primes under which $\Phi_K>0$ for every d that satisfies them, a positive counterpart of Theorem 17.18?

17.10 NOTES

The question and both answers are in [383]. The results on the size of $\Delta(x)$, unconditional and on RH, are credited where they are used, in and after Corollary 17.9 and at the end of Section 17.3. Section 13.2.1 of [401] sets as exercises the exponents $\frac{2}{5}$ and $\frac{1}{3}$ on RH and, for the Cesàro weight $1-n/x$, the exponent $\frac{1}{4}+\varepsilon$, a smoothed statement like one direction of Theorem 17.7. Meng [392] studies the distribution of $x^{-1/4}\Delta(x)$ under RH and further hypotheses, and Gorodetsky, Matomäki, Radziwiłł and Rodgers [215] prove that RH is equivalent to an essentially sharp bound for the variance of squarefree integers in short intervals. Theorems 17.7 and 17.8 follow in a few lines from Mellin inversion, the Paley Wiener theorem and Conrad's descent; we know no source that states them as equivalences.

Pólya's kernel Φ goes back to Riemann [463]; Pólya [444] studied it with its series, evenness and decay. Its positivity, immediate from the series for $u\geq 0$ and evenness, is credited by Biane, Pitman and Yor [82, Section 2.2] to Chung and to Newman [417]. They also discuss the Gamma(2) structure of the Bessel hitting times of Section 17.5 [82, (6) to (8) and Table 2], through which Williams [583, Section 5] represents ξ. The family φ_σ and its Thorin densities are Theorem 18 and Remark 21 of [442] and Theorem 16.8, and the positivity criterion of Section 17.6 is due to Hinkkanen [261] and Lagarias [339]. Polson [443] studies the Thorin

density of $\xi(\sigma)/\xi(\sigma+s)$ further, with an explicit formula representation, the exact strength of its pointwise positivity and a quasi infinitely divisible factorisation. Stopple [529] studies the Pólya kernels of quadratic Dirichlet L functions under the heat flow.

As far as we know, Theorems 17.11, 17.14, 17.17 and 17.18 are new. The equivalences with RH in this chapter are reformulations; none of them proves a case of it.

17.11 Exercises

1. Show that $\kappa = \gamma_{\mathrm{E}} - 2\zeta'(2)/\zeta(2)$, and deduce $\kappa = 24\log A - \gamma_{\mathrm{E}} - 2\log 2\pi$ from the functional equation and $\zeta'(-1) = \frac{1}{12} - \log A$, where A is Glaisher's constant.

2. Prove (17.9) and $\int_0^\infty g(\theta)\theta^{s-1}d\theta = -(1+s)\Gamma(s)\zeta(s)$ for $0 < \operatorname{Re} s < 1$, and deduce (17.10).

3. Show that $(1+s)\Gamma(s)2^{-s}$ is regular at $s = -1$, and deduce the term $-\pi^2/(3\zeta(3)X)$ in Corollary 17.12(iii). Compare it with Table 17.2 at $X = 10^3$, and explain why exponential smoothing (Theorem 17.7) gives a term in $\log x/x$ instead.

4. Derive the closed forms of R_1 and R_2. Show, by comparing the Taylor coefficients of $\sinh^3 x = \frac{1}{4}(\sinh 3x - 3\sinh x)$ with those of $x^3\cosh x$, that $(\sinh x/x)^p \geq \cosh x$ for all real x if and only if $p \geq 3$, and that $f_V(v) = v^2/60 + O(v^4)$ as $v \to 0$.

5. Show that $h_c^{(\delta)}$ is a probability density, that $2t\,h_c^{(\delta)}(t^2)$ is the sum of two Poisson kernels, and that near $r = c$ it is a Cauchy density of half width $2\delta\sqrt{c}$ up to a relative error $O(\delta/\sqrt{c})$.

6. Show that for no $\sigma > \frac{1}{2}$ is $u \mapsto 1/D(\sigma+\sqrt{u})$ the Laplace transform of a positive measure: for $\sigma > 1$ it increases from $1/D(\sigma)$ to 1, for $\sigma = 1$ it vanishes at $u = 0$, and for $\frac{1}{2} < \sigma < 1$ it is negative at $u = 0$ and vanishes at $u = (1-\sigma)^2$. For $\sigma > 1$, show that $D(\sigma+\sqrt{u})/D(\sigma)$ is the Laplace transform of the first passage time of a Brownian motion to the independent level $\log N/\sqrt{2}$, with N as in Section 17.9, and that it is not a GGC transform, since it does not tend to 0 as $u \to \infty$.

7. Show that a zero of ζ_K at $s = \frac{1}{2}$ of order $2m$ adds $\frac{4m}{3}$ to the zero sum of Theorem 17.16. For $K = \mathbb{Q}(i)$, show with the duplication formula that $\xi_K = \frac{2}{\sqrt{\pi}}\,\xi\,\Lambda$ (with ξ_K as in this chapter, four times that of Chapter 18), with Λ the completed β function of Chapter 18, and deduce from Proposition 18.2 and the positivity of Pólya's kernel that $\Phi_K > 0$.

8. Show that 2 and 3 are both inert in $\mathbb{Q}(\sqrt{d})$ if and only if $d \equiv 5 \bmod 24$ and that $a_n \le a_n^*$ in the proof of Theorem 17.18, and check by machine that $F(1.42) = -0.19908$ and $\sum_{n\ge 4} a_n^* F(1.42\,n) = 0.13918$.

17.12 Open computational problems

Computational Problem 17.1 (The squarefree error beyond 10^{18}). Objective: extend by one decade the computation of Mossinghoff, Oliveira e Silva and Trudgian [405], who found $-1.12543\,x^{1/4} < \Delta(x) < 1.11653\,x^{1/4}$ for all $0 < x \le 10^{18}$ (their sparse samples up to 10^{28} stay below 1.2), and compare the extremes with the atom sums of Figure 17.1. Objects and scale: the extremes of $\Delta(x)/x^{1/4}$ for $10^{18} < x \le 10^{19}$, by a segmented sieve of μ^2, and the sums $1 + 2\,\mathrm{Re}\sum_{k\le K} R_{\rho_k} x^{\rho_k/2}$ of (17.8) with $K = 10^4$ at the extreme points. Route: integer arithmetic for Q in two independent implementations, checked at sparse points against Pawlewicz's algorithm [434]; zeros from the LMFDB [371] or PARI/GP [432], and R_ρ in ball arithmetic. Deliverable: the record values of $|\Delta(x)|/x^{1/4}$ with their locations, and the agreement of the atom sums with $\Delta(x)/x^{1/4}$ there as a function of K. Why it matters: up to 10^{18} the values of $|\Delta(x)|/x^{1/4}$ stay below 1.12543, far below the proved $\limsup \Delta(x)/x^{1/4} > 3$; the computation measures how slowly the oscillations forced by the zeros appear, against the limiting distribution studied by Meng [392].

Computational Problem 17.2 (Oscillations of squarefree ideals). Objective: for $K = \mathbb{Q}(i)$ and $\mathbb{Q}(\sqrt{-3})$ let $\Delta_K(x) = Q_0(x) - c_K x/\zeta_K(2)$ be the error in the count of squarefree ideals, the case $\alpha = 0$ of (17.2). Landau's argument of Corollary 17.9, applied to $\zeta_K(s)/\zeta_K(2s)$, whose poles at $\rho/2$ come from the zeros of ζ and of $L(s,\chi_d)$, gives $\Delta_K(x) = \Omega(x^{1/4})$; prove explicit bounds $\limsup \Delta_K(x)/x^{1/4} > b_K$ and $\liminf \Delta_K(x)/x^{1/4} < -b_K$ and find the extreme values. Objects and scale: the residues of $\zeta_K(s)x^s/(s\,\zeta_K(2s))$ at the poles $\rho/2$ given by the first 2000 zeros of ζ and of $L(s,\chi_d)$; counts of squarefree ideals of norm up to 10^{14}. Route: the method of [405], that is Ingham's argument with many zeros and lattice reduction to establish a weak independence of the pooled ordinates; for the counts, a segmented sieve on norms, the number of squarefree ideals of norm n being determined by the factorisation of n and the splitting of its primes. Deliverable: constants b_K with the computations that prove them, and the extreme values of $\Delta_K(x)/x^{1/4}$ up to 10^{14} with their locations. Why it matters: the zeros of ζ and of $L(s,\chi_d)$ pool their atoms (Theorem 17.16), and the oscillations test whether the pooled ordinates behave independently; we found no Omega result and no computation of the extremes of Δ_K for these fields in the literature.

Computational Problem 17.3 (Pólya kernels of imaginary quadratic fields). Objective: extend Table 17.3 and prove its entries. Objects and scale: all fundamen-

tal discriminants $-10^7 \le d < 0$. Route: a_n by a divisor sieve; H_K on $[1, \sqrt{|d|}/\pi]$, beyond which every term is positive and below which evenness applies, in interval arithmetic with a bound on the derivative between grid points, so that positivity is proved and not sampled. Deliverable: proved counts of the fields with $\Phi_K > 0$ in dyadic ranges of $|d|$ and by the splitting of 2, 3, 5 and 7, and the splitting types, if any, for which every field in range is positive. Why it matters: positivity of Φ_K makes $\xi_K(\frac{1}{2}+w)/\xi_K(\frac{1}{2})$ a moment generating function (Theorem 17.17); Theorem 17.18 gives infinitely many fields with sign changes, and whether the positive fields have positive density is Open Problem 17.2. Problem 21.3 asks the same question for the theta functions of the characters χ_d, from which Φ_K is obtained by convolution with Pólya's kernel.

COMPUTATIONAL PROBLEM 17.4 (Pólya kernels of real quadratic fields). Objective: construct the analogue of Theorem 17.17 for real quadratic fields and compute its signs. Objects and scale: fundamental discriminants $0 < d \le 10^5$. Route: since $\int_0^\infty K_0(2y)y^{s-1}dy = \Gamma(s/2)^2/4$, the completed function $d^{s/2}\Gamma_{\mathbb{R}}(s)^2\zeta_K(s)$ is the Mellin transform of $\psi_K(x) = 4\sum_n a_n K_0(2\pi n x/\sqrt{d})$; put $H_K = (x^2\psi_K')'$ and $\Phi_K(v) = e^{v/2}H_K(e^v)$, evaluate K_0 by its series and asymptotic expansion, and decide signs in interval arithmetic. As in Section 17.7, Φ_K is also the convolution of Pólya's kernel with the theta kernel of the even character χ_d (proof of Theorem 18.25), whose sign can be computed separately; positive real zeros of such theta functions are studied in [372]. Deliverable: the set of d with $\Phi_K > 0$ sorted by the splitting of 2 and 3, the first d with a negative value, and, if sign changes persist when 2 and 3 are inert, an explicit point and bound as in the proof of Theorem 17.18. Why it matters: the Bessel kernel changes the archimedean side and keeps the arithmetic one, and a negative value of the theta kernel can be smoothed away by the convolution, so the computation separates the sources of sign changes.

CHAPTER 18

THE DIRICHLET BETA FUNCTION AND THE GAUSSIAN FIELD

The unitary automorphic representations of $\mathrm{GL}(1)$ over $\mathbb{Q}$ are, up to twists by $|\cdot|^{it}$, the primitive Dirichlet characters, and one of the simplest nontrivial ones is the character χ_{-4} of the Gaussian field $K = \mathbb{Q}(i)$: $\chi_{-4}(n) = 0$ for even n, $\chi_{-4}(n) = 1$ for $n \equiv 1$ and $\chi_{-4}(n) = -1$ for $n \equiv 3 \pmod 4$. Its L function is the Dirichlet beta function

$$\beta(s) = \sum_{n \geq 0} \frac{(-1)^n}{(2n+1)^s} = \sum_{n \geq 1} \frac{\chi_{-4}(n)}{n^s} = \prod_p \bigl(1 - \chi_{-4}(p) p^{-s}\bigr)^{-1}, \qquad \operatorname{Re} s > 1, \tag{18.1}$$

with $\beta(1) = \pi/4$ and $\beta(2) = 0.9159656$, Catalan's constant. Since χ_{-4} is real, odd and primitive with root number one, the completed function

$$\Lambda(s) = \Big(\frac{4}{\pi}\Big)^{s/2} \Gamma\Big(\frac{s+1}{2}\Big) \beta(s) \tag{18.2}$$

is entire of order one with $\Lambda(s) = \Lambda(1-s)$. The generalized Riemann hypothesis (GRH) for β asserts that every zero of Λ lies on $\operatorname{Re} s = \frac{1}{2}$.

This chapter works out for β the Thorin framework of Chapters 14 and 16 [441, 442]. For real primitive characters the closed form of the Thorin density, the GGC property to the right of the critical strip and the equivalence of GRH with the GGC property at every basepoint $\alpha > \frac{1}{2}$ are Theorem 30 and Remark 32 of [442], and in this book they are the case of χ_{-4} of Theorem 16.8; what β adds is an explicit positive kernel, a signed Lévy measure, and the question of what the signs of that measure can prove. The framework has three parts: a Lévy form of the completed function to the right of the critical strip; the $2\sin^2$ lemma of Chapter 16 (Lemma 16.7, from [442]), which turns the Lévy measure of the primal function into the Thorin measure of its reciprocal; and Thorin's condition, that $\Lambda(\frac{1}{2})/\Lambda(\frac{1}{2} + \sqrt{v})$ is the Laplace transform of a generalized gamma convolution (GGC, Definition 13.2), whose analyticity in the cut plane excludes zeros off the critical line (Theorem 14.8).

Three features make β the natural first case beyond ζ. Its Mellin kernel is the weight $3/2$ theta series η^3, positive by the product formula. Its prime atoms carry the weights $\chi_{-4}(n)\Lambda(n)$, negative at $3, 7, 11, 19, \ldots$, so Thorin positivity is no longer termwise but arithmetic. Finally $\zeta_K = \zeta \cdot \beta$: the trivial character of

$\mathrm{GL}(1)$ over K induces to the isobaric representation $1 \boxplus \chi_{-4}$ of $\mathrm{GL}(2)$ over $\mathbb{Q}$, and Thorin densities respect this decomposition.

Sections 18.1 to 18.3 construct the primal law, the Lévy form and the closed form $T_\alpha(y) = \mathrm{Re}(\Lambda'/\Lambda)(\alpha + i\sqrt{y})/(2\pi\sqrt{y})$ of the Thorin density. Sections 18.4 and 18.5 recall the GGC property for $\alpha \geq 1$ and the equivalence of GRH with the GGC property for all $\alpha > \frac{1}{2}$. Section 18.6 treats the Gaussian field and the limits of termwise bounds; Section 18.7 shows that positivity is free on the real axis, can fail only in short windows, and that GRH is equivalent to the density on each line $\frac{1}{2} < \mathrm{Re}\, s \leq 1$ being smallest on the real axis; Section 18.8 identifies the Thorin measure with the argument of Λ. Sections 18.9 and 18.10 treat the centre and the pair (η, χ_{12}).

18.1 The completed beta function and its primal law

18.1.1 The eta cube kernel

Jacobi's identity gives, with $q = e^{2\pi i\tau}$,

$$\eta(\tau)^3 = q^{1/8} \prod_{m\geq 1} (1-q^m)^3 = \sum_{n\geq 0} (-1)^n (2n+1) q^{(2n+1)^2/8}. \tag{18.3}$$

At $\tau = ix$ this is the theta series of χ_{-4},

$$\vartheta_\beta(x) := \eta(ix)^3 = \sum_{n\geq 1} \chi_{-4}(n)\, n\, e^{-\pi n^2 x/4}, \tag{18.4}$$

and $\eta(i/x) = \sqrt{x}\,\eta(ix)$ gives $\vartheta_\beta(1/x) = x^{3/2}\vartheta_\beta(x)$.

Proposition 18.1 (Mellin form; classical). *For all $s \in \mathbb{C}$,*

$$\Lambda(s) = \frac{\sqrt{\pi}}{2} \int_0^\infty x^{(s-1)/2}\, \eta(ix)^3\, dx. \tag{18.5}$$

Proof. For $\mathrm{Re}\, s > 1$ integrate (18.4) term by term, using

$$\int_0^\infty x^{(s+1)/2-1} e^{-\pi n^2 x/4}\, dx = \Gamma\Big(\frac{s+1}{2}\Big)\Big(\frac{4}{\pi n^2}\Big)^{(s+1)/2}$$

and $\sum_n \chi_{-4}(n)\, n^{-s} = \beta(s)$; the result is $(4/\pi)^{(s+1)/2}\Gamma(\frac{s+1}{2})\beta(s)$, which equals $(2/\sqrt{\pi})\Lambda(s)$. Since ϑ_β decays like $e^{-\pi x/4}$ at infinity and like $x^{-3/2}e^{-\pi/(4x)}$ at zero, the integral is entire. □

Proposition 18.2 (Primal kernel). *The function $\Phi_\beta(u) = e^{3u/2}\,\eta(ie^{2u})^3$ is even, strictly positive, and decays like $\exp(\frac{3}{2}|u| - \frac{\pi}{4}e^{2|u|})$. For all $w \in \mathbb{C}$,*

$$\Lambda\big(\tfrac{1}{2}+w\big) = \sqrt{\pi}\int_{-\infty}^{\infty} e^{wu}\Phi_\beta(u)\, du = 2\sqrt{\pi}\int_0^\infty \cosh(wu)\,\Phi_\beta(u)\, du. \tag{18.6}$$

PROOF. Evenness is the transformation law, and positivity is the product in (18.3), every factor $1-q^m$ being positive for $q=e^{-2\pi x}\in(0,1)$. The identity is (18.5) after $x=e^{2u}$. □

This is the theta series proof of the functional equation for odd characters [276]; the eta cube form is in [442, Section 8.1], and the theta identity is recalled in Chapter 27. Up to normalisation, Φ_β is the Pólya kernel of Lemma 14.12 for Ξ^{sd}, here $\Lambda(\frac12+z)$ up to a constant factor, and it is positive for the reason given after that lemma for eta products, whereas the positivity of the Riemann Pólya kernel takes work; and unlike ξ, Λ has no polar factor. The normalised kernel $\Phi_\beta/\int\Phi_\beta$ is the density of a symmetric random variable X_β, the Pólya variable of Ξ^{sd}, which we call the *primal law*, with

$$\mathbb{E}\big[e^{wX_\beta}\big]=\frac{\Lambda(\frac12+w)}{\Lambda(\frac12)},\qquad \mathbb{E}\big[e^{itX_\beta}\big]=\frac{\Lambda(\frac12+it)}{\Lambda(\frac12)}. \tag{18.7}$$

Numerically $\Lambda(\frac12)=0.86913481$, $\beta(\frac12)=0.66769146$ and $\Phi_\beta(0)=0.45338383$.

18.1.2 Zeros and the tilt

LEMMA 18.3 (No real zeros). *$\beta(\sigma)\ge 1-3^{-\sigma}>0$ for real $\sigma>0$. Hence Λ has no zeros in $(0,1)$.*

PROOF. For $\sigma>0$ the series (18.1) converges, alternates and has strictly decreasing terms. □

The zeros ρ of Λ are symmetric under $\rho\mapsto 1-\rho$ and $\rho\mapsto\bar\rho$, and every zero satisfies $0<\operatorname{Re}\rho<1$ (the Euler product excludes $\operatorname{Re}s>1$, and the nonvanishing of β on $\operatorname{Re}s=1$ with the functional equation excludes the edges). Writing $\rho=\frac12+i\theta$ (by Lemma 18.3 no θ is purely imaginary, so the condition $\operatorname{Re}\theta>0$ picks one of each pair $\pm\theta$), the functional equation gives $\Lambda'(\frac12)=0$ and kills the exponential factor of the Hadamard product, so

$$\frac{\Lambda(\frac12+w)}{\Lambda(\frac12)}=\prod_{\operatorname{Re}\theta>0}\Big(1+\frac{w^2}{\theta^2}\Big),\qquad \operatorname{Re}\frac{\Lambda'}{\Lambda}(s)=\sum_\rho\frac{\operatorname{Re}s-\operatorname{Re}\rho}{|s-\rho|^2}, \tag{18.8}$$

the sum converging absolutely since $\sum|\rho|^{-2}<\infty$. Under GRH every θ is a real ordinate γ and $\Lambda(\frac12)/\Lambda(\frac12+w)=\prod_{\gamma>0}\gamma^2/(\gamma^2+w^2)$, the product (14.3) of Chapter 14 for Ξ^{sd}. The first ordinates γ are 6.02094890, 10.24377030, 12.98809801, 16.34260710 and 18.29199320. We write

$$c_\alpha:=\frac{\Lambda'}{\Lambda}(\alpha)=\tfrac12\log\frac{4}{\pi}+\tfrac12\psi\Big(\frac{\alpha+1}{2}\Big)+\frac{\beta'}{\beta}(\alpha) \tag{18.9}$$

and $V:=(\log\Lambda)''(\frac12)=\operatorname{Var}(X_\beta)=0.156030$, twice the sum $S_1=0.078015$ of Table 14.1. By (18.8), $c_\alpha>0$ for $\alpha\ge1$, and $c_{1/2}=0$.

18.2 The Lévy form for $\alpha > 1$

THEOREM 18.4 (Lévy form; as in Section 16.4.3). *For $\alpha > 1$ and $\operatorname{Re} s \geq 0$,*

$$\log \frac{\Lambda(\alpha+s)}{\Lambda(\alpha)} - c_\alpha s = \int_0^\infty \bigl(e^{-sx} - 1 + sx\bigr) e^{-\alpha x}\, \frac{\mu_\beta(dx)}{x}, \tag{18.10}$$

with the signed measure

$$\mu_\beta(dx) = \frac{dx}{2\sinh x} + \sum_{n\geq 2} \chi_{-4}(n)\,\Lambda(n)\,\delta_{\log n}(dx). \tag{18.11}$$

PROOF. The factor $(4/\pi)^{s/2}$ is absorbed by the tilt. For $\operatorname{Re} z > 0$,

$$\log \frac{\Gamma(z+v)}{\Gamma(z)} - v\psi(z) = \int_0^\infty \bigl(e^{-vx} - 1 + vx\bigr)\, \frac{e^{-zx}}{x(1-e^{-x})}\, dx,$$

since both sides and their first derivatives vanish at $v = 0$ and both have second derivative $\psi'(z+v)$. Taking $z = \frac{\alpha+1}{2}$, $v = \frac{s}{2}$ and $x \mapsto 2x$ turns the kernel into $e^{-\alpha x}/(2x\sinh x)$. For the Dirichlet series, on $\operatorname{Re} s > 1$ the function $\log \beta(s)$ is $\sum_{n\geq 2} \chi_{-4}(n)\Lambda(n)(\log n)^{-1} n^{-s}$, which gives the atoms. Absolute convergence follows from $|e^{-sx} - 1 + sx| \leq \min(2 + |s|x, \frac{1}{2}|s|^2 x^2)$. □

With the prime sum truncated at $n \leq 2 \times 10^5$ the two sides of (18.10) agree to seven digits at $(\alpha, s) = (2, 2)$ and to twelve at $(3, 2)$, where both equal 0.232881325700.

REMARK 18.5 (Comparison with ξ). For $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s)$ the same computation gives the measure $-e^x\,dx + dx/(e^{2x}-1) + \sum_{n\geq 2}\Lambda(n)\delta_{\log n}$. The polar factor $\alpha - 1 + s$ is log concave in s, so it enters with a negative sign (Section 16.4.3). For β there is no polar term, the archimedean density is $1/(2\sinh x)$ (an odd character shifts the gamma argument by one half), and the prime part is the difference of two positive measures, carried by the prime powers $n \equiv 1$ and $n \equiv 3 \pmod 4$ respectively.

18.3 The $2\sin^2$ lemma and the Thorin density

Lemma 16.7 transports a positive Lévy measure. The transport is linear in the measure, and it holds for a signed measure once the positivity conclusion is dropped; we state it in the Laplace variable $v = s^2$ and prove it by Fourier inversion, which also gives (18.17).

LEMMA 18.6 ($2\sin^2$ lemma, signed form; after Lemma 16.7). *Let f be analytic and nonvanishing near $[\alpha, \infty)$, let $c = f'(\alpha)/f(\alpha)$, and suppose that for $s \geq 0$*

$$\log \frac{f(\alpha+s)}{f(\alpha)} - cs = \int_0^\infty \bigl(e^{-sx} - 1 + sx\bigr)\, m(dx)$$

for a signed measure m with $\int \min(x,x^2)\,|m|(dx)<\infty$. Put

$$T(y)=\frac{1}{2\pi\sqrt{y}}\Big[c+\int_0^\infty 2\sin^2\Big(\frac{x\sqrt{y}}{2}\Big)\,x\,m(dx)\Big],\qquad y>0. \tag{18.12}$$

Then for every $v>0$,

$$\frac{f(\alpha)}{f(\alpha+\sqrt{v})}=\exp\Big(-\int_0^\infty \log\Big(1+\frac{v}{y}\Big)T(y)\,dy\Big). \tag{18.13}$$

The map $m\mapsto T$ is linear and positivity preserving, and $c\ge 0$ contributes the positive stable part $c/(2\pi\sqrt{y})$.

PROOF. With $y=t^2$, $T(y)\,dy=\pi^{-1}\big[c+\int x(1-\cos tx)\,m(dx)\big]dt$. The integrals

$$\int_0^\infty \log\Big(1+\frac{v}{t^2}\Big)dt=\pi\sqrt{v},\qquad \int_0^\infty \log\Big(1+\frac{v}{t^2}\Big)\cos(xt)\,dt=\frac{\pi}{x}\big(1-e^{-x\sqrt{v}}\big),$$

the second by parts from $\int_0^\infty \sin(xt)\,dt/(t(t^2+v))=\pi(1-e^{-x\sqrt{v}})/(2v)$, give

$$\begin{aligned}\int_0^\infty \log\Big(1+\frac{v}{y}\Big)T(y)\,dy&=c\sqrt{v}+\int_0^\infty\big(e^{-x\sqrt{v}}-1+x\sqrt{v}\big)m(dx)\\&=\log\frac{f(\alpha+\sqrt{v})}{f(\alpha)},\end{aligned}$$

Fubini being justified by $|1-\cos tx|\le\min(2,t^2x^2/2)$. □

If f is analytic on $\operatorname{Re}s\ge\alpha$, differentiating the Lévy form at $s=it$ gives $\operatorname{Re}(f'/f)(\alpha+it)-c=\int x(1-\cos tx)\,m(dx)$, so the bracket in (18.12) is $\operatorname{Re}(f'/f)(\alpha+i\sqrt{y})$. For $f=\Lambda$ this is the main formula of the chapter.

THEOREM 18.7 (Thorin density of the reciprocal beta function; [442], Theorem 30). *For $\alpha>1$ and $v>0$,*

$$\frac{\Lambda(\alpha)}{\Lambda(\alpha+\sqrt{v})}=\exp\Big(-\int_0^\infty \log\Big(1+\frac{v}{y}\Big)T_\alpha(y)\,dy\Big), \tag{18.14}$$

where

$$T_\alpha(y)=\frac{1}{2\pi\sqrt{y}}\Big[c_\alpha+\int_0^\infty 2\sin^2\Big(\frac{x\sqrt{y}}{2}\Big)e^{-\alpha x}\mu_\beta(dx)\Big] \tag{18.15}$$

$$=\frac{c_\alpha+G_\alpha(\sqrt{y})+P_\alpha(\sqrt{y})}{2\pi\sqrt{y}}=\frac{1}{2\pi\sqrt{y}}\operatorname{Re}\frac{\Lambda'}{\Lambda}\big(\alpha+i\sqrt{y}\big), \tag{18.16}$$

with the archimedean and prime parts

$$G_\alpha(t) = \tfrac{1}{2}\Big[\operatorname{Re}\,\psi\Big(\frac{\alpha+1+it}{2}\Big) - \psi\Big(\frac{\alpha+1}{2}\Big)\Big] \geq 0,$$
$$P_\alpha(t) = \sum_{n\geq 2} \chi_{-4}(n)\Lambda(n)\,n^{-\alpha}\big(1-\cos(t\log n)\big).$$

Equivalently,

$$\log\frac{\Lambda(\alpha+\sqrt{v})}{\Lambda(\alpha)} = \frac{1}{\pi}\int_0^\infty \log\Big(1+\frac{v}{t^2}\Big)\,\operatorname{Re}\frac{\Lambda'}{\Lambda}(\alpha+it)\,dt. \tag{18.17}$$

PROOF. Apply Lemma 18.6 to Theorem 18.4. The density satisfies $T_\alpha(y) \sim c_\alpha/(2\pi\sqrt{y})$ as $y\to 0$, and $T_\alpha(y) = O(\log y/\sqrt{y})$ as $y\to\infty$ because P_α is bounded and $\operatorname{Re}\,\psi(\frac{\alpha+1+it}{2}) \sim \log(t/2)$. □

In the notation of Theorem 16.8, whose Ξ_π is $\Lambda(\frac{1}{2}+z)^2$ up to a constant factor, $2T_\alpha = V_{\alpha-1/2}$, and $T_\alpha(y)\,dy$ is the Thorin measure τ (Definition 13.2) of $\Lambda(\alpha)/\Lambda(\alpha+\sqrt{v})$ when every zero of Λ has real part less than α (for every $\alpha\geq 1$ by part (iii) of that theorem, and for every $\alpha > \frac{1}{2}$ under GRH).

The sine squared form (18.15), with primes to 2×10^5, and the closed form (18.16) give the same values of $2\pi t\,T_\alpha(t^2)$, namely 0.2353511, 0.7691193 and 0.32628 at $(\alpha,t) = (2,1),(2,7),(\frac{3}{2},4)$. Formula (18.17) at $(\alpha,v) = (2,1)$, by quadrature to $t = 150$ plus the archimedean tail $v(\log(2T/\pi)+1)/(2\pi T)$ at $T = 150$, gives 0.2977961 against $\log\Lambda(3)/\Lambda(2) = 0.2977947$; the residual comes from the oscillating prime part beyond $t = 150$.

18.4 WHAT HOLDS UNCONDITIONALLY

THEOREM 18.8 (GGC for $\alpha\geq 1$; Theorem 16.8(iii)). *For every $\alpha\geq 1$ there is a GGC random variable $H_\alpha^\beta \geq 0$, with drift zero and Thorin density $T_\alpha > 0$ given by* (18.16)*, such that*

$$\frac{\Lambda(\alpha)}{\Lambda(\alpha+\sqrt{v})} = \mathbb{E}\big[e^{-vH_\alpha^\beta}\big], \qquad v>0.$$

In particular $\frac{\sqrt{\pi}}{2}/\Lambda(1+\sqrt{v})$ is a GGC Laplace transform.

PROOF. Since every zero has $\operatorname{Re}\rho < 1$, each term of (18.8) is positive on $\operatorname{Re} s = \alpha \geq 1$, so $T_\alpha > 0$. By Theorem 16.8(iii) with $a = \alpha - \frac{1}{2}$, the square $(\Lambda(\alpha)/\Lambda(\alpha+\sqrt{v}))^2$ is the transform of a GGC with drift zero and Thorin density $V_{\alpha-1/2} = 2T_\alpha$, so by Definition 13.2 its square root is the transform of a GGC with drift zero and Thorin density T_α. For $\alpha > 1$ this is also Theorem 18.7. □

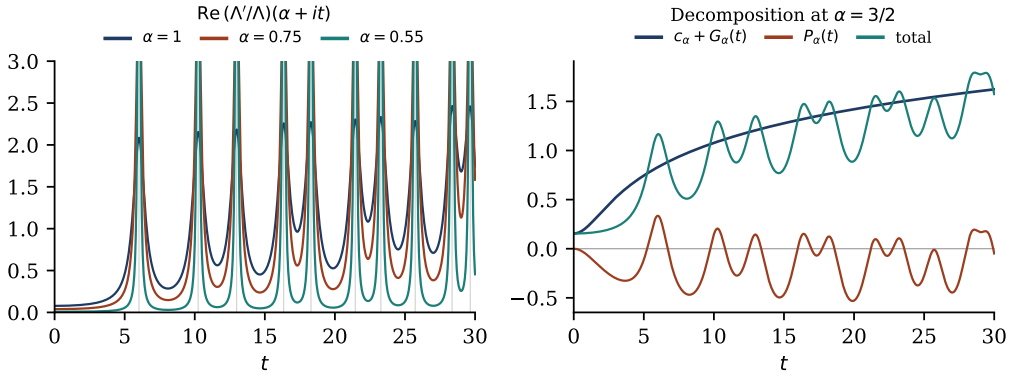


Figure 18.1: Left: $\mathrm{Re}(\Lambda'/\Lambda)(\alpha+it) = 2\pi t\, T_\alpha(t^2)$ for $\alpha = 1$, 0.75, 0.55 and $0 \le t \le 30$, with the vertical axis cut at 3; grey lines mark the ordinates of the zeros of β. As $\alpha \downarrow \frac{1}{2}$ the density concentrates on the zeros. Right: at $\alpha = \frac{3}{2}$, the archimedean part with the constant, $c_\alpha + G_\alpha$, the signed prime part P_α, and their sum (labelled total).

The proof uses the zeros rather than the Lévy measure. On the Lévy side the sine squared integrand in (18.15) is signed, and bounding the atoms by their absolute values proves positivity at every height only above a threshold α_c. Write

$$A_\alpha(t) = \tfrac{1}{2}\log\frac{4}{\pi} + \tfrac{1}{2}\,\mathrm{Re}\;\psi\Big(\frac{\alpha+1+it}{2}\Big),$$
$$S(\alpha) = \sum_{n\ \mathrm{odd}} \Lambda(n)n^{-\alpha} = -\frac{\zeta'}{\zeta}(\alpha) - \frac{\log 2}{2^\alpha - 1}, \tag{18.18}$$

so that $\mathrm{Re}\;\Lambda'/\Lambda(\alpha+it) = A_\alpha(t) + \mathrm{Re}\;\beta'/\beta(\alpha+it)$ and $|\beta'/\beta(\alpha+it)| \le S(\alpha)$ for $\alpha > 1$.

PROPOSITION 18.9 (Termwise positivity). *$A_\alpha(t)$ increases in t, $A_\alpha(0)$ increases in α, and $S(\alpha)$ decreases. Let α_c be the root of $A_\alpha(0) = S(\alpha)$, numerically $\alpha_c = 2.28231064$. For $\alpha \ge \alpha_c$ the bound $A_\alpha(t) - S(\alpha)$ proves $T_\alpha(t^2) > 0$ for all $t > 0$; for $1 < \alpha < \alpha_c$ it fails near $t = 0$, although $T_\alpha > 0$ by Theorem 18.8.*

PROOF. Monotonicity in t follows from $\mathrm{Re}\;\psi(x+iy) - \psi(x) = \sum_{k\ge0} y^2/\big((x+k)((x+k)^2+y^2)\big)$; ψ increases on $(0,\infty)$, and S is a Dirichlet series with positive coefficients. □

At $\alpha = \frac{3}{2}$ (Figure 18.1, right) the prime part P_α is negative over most of $[0,30]$, with minimum -0.532 near $t = 19.9$, where $c_\alpha + G_\alpha = 1.418$. For ζ the corresponding prime part is a positive sum.

18.5 GRH AS THORIN POSITIVITY

THEOREM 18.10 (GRH as Thorin positivity; Theorem 16.8(iv) and [442], Remark 32). *The following are equivalent.*

(i) GRH for β.

(ii) For every $\alpha > \frac{1}{2}$, $\Lambda(\alpha)/\Lambda(\alpha+\sqrt{v})$ is the Laplace transform of a GGC.

(iii) $\operatorname{Re}(\Lambda'/\Lambda)(s) > 0$ *for* $\operatorname{Re} s > \frac{1}{2}$.

In this case the Thorin measures $T_\alpha(y)\,dy$ converge vaguely on $(0,\infty)$, as $\alpha \downarrow \frac{1}{2}$, to $\sum_{\gamma>0}\delta_{\gamma^2}$ counted with multiplicity, the Thorin measure of $\Lambda(\frac{1}{2})/\Lambda(\frac{1}{2}+\sqrt{v})$.

PROOF. Up to a constant factor $\Lambda(\frac{1}{2}+z)^2$ is the function Ξ_π of Theorem 16.8, and by Definition 13.2 a GGC transform has GGC square and square root. So (i)⇒(ii) is part (i) of that theorem for every $a = \alpha - \frac{1}{2} > 0$, and (ii)⇒(i) is its part (ii), the cut plane argument of Lemma 13.3, for every $a > 0$. (i)⇒(iii) is (18.8). (iii)⇒(i): a zero ρ of multiplicity m with $\operatorname{Re}\rho > \frac{1}{2}$ is a pole of Λ'/Λ with residue m, so $\operatorname{Re}\,\Lambda'/\Lambda(s) = m\operatorname{Re}(s-\rho)^{-1} + O(1)$ takes negative values for s near ρ with $\operatorname{Re} s < \operatorname{Re}\rho$. The limit is part (iv) of the theorem: under (i), with $a = \alpha - \frac{1}{2}$ and $y = t^2$, $T_\alpha(y)\,dy = \pi^{-1}\sum_\gamma a\,(a^2+(t-\gamma)^2)^{-1}dt$ is a sum of Poisson kernels, which converges to $\sum_{\gamma>0}\delta_\gamma(dt)$ on $t > 0$. □

The equivalence of (i) and (iii) is the criterion of Hinkkanen [261] and Lagarias [339] for χ_{-4} (Lagarias's Theorem 1.1 is stated for any admissible set of zeros), and Theorem 16.8(iv) is its Thorin form. The two conditions are one statement in two variables: the Thorin density at $y = t^2$ is $\operatorname{Re}\,\Lambda'/\Lambda$ at height t divided by $2\pi t$. For ξ more is known: the minimum of $\operatorname{Re}\,\xi'/\xi$ on the line $\operatorname{Re} s = \alpha$ is attained only at $t = 0$ under RH for every $\alpha > \frac{1}{2}$ (Lagarias's Theorem 1.3 and its proof in Section 3), and at $t = 0$ unconditionally for every $\alpha \geq 1$ (Garunkštis [208]). Numerically the same holds for β: over $t \in [0,50]$ the minimum of $\operatorname{Re}\,\Lambda'/\Lambda(\alpha+it)$ is c_α, attained at $t = 0$:

α	1	0.75	0.6	0.55
c_α	0.077784	0.038978	0.015601	0.007801
$c_\alpha/(\alpha-\frac{1}{2})$	0.1556	0.1559	0.1560	0.1560

The margin at $t = 0$ shrinks like $(\alpha-\frac{1}{2})V$, while near each ordinate the density grows like $(\alpha-\frac{1}{2})^{-1}$. Section 18.7 shows (Theorem 18.21) that for $\frac{1}{2} < \alpha \leq 3.15$ the real axis is the minimum, unconditionally, for $|t| \leq H_0 - 1.21$, where H_0 is the height to which GRH for β has been verified.

18.6 The Gaussian field and termwise bounds

18.6.1 Automorphic induction and Thorin densities

Let $K = \mathbb{Q}(i)$, with $\zeta_K(s) = \sum_{\mathfrak{a}} N(\mathfrak{a})^{-s}$ over the nonzero ideals of $\mathbb{Z}[i]$. A prime $p \equiv 1 \pmod 4$ splits into two Gaussian primes of norm p, a prime $p \equiv 3 \pmod 4$ is inert with norm p^2, and 2 ramifies; this splitting law is the identity $\zeta_K = \zeta \cdot \beta$, and by class field theory χ_{-4} is the quadratic character of $\mathrm{Gal}(K/\mathbb{Q})$. In the Langlands picture the trivial character of $\mathrm{GL}(1)$ over K induces to the isobaric representation $1 \boxplus \chi_{-4}$ of $\mathrm{GL}(2)$ over $\mathbb{Q}$, whose standard L function is $\zeta\beta$. The pairing of the atoms at $p \equiv 1$ and $p \equiv 3 \pmod 4$ is exactly this splitting.

Remark 18.11 (Isobaric sums). The closed form (18.16) is linear in the logarithmic derivative, so if $\Lambda_\pi = \Lambda_{\pi_1}\Lambda_{\pi_2}$ for an isobaric sum $\pi = \pi_1 \boxplus \pi_2$, then $T_\alpha^{\pi} = T_\alpha^{\pi_1} + T_\alpha^{\pi_2}$ on every vertical line and, on lines $\mathrm{Re}\, s = \alpha$ to the right of every zero of both functions (every $\alpha \geq 1$, and every $\alpha > \frac{1}{2}$ under GRH for both), the GGC variables of the reciprocals are independent summands. Positivity of the summands implies positivity of the sum but not conversely, which is why Thorin positivity for a component becomes a domination statement inside the sum.

18.6.2 Thorin domination

Put $\xi_K(s) = \frac{1}{2}s(s-1)\pi^{-s}\Gamma(s)\zeta_K(s)$.

Proposition 18.12 (Factorisation). $\Lambda(s) = 2\sqrt{\pi}\,\xi_K(s)/\xi(s)$. *For $\alpha > 1$ the Lévy measures satisfy $\mu_\beta = \mu_K - \mu_\xi$, where*

$$\mu_K(dx) = -e^x\,dx + \frac{dx}{e^x - 1} + \sum_{n \geq 2} \Lambda(n)\big(1 + \chi_{-4}(n)\big)\delta_{\log n}(dx),$$

$$\mu_\xi(dx) = -e^x\,dx + \frac{dx}{e^{2x} - 1} + \sum_{n \geq 2} \Lambda(n)\,\delta_{\log n}(dx).$$

Apart from the common polar term, both measures are positive.

Proof. The duplication formula $\Gamma(s) = 2^{s-1}\pi^{-1/2}\Gamma(\frac{s}{2})\Gamma(\frac{s+1}{2})$ gives

$$\frac{\pi^{-s}\Gamma(s)}{\pi^{-s/2}\Gamma(\frac{s}{2})} = \frac{1}{2\sqrt{\pi}}\Big(\frac{4}{\pi}\Big)^{s/2}\Gamma\Big(\frac{s+1}{2}\Big).$$

The measures follow as in Theorem 18.4 and Remark 18.5: $\frac{1}{2}s(s-1)\Gamma(s) = (s-1)\Gamma(s+1)/2$ gives $-e^x\,dx$ and $1/(e^x-1)$, and $-\zeta_K'/\zeta_K$ has coefficients $\Lambda(n)(1 + \chi_{-4}(n)) \geq 0$. The polar terms cancel and $1/(e^x-1) - 1/(e^{2x}-1) = 1/(2\sinh x)$. $\square$

THEOREM 18.13 (Thorin domination; positivity and equivalence by Theorem 16.8). *Let T_α^K and T_α^ξ be the Thorin densities given by* (18.16) *with ξ_K and ξ in place of Λ, that is, those of $\xi_K(\alpha)/\xi_K(\alpha+\sqrt{v})$ and $\xi(\alpha)/\xi(\alpha+\sqrt{v})$. Then $T_\alpha^\beta = T_\alpha^K - T_\alpha^\xi$. For $\alpha \geq 1$ all three densities are positive and there are independent GGC variables with $H_\alpha^K \stackrel{d}{=} H_\alpha^\xi + H_\alpha^\beta$. GRH for β holds if and only if $T_\alpha^K \geq T_\alpha^\xi$ for every $\alpha > \frac{1}{2}$; if Thorin's condition holds for ξ, this says that H_α^ξ is a GGC factor of H_α^K for every $\alpha > \frac{1}{2}$.*

PROOF. By Proposition 18.12, $\operatorname{Re}\,\Lambda'/\Lambda = \operatorname{Re}\,\xi_K'/\xi_K - \operatorname{Re}\,\xi'/\xi$ on every vertical line. Positivity for $\alpha \geq 1$ (strict, since every term of the Hadamard sums is then positive, as in the proof of Theorem 18.8) and the equivalence are parts (iii) and (iv) of Theorem 16.8 for ζ, where $\Xi_\zeta(z) = \xi(\frac{1}{2}+z)$, and for χ_{-4}, where $\Xi_\pi(z)$ is $\Lambda(\frac{1}{2}+z)^2$ up to a constant factor and has density $2T_\alpha$ (see also [442, Theorem 30 and Remark 32]). □

At $\alpha = \frac{3}{2}$ and $t = 0, 2, 5, 10, 20$:

t	0	2	5	10	20
$\operatorname{Re}\,\xi_K'/\xi_K$	0.20035	0.22524	0.68519	1.32766	1.49863
$\operatorname{Re}\,\xi'/\xi$	0.04614	0.04704	0.05267	0.10289	0.60611
$\operatorname{Re}\,\Lambda'/\Lambda$	0.15422	0.17820	0.63252	1.22477	0.89252

18.6.3 Kronecker sharpness

The factorisation is the right language for pairing, but it cannot lower α_c on its own, because the termwise bound is sharp.

PROPOSITION 18.14 (Kronecker sharpness). *For $\alpha > 1$ and every $T > 0$,*

$$\inf_{t \geq T} \operatorname{Re} \frac{\beta'}{\beta}(\alpha + it) = -S(\alpha).$$

Hence a bound on the prime part that uses only $|\chi_{-4}(p)|$ proves $T_\alpha(t^2) > 0$ exactly for $t > t_0(\alpha)$, where $A_\alpha(t_0) = S(\alpha)$ (and $t_0 = 0$ for $\alpha \geq \alpha_c$), and $\log t_0(\alpha) = 2/(\alpha-1) + O(1)$ as $\alpha \downarrow 1$.

PROOF. From the Euler product, $\operatorname{Re}\,\beta'/\beta(\alpha+it) = -\sum_p \log p \operatorname{Re}\, w_p/(1-w_p)$ with $w_p = \chi_{-4}(p)p^{-\alpha-it}$, and $\operatorname{Re}\, w/(1-w) \leq r/(1-r)$ on $|w| = r < 1$ with equality at $w = r$. As in the proof of Proposition 11.3, the $\log p$ are linearly independent over $\mathbb{Q}$, so Kronecker's theorem gives $t \geq T$ with $\chi_{-4}(p)p^{-it}$ arbitrarily close to 1 for all $p \leq P$, and the tail over $p > P$ is small. The asymptotic uses $S(\alpha) = (\alpha-1)^{-1} + O(1)$ and $A_\alpha(t) = \frac{1}{2}\log t + O(1)$. □

α	2	1.75	1.5	1.25	1.1	1.05
$S(\alpha)$	0.339	0.580	1.126	2.964	8.835	18.78
$t_0(\alpha)$	2.37	4.72	14.9	589	7.4×10^7	3.3×10^{16}

For each $\alpha \in (1, \alpha_c)$ the elementary route thus reduces positivity to a finite computation with error control on $[0, t_0(\alpha)]$, but the range grows like $e^{2/(\alpha-1)}$. The route does not reach $\alpha = 1$. There the GGC property itself follows from $\alpha > 1$, since GGC laws are closed under weak limits; what needs the nonvanishing of β on $\operatorname{Re} s = 1$, that is the prime number theorem for primes modulo 4, is that the Thorin measure at $\alpha = 1$ is $T_1(y)\,dy$, with no atom at the square of the ordinate of a zero on that line (for ξ, [442, Remark 21]). The sizes of the atoms do not contain this information. Dirichlet's theorem for the modulus 4 needs only $\beta(1) = \pi/4 \neq 0$.

18.7 The real axis, localisation and comparison

18.7.1 Positivity on the real axis and localisation

THEOREM 18.15 (Real axis). *For every $\alpha > \frac{1}{2}$, unconditionally, $c_\alpha > 0$, and hence $T_\alpha(y) \sim c_\alpha/(2\pi\sqrt{y}) > 0$ as $y \downarrow 0$.*

PROOF. By (18.7), $C(a) = \log \Lambda(\frac{1}{2} + a) - \log \Lambda(\frac{1}{2})$ is the cumulant generating function of X_β, which has a strictly positive density. So C is strictly convex, $C'(0) = 0$ by symmetry, and $c_{1/2+a} = C'(a) > 0$ for $a > 0$. The asymptotic follows from (18.16), Λ having no zeros near the real axis. □

So the place where Thorin positivity is weakest under GRH is exactly where it holds for free. Away from the real axis it can fail only in short windows.

THEOREM 18.16 (Localisation; for ξ, Levinson and Montgomery [362]). *Group the zeros into those on the line and the pairs $\{b + i\gamma,\ 1 - b + i\gamma\}$ with $b > \frac{1}{2}$. For $\alpha > \frac{1}{2}$, $t \in \mathbb{R}$ and $u = t - \gamma$,*

$$\operatorname{Re} \frac{\Lambda'}{\Lambda}(\alpha + it) = \sum_{\operatorname{Re}\rho = \frac{1}{2}} \frac{\alpha - \frac{1}{2}}{|\alpha + it - \rho|^2} + \sum_{pairs} \frac{(2\alpha - 1)\big(u^2 - (b - \frac{1}{2})^2 + (\alpha - \frac{1}{2})^2\big)}{\big((\alpha - b)^2 + u^2\big)\big((\alpha + b - 1)^2 + u^2\big)}, \tag{18.19}$$

both series converging absolutely. Hence $T_\alpha(t^2) > 0$ unless some zero $b + i\gamma$ with $b > \alpha$ satisfies $|t - \gamma|^2 < (b - \frac{1}{2})^2 - (\alpha - \frac{1}{2})^2 < \frac{1}{4}$.

PROOF. Zeros $b + i\gamma$ and $1 - b + i\gamma$ have equal multiplicity by the functional equation and the reality of Λ on $\mathbb{R}$, and absolute convergence permits regrouping.

With $x_1 = \alpha - b$ and $x_2 = \alpha + b - 1$,

$$\frac{x_1}{x_1^2 + u^2} + \frac{x_2}{x_2^2 + u^2} = \frac{(x_1 + x_2)(x_1 x_2 + u^2)}{(x_1^2 + u^2)(x_2^2 + u^2)},$$

with $x_1 + x_2 = 2\alpha - 1$ and $x_1 x_2 = (\alpha - \tfrac12)^2 - (b - \tfrac12)^2$. Every term on the line is positive, and a pair term is negative only if $u^2 < (b - \tfrac12)^2 - (\alpha - \tfrac12)^2$, which requires $b > \alpha$. □

The proof uses only the functional equation and the Hadamard product, so it holds for ξ and for every completed L function of a real primitive character. Platt [439] verified GRH for conductor 4 to height $H_0 = 2.5 \times 10^7$ (his height is $\max(10^8/q,\, 7.5 \cdot 10^7/q + 200)$ for even moduli q, which gives 8.3×10^6 for $q = 12$; the computation is rigorous), and RH is known to height 3×10^{12} [440].

COROLLARY 18.17. *For every $\alpha > \tfrac12$, $T_\alpha(y) > 0$ for $0 < \sqrt{y} < H_0 - \tfrac12$. If GRH fails, the set where $T_\alpha < 0$ lies in windows of half width less than $\tfrac12$ in $\sqrt{y}$, centred at ordinates of zeros with real part exceeding α, all above $H_0 - \tfrac12$. The same holds for ξ below 3×10^{12}.*

18.7.2 A comparison lemma

At $a = \alpha - \tfrac12$ and $u = t - \gamma$, a zero on the line contributes $a/(a^2 + u^2)$ to $\operatorname{Re} \Lambda'/\Lambda(\alpha + it)$, and an off line pair $\tfrac12 \pm B + i\gamma$ with $0 < B < \tfrac12$ contributes

$$Q_B(a, u) = \frac{a - B}{(a - B)^2 + u^2} + \frac{a + B}{(a + B)^2 + u^2}.$$

LEMMA 18.18 (Comparison). *Let $0 \le B < \tfrac12$ and $|u| \ge (1 + \sqrt{2})B$. Then $a \mapsto Q_B(a, u)/a$ is nonincreasing on $a > 0$. The constant $1 + \sqrt{2}$ is sharp.*

PROOF. With $w = a^2$, $c = u^2 - B^2$ and $c' = u^2 + B^2$, one finds $Q_B(a, u)/(2a) = g(w) = (w + c)/(w^2 + 2cw + c'^2)$ and

$$g'(w)\,(w^2 + 2cw + c'^2)^2 = c'^2 - 2c^2 - 2cw - w^2.$$

The hypothesis gives $c \ge 0$, so for $w \ge 0$ the right side is at most $c'^2 - 2c^2$, which is nonpositive exactly when $u^2 + B^2 \le \sqrt{2}(u^2 - B^2)$, that is $|u| \ge (1 + \sqrt{2})B$. If $(\sqrt{2} - 1)B < |u| < (1 + \sqrt{2})B$ then $g'(0) > 0$, so the constant $1 + \sqrt{2}$ cannot be lowered. □

THEOREM 18.19 (Comparison across vertical lines). *Fix t, and suppose every zero $\tfrac12 + B + i\gamma$ with $B > 0$ satisfies $|t - \gamma| \ge (1 + \sqrt{2})B$. Then*

$$\alpha \;\mapsto\; \frac{1}{\alpha - \tfrac12} \operatorname{Re} \frac{\Lambda'}{\Lambda}(\alpha + it) \quad \text{is nonincreasing on } \alpha > \tfrac12.$$

The hypothesis holds for every t under GRH, and unconditionally when t is at distance at least $(1+\sqrt{2})/2 = 1.2071$ from every off line ordinate, in particular for $|t| \le H_0 - 1.2071$. The same holds for ξ and for every completed L function of a real primitive character.

PROOF. Apply Lemma 18.18 to each group in (18.19); zeros on the line are the case $B = 0$. □

For β the ratio at $\alpha = 0.55,\ 0.6,\ 0.75,\ 1,\ 1.5,\ 2.5$ equals 0.15603, 0.15601, 0.15591, 0.15557, 0.15422, 0.14924 at $t = 0$, and 10.464, 9.846, 7.069, 3.840, 1.770, 0.838 at $t = 40$.

COROLLARY 18.20 (Explicit lower bound). *Under the hypothesis of Theorem 18.19, for $\delta > 0$ and $\frac{1}{2} < \alpha \le 1 + \delta$,*

$$2\pi t\, T_\alpha(t^2) = \operatorname{Re} \frac{\Lambda'}{\Lambda}(\alpha + it) \ \ge\ \frac{\alpha - \frac{1}{2}}{\frac{1}{2} + \delta}\Big(A_{1+\delta}(t) - S(1+\delta) \Big),$$

and the right side is $\frac{\alpha - \frac{1}{2}}{1+2\delta} \log t + O(1)$ as $t \to \infty$.

PROOF. Compare the lines α and $1 + \delta$ by Theorem 18.19, and bound the second by $A_{1+\delta}(t) - S(1+\delta)$. □

18.7.3 The minimum is on the real axis

THEOREM 18.21 (Minimum on the real axis). *(i) For every $\alpha > \frac{1}{2}$, unconditionally, $0 < c_\alpha \le (\alpha - \frac{1}{2})V$.*

(ii) For $\frac{1}{2} < \alpha \le 3.15$, unconditionally, $\operatorname{Re} \Lambda'/\Lambda(\alpha + it) > c_\alpha$ for $0 < |t| \le H_0 - 1.2071$.

(iii) GRH for β holds if and only if, for every $\alpha \in (\frac{1}{2}, 1]$, the minimum over $t \in \mathbb{R}$ of $\operatorname{Re} \Lambda'/\Lambda(\alpha + it)$ is c_α, attained only at $t = 0$; equivalently, $T_\alpha(y) > c_\alpha/(2\pi\sqrt{y})$ for all $y > 0$, so the Thorin density dominates its own stable part: the density U of (16.4) with $f = \Lambda$ and $b = c_\alpha$ is positive. For $\Lambda(s, \chi)$ alone with χ complex the analogue can fail [442, Remark 31].

PROOF. (i) Let $a = \alpha - \frac{1}{2}$. By (18.8), c_α is the sum of $a/(a^2 + \gamma^2)$ over the zeros on the line and of $Q_B(a, \gamma)$ over the off line pairs. Every ordinate has $|\gamma| \ge 6.02 > (1+\sqrt{2})B$, so by Lemma 18.18 each term divided by a is nonincreasing in a. Since $c_{1/2} = 0$, $V = \lim_{a \downarrow 0} c_{1/2+a}/a$, and by monotone convergence V is the sum of the termwise limits, each of which dominates the corresponding term of c_α/a.

(ii) Take $\delta = 2.15$. Since $A_{3.15}$ increases in t, $A_{3.15}(t) - S(3.15) > 2.65\,V$ exactly for $t > t_1 = 2.438$, so Corollary 18.20 and (i) give $\operatorname{Re} \Lambda'/\Lambda(\alpha + it) > aV \ge c_\alpha$ for $t_1 < |t| \le H_0 - 1.2071$ and $\alpha \le 3.15$. For $|t| \le t_1$, each symmetric pair of zeros

$\frac12 \pm i\gamma$ on the line contributes $f(t-\gamma)+f(t+\gamma)$ with $f(x)=a/(a^2+x^2)$, which is even and strictly convex on $|t|<\gamma-a/\sqrt3$ (compare Lemma 1 of [207]); this interval contains $[-t_1,t_1]$ because $\gamma\ge 6.0209 > t_1+2.65/\sqrt3 = 3.968$. Off line pairs, if any, have $|\gamma|>H_0$; the groups at γ and $-\gamma$ together contribute an even function of t, which is convex for $|t|\le t_1$ since $\partial_u^2 Q_B(a,u) = 12a\,u^{-4}\bigl(1+O(u^{-2})\bigr)$ uniformly for $B<\frac12$ and $a\le 2.65$. So the minimum over $|t|\le t_1$ is attained only at $t=0$.

(iii) Under GRH the hypothesis of Theorem 18.19 holds for every t, so (ii) holds with $H_0=\infty$. Conversely, the condition gives $\operatorname{Re}\,\Lambda'/\Lambda \ge c_\alpha > 0$ on $\frac12 < \operatorname{Re} s \le 1$, which excludes zeros there by the argument for (iii)⇒(i) in Theorem 18.10; $\operatorname{Re} s>1$ is zero free. □

The same argument applies to the other functions of this chapter, for $\frac12<\alpha\le 1+\delta$:

function	V	δ	t_1	$t_1+\frac{1/2+\delta}{\sqrt3}$	first ordinate	H_0
Λ	0.15603	2.15	2.438	3.968	6.0209	2.5×10^{7}
Λ_{12}	0.33292	1.30	1.684	2.724	3.8046	8.3×10^{6}
ξ	0.04621	3.75	9.235	11.689	14.1347	3×10^{12}

For Λ_{12} (Section 18.10) the archimedean term is $\frac12\log\frac{12}{\pi}+\frac12\operatorname{Re}\,\psi(\frac{\alpha+it}{2})$ and the prime sum runs over $(n,6)=1$. For ξ we use $\operatorname{Re}\,\xi'/\xi(\alpha+it)\ge -\frac12\log\pi+\frac12\operatorname{Re}\,\psi(\frac{\alpha+it}{2})+\frac{\zeta'}{\zeta}(\alpha)$ for $\alpha>1$, dropping $\operatorname{Re}(1/s+1/(s-1))\ge 0$. In each case the fifth column lies below the first ordinate. So, unconditionally, $\operatorname{Re}\,\xi'/\xi(\alpha+it)>\xi'/\xi(\alpha)$ for $\frac12<\alpha\le 4.75$ and $0<|t|\le 3\times10^{12}-1.21$. For $\alpha\ge1$ the minimum at $t=0$ is the theorem of Garunkštis [208] recalled in Section 18.5, stated for the infimum and valid at every height, so what the argument adds for ξ is the range $\frac12<\alpha<1$ below height 3×10^{12}. By Lagarias's theorem under RH [339] and the criterion of Hinkkanen [261] and Lagarias, RH is equivalent to the analogue of part (iii) for ξ.

18.8 The Thorin measure is the argument

THEOREM 18.22 (Argument form; for ξ, [441], Theorem 8.2). *Let $\alpha>\frac12$ with Λ zero free on $\operatorname{Re} s=\alpha$, and let $\tau_\alpha(dy)=T_\alpha(y)\,dy$, the Thorin measure when every zero of Λ has real part less than α and a signed measure in general. Then*

$$\tau_\alpha\bigl((0,t^2]\bigr)=\frac{1}{\pi}\arg\Lambda(\alpha+it),$$

the argument being continuous along $[\alpha,\alpha+it]$ and zero at α. As $t\to\infty$,

$$\tau_\alpha\bigl((0,t^2]\bigr)=\frac{t}{2\pi}\log\frac{2t}{\pi e}+\frac{\alpha}{4}+\frac{1}{\pi}\arg\beta(\alpha+it)+O(t^{-1}),$$

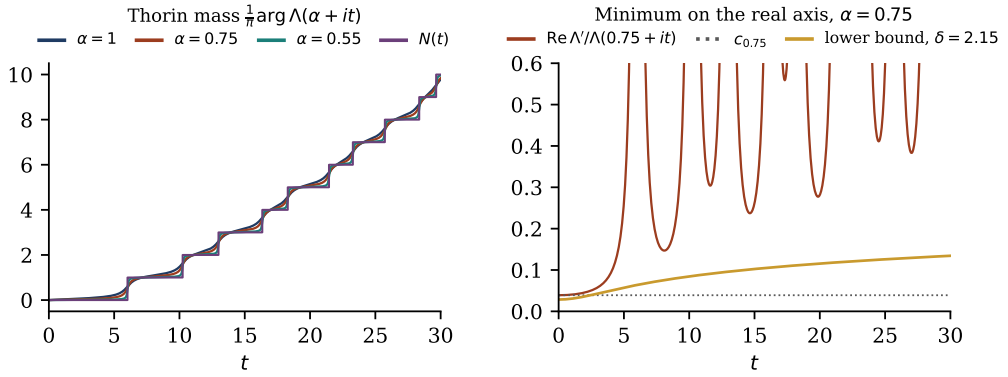


Figure 18.2: Left: cumulative Thorin mass $\tau_\alpha((0,t^2]) = \pi^{-1}\arg\Lambda(\alpha+it)$ for $\alpha = 1,\ 0.75,\ 0.55$ against the zero counting function $N(t)$ of β. Right: $\mathrm{Re}\ \Lambda'/\Lambda(0.75+it)$, its value $c_{0.75}$ at $t=0$, and the bound of Corollary 18.20 with $\delta = 2.15$, which exceeds $c_{0.75}$ for $t > 2.434$.

and under GRH $\tau_\alpha((0,t^2]) \to N(t)$ *as* $\alpha \downarrow \frac{1}{2}$, *where* $N(t)$ *counts zeros with* $0 < \gamma \le t$, *with half weight at an ordinate.*

PROOF. By (18.16), $T_\alpha(y)\,dy = \pi^{-1}\,\mathrm{Re}\ \Lambda'/\Lambda(\alpha+iu)\,du$ with $y = u^2$, and $\frac{d}{du}\arg\Lambda(\alpha+iu) = \mathrm{Re}\ \Lambda'/\Lambda(\alpha+iu)$. The asymptotic is Stirling's formula for $\mathrm{Im}\log\Gamma(\frac{\alpha+1+it}{2})$ plus $\frac{t}{2}\log\frac{4}{\pi}$, and the limit is that of Theorem 16.8(iv), whose Poisson kernels also give the half weight at an ordinate. □

Letting $\alpha \downarrow \frac{1}{2}$ under GRH gives $N(t) = \frac{t}{2\pi}\log\frac{2t}{\pi e} + \frac{1}{8} + \frac{1}{\pi}\arg\beta(\frac{1}{2}+it) + O(t^{-1})$, the Riemann von Mangoldt formula for $L(s,\chi_{-4})$ [276], which holds unconditionally when $\arg\beta(\frac{1}{2}+it)$ is defined as usual by continuous variation from $\arg\beta(2) = 0$ along the segments from 2 to $2+it$ to $\frac{1}{2}+it$. Under GRH the Thorin measure on the line α is the zero counting function smoothed by the Poisson kernel of width $\alpha - \frac{1}{2}$, and on a line to the right of every zero each zero ρ is smoothed with width $\alpha - \mathrm{Re}\,\rho$ (Figure 18.2, left). The two sides of the theorem agree to sixteen digits at $(\alpha,t) = (1.5,10), (1.5,25), (2,30)$, with values 1.65731, 7.40150, 9.85154.

COROLLARY 18.23. *GRH for* β *holds if and only if* $t \mapsto \arg\Lambda(\alpha+it)$ *is nondecreasing on every line* $\mathrm{Re}\,s = \alpha > \frac{1}{2}$ *on which* Λ *has no zero. The argument decreases exactly where* T_α *is negative, and by Theorem 18.16 that set is contained in a union of intervals of half width less than* $\frac{1}{2}$ *centred at ordinates of zeros with real part exceeding* α.

For ξ this is part of [441, Theorem 8.4].

18.9 THE CENTRE

An unconditional input at the centre would be a positive structure of the primal law that does not come from the zeros. One candidate is a scale mixture of $\Gamma(2)$ densities with positive mixing measure, which is infinitely divisible by Kristiansen's theorem [333] (Steutel's conjecture). For β neither the two sided nor the one sided form of this holds.

By Remark 16.4, whose argument is that of Roynette and Yor [467] for ξ, X_β is not infinitely divisible: its characteristic function $\Lambda(\frac12+it)/\Lambda(\frac12)$ vanishes at $t = 6.0209489$ (it is $+0.006077$ at $t = 5.9$ and -0.003508 at $t = 6.1$). Nor is $\Lambda(\frac12+w)/\Lambda(\frac12)$ a positive mixture $\int M(d\lambda)/(w+\lambda)^2$ of $\Gamma(2)$ transforms, since such a mixture decreases on $w > 0$ while $\Lambda(\frac12+w)$ increases there.

For β the one sided object is $\Phi_\beta 1_{\{u>0\}}$, whose Laplace transform

$$\int_0^\infty e^{-wu}\Phi_\beta(u)\,du = \frac12\int_1^\infty x^{(s-1)/2}\eta(ix)^3\,dx, \qquad s = \tfrac12 - w,$$

is an incomplete Mellin transform of η^3, the analogue of the piece from $x = 1$ to ∞ in Riemann's theta function proof of the functional equation of ζ [550]. It is not a $\Gamma(2)$ mixture either.

PROPOSITION 18.24. *Neither $\Phi_\beta(u)/u$ on $u > 0$ nor $\eta(ix)^3$ on $x > 0$ is completely monotone; in particular $\Phi_\beta 1_{\{u>0\}}$ is not a positive mixture of $\Gamma(2)$ densities.*

PROOF. A completely monotone $g = \int e^{-\lambda u}M(d\lambda)$ with $M \neq 0$ satisfies $g(u) \geq M([0,L])e^{-Lu}$ for any L with $M([0,L]) > 0$, so it decays at most exponentially, whereas $\Phi_\beta(u)/u$ decays like $\exp(\frac32 u - \frac{\pi}{4}e^{2u})$. By (18.3) and uniqueness of the Laplace transform, the representing measure of $\eta(ix)^3$ has the signed weights $(-1)^n(2n+1)$ at $\pi(2n+1)^2/4$. □

The decay argument applies to every degree one kernel, since all decay like $\exp(-ce^{2u})$: no Pólya kernel of a degree one L function, Riemann's kernel Φ of (13.1) included, is of the form $\int u^{a-1}e^{-\lambda u}M(d\lambda)$ on a half line with $M \geq 0$ and $a > 0$.

The positivity sits in the reciprocal of the product: with $q = e^{-2\pi x}$ and $\sigma_{-1}(N) = \sum_{d|N} d^{-1}$,

$$\frac{e^{-\pi x/4}}{\eta(ix)^3} = \prod_{m\geq1}(1-q^m)^{-3} = \exp\Big(3\sum_{N\geq1}\sigma_{-1}(N)\,e^{-2\pi Nx}\Big) \tag{18.20}$$

is the exponential of a completely monotone function, hence completely monotone. As in Thorin's theory, the representing measure of $\eta(ix)^3$ is signed while the reciprocal of its product part is the exponential of a positive mixture.

18.10 The Dedekind eta function and χ_{12}

Euler's pentagonal identity gives $\eta(ix) = \sum_{n\ge1}\chi_{12}(n)e^{-\pi n^2x/12}$, where χ_{12} is the real even primitive character modulo 12 with $\chi_{12}(\pm1)=1$ and $\chi_{12}(\pm5)=-1$. As in Section 18.1,

$$\Lambda_{12}(s) := \Big(\frac{12}{\pi}\Big)^{s/2}\Gamma\Big(\frac{s}{2}\Big)L(s,\chi_{12}) = \int_0^\infty x^{s/2-1}\eta(ix)\,dx,$$

so $\Lambda_{12}(\frac12+w) = 2\int_{\mathbb R} e^{wu}\Phi_{12}(u)\,du$ with the even, strictly positive kernel $\Phi_{12}(u) = e^{u/2}\eta(ie^{2u})$, and $\Lambda_{12}(s)=\Lambda_{12}(1-s)$. The Lévy measure for $\alpha>1$ is $dx/(1-e^{-2x}) + \sum_{n\ge2}\chi_{12}(n)\Lambda(n)\delta_{\log n}$, again with positive archimedean and signed prime parts, and with $K=\mathbb Q(\sqrt3)$ one has $\zeta_K = \zeta\cdot L(s,\chi_{12})$.

Theorem 18.25 (Real characters; after [442], Theorem 30 and Remark 32). *Let χ be a real primitive character with quadratic field K. Then $T_\alpha^\chi(y) = \mathrm{Re}(\Lambda_\chi'/\Lambda_\chi)(\alpha + i\sqrt y)/(2\pi\sqrt y) = T_\alpha^K(y) - T_\alpha^\zeta(y)$; by Theorem 16.8 the reciprocal $\Lambda_\chi(\alpha)/\Lambda_\chi(\alpha+\sqrt v)$ is a GGC transform for $\alpha\ge1$, and GRH for χ is equivalent to $T_\alpha^K \ge T_\alpha^\zeta$ for every $\alpha>\frac12$; and Theorems 18.16 and 18.19 hold for Λ_χ, as does Theorem 18.15 when the theta kernel of χ is positive.*

Proof. With d_K the discriminant, put

$$\xi_K(s) = \tfrac12 s(s-1)|d_K|^{s/2}(2\pi)^{-s}\Gamma(s)\zeta_K(s) \qquad \text{for imaginary } K,$$
$$\xi_K(s) = \tfrac12 s(s-1)|d_K|^{s/2}\pi^{-s}\Gamma(\tfrac s2)^2\zeta_K(s) \qquad \text{for real } K;$$

for $K=\mathbb Q(i)$ the first agrees with the definition above, since $4^{s/2}(2\pi)^{-s}=\pi^{-s}$. Then $\xi_K = c\,\xi\,\Lambda_\chi$ for a constant c (by the duplication formula for imaginary K, directly for real K); the factor $|d_K|^{s/2}$ is essential, since without it the ratio acquires a factor e^{ks} that shifts $\mathrm{Re}\,\xi_K'/\xi_K$ by a constant. The proofs above then apply to logarithmic derivatives. □

For χ_{12}, $L(\sigma,\chi_{12})$ is 0.125, 0.334, 0.499, 0.625, 0.721 at $\sigma=0.1,0.3,0.5,0.7,0.9$; the first ordinates are 3.8046, 6.6922, 8.8906, 11.1884; $V_{12}=0.33292$; the termwise threshold $\alpha_c(\chi_{12})=1.8349$ is the root of

$$\tfrac12\log\tfrac{12}{\pi} + \tfrac12\psi(\tfrac\alpha2) = -\frac{\zeta'}{\zeta}(\alpha) - \frac{\log2}{2^\alpha-1} - \frac{\log3}{3^\alpha-1};$$

and over $t\in[0,30]$ the minimum of $\mathrm{Re}\,\Lambda_{12}'/\Lambda_{12}(\alpha+it)$ is c_α, equal to 0.16508, 0.08306, 0.03328 at $\alpha=1,\,0.75,\,0.6$. The smaller threshold reflects the larger conductor term $\frac12\log\frac{12}{\pi}$ and the removal of the primes 2 and 3.

18.10.1 Open problems

OPEN PROBLEM 18.1 (A coupling below $\alpha = 1$). Construct the coupling $H_\alpha^K \stackrel{d}{=} H_\alpha^\zeta + H_\alpha^\beta$ of Theorem 18.13 from the Gaussian primes in a form that could survive below $\alpha = 1$. By Proposition 18.14 it must use the phases $\chi_{-4}(p)p^{-it}$, not only the sizes of the atoms. A first computation is the smallest margin of Re $\xi_K'/\xi_K -$ Re ζ'/ζ on lines $\frac{1}{2} < \alpha < 1$.

OPEN PROBLEM 18.2 (Negative windows). Find a lower bound for the decrease of $\arg \Lambda(\alpha + it)$ across the window of Theorem 18.16 caused by a pair $\frac{1}{2} \pm B + i\gamma$, in terms of B, α and the local density of zeros, that a mean value estimate for Re Λ'/Λ over intervals of length one could contradict.

OPEN PROBLEM 18.3 (All lines to the right). Theorem 18.21(ii) places the minimum over t of Re $\Lambda'/\Lambda(\alpha + it)$ at $t = 0$ for $\frac{1}{2} < \alpha \le 3.15$ and $|t| \le H_0 - 1.2071$. Extend it to every $\alpha > \frac{1}{2}$ at these heights, either by optimising δ as a function of α or by a direct argument for large α, where the prime part is small, and to every height for $\alpha \ge 1$. Numerically the minimum over $t \in [0, 40]$ is at $t = 0$ for $\alpha = 1.5$, 2, 3, 3.15 and 5. For ζ the statement at every height is Lagarias's theorem under RH [339], and unconditionally it is known for $\alpha \ge 1$ [208]; the arguments of Lagarias for $\alpha \ge 10$, of Garunkštis [207, 208] and of Broughan [86] near $\alpha = 1$ are natural routes for β.

OPEN PROBLEM 18.4 (Positivity on the reciprocal side). By (18.20) and its analogue for η, the functions $e^{-\pi x/4}/\eta(ix)^3$ and $e^{-\pi x/12}/\eta(ix)$ are completely monotone, with representing measures at the points $2\pi n$, $n \ge 0$, whose weights are the coefficients of $\prod_m (1 - q^m)^{-3}$ and $\prod_m (1 - q^m)^{-1}$. Decide whether this positivity passes, through incomplete Mellin transforms, to a positive representation on the reciprocal side of Λ or of Λ_{12}.

18.11 NOTES

The Thorin framework of this chapter is that of Chapters 14 and 16. For real primitive characters, χ_{-4} among them, the closed form of the Thorin density, the GGC property for $\alpha > 1$ and the equivalence of GRH with the GGC property at every $\alpha > \frac{1}{2}$ are Theorem 30 and Remark 32 of [442], and Theorem 16.8 gives them, with $\alpha \ge 1$, for every L function of Chapter 14; Theorems 18.7, 18.8 and 18.10, and the corresponding parts of Theorems 18.13 and 18.25, are their cases; the Lévy form of Theorem 18.4 is the computation of Section 16.4.3 for an odd character. The Mellin form of Proposition 18.1 is the theta series proof of the functional equation for odd characters [276], [442, Section 8.1] relates η^3 to β through its Laplace transform, and Jacobi's identity for η^3 and Euler's pentagonal identity for η are classical. Lemma 18.6 is the signed form of Lemma 16.7, which is Lemma 14

of [442] with a first step of Roynette and Yor [467]. The splitting $\zeta_K = \zeta\beta$ for $K = \mathbb{Q}(i)$ is classical, and the induction of the trivial character of K to $1 \boxplus \chi_{-4}$ is the simplest case of automorphic induction.

Proposition 18.14 rests on Kronecker's theorem and the linear independence of the $\log p$, as in Proposition 11.3 and in the value distribution theory of Bohr and Jessen [70]. For $1 < \alpha < \alpha_c \approx 2.28$ the atoms of β prove positivity only above the height $t_0(\alpha)$, and no bound that uses only their sizes does better; at $\alpha = 1$ the nonvanishing of β on $\operatorname{Re} s = 1$ is what excludes atoms from the Thorin measure, as for ξ in [442, Remark 21].

For ξ the minimum of $\operatorname{Re} \xi'/\xi$ on the line $\operatorname{Re} s = \alpha$ lies at $t = 0$ unconditionally for $\alpha \geq 10$ and, under RH, for every $\alpha > \frac{1}{2}$, attained only at $t = 0$ (Lagarias [339], Theorems 1.2 and 1.3 and Section 3); Garunkštis proved it unconditionally for $\alpha > 4.5$ [207] and for every $\alpha \geq 1$ [208], both stated for the infimum, and Lemma 2.5 and Theorem 3.2 of Broughan [86], with the verification of RH to height 3×10^{12} [440], give the strict inequality for every $t \neq 0$ when $\alpha \geq 1.043$. The criterion that RH is equivalent to $\operatorname{Re} \xi'/\xi > 0$ on $\operatorname{Re} s > \frac{1}{2}$ is due to Hinkkanen [261] and Lagarias [339]. In the notation of (16.4) the minimum at $t = 0$ is the positivity of the Thorin density U that remains when the $\frac{1}{2}$ stable part is removed, which [442, Remark 31] shows can fail for $\Lambda(s, \chi)$ alone with χ a complex character modulo 5. As far as we know the following are new in this form: Theorem 18.21 for β and its analogue for χ_{12}; for ξ, the unconditional range $\frac{1}{2} < \alpha < 1$ below height 3×10^{12}; and the comparison lemma with its sharp constant $1 + \sqrt{2}$. They are elementary consequences of the Hadamard product, the functional equation and the rigorous verifications of GRH for small conductors by Platt [439] and of RH by Platt and Trudgian [440], and none of them proves a Riemann hypothesis. The grouping (18.19) and its consequence, that $\operatorname{Re} \xi'/\xi(\sigma + it)$ can be negative only within distance $\frac{1}{2}$ of the ordinate of a zero off the line, are due to Levinson and Montgomery [362, Section 2], and Broughan [86] uses the same grouping; Corollary 18.17 reads them as a localisation of negative Thorin density, a quantitative form of the negative dip noted after Theorem 16.8.

For ξ, Theorem 18.22 and Corollary 18.23 are Theorem 8.2 and part of Theorem 8.4 of [441]. That X_β is not infinitely divisible is Remark 16.4, after Roynette and Yor [467]. The completely monotone reciprocal product (18.20) is a candidate for an unconditional input at the centre on the reciprocal side (Open Problem 18.4), not a proof that one exists.

Theorem 18.25 lists what extends to every real primitive character; the minimum on the real axis, Theorem 18.21, is established here only for χ_{-4}, χ_{12} and ξ, and Problem 18.1 asks how far it extends. Problem 18.2 poses the same questions for the Hecke characters of $\mathbb{Q}(i)$, whose L functions are those of $\mathrm{GL}(2)$ newforms with complex multiplication, to which Theorem 16.8 already applies.

18.12 Exercises

1. Prove Re $w/(1-w) \le r/(1-r)$ for $|w| = r < 1$, with equality only at $w = r$. Deduce Re $\beta'/\beta(\alpha+it) \ge -S(\alpha)$ for $\alpha > 1$, and check $S(2) = -\zeta'/\zeta(2) - \frac{1}{3}\log 2 = 0.339$.

2. Prove Lemma 18.18 from the identity for $g'(w)$ in its proof. Show that $g'(0) > 0$ exactly when $(\sqrt{2}-1)B < |u| < (1+\sqrt{2})B$, that $g'(B^2-u^2) > 0$ when $0 < |u| < B$, and that $Q_B(a,0)/a$ has a pole at $a = B$; conclude that $a \mapsto Q_B(a,u)/a$ is nonincreasing on $a > 0$ if and only if $|u| \ge (1+\sqrt{2})B$.

3. Show that $x \mapsto a/(a^2+x^2)$ is strictly convex exactly for $|x| > a/\sqrt{3}$, that $Q_B(a,u)$ is odd in a, and that $\partial_u^2 Q_B(a,u) = 12a\,u^{-4} - 40a(a^2+3B^2)u^{-6} + O(u^{-8})$. Where does the proof of Theorem 18.21 use each fact?

4. From Stirling's formula derive the asymptotic of Theorem 18.22 and the constant $\frac{1}{8}$ of the zero counting formula after it. What is the constant for $L(s,\chi_{12})$, whose gamma factor is $\Gamma(\frac{s}{2})$?

5. Show that a completely monotone $g = \int e^{-\lambda u} M(d\lambda)$ with $M \ne 0$ satisfies $g(u) \ge M([0,L])e^{-Lu}$ whenever $M([0,L]) > 0$, and deduce that no Pólya kernel of a degree one L function is a positive mixture of $\Gamma(a)$ densities on a half line, for any $a > 0$. Prove (18.20), compute the first five coefficients of $\prod_m (1-q^m)^{-3}$, and explain why the decay argument does not apply to $e^{-\pi x/4}/\eta(ix)^3$.

6. From Jacobi's identity (18.3) show that the theta series of χ_{-4} is $\eta(ix)^3$, and deduce from $\eta(i/x) = \sqrt{x}\,\eta(ix)$ that Φ_β is even and positive. Do the same for Λ_{12} and Φ_{12} with Euler's pentagonal identity.

7. For a real primitive character χ with quadratic field K verify $\xi_K = c\,\xi\,\Lambda_\chi$, with $c = 1$ for real K and $c = 1/(2\sqrt{\pi})$ for imaginary K, and show that omitting the factor $|d_K|^{s/2}$ changes ξ_K'/ξ_K by the constant $\frac{1}{2}\log|d_K|$. Which statements of this chapter need the theta kernel of χ to be positive?

8. Show that $\Lambda(1)/\Lambda(1+\sqrt{v})$ is a GGC transform, as the weak limit of the transforms at $\alpha \downarrow 1$, without using the nonvanishing of β on Re $s = 1$, and that a zero of β at $1+i\gamma$ would give the Thorin measure at $\alpha = 1$ an atom at γ^2.

18.13 Open computational problems

Each problem can be attacked numerically first, with the closed form (18.16), the argument form of Theorem 18.22 and the comparison bound of Corollary 18.20; a sign found numerically is proved only by a computation with error control.

COMPUTATIONAL PROBLEM 18.1 (Minimum on the real axis for real characters). Objective: for every real primitive character χ of conductor at most 10^3, compute $V_\chi = (\log \Lambda_\chi)''(\frac{1}{2})$, the termwise threshold $\alpha_c(\chi)$ (for χ_{-4} and χ_{12} it is 2.2823 and 1.8349) with the heights $t_0(\alpha)$ of Proposition 18.14, the comparison height $t_1(\chi)$ of Theorem 18.21 for the best δ, and the lowest ordinate $\gamma_1(\chi)$, and record whether the theta kernel of χ is positive, which Theorem 18.15 needs. List the characters with $t_1(\chi) + (\frac{1}{2} + \delta)/\sqrt{3} \geq \gamma_1(\chi)$, for which the convexity step in the proof of part (ii) fails near the real axis; χ_{-163} is one, since its lowest ordinate is $0.2029 < 1/(2\sqrt{3})$. Then test whether the minimum over t of $\operatorname{Re} \Lambda_\chi'/\Lambda_\chi(\alpha + it)$ is attained at $t = 0$ on every line $\frac{1}{2} < \alpha \leq 10$. Objects and scale: the real primitive characters of conductor at most 10^3, about six hundred; α on a grid of step 0.01; $|t| \leq 100$. Route: PARI/GP (`lfuninit`, `lfunlambda`, `lfunzeros`) for Λ_χ, its logarithmic derivative and its zeros, with the number of zeros checked by the argument principle; $\alpha_c(\chi)$ by root finding as in Section 18.10; ball arithmetic to prove the signs near the minima and for the characters near the failure boundary. Deliverable: a table by conductor and parity of V_χ, $\alpha_c(\chi)$, $t_1(\chi)$, $\gamma_1(\chi)$, the sign of the kernel and the smallest margin over t on each line; the exceptional characters, with a separate argument for part (ii) where one can be found; and fits of V_χ, $\gamma_1(\chi)$ and $\alpha_c(\chi)$ against $\log q$. Why it matters: it shows how far Theorem 18.21, the analogue for β of Lagarias's theorem [339], reaches among real characters as the conductor grows and the lowest zero comes down, on average, towards the real axis; only the signs proved in ball arithmetic are verified.

COMPUTATIONAL PROBLEM 18.2 (Hecke characters of the Gaussian field). Objective: for the Hecke characters [248] $\lambda_k((a)) = (a/|a|)^{4k}$ of $\mathbb{Q}(i)$, whose L functions are those of the newforms with complex multiplication of weight $4k+1$ and level 4, compute on lines $\alpha > \frac{1}{2}$ the Thorin density of Theorem 16.8, the termwise threshold $\alpha_c(\lambda_k)$ and the analogue of Proposition 18.14, the constants V and t_1 and the first ordinate, and test the minimum on the real axis of Theorem 18.21, after checking the root number and the sign of the Pólya kernel. Objects and scale: $k = 1, \ldots, 5$, weights 5 to 21; ordinates to height 10^3. Route: PARI/GP `mfinit` and `lfuncreate` for the newforms, or `gcharinit` for the characters, with `lfunzeros`; archimedean factor $\Gamma_{\mathbb{C}}(s + 2k)$ in the unitary normalisation; atoms $2\cos(4km\theta_p)\log p$ at p^m for split $p = \pi\bar{\pi}$ with $\theta_p = \arg \pi$, $2\log p$ at the even powers of inert p, and $(-1)^{km}\log 2$ at 2^m; ball arithmetic near the minima. Deliverable: for each k the threshold, V, t_1, the first ordinate and the minimum of $\operatorname{Re} \Lambda'/\Lambda(\alpha + it) - c_\alpha$ for $\alpha \in \{0.6, 0.75, 1\}$ over the scanned range, with the lines on which the minimum leaves the real axis, compared with the real characters of Problem 18.1. Why it matters: the atoms are cosines of the angles of Gaussian primes rather than signs, so Kronecker's theorem acts on angles, and the problem tests whether the minimum on the real axis is special to Dirichlet characters.

COMPUTATIONAL PROBLEM 18.3 (Synthetic off line zeros). Objective: insert a hypothetical pair $\frac{1}{2} \pm B + i\gamma$, with its conjugates, into the true zeros of β and measure (a) the decrease of $\arg \Lambda(\alpha + it)$ across the window of Theorem 18.16; (b) the distance from the off line ordinate beyond which complete monotonicity in the width survives: under GRH, $(\alpha - \frac{1}{2})^{-1} \operatorname{Re} \Lambda'/\Lambda(\alpha + it) = \sum_{\gamma'} 1/(w + (t - \gamma')^2)$ over the true ordinates γ', with $w = (\alpha - \frac{1}{2})^2$, is a Stieltjes function of w with representing function $\rho_t(x) = \sum_{\gamma'} e^{-x(t-\gamma')^2}$, while the pair adds $2e^{-cx} \cos \kappa x$ with $u = t - \gamma$, $c = u^2 - B^2$ and $\kappa = 2|u|B$, which changes sign, so the distance is the analogue of the constant $1 + \sqrt{2}$ of Lemma 18.18; and (c) the depth of the negative lobe of the signed density u_ε of Proposition 19.1 (whose A and B are $\gamma^2 - B^2$ and $2\gamma B$ in the notation here), and whether the neighbouring zeros mask it. Objects and scale: $L(s, \chi_{-4})$, its zeros to height 10^3, B from 10^{-3} to 0.4, ten values of γ, lines $\frac{1}{2} < \alpha \leq 1$. Route: the true Λ from PARI/GP `lfunlambda` times the Hadamard factors of the synthetic zeros; the closed forms of Theorem 18.22 and Lemma 18.18; $\rho_t(x)$ on a logarithmic grid $10^{-4} \leq x \leq 10^4$, with the tail of the zero sum bounded through the zero counting formula; ball arithmetic wherever a sign is claimed. Deliverable: the three quantities as functions of B, α (or ε) and the local density of zeros, and the smallest B detected at each ε. Why it matters: it measures how visible a zero off the line is in the Thorin measure, in its argument and in the signed densities of Chapter 19, which is what Open Problems 18.2 and 19.2 ask to turn into bounds; an experiment with synthetic zeros proves nothing about the true ones.

CHAPTER 19

SIGNED THORIN MEASURES AND THE ARGUMENT PRINCIPLE

By Theorem 14.8, the generalized Riemann hypothesis (GRH) for an automorphic L function holds if and only if its Thorin measure, with unit atoms at the squared ordinates γ_k^2 of its zeros, is a positive measure on $(0,\infty)$. This chapter computes a smoothed signed version of that measure, with no hypothesis on the zeros, by Stieltjes inversion of the logarithmic derivative of the clock $\Lambda(c+\sqrt{s})/\Lambda(c)$ at distance ε above its cut; positivity of the resulting density u_ε for every ε, together with the absence of real zeros, which u_ε does not see, is the Herglotz criterion of Theorem 14.8(e). We compute u_ε for ζ, $L(s,\chi_{12})$ and $L(s,\Delta)$, show how an off line pair of zeros appears in it and why one value of ε cannot exclude such a pair, and count zeros by the argument principle, which sees every zero at once. Such a positivity statement contains GRH and is not evidence for it; a finite computation tests it to a height, and none below is certified, since none has error control.

19.1 WHERE SCHOLZE'S WORK ENTERS

The L functions of this chapter are classical. For a cuspidal automorphic representation π of GL_n over $\mathbb{Q}$, the completed function $\Lambda(s,\pi)$, its continuation to an entire function of order one and its functional equation are due to Godement and Jacquet [217], after Hecke and Tate for $n=1$ [248, 539] and Hecke for modular forms [250, 251]; the conductor is that of Jacquet, Piatetski Shapiro and Shalika [285]. The archimedean factor is determined by the component π_∞ through Langlands's archimedean correspondence [352, 324]; for the L function of a motive it is fixed by the Hodge numbers and the action of complex conjugation [501].

Scholze's work enters on the Galois side. He attached Galois representations to torsion classes in the cohomology of the locally symmetric spaces of GL_n over totally real and CM fields [490], and gave a new proof [489] of the local Langlands correspondence for GL_n of Harris and Taylor and of Henniart [241, 257]. The torsion representations feed the potential automorphy theorems of Allen, Calegari, Caraiani, Gee, Helm, Le Hung, Newton, Scholze, Taylor and Thorne [16]: for an elliptic curve over a CM field, and more generally for a rank two compatible system of weight zero, every symmetric power L function has meromorphic continuation to $\mathbb{C}$ and a functional equation (Corollary 7.1.13 there).

A meromorphic continuation does not give the entire completed function that Theorem 14.8 needs, and a torsion class has no complex L function at all. Fargues and Scholze [183] attach to every irreducible smooth representation of a reductive group over a nonarchimedean local field a semisimple L parameter; for GL_n it is the semisimplification of the parameter of the local correspondence, and it does not determine the local L and ε factors (the trivial and the Steinberg representations of $\mathrm{GL}_2(\mathbb{Q}_p)$ share one).

None of this enters the computations below, nor does Deligne's Galois representation of Δ [139]. What the Galois side supplies is new L functions: once a representation of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ is shown to be cuspidal automorphic, its L function falls under Theorem 14.8 and the methods of this chapter.

19.2 The signed density of a clock

Let π be self dual, $\pi \cong \tilde{\pi}$, with $\Lambda(s,\pi)$ completed as in Section 14.2, the conductor factor $N_\pi^{s/2}$ included; here $c = \frac{1}{2}$, except that for Δ we keep the classical normalisation of Chapter 13, with centre $c = 6$. Let r be the order of $\Lambda(s,\pi)$ at $s = c$ and $\Xi_\pi^{\mathrm{sd}}(z) = \Lambda(c+z,\pi)/z^r$ as in Definition 14.5 (for Δ the definition carries a factor 2, which cancels in G_π); it is even, since $\varepsilon(\pi) = (-1)^r$ (Lemma 14.6(v)). The *clock* of π is

$$G_\pi(s) = \frac{\Xi_\pi^{\mathrm{sd}}(\sqrt{s})}{\Xi_\pi^{\mathrm{sd}}(0)},$$

an entire function of order at most $\frac{1}{2}$ with $G_\pi(0) = 1$ and $G_\pi^2 = F_\pi$; for ζ, with $\Lambda = \xi$, it is F_ζ itself. The three L functions of this chapter have root number $+1$ and $\Lambda(c,\pi) \neq 0$, so $r = 0$ and $G_\pi(s) = \Lambda(c+\sqrt{s},\pi)/\Lambda(c,\pi)$; we assume $r = 0$ from now on.

List the zeros of Ξ_π^{sd} in pairs $\pm i\gamma_k$ with multiplicity and put $t_k = \gamma_k^2$, as in (14.3). Then $G_\pi(s) = \prod_k (1 + s/t_k)$ and

$$\frac{G_\pi'}{G_\pi}(s) = \sum_k \frac{1}{s+t_k} = \frac{1}{2\sqrt{s}}\,\frac{\Lambda'}{\Lambda}(c+\sqrt{s},\pi). \tag{19.1}$$

Every zero $c + \delta + i\gamma$ of $\Lambda(s,\pi)$ has $|\delta| < \frac{1}{2}$ (Lemma 14.6(iii)). A zero on the line, $\delta = 0 < \gamma$, gives the atom $\gamma^2 > 0$; a zero with $\gamma > 0$ and $\delta \neq 0$, with its reflection $c - \delta + i\gamma$, gives the conjugate atoms $A \pm iB$, where $A = \gamma^2 - \delta^2$ and $B = 2\gamma|\delta|$; and a real zero $c \pm \beta$ gives the atom $-\beta^2$. By Theorem 14.8, with Ξ_π^{sd} for Ξ_π, GRH for π holds if and only if $1/G_\pi$ is the Laplace transform of a generalized gamma convolution (Definition 13.2); its Thorin measure is then $\tau_\pi = \sum_k \delta_{t_k}$, and $(G_\pi'/G_\pi)(s) = \int \tau_\pi(dt)/(s+t)$.

For $\varepsilon > 0$ and real z the *signed density* of π is

$$u_\varepsilon(z) = -\frac{1}{\pi}\,\mathrm{Im}\,\frac{G_\pi'}{G_\pi}(-z + i\varepsilon).$$

By (19.1) it is computed from Λ'/Λ alone, at points to the right of the critical line, and needs no hypothesis. Under GRH it is the Lorentzian smoothing of τ_π at width ε, which tends to τ_π as $\varepsilon \to 0$ (Stieltjes inversion [10]). The condition $u_\varepsilon \geq 0$ for all $\varepsilon > 0$ and real z says $\operatorname{Im}(G_\pi'/G_\pi) \leq 0$ on the upper half plane: it is the inequality of the Herglotz criterion of Theorem 14.8(e), with G_π for F_π, read on the lines $\operatorname{Im} s = \varepsilon$ (Problem 14.2 maps it on a grid); the criterion also excludes real zeros, which give positive Lorentzians (Proposition 19.1(iii)). Write $\sqrt{-z+i\varepsilon} = a + iy$ with $a, y > 0$, so that $z = y^2 - a^2$ and $\varepsilon = 2ay$, and let $L = (\Lambda'/\Lambda)(c + a + iy, \pi)$. Then

$$u_\varepsilon(z) = \frac{y \operatorname{Re} L - a \operatorname{Im} L}{2\pi(a^2+y^2)}, \qquad V_a(y^2) = \frac{\operatorname{Re} L}{2\pi y},$$

where V_a is the density of Theorem 16.8, formed with Ξ_π^{sd} for Ξ_π. So u_ε and V_a read Λ'/Λ on the same vertical line, at distance $a = \varepsilon/(2y)$ to the right of the centre; u_ε adds the term in $\operatorname{Im} L$ and the factor $y^2/(a^2+y^2)$. At $z = A$ the threshold $\varepsilon = B$ of the next proposition is the point $a + iy = |\delta| + i\gamma$, the condition $|\operatorname{Re} z_k| = a$ of Theorem 16.8, and the lobe is the dip of V_a described after it.

PROPOSITION 19.1 (The negative lobe). *Let π be as above, with $r = 0$, and $\varepsilon > 0$.*

(i) A zero $c + i\gamma$, $\gamma > 0$, of multiplicity m contributes to u_ε the Lorentzian $\frac{m}{\pi}\,\varepsilon/((z-\gamma^2)^2+\varepsilon^2)$, of mass m and height $m/(\pi\varepsilon)$, centred at γ^2.

(ii) A zero $c + \delta + i\gamma$ with $\gamma > 0$ and $\delta \neq 0$, together with its reflection, contributes

$$\frac{1}{\pi}\,\frac{B+\varepsilon}{(z-A)^2+(B+\varepsilon)^2} - \frac{1}{\pi}\,\frac{B-\varepsilon}{(z-A)^2+(B-\varepsilon)^2}.$$

For $\varepsilon < B$ this is negative exactly on $|z-A| < (B^2-\varepsilon^2)^{1/2}$ and equals $-2\varepsilon/(\pi(B^2-\varepsilon^2))$ at $z = A$; its integral over z is 0, and its absolute value is at most $2\varepsilon/(\pi(B-\varepsilon)^2)$.

(iii) A real zero $c \pm \beta$, $0 < \beta < \frac{1}{2}$, contributes a Lorentzian centred at $-\beta^2$, positive at every z.

(iv) If $\Lambda(\sigma, \pi) \neq 0$ for every real σ, then $u_\varepsilon(z) \geq 0$ for all real z and all $\varepsilon > 0$ (away from the poles of G_π'/G_π) if and only if every zero lies on the line. If moreover no zero has ordinate in $(0, \frac{1}{2}]$, the same holds with z restricted to $(0, \infty)$.

PROOF. An atom $t = A' + iB'$ contributes $-\pi^{-1}\operatorname{Im}(-z+i\varepsilon+t)^{-1} = \pi^{-1}(B'+\varepsilon)/((z-A')^2+(B'+\varepsilon)^2)$, a Lorentzian centred at A' of mass $\operatorname{sign}(B'+\varepsilon)$; with $B' = 0$ this gives (i) and (iii). In (ii) put $x = z - A$ and $f(b) = b/(x^2+b^2)$, so that the pair contributes $\pi^{-1}(f(b_1) - f(b_2))$ with $b_1 = B + \varepsilon$ and $b_2 = B - \varepsilon$. Since $f(b_1) - f(b_2) = (b_1 - b_2)(x^2 - b_1 b_2)/((x^2+b_1^2)(x^2+b_2^2))$, it has the sign of $x^2 - (B^2 - \varepsilon^2)$; its value at $x = 0$ is $\pi^{-1}(1/b_1 - 1/b_2)$, its two terms have masses 1 and -1, and $|f'(b)| \leq b^{-2}$ with the mean value theorem gives the bound. In (iv),

if every zero is on the line every term is positive. Conversely let $c+\delta+i\gamma$ be a zero of multiplicity m with $\delta \neq 0$; by hypothesis $\gamma \neq 0$, and we may take $\gamma > 0$. As $\varepsilon \uparrow B$ its pair contributes $-2m\varepsilon/(\pi(B^2-\varepsilon^2)) \to -\infty$ at $z = A$, while the other terms stay bounded near $(z,\varepsilon) = (A,B)$: a term is unbounded there only if its atom is $A - iB$, and the series converges uniformly there because $\sum_k |t_k|^{-1} < \infty$. So $u_\varepsilon(A) < 0$ for ε close to B. If no ordinate lies in $(0, \frac{1}{2}]$, then $\gamma > \frac{1}{2} > |\delta|$ and $A > 0$. □

At a fixed ε the sign of u_ε excludes nothing by itself. A pair with $B < \varepsilon$ contributes two positive Lorentzians, and a pair with $B > \varepsilon$ shows only if the other atoms contribute less than $2\varepsilon/(\pi(B^2-\varepsilon^2))$ near $z = A$. An atom on the line at A adds $1/(\pi\varepsilon)$ there, and for $B = 3\varepsilon$ the three Lorentzians sum to

$$\frac{3\varepsilon(x^4+2\varepsilon^2x^2+16\varepsilon^4)}{\pi(x^2+16\varepsilon^2)(x^2+4\varepsilon^2)(x^2+\varepsilon^2)} > 0, \qquad x = z - A,$$

so the pair is masked (Open Problem 19.2). By (ii) the contribution of each off line pair tends to 0 as $\varepsilon \to 0$: in the limit u_ε sees only the real atoms, whether or not GRH holds, and off line zeros are seen only at positive ε.

19.3 THREE CLOCKS

The computations use ζ with $\Lambda = \xi$; $L(s,\chi_{12})$, where χ_{12} is the even real character of $\mathbb{Q}(\sqrt{3})$, with $\Lambda(s,\chi_{12}) = (12/\pi)^{s/2}\Gamma(s/2)L(s,\chi_{12})$; and $L(s,\Delta)$, with $\Lambda(s,\Delta) = (2\pi)^{-s}\Gamma(s)L(s,\Delta)$ and $c = 6$. Their Pólya kernels are positive: Pólya's kernel (13.1); the kernel $\Phi_{12}(u) = e^{u/2}\eta(ie^{2u})$ of Section 18.10, built from η, the theta function of χ_{12} by Euler's pentagonal identity; and $\Delta(ie^u)e^{6u}$ (Proposition 13.4). So $\Lambda(c+x,\pi)$, a positive multiple of $\int \Phi(u)e^{xu}\,du$, is positive for real x, the hypothesis of Proposition 19.1(iv) holds, and the lowest ordinates, 14.13, 3.80 and 9.22, exceed $\frac{1}{2}$.

Positivity of the kernel holds for ζ, Δ and every multiplicative eta product (the paragraph after Lemma 14.12) and for χ_{-4} and χ_{12} (Chapter 18), but not in general. For the even real character χ of conductor 53 the kernel is negative at $u = 0$, since $\sum_{n\geq 1}\chi(n)e^{-\pi n^2/53} = -0.111$, and infinitely many even real characters have a theta function with a positive real zero [372]; the kernels of infinitely many imaginary quadratic fields change sign (Theorem 17.18); and for the eigenform f of weight 24 and level one with $a(2) = 540 - 12\sqrt{144169}$, whose root number is $+1$, the kernel $f(ie^u)e^{12u}$ is negative at $u = 0$, where $f(i) = -0.0095$. Open Problem 19.1 asks which kernels are positive.

We computed u_2 on a grid in z with mpmath at twenty digits: for ζ from the logarithmic derivative of ξ, for χ_{12} from the Hurwitz zeta functions $\zeta(s,b/12)$, and for Δ from Hecke's integral of Proposition 13.4, folded at $y = 1$, and its derivative

[250], with grid steps 0.875 for $T = 35$ and 1 for Δ with $T = 30$. Independently, sign changes of $t \mapsto \Lambda(c + it, \pi)$, refined by bisection, located the zeros on the line (Table 19.1).

Table 19.1: Zeros on the line below height T, and the minimum and mass of u_2 on $(0, T^2]$ (numerical, not certified).

clock	T	zeros on the line	$\min u_2$	mass of u_2
ζ, $c = \frac{1}{2}$	35	5	$2.4 \cdot 10^{-5}$	4.99
$L(s, \chi_{12})$, $c = \frac{1}{2}$	35	17	$4.0 \cdot 10^{-4}$	17.2
$L(s, \Delta)$, $c = 6$	30	8	$1.3 \cdot 10^{-4}$	7.98

On each range u_2 is positive. For ζ and Δ its minimum sits at the first grid point, since u_2 increases on $(0, t_1)$; for χ_{12} it is the trough at $z = 1158.5$ between the last atom below 35^2 and the first above. Every sampled peak lies within six per cent of $1/(2\pi) = 0.159$ and within half a grid step of an atom. The excess mass 17.2 for χ_{12} is leakage from the Lorentzians of the atoms above 35^2, mostly that of the zero at 35.027, and shrinks with ε (the mass is 17.14 at $\varepsilon = 1$). The ordinates found, whose squares are the atoms of τ_ζ, $\tau_{\chi_{12}}$ and τ_Δ, agree with PARI/GP [432] to 10^{-13}; the first of them are

$$\begin{aligned}
\tau_\zeta &: 14.134725,\ 21.022040,\ 25.010858,\ 30.424876,\ 32.935062,\\
\tau_{\chi_{12}} &: 3.804628,\ 6.692223,\ 8.890593,\ 11.188393,\ 12.966179,\ \ldots\\
\tau_\Delta &: 9.222379,\ 13.907550,\ 17.442777,\ 19.656513,\\
&\quad 22.336104,\ 25.274637,\ 26.804391,\ 28.831683.
\end{aligned}$$

To see the lobe of Proposition 19.1(ii), take the first five atoms of τ_ζ and move the third zero off the line by $\delta = 0.3$, so that $A = 625.45$ and $B = 15.01$. At $\varepsilon = 2$ the signed density of this toy clock has a negative lobe of depth $-5.7 \cdot 10^{-3}$ centred at $z = A$, close to the value $-2\varepsilon/(\pi(B^2 - \varepsilon^2))$ of the pair alone, and at $\varepsilon = 20 > B$ the lobe is gone. The test fires on this isolated pair; by the masking example it need not fire on a pair with an atom on the line near A.

For ζ the total variation distance on $(0, 35^2]$ between u_ε and the Lorentzian smoothing of the five atoms below 35^2 is 0.0062, 0.0031 and 0.0015 at $\varepsilon = 2$, 1 and $\frac{1}{2}$. It is the leakage into the window of the atoms $t_k > T^2$, since those below contribute equally to both,

$$\begin{aligned}
&\frac{1}{\pi} \sum_{t_k > T^2} \Big[\arctan \frac{t_k}{\varepsilon} - \arctan \frac{t_k - T^2}{\varepsilon} \Big] \\
&\qquad = \frac{\varepsilon}{\pi} \sum_{t_k > T^2} \Big[\frac{1}{t_k - T^2} - \frac{1}{t_k} \Big] + O(\varepsilon^3),
\end{aligned}$$

which for ζ at $T = 35$ is 0.00619, 0.00310 and 0.00155. So the two measures agree as $\varepsilon \to 0$, as they must once every zero below height 35 is on the line. The largest sampled values, 0.3175 at $\varepsilon = 1$ and 0.6300 at $\varepsilon = \frac{1}{2}$, are within about one per cent of $1/(\pi\varepsilon)$, so no peak carries a double atom.

19.4 Zero counts by the argument principle

The smoothed mass only approximates a count, and by Proposition 19.1 one value of ε can miss an off line pair; the argument principle sees every zero. Suppose that $\Lambda(s, \pi)$ has no real zero, and let $N_\pi(T)$ be the number of zeros $c + \delta + i\gamma$ with $0 < \gamma \le T$, on the line or not, counted with multiplicity. For $a > \frac{1}{2}$ and T not an ordinate these are the zeros inside the rectangle $R = c + [-a, a] \times i[0, T]$, and

$$N_\pi(T) = \frac{1}{2\pi} \operatorname{var}_{\partial R} \arg \Lambda(s, \pi),$$

the change of $\arg \Lambda$ around ∂R divided by 2π, which needs no derivative. Computed from samples, it is guaranteed to be right if $\arg \Lambda$ moves by less than π between consecutive samples: the phase steps, each reduced to $(-\pi, \pi]$, add up around a closed contour to 2π times an integer whatever the sampling, so an integer output is no check by itself. If $N_\pi(T)$ equals the number of sign changes of the real function $t \mapsto \Lambda(c + it, \pi)$ on $(0, T]$, every zero of height at most T is on the line and simple. Table 19.2 gives the counts for $a = 0.9$.

Table 19.2: Sign changes of $\Lambda(c + it)$ on $(0, T]$ and winding numbers of Λ around ∂R, from 400 samples per side at twelve digits (numerical, not certified).

clock	T	sign changes of $\Lambda(c + it)$	winding number
ζ	35	5	5
ζ	200	79	79
$L(s, \chi_{12})$	35	17	17
$L(s, \chi_{12})$	100	67	67
$L(s, \Delta)$	30	8	8

In every row the two counts agree, and the largest phase step between samples is 0.39π, for ζ at $T = 200$. With 100, 50 and 25 samples per side the same code returns the integers 35, -5 and -1 there. So, numerically, every zero of ζ below height 200, of $L(s, \chi_{12})$ below 100 and of $L(s, \Delta)$ below 30 is on the line and simple. This is not a certificate, which would need interval arithmetic with a bound for $|\Lambda'/\Lambda|$ on each side (Problem 19.2) or Turing's method [554, 75]. That these zeros lie on the line is known: Backlund reported that the 79 zeros of ζ below height 200 lie on the line [30], a statement contained in later verifications [440], Rumely verified GRH for the primitive characters of conductor at most 13 to height 10^4

[472], and Hiary, Ireland and Kyi verified that every zero of $L(s,\Delta)$ up to height 84 is on the line [259, Section 7.3]. The counts agree with the main terms of the Riemann von Mangoldt formula and its analogues [550, 276]: 79.19, 67.55 and 7.97 for $T = 200$, 100 and 30.

The Δ row needs care. On the line $|\Lambda(6+it,\Delta)|$ carries the factor $t^{11/2}e^{-\pi t/2}$ and falls to $5\cdot 10^{-17}$ at $t = 30$ and $2.6\cdot 10^{-36}$ at $t = 60$, against an integrand of size 10^{-3} in Hecke's integral; at twelve digits about two correct digits remain at $t = 30$ and none from $t = 35$ on. The winding number is still 8, but the zero search is not reliable: at twelve digits, grids of 100 to 400 points on $(0, 30]$ find 8 sign changes and grids of 600 to 1200 points find two spurious ones between the zeros at 25.27 and 26.80. The grid of the run in the table is not recorded; the eight ordinates of τ_Δ come from the run at twenty digits, where each of seventeen grids of 100 to 3000 points gives them. At fifty digits one evaluation at $t = 60$ takes about 0.57 seconds, about 15 minutes for a contour of 1600 points. Beyond $T \approx 30$ the clock of Δ is better computed from the Ramanujan coefficients through a smoothed approximate functional equation, with about $t/(2\pi)$ terms at height t, for instance as in Rubinstein's method [469, Section 3.3], where a factor w^{-s} with $|w| = 1$ offsets the exponential decay of the gamma factor; Problem 13.1 asks for the zeros of $\Lambda(6+it,\Delta)$ to height 10^5, with the count certified by Turing's method.

19.4.1 Open problems

OPEN PROBLEM 19.1 (Positive Pólya kernels). Characterise the self dual L functions with root number $+1$ whose Pólya kernel is positive (Section 19.3); for a form of level one, positivity fails as soon as the form has a zero of odd order on the imaginary axis. Is positivity, at fixed degree, decided by finitely many coefficients and the conductor?

OPEN PROBLEM 19.2 (Masking of the negative lobe). Find conditions on the zeros near height γ under which $u_\varepsilon \geq 0$ at one ε excludes a zero $c+\delta+i\gamma$ with $2\gamma|\delta| > \varepsilon$; equivalently, compare the depth $2\varepsilon/(\pi(B^2-\varepsilon^2))$ of the lobe of an off line pair with the contribution of the neighbouring atoms near $z = A$ (compare the dip of V_a after Theorem 16.8).

19.5 NOTES

Theorem 14.8 extends [442], and the densities V_a of Theorem 16.8 are, for ξ, its Theorem 18; Stieltjes inversion is classical [528, 10]. Proposition 19.1, the masking identity and the leakage formula are elementary consequences of (19.1); for ξ the Poisson kernel smoothing of the ordinates and its limit are in [442], and [443] shows that pointwise positivity of the Thorin density of $\xi(\sigma)/\xi(\sigma+s)$ forces every extremal zero onto the critical line, with an explicit negative value of the density

when the first zero is off the line. The computations of Sections 19.3 and 19.4 were made for this book. Pólya's kernel is in [444], the kernel of χ_{12} comes from η through Euler's pentagonal identity (Section 18.10), and that of Δ is Hecke's Mellin transform [250]; the values for the character of conductor 53 and the eigenform of weight 24 were computed for this book. RH and GRH are verified far beyond the heights of Table 19.2 for ζ and χ_{12} [440, 439], and for Δ to height 84 [259]; Turing's method is in [554], and its extension to other L functions in [75]. The scripts are in `scholze/` (Appendix B); PARI/GP [432] gave independent zero lists and values.

19.6 Exercises

1. Derive the formulas for $u_\varepsilon(z)$ and $V_a(y^2)$ in Section 19.2, and show that $u_\varepsilon(z) - V_a(y^2) = -a(a \operatorname{Re} L + y \operatorname{Im} L)/(2\pi y(a^2+y^2))$.
2. Add an atom on the line at A to the pair of Proposition 19.1(ii), and show that the sum is negative at $z = A$ if and only if $B/\sqrt{3} < \varepsilon < B$. Verify the masking identity for $B = 3\varepsilon$.
3. Derive the closed form of the leakage in Section 19.3, and show that its expansion in ε has no term of order ε^2.
4. Let Λ be holomorphic and zero free near a closed polygon P, with $|\Lambda'/\Lambda| \le M$ on P. Show that sampled phase differences, reduced to $(-\pi, \pi]$, add up to 2π times an integer, and that this integer is the winding number of Λ around P when consecutive samples are less than π/M apart along P.
5. Let f be the eigenform of weight 18 and level one. From $f(-1/\tau) = \tau^{18} f(\tau)$ show that $f(i) = 0$, that the kernel $f(ie^u)e^{9u}$ is odd, and that $\Lambda(9+z, f)/z$ is even.

19.7 Open computational problems

Computational Problem 19.1 (Signs of Pólya kernels across degrees). Objective: decide for self dual L functions of degree at most four with root number $+1$ whether the Pólya kernel of Ξ_π^{sd} (Lemma 14.12) is positive, and where it first changes sign. Objects and scale: the L functions of degree at most 4 and conductor at most 10^3 in the LMFDB [371] (newforms, symmetric squares, Rankin Selberg products, Hilbert modular forms over $\mathbb{Q}(\sqrt{5})$), and the eigenforms of level one and weight up to 100 with root number $+1$. Route: the kernel as the inverse Fourier transform of $\Lambda(c+it, \pi)/\Lambda(c, \pi)$ from PARI/GP `lfun` [432], or from a theta series where one exists; signs proved in ball arithmetic, positivity on a grid with bounds for the derivative and the tails. Deliverable: a table by degree, weight and conductor giving positivity proved on $\mathbb{R}$, or a negative value with

its location and error bound. Why it matters: a positive kernel is the hypothesis of Proposition 14.13 and Theorem 16.6, and makes Λ positive on the real axis, as Proposition 19.1(iv) needs; Problems 17.3 and 17.4 ask the same for quadratic fields.

COMPUTATIONAL PROBLEM 19.2 (Zero counts by the argument principle in ball arithmetic). Objective: turn the sampled counts of Table 19.2 into proved counts. Objects and scale: ζ, $L(s, \chi_{12})$ and $L(s, \Delta)$ to height $T = 200$, the last through the approximate functional equation. Route: Λ on ∂R in ball arithmetic, with adaptive subdivision until the change of $\arg \Lambda$ between consecutive points is provably below π; Turing's method [554, 75] as an independent check. Deliverable: proved values of $N_\pi(T)$ for $T \leq 200$ and the number of evaluations each needs. Why it matters: it replaces the sampled counts of this chapter by proofs; the counts for ζ and χ_{12}, and for Δ to height 84, are known [440, 472, 259], so the value lies in a method that needs only values of Λ and carries over to the L functions of Problem 19.1.

COMPUTATIONAL PROBLEM 19.3 (Window leakage of signed Thorin densities). Objective: test the leakage formula of Section 19.3, and use the leakage of u_ε near the edge of a window to recover the first atoms above it. Objects and scale: ζ and $L(s, \chi_{12})$ with $T \leq 35$, $L(s, \Delta)$ with $T \leq 30$, and $\varepsilon \in \{2, 1, \frac{1}{2}, \frac{1}{4}\}$. Route: u_ε from `scholze/signed_thorin.py`, minus the Lorentzian smoothing of the atoms below T^2, fitted on $(0, T^2]$ by least squares with Lorentzians at unknown atoms above T^2 and the smooth density of zeros for the far tail. Deliverable: the measured leakage against the closed form for each ε, the recovered atoms against the true ones, and how many atoms above T^2 the fit determines to a given accuracy. Why it matters: to first order in ε the total leakage is one weighted sum over the atoms above T^2, so single atoms must come from the shape of the remainder; the answer says how far a signed density on a finite window sees beyond it.

CHAPTER 20

RECIPROCITY THROUGH THE SWAP: THORIN MEASURES, WALD COUPLES AND VAN DANTZIG DUALITY

Langlands reciprocity asserts that two very different kinds of object carry the same information: a Galois representation ρ, known through the conjugacy classes of its Frobenius elements, one prime at a time, and an automorphic representation π, a harmonic object on an adelic quotient, known through Hecke eigenvalues, Fourier coefficients and spectral data. Reciprocity says that $L(s,\rho) = L(s,\pi)$. The difficulty is that Frobenius data are local and arithmetic, while automorphy is a global symmetry visible only to harmonic analysis.

This chapter proposes to bring the Thorin framework of Chapters 14 to 16 to bear on reciprocity, starting from one observation: the completed L function is a single function with two faces. On the Euler half line $\operatorname{Re} s > 1$ it is an Euler product, a statement about primes, the Galois side; on the critical line it is the Mellin transform of an automorphic form, the automorphic side. Three devices from those chapters pass between the faces: the explicit formula with the $2\sin^2$ transport of Lemma 16.7, which swaps the primes for the zeros; the van Dantzig involution, which exchanges the critical line with the real axis; and the Wald couple of Definition 16.3, which under GRH pairs a variable built from the automorphic form with a variable built from the Euler product. Read through them, modularity at level one becomes a symmetry statement about a Pólya kernel, and holomorphy together with GRH becomes a positivity statement about a measure and a continuation statement about generalized gamma convolutions.

Before the proposal, three sections restate parts of reciprocity in the language of probability. Section 20.2 derives quadratic reciprocity, Gauss's sign and the functional equations of even Dirichlet L functions from the characteristic function of the normal law, and proves the theta transformation a second time by Fejér positivity: the swap for the circle, the model for everything after it. Section 20.3 puts Galois and automorphic representations on one probability space, where reciprocity is almost sure equality and its uniqueness statements are Hilbert space inequalities. Section 20.4 replaces that space by the zeta distributed random integer, whose Frobenius labelled factorisation converges to a Gamma process; reciprocity laws become the vanishing of Thorin measures, and the drift of the split primes is Ihara's Euler Kronecker constant over the degree, up to finitely

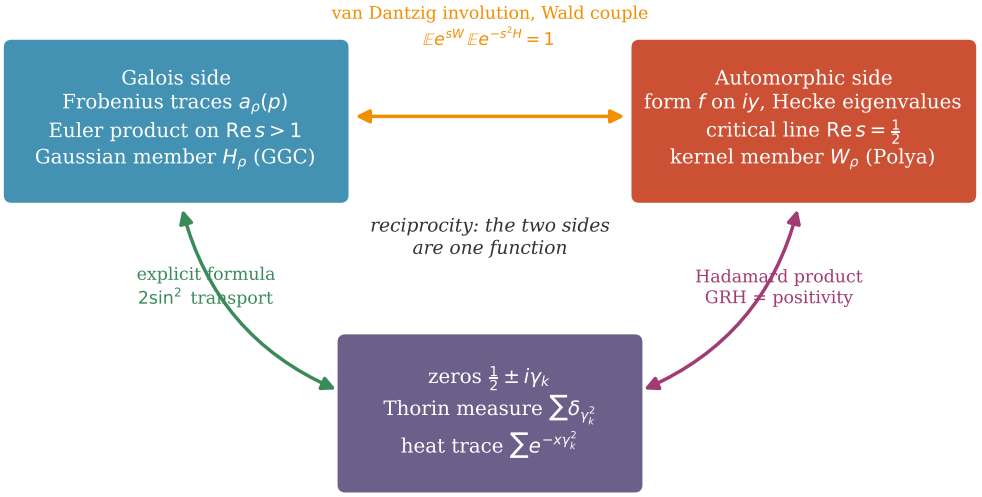


Figure 20.1: One function, two faces. The Euler product on $\mathrm{Re}\,s > 1$ carries the Galois data, the Mellin transform on the critical line carries the automorphic form, and the zeros sit between them. The explicit formula with the $2\sin^2$ transport, the Hadamard product, and the van Dantzig involution with its Wald couple pass between the three.

many Euler factors and a prime power term.

We do not prove new cases of reciprocity. We prove the results that make the proposal precise: a symmetry criterion for modularity at level one; a Gamma limit law that detects irreducibility; a complete monotonicity criterion for holomorphy together with GRH, computed from Frobenius traces, the conductor and the gamma factor, with which finitely many certified inequalities can disprove them but are not known to verify them to any finite height; a rigidity statement showing that the swap sees every single prime; and a Thorin form of Hecke's converse theorem. We test the criteria numerically on a two dimensional Artin representation of conductor 23 (Example 3.3) and on Ramanujan's Δ. Proposition 1.5 showed that no argument which uses only the limiting distribution of Frobenius data can prove reciprocity; the criteria of Sections 20.5, 20.7 and 20.8 see the individual primes, while the limit laws of Sections 20.3, 20.4 and 20.6 do not. Figure 20.1 summarises the scheme.

20.1 THE RECIPROCITY PROBLEM

Let ρ be an Artin representation of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ on an n dimensional complex vector space V or, more generally, a geometric ℓ adic representation on an n dimensional $\overline{\mathbb{Q}}_\ell$ vector space V, pure of weight w, made complex through a fixed isomorphism $\overline{\mathbb{Q}}_\ell \cong \mathbb{C}$. Its gamma factor $\gamma(s,\rho)$ is determined by the Hodge numbers and the action of complex conjugation, and N_ρ is its conductor. For p unramified put $a_\rho(p^m) = \operatorname{tr}\rho(\mathrm{Frob}_p^m)$. In the unitary normalisation (s shifted by $w/2$; at $p=\ell$ the local factor is that of the associated Weil Deligne representation) the Euler product

$$L(s,\rho) = \prod_p \det\bigl(1-\rho(\mathrm{Frob}_p)p^{-s} \mid V^{I_p}\bigr)^{-1}$$

converges absolutely for $\operatorname{Re} s > 1$, and $\Lambda(s,\rho) = N_\rho^{s/2}\gamma(s,\rho)L(s,\rho)$ is its completion (the factor $N_\rho^{s/2}$ is kept out of $\gamma(s,\rho)$, unlike in Appendix A). Everything in $\Lambda(s,\rho)$ for $\operatorname{Re} s > 1$ is Galois data: Frobenius traces, the conductor and the gamma factor. For Artin representations, Brauer's induction theorem [84] gives meromorphic continuation and the functional equation $\Lambda(s,\rho) = \varepsilon(\rho)\Lambda(1-s,\check\rho)$; Artin's conjecture [27] asserts that $\Lambda(s,\rho)$ is entire when ρ has no trivial constituent.

CONJECTURE 20.1 (Langlands reciprocity [350]). *For every irreducible ρ as above there is a cuspidal automorphic representation π of $\mathrm{GL}_n(\mathbb{A}_\mathbb{Q})$ with $L(s,\rho_p) = L(s,\pi_p)$ for every place p.*

For $n=1$ this is class field theory (Artin's reciprocity law [28]). For $n=2$ it is known for Artin representations with solvable image, odd or even, by Hecke, Langlands and Tunnell [351, 553], for all odd ones as a consequence of Serre's conjecture, proved by Khare and Wintenberger [311, 312] with work of Kisin [320], and for elliptic curves over $\mathbb{Q}$ by Wiles, Taylor and Wiles, and Breuil, Conrad, Diamond and Taylor [581, 541, 41]. Potential automorphy [266, 67] gave meromorphic continuation and functional equations for the symmetric powers of holomorphic newforms, and Newton and Thorne [419, 420] proved them automorphic for newforms of weight at least two without complex multiplication (Chapter 3). In dimension two the even icosahedral case is open.

The analytic route to reciprocity runs through converse theorems. Hecke [250] showed that a Dirichlet series whose completion is entire, bounded in vertical strips and satisfies the level one functional equation comes from a modular form for $\mathrm{SL}_2(\mathbb{Z})$ (Section 2.3). Weil [571] added twists by Dirichlet characters to reach $\Gamma_0(N)$; Jacquet and Langlands [277] recast this for GL_2, and Cogdell and Piatetski Shapiro [108] extended it to GL_n, with twists by cuspidal representations π' of GL_m, $m \le n-1$. So, once the local Langlands correspondence has fixed the local components of the candidate π, reciprocity for ρ follows from *analytic* properties of the twisted L functions $L(s,\rho\times\pi')$: continuation, holomorphy, boundedness

and functional equations (Open Problem 30.4 asks for a probabilistic counterpart of these theorems). Booker [75] turned this into a computation, verifying Artin's conjecture up to a finite height for some nonmonomial representations with A_5 and S_5 image (Section 20.7).

20.2 Quadratic reciprocity from the Gaussian law

For quadratic characters the analytic content of reciprocity is one probabilistic fact: the characteristic function of a normal law is a Gaussian. Here it is applied to a single random variable, the position at a fixed time of a Brownian motion wrapped around a circle. None of the theorems of this section is new, apart from the explicit bounds (i) and (ii) of Theorem 20.10, which are elementary; the route shows which part of the classical reciprocity laws is probability.

Lemma 20.2 (wrapped Gaussian). *For $x > 0$ and real y,*

$$\sum_{k\in\mathbb{Z}} e^{-xk^2+iky} = \sqrt{\frac{\pi}{x}} \sum_{n\in\mathbb{Z}} e^{-(y+2\pi n)^2/(4x)}.$$

Proof. Let $Y \sim N(0, 2x)$, with density $\varphi(y) = (4\pi x)^{-1/2} e^{-y^2/(4x)}$, and let $\varphi_{2\pi}(y) = \sum_n \varphi(y + 2\pi n)$ be the density of Y reduced modulo 2π. The series converges uniformly on compact sets, so $\varphi_{2\pi}$ is continuous and 2π periodic, and its Fourier coefficients are $\int_0^{2\pi} \varphi_{2\pi}(y) e^{-iky} dy = \mathbb{E}e^{-ikY} = e^{-xk^2}$. These are absolutely summable, so the Fourier series of $\varphi_{2\pi}$ converges uniformly, and by Fejér's theorem its sum is $\varphi_{2\pi}$: $\varphi_{2\pi}(y) = (2\pi)^{-1} \sum_k e^{-xk^2} e^{iky}$ for every y. Multiplying by 2π gives the lemma. □

Put $\theta(\tau) = \sum_{n\in\mathbb{Z}} e^{\pi i n^2 \tau}$ for $\operatorname{Im}\tau > 0$.

Theorem 20.3 (theta transformation). *$\theta(-1/\tau) = \sqrt{\tau/i}\,\theta(\tau)$, with the principal branch of the square root.*

Proof. Lemma 20.2 with $y = 0$ and $x = \pi t$ gives $\theta(it) = t^{-1/2}\theta(i/t)$ for $t > 0$, which is the claim on the imaginary axis, since $-1/(it) = i/t$ and $\sqrt{it/i} = \sqrt{t}$. Both sides are holomorphic on the upper half plane, because τ/i has positive real part there, so they agree everywhere. □

We need one elementary estimate. If $h \in C^1(\mathbb{R})$ with h and h' integrable, then for $t_0 \in \mathbb{R}$ and $\eta > 0$,

$$\Big|\eta \sum_{m\in\mathbb{Z}} h(t_0 + m\eta) - \int_{\mathbb{R}} h\Big| \le \eta \int_{\mathbb{R}} |h'|, \tag{20.1}$$

because on each interval $[t_0 + m\eta, t_0 + (m+1)\eta]$ the difference between $\eta h(t_0 + m\eta)$ and the integral of h is at most η times the integral of $|h'|$ over that interval.

For integers $a, c \geq 1$ with ac even put

$$S(a,c) = \sum_{n \bmod c} e^{\pi i a n^2/c} = c\,\mathbb{E}e^{\pi i a U^2/c}, \qquad U \text{ uniform on } \mathbb{Z}/c\mathbb{Z}.$$

The summand depends only on n modulo c, because $a(n+c)^2/c = an^2/c + 2an + ac$ and ac is even.

THEOREM 20.4 (reciprocity for Gauss sums; Schaar, Landsberg). *For integers $a, c \geq 1$ with ac even,*

$$\frac{1}{\sqrt{c}}\,S(a,c) = \frac{e^{\pi i/4}}{\sqrt{a}}\,\overline{S(c,a)}.$$

PROOF. Let $\tau = a/c + i\varepsilon$ with $\varepsilon > 0$; we let $\varepsilon \downarrow 0$, so that $\theta(\tau)/\theta(i\varepsilon)$ is the characteristic function of the square of a discrete Gaussian of variance of order ε^{-1}, evaluated at $\pi a/c$. Grouping n by its residue r modulo c,

$$\theta(\tau) = \sum_{r \bmod c} e^{\pi i a r^2/c} \sum_{m \in \mathbb{Z}} e^{-\pi\varepsilon(r+mc)^2}.$$

By (20.1) with $h(u) = e^{-\pi u^2}$ and $\eta = c\sqrt{\varepsilon}$, $\sqrt{\varepsilon}\sum_m e^{-\pi\varepsilon(r+mc)^2} \to 1/c$, so $\sqrt{\varepsilon}\,\theta(\tau) \to S(a,c)/c$: the discrete Gaussian reduced modulo c becomes uniform.

On the other side, $-1/\tau = -c/a + w$ with $w = i\delta/(1+i\kappa)$, $\delta = c^2\varepsilon/a^2$ and $\kappa = c\varepsilon/a$. Hence $e^{\pi i n^2(-1/\tau)} = e^{-\pi i c n^2/a} e^{-\pi\delta\beta n^2}$ with $\beta = (1+i\kappa)^{-1}$, and the first factor depends only on n modulo a. Grouping by residues modulo a and applying (20.1) to $h_\beta(u) = e^{-\pi\beta u^2}$ with $\eta = a\sqrt{\delta}$, where $\int |h_\beta'|$ is bounded for $|\kappa| \leq 1$ and $\int h_\beta = \beta^{-1/2} \to 1$, we get $\sqrt{\delta}\,\theta(-1/\tau) \to \overline{S(c,a)}/a$. Since $\sqrt{\delta} = (c/a)\sqrt{\varepsilon}$, this says $\sqrt{\varepsilon}\,\theta(-1/\tau) \to \overline{S(c,a)}/c$.

Now multiply the identity of Theorem 20.3 by $\sqrt{\varepsilon}$ and let $\varepsilon \downarrow 0$. Since $\sqrt{\tau/i} \to \sqrt{a/c}\,e^{-\pi i/4}$, we get $\overline{S(c,a)}/c = \sqrt{a/c}\,e^{-\pi i/4} S(a,c)/c$, which is the theorem. □

COROLLARY 20.5 (Gauss's sign). *For odd $m \geq 1$, $g(m) = \sum_{n \bmod m} e^{2\pi i n^2/m}$ equals $\sqrt{m}$ if $m \equiv 1 \bmod 4$ and $i\sqrt{m}$ if $m \equiv 3 \bmod 4$.*

PROOF. Take $a = 2$, $c = m$: $g(m) = S(2,m) = \sqrt{m/2}\,e^{\pi i/4}(1 + e^{-\pi i m/2})$. For $m \equiv 1 \bmod 4$, $1 + e^{-\pi i m/2} = 1 - i = \sqrt{2}e^{-\pi i/4}$; for $m \equiv 3 \bmod 4$ it is $1 + i = \sqrt{2}e^{\pi i/4}$. □

Gauss conjectured the sign in 1801, found a proof only in 1805 and published it in 1811 [210]. Here it is the limit point of the curve $\varepsilon \mapsto \sqrt{\varepsilon}\,\theta(2/m + i\varepsilon)$ (Figure 20.2, left).

COROLLARY 20.6 (quadratic reciprocity). *For distinct odd primes p, q,*

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}\frac{q-1}{2}}, \qquad \left(\frac{-1}{p}\right) = (-1)^{\frac{p-1}{2}}, \qquad \left(\frac{2}{p}\right) = (-1)^{\frac{p^2-1}{8}}.$$

PROOF. For $p \nmid k$ put $g_k(p) = \sum_{n \bmod p} e^{2\pi i k n^2/p}$. Since $n^2 \equiv b$ has $1 + (\frac{b}{p})$ solutions, $g_k(p) = \sum_b (\frac{b}{p}) e^{2\pi i k b/p} = (\frac{k}{p}) g(p)$ after substituting $b \mapsto k^{-1}b$. By Proposition 20.7 below, $g(pq) = g_q(p) g_p(q) = (\frac{q}{p})(\frac{p}{q}) g(p) g(q)$. Write $g(m) = \omega_m \sqrt{m}$ with $\omega_m \in \{1, i\}$ from Corollary 20.5, which applies to the composite $m = pq$. Then $(\frac{p}{q})(\frac{q}{p}) = \omega_{pq}/(\omega_p \omega_q)$, which is $1/i^2 = -1$ when $p \equiv q \equiv 3 \bmod 4$ and 1 otherwise.

For the first supplement, $\overline{g(p)} = g_{-1}(p) = (\frac{-1}{p}) g(p)$, and $g(p)$ is real exactly when $p \equiv 1 \bmod 4$. For the second, Theorem 20.4 with $a = 4$, $c = p$ gives $g_2(p) = S(4, p) = \frac{\sqrt{p}}{2} e^{\pi i/4} \sum_{n \bmod 4} e^{-\pi i p n^2/4} = \frac{\sqrt{p}}{2} e^{\pi i/4} (1 + 2e^{-\pi i p/4} + e^{-\pi i p}) = \sqrt{p}\, e^{\pi i (1-p)/4}$, so $(\frac{2}{p}) = e^{\pi i (1-p)/4}/\omega_p$, which is $1, -1, -1, 1$ for $p \equiv 1, 3, 5, 7 \bmod 8$. □

The one step of the proof that does not use the Gaussian is finite Fourier analysis, and it has a statistical origin. For $m \geq 1$ and $\gcd(k, m) = 1$ let $\mathsf{F}_m^{(k)} = (e^{2\pi i k j l/m})_{j,l \bmod m}$, so that $\mathsf{F}_m = \mathsf{F}_m^{(1)}$ is the discrete Fourier transform matrix. Then $\operatorname{tr} \mathsf{F}_m^{(k)} = \sum_{j \bmod m} e^{2\pi i k j^2/m}$ is a Gauss sum, written $g_k(m)$, with $g_1(m) = g(m)$.

PROPOSITION 20.7 (Good's factorisation). *Let $m = pq$ with $\gcd(p, q) = 1$. Index rows and columns of F_m by $(x, y) \in \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/q\mathbb{Z}$ through $j = xq + yp$. Then F_m becomes the tensor product $\mathsf{F}_p^{(q)} \otimes \mathsf{F}_q^{(p)}$, and taking traces, $g(pq) = g_q(p)\, g_p(q)$.*

PROOF. The map is a bijection by the Chinese remainder theorem, and for $j = xq + yp$, $l = x'q + y'p$ we have $jl/(pq) \equiv xx'q/p + yy'p/q \bmod 1$. The trace is invariant under the simultaneous permutation of rows and columns and is multiplicative on tensor products. □

This index map is one of the two maps of Good's prime factor algorithm for the fast Fourier transform [220], which grew out of Yates's interaction algorithm for factorial experiments. In [221] he set the discrete Fourier transform beside Fourier series and Fourier integrals, as the third member of a family of analogues, and proved a discrete Poisson summation formula: for f on $\mathbb{Z}/N\mathbb{Z}$ with $N = mn$ and $f^*(r) = \sum_j f(j) e^{-2\pi i j r/N}$,

$$\sum_{k \bmod n} f(km) = \frac{1}{m} \sum_{l \bmod m} f^*(ln).$$

The proof of Corollary 20.6 uses all three of Good's analogues in turn: the Fourier integral is the normal characteristic function, the Fourier series the wrapped normal law of Lemma 20.2, and the discrete transform the Gauss sum and its factorisation. The Brownian limit of Theorem 20.4 is the passage from the second to the third, turning the theta transformation into a statement about F_c. Schur [492] gave a proof of Gauss's sign that stays entirely within the third analogue, by computing the multiplicities of the eigenvalues $\pm 1, \pm i$ of $\mathsf{F}_m/\sqrt{m}$, whose trace is

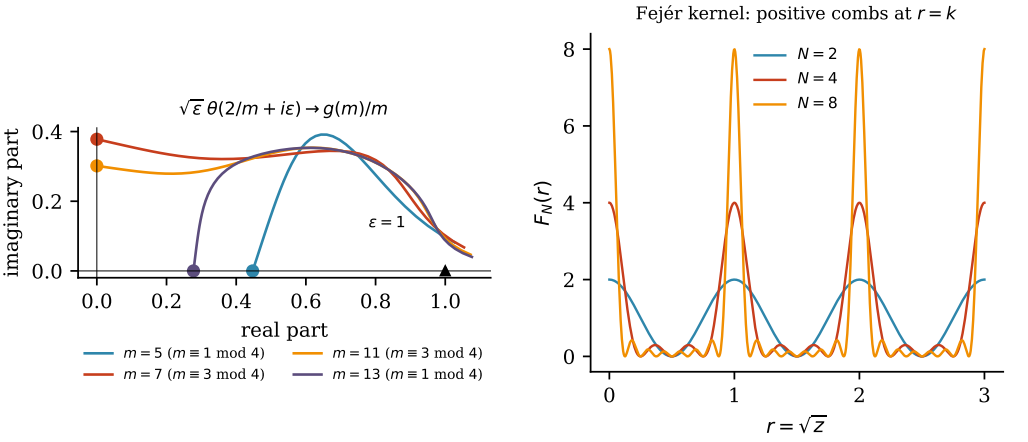


Figure 20.2: Left: Theorem 20.4 as a Brownian limit. The curves $\varepsilon \mapsto \sqrt{\varepsilon}\,\theta(2/m + i\varepsilon)$ for $m = 5, 7, 11, 13$, $10^{-5} \le \varepsilon \le 1$, start near 1 (triangle) and end at $g(m)/m$, which is real for $m \equiv 1 \bmod 4$ and imaginary for $m \equiv 3 \bmod 4$ (Corollary 20.5). Right: the Fejér kernel $F_N(r) = \sin^2(\pi N r)/(N \sin^2 \pi r)$ for $N = 2, 4, 8$, the positive combination of winding number cosines in Proposition 20.9; as N grows it concentrates at the integers $r = k$, that is at the eigenvalues $z = k^2$.

$g(m)/\sqrt{m}$. The Gaussian is the basic eigenfunction of the Fourier transform, with eigenvalue 1, and the Brownian limit is in this sense the continuous shadow of Schur's argument.

In the language of Section 20.1 the corollary is reciprocity for quadratic Galois characters: for $q \equiv 1 \bmod 4$ the Frobenius character $p \mapsto (\frac{q}{p})$ of $\mathbb{Q}(\sqrt{q})$, a Galois object, equals the Dirichlet character $p \mapsto (\frac{p}{q})$, an automorphic object for GL_1. This is the simplest case of Conjecture 20.1, a theorem since Gauss; what the Gaussian supplies is its analytic part.

The same lemma gives the functional equation of the L function of every even primitive Dirichlet character (stated in Chapter 2), with its root number.

THEOREM 20.8 (the Dirichlet functional equation as a kernel symmetry). *Let χ be a primitive even character modulo $q > 1$, $g(\chi) = \sum_{m \bmod q} \chi(m) e^{2\pi i m/q}$ its Gauss sum, and $\theta_\chi(t) = \sum_{n \in \mathbb{Z}} \chi(n) e^{-\pi n^2 t/q}$ for $t > 0$. Then $\theta_\chi(1/t) = \frac{g(\chi)}{\sqrt{q}} \sqrt{t}\, \theta_{\bar\chi}(t)$. Equivalently, the Pólya kernel $\Phi_\chi(u) = \theta_\chi(e^{2u}) e^{u/2}$, which satisfies $\Lambda(\frac{1}{2} + z, \chi) = \int \Phi_\chi(u) e^{zu} du$ for every complex z, where $\Lambda(s, \chi) = (q/\pi)^{s/2} \Gamma(s/2) L(s, \chi)$, obeys $\Phi_\chi(-u) = \frac{g(\chi)}{\sqrt{q}} \Phi_{\bar\chi}(u)$; that is, $\Lambda(s, \chi) = \frac{g(\chi)}{\sqrt{q}} \Lambda(1 - s, \bar\chi)$.*

PROOF. Primitivity gives $\chi(n) = g(\bar\chi)^{-1} \sum_{m \bmod q} \bar\chi(m) e^{2\pi i m n/q}$ for all n. Insert this into $\theta_\chi(t)$ and apply Lemma 20.2 with $x = \pi t/q$ and $y = 2\pi m/q$. The expo-

nent becomes $(y+2\pi k)^2/(4x) = \pi(m+kq)^2/(qt)$, and $\ell = m+kq$ runs over the class of m, so $\theta_\chi(t) = g(\bar\chi)^{-1}\sqrt{q/t}\,\theta_{\bar\chi}(1/t)$. Since $g(\chi)g(\bar\chi) = \chi(-1)q = q$, this is the claim with t and $1/t$ exchanged. The Mellin formula $\int_0^\infty e^{-\pi n^2 t/q} t^{s/2} dt/t = \Gamma(s/2)(q/\pi n^2)^{s/2}$ gives $\Lambda(s,\chi) = \frac{1}{2}\int_0^\infty \theta_\chi(t) t^{s/2} dt/t$, and $t = e^{2u}$ gives the kernel form. □

For a real even primitive χ the root number $g(\chi)/\sqrt{q}$ is ± 1, since $g(\chi)^2 = q$. Whether the Pólya kernel is even or odd is therefore a sign problem, and for $\chi = (\frac{\cdot}{q})$ with $q \equiv 1 \bmod 4$ prime, $g(\chi) = g(q) = \sqrt{q}$ by Corollary 20.5. So *for primes $q \equiv 1 \bmod 4$, Gauss's sign theorem is the statement that a Pólya kernel is even*, the GL_1 analogue of Proposition 20.27, and the Brownian limit of Figure 20.2 decides it.

20.2.1 The swap for the circle

The theta transformation can also be proved by a positivity argument, with the Laplace pair of the $2\sin^2$ lemma (compare Section 20.7). Write $\cos u\sqrt{z} = 1 - 2\sin^2(u\sqrt{z}/2)$ in (16.5) with $a = u^2/4$; since $\int_0^\infty e^{-xz}(\pi z)^{-1/2} dz = x^{-1/2}$ this gives

$$\sqrt{\frac{\pi}{x}}\, e^{-u^2/(4x)} = \int_0^\infty e^{-xz} \cos(u\sqrt{z})\, \frac{dz}{\sqrt{z}}. \tag{20.2}$$

The circle $\mathbb{R}/2\pi\mathbb{Z}$ has Laplace eigenvalues k^2, $k \in \mathbb{Z}$, playing the part of the γ^2, and closed geodesics of lengths $2\pi|n|$, playing the part of the prime powers. Its heat trace is $\sum_k e^{-xk^2}$, the Laplace transform of the comb $\sum_{k\in\mathbb{Z}} \delta_{k^2}$, which is the local Thorin comb $\tau_{0,2\pi}$ of Theorem 15.3 in its degenerate case $\theta = 0$ (Exercise 2); (20.2) with $u = 2\pi n$ attaches to each winding number n the cosine density $\cos(2\pi n\sqrt{z})z^{-1/2}$, a Fourier mode of this comb. In the same way the explicit formula attaches to each prime power p^m a multiple of the cosine density $\cos(m\sqrt{z}\log p)z^{-1/2}$ (Proposition 20.31), which by Proposition 15.8 is, up to sign and the factor $p^{-m/2}$, the mth Fourier mode of the local Thorin comb of p.

PROPOSITION 20.9 (Fejér positivity for the circle). *For $N \geq 1$ and $x > 0$ let $\mathcal{W}_N(x) = \sqrt{\pi/x}\sum_{|n|<N}(1-|n|/N)e^{-\pi^2 n^2/x}$. Then*

$$\mathcal{W}_N(x) = \int_0^\infty e^{-xz} F_N(\sqrt{z})\frac{dz}{\sqrt{z}}, \qquad F_N(r) = \sum_{|n|<N}\Big(1-\frac{|n|}{N}\Big)\cos 2\pi n r,$$

where $F_N(r) = \sin^2(\pi N r)/(N\sin^2\pi r) \geq 0$ is Fejér's kernel, so every $\mathcal{W}_N$ is completely monotone. As $N\to\infty$ the measures $F_N(\sqrt{z})z^{-1/2}dz$ converge vaguely to the comb $\sum_{k\in\mathbb{Z}}\delta_{k^2} = \delta_0 + 2\sum_{k\geq 1}\delta_{k^2}$, and consequently $\sqrt{\pi/x}\sum_n e^{-\pi^2 n^2/x} = \sum_k e^{-xk^2}$.

PROOF. The integral formula is (20.2) summed with the Fejér weights, and the closed form of F_N is classical. In the variable $r = \sqrt{z}$ the measure is $2F_N(r)dr$ on

$[0,\infty)$. The Fejér kernel is 1 periodic, nonnegative, of mass 1 over a period, and tends to zero uniformly away from the integers; so $2F_N(r)dr$ converges vaguely to $2\sum_{k\geq 1}\delta_k+\delta_0$, the half mass at 0 coming from the half bump on $[0,\frac{1}{2}]$. The tails are uniformly small, since $\int_{r>R}e^{-xr^2}2F_N(r)dr\leq 2\sum_{k\geq\lfloor R\rfloor}e^{-xk^2}$, so the Laplace transforms converge: $\mathcal{W}_N(x)\to 1+2\sum_{k\geq 1}e^{-xk^2}$. The left side converges to $\sqrt{\pi/x}\sum_n e^{-\pi^2n^2/x}$ by dominated convergence. □

In the variable $r=\sqrt{z}$, (20.2) is again the normal characteristic function, and the proof is Fejér's proof of Poisson summation written as a Laplace transform; it proves Theorem 20.3, hence Corollaries 20.5 and 20.6, a second time. Its shape is that of Theorem 20.30: the geodesic side, summed with positive weights, is the Laplace transform of a positive measure, which turns out to be the spectral measure. For the circle positivity is free, because the Fejér kernel is a square. For an L function the prime powers enter with the signed weights $-\Lambda_\rho(n)n^{-1/2}$ of Definition 20.29, and the archimedean term is not positive either: its density $\log N_\rho+2\,\mathrm{Re}\,\frac{\gamma'}{\gamma}(\frac{1}{2}+it,\rho)$ is smallest at $t=0$, where it is negative for ζ, Δ and the Artin representation of conductor 23. A Fejér type positive summation of the right side of Definition 20.29 would prove, through Theorem 20.30, holomorphy and GRH at once (Open Problem 20.4).

20.2.2 Heat flow and generalized Gauss sums

For integers $a,c\geq 1$ with ac even and $k\in\mathbb{Z}$ put

$$S(a,c;k)=\sum_{r\bmod c}e^{\pi i(ar^2+2kr)/c},$$

so that $S(a,c;0)$ is the Gauss sum of Theorem 20.4. For $\varepsilon>0$ let X_ε be the discrete Gaussian, $\mathbb{P}(X_\varepsilon=n)=e^{-\pi\varepsilon n^2}/\theta(i\varepsilon)$.

THEOREM 20.10 (heat flow reciprocity). *Let $u=e^{-\pi/(c^2\varepsilon)}$.*

(i) For every r,

$$\mathbb{P}(X_\varepsilon\equiv r\bmod c)=\frac{\sum_k e^{-\pi k^2/(c^2\varepsilon)}e^{2\pi ikr/c}}{c\sum_k e^{-\pi k^2/\varepsilon}},$$

and the total variation distance from the uniform law on $\mathbb{Z}/c\mathbb{Z}$ is at most $2u/(1-u)$.

(ii) $\left|\mathbb{E}e^{\pi iaX_\varepsilon^2/c}-S(a,c;0)/c\right|\leq 4u/(1-u)$.

(iii) With $\tau=a/c+i\varepsilon$ and $\kappa=c\varepsilon/a$, for every $\varepsilon>0$,

$$\sqrt{\tau/i}\sum_{k\in\mathbb{Z}}e^{-\pi k^2/(c^2\varepsilon)}S(a,c;k)=\sqrt{1+i\kappa}\sum_{k\in\mathbb{Z}}e^{-\pi k^2(1+i\kappa)/(c^2\varepsilon)}\,\overline{S(c,a;k)}.$$

(iv) Consequently, for every k,

$$S(a,c;k) = \sqrt{c/a}\, e^{\pi i/4}\, e^{-\pi i k^2/(ac)}\, \overline{S(c,a;k)}.$$

PROOF. (i) Group n by residue and apply Lemma 20.2 (Poisson summation) to

$$\sum_m e^{-\pi\varepsilon(r+mc)^2} = (c\sqrt{\varepsilon})^{-1} \sum_k e^{-\pi k^2/(c^2\varepsilon)} e^{2\pi i k r/c},$$

and to the normaliser $\theta(i\varepsilon) = \varepsilon^{-1/2} \sum_k e^{-\pi k^2/\varepsilon}$. The difference from $1/c$ is at most

$$c^{-1}\Big(2\sum_{k\geq 1} u^{k^2} + 2\sum_{k\geq 1} e^{-\pi k^2/\varepsilon}\Big) \leq 4c^{-1} \sum_{k\geq 1} u^{k^2}$$

since the denominator is at least 1, and summing over r and halving gives the bound. (ii) $|e^{\pi i a r^2/c}| = 1$ and $e^{\pi i a r^2/c}$ depends only on r mod c, so the difference is at most twice the total variation distance. (iii) As in (i), $\sqrt{\varepsilon}\,\theta(\tau) = c^{-1} \sum_k e^{-\pi k^2/(c^2\varepsilon)} S(a,c;k)$. On the other side $-1/\tau = -c/a + i\delta/(1+i\kappa)$ with $\delta = c^2\varepsilon/a^2$, and Poisson summation with the complex Gaussian weights $e^{-\pi\delta(r+ma)^2/(1+i\kappa)}$ (Lemma 20.2 continued analytically to $\operatorname{Re} x > 0$, both sides being holomorphic in x) gives

$$\theta(-1/\tau) = \frac{\sqrt{1+i\kappa}}{c\sqrt{\varepsilon}} \sum_k e^{-\pi k^2(1+i\kappa)/(c^2\varepsilon)}\, \overline{S(c,a;-k)}.$$

The theta transformation $\theta(-1/\tau) = \sqrt{\tau/i}\,\theta(\tau)$ and $S(c,a;-k) = S(c,a;k)$ give (iii). (iv) Since $\tau = (a/c)(1+i\kappa)$ and $\kappa/(c^2\varepsilon) = 1/(ac)$, (iii) reads

$$\sum_k w_k \Big[\sqrt{a/c}\, e^{-\pi i/4} S(a,c;k) - e^{-\pi i k^2/(ac)}\, \overline{S(c,a;k)}\Big] = 0$$

with $w_k = e^{-\pi k^2/(c^2\varepsilon)}$, for every $\varepsilon > 0$. The terms for k and $-k$ coincide, and the functions $\varepsilon \mapsto e^{-\pi k^2/(c^2\varepsilon)}$, $k \geq 0$, are Laplace transforms in $1/\varepsilon$ of distinct point masses, so letting $1/\varepsilon \to \infty$ and peeling off the terms one exponent k^2 at a time, using that the brackets are bounded, shows that every bracket vanishes. □

Part (iv) is the reciprocity law for generalized quadratic Gauss sums with even linear term (Berndt, Evans and Williams [52], Theorem 1.2.2). The proof shows that the whole family is a single identity, (iii), between two Gaussian mixtures, which the theta transformation and uniqueness of Laplace transforms separate into its components. At $\varepsilon \downarrow 0$ only $k = 0$ survives and (iii) becomes the reciprocity of Schaar and Landsberg. Part (i) quantifies the Brownian limit of Theorem 20.4: the residue of a discrete Gaussian is uniform up to $e^{-\pi/(c^2\varepsilon)}$ (Figure 20.5, right: for the plotted $c = 5$, 7 and 12, wherever the bound is below 0.1 it exceeds the distance by a factor between 3 and 3.4). Numerically, (iii) holds to $3 \cdot 10^{-40}$ at forty digit precision for the four parameter sets tested, and (iv) to 10^{-47} at fifty digit precision for all $a, c \leq 24$ with ac even and $|k| \leq 6$.

20.3 Reciprocity as an identity between random variables

This section restates reciprocity and its first consequences in the language of probability. The analytic inputs are classical (Brauer's induction theorem, Hecke L functions, the Rankin Selberg method); granted them, the proofs are short and show which part of each statement is probability and which is arithmetic.

20.3.1 The Frobenius probability space

Fix a finite set S of primes and $\sigma > 1$. Let Ω_S be the set of prime powers p^m with $p \notin S$, $m \geq 1$, and let $\mathbb{P}_\sigma$, Dirichlet's measure, be the probability measure

$$\mathbb{P}_\sigma(\{p^m\}) = \frac{p^{-m\sigma}}{m\, Z_\sigma}, \qquad Z_\sigma = \sum_{p \notin S,\, m \geq 1} \frac{p^{-m\sigma}}{m} = \log \zeta^S(\sigma) = \log \frac{1}{\sigma - 1} + O(1),$$

where ζ^S, like L^S below, omits the Euler factors at S, and $O(1)$ refers to $\sigma \downarrow 1$. A Galois representation ρ as in Section 20.1, unramified outside S and in the unitary normalisation, so that its Frobenius eigenvalues have absolute value 1, defines the random variable $a_\rho(p^m) = \operatorname{tr} \rho(\mathrm{Frob}_p^m)$, and when ρ factors through a finite group $G = \mathrm{Gal}(K/\mathbb{Q})$ it defines the random conjugacy class $F(p^m) = \mathrm{Frob}_p^m$ of G. A unitary cuspidal automorphic representation π of $\mathrm{GL}_n(\mathbb{A}_\mathbb{Q})$ unramified outside S, with Satake parameters $\{\alpha_{1,p}, \ldots, \alpha_{n,p}\}$, defines $a_\pi(p^m) = \sum_i \alpha_{i,p}^m$. The definitions are made so that

$$\mathbb{E}_\sigma\big[a\, \bar{b}\,\big] = \frac{1}{Z_\sigma} \log L^S(\sigma, \cdot) \tag{20.3}$$

for the Rankin Selberg L function of the pair: $L^S(\sigma, \rho \otimes \check{\rho}')$ for two Galois representations, $L^S(\sigma, \pi \times \tilde{\pi}')$ for two automorphic ones, the logarithm being the one given by the Dirichlet series. This uses $\overline{\operatorname{tr} \rho'(\mathrm{Frob}_p^m)} = \operatorname{tr} \check{\rho}'(\mathrm{Frob}_p^m)$ and, for unitary π', that the Satake parameters of $\tilde{\pi}'$ are the conjugates of those of π'. For $\pi \times \tilde{\pi}$ the coefficients $|a_\pi(p^m)|^2$ are nonnegative and the series converges for $\sigma > 1$ by Jacquet and Shalika [279] (Theorem 11.4); Cauchy Schwarz then gives absolute convergence for every pair. In particular all these variables are square integrable under $\mathbb{P}_\sigma$.

Proposition 20.11 (reciprocity is almost sure equality). *Let ρ be an n dimensional Galois representation and π a cuspidal automorphic representation of $\mathrm{GL}_n(\mathbb{A}_\mathbb{Q})$, both unramified outside S. Then $L(s, \rho_p) = L(s, \pi_p)$ for all $p \notin S$ if and only if $a_\rho = a_\pi$ $\mathbb{P}_\sigma$ almost surely, for one and then every $\sigma > 1$.*

Proof. $\mathbb{P}_\sigma$ charges every point of Ω_S, so almost sure equality is equality at every p^m. The power sums $a(p^m)$, $m \geq 1$, of n numbers determine their elementary symmetric functions by Newton's identities, hence the characteristic polynomial of $\rho(\mathrm{Frob}_p)$ and the multiset of Satake parameters, hence the local factor; the converse is immediate. □

So Conjecture 20.1, weakened to agreement at the primes outside a set S containing those where ρ ramifies, is an existence statement about random variables on $(\Omega_S, \mathbb{P}_\sigma)$: every Frobenius trace variable a_ρ of an irreducible ρ is almost surely equal to a Hecke eigenvalue variable a_π of a cuspidal π.

20.3.2 Chebotarev as convergence in law

Everything on the Galois side rests on one lemma about Artin L functions of virtual characters, a consequence of Brauer's induction theorem (Section 3.2).

LEMMA 20.12. *Let $K/\mathbb{Q}$ be finite Galois with group G, unramified outside S, let ψ be a virtual character of G, and let $e = \langle \psi, 1 \rangle_G$. Then $(s-1)^e L(s,\psi)$ is holomorphic and nonzero near $s = 1$, and $\log L^S(\sigma,\psi) = e \log \frac{1}{\sigma-1} + O(1)$ as $\sigma \downarrow 1$.*

PROOF. By Brauer $\psi = \sum_i n_i \operatorname{Ind}_{H_i}^G \chi_i$ with one dimensional χ_i, so $L(s,\psi) = \prod_i L(s,\chi_i)^{n_i}$ with Hecke L functions, each holomorphic and nonzero at $s = 1$ except for a simple pole when χ_i is trivial. The order of the pole is $\sum_i n_i \langle \chi_i, 1 \rangle_{H_i}$, which is $\langle \psi, 1 \rangle_G$ by Frobenius reciprocity. Removing finitely many Euler factors changes nothing near 1, and the logarithm defined by the Dirichlet series is a continuous branch on $(1,2]$, so it differs from $e \log \frac{1}{\sigma-1}$ by the logarithm of a continuous nonvanishing function with a limit at 1. □

THEOREM 20.13 (Chebotarev in law [102] and the Frobenius distance). *Let $K/\mathbb{Q}$ be finite Galois with group G, unramified outside S. Under $\mathbb{P}_\sigma$ the random class F converges in law, as $\sigma \downarrow 1$, to the image of Haar measure μ_G on conjugacy classes. Consequently, for Artin representations ρ, ρ' factoring through G,*

$$\mathbb{E}_\sigma \left[a_\rho \overline{a_{\rho'}} \right] \to \langle \chi_\rho, \chi_{\rho'} \rangle_G,$$
$$d(\rho,\rho')^2 := \lim_{\sigma \downarrow 1} \mathbb{E}_\sigma |a_\rho - a_{\rho'}|^2 = \int_G |\chi_\rho - \chi_{\rho'}|^2 \, d\mu_G.$$

In particular $d(\rho,\rho') = 0$ if and only if $\rho \cong \rho'$, and $d(\rho,\rho')^2 = 2$ for nonisomorphic irreducible ρ, ρ'.

PROOF. By (20.3) and Lemma 20.12, $\mathbb{E}_\sigma \chi(F) = \log L^S(\sigma,\chi)/Z_\sigma \to \langle \chi, 1 \rangle_G = \int \chi \, d\mu_G$ for every irreducible character χ. Irreducible characters span the class functions, so $\mathbb{P}_\sigma(F \in C) \to |C|/|G|$ for every class C, which is convergence in law on the finite set of classes. The remaining statements apply this to the class functions $\chi_\rho \overline{\chi_{\rho'}}$ and $|\chi_\rho - \chi_{\rho'}|^2$, and use that a representation is determined by its character and that irreducible characters are orthonormal. □

The theorem is the Dirichlet density form of Chebotarev's theorem [102], with the standard proof through Artin L functions rather than Chebotarev's own (Section 3.1); the natural density form is Corollary 3.2. Only the language is

new, and it is worth having for three reasons. First, d is a pseudometric, and by the theorem a metric on isomorphism classes of Artin representations: $\|f\|_\sigma = (\mathbb{E}_\sigma |f|^2)^{1/2}$ is a seminorm, so Minkowski's inequality gives the triangle inequality for d (with lim sup in place of lim when the limit is not known to exist). Second, the theorem is a statement about couplings: for Artin representations $\rho : G \to \mathrm{GL}_n$, $\rho' : G' \to \mathrm{GL}_{n'}$ the pair (F, F') of Frobenius classes lives on one space, and Theorem 20.13 applied to the compositum says that it converges in law to the image of Haar measure of the fibre product $H = \mathrm{Gal}(KK'/\mathbb{Q})$, embedded in $G \times G'$ by restriction. Third, the theorem separates two levels of information that are easily confused: the *marginal* law of a_ρ (the Sato Tate law of ρ, the image of μ_G under χ_ρ), and the *joint* law of $(a_\rho, a_{\rho'})$.

Marginal laws cannot detect reciprocity (Proposition 1.5), but joint laws see more: the limiting joint law of $(a_\rho, a_{\rho'})$ separates nonisomorphic Artin representations with the same marginal law, and at each fixed σ reciprocity says that the joint law of (a_ρ, a_π) is carried by the diagonal (Proposition 20.11). Take ρ the S_3 representation of $x^3 - x - 1$ and ρ_{31} the S_3 representation of $x^3 + x + 1$, of conductor 31. Their limiting marginal laws are identical (compare Example 3.3): $a = 2, 0, -1$ with probabilities $\frac{1}{6}, \frac{1}{2}, \frac{1}{3}$. The splitting fields are linearly disjoint, since their quadratic subfields $\mathbb{Q}(\sqrt{-23})$ and $\mathbb{Q}(\sqrt{-31})$ differ, so $H = S_3 \times S_3$, the coupling is the independent one, and $d(\rho, \rho_{31})^2 = 2$. Over the 78496 primes below 10^6 other than 23 and 31 the empirical joint law of $(a_\rho(p), a_{\rho_{31}}(p))$ agrees with the product law $\frac{1}{36}, \frac{1}{12}, \ldots, \frac{1}{9}$ to within 0.0007 in every cell.

20.3.3 Cuspidal orthonormality on the automorphic side

The automorphic analogue of Theorem 20.13 is the Rankin Selberg method.

THEOREM 20.14 (orthonormality of cuspidal representations [286, 279, 397, 510]). *Let π, π' be unitary cuspidal automorphic representations of $\mathrm{GL}_n(\mathbb{A}_\mathbb{Q})$ and $\mathrm{GL}_{n'}(\mathbb{A}_\mathbb{Q})$, unramified outside S. Then $\mathbb{E}_\sigma[a_\pi \overline{a_{\pi'}}] \to \delta_{\pi \cong \pi'}$ as $\sigma \downarrow 1$. Hence $d(\pi, \pi')^2 = 2$ when $\pi \ncong \pi'$, and $d(\pi, \pi') = 0$ forces $\pi \cong \pi'$.*

PROOF. By Jacquet, Piatetski Shapiro and Shalika for $n \neq n'$, and by Jacquet and Shalika and Moeglin and Waldspurger for $n = n'$, $L(s, \pi \times \tilde{\pi}')$ is holomorphic at $s = 1$ unless $\pi' \cong \pi$, when it has a simple pole, and by Shahidi it does not vanish on $\mathrm{Re}\, s = 1$. The argument of Lemma 20.12 gives $\log L^S(\sigma, \pi \times \tilde{\pi}') = \delta_{\pi \cong \pi'} \log \frac{1}{\sigma - 1} + O(1)$, and (20.3) finishes the proof. For the distance expand $|a_\pi - a_{\pi'}|^2$. □

Theorems 20.13 and 20.14 are one statement on two sides: irreducible Artin representations and cuspidal automorphic representations are both orthonormal systems of random variables in the limit $\sigma \downarrow 1$, with unit second moment (the exponential law of Theorem 20.28 below). For irreducible ℓ adic representations

orthonormality is a conjecture of Tate type [538], on the pole of $L(s, \rho \otimes \check{\rho}')$ at $s = 1$. Strong multiplicity one follows in one line, by the argument of Jacquet and Shalika [279]: if the local factors of two cuspidal representations agree outside a finite set of primes, enlarge S to contain it; then $a_\pi = a_{\pi'}$ on Ω_S, so $d(\pi, \pi') = 0$ and $\pi \cong \pi'$.

20.3.4 Reciprocity in law

Call ρ and π *reciprocal in law* if $d(\rho, \pi) = 0$, that is $\mathbb{E}_\sigma |a_\rho - a_\pi|^2 \to 0$ as $\sigma \downarrow 1$.

COROLLARY 20.15 (this book). *Let ρ, ρ' be Artin representations and π, π' unitary cuspidal automorphic representations, all unramified outside S.*

(i) If ρ and π correspond in the sense of Conjecture 20.1, they are reciprocal in law.

(ii) If ρ is reciprocal in law with π and with π', then $\pi \cong \pi'$. If ρ and ρ' are both reciprocal in law with π, then $\rho \cong \rho'$.

(iii) If ρ is reciprocal in law with a cuspidal π, then ρ is irreducible.

(iv) If ρ is reciprocal in law with π, then for every $\epsilon > 0$ the primes with $|a_\rho(p) - a_\pi(p)| > \epsilon$ have Dirichlet density zero.

(v) If ρ is automorphic, corresponding to π_0 in the sense of Conjecture 20.1, then ρ is reciprocal in law with π only for $\pi \cong \pi_0$.

PROOF. (i) is Proposition 20.11 after enlarging S. For (ii), the triangle inequality gives $d(\pi, \pi') \le d(\pi, \rho) + d(\rho, \pi') = 0$, and Theorem 20.14 applies; likewise $d(\rho, \rho') = 0$ and Theorem 20.13 applies. For (iii), Minkowski gives $|\|a_\rho\|_\sigma - \|a_\pi\|_\sigma| \le \|a_\rho - a_\pi\|_\sigma \to 0$, while $\|a_\rho\|_\sigma^2 \to \langle \rho, \rho \rangle$ and $\|a_\pi\|_\sigma^2 \to 1$; so $\langle \rho, \rho \rangle = 1$. For (iv), with E the set of such primes, $\sum_{p \in E} p^{-\sigma} / Z_\sigma \le \mathbb{P}_\sigma(|a_\rho - a_\pi| > \epsilon) \le \epsilon^{-2} \mathbb{E}_\sigma |a_\rho - a_\pi|^2 \to 0$ by Chebyshev's inequality. (v) is (i) and (ii). □

The corollary is the probabilistic skeleton of the reciprocity dictionary, each item a Hilbert space inequality from the two orthonormality theorems. The agreement in (iv) is only approximate: exact agreement, even on a set of density one, is open in general, and Open Problem 20.5 asks for it at all but finitely many primes. Nor does the corollary give existence, and it cannot: the powers of a set of primes of Dirichlet density zero carry $\mathbb{P}_\sigma$ mass $o(1)$, so reciprocity in law, like the laws of Proposition 1.5, is blind to bounded changes on such a set, while reciprocity itself is not. Figure 20.3 shows the natural density version $\pi(X)^{-1} \sum_{p \le X} |a(p) - b(p)|^2$ of d^2 for six pairs. For ρ and the weight one form $\eta(z)\eta(23z)$, whose coefficients were computed independently from the product expansion, it is exactly 0 for all $X \le 10^6$, a pointwise check of the reciprocity for ρ proved by Hecke (Example 3.3). For ρ and the data a' obtained from a_ρ by moving Frob_2 from a three cycle to the identity, as in Section 20.5, it is $1.1 \cdot 10^{-4}$ and tends to 0: in law a' is reciprocal

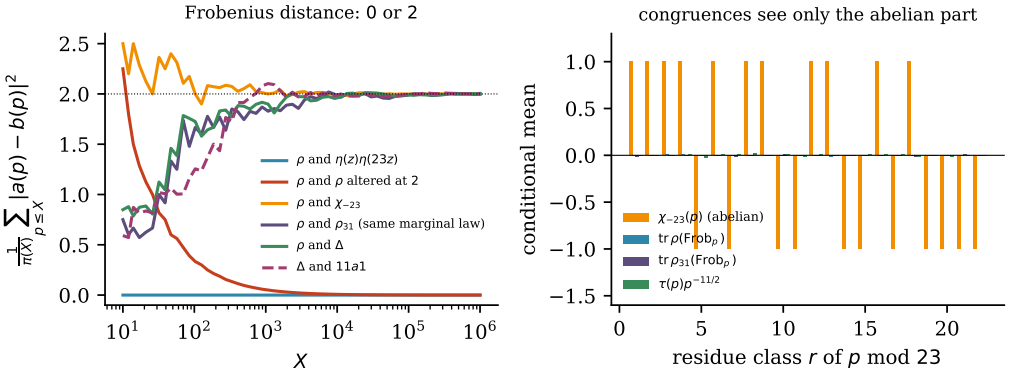


Figure 20.3: Left: the natural density version $\pi(X)^{-1}\sum_{p\le X}|a(p)-b(p)|^2$ of the squared Frobenius distance. Distinct irreducible or cuspidal objects sit at distance $\sqrt{2}$ (Theorems 20.13 and 20.14), including ρ and ρ_{31}, which have the same marginal law; ρ and $\eta(z)\eta(23z)$ agree at every prime, and ρ altered at 2 is at distance 0 from ρ in the limit. Right: means of four Frobenius variables conditional on p mod 23, primes below 10^6. Only the abelian χ_{-23} is a function of the residue; the other three are conditionally centred (Proposition 20.16 for the two Artin traces, Theorem 20.14 with π' a Dirichlet character modulo 23 for Δ).

to $\eta(z)\eta(23z)$. Yet no cuspidal representation has the local factors of a' at every prime: it would be at distance 0 from $\eta(z)\eta(23z)$, hence isomorphic to it by Theorem 20.14, and the Euler factors at 2 differ. The Pólya kernel and the Galois heat trace of the later sections do see this one change (Figures 20.7 and 20.9). For the other four pairs, (ρ,χ_{-23}), (ρ,ρ_{31}), (ρ,Δ) and $(\Delta,11a1)$, it is 2.000, 1.998, 2.005 and 2.002 at $X=10^6$, the value 2 of two orthonormal vectors; over the same primes the mean squares of a_ρ and of $\tau(p)p^{-11/2}$ are 0.997 and 0.999.

20.3.5 Class field theory as measurability

For $N\ge 1$ let $\mathcal{F}_N$ be the sigma algebra on Ω_S generated by the residue of p^m modulo N (take S to contain the primes dividing N). Since Frob_p^m acts on $\mathbb{Q}(\zeta_N)$ as $\zeta\mapsto\zeta^{p^m}$, $\mathcal{F}_N$ is generated by the Frobenius class in $\mathrm{Gal}(\mathbb{Q}(\zeta_N)/\mathbb{Q})$. The conditional law given $\mathcal{F}_N$ is the probabilistic form of a congruence condition.

PROPOSITION 20.16 (congruences see exactly the abelian part; this book, in this form). *Let ρ be an irreducible Artin representation and $N\ge 1$. For every residue r prime to N,*

$$\lim_{\sigma\downarrow 1}\mathbb{E}_\sigma\big[a_\rho\;\big|\;p^m\equiv r \bmod N\big]=\begin{cases}\rho(r) & \text{if } \rho \text{ is inflated from } A_N,\\ 0 & \text{otherwise,}\end{cases}$$

where $A_N = \mathrm{Gal}(\mathbb{Q}(\zeta_N)/\mathbb{Q}) \cong (\mathbb{Z}/N\mathbb{Z})^\times$*, the first case occurs only when* $\dim \rho = 1$*, and in it* a_ρ *is itself* $\mathcal{F}_N$ *measurable. In particular the Kronecker Weber theorem is equivalent to: every one dimensional Artin representation is* $\mathcal{F}_N$ *measurable for some* N*; and no irreducible Artin representation of dimension at least two has a nonzero limiting conditional mean given any congruence.*

PROOF. Let K be the field cut out by ρ and Γ the Galois group of the compositum $L = K(\zeta_N)$, write A for A_N, and let Γ_0 be the subgroup fixing $\mathbb{Q}(\zeta_N)$, so that $A = \Gamma/\Gamma_0$ is abelian. By Theorem 20.13 for L, the Frobenius class in Γ converges in law to Haar measure, so the conditional mean of a_ρ given that its image in A is r tends to the average of χ_ρ over the coset $g\Gamma_0$ lying above r. Averaging over cosets of Γ_0 is the orthogonal projection of class functions onto the span of the characters of A, so the average equals $\sum_{\psi \in \hat{A}} \langle \chi_\rho, \psi \rangle_\Gamma \psi(r)$. For irreducible ρ of dimension at least two every $\langle \chi_\rho, \psi \rangle$ vanishes. For one dimensional ρ it is 1 for $\psi = \rho$ if ρ is inflated from A, and 0 for every ψ otherwise. In the first case $\rho(\mathrm{Frob}_p^m) = \rho(r)$ exactly, which is measurability; here $\rho(r)$ means ρ evaluated at the element $\zeta \mapsto \zeta^r$ of A_N. □

This is why reciprocity for $n \geq 2$ cannot be a congruence condition, and why it needs automorphic forms: the Hecke eigenvalue $a_\pi(p)$ is the new measurable object that replaces the residue of p. Higher conditional moments do see the abelian quotient: for the S_3 representation, given $(\frac{p^m}{23}) = -1$ the trace is 0 almost surely, and given $(\frac{p^m}{23}) = 1$ it is 2 or -1 with limiting conditional probabilities $\frac{1}{3}, \frac{2}{3}$, so the limiting conditional mean is 0 in both cases (Figure 20.3, right).

20.3.6 A dictionary

Arithmetic	Probability
Frobenius at p, Hecke eigenvalue at p	random variables a_ρ, a_π on $(\Omega_S, \mathbb{P}_\sigma)$
reciprocity $\rho \leftrightarrow \pi$	$a_\rho = a_\pi$ almost surely
Chebotarev density theorem	convergence in law to Haar measure
Schur orthogonality; Rankin Selberg	orthonormality as $\sigma \downarrow 1$
irreducible Artin; cuspidal	unit second moment; exponential Gamma limit
strong multiplicity one	distance zero implies isomorphism
class field theory	abelian traces are $\mathcal{F}_N$ measurable
quadratic reciprocity, Gauss's sign	self duality of the normal law
functional equation	symmetry of the Pólya kernel
Artin's conjecture and GRH, self dual ρ (Theorem 20.30)	complete monotonicity (Bernstein)
converse theorems	a law determined by symmetries of its twists

The last line is an analogy rather than a theorem. Twisting by a Dirichlet character is conditioning: since $1_{p^m \equiv r \bmod N} = \varphi(N)^{-1} \sum_\chi \bar\chi(r)\chi(p^m)$, the twisted series $\sum a(p^m)\chi(p^m)m^{-1}p^{-ms}$ for all χ modulo N carry the same information as the conditional series given $\mathcal{F}_N$, and Weil's converse theorem asks for analytic properties of the twists for all moduli in a suitable infinite set, much as the Cramér Wold device determines a law on $\mathbb{R}^n$ from its one dimensional projections.

20.4 Reciprocity in the Gamma process of a random integer

This section replaces Dirichlet's measure of Section 20.3 by the random integer it comes from, and the random variables by a random measure: label each prime factor of a zeta distributed random integer by its Frobenius class, or by a Satake parameter, and let $\sigma \downarrow 1$.

20.4.1 The zeta random integer

For $\sigma > 1$ let $N_\sigma = \prod_p p^{e_p}$ be a zeta distributed random integer, $\mathbb{P}(N_\sigma = n) = n^{-\sigma}/\zeta(\sigma)$, as in Chapter 11, where e_p is written G_p. By the compound Poisson form of the geometric law (Exercise 1 of that chapter), $(e_p)_p$ has the law of $(\sum_m mM(p^m))_p$ for independent Poisson variables $M(p^m)$ with means $p^{-m\sigma}/m$, and $\log N_\sigma$ is the zeta subordinator of Proposition 16.10 at time 1, with Lévy measure Π_σ of Chapter 16.

Lemma 20.17. *The exponents e_p are independent and geometric, $\mathbb{P}(e_p = k) = (1 - p^{-\sigma})p^{-k\sigma}$ for $k \geq 0$, and $(\sigma - 1)\log N_\sigma$ converges in law to the standard exponential law as $\sigma \downarrow 1$.*

Proof. The Euler product factorises: for finitely many t_p with $|t_p| \leq 1$, and $t_p = 1$ for the other primes, $\mathbb{E}\prod_p t_p^{e_p} = \zeta(\sigma)^{-1}\prod_p \sum_{k\geq 0} t_p^k p^{-k\sigma} = \prod_p (1 - p^{-\sigma})/(1 - t_p p^{-\sigma})$, a product of geometric generating functions. For the limit, $\mathbb{E}e^{-\lambda(\sigma-1)\log N_\sigma} = \zeta(\sigma + \lambda(\sigma-1))/\zeta(\sigma) \to (1+\lambda)^{-1}$ by the simple pole of ζ at 1. □

20.4.2 Equidistribution is a Gamma process

Let Y be a compact metric space, S a finite set of primes, and $(\theta_p)_{p\notin S}$ points of Y. We say (θ_p) is equidistributed with respect to a Borel probability measure μ on Y if $\pi(t)^{-1}\sum_{p\leq t,\, p\notin S} f(\theta_p) \to \int f\, d\mu$ for every continuous f. Define the random measure and the marked point process

$$\Xi_\sigma = \sum_{p\notin S} (\sigma - 1)e_p \log p\ \delta_{\theta_p}, \qquad \mathcal{M}_\sigma = \sum_{p\notin S,\, e_p \geq 1} \delta_{((\sigma-1)e_p \log p,\, \theta_p)}.$$

So Ξ_σ (a random measure, unrelated to Riemann's Ξ and to the completions Ξ_π of Chapter 14) spreads the rescaled logarithm of N_σ over the label space, and $\mathcal{M}_\sigma$ records the rescaled sizes of the prime factors together with their labels. The Gamma random measure Γ_μ with shape measure μ is the random measure on Y that takes independent values on disjoint Borel sets, with $\Gamma_\mu(B) \sim \text{Gamma}(\mu(B), 1)$. Writing $\Gamma_\mu(\lambda) = \int \lambda \, d\Gamma_\mu$, and $\Xi_\sigma(\lambda)$ likewise, its Laplace functional is $\mathbb{E}e^{-\Gamma_\mu(\lambda)} = \exp(-\int \log(1+\lambda)\, d\mu)$ for continuous $\lambda \geq 0$, and its atoms form a Poisson process with intensity $x^{-1}e^{-x}dx\, \mu(d\theta)$ [318].

THEOREM 20.18 (equidistribution is a Gamma process; this book). *Suppose (θ_p) is equidistributed with respect to μ. As $\sigma \downarrow 1$:*

(i) $\mathcal{M}_\sigma$ converges in law, vaguely on $(0,\infty) \times Y$, to the Poisson process with intensity $x^{-1}e^{-x}dx \otimes \mu$: the rescaled sizes of the prime factors of N_σ are asymptotically the jumps of a standard Gamma subordinator on $[0,1]$, and their labels are asymptotically independent of the sizes and independent of one another, with law μ;

(ii) Ξ_σ converges in law to Γ_μ: for every continuous $\lambda \geq 0$ on Y, $\mathbb{E}e^{-\Xi_\sigma(\lambda)} \to \exp(-\int_Y \log(1+\lambda)\, d\mu)$.

For one point Y, part (i) with the sizes normalised by their sum is the Poisson Dirichlet law of the large prime factors of a random integer, due to Billingsley [61] for uniform integers; Lloyd [370] obtained the limit laws of the largest prime factors of a harmonically distributed integer through the zeta distribution and Poisson variables. Chapter 24 develops this: normalised, Ξ_σ converges in law to Ferguson's Dirichlet process with base measure μ, and the random integers weighted by the coefficients of the powers of ζ give the whole Ewens family. The proof of the theorem rests on one estimate.

LEMMA 20.19. *Let $\nu_\sigma = (\sigma-1)\sum_{p\notin S}(\log p)\, p^{-\sigma}\, \delta_{((\sigma-1)\log p,\, \theta_p)}$, a finite measure on $[0,\infty] \times Y$. If (θ_p) is equidistributed with respect to μ, then ν_σ converges weakly to $e^{-x}dx \otimes \mu$.*

PROOF. For continuous $f \geq 0$ on Y put $A_f(t) = \sum_{p\leq t, p\notin S} f(\theta_p)\log p$. Equidistribution and the prime number theorem give $A_f(t) = \mu(f)t + o(t)$ by partial summation, and then the Abelian theorem for Dirichlet series gives

$$\sum_p f(\theta_p)(\log p)p^{-w} = \frac{\mu(f)}{w-1} + o\Big(\frac{1}{w-1}\Big), \qquad w \downarrow 1.$$

With $w = \sigma + (\sigma-1)s$, $s \geq 0$,

$$\begin{aligned}\int e^{-sx} f(\theta)\, \nu_\sigma(dx, d\theta) &= (\sigma-1)\sum_p f(\theta_p)(\log p)p^{-\sigma-(\sigma-1)s} \\ &\to \frac{\mu(f)}{1+s} = \int_0^\infty e^{-sx}e^{-x}dx\ \mu(f).\end{aligned}$$

By linearity this holds for every continuous f. The functions $e^{-sx}f(\theta)$, $s \geq 0$, $f \in C(Y)$, span an algebra that contains the constants and separates the points of the compact space $[0,\infty] \times Y$, so it is dense in the continuous functions by Stone and Weierstrass. The total masses converge (take $s = 0$, $f = 1$), so the convergence extends to all continuous functions. □

Proof of Theorem 20.18. (ii) By Lemma 20.17,

$$\begin{aligned}\log \mathbb{E}e^{-\Xi_\sigma(\lambda)} &= \sum_{p\notin S}\Big[\log(1-p^{-\sigma}) - \log\big(1-p^{-\sigma}e^{-\lambda_p u_p}\big)\Big] \\ &= -\sum_{p\notin S} p^{-\sigma}\big(1-e^{-\lambda_p u_p}\big) + R_\sigma,\end{aligned}$$

where $u_p = (\sigma-1)\log p$, $\lambda_p = \lambda(\theta_p)$ and $R_\sigma = -\sum_{p\notin S}\sum_{m\geq 2} m^{-1}p^{-m\sigma}(1 - e^{-m\lambda_p u_p})$. Each term of R_σ tends to 0 and is bounded by p^{-2} times a constant, so $R_\sigma \to 0$ by dominated convergence. The main term is $-\int h\,d\nu_\sigma$ with $h(x,\theta) = (1-e^{-\lambda(\theta)x})/x$, extended by $\lambda(\theta)$ at $x = 0$ and by 0 at $x = \infty$; h is continuous on $[0,\infty]\times Y$. By Lemma 20.19 and Frullani's integral,

$$\int h\,d\nu_\sigma \to \int_Y\int_0^\infty \frac{1-e^{-\lambda(\theta)x}}{x}\,e^{-x}dx\,\mu(d\theta) = \int_Y \log(1+\lambda)\,d\mu.$$

(i) Let $F \geq 0$ be continuous with compact support in $(0,\infty)\times Y$, say in $[a,b]\times Y$ with $a > 0$. Then $-\log \mathbb{E}e^{-\mathcal{M}_\sigma(F)} = -\sum_p \log(1-y_p)$ with $y_p = \sum_{k\geq 1}\mathbb{P}(e_p = k)(1-e^{-F(ku_p,\theta_p)})$. The terms with $k \geq 2$ contribute at most $\sum_p\sum_{k\geq 2} p^{-k}1\{ku_p \geq a\}$, which tends to 0 by dominated convergence, and $\sum_p y_p^2 \to 0$ in the same way; replacing $\mathbb{P}(e_p = 1)$ by $p^{-\sigma}$ costs at most $\sum_p p^{-2}1\{u_p \geq a\} \to 0$, so

$$-\log \mathbb{E}e^{-\mathcal{M}_\sigma(F)} = \sum_p p^{-\sigma}(1-e^{-F(u_p,\theta_p)}) + o(1) = \int g\,d\nu_\sigma + o(1)$$

with $g = (1-e^{-F})/x$, which is continuous on $[0,\infty]\times Y$ because F vanishes near $x = 0$. Lemma 20.19 gives the limit $\int(1-e^{-F})x^{-1}e^{-x}dx\,d\mu$, the Laplace functional of the Poisson process. □

The hypothesis holds for the following label sequences. For a finite Galois extension $K/\mathbb{Q}$ with group G, θ_p is the class Frob_p in the set $G^\natural$ of conjugacy classes and $\mu = \mu_G$ is the image of Haar measure, by Chebotarev's theorem in its natural density form (Corollary 3.2) [102]. For a holomorphic newform of weight $k \geq 2$ with trivial character and without complex multiplication, $\theta_p \in [0,\pi]$ is the angle with $x_p = a_p/p^{(k-1)/2} = 2\cos\theta_p$, and μ is the Sato Tate law $\frac{2}{\pi}\sin^2\theta\,d\theta$, whose image under $\theta \mapsto 2\cos\theta$ is μ_{ST}, by Theorem 1.4 [67]. For two elliptic curves over $\mathbb{Q}$ without complex multiplication and not geometrically isogenous, θ_p is the pair of angles and μ the product of two Sato Tate laws, by Theorem 4.11 [240].

COROLLARY 20.20 (Chebotarev as a Gamma process; this book). *Let $K/\mathbb{Q}$ be finite Galois with group G, unramified outside S, and for a conjugacy class C let $L_C(\sigma) = (\sigma - 1)\sum_{p\notin S,\,\mathrm{Frob}_p\in C} e_p \log p$ be the rescaled part of $\log N_\sigma$ carried by primes with Frobenius in C. As $\sigma \downarrow 1$:*

(i) the $L_C(\sigma)$, $C \in G^\natural$, are independent for every σ, and $L_C(\sigma)$ converges in law to $\mathrm{Gamma}(|C|/|G|, 1)$*;*

(ii) the shares $L_C / \sum_{C'} L_{C'}$ have a Dirichlet limit law with parameters $(|C|/|G|)_C$, independent of the total, which is standard exponential;

(iii) for a quadratic character χ, the share of $\log N_\sigma$ carried by the primes with $\chi(p) = -1$ converges to the arcsine law $\pi^{-1}(u(1-u))^{-1/2}du$ on $(0,1)$.

PROOF. (i) The $L_C(\sigma)$ depend on disjoint sets of the independent e_p, and the limit is Theorem 20.18(ii) on the finite space $G^\natural$, with λ a function of the class. (ii) is the beta gamma algebra (independent gamma variables normalised by their sum are Dirichlet and independent of the sum) and the continuous mapping theorem, the total being positive almost surely. (iii) is (ii) for $G = \{\pm 1\}$, where the Dirichlet law is $\mathrm{Beta}(\frac{1}{2}, \frac{1}{2})$. □

The mean of the share in (ii) is $|C|/|G|$, which is Chebotarev's density. What the Gamma process adds is the fluctuation: the share does not concentrate, because $\log N_\sigma$ is dominated by a few large prime factors, and their labels are independent Haar samples. In the language of Chapter 24, the limit shares are the masses of the classes under a Dirichlet process with base measure μ, the limit of the normalised labelled factorisation (Theorem 24.3). Part (iii) is an arithmetic arcsine law, in the same sense as Lévy's arcsine law for the time Brownian motion spends positive: a Gamma subordinator split into two halves of equal intensity.

For the S_3 field of $x^3 - x - 1$, ramified only at 23 (Example 3.3; Figure 3.1 shows its Chebotarev frequencies in natural density), the Laplace transforms $\mathbb{E}e^{-\lambda L_C(\sigma)}$ were computed from the Euler product over the primes below 10^7, with the remaining primes accounted for by the prime number theorem. Figure 20.4, left, shows them converging to $(1+\lambda)^{-1/6}$, $(1+\lambda)^{-1/2}$ and $(1+\lambda)^{-1/3}$ for the identity class, the transpositions and the three cycles. At $\lambda = 1$ the transform differs from its limit by a factor of about $1 + \kappa_C(\sigma - 1)$, with κ_C between 0.38 and 1.07 (Theorem 20.26).

20.4.3 Frobenius functionals are generalized gamma convolutions

Recall from Definition 13.2 that a GGC is a law on $[0,\infty)$ with Laplace transform

$$\exp\Big(-a\lambda - \int_{(0,\infty)} \log(1+\lambda/t)\,\tau(dt)\Big), \qquad \lambda \geq 0,$$

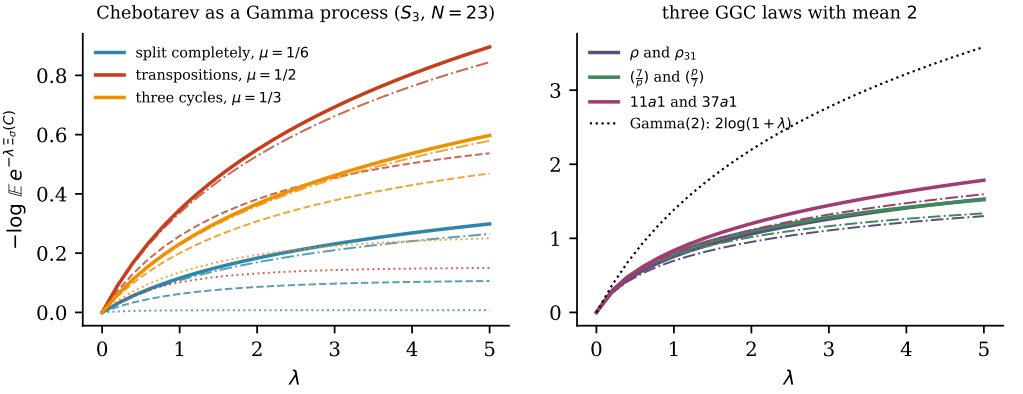


Figure 20.4: Left: Corollary 20.20 for the S_3 field of $x^3 - x - 1$. Laplace exponents of $L_C(\sigma) = \Xi_\sigma(C)$, from Euler products over $p < 10^7$ with a prime number theorem tail, for $\sigma = 1.5$ (dotted), 1.1 (dashed) and 1.01 (dash dotted) against the Gamma limits $\mu(C)\log(1+\lambda)$. Right: three GGC limits from Theorems 20.21 and 20.22, all with mean 2 and with different Thorin measures, against the Gamma(2) limit of the intensity scaling of Theorem 20.23; dash dotted curves are $\sigma = 1.01$.

where $a \geq 0$ is the drift and τ the Thorin measure [546, 73]. Its mean is $a + \int t^{-1}\,\tau(dt)$, and we call $\tau((0,\infty))$ its Thorin mass. We write Gamma(α) for the gamma law of shape α and rate 1, whose Thorin measure is $\alpha\delta_1$, with Gamma$(0) = \delta_0$.

THEOREM 20.21 (Frobenius functionals are GGCs; this book). *Suppose (θ_p) is equidistributed with respect to μ and let $Q \geq 0$ be continuous on Y. Then*

$$\Xi_\sigma(Q) = (\sigma - 1)\sum_{p \notin S} e_p \log p\; Q(\theta_p)$$

converges in law to a GGC with drift zero and Thorin measure τ_Q, the image of μ restricted to $\{Q > 0\}$ under $\theta \mapsto 1/Q(\theta)$. Its mean is $\int Q\, d\mu$ and its Thorin mass is $\mu(Q > 0)$. Moreover, if $\{Q > 0\}$ is a μ continuity set,

$$\log \mathbb{P}\big(\Xi_\sigma(Q) = 0\big) = -\mu(Q > 0)\log\frac{1}{\sigma - 1} + o\Big(\log\frac{1}{\sigma - 1}\Big).$$

PROOF. Apply Theorem 20.18(ii) to λQ: the limit Laplace transform is

$$\exp\Big(-\int \log(1 + \lambda Q)\, d\mu\Big) = \exp\Big(-\int_{\{Q>0\}} \log\big(1 + \lambda/(1/Q)\big)\, d\mu\Big),$$

which is the GGC transform with Thorin measure τ_Q. The mean is $\int t^{-1}\tau_Q(dt)$, which is $\int Q\, d\mu$. For the last statement, $\Xi_\sigma(Q) = 0$ exactly when $e_p = 0$ for every

p with $Q(\theta_p) > 0$, so

$$\log \mathbb{P}(\Xi_\sigma(Q) = 0) = \sum_{Q(\theta_p)>0} \log(1 - p^{-\sigma}) = -\sum_{Q(\theta_p)>0} p^{-\sigma} + O(1),$$

and equidistribution on the continuity set $\{Q > 0\}$ gives $\sum_{Q(\theta_p)>0} p^{-\sigma} \sim \mu(Q > 0) \log \frac{1}{\sigma-1}$ by partial summation. □

So the Thorin mass is visible as an exponent, and for Chebotarev labels, where the error term is $O(1)$ by Lemma 20.12, the probability that no prime factor of N_σ has a label where $Q > 0$ is of exact order $(\sigma - 1)^{\mu(Q>0)}$. Figure 20.5, left, shows the fitted slopes 0.608, 0.497 and 0.989, against Thorin masses $\frac{11}{18}$, $\frac{1}{2}$ and 1 for the three examples of the next section. For $Q = 1_C$ Chapter 24 computes the constant of this order, and its counterpart for uniform random integers, a case of Serre's theorem on frobenian sets (Theorem 24.8).

20.4.4 Reciprocity as vanishing of a Thorin measure

We now take Q to be a squared discrepancy between two kinds of Frobenius data. The limit is δ_0 exactly when τ_Q vanishes, so reciprocity laws become vanishing statements, and their failures have computable Thorin measures.

THEOREM 20.22 (Thorin criteria for reciprocity; this book). *In each case S contains the primes at which the data are ramified (in (b) those ramified in $K(\zeta_N)$, in (c) 2 and q, in (d) the primes of bad reduction of E and E'), and $\Xi_\sigma(Q)$ converges in law to the GGC with Thorin measure τ_Q of Theorem 20.21.*

(a) Artin representations. Let ρ, ρ' be Artin representations unramified outside S, with fields K, K' (the fixed fields of their kernels), and put $H = \mathrm{Gal}(KK'/\mathbb{Q})$ and $Q = |\chi_\rho - \chi_{\rho'}|^2$ on $H^\natural$. Then τ_Q is the image of Haar probability measure on $\{\chi_\rho \neq \chi_{\rho'}\} \subset H$ under $1/|\chi_\rho - \chi_{\rho'}|^2$. The limit law has mean $\|\chi_\rho - \chi_{\rho'}\|^2_{L^2(H)}$, which is 2 for nonisomorphic irreducible ρ, ρ', and $\tau_Q = 0$ if and only if $\rho \cong \rho'$.

(b) Abelian reciprocity. Let ψ be a character of finite order of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, χ a Dirichlet character modulo N, and $Q = |\psi(\mathrm{Frob}_p) - \chi(p)|^2$. If $\psi \neq \chi$ as characters of $\mathrm{Gal}(K(\zeta_N)/\mathbb{Q})$, where K is the field of ψ, and $\psi\bar{\chi}$ has order d, then

$$\tau_Q = \frac{1}{d} \sum_{j=1}^{d-1} \delta_{1/(4\sin^2(\pi j/d))},$$

and the limit law has mean 2 and Thorin mass $1 - \frac{1}{d}$; otherwise $\tau_Q = 0$. The Kronecker Weber theorem is equivalent to the statement that for every ψ there are N and χ with $\tau_Q = 0$.

(c) *Quadratic reciprocity. Let $q \equiv 3 \bmod 4$ be prime. For $Q = |(\frac{q}{p}) - (\frac{p}{q})|^2$ the limit law is the gamma law of shape $\frac{1}{2}$ and rate $\frac{1}{4}$, that is $2\chi_1^2$, with Thorin measure $\frac{1}{2}\delta_{1/4}$; for $Q = |(\frac{q}{p}) - (\frac{p}{q})(\frac{-1}{p})|^2$ the limit is δ_0. For $q \equiv 1 \bmod 4$ the first Q already gives δ_0.*

(d) *Elliptic curves. Let E, E' be elliptic curves over $\mathbb{Q}$ without complex multiplication and not geometrically isogenous, with normalised traces x_p, x'_p, and $Q = (x_p - x'_p)^2$. Then τ_Q is the law of $1/(X - X')^2$ for independent X, X' with law μ_{ST}, so the Thorin mass is 1 and the limit law has mean 2. If instead E' is isogenous to E, then $Q = 0$ for all good p and the limit is δ_0.*

PROOF. (a) Chebotarev for KK' gives equidistribution of Frob_p in $H^\natural$ with respect to Haar measure, and Theorem 20.21 applies with $Y = H^\natural$. The mean is $\int |\chi_\rho - \chi_{\rho'}|^2$; it is $\langle\chi_\rho, \chi_\rho\rangle + \langle\chi_{\rho'}, \chi_{\rho'}\rangle - 2\,\mathrm{Re}\langle\chi_\rho, \chi_{\rho'}\rangle = 2$ for nonisomorphic irreducibles. Every class of the finite group H has positive Haar measure, so $\tau_Q = 0$ forces $\chi_\rho = \chi_{\rho'}$ on H, hence $\rho \cong \rho'$.

(b) Here ψ and χ are both characters of $G_N = \mathrm{Gal}(K(\zeta_N)/\mathbb{Q})$, χ through the cyclotomic identification $\mathrm{Frob}_p \mapsto p \bmod N$. Then $Q(g) = |1 - (\psi\bar{\chi})(g)|^2$. If $\psi\bar{\chi}$ is nontrivial of order d, its values are equidistributed over the dth roots of unity under Haar measure on G_N, and $|1 - e^{2\pi ij/d}|^2 = 4\sin^2(\pi j/d)$. The mean is $\frac{1}{d}\sum_j 4\sin^2(\pi j/d) = 2$. If $\psi\bar{\chi}$ is trivial, Q vanishes identically. Kronecker Weber says every ψ factors through some $\mathrm{Gal}(\mathbb{Q}(\zeta_N)/\mathbb{Q})$, which is the existence of χ with $\psi = \chi$ on G_N; conversely $\tau_Q = 0$ gives $\psi = \chi$ on G_N.

(c) For odd $p \neq q$, $(\frac{q}{p})$ is the Frobenius character of $\mathbb{Q}(\sqrt{q})$, and the law of quadratic reciprocity is $(\frac{q}{p}) = (\frac{p}{q})(\frac{-1}{p})^{(q-1)/2}$. So for $q \equiv 3 \bmod 4$ the character $\psi\bar{\chi}$ in (b) is $(\frac{-1}{\cdot})$, of order $d = 2$, and $\tau_Q = \frac{1}{2}\delta_{1/4}$, whose GGC is $(1 + 4\lambda)^{-1/2}$, the transform of $2\chi_1^2$.

(d) Theorem 4.11 gives equidistribution of (x_p, x'_p) with respect to $\mu_{\mathrm{ST}} \otimes \mu_{\mathrm{ST}}$, and the diagonal has measure zero, so the mass is 1; the mean is $\mathbb{E}(X - X')^2 = 2$. For isogenous curves $a_p = a'_p$ for all good p. □

Parts (b) and (c) restate Kronecker Weber and quadratic reciprocity rather than reprove them, since the vanishing of τ_Q is equivalent to the equality of two characters; what is new is the shape of failure. A mismatched abelian reciprocity is a GGC whose Thorin atoms $1/(4\sin^2(\pi j/d))$ are reciprocals of the squared sines of the $2\sin^2$ lemma (Lemma 16.7), and omitting the sign factor $(\frac{-1}{p})$ from quadratic reciprocity leaves precisely a scaled chi squared variable with one degree of freedom. Numerically (Figure 20.4, right), the three examples (ρ, ρ_{31}), $((\frac{7}{p}), (\frac{p}{7}))$ and $(11a1, 37a1)$ all have mean 2 and different limit laws: orthonormality fixes the mean, and the Thorin measure records the law of the squared discrepancy Q, which depends on the joint law of the two kinds of Frobenius data.

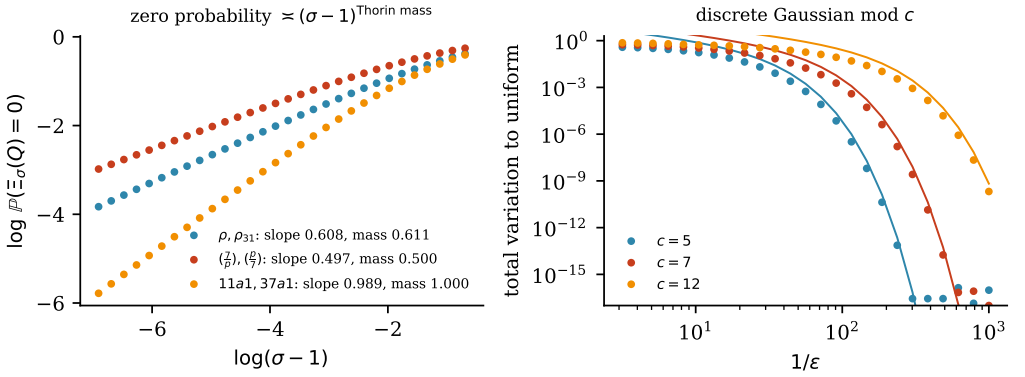


Figure 20.5: Left: Theorem 20.21. The probability that no prime factor of N_σ has a label where $Q > 0$, computed from the Euler product over primes below 10^7 with a prime number theorem tail, decays like $(\sigma-1)^m$, with m the Thorin mass. Right: Theorem 20.10. Total variation distance between a discrete Gaussian with parameter ε reduced modulo c and the uniform law (dots), against the bound of part (i) (lines). For $c = 5$ and 7 the dots that level off near 10^{-16}, above the lines, are rounding errors of the double precision computation, not distances.

20.4.5 A second scaling and a zero two law

Theorem 20.21 uses Q as a mark on the jumps of a fixed random integer. One can instead let Q change the intensity: take two of the Frobenius variables a, b of Section 20.3 (each an a_ρ or an a_π, fixed in the next theorem), put $Q_m(p) = |a(p^m) - b(p^m)|^2 \geq 0$, and replace the zeta integer by the random integer $N_{Q,\sigma}$ with $\mathbb{P}(N_{Q,\sigma} = n) \propto d_Q(n) n^{-\sigma}$, where $\sum_n d_Q(n) n^{-s} = D_Q(s)$ is the Dirichlet series with $\log D_Q(s) = \sum_{p \notin S, m} Q_m(p) m^{-1} p^{-ms}$. Its logarithm $X_\sigma = \log N_{Q,\sigma}$ is the discrepancy subordinator; for $b = 0$ it is the Rankin Selberg subordinator of Theorem 20.28, which is the case $\rho' = 0$ of the next theorem.

THEOREM 20.23 (zero two law; this book). *(a) For Artin representations ρ, ρ' with $a = \operatorname{tr}\rho$, $b = \operatorname{tr}\rho'$, $(\sigma-1)X_\sigma$ converges in law to* Gamma$(\|\chi_\rho - \chi_{\rho'}\|^2)$*, a GGC with Thorin measure $\|\chi_\rho - \chi_{\rho'}\|^2 \delta_1$ and integer shape; for irreducible ρ, ρ' the shape is* 0 *or* 2*. (b) For unitary cuspidal automorphic representations π, π' of* $\mathrm{GL}_n(\mathbb{A}_\mathbb{Q})$, $\mathrm{GL}_{n'}(\mathbb{A}_\mathbb{Q})$ *with $a = a_\pi$, $b = a_{\pi'}$, the limit is δ_0 if $\pi \cong \pi'$ and* Gamma(2) *otherwise.*

PROOF. $D_Q = L(\rho \otimes \check{\rho}) L(\rho' \otimes \check{\rho}') / (L(\rho \otimes \check{\rho}') L(\rho' \otimes \check{\rho}))$ up to finitely many Euler factors, the L function of the virtual character $|\chi_\rho - \chi_{\rho'}|^2$. By Lemma 20.12 it has a pole of order $e = \langle |\chi_\rho - \chi_{\rho'}|^2, 1 \rangle = \|\chi_\rho - \chi_{\rho'}\|^2$ at $s = 1$ and is otherwise holomorphic and nonzero near 1, so $\mathbb{E} e^{-\lambda(\sigma-1)X_\sigma} = D_Q(\sigma + \lambda(\sigma-1)) / D_Q(\sigma) \to (1+\lambda)^{-e}$. For (b), the same computation with $L(s, \pi \times \tilde{\pi}')$, which by Jacquet,

Shalika, Moeglin, Waldspurger and Shahidi [279, 397, 510] has a simple pole at 1 exactly when $\pi' \cong \pi$ and no zero on $\operatorname{Re} s = 1$, gives $e = 2 - 2\delta_{\pi \cong \pi'}$. □

The two scalings give two GGCs with the same mean $\int Q\, d\mu$: in Theorem 20.23 the Thorin measure τ_Q of Theorem 20.21 collapses to a single atom at 1 carrying the mean. Integrality of the collapsed shape is Schur orthogonality, and the value 2 for distinct irreducibles is the orthonormality of characters; for cuspidal pairs it is Rankin Selberg. The natural extension to the mixed case is the following conjecture.

CONJECTURE 20.24 (mixed zero two law). *Let ρ be an irreducible Artin representation of dimension n and π a unitary cuspidal automorphic representation of $\mathrm{GL}_n(\mathbb{A}_{\mathbb{Q}})$, with $a = \operatorname{tr}\rho$ and $b = a_\pi$. Then $(\sigma - 1)X_\sigma$ converges in law to δ_0 or to $\mathrm{Gamma}(2)$, and to δ_0 exactly when $L(s, \rho_p) = L(s, \pi_p)$ for almost all p.*

PROPOSITION 20.25 (this book). *(i) If ρ is automorphic, Conjecture 20.24 holds for ρ and every π. Hence it holds for $n = 1$, for odd two dimensional ρ, and for two dimensional ρ with solvable image. (ii) Unconditionally, $\limsup_{\sigma \downarrow 1} \mathbb{E}[(\sigma - 1)X_\sigma] \le 4$ and $\mathbb{E}[((\sigma - 1)X_\sigma)^2]$ stays bounded, so under the conjecture $\mathbb{E}[(\sigma - 1)X_\sigma]$ tends to 0 or 2. (iii) If the limit is δ_0 then ρ and π are reciprocal in law, so π is unique up to isomorphism and, for every $\epsilon > 0$, $|a_\rho(p) - a_\pi(p)| \le \epsilon$ outside a set of primes of Dirichlet density zero (Corollary 20.15(ii) and (iv)).*

PROOF. (i) If ρ corresponds to π_ρ, then $a = a_{\pi_\rho}$ off a finite set and Theorem 20.23(b) applies to (π_ρ, π). The listed cases are class field theory, Khare and Wintenberger [311, 312] with work of Kisin [320], and Langlands and Tunnell [351, 553]. (ii) The mean is $(\sigma - 1)\sum_{p,m} Q_m(p)(\log p)p^{-m\sigma}$. Since $Q_m \le 2|a(p^m)|^2 + 2|b(p^m)|^2$, it is at most $2(\sigma - 1)(-L'/L)(\sigma, \rho \otimes \check{\rho}) + 2(\sigma - 1)(-L'/L)(\sigma, \pi \times \tilde{\pi})$ up to finitely many Euler factors, and both terms tend to 2 because each L function has a simple pole at 1. The same bound for $(\log D_Q)''(\sigma)$, the variance of X_σ, gives $\operatorname{Var}((\sigma - 1)X_\sigma) \le 4 + o(1)$, since $(\sigma - 1)^2(\log L)''(\sigma) \to 1$ at a simple pole; so $\mathbb{E}[((\sigma - 1)X_\sigma)^2] \le 20 + o(1)$, and the means converge whenever the laws do. (iii) Put

$$\eta_\sigma = (\sigma - 1)\sum_{p,m} Q_m(p)(\log p)p^{-m\sigma}\delta_{(\sigma-1)m\log p}.$$

The Laplace exponent at $\lambda = 1$ is

$$\int \frac{1 - e^{-x}}{x}\,\eta_\sigma(dx) \ge \frac{1 - e^{-K}}{K}\,\eta_\sigma([0, K]),$$

so $\eta_\sigma([0, K]) \to 0$ for every K; the tails $\eta_\sigma([K, \infty))$ are at most $4e^{-K} + o(1)$ by the same bound as in (ii) applied to the Gamma limits of $|a|^2$ and $|b|^2$. Hence $(\sigma - 1)\sum Q_m(p)(\log p)p^{-m\sigma} \to 0$, and integrating in σ, $\log D_Q(\sigma) =$

$\int_\sigma^2 \sum Q_m(p)(\log p)p^{-mw}dw + O(1) = o(\log \frac{1}{\sigma-1})$, which is the definition of reciprocity in law. □

Conjecture 20.24 follows from reciprocity (Proposition 20.25(i)) but does not imply it, because it asserts no existence: for a nonautomorphic ρ it predicts the limit Gamma(2) for every π. It is the orthogonality half of reciprocity, and existence is the other half: by its last clause, reciprocity for ρ is equivalent to the conjecture together with the existence of one π whose limit is δ_0. If $L(s,\rho_p) = L(s,\pi_p)$ for almost all p, then $Q_m(p) = 0$ outside a finite set of primes and the limit is δ_0 unconditionally; conversely, without the conjecture, a δ_0 limit makes ρ and π reciprocal in law (Proposition 20.25(iii)), and whether that forces $a_\rho(p) = a_\pi(p)$ for all but finitely many p is Open Problem 20.5.

20.4.6 Second order Chebotarev: the Euler Kronecker drift

The Gamma limit of Corollary 20.20 is approached at rate $\sigma - 1$ (Figure 20.6), with a constant given by logarithmic derivatives of Artin L functions at $s = 1$ and a prime power term.

For a character χ of $G = \mathrm{Gal}(K/\mathbb{Q})$ and the finite set S put

$$\gamma^S(\chi) = \lim_{s\to 1}\Big(\frac{(L^S)'}{L^S}(s,\chi) + \frac{\delta_{\chi=1}}{s-1}\Big),$$

where L^S omits the Euler factors at S. For $\chi \neq 1$ this is

$$\frac{L'}{L}(1,\chi) + \sum_{p\in S}\sum_{m\geq 1} \mathrm{tr}(\mathrm{Frob}_p^m \mid V_\chi^{I_p})\,(\log p)\,p^{-m},$$

where V_χ is a representation with character χ and I_p an inertia group at p, and for $\chi = 1$ it is $\gamma_{\mathrm{E}} + \sum_{p\in S}\frac{\log p}{p-1}$ with γ_{E} Euler's constant. The Euler Kronecker constant of K in Ihara's sense [270] is

$$\gamma_K = \lim_{s\to 1}\Big(\frac{\zeta_K'}{\zeta_K}(s) + \frac{1}{s-1}\Big) = \sum_\chi \chi(1)\gamma^{\emptyset}(\chi),$$

and γ_K^S denotes the same constant for ζ_K with the Euler factors above S removed.

THEOREM 20.26 (Euler Kronecker drift; this book). *Let $K/\mathbb{Q}$ be finite Galois with group G, unramified outside S, and C a conjugacy class. Uniformly for λ in compact subsets of $[0,\infty)$,*

$$\log \mathbb{E}e^{-\lambda L_C(\sigma)} = -\mu(C)\log(1+\lambda) + \kappa_C\,\lambda\,(\sigma-1) + O\big((\sigma-1)^2\big),$$

with

$$\kappa_C = \mu(C)\sum_\chi \overline{\chi(C)}\,\gamma^S(\chi) \;-\; \sum_{p\notin S}\sum_{m\geq 2}\big[1_C(\mathrm{Frob}_p) - 1_C(\mathrm{Frob}_p^m)\big]\,(\log p)\,p^{-m},$$

the sum over irreducible characters. In particular $\mathbb{E}L_C(\sigma) = \mu(C) - \kappa_C(\sigma-1) + O((\sigma-1)^2)$*: to first order the Gamma limit is shifted by the deterministic amount* $-\kappa_C(\sigma-1)$*. For the class of primes that split completely,*

$$\kappa_{\{1\}} = \frac{\gamma_K^S}{|G|} + \sum_{p\notin S,\ \mathrm{Frob}_p\neq 1}\ \sum_{m\geq 2,\ \mathrm{Frob}_p^m=1} (\log p)\, p^{-m},$$

the Euler Kronecker constant of K with the Euler factors above S removed, divided by the degree, plus a nonnegative prime power term. For a class function $Q \geq 0$*, independence of the* $L_C(\sigma)$ *gives* $\log \mathbb{E}e^{-\lambda \Xi_\sigma(Q)} = -\sum_C \mu(C)\log(1+\lambda Q_C) + \kappa_Q \lambda(\sigma-1) + O((\sigma-1)^2)$ *with* $\kappa_Q = \sum_C Q_C\, \kappa_C$*.*

PROOF. By Lemma 20.17, $\log \mathbb{E}e^{-\lambda L_C(\sigma)} = P_C(\sigma') - P_C(\sigma)$ with $\sigma' = \sigma + \lambda(\sigma-1)$ and $P_C(s) = \sum_{p\notin S, \mathrm{Frob}_p\in C} \sum_m m^{-1}p^{-ms}$. Expand $1_C = \sum_\chi c_\chi \chi$ with $c_\chi = \mu(C)\overline{\chi(C)}$. Since $\sum_\chi c_\chi \chi(g^m) = 1_C(g^m)$,

$$\sum_\chi c_\chi \log L^S(s,\chi) = \sum_{p\notin S}\sum_m \frac{1_C(\mathrm{Frob}_p^m)}{m\, p^{ms}},$$

so that

$$P_C(s) = \sum_\chi c_\chi \log L^S(s,\chi) + R_C(s),$$

with

$$R_C(s) = \sum_{p\notin S}\sum_{m\geq 2} \left[1_C(\mathrm{Frob}_p) - 1_C(\mathrm{Frob}_p^m)\right] m^{-1}p^{-ms}$$

holomorphic on $\mathrm{Re}\, s > \frac{1}{2}$. By Brauer's theorem each $L^S(s,\chi)$, $\chi \neq 1$, is holomorphic and nonzero near $s=1$, and $(s-1)\zeta^S(s)$ is too. So $h_C(s) = P_C(s) - \mu(C)\log\frac{1}{s-1}$ is real analytic near $s=1$, with $h_C'(1) = \sum_\chi c_\chi \gamma^S(\chi) + R_C'(1) = \kappa_C$. Then $P_C(\sigma') - P_C(\sigma) = -\mu(C)\log(1+\lambda) + h_C(\sigma') - h_C(\sigma)$, and Taylor's theorem gives the expansion. The mean is minus the derivative in λ at 0. For $C=\{1\}$, $c_\chi = \chi(1)/|G|$ and $\sum_\chi \chi(1)\gamma^S(\chi) = \gamma_K^S$ because $\zeta_K^S = \prod_\chi L^S(s,\chi)^{\chi(1)}$; the prime power term is nonnegative because a power of the identity is the identity. The statement for Q follows by summing over classes. □

For the S_3 field of $x^3 - x - 1$ the set S consists of 2, 7, 11, 23, 31 and 37, the primes that are bad for some example of this section. PARI/GP gives $\frac{L'}{L}(1,\chi_{-23}) = -0.42212$ and $\frac{L'}{L}(1,\rho) = 1.01706$, so $\gamma_K = \gamma_{\mathrm{E}} - 0.42212 + 2(1.01706) = 2.18922$; with the Euler factors above S removed, $\gamma_K^S = 3.59058$, and the prime power term of Theorem 20.26 is 0.14171, so $\kappa_{\{1\}} = 3.59058/6 + 0.14171 = 0.74014$. Theorem 20.26 predicts the drifts in the first row below; the second row is the intercept of a linear fit of $[\log \mathbb{E}e^{-L_C(\sigma)} + \mu(C)\log 2]/(\sigma-1)$ over $\sigma - 1 \in [0.00125, 0.04]$ (for Q, with $\frac{1}{2}\log 5$ in place of $\mu(C)\log 2$), from the Euler products used for Figure 20.4 (Figure 20.6).

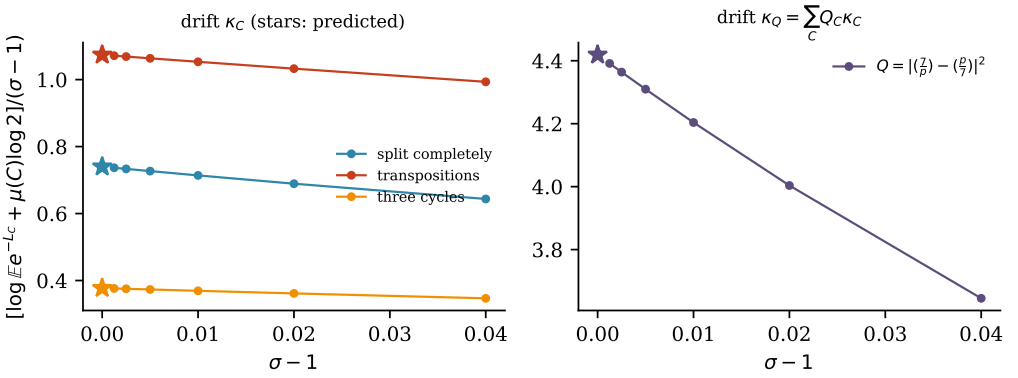


Figure 20.6: Theorem 20.26. The normalised deviation of the logarithm of the Laplace transform from its Gamma limit, at $\lambda = 1$, as $\sigma \downarrow 1$, for the three classes of the S_3 field (left) and for the quadratic reciprocity discrepancy with the sign factor omitted (right). Stars are the predicted drifts.

	identity	transpositions	three cycles	$Q = \lvert(\frac{7}{p}) - (\frac{p}{7})\rvert^2$
κ predicted	0.74014	1.07416	0.37746	4.4189
fitted	0.73881	1.07365	0.37710	4.4058

The three class drifts add up to $\gamma_{\mathrm{E}} + \sum_{p\in S} \frac{\log p}{p-1} = 2.19176$, the drift of $(\sigma - 1)\sum_{p\notin S} e_p \log p$, the part of $(\sigma-1)\log N_\sigma$ carried by the primes outside S; for $(\sigma-1)\log N_\sigma$ itself the drift is γ_{E}. The prime power term of κ_C is the Dirichlet density counterpart of the prime square bias that Section 3.1 measures for the same field. For uniform random integers below x, with S the set of ramified primes, a size biased prime factor has Frobenius in C with probability $\mu(C) + (\gamma_{\mathrm{E}}\mu(C) - \kappa_C)/(\log x - 1) + O((\log x)^{-2})$ (Theorem 24.5).

Under GRH Ihara showed that γ_K is at most about $2\log\log\sqrt{|d_K|}$ [270]. Since $(\gamma_K^S - \gamma_K)/|G| \le \sum_{p\in S}\frac{\log p}{p-1}$ and the prime power term is at most $\sum_p \frac{\log p}{p(p-1)} < 0.76$, GRH bounds the drift of the split class from above by about $2\log\log\sqrt{|d_K|}/|G| + \sum_{p\in S}\frac{\log p}{p-1} + 0.76$. No lower bound of this form holds for all K: γ_K is negative when K has many primes of small norm, and Ihara's lower bound, which needs no hypothesis, is only of order $-\log\sqrt{|d_K|}$ [270]. Ihara's bounds say nothing about the other classes, whose drifts involve the individual values $\frac{L'}{L}(1,\chi)$, not γ_K alone.

20.5 One function with two faces

Let $f(z) = \sum_{n\ge 1} a_n e^{2\pi i n z}$ with $a_n = O(n^{(k-1)/2+\epsilon})$, $k \ge 1$ an integer, $N \ge 1$, and put $\Lambda(s) = (\sqrt{N}/2\pi)^s \Gamma(s) \sum a_n n^{-s}$. As in Section 2.3, termwise integration gives,

for $\operatorname{Re} s$ large,

$$\Lambda(s) = \int_0^\infty f\Big(\frac{iy}{\sqrt{N}}\Big) y^s \frac{dy}{y}, \qquad \Lambda\Big(\frac{k}{2}+t\Big) = \int_{\mathbb{R}} \Phi_f(u) e^{tu}\, du, \qquad \Phi_f(u) = f\Big(\frac{ie^u}{\sqrt{N}}\Big) e^{ku/2}. \tag{20.4}$$

We call Φ_f the *Pólya kernel* of the coefficient sequence (for Δ see Exercise 1); when it is nonnegative and integrable it is, up to normalisation, the density of the Pólya variable W_f with $\mathbb{E}e^{tW_f} = \Lambda(\frac{k}{2}+t)/\Lambda(\frac{k}{2})$. The kernel is built from the coefficients alone: if they are the Dirichlet coefficients of a Galois L function, Φ_f is a Galois object, and if f is a modular form, Φ_f is the restriction of an automorphic form to the geodesic $i\mathbb{R}_{>0}$. Reciprocity asks when the two descriptions coincide.

PROPOSITION 20.27 (Hecke's criterion as a symmetry; this book). *With f and Φ_f as above and $\varepsilon \in \{\pm 1\}$, the following are equivalent.*

(i) $\Phi_f(-u) = \varepsilon\, \Phi_f(u)$ *for all real u.*

(ii) $f(-1/(Nz)) = \varepsilon\, i^{-k} N^{k/2} z^k f(z)$ *for all z in the upper half plane.*

(iii) Λ *extends to an entire function, bounded in vertical strips, with* $\Lambda(k-s) = \varepsilon\Lambda(s)$.

For $N = 1$, k even and $\varepsilon = (-1)^{k/2}$ these are equivalent to: f is a cusp form of weight k for $\mathrm{SL}_2(\mathbb{Z})$.

PROOF. Put $y = e^u/\sqrt{N}$, so that $ie^{-u}/\sqrt{N}$ is $i/(Ny)$. Then (i) reads

$$f\big(i/(Ny)\big) = \varepsilon N^{k/2} y^k f(iy),$$

which is (ii) on the imaginary axis, since $-1/(N \cdot iy)$ equals $i/(Ny)$ and $z^k = i^k y^k$. Both sides of (ii) are holomorphic in the upper half plane, so agreement on the imaginary axis gives agreement everywhere; this proves (i) if and only if (ii). The equivalence of (ii) and (iii) is Hecke's theorem [250] (Section 2.3): split (20.4) at $y = 1$, the fixed point of the Fricke involution $y \mapsto 1/y$ in this variable, and use (ii) to fold the lower range onto the upper one, which gives the continuation, the bound and the functional equation; conversely Mellin inversion and a contour shift recover (ii) from (iii). For $N = 1$, the group $\mathrm{SL}_2(\mathbb{Z})$ is generated by $z \mapsto z+1$, which every q series respects, and $z \mapsto -1/z$, which is (ii) with $\varepsilon i^{-k} = 1$; cuspidality follows from $a_0 = 0$ and the growth of the coefficients. □

So at level one, reciprocity for a Galois representation of conductor 1 with the Hodge numbers of weight k is the statement that one function on the line, built from its Frobenius traces, is even (odd when $k \equiv 2 \bmod 4$). When the kernel is a density, it says that a random variable built from primes is symmetric. For level N the symmetry (i) is only the Fricke involution; Weil's converse theorem asks for

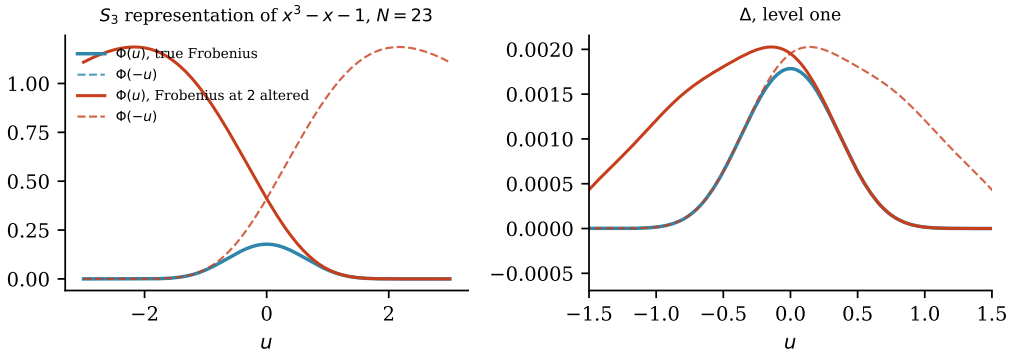


Figure 20.7: Reciprocity as a symmetry. The Pólya kernel $\Phi_f(u)$ built from Frobenius traces (blue) coincides with $\Phi_f(-u)$ (dashed). Changing the Frobenius data at the single prime 2 (red) destroys the symmetry. Left: the S_3 representation of conductor 23; right: Δ.

the analogous symmetries of the twisted kernels $\Phi_{f\otimes\chi}$, which pair χ with $\bar{\chi}$. Chapter 22 reads the modularity of elliptic curves in this form. Chapter 23 computes the kernel of $\eta(z)\eta(23z)$, there normalised as $2\Phi_f$, from the Artin L function on the critical line rather than from the q expansion, by Bromwich inversion, which uses the holomorphy that Hecke's theorem supplies (Proposition 23.8).

The symmetry is extremely sensitive to the Frobenius data. Figure 20.7 shows Φ_f for the weight one form of level 23 attached to the S_3 representation ρ of $x^3 - x - 1$, and for Δ. With the true Frobenius traces the kernels are symmetric to $2 \cdot 10^{-16}$ and $2 \cdot 10^{-18}$. If the Frobenius data at 2 alone are changed (Frob_2 moved from order three to the identity for ρ, the sign of $\tau(2)$ reversed for Δ), and all coefficients a_n are rebuilt multiplicatively, the maximal asymmetry becomes 1.19 for the Artin representation, whose true kernel has maximum 0.178, and $1.4 \cdot 10^{-3}$ for Δ, whose true kernel has maximum $1.8 \cdot 10^{-3}$.

20.6 The Galois side: Rankin Selberg subordinators

Infinite divisibility on the Galois side comes for free once one passes to $\rho \otimes \check{\rho}$. Let S be the set of ramified primes and

$$D^S(s) = L^S(s, \rho \otimes \check{\rho}) = \prod_{p \notin S} \det\bigl(1 - (\rho \otimes \check{\rho})(\mathrm{Frob}_p)\, p^{-s}\bigr)^{-1},$$

$$\log D^S(s) = \sum_{p \notin S} \sum_{m \geq 1} \frac{|a_\rho(p^m)|^2}{m\, p^{ms}}.$$

All logarithmic coefficients are nonnegative, so for $\sigma > 1$ the function $\lambda \mapsto D^S(\sigma + \lambda)/D^S(\sigma)$ is the Laplace transform of a compound Poisson subordinator X_σ with Lévy measure $\sum_{p,m} |a_\rho(p^m)|^2 m^{-1} p^{-m\sigma} \delta_{m \log p}$. This is the Galois counterpart of the Rankin Selberg laws of Theorem 11.4 and of the zeta subordinators of Chapter 16.

THEOREM 20.28 (Gamma shape detects irreducibility; this book). *Let ρ be an Artin representation and $m = \langle \rho, \rho \rangle$ the sum of the squares of the multiplicities of its irreducible constituents. As $\sigma \downarrow 1$, $(\sigma - 1) X_\sigma$ converges in law to the Gamma law with shape m and scale 1. In particular ρ is irreducible if and only if the limit is the standard exponential law.*

PROOF. By Schur's lemma the trivial representation occurs in $\rho \otimes \check{\rho} = \mathrm{End}(\rho)$ with multiplicity $\langle \rho, \rho \rangle = m$, so by Lemma 20.12 for $\psi = \rho \otimes \check{\rho}$ (finitely many Euler factors change nothing near $s = 1$), $D^S(s) = (s-1)^{-m} G(s)$ with G holomorphic and nonzero near 1, and

$$\mathbb{E} e^{-\lambda(\sigma-1)X_\sigma} = \frac{D^S(\sigma + \lambda(\sigma-1))}{D^S(\sigma)} = \frac{G(\sigma + \lambda(\sigma-1))}{(1+\lambda)^m G(\sigma)} \to (1+\lambda)^{-m}$$

for every $\lambda \geq 0$. The continuity theorem for Laplace transforms finishes the proof. □

The Gamma law with shape m is a generalized gamma convolution with Thorin measure $m\,\delta_1$, so the multiplicity $\langle \rho, \rho \rangle$ appears as a Thorin atom of mass m, much as the order of central vanishing appears in Chapter 14 as a degenerate Thorin atom at the origin. On the automorphic side the Rankin Selberg theory of Jacquet, Shalika, Moeglin and Waldspurger [279, 397] gives the same limit with $m = 1$ precisely for cuspidal π, so "irreducible Artin representation corresponds to cuspidal automorphic representation" has a probabilistic shadow: *both are the exponential law*. Figure 20.8 shows the convergence for the S_3 representation ($m = 1$) and for $\rho \oplus \rho$ ($m = 4$), computed from the complete product $L(s, \rho \otimes \rho) = \zeta(s) L(s, \chi_{-23}) L(s, \rho)$, which keeps the Euler factor $(1 - 23^{-s})^{-2}$ that D^S omits and has the same limit. At $\sigma = 1.001$ the Laplace transforms at $\lambda = 1$ are 0.50059 and 0.06279, against $\frac{1}{2}$ and $\frac{1}{16}$.

20.7 THE SWAP: PRIMES TO ZEROS, COMPUTED FROM FROBENIUS

The heat trace $\mathcal{W}_\pi = k_\pi$ of Chapter 16, a sum of $e^{-x\gamma^2}$ over the pairs of zeros $\frac{1}{2} \pm i\gamma$ (over $\gamma > 0$ under GRH), is a function of the zeros, and for automorphic π the explicit formula (16.6) expresses it through the primes, as Table 16.1 does for Δ. The key point for reciprocity is that the prime expression makes sense for any Galois representation, automorphic or not.

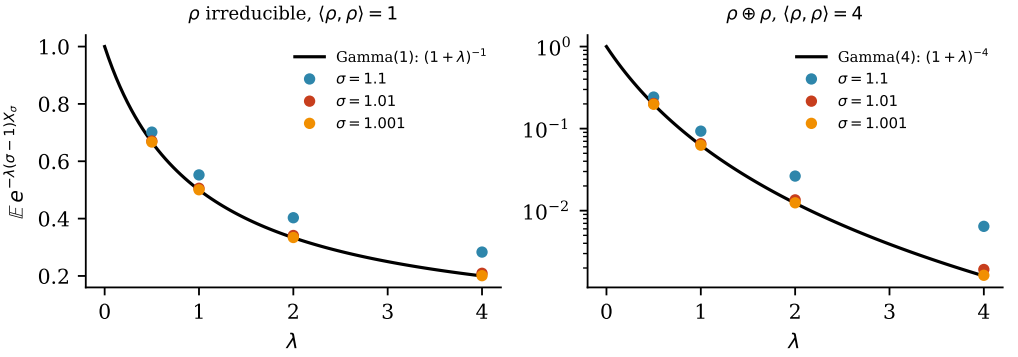


Figure 20.8: Theorem 20.28: Laplace transforms of $(\sigma-1)X_\sigma$, computed from the complete product $L(s,\rho\otimes\rho)$, for the S_3 representation and for $\rho\oplus\rho$, against the Gamma laws with shapes 1 and 4.

DEFINITION 20.29. Let ρ be self dual, $\Lambda_\rho(n)$ the coefficients of $-L'/L(s,\rho)$ (so $\Lambda_\rho(p^m)=a_\rho(p^m)\log p$ at unramified p), and $\gamma(s,\rho)$, N_ρ its gamma factor and conductor. For $x>0$ the *Galois heat trace* is

$$\mathcal{W}_\rho(x)=\frac{1}{4\pi}\int_{\mathbb{R}} e^{-xt^2}\Big(\log N_\rho+2\,\mathrm{Re}\,\frac{\gamma'}{\gamma}\big(\tfrac12+it,\rho\big)\Big)\,dt \\ -\frac{1}{2\pi}\sum_{n\geq 2}\frac{\Lambda_\rho(n)}{\sqrt{n}}\sqrt{\frac{\pi}{x}}\,e^{-(\log n)^2/(4x)}.$$

The series converges absolutely for every $x>0$, since $|\Lambda_\rho(n)|\leq \dim\rho\cdot\log n$ for an Artin representation (more generally $|\Lambda_\rho(n)|\leq\dim\rho\cdot n^\theta\log n$ when the Satake parameters at every p are bounded by p^θ) and the Gaussian factor decays faster than any power of n. So $\mathcal{W}_\rho$ is defined from Frobenius traces, the conductor and the gamma factor alone, with no analytic continuation. To accuracy about e^{-A} only the prime powers with $\log n\leq x+\sqrt{x^2+4xA}$, about $\sqrt{4xA}$ for small x, matter, so small x needs few primes.

THEOREM 20.30 (Holomorphy and GRH from the Galois heat trace; this book, after Weil [569, 572]). *Let ρ be self dual, and suppose $\Lambda(s,\rho)$ has meromorphic continuation of finite order with $\Lambda(s,\rho)=\pm\Lambda(1-s,\rho)$. Write its zeros and poles as $\frac12\pm i\gamma$. Then:*

(i) $\mathcal{W}_\rho(x)=\frac12\sum_{zeros}e^{-x\gamma^2}-\frac12\sum_{poles}e^{-x\gamma^2}$, *with multiplicity, the sums absolutely convergent for $x>0$.*

(ii) $\mathcal{W}_\rho$ *is completely monotone on $(0,\infty)$ if and only if $\Lambda(s,\rho)$ is entire and all its zeros lie on* $\mathrm{Re}\,s=\frac12$.

In particular, for an Artin representation with real character and no trivial constituent (a trivial constituent of multiplicity r adds the pole term $-re^{x/4}$), Artin's conjecture together

with GRH for $L(s,\rho)$ is equivalent to the complete monotonicity of a function computed from Frobenius traces.

PROOF. (i) is Weil's explicit formula [569] (see [276], Theorem 5.12) for the even test function $h(t) = e^{-xt^2}$, whose Fourier transform is $\sqrt{\pi/x}\, e^{-u^2/(4x)}$; h is entire and decays in every horizontal strip, so the contour shifts are justified, and poles enter with the opposite sign to zeros. Self duality makes the prime side real and equal to $-\frac{1}{\pi}\sum \Lambda_\rho(n) n^{-1/2} \hat{h}(\log n)$.

(ii) If Λ is entire with all γ real, (i) exhibits $\mathcal{W}_\rho$ as the Laplace transform of the positive measure $\frac{1}{2}\sum \delta_{\gamma^2}$, as in Theorem 14.8(d), which is the automorphic case. Conversely let $\mathcal{W}_\rho$ be completely monotone, so $\mathcal{W}_\rho(x) = \int_{[0,\infty)} e^{-xu}\mu(du)$ with $\mu \geq 0$ by Bernstein's theorem. Write (i) as $\sum_k c_k e^{-xw_k}$ with distinct $w_k = \gamma_k^2 \in \mathbb{C}$ and nonzero weights c_k, half the multiplicity, negative for poles. A pole and a zero cannot share a value of w: by the functional equation the zero and pole sets are each symmetric under $\gamma \mapsto -\gamma$, and γ^2 determines γ up to sign. All zeros and poles lie in $0 \leq \operatorname{Re} s \leq 1$, by the Euler product and the functional equation, so $|\operatorname{Im}\gamma_k| \leq \frac{1}{2}$ and $\operatorname{Re} w_k \geq -\frac{1}{4}$. Since Λ has finite order, the numbers of zeros and poles with $|\gamma| \leq T$ grow at most polynomially in T, so $\sum_k |c_k| e^{-x \operatorname{Re} w_k} \ll x^{-A}$ as $x \to 0$ for some A. Fix an integer $M > A + 1$. The transform

$$J(s) = \int_0^\infty e^{-sx} x^M \mathcal{W}_\rho(x)\, dx$$

converges for $\operatorname{Re} s$ large, and integrating term by term in each representation,

$$J(s) = \sum_k \frac{M!\, c_k}{(s+w_k)^{M+1}} = \int_{[0,\infty)} \frac{M!\, \mu(du)}{(s+u)^{M+1}}.$$

The series converges absolutely and locally uniformly off the points $-w_k$, because $|w_k| \asymp |\gamma_k|^2$ and the counting function grows polynomially, so it defines a meromorphic function on $\mathbb{C}$ with a pole of order $M+1$ at each $-w_k$. The integral is holomorphic on $\mathbb{C} \setminus (-\infty, 0]$. Since the two agree for $\operatorname{Re} s$ large, they agree everywhere, so every $-w_k$ lies in $(-\infty, 0]$: every w_k is real and nonnegative. With real exponents, uniqueness of the Laplace transform of measures on $[0,\infty)$ gives $\mu = \sum c_k \delta_{w_k}$, so every c_k is positive. Hence there are no poles, so Λ is entire, and every γ_k is real, since $\gamma_k^2 = w_k \geq 0$. □

Complete monotonicity is a positivity condition on infinitely many derivatives, but each instance $(-1)^j \mathcal{W}_\rho^{(j)}(x) \geq 0$ is a finite computation with Frobenius traces. A pair of poles of Λ at $\frac{1}{2} \pm iT$ contributes $-e^{-xT^2}$, but it is invisible near $x = T^{-2}$: there the prime sum is of order $Te^{-(T\log 2)^2/4}$, so $\mathcal{W}_\rho(T^{-2})$ is fixed by the conductor and the gamma factor, and as soon as there are ordinates in $(\frac{1}{2}T, T)$ and in $(T, \frac{3}{2}T)$ they outweigh the pole term in every inequality $(-1)^j \mathcal{W}_\rho^{(j)}(T^{-2}) \geq 0$. A pole

midway between consecutive ordinates 2δ apart first violates an inequality near $x = j/T^2$ with $j \approx \frac{\log 2}{2}(T/\delta)^2$ (Exercise 4). So the inequalities can disprove Artin's conjecture or GRH for ρ, one violation at a time, but no finite set of them is known to verify either to any finite height. Booker's verification of Artin's conjecture to finite height [75] combines a group theoretic criterion with the location of zeros by the explicit formula and Turing's method.

The swap is also rigid.

PROPOSITION 20.31 (The swap sees every prime; this book). *Let ρ satisfy the conclusion of Theorem 20.30(ii), and let ρ' be self dual Galois data with the same conductor and the same gamma factor, whose local factors differ from those of ρ at a nonempty set P of primes with $\sum_{p\in P} p^{-1/2}\log p < \infty$; suppose that the Satake parameters of ρ and ρ' have absolute value at most 1. Then $\mathcal{W}_{\rho'}$ is not completely monotone. In particular, if $\Lambda(s,\rho')$ has meromorphic continuation of finite order with $\Lambda(s,\rho') = \pm\Lambda(1-s,\rho')$, it has a pole or a zero off the critical line.*

PROOF. By the cosine form (20.2) of the Laplace pair (16.5),

$$\mathcal{W}_{\rho'}(x) - \mathcal{W}_{\rho}(x) = \int_0^\infty e^{-xz} g(z)\, dz,$$

where $c_{p,m} = \Lambda_{\rho'}(p^m) - \Lambda_\rho(p^m)$ and

$$g(z) = -\frac{1}{2\pi\sqrt{z}} \sum_{p\in P}\sum_{m\geq 1} \frac{c_{p,m}}{p^{m/2}} \cos(\sqrt{z}\, m \log p).$$

By hypothesis the coefficients are absolutely summable, so $\sqrt{z}\, g(z)$ is a nonzero uniformly almost periodic function of $r = \sqrt{z}$ with distinct frequencies $m\log p$ (unique factorisation) and no constant term. If $\mathcal{W}_{\rho'}$ were completely monotone it would be the Laplace transform of $\mu' \geq 0$, and by uniqueness $\mu' = \frac{1}{2}\sum \delta_{\gamma^2} + g(z)dz$. The absolutely continuous part of a positive measure is nonnegative, so $g \geq 0$ almost everywhere. But a nonzero almost periodic function with mean zero takes negative values on a set of positive measure. □

Compare Proposition 1.5: altering Frobenius data on a set of primes of density zero changes no distributional statement, but by Proposition 20.31 it destroys the Galois heat trace whenever $\sum_{p\in P} p^{-1/2}\log p < \infty$; a single prime suffices.

Figure 20.9 tests this on the S_3 representation of conductor 23, which is automorphic (it is the weight one form $\eta(z)\eta(23z)$; its first zero, $\gamma_1 = 5.1157$, and its Thorin data are in Table 14.1). The Galois heat trace was computed from Frobenius classes obtained by factoring $x^3 - x - 1$ modulo primes below 60000, with no use of the modular form. It agrees with the sum over the zeros of $L(s,\rho)$ to all 16 digits at $x = 0.02$ (both give 1.4102560772724), to an absolute error of $1.5\cdot 10^{-16}$ at $x = 0.3$ and to 10^{-8} relative at $x = 0.7$, where double precision runs out. With the

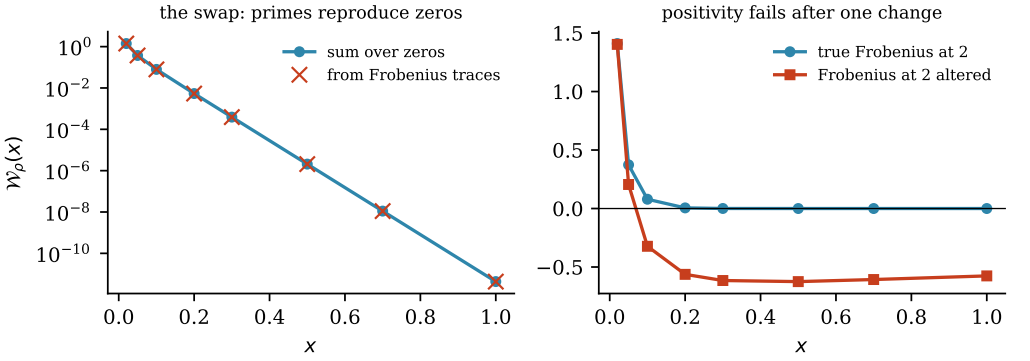


Figure 20.9: Left: the Galois heat trace of Definition 20.29 for the S_3 representation, computed from Frobenius classes alone (crosses), against the sum over zeros (line). Right: changing the Frobenius class at 2 makes it negative.

class of Frob_2 changed from order three to the identity, the same function becomes negative from $x = 0.1$ on (-0.32 at $x = 0.1$, -0.62 at $x = 0.5$), so it is not even positive, let alone completely monotone.

20.8 Van Dantzig duality and the Euler half line

For a self dual Λ with centre $\frac{1}{2}$, $\Lambda(\frac{1}{2}) \neq 0$ and a nonnegative Pólya kernel, the van Dantzig pair of Theorem 16.6, which exists exactly under GRH, is

$$f_\Lambda(t) = \frac{\Lambda(\frac{1}{2} + it)}{\Lambda(\frac{1}{2})}, \qquad g_\Lambda(t) = \frac{\Lambda(\frac{1}{2})}{\Lambda(\frac{1}{2} + t)}.$$

The first member lives on the critical line, where by (20.4) it is the Fourier transform of the Pólya kernel, that is of the automorphic form on $i\mathbb{R}_{>0}$. The second lives on the real axis, and for $t > \frac{1}{2}$ it is the reciprocal of an Euler product times the central value $\Lambda(\frac{1}{2})$, that is Galois data up to one constant. The van Dantzig involution $F \mapsto 1/F(i\,\cdot)$ passes between the two faces, and the Wald couple (W, H) of Theorem 16.6(c), with $\mathbb{E}e^{sW} = \Lambda(\frac{1}{2} + s)/\Lambda(\frac{1}{2})$ and $\mathbb{E}e^{-s^2 H} = \Lambda(\frac{1}{2})/\Lambda(\frac{1}{2} + s)$, pairs a variable whose density is the automorphic form with a generalized gamma convolution whose transform is the Euler product. This suggests reading reciprocity as the existence of a Wald couple whose two members are built on different sides; the next theorem makes one direction precise, using only the Euler half line.

Theorem 20.32 (Euler half line criterion; this book). *Let ρ be self dual with completion $\Lambda(s, \rho)$, defined for $\mathrm{Re}\, s > 1$ by its Euler product, and let $s_0 > \frac{1}{4}$. Suppose that for*

$s > \frac{1}{4}$,

$$\frac{\Lambda(\frac{1}{2}+\sqrt{s_0},\rho)}{\Lambda(\frac{1}{2}+\sqrt{s},\rho)} = C\,\mathbb{E}e^{-sH}$$

for a constant $C > 0$ and a generalized gamma convolution H whose Thorin measure is $\sum_k c_k\delta_{u_k}$ with positive integers c_k and $u_k > 0$. Then $\Lambda(s,\rho)$ extends to an entire function of finite order, $\Lambda(s,\rho) = \Lambda(1-s,\rho)$, every zero lies on $\operatorname{Re} s = \frac{1}{2}$, and Λ is bounded in vertical strips. Conversely, if $\Lambda(s,\rho)$ is entire of order one with this functional equation, $\Lambda(\frac{1}{2},\rho) \neq 0$ and all zeros on the critical line, then the hypothesis holds with $u_k = \gamma_k^2$.

PROOF. A GGC with drift $a \geq 0$ and Thorin measure $\sum c_k\delta_{u_k}$ has $\mathbb{E}e^{-sH} = e^{-as}\prod_k(1+s/u_k)^{-c_k}$, with $\sum_k c_k/u_k < \infty$. Since the c_k are positive integers, $E(z) = e^{az^2}\prod_k(1+z^2/u_k)^{c_k}$ is an even entire function of z of finite order. For real $z > \frac{1}{2}$ we have $\Lambda(\frac{1}{2}+z,\rho) = C^{-1}\Lambda(\frac{1}{2}+\sqrt{s_0},\rho)E(z)$, and both sides are holomorphic on $\operatorname{Re} z > \frac{1}{2}$, so they agree there; the right side provides the continuation of $\Lambda(\frac{1}{2}+z,\rho)$ to an entire function. Evenness of E is the functional equation, and the zeros of E are at $z = \pm i\sqrt{u_k}$, that is $s = \frac{1}{2} \pm i\sqrt{u_k}$. On $\operatorname{Re} s = 1+\delta$ the completion is bounded because the gamma factor decays and the Euler product is bounded; by the functional equation the same holds on $\operatorname{Re} s = -\delta$; finite order and the Phragmén Lindelöf principle give boundedness in the strip between. The converse is Theorem 16.5 applied to $\Lambda(\frac{1}{2}+z,\rho)$, normalised at $z = \sqrt{s_0}$ instead of 0. □

Combining Theorem 20.32 with Proposition 20.27, we obtain a Thorin form of Hecke's converse theorem.

COROLLARY 20.33 (Thorin converse theorem at level one; this book). *Let $k \equiv 0 \bmod 4$ and let (a_p) be the Frobenius traces of a self dual Galois representation of conductor 1 with the gamma factor $\Gamma_{\mathbb{C}}(s+\frac{k-1}{2})$, where $\Gamma_{\mathbb{C}}(s) = \Gamma_{\mathbb{R}}(s)\Gamma_{\mathbb{R}}(s+1) = 2(2\pi)^{-s}\Gamma(s)$, and Ramanujan bound, with a_n its Dirichlet coefficients. If the normalised Euler product $\Lambda(\frac{1}{2}+\sqrt{s_0})/\Lambda(\frac{1}{2}+\sqrt{s})$, $s > \frac{1}{4}$, is a constant times the Laplace transform of a GGC whose Thorin measure is a sum of point masses with positive integer weights, then $\sum a_n n^{(k-1)/2}e^{2\pi inz}$ is a cusp form of weight k for $\mathrm{SL}_2(\mathbb{Z})$, and it satisfies GRH.*

PROOF. By Theorem 20.32 the completion is entire, bounded in vertical strips and even about the centre. In the classical normalisation (centre $\frac{k}{2}$) this is $\Lambda(k-s) = \Lambda(s) = (-1)^{k/2}\Lambda(s)$, which is (iii) of Proposition 20.27 with $N = 1$, so f is a cusp form. □

The hypothesis of Corollary 20.33 is stated entirely on the Euler half line, so it is a statement about Galois data; it is stronger than reciprocity, since it includes GRH. Its natural weakening, with nonzero integer weights and complex atoms, always holds for the Artin representations of Open Problem 20.3, where Artin's conjecture is the positivity of the weights.

20.8.1 An unconditional Galois family

Positivity is available unconditionally on the Galois side whenever the logarithmic coefficients are nonnegative, as for Dedekind zeta functions; the case used here is the Rankin Selberg square. For $\alpha > 1$ let $F(s) = \Lambda^S(s, \rho \otimes \check{\rho})$, the completion of D^S of Section 20.6.

PROPOSITION 20.34 (this book). *For every Artin representation ρ and every $\alpha > 1$ there is a GGC H_α with*

$$\frac{F(\alpha)}{F(\alpha + \sqrt{s})} = e^{-b_\alpha \sqrt{s}}\, \mathbb{E} e^{-sH_\alpha}, \qquad b_\alpha = \frac{F'(\alpha)}{F(\alpha)},$$

whose Thorin density is the $2\sin^2$ *transport (Lemma 16.7) of the measure*

$$\sum_{p \notin S,\, m \geq 1} |a_\rho(p^m)|^2 (\log p)\, p^{-m\alpha} \delta_{m \log p}$$

(x times the Lévy measure of X_α) together with the gamma factor contribution. It is computed from Frobenius traces alone.

PROOF. The function F is holomorphic and zero free on $\operatorname{Re} s > 1$. The second derivative of $\log F$ along the real axis is the sum of the trigamma terms of the gamma factor and of $\sum |a_\rho(p^m)|^2 (m \log p)^2 m^{-1} p^{-ms}$, both completely monotone and tending to zero. Lemma 16.7 applies. □

As α decreases to 1 the pole of order $\langle \rho, \rho \rangle$ appears (Theorem 20.28). Below 1 the Euler product no longer defines F. Meromorphic continuation of $L(s, \rho \otimes \check{\rho})$ is known for every Artin ρ by Brauer's theorem, and holomorphy away from $s = 1$ follows from automorphy of ρ by the Rankin Selberg theory; neither gives positivity. For automorphic ρ, apply the argument of Theorem 16.8 to the complete function $\Lambda(s, \rho \otimes \check{\rho})$, all Euler factors included and the poles at $s = 0, 1$ removed: its transforms at the base points $\alpha \in (\frac{1}{2}, 1]$ are GGCs, with nonnegative Thorin densities $V_{\alpha - 1/2}$, for every such α exactly when GRH holds for $L(s, \rho \otimes \check{\rho})$, and the densities then converge to the Thorin measure of the zeros as $\alpha \downarrow \frac{1}{2}$. For irreducible two dimensional ρ this GRH contains the Riemann hypothesis, since $L(s, \rho \otimes \check{\rho}) = \zeta(s) L(s, \operatorname{Ad} \rho)$ with $L(s, \operatorname{Ad} \rho)$ entire [212]. Splitting off the factor $e^{-b_\alpha \sqrt{s}}$ as in Proposition 20.34, with b_α now the logarithmic derivative of this complete function at α, leaves the densities $V_{\alpha - 1/2} - b_\alpha / (2\pi\sqrt{z})$ (Lemma 16.7), and their positivity below $\alpha = 1$ is not known to follow from GRH; for ξ it follows from the Riemann hypothesis by a theorem of Lagarias, and is not known unconditionally (Section 16.4.3). *Continuing a family of positive measures, defined from Frobenius traces above the line* $\operatorname{Re} s = 1$*, down to the centre while preserving positivity* is the probabilistic form of the problem that this chapter proposes; for the Rankin Selberg square it is holomorphy away from $s = 1$ together with GRH for $L(s, \rho \otimes \check{\rho})$, which does not by itself give reciprocity.

20.9 THE PROGRAMME

The equivalences of this chapter prove nothing new about reciprocity; like the Thorin forms of the Riemann hypothesis, they move the difficulty rather than remove it. They offer positivity and symmetry statements about Galois data, to which probabilistic tools apply and which can be tested one prime at a time.

1. Symmetry. At level one, reciprocity is the evenness (oddness when $k \equiv 2 \bmod 4$) of the Pólya kernel built from Frobenius traces (Proposition 20.27). At level N it is the pairing of twisted kernels required by Weil's converse theorem.

2. Positivity. For self dual ρ with meromorphic continuation of finite order and a functional equation, holomorphy together with GRH is the complete monotonicity of the Galois heat trace (Theorem 20.30), a function of Frobenius traces each of whose derivatives is computable at each x to any accuracy in finitely many steps. One certified violation disproves holomorphy or GRH; no finite set of verified inequalities is known to prove them to any finite height.

3. Rigidity. The Galois heat trace detects a change of Frobenius data at a single prime (Proposition 20.31), in contrast with every distributional statement (Proposition 1.5).

4. Continuation. On the Euler half line the Galois side produces GGCs, from the Rankin Selberg square unconditionally (Proposition 20.34) and from ρ itself under the Thorin hypothesis (Theorem 20.32). Holomorphy together with GRH, for $L(s, \rho \otimes \check{\rho})$ in the first case and for $L(s, \rho)$ in the second, is the continuation to the centre of the corresponding GGCs (for the Rankin Selberg square with its poles removed and the drift kept inside, as in Theorem 16.8); at level one the second gives reciprocity (Corollary 20.33).

5. Laws. On the Frobenius probability space reciprocity is almost sure equality, and its uniqueness statements, the correspondence of irreducible with cuspidal and, for every $\epsilon > 0$, agreement within ϵ outside a set of primes of Dirichlet density zero follow from two orthonormality theorems (Corollary 20.15); class field theory is measurability with respect to congruences (Proposition 20.16). Laws cannot give existence; the objects of items 1 to 3 see individual primes, but they too give existence only through a converse theorem.

6. Gamma processes. Equidistribution of Frobenius data is a Gamma process limit of the labelled factorisation of a random integer (Theorem 20.18); reciprocity laws are vanishing statements for Thorin measures (Theorem 20.22); and comparing an Artin representation with a cuspidal one gives a conjec-

tured zero two law (Conjecture 20.24), the orthogonality half of reciprocity: it is implied by reciprocity and asserts no existence, and the existence of a partner is the other half.

7. Gaussian model. For GL_1 the symmetry part of the programme runs, and the circle models the positivity part: the theta transformation, proved from the normal characteristic function or by Fejér positivity of the winding number terms, gives quadratic reciprocity and Gauss's sign (Section 20.2).

8. Irreducibility. The Gamma shape of the Rankin Selberg subordinator is $\langle \rho, \rho \rangle$ (Theorem 20.28), the probabilistic shadow of "irreducible corresponds to cuspidal".

The most concrete target is the even icosahedral case, where reciprocity is open and the Galois data are explicit. For such ρ every quantity above can be computed from the Frobenius classes. Factorisation patterns modulo p give these only up to Galois conjugacy, so they cannot tell apart the two classes of elements of order 5 in the projective image A_5, on whose preimages the traces of ρ differ; a Frobenius resolvent separates them [161]. When $\det \rho$ is trivial, so that ρ is self dual, item 2 tests Artin's conjecture together with GRH (Problem 20.1). The analogue of Proposition 20.27 for the gamma factor $\Gamma_{\mathbb{R}}(s)^2$ or $\Gamma_{\mathbb{R}}(s+1)^2$ of an even representation (as complex conjugation acts by 1 or -1), with a K Bessel kernel in place of the exponential, gives symmetry tests for the twisted kernels.

20.9.1 Open problems

OPEN PROBLEM 20.1 (The Thorin converse theorem in general). Formulate and prove a version of Theorem 20.32 for twists $\rho \times \pi'$ that are not self dual, strong enough to combine with the converse theorem of Cogdell and Piatetski Shapiro. The obstacle is that the completion Ξ_ρ of Definition 14.5, built from $\Lambda(\frac{1}{2}+z,\rho)\Lambda(\frac{1}{2}+z,\check{\rho})$, controls the product of two L functions, whereas converse theorems need each one.

OPEN PROBLEM 20.2 (Descending the base point). For automorphic ρ, positivity of the densities $V_{\alpha-1/2}$, with the drift kept inside, for every $\alpha \in (\frac{1}{2}, 1]$ is GRH for $L(s, \rho \otimes \check{\rho})$ (end of Section 20.8). Decide whether the densities $V_{\alpha-1/2} - b_\alpha/(2\pi\sqrt{z})$ of that section, with the drift split off, also stay nonnegative there, which for ξ holds under the Riemann hypothesis and is proved unconditionally for $\alpha \geq 1$ (Section 16.4.3) and for $\frac{1}{2} < \alpha < 1$ below height 3×10^{12} (Chapter 18), and identify the obstruction to positivity below $\alpha = 1$ in terms of Frobenius data alone: which features of the Lévy measure of X_α at $\alpha > 1$ control the sign of the transported density at $\alpha < 1$? For ξ and Δ the computational counterpart is the problem on transported Thorin densities in the strip in Chapter 16.

OPEN PROBLEM 20.3 (Signed Thorin measures). For an Artin representation ρ with real character, no trivial constituent, root number 1 and $\Lambda(\frac{1}{2},\rho)$ finite and nonzero, Brauer's theorem and Hadamard factorisation give, for a fixed $s_0 > \frac{1}{4}$, $\Lambda(\frac{1}{2}+\sqrt{s_0},\rho)/\Lambda(\frac{1}{2}+\sqrt{s},\rho) = C\prod_k (1+s/u_k)^{-c_k}$ with nonzero integers c_k and complex $u_k = \gamma_k^2$, where $c_k < 0$ exactly at poles, and $u_k \notin (0,\infty)$ exactly at zeros and poles off the critical line. Artin's conjecture for ρ is the statement that every c_k is positive. Find a property of the signed Lévy measure of $\log\Lambda(s,\rho)$ on the Euler half line, that is of Frobenius data, which forces every c_k to be positive without forcing the u_k to be real, so that holomorphy is separated from GRH (compare Section 14.6).

OPEN PROBLEM 20.4 (A Fejér summation for primes). Find positive weights $w_N(n) \le 1$ with $w_N(n) \to 1$ such that the prime side of Definition 20.29, with $\Lambda_\rho(n)$ replaced by $w_N(n)\Lambda_\rho(n)$ and the archimedean term adjusted accordingly, is the Laplace transform of a positive measure, and such that positivity for each N follows from Frobenius data alone, as Fejér positivity does for the circle (Proposition 20.9); otherwise $w_N \equiv 1$ works whenever holomorphy and GRH hold. By Theorem 20.30 this would prove holomorphy and GRH for $L(s,\rho)$; the problem is to see what positivity of this kind forces on Frobenius data (compare Open Problem 30.8).

OPEN PROBLEM 20.5 (Reciprocity in law versus reciprocity). Show that if an irreducible Artin representation ρ is reciprocal in law with a cuspidal π (Corollary 20.15), then $a_\rho(p) = a_\pi(p)$ for all but finitely many p (compare Open Problem 30.4 and Corollary 1.6). It holds whenever ρ is known to be automorphic, by Corollary 20.15(v); for $n = 2$, where this leaves the even icosahedral ρ, it would follow from a refinement of Ramakrishnan's theorem [451] to the case in which one member is Galois and the agreement is only approximate, as in Corollary 20.15(iv); in general it is a strong multiplicity one statement across the two sides.

OPEN PROBLEM 20.6 (The mixed zero two law). Show that meromorphy of $L(s,\rho\times\tilde{\pi})$ near $\operatorname{Re} s = 1$, with at most a simple pole at $s = 1$ and no zeros on the line, gives the dichotomy of Conjecture 20.24 by the argument of Theorem 20.23 and reduces the rest of the conjecture to a form of Open Problem 20.5, and prove the conjecture for one even icosahedral Artin representation.

OPEN PROBLEM 20.7 (Joint marked processes). Show that the joint limit of the marked processes of Theorem 20.18 exists for an Artin representation ρ and a cuspidal π of the same dimension; it is not determined by the two marginal limits (compare Exercise 7 and Theorem 4.11). Is the joint label law either supported on $\{a_\rho = a_\pi\}$, or such that the Thorin measure of $|a_\rho - a_\pi|^2$ has mass bounded below by a constant depending only on n?

OPEN PROBLEM 20.8 (Signs of drifts). When does the split class approach its

Gamma limit from above, $\kappa_{\{1\}} < 0$ in Theorem 20.26? This requires $\gamma_K^S < 0$. Characterise such fields and relate the signs of the κ_C to the bias between classes that Section 3.1 measures for the field of Example 3.3 (Exercise 5), and to the Chebotarev races of the computational problems of Chapter 3.

20.10 NOTES

Reciprocity in the form used here is from Langlands' letter and lectures [349, 350]; the known cases cited are [351, 553, 581, 541, 41, 311, 312, 320]; potential automorphy is from [266, 67] and symmetric power functoriality from [419, 420]. Artin's conjecture and Brauer's theorem are [27, 84], and Artin's reciprocity law is [28]. Converse theorems are due to Hecke [250], Weil [571], Jacquet and Langlands [277] and Cogdell and Piatetski Shapiro [108]. The Rankin Selberg pole is from Jacquet and Shalika [279] and Moeglin and Waldspurger [397]. Theorem 20.30(ii) is Weil's positivity criterion [569, 572] restricted to the test functions $t^{2j}e^{-xt^2}$; Weil's extension of the explicit formula to Artin Hecke L functions [572] already ties positivity to Artin's conjecture together with GRH. The Thorin framework for automorphic L functions of Chapters 14 to 16 extends the first author's work [441, 442]; the sine squared lemma is Lemma 14 of [442], with its first step from Roynette and Yor [467], who also introduced Wald couples with infinitely divisible members; van Dantzig pairs and Wald couples for ξ are in [467, 442, 330]. The theta transformation goes back to Jacobi and Poisson; reciprocity for Gauss sums is due to Schaar [486] and Landsberg [346], and the derivation of quadratic reciprocity from it is in Dym and McKean [168]. Our proofs in Section 20.2 differ only in reading Poisson summation as the Fourier series of a wrapped normal law and Fejér's proof of it as a Laplace transform through the pair of the $2\sin^2$ lemma (Proposition 20.9). Zolotarev's proof [595], through signs of permutations, is another short route to Corollary 20.6. Dedekind sums and their reciprocity law are treated in [449]. Poisson random measures and the Gamma process are in Kingman [318], the Dirichlet process in Ferguson [186], and generalized gamma convolutions in Thorin [546] and Bondesson [73]. Sato Tate for newforms and its joint form are [67, 240]. Euler Kronecker constants are Ihara's [270]; negative values for cyclotomic fields are in [196]. Ours are Theorems 20.18, 20.21, 20.22, 20.23 and 20.26 in the forms stated, the bounds (i) and (ii) of Theorem 20.10, Corollary 20.20, Conjecture 20.24 with Proposition 20.25, the probabilistic formulations of Section 20.3, Corollary 20.15, Proposition 20.16 (a reformulation of Chebotarev's theorem for $K(\zeta_N)$), Proposition 20.27 in this form, Theorem 20.28, Definition 20.29, Theorem 20.30, Proposition 20.31, Theorem 20.32, Corollary 20.33, Proposition 20.34 and the computations; each rests on the cited theory.

20.11 EXERCISES

1. Show that the kernel Φ_f of (20.4) for $f = \Delta$ is $\Lambda(6,\Delta)\Phi_\Delta$, with Φ_Δ the normalised kernel of Chapters 2 and 13, so that its evenness, proved there, is the case $N = 1$, $k = 12$ of Proposition 20.27. For a newform f of level N and trivial character, and a primitive character χ modulo q with $(q, N) = 1$, write the relation between $\Phi_{f\otimes\chi}(-u)$ and $\Phi_{f\otimes\bar\chi}(u)$ that Weil's converse theorem requires, with its constant.

2. For the S_3 representation ρ of $x^3 - x - 1$, show that $\rho \otimes \rho \cong 1 \oplus \mathrm{sgn} \oplus \rho$ and $L(s, \rho\otimes\rho) = \zeta(s)L(s,\chi_{-23})L(s,\rho)$, and deduce the shape 1 in Theorem 20.28. What is the shape for $1 \oplus \mathrm{sgn}$?

3. Identify, prime by prime, the Laplace transform in Proposition 20.31 with the oscillatory part of a local Thorin comb (Proposition 15.8). Compare it with the winding number terms of Proposition 20.9: why do the cosine densities of all winding numbers, summed with Fejér weights, give point masses, while those of the powers of a single prime give a signed density?

4. Suppose $\Lambda(s,\rho)$ has simple poles at $\frac12 \pm iT$, $\Lambda(\frac12,\rho) \neq 0$, and is otherwise as in GRH, with positive ordinates $\gamma_1 < \gamma_2 < \cdots$. Show that $\mathcal{W}_\rho(x) < 0$ for all large x if $T < \gamma_1$. If $\frac12 T < \gamma_n < T < \gamma_{n+1} < \frac32 T$, show that every inequality $(-1)^j \mathcal{W}_\rho^{(j)}(T^{-2}) \geq 0$ holds, but that $(-1)^j \mathcal{W}_\rho^{(j)}(j/T^2) < 0$ for all large j; when T is the midpoint and $2\delta = \gamma_{n+1} - \gamma_n$ is small compared with T, the least such j is about $\frac{\log 2}{2}(T/\delta)^2$. (For the S_3 representation of conductor 23 and T midway between γ_1 and γ_2 the first violation is at $j = 13$, near $x = 13.5/T^2$.)

5. Apply Lemma 20.2 with $y \neq 0$ in the proof of Theorem 20.4 to obtain reciprocity for $\sum_{n \bmod c} e^{\pi i(an^2+bn)/c}$ when $ac + b$ is even.

6. Let K be uniform on $\mathbb{Z}/c\mathbb{Z}$, let $((x)) = x - \lfloor x\rfloor - \frac12$ for $x \notin \mathbb{Z}$ and $((x)) = 0$ otherwise, and let $s(b,c) = \sum_{k \bmod c}((k/c))((bk/c))$ be the Dedekind sum. Show that $s(b,c)$ is c times the covariance of $((K/c))$ and $((bK/c))$, and that for U uniform on $[0,1)$ the covariance of $((bU))$ and $((cU))$ is $1/(12bc)$ when $\gcd(b,c) = 1$. Explain why this does not yet give Dedekind's reciprocity law $s(b,c) + s(c,b) = \frac{b^2+c^2+1}{12bc} - \frac14$. A probabilistic proof of the reciprocity law, and an extension of Theorem 20.10 to the Dedekind eta function, whose multiplier involves Dedekind sums, are left open.

7. For the S_3 representations ρ and ρ_{31} of the text, compute the limiting joint law of $(a_\rho, a_{\rho_{31}})$ and check that $d(\rho,\rho_{31})^2 = 2$. What is the joint law for ρ and $\rho \otimes \chi_{-23}$, and what is d?

8. Show that for $s > \frac{1}{2}$ the Wald couple identity $\mathbb{E}e^{sW}\mathbb{E}e^{-s^2H} = 1$ involves Λ only through the Euler product values $\Lambda(\frac{1}{2}+s)$ and the single central value $\Lambda(\frac{1}{2})$. Show that a random variable W with $\mathbb{E}e^{sW} = \Lambda(\frac{1}{2}+s)/\Lambda(\frac{1}{2})$ for all $s > \frac{1}{2}$, if it exists, has its law determined by these values (tilt by $e^{s_0 W}$ with $s_0 > \frac{1}{2}$), and explain why the identity on $s > \frac{1}{2}$ cannot by itself show that such a W exists.

20.12 Open computational problems

Computational Problem 20.1 (Even icosahedral representations). Objective: for an icosahedral Artin representation ρ with trivial determinant (hence self dual and even) from the LMFDB, compute the Galois heat trace of Definition 20.29 and its derivatives in ball arithmetic and search for a certified violation of $(-1)^j \mathcal{W}_\rho^{(j)}(x) \geq 0$, which by Theorem 20.30 would disprove Artin's conjecture or GRH for ρ. Objects and scale: one representation, prime powers up to 10^{10}, orders $j \leq 10^3$ and $x \in [0.01, 4]$, where these primes suffice to certify signs (beyond $x \approx 5$ the omitted prime powers can outweigh the inequalities); a pole at height T midway between ordinates 2δ apart would first show near $x = j/T^2$ with $j \approx \frac{\log 2}{2}(T/\delta)^2$, that is near $x = \frac{\log 2}{2\delta^2}$, so this range reaches it only for δ above about 0.3 and $T \leq 54\delta$, which at conductors of a few million, as in the LMFDB examples, leaves the lowest gaps and unusually wide ones. Route: Frobenius classes from the factorisation of the defining polynomial modulo primes, with a Frobenius resolvent [161] separating the two Galois conjugate classes of order 5 and the two of order 10, and archimedean integrals in Arb; the same computation for newforms is the problem on monotonicity across weights and levels in Chapter 13. Deliverable: certified signs on the grid, and the region of (T, δ) in which a pole would have been detected, compared with Booker's finite height verifications [75]. Why it matters: it evaluates the swap on a representation whose automorphy is unknown; satisfied inequalities are not known to verify Artin's conjecture or GRH.

Computational Problem 20.2 (Kernel symmetry at level N). Objective: test the twisted kernel symmetries required by Weil's converse theorem for Galois data of conductor up to 10^4 (Artin representations and elliptic curves from the LMFDB), and measure how quickly a change of Frobenius at one prime is detected as a function of that prime. This extends the computational problems on functional equations as a test statistic (Chapter 2) and on detecting impostors (Chapter 1) to twisted kernels and Artin representations. Route: twisted kernels $\Phi_{f\otimes\chi}$ from q expansions (for even Artin representations, with the K Bessel kernel of their gamma factor), with coefficients from the LMFDB or from Frobenius data, for primitive χ of conductor up to 50 prime to N, checked against PARI/GP `lfun`. For even representations the kernels can also be computed from the L function alone,

by Bromwich inversion of $\Lambda(s, \rho \otimes \chi)$ on the critical line as in Proposition 23.8, with the gamma factor $\Gamma_{\mathbb{R}}(s)^2$ or $\Gamma_{\mathbb{R}}(s+1)^2$; this assumes Artin's conjecture for $\rho \otimes \chi$, and reciprocity predicts that the result is built from a Maass form u of eigenvalue $\frac{1}{4}$ on the imaginary axis: from u itself for $\Gamma_{\mathbb{R}}(s)^2$, and from $\partial u/\partial x$, u being odd, for $\Gamma_{\mathbb{R}}(s+1)^2$. Deliverable: asymmetry tables and detection thresholds. Why it matters: it measures how much of Weil's converse theorem a finite computation sees.

COMPUTATIONAL PROBLEM 20.3 (Gamma process tests at scale). Objective: compute the Thorin measures of Theorem 20.22 and the drifts of Theorem 20.26 for every Galois field of degree at most 10 in the LMFDB with absolute discriminant below 10^6, and for pairs of elliptic curves of conductor up to 10^4. Route: the scripts `laws.py` and `second.py` of Table B.1, with Frobenius elements from `idealfrobenius` after `galoisinit` (which handles every group of order at most 10) and a_p from `ellap` in PARI/GP, for primes up to 10^8. Deliverable: tables of drifts against γ_K, and of zero probability exponents against Thorin masses. Why it matters: it tests the drift of the split class against Ihara's Euler Kronecker constant and its GRH bound (Theorem 20.26) on many fields at once.

COMPUTATIONAL PROBLEM 20.4 (Gamma shapes for ℓ adic representations). Objective: estimate the Gamma shape of Theorem 20.28 from Frobenius traces alone, for symmetric powers of elliptic curves and for GL_3 motives, and use it as a statistical test of irreducibility. It is the Dirichlet density counterpart of the Weyl average problems of Chapter 4. Route: $a(p)$ from `ellap` and the symmetric power recursions in PARI/GP for $p \le X = 10^9$. Deliverable: shape estimates with confidence intervals at $\sigma - 1 \ge 5/\log X$, after correcting the truncation bias $mE_1((\sigma-1)\log X)$, compared with the direct estimate $\pi(X)^{-1}\sum_{p\le X}|a(p)|^2$ of m. Why it matters: the Gamma shape is the probabilistic form of the Rankin Selberg pole order, and the comparison shows how much of it is visible at finite X.

CHAPTER 21

HITTING TIMES AND RECIPROCITY: ISOGENY WALKS, PRIME FACTORS, EXPLOSION AND EXIT

A hitting time converts spectral data into an expectation. If a reversible chain has eigenvalues μ_j, its mean hitting times are weighted sums of the terms $1/(1-\mu_j)$, and Kemeny's constant is their plain sum; if a birth process has rates u_k, its explosion time has Laplace transform $\prod_k (1+s/u_k)^{-1}$; if a Brownian particle is killed at the ends of an interval, its exit density has two expansions, one over eigenvalues and one over reflected images. This chapter uses the three mechanisms to read reciprocity laws as identities between hitting times, and adds a fourth in which the hitting rate of a set of primes is a Chebotarev density.

The first section is about a reciprocity between two primes, whose simplest instance, the count of spanning trees, is a formula of Hashimoto [246, 364], read as a reciprocity law by Munier and Shnidman [408]. A random walk on supersingular elliptic curves in characteristic N, stepping along isogenies of degree ℓ, is a geometric object attached to N. Its Kemeny constant, its number of spanning trees and the sum of its resistances are computed by the reduction modulo ℓ of the elliptic curves and abelian varieties attached to the newforms of level N (Theorems 21.1, 21.2 and 21.6). The two primes exchange roles: the walk lives modulo N and moves by ℓ; the answer counts points modulo ℓ on objects of level N. The bridge is the Jacquet Langlands correspondence of Chapter 10, and the Eichler Shimura congruence relation. The denominator in the tree count is the rational torsion of $J_0(N)$ (Corollary 21.3), the critical group of the graph has the same order as $J_0(N)(\mathbb{F}_\ell)$ modulo its cuspidal subgroup but in general not the same structure (Propositions 21.4 and 21.5), and as $N \to \infty$ the exponential growth rate of $|J_0(N)(\mathbb{F}_\ell)|$ per unit of genus, an asymptotic class number given by a theorem of Tsfasman and Vlăduţ [552], is the spanning tree entropy of the $(\ell+1)$ regular tree (Theorem 21.8).

The other three sections give hitting time forms of the results of Chapter 20. Chebotarev densities are hitting rates for the largest prime factors of a random integer (Theorem 21.10), so that reciprocity laws are statements that a discrepancy set is hit at rate zero. The Thorin converse theorem of Chapter 20 becomes a statement about the explosion time of a pure birth process whose rates are the squared ordinates of the zeros (Theorem 21.11 and Corollary 21.12); under GRH

the mean explosion time is the integral of the Galois heat trace. Finally the functional equation of every odd primitive Dirichlet L function, with its root number, is the identity between the image and the eigenfunction expansions of a signed exit law of Brownian motion from an interval (Theorem 21.13). No new case of reciprocity is proved.

21.1 Two prime reciprocity on isogeny graphs

Let N be prime and $\ell \neq N$ another prime. As in Chapter 10 let $E_1, \dots, E_h$ be the supersingular elliptic curves over $\overline{\mathbb{F}}_N$ up to isomorphism, and $w_i = |\operatorname{Aut}(E_i)|/2$. The Brandt matrix $B(\ell)$ has entries $B(\ell)_{ij}$ equal to the number of cyclic subgroups $C \subset E_i$ of order ℓ with $E_i/C \cong E_j$. Each row sums to $\ell + 1$, so

$$P = \frac{B(\ell)}{\ell + 1}$$

is a stochastic matrix: the *isogeny walk* moves from a curve to its quotient by a uniformly chosen subgroup of order ℓ. It is reversible with respect to $\pi_i \propto 1/w_i$, because $B(\ell)_{ij}/w_i = B(\ell)_{ji}/w_j$ [227], and $\sum_i 1/w_i = (N-1)/12$ by Eichler's mass formula. For $N \equiv 1 \bmod 12$ all $w_i = 1$, $h = (N-1)/12$, π is uniform and the walk is simple random walk on the $(\ell+1)$ regular multigraph $G_N(\ell)$ of Chapter 10.

By Eichler and Gross [171, 227] (Theorem 10.4 for $N \equiv 1 \bmod 12$), the eigenvalues of $B(\ell)$ are $\ell + 1$, once, and the eigenvalues $a_\ell(f)$ of T_ℓ on $S_2(\Gamma_0(N))$, with multiplicity, where f runs over the $g = \dim S_2(\Gamma_0(N))$ normalised Hecke eigenforms; g is the genus of $X_0(N)$ and $h = g + 1$. By Eichler and Shimura together with Weil's Riemann hypothesis for curves, $|a_\ell(f)| \leq 2\sqrt{\ell}$. The arithmetic input that turns spectra into point counts is the Eichler Shimura congruence relation [514]: for $\ell \nmid N$ the characteristic polynomial of Frobenius on the Tate module of the reduction of $J_0(N)$ modulo ℓ is $\prod_f (x^2 - a_\ell(f)x + \ell)$, so

$$|J_0(N)(\mathbb{F}_\ell)| = \prod_f \bigl(\ell + 1 - a_\ell(f)\bigr), \tag{21.1}$$

and if f has rational coefficients, attached to an elliptic curve E_f of conductor N, then $\ell + 1 - a_\ell(f) = |E_f(\mathbb{F}_\ell)|$.

Theorem 21.1 (two prime Kemeny formula; this book). *For every prime N and prime $\ell \neq N$, Kemeny's constant of the isogeny walk is*

$$K_N(\ell) = \sum_f \frac{\ell + 1}{\ell + 1 - a_\ell(f)}.$$

In particular, if every newform of level N has rational coefficients,

$$K_N(\ell) = (\ell + 1) \sum_E \frac{1}{|E(\mathbb{F}_\ell)|},$$

the sum over isogeny classes of elliptic curves over $\mathbb{Q}$ of conductor N. In general the contribution of a Galois orbit of forms, attached to an abelian variety A_f, is $(\ell+1)\,\mathrm{Tr}_{K_f/\mathbb{Q}}\big(1/(\ell+1-a_\ell(f))\big)$.

PROOF. Theorem A.4 gives $K=\sum_{j\ge 2}(1-\mu_j)^{-1}$ for a reversible chain, with μ_j the nontrivial eigenvalues of P. These are $a_\ell(f)/(\ell+1)$, and $1-a_\ell(f)/(\ell+1)=(\ell+1-a_\ell(f))/(\ell+1)$. By Eichler and Shimura and the modularity of semistable elliptic curves [581, 541], the rational newforms of level N correspond to the isogeny classes of elliptic curves over $\mathbb{Q}$ of conductor N, and isogenous curves have equally many points over $\mathbb{F}_\ell$, so the second statement follows from $\ell+1-a_\ell(f)=|E_f(\mathbb{F}_\ell)|$. □

So the expected time for the walk in characteristic N to reach a stationary target is a harmonic sum of the factors $\ell+1-a_\ell(f)$ of $|J_0(N)(\mathbb{F}_\ell)|$, which are point counts modulo ℓ when f is rational. For $N\equiv 1 \bmod 12$ the first formula is the expression in the proof of Theorem 10.6. For $N=11$ there are two supersingular curves, $g=1$, and the curve $11a1$ has $|11a1(\mathbb{F}_2)|=5$, so $K_{11}(2)=\frac{3}{5}$. For $N=37$, with the two curves $37a1$ and $37b1$, $|37a1(\mathbb{F}_2)|=5$ and $|37b1(\mathbb{F}_2)|=3$, so $K_{37}(2)=3(\frac{1}{5}+\frac{1}{3})=\frac{8}{5}$. Both values agree with the walk computed from modular polynomials. At $N=2$, and likewise at $N=3,5,7,13$, there is a single supersingular curve and $K_N(\ell)=0$; Chapter 22 reads the case $N=2$ as the last step of the proof of Fermat's last theorem.

THEOREM 21.2 (two prime matrix tree theorem; Hashimoto [246, 364, 408]). *Let $N\equiv 1 \bmod 12$ be prime and $\ell\neq N$ prime. The number of spanning trees of the supersingular ℓ isogeny graph $G_N(\ell)$ is*

$$\tau\big(G_N(\ell)\big)=\frac{|J_0(N)(\mathbb{F}_\ell)|}{h},\qquad h=\frac{N-1}{12}.$$

Equivalently, the critical group (sandpile group) of $G_N(\ell)$ has order $|J_0(N)(\mathbb{F}_\ell)|/h$.

PROOF. For a connected multigraph on h vertices with Laplacian $L=(\ell+1)I-B(\ell)$, Kirchhoff's matrix tree theorem gives $\tau=h^{-1}\prod_{\mu\neq 0}\mu$, the product over the nonzero eigenvalues of L; loops do not affect L. These eigenvalues are $\ell+1-a_\ell(f)$, and (21.1) gives the product. The critical group is the cokernel of the reduced Laplacian, whose order is τ [59]. □

For $N=37$ and $\ell=2$ the graph has three vertices and $|J_0(37)(\mathbb{F}_2)|=5\cdot 3=15$, so it has 5 spanning trees. For $N=61$, $\ell=7$, $|J_0(61)(\mathbb{F}_7)|=4865$ and the graph on 5 vertices has 973 spanning trees.

The factor h has a meaning. By Mazur's theorem [384], conjectured by Ogg [428], the rational torsion subgroup of $J_0(N)$ for prime N is the cuspidal subgroup C_N, cyclic of order the numerator of $(N-1)/12$, which is h when $N\equiv 1 \bmod 12$.

For odd $\ell \neq N$ reduction modulo ℓ is injective on the rational torsion of an abelian variety with good reduction at ℓ [295].

COROLLARY 21.3. *For $N \equiv 1 \bmod 12$ prime and $\ell \neq N$ prime,*

$$\tau\big(G_N(\ell)\big) = \frac{|J_0(N)(\mathbb{F}_\ell)|}{|J_0(N)(\mathbb{Q})_{\text{tors}}|},$$

and for odd ℓ this is the order of the quotient group $J_0(N)(\mathbb{F}_\ell)/C_N$.

For $N \equiv 1 \bmod 12$ the structure of the two groups can be compared. Let $X = \mathbb{Z}^h$ be the divisors on the supersingular curves, X^0 those of degree zero, and $L = (\ell+1)I - B(\ell)$ the Laplacian. The critical group of the graph is $X^0/L(X)$.

PROPOSITION 21.4 (Munier and Shnidman [408]). *L maps X^0 injectively into itself, $|X^0/L(X^0)| = |J_0(N)(\mathbb{F}_\ell)|$, and the critical group of $G_N(\ell)$ is the quotient of $X^0/L(X^0)$ by the cyclic subgroup of order h generated by the class of $L\delta_E$, for any vertex E.*

PROOF. Rows and columns of L sum to zero, so $L(X) \subset X^0$, and the kernel of L is the constants, which meet X^0 only in 0. The determinant of L on the lattice X^0 is the product of its eigenvalues on the orthogonal complement of the constants, that is $\prod_f (\ell + 1 - a_\ell(f)) = |J_0(N)(\mathbb{F}_\ell)|$ by (21.1). Finally L induces an isomorphism of $X/(X^0 + \mathbb{Z}1)$, which is cyclic of order h through the degree, onto $L(X)/L(X^0)$, and $L\delta_E$ is the image of a generator. □

So both sides have the same order: a group of order $|J_0(N)(\mathbb{F}_\ell)|$ modulo a cyclic subgroup of order h. They do not have the same structure in general, as Munier and Shnidman observe [408]: the critical group is a quotient of X^0, a module of rank one over the Hecke algebra, while $J_0(N)(\mathbb{F}_\ell)$ comes from a Tate module, of rank two. Let $\mathbb{T}$ be the ring generated by the Hecke operators T_m, $N \nmid m$, on $S_2(\Gamma_0(N))$, which acts on $J_0(N)$ and, through the Brandt matrices, on X^0, and for a prime r put $\mathbb{T}_r = \mathbb{T} \otimes \mathbb{Z}_r$; for $r \neq \ell$ let F be the action of Frobenius at ℓ on the r adic Tate module V_r of $J_0(N)$.

PROPOSITION 21.5 (after Munier and Shnidman [408]). *Let $N \equiv 1 \bmod 12$ and $\ell \neq N$ be primes, and let $r \neq \ell$ be a prime that divides neither hN nor the discriminant of $\mathbb{T}$. Then the r part of the critical group of $G_N(\ell)$ is isomorphic to $\mathbb{T}_r/(\ell + 1 - T_\ell)\mathbb{T}_r$. It is isomorphic to the r part of $J_0(N)(\mathbb{F}_\ell)$ if and only if Frobenius at ℓ acts nontrivially on $J_0(N)[\lambda]$ for every maximal ideal λ of $\mathbb{T}$ above r; otherwise the two are not isomorphic even as groups.*

PROOF. Since r does not divide the discriminant, $\mathbb{T}_r$ is a product of unramified discrete valuation rings $\mathcal{O}_\lambda$, one for each λ above r, each with uniformiser r, and every $\mathbb{T}_r$ module is the sum of its components over the $\mathcal{O}_\lambda$. The module $X^0 \otimes \mathbb{Z}_r$ has no torsion and, by multiplicity one, rank one over each $\mathcal{O}_\lambda$, so it is free; L

acts on it as $\ell+1-T_\ell$, and since $r\nmid h$, Proposition 21.4 gives the first statement [408, Proposition 3.4]. The component V_λ of V_r is free of rank two over $\mathcal{O}_\lambda$, and $\det(F-1)=\ell+1-T_\ell$ on it [408, Proposition 3.6]. This is not zero, so the r part of $J_0(N)(\mathbb{F}_\ell)$ is the cokernel of $F-1$ on V_r. On V_λ the matrix of $F-1$ has Smith form $\operatorname{diag}(r^a,r^b)$ with $a\le b$, so the λ component of the Jacobian side is $\mathcal{O}_\lambda/r^a\oplus\mathcal{O}_\lambda/r^b$, while that of the graph side is $\mathcal{O}_\lambda/(\ell+1-T_\ell)=\mathcal{O}_\lambda/r^{a+b}$. They agree if and only if $a=0$, that is, if and only if F is not the identity on V_λ/rV_λ, which is $J_0(N)[\lambda]$. If $a\ge 1$ for some λ, the Jacobian side has the larger r rank. □

By Chebotarev's theorem Frobenius at ℓ fixes some $J_0(N)[\lambda]$ pointwise for a set of primes ℓ of positive density. A cruder count works for every prime r: if $\ell\ne r$ splits completely in the field generated by $J_0(N)[r]$, then $J_0(N)[r]\cong(\mathbb{Z}/r)^{2g}$ lies in $J_0(N)(\mathbb{F}_\ell)$, so $J_0(N)(\mathbb{F}_\ell)/C_N$ has r rank at least $2g-1$, while the critical group, a quotient of $X^0\cong\mathbb{Z}^g$, has r rank at most g. Since $g\ge 2$ for every prime $N\equiv 1 \bmod 12$ from 37 on, the critical group and $J_0(N)(\mathbb{F}_\ell)/C_N$ have the same order (Corollary 21.3) but are not isomorphic for a set of primes ℓ of positive density. Munier and Shnidman find the law of the critical groups: for fixed N and r prime to $6h$ and to the discriminant of $\mathbb{T}$, the r parts of the critical groups of $G_N(\ell)$, as ℓ runs through the primes, are distributed like $\mathbb{T}_r/\det(\gamma-1)\mathbb{T}_r$ for γ Haar random in the subgroup of $\mathrm{GL}_2(\mathbb{T}_r)$ where $\det\gamma\in\mathbb{Z}_r^\times$, which is not the Cohen Lenstra law of Chapter 8.

The critical groups are often not cyclic: of the 192 cases with $37\le N<1300$ and $\ell\in\{2,3,5,7\}$, 153 critical groups are not cyclic, for instance $\mathbb{Z}/12\oplus\mathbb{Z}/2$ for $(N,\ell)=(61,3)$, $\mathbb{Z}/9\oplus\mathbb{Z}/3$ for $(37,7)$ and $\mathbb{Z}/4542148464\oplus(\mathbb{Z}/4)^2\oplus\mathbb{Z}/2$ for $(433,2)$, and 67 have a noncyclic odd part, as at $(37,7)$. At $N=37$ the comparison can be made explicit. Here $h=3$, the discriminant of $\mathbb{T}$ is 4, since the newforms attached to $37a1$ and $37b1$ are congruent modulo 2, and $J_0(37)$ is isogenous to $37a1\times 37b1$ by an isogeny of 2 power degree, so the odd part of $J_0(37)(\mathbb{F}_\ell)$ is the odd part of $37a1(\mathbb{F}_\ell)\times 37b1(\mathbb{F}_\ell)$, computed with PARI/GP. The Brandt matrices have common eigenvectors, so the critical group is determined by $a_\ell(37a1)$ and $a_\ell(37b1)$. At $\ell=7$, where $37a1(\mathbb{F}_7)\cong\mathbb{Z}/9$ and $37b1(\mathbb{F}_7)\cong(\mathbb{Z}/3)^2$, the critical group $\mathbb{Z}/9\oplus\mathbb{Z}/3$ is a quotient of $J_0(37)(\mathbb{F}_7)\cong\mathbb{Z}/9\oplus(\mathbb{Z}/3)^2$ by a subgroup of order 3, but $X^0/L(X^0)\cong\mathbb{Z}/9\oplus\mathbb{Z}/9$ is not isomorphic to $J_0(37)(\mathbb{F}_7)$. Of the 78496 odd primes $\ell<10^6$, $\ell\ne 37$, there are 3305 at which the odd part of the critical group is not the quotient of the odd part of $J_0(37)(\mathbb{F}_\ell)$ by any subgroup of order 3. The first is $\ell=433$: $37a1(\mathbb{F}_{433})\cong\mathbb{Z}/425$ and $37b1(\mathbb{F}_{433})\cong\mathbb{Z}/45\oplus\mathbb{Z}/9$, so the 3 part of $J_0(37)(\mathbb{F}_{433})/C_{37}$ is $\mathbb{Z}/9\oplus\mathbb{Z}/3$, while the critical group is $\mathbb{Z}/11475\oplus\mathbb{Z}/5$, with 3 part $\mathbb{Z}/27$. At $\ell=1091$, where $37b1(\mathbb{F}_{1091})\cong\mathbb{Z}/210\oplus\mathbb{Z}/5$ contains all the 5 torsion, the critical group is $\mathbb{Z}/185850\oplus\mathbb{Z}/2$, with 5 part $\mathbb{Z}/25$, while the 5 part of $J_0(37)(\mathbb{F}_{1091})$ is $(\mathbb{Z}/5)^2$, as Proposition 21.5 predicts.

Spanning trees are themselves hitting time objects. By the Aldous Broder

algorithm [12, 85], the first entrance edges of a simple random walk run until it has covered the graph form a uniform spanning tree; by Wilson's algorithm [584] loop erased walks stopped on hitting the current tree do the same. Theorem 21.2 therefore says that the random walk in characteristic N samples uniformly from a set of size $|J_0(N)(\mathbb{F}_\ell)|/h$, a point count modulo ℓ divided by the rational torsion (Corollary 21.3).

THEOREM 21.6 (resistances and commute times; this book). *Let $N \equiv 1 \bmod 12$ and $\ell \neq N$ be primes, and let $R(x,y)$ be the effective resistance between vertices of $G_N(\ell)$ with unit conductance on each edge. Then*

$$\sum_{\{x,y\}} R(x,y) = h \sum_f \frac{1}{\ell + 1 - a_\ell(f)} = \frac{h\, K_N(\ell)}{\ell + 1},$$

and the commute time $\kappa(x,y) = \mathbb{E}_x \tau_y + \mathbb{E}_y \tau_x = h(\ell+1)R(x,y)$ has mean $2hK_N(\ell)/(h-1)$ over unordered pairs of distinct vertices.

PROOF. The sum of all effective resistances, Kirchhoff's index, is $h \sum_{\mu \neq 0} \mu^{-1}$ over the nonzero Laplacian eigenvalues, by Gutman and Mohar and by Zhu, Klein and Lukovits [230, 594], whose proof applies to multigraphs. The commute time identity of Chandra, Raghavan, Ruzzo, Smolensky and Tiwari [99] gives $\kappa(x,y) = c_{\text{tot}} R(x,y)$, where $c_{\text{tot}} = \sum_{x,y} B(\ell)_{xy} = h(\ell+1)$ is the total conductance, the diagonal included, so that a loop counts twice and a half loop once, as in Chapter 10 (loops change the walk but not the resistances); this gives the rest. □

The Weil bounds are hitting time bounds. The Ramanujan property, $|a_\ell(f)|$ at most $2\sqrt{\ell}$, is the Hasse Weil interval

$$(\sqrt{\ell} - 1)^2 \leq \ell + 1 - a_\ell(f) \leq (\sqrt{\ell} + 1)^2$$

for each factor.

COROLLARY 21.7. *For every prime N and $\ell \neq N$,*

$$\frac{g(\ell+1)}{(\sqrt{\ell}+1)^2} \leq K_N(\ell) \leq \frac{g(\ell+1)}{(\sqrt{\ell}-1)^2},$$

and for $N \equiv 1 \bmod 12$, $(\sqrt{\ell}-1)^{2g} \leq h\,\tau(G_N(\ell)) \leq (\sqrt{\ell}+1)^{2g}$.

For $N \equiv 1 \bmod 12$ the bounds on $K_N(\ell)$ are the spectral bounds of Chapter 10, since $(h-1)/(1 \pm \rho_\ell) = g(\ell+1)/(\sqrt{\ell} \pm 1)^2$ for $\rho_\ell = 2\sqrt{\ell}/(\ell+1)$. The bound on $h\,\tau(G_N(\ell))$ is the Weil bound for $J_0(N)$ over $\mathbb{F}_\ell$, read as a bound on the number of spanning trees. The exact growth rate comes from the vertical Sato Tate theorem.

THEOREM 21.8 (entropy of $J_0(N)$ modulo ℓ; after Tsfasman and Vlăduţ [552]). *Fix a prime ℓ. As $N \to \infty$ through primes,*

$$\frac{1}{g}\log|J_0(N)(\mathbb{F}_\ell)| \longrightarrow \log\frac{\ell^\ell}{(\ell^2-1)^{(\ell-1)/2}}.$$

The limit is the spanning tree entropy of the $(\ell+1)$ regular tree, so for $N \equiv 1 \bmod 12$ also $h^{-1}\log\tau(G_N(\ell))$ converges to it.

PROOF. By (21.1), $g^{-1}\log|J_0(N)(\mathbb{F}_\ell)| = g^{-1}\sum_f \log(\ell+1-\sqrt{\ell}\,x_f)$ with $x_f = a_\ell(f)/\sqrt{\ell} \in [-2,2]$. The function $x \mapsto \log(\ell+1-\sqrt{\ell}x)$ is continuous on $[-2,2]$ because $\ell+1-2\sqrt{\ell} > 0$. By Serre's theorem (Theorem 9.4) the x_f are equidistributed with respect to the Kesten McKay law μ_ℓ as $N \to \infty$, so the average converges to $\int \log(\ell+1-\sqrt{\ell}x)\,d\mu_\ell(x)$. Expanding $\log(\ell+1-\sqrt{\ell}x) = \log(\ell+1) - \sum_{t\ge1} t^{-1}(\sqrt{\ell}x/(\ell+1))^t$ and using Theorem 9.1 as in the proof of Theorem 10.6, the integral equals $\log(\ell+1) - \sum_{t\ge1} p_t/t$, where p_t is the probability that simple random walk on the $(\ell+1)$ regular tree is back at its start at time t. McKay [388] evaluated this as $\log[(d-1)^{d-1}/(d^2-2d)^{d/2-1}]$ with $d = \ell+1$, which is the stated value; Lyons [377] identified it as the tree entropy of the regular tree. For $N \equiv 1 \bmod 12$, $\log\tau = \log|J_0(N)(\mathbb{F}_\ell)| - \log h$ by Theorem 21.2, and $g/h \to 1$. □

The limits are 0.83699 for $\ell = 2$, 1.21640 for $\ell = 3$, 1.69108 for $\ell = 5$ and 2.00777 for $\ell = 7$; the integral was also checked by quadrature to 10^{-12}.

The limit is also the asymptotic class number of the curves $X_0(N)$ over $\mathbb{F}_\ell$. These have about $(\ell-1)g$ supersingular points over $\mathbb{F}_{\ell^2}$, so they attain the Drinfeld Vlăduţ bound there, and the generalized Brauer Siegel theorem for function fields of Tsfasman and Vlăduţ [552, Corollary D′], applied with about $\frac{\ell-1}{2}g$ points of degree 2 and $o(g)$ of each other degree, gives, in natural logarithms, $\log\ell + \frac{\ell-1}{2}\log\frac{\ell^2}{\ell^2-1}$, the value above. What Theorem 21.8 adds, as far as we know, is its reading as the tree entropy.

PROPOSITION 21.9 (vertex hitting times). *Let ϕ_f, f ranging over the eigenforms, be common eigenvectors of the Brandt matrices $B(m)$, $N \nmid m$, orthonormal in $L^2(\pi)$, so that ϕ_f is the quaternionic form attached to f by the Jacquet Langlands correspondence. For every curve E_y,*

$$\mathbb{E}_\pi\tau_y = \sum_f \frac{(\ell+1)\,|\phi_f(y)|^2}{\ell+1-a_\ell(f)}, \qquad \sum_y \pi(y)\mathbb{E}_\pi\tau_y = K_N(\ell).$$

PROOF. For a reversible chain $\mathbb{E}_\pi\tau_y = Z(y,y)/\pi(y)$ with Z the fundamental matrix of Theorem A.4, and $Z(y,y) = \pi(y)\sum_{j\ge2}|\phi_j(y)|^2/(1-\mu_j)$. Averaging over y with weights π uses $\sum_y \pi(y)|\phi_f(y)|^2 = 1$. For $N \equiv 1 \bmod 12$, where π is

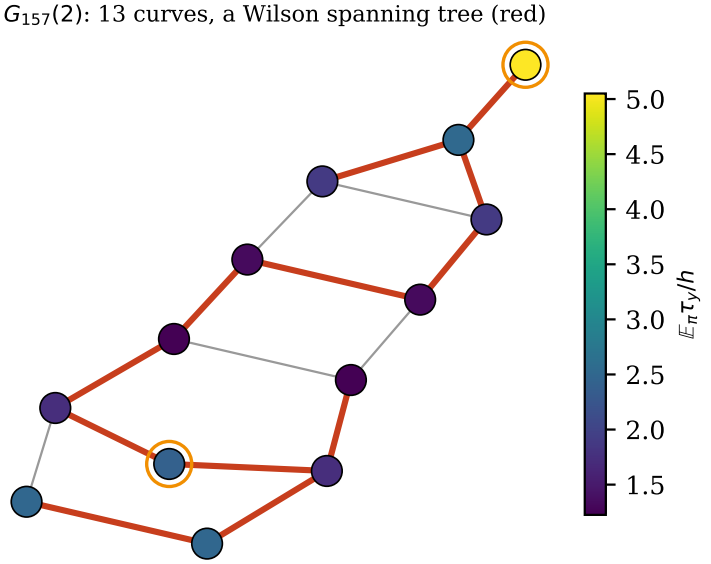


Figure 21.1: The supersingular 2 isogeny graph $G_{157}(2)$ (edge width is multiplicity, circles mark the two curves with loops), coloured by $\mathbb{E}_\pi \tau_y / h$ (Proposition 21.9), with a uniform spanning tree sampled by Wilson's algorithm in red. The graph has $|J_0(157)(\mathbb{F}_2)|/13 = 1418$ spanning trees.

uniform, the first step is Exercise 5 of Chapter 10, and Figure 10.1 colours $G_{1009}(2)$ by $\mathbb{E}_\pi \tau_y / h$. □

So the mean hitting time of a single supersingular curve is a weighted harmonic sum of the factors $\ell + 1 - a_\ell(f)$, with weights the squared values at that curve of the quaternionic newforms. Gross's formula [227] expresses $L(f,1)L(f \otimes \chi_D, 1)$ through the square of a linear form in the same values, whose coefficients count the curves with complex multiplication by the order of discriminant D that reduce to each E_y; Open Problem 21.2 asks what the hitting times of the walk say about these L values.

Figure 21.1 shows $G_{157}(2)$: 13 curves, $|J_0(157)(\mathbb{F}_2)| = 18434$ and so 1418 spanning trees, one of them drawn as sampled by Wilson's algorithm. The mean hitting times $\mathbb{E}_\pi \tau_y$ range from $1.23h$ to $5.05h$. The largest, $5.05h$, belongs to the curve $j = -3375$, whose endomorphisms $(1 \pm \sqrt{-7})/2$ of degree 2 form a loop counted twice; the next values, about $2.5h$, are at curves without loops, and the curve $j = 8000$, with a half loop, has $2.40h$.

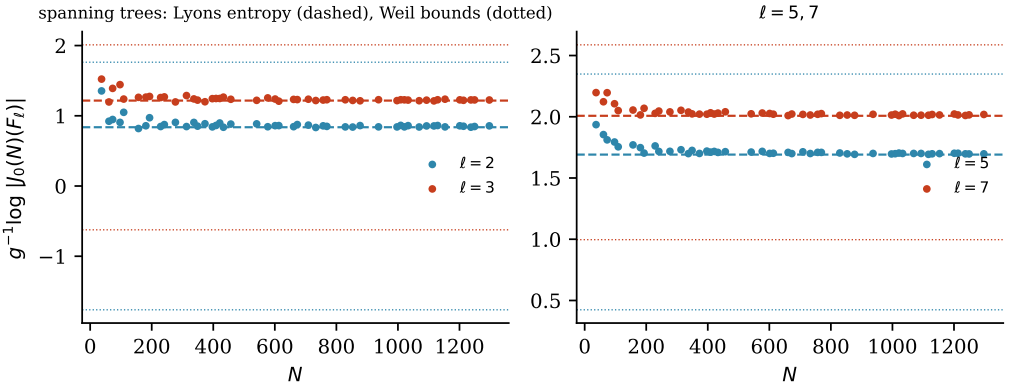


Figure 21.2: Theorems 21.2 and 21.8: $g^{-1}\log|J_0(N)(\mathbb{F}_\ell)|$, which equals $g^{-1}\log(h\,\tau(G_N(\ell)))$, for primes $N \equiv 1 \bmod 12$ below 1300, against the spanning tree entropy of the $(\ell+1)$ regular tree (dashed) and the Weil bounds $2\log(\sqrt{\ell}\pm 1)$ (dotted).

21.1.1 Data

We built the isogeny walks from modular polynomials over $\mathbb{F}_{N^2}$, as in Section B.2, for $\ell \in \{2,3,5,7\}$, the 49 primes $N \equiv 1 \bmod 12$ below 1300, and the nine primes $11, 23, 43, 47, 59, 71, 83, 107, 131$, which are not 1 modulo 12, and computed T_ℓ on $S_2(\Gamma_0(N))$ independently with modular symbols in PARI/GP. Kemeny's constant of the walk, computed from the spectrum of P, agrees with the sum in Theorem 21.1 to relative error below $2 \cdot 10^{-14}$ in all 228 cases with $g \geq 1$, including the levels with extra automorphisms; computed in rational arithmetic from the Laplacian, it equals the sum exactly in the 108 cases with $N \equiv 1 \bmod 12$ below 700. In all 196 cases with $N \equiv 1 \bmod 12$ the exact integer determinant of the reduced Laplacian times h equals $|J_0(N)(\mathbb{F}_\ell)|$, computed as the characteristic polynomial of T_ℓ evaluated at $\ell+1$; the largest, for $N = 1297$ and $\ell = 7$, has 94 digits. The Kirchhoff index agrees with Theorem 21.6 to $6 \cdot 10^{-16}$. Figure 21.2 shows $g^{-1}\log|J_0(N)(\mathbb{F}_\ell)|$ approaching the tree entropy, from above for $\ell = 5$ and 7; for $\ell = 2$ three of the 48 levels with $g \geq 2$, and for $\ell = 3$ eight, lie below it, by at most 0.02. At $N = 1297$ the values are 0.8566, 1.2249, 1.6973 and 2.0187.

21.2 Chebotarev densities are hitting rates

Return to the zeta random integer N_σ of Section 20.4, with prime factors labelled by Frobenius classes or Satake parameters $\theta_p \in Y$, equidistributed with respect to μ. Scan the prime factors from the largest down. For a Borel set $B \subset Y$ let P_B^+ be the largest prime factor of N_σ with label in B, and $V_B = (\sigma - 1)\log P_B^+$ ($V_B = 0$ if

there is none). Write $E_1(v) = \int_v^\infty e^{-x}x^{-1}dx$ for the exponential integral.

THEOREM 21.10 (hitting rates; after Lloyd [370]). *Let B be a μ continuity set. As $\sigma \downarrow 1$:*

(i) $\mathbb{P}(V_B \le v) \to \exp(-\mu(B)E_1(v))$ *for every $v > 0$; if $\mu(B) > 0$, equivalently, $E_1(V_B)$ converges in law to the exponential law with rate $\mu(B)$, and if $\mu(B) = 0$, then $V_B \to 0$ in probability;*

(ii) the label of the largest prime factor of N_σ lies in B with probability tending to $\mu(B)$, and the labels of the k largest prime factors are asymptotically independent with law μ, independent of their sizes;

(iii) for two continuous functions a and b on Y, such as the characters χ_ρ and $\chi_{\rho'}$ of Theorem 20.22(a), whose discrepancy set $D = \{a \ne b\}$ is a μ continuity set, (i) holds for $B = D$, and $\mu(D)$ is the mass of the Thorin measure τ_Q of Theorem 20.21 for $Q = |a - b|^2$; in particular $V_D \to 0$ in probability if and only if $\tau_Q = 0$.

PROOF. By Theorem 20.18(i) the rescaled sizes and labels of the prime factors converge to the Poisson process with intensity $x^{-1}e^{-x}dx \otimes \mu$. Prime factors with $e_p \ge 2$ give no point above a fixed level $v > 0$ in the limit, since $\sum_{e\ge2}\sum_{p^e>\exp(v/(\sigma-1))} p^{-e\sigma} \to 0$, so the points $((\sigma-1)\log p, \theta_p)$ have the same limit. The event $\{V_B \le v\}$ is the event that this process has no point in $(v,\infty)\times B$, whose mean number of points is $\mu(B)E_1(v)$; points beyond a large level M are excluded with probability at least $1 - E_1(M) - o(1)$, uniformly in σ, because the expected number of prime factors p of N_σ with $(\sigma-1)\log p > M$ is at most $\sum_{(\sigma-1)\log p>M} p^{-\sigma}(1-p^{-\sigma})^{-1} \to E_1(M)$. This gives (i), and (ii) follows from the independence of labels and sizes in the limiting Poisson process. For (iii) apply (i) to $B = D$; the Thorin mass is $\mu(D)$ by Theorem 20.21. □

So Chebotarev's density is literally a hitting rate: in the logarithmic integral scale E_1, the first prime factor (from the top) whose Frobenius lies in C arrives after an exponential time with rate $|C|/|G|$. Reciprocity laws are statements that a discrepancy set is hit at rate zero. For a prime $q \equiv 3 \bmod 4$, omitting the factor $(\frac{-1}{p})$ from quadratic reciprocity gives a discrepancy set that is hit at rate $\frac12$; the pair (ρ,ρ_{31}) of S_3 representations of Chapter 20 has discrepancy rate $\frac{11}{18}$.

For the S_3 field of $x^3 - x - 1$ the probability that the largest prime factor of N_σ outside the set $S = \{2,7,11,23,31,37\}$ of the example after Theorem 20.26 has Frobenius in a given class, conditional on there being such a factor, was computed exactly from the primes below 10^7, with larger primes from the prime number theorem (at $\sigma = 1.005$ the largest prime factor exceeds 10^7 with probability 0.87, and its class is then given the limit law). At $\sigma = 1.005$ it is 0.1626, 0.5000 and 0.3375 for the identity, the transpositions and the three cycles, against $\frac16$, $\frac12$, $\frac13$; at $\sigma = 1.5$ it is 0.021, 0.392 and 0.587, so at moderate σ small primes still

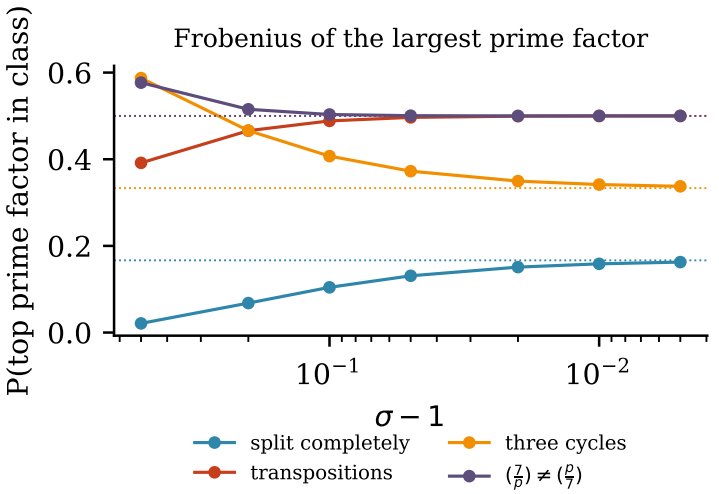


Figure 21.3: Theorem 21.10 for the S_3 field of $x^3 - x - 1$: the probability that the largest prime factor of N_σ outside S has Frobenius in a given class, conditional on one existing, and that it lies in the discrepancy set $\{p : (\frac{7}{p}) \neq (\frac{p}{7})\}$, against $\sigma - 1$, with their limits (dotted).

dominate (Figure 21.3). To first order the approach is linear in $\sigma - 1$, with slope $s_C = \sum_{p \notin S} \big(1_C(\mathrm{Frob}_p) - \mu(C)\big) p^{-1} \prod_{p' \leq p \text{ or } p' \in S} (1 - 1/p')^{-1}$, a Chebotarev bias weighted by Mertens products; for the primes below 10^7, $s_C = -0.835$, 0.003 and 0.832, so the transpositions converge fastest. For Chebotarev labels and a uniform random integer below x, the class of a size biased prime factor approaches Haar measure at rate $1/\log x$, the class of the largest one faster still, but the classes of the distinct prime factors only at rate $1/\log\log x$ (Chapter 24).

21.3 Explosion Times and the Thorin Converse Theorem

A pure birth process on $\{0, 1, 2, \dots\}$ started at 0 with rates $u_1 \leq u_2 \leq \dots$, repetitions allowed, spends independent exponential times with means $1/u_k$ in successive states. If $\sum_k 1/u_k < \infty$ it explodes at the finite time

$$T = \sum_k \frac{\mathcal{E}_k}{u_k}, \qquad \mathbb{E}e^{-sT} = \prod_k \Big(1 + \frac{s}{u_k}\Big)^{-1},$$

with $\mathcal{E}_k$ independent standard exponentials. So explosion laws are exactly the GGCs with zero drift whose Thorin measure is a sum of unit point masses: atomic Thorin measures with *integer* masses. This is the integrality condition in Theorem 20.32, now with a meaning: multiplicities of zeros are repeated birth rates.

THEOREM 21.11 (explosion times; after Theorem 16.5). *Let $\Lambda(s)$ be entire of order at most one with $\Lambda(s) = \Lambda(1-s)$ and $\Lambda(\frac12) \neq 0$, and write its zeros as $\frac12 \pm i\gamma_k$, with multiplicity. The following are equivalent.*

(i) Every γ_k is real.

(ii) There is a generalized gamma convolution T with $\mathbb{E}e^{-sT} = \Lambda(\frac12)/\Lambda(\frac12 + \sqrt{s})$ for all $s > \frac14$.

(iii) $\Lambda(\frac12)/\Lambda(\frac12 + \sqrt{s})$, $s > \frac14$, is the Laplace transform of the explosion time of a pure birth process.

In that case the birth rates are the γ_k^2, T is infinitely divisible with Lévy density $\mathcal{W}(x)/x$, where $\mathcal{W}(x) = \sum_{\gamma_k>0} e^{-x\gamma_k^2}$ is the heat trace, and the cumulants of T are

$$\kappa_m(T) = (m-1)! \sum_k \gamma_k^{-2m} = \int_0^\infty x^{m-1}\mathcal{W}(x)\,dx,$$
$$\mathbb{E}T = \sum_k \frac{1}{\gamma_k^2} = \frac12 \frac{\Lambda''}{\Lambda}\Big(\frac12\Big).$$

PROOF. For real Λ with $\Lambda(\frac12) > 0$, the equivalence of (i) and (ii) and the Thorin measure $\sum_k \delta_{\gamma_k^2}$ are Theorem 16.5 with $f = \Lambda$ and $\alpha = \frac12$, and the variable in (iii) is that of Theorem 14.7(c); the argument is short, and we repeat it without these hypotheses. By Hadamard, $\Lambda(\frac12 + z)/\Lambda(\frac12)$ is an even entire function of order at most one, so it equals $\prod_k (1 + z^2/\gamma_k^2)$ with $\sum |\gamma_k|^{-2} < \infty$, the exponential factor being trivial by evenness. If (i) holds, (iii) holds with rates γ_k^2, and an explosion time is a GGC with Thorin measure $\sum_k \delta_{\gamma_k^2}$, so (iii) implies (ii). Conversely let T be a GGC with drift a and Thorin measure τ as in (ii). Taking logarithms and differentiating in s,

$$a + \int \frac{\tau(dt)}{t+s} = \sum_k \frac{1}{\gamma_k^2 + s}, \qquad s > \tfrac14.$$

The left side is analytic on $\mathbb{C} \setminus (-\infty, 0]$, and the right side is meromorphic on $\mathbb{C}$ with a pole at each $-\gamma_k^2$. So every γ_k^2 lies in $[0,\infty)$, and $\gamma_k^2 \neq 0$ because $\Lambda(\frac12) \neq 0$: every γ_k is real. (Letting $s \to \infty$ shows $a = 0$, and comparing residues shows $\tau = \sum_k \delta_{\gamma_k^2}$, as in Theorem 20.32.) For the Lévy density, Frullani's integral gives $\sum_k \log(1 + s/\gamma_k^2) = \int_0^\infty (1 - e^{-sx})\mathcal{W}(x)x^{-1}dx$. The cumulants follow by differentiating at $s = 0$, or from $\int_0^\infty x^{m-1}e^{-x\gamma^2}dx = (m-1)!\gamma^{-2m}$, and $\Lambda''/\Lambda(\frac12) = 2\sum_k \gamma_k^{-2}$ from the product. □

The hypothesis in (ii) cannot be weakened to positivity of T. With $w = 2 + i$, the function $1/((1+s)(1+s/w)(1+s/\bar w))$ is the Laplace transform of the probability density $\frac52 e^{-t}\big(1 - e^{-t}(\cos t + \sin t)\big)$ on $(0,\infty)$, which is nonnegative

because $e^{-t}(\cos t + \sin t)$ decreases on $[0, \pi]$ and is at most $\sqrt{2}e^{-\pi} < 1$ beyond, although two of its poles are not real. What the GGC property adds is positivity of the Thorin measure, which forces the poles onto the negative axis.

The mean explosion time is a number computed from Galois data. By Definition 20.29 the heat trace $\mathcal{W}$ is given by the explicit formula from the prime powers, the conductor and the gamma factor, and $\mathbb{E}T = \int_0^\infty \mathcal{W}(x)\,dx$. Under GRH this integral is the mean of a positive random variable, the explosion time whose birth rates are the squared ordinates γ_k^2 of the zeros. For ξ, for $L(s, \Delta)$ in the unitary normalisation and for the S_3 representation ρ of $x^3 - x - 1$, of conductor 23, the sum $\sum_k \gamma_k^{-2}$ is the constant S_1 of Table 14.1, 0.023105, 0.0561759 and 0.170287, computed there from the central Taylor coefficients and checked against the zeros.

The mean explosion time can also be computed from the Frobenius classes, the conductor and the gamma factor, with no analytic continuation; only the tail of the integral needs a bound on the zeros. For the S_3 representation ρ we evaluated the Galois heat trace of Definition 20.29 from the Frobenius classes of $x^3 - x - 1$ at the 664579 primes below 10^7 and their powers below 10^7, with the archimedean integral by quadrature. It agrees with $\sum e^{-x\gamma^2}$ over the zeros to fifteen digits at $x = 0.001$, 0.005, 0.02 and 0.1. Integrating over $e^{-60} \le x \le 1.1$ (beyond which $\mathcal{W} < e^{-28}$, by the first zero $\gamma_1 = 5.1157$) gives

$$\int_{e^{-60}}^{1.1} \mathcal{W}(x)\,dx = 0.17028731076215,$$

against 0.17028731076373 from the Hadamard product, a difference of $1.6 \cdot 10^{-12}$ that the integral over the two omitted ends accounts for to $3 \cdot 10^{-17}$. Since $L(s, \rho)$ is entire, ρ being dihedral, the explicit formula gives $\mathcal{W}(x) = \sum_k e^{-x\gamma_k^2}$ unconditionally. If GRH holds for $L(s, \rho)$, which is not known, $\sum_k \gamma_k^{-2}$ is the mean explosion time of a birth process whose rates are the squared ordinates of the zeros, and it has been computed from the factorisation of $x^3 - x - 1$ modulo primes (which determines the Frobenius class in S_3), the conductor 23 and the gamma factor. The agreement does not verify GRH: a finite computation of this kind can disprove it and is not known to verify it.

Combining Theorem 21.11 with Corollary 20.33 gives an explosion form of the Thorin converse theorem.

COROLLARY 21.12 (Thorin converse theorem as explosion; this book). *Let $k \equiv 0 \bmod 4$ and let (a_p) be the Frobenius traces of a self dual Galois representation of conductor 1 with the gamma factor $\Gamma_{\mathbb{C}}(s + \frac{k-1}{2})$ of Corollary 20.33 and the Ramanujan bound, and Λ its completion, defined on $\operatorname{Re} s > 1$ by the Euler product. If for some constant $c > 0$ the function $c/\Lambda(\frac{1}{2} + \sqrt{s})$, $s > \frac{1}{4}$, is the Laplace transform of the explosion time T of a pure birth process, then $f = \sum a_n n^{(k-1)/2} q^n$ is a cusp form for $\mathrm{SL}_2(\mathbb{Z})$ satisfying GRH, and $c = \Lambda(\frac{1}{2}) \neq 0$. Conversely, if f is such a cusp form satisfying GRH with $\Lambda(\frac{1}{2}) \neq 0$, this holds with $c = \Lambda(\frac{1}{2})$. If moreover the Pólya kernel*

Φ_f is nonnegative, as for Δ (but not for one of the two eigenforms of weight 24), then the Pólya variable W_f and T form a Wald couple: $\mathbb{E}e^{tW_f}\,\mathbb{E}e^{-t^2T}=1$ for all real t.

PROOF. Explosion laws are exactly the GGCs with integer atomic Thorin measures. Under the hypothesis, $\Lambda(\frac12+\sqrt{s_0})/\Lambda(\frac12+\sqrt{s})$ is $c^{-1}\Lambda(\frac12+\sqrt{s_0})$ times the Laplace transform of such a GGC, so Corollary 20.33 gives the cusp form and GRH; then $\Lambda(\frac12+z)=c\prod_k(1+z^2/u_k)$ and $z=0$ gives $c=\Lambda(\frac12)$. The converse is Theorem 21.11. For the Wald couple, $\mathbb{E}e^{tW_f}=\Lambda(\frac12+t)/\Lambda(\frac12)$ for every real t, by (20.4) (whose classical completion at $\frac{k}{2}+t$ is a constant times $\Lambda(\frac12+t)$) and the doubly exponential decay of Φ_f, and $\mathbb{E}e^{-t^2T}=\prod_k(1+t^2/\gamma_k^2)^{-1}=\Lambda(\frac12)/\Lambda(\frac12+t)$ by the Hadamard product; this is the Wald couple of Theorem 16.6(c). □

As with Corollary 20.33, the hypothesis contains GRH, so it is stronger than reciprocity. Keilson's theorem (Theorem 15.14) and Kent's extension to diffusions [308] go the other way: first passage times of birth death chains and of diffusions have laws of this form, with rates from their spectra, and whether a given law is such a passage time is an inverse spectral question. The pure birth realisation needs no simplicity of the zeros and is the one that matches the integrality in the Thorin converse theorem.

21.4 BROWNIAN EXIT AND THE FUNCTIONAL EQUATION OF ODD CHARACTERS

The functional equations of even Dirichlet L functions came in Section 20.2 (Theorem 20.8) from the wrapped normal law. For odd characters the natural object is the exit of Brownian motion from an interval, and the signed starting distribution is the character itself.

Let $q\ge3$, let B be standard Brownian motion started at $r\in(0,q)$, let τ be its exit time from $(0,q)$, and let $f_r^0(t)$ and $f_r^q(t)$ be the densities of τ on the events of exit at 0 and at q.

THEOREM 21.13 (signed exit law; after Biane, Pitman and Yor [82]). *Let χ be an odd primitive character modulo q and $g(\chi)=\sum_r\chi(r)e^{2\pi ir/q}$. For every $t>0$,*

$$\sum_{r=1}^{q-1}\chi(r)\big(f_r^0(t)-f_r^q(t)\big)=\frac{2}{\sqrt{2\pi t^3}}\sum_{m\ge1}\chi(m)\,m\,e^{-m^2/(2t)}$$
$$=-\frac{4\pi i\,g(\chi)}{q^2}\sum_{j\ge1}\bar\chi(j)\,j\,e^{-2\pi^2j^2t/q^2}.$$

Consequently $\vartheta_\chi(x)=\sum_{n\ge1}\chi(n)\,n\,e^{-\pi n^2x/q}$ satisfies

$$\vartheta_\chi(1/x)=\frac{g(\chi)}{i\sqrt{q}}\,x^{3/2}\,\vartheta_{\bar\chi}(x),$$

and $\Lambda(s,\chi) = (q/\pi)^{(s+1)/2}\Gamma(\frac{s+1}{2})L(s,\chi)$ *satisfies* $\Lambda(s,\chi) = \frac{g(\chi)}{i\sqrt{q}}\Lambda(1-s,\bar\chi)$.

PROOF. The method of images (the reflection principle) gives

$$f_r^0(t) = \sum_{n\in\mathbb{Z}} \frac{r+2nq}{\sqrt{2\pi t^3}} e^{-(r+2nq)^2/(2t)},$$
$$f_r^q(t) = \sum_{n\in\mathbb{Z}} \frac{q-r+2nq}{\sqrt{2\pi t^3}} e^{-(q-r+2nq)^2/(2t)}.$$

In the first sum $m = r+2nq$ runs over the integers $\equiv r \bmod 2q$ and $\chi(r) = \chi(m)$. In the second $m = q-r+2nq \equiv -r \bmod q$, so $\chi(r) = -\chi(m)$ because χ is odd. Hence $\sum_r \chi(r)(f_r^0 - f_r^q) = 2(2\pi t^3)^{-1/2}\sum' \chi(m) m e^{-m^2/2t}$, the sum over all integers m whose residue modulo $2q$ lies in $(0,q)$. For negative m put $m' = -m$; then $m' \bmod 2q \in (q,2q)$ and $\chi(m)m = \chi(m')m'$. The two ranges together cover every positive integer prime to q once, which is the first identity. The eigenfunction expansion of the killed heat kernel gives

$$f_r^0(t) - f_r^q(t) = \sum_{k\ge 1} \frac{k\pi}{q^2}\bigl(1+(-1)^k\bigr)\sin\frac{k\pi r}{q}\, e^{-k^2\pi^2 t/(2q^2)},$$

since exit at q from r is exit at 0 from $q-r$, and $\sin(k\pi(q-r)/q)$ equals $(-1)^{k+1}\sin(k\pi r/q)$. Only $k = 2j$ survives, and for odd primitive χ,

$$\sum_r \chi(r)\sin\frac{2\pi jr}{q} = \frac{1}{2i}\bigl(\bar\chi(j) - \bar\chi(-j)\bigr)g(\chi) = -i\,\bar\chi(j)g(\chi).$$

This is the second identity. Putting $t = qx/(2\pi)$ and comparing gives the transformation of ϑ_χ, and the Mellin transform $\int_0^\infty \vartheta_\chi(x)x^{(s+1)/2}dx/x = \Lambda(s,\chi)$ gives the functional equation in the usual way. □

The root number $g(\chi)/(i\sqrt{q})$ is the eigenfunction weight of the signed starting law, the Fourier coefficient of $r \mapsto \chi(r)$ against the sine modes of the interval. For a real odd character modulo a prime $q \equiv 3 \bmod 4$ it is 1, which is Gauss's sign $g(\chi) = g(q) = i\sqrt{q}$ of Corollary 20.5. For $q = 4$, $\chi = \chi_{-4}$, the theorem is the reciprocal law of the exit time of Brownian motion started at the centre of an interval, which Biane, Pitman and Yor [82] credit to Ciesielski and Taylor [104] and from which they obtained the functional equation of Dirichlet's β function; Salminen and Vignat [474] derive it by comparing the image and eigenfunction expansions, as here. For real odd characters Biane, Pitman and Yor wrote the Laplace transform of ϑ_χ, using the classical transformation formula, as a combination with weights $\chi(n)$ of ratios of hyperbolic sines, which are Laplace transforms of Brownian exit laws from an interval, and observed, after Pólya, that ϑ_χ need not be positive, while its positivity would exclude a real zero of $L(s,\chi)$ ([82], Section 6.1).

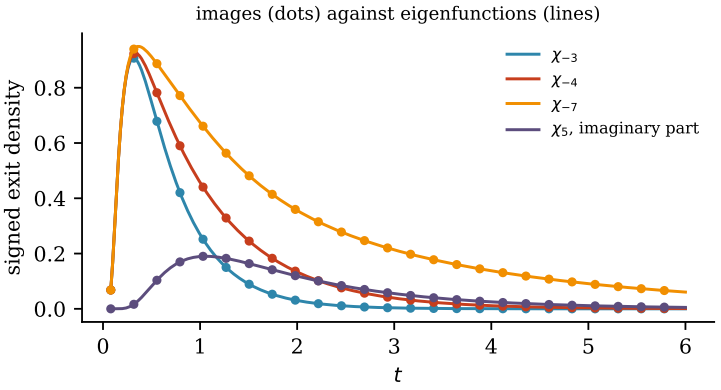


Figure 21.4: Theorem 21.13 for χ_{-3}, χ_{-4}, χ_{-7} and the odd character modulo 5 with $\chi(2) = i$ (imaginary part, labelled χ_5): the two closed forms of the signed exit density, the image sum (dots) and the eigenfunction sum with the Gauss sum as weight (lines).

Theorem 21.13 reads that combination as an exit law whose starting point is spread over the interval with the character as a signed weight, and reverses the argument: the transformation formula follows by comparing the image and eigenfunction expansions. For χ_{-4} the function ϑ_χ is positive, since $\vartheta_{\chi_{-4}}(x) = \eta(ix)^3$ by Jacobi's identity (Proposition 18.2). For the character χ_d of an imaginary quadratic field K of discriminant d, the completed Dedekind zeta function of K is a constant multiple of $\xi(s)\Lambda(s, \chi_d)$, so its Pólya kernel (Theorem 17.17) is a positive multiple of the convolution of Pólya's kernel of ξ with $v \mapsto e^{3v/2}\vartheta_{\chi_d}(e^{2v})$; hence, by Theorem 17.18, ϑ_{χ_d} takes negative values whenever 2 and 3 are inert in K, that is whenever $d \equiv 5 \bmod 24$. In 40 digit arithmetic, for χ_{-3}, χ_{-4}, χ_{-7}, χ_{-8}, the odd character modulo 5 with $\chi(2) = i$ and an odd character of order 6 modulo 7 at $t = 0.3$, 1 and 3, the image and eigenfunction sums agree with each other and with both closed forms to 10^{-40}; Figure 21.4 plots four of these characters.

21.5 A DICTIONARY

The correspondences of the chapter, side by side.

Hitting time	Reciprocity
Kemeny constant of the isogeny walk mod N	harmonic sum of the factors $\ell+1-a_\ell(f)$ of $\lvert J_0(N)(\mathbb{F}_\ell)\rvert$
spanning trees ($N\equiv 1 \bmod 12$)	$\lvert J_0(N)(\mathbb{F}_\ell)\rvert/\lvert J_0(N)(\mathbb{Q})_{\mathrm{tors}}\rvert$
critical group ($N\equiv 1 \bmod 12$)	order $\lvert J_0(N)(\mathbb{F}_\ell)\rvert/h$; r part $\mathbb{T}_r/(\ell+1-T_\ell)\mathbb{T}_r$ for $r\nmid \ell hN$ prime to the discriminant of $\mathbb{T}$ (Proposition 21.5)
tree entropy of $\mathcal{T}_\ell$	growth of $\lvert J_0(N)(\mathbb{F}_\ell)\rvert$
bounds on hitting times and spanning trees	Weil bound for $a_\ell(f)$
hitting rate of a class by top prime factors	Chebotarev density
discrepancy set hit at rate zero	reciprocity, vanishing Thorin mass
explosion time of a pure birth process	Thorin converse theorem, GRH
mean explosion time	$\int \mathcal{W}$, computable from Frobenius data, conductor and gamma factor
signed exit law, images against eigenfunctions	functional equation of odd $L(s,\chi)$

21.5.1 Open problems

OPEN PROBLEM 21.1 (Sandpile groups and Jacobians). Fix ℓ and let $N\to\infty$ through primes $N\equiv 1 \bmod 12$. Find the limiting law of the r part of the critical group of $G_N(\ell)$, a question Munier and Shnidman raise [408]. By Proposition 21.5, for $r\neq\ell$ prime to hN and to the discriminant of $\mathbb{T}$, it is governed by the valuations of $\ell+1-a_\ell(f)$ at the primes above r of the Hecke fields, so, as Munier and Shnidman note, it needs an r adic analogue of the vertical equidistribution of Theorem 9.4. For the primes r dividing h or the discriminant of $\mathbb{T}$, which Proposition 21.5 excludes, describe the critical group, and decide when the class of $L\delta_E$ corresponds to the cuspidal subgroup C_N. Extend the comparison to $\ell=2$ and to levels $N\not\equiv 1 \bmod 12$, where the graph is weighted by automorphisms.

OPEN PROBLEM 21.2 (Vertex hitting times and Gross's formula). By Exercise 6, Gross's formula turns the mean $\sum_{x,y} c_x c_y \mathbb{E}_x\tau_y$ of hitting times between the reductions of the curves with complex multiplication by discriminant D, less its stationary part $(\deg e_D)\sum_y c_y\mathbb{E}_\pi\tau_y$, into a combination of the central values $L(f,1)L(f\otimes\chi_D,1)$ over the newforms of level N. For fixed N and ℓ, find the distribution of this combination over D, a question about the central values of the quadratic twists of the newforms of level N. Find also the law of the hitting time of the support of e_D from stationarity, and decide which arithmetic quantities it computes: its Laplace transform involves the values $\phi_f(x)\overline{\phi_f(y)}$ at pairs of points of the support, not only the linear forms $\sum_y c_y\phi_f(y)$ of Gross's formula.

21.6 NOTES

The supersingular isogeny graph and Brandt matrices are in Deuring, Eichler, Mestre, Gross and Pizer [148, 171, 394, 227, 438]; the congruence relation is Eichler and Shimura [514]; rational torsion of $J_0(N)$ is Ogg's conjecture proved by Mazur [428, 384], and injectivity of reduction on torsion is in Katz [295]. Kemeny's constant is from [304], the matrix tree theorem is Kirchhoff's [319], critical groups are in Biggs [59] and Baker and Norine [33], uniform spanning trees from random walks in Aldous, Broder and Wilson [12, 85, 584], resistances in Doyle and Snell [164], the Kirchhoff index in [230, 594] and the commute time identity in [99]. McKay [388] computed the asymptotic number of spanning trees of regular graphs with large girth and Lyons [377] the tree entropy. The vertical Sato Tate theorem is Serre's [508]. Explosion of pure birth processes is classical (Feller [184], Section XVII.4), Keilson's theorem is [303], Kent's is [308]. The tree count of Theorem 21.2 is Hashimoto's formula [246, (7.14)], presented by Li [364, Section 4.3]; Munier and Shnidman [408] sketch the proof given here, call the formula a kind of reciprocity law between the Jacobian of the graph and that of $X_0(N)$, prove Proposition 21.4 and the first statement of Proposition 21.5, note the cuspidal point behind Corollary 21.3, and find the laws of the r parts of the critical groups for fixed N and $r \nmid 6h$ prime to the discriminant of $\mathbb{T}$, as $\ell \to \infty$. The limit in Theorem 21.8 is an asymptotic class number of Tsfasman and Vlăduţ [552]. For one label, Theorem 21.10(i) is Lloyd's limit law for the largest prime factor [370], and Theorem 21.11 restates Theorems 14.7 and 16.5. Theorem 21.13 is after Biane, Pitman and Yor [82], as explained after it, and the transformation formula is classical ([92], Section 1.1). Deriving theta transformations from the heat equation goes back at least to Bellman [42], for ϑ_3, and Salminen and Vignat [474] derive the modular identities of the Jacobi theta functions by comparing the image and eigenfunction expansions of the heat kernels of an interval, among them the case $q = 4$ of Theorem 21.13, the exit time from the centre. Ours are Theorems 21.1 and 21.6 in the forms stated (for $N \equiv 1 \bmod 12$ the Kemeny formula is the expression in the proof of Theorem 10.6), the isomorphism criterion of Proposition 21.5, the reading of the limit in Theorem 21.8 as a tree entropy, the labelled form of Theorem 21.10, the explosion reading of Theorem 21.11, the extension of the image and eigenfunction derivation to every odd primitive character through the signed exit law of Theorem 21.13, Corollaries 21.7 and 21.12, and the computations; each is a short consequence of the cited theory, and none proves a new case of reciprocity, of Artin's conjecture or of GRH.

21.7 EXERCISES

1. Compute $G_{11}(2)$ and $G_{11}(3)$ from Φ_2 and Φ_3 over $\mathbb{F}_{121}$, with the weights w of $j=0$ and $j=1728$, and check $K_{11}(2)=\frac{3}{5}$, $K_{11}(3)=\frac{4}{5}$ against $|11a1(\mathbb{F}_2)|=|11a1(\mathbb{F}_3)|=5$.

2. For $N=37$ and $\ell=2$, draw the three vertex graph, list its five spanning trees and compute all hitting times. Compare with Proposition 21.9.

3. Show that the Laplace transform of the explosion time of a pure birth process determines the multiset of rates, and deduce that Corollary 21.12 recovers the zeros with multiplicity.

4. Check Theorem 21.13 for χ_{-3} by computing the image sum and the eigenfunction sum at $t=1$ to ten digits, verify $g(\chi_{-3})=i\sqrt{3}$, and derive the functional equation of $L(s,\chi_{-3})$ from the transformation of $\vartheta_{\chi_{-3}}$.

5. Show that $\mathbb{E}[(\sigma-1)\log P^+(N_\sigma)]\to\int_0^\infty x\cdot\frac{d}{dx}e^{-E_1(x)}dx$, where P^+ is the largest prime factor ($P^+(1)=1$), and that the limit is the Golomb Dickman constant $\int_0^\infty e^{-x-E_1(x)}dx=0.62433\ldots$ (Lloyd [370]; for uniform integers, Theorem 24.1). To pass from convergence in law to convergence of means, bound $\mathbb{E}[V;V>M]$, for $V=(\sigma-1)\log P^+(N_\sigma)$, by the expected sum of $(\sigma-1)\log p$ over the prime factors p with $(\sigma-1)\log p>M$, which tends to e^{-M}.

6. Let $N\equiv 1 \bmod 12$, let $D<0$ be a fundamental discriminant with N inert in $\mathbb{Q}(\sqrt{D})$, and let $e_D=\sum_y c_y[E_y]$ be the divisor of Gross's formula [227], whose coefficient c_y counts the curves with complex multiplication by the ring of integers of $\mathbb{Q}(\sqrt{D})$ that reduce to E_y. With ϕ_f as in Proposition 21.9, show that

$$\sum_{x,y}c_xc_y\mathbb{E}_x\tau_y=(\deg e_D)\sum_y c_y\mathbb{E}_\pi\tau_y-(\ell+1)\sum_f\frac{\left|\sum_y c_y\phi_f(y)\right|^2}{\ell+1-a_\ell(f)},$$

   using $\mathbb{E}_x\tau_y=(Z(y,y)-Z(x,y))/\pi(y)$ (Theorem A.4), and deduce from Gross's formula that the last sum is a combination of the $L(f,1)L(f\otimes\chi_D,1)/(\ell+1-a_\ell(f))$ with positive coefficients that do not depend on ℓ.

7. Show that the nonbacktracking walk on a connected $(q+1)$ regular graph on n vertices without half loops, $q\geq 2$, has Kemeny constant $(n-1)+\frac{q-1}{q+1}K+\frac{q}{q-1}+\frac{nq^2}{q+1}$, where K is the Kemeny constant of the simple walk (use the Ihara Bass formula of Chapter 10). When $G_N(\ell)$ has no half loops, write it with Theorem 21.1 as a sum over the factors $\ell+1-a_\ell(f)$ of $|J_0(N)(\mathbb{F}_\ell)|$.

21.8 Open computational problems

Computational Problem 21.1 (Two prime tables). Objective: for fixed ℓ and small primes $r \neq \ell$, the empirical law of the r part of the critical group of $G_N(\ell)$ over the primes $N \equiv 1 \bmod 12$, against the Haar model that Munier and Shnidman prove for fixed N and $\ell \to \infty$ and against the Cohen Lenstra law of Chapter 8 (Open Problem 21.1), with the frequency of noncyclic r parts of $J_0(N)(\mathbb{F}_\ell)$. Objects and scale: the primes $N \equiv 1 \bmod 12$ below 10^5 and $\ell \leq 13$, with graphs of up to about 8300 vertices. Route: for r prime to hN and to the discriminant of $\mathbb{T}$, the valuations of $\ell + 1 - a_\ell(f)$ in the Hecke fields (Proposition 21.5), from modular symbols in PARI/GP or Magma; Smith normal forms of the reduced Laplacians for the other r. Deliverable: tables of the laws and of the proportion of cyclic critical groups, and the second order term $\log |J_0(N)(\mathbb{F}_\ell)| - g \log \frac{\ell^\ell}{(\ell^2-1)^{(\ell-1)/2}}$ of Theorem 21.8, which the trace formula writes formally as a series in Legendre symbols at N, like the Kemeny correction of Problem 10.1. Why it matters: it tests a random matrix model for arithmetic critical groups in the level aspect, where no theorem is known. Kemeny constants at these degrees are Problem 10.1 and the problem on other degrees in Chapter 10.

Computational Problem 21.2 (Explosion cumulants from Frobenius). Objective: for the even icosahedral representation ρ of Problem 20.1, compute $\int_0^\infty x^{m-1} \mathcal{W}_\rho(x)\, dx$ for $m = 1, 2$ via Definition 20.29, as Section 21.3 does for the S_3 representation, from the Frobenius classes (factorisation patterns, with a Frobenius resolvent [161] to separate the two classes of order 5 and the two of order 10), the conductor and the gamma factor. If $\Lambda(s, \rho)$ is entire and satisfies GRH, neither of which is known, these are $(m-1)! \sum_k \gamma_k^{-2m}$, the mean and the variance of the explosion time. Objects and scale: prime powers up to 10^{10}, which at conductors of a few million control $\mathcal{W}_\rho$ only for x up to about 4 (Problem 20.1). Route: ball arithmetic for $x \leq 4$; for larger x, the explicit formula with the lowest zeros located numerically, since the range of x grows only like $(\log P)^2$ with the prime range P. Deliverable: the integrals over $x \leq 4$ with rigorous error bounds, compared with the same integrals over the zeros, and the full integrals, whose part beyond $x = 4$ rests on the computed zeros, compared with the power sums S_1 and S_2 of (14.4) computed from central derivatives as in Chapter 14. Why it matters: it checks the Galois data against the zeros on the range that the primes control; a disagreement beyond the error bounds would point to a pole or a missed zero, while agreement verifies neither holomorphy nor GRH.

Computational Problem 21.3 (Signed exit laws and real zeros). Objective: for every real odd primitive character χ of conductor $q < 10^6$, decide whether the signed exit density of Theorem 21.13, here $(4\pi/q^{3/2})\vartheta_\chi(2\pi t/q)$, is positive for all $t > 0$; positivity excludes a real zero of $L(s, \chi)$ [82]. For $\chi = \chi_D$ with

$D \equiv 5 \bmod 24$ the density takes negative values, by Theorem 17.18 (Section 21.4). Objects and scale: the characters χ_D of the fundamental discriminants $-10^6 < D < 0$, about $3 \cdot 10^5$ of them. Route: the root number is 1, so $\vartheta_\chi(1/x) = x^{3/2}\vartheta_\chi(x)$ and it suffices to take $x \geq 1$; lower bounds in ball arithmetic (Arb or PARI/GP) on a grid, with bounds on the derivative between grid points. Deliverable: the list of D with positive density and their proportion in dyadic ranges of $|D|$, the first sign change for the others, and a comparison with the positivity on $(0, 1)$ of the Fekete polynomial $\sum_{n<q} \chi(n) y^n$, a weaker condition implied by positivity of the density (its Laplace transform is a positive multiple of a value of this polynomial), which already fails for $q = 43$ ([82], Section 6.1). Why it matters: it measures how far the Brownian route to excluding real zeros goes, and on which characters it fails first.

CHAPTER 22

WEIGHT ONE FORMS, WILES'S THEOREM AND FERMAT: IS THERE A THORIN MEASURE FORM OF FERMAT'S LAST THEOREM?

Two kinds of "one forms" carry the proof of Fermat's last theorem. Weight one modular forms enter through the Langlands Tunnell theorem, which makes the mod 3 representation of a semistable elliptic curve modular when it is irreducible, and so starts Wiles's argument. Weight two forms are holomorphic differential 1 forms $f(z)\,dz$ on the modular curve $X_0(N)$, and the proof ends with the fact that $X_0(2)$ has none. This chapter reads both through the probabilistic language of Chapters 20 and 21, and answers the question in the title. No step of the proof is reproved here, and no new case of modularity is proved.

Weight one newforms are exactly the newforms whose Frobenius label law is atomic (Proposition 22.3). The congruence of Wilton, which says that $\tau(p)$ is congruent modulo 23 to the pth coefficient of $\eta(z)\eta(23z)$, and which we verify at all 25996 primes $p < 3 \cdot 10^5$, $p \neq 23$, is a classical instance of the congruences between forms of different weights that Wiles needed. Modularity implies a symmetry in law: for $11a1$, whose form is $\eta(z)^2\eta(11z)^2$, the Pólya kernel is a positive multiple of a probability density, and modularity (here Eichler and Shimura) makes the corresponding random variable symmetric; at this level, given the Euler product, the symmetry also implies modularity, by a converse theorem of Conrey and Farmer. For $37a1$ the kernel is odd and the root number is -1 (Proposition 22.4). An elliptic curve carries three Thorin objects: the Thorin measure of its Frobenius functionals, the unconditional Rankin Selberg generalized gamma convolution, and the counting measure of its squared zeros, which is positive exactly under GRH. We compute the heat trace of $11a1$ from its point counts modulo primes, its conductor and its gamma factor, matching the sum over its zeros to a relative error below $2 \cdot 10^{-13}$, and its integral over $x \geq e^{-60}$, 0.127589010327, against $\frac{1}{2}\sum \gamma^{-2} = 0.127589010329$, summed over all zeros, from the Hadamard product. Under GRH this constant is the mean explosion time of the birth process of Chapter 21 whose rates are the squared ordinates of the zeros, and that explosion time and the Pólya variable form an explicit Wald couple (Proposition 22.5).

The answer to the title question has three parts. There is no Thorin measure statement equivalent to Fermat's last theorem in a meaningful sense: Fermat is a statement about rational points, Thorin measures see L functions, and the bridge

used in the proof is modularity. What does exist is a Thorin form of the last step. Attach to each prime level N and auxiliary prime ℓ the generalized gamma convolution whose Thorin atoms are the numbers $\ell + 1 - a_\ell(f)$ for the weight two newforms f of level N. Its Thorin measure is empty exactly when $X_0(N)$ has genus zero, when there is one supersingular elliptic curve in characteristic N, and when the isogeny walk of Chapter 21 has nowhere to go (Theorem 22.7). The conclusion of Ribet's level lowering is the vanishing of a mod p discrepancy Thorin mass (Proposition 22.8), illustrated by a newform of level 77 congruent to $11a1$ modulo 3 at all 2259 primes tested. Given the theorems of Wiles, Taylor and Wiles, Ribet and Mazur, Fermat's last theorem for prime exponents $p \geq 5$ follows from the emptiness of the level 2 Thorin measure (Theorem 22.9), which is the class number one of the Hurwitz quaternions. Section 22.7 makes the connection explicit. The Frobenius functional that counts primes ℓ with $p \nmid |E(\mathbb{F}_\ell)|$ converges to a Gamma law whose shape $c(E,p)$, a local density of Koblitz's conjecture [326, 596], we call the Eisenstein Thorin shape; it equals $1 - (p^2 - 2)/((p^2-1)(p-1))$ when the image is $\mathrm{GL}_2(\mathbb{F}_p)$, and it vanishes exactly when $\bar{\rho}_{E,p}$ is Eisenstein (Proposition 22.11). By Serre's argument from Mazur's torsion theorem, a Frey curve has $c(E,p) > 0$ for $p \geq 5$, while its rational 2 torsion forces $c(E,2) = 0$ (Proposition 22.12); the Frey type curve of $3 + 5^3 = 2^7$ has measured shapes $0, 0, 0.762, 0.837$ at $p = 2, 3, 5, 7$ over the primes $\ell < 2 \cdot 10^6$, against $73/96 = 0.760$ and $241/288 = 0.837$ for image $\mathrm{GL}_2(\mathbb{F}_p)$. Given the same theorems, Fermat's last theorem is the incompatibility of two Thorin statements: a positive Eisenstein shape and an empty cuspidal Thorin measure at level 2 (Theorem 22.13). The deep content is in the theorems of Langlands and Tunnell, Wiles, Taylor and Wiles, Ribet and Mazur, not in the probabilistic restatement.

22.1 The architecture of the proof

Fermat's last theorem asserts that $a^n + b^n = c^n$ has no solution in nonzero integers for $n \geq 3$. Its proof, completed by Wiles [581] with Taylor and Wiles [541], runs through a chain of reductions due to Hellegouarch, Frey, Serre and Ribet [255, 202, 507, 460]. It suffices to treat $n = 4$, done by Fermat, and odd primes $n = p$, the case $p = 3$ being Euler's. In outline: a solution with $n = p \geq 5$ prime produces a semistable elliptic curve E, the Frey curve; E is modular by Theorem 22.1; Ribet's theorem then forces the mod p Galois representation of E to come from a weight two newform of level 2; and there are no such forms, because the modular curve $X_0(2)$ has genus zero.

Suppose $a^p + b^p = c^p$ with $abc \neq 0$, $p \geq 5$ prime, $\gcd(a,b,c) = 1$, and after permuting and changing signs $a \equiv -1 \bmod 4$ and b even. The Frey curve

$$E : \; y^2 = x(x - a^p)(x + b^p)$$

is semistable [202, 507]. Its conductor is the product N_E of the primes dividing abc, and its minimal discriminant is $2^{-8}(abc)^{2p}$. Its mod p representation

$$\bar{\rho}_{E,p} : \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \to \mathrm{GL}_2(\mathbb{F}_p)$$

is irreducible, by Serre's argument [507] from Mazur's torsion theorem [384], since E is semistable with all its 2 torsion rational [131, Theorem 2.9(c)]. It is unramified at every odd prime $q \neq p$ of multiplicative reduction, because the exponent of q in the minimal discriminant is divisible by p, and it is finite flat at p, by the same divisibility if $p \mid abc$ and because E has good reduction at p otherwise. At 2 the exponent $2pv_2(b) - 8$ of the minimal discriminant is not divisible by p, so $\bar{\rho}_{E,p}$ is ramified at 2.

THEOREM 22.1 (Wiles; Taylor and Wiles [581, 541]). *Every semistable elliptic curve over $\mathbb{Q}$ is modular: there is a weight two newform f of level N_E with $a_\ell(f) = \ell + 1 - |E(\mathbb{F}_\ell)|$ for all $\ell \nmid N_E$.*

Breuil, Conrad, Diamond and Taylor [41] removed the semistability hypothesis (Theorem 2.7).

THEOREM 22.2 (Ribet [460], Theorem 1.1; see [131]). *Let p be an odd prime, f a weight two newform of level N with trivial character, $\mathfrak{p}$ a prime above p of its coefficient field with $\bar{\rho}_{f,\mathfrak{p}}$ irreducible, and $q \| N$ a prime at which $\bar{\rho}_{f,\mathfrak{p}}$ is unramified, or finite flat if $q = p$. If $q \not\equiv 1 \bmod p$ or $p \nmid N$, there are a weight two newform g of level dividing N/q with trivial character and a prime $\mathfrak{p}'$ above p of its coefficient field with $\bar{\rho}_{g,\mathfrak{p}'} \cong \bar{\rho}_{f,\mathfrak{p}}$ over $\overline{\mathbb{F}}_p$.*

Applied to the Frey curve, first at $q = p$ if $p \mid N_E$, where $q \not\equiv 1 \bmod p$ holds trivially, and then one at a time at the odd primes $q \neq p$ of the level, which is now prime to p, Theorems 22.1 and 22.2 give a weight two newform of level dividing 2. There is none, since $S_2(\Gamma_0(1)) = 0$ and $\dim S_2(\Gamma_0(2))$ is the genus of $X_0(2)$, which is 0. This is the whole skeleton; the depth is in Theorems 22.1 and 22.2 and in Mazur's torsion theorem.

22.2 WEIGHT ONE FORMS

A newform f of weight one and level N gives, by Deligne and Serre [142], an odd irreducible two dimensional Artin representation ρ with $\mathrm{tr}\,\rho(\mathrm{Frob}_p) = a_p(f)$ for $p \nmid N$; its character is $\det\rho$. The image of ρ is finite, its projective image is dihedral, A_4, S_4 or A_5, and the Ramanujan bound $|a_p| \leq 2$ holds trivially. Conversely every odd irreducible two dimensional Artin representation arises in this way, by Hecke, Langlands and Tunnell [351, 553] when the projective image is solvable, and in general by Khare and Wintenberger [311, 312], with work of Kisin [320]. The book's running example (Example 3.3) is $\eta(z)\eta(23z)$, of level

23 and character $(\frac{-23}{\cdot})$, attached to the S_3 representation of the splitting field of $x^3 - x - 1$: $a_p = 2, 0, -1$ as $x^3 - x - 1$ has 3, 1 or 0 roots modulo p, for $p \neq 23$.

22.2.1 Weight one forms in Wiles's proof

Wiles's argument starts from a modular residual representation. Suppose that $\bar{\rho}_{E,3}$ is irreducible. Its image lies in $\mathrm{GL}_2(\mathbb{F}_3)$, which is solvable, and $\mathrm{GL}_2(\mathbb{F}_3)$ embeds in $\mathrm{GL}_2(\mathbb{Z}[\sqrt{-2}])$ compatibly with reduction modulo $1 + \sqrt{-2}$. Composing gives an odd Artin representation with octahedral or smaller projective image, which by Langlands and Tunnell [351, 553] comes from a weight one form g. The character of g is $\chi_{-3} = (\frac{-3}{\cdot})$, which lifts $\det \bar{\rho}_{E,3}$. Multiplying g by the weight one Eisenstein series $E_{1,\chi_{-3}}$, the theta series of $x^2 + xy + y^2$, which is congruent to 1 modulo 3, gives a weight two form with trivial character congruent to g, and the lemma of Deligne and Serre [142] gives a weight two eigenform with the same congruence, so $\bar{\rho}_{E,3}$ is modular of weight two [581, 131]. When $\bar{\rho}_{E,3}$ is reducible, which for semistable E is the same as reducibility of its restriction to $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}(\sqrt{-3}))$ [131], Wiles passes to $p = 5$ through an auxiliary curve, the 3 to 5 switch. So a weight one form, and through it the Langlands Tunnell theorem for solvable groups, is the base of the proof.

The mechanism, a weight one form congruent to a form of higher weight, has a classical prototype. Wilton [585] proved, and Serre [500] explained through the mod 23 representation of Δ (see also Swinnerton Dyer [533]), that Ramanujan's τ is controlled modulo 23 by the S_3 field of $x^3 - x - 1$:

$$\tau(p) \equiv a_p\big(\eta(z)\eta(23z)\big) \pmod{23} \qquad (p \neq 23),$$

so $\tau(p) \equiv 0$, 2 or -1 modulo 23 according as p is inert in $\mathbb{Q}(\sqrt{-23})$, splits completely in the S_3 field, or neither. Equivalently, the mod 23 representation of Δ is the reduction of the S_3 representation, so its projective image is S_3, and away from 23 the weight 12 form Δ is congruent to the weight one form $\eta(z)\eta(23z)$. We checked the congruence at all 25996 primes $p < 3 \cdot 10^5$, $p \neq 23$, with no exception.

22.2.2 Weight one is an atomic label law

In the Gamma process of Section 20.4 the prime factors of a zeta random integer carry labels, here the normalised coefficients $x_p = a_p(f)/p^{(k-1)/2}$ of a newform f of weight k. The label law μ_f is the limit of their empirical distribution.

PROPOSITION 22.3 (atomic labels characterise weight one). *For a newform f of weight $k \geq 1$ the following are equivalent: (i) $k = 1$; (ii) x_p takes finitely many values; (iii) the label law μ_f is finitely supported. In that case μ_f is the image of Haar measure on the finite image of the Deligne Serre representation under the trace.*

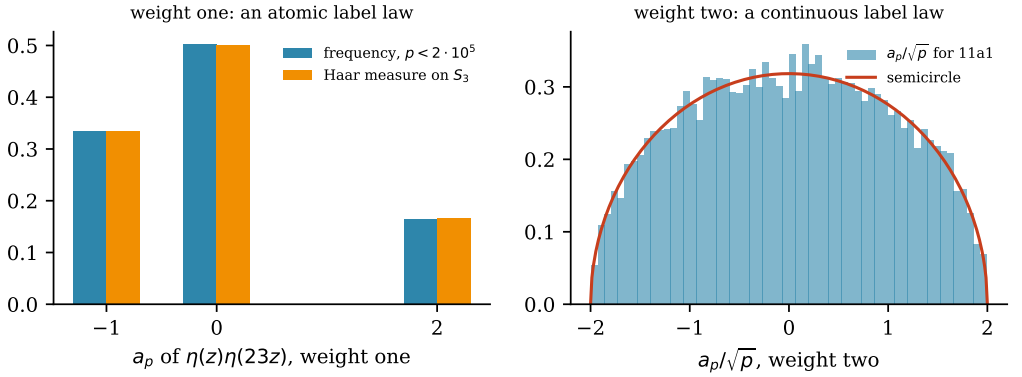


Figure 22.1: Proposition 22.3. Left: the coefficients of the weight one form $\eta(z)\eta(23z)$ over primes below $2 \cdot 10^5$, against Haar measure on S_3. Right: the normalised coefficients of the weight two form of $11a1$, against the semicircle.

PROOF. If $k = 1$, Deligne and Serre give a finite image G, so $x_p = \operatorname{tr}\rho(\mathrm{Frob}_p)$ takes finitely many values and Chebotarev gives μ_f as the image of Haar measure on G. If $k \geq 2$ and f has complex multiplication, Hecke's theorem gives $\mu_f = \mu_{\mathrm{CM}} = \frac{1}{2}\delta_0 + \frac{1}{2}\nu$ (Section 4.6), with ν the continuous law of $2\cos\theta$, θ uniform; if f has no complex multiplication and trivial character, μ_f is the semicircle law by the Sato Tate theorem [67]. With a character ψ, x_p/ζ is equidistributed for the semicircle law on the primes with $\psi(p) = \zeta^2$ [67, Theorem B], so μ_f is a mixture of rotated semicircle laws, and with complex multiplication the law above is rotated in the same way. None of these laws is finitely supported, and in every case x_p takes infinitely many values. □

Figure 22.1 shows the two kinds of label law. Among the primes $p \neq 23$ below $2 \cdot 10^5$ the coefficients of $\eta(z)\eta(23z)$ take the values $-1, 0, 2$ with frequencies $0.334, 0.502, 0.165$ (and $0.33336, 0.50027, 0.16637$ below 10^7, Example 3.3), against Haar measure $\frac{1}{3}, \frac{1}{2}, \frac{1}{6}$ on S_3, while $a_p/\sqrt{p}$ for the weight two form of $11a1$ follows the semicircle. In Thorin terms a Frobenius functional of a weight one form is a finite gamma convolution with atomic Thorin measure (Theorem 20.21 with a finite label space), while for weight two and $Q(x) = x^2$ (Figure 22.2, right) it is a generalized gamma convolution with a continuous Thorin measure.

22.3 MODULARITY AND SYMMETRY IN LAW

Let E have conductor N and L series $\sum a_n n^{-s}$, and put $f(z) = \sum a_n e^{2\pi i n z}$. The Pólya kernel of Chapter 20,

$$\Phi_E(u) = f\Big(\frac{ie^u}{\sqrt{N}}\Big)e^u, \qquad \Lambda(1+t,E) = \int_{\mathbb{R}} \Phi_E(u)e^{tu}\,du,$$

with $\Lambda(s,E) = N^{s/2}(2\pi)^{-s}\Gamma(s)L(s,E)$, is built from the point counts $a_p = p+1-|E(\mathbb{F}_p)|$ and the conductor. By Hecke's criterion (Proposition 20.27), f satisfies the Fricke transformation with root number ε if and only if $\Phi_E(-u) = \varepsilon\,\Phi_E(u)$. Weil's converse theorem [571] upgrades the corresponding reflection identities for the twisted kernels $\Phi_{E\otimes\chi}$, which pair χ with $\bar\chi$ and are complex valued, to modularity; Problem 20.2 poses their numerical test. For $11a1$ the reflection of Φ_E alone suffices, given the Euler product: the converse theorem of Conrey and Farmer [115] replaces the twists by the shape of the Euler factors at the levels $5 \le N \le 12$, $14 \le N \le 17$ and 23. So Theorem 22.1 says, for a semistable E, that a family of functions built from point counts and the conductor has the reflection symmetries of a modular form.

PROPOSITION 22.4 (the Pólya variable of an elliptic curve; after Proposition 20.27). *Let f be the newform of an elliptic curve E and suppose $\Phi_E \ge 0$. Then $\Phi_E/\Lambda(1,E)$ is the density of a random variable W_E with $\mathbb{E}e^{tW_E} = \Lambda(1+t,E)/\Lambda(1,E)$ for all real t, and W_E is symmetric in law; positivity forces the root number to be $+1$, since a kernel with $\varepsilon = -1$ is odd. For $E = 11a1$, $f(z) = \eta(z)^2\eta(11z)^2$ is positive on the imaginary axis, so W_E exists and is symmetric.*

PROOF. The Mellin formula gives the transform, which converges for all real t because $f(iy)$ decays exponentially as $y \to \infty$ and, by the Fricke symmetry, as $y \to 0$. Symmetry is Hecke's criterion with $\varepsilon = +1$. The product formula $\eta(z) = q^{1/24}\prod(1-q^n)$ is positive on $z = iy$, and the root number of $11a1$ is $+1$. □

We computed Φ_E from the first 4000 coefficients for $11a1$ and $37a1$ (Figure 22.2, left). For $11a1$ the kernel is positive, with minimum $1.2\cdot10^{-9}$ on $|u| \le 2.5$, and $\max|\Phi_E(u) - \Phi_E(-u)| = 9\cdot10^{-17}$. For $37a1$, the curve of smallest conductor with positive rank, the root number is -1 and the kernel is odd to $4\cdot10^{-16}$: the parity of the analytic rank is the parity of a Pólya kernel. A Frey curve, if it existed, would by Theorem 22.1 have kernels with all the twisted reflection symmetries; by Ribet its mod p labels would agree with those of a level 2 form; there is none.

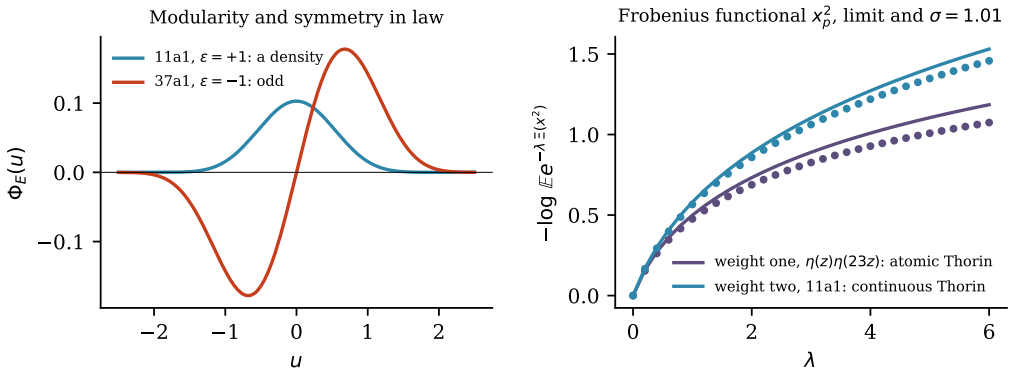


Figure 22.2: Left: Pólya kernels $\Phi_E(u)$ (solid) and $\varepsilon\Phi_E(-u)$ (dashed, hidden under the solid curves) computed from point counts and the conductor: $11a1$, a positive multiple of a symmetric probability density, and $37a1$, odd. Right: the Frobenius functional $\Xi_\sigma(Q)$, $Q(x) = x^2$, of the zeta random integer for the weight one form $\eta(z)\eta(23z)$ and the weight two form of $11a1$, shown as $-\log \mathbb{E}e^{-\lambda\,\Xi_\sigma(Q)}$: the limits $\int \log(1+\lambda x^2)\,d\mu$ (lines) and the values at $\sigma = 1.01$ from the Euler product over the primes below 10^7, the rest estimated by the prime number theorem (dots). Both have mean 1; their Thorin measures differ.

22.4 The Thorin measures of an elliptic curve

For an elliptic curve, or any weight two newform, there are three Thorin objects, of increasing depth.

Frobenius functionals. By Theorem 20.21 the functional $\Xi_\sigma(Q)$ of the zeta random integer, with labels $x_p = a_p/\sqrt{p}$, converges to the GGC whose Thorin measure is the image of the label law, restricted to $\{Q > 0\}$, under $x \mapsto 1/Q(x)$. For $Q(x) = x^2$ and $11a1$ this is the law of $1/X^2$ for X semicircular, a continuous Thorin measure; for the weight one form $\eta(z)\eta(23z)$ it is atomic, $\frac{1}{6}\delta_{1/4} + \frac{1}{3}\delta_1$. Both GGCs have mean $\int x^2\,d\mu = 1$, which is the orthonormality of Theorem 20.14 in the form *unit second moment*; the Thorin measures, images of the label laws, are what distinguish weight one from weight two (Figure 22.2, right; Proposition 22.3). These statements are unconditional.

The Rankin Selberg GGC. Let $D(s)$ be the Rankin Selberg Euler product of E in the unitary normalisation, with $\alpha_p = a_p/\sqrt{p}$ and $\beta_p = 0$ at the primes $p \mid N$, so that $-D'/D(s) = \sum_{p,m} |\alpha_p^m + \beta_p^m|^2 (\log p) p^{-ms}$ with nonnegative coefficients and D has a pole at $s = 1$; for semistable E, D is $L(s, E \times E)$ without its factors $(1 - p^{-s})^{-1}$ at the primes $p \mid N$. For $\alpha > 1$ put

$$U_\alpha(z) = \frac{1}{2\pi\sqrt{z}} \sum_{p,m} |\alpha_p^m + \beta_p^m|^2 (\log p)\, p^{-m\alpha}\, 2\sin^2\Big(\frac{m \log p\, \sqrt{z}}{2}\Big).$$

By Lemma 16.7 with $f = D$, as for the density U_{ar} of Section 16.4.3 (Proposition 20.34 is the analogue for Artin representations),

$$e^{\sqrt{s}\,D'/D(\alpha)}\,\frac{D(\alpha)}{D(\alpha+\sqrt{s})}$$

is the Laplace transform of a GGC with Thorin density U_α. It is computed from point counts and is positive unconditionally. Figure 22.3, right, shows U_α for $11a1$ at $\alpha = 1.5$ and 1.2, with the bad prime 11 entering through $|\alpha_{11}^m|^2 = 11^{-m}$; it grows as α decreases towards the pole at $s = 1$.

The zeros. The deepest object is the Galois heat trace $\mathcal{W}_E$ of Definition 20.29, given by the explicit formula from the point counts, the conductor and the gamma factor. Since $L(s,E)$ is entire by modularity (Theorem 2.7), Theorem 20.30 gives $\mathcal{W}_E(x) = \frac{1}{2}\sum e^{-x\gamma^2}$, the sum over the zeros $1+i\gamma$ of $\Lambda(s,E)$ with multiplicity, which are the zeros $\frac{1}{2}+i\gamma$ in the unitary normalisation of Chapter 20, and $\mathcal{W}_E$ is completely monotone if and only if GRH holds for $L(s,E)$. Its Bernstein measure is then $\frac{r}{2}\delta_0 + \sum_{\gamma>0}\delta_{\gamma^2}$, where r is the order of the zero at the centre, which is odd when the root number is -1; for $11a1$, $r = 0$. If $r = 0$ and GRH holds, Theorem 21.11 makes $\mathcal{W}_E(x)/x$ the Lévy density of the explosion time T_E of a pure birth process with rates γ^2, with $\mathbb{E}T_E = \int_0^\infty \mathcal{W}_E(x)\,dx = \sum_{\gamma>0}\gamma^{-2}$; by the same theorem T_E exists only if GRH holds for $L(s,E)$, which is not known.

For $11a1$ we computed $\mathcal{W}_E$, as Chapter 21 does for the S_3 representation, from the conductor 11, the gamma factor and the point counts at the 664579 primes below 10^7 and their powers, with the bad prime 11 entering through $a_{11} = 1$. It agrees with the sum over the zeros to fifteen digits at $x = 0.002$, 0.01 and 0.05 and to thirteen digits at $x = 0.2$ (Figure 22.3, left), finite differences give its first two derivatives the signs required by complete monotonicity at $x = 0.05, 0.1, 0.2, 0.4$, and

$$\int_{e^{-60}}^{2} \mathcal{W}_E(x)\,dx = 0.127589010327$$

from point counts, against 0.127589010329 computed by PARI for $\frac{1}{2}\sum\gamma^{-2} = \frac{1}{2}\Lambda''(1,E)/\Lambda(1,E)$, an identity of the Hadamard product that needs no hypothesis; this is the constant S_1 of $11a1$ in Table 14.1. The first zero is $\gamma_1 = 6.3626$, so $\mathcal{W}_E(x) < e^{-80}$ for $x \geq 2$, and the difference $1.6\cdot10^{-12}$ is the part of the integral over $x < e^{-60}$, which the quadrature leaves out and which the density of the zeros gives as $1.55\cdot10^{-12}$; with it the two values differ by $6\cdot10^{-17}$. As in Chapter 21, the agreement does not verify GRH: a finite computation of this kind can disprove it and is not known to verify it.

PROPOSITION 22.5 (an explicit Wald couple; after Theorem 16.6). *Assume GRH for $L(s,11a1)$. Then the Pólya variable W_E of Proposition 22.4 and the explosion time T_E satisfy*

$$\mathbb{E}e^{tW_E}\,\mathbb{E}e^{-t^2T_E} = 1 \qquad \textit{for every real } t:$$

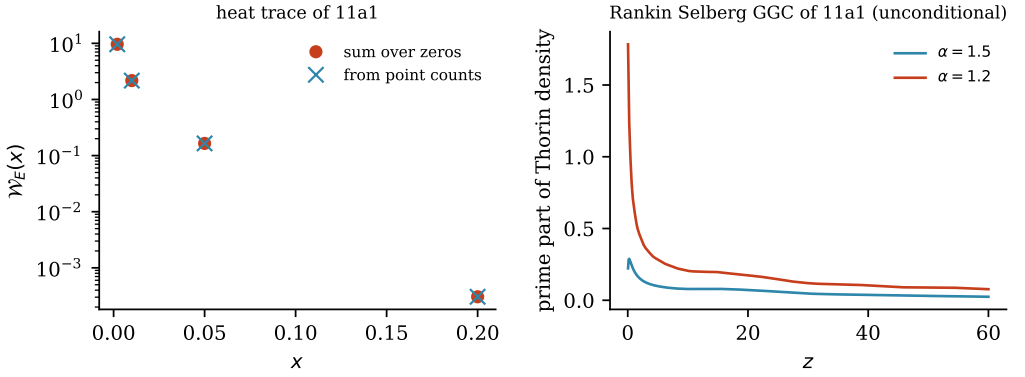


Figure 22.3: Left: the heat trace of $11a1$ computed from the point counts modulo the primes below 10^7, the conductor and the gamma factor (crosses), against the sum over its zeros (dots). Right: the Rankin Selberg Thorin density U_α of $11a1$ at $\alpha = 1.5$ and 1.2, positive unconditionally, summed over the primes below 10^7; the larger primes would add under 0.4 percent at $\alpha = 1.5$ and between 3.5 and 9.3 percent at $\alpha = 1.2$.

they form a Wald couple in the sense of Definition 16.3, built respectively from the modular form $\eta(z)^2\eta(11z)^2$ and from the zeros of its L function.

PROOF. This is the implication from (a) to (c) of Theorem 16.6 for $\eta(z)^2\eta(11z)^2$, with $Z_\pi = T_E$. Directly: $\mathbb{E}e^{tW_E} = \Lambda(1+t,E)/\Lambda(1,E)$, and under GRH

$$\mathbb{E}e^{-t^2T_E} = \prod_{\gamma>0}\Big(1+\frac{t^2}{\gamma^2}\Big)^{-1} = \frac{\Lambda(1,E)}{\Lambda(1+t,E)}$$

by the Hadamard product, since $\Lambda(1,E) \neq 0$ and $\Lambda(1+z,E)$ is even of order one, so its exponential factors cancel in pairs. □

22.5 WEIGHT TWO FORMS ARE ONE FORMS

For $f \in S_2(\Gamma_0(N))$ the differential $f(z)\,dz$ is invariant under $\Gamma_0(N)$ and holomorphic at the cusps, and every holomorphic 1 form on $X_0(N)$ arises this way, so $\dim S_2(\Gamma_0(N)) = g(X_0(N))$. For prime N, by Deuring [148] and Eichler, the number of supersingular elliptic curves over $\overline{\mathbb{F}}_N$ is $g(X_0(N))+1$, and by Eichler and Gross the 1 forms on $X_0(N)$ correspond to the nonconstant eigenfunctions of the isogeny walk of Section 21.1 [171, 173, 227] (Theorem 10.4 for $N \equiv 1 \bmod 12$). The genus is zero exactly for $N \in \{2,3,5,7,13\}$ among primes; PARI/GP gives $\dim S_2(\Gamma_0(N)) = 0,0,0,0,1,0,1,1,2$ for $N = 2,3,5,7,11,13,17,19,23$.

At $N = 2$ there is one supersingular curve, $j = 0 = 1728$ in characteristic 2, with automorphism group of order 24; with $w = |\operatorname{Aut} E|/2 = 12$, Eichler's mass formula $\sum 1/w = (N-1)/12$ of Chapter 21 reads $1/12 = 1/12$. Its endomorphism ring is the Hurwitz order of quaternions $\frac{1}{2}(a + bi + cj + dk)$ with a, b, c, d all even or all odd, which has class number one. The theta series of its norm form lies in $M_2(\Gamma_0(2))$, which is spanned by an Eisenstein series because $S_2(\Gamma_0(2)) = 0$, and so counts exactly: the number of Hurwitz quaternions of norm n is $24 \sum_{d \mid n,\, d \text{ odd}} d$. We checked this by enumeration for $n \le 60$; the first values are $24, 24, 96, 24, 144, 96, 192, 24$. The isogeny walk has a single state, and every ℓ isogeny returns to the same curve.

22.6 Is there a Thorin measure form of Fermat's last theorem?

The answer has three parts.

First, no Thorin statement is *equivalent* to Fermat's last theorem in a useful sense. Fermat is about rational points on the curves $x^p + y^p = 1$; Thorin measures are attached to L functions and Frobenius statistics. The bridge used in the proof, and the only one known for all exponents, is modularity, a deep theorem. Any true statement is formally equivalent to any other, so the question is what Fermat *reduces to*, after the deep theorems.

Second, it reduces to the emptiness of a Thorin measure.

Definition 22.6. For a prime N and a prime $\ell \neq N$, the *Hecke GGC* $H_N(\ell)$ is the generalized gamma convolution with Thorin measure $\sum_f \delta_{\ell+1-a_\ell(f)}$, the sum over the normalised eigenforms of $S_2(\Gamma_0(N))$.

The atoms are positive because $a_\ell(f) < \ell + 1$, which Perron Frobenius gives on the connected supersingular isogeny graph (the bound $|a_\ell(f)| \le 2\sqrt{\ell}$ of Eichler and Shimura with Weil, as in Chapter 21, gives more), so $H_N(\ell)$ is a sum of independent exponential variables with rates $\ell + 1 - a_\ell(f)$, the time that a pure birth chain with these rates takes to pass through its states. For a rational f attached to an elliptic curve E_f the rate is $|E_f(\mathbb{F}_\ell)|$. By (21.1) and Theorem 21.1,

$$\mathbb{E}e^{-sH_N(\ell)} = \prod_f \frac{\ell + 1 - a_\ell(f)}{\ell + 1 + s - a_\ell(f)} = \frac{|J_0(N)(\mathbb{F}_\ell)|}{\det\big((\ell + 1 + s) - T_\ell \mid S_2(\Gamma_0(N))\big)},$$

and $\mathbb{E}H_N(\ell) = K_N(\ell)/(\ell + 1)$, with $K_N(\ell)$ the Kemeny constant of the supersingular ℓ isogeny walk in characteristic N. Chapter 23 inverts this transform through the B spline whose knots are the atoms $\ell + 1 - a_\ell(f)$ (Theorem 23.11).

Theorem 22.7 (empty Thorin measures). *For a prime N and a prime $\ell \ge 5$, $\ell \neq N$, the following are equivalent.*

(a) The curve $X_0(N)$ has genus zero, so it carries no nonzero holomorphic 1 form and $S_2(\Gamma_0(N)) = 0$.

(b) The Thorin measure of $H_N(\ell)$ is zero, that is $H_N(\ell) = 0$ almost surely.

(c) $|J_0(N)(\mathbb{F}_\ell)| = 1$.

(d) There is exactly one supersingular elliptic curve over $\overline{\mathbb{F}}_N$, and the isogeny walk has a single state.

(e) The Kemeny constant $K_N(\ell)$ vanishes.

PROOF. (a) and (b) are the same statement, since the atoms are indexed by eigenforms. (a) implies (c) because $J_0(N) = 0$. If $g \geq 1$, the Weil bound gives $|J_0(N)(\mathbb{F}_\ell)| = \prod_f (\ell + 1 - a_\ell(f)) \geq (\sqrt{\ell} - 1)^{2g}$, as in Corollary 21.7, and $(\sqrt{5} - 1)^2 > 1$, so (c) implies (a). (a) is equivalent to (d) by the count $g + 1$ of supersingular curves, and to (e) by Theorem 21.1, which writes $K_N(\ell)$ as a sum of g positive terms. □

Third, Ribet's theorem produces a newform whose mod p discrepancy Thorin mass against the given one vanishes; equivalently, by Theorem 21.10, the discrepancy set of primes is hit by the prime factors of a random integer at rate zero. Label the primes by the pair of mod p Frobenius classes of two Galois representations; by Chebotarev these labels are equidistributed on a finite group, and Theorem 20.21 applies with a finite label space, as in Theorem 20.22(a) for Artin representations.

PROPOSITION 22.8 (mod p reciprocity as a vanishing Thorin mass; compare Theorem 20.22(a)). *Let ρ_1, ρ_2 be the representations, at primes above p of the coefficient fields, of two weight two newforms with trivial character, or of an elliptic curve and such a newform, with semisimple reductions $\bar{\rho}_1, \bar{\rho}_2$ over a common finite field k of characteristic p. Let D_p be the set of primes $\ell \nmid pN_1N_2$ with $\operatorname{tr}\bar{\rho}_1(\mathrm{Frob}_\ell) \neq \operatorname{tr}\bar{\rho}_2(\mathrm{Frob}_\ell)$ in k, and $Q = 1_{D_p}$. The following are equivalent: (i) $\bar{\rho}_1 \cong \bar{\rho}_2$; (ii) D_p has Dirichlet density zero; (iii) the Frobenius functional $\Xi_\sigma(Q)$ of the zeta random integer converges to 0, its Thorin measure being zero.*

PROOF. Both representations factor through the finite group $G = \mathrm{Gal}(K/\mathbb{Q})$ cut out by $\bar{\rho}_1 \oplus \bar{\rho}_2$, and D_p is the set of primes whose Frobenius lies in the union of classes where $\operatorname{tr}\bar{\rho}_1 \neq \operatorname{tr}\bar{\rho}_2$. By Chebotarev its density is the Haar measure of that union, which is also the Thorin mass of the limit of $\Xi_\sigma(Q)$ by Theorem 20.21. If the density is zero the traces agree on all of G; both determinants are the mod p cyclotomic character, so the characteristic polynomials agree, and the Brauer Nesbitt theorem gives (i). The converse is clear. □

Figure 22.4 illustrates Proposition 22.8 with a congruence between levels. The curve $11a1$ has $a_7 = -2 \equiv -(7 + 1) \bmod 3$, which is Ribet's condition [459] for level raising at $q = 7$ modulo 3, and indeed one of the three rational newforms of level 77 agrees with $11a1$ modulo 3 at all 2259 primes $\ell < 2 \cdot 10^4$ prime to $3 \cdot 7 \cdot 11$, consistent with zero discrepancy Thorin mass; Ribet's level raising

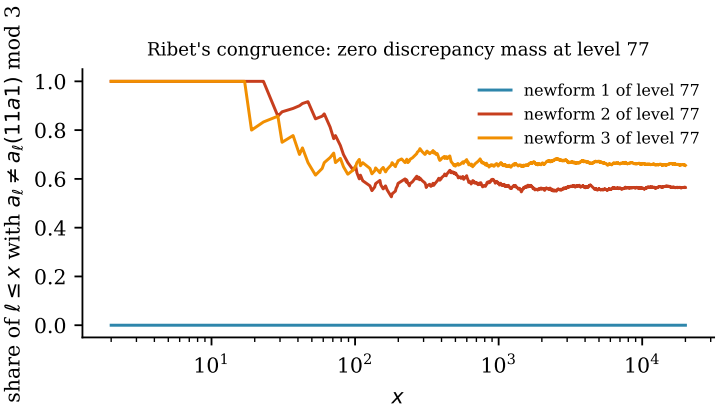


Figure 22.4: Proposition 22.8: the share of primes $\ell \le x$ at which the coefficients of each rational newform of level 77 differ modulo 3 from those of $11a1$. One newform has no discrepancy prime below $2 \cdot 10^4$, consistent with Ribet's level raising theorem [459], which predicts from the condition $a_7(11a1) \equiv -(7+1) \bmod 3$ a newform of level 77 congruent to $11a1$ modulo a prime above 3.

theorem predicts a newform of level 77 congruent to $11a1$ modulo a prime above 3, and here it has rational coefficients. The other two disagree at 57 and 66 percent of these primes. Read from level 77 down to level 11, this is level lowering at $q = 7$, the step that Ribet's theorem performs for the Frey curve.

THEOREM 22.9 (Fermat's last theorem in Thorin form). *Assume Theorems 22.1 and 22.2 and Mazur's torsion theorem [384]. Then for every prime $p \ge 5$ a solution of $a^p + b^p = c^p$ in nonzero integers would produce a weight two newform g of level dividing 2 whose mod p discrepancy Thorin mass against the Frey curve is zero. Since $S_2(\Gamma_0(1)) = 0$ and the Thorin measure of $H_2(\ell)$ is empty (Theorem 22.7), no such g exists, and Fermat's last theorem holds for $p \ge 5$.*

PROOF. The Frey curve E is semistable, hence modular by Theorem 22.1, with $\bar{\rho}_{E,p}$ irreducible by Serre's argument from Mazur's torsion theorem [507] (see Proposition 22.12), unramified at every odd prime dividing N_E other than p, and finite flat at p. Apply Theorem 22.2 first at $q = p$ if $p \mid N_E$, where $q \not\equiv 1 \bmod p$ holds trivially, and then one at a time at the odd primes $q \neq p$ of the level, which is now prime to p. This gives a newform g of level dividing 2 and a prime $\mathfrak{p}'$ above p with $\bar{\rho}_{g,\mathfrak{p}'} \cong \bar{\rho}_{E,p}$ over $\overline{\mathbb{F}}_p$, which by Proposition 22.8 is the vanishing of the discrepancy Thorin mass. But $S_2(\Gamma_0(1)) = 0$, and $X_0(2)$ has genus zero, so by Theorem 22.7 there are no weight two newforms of level 2. □

With the exponents 3 (Euler) and 4 (Fermat) this gives every $n \ge 3$, since each

is divisible by 4 or by an odd prime. So the last line of the proof of Fermat's last theorem, $S_2(\Gamma_0(2)) = 0$, is elementary, and in the language of this chapter it reads: in characteristic 2 there is one supersingular elliptic curve, so the isogeny walk is stuck, its Kemeny constant is zero, the Hecke GGC is the zero variable, and its Thorin measure is empty. Equivalently, the Hurwitz quaternions have class number one. Everything before that line is the work of Hellegouarch and Frey (the curve), Frey and Serre (the reduction), Mazur, Ribet, Langlands and Tunnell, Wiles, and Taylor and Wiles (Sections 22.1 and 22.2).

22.7 Fermat's last theorem, one Thorin measure at a time

The previous section answered the title question in outline. This section makes every Thorin measure in the argument explicit: on which space it lives, what it is, which theorem says that it vanishes or does not, and what it is numerically. Two Thorin measures appear. The first, the *Eisenstein Thorin shape* of an elliptic curve, is a single number $c(E,p) \in [0,1]$: the density of the primes ℓ with $p \nmid |E(\mathbb{F}_\ell)|$, which is the local density at p in Koblitz's conjecture [326], in Zywina's form for any image [596], read here as the shape of a Gamma law. Mazur's torsion theorem, through Serre's argument, makes it positive for Frey curves when $p \geq 5$. The second is the Hecke Thorin measure of Definition 22.6, which is empty at level 2. Given Theorems 22.1 and 22.2 and Mazur's torsion theorem, a Frey curve, whose shape is positive, would give the second measure an atom; since it has none, there is no Frey curve (Theorem 22.13).

22.7.1 The Eisenstein Thorin shape

The weight two Eisenstein series of prime level N has Hecke eigenvalues $a_\ell = 1+\ell$ for $\ell \neq N$. For an elliptic curve the difference from Eisenstein is a point count: $1 + \ell - a_\ell(E) = |E(\mathbb{F}_\ell)|$. So "$E$ agrees with Eisenstein modulo p at ℓ" means exactly that p divides $|E(\mathbb{F}_\ell)|$.

Definition 22.10. Let $E/\mathbb{Q}$ have conductor N and let p be prime. The *Eisenstein discrepancy* of (E,p) is $Q_{E,p}(\ell) = 1\{p \nmid |E(\mathbb{F}_\ell)|\}$ for primes $\ell \nmid pN$, and its *Eisenstein Thorin shape* is

$$c(E,p) = \mu_{G_{E,p}}\{g \in G_{E,p} : \det(1-g) \neq 0\},$$

where $G_{E,p} \subset \mathrm{GL}_2(\mathbb{F}_p)$ is the image of $\bar{\rho}_{E,p}$, with Haar probability measure $\mu_{G_{E,p}}$.

Proposition 22.11 (Gamma law of the Eisenstein discrepancy; (a) this book). *Let $E/\mathbb{Q}$ be an elliptic curve and p a prime.*

(a) The Frobenius functional $\Xi_\sigma(Q_{E,p})$ of the zeta random integer converges in law, as $\sigma \downarrow 1$, to the Gamma law with shape $c(E,p)$ and scale 1. Its Thorin measure is

$c(E,p)\,\delta_1$, and $c(E,p)$ is the rate at which the prime factors of a random integer hit the primes ℓ with $p \nmid |E(\mathbb{F}_\ell)|$ (Theorem 21.10).

(b) $c(E,p) = 0$ if and only if $\bar\rho^{\mathrm{ss}}_{E,p} \cong 1 \oplus \chi_p$, with χ_p the mod p cyclotomic character, if and only if p divides $|E(\mathbb{F}_\ell)|$ for every $\ell \nmid pN$. In that case E has a rational isogeny of degree p. If E is semistable, $c(E,p) = 0$ if and only if $\bar\rho_{E,p}$ is reducible.

(c) If $G_{E,p} = \mathrm{GL}_2(\mathbb{F}_p)$, then

$$c(E,p) = c_p := 1 - \frac{p^2-2}{(p^2-1)(p-1)},$$

so that $c_2 = \frac{1}{3}$, $c_3 = \frac{9}{16}$, $c_5 = \frac{73}{96}$ and $c_7 = \frac{241}{288}$.

PROOF. (a) The label of ℓ is the conjugacy class of $\bar\rho_{E,p}(\mathrm{Frob}_\ell)$ in the finite group $G_{E,p}$, and $Q_{E,p}(\ell) = 1$ exactly when $\det\big(1 - \bar\rho_{E,p}(\mathrm{Frob}_\ell)\big)$, which is $|E(\mathbb{F}_\ell)|$ modulo p, is nonzero. By Chebotarev the labels are equidistributed for Haar measure, and Theorem 20.21 with $Q \in \{0,1\}$ gives the GGC whose Thorin measure is $\mu(Q=1)\,\delta_1 = c(E,p)\,\delta_1$, the Gamma law with shape $c(E,p)$. The hitting rate is Theorem 21.10 for the same set of primes.

(b) If $c(E,p) = 0$ then every $g \in G_{E,p}$ has eigenvalue 1, so its characteristic polynomial is $(x-1)(x - \det g)$, which is that of $1 \oplus \chi_p$ since $\det \bar\rho_{E,p} = \chi_p$. By Brauer Nesbitt $\bar\rho^{\mathrm{ss}}_{E,p} \cong 1 \oplus \chi_p$, so $\bar\rho_{E,p}$ is reducible and its invariant line gives a rational p isogeny. The converse and the point count form are immediate. For semistable E the two characters of a reducible $\bar\rho_{E,p}$ are 1 and χ_p [502], so reducibility forces $c(E,p) = 0$.

(c) In $\mathrm{GL}_2(\mathbb{F}_p)$ the elements with eigenvalue 1 have characteristic polynomial $(x-1)(x-d)$. For each of the $p-2$ values $d \neq 0,1$ these form one split semisimple class of size $p(p+1)$; for $d = 1$ there are the identity and the p^2-1 nontrivial unipotents. The count is $(p-2)p(p+1) + p^2 = p^3 - 2p$, and dividing by $|\mathrm{GL}_2(\mathbb{F}_p)| = (p^2-1)(p^2-p)$ gives $(p^2-2)/((p^2-1)(p-1))$. □

Part (c) is the class count behind the factor at p of the constant in Koblitz's conjecture on the primality of $|E(\mathbb{F}_\ell)|$ [326, 596]: $c_p/(1-1/p) = 1 - (p^2-p-1)/((p-1)^3(p+1))$. By a theorem of Katz [295], the curves with $c(E,p) = 0$ in (b) are those isogenous to a curve with a rational point of order p. What (a) adds is the reading of this density as the shape of a Gamma law.

So the Eisenstein shape is a Thorin measure with one atom, and the whole content is in the mass of that atom. Two of its behaviours appear in the book's running examples (Figure 22.5). For $11a1$ and $p = 5$ the shape is exactly 0: $11a1$ has a rational point of order 5, so 5 divides $|11a1(\mathbb{F}_\ell)|$ for all 664577 primes $\ell < 10^7$ with $\ell \neq 5, 11$, and the Laplace transform of $\Xi_\sigma(Q)$ is identically 1. This is Mazur's Eisenstein congruence [384]: $J_0(11)$ is the curve $11a1$, whose rational torsion, the

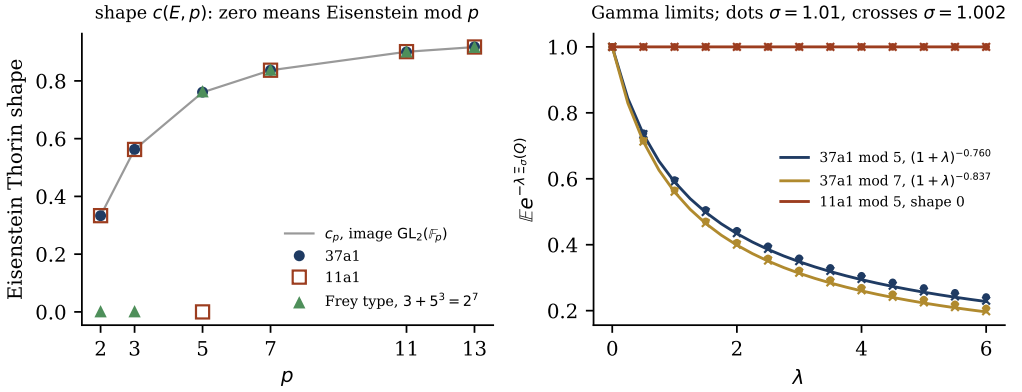


Figure 22.5: Proposition 22.11. Left: the Eisenstein Thorin shape $c(E,p)$, measured as the share of primes $\ell < 10^7$ with $p \nmid |E(\mathbb{F}_\ell)|$ for $37a1$ and $11a1$, and of primes $\ell < 2\cdot 10^6$ for the semistable Frey type curve $y^2 = x(x+125)(x+128)$ of $3+5^3 = 2^7$, against c_p. Right: Laplace transforms of $\Xi_\sigma(Q_{E,p})$ for $37a1$ modulo 5 and 7 and $11a1$ modulo 5, at $\sigma = 1.01$ (dots) and $\sigma = 1.002$ (crosses), against the Gamma limits $(1+\lambda)^{-c(E,p)}$.

cuspidal subgroup, has order 5, the numerator of $(11-1)/12$, and injects into $J_0(11)(\mathbb{F}_\ell)$ for $\ell \neq 11$ (for $\ell \geq 3$ by the appendix of [295], and for $\ell = 2$ because its order is odd), so 5 divides the product $|J_0(11)(\mathbb{F}_\ell)|$ of the Thorin atoms of $H_{11}(\ell)$. For $37a1$, whose mod p images are all of $\mathrm{GL}_2(\mathbb{F}_p)$ [502], the observed shares of discrepancy primes below 10^7 are 0.3332, 0.5628, 0.7611, 0.8370, 0.9002, 0.9178 for $p = 2,3,5,7,11,13$, against $c_p = 0.3333, 0.5625, 0.7604, 0.8368, 0.9008, 0.9172$, and the Laplace transforms at $\sigma = 1.01$ and $\sigma = 1.002$ approach $(1+\lambda)^{-c_p}$.

22.7.2 *Mazur's theorem and the Eisenstein shape of a Frey curve*

A Frey curve $y^2 = x(x-a^p)(x+b^p)$ is semistable and has all its 2 torsion rational. The second fact gives the Eisenstein shape $c(E,2) = 0$, and with Mazur's torsion theorem it gives positive shapes at every $p \geq 5$.

PROPOSITION 22.12 (Eisenstein shapes of a Frey curve; after Serre [507] and Mazur [384]). *Let E be an elliptic curve over $\mathbb{Q}$ with $E[2] \subset E(\mathbb{Q})$, for example a Frey curve. Then $c(E,2) = 0$, and $c(E,p) > 0$ for every prime $p \geq 5$.*

PROOF. For odd ℓ of good reduction, reduction is injective on $E(\mathbb{Q})[2] \cong (\mathbb{Z}/2)^2$, so 4 divides $|E(\mathbb{F}_\ell)|$ and $Q_{E,2} \equiv 0$. If $c(E,p) = 0$ for some $p \geq 5$, then $\bar{\rho}^{\mathrm{ss}}_{E,p} \cong 1 \oplus \chi_p$ by Proposition 22.11(b), so E, or its quotient E' by the rational subgroup of order p, has a rational point of order p. An isogeny of odd degree maps $E[2]$ isomorphically onto $E'[2]$, so that curve has rational torsion containing $\mathbb{Z}/2 \times \mathbb{Z}/2p$, which

Mazur's torsion theorem [384] excludes for $p \geq 5$. This is Serre's argument [507]; see also [131, Theorem 2.9(c)]. □

So the irreducibility of $\bar{\rho}_{E,p}$ for a Frey curve and $p \geq 5$, the hypothesis of Ribet's theorem that Serre deduced from Mazur's torsion theorem, reads in Thorin form: the Gamma law of the Eisenstein discrepancy has positive shape; equivalently, p fails to divide $|E(\mathbb{F}_\ell)|$ for a positive proportion $c(E,p)$ of primes. The hypothesis $p \geq 5$ is sharp, and a real Frey type curve shows why. The *abc* triple $3 + 5^3 = 2^7$, written as $-125 + 128 = 3$ with $-125 \equiv -1 \bmod 4$ and $32 \mid 128$, gives the Frey type curve

$$E_0 : \; y^2 = x(x+125)(x+128),$$

which is semistable of conductor 30 with minimal discriminant $2^6 3^2 5^6$. Its shapes, measured over the primes below $2 \cdot 10^6$, are $c(E_0,2) = 0$ and $c(E_0,3) = 0$, but $c(E_0,5) = 0.762$ and $c(E_0,7) = 0.837$, against $c_5 = 0.760$ and $c_7 = 0.837$ (Figure 22.5, left). At $p = 3$ the exponents 6 and 6 of the primes 2 and 5 are divisible by 3, so $\bar{\rho}_{E_0,3}$ is unramified at 2 and 5. A genuine Frey curve differs at 2: there the exponent is $2pv_2(b) - 8$, never divisible by an odd p, which is why level 2 remains. And E_0 has a rational 3 isogeny, its Eisenstein shape at 3 is 0, and Ribet's theorem, which needs an irreducible $\bar{\rho}$, does not apply. At $p \geq 5$, where Mazur's torsion theorem forces the shape to be positive, the exponents are not divisible by p and there is nothing to lower. A genuine Frey curve would need both at once.

22.7.3 The level two Brandt module

At level $N = 2$ the isogeny walk has the single state of Section 22.5, and its Brandt matrix is the 1×1 matrix $B(\ell) = (\ell + 1)$. By Eichler and Gross [171, 173, 227] the constants carry the Eisenstein eigenvalues $\ell + 1$, and their orthogonal complement in L^2 of the stationary law carries $S_2(\Gamma_0(2))$, which is 0 here. So every weight two Hecke eigensystem of level 2, cuspidal or not, is visible on this one point space, and a mod p eigensystem there could be of two kinds, of which only the first occurs:

(E) the Eisenstein system $a_\ell = \ell + 1$ of the stationary law, against which a curve E has discrepancy Thorin shape $c(E,p)$;

(C) a cuspidal system, an atom of the Hecke Thorin measure of $H_2(\ell)$, which is empty by Theorem 22.7.

THEOREM 22.13 (Fermat's last theorem as two Thorin statements). *Assume Theorems 22.1 and 22.2 and Mazur's torsion theorem [384], used through Proposition 22.12. Let $p \geq 5$ and suppose $a^p + b^p = c^p$ with $abc \neq 0$, and let E be its Frey curve. Then $\bar{\rho}_{E,p}$ would have zero discrepancy Thorin mass against some weight two eigensystem of level 2. But:*

(i) against the Eisenstein system the discrepancy is the Gamma law of shape $c(E,p) > 0$ *(Propositions 22.11 and 22.12);*

(ii) there is no cuspidal system, because the Hecke Thorin measure of level 2 *is empty (Theorem 22.7).*

Hence there is no such solution.

PROOF. By Proposition 22.12, $c(E,p) > 0$, so $\bar{\rho}_{E,p}$ is irreducible by Proposition 22.11(b), since E is semistable. It is unramified at the odd primes $q \neq p$ dividing N_E and finite flat at p (Section 22.1). By Theorem 22.1 it comes from a newform of level N_E, and Theorem 22.2, applied in the order of the proof of Theorem 22.9, gives a newform g of level dividing 2, hence of level 2 since $S_2(\Gamma_0(1)) = 0$, and a prime $\mathfrak{p}'$ above p with $\bar{\rho}_{g,\mathfrak{p}'} \cong \bar{\rho}_{E,p}$ over $\overline{\mathbb{F}}_p$, that is (Proposition 22.8) zero discrepancy Thorin mass between E and a cuspidal system of level 2. This contradicts (ii). Statement (i) shows that the Eisenstein system cannot take its place: it is the same positivity that made Ribet's theorem applicable. □

The theorem uses two Thorin measures in opposite directions. The Eisenstein shape of the Frey curve must be *positive*, a statement about Frobenius statistics that follows, by Serre's argument, from Mazur's theorem on rational torsion, proved through the Eisenstein ideal; the Hecke Thorin measure of level 2 must be *empty*, a statement about the one state isogeny walk that is equivalent to the class number one of the Hurwitz quaternions. Modularity (Wiles; Taylor and Wiles) and Ribet's theorem connect the two by transporting the Frobenius labels of the Frey curve, with zero discrepancy mass, from level N_E to level 2.

22.7.4 The proof of Fermat's last theorem, step by step

Step	Probabilistic or Thorin form	Status
Langlands Tunnell, weight one	zero discrepancy Thorin mass between the Artin representation and a weight one form (Theorem 20.22(a)); both have the atomic label law of Proposition 22.3, which does not produce the form (Corollary 20.15)	theorem (Langlands, Tunnell)
mod 3 to weight two	Wilton type congruence, zero discrepancy mass	theorem (Wiles)
modularity of semistable curves	reflection symmetry of twisted Pólya kernels	theorem (Wiles; Taylor and Wiles)
Frey curve (Hellegouarch, Frey), full 2 torsion	implies Eisenstein shape $c(E,2)=0$ (Proposition 22.12)	elementary
irreducibility of $\bar{\rho}_{E,p}$, $p \geq 5$	Eisenstein shape $c(E,p)>0$: Gamma law of the discrepancy, positive hitting rate	theorem (Mazur; Serre)
Ribet's level lowering	zero mod p discrepancy Thorin mass against a newform of level dividing N/q, zero hitting rate	theorem (Ribet)
$g(X_0(2))=0$	empty Hecke Thorin measure, Kemeny constant 0; only the Eisenstein system lives at level 2	elementary

The table shows what the Thorin framework restates. It restates the last four steps as statements about positive measures, Gamma shapes and hitting times, and modularity as a symmetry in law that can be tested numerically, though not verified, from point counts and the conductor. The Thorin form of the analytic theory of $L(s,E)$, complete monotonicity of the heat trace, is GRH, which Fermat's last theorem does not need and which remains open.

22.7.5 Where probability meets the deep part

Two points of the argument of Wiles and of Taylor and Wiles have a direct probabilistic reading. The Taylor Wiles method needs, for each n, auxiliary primes $q \equiv 1 \bmod p^n$ at which $\bar{\rho}(\mathrm{Frob}_q)$ has distinct eigenvalues; their existence, with positive density, once $\bar{\rho}$ is absolutely irreducible on the absolute Galois group of $\mathbb{Q}(\sqrt{p^*})$, where $p^* = (-1)^{(p-1)/2}p$, is a Chebotarev statement, which by Theorem 21.10 is a positive hitting rate of a class by the prime factors of a random integer. And the modularity of the Frey curve itself, if one wanted it in the probabilistic forms of Chapter 20, is the reflection symmetry of the twisted Pólya kernels required by Weil's converse theorem. A Thorin converse theorem of the kind

proved at level one in Corollary 20.33, not known at level N (Open Problem 20.1), would take as its hypothesis a positivity statement that contains GRH for $L(s, E)$, which is open; without GRH it could give neither modularity nor GRH. The Thorin framework restates modularity; it does not supply it.

22.7.6 Open problems

OPEN PROBLEM 22.1 (Hasse Weil from hitting times). Give a probabilistic proof of the bound $|a_\ell(f)| \leq 2\sqrt{\ell}$ of Eichler and Shimura with Weil for weight two newforms of prime level N, for instance through hitting times or a heat kernel on the supersingular isogeny graph. The positivity of the Thorin atoms of $H_N(\ell)$ needs only $a_\ell(f) < \ell + 1$, which Perron Frobenius gives; the bound itself gives the estimates of Corollary 21.7, and its local Thorin form is that of Chapter 15. Compare Open Problem 30.1, where no geometric proof is available.

OPEN PROBLEM 22.2 (Patching as a Gamma process). The Taylor Wiles method chooses, for each n, finite sets of auxiliary primes $q \equiv 1 \bmod p^n$ at which $\bar{\rho}(\mathrm{Frob}_q)$ has distinct eigenvalues; by Chebotarev and Theorem 21.10 such primes occur among the large prime factors of a zeta random integer at a positive rate. Is there a formulation of the patching argument in which the auxiliary sets are the atoms of the Gamma process of Section 20.4 in a fixed Chebotarev class, and the patched objects are limits in law as $\sigma \downarrow 1$?

OPEN PROBLEM 22.3 (A probabilistic Mazur theorem). Proposition 22.12 deduces $c(E, p) > 0$ for $p \geq 5$ and every curve with $E[2] \subset E(\mathbb{Q})$ from Mazur's torsion theorem. One rational point of order 2 is not enough: the semistable curve $y^2 + xy = x^3 - 45x + 81$ of conductor 66 has torsion $\mathbb{Z}/10$, so $c(E, 5) = 0$. Is there a proof of Proposition 22.12 through the Gamma law of the Eisenstein discrepancy and the isogeny walks of Chapter 21? Such a proof must use an input special to $\mathbb{Q}$: over some quadratic fields there are curves with torsion $\mathbb{Z}/2 \times \mathbb{Z}/10$ [306, 290, 291], whose analogue of the Eisenstein shape at 5, over the primes of the field, vanishes.

22.8 NOTES

The curve goes back to Hellegouarch and Frey [255, 202], and the reduction of Fermat's last theorem to modularity is due to Frey and Serre [202, 507], with level lowering by Ribet [460], and the modularity of semistable curves to Wiles and to Taylor and Wiles [581, 541]; Darmon, Diamond and Taylor [131] give a detailed account, including the use of Langlands and Tunnell [351, 553] and the 3 to 5 switch. Mazur's theorems on rational torsion and on rational isogenies of prime degree are in [384, 385]; Serre deduced from the first the irreducibility of $\bar{\rho}_{E,p}$ for Frey curves and $p \geq 5$ [507], which Proposition 22.12 restates as

a positive shape. Level raising is due to Ribet [459, 461], and the removal of semistability to Breuil, Conrad, Diamond and Taylor [41]. Deligne and Serre [142] attach an odd irreducible two dimensional Artin representation to each weight one newform, and by Khare and Wintenberger, with Kisin [311, 312, 320], every such representation arises in this way. The label law does not see parity: the S_3 representations of the splitting fields of $x^3 - x - 1$ (odd) and of $x^3 - 4x - 1$ (even) have the same label law; for the even representations see Open Problem 20.5 and Problem 20.1. Wilton's congruence is [585], and Serre [500] explained it by the mod 23 representation of Δ (see also Swinnerton Dyer [533]). Supersingular curves and their endomorphism rings go back to Deuring [148]; quaternion orders and Brandt matrices are in Eichler and Gross [171, 173, 227].

The Pólya kernel is Pólya's (Chapter 13), in the form of Chapter 20, from which the Frobenius functionals, the Galois heat trace and the Thorin converse theorem also come; the Rankin Selberg density U_α is the density U_{ar} of Chapter 16, Proposition 22.5 is the case $\eta(z)^2\eta(11z)^2$ of Theorem 16.6, and the Kemeny constant, hitting rates and explosion times are from Chapter 21. The density $c(E,p)$ of Definition 22.10 is classical: it is the local density at p in Koblitz's conjecture [326] that, for a curve without complex multiplication and not isogenous to one with a rational torsion point, the number of primes $\ell \le x$ for which $|E(\mathbb{F}_\ell)|$ is prime is asymptotic to a constant times $x/(\log x)^2$; Zywina [596] built the constant from the Haar measures of the images of E modulo all integers, which corrects for their entanglement, and no case with a positive constant is known. The vanishing of $c(E,p)$ is characterised by Katz [295] and, for semistable curves, by Serre [502]. New here are the readings of these statements as Thorin statements (Definition 22.10, Proposition 22.11(a) and Theorem 22.13) and the numerical checks; Propositions 22.4, 22.5 and 22.8 and Theorems 22.7 and 22.9 are restatements in this book's language, and Proposition 22.3 combines Deligne and Serre with the Sato Tate theorem.

22.9 Exercises

1. Show that $H_{11}(\ell)$ is exponential with rate $|11a1(\mathbb{F}_\ell)|$, and check that $\mathbb{E}H_{11}(2) = \frac{1}{5}$ agrees with $K_{11}(2) = \frac{3}{5}$ of Chapter 21.
2. Verify Wilton's congruence for $p < 1000$ by computing $\tau(p)$ from $\Delta = q\prod(1-q^n)^{24}$ and the number of roots of $x^3 - x - 1$ modulo p.
3. Prove from the Laplace pair $\int_0^\infty \log(1 + b^2/y^2)\cos(ay)\,dy = \frac{\pi}{a}(1 - e^{-ab})$ that

$$\int_0^\infty \log\Big(1 + \frac{s}{z}\Big) U_\alpha(z)\,dz = -\sqrt{s}\,\frac{D'}{D}(\alpha) - \log D(\alpha) + \log D(\alpha + \sqrt{s}).$$

   Compare Lemma 16.7, which proves it by another route.

4. Count the elements of $\mathrm{GL}_2(\mathbb{F}_p)$ without eigenvalue 1 for $p = 3, 5, 7$ by machine and check c_p of Proposition 22.11. Show that $c_p = 1 - \frac{1}{p} + O(p^{-2})$, so that, for a curve with image $\mathrm{GL}_2(\mathbb{F}_p)$ and large p, almost every prime is an Eisenstein discrepancy prime.

5. For $E_0 : y^2 = x(x+125)(x+128)$ find the rational 3 isogeny, and show that 3 divides $|E_0(\mathbb{F}_\ell)|$ for every $\ell \geq 7$.

6. Count the Hurwitz quaternions of norm $n \leq 20$ by enumeration and compare with $24 \sum_{d \mid n,\, d \text{ odd}} d$. Explain why the formula fails for the Lipschitz quaternions $a + bi + cj + dk$ with $a, b, c, d \in \mathbb{Z}$.

7. For $37a1$ show that Φ_E is odd and compute $\int_0^\infty u\, \Phi_E(u)\, du$ numerically; relate it to $\Lambda'(1, E)$ and hence to the Birch and Swinnerton Dyer leading term.

22.10 Open computational problems

Computational Problem 22.1 (Explosion cumulants of elliptic curves). Objective: for elliptic curves of both root numbers, compute $\int_0^\infty x^{m-1} \big(\mathcal{W}_E(x) - \frac{r}{2}\big)\, dx$ for $m = 1, 2, 3$, with r the order of vanishing at the centre, from the point counts, the conductor and the gamma factor through Definition 20.29, as Section 22.4 does for $11a1$. If GRH holds for $L(s, E)$, which is not known, these are $(m-1)! \sum_{\gamma > 0} \gamma^{-2m}$, and for $r = 0$ they are the first three cumulants of the explosion time T_E (Theorem 21.11). Objects and scale: one curve from each isogeny class of conductor below 10^3, which includes curves of rank 0, 1 and 2; prime powers up to 10^8. Route: `thorin_ec.py`, with point counts from the pipeline of Chapter 5 or from `ellap`; ball arithmetic for the archimedean term and on the range of x that the primes control, as in Problem 21.2; beyond it, the sum over the zeros located by `lfunzeros`; r from `ellanalyticrank`. Deliverable: a table by conductor, root number and rank of the integrals over the controlled range, with rigorous error bounds, against the same integrals over the zeros, and of the full integrals against the power sums S_1, S_2, S_3 of (14.4) computed from central Taylor coefficients as in Chapter 14. Why it matters: it is the family version of Problem 21.2, and it covers the curves with a central zero, which the problem on monotonicity across weights and levels in Chapter 13 leaves out; a disagreement beyond the error bounds would point to a zero off the line or to an error in the data, while agreement does not verify GRH.

Computational Problem 22.2 (Koblitz's conjecture and Eisenstein shapes). Objective: measure the Eisenstein shapes $c(E, p)$ of Definition 22.10 for $p \leq 37$ and the number of primes $\ell \leq X$ for which $|E(\mathbb{F}_\ell)|$ is prime, and compare the counts with Koblitz's conjecture [326] in the form refined by Zywina [596], whose

constant is built from the same Haar measures as the shapes, corrected for the entanglement of the images modulo different primes. Objects and scale: about 20 curves with different images, among them $37a1$, whose mod p images are all of $\mathrm{GL}_2(\mathbb{F}_p)$, curves with a rational isogeny of degree 2, 3, 5 or 7 but no rational point of that order, $11a1$ and the curve E_0 of Section 22.7, whose constants vanish, and one curve with complex multiplication; primes $\ell \leq 10^{10}$. Route: point counts from the pipeline of Chapter 5; the images from the LMFDB [371] and the classifications of [468, 598]; $c(E,p)$ by counting the elements of the image without eigenvalue 1; the Laplace transforms of $\Xi_\sigma(Q_{E,p})$ against $(1+\lambda)^{-c(E,p)}$ with $\sigma - 1 \geq 5/\log X$, as in Problem 20.4. Deliverable: for each curve the constant, the counts against the prediction with normalised fluctuations, and the measured shapes and hitting rates against their Haar values. Why it matters: the shapes give the local factors of Koblitz's constant, so the Gamma laws of Proposition 22.11 meet a conjecture that is open for every curve with a positive constant; Problem 5.2 tests the Lang Trotter constants, which are built from the same images.

COMPUTATIONAL PROBLEM 22.3 (Level raising congruences). Objective: test Proposition 22.8 and Ribet's level raising condition [459] at scale. For each rational newform g of weight two and level M and each prime p with $\bar{\rho}_{g,p}$ irreducible, list the primes $q \nmid Mp$ with $a_q(g) \equiv \pm(q+1) \bmod p$, find the newforms f of level dividing Mq and divisible by q whose coefficients are congruent to those of g modulo a prime above p, and compute the discrepancy masses of all pairs (g,f). Objects and scale: rational newforms of level $M \leq 200$, $p \in \{3,5,7\}$, primes $q < 50$, so levels below 10^4. Route: Hecke eigenvalues from the LMFDB [371] or from modular symbols (`mfinit` in PARI/GP, or Sage); congruences proved by comparing coefficients up to the Sturm bound; discrepancy masses from the joint mod p Frobenius classes, as in Figure 22.4, against their Chebotarev values computed from the images. Deliverable: a table of the predicted congruences against those found, with the residue fields and the coefficient fields of f, and the distribution of the discrepancy masses of the noncongruent pairs. Why it matters: the chapter shows one congruence, between $11a1$ and a newform of level 77; Ribet's theorem [459, 461] predicts all of them, and the masses of the noncongruent pairs show how far apart in law distinct mod p representations are, the mod p counterpart of the problem on near coincidences of newforms in Chapter 1 and of the pairs of curves of Problem 20.3.

Chapter 23

Inverse Laplace Transforms: Frobenius Traces, Pólya Kernels and Zeros

Langlands proposed that two different kinds of mathematics describe the same objects. On the Galois side an L function is an Euler product built from Frobenius traces, one prime at a time, and it lives on the half plane $\operatorname{Re} s > 1$ where the product converges. On the automorphic side the same L function is a Mellin transform of a modular form read along the imaginary axis, it continues to the whole plane, it has a functional equation, and its nontrivial zeros lie in the critical strip, on the critical line if GRH holds. Reciprocity is the statement that these two descriptions agree.

Part IV has used Laplace transforms mostly in the forward direction: the logarithm of an Euler product is the Laplace exponent of a Lévy measure carried by the logarithms of prime powers (Chapter 11); a completed L function is the two sided Laplace transform of a Pólya kernel (Chapter 13, Lemma 14.12); the $2\sin^2$ lemma turns the Lévy measure into a Thorin density (Lemma 16.7); and the heat trace is the Laplace transform of the counting measure of the squared ordinates of the zeros (Chapter 16), which Chapter 20 computes from Frobenius traces (Definition 20.29). Some inversions have already appeared: Stieltjes inversion gives the closed form of the sine squared lemma (Chapter 16) and the signed Thorin densities of Chapter 19, Mellin inversion proves Hecke's criterion (Proposition 20.27), Fourier inversion computes the van Dantzig partner of Δ (Figure 16.1), and the zeros are recovered from the primes (Figure 16.4). This chapter collects the inverse transforms and uses them to pass between the faces of one L function. The passage from the Galois side to the automorphic side still needs the analytic continuation that reciprocity supplies: it assumes that $\Lambda(s, \rho)$ is entire and bounded in vertical strips, which is Artin's conjecture for ρ; for the running example, the two dimensional representation ρ_{23} of the Galois group of $x^3 - x - 1$, this is Hecke's theorem (Example 3.3).

We organise an L function, say $L(s, \rho)$, into four faces (Figure 23.1): the *Galois face*, the measure $\nu_\alpha = \sum_n \Lambda_\rho(n) n^{-\alpha} \delta_{\log n}$, which is x times the Lévy measure of $\log L$ at α, as in Lemma 16.7, and whose atoms carry the Frobenius traces; the *transport face*, the $2\sin^2$ transform U_α of ν_α, which is the Thorin density of Lemma 16.7 when $\nu_\alpha \geq 0$; the *automorphic face*, the Pólya kernel Φ_ρ, which is

a modular form on the imaginary axis when ρ is modular; and the *zero face*, the family of Thorin densities V_a of the completed function, which under GRH concentrates on the squared ordinates of the zeros as $a \downarrow 0$ (Theorem 16.8(iv)). For this face Figure 23.1 shows the zeros themselves, as sign changes, and Figure 23.2 shows the densities V_a for ζ. Schematically, with the Frobenius data at the top left, the transport below it, the zeros at the bottom centre and the kernel at the bottom right,

$$\begin{array}{ccccc} \Lambda_\rho(n) & \xrightarrow{\text{Laplace}} & \log L(\alpha+s,\rho) & \xrightarrow{\text{continue}} & \Lambda(\frac{1}{2}+it,\rho) \\ {\scriptstyle 2\sin^2}\downarrow\uparrow{\scriptstyle \text{Bohr}} & & & & \downarrow{\scriptstyle \text{Bromwich}} \\ U_\alpha & \xrightarrow{\text{Cauchy}} & V_a & & \Phi_\rho(u) \end{array}$$

The bottom arrow completes the L function, which adds explicit terms to U_α and gives V_a at $a = \alpha - \frac{1}{2}$, and then moves the basepoint, which adds or removes Cauchy noise. The inversions are classical (Section 23.1), and of the results below only the explicit error in (1), the readings of the Hecke GGC and of the Sato Tate law in (5), and the computations are new, as far as we know; (3) restates, through folded Cauchy variables and for every self dual π, what [441] proves for ξ (see the Notes).

(1) The $2\sin^2$ transport is injective, and Bohr's mean value formula inverts it, with an explicit error for a Gaussian window (Theorem 23.5). Frobenius traces, times $-\frac{1}{2}p^{-m\alpha}\log p$, are the Bohr coefficients of the transported density, and from the transported density of the weight one form $\eta(z)\eta(23z)$ we recover $\Lambda_\rho(n)$, $\rho = \rho_{23}$, for every $n \le 64$ to $8.6 \cdot 10^{-9}$, and with it the Frobenius class at every unramified prime $p \le 61$.

(2) For an odd two dimensional ρ with $\Lambda(s,\rho)$ entire and bounded in vertical strips, Bromwich inversion on the critical line gives a kernel Φ_ρ with $\int \Phi_\rho(u)e^{tu}\,du = \Lambda(\frac{1}{2}+t,\rho)$, and if ρ comes from a weight one form f of level N, then $\Phi_\rho(u) = 2f(ie^u/\sqrt{N})e^{u/2}$: this is Hecke's correspondence read backwards (Proposition 23.8). For ρ_{23} both hold by Hecke's theorem, with $f = \eta(z)\eta(23z)$. From $\Lambda(\frac{1}{2}+it,\rho_{23})$, computed as ζ_K/ζ through its continuation, K the cubic field of $x^3 - x - 1$, the inversion gives a kernel that agrees with $2f(ie^u/\sqrt{23})e^{u/2}$ on $|u| \le 3$ to $2.7 \cdot 10^{-16}$ in absolute value, the level of rounding.

(3) Moving the basepoint is a Cauchy semigroup (Theorem 23.9), as in the proof of Theorem 16.8: in the variable r, the square root of the Thorin variable, the Thorin measure at basepoint $\frac{1}{2}+a$ is half the sum, over the zeros $\frac{1}{2}+\beta+i\gamma$ of the completed function, of the laws of $|\gamma + (a-\beta)C|$ with C standard Cauchy, and under GRH all the scales equal a. We check this against 1000 zeros of ζ to 10^{-7}.

(4) Inverting the Thorin law gives the universal tail found by Roynette and Yor [467] (Theorem 23.10). If L has nonnegative coefficients $\Lambda_L(n)$, then for every $\alpha > 1$ the Thorin variable H of Lemma 16.7 has mean $\frac{1}{2}(\log L)''(\alpha)$, infinite variance, and a Pareto tail of index $\frac{3}{2}$, $\mathbb{P}(H > t) \sim -(\log L)'''(\alpha)\, t^{-3/2}/(12\sqrt{\pi})$.

(5) Two classical papers fit the same picture (Section 23.7). Inverting the Laplace transform of a GGC with finitely many Thorin atoms gives a Gamma variable times a Curry Schoenberg B spline variable, the atomic case of a factorisation of James, Roynette and Yor [280] (Theorem 23.11); so the Hecke GGC of Definition 22.6 is a Gamma variable times a B spline variable with knots $1/(\ell + 1 - a_\ell(f))$, and its density is proportional to t^{g-1} times the Laplace transform of a B spline whose knots are the numbers $\ell + 1 - a_\ell(f)$, point counts when f is rational. Among densities whose square root vanishes on the walls of the alcove, the Good and Gaskins roughness penalty is minimised exactly by the Sato Tate law, by an inequality of Huber for $\mathrm{SU}(2)$ and the Rayleigh principle for the alcove in general [267, 44]; the penalty is diagonal in the basis of irreducible characters, symmetric powers for $\mathrm{SU}(2)$, with weights the Laplace eigenvalues $|\lambda + \rho_G|^2$ (Theorem 23.12).

Section 23.1 collects the classical inversion theorems in the form we use, with Feller's probabilistic proof of the Post Widder formula.

23.1 Inverse Laplace transforms for probabilists

Four inversion theorems do all the work. Lerch's theorem says that the forward transform loses nothing; Bromwich's formula inverts it along a vertical line; the Post Widder formula inverts it on the real axis; and Stieltjes inversion reads a Thorin measure off its boundary values.

THEOREM 23.1 (Lerch; uniqueness). *If two positive or signed measures on $[0, \infty)$ have Laplace transforms that converge absolutely for $s > c$ and agree on a set with an accumulation point $s_0 > c$, or on an arithmetic progression $c + kh$, $k \geq 1$, with $h > 0$, then the measures are equal. In particular two Dirichlet series $\sum a_n n^{-s}$ and $\sum b_n n^{-s}$ that converge absolutely and agree on such a set have $a_n = b_n$ for all n.*

The second statement is the first applied to the measures $\sum a_n \delta_{\log n}$ and $\sum b_n \delta_{\log n}$, since $\sum a_n n^{-s}$ is their Laplace transform ([578], Chapter II, Theorems 6.2 and 6.3). For the Langlands programme it means that an identity $L(s, \rho) = L(s, \pi)$ on any interval of the half plane of absolute convergence is an identity of every Euler factor.

THEOREM 23.2 (Bromwich). *Let $F(s) = \int_{\mathbb{R}} \Phi(u) e^{su}\, du$ converge absolutely on a vertical strip containing the line $\operatorname{Re} s = c$, and suppose $t \mapsto F(c + it)$ is integrable. Then*

$\Phi(u) = \frac{1}{2\pi}\int_{\mathbb{R}} F(c+it)e^{-(c+it)u}\,dt$ *at every point of continuity of* Φ.

This is Fourier inversion applied to $\Phi(u)e^{cu}$ ([578], Chapter VI, Theorem 5a). For a Dirichlet series the transform is not integrable on vertical lines, and the same inversion, taken as a symmetric principal value, is Perron's formula, to which we return in Section 23.6.

THEOREM 23.3 (Post Widder as a law of large numbers). *Let g be bounded and let $F(s) = \int_0^\infty e^{-sx}g(x)\,dx$ be its Laplace transform. For $k \geq 1$ and $t > 0$ let Y_{k+1} be a Gamma variable with shape $k+1$ and scale* 1. *Then*

$$\frac{(-1)^k}{k!}\Big(\frac{k}{t}\Big)^{k+1}F^{(k)}\Big(\frac{k}{t}\Big) = \mathbb{E}\,g\Big(\frac{t\,Y_{k+1}}{k}\Big) \;\longrightarrow\; g(t)$$

as $k \to \infty$ at every point t where g is continuous.

PROOF. Differentiating under the integral, $(-1)^kF^{(k)}(s) = \int_0^\infty x^k e^{-sx}g(x)\,dx$. With $s = k/t$ the function $x \mapsto (k/t)^{k+1}x^k e^{-kx/t}/k!$ is the density of tY_{k+1}/k, which gives the identity. By the weak law of large numbers $Y_{k+1}/k \to 1$ in probability, so $tY_{k+1}/k \to t$, and since g is bounded and continuous at t, $\mathbb{E}g(tY_{k+1}/k) \to g(t)$. □

Post found the formula and Widder extended it ([578], Chapter VII, Section 1); the reading as a law of large numbers is Feller's [185]. It shows what the inversion costs: the error is the Gamma smoothing of g at relative width $k^{-1/2}$, so Post Widder converges like $1/k$ at smooth points, and in practice one uses contour methods [535, 1]. Our computations use Talbot's contour.

THEOREM 23.4 (Stieltjes inversion of Thorin measures; [528, 10]). *Let φ be the Laplace transform of a GGC with drift a and Thorin measure τ. On any open interval of $(0,\infty)$ where τ has a continuous density,*

$$\tau(z) = \frac{1}{\pi}\lim_{\varepsilon\downarrow 0}\operatorname{Im}\frac{\varphi'}{\varphi}(-z+i\varepsilon).$$

PROOF. By the discussion after Definition 14.2, $-\varphi'/\varphi(s) = a + \int \tau(dt)/(s+t)$, holomorphic off $(-\infty, 0]$ by Lemma 13.3. For $s = -z + i\varepsilon$, $\operatorname{Im}(t - z + i\varepsilon)^{-1} = -\varepsilon/((t-z)^2+\varepsilon^2)$, and $\frac{1}{\pi}\varepsilon/((t-z)^2+\varepsilon^2)$ is an approximate identity, so $\operatorname{Im}(-\varphi'/\varphi)(-z+i\varepsilon) \to -\pi\tau(z)$. □

As noted after Lemma 16.7, the theorem applied to $\varphi(s) = e^{b\sqrt{s}}f(\alpha)/f(\alpha+\sqrt{s})$, with $\sqrt{s} = i\sqrt{z}$ on the upper side of the cut, gives the closed form (16.4). For $\varphi = 1/G_\pi$ the right side before the limit is the signed density u_ε of Chapter 19.

23.2 Inverting the sine squared transport

Lemma 16.7 goes from the Galois face to the transport face. Its arithmetic content is in the closed form: for an L function with $-L'/L(s) = \sum_n \Lambda_L(n) n^{-s}$ and for $\alpha > 1$, by (16.4),

$$g_\alpha(r) := 2\pi r\, U_\alpha(r^2) = \operatorname{Re} \frac{L'}{L}(\alpha + ir) - \frac{L'}{L}(\alpha) = \sum_{n \geq 2} \frac{\Lambda_L(n)}{n^\alpha} \big(1 - \cos(r \log n)\big), \tag{23.1}$$

which makes sense whether or not $\Lambda_L(n) \geq 0$ (the middle expression assumes the $\Lambda_L(n)$ real, as for a self dual L; in general it reads $\frac{1}{2}\big(\frac{L'}{L}(\alpha+ir) + \frac{L'}{L}(\alpha - ir)\big) - \frac{L'}{L}(\alpha)$); U_α is the density U_{ar} of Section 16.4.3, and g_α is unrelated to the function of that name in the van Dantzig pairs of Chapter 16. When $\Lambda_L \geq 0$, U_α is a Thorin density; when Λ_L is signed, as for $\eta(z)\eta(23z)$, it is a signed density, the transport of a signed Lévy measure. A mean value inverts it.

Theorem 23.5 (inverse of the sine squared transport; after Bohr [51]). *Let $\nu = \sum_j c_j \delta_{x_j}$ with distinct $x_j > 0$ and $\sum_j |c_j| < \infty$, and let $g(r) = \int (1 - \cos rx)\, \nu(dx)$. For $R > 0$ write $\mathcal{G}_R[h] = \int_0^\infty h(r) e^{-r^2/(2R^2)} dr \,/\, \int_0^\infty e^{-r^2/(2R^2)} dr$. Then*

$$\mathcal{G}_R\big[g(r)\cos(rx_k)\big] = \sum_j c_j \Big[e^{-x_k^2 R^2/2} - \tfrac{1}{2} e^{-(x_j - x_k)^2 R^2/2} - \tfrac{1}{2} e^{-(x_j + x_k)^2 R^2/2} \Big],$$

so that, with $\delta_k = \min(x_k, \inf_{j \neq k} |x_j - x_k|)$, assumed positive,

$$\Big| c_k + 2\mathcal{G}_R\big[g(r)\cos(rx_k)\big] \Big| \leq 2\Big(\sum_j |c_j|\Big) e^{-\delta_k^2 R^2/2}.$$

In particular $c_k = -2 \lim_{R\to\infty} \mathcal{G}_R[g(r)\cos(rx_k)]$, the same limit holds for the plain Bohr mean $\lim \frac{1}{R}\int_0^R$, and the transport $\nu \mapsto g$ is injective.

Proof. The Gaussian characteristic function gives $\mathcal{G}_R[\cos(ar)] = e^{-a^2R^2/2}$ for real a. Expand

$$g(r)\cos(rx_k) = \sum_j c_j \Big[\cos(rx_k) - \tfrac{1}{2}\cos\big(r(x_j - x_k)\big) - \tfrac{1}{2}\cos\big(r(x_j + x_k)\big) \Big];$$

the series converges uniformly, so $\mathcal{G}_R$ may be applied term by term, which gives the identity. The term $j = k$ contributes $c_k[e^{-x_k^2R^2/2} - \frac{1}{2} - \frac{1}{2}e^{-2x_k^2R^2}]$, so $c_k + 2\mathcal{G}_R[\cdot]$ is the sum of

$$c_k\big(2e^{-x_k^2R^2/2} - e^{-2x_k^2R^2}\big),$$

of size at most $2|c_k| e^{-\delta_k^2R^2/2}$, and of the terms $c_j(2e_0 - e_j^- - e_j^+)$ for $j \neq k$. Here $e_0 = e^{-x_k^2R^2/2}$ and $e_j^\pm = e^{-(x_j \pm x_k)^2R^2/2}$ lie in $[0, e^{-\delta_k^2R^2/2}]$ because x_k and $|x_j \pm x_k|$ are

at least δ_k, so each such term has size at most $2|c_j|e^{-\delta_k^2 R^2/2}$. For the plain mean, $\frac{1}{R}\int_0^R \cos(ar)\,dr \to \delta_{a,0}$ and uniform convergence gives the same limit. If $g = 0$ every c_k vanishes, which is injectivity. □

The limit is Bohr's formula for the Fourier coefficients of an almost periodic function, and injectivity is his uniqueness theorem ([51], Chapter I, Sections 3 and 4; for Dirichlet series, Chapter III, Section 3); the Gaussian window makes the error explicit.

COROLLARY 23.6 (Frobenius traces from the transported density). *Let $L(s) = \sum a_n n^{-s}$ have an Euler product with $-L'/L(s) = \sum \Lambda_L(n) n^{-s}$ absolutely convergent for $\operatorname{Re} s > 1$, as for Artin L functions and, by the Rankin Selberg method of Jacquet and Shalika [279] and the Cauchy Schwarz inequality, for cuspidal automorphic ones. For $\alpha > 1$ and $n \geq 2$,*

$$\Lambda_L(n) = -2n^{\alpha} \lim_{R\to\infty} \mathcal{G}_R\big[g_\alpha(r)\cos(r\log n)\big],$$

and the error at a finite R is at most

$$2n^{\alpha}\Big(\sum_m |\Lambda_L(m)| m^{-\alpha}\Big) e^{-\delta_n^2 R^2/2}, \qquad \delta_n = \log(1+1/n).$$

If $L = L(s,\rho)$ for an Artin representation unramified at p, then $\Lambda_L = \Lambda_\rho$ and $\operatorname{tr}\rho(\mathrm{Frob}_p^m) = \Lambda_\rho(p^m)/\log p$, so by Theorem 23.7 the transported density on any interval of r determines every Frobenius trace.

PROOF. Apply Theorem 23.5 to $\nu_\alpha = \sum \Lambda_L(n) n^{-\alpha}\delta_{\log n}$, with $x_n = \log n$, of weight 0 when n is not a prime power; every other atom lies at distance at least $\log(1+1/n)$ from $\log n$, and $\log(1+1/n) < \log n$. □

THEOREM 23.7 (reciprocity on any face). *Let L_1, L_2 be Euler products as in Corollary 23.6, for instance $L^S(s,\rho)$ and $L^S(s,\pi)$ with the Euler factors at a finite set S removed. Fix $\alpha > 1$. The following are equivalent.*

(i) $L_1(s) = L_2(s)$ for s in some interval of $(1,\infty)$.

(ii) The Galois faces agree: $\Lambda_{L_1}(n) = \Lambda_{L_2}(n)$ for all n, that is $\nu_\alpha^{(1)} = \nu_\alpha^{(2)}$.

(iii) The transport faces agree: $g_\alpha^{(1)} = g_\alpha^{(2)}$ on some interval $[0, r_0)$.

PROOF. (i) implies (ii) by Theorem 23.1 applied to $\log L_j = \sum \Lambda_{L_j}(n) n^{-s}/\log n$, and (ii) implies (iii) by (23.1). If (iii) holds, both $g_\alpha^{(j)}$ extend holomorphically to the strip $|\operatorname{Im} r| < \alpha - 1$, since $\sum |\Lambda_{L_j}(n)| n^{-\alpha}\cosh(y\log n) < \infty$ for $|y| < \alpha - 1$; they agree on an interval, hence on all of $\mathbb{R}$, and Theorem 23.5 gives (ii). Finally (ii) gives $\log L_1 = \log L_2$ on $\operatorname{Re} s > 1$. □

So a reciprocity law may be tested on either face, and a failure shows up on both. Figure 23.1 shows this for the running example of the book (Example 3.3). The q expansion of $\eta(z)\eta(23z)$ was computed from Euler's pentagonal series, and the Frobenius data of $x^3 - x - 1$ from the splitting of primes in its cubic field. They agree at all 17984 primes and all 18120 prime powers below $2 \cdot 10^5$, including $a_{23} = 1$. So the two transported densities at $\alpha = 1.5$ coincide exactly, while a single wrong coefficient, a_{59} changed from 2 to 0, is visible at every r except the points $r = 2\pi k/\log 59$.

The inverse arrow is the substantial one. We summed $L(s)$ and $L'(s)$ for the modular form from its first 30000 coefficients at $s = 2 + ir$, $0 \le r \le 2700$, formed $g_2(r)$ from (23.1), and applied Corollary 23.6 with $R = 450$. The recovered $\Lambda_\rho(n)$ agree with the Frobenius values for every $n \le 64$ to $8.6 \cdot 10^{-9}$; the error comes from the truncation of the window at $r = 6R$, not from the theorem. Dividing by $\log p$ returns the Frobenius labels: -1 at $p = 2, 3, 13, 29, 31, 41, 47$, where $x^3 - x - 1$ has no root, 0 at $p = 5, 7, 11, 17, 19, 37, 43, 53, 61$, the primes inert in $\mathbb{Q}(\sqrt{-23})$, 2 at $p = 59$, the first prime that splits completely, and 1 at the ramified prime 23. For $n = 59$ the bound of Corollary 23.6 is $1.1 \cdot 10^{-9}$, below the observed error $2.9 \cdot 10^{-9}$, and the observed errors exceed the bound at every $n \le 60$; but the bound allows for atoms at $n \pm 1$, while the nearest prime powers to 59 are 53 and 61, so the exact Gaussian mean is far more accurate than the bound, and the observed errors come from the window. With exact atoms and the same truncated window the error is again $8.6 \cdot 10^{-9}$, largest at $n = 60$; with the window cut at $r = 8R$ it falls to $1.7 \cdot 10^{-11}$. In this sense the transported density of the modular form *contains* the Galois representation, and Corollary 23.6 reads it off.

23.3 Bromwich inversion: from the Artin L function to the modular form

The automorphic face of a degree two L function is a function on the imaginary axis, and the Mellin transform makes it a two sided Laplace transform, as in Section 2.3 and (20.4). Write $\Gamma_{\mathbb{C}}(s) = 2(2\pi)^{-s}\Gamma(s) = \Gamma_{\mathbb{R}}(s)\Gamma_{\mathbb{R}}(s+1)$.

Proposition 23.8 (Mellin inversion of an Artin L function; after Hecke [250]). *Let ρ be an odd two dimensional Artin representation of conductor N, and assume that its completed L function $\Lambda(s,\rho) = N^{s/2}\Gamma_{\mathbb{C}}(s)L(s,\rho)$ is entire and bounded in vertical strips (Artin's conjecture for ρ). Put*

$$\Phi_\rho(u) = \frac{1}{2\pi}\int_{\mathbb{R}} \Lambda(\tfrac{1}{2} + it, \rho)e^{-itu}\,dt.$$

Then $\Lambda(\frac{1}{2} + t, \rho) = \int_{\mathbb{R}} \Phi_\rho(u)e^{tu}\,du$ for real t. If ρ is attached to a weight one newform f, then $\Phi_\rho(u) = 2f(ie^u/\sqrt{N})e^{u/2}$. If moreover $\Lambda(s,\rho) = \Lambda(1-s,\rho)$, then Φ_ρ is even,

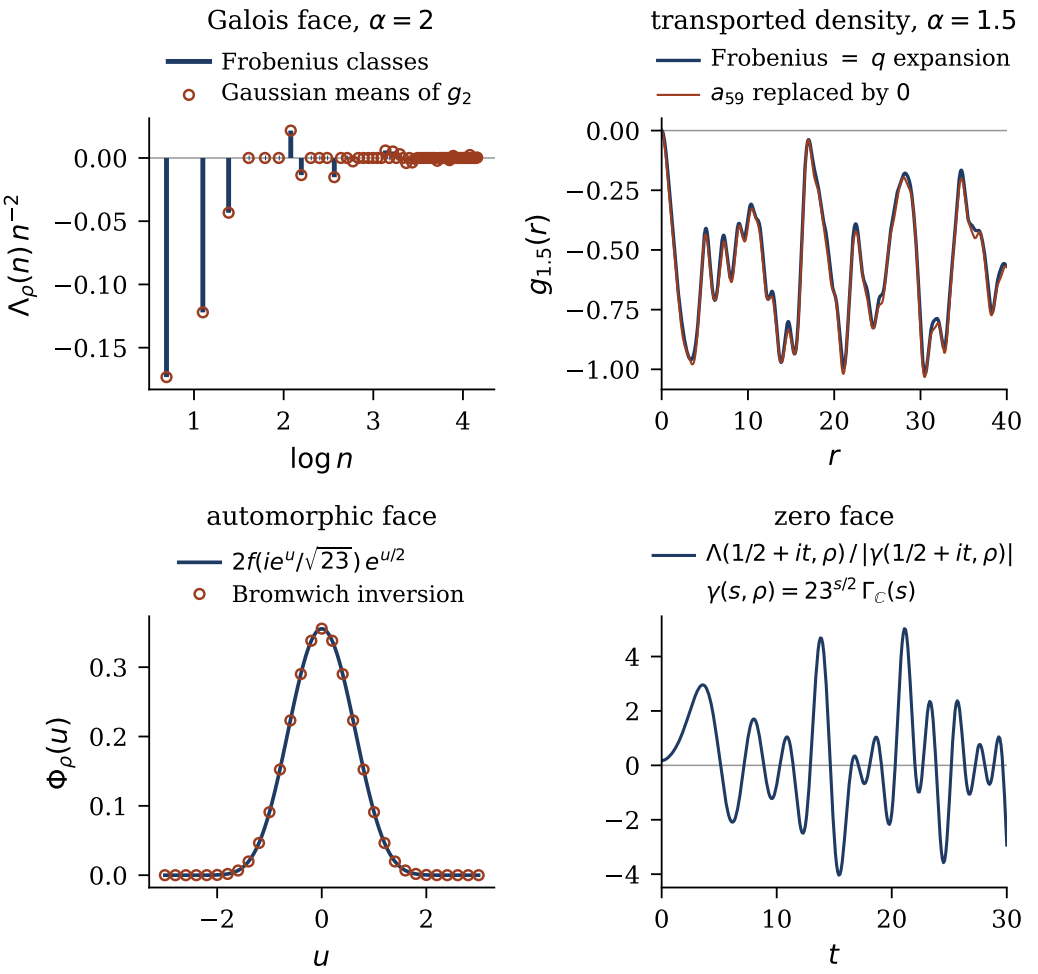


Figure 23.1: Four faces of $L(s,\rho) = L(s,\eta(z)\eta(23z))$, ρ the two dimensional representation of the Galois group of the splitting field of $x^3 - x - 1$. Top left: the Galois face at $\alpha = 2$, the atoms $\Lambda_\rho(n)n^{-2}$, $n \leq 64$, from Frobenius classes (bars) and as Gaussian means of the transported density g_2 computed from the modular form (circles), Corollary 23.6. Top right: the transport face, the transported density $g_{1.5}$ from Frobenius and from the q expansion (one curve, the two being identical), and with a_{59} replaced by 0. Bottom left: the automorphic face, the kernel Φ_ρ computed by Bromwich inversion of $\Lambda(\frac{1}{2}+it,\rho)$ (circles) against $2f(ie^u/\sqrt{23})e^{u/2}$ for $f = \eta(z)\eta(23z)$ (line), Proposition 23.8. Bottom right: for the zero face, the zeros themselves, as the sign changes of the real function $\Lambda(\frac{1}{2}+it,\rho)/|\gamma(\frac{1}{2}+it,\rho)|$, with $\gamma(s,\rho) = 23^{s/2}\Gamma_{\mathbb{C}}(s)$ the gamma factor of Section 14.2, of absolute value $|L(\frac{1}{2}+it,\rho)|$.

and if f is positive on the imaginary axis then $\Phi_\rho / \Lambda(\frac{1}{2}, \rho)$ is the density of a symmetric random variable.

PROOF. By the functional equation $\Lambda(s,\rho) = W(\rho)\Lambda(1-s,\bar\rho)$ and the Phragmén Lindelöf principle, boundedness in vertical strips gives polynomial growth of $L(\sigma+it,\rho)$ in $|t|$, uniformly for σ in compact sets, and Stirling's formula then gives exponential decay of $\Lambda(\sigma+it,\rho)$ in $|t|$. As in the proof of Lemma 14.12, $\frac{1}{2\pi}\int_{\mathbb{R}} \Lambda(\frac{1}{2}+x+it,\rho)e^{-(x+it)u}\,dt$ does not depend on x, so it equals $\Phi_\rho(u)$, it decays faster than any exponential as $u \to \pm\infty$, and $\int_{\mathbb{R}} \Phi_\rho(u)e^{zu}\,du = \Lambda(\frac{1}{2}+z,\rho)$ for every complex z, which at real $z = t$ is the first statement. If ρ corresponds to f, then $\Lambda(s,\rho)$ is twice the function $\Lambda(s)$ of Section 20.5 with $k = 1$, because $\Gamma_{\mathbb{C}}(s) = 2(2\pi)^{-s}\Gamma(s)$. Both sides of the second identity in (20.4) are entire in t, because $f(iy)$ decays exponentially as $y \to \infty$ and, by the Fricke involution, as $y \to 0$, so that identity and the uniqueness of the Fourier transform give $\Phi_\rho = 2\Phi_f$, with $\Phi_f(u) = f(ie^u/\sqrt{N})e^{u/2}$ the Pólya kernel of Chapter 20. The last statements are Hecke's criterion (Proposition 20.27) and positivity. □

The input to the integral is Galois data together with analytic continuation. An Artin L function is determined by the Frobenius classes, and its meromorphic continuation and functional equation follow from Brauer's induction theorem [84]; that $\Lambda(s,\rho_{23})$ is entire and bounded in strips is Hecke's theorem, since $L(s,\rho_{23}) = \zeta_K(s)/\zeta(s)$, $K = \mathbb{Q}[x]/(x^3-x-1)$, is the L function of the weight one form $\eta(z)\eta(23z)$ (Example 3.3). PARI computes $\Lambda(s,\rho_{23})$ as this quotient, using the functional equations of ζ_K and ζ. The output is a function on the imaginary axis, and reciprocity says that it is a modular form. We evaluated $\Lambda(\frac{1}{2}+it,\rho)$ for $0 \le t \le 60$ in steps of 0.005 (it is real there, and below 10^{-26} in absolute value beyond $t = 40$), integrated, and compared Φ_ρ with $2f(ie^u/\sqrt{23})e^{u/2}$ for $f = \eta(z)\eta(23z)$, summed from 4000 terms of its q expansion. On $|u| \le 3$ the two agree to $2.7 \cdot 10^{-16}$ in absolute value, the level of rounding (Figure 23.1, bottom left). The kernel is positive, with $\int \Phi_\rho = \Lambda(\frac{1}{2},\rho) = 0.5390$, so $\Phi_\rho/\Lambda(\frac{1}{2},\rho)$, the Pólya kernel Φ^{sd} of Lemma 14.12 for the weight one form, is the density of a symmetric random variable, its Pólya variable (Chapter 20 uses $\Phi_f = \frac{1}{2}\Phi_\rho$).

The proposition also says where the deep content of reciprocity sits. The inverse transform exists once $\Lambda(s,\rho)$ is entire and bounded in strips. That the result is modular, transforming correctly under all of $\Gamma_0(N)$ with its character and not merely under the Fricke involution, follows from Weil's converse theorem ([571], Satz 2) once the twists $\Lambda(s,\rho\otimes\chi)$ are also entire, bounded in strips and satisfy their functional equations; for two dimensional Artin representations this is Théorème 4.10 of Deligne and Serre [142]. For odd irreducible ρ these hypotheses are now known, by Hecke, Langlands and Tunnell [351, 553] when the projective image is solvable and in general through Serre's conjecture, proved by Khare and Wintenberger [311, 312] with work of Kisin [320] (Chapter 22).

23.4 Moving the basepoint: the Cauchy semigroup

The Galois face lives at basepoints $\alpha > 1$, the zeros at the centre. Theorem 16.8 gives a Thorin representation at every basepoint to the right of all zeros, and its proof writes the Thorin density as a sum of Poisson kernels over the zeros (see also Proposition 16.9 and Section 17.6); we now read that sum as a law. Let π be self dual and let $\Xi_\pi^{\mathrm{sd}}(w)$ be $\Lambda(\frac{1}{2}+w,\pi)$ divided by the power of w that removes any zero at $w=0$, as in Definition 14.5; for ζ use $\Xi_\zeta(w)=\xi(\frac{1}{2}+w)$ in its place below. Since $\Xi_\pi=(\Xi_\pi^{\mathrm{sd}})^2$, Theorem 16.8 holds for Ξ_π^{sd} with every Thorin density halved, as in the last sentence of Theorem 14.8, and below V_a and H_a denote this version. The function Ξ_π^{sd} is even and entire of order at most one (Lemma 14.6(iii) and (v)); its zeros $z=\beta+i\gamma$, counted with multiplicity, form a multiset $\mathcal{Z}$ closed under $z\mapsto -z$ and $z\mapsto \bar z$, and $|\beta|<\frac{1}{2}$. Let C denote a standard Cauchy variable, with density $1/(\pi(1+x^2))$.

Theorem 23.9 (zeros plus Cauchy noise; after [441, 442]). *Let $a>|\beta|$ for every $z=\beta+i\gamma$ in $\mathcal{Z}$; this holds unconditionally for $a\ge\frac{1}{2}$. In the variable $r>0$, the square root of the Thorin variable, the Thorin measure of the GGC H_a of Theorem 16.8, $\Xi_\pi^{\mathrm{sd}}(a)/\Xi_\pi^{\mathrm{sd}}(a+\sqrt{s})=\mathbb{E}e^{-sH_a}$, is*

$$V_a(r^2)\,2r\,dr=\frac{1}{2}\sum_{z\in\mathcal{Z}}\mathrm{Law}\big(|\gamma+(a-\beta)C|\big)(dr),\qquad r>0.$$

Under GRH this is $\sum_{\gamma>0}\mathrm{Law}(|\gamma+aC|)$ for every $a>0$, a superposition of folded Cauchy laws centred at the ordinates with common scale a. For $b>0$, V_{a+b} is obtained from V_a by adding independent Cauchy noise of scale b before folding, and on the Galois face, for $a>\frac{1}{2}$, this is multiplication by e^{-bx} of the whole Lévy measure of $\log\Xi_\pi^{\mathrm{sd}}$ at basepoint a, archimedean and polynomial factors included.

Proof. By the definition of V_a in Theorem 16.8, $V_a(r^2)\,2r=\frac{1}{\pi}\operatorname{Re}(\log\Xi_\pi^{\mathrm{sd}})'(a+ir)$, and by the Hadamard product $(\log\Xi_\pi^{\mathrm{sd}})'(w)$ is the sum of $(w-z)^{-1}$ over $z\in\mathcal{Z}$, taken in pairs $\pm z$. For $z=\beta+i\gamma$,

$$\frac{1}{\pi}\operatorname{Re}\frac{1}{a+ir-z}=\frac{1}{\pi}\frac{a-\beta}{(a-\beta)^2+(r-\gamma)^2},$$

the density at r of $\gamma+(a-\beta)C$, positive because $a>|\beta|$. The symmetries of $\mathcal{Z}$ make the sum even in r, so on $r>0$ it is half the sum of the folded densities, as stated. Under GRH, $\mathcal{Z}=\{\pm i\gamma\}$, $\beta=0$, and the two zeros $\pm i\gamma$ give the two halves of $\mathrm{Law}(|\gamma+aC|)$. Cauchy laws are stable, $(a-\beta)C+bC'$ having the law of $(a+b-\beta)C$ for independent C,C', which is the semigroup. On the Galois face, for $a>\frac{1}{2}$ the Lévy measure of $\log\Xi_\pi^{\mathrm{sd}}$ at basepoint a is $e^{-ax}\mu(dx)/x$ with μ independent of a, as in (16.3) with ν now signed, so moving to $a+b$ multiplies it by e^{-bx}; and $e^{-b|x|}$ is the Fourier transform of the Cauchy density of scale b. □

The Poisson kernel form of V_a is in the proof of Theorem 16.8 and in Proposition 16.9, for ξ in [442] (Theorem 18 and Remark 21), and Chapter 17 writes the density contributed by a pair of zeros off the line as a sum of two Poisson kernels (Section 17.6); the semigroup in the basepoint is the backward Cauchy flow of [441] (Section 8 and Theorem 9.2). The theorem states these facts through folded Cauchy variables and for every self dual π; for ξ, its distribution function is the sum of arctangents of Theorem 8.2 of [441]. We check it below against 1000 zeros of ζ. Chapter 19 smooths the atoms at the squared ordinates by Lorentzians of width ε, through Stieltjes inversion; here the smoothing is in the variable r, its scale is $a-\beta$, and it comes from moving the basepoint.

The theorem gives a probabilistic picture of the passage from the Galois face to the zeros. At $a \geq \frac{1}{2}$, where the Thorin measure is positive unconditionally, each zero is seen through Cauchy noise of scale between $a-\frac{1}{2}$ and $a+\frac{1}{2}$. As a decreases the noise shrinks, and GRH is the statement that it can be shrunk to zero for every zero at once; a zero with $\beta > 0$ would make the scale $a-\beta$ negative for $a < \beta$, a negative Cauchy density. The proof of Theorem 16.8(iv) and the remark after it use this sign change, and the criterion of Hinkkanen and Lagarias [261, 339], that the Riemann hypothesis holds if and only if $\operatorname{Re}\xi'/\xi(s) > 0$ for $\operatorname{Re} s > \frac{1}{2}$, is the same statement about scales: its proof ([339], Section 2) adds the same Poisson kernels term by term.

For ζ we compared $\frac{1}{\pi}\operatorname{Re}(\Xi_\zeta'/\Xi_\zeta)(a+ir)$ on $0 < r \leq 50$, computed from ζ, Γ and the polynomial factor, with the Cauchy superposition over the first 1000 zeros plus a smooth tail beyond $\gamma_{1000} = 1419.42$. The maximal differences are $9.2 \cdot 10^{-8}$, $3.7 \cdot 10^{-8}$ and $9.2 \cdot 10^{-9}$ at $a = 0.5$, 0.2 and 0.05, against peak values 0.72, 1.63 and 6.37 (Figure 23.2, left).

23.5 Inverting the Thorin law: a universal tail

The Thorin variable H of Lemma 16.7 is the second member of a Wald couple of Roynette and Yor [467], who found the tail of its density. Inverting the Laplace transform recovers their result, and the inversion has a universal feature.

THEOREM 23.10 (the tail of the Thorin variable; Roynette and Yor [467]). *Let $\Lambda_f(n) \geq 0$, not all zero, and let $f(s) = \exp\bigl(\sum_{n\geq 2}\Lambda_f(n)n^{-s}/\log n\bigr)$ converge absolutely for $\operatorname{Re} s > \alpha_0$. For $\alpha > \alpha_0$ let H be the GGC of Lemma 16.7 for f at the basepoint α, with $\nu_\alpha = \sum_n \Lambda_f(n)n^{-\alpha}\delta_{\log n}$, so that*

$$\mathbb{E}e^{-sH} = e^{b\sqrt{s}}\,\frac{f(\alpha)}{f(\alpha+\sqrt{s})}, \qquad b = \frac{f'}{f}(\alpha) < 0.$$

Then

$$\mathbb{E}H = \tfrac{1}{2}(\log f)''(\alpha), \qquad \mathbb{E}H^2 = \infty,$$

and the density p of H satisfies

$$p(t) \sim A\, t^{-5/2}, \qquad \mathbb{P}(H > t) \sim \tfrac{2}{3} A\, t^{-3/2},$$

where

$$A = \frac{-(\log f)'''(\alpha)}{8\sqrt{\pi}} = \frac{1}{8\sqrt{\pi}} \sum_n \frac{\Lambda_f(n)(\log n)^2}{n^\alpha}.$$

This is Proposition I.3.1 and Remarque I.3.2 of [467] for $c(dx) = \nu_\alpha(dx)/x$, their H being $2H$ here, with the density given by part (1) of that proposition; for ζ the variable and its mean are their Théorème V.1.1. They reach the constant through a Tauberian theorem; the proof below uses Doetsch's transfer theorem, which also gives the next terms.

PROOF. Write $\varphi(s) = \mathbb{E}e^{-sH}$ and $w = \sqrt{s}$. Since $\log f$ is holomorphic near α, Taylor's formula gives $\log f(\alpha + w) = \log f(\alpha) + bw + \frac{1}{2} f_2 w^2 + \frac{1}{6} f_3 w^3 + O(w^4)$ with $f_k = (\log f)^{(k)}(\alpha)$, so

$$\log \varphi(s) = -\tfrac{1}{2} f_2\, s - \tfrac{1}{6} f_3\, s^{3/2} + O(s^2),$$

uniformly as $s \to 0$ in the slit plane. Differentiating the series of $\log f$ term by term gives $f_2 = \sum \Lambda_f(n) \log n\, n^{-\alpha}$ and $f_3 = -\sum \Lambda_f(n)(\log n)^2 n^{-\alpha} < 0$. The mean is the coefficient of $-s$. The term $s^{3/2}$ is not twice differentiable at 0, so $\mathbb{E}H^2 = \infty$. The function φ is holomorphic on $\mathbb{C} \setminus (-\infty, 0]$, because $\operatorname{Re}\sqrt{s} > 0$ there and f has no zeros on $\operatorname{Re} s > \alpha_0$, and $\varphi(s) \to 0$ as $|s| \to \infty$, uniformly in every sector $|\arg s| \le \pi - \delta$, since $f(\alpha + \sqrt{s}) \to 1$ and $b < 0$. Since $\log f(\alpha + w)$ is analytic in w, the expansion of φ at 0 is a power series in $\sqrt{s}$, so after $s^{3/2}$ the next nonintegral power is $s^{5/2}$. Doetsch's transfer theorem for Laplace transforms with algebraic singularities [159], whose hypotheses are these two properties, turns each term cs^λ, $\lambda \notin \{0, 1, 2, \dots\}$, of the expansion at 0 into a term $ct^{-\lambda-1}/\Gamma(-\lambda)$ of the density at infinity. With $c = -f_3/6$, $\lambda = \frac{3}{2}$ and $\Gamma(-\frac{3}{2}) = \frac{4}{3}\sqrt{\pi}$ this is A. Integrating gives the tail of the distribution function. □

The index $\frac{3}{2}$ does not depend on f or on α. By Step 3 of the proof of Lemma 16.7, H has the Lévy density

$$\frac{k(y)}{y} = \frac{1}{2\sqrt{\pi}\, y^{3/2}} \int_0^\infty \left(1 - e^{-x^2/(4y)}\right) \nu_\alpha(dx) \sim A y^{-5/2}, \qquad y \to \infty,$$

since $1 - e^{-x^2/(4y)} \sim x^2/(4y)$: the Lévy measure already has the tail of the theorem, the power coming from the stable law of index $\frac{1}{2}$ in Step 1, and the theorem says that the law inherits it. What depends on the arithmetic is the constant, $(8\sqrt{\pi})^{-1}$ times the second moment $\int x^2 \nu_\alpha(dx)$ of the Galois face. Since

$8\sqrt{\pi}A(\alpha) = \sum_n \Lambda_f(n)(\log n)^2 n^{-\alpha}$ is a Dirichlet series, the function $\alpha \mapsto A(\alpha)$ on an interval determines Λ_f, by Theorem 23.1.

The variable H_a of Theorem 23.9, for $\Xi = \Xi_\pi^{\mathrm{sd}}$ or Ξ_ζ, is not compensated: its transform $\Xi(a)/\Xi(a+\sqrt{s})$ contains the factor $e^{-b_a\sqrt{s}}$ of a positive stable law of index $\frac{1}{2}$, with $b_a = (\Xi'/\Xi)(a) > 0$. So $\mathbb{P}(H_a > t) \sim b_a/\sqrt{\pi t}$ for every a to which Theorem 16.8(i) applies (every $a \geq \frac{1}{2}$, and every $a > 0$ under GRH), while under GRH the limit at $a = 0$, the explosion time of Theorem 21.11, has an exponential tail. With that factor removed, $e^{b_a\sqrt{s}}\,\Xi(a)/\Xi(a+\sqrt{s})$ is a GGC transform exactly when the density $V_a - b_a/(2\pi\sqrt{z})$ is nonnegative, which for ξ holds under the Riemann hypothesis, is known unconditionally for $\frac{1}{2} + a \geq 1$, and is open below (Section 16.4.3, Open Problem 20.2). Where it holds, the proof of Theorem 23.10 applies with $f = \Xi$ at the basepoint a, the decay of the transform in sectors now coming from Stirling's formula: the compensated variable has the density tail $A_a t^{-5/2}$, $A_a = -(\log\Xi)'''(a)/(8\sqrt{\pi})$, which for ξ is positive whenever $\frac{1}{2} + a < 24$.

For ζ at $\alpha = 2$ we inverted φ by Talbot's method on $0.0005 \leq t \leq 400$ (Figure 23.2, right). Here $b = -0.569961$, $\mathbb{E}H = \frac{1}{2}(\log\zeta)''(2) = 0.442241$ and $A = 0.137531$. The computed density has mass 0.99999 on $[0.0005, 400]$, the rest being $\mathbb{P}(H > 400) \approx \frac{2}{3}A \cdot 400^{-3/2} = 1.15 \cdot 10^{-5}$, and mean 0.4285 there, which with 0.0137 from the tail $At^{-5/2}$ beyond 400 gives $\mathbb{E}H = 0.4422$. The ratios $p(t)t^{5/2}/A$ are 0.991, 0.996, 0.998, 0.999 at $t \approx 50, 100, 200, 400$, approaching 1 like $1/t$. The Post Widder approximations of Theorem 23.3 at $t = 0.05$ are 4.025, 4.455 and 4.551 for $k = 4$, 16 and 32, against the Talbot value 4.657, an error decreasing roughly like $1/k$.

23.6 Perron's formula as Bromwich inversion

The oldest passage between the two descriptions is classical, and we record it in the language of this chapter; Chapter 17 uses the same contour shift in (17.2), for the squarefree ideals of a number field. The undamped Galois face of ζ is the measure $\nu_0 = \sum_n \Lambda(n)\delta_{\log n}$, whose distribution function is Chebyshev's $\psi(e^u) = \sum_{n \leq e^u} \Lambda(n)$, and $-\zeta'/\zeta(s) = \int e^{-su}\, d\psi(e^u)$ is its Laplace Stieltjes transform. Bromwich inversion on a line $\operatorname{Re} s = c > 1$, taken as a symmetric principal value since x^s/s is not integrable on the line, is Perron's formula [435],

$$\psi(x) = \frac{1}{2\pi i}\int_{c-i\infty}^{c+i\infty} \Big(-\frac{\zeta'}{\zeta}(s)\Big)\frac{x^s}{s}\, ds, \qquad x > 1 \text{ not a prime power},$$

in which only Galois data enter. Moving the contour to the left across the critical strip, the integrand meets the pole at $s = 1$, the zeros ρ, the pole at $s = 0$, which gives $-\log 2\pi$, and the trivial zeros, which give $-\frac{1}{2}\log(1 - x^{-2})$, and the residues give the explicit formula ([125], Chapter 17), the sum over ρ being the limit of the

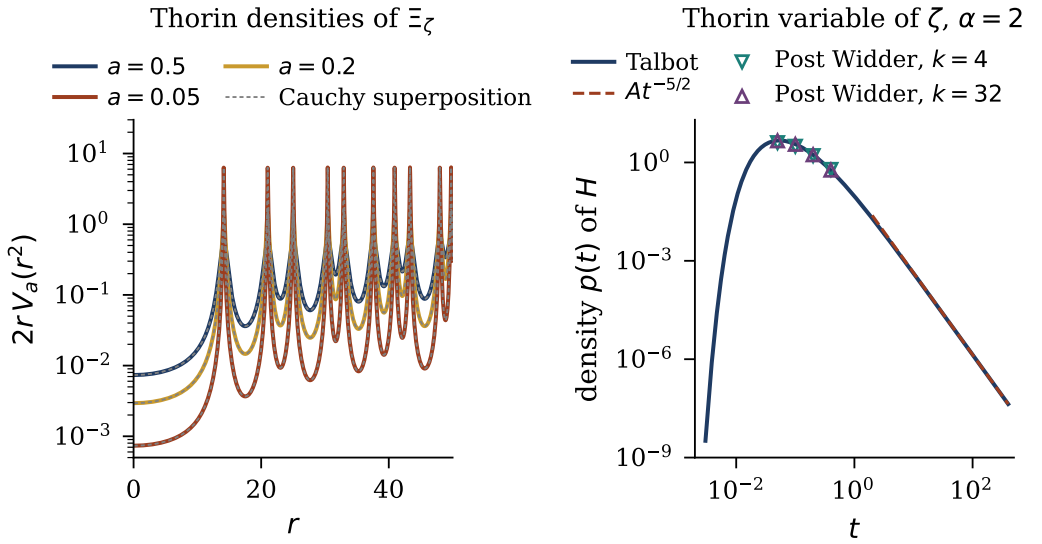


Figure 23.2: Left: Theorem 23.9 for ζ, the densities $2rV_a(r^2)$ of Ξ_ζ (Thorin densities for $a \geq \frac{1}{2}$, and under the Riemann hypothesis below) at basepoints $a = 0.5$, 0.2, 0.05 (those of ξ at $\frac{1}{2} + a$), with the Cauchy superpositions over the zeros drawn on top in grey dashed lines; the two agree to 10^{-7}. Right: Theorem 23.10, the density of the Thorin variable of ζ at $\alpha = 2$ by Talbot inversion, its tail $At^{-5/2}$, and the Post Widder approximations with $k = 4$ and 32, on logarithmic axes.

sums over $|\gamma| < T$ as $T \to \infty$:

$$\psi(x) = x - \sum_{\rho} \frac{x^{\rho}}{\rho} - \log 2\pi - \tfrac{1}{2}\log(1 - x^{-2}).$$

The contour shift is the passage: on $\operatorname{Re} s > 1$ the integrand is an Euler product, and to the left of the critical strip it is controlled by the functional equation, which for ζ is the modularity of the theta function. The Cauchy semigroup of Section 23.4 reaches the zeros by moving a basepoint instead of a contour, with positive measures wherever Theorem 16.8(i) applies. The explicit formula is also used in Section 5.4 and Chapter 20.

23.7 Splines, Pólya frequency functions and roughness penalties

Two classical papers sit exactly on the arrows of this chapter. Curry and Schoenberg [121], in the full version of their 1947 announcement, developed the fundamental splines, now called B splines, proved that they are probability densities, the volumes of the sections of a simplex, and studied their limits, which are Schoenberg's Pólya frequency functions [488], totally positive of every order. Good and Gaskins [222] estimated a probability density f by maximising the log likelihood minus a roughness penalty $\mathcal{R}(f) = \omega_1 \int (\sqrt{f})'^2 + \omega_2 \int (\sqrt{f})''^2$, whose first term is $\frac{\omega_1}{4}$ times the Fisher information for location, $\int f'^2/f = 4\int (\sqrt{f})'^2$; they read $e^{-\mathcal{R}}$ as a prior [222, 223], and their second paper used the method to hunt for bumps. De Montricher, Tapia and Thompson [144] showed (Proposition 3.5) that with the first penalty alone the estimate is an exponential spline with knots at the observations. So the two papers meet through splines, and both meet the inverse Laplace transforms of this chapter. Inverting a GGC with finitely many Thorin atoms gives a Gamma variable times a B spline variable, Schoenberg's theorem turns GRH into a Pólya frequency statement (Chapter 16), and among densities vanishing on the walls the Good and Gaskins penalty singles out the Sato Tate law.

23.7.1 Atomic Thorin laws invert to B splines

For $k \geq 1$ and knots $u_0 \leq \cdots \leq u_k$ with $u_0 < u_k$ the Curry Schoenberg B spline is $M(x; u_0, \ldots, u_k) = k\,[u_0, \ldots, u_k](\cdot - x)_+^{k-1}$, the divided difference in the first variable, read as a confluent one at repeated knots. It is a probability density on $[u_0, u_k]$, and it is the density of $\sum_i u_i P_i$ with $(P_0, \ldots, P_k)$ uniform on the simplex, that is Dirichlet with all parameters 1: the spline is the shadow of a simplex [121]. Both facts follow from the classical Hermite Genocchi formula $[u_0, \ldots, u_k]\phi = \frac{1}{k!}\mathbb{E}\phi^{(k)}(\sum_i u_i P_i)$ [132].

THEOREM 23.11 (atomic Thorin laws are Gamma times B spline; after [121, 280]). *Let $k \geq 1$, $0 < c_0 \leq \cdots \leq c_k$ with $c_0 < c_k$, and $H = \sum_i E_i/c_i$ with independent standard exponentials E_i, the GGC with Thorin measure $\sum_i \delta_{c_i}$.*

(a) H has the law of $Y_{k+1}D$, with Y_{k+1} Gamma of shape $k+1$ independent of D, whose density is the B spline with knots $1/c_0, \dots, 1/c_k$.

(b) The inverse Laplace transform of $\prod_i (1 + s/c_i)^{-1}$ is

$$p_H(t) = \Big(\prod_i c_i\Big)(-1)^k [c_0, \dots, c_k] e^{-t\cdot}$$
$$= \Big(\prod_i c_i\Big) \frac{t^k}{k!} \int_{c_0}^{c_k} e^{-tx} M(x; c_0, \dots, c_k)\, dx,$$

that is $(\prod_i c_i) t^k/k!$ times the Laplace transform of the B spline with knots $c_0, \dots, c_k$. Repeated atoms give repeated knots, with the divided difference read as a confluent one.

PROOF. (a) With $Y_{k+1} = \sum_i E_i$ and $P_i = E_i/Y_{k+1}$, Y_{k+1} is Gamma of shape $k+1$, $(P_0, \dots, P_k)$ is Dirichlet with all parameters 1, and the two are independent by the beta gamma algebra ([280], Section 1.4). Then $H = Y_{k+1}D$ with $D = \sum_i P_i/c_i$, which has the B spline density with knots $1/c_i$. (b) For distinct atoms, partial fractions give $p_H(t) = (\prod_j c_j) \sum_i e^{-c_i t} / \prod_{j \neq i} (c_j - c_i)$, which is the divided difference written; repeated atoms follow by continuity in the c_i. Hermite Genocchi with $\phi(x) = e^{-tx}$, $\phi^{(k)} = (-t)^k e^{-tx}$, gives the second form. □

Part (a) is the atomic case of the factorisation of James, Roynette and Yor [280] (Section 1.4, equation (33)): a GGC whose Thorin measure has finite mass m is a Gamma variable of shape m times an independent Dirichlet mean, by the beta gamma algebra. In part (b) the first form is the classical density of a sum of independent exponentials, written as a divided difference, and the second is its Peano kernel form [132]. So inverting the Laplace transform of an atomic Thorin law is a divided difference, and a divided difference is an expectation over a simplex. When the Thorin measure is the image under $1/Q$ of a probability measure μ restricted to $\{Q > 0\}$, as for the Frobenius functionals of Chapter 20, the factorisation can also be written with a standard exponential variable and the Dirichlet mean $\int Q\, dP$ of a Dirichlet process P with base measure μ, and Chapter 24 identifies P with the limit of the normalised labelled factorisation of a random integer (Theorem 24.3).

For the Hecke GGC $H_N(\ell)$ of Definition 22.6, whose g atoms $\ell + 1 - a_\ell(f)$ have product $|J_0(N)(\mathbb{F}_\ell)|$ (Chapter 22), part (b) gives

$$p_{H_N(\ell)}(t) = |J_0(N)(\mathbb{F}_\ell)| \, \frac{t^{g-1}}{(g-1)!} \int e^{-tx} M\big(x; \{\ell + 1 - a_\ell(f)\}_f\big)\, dx :$$

the density of the Hecke variable is t^{g-1} times the Laplace transform of a B spline whose knots are the atoms. The atom of a rational f, attached to an elliptic curve E_f, is the point count $|E_f(\mathbb{F}_\ell)|$, but only the product of all the atoms is always a point count. Hecke atoms can repeat, for example at $(N, \ell) = (37, 7)$ and $(113, 2)$, and then the knots repeat; at $(37, 7)$ both atoms are 9, the B spline degenerates to the point mass at 9, and $H_{37}(7)$ is a Gamma variable of shape 2 and rate 9. For $N = 67$, $\ell = 2$, the five atoms are 1, 2.382, 3.382, 4.618, 5.618 with product $209 = |J_0(67)(\mathbb{F}_2)|$. Computed exactly, the B spline formula agrees with the partial fraction density to within $6 \cdot 10^{-15}$ on $[0, 4]$, and a histogram of 400000 samples of $Y_5 D$ agrees with both within sampling error (Figure 23.3, right).

23.7.2 Pólya frequency functions and GRH

Schoenberg's theorem (Theorem 13.1) characterises the Pólya frequency functions, totally positive of every order, by their bilateral Laplace transforms, the reciprocals of functions of the Laguerre Pólya class, and Curry and Schoenberg showed that these functions arise as limits of their B splines [121]. Through Schoenberg's theorem, Theorem 16.6 and the remark after its proof show that GRH for π holds if and only if $t \mapsto \Xi_\pi(0)/\Xi_\pi(t)$ is the characteristic function of a symmetric Pólya frequency density, that of $V_\pi = \sum_k L_k/\gamma_k$, with independent standard Laplace variables L_k, which has the law of $\sqrt{2Z_\pi}\, N$, where Z_π is the explosion time of Theorem 21.11 and N is standard normal and independent of Z_π (for ξ, [467, 442, 330]). Under GRH the density of V_π is therefore a limit of B spline densities, and its truncation to the first K zeros is a difference $\sum_{k \le K} E_k/\gamma_k - \sum_{k \le K} E'_k/\gamma_k$ of two independent hypoexponential variables, each a Gamma variable times a B spline variable by Theorem 23.11. The bilateral Laplace transform of V_π is $\mathbb{E}e^{sV_\pi} = \mathbb{E}e^{s^2 Z_\pi}$, the transform of the explosion time continued to negative argument.

23.7.3 The Sato Tate law minimises the Good and Gaskins penalty

Let G be a compact connected simply connected Lie group, such as $\mathrm{SU}(2)$ or $\mathrm{USp}(2n)$, with alcove $\mathcal{A}$, positive roots α and half sum ρ_G, and give the Lie algebra of a maximal torus the invariant inner product in which, for $\mathrm{SU}(2)$ and $\mathrm{USp}(2n)$, the eigenvalue angles are orthonormal coordinates. By Weyl's integration formula (Theorem 6.1) the image of Haar measure on conjugacy classes, the Sato Tate law of Chapter 4 for $\mathrm{SU}(2)$ and of Chapter 29 for $\mathrm{USp}(4)$, has density $f_G = c|\Delta_G|^2$ on $\mathcal{A}$, with $\Delta_G = \prod_{\alpha > 0}(e^{i\alpha/2} - e^{-i\alpha/2})$. By the Weyl character formula $\Delta_G \chi_\lambda = \sum_w \mathrm{sgn}(w) e^{iw(\lambda + \rho_G)}$, an eigenfunction of the Laplacian with eigenvalue $|\lambda + \rho_G|^2$ that vanishes on the walls of $\mathcal{A}$; for $\lambda = 0$ it has constant sign inside $\mathcal{A}$.

THEOREM 23.12 (Sato Tate is the Good and Gaskins ground state; this book, after [267, 44]). *Among densities f on $\mathcal{A}$ such that $\sqrt{f}$ vanishes on the boundary $\partial\mathcal{A}$ and*

$\int_{\mathcal{A}} |\nabla\sqrt{f}|^2 < \infty$, the Good and Gaskins roughness $\int_{\mathcal{A}} |\nabla\sqrt{f}|^2$, one quarter of the trace of the Fisher information for location, is at least $|\rho_G|^2$, with equality exactly for the Sato Tate law f_G. For $G = \mathrm{SU}(2)$, with $\theta \in [0,\pi]$ and $\sqrt{f} = \sqrt{2/\pi}\sum_j b_j \sin j\theta$, $\sum_j b_j^2 = 1$,

$$\int_0^\pi \big((\sqrt{f})'\big)^2 d\theta = \sum_j j^2 b_j^2, \qquad \frac{\sin j\theta}{\sin\theta} = \operatorname{tr}\operatorname{Sym}^{j-1}\begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta}\end{pmatrix},$$

so the roughness is the Laplace eigenvalue energy $\sum_\lambda |\lambda+\rho_G|^2 b_\lambda^2$, which is $|\rho_G|^2$ plus the Casimir energy, in the basis of symmetric power characters, and the Sato Tate law is $b = e_1$. Given angles $\theta_1,\dots,\theta_n$, put $\hat m_j = \frac{1}{n}\sum_{i=1}^n \operatorname{tr}\operatorname{Sym}^{j-1}(\theta_i)$, the trace being that of the matrix above. Near the Sato Tate law, the maximiser of $\frac{1}{n}\sum_i \log f(\theta_i) - \omega\int_0^\pi(\sqrt{f})'^2$ over $\sqrt{f} = \sqrt{2/\pi}\sum_{j\le J} b_j \sin j\theta$ satisfies

$$b_j = \frac{\hat m_j}{2+\omega(j^2-1)} + O\Big(\max_{2\le l\le 2J-1} \hat m_l^2\Big), \qquad 2 \le j \le J.$$

PROOF. The functions $\Delta_G\chi_\lambda$ restricted to $\mathcal{A}$, with λ dominant, are a complete orthogonal system of Dirichlet eigenfunctions of the Laplacian on $\mathcal{A}$, by the Weyl character formula and the Peter Weyl theorem (Theorem A.3), with eigenvalues $|\lambda+\rho_G|^2$; the smallest is $|\rho_G|^2$, at $\lambda = 0$, and it is simple. The Rayleigh quotient of $\sqrt{f}$, with $\int f = 1$, is $\int|\nabla\sqrt{f}|^2 \ge |\rho_G|^2$, with equality only for $\sqrt{f} = c|\Delta_G|$, which is $f = f_G$. For $\mathrm{SU}(2)$, $\mathcal{A} = [0,\pi]$, $\rho_G = 1$, $\Delta_G\chi_{j-1} = 2i\sin j\theta$, and $\int(\sum_j b_j\sqrt{2/\pi}\sin j\theta)'^2 = \sum_j j^2 b_j^2$. For the maximiser write $b = (\sqrt{1-|\epsilon|^2}, \epsilon_2, \dots, \epsilon_J)$ and $\sin j\theta/\sin\theta = U_{j-1}(\cos\theta)$. Then $\frac{1}{n}\sum_i \log f(\theta_i)$ is a constant plus $2\sum_j \epsilon_j\hat m_j - |\epsilon|^2 - \sum_{j,k} Q_{jk}\epsilon_j\epsilon_k + O(|\epsilon|^3)$, where $Q_{jk} = \frac{1}{n}\sum_i U_{j-1}(\cos\theta_i)U_{k-1}(\cos\theta_i)$ is the sum of the $\hat m_l$ over $l = |j-k|+1, |j-k|+3, \dots, j+k-1$ by the Clebsch Gordan rule, with $\hat m_1 = 1$, so that $Q_{jk} = \delta_{jk} + O(\max_l|\hat m_l|)$. The penalty is $\omega(1+\sum_j(j^2-1)\epsilon_j^2)$. At the stationary point $(2+\omega(j^2-1))\epsilon_j = \hat m_j + O(|\epsilon|\max_l|\hat m_l| + |\epsilon|^2)$, and $\epsilon = O(\max_l|\hat m_l|)$ gives the formula: the cross terms $\epsilon_j\epsilon_k$ have coefficients $O(\max_l|\hat m_l|)$, so they change b only at second order. □

For $\mathrm{SU}(2)$ the inequality is Huber's theorem that among laws on an interval the sine squared arch has least Fisher information [267]; in general it is the Rayleigh principle for the Dirichlet Laplacian of the alcove, whose spectrum Bérard computed [44] (Proposition 9).

For $\mathrm{USp}(4)$, with $\sqrt{f} \propto (\cos\theta_1 - \cos\theta_2)\sin\theta_1\sin\theta_2 = \frac{1}{2}(\sin 2\theta_1\sin\theta_2 - \sin\theta_1\sin 2\theta_2)$ on $0 < \theta_2 < \theta_1 < \pi$, the eigenvalue is $5 = |\rho_G|^2$, $\rho_G = (2,1)$, and the normalising constant is $16/\pi^2$, twice the constant of Proposition 29.4, which integrates over the square $[0,\pi]^2$. In the Bayesian reading of Good and Gaskins, with the prior $e^{-n\omega\int(\sqrt{f})'^2}$ for n angles, the Sato Tate law is, formally, the mode

of the prior among densities whose square root vanishes on the walls, and the posterior mode is the maximiser of the theorem. Since the fitted value of $\hat{m}_j$, the mean of $\operatorname{tr}\operatorname{Sym}^{j-1}$, is about $2b_j$, the posterior mode shrinks the empirical moment $\hat{m}_j$ by the factor $(1+\omega(j^2-1)/2)^{-1}$: the larger the representation, the stronger the shrinkage. The $\hat{m}_j$ are the Weyl averages $W_\rho(n)$, $\rho=\operatorname{Sym}^{j-1}$, of Chapter 29, and the roughness prior is a nonparametric alternative of the kind discussed there, where the Bayesian identification compares finitely many candidate groups.

Figure 23.3 fits the Good and Gaskins estimate with twelve sine terms to the Frobenius angles of two curves. For $11a1$, with the 664578 primes below 10^7 other than 11, the moments $\hat{m}_2,\dots,\hat{m}_7$ are of size $4\cdot10^{-4}$, the estimate is indistinguishable from the Sato Tate law (roughness 1.0001 at $\omega=0.01$), and the exact maximiser agrees with the first order formula to within 0.7 percent for $j\le6$ and $\omega=0.01,0.1,1$. For the curve $E_2\colon y^2=x^3-x$ of Chapter 1 (conductor 32, complex multiplication by $\mathbb{Z}[i]$), with the 78497 odd primes below 10^6, the angle law is μ_{CM}, $\frac12\delta_{\pi/2}$ plus half the uniform law, so $\mathbb{E}\operatorname{tr}\operatorname{Sym}^{2m}=\frac12((-1)^m+1)$ (Proposition 4.10). The data give $\hat{m}_3=-0.0033$ and $\hat{m}_5=0.998$, against 0 and 1, and the estimate at $\omega=0.01$ has $b_5=0.334$ and a spike at $\pi/2$; at $\omega=1$ the prior pulls it back to $b_5=0.035$, close to the Sato Tate shape. Here the first order formula is only qualitative, because this curve is far from Sato Tate.

23.7.4 Open problems

OPEN PROBLEM 23.1 (Stable recovery of Frobenius from a window). By Theorem 23.7(iii) the transported density g_α on any interval $[0,r_0)$ determines every $\Lambda_L(n)$, but only through analytic continuation, which is unstable. Suppose g_α is known on $[0,r_1]$ to within η. Corollary 23.6, with the Gaussian mean truncated at r_1, recovers $\Lambda_L(n)$ to within ε once η is of order $\varepsilon n^{-\alpha}$ and r_1 of order $n\log(n^\alpha/\varepsilon)$. Is an interval of length of order n, the reciprocal of the gap $\log(1+1/n)$ between the frequencies $\log n$ and $\log(n+1)$, necessary for every stable method, as Ingham's inequality for nonharmonic Fourier series [272] suggests? A positive answer would be an uncertainty principle for the Galois face, comparable with the bounds in effective Chebotarev theorems [340].

OPEN PROBLEM 23.2 (Good and Gaskins for Thorin densities). The inversions of this chapter are exact but need exact input. Treat the ordinates $0<\gamma_1\le\cdots\le\gamma_K$ as a sample from the normalised counting measure of the zeros, or the transported density on a grid as noisy data; define the penalised likelihood of Good and Gaskins for the square root of the Thorin density U_{ar} of Section 16.4.3, or for the density of H; identify its maximiser as de Montricher, Tapia and Thompson did; and compare its error with that of the Gaussian means of Theorem 23.5 on the same input.

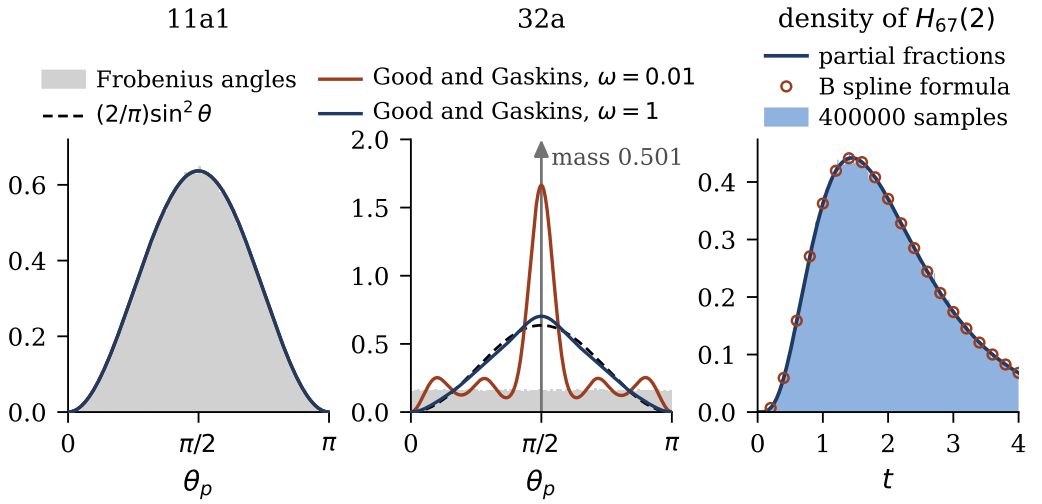


Figure 23.3: Left and centre: Theorem 23.12. Good and Gaskins estimates of the Frobenius angle density with $\omega = 0.01$ and $\omega = 1$, against the Sato Tate law $\frac{2}{\pi}\sin^2\theta$ and over histograms of the angles, for $11a1$ (primes below 10^7 other than 11) and for E_2, of class $32a$ (odd primes below 10^6). The atom of E_2 at $\pi/2$, from the primes with $a_p = 0$, is drawn as an arrow labelled with its mass, and the bars show the other angles; the estimate with $\omega = 0.01$ has a spike there. Right: Theorem 23.11. The density of the Hecke GGC $H_{67}(2)$ by partial fractions and by the B spline formula computed exactly, over a histogram of 400000 samples of a Gamma variable times a B spline variable.

23.8 NOTES

Lerch's theorem (Chapter II, Theorems 6.2 and 6.3), the complex inversion formula (Chapter II, Theorem 7.3, and Chapter VI, Theorem 5a) and the real inversion formula, found by Post and extended by Widder (Chapter VII), are in Widder's book [578]; Feller [185] reads the last as a law of large numbers. Talbot's contour [535] and the framework of Abate and Whitt [1] are the standard numerical inversions, and Stieltjes inversion is classical [528, 10]. Theorem 23.5 and Corollary 23.6 are Bohr's formula for the Fourier coefficients of an almost periodic function and its uniqueness theorem, for Dirichlet series in Besicovitch [51] (Chapter I, Sections 3 and 4; Chapter III, Section 3), and Theorem 23.7 combines them with the uniqueness of Dirichlet series. Proposition 23.8 is Hecke's Mellin integral ([250], Section 2), which goes from the L function to the form, the reverse of Section 2.3 and (20.4); that the inverse transform is modular for all of $\Gamma_0(N)$ is Weil's converse theorem [571] (Satz 2), and for two dimensional Artin representations it is Théorème 4.10 of Deligne and Serre [142]. Perron's formula is [435], and the explicit formula for ψ is in [125].

The $2\sin^2$ lemma is Lemma 14 of [442] (see also [441], Section 5), with its first step from Roynette and Yor [467], Théorème I.1.1, as recorded in Chapter 16, where its Thorin consequences are developed. The Poisson kernel form of Theorem 23.9 is in the proof of Theorem 16.8, in Proposition 16.9 and in [442] (Theorem 18 and Remark 21), and the backward Cauchy flow is in [441] (Section 8 and Theorem 9.2), whose Theorem 8.2 writes, for ξ, the distribution function of the measure of Theorem 23.9 as a sum of arctangents; Theorem 23.9 restates it through folded Cauchy variables and for every self dual π. The criterion of Hinkkanen and Lagarias [261, 339] adds the same Poisson kernels. The Pólya frequency form of GRH is Theorem 16.6, for ξ in [467] (Théorème V.3.2), [442] and [330]. Theorem 23.10 is due to Roynette and Yor [467], whose proof is Tauberian; ours uses Doetsch's transfer theorem for Laplace transforms with algebraic singularities [159].

The B splines and their limits are from Curry and Schoenberg [121], the Peano kernel and Hermite Genocchi forms of a divided difference are in de Boor's survey [132], Pólya frequency functions are from Schoenberg [488] and Karlin [293], and the Gamma times Dirichlet mean structure of GGCs, whose atomic case is Theorem 23.11(a), is from James, Roynette and Yor [280] (Section 1.4). The roughness penalty and its Bayesian reading are from Good and Gaskins [222, 223], its exponential spline solutions from de Montricher, Tapia and Thompson [144]; Good's earlier work on Poisson summation [221] appears in Chapter 20. The inequality of Theorem 23.12 is Huber's for $\mathrm{SU}(2)$ [267] and the Rayleigh principle for the alcove in general [44], the ground state being the Weyl denominator. New here as formulated, as far as we know, are the Gaussian error bounds of Theorem 23.5 and Corollary 23.6, the Hecke reading of Theorem 23.11, the Good

and Gaskins reading of the Sato Tate law in Theorem 23.12, with its expansion in symmetric power characters and its shrinkage formula, and the computations.

23.9 Exercises

1. Apply Theorem 23.3 to $g(x) = e^{-x}$ and show that the kth Post Widder approximation is $(1 + t/k)^{-k-1}$.
2. Use the weak form of Theorem 23.4, integrating $\frac{1}{\pi}\operatorname{Im}(\varphi'/\varphi)(-z + i\epsilon)$ against test functions, to show that the Gamma law with shape m, $\varphi(s) = (1+s)^{-m}$, has Thorin measure $m\delta_1$.
3. Derive the identity of Theorem 23.5 for a single atom $\nu = c\delta_x$, and, for $\nu = \delta_1 + \delta_{1.01}$, find the smallest window R for which the Gaussian mean $\mathcal{G}_R$ recovers the weight at $x = 1$ to 10^{-6}.
4. Show that the quadruple $z, -z, \bar{z}, -\bar{z}$ of zeros of Ξ^{sd}_π, $z = \beta + i\gamma$, $0 < \beta < a$, contributes $\mathrm{Law}(|\gamma + (a-\beta)C|) + \mathrm{Law}(|\gamma + (a+\beta)C|)$ to the Thorin measure of Theorem 23.9, of total mass 2, the same as two zeros of $L(s, \pi)$ on the critical line in the upper half plane.
5. For the Dedekind zeta function of $\mathbb{Q}(i)$, compute numerically A of Theorem 23.10 at $\alpha = 2$ and compare it with the constant for ζ.
6. Show that the B spline with two knots $u_0 < u_1$ is the uniform density on $[u_0, u_1]$, and deduce from Theorem 23.11 the density of $E_0/c_0 + E_1/c_1$.
7. Show that the Fisher information for location of the Sato Tate angle law $\frac{2}{\pi}\sin^2\theta$ on $[0, \pi]$ is 4, and that no density on $[0, \pi]$ vanishing at the ends has smaller Fisher information.
8. Check Perron's formula numerically for $x = 100.5$ by integrating along $\operatorname{Re} s = 2$, $|\operatorname{Im} s| \le T$, and observe the convergence in T.

23.10 Open computational problems

The scripts named below are in `laplace/code/` (Section B.7).

Computational Problem 23.1 (Four faces of elliptic curves). Objective: carry out the inversions of this chapter for elliptic curves of both root numbers. Recover $\Lambda_E(n)$ for $n \le 100$ from the transported density g_α by the Gaussian means of Corollary 23.6, with the smallest window that gives each value to 10^{-6}; rebuild the Pólya kernel Φ_E of Proposition 22.4 from $\Lambda(1 + it, E)$ by Bromwich inversion and compare it with the kernel summed from the q expansion; and compare the densities V_a of Ξ^{sd}_E at $a = 0.5$, 0.2 and 0.05 with the Cauchy superpositions of Theorem 23.9 over the zeros. Objects and scale: $11a1$ and $37a1$, as Figure 23.1 does

for the weight one form, then one curve from each isogeny class of conductor below 100; Dirichlet coefficients to 10^5; basepoints $\alpha \in \{1.5, 2\}$ for $L(s+\frac{1}{2}, E)$, whose coefficients are $\Lambda_E(n)n^{-1/2}$; windows up to 10^3; zeros to height 100. Route: `switch23.py` and `switch23b.py`, with the eta quotient replaced by the L function of the curve (PARI/GP `lfuncreate`, `lfunlambda` on the critical line, `lfunzeros`); Gaussian means by quadrature truncated at six windows, as in Section 23.2, with the bound of Corollary 23.6 for a finite window and a separate bound for the truncation. Deliverable: tables of the recovered $\Lambda_E(n)$, their errors and the minimal windows against n, with a fitted law for the window, the numerical side of Open Problem 23.1; the maximal difference of the two kernels on $|u| \le 3$, the kernel of $37a1$ being odd; and the residuals of the Cauchy superpositions. Why it matters: the Frobenius traces of the weight one form take three values at the unramified primes, while for an elliptic curve they follow the Sato Tate law and the kernels have both parities, so the problem tests the inverse arrows on the forms that carry the modularity of Chapter 22, and it measures how pessimistic the bound of Corollary 23.6 is.

COMPUTATIONAL PROBLEM 23.2 (Zeros from the heat trace by Post Widder). Objective: invert the heat trace by the Post Widder formula. With $N(t)$ the number of ordinates $\gamma > 0$ with $\gamma^2 \le t$, $\mathcal{W}(x)/x = \int_0^\infty e^{-xt} N(t)\,dt$, and the identity of Theorem 23.3, which holds for g of polynomial growth such as N, writes the kth approximation as a Gamma smoothing of N of relative width about $k^{-1/2}$; find the order k and the working precision at which the step of N at the jth zero is resolved. Objects and scale: $L(s, \Delta)$, ζ and $L(s, \rho)$ for the S_3 representation of $x^3 - x - 1$; the first 50 ordinates; orders $k \le 10^3$. Route: $\mathcal{W}$ and its derivatives from the primes through the explicit formula (16.6), or from Frobenius traces through Definition 20.29, in ball arithmetic, the pipeline of the derivative tests of Problem 20.1; the sums over the zeros as a check. Deliverable: for each function and each $j \le 50$ the smallest k and the precision that resolve the jth step, compared, at equal cost, with the Gaussian bump of Figure 16.4 and the Cauchy smoothing of Theorem 23.9. Why it matters: the derivatives $(-1)^k \mathcal{W}^{(k)}$ are the quantities whose signs the complete monotonicity criteria of Theorems 14.8 and 20.30 concern; the problem measures how much of the zero set a given number of them determines, and whether the slow convergence of Post Widder, an error of order $1/k$, can be paid for by the accuracy of the explicit formula at small x.

COMPUTATIONAL PROBLEM 23.3 (Thorin laws in the strip by Talbot inversion). Objective: for a completed L function f and basepoints α inside and to the right of the critical strip, invert by Talbot's contour the transform $\varphi_\alpha(s) = e^{b\sqrt{s}} f(\alpha)/f(\alpha + \sqrt{s})$, $b = (f'/f)(\alpha)$, whose Thorin density is the density U of (16.4); record where the inverted density is negative, and compare its mean and its tail with $\frac{1}{2}(\log f)''(\alpha)$ and with the constant $-(\log f)'''(\alpha)/(8\sqrt{\pi})$ of the proof of Theo-

rem 23.10. Objects and scale: ξ, $\Lambda(s, \Delta)$ at the same offsets $\alpha - \frac{1}{2}$ from its centre, and $\Lambda(s, \chi_{-4})$; the basepoints 0.55, 0.75, 0.99, 1.25 and 2 of Section 16.4.3; $5 \cdot 10^{-4} \leq t \leq 10^4$. Route: `zeta_tail.py`, which does this for ζ at $\alpha = 2$, with ζ replaced by the completed function, f and f'/f from mpmath or PARI/GP `lfunlambda` at 30 digits; Talbot's contour checked against the Euler algorithm of Abate and Whitt [1]; error bounds from the size of φ_α on the contour. Deliverable: the densities with their minima, total masses and means, fitted tail constants with standard errors against the predicted ones, and for each function the smallest basepoint at which the computed density is positive on the grid. Why it matters: Theorem 23.10 needs nonnegative Dirichlet coefficients; for a completed function φ_α is a GGC transform exactly where $U \geq 0$, which for ξ holds under the Riemann hypothesis, is known for $\alpha \geq 1$ and is open below (Section 16.4.3, Open Problem 20.2); the inverted laws are the probabilistic content of that positivity, and a density positive on a grid verifies nothing about the zeros.

Chapter 24

Random Integers, the Poisson Dirichlet Law and Chebotarev as a Dirichlet Process

Chapter 20 built a Gamma process from the prime factors of a zeta distributed random integer, and Chapter 21 computed the laws of the labels of its largest prime factors. Both are known objects under other names, some of which Chapter 20 recalled (Billingsley, Lloyd [370]), and this chapter develops them. The normalised sizes of the prime factors of a random integer converge to the Poisson Dirichlet law with parameter 1, the limit law of the normalised cycle lengths of a uniform random permutation (Billingsley; Donnelly and Grimmett; Arratia, Barbour and Tavaré); the share of the largest prime factor follows Dickman's law; the number of prime factors is asymptotically normal by the theorem of Erdős and Kac; and the Gamma process of Chapter 20, normalised, is Ferguson's Dirichlet process. Chebotarev's theorem, in its Dirichlet density form, is then the statement that the base measure is Haar measure, and the limits of the Frobenius functionals of Chapter 20 are a standard exponential variable times a Dirichlet mean, so that the Gamma times Dirichlet mean factorisation of James, Roynette and Yor, whose atomic case is Theorem 23.11, and Theorem 20.21 describe the same laws.

The results are of three kinds, and the Notes say which are known. Theorem 24.3 extends the Dirichlet law of the shares of Chapter 20 to the Dirichlet process limit and identifies the limit of a Frobenius functional as a standard exponential variable times a Dirichlet mean. Theorem 24.4 shows that for the random integers weighted by the coefficients of ζ^{ϑ}, the ones that drive the zeta subordinator of Chapter 16, the normalised sizes of the prime factors converge to the Poisson Dirichlet law with parameter ϑ, so that powers of ζ give the whole Ewens family. Theorem 24.5 gives the rates at which three natural colourings of the prime factors of a uniform random integer below x approach Haar measure: the largest factor faster than any power of $1/\log x$ (Dawsey's theorem), the size biased factor at rate $1/\log x$, and the count of distinct factors only at rate $1/\log\log x$, the last two with explicit constants which are Mertens constants per Chebotarev class. The last rate shows in the data: among the distinct prime factors of random integers below 10^7, the identity class of S_3 for $x^3 - x - 1$ accounts for 7.2 percent of them against its Haar measure of 16.7 percent, because no prime below 59 lies in that class. Section 24.5 then answers two questions and bounds a third from one side:

the Mertens constants per class are explicit in $L(1,\chi)$ and $\frac{L'}{L}(1,\chi)$ (Theorem 24.7); for an Euler product whose logarithm has nonnegative coefficients, given at the primes by a continuous function of equidistributed labels, the Ewens parameter is the exponent of the singularity at $s=1$ (the order of the pole when there is one), and the colouring of its random integer is the equidistribution measure weighted by that function (Theorem 24.9); and in the Kolmogorov distance of the shares the Dirichlet process limit is approached no faster than $(\sigma-1)^{\delta_{\min}}$, because the smallest class is empty with probability of that order, which Chapter 20 found and Theorem 24.8 gives with its constant.

24.1 The Poisson Dirichlet law

Let $U_1, U_2, \dots$ be independent uniform variables on $(0,1)$ and define the stick breaking sequence $W_1 = U_1$, $W_k = (1-U_1)\cdots(1-U_{k-1})U_k$. The sequence (W_k) sums to 1; its decreasing rearrangement $V_1 \geq V_2 \geq \dots$ has the *Poisson Dirichlet law* $\mathrm{PD}(1)$, and (W_k) itself is its size biased permutation, the GEM law [318, 437]. Two descriptions matter here. First, if Γ is a standard Gamma subordinator on $[0,1]$, with Lévy measure $x^{-1}e^{-x}dx$, then the ranked jumps of Γ divided by $\Gamma(1)$ have the law $\mathrm{PD}(1)$ [317], and the quotient is independent of $\Gamma(1)$, which is standard exponential. Second, if π_n is a uniform random permutation of n letters, the ranked cycle lengths divided by n converge in law to $\mathrm{PD}(1)$, a fact that goes back to Shepp and Lloyd [513]. Its analogue for prime factorisations is Billingsley's theorem [61], found independently by Knuth and Trabb Pardo [325]; Donnelly and Grimmett [162] gave a short proof by size biased sampling, and Arratia, Barbour and Tavaré [22] treat integers and permutations together as logarithmic combinatorial structures.

Theorem 24.1 (Billingsley [61]). *Let n be uniform on $\{2,\dots,x\}$ with prime factorisation $n=\prod p_i^{e_i}$, and let $V_1(n)\geq V_2(n)\geq\dots$ be the decreasing rearrangement of the numbers $e_i\log p_i/\log n$, padded with zeros. As $x\to\infty$, $(V_1(n),V_2(n),\dots)$ converges in law to $\mathrm{PD}(1)$. In particular $\log P^+(n)/\log n$, with P^+ the largest prime factor, converges in law to V_1, whose distribution function is Dickman's function [156], $\mathbb{P}(V_1\leq v)=\varrho(1/v)$, where $\varrho(u)=1$ on $[0,1]$ and $u\varrho'(u)=-\varrho(u-1)$ for $u>1$; its mean is the Golomb Dickman constant* 0.62433.

The parameter 1 is the order of the pole of ζ at $s=1$, and the error in the convergence is at most of order $1/\log x$ (Tenenbaum [542]). The number of prime factors has its own limit law.

Theorem 24.2 (Erdős and Kac; Sathe and Selberg). *Let $\omega(n)$ be the number of distinct prime factors of n uniform on $\{1,\dots,x\}$. Then $(\omega(n)-\log\log x)/\sqrt{\log\log x}$ converges in law to the standard normal law [177]. More precisely [484, 495], uniformly*

for $1 \le k \le A \log\log x$*, for any fixed* $A > 0$*,*

$$\mathbb{P}(\omega(n) = k) = \frac{(\log\log x)^{k-1}}{(k-1)!\,\log x}\Big(G\Big(\frac{k-1}{\log\log x}\Big) + O\Big(\frac{1}{\log\log x}\Big)\Big),$$

where

$$G(z) = \frac{1}{\Gamma(1+z)}\prod_p \Big(1 + \frac{z}{p-1}\Big)\Big(1 - \frac{1}{p}\Big)^z,$$

and $\mathbb{E}\,\omega(n) = \log\log x + B + O(1/\log x)$ *with* $B = 0.26150$ *the Mertens constant [543], while* $\operatorname{Var}\omega(n) = \log\log x + B - \frac{\pi^2}{6} - \sum_p p^{-2} + o(1)$ *(Diaconis, Mosteller and Onishi [153]).*

Figure 24.1 shows both theorems at $x = 10^7$ for 400000 random integers, factorised with a sieve of smallest prime factors. The empirical distribution of $\log P^+ / \log n$ differs from Dickman's $\varrho(1/v)$ by at most 0.066, the supremum being approached as $v \uparrow 1$, where the 6.6 percent of integers below 10^7 that are prime form an atom at share 1, and this is the $1/\log x$ term with $\log x = 16.1$. The mean is 0.6463 against 0.6243, and the second largest share has mean 0.2255 against the PD(1) value 0.2096. The mean of $\omega(n)$ is 3.014, against $\log\log x + B + (\gamma_{\mathrm{E}} - 1)/\log x = 3.015$, but the variance is 1.10, far below the mean (the formula of Theorem 24.2 without its $o(1)$ term gives 0.94), and neither the Poisson law with mean 3.04 nor the normal law of Erdős and Kac fits the histogram at this x, where $\log\log x = 2.78$. The mean of $\Omega(n) - \omega(n)$, which counts repeated factors, is 0.7723 against $\sum_p 1/(p(p-1)) = 0.7732$.

24.2 The Dirichlet process of a random integer

Ferguson's Dirichlet process with base measure $\vartheta\mu$, for $\vartheta > 0$ and a probability measure μ on a compact metric space Y, is the random probability measure $P = \Gamma_{\vartheta\mu}/\Gamma_{\vartheta\mu}(Y)$ obtained by normalising the Gamma random measure with shape measure $\vartheta\mu$ ([186], Section 4); for every partition $B_1, \dots, B_k$ of Y the vector $(P(B_1), \dots, P(B_k))$ has the Dirichlet law with parameters $(\vartheta\mu(B_1), \dots, \vartheta\mu(B_k))$, and P is independent of the total mass $\Gamma_{\vartheta\mu}(Y)$, which is Gamma with shape ϑ ([280], Section 1.4; [437], Section 4.2). Written as $P = \sum_i V_i \delta_{y_i}$ with $V_1 \ge V_2 \ge \dots$, its weights have the law PD(ϑ) and its labels y_i are independent of the weights and of one another, with law μ (several labels coincide when μ has atoms); PD(1) is the case $\vartheta = 1$ of Section 24.1.

Let N_σ be the zeta random integer of Chapter 20, $\mathbb{P}(N_\sigma = n) = n^{-\sigma}/\zeta(\sigma)$, with prime exponents e_p, and let (θ_p) be labels in Y, equidistributed with respect to μ outside a finite set S of primes. The labelled factorisation is the random measure

$$\Xi_\sigma = \sum_{p \notin S} (\sigma - 1) e_p \log p \; \delta_{\theta_p},$$

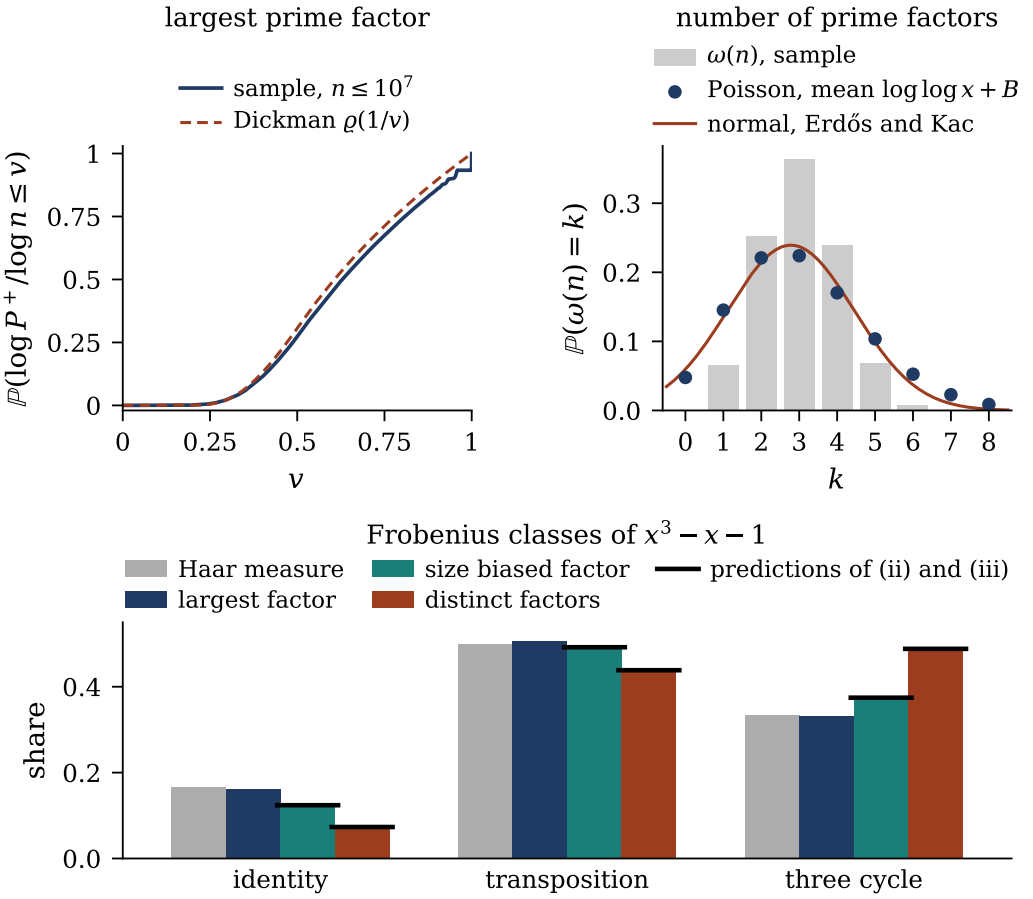


Figure 24.1: A sample of 400000 uniform random integers below 10^7. Top left: the empirical distribution function of $\log P^+ / \log n$ against Dickman's $\varrho(1/v)$ (Theorem 24.1). Top right: the histogram of the number $\omega(n)$ of distinct prime factors, whose mean is close to $\log\log x + B$, with the Poisson law of mean $\log\log x + B$ and the normal law of Erdős and Kac, of mean and variance $\log\log x$, neither of which fits at this x (Theorem 24.2). Bottom: three colourings of the prime factors by their Frobenius class in the S_3 field of $x^3 - x - 1$ (the class of the largest factor, of a size biased factor, and of the distinct factors pooled over the sample), against Haar measure; the short bars are the predictions of Theorem 24.5(ii) and (iii) with the constants of Table 24.1.

and Theorem 20.18 says that Ξ_σ converges in law to the Gamma random measure Γ_μ as $\sigma \downarrow 1$.

THEOREM 24.3 (Chebotarev as a Dirichlet process; after [186, 105]). *As $\sigma \downarrow 1$:*

(i) the normalised labelled factorisation $\mathcal{P}_\sigma = \Xi_\sigma / \Xi_\sigma(Y)$, set equal to μ on the event A_σ that N_σ has no prime factor outside S, converges in law, as a random probability measure on Y with the weak topology, to the Dirichlet process P with base measure μ, and jointly with it the total mass $\Xi_\sigma(Y) = (\sigma - 1)\sum_{p \notin S} e_p \log p$ converges to a standard exponential variable E independent of P;

(ii) for every continuous $Q \geq 0$ on Y the Frobenius functional $\Xi_\sigma(Q)$ converges in law to $E \int_Y Q\, dP$, a standard exponential variable times a Dirichlet mean, and

$$\mathbb{E} \exp\Big(-\lambda E \int_Y Q\, dP\Big) = \mathbb{E}\Big(1 + \lambda \int_Y Q\, dP\Big)^{-1} = \exp\Big(-\int_Y \log(1 + \lambda Q)\, d\mu\Big);$$

(iii) for a partition of Y into μ continuity sets $B_1, \ldots, B_k$ with $\mu(B_i) > 0$, the shares $\mathcal{P}_\sigma(B_i)$ converge jointly in law to the Dirichlet law with parameters $\mu(B_i)$, so that each share $\mathcal{P}_\sigma(B)$ converges to a Beta variable with parameters $(\mu(B), 1 - \mu(B))$, of mean $\mu(B)$.

PROOF. (i) The map $\xi \mapsto (\xi/\xi(Y), \xi(Y))$ is continuous, in the weak topology, at every finite measure with $\xi(Y) > 0$, a set of full Γ_μ probability, so the continuous mapping theorem and Theorem 20.18(ii) give convergence of $(\mathcal{P}_\sigma, \Xi_\sigma(Y))$ to $(\Gamma_\mu/\Gamma_\mu(Y), \Gamma_\mu(Y))$, which is Ferguson's construction of the Dirichlet process with base measure μ, independent of the Gamma total mass of shape $\mu(Y) = 1$. (ii) $\Xi_\sigma(Q) = \Xi_\sigma(Y) \int Q\, d\mathcal{P}_\sigma$ converges by (i) to $E \int Q\, dP$. The first equality is the Gamma integral $\mathbb{E}e^{-\lambda E m} = (1 + \lambda m)^{-1}$, conditioned on $\int Q\, dP = m$ and using independence; the second is Theorem 20.21, since $\Xi_\sigma(Q)$ converges in law to the GGC whose Laplace transform is the right hand side. (iii) follows from (i): $\mathbb{E}P(\partial B_i) = \mu(\partial B_i) = 0$, so $\nu \mapsto (\nu(B_1), \ldots, \nu(B_k))$ is continuous at almost every P, and the shares of a Dirichlet process are Dirichlet. □

For Frobenius classes in a finite Galois group G, with μ the image of Haar measure on the set of classes, Chebotarev's theorem supplies the hypothesis, and the proof of Lemma 20.19 uses only its Dirichlet density form $\sum_{p \in C} (\log p) p^{-s} \sim \delta(C)/(s-1)$ as $s \downarrow 1$, with $\delta(C) = |C|/|G|$. Then (i) and (iii) are Corollary 20.20(ii) of Chapter 20 read as a statement about the Dirichlet process: the share $\mathcal{P}_\sigma(C)$ carried by the primes with Frobenius in a class C is asymptotically Beta$(\delta(C), 1 - \delta(C))$, with mean Chebotarev's density $\delta(C)$.

The identity between the last two expressions in (ii), $\mathbb{E}(1 + \lambda \int Q\, dP)^{-\vartheta} = \exp(-\vartheta \int \log(1 + \lambda Q)\, d\mu)$ for a Dirichlet process with base measure $\vartheta\mu$, is Cifarelli

and Regazzini's identity for Dirichlet means [105], the Markov Krein identity when $\vartheta = 1$ (Diaconis and Kemperman [152]), in the form of James, Roynette and Yor ([280], Section 1.4, (33)): a GGC of finite Thorin mass is a Gamma variable times an independent Dirichlet mean, whose atomic case, Theorem 23.11, writes atomic Thorin laws as Gamma variables times B spline variables. Here it says what the Thorin measure of a Frobenius functional is: the limit of $\Xi_\sigma(Q)$ is a GGC with Thorin mass $\mu(Q > 0)$ and Thorin measure the image of μ restricted to $\{Q > 0\}$ under $1/Q$ (Theorem 20.21), and it is also a Gamma variable of shape $\mu(Q > 0)$ times the Dirichlet mean $\int Q\, dP'$ of an independent Dirichlet process P' with base measure μ restricted to $\{Q > 0\}$. Figure 24.2, left, checks the identity in the simplest case, $Q = \mathrm{tr}^2$ for the two dimensional representation of S_3 (traces $2, 0, -1$ on the three classes) with Haar measure, where $\int Q\, dP = 4P(\{1\}) + P(C_3)$, with $\{1\}$, C_2 and C_3 the identity, the transpositions and the three cycles, and $(P(\{1\}), P(C_2), P(C_3))$ Dirichlet with parameters $(\frac{1}{6}, \frac{1}{2}, \frac{1}{3})$: two million samples of $E(4P(\{1\}) + P(C_3))$ reproduce $\exp(-\frac{1}{6}\log(1+4\lambda) - \frac{1}{3}\log(1+\lambda))$ to $2.8 \cdot 10^{-4}$. The Laplace transform at $\sigma = 1.01$, computed from the Euler product over the primes below 10^7, with the prime number theorem for the larger primes, lies 0.036 above the limit at $\lambda = 6$; this is the Euler Kronecker drift of Theorem 20.26, of order $\sigma - 1$ (its first order term $\kappa_Q \lambda(\sigma - 1)$, with $\kappa_Q = \sum_C Q_C \kappa_C = 2.53$ for $S = \{23\}$, is 0.15 in the logarithm, against the observed 0.11, the identity class being evaluated at $\sigma + 24(\sigma - 1) = 1.25$), not a defect of the identity.

Theorem 24.3 has two readings. The first is Bayesian: in Chapter 29 the Dirichlet process is a nonparametric prior on the trace law, set aside there because its draws are discrete; here it is the limit law of the labelled factorisation, with the equidistribution measure (Haar measure for Frobenius classes) as base measure, and Ferguson's conjugacy ([186], Section 3, Theorem 1), which turns a prior with base measure $\vartheta\mu$ into a posterior with base measure $\vartheta\mu + \sum_i \delta_{y_i}$ after labels $y_1, \ldots, y_n$, applies to it unchanged. The second is the Chinese restaurant process. Sampling points from a Dirichlet process with base measure $\vartheta\mu$ produces, by the Pólya urn of Blackwell and MacQueen [66], a random partition: the nth point joins an existing cluster of size m with probability $m/(n - 1 + \vartheta)$ and opens a new cluster, with a fresh label drawn from μ, with probability $\vartheta/(n - 1 + \vartheta)$. The cluster sizes follow the Ewens sampling formula with parameter ϑ [181, 19, 264], and the cluster labels are independent with law μ. For $\vartheta = 1$ the partition of n sampled points has the law of the cycle type of a uniform permutation of n letters, which is the sense in which the large prime factors of a random integer and the long cycles of a random permutation have the same limit.

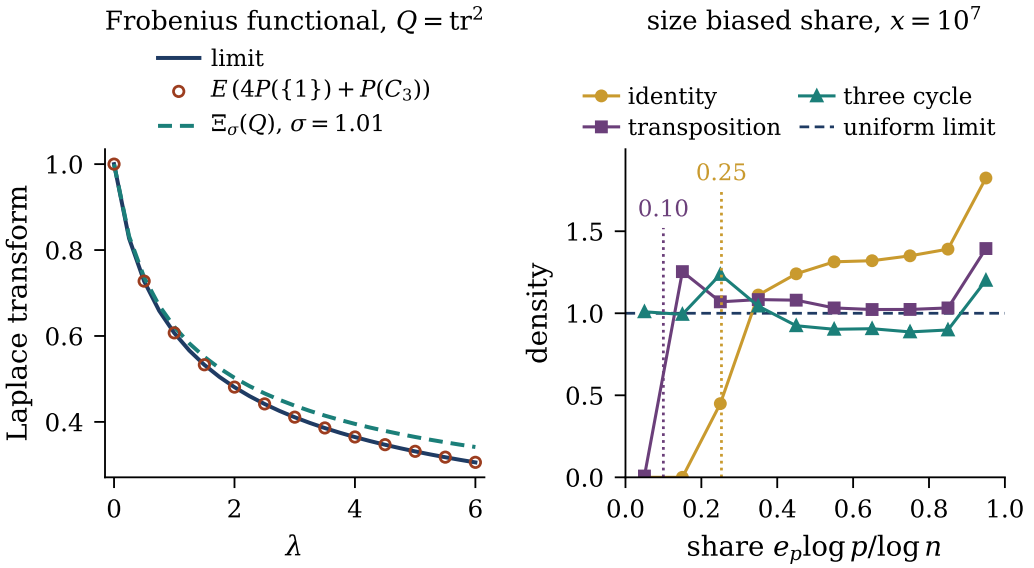


Figure 24.2: Left: Theorem 24.3(ii) for $Q = \mathrm{tr}^2$, the squared trace of the two dimensional representation of the S_3 field of $x^3 - x - 1$: the Laplace transform of the limit, $\exp(-\frac{1}{6}\log(1+4\lambda) - \frac{1}{3}\log(1+\lambda))$, that of two million samples of $E(4P(\{1\}) + P(C_3))$, a standard exponential variable times a Dirichlet mean, and that of $\Xi_\sigma(Q)$ at $\sigma = 1.01$ (primes below 10^7, prime number theorem beyond). Right: the size biased share $e_p \log p / \log n$ of the 400000 uniform random integers below 10^7 of Figure 24.1, by Frobenius class of the chosen prime p. The limit is uniform for every class (Theorem 24.6); at $x = 10^7$ the share is at least $\log 59 / \log 10^7 = 0.25$ when p is in the identity class, whose smallest prime is 59, and at least $\log 5 / \log 10^7 = 0.10$ when p is in the class of transpositions, whose smallest prime is 5 (both floors marked). The rise in the last bin is the atom at share 1 of the prime powers $n = p^k$, nearly all of them primes.

24.3 THE EWENS FAMILY FROM THE COEFFICIENTS OF ζ^ϑ

The zeta subordinator X_t of Chapter 16 runs, by Proposition 16.10, through the random integers whose law is $d_t(n)n^{-\sigma}/\zeta(\sigma)^t$, where d_t are the Dirichlet coefficients of ζ^t, multiplicative with $d_t(p^k) = \binom{t+k-1}{k}$. For $t = 1$ this is N_σ; for $t = 2$ the weight is the divisor function. These integers, with $t = \vartheta$, give the whole Ewens family.

THEOREM 24.4 (the prime factors of the ζ^ϑ random integer are Poisson Dirichlet with parameter ϑ; after [174, 530]). *Fix $\vartheta > 0$ and let the random integer $N_\sigma^{(\vartheta)}$ have the law $d_\vartheta(n)n^{-\sigma}/\zeta(\sigma)^\vartheta$, with labels (θ_p) equidistributed with respect to μ outside S, and let $\Xi_\sigma^{(\vartheta)} = \sum_{p\notin S}(\sigma-1)e_p \log p\, \delta_{\theta_p}$ be its labelled factorisation. Then the prime exponents are independent negative binomial variables, $\mathbb{P}(e_p = k) = \binom{\vartheta+k-1}{k}(1-p^{-\sigma})^\vartheta p^{-k\sigma}$, and as $\sigma \downarrow 1$:*

(i) the marked process $\sum_{p\notin S, e_p\geq 1}\delta_{((\sigma-1)e_p\log p,\,\theta_p)}$ converges in law, vaguely on $(0,\infty)\times Y$, to the Poisson process with intensity $\vartheta x^{-1}e^{-x}dx\otimes\mu$;

(ii) the labelled factorisation $\Xi_\sigma^{(\vartheta)}$ converges in law to the Gamma random measure with shape measure $\vartheta\mu$, its total mass, which is $(\sigma-1)\log N_\sigma^{(\vartheta)}$ up to the primes in S, to a Gamma variable with shape ϑ, and the normalised factorisation to the Dirichlet process with base measure $\vartheta\mu$;

(iii) the ranked normalised sizes of the prime factors converge to $\mathrm{PD}(\vartheta)$, and for every continuous $Q \geq 0$ the functional $\Xi_\sigma^{(\vartheta)}(Q)$ converges to a GGC with Thorin measure ϑ times the image of $\mu|_{\{Q>0\}}$ under $1/Q$.

PROOF. The generating function of the law is $\zeta(\sigma)^{-\vartheta}\sum_n d_\vartheta(n)n^{-\sigma}\prod_p t_p^{e_p} = \prod_p\big((1-p^{-\sigma})/(1-t_pp^{-\sigma})\big)^\vartheta$, a product over primes, so the exponents are independent and negative binomial. It is the ϑth power of the generating function of N_σ (Lemma 20.17), so $\mathbb{E}e^{-\Xi_\sigma^{(\vartheta)}(\lambda)} = (\mathbb{E}e^{-\Xi_\sigma(\lambda)})^\vartheta$ for continuous $\lambda \geq 0$, and Theorem 20.18(ii) gives the first two parts of (ii). The proof of Theorem 20.18(i) gives (i), with $\mathbb{P}(e_p = k) \leq \binom{\vartheta+k-1}{k}p^{-k}$, whose sum over $k \geq 2$ is $O(p^{-2})$, in place of p^{-k}, and $\vartheta p^{-\sigma}$ in place of $p^{-\sigma}$. The atoms of size below a carry expected mass at most $\vartheta\nu_\sigma([0,a)\times Y) + O(\sigma-1)$, with ν_σ as in Lemma 20.19, which tends to $\vartheta(1-e^{-a})$, uniformly small as $a \to 0$; so the total mass converges jointly with the marked process of (i), and the ranked sizes divided by the total converge to the ranked jumps of a Gamma subordinator on $[0,\vartheta]$ divided by their sum, which is $\mathrm{PD}(\vartheta)$ [317]. The third part of (ii) follows as in Theorem 24.3, with $\Gamma_{\vartheta\mu}(Y)$ Gamma of shape ϑ, and the GGC limit in (iii) is (ii) applied to λQ, as in the proof of Theorem 20.21. □

So the parameter of the Poisson Dirichlet law is the exponent ϑ of the singularity $(s-1)^{-\vartheta}$ of the generating Dirichlet series at $s=1$, the order of a pole when ϑ is an integer. For the Rankin Selberg random integer of an Artin representation ρ, with law proportional to the coefficients, which are nonnegative, of $L^S(s, \rho \otimes \check{\rho})$, the L function without its Euler factors at the ramified primes, the exponents are $e_p = \sum_m mM(p^m)$ with independent Poisson variables $M(p^m)$ of means $|\chi_\rho(\mathrm{Frob}_p^m)|^2 p^{-m\sigma}/m$, and Theorem 24.9 below gives a marked limit with intensity $x^{-1}e^{-x}dx \otimes |\chi_\rho|^2\mu$, of total mass $\langle \rho, \rho \rangle$. So the Gamma shape of Theorem 20.28 is an Ewens parameter, the normalised factorisation converges to the Dirichlet process with base measure $|\chi_\rho|^2\mu$ rather than a multiple of Haar measure, and the ranked sizes are $\mathrm{PD}(\langle \rho, \rho \rangle)$: $\mathrm{PD}(1)$ for irreducible ρ and $\mathrm{PD}(4)$ for $\rho \oplus \rho$, with correspondingly more and smaller prime factors.

24.4 Three colourings and their rates

Theorem 24.3 is a statement about the zeta random integer as $\sigma \downarrow 1$. For a uniform random integer below x the sizes have the same limit (Theorem 24.1), but the rates differ sharply between three natural ways of colouring the prime factors by their Frobenius classes, and the differences are visible at $x = 10^7$. Let $K/\mathbb{Q}$ be Galois with group G, $C \subset G$ a conjugacy class with Haar measure $\delta(C) = |C|/|G|$, which is $\mu(C)$ in the notation of Chapter 20, and write $p \in C$ when the Frobenius class of p is C. The three colourings of n are: the class of the largest prime factor $P^+(n)$; the class of a *size biased* prime factor, chosen with probability $e_p \log p / \log n$; and the classes of the distinct prime factors, pooled over n, that is the ratio $\mathbb{E}\omega_C(n) / \sum_{C'} \mathbb{E}\omega_{C'}(n)$, where $\omega_C(n)$ counts the distinct prime factors of n in C.

Theorem 24.5 (three colourings; (i) Dawsey [130], (ii) this book, (iii) Mertens per class [35]). *Let n be uniform on $\{2, \dots, x\}$, and let the ramified primes be left out of every class. By the prime number theorem for K in the form [340]*

$$\pi_C(t) = \delta(C)\,\mathrm{li}(t) + O\big(t \exp(-a_K\sqrt{\log t})\big),$$

the limits

$$b_C = \lim_{t\to\infty}\Big(\sum_{p\le t,\, p\in C} \frac{1}{p} - \delta(C)\log\log t\Big),$$

$$c_C = \lim_{t\to\infty}\Big(\sum_{p\le t,\, p\in C} \frac{\log p}{p} - \delta(C)\log t\Big)$$

exist, with $\sum_C b_C = B - \sum_{p\ ram.} 1/p$ and $\sum_C c_C = c - \sum_{p\ ram.} \log p/p$, where $B = 0.26150$ and $c = -\gamma_{\mathrm{E}} - \sum_p \log p/(p(p-1)) = -1.33258$ are the two Mertens constants. Let $\kappa_C = -c_C - \sum_{p\in C} \log p/(p(p-1))$, which by the Abelian theorem is

the Euler Kronecker drift of Theorem 20.26 for S the set of ramified primes, so that $\sum_C(\gamma_{\mathrm{E}}\delta(C)-\kappa_C) = -\sum_{p\ ram.}\log p/(p-1)$. Then, as $x\to\infty$:

(i) $\mathbb{P}(P^+(n)\in C) = \delta(C) + O\big(\exp(-a'_K(\log x)^{1/3})\big)$ *for some* $a'_K>0$;

(ii) the size biased prime factor lies in C with probability

$$\delta(C) + \frac{\gamma_{\mathrm{E}}\delta(C)-\kappa_C}{\log x - 1} + O\big((\log x)^{-2}\big);$$

(iii) $\mathbb{E}\,\omega_C(n) = \delta(C)\log\log x + b_C + O(1/\log x)$, *so that with* $B' = B - \sum_{p\ ram.} 1/p$ *the ratio of means* $\mathbb{E}\omega_C(n)/\sum_{C'}\mathbb{E}\omega_{C'}(n)$, *the share of the unramified distinct prime factors that lie in C, equals*

$$\delta(C) + \frac{b_C - \delta(C)B'}{\log\log x + B'} + O\Big(\frac{1}{\log x\,\log\log x}\Big).$$

PROOF. The constants exist by partial summation against $\pi_C(t)-\delta(C)\,\mathrm{li}(t)$, exactly as in Mertens's theorems for $\pi(t)$ [393, 543]; the sums over all classes are the classical constants. (i) is Dawsey's theorem ([130], Theorem 2), which extends Alladi's for progressions [15]. The error is so small because the unlabelled count is exact: every n with $2\le n\le x$ is pm with $p=P^+(n)$ and $m\le x/p$ free of primes above p, so $\sum_{p\le x}\Psi(x/p,p) = \lfloor x\rfloor - 1$, with $\Psi(y,p)$ the number of such $m\le y$, and only the equidistribution of the primes of C, weighted by $\Psi(x/p,p)$, enters. (iii) By Mertens's theorem for the class ([35], Theorems 8.3 and 8.5),

$$\begin{aligned}\mathbb{E}\omega_C(n) &= \frac{1}{\lfloor x\rfloor - 1}\sum_{p\le x,\,p\in C}\Big\lfloor\frac{x}{p}\Big\rfloor = \sum_{p\le x,\,p\in C}\frac{1}{p} + O\Big(\frac{1}{\log x}\Big)\\ &= \delta(C)\log\log x + b_C + O\Big(\frac{1}{\log x}\Big);\end{aligned}$$

summing over the classes gives $\log\log x + B' + O(1/\log x)$, and the ratio of the two expansions is the display. (ii) Write $\ell_C(n) = \sum_{p\in C} e_p\log p$, so that the size biased factor lies in C with probability $\mathbb{E}[\ell_C(n)/\log n]$. Replacing $1/\log n$ by $1/(\log x-1)$, where $\log x - 1 + O(\log x/x)$ is the mean of $\log n$, costs $O((\log x)^{-2})$: for $n\ge\sqrt{x}$, writing $u=\log(n/x)$,

$$\frac{1}{\log n} = \frac{1}{\log x - 1} - \frac{u+1}{(\log x-1)^2} + O\Big(\frac{(u+1)^2}{(\log x)^3}\Big);$$

the first order term is handled by $\mathbb{E}[\ell_C(n)(u+1)] = O(1)$, because for each prime power $y = x/p^k$ the inner sum $\sum_{m\le y}(\log(m/y)+1)$ is $O(\log y+1)$, the second by $\mathbb{E}[\ell_C(n)(u+1)^2] \le \log x\,\mathbb{E}(u+1)^2 = O(\log x)$, and the $n<\sqrt{x}$ have probability $O(x^{-1/2})$. Now $(\lfloor x\rfloor-1)\mathbb{E}e_p = \sum_{k\ge1}\lfloor x/p^k\rfloor = x/(p-1) -$

$\sum_{k\ge1}\{x/p^k\}$. Summing over $p \le x$ in C, with $x/(\lfloor x\rfloor - 1) = 1 + O(1/x)$, the main term is

$$\sum_{p\le x,\, p\in C} \frac{\log p}{p-1} = \delta(C)\log x + c_C + \sum_{p\in C} \frac{\log p}{p(p-1)} + O\Big(\frac{1}{\log x}\Big)$$

by the definition of c_C, and the fractional parts contribute

$$-\frac{1}{x}\sum_{p\le x,\, p\in C} \log p \,\Big\{\frac{x}{p}\Big\} + O\Big(\frac{1}{\log x}\Big) = -\delta(C)(1-\gamma_{\mathrm{E}}) + O\Big(\frac{1}{\log x}\Big).$$

The last estimate is the class version of

$$\sum_{n\le x} \Lambda(n)\Big\{\frac{x}{n}\Big\} = (1-\gamma_{\mathrm{E}})x + O\Big(\frac{x}{\log x}\Big),$$

which follows from $\log\lfloor x\rfloor! = \sum_{n\le x}\Lambda(n)\lfloor x/n\rfloor$ and $\sum_{n\le x}\Lambda(n)/n = \log x - \gamma_{\mathrm{E}} + o(1/\log x)$: on each interval $x/(j+1) < p \le x/j$ with $j < \log x$ the function $\{x/p\} = x/p - j$ is monotone, and partial summation against $\sum_{p\le t,\, p\in C}\log p - \delta(C)\sum_{p\le t}\log p = O(t\exp(-a_K\sqrt{\log t}))$ replaces the sum over $p \in C$ by $\delta(C)$ times the sum over all primes, with an error $O(x/\log x)$, the primes $p \le x/\log x$ contributing $O(x/\log x)$. This term $-\delta(C)(1-\gamma_{\mathrm{E}})$ is why the constant carries the term $\gamma_{\mathrm{E}}\delta(C)$. Hence $\mathbb{E}\ell_C(n) = \delta(C)(\log x - 1) + \gamma_{\mathrm{E}}\delta(C) - \kappa_C + O(1/\log x)$, and dividing by $\log x - 1$ gives (ii). □

For the zeta integer the same constant is, by Theorem 20.26, the first order term of the ratio of means $\mathbb{E}L_C(\sigma)/\mathbb{E}[(\sigma-1)\log N_\sigma] = \delta(C) + (\gamma_{\mathrm{E}}\delta(C) - \kappa_C)(\sigma-1) + O((\sigma-1)^2)$, with $L_C(\sigma)$ the rescaled class mass of Corollary 20.20 and $\sigma - 1$ in the place of $1/(\log x - 1)$.

class C of $x^3 - x - 1$	$\delta(C)$	b_C	c_C	$\gamma_{\mathrm{E}}\delta(C) - \kappa_C$	smallest prime
identity (3 roots)	$1/6$	-0.2436	-0.7429	-0.6439	59
transposition (1 root)	$1/2$	-0.0755	-0.5931	-0.1211	5
three cycle (0 roots)	$1/3$	0.5372	-0.1327	0.6226	2

Table 24.1: Mertens constants per Frobenius class, computed from the unramified primes below 10^7, for the S_3 extension of $x^3 - x - 1$; the ramified prime 23 is left out, so the b_C sum to $0.2180 = B - \frac{1}{23}$ and the c_C to -1.4687, against the limit $c - \frac{\log 23}{23} = -1.4689$. The identity class has a large negative b_C because its first prime is 59; the three cycle class, which contains 2, 3 and 13, has a large positive one.

Table 24.1 gives the constants for the running example, and Figure 24.1, bottom, compares the three colourings at $x = 10^7$ with Haar measure and with the predictions of the theorem. The largest prime factor is already close to Haar: shares

0.161, 0.506, 0.332 against $\frac{1}{6}, \frac{1}{2}, \frac{1}{3}$, the deviation at this x coming from the small primes. For the zeta integer this is Theorem 21.10(ii), and Chapter 21 computes its approach, linear in $\sigma - 1$, for the same field. The size biased factor has shares 0.124, 0.492, 0.374, against the prediction 0.124, 0.492, 0.375 of (ii), and the remaining 0.93 percent of picks fall on the ramified prime 23, against $\log 23/(22(\log x - 1))$, that is 0.94 percent. The distinct prime factors, pooled over all 400000 integers, have shares 0.072, 0.438, 0.489, against the prediction 0.073, 0.438, 0.488 of (iii), and the identity class is at less than half its Haar measure. The reason is in the last column of the table: the count of distinct prime factors is dominated by small primes, whose classes are fixed, and the identity class has no prime below 59. The deficit of its share, $(b_C - \delta(C)B')/(\log\log x + B')$ with $b_C - \delta(C)B' = -0.280$, decays only like $1/\log\log x$: it is -0.093 at $x = 10^7$ and still -0.059 at $x = 10^{40}$. Chebotarev's theorem holds for all three colourings, but for the third the identity class is still more than a third below its Haar measure at $x = 10^{40}$. The same effect sits inside Theorem 24.6 below: the Frobenius class and the size of a size biased factor are asymptotically independent, but at $x = 10^7$ a size biased factor in the identity class carries at least $\log 59/\log 10^7 = 0.25$ of $\log n$ (Figure 24.2, right). So the statistics of a random integer that see Chebotarev's theorem at a computable scale are those that select or weight prime factors by their size, as the Frobenius functionals of Chapter 20 do with the weight $e_p \log p$; counting distinct prime factors, where the small primes dominate, does not.

THEOREM 24.6 (size biased factors are uniform, whatever their class; after Donnelly and Grimmett [162]). *Let n be uniform on $\{2, \dots, x\}$, let p^* be a size biased prime factor of n, and let $F^* = e_{p^*} \log p^*/\log n$ be its share. As $x \to \infty$, p^* is unramified with probability tending to 1, and $(F^*, \text{class of } p^*)$ converges in law to the product of the uniform law on $(0,1)$ and the law $C \mapsto \delta(C)$ on the classes. The same holds for the zeta random integer as $\sigma \downarrow 1$, with p^* picked, off A_σ, among the prime factors outside S and its share taken relative to their part of $\log N_\sigma$.*

PROOF. For the uniform integer the statement is a first moment computation:

$$\mathbb{P}(p^* \in C,\ F^* \le f) = \mathbb{E} \sum_{p \in C} \frac{e_p \log p}{\log n} 1\{e_p \log p \le f \log n\}.$$

The terms with $e_p \ge 2$ contribute $O(1/\log x)$, the integers $n \le x/(\log x)^2$ contribute $O((\log x)^{-2})$, and for the others $\log n = \log x + O(\log\log x)$, so the probability is $(\log x)^{-1} \sum_{p \le x^f,\, p \in C} \log p/p + O(\log\log x/\log x) = \delta(C) f + o(1)$ by Mertens's first theorem for the class, which follows from the existence of c_C in Theorem 24.5. For the zeta integer, the proof of Theorem 24.4 with $\vartheta = 1$ shows that the rescaled sizes with their labels and the total mass converge jointly, so that the normalised sizes converge to $\mathrm{PD}(1)$, a limit that goes back to Lloyd [370], with labels independent of the sizes and of law μ (Theorem 20.18(i)). Since picking

in proportion to size is a continuous operation outside a null set, the share of p^* converges to a size biased pick from PD(1), which is uniform ([437], Section 3.1), and its class to an independent label of law μ. □

24.5 Mertens constants, empty classes and Ewens parameters

This section answers two questions about the limits above and bounds a third from one side. The Mertens constants per class are explicit in $L(1,\chi)$ and $\frac{L'}{L}(1,\chi)$ (Theorem 24.7). The Ewens parameter of an Euler product whose logarithm has nonnegative coefficients, given at the primes by a continuous function of equidistributed labels, is the exponent of its singularity at $s=1$, the order of its pole when it continues meromorphically, and the colouring of its random integer is the equidistribution measure weighted by that function (Theorem 24.9). And while the Euler Kronecker drifts of Chapter 20 govern the means, the Kolmogorov distance between the law of the share of a class C and its Beta limit is at least of order $(\sigma-1)^{\delta(C)}$, because at any $\sigma>1$ the class is empty with probability of that order, which Chapter 20 found (Theorem 20.21 and the paragraph after it) and Theorem 24.8 gives with its constant. The constants of the first and last answers come from one function, the regular part h_C of the partial Euler product over a class: b_C and the constant of the last answer from its value at $s=1$, and c_C from its derivative there.

Throughout, $K/\mathbb{Q}$ is Galois with group G, S is the set of ramified primes and C a class with $\delta(C)=|C|/|G|$. As in the proof of Theorem 20.26,

$$P_C(s)=\sum_{p\notin S,\ \mathrm{Frob}_p\in C}\ \sum_{m\geq 1}\frac{p^{-ms}}{m}=\sum_{\chi}c_\chi\log L^S(s,\chi)+R_C(s),$$

with $c_\chi=\delta(C)\overline{\chi(C)}$ for each irreducible character χ, L^S the L function without its Euler factors at S, and a prime power term R_C holomorphic on $\mathrm{Re}\,s>\frac12$. By Brauer's theorem and the nonvanishing of Artin L functions at $s=1$, $h_C(s)=P_C(s)-\delta(C)\log\frac{1}{s-1}$ is real analytic near $s=1$, and $h_C'(1)$ is the drift κ_C of that theorem for this S.

24.5.1 Mertens constants from $L(1,\chi)$

THEOREM 24.7 (Mertens constants per class; after [582, 35]). *With b_C and c_C as in Theorem 24.5 and B, c the Mertens constants,*

$$b_C = \delta(C)\Big(B - \sum_{p\in S}\frac{1}{p}\Big) + \sum_{\chi\neq 1} c_\chi\Big[\log L^S(1,\chi) - \sum_{p\notin S}\sum_{m\geq 2}\frac{\chi(\mathrm{Frob}_p^m)}{m\,p^m}\Big],$$

$$c_C = \delta(C)\Big(c - \sum_{p\in S}\frac{\log p}{p}\Big) - \sum_{\chi\neq 1} c_\chi\Big[\frac{(L^S)'}{L^S}(1,\chi) + \sum_{p\notin S}\sum_{m\geq 2}\frac{\chi(\mathrm{Frob}_p^m)\log p}{p^m}\Big],$$

and the constant term of P_C at $s=1$ is

$$h_C(1) = b_C - \gamma_{\mathrm{E}}\,\delta(C) + \sum_{p\notin S,\,\mathrm{Frob}_p\in C}\ \sum_{m\geq 2}\frac{p^{-m}}{m}.$$

PROOF. Expand $1_C(\mathrm{Frob}_p) = \sum_\chi c_\chi\chi(\mathrm{Frob}_p)$. The term $\chi = 1$ gives, by Mertens, $\delta(C)$ times $\sum_{p\leq t,\,p\notin S} 1/p$, which is $\log\log t + B - \sum_{p\in S} 1/p + o(1)$. For $\chi\neq 1$ the partial sums $\sum_{p\leq t,\,p\notin S}\chi(\mathrm{Frob}_p)/p$ converge to the bracket in the theorem: the Dirichlet series $\sum_{p\notin S}\chi(\mathrm{Frob}_p)p^{-s}$ equals $\log L^S(s,\chi)$ minus the prime power sum, which is holomorphic on $\mathrm{Re}\,s > \frac12$, and $L^S(s,\chi)$ is holomorphic and nonzero on $\mathrm{Re}\,s\geq 1$, so the partial sums converge to the value at $s=1$ by the prime number theorem for K and partial summation, exactly as for Dirichlet characters [543]; for Chebotarev classes see [35], Theorem 8.3. The formula for c_C is the same argument with $\log p/p$ in place of $1/p$, the $\chi = 1$ term being the prime number theorem in the form $\sum_{p\leq t}\log p/p = \log t + c + o(1)$ and the others the derivative of $-\log L^S(s,\chi)$ at 1. For $h_C(1)$, the Abelian theorem $\sum_{p\in C} p^{-s} = \delta(C)\log\frac{1}{s-1} + b_C - \gamma_{\mathrm{E}}\delta(C) + o(1)$ follows from $\sum_{p\leq t,\,p\in C} 1/p = \delta(C)\log\log t + b_C + o(1)$ by partial summation, and the prime powers add their convergent sum. □

The formula for c_C is Theorem 20.26 in other terms: by the Abelian theorem $c_C + \sum_{p\in C}\log p/(p(p-1)) = -\kappa_C$, minus the drift of the class C. The formula for b_C, its analogue for the weights $1/p$ with $\log L(1,\chi)$ in place of $\frac{L'}{L}(1,\chi)$, is (8.1), (8.2) and Theorem 8.3 of Bardestani and Freiberg [35], and the asymptotic $\prod_{p\leq x,\,p\in C}(1-1/p)\sim e^{-h_C(1)-\gamma_{\mathrm{E}}\delta(C)}(\log x)^{-\delta(C)}$ is (8.8) of the same theorem, and with an error term Theorem A of Arango Piñeros, Keliher and Keyes [20].

For the S_3 field of $x^3 - x - 1$, with $S = \{23\}$, the characters are the trivial one, the sign χ_{-23} and the two dimensional ρ, with $L(1,\chi_{-23}) = 3\pi/\sqrt{23} = 1.965202$, $L^{(23)}(1,\rho) = 0.352392$, $\frac{L'}{L}(1,\chi_{-23}) = -0.422121$ and $\frac{(L^{(23)})'}{L^{(23)}}(1,\rho) = 1.159587$ from PARI. The theorem gives $b_C = -0.243641$, -0.075520, 0.537179 and $c_C = -0.742978$, -0.593191, -0.132738 for the identity, the transpositions and the three cycles, against the values -0.243638, -0.075516, 0.537183 and -0.742913,

-0.593134, -0.132674 read from the primes below 10^7 in Table 24.1; the differences, below $7 \cdot 10^{-5}$, are the tails beyond 10^7, and with the primes below 10^9 they fall below $3 \cdot 10^{-5}$. The constants $h_C(1)$ are -0.33954, -0.32024 and 0.61533. What the theorem leaves open is the sign of the deficit of the identity class as the field varies, Open Problem 24.2.

24.5.2 *Atoms at zero: a class is empty with probability of order* $(\sigma-1)^{\delta(C)}$

For $Q = 1_C$ the probability in Theorem 20.21 that no prime factor of N_σ has a label where $Q > 0$ is the probability that the class C is empty, of exact order $(\sigma-1)^{\delta(C)}$ by the paragraph after that theorem. The next theorem gives its constant, and the atom at zero that it describes bounds from below the Kolmogorov distance between the law of a share and its Beta limit.

THEOREM 24.8 (atoms at zero and a lower bound for the rate; (i) refines Theorem 20.21, (iii) after Serre [504]). *Let G be nontrivial, let $L_C(\sigma)$ be the mass of the class C, as in Corollary 20.20, let A_σ be the event that N_σ has no prime factor outside S, and let $\mathcal{P}_\sigma(C)$ be the share of C, equal to $L_C(\sigma)/\sum_{C'} L_{C'}(\sigma)$ off A_σ and to $\delta(C)$ on A_σ, as in Theorem 24.3.*

(i) $\mathbb{P}(L_C(\sigma)=0) = e^{-P_C(\sigma)} = e^{-h_C(1)}(\sigma-1)^{\delta(C)}\big(1+O(\sigma-1)\big)$, *with $h_C(1)$ as in Theorem 24.7.*

(ii) The Kolmogorov distance between the law of $\mathcal{P}_\sigma(C)$ and its limit $\mathrm{Beta}(\delta(C), 1-\delta(C))$ *is at least*

$$\mathbb{P}(L_C(\sigma)=0) - \mathbb{P}(A_\sigma) = e^{-h_C(1)}(\sigma-1)^{\delta(C)}\big(1+o(1)\big),$$

hence at least of order $(\sigma-1)^{\delta_{\min}}$ for the smallest class, $\delta_{\min} = \min_C \delta(C)$.

(iii) For the uniform random integer below x the same exponent appears in the density of the integers free of C: the proportion of $n \le x$ with no prime factor in C equals $K_C(\log x)^{-\delta(C)}(1+O(1/\log x))$, where

$$K_C = \frac{e^{\gamma_{\mathrm{E}}\delta(C)}}{\Gamma(1-\delta(C))}\lim_{t\to\infty}(\log t)^{\delta(C)}\prod_{p\le t,\,p\in C}\Big(1-\frac{1}{p}\Big) = \frac{e^{-h_C(1)}}{\Gamma(1-\delta(C))}\,:$$

the same constant $e^{-h_C(1)}$ governs the zeta integer, through $e^{-h_C(1)}(\sigma-1)^{\delta(C)}$, and the uniform integer, through $e^{-h_C(1)}(\log x)^{-\delta(C)}/\Gamma(1-\delta(C))$.

PROOF. (i) By Lemma 20.17 the exponents are independent and $\mathbb{P}(e_p=0) = 1-p^{-\sigma}$, so $\mathbb{P}(L_C(\sigma)=0) = \prod_{p\in C}(1-p^{-\sigma}) = \exp(-P_C(\sigma))$, and $P_C(\sigma) = \delta(C)\log\frac{1}{\sigma-1} + h_C(1) + O(\sigma-1)$ by the analyticity of h_C. (ii) The Beta law has no atom at 0, while $\mathcal{P}_\sigma(C)=0$ exactly when $L_C(\sigma)=0$ and A_σ fails, and $A_\sigma \subset \{L_C(\sigma)=0\}$; the distribution functions therefore differ at 0 by $\mathbb{P}(L_C(\sigma)=0) -$

$\mathbb{P}(A_\sigma)$, and $\mathbb{P}(A_\sigma) = \prod_{p\notin S}(1-p^{-\sigma}) = O(\sigma-1)$, which is $o((\sigma-1)^{\delta(C)})$ because $\delta(C) < 1$. (iii) is a case of Serre's theorem on frobenian sets [504], proved by Landau's method [345], which gives an expansion in powers of $1/\log x$; Wirsing's theorem [587] gives its leading term without the rate. For the multiplicative indicator f of the integers free of primes in C, whose mean value over the primes is $1-\delta$, with $\delta = \delta(C)$, Wirsing's theorem reads

$$\sum_{n\le x} f(n) \sim \frac{e^{-\gamma_{\mathrm{E}}(1-\delta)}\, x}{\Gamma(1-\delta)\log x} \prod_{p\le x,\, p\notin C} \Big(1-\frac{1}{p}\Big)^{-1},$$

and Mertens's third theorem turns the product into $e^{\gamma_{\mathrm{E}}}\log x \prod_{p\le x,\, p\in C}(1-1/p)$. The second form of K_C follows from Theorem 24.7, which gives

$$\sum_{p\le t,\, p\in C} \log\Big(1-\frac{1}{p}\Big) = -\delta\log\log t - h_C(1) - \gamma_{\mathrm{E}}\delta + o(1). \qquad \square$$

In the Kolmogorov distance of the shares the theorem gives a lower bound of order $(\sigma-1)^{\delta_{\min}}$, far above the order $(\sigma-1)\log\frac{1}{\sigma-1}$ at which the means converge (below); the matching upper bound is Open Problem 24.1. For the S_3 field, $\delta_{\min} = \frac{1}{6}$, and the identity class, the primes that split completely, is empty with probability 0.647, 0.444 and 0.303 at $\sigma = 1.01$, 1.001 and 1.0001, against 0.652, 0.444 and 0.303 from (i) with $h_C(1) = -0.33954$; the transpositions are empty with probability 0.137, 0.044, 0.014 and the three cycles with probability 0.117, 0.054, 0.025. At $\sigma = 1.0001$ the Kolmogorov distance between the law of the share of the identity class and its Beta limit is still at least 0.30. Since $\mathrm{Beta}(\delta, 1-\delta)$ puts mass about $\varepsilon^\delta \sin(\pi\delta)/(\pi\delta)$ on $[0,\varepsilon]$, the same atom forces only a Lévy distance of order $\sigma-1$, about 10^{-3} at $\sigma = 1.0001$. The means behave differently. The share means $\mathbb{E}[\mathcal{P}_\sigma(C);\, A_\sigma^c]$, computed as $\int_0^\infty (-\phi_C')\prod_{C'\neq C}\phi_{C'}\,dt$ from the Laplace transforms $\phi_C(t) = \mathbb{E}e^{-tL_C(\sigma)}$, are 0.1427, 0.4871, 0.3599 at $\sigma = 1.01$ and 0.1628, 0.4986, 0.3375 at $\sigma = 1.001$. For the identity class they fall short of $\delta(C)(1-\mathbb{P}(A_\sigma))$, the value the mean would have if the shares were exactly Dirichlet off A_σ, by 0.022 and 0.0037. The deficit has an exact form: since $\phi_C(t) = e^{P_C(s)-P_C(\sigma)}$ with $s = \sigma + t(\sigma-1)$, by Lemma 20.17, it equals $(\sigma-1)\int_\sigma^\infty (h_C'(s) - \delta(C)h'(s))e^{h(s)-h(\sigma)}\,ds/(s-1)$ with $h = \sum_{C'} h_{C'}$, which is $(\kappa_C - \delta(C)\sum_{C'}\kappa_{C'})(\sigma-1)\log\frac{1}{\sigma-1} + O(\sigma-1)$, the drifts of Theorem 20.26 entering through $h_C'(1) = \kappa_C$; for the identity class the coefficient is 0.620. So the Euler Kronecker constants govern the convergence of the means, and the constants $h_C(1)$ govern the atom at zero, which keeps the Kolmogorov distance of the shares at least of order $(\sigma-1)^{\delta(C)}$.

The exponent is the same one that governs the counting function of the uniform integers: the number of integers below x with no prime factor in C is of order $x(\log x)^{-\delta(C)}$, which for the inert primes of $\mathbb{Q}(i)$ is the order $x/\sqrt{\log x}$ of Landau's count of sums of two squares [345]. Among the 400000 integers below 10^7 of

Section 24.4, 79.3 percent have no prime factor in the identity class, 19.8 percent none among the transpositions and 15.7 percent none among the three cycles; and the asymptotic of (iii) with the constants $K_C = e^{-h_C(1)}/\Gamma(1-\delta(C))$ gives 78.3, 19.4 and 15.8 percent, the remaining gaps, 1.1, 0.4 and -0.1 points, being of the size of the $1/\log x$ term at $\log x = 16.1$.

24.5.3 The Ewens parameter of an L function

THEOREM 24.9 (Ewens parameters are pole orders; after [23, 174]). *Let $D(s) = \sum_n a(n)n^{-s} = \exp\bigl(\sum_p \sum_{m\ge1} w_m(p)p^{-ms}/m\bigr)$ be an Euler product with $w_m(p) \ge 0$ bounded, and suppose the labels (θ_p) are equidistributed with respect to μ on the compact space Y and $w_1(p) = w(\theta_p)$ outside a finite set of primes, for a continuous $w \ge 0$ on Y with $\int w\,d\mu > 0$. Let N_σ^D have the law $a(n)n^{-\sigma}/D(\sigma)$, $\sigma > 1$. Then the prime exponents of N_σ^D are independent, $e_p = \sum_m mM(p^m)$ with independent Poisson variables of means $w_m(p)p^{-m\sigma}/m$, and as $\sigma \downarrow 1$ the labelled factorisation converges in law to the Gamma random measure with shape measure $w\mu$, the normalised factorisation to the Dirichlet process with base measure $w\mu$, and the ranked normalised sizes to* $\mathrm{PD}(\vartheta)$ *with*

$$\vartheta = \int_Y w\,d\mu = \lim_{s\downarrow1} \frac{\log D(s)}{\log \frac{1}{s-1}},$$

the order of the pole of D at $s = 1$ when D continues meromorphically. In particular: for $D = L(s, \rho\otimes\check\rho)$, $w = |\chi_\rho|^2$ and $\vartheta = \langle\rho,\rho\rangle$, the Gamma shape of Theorem 20.28; for $D = \zeta_K$ with $K/\mathbb{Q}$ of degree n and Galois closure of group G acting on the n embeddings, $w(g)$ is the number of fixed points of g and $\vartheta = 1$ by Burnside's lemma.

PROOF. The generating function of the exponents factorises over primes as $\prod_p \exp(\sum_m w_m(p)p^{-m\sigma}(t_p^m - 1)/m)$, which is the Poisson representation. The proof of Theorem 20.18 uses only this structure and Lemma 20.19 with the weights $w_1(p)$: the measure $(\sigma-1)\sum_p w(\theta_p)(\log p)p^{-\sigma}\delta_{((\sigma-1)\log p,\,\theta_p)}$ converges to $e^{-x}dx \otimes w\mu$ because w is continuous and (θ_p) is equidistributed, and the prime powers $m \ge 2$ contribute a bounded expected number of atoms, of total expected rescaled mass $O(\sigma-1)$, which vanish in the limit. The rest is Theorem 24.3 with μ replaced by $w\mu$ and the Gamma total mass of shape $\int w\,d\mu$, and the ranked sizes follow as in the proof of Theorem 24.4, with the intensity $x^{-1}e^{-x}dx \otimes w\mu$ in place of $\vartheta x^{-1}e^{-x}dx \otimes \mu$. The identification of ϑ with the pole order extends Lemma 20.12, which gives it for Artin L functions: $\log D(s) = \sum_p w(\theta_p)p^{-s} + O(1)$ and $\sum_p w(\theta_p)p^{-s} = (\int w\,d\mu + o(1))\log\frac{1}{s-1}$ by equidistribution and partial summation, which is all the limit needs. For $\rho\otimes\check\rho$ the coefficients $w_m(p) = |\chi_\rho(\mathrm{Frob}_p^m)|^2$ are nonnegative (at a ramified prime, the average of $|\chi_\rho|^2$ over the inertia coset); for ζ_K the Euler factor at p is $\prod_i(1-p^{-f_i s})^{-1}$ over the primes above p, so $w_m(p) = \sum_{f_i\mid m} f_i$ is the number of fixed points of Frob_p^m on the embeddings, and $\int w\,d\mu = 1$ is Burnside's count of orbits. □

Theorem 24.4 is the case $w_m(p) = \vartheta$, that is $D = \zeta^\vartheta$. Theorem 24.9 also says something the unlabelled theory does not: the colouring of the Rankin Selberg integer is not Haar measure but $|\chi_\rho|^2\mu$. For the two dimensional representation of S_3, $|\chi_\rho|^2$ is 4, 0, 1 on the identity, the transpositions and the three cycles, so the base measure puts mass $\frac{2}{3}$ on the identity and $\frac{1}{3}$ on the three cycles: primes with Frobenius a transposition occur in the Rankin Selberg integer only to even powers, and their share tends to zero. We checked this on the Laplace transforms of the three class masses, computed from the primes below 10^7 with the prime number theorem for the tail: at $\sigma = 1.001$ and $\lambda = 4$ they are 0.346, 0.9975 and 0.583 against the limits $(1+\lambda)^{-2/3} = 0.342$, 1 and $(1+\lambda)^{-1/3} = 0.585$.

24.6 Discussion: what Bondesson's theory contributes

The generalized gamma convolutions were introduced by Thorin [546, 547] to prove that the Pareto and lognormal laws are infinitely divisible, and the theory was built by Bondesson [73]. The structure theorem (an infinitely divisible law on $[0,\infty)$ is a GGC if and only if its Lévy density is $k(x)/x$ with k completely monotone, equivalently $-\varphi'/\varphi$ is a Stieltjes function with no pole at the origin, integrable at 0), self decomposability, unimodality and closure under convolution and weak limits were recalled after Definitions 13.2 and 14.2 (unimodality of self decomposable laws is Yamazato's theorem [590]), and Stieltjes inversion is Theorem 23.4. Since much of Part IV takes place in this class, it is worth saying which further results of Bondesson do arithmetic work here and which do not.

Mass and shape. If the drift vanishes and the Thorin mass $m = \tau((0,\infty))$ is finite, the density is $f(x) = x^{m-1}g(x)/\Gamma(m)$ with g completely monotone and $g(0+) = \exp\int \log t\,\tau(dt)$, finite when $\int_1^\infty \log t\,\tau(dt) < \infty$, for instance when τ has bounded support; in every case $m = \sup\{\alpha : f(x)x^{1-\alpha} \to 0 \text{ as } x \downarrow 0\}$ is the exponent of small values ([73], Chapter 4; [280], Section 1.3). If moreover $m \le 1$, the density is completely monotone, that is the law is a mixture of exponential laws. The Thorin mass of a Frobenius functional is $\mu(Q > 0)$ (Theorem 20.21), and the atom at zero of Theorem 24.8, of order $(\sigma-1)^{\delta(C)}$, has the same exponent as the small values of the limit law, $m = \delta(C)$; the order of the pole of $L(s, \rho\otimes\check\rho)$ at $s = 1$ is the Thorin mass of the Gamma limit of Theorem 20.28 and the Ewens parameter of Theorem 24.9. Every class has $\delta(C) \le 1$, so the limit law $\mathrm{Gamma}(\delta(C))$ of the mass L_C of a Chebotarev class is a mixture of exponentials, and its normalisation is the Beta law of Theorem 24.3; only when $\langle\rho,\rho\rangle \ge 2$, that is for reducible ρ, does the Gamma limit of Theorem 20.28 leave the mixtures of exponentials, and the loss of complete monotonicity is a probabilistic reading of reducibility.

Hyperbolic complete monotonicity. A density f on $(0,\infty)$ is HCM if, for every $u > 0$, the function $f(uv)f(u/v)$ is completely monotone as a function of $v + 1/v$. Every HCM law is a GGC; the HCM class is closed under products, quotients

and powers x^q, $|q| \geq 1$, of independent variables; and it contains the gamma, lognormal and beta of the second kind laws ([73], Chapter 5) and the positive stable laws of index at most $\frac{1}{2}$ [79]. This book does not use it. The factorisation of a GGC of finite Thorin mass as a Gamma variable times an independent Dirichlet mean, in Section 24.2 and Theorem 23.11, is Cifarelli and Regazzini's identity [105] and holds for every Dirichlet mean; HCM variables enter it through Bondesson's theorem that a Gamma variable times an independent HCM variable is a GGC, which makes every HCM variable with a finite negative moment of order m a Dirichlet mean ([280], Proposition 1.5). Whether the Thorin variable of Theorem 23.10 is HCM is not known to us.

Closure and arithmetic. Multiplying L functions, that is taking direct or isobaric sums, convolves the corresponding laws and adds Thorin measures. The passage $\sigma \downarrow 1$ does not come from closure under weak limits: at $\sigma > 1$ a Frobenius functional is a compound Poisson law with an atom at 0, not a GGC ([485], Section 27), and its GGC limit, with Thorin measure the image of Haar measure restricted to $\{Q > 0\}$ under $1/Q$, comes from the Poisson structure of the prime exponents (Theorem 20.18). By a later theorem of Bondesson [74] the product of two independent GGC variables is again a GGC; this multiplicative closure has no arithmetic counterpart that we know of, and Open Problem 24.3 asks for one. The prime side need not stay in the class under the operations of the Langlands programme: the sine squared lemma (Lemma 16.7) turns a Lévy measure into a Thorin density when the logarithmic coefficients are nonnegative, as for $L(s, \pi \times \tilde{\pi})$, a sufficient condition and not a necessary one; symmetric powers and base change have signed coefficients in general, their transported densities can be signed (Section 16.4.3), and the Thorin representation of such an L function to the right of its zeros comes from its Hadamard product (Theorem 16.8), that is from its automorphy.

The limits. Bondesson's theorems are structural, not arithmetic. They translate conjectures and theorems (GRH into membership of the class, Theorem 14.8; irreducibility into Thorin mass one; Chebotarev's theorem into the base measure of a Dirichlet process), but they do not produce the automorphic partner of a Galois representation or the Galois partner of an automorphic one. For a self dual automorphic π the Thorin measure at a basepoint $\frac{1}{2} + a$ with $a \geq \frac{1}{2}$ exists whatever the zeros, each zero $\beta + i\gamma$ entering through Cauchy noise of scale $a - \beta$ (Theorem 23.9), and at $a = 0$ its existence is GRH. For an Artin representation ρ the Thorin representation at a basepoint $\alpha > 1$ is unconditional when the logarithmic coefficients are nonnegative, as for $\rho \otimes \check{\rho}$ (Proposition 20.34); for ρ itself (with $\check{\rho}$ when ρ is not self dual) it is known when ρ is automorphic (Theorem 16.8(iii)), and at the centre it is the holomorphy of $\Lambda(s, \rho)$ together with GRH, for self dual ρ the complete monotonicity of the Galois heat trace (Theorem 20.30). The Thorin measures of a Galois representation and of an automorphic representation are each

computable to any precision, and reciprocity is the assertion that they coincide, which no closure property will supply. What the theory supplies is a class of positive objects in which the two sides can be compared.

24.6.1 Open problems

OPEN PROBLEM 24.1 (The upper bound for the rate). Theorem 24.8 shows that, in the Kolmogorov distance of the shares, the Dirichlet process limit is approached no faster than $(\sigma-1)^{\delta_{\min}}$. Prove the matching upper bound: that the Kolmogorov distance between the law of $\mathcal{P}_\sigma(C)$ and $\mathrm{Beta}(\delta(C), 1-\delta(C))$ is $O((\sigma-1)^{\delta(C)})$, and that, for the Euclidean distance between share vectors, the Wasserstein distance between the laws of $(\mathcal{P}_\sigma(C))_C$ and of its Dirichlet limit is $O((\sigma-1)\log\frac{1}{\sigma-1})$, the order at which the means converge.

OPEN PROBLEM 24.2 (Signs of the Mertens deficits). Theorem 24.7 expresses b_C and c_C through $L(1,\chi)$ and $\frac{L'}{L}(1,\chi)$. For the splitting field of a polynomial of degree n with group S_n, the deficit of the identity class in Theorem 24.5(iii) is $b_{\{1\}} - \delta(\{1\})B' = \lim_{t\to\infty}\sum_{p\le t,\, p\notin S}(1_{\{1\}}(\mathrm{Frob}_p) - \delta(\{1\}))/p$; it is -0.280 for $x^3 - x - 1$, whose smallest completely split prime is 59. A prime $p < n$ splits completely only if it divides the index of $\mathbb{Z}[\alpha]$, with α a root of the polynomial, and this happens: in Dedekind's field of $x^3 - x^2 - 2x - 8$ the prime 2 splits completely and adds $\frac{5}{12}$ to the sum. Read from the primes below 10^7, the deficit is 0.224 for Dedekind's field and 0.245 for $x^3 - x + 15$, in which 3 and 5 split completely, so both signs occur already with group S_3. Determine the distribution of the sign of the deficit over the fields with group S_n ordered by discriminant, and its sign along a family such as $x^n - x - 1$. This is the analogue, for the weights $1/p$, of Open Problem 20.8; Problem 24.2 computes the constants.

OPEN PROBLEM 24.3 (An arithmetic product). By Bondesson [74] the product of two independent GGC variables is a GGC, so the product of independent copies of the limits of $\Xi_\sigma(Q_1)$ and $\Xi_\sigma(Q_2)$ in Theorem 20.21 is a GGC, with a Thorin measure τ_{Q_1,Q_2}. Multiplying Dirichlet series realises the additive closure, the Frobenius functional of the product of two independent random integers being the sum of two independent ones, and tensor products, as in Rankin Selberg convolution, give the functional of Q_1Q_2. Find the arithmetic operation, if any, that realises the multiplicative closure: a representation or an L function whose Frobenius functional has Thorin measure τ_{Q_1,Q_2}; and compute τ_{Q_1,Q_2} for $Q_1 = Q_2 = \mathrm{tr}^2$ on S_3. For a finite group the Thorin measure of a limiting Frobenius functional is finitely supported and that of such a product is not, so the operation, if any, leads to an infinite label space, such as a Sato Tate group.

24.7 NOTES

The Poisson Dirichlet law is Kingman's [317], and the Gamma process description is in his book ([318], Chapter 9). For permutations the limit goes back to Shepp and Lloyd [513]; for prime factorisations it is Billingsley's theorem [61], found independently by Knuth and Trabb Pardo [325], and Tenenbaum [542] proved the rate $1/\log x$ as an expansion in powers of $1/\log x$ whose term in $1/\log x$ is, for each prime factor separately, Knuth and Trabb Pardo's. Donnelly and Grimmett [162] proved Billingsley's theorem by size biased sampling, which gives the uniform share of Theorem 24.6; Arratia, Barbour and Tavaré [22] treat integers and permutations together; Pitman's lectures [437] cover stick breaking, size biased sampling, the GEM law and the Chinese restaurant process (Sections 3.1, 3.2 and 4.1). The route through the zeta random integer is Lloyd's [370], as recalled in Chapter 20.

Dickman's function is from [156], the Erdős Kac theorem from [177], the Sathe Selberg formula from [484, 495], and the variance of $\omega(n)$ with its constant from Diaconis, Mosteller and Onishi [153]. Mertens's theorems are in [393], which already treats arithmetic progressions; for progressions the constants are in Williams [582] and Languasco and Zaccagnini [354]. For a Chebotarev class, Bardestani and Freiberg ([35], Section 8, Theorem 8.3) give b_C in the form of Theorem 24.7, and Arango Piñeros, Keliher and Keyes ([20], Theorem A) give the asymptotic of $\prod_{p\le x,\, p\in C}(1-1/p)$ with an error term; the expression of c_C through $\frac{L'}{L}(1,\chi)$ is the drift of Theorem 20.26. Dawsey ([130], Theorem 2) proved that the class of the largest prime factor is equidistributed with error $O(\exp(-a'_K(\log x)^{1/3}))$, extending Alladi's theorem for progressions [15]; this is Theorem 24.5(i). The counts $\omega_C(n)$ are asymptotically independent Poisson variables with means $\delta(C)\log\log x + b_C$ (Delange [135], Section 6.5.3, for two classes; Tenenbaum [544], for any number of classes, with a rate; see Ford [195]), which refines Theorem 24.5(iii). The density of the integers free of the primes of a class, with an expansion in powers of $1/\log x$ and the constant K_C, is a case of Serre's theorem on frobenian sets [504], proved by Landau's method [345]; Odoni [427] treated norms, and Wirsing's theorem [587] gives the asymptotic without the rate.

The Dirichlet process is Ferguson's ([186], Section 3, Theorem 1, for the posterior and Section 4 for the normalised Gamma process); the urn for its samples is Blackwell and MacQueen's [66], and the Ewens sampling formula [181] for its clusters is due to Antoniak [19], with Hoppe's urn [264]. The identity $\mathbb{E}(1+\lambda\int Q\,dP)^{-\vartheta} = \exp(-\vartheta\int\log(1+\lambda Q)\,d\mu)$ for the means of a Dirichlet process with base measure $\vartheta\mu$ is Cifarelli and Regazzini's [105], the Markov Krein identity when $\vartheta = 1$ (Diaconis and Kemperman [152], Corollary 2 and Section 3), in the form of James, Roynette and Yor ([280], Section 1.4, (33)) that Theorem 23.11 uses. For integers weighted by $d_\vartheta(n)$, and more generally by multiplicative

weights of mean ϑ on the primes, the Poisson Dirichlet law with parameter ϑ is due to Elboim and Gorodetsky, who also prove the Erdős Kac law with mean and variance $\vartheta \log\log x$ in this generality ([174], Theorem 1.1; for weights bounded on the primes, such as d_ϑ, that law is in [544], Corollary 2.5), and for normed arithmetic semigroups whose primes have density ϑ, such as the integers composed of primes in a set of progressions or the Gaussian integers counted by norm, the Poisson Dirichlet law is due to Arratia, Kochman and Miller ([23], Theorem 2 and Section 5). For the ζ^ϑ random integer itself, the unlabelled limits of Theorem 24.4 are Sun's ([530], Propositions 4.2 and 4.3).

The generalized gamma convolutions are Thorin's [546, 547]; Bondesson's monograph [73] has the structure, density and HCM results recalled in Section 24.6 (Chapters 3 to 5), and the product closure is [74]. Self decomposable laws are unimodal by Yamazato's theorem [590], and the positive stable law of index α is HCM if and only if $\alpha \leq \frac{1}{2}$ (Bosch and Simon [79]).

What is ours, as far as we know, is the labelled form at $\sigma \downarrow 1$: the normalised labelled factorisation of the zeta random integer converges to the Dirichlet process with base measure μ, Haar measure for Frobenius labels (Theorem 24.3, whose part (iii) for finitely many labels is Corollary 20.20(ii)), with its versions for the ζ^ϑ integer and for Euler products whose logarithm has nonnegative coefficients, of base measures $\vartheta\mu$ and $w\mu$ (Theorems 24.4 and 24.9); the expansion of the size biased colouring with the constant $\gamma_{\mathrm{E}}\delta(C) - \kappa_C$ (Theorem 24.5(ii)) and the independence of class and share (Theorem 24.6); and the reading of the classical constant $e^{-h_C(1)}$ as the atom at zero of the zeta integer, whose order is that of Theorem 20.21, with the Kolmogorov bound it gives (Theorem 24.8(i) and (ii)). Each follows from Theorem 20.18, Ferguson's construction and the theorems above, and through them everything known about PD(ϑ), the Ewens sampling formula and Dirichlet means applies to the Frobenius labelled factorisations of Chapter 20.

24.8 Exercises

1. Show that the stick breaking sequence $W_k = (1-U_1)\cdots(1-U_{k-1})U_k$ has $\mathbb{E}W_k = 2^{-k}$, $\mathbb{E}W_k^2 = 3^{-k}$ and $\sum_k \mathbb{E}W_k^2 = \frac{1}{2}$. Deduce $\mathbb{E}\sum_k V_k^2 = \frac{1}{2}$ for PD(1), the case $\vartheta = 1$ of $1/(1+\vartheta)$, and check it on the random integers of Section 24.1.

2. Compute Dickman's ϱ on $[1,3]$ from its delay equation and verify $\varrho(2) = 1 - \log 2 = 0.30685$ and $\varrho(3) = 0.04861$.

3. Show that for a Dirichlet process P with base measure μ on the conjugacy classes of a finite group and $Q = 1_C$ the Dirichlet mean $\int Q\,dP$ is Beta$(\mu(C), 1-\mu(C))$, and that $E\int Q\,dP$, with E standard exponential and

independent of P, is then Gamma with shape $\mu(C)$; compare with Corollary 20.20(i).

4. Derive the Ewens sampling formula with parameter ϑ from the Chinese restaurant process, and show that for $\vartheta = 1$ it gives the law of the cycle type of a uniform permutation.

5. Prove that $\sum_C c_C = c - \sum_{p \text{ ram.}} \log p/p$, which is -1.4689 for $x^3 - x - 1$, where $c = \lim(\sum_{p \le t} \log p/p - \log t) = -\gamma_{\mathrm{E}} - \sum_p \log p/(p(p-1)) = -1.33258$, and compute $\sum_p \log p/(p(p-1))$ to six digits.

6. For the Rankin Selberg random integer of $\rho \oplus \rho$, with $\vartheta = 4$, simulate the ranked normalised prime factors at $\sigma = 1.01$ and compare the mean of the largest with the PD(4) value (for PD(ϑ), $\mathbb{E}V_1 = \vartheta^{-1} \int_0^\infty (1 - e^{-\vartheta E_1(x)})\, dx$, with $E_1(x) = \int_x^\infty e^{-t} t^{-1} dt$).

7. Show that the number of prime factors of $N_\sigma^{(\vartheta)}$ counted with multiplicity has mean $\vartheta \sum_p p^{-\sigma}/(1 - p^{-\sigma})$ and is asymptotically normal as $\sigma \downarrow 1$ with mean and variance $\vartheta \log \frac{1}{\sigma - 1}$; this is the Erdős Kac theorem for the Ewens family.

24.9 Open computational problems

The scripts named below are in `pd/code/` (Section B.7).

Computational Problem 24.1 (Colourings at scale). Objective: for uniform random integers below x, compute the three colourings of Theorem 24.5 (the class of the largest prime factor, the class of a size biased prime factor, and the pooled classes of the distinct prime factors) and the proportion of integers with no prime factor in each class, and compare them with the predictions: Haar measure up to Dawsey's error $O(\exp(-a'_K(\log x)^{1/3}))$ for the largest factor, Theorem 24.5(ii) and (iii) for the other two and Theorem 24.8(iii) for the proportions, with the constants computed through Theorem 24.7. Objects and scale: the S_3 field of $x^3 - x - 1$, the S_4 field of $x^4 - x - 1$, and as an abelian control the cyclic cubic field of conductor 7; 10^6 integers at each of $x = 10^9$, 10^{12} and 10^{15}. Route: `pd1.py` and `pd2.py`, with the sieve replaced by `factorint` in PARI/GP for the sampled integers; the class of a prime p is the factorisation pattern of the polynomial modulo p (`factormod`), which determines it in S_3 and S_4, and p mod 7 in the cyclic field; $L(1, \chi)$ and $\frac{L'}{L}(1, \chi)$ from Dedekind zeta functions of subfields (`lfuncreate` and `lfundiv`, as `op_pd.py` does for S_3), checked against prime sums to 10^9. Deliverable: for each field, class and x, the three shares and the proportions of integers with an empty class, with binomial standard errors, against the predictions with and without their second order terms, and the scale at which each colouring is within one standard error of Haar measure. Why it matters: at $x = 10^7$ the distinct prime factors put the identity class of $x^3 - x - 1$ at less than half its Haar measure, and

Theorem 24.5(iii) says the deficit decays only like $1/\log\log x$; the table shows how much of Chebotarev's theorem each colouring sees at computable scales, and the identity class of the S_4 field, of Haar measure $\frac{1}{24}$, is a sharper test than that of S_3.

COMPUTATIONAL PROBLEM 24.2 (Mertens constants of symmetric fields). Objective: compute the Mertens constants b_C and c_C of Theorem 24.5 and the constants $h_C(1)$ and K_C of Theorem 24.8 for every class of the splitting field of $x^n - x - 1$, and determine the sign of the deficit $b_{\{1\}} - \delta(\{1\})B'$ of the identity class that Open Problem 24.2 asks about. Objects and scale: $3 \le n \le 7$; primes up to 10^{10}. Route: confirm with `polgalois` that the group is S_n; form the partial sums over the primes up to 10^{10} by a segmented sieve, each prime not dividing the discriminant of $x^n - x - 1$ classed by the factorisation pattern of $x^n - x - 1$ modulo p, which is its cycle type and so determines its class in S_n, and the other unramified primes by `idealprimedec`; take the tail beyond 10^{10} from the prime number theorem for the field, with an error estimate under GRH; for $n \le 5$ obtain an independent value through Theorem 24.7, the characters of S_n being integer combinations of the permutation characters of Young subgroups (Young's rule), whose L functions are Dedekind zeta functions of subfields of the splitting field (`lfuncreate` and `lfundiv` in PARI/GP). Deliverable: tables of b_C, c_C, $h_C(1)$ and K_C for all classes with error estimates, the two routes compared where both apply, and the deficit of the identity class against n. Why it matters: Open Problem 24.2 rests on S_3 alone, and Problem 20.3, which computes drifts of Galois fields of degree at most 10, does not reach the splitting fields of degree 24 and more. The deficit is -0.280 for $x^3 - x - 1$, whose first completely split prime is 59, while a completely split prime p adds $(1 - \delta(\{1\}))/p$ to the sum that defines it, $\frac{5}{12}$ for the prime 2 of Dedekind's field in Open Problem 24.2; so its sign along $x^n - x - 1$ is the question.

COMPUTATIONAL PROBLEM 24.3 (Laws of class shares near $\sigma = 1$). Objective: compute the laws of the class masses $L_C(\sigma)$ of the zeta random integer and of the shares $\mathcal{P}_\sigma(C)$, the Kolmogorov distance of each share from its limit $\mathrm{Beta}(\delta(C), 1 - \delta(C))$, and the means of the shares, and fit the rates in $\sigma - 1$ that Theorem 24.8 and Open Problem 24.1 concern. Objects and scale: the S_3 field of $x^3 - x - 1$ and the S_4 field of $x^4 - x - 1$; $10^{-5} \le \sigma - 1 \le 10^{-1}$. Route: the Laplace transforms $\mathbb{E}e^{-tL_C(\sigma)}$ from the Euler products over the primes up to 10^{10} and the prime number theorem for the field beyond, or from the L functions through the decomposition of P_C at the start of Section 24.5, as `op_pd.py` does; off A_σ the event $\mathcal{P}_\sigma(C) \le u$ is $(1-u)L_C(\sigma) \le u\sum_{C' \ne C} L_{C'}(\sigma)$, whose probability follows by Gil Pelaez inversion of characteristic functions, checked by Monte Carlo at $\sigma - 1 = 10^{-2}$. Deliverable: for each class, the Kolmogorov distance and the defect of the mean against $\sigma - 1$ on log log axes, with fitted exponents and constants, compared with $e^{-h_C(1)}(\sigma-1)^{\delta(C)}$ for the distance and $(\sigma-1)\log\frac{1}{\sigma-1}$ for the means. Why it matters: Theorem 24.8 proves the lower bound only; a fitted exponent equal to $\delta(C)$ with a constant

near $e^{-h_C(1)}$, and a defect of the means of order $(\sigma - 1) \log \frac{1}{\sigma - 1}$, are the numerical content of Open Problem 24.1, which the zero probability exponents of Problem 20.3 do not decide.

CHAPTER 25

DEGREE ONE: DIRICHLET CHARACTERS, HECKE CHARACTERS AND COMPLEX MULTIPLICATION

Reciprocity for GL_1 is a theorem. Over $\mathbb{Q}$ it is the theorem of Kronecker and Weber, that every one dimensional representation of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ is a Dirichlet character; over a number field K it is Artin's reciprocity law, that the one dimensional representations are the characters of ray class groups, which Hecke extended to characters of infinite order with entire L functions. Everything the book does for GL_2 under hypotheses it can do for GL_1 without them, and this chapter does it: Dirichlet's theorem on primes in progressions becomes a Dirichlet process on the classes mod q, with Mertens constants computed from $L(1,\chi)$ and with Landau's theorem on integers without prime factors in a class as the atom at zero of Chapter 24; Hecke's theorem on the angles of Gaussian primes becomes a Dirichlet process on the circle; and the Thorin measure of a degree one L function is checked against its zeros to $2 \cdot 10^{-8}$.

The chapter then treats the one case of GL_2 reciprocity that reduces to GL_1, complex multiplication. Deuring's theorem says that the L function of an elliptic curve with complex multiplication by K is the L function of a Hecke character of K, so that a degree two Euler product over $\mathbb{Q}$ is a degree one Euler product over K. The Frobenius angles of such a curve are the arguments of Gaussian integers, their law is half uniform and half an atom, and the Rankin Selberg subordinator of Chapter 20 has an exact property that no curve without complex multiplication shares: seen from K it is the sum of two independent copies of itself seen from $\mathbb{Q}$, so that its Gamma shape is 2 rather than 1. This is the Gamma shape test of Theorem 20.28 detecting complex multiplication, and it is proved, not conjectured. In the logic of the book the chapter belongs early, after the Galois representations of Chapter 3 and before the equidistribution of Part II; it is placed here, after Chapter 24, because it is the one place where the reader can see every object of Chapters 20 to 24 with nothing left to prove.

25.1 DIRICHLET CHARACTERS AND THE RANDOM INTEGER MOD q

Fix $q \geq 3$. The cyclotomic field $\mathbb{Q}(\zeta_q)$ has Galois group $(\mathbb{Z}/q\mathbb{Z})^\times$, and the Frobenius class of a prime $p \nmid q$ is the residue of p modulo q. Chebotarev's theorem for

this field is Dirichlet's theorem: each residue class a prime to q receives primes with density $1/\varphi(q)$. The Haar measure of Chapter 20 is counting measure on $(\mathbb{Z}/q\mathbb{Z})^\times$ divided by $\varphi(q)$, and the characters of the group are the Dirichlet characters modulo q.

Let N_σ be the zeta distributed random integer of Chapter 20, $\mathbb{P}(N_\sigma = n) = n^{-\sigma}/\zeta(\sigma)$, with the primes dividing q removed as in Section 24.5, and colour each prime factor by its residue class. Everything in Chapters 20 and 24 applies with $G = (\mathbb{Z}/q\mathbb{Z})^\times$ and every class of size one.

THEOREM 25.1 (Dirichlet's theorem in law). *Let $q \geq 3$ and $\sigma \downarrow 1$.*

(i) The vector of shares $(P_\sigma(a))_{a \in (\mathbb{Z}/q\mathbb{Z})^\times}$, $P_\sigma(a)$ the fraction of $\log N_\sigma$ carried by the prime factors congruent to a, converges in law to the Dirichlet law with all parameters $1/\varphi(q)$; the labelled factorisation converges to Ferguson's Dirichlet process with base measure the uniform law on $(\mathbb{Z}/q\mathbb{Z})^\times$.

(ii) For every a the probability that N_σ has no prime factor congruent to a is $e^{-h_a(1)}(\sigma-1)^{1/\varphi(q)}(1+o(1))$, where $e^{h_a(1)}$ is the regular part at $s=1$ of $\prod_{p\equiv a}(1-p^{-s})^{-1}(s-1)^{1/\varphi(q)}$.

(iii) (Mertens for progressions.) $\sum_{p\leq x,\, p\equiv a} p^{-1} = \varphi(q)^{-1}\log\log x + M(q,a) + o(1)$ with

$$M(q,a) = \frac{1}{\varphi(q)}\Big[M - \sum_{p\mid q}\frac{1}{p} + \sum_{\chi\neq\chi_0}\bar\chi(a)S(\chi)\Big],$$

where $S(\chi) = \log L(1,\chi) - \sum_{k\geq 2}\sum_p \chi(p)^k(kp^k)^{-1}$, $M = 0.2614972\ldots$ the Mertens constant, and

$$h_a(1) = M(q,a) - \frac{\gamma}{\varphi(q)} + \sum_{k\geq 2}\sum_{p\equiv a}\frac{1}{kp^k}.$$

PROOF. (i) is Theorem 24.3 with μ uniform on $(\mathbb{Z}/q\mathbb{Z})^\times$, the equidistribution hypothesis being Dirichlet's theorem. (ii) is Theorem 24.8 with $\delta(C) = 1/\varphi(q)$. (iii) is Theorem 24.7 written out: by orthogonality of characters,

$$\sum_{p\equiv a} p^{-s} = \frac{1}{\varphi(q)}\sum_\chi \bar\chi(a)\sum_{p\nmid q}\chi(p)p^{-s},$$

and $\sum_p \chi(p)p^{-s} = \log L(s,\chi) - \sum_{k\geq 2}\sum_p \chi(p)^k(kp^{ks})^{-1}$ for $\operatorname{Re} s > 1$, continuous up to $s=1$ for $\chi \neq \chi_0$ since $L(1,\chi)\neq 0$. The principal character contributes $\sum_{p\leq x} p^{-1} - \sum_{p\mid q} p^{-1}$ and Mertens' theorem. The formula for $h_a(1)$ compares two products. By (iii),

$$\prod_{p\equiv a,\, p\leq x}\Big(1-\frac{1}{p}\Big)^{-1} = \exp\Big(\frac{\log\log x}{\varphi(q)} + M(q,a) + \sum_{k\geq 2}\sum_{p\equiv a}\frac{1}{kp^k} + o(1)\Big),$$

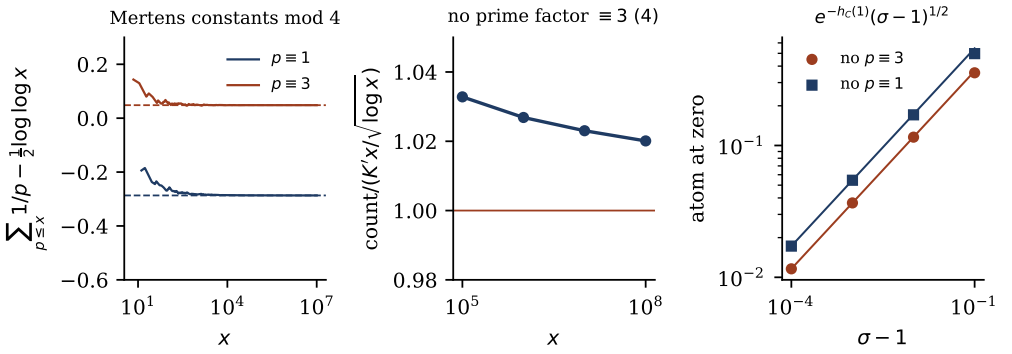


Figure 25.1: Dirichlet's theorem in law for $q = 4$. Left: the running sums $\sum_{p \le x,\, p \equiv a} p^{-1} - \frac{1}{2}\log\log x$ for the two classes, converging to the Mertens constants $M(4,1) = -0.2867$ and $M(4,3) = 0.0482$ computed from $L(1,\chi_4)$ (dashed). Middle: the number of integers below x with no prime factor congruent to 3 modulo 4, divided by $K'x/\sqrt{\log x}$ with $K' = 0.654259$; the excess decays like $1/\log x$. Right: the probability that the zeta distributed integer has no prime factor in a class modulo 4, against $e^{-h_C(1)}(\sigma-1)^{1/2}$.

while $\prod_{p \equiv a}(1-p^{-s})^{-1} \sim e^{h_a(1)}(s-1)^{-1/\varphi(q)}$ by definition. Mertens' theorem $\prod_{p \le x}(1-p^{-1})^{-1} \sim e^{\gamma}\log x$ and $\zeta(s) \sim (s-1)^{-1}$ relate the two scales, and the class carries the fraction $1/\varphi(q)$ of the exponent, which is the factor $e^{-\gamma/\varphi(q)}$, as in the proof of Theorem 24.7. □

For $q = 4$ the only nonprincipal character is χ_4 with $L(1,\chi_4) = \pi/4$, and

$$M(4,1) = -0.286742, \qquad M(4,3) = 0.048239,$$

against the sums over the primes below 10^7, which give -0.286728 and 0.048235, the difference being the tail beyond 10^7. The atoms at zero are

$$\mathbb{P}(\text{no factor } \equiv 3) \sim e^{-h_3(1)}\sqrt{\sigma-1}, \qquad e^{-2h_3(1)} = 2L(1,\chi_4)\prod_{p \equiv 3\,(4)}(1-p^{-2}),$$

with $e^{-h_3(1)} = 1.159643$ and $e^{-h_1(1)} = 2/e^{-h_3(1)} = 1.724668$, since the two classes together exhaust the odd primes and $\prod_{p\ \mathrm{odd}}(1-p^{-\sigma}) \sim 2(\sigma-1)$. The exact values $\prod_{p \equiv 3}(1-p^{-\sigma})/\sqrt{\sigma-1}$ are 1.1255, 1.1563, 1.1593, 1.1596 at $\sigma - 1 = 10^{-1}, \dots, 10^{-4}$ (Figure 25.1, right). The closed forms come from $\zeta^{(2)}(s)L(s,\chi_4) = \prod_{p \equiv 1}(1-p^{-s})^{-2}\prod_{p \equiv 3}(1-p^{-2s})^{-1}$ and its companion with L in the denominator, which express each partial Euler product through ζ, $L(s,\chi_4)$ and a rapidly convergent product over squares.

The atom at zero has a classical counterpart for the uniform integer, due to Landau, which Theorem 24.8 identifies with its constant.

THEOREM 25.2 (Landau's theorem with Wirsing's constant). *The number of $n \le x$ all of whose odd prime factors are congruent to 1 modulo 4 is*

$$K' \frac{x}{\sqrt{\log x}}(1+o(1)), \qquad K' = \Big(\tfrac{1}{2} \prod_{p \equiv 3\,(4)} (1-p^{-2})\Big)^{1/2} = 0.654259\ldots$$

By comparison the number of $n \le x$ that are sums of two squares is $Kx/\sqrt{\log x}$ with the Landau Ramanujan constant $K = (2\prod_{p\equiv 3}(1-p^{-2}))^{-1/2} = 0.764224\ldots$, and $K'/K = \prod_{p\equiv 3\,(4)}(1-p^{-2}) = 0.856109$, the probability that an integer is not divisible by the square of any prime congruent to 3 modulo 4.

PROOF. The generating Dirichlet series is $F(s) = (1-2^{-s})^{-1}\prod_{p\equiv 1}(1-p^{-s})^{-1}$ and, by the identity above, $F(s)^2 = (1-2^{-s})^{-1}\zeta(s)L(s,\chi_4)\prod_{p\equiv 3}(1-p^{-2s})$, so $F(s) \sim C(s-1)^{-1/2}$ with $C^2 = 2L(1,\chi_4)\prod_{p\equiv 3}(1-p^{-2}) = \frac{\pi}{2}\prod_{p\equiv 3}(1-p^{-2})$. The coefficients of F are the indicator of the set, a multiplicative function with values in $\{0,1\}$ whose sum over primes $p \le x$ is $\frac{1}{2}x/\log x$, so Wirsing's theorem [587], in the form used in Theorem 24.8, gives the count $Cx(\log x)^{-1/2}/\Gamma(\frac{1}{2})$, and $C/\sqrt{\pi} = K'$. In the notation of Theorem 24.8, $K' = e^{-h_3(1)}/\Gamma(\frac{1}{2})$ with $e^{-h_3(1)} = 1.159643$ as computed above, since the integers counted are exactly those with no prime factor in the class of 3. The sums of two squares have generating series $\big(\zeta(s)L(s,\chi_4)(1-2^{-s})^{-1}\prod_{p\equiv 3}(1-p^{-2s})^{-1}\big)^{1/2}$, which gives K [345]. □

A sieve to 10^8 gives counts 19915, 180751, 1667197 and 15550333 at $x = 10^5,\ldots,10^8$, which are 1.033, 1.027, 1.023 and 1.020 times $K'x/\sqrt{\log x}$ (Figure 25.1, left); the excess is the secondary term $c/\log x$ of the Selberg Delange expansion, with $c \approx 0.37$.

25.2 RECIPROCITY FOR GL_1: CLASS FIELD THEORY AS MEASURABILITY

The statements of reciprocity in degree one are these.

THEOREM 25.3 (Kronecker, Weber; Artin; Hecke). *(i) Every continuous one dimensional representation of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ factors through $\mathrm{Gal}(\mathbb{Q}(\zeta_q)/\mathbb{Q})$ for some q, hence is a Dirichlet character.*

(ii) For a number field K and a modulus $\mathfrak{f}$, the Artin map $\mathfrak{p} \mapsto \mathrm{Frob}_\mathfrak{p}$ induces an isomorphism from the ray class group modulo $\mathfrak{f}$ onto $\mathrm{Gal}(K_\mathfrak{f}/K)$, $K_\mathfrak{f}$ the ray class field, and every abelian extension of K lies in some $K_\mathfrak{f}$. Hence the one dimensional representations of $\mathrm{Gal}(\overline{K}/K)$ are the characters of ray class groups.

(iii) A Hecke character of K is a character ψ of the group of fractional ideals prime to $\mathfrak{f}$ for which there is a continuous character ψ_∞ of $(K \otimes \mathbb{R})^\times$ with $\psi((\alpha)) = \psi_\infty(\alpha)^{-1}$, or $\psi_\infty(\alpha)$ in the convention of this book, for $\alpha \equiv 1$ modulo $\mathfrak{f}$. Its L function $L(s,\psi) = \prod_{\mathfrak{p}\nmid\mathfrak{f}}(1-\psi(\mathfrak{p})N\mathfrak{p}^{-s})^{-1}$ has meromorphic continuation and a functional equation, and is entire unless ψ is a power of the norm.

The characters of finite order in (iii) are the ray class characters of (ii); the characters of infinite order, Hecke's Grössencharaktere [249], are the automorphic representations of $\mathrm{GL}_1(\mathbb{A}_K)$ that are not Artin representations (the algebraic ones among them are ℓ adic characters of $\mathrm{Gal}(\overline{K}/K)$, which is how ψ appears in Section 25.4), and they are the subject of Sections 25.3 and 25.4.

The probabilistic reading is Proposition 20.16: a Frobenius random variable $p \mapsto \rho(\mathrm{Frob}_p)$ on the probability space of primes is measurable with respect to the congruence σ algebra, the one generated by the events $\{p \equiv a\ (q)\}$, if and only if ρ is abelian. Theorem 25.3(i) is the statement that for one dimensional ρ this is always so. The Dirichlet process of Theorem 25.1 is the normalised Gamma process of Theorem 20.18 seen through that σ algebra, and its base measure is uniform because the Frobenius elements of an abelian extension of $\mathbb{Q}$ are equidistributed in the Galois group, which is Dirichlet's theorem again.

25.3 Hecke characters of $\mathbb{Q}(i)$ and the angles of Gaussian primes

Let $K = \mathbb{Q}(i)$, with ring of integers $\mathbb{Z}[i]$, units $\{\pm 1, \pm i\}$, and class number one. A prime $p \equiv 1$ modulo 4 splits as $p = \pi\bar{\pi}$ with $\pi = a + bi$, $p = a^2 + b^2$; a prime $p \equiv 3$ stays prime, with residue degree two; and $2 = -i(1+i)^2$ ramifies. Among the four associates of π exactly one is *primary*, $\pi \equiv 1$ modulo $(2+2i)$, which for $a + bi$ means a odd, b even and $a + b \equiv 1$ modulo 4; the primary generator is the one used by Gauss for biquadratic reciprocity. Both π and $\bar{\pi}$ are primary. Write $\theta_p = |\arg \pi| \in [0, \pi]$, which is the same for both, and note that the argument of any generator is $\pm\theta_p$ plus a multiple of $\pi/2$, so $4\theta_p$ modulo 2π is well defined up to sign.

For $k \in \mathbb{Z}$ the function $\lambda_k((\alpha)) = (\alpha/|\alpha|)^{4k}$ is well defined on all nonzero ideals, since the units are fourth roots of unity, and it is a Hecke character of K with trivial conductor and infinity type $z \mapsto (z/|z|)^{4k}$. Its L function $L(s, \lambda_k)$ is entire for $k \neq 0$ and does not vanish on $\mathrm{Re}\, s = 1$ (Hecke).

THEOREM 25.4 (Hecke [249]). *The angles $4\theta_p$, $p \equiv 1$ modulo 4, are equidistributed modulo 2π: for every continuous f on the circle, $\frac{1}{\pi_1(x)}\sum_{p\le x,\, p\equiv 1} f(4\theta_p) \to \frac{1}{2\pi}\int_0^{2\pi} f$. More precisely the arguments of the primary Gaussian primes, both π and $\bar{\pi}$ for each p, are equidistributed on the circle, so that θ_p is equidistributed on $[0, \pi]$.*

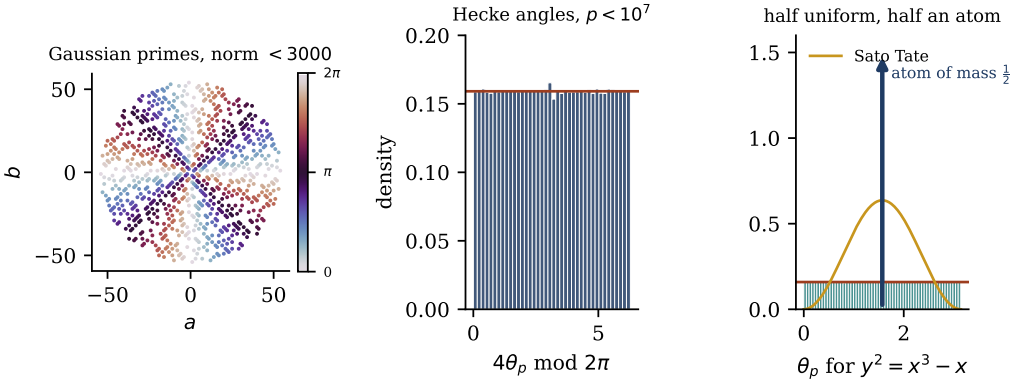


Figure 25.2: Left: the Gaussian primes of norm below 3000 with their four associates, coloured by the Hecke angle $4\theta_p$. Middle: the angles $4\theta_p$ of the Gaussian primes below 10^7 against the uniform law. Right: the Frobenius angles θ_p of $y^2 = x^3 - x$, uniform on $[0, \pi]$ for the split primes and equal to $\pi/2$ for the inert primes, against the Sato Tate density of a curve without complex multiplication.

PROOF. By Weyl's criterion it suffices that $\sum_{p \le x, p \equiv 1} e^{4ik\theta_p} = o(x/\log x)$ for $k \neq 0$. The sum is $\sum_{N\mathfrak{p} \le x} \lambda_k(\mathfrak{p})$ over the degree one primes of K, up to the $O(\sqrt{x})$ primes of degree two, and the prime ideal theorem for $L(s, \lambda_k)$, that is the nonvanishing on $\operatorname{Re} s = 1$ with the Wiener Ikehara theorem applied to $-L'/L$, gives $\sum_{N\mathfrak{p} \le x} \lambda_k(\mathfrak{p}) = o(x/\log x)$. For the primary arguments the same argument applies to the Hecke characters $\lambda'_m((\alpha)) = (\alpha/|\alpha|)^m$, $\alpha \equiv 1$ modulo $(2+2i)$, which are well defined for every m because 1 is the only unit congruent to 1 modulo $(2+2i)$, and whose L functions are entire and nonvanishing on $\operatorname{Re} s = 1$ for $m \neq 0$; the sum over the degree one primes of K runs over both π and $\bar{\pi}$, and the symmetric law of the pair $\{\theta_p, -\theta_p\}$ is uniform on the circle if and only if θ_p is uniform on $[0, \pi]$. □

The angles of the 332180 Gaussian primes below 10^7 have discrepancy $D = 0.00145$ from the uniform law, $\sqrt{n}D = 0.84$, and Weyl sums $|\frac{1}{n}\sum e^{4ik\theta_p}|$ between $6 \cdot 10^{-4}$ and $1.1 \cdot 10^{-3}$ for $k \le 5$, against $n^{-1/2} = 1.7 \cdot 10^{-3}$ for independent uniform angles (Figure 25.2, left). Under GRH for the $L(s, \lambda_k)$ the number of Gaussian primes below x in an arc is its expected value up to $O(\sqrt{x}\log^2 x)$, and the Weyl sums are of the size of a random walk, as here.

Hecke's theorem is Chebotarev's theorem for an infinite extension: the characters λ_k do not factor through any finite Galois group, and the "Galois group" in which the Frobenius elements equidistribute is the circle, the connected component of the Sato Tate group of Section 25.4. The Gamma process of Theorem 20.18 applies to the random integer coloured by $4\theta_p$ with Y the circle and μ uniform, so that for a continuous $Q \ge 0$ on the circle the Frobenius func-

tional $\Xi_\sigma(Q) = (\sigma-1)\sum_{p\equiv 1} e_p \log p\, Q(4\theta_p)$ converges to a GGC with Thorin measure $\frac12$ times the image of the uniform law under $1/Q$, the factor $\frac12$ being the density of the split primes, and the labelled factorisation converges to a Dirichlet process whose base measure is $\frac12$ times the uniform probability on the circle. The Mertens constants are given by Theorem 24.7 with the $L(1,\lambda_k)$ in place of the $L(1,\chi)$: for a trigonometric polynomial $Q(y) = \sum_{|k|\le m} c_k e^{iky}$, $\sum_{p\le x,\, p\equiv 1} Q(4\theta_p)p^{-1} = \frac12 c_0 \log\log x + M(Q) + o(1)$ with $M(Q) = \frac12 c_0 M(4,1) + \frac12 \sum_{0<|k|\le m} c_k S(\lambda_k)$, $S(\lambda_k) = \log L(1,\lambda_k) - \sum_{j\ge 2}\sum_{\mathfrak{p}} \lambda_k(\mathfrak{p})^j (jN\mathfrak{p}^j)^{-1}$; for the indicator of an arc the Fourier series of M converges only conditionally and is better handled by smoothing.

25.4 Complex multiplication

Let E be the curve $y^2 = x^3 - x$, of conductor 32, with the automorphism $(x,y)\mapsto(-x,iy)$ giving complex multiplication by $\mathbb{Z}[i]$. Let ψ be the Hecke character of $K=\mathbb{Q}(i)$ of conductor $\mathfrak{f} = (2+2i)$ with $\psi((\alpha)) = \alpha$ for $\alpha \equiv 1$ modulo $\mathfrak{f}$; it is well defined because 1 is the only unit congruent to 1 modulo $\mathfrak{f}$, and its infinity type is the identity. At the inert prime (p), $p\equiv 3$ modulo 4, the primary generator is $-p$, since $4\in\mathfrak{f}$, so $\psi((p)) = -p$.

Theorem 25.5 (Deuring [149]). *$L(E,s) = L(s,\psi)$. Explicitly, for $p\equiv 1$ modulo 4 with primary $\pi = a+bi$, $a_p(E) = \pi+\bar\pi = 2a$; for $p\equiv 3$, $a_p(E)=0$; and the Euler factor at $p\equiv 3$ is $(1+p^{1-2s})^{-1}$ on both sides.*

The identity $a_p = 2a$ with the primary normalisation was checked for all 332180 primes $p\equiv 1$ modulo 4 below 10^7 against the point counts of PARI; it holds without exception, and the other associates $-\pi$, $i\pi$, $-i\pi$ fail at every prime. The general theorem is that for $E/\mathbb{Q}$ with complex multiplication by an imaginary quadratic K, $L(E,s) = L(s,\psi)$ for a Hecke character ψ of K of infinity type the identity and conductor $\mathfrak{f}$ with $N_E = |d_K| N\mathfrak{f}$; here $32 = 4\cdot 8$.

Deuring's theorem is reciprocity for GL_2 reduced to GL_1: the two dimensional representation ρ_E of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ is induced from the one dimensional representation ψ of $\mathrm{Gal}(\overline{\mathbb{Q}}/K)$, which is abelian, so that ρ_E is *dihedral*, and its automorphic partner is the automorphic induction of ψ, the theta series $\sum_{\mathfrak{a}} \psi(\mathfrak{a}) q^{N\mathfrak{a}}$, a weight two form of level 32 (Hecke). In the language of Chapter 20, reciprocity for E is the almost sure equality of $p\mapsto a_p(E)$ and $p\mapsto \pi+\bar\pi$, which Theorem 25.5 proves prime by prime.

Proposition 25.6 (the angle law of a CM curve). *Write $a_p(E) = 2\sqrt{p}\cos\theta_p$ with $\theta_p\in[0,\pi]$. Then θ_p is the angle of Section 25.3 for split p and $\theta_p = \pi/2$ for inert p, and the law of θ_p is $\frac12\,\mathrm{Unif}[0,\pi] + \frac12\delta_{\pi/2}$. The moments are $\mathbb{E}\cos^2\theta_p = \frac14$ and $\mathbb{E}\cos^4\theta_p = \frac{3}{16}$, against $\frac14$ and $\frac18$ for the Sato Tate law; below 10^7 the sample values are*

0.24988 *and* 0.18738.

PROOF. For split p, $\cos\theta_p = a/\sqrt{p} = \cos\arg\pi$ by Theorem 25.5, and $|\arg\pi|$ is uniform on $[0,\pi]$ by the second part of Theorem 25.4. The two classes have density $\frac{1}{2}$ each by Dirichlet. □

The law in Proposition 25.6 is the law of the trace of a Haar distributed element of the normaliser $N(\mathrm{U}(1))$ of the maximal torus $\mathrm{U}(1)$ in $\mathrm{SU}(2)$: the identity component contributes the uniform angle, the other component consists of matrices of trace zero. This is the Sato Tate group of a CM curve, and Chapter 29 identifies it from the moments.

Now take the Rankin Selberg subordinator X_σ of Section 20.6, with Laplace transform $D^S(\sigma+\lambda)/D^S(\sigma)$, $D^S(s) = L^S(s, \rho\otimes\check\rho)$, built over $\mathbb{Q}$, and the same subordinator X_σ^K built over K from the restriction of ρ to $\mathrm{Gal}(\overline{\mathbb{Q}}/K)$, with $D_K^S(s) = L_K^S(s, \rho|_K\otimes\check\rho|_K)$ an Euler product over the primes of K.

THEOREM 25.7 (complex multiplication doubles the Gamma shape; this book). *Let $E/\mathbb{Q}$ be an elliptic curve, ρ its normalised ℓ adic representation, K an imaginary quadratic field and χ_K its character.*

(i) $D_K^S(s) = D^S(s)\,L^S(s, \rho\otimes\check\rho\otimes\chi_K)$, *the partial products being over the primes not dividing $N_E d_K$.*

(ii) If E has complex multiplication by K then $D_K^S(s) = D^S(s)^2$ exactly, so that X_σ^K has the law of the sum of two independent copies of X_σ for every $\sigma > 1$, and $(\sigma-1)X_\sigma^K$ converges to the Gamma law of shape 2.

(iii) If E has no complex multiplication by K then $(\sigma-1)X_\sigma^K$ converges to the standard exponential law, the Gamma shape being 1.

PROOF. (i) Restriction followed by induction multiplies by the regular representation of $\mathrm{Gal}(K/\mathbb{Q})$: $\mathrm{Ind}_K^{\mathbb{Q}}\mathrm{Res}_K(\rho\otimes\check\rho) = (\rho\otimes\check\rho)\otimes(1\oplus\chi_K)$, and the L function of an induced representation is the L function over the smaller field of the representation induced from. At the level of Euler factors this is the statement that a prime p of $\mathbb{Q}$ split in K contributes two identical factors over K, while an inert p contributes one factor at p^2, which equals the product of the factors of $\rho\otimes\check\rho$ and $\rho\otimes\check\rho\otimes\chi_K$ at p.

(ii) If E has complex multiplication by K then $a_p = 0$ for every inert p, so $\mathrm{tr}\,\rho(\mathrm{Frob}_p)\chi_K(p) = \mathrm{tr}\,\rho(\mathrm{Frob}_p)$ for all $p\nmid N_E d_K$, and $\rho\otimes\chi_K \cong \rho$ by Brauer and Nesbitt, both being semisimple. Hence $\rho\otimes\check\rho\otimes\chi_K \cong \rho\otimes\check\rho$, the two factors in (i) coincide, and $D_K^S = (D^S)^2$. A squared Laplace transform is the transform of a sum of two independent copies. By Theorem 20.28 and its proof, which use only the order of the pole of D^S at $s=1$, the Gamma shape is twice the shape over $\mathbb{Q}$, which is 1 because ρ is irreducible.

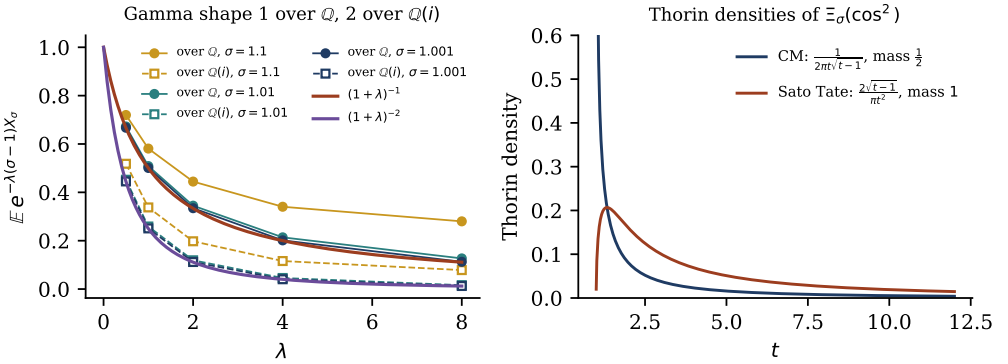


Figure 25.3: Complex multiplication doubles the Gamma shape. Left: the Laplace transforms of $(\sigma-1)X_\sigma$ for $y^2 = x^3 - x$ over $\mathbb{Q}$ and over $\mathbb{Q}(i)$ at three values of σ, against the exponential and the Gamma(2) limits of Theorem 25.7. Right: the Thorin densities of the limiting Frobenius functional $\Xi_\sigma(\cos^2)$ for a CM curve, the arcsine density of mass $\frac{1}{2}$, and for a curve without complex multiplication, the image of the Sato Tate law, of mass 1.

(iii) Over K the Gamma shape is the order of the pole of D_K^S at 1, which by (i) is $1+m$, m the order of the pole of $L(s, \rho \otimes \check{\rho} \otimes \chi_K)$. By modularity and the theory of Rankin and Selberg, $L(s, \rho \otimes \check{\rho} \otimes \chi_K) = L(s, f \times (f \otimes \chi_K))$ for the newform f of E, and this has a pole at $s = 1$ if and only if $f \otimes \chi_K = f$, that is $\rho \otimes \chi_K \cong \rho$, which forces $a_p = 0$ for all inert p of K, a set of density $\frac{1}{2}$. For a curve without complex multiplication the primes with $a_p = 0$ have density zero (Serre), and for a curve with complex multiplication by a field $K' \neq K$ they are, up to finitely many, the primes inert in K', which exclude the primes split in K' and inert in K, a set of density $\frac{1}{4}$. So $f \otimes \chi_K = f$ forces complex multiplication by K, hence $m = 0$ and the shape is 1. □

For the curve $y^2 = x^3 - x$ the Laplace transforms were computed exactly from $D^S(s) = \zeta^{(2)}(s) L(s+1, \mathrm{Sym}^2 E)$, the symmetric square from PARI, whose Euler factor at 2 is trivial; this identity is the decomposition $\rho \otimes \check{\rho} = 1 \oplus \chi_4 \oplus \mathrm{Ind}\, \tilde{\psi}^2$ together with $\mathrm{Sym}^2 \rho = \mathrm{Ind}\, \tilde{\psi}^2 \oplus \chi_4$. At $\sigma = 1.001$ the transform of $(\sigma-1)X_\sigma$ at $\lambda = \frac{1}{2}, 1, 2, 4, 8$ is 0.6672, 0.5009, 0.3345, 0.2014, 0.1127 against $(1+\lambda)^{-1} = 0.6667$, 0.5, 0.3333, 0.2, 0.1111, and over K the squares 0.4452, 0.2509, 0.1119, 0.0406, 0.0127 against $(1+\lambda)^{-2}$. The approach is at the rate $\sigma - 1$ of the Euler Kronecker drift (Theorem 20.26): the deviations at $\lambda = 1$ are 0.0810, 0.0086 and 0.0009 at $\sigma - 1 = 10^{-1}, 10^{-2}, 10^{-3}$ (Figure 25.3, left).

Theorem 25.7 says what the Gamma shape of Chapter 20 measures when the base field changes: it counts the irreducible constituents with multiplicity, and

restriction to K splits ρ exactly when ρ is induced from K. For a CM curve the restriction is $\tilde{\psi} \oplus \tilde{\psi}^c$, two distinct characters, and the shape is $1^2 + 1^2 = 2$; a curve without complex multiplication by K stays irreducible over K and keeps shape 1. The test is unconditional because both pole orders are theorems.

Finally the Frobenius functionals. Colour the prime factors of N_σ by $y_p = 4\theta_p$ on the circle for split p and by a point $\{\text{in}\}$ for inert p, so $Y = S^1 \sqcup \{\text{in}\}$ with $\mu = \frac{1}{2}\text{Unif} + \frac{1}{2}\delta_{\text{in}}$. Theorem 20.21 gives, for continuous $Q \geq 0$ on Y, the GGC limit of $\Xi_\sigma(Q)$ with Thorin measure the image of μ under $1/Q$ on $\{Q > 0\}$. Two cases are instructive. For $Q = 1_{\text{in}}$ the limit is the Gamma law of shape $\frac{1}{2}$, with Thorin measure $\frac{1}{2}\delta_1$, and the atom at zero $e^{-h_3(1)}(\sigma - 1)^{1/2}$ is the Landau phenomenon of Theorem 25.2. For $Q = \cos^2\theta_p = a^2/p$, a quarter of the squared normalised trace, the limit has Thorin measure $\frac{1}{2}$ times the image of the uniform law on $[0, \pi]$ under $\theta \mapsto 1/\cos^2\theta$, that is the measure $\frac{dt}{2\pi t\sqrt{t-1}}$ on $(1, \infty)$, of total mass $\frac{1}{2}$ (Figure 25.3, right): the Frobenius functional $\Xi_\sigma(\cos^2)$ of a CM curve is asymptotically a GGC with an explicit arcsine Thorin density. For a curve without complex multiplication the same functional has the Thorin density $\frac{2}{\pi t^2}\sqrt{t-1}$ of total mass one on $(1, \infty)$, the image of the Sato Tate law.

25.5 The Thorin measure of a degree one L function

Let χ be a primitive real character modulo q and $\kappa \in \{0, 1\}$ its parity, so that the completed function $\Lambda(s, \chi) = (q/\pi)^{s/2}\Gamma(\frac{s+\kappa}{2})L(s, \chi)$ satisfies $\Lambda(s, \chi) = \Lambda(1 - s, \chi)$ with root number $+1$. Put $\Xi_\chi(w) = \Lambda(\frac{1}{2} + w, \chi)$, an even entire function of order one, and write its zeros as $\pm i\gamma_k$, with multiplicity, choosing one γ_k from each pair; only γ_k^2 enters below.

THEOREM 25.8 (Thorin form in degree one). *Suppose $L(\frac{1}{2}, \chi) \neq 0$. Then $\Xi_\chi(\sqrt{\lambda})/\Xi_\chi(0) = \prod_k (1 + \lambda/\gamma_k^2)$ for $\lambda \geq 0$, the product converging absolutely, and GRH for χ holds if and only if $\Xi_\chi(0)/\Xi_\chi(\sqrt{\lambda})$ is the Laplace transform of a GGC, in which case the Thorin measure is $\sum_k \delta_{\gamma_k^2}$ and the GGC is $\sum_k E_k/\gamma_k^2$ with E_k independent standard exponentials. For every basepoint $\frac{1}{2} + a$ with $a \geq \frac{1}{2}$ the shifted representation of Theorem 16.8 holds unconditionally, with the Thorin density V_a positive.*

PROOF. This is Theorem 14.8 for $n = 1$, which Section 14.2 admits; the proof reads: Ξ_χ is even and entire of order one, so Hadamard's product has the form $e^{bw}\prod(1 + w^2/\gamma_k^2)$ with $b = 0$ by evenness, and $\Xi_\chi(0) \neq 0$ by hypothesis. The nonvanishing hypothesis holds for χ_4, where $L(\frac{1}{2}, \chi_4) = 0.6677$, and is Chowla's conjecture for real χ in general; a zero of order $2m$ at $w = 0$ is removed by dividing by w^{2m} and changes nothing else. The last assertion is Theorem 16.8(iii). □

For χ_4 the first zeros are at $\gamma = 6.020949, 10.243770, 12.988098, 16.342607, 18.291993$, and there are 3130 below height 3000, where the counting function

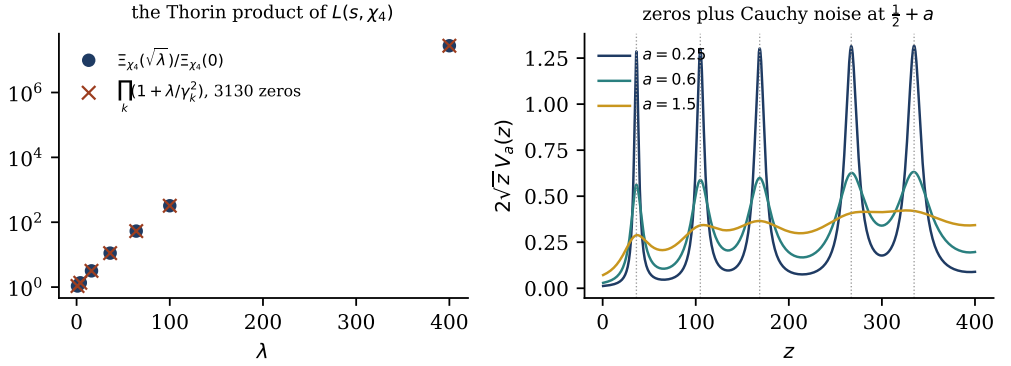


Figure 25.4: The Thorin measure of $L(s,\chi_4)$. Left: $\Xi_{\chi_4}(\sqrt{\lambda})/\Xi_{\chi_4}(0)$ against the product over the 3130 zeros below height 3000 with the tail correction; the two agree to between $2\cdot10^{-8}$ and $9\cdot10^{-6}$. Right: the shifted Thorin density $V_a(z)$ of Theorem 16.8, drawn as $2\sqrt{z}V_a(z)=\frac{1}{\pi}\sum_k\big[\frac{a}{a^2+(\sqrt{z}-\gamma_k)^2}+\frac{a}{a^2+(\sqrt{z}+\gamma_k)^2}\big]$, for $a=0.25$, 0.6 and 1.5; the vertical lines mark γ_k^2 for the first five zeros.

$\frac{T}{2\pi}\log\frac{4T}{2\pi e}$ predicts 3129.7. With these zeros and the tail $\int_T^\infty\log(1+\lambda/t^2)\,dN(t)$ from the smooth counting function, the product $\prod_k(1+\lambda/\gamma_k^2)$ agrees with $\Xi_{\chi_4}(\sqrt{\lambda})/\Xi_{\chi_4}(0)$ to relative error $2\cdot10^{-8}$ at $\lambda=1$, $3\cdot10^{-7}$ at $\lambda=16$, $2\cdot10^{-6}$ at $\lambda=100$ and $9\cdot10^{-6}$ at $\lambda=400$, where the ratio is $2.8\cdot10^7$; the residual error is the tail approximation, which grows with λ (Figure 25.4, left). The right panel of the figure shows the Thorin density of Theorem 16.8 at three basepoints $\frac{1}{2}+a$: by Theorem 23.9 it is the measure $\sum_k\delta_{\gamma_k}$ on the $\sqrt{z}$ axis smoothed by Cauchy noise of scale a, and as $a\downarrow0$ the peaks sharpen into the atoms $\delta_{\gamma_k^2}$ of Theorem 25.8.

The Galois face is different in degree one from degree two, and the difference is worth recording. The sine squared lemma (Lemma 16.7) represents $f(\alpha)/f(\alpha+\sqrt{\lambda})$ through a GGC when $\log f(\alpha+s)$ is the Laplace transform of a positive measure, as it is for ζ, for ζ_K, and for every Rankin Selberg L function. For $L(s,\chi)$ with χ real the logarithmic coefficients $\chi(p)^k\log p/k$ change sign, and the formula of the lemma produces a signed density: for small z it is $-\frac{\sqrt{z}}{4\pi}(\log L)'''(\alpha)=\frac{\sqrt{z}}{4\pi}\sum_n\Lambda_\chi(n)(\log n)^2n^{-\alpha}$, which for χ_4 equals -0.257, -0.156, -0.090 at $\alpha=1.1,1.5,2$, whereas for $\zeta_{\mathbb{Q}(i)}=\zeta L(\cdot,\chi_4)$ the same quantity is 2000, 15.8, 1.86. So on the Galois face a real character is not by itself a GGC object; its Dedekind zeta function is, and so is its completed L function at every basepoint $\alpha\geq1$ by Theorem 25.8, the gamma factor supplying the positivity that the Euler product lacks. This is the degree one instance of the remark in Chapter 23 that the two faces carry the same measure but not the same positivity.

25.6 What this adds to the programme

The chapter proves, in degree one, the three things that the rest of the book conjectures or assumes in degree two: equidistribution with the right base measure (Dirichlet, Hecke), reciprocity (Kronecker Weber, Artin, Deuring), and the Thorin structure at every basepoint to the right of the zeros. Two of its theorems are new in substance. Theorem 25.2, whose constant is a standard application of Landau and Wirsing, identifies the Wirsing constant of the atom at zero of Chapter 24 with Landau's constant, corrected by the probability $\prod_{p\equiv 3}(1-p^{-2})$ of squarefreeness in the class, and locates Landau's theorem inside the Dirichlet process. Theorem 25.7 shows that the Gamma shape of Theorem 20.28, which detects irreducibility over $\mathbb{Q}$, detects complex multiplication when the base field is changed, with an exact identity $D_K^S = (D^S)^2$ rather than an asymptotic one. It also shows what the Gamma shape cannot do: over $\mathbb{Q}$ a CM curve and a non CM curve both have shape 1, and the difference between them is visible only in the Thorin measure of the Frobenius functionals (an arcsine density of mass $\frac{1}{2}$ against a Sato Tate density of mass 1) or after base change.

25.7 Notes

The theorem of Kronecker and Weber, Artin's reciprocity law and Hecke's Grössencharaktere are in any account of class field theory; Neukirch [415] is the reference used here, and Tate's thesis [537] is the adelic form of Theorem 25.3(iii). Hecke's equidistribution theorem is [249]; its discrepancy under GRH is treated by Kubilius [334]. Mertens' theorem for progressions is due to Williams [582]. Landau's theorem on sums of two squares is [345]; the constant is often attributed jointly to Ramanujan, and Wirsing's theorem [587] is the general form. Deuring's theorem is [149]; Silverman's second volume [519] gives the proof and the computation of the Hecke character of $y^2 = x^3 - x$, and Ireland and Rosen [273] the elementary count $a_p = 2a$ with the primary normalisation. The Rankin Selberg criterion for $f \otimes \chi = f$ used in Theorem 25.7 is in [276]. Theorem 25.1(iii) is Williams' theorem, and (i) and (ii) are its Dirichlet process reading; Theorem 25.2 is classical apart from the identification with $h_3(1)$; Theorems 25.7 and 25.8 are new as formulated. Their ingredients are the theorems cited and Chapters 14, 20 and 24.

25.8 Exercises

1. Show that among the four associates $\pm\pi, \pm i\pi$ of a Gaussian prime exactly two, π and $-\pi$, have odd real part, and that $a_p = 2\operatorname{Re}\pi$ holds for the primary one only. Determine the proportion of primes at which the normalisation $a \equiv 1$ modulo 4 alone gives the right sign.

2. Compute $M(8,a)$ for the four classes modulo 8 from $L(1,\chi)$ for the three nonprincipal characters, and check against the primes below 10^7.

3. Show that the number of $n \le x$ all of whose prime factors are congruent to 1 modulo q is asymptotic to $c_q x (\log x)^{1/\varphi(q)-1}$, and express c_q through the $L(1,\chi)$.

4. For E with complex multiplication by K, show that the Sato Tate group is $N(\mathrm{U}(1))$ and that the moments of $\cos\theta_p$ are $\mathbb{E}\cos^{2m}\theta_p = \frac{1}{2}\binom{2m}{m}4^{-m}$. Verify the values $\frac{1}{4}$ and $\frac{3}{16}$.

5. Show that for $Q = \cos^2(y/4)$ on the circle the image of the uniform law under $1/Q$ has density $1/(\pi t\sqrt{t-1})$ on $(1,\infty)$, and that the corresponding GGC has Laplace transform $\exp(-\frac{1}{2}\int_1^\infty \log(1+\lambda/t)\frac{dt}{\pi t\sqrt{t-1}})$. Compute this integral in closed form.

6. Prove Theorem 25.7(i) at the level of Euler factors for a split and for an inert prime.

7. Let χ be a real character with $L(\frac{1}{2},\chi) = 0$ of order $2m$. Show that $\Xi_\chi(\sqrt{\lambda})/(\lambda^m \Xi_\chi^{(2m)}(0)/(2m)!)$ is again $\prod(1+\lambda/\gamma_k^2)$ over the remaining zeros.

25.9 Open problems

Open Problem 25.1 (the discrepancy of Hecke angles). For the Gaussian primes below x the discrepancy of $4\theta_p$ satisfies $\sqrt{\pi_1(x)}D = 1.07$ at $x = 10^6$ and 0.84 at $x = 10^7$. Determine whether $\sqrt{\pi_1(x)}D$ has a limit law as $x \to \infty$ (the Kolmogorov law would be the Brownian bridge prediction, as in Open Problem 30.3 for Sato Tate), or whether the arithmetic of the $L(s,\lambda_k)$ forces a different law; the same question for the imaginary quadratic fields of class number one.

Open Problem 25.2 (Euler Kronecker constants of Hecke characters). The rate of convergence of the Dirichlet process of Section 25.3 is governed, for the means, by the constants $L'/L(1,\lambda_k)$ in the same way that Theorem 20.26 is governed by Euler Kronecker constants. Compute these constants for $\mathbb{Q}(i)$ and $k \le 20$, determine the sign pattern of the deficits of the arcs, and find the analogue of Theorem 24.8 for the circle, where no arc is a finite union of classes.

Open Problem 25.3 (an effective Gamma shape test). Theorem 25.7 distinguishes complex multiplication by K through a limit as $\sigma \downarrow 1$. By Faltings' theorem finitely many a_p determine ρ up to isomorphism, so finitely many primes decide whether $\rho \otimes \chi_K \cong \rho$. Find an explicit bound, in terms of N_E and d_K, on the primes needed so that the empirical Laplace transform of $(\sigma-1)X_\sigma^K$ at a suitable σ separates

shape 1 from shape 2 with a proof, and compare with the bound from the effective Chebotarev theorem.

COMPUTATIONAL PROBLEM 25.1 (class number one fields). Objective: for each of the nine imaginary quadratic fields of class number one, compute the Hecke angles of the split primes below 10^9, their discrepancy, and the Mertens constants of the arcs of length $\pi/2$, with the predicted constants from $L(1, \lambda_k)$. For the CM curves of conductor 27, 32, 36, 49, 121 and 256 verify Theorem 25.7(ii) from the exact Laplace transforms at $\sigma = 1.0001$. Starting point: `g1.py`, `g3.py` and `g4.py`.

CHAPTER 26

MAASS FORMS, THE SELBERG TRACE FORMULA AND BROWNIAN MOTION ON THE MODULAR SURFACE

The holomorphic modular forms of Chapter 2 are half of the automorphic spectrum of $\mathrm{SL}_2(\mathbb{Z})$. The other half consists of the Maass forms, the eigenfunctions of the hyperbolic Laplacian on the modular surface $X = \mathrm{SL}_2(\mathbb{Z})\backslash\mathbb{H}$, and the book has so far mentioned them only in passing. They matter to its programme for two reasons. The first is that the Galois representations on the Maass side are the even ones, which the gamma factor of the Galois heat trace of Chapter 20 tells apart from the odd ones (Section 26.5): an even two dimensional Artin representation corresponds, when it corresponds to anything, to a Maass form of eigenvalue exactly $\frac{1}{4}$, and the even icosahedral case is the one two dimensional case of reciprocity that is open. The second is that the Selberg trace formula, the fundamental identity of the Maass side, is a heat trace identity of precisely the shape of the Galois heat trace of Chapter 20, with closed geodesics in the place of primes, and its probabilistic content is Brownian motion on X: the sum of $e^{-t\lambda_j}$ over the Laplace spectrum is the return density of Brownian motion integrated over the surface, decomposed by the homotopy class of the loop the path has made.

This chapter sets out the spectral theory of X in the language of the book. The modular surface is a probability space with the normalised hyperbolic area, Brownian motion on it is ergodic with that law, and the heat kernel decomposes over Maass forms and Eisenstein series. The Selberg trace formula is stated for compact quotients, where it is clean, and read through Brownian loops. Two theorems transcribe the book's dictionary to this setting: Selberg's eigenvalue conjecture is the complete monotonicity of the heat trace, exactly as GRH is the complete monotonicity of the Galois heat trace, and is a theorem for $\mathrm{SL}_2(\mathbb{Z})$ and within $(\frac{7}{64})^2$ of a theorem in general; and the closed geodesics, ordered by length, form a Beurling system of primes to which the Poisson Dirichlet theory of Chapter 24 and the sine squared lemma of Chapter 16 apply without change, so that the geodesic Euler product $Z(s+1)/Z(s)$ is represented unconditionally through a GGC at every basepoint to the right of 1. The chapter also records an obstruction: Weyl's law places the Laplace spectrum just outside the Thorin class at the centre, because $\sum_j r_j^{-2}$ diverges, whereas $\sum_\gamma \gamma^{-2}$ over the zeros of

an L function converges. The last section states and checks the Eichler Selberg trace formula, the holomorphic analogue that Chapter 9 uses, and reads it as the moment identity behind Serre's vertical Sato Tate law.

26.1 THE MODULAR SURFACE AS A PROBABILITY SPACE

Let $\mathbb{H} = \{z = x + iy : y > 0\}$ with the hyperbolic metric $ds^2 = (dx^2 + dy^2)/y^2$, area element $d\mu = dx\,dy/y^2$, and Laplacian

$$\Delta = -y^2\Big(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\Big),$$

a nonnegative operator in the sign convention of this chapter. The group $\mathrm{SL}_2(\mathbb{R})$ acts by isometries, and $\Gamma = \mathrm{SL}_2(\mathbb{Z})$ acts with the fundamental domain $F = \{|x| \le \frac{1}{2},\ |z| \ge 1\}$ of area $\pi/3$. The modular surface $X = \Gamma\backslash\mathbb{H}$ carries the probability measure $\frac{3}{\pi}d\mu$; this is the probability space of the chapter. Under it the height y has a heavy tail, $\mathbb{P}(y > Y) = 3/(\pi Y)$ for $Y \ge 1$, so that y has no mean: the cusp carries a vanishing fraction of the area at great height, but not a summable one.

A *Maass form* is a Γ invariant eigenfunction of Δ of moderate growth; a Maass cusp form has vanishing constant term in its Fourier expansion, hence is square integrable and decays exponentially in the cusp. The spectral theorem for $L^2(X)$ (Selberg) says that $L^2(X)$ is the orthogonal sum of the constants, of the closed span of the Maass cusp forms u_j with eigenvalues $\lambda_j = \frac{1}{4} + r_j^2$, $0 < \lambda_1 \le \lambda_2 \le \cdots$, and of the continuous spectrum $[\frac{1}{4}, \infty)$ spanned by the Eisenstein series $E(z, \frac{1}{2} + ir)$, $r \in \mathbb{R}$. For $\mathrm{SL}_2(\mathbb{Z})$ every cuspidal eigenvalue exceeds $\frac{1}{4}$, so every r_j is real, and the first few are

$$r_j = 9.533695,\ 12.173008,\ 13.779751,\ 14.358510,$$
$$16.138073,\ 16.644259,\ 17.738563,\ 18.180918,$$

with $\lambda_1 = 91.141$; the first form is odd and the first even form, invariant under $x \mapsto -x$, is the third. The eigenvalues are known to many digits [254, 371] and have been certified rigorously [76]; the first hundred, 61 odd and 39 even, are shown in Figure 26.1. Weyl's law for X, due to Selberg, is

$$N(T) = \#\{j : r_j \le T\} = \frac{T^2}{12} - \frac{2T}{\pi}\log T + O(T), \tag{26.1}$$

the leading term being $\frac{\mathrm{area}(X)}{4\pi}T^2$, so that the cusp forms are as numerous as the eigenfunctions of a compact surface of the same area: the continuous spectrum takes up none of Weyl's count. This is special to congruence groups and is the analytic reason the spectral theory of X is arithmetic. The first hundred

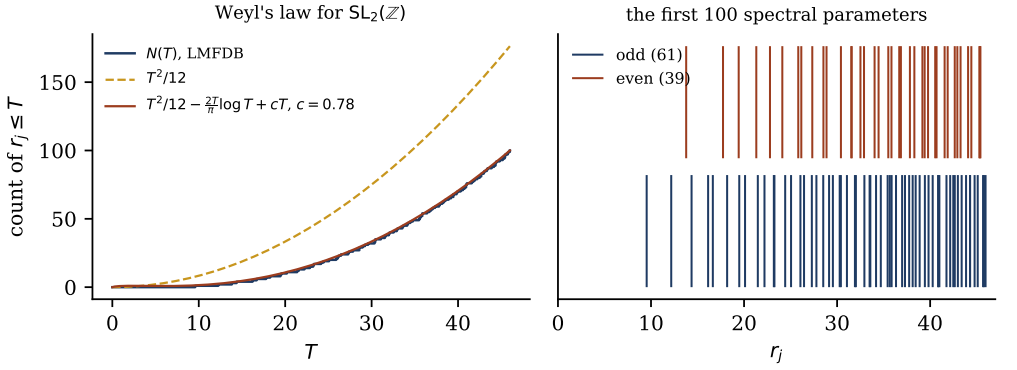


Figure 26.1: The cuspidal spectrum of $\mathrm{SL}_2(\mathbb{Z})$. Left: the counting function $N(T)$ of the first hundred spectral parameters against $T^2/12$ and against Weyl's law with its logarithmic term and a fitted linear term. Right: the spectral parameters themselves, odd forms below and even forms above.

eigenvalues reach $r = 46$, where $T^2/12 = 176$ and the logarithmic term has removed 112; the remaining linear term is about $0.78T$ at this height (Figure 26.1).

The Hecke operators T_n commute with Δ and with each other, so the cusp forms can be taken to be Hecke eigenforms, $T_n u_j = \lambda_j(n) u_j$, with Fourier expansions $u_j(z) = \sum_{n \neq 0} \rho_j(n) \sqrt{y}\, K_{ir_j}(2\pi|n|y) e^{2\pi i n x}$ and $\rho_j(n) = \rho_j(1)\lambda_j(n)$ for $n \geq 1$. The L function $L(s, u_j) = \sum_{n \geq 1} \lambda_j(n) n^{-s}$ has the Euler product $\prod_p (1 - \lambda_j(p)p^{-s} + p^{-2s})^{-1}$, Satake parameters α_p, α_p^{-1} with $\alpha_p + \alpha_p^{-1} = \lambda_j(p)$, and the completed function

$$\Lambda(s, u_j) = \Gamma_{\mathbb{R}}(s + \epsilon_j + ir_j)\Gamma_{\mathbb{R}}(s + \epsilon_j - ir_j) L(s, u_j) = (-1)^{\epsilon_j} \Lambda(1 - s, u_j),$$

$\epsilon_j \in \{0, 1\}$ the parity, is entire. In the normalisation of Section 14.2 the archimedean parameters are $\mu = \epsilon_j \pm i r_j$, and u_j generates a cuspidal automorphic representation of $\mathrm{GL}_2(\mathbb{A}_{\mathbb{Q}})$, so Theorem 14.8 applies: GRH for $L(s, u_j)$ is equivalent to $\Xi_{u_j}(0)/\Xi_{u_j}(\sqrt{\lambda})$ being a GGC transform, and Theorem 16.8 gives the Thorin representation at every basepoint $\alpha \geq 1$ unconditionally.

Two conjectures govern the Maass side and both are statements about the Thorin framework. The *Ramanujan conjecture* for Maass forms is $|\lambda_j(p)| \leq 2$, that is $|\alpha_p| = 1$; for holomorphic forms this is Deligne's theorem, for Maass forms the best known bound is Kim and Sarnak's $|\alpha_p| \leq p^{7/64}$ [315]. *Selberg's eigenvalue conjecture* is that $\lambda_1 \geq \frac{1}{4}$ for every congruence subgroup, that is that no r_j is imaginary; it is the archimedean Ramanujan conjecture, and Kim and Sarnak's bound gives $\lambda_1 \geq \frac{1}{4} - (\frac{7}{64})^2 = \frac{975}{4096}$. For $\mathrm{SL}_2(\mathbb{Z})$ itself $\lambda_1 = 91.14$ and the conjecture is trivially true; the content is in the groups $\Gamma_0(N)$ and $\Gamma(N)$ as N grows.

COROLLARY 26.1 (Ramanujan at p is local Thorin positivity). *Let u be a Hecke Maass eigenform and p a prime. The function*

$$\lambda \mapsto \frac{2-\lambda(p)}{p^{\sqrt{\lambda}}+p^{-\sqrt{\lambda}}-\lambda(p)} = (2-\lambda(p))\,p^{-\sqrt{\lambda}}L_p(\sqrt{\lambda},u),$$

where $L_p(s,u)=(1-\lambda(p)p^{-s}+p^{-2s})^{-1}$ is the Euler factor, is the Laplace transform of a probability law on $[0,\infty)$ if and only if $-2\le\lambda(p)<2$, in which case the law is the GGC with Thorin comb $\sum_{k\in\mathbb{Z}}\delta_{t_k}$, $t_k=((\theta_p+2\pi k)/\log p)^2$, $\lambda(p)=2\cos\theta_p$. In particular the Ramanujan conjecture at p (with $\lambda(p)\neq 2$) holds if and only if the local factor of u at p is a GGC transform in the variable $\sqrt{\lambda}$.

PROOF. $L_p(s,u)^{-1}=1-\lambda(p)p^{-s}+p^{-2s}=p^{-s}(p^s+p^{-s}-\lambda(p))$, so the displayed function is f_x of Theorem 15.3 with $x=\lambda(p)$ and $L=\log p$. □

For holomorphic forms the corollary is a theorem at every p, by Deligne; for Maass forms it is known only that $\lambda(p)\in[-p^{7/64}-p^{-7/64},p^{7/64}+p^{-7/64}]$, and whether every local factor of every Maass form is a GGC transform is the Ramanujan conjecture itself. Numerically every Hecke eigenvalue ever computed for a Maass form satisfies the bound.

26.2 BROWNIAN MOTION AND THE HEAT KERNEL

Brownian motion on $\mathbb{H}$ is the diffusion with generator $-\Delta=y^2(\partial_x^2+\partial_y^2)$, so that its transition density solves $\partial_t u=-\Delta u$; it is the standard Brownian motion of the metric run at twice the usual speed, which is the convention that makes the heat kernel $e^{-t\Delta}$. In coordinates it solves

$$dx_t=\sqrt{2}\,y_t\,dB_t^1,\qquad d\log y_t=\sqrt{2}\,dB_t^2-dt,$$

so that $\log y$ is a Brownian motion with drift -1 and the process drifts towards the real axis in the half plane, while on X the identification by Γ returns it from the cusp. The heat kernel on $\mathbb{H}$ depends only on the distance $\rho=d(z,w)$ and is (McKean [389])

$$p_t(\rho)=\frac{\sqrt{2}\,e^{-t/4}}{(4\pi t)^{3/2}}\int_\rho^\infty \frac{s\,e^{-s^2/4t}}{\sqrt{\cosh s-\cosh\rho}}\,ds, \tag{26.2}$$

with $p_t(0)\sim(4\pi t)^{-1}$ as $t\to 0$, the Euclidean value in two dimensions, $p_t(0)\sim\sqrt{\pi}e^{-t/4}/(8t^{3/2})$ as $t\to\infty$ (Figure 26.2), and $p_t(\rho)\asymp t^{-3/2}e^{-t/4-\rho^2/4t-\rho/2}$ up to polynomial factors in ρ and t at large t and ρ, the factor $e^{-t/4}$ being the bottom of the spectrum of Δ on $\mathbb{H}$. The heat kernel on X is the sum over the group,

$$K_t(z,w)=\sum_{\gamma\in\bar\Gamma}p_t(d(z,\gamma w)),\qquad \bar\Gamma=\Gamma/\{\pm1\}, \tag{26.3}$$

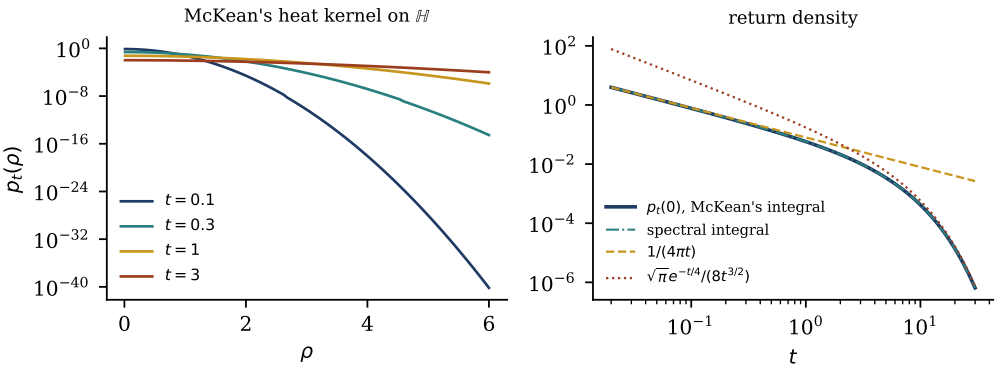


Figure 26.2: McKean's heat kernel on $\mathbb{H}$. Left: $p_t(\rho)$ as a function of the distance ρ for four values of t. Right: the return density $p_t(0)$ from (26.2) and from the spectral integral $\frac{1}{4\pi}\int_{\mathbb{R}} e^{-t(1/4+r^2)} r \tanh(\pi r)\, dr$, which agree to 10^{-4}, between the small time asymptote $1/(4\pi t)$ and the large time asymptote $\sqrt{\pi}e^{-t/4}/(8t^{3/2})$, whose exponential rate $\frac{1}{4}$ is the bottom of the spectrum.

and has the spectral expansion

$$K_t(z,w) = \frac{3}{\pi} + \sum_{j\geq 1} e^{-t\lambda_j} u_j(z)\overline{u_j(w)} + \frac{1}{4\pi}\int_{-\infty}^{\infty} e^{-t(\frac{1}{4}+r^2)} E(z,\tfrac{1}{2}+ir)\overline{E(w,\tfrac{1}{2}+ir)}\, dr. \tag{26.4}$$

PROPOSITION 26.2 (Brownian motion on X). *Brownian motion on X is a positive recurrent diffusion with invariant probability $\frac{3}{\pi}d\mu$, and for every $f \in L^1(X,\mu)$, $\frac{1}{T}\int_0^T f(B_s)\, ds \to \frac{3}{\pi}\int_X f\, d\mu$ almost surely. The law of B_t started anywhere converges to $\frac{3}{\pi}d\mu$ in total variation, at the rate $e^{-t\lambda_1}$ in L^2 with $\lambda_1 = 91.14$ for the cuspidal part, and at the rate of the continuous spectrum, $e^{-t/4}$ up to a power of t, for the Eisenstein part, which is the time it takes the process to come back from the cusp.*

PROOF. Positive recurrence follows from finiteness of the area and the spectral gap of Δ on cusp forms together with the explicit form of the continuous spectrum: $\int_X K_t(z,w) f(w)\, d\mu(w) - \frac{3}{\pi}\int f\, d\mu$ is bounded in L^2 by $e^{-t\lambda_1}\|f\|_2$ for the cuspidal projection and by $e^{-t/4}$ times a power of t for the Eisenstein projection, from (26.4). The ergodic theorem is Birkhoff's for a positive recurrent diffusion. □

The Euler scheme for the stochastic differential equation, with step 10^{-3}, each step followed by reduction to F, gives for 64 paths run to time $2\cdot 10^4$ the time fractions $\mathbb{P}(y > 1) = 0.9551$, $\mathbb{P}(y > 2) = 0.4784$, $\mathbb{P}(y > 4) = 0.2397$, $\mathbb{P}(y > 8) = 0.1201$ against $3/(\pi Y) = 0.9549, 0.4775, 0.2387, 0.1194$, and the

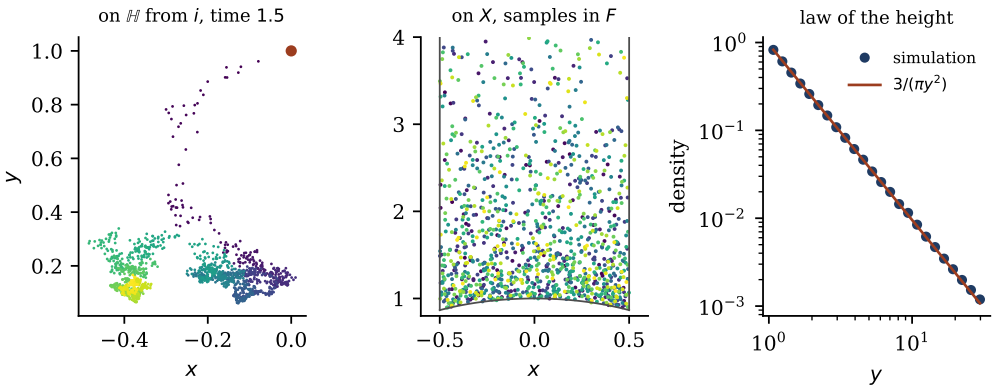


Figure 26.3: Brownian motion on the modular surface. Left: a path of the hyperbolic Brownian motion in $\mathbb{H}$ started at i, run to time 1.5; the drift of $\log y$ carries it towards the real axis. Middle: the process on X, every fiftieth sample point in the fundamental domain, coloured by time, filling the domain according to the hyperbolic area. Right: the empirical law of the height y along the path against the density $3/(\pi y^2)$ of the normalised hyperbolic area for $y \geq 1$.

empirical density of y on $[1, 32]$ agrees with $3/(\pi y^2)$ to within 1.1 percent in every one of twenty logarithmic bins (Figure 26.3). The time average of $y^{-1}1_{y\geq 1}$ is 0.4769 against $3/2\pi = 0.4775$.

26.3 The Selberg trace formula as a heat trace

For a compact quotient the trace of the heat kernel is both a spectral and a geometric sum, and that is the trace formula. Let Γ be a discrete subgroup of $\mathrm{SL}_2(\mathbb{R})$ acting without fixed points with compact quotient X_Γ of genus $g \geq 2$, area $4\pi(g-1)$, and Laplace eigenvalues $0 = \lambda_0 < \lambda_1 \leq \cdots$, $\lambda_j = \frac{1}{4} + r_j^2$. Every nontrivial conjugacy class in Γ is hyperbolic and is a power P^k of a primitive class P, which corresponds to a primitive closed geodesic of length $\ell(P)$; put $N(P) = e^{\ell(P)}$, the norm.

Theorem 26.3 (Selberg [496]; heat kernel form). *For every $t > 0$,*

$$\sum_{j\geq 0} e^{-t\lambda_j} = \frac{\operatorname{area}(X_\Gamma)}{4\pi} \int_{-\infty}^{\infty} e^{-t(\frac{1}{4}+r^2)}\, r \tanh(\pi r)\, dr + \sum_{P} \sum_{k\geq 1} \frac{\ell(P)}{2\sinh(k\ell(P)/2)} \frac{e^{-t/4} e^{-(k\ell(P))^2/4t}}{\sqrt{4\pi t}}. \tag{26.5}$$

This is the general trace formula

$$\sum_j h(r_j) = \frac{\text{area}}{4\pi}\int h(r)\, r\tanh(\pi r)\, dr + \sum_{P,k} \frac{\ell(P)}{2\sinh(k\ell(P)/2)}\, g(k\ell(P)),$$

with $g(u) = \frac{1}{2\pi}\int h(r)e^{-iru}\, dr$, applied to $h(r) = e^{-t(1/4+r^2)}$, for which $g(u) = e^{-t/4}e^{-u^2/4t}/\sqrt{4\pi t}$. For the modular surface the same identity holds with elliptic terms for the two elliptic points, parabolic terms for the cusp, and the spectral side carrying the continuous spectrum through the scattering determinant; the statement is in Hejhal [253] and Iwaniec [275], and the structure is the same.

The probabilistic reading is McKean's [389], and the book's dictionary makes it exact. By (26.3) and (26.4),

$$\sum_j e^{-t\lambda_j} = \int_{X_\Gamma} K_t(z,z)\, d\mu(z) = \int_{X_\Gamma} \sum_\gamma p_t(d(z,\gamma z))\, d\mu(z).$$

The term $\gamma = 1$ is area $\cdot\, p_t(0)$, the return density of free Brownian motion on $\mathbb{H}$ integrated over the surface, and (26.2) integrated against tanh is the statement $p_t(0) = \frac{1}{4\pi}\int e^{-t(1/4+r^2)} r\tanh(\pi r)\, dr$, the spectral density of Δ on $\mathbb{H}$. The terms $\gamma \sim P^k$ collect, for each z, the density at z of Brownian paths from z that return to z in the free homotopy class of the closed geodesic P^k: a path from z to γz in $\mathbb{H}$ is a loop on X_Γ in the class of γ. Integrating over a fundamental domain of the centraliser of P gives the factor $\ell(P)/(2\sinh(k\ell(P)/2))$, and $g(k\ell(P))$ is the one dimensional Gaussian density at distance $k\ell(P)$ with the spectral shift $e^{-t/4}$. So (26.5) says: the total return probability of Brownian motion on X_Γ equals the free return probability plus the sum over closed geodesics of the probability of returning around that geodesic. Compare the Galois heat trace of Definition 20.29,

$$\mathcal{W}_\rho(x) = \frac{1}{4\pi}\int e^{-xr^2}\Big(\log N_\rho + 2\,\mathrm{Re}\,\frac{\gamma'}{\gamma}(\tfrac12 + ir, \rho)\Big)dr \\ - \frac{1}{2\sqrt{\pi x}}\sum_{n\geq 2}\frac{\Lambda_\rho(n)}{\sqrt{n}}\, e^{-(\log n)^2/4x},$$

which equals $\frac12\sum e^{-x\gamma^2}$ over the zeros $\frac12 + i\gamma$ (Theorem 20.30). The two formulas have the same three parts: a spectral sum ($e^{-tr_j^2}$ over eigenvalues, $e^{-x\gamma^2}$ over zeros), an archimedean integral ($r\tanh\pi r$, γ'/γ), and a sum over primes ($\log N(P) = \ell(P)$ with the Gaussian $e^{-(k\ell)^2/4t}$, $\Lambda(n) = \log p$ with $e^{-(\log n)^2/4x}$). The one difference is the sign of the prime sum, and it is explained by the Selberg zeta function below: Z is built like $1/\zeta$, so its logarithmic coefficients are $-\ell(P)$ where those of ζ are $+\log p$.

THEOREM 26.4 (Selberg's conjecture as complete monotonicity; this book). *Let Γ be a congruence subgroup of $\mathrm{SL}_2(\mathbb{Z})$ and $\Theta_\Gamma(t) = \sum_{j\geq 1} e^{-t\lambda_j}$ the heat trace over the Maass cusp forms, $\lambda_j = \frac14 + r_j^2$.*

(i) $e^{t/4}\Theta_\Gamma(t) = \sum_j e^{-tr_j^2}$ *is completely monotone on* $(0,\infty)$ *if and only if* $\lambda_j \geq \frac{1}{4}$ *for every* j*, that is if and only if Selberg's eigenvalue conjecture holds for* Γ*; in that case its Bernstein measure is* $\sum_j \delta_{r_j^2}$.

(ii) Unconditionally, $e^{t(\frac{1}{4}-(7/64)^2)}\Theta_\Gamma(t)$ *is completely monotone.*

(iii) For $\Gamma = \mathrm{SL}_2(\mathbb{Z})$*, (i) holds, and* $\sum_j e^{-tr_j^2} \sim \frac{1}{12t}$ *as* $t \to 0$.

(iv) The measure $\sum_j \delta_{r_j^2}$ *is not a Thorin measure:* $\sum_j r_j^{-2} = \infty$.

PROOF. (i) By Weyl's law the series converges for $t > 0$. If every $r_j^2 \geq 0$ the function is a Laplace transform of a positive measure, hence completely monotone by Bernstein's theorem, with the stated Bernstein measure by uniqueness. If some $\lambda_j < \frac{1}{4}$ then $r_j^2 < 0$ and the corresponding term grows exponentially, so the function is not bounded as $t \to \infty$, while a completely monotone function is decreasing. (ii) is (i) applied with the shift $\frac{1}{4} - (\frac{7}{64})^2$, since $\lambda_j \geq \frac{975}{4096}$ by Kim and Sarnak. (iii) The first statement is $\lambda_1 = 91.14$; the asymptotic is Karamata's Tauberian theorem applied to (26.1), or directly the identity term of the trace formula, $\frac{\text{area}}{4\pi}\int e^{-tr^2} r \tanh(\pi r)\, dr \sim \frac{\text{area}}{4\pi t} = \frac{1}{12t}$. (iv) By Weyl's law for Γ, $N(T) \sim \frac{\text{area}}{4\pi}T^2$, of which (26.1) is the case $\mathrm{SL}_2(\mathbb{Z})$, the sum $\sum_j r_j^{-2} = \int_1^\infty T^{-2}\, dN(T)$ diverges like $\frac{\text{area}}{2\pi}\log T$ (like $\frac{1}{6}\log T$ for $\mathrm{SL}_2(\mathbb{Z})$), so the second integrability condition of Definition 13.2, $\int_{(1,\infty)} t^{-1}\tau(dt) < \infty$, fails. □

Theorem 26.4 is Theorem 20.30 with eigenvalues for zeros, and on the Maass side much of the analogue of the Riemann hypothesis comes free: the Laplacian is self adjoint and the Hecke operators commute with it, so every r_j^2 is real and the only possible exceptions are finitely many eigenvalues $\lambda_j < \frac{1}{4}$, which is the whole content of the Maass side being spectral. Selberg's conjecture, that there are none, is open in general; it holds for $\mathrm{SL}_2(\mathbb{Z})$, which gives part (iii), and part (ii) is what the bound of Kim and Sarnak gives for every congruence subgroup. Part (iv) is the limit of the analogy. The zeros of an L function of degree n have counting function $\frac{nT}{2\pi}\log T$, a one dimensional spectrum, and $\sum \gamma^{-2}$ converges, so Theorem 14.8 can place them in a Thorin measure. The eigenvalues of a surface have a two dimensional count, $\sum r_j^{-2}$ diverges logarithmically, and the function $\prod_j(1+\lambda/r_j^2)$ does not converge; the Selberg zeta function is of order two, and no GGC has it for a Laplace transform. The heat trace is the right object, and the Thorin framework enters, as the next section shows, through the Euler product and not through the spectrum.

26.4 Closed geodesics as primes

The Selberg zeta function of Γ is

$$Z(s) = \prod_P \prod_{k \geq 0} \bigl(1 - N(P)^{-s-k}\bigr), \qquad \operatorname{Re} s > 1,$$

the product over primitive hyperbolic classes. It continues to an entire function of order two (compact case; meromorphic for $\mathrm{SL}_2(\mathbb{Z})$), with zeros at $s_j = \frac{1}{2} \pm i r_j$ for each eigenvalue $\lambda_j = \frac{1}{4} + r_j^2$, a simple zero at $s = 1$ from $\lambda_0 = 0$, and zeros at the nonpositive integers of multiplicity growing linearly; for $\mathrm{SL}_2(\mathbb{Z})$ there are in addition zeros at $s = \rho/2$ for the nontrivial zeros ρ of ζ, coming from the scattering determinant $\varphi(s) = \sqrt{\pi}\,\Gamma(s - \frac{1}{2})\zeta(2s-1)/(\Gamma(s)\zeta(2s))$. For a congruence subgroup the zeros s_j lie on $\operatorname{Re} s = \frac{1}{2}$ if and only if Selberg's eigenvalue conjecture holds, and for $\mathrm{SL}_2(\mathbb{Z})$ the additional zeros $\rho/2$ lie on $\operatorname{Re} s = \frac{1}{4}$ if and only if the Riemann hypothesis holds; a compact surface may have eigenvalues below $\frac{1}{4}$, and then Z has real zeros in $(\frac{1}{2}, 1)$.

The ratio

$$R(s) = \frac{Z(s+1)}{Z(s)} = \prod_P \bigl(1 - N(P)^{-s}\bigr)^{-1} = \sum_{\mathfrak{n}} N(\mathfrak{n})^{-s} \tag{26.6}$$

is an Euler product of the shape of ζ, the sum running over the free abelian monoid generated by the primitive geodesics, whose elements we call *geodesic integers* and whose norms are multiplicative. The prime geodesic theorem,

$$\pi_\Gamma(x) = \#\{P : \ N(P) \leq x\} = \operatorname{li}(x) + O(x^{3/4+\epsilon})$$

for the modular surface, is due to Huber and Selberg for compact surfaces, where the terms $\operatorname{li}(x^{s_j})$ of the eigenvalues $\lambda_j < \frac{1}{4}$ are added, and to Sarnak for $\mathrm{SL}_2(\mathbb{Z})$ [475], where the exponent has since been lowered to $\frac{25}{36}$ [523]; the primitive geodesics of $\mathrm{SL}_2(\mathbb{Z})$ correspond to the narrow classes of primitive binary quadratic forms of positive nonsquare discriminant d, with $N(P) = \epsilon_d^2$ for the smallest unit $\epsilon_d > 1$ of norm $+1$ in the order of discriminant d, so that $\pi_\Gamma(x) = \sum_{\epsilon_d \leq \sqrt{x}} h^+(d)$ and the theorem is a statement about class numbers of indefinite forms. Enumerating the discriminants to $4 \cdot 10^6$ with their units and class numbers gives 78441 primitive geodesics of norm at most 10^6, against $\operatorname{li}(10^6) = 78628$, ratio 0.9976, and $\sum_{N(P)^k \leq x} \ell(P) = 0.9991x$ at $x = 10^6$ (Figure 26.4, left); the shortest geodesic has $N(P) = \varphi^4 = 6.854$, φ the golden ratio, from the discriminant 5, and the second shortest has norm 13.93 with multiplicity two, from the discriminant 12 whose order has no unit of norm -1.

The eigenvalues are visible in the geodesic count. The explicit formula for the modular surface, which is the trace formula with a sharp cutoff for h, gives

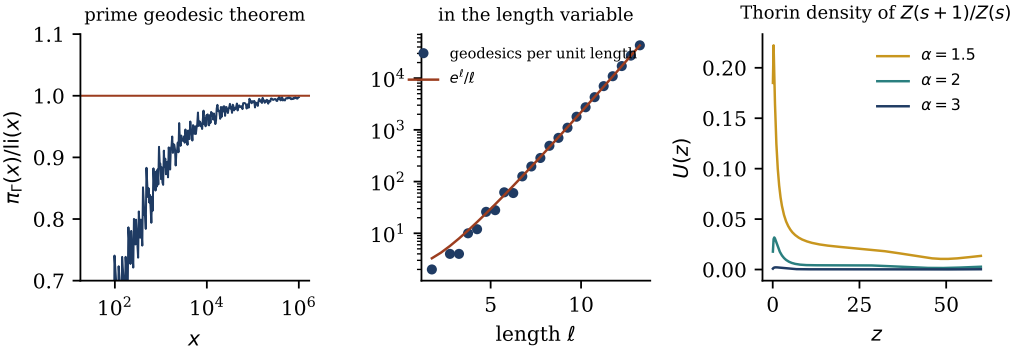


Figure 26.4: Left: the prime geodesic theorem on the modular surface, $\pi_\Gamma(x)/\operatorname{li}(x)$ for $x \le 10^6$ from the class numbers and units of the $2 \cdot 10^6$ discriminants below $4 \cdot 10^6$. Middle: the number of primitive geodesics per unit length against e^ℓ/ℓ, the prime geodesic theorem in the length variable. Right: the Thorin density $U(z)$ of the geodesic Euler product $Z(s+1)/Z(s)$ at the basepoints $\alpha = 1.5$, 2 and 3, from Theorem 26.5; it is positive and integrable, and at $\alpha = 2$ its integral against $\log(1+z/t)$ reproduces $\log(R(2)/R(2+\sqrt{z}))$ to between 10^{-7} and 10^{-5}.

$\psi_\Gamma(x) = \sum_{N(P)^k \le x} \ell(P) = x + \sum_j x^{s_j}/s_j + \cdots$ with $s_j = \frac{1}{2} \pm i r_j$, so that $(\psi_\Gamma(x) - x)/\sqrt{x}$ is, up to smaller terms, the almost periodic function $\sum_j 2\operatorname{Re}(x^{ir_j}/s_j)$ of $u = \log x$, with frequencies the r_j and amplitudes about $2/r_j$. Figure 26.5 shows this function on $50 \le x \le 10^6$ and its periodogram in u: the largest peak below $r = 19$ is at $r = 9.53$, the first eigenvalue, and the next peaks are at 12.13, 14.11 and 16.27, where the resolution $2\pi/\log(2 \cdot 10^4) \approx 0.6$ of a window of length $\log(2 \cdot 10^4)$ does not separate the pairs 13.78, 14.36 and 16.14, 16.64. This is the prime side hearing the spectral side, Weil's explicit formula read backwards, and it is the arithmetic content of Theorem 26.3. The right panel of the figure shows the two terms of the heat trace formula as functions of t, the identity term $\sim 1/(12t)$ at small t and the geodesic sum, which is $O(e^{-\ell_1^2/4t})$ at small t and overtakes the identity term only for t of order one, where the whole trace is dominated by $\lambda_0 = 0$.

With these facts the geodesics are a Beurling system of generalised primes satisfying the prime number theorem, and the two tools of Chapters 16 and 24 that use nothing else apply to them.

THEOREM 26.5 (Thorin representation of the geodesic Euler product; this book). *Let $\alpha > 1$. Then for $\lambda > 0$*

$$\frac{R(\alpha)}{R(\alpha+\sqrt{\lambda})} = e^{-b\sqrt{\lambda}}\,\mathbb{E}e^{-\lambda H}, \qquad b = \frac{R'}{R}(\alpha) = -\sum_{P,k} \ell(P) N(P)^{-k\alpha},$$

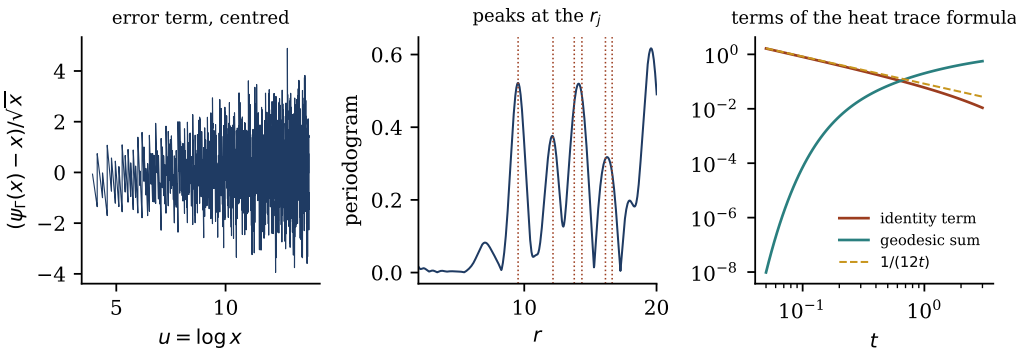


Figure 26.5: The spectrum in the geodesics. Left: the error term $(\psi_\Gamma(x)-x)/\sqrt{x}$ of the prime geodesic theorem, centred, as a function of $u=\log x$. Middle: its periodogram in u, with the first six eigenvalue parameters r_j marked; the peak at 9.53 is the first Maass form. Right: the identity term and the geodesic sum of the heat trace formula (26.5) for the modular surface as functions of t.

where H is a GGC with drift zero and Thorin density

$$U(z)=\frac{1}{2\pi\sqrt{z}}\sum_P\sum_{k\ge1}\ell(P)\,N(P)^{-k\alpha}\,2\sin^2\Big(\frac{k\ell(P)\sqrt{z}}{2}\Big)\ge0,$$

equal to $(\mathrm{Re}(R'/R)(\alpha+i\sqrt{z})-b)/(2\pi\sqrt{z})$. The same holds for the modular surface, and for the Dedekind zeta function of any number field and the Rankin Selberg L function $L(s,\pi\times\tilde\pi)$ of an automorphic representation and its contragredient, whose logarithmic coefficients $|a_\pi(p^k)|^2\log p$ are nonnegative, with the primes of the field or the Rankin Selberg coefficients in place of the geodesics.

PROOF. $\log R(\alpha+s)=\sum_{P,k}k^{-1}N(P)^{-k\alpha}e^{-sk\ell(P)}$, so (16.3) holds with the positive measure $\nu=\sum_{P,k}\ell(P)N(P)^{-k\alpha}\delta_{k\ell(P)}$, which is finite by the prime geodesic theorem, and Lemma 16.7 applies. The formula for U is (16.4). □

At $\alpha=2$ the density is dominated by the shortest geodesics and is shown in Figure 26.4; the identity

$$\log\frac{R(\alpha)}{R(\alpha+\sqrt{z})}=-b\sqrt{z}-\int_0^\infty\log\Big(1+\frac{z}{t}\Big)U(t)\,dt$$

checks to $9\cdot10^{-8}$, $5\cdot10^{-7}$ and $4\cdot10^{-6}$ at $z=\frac12,2,10$, the residual being the numerical integration. The drift b is negative, so $R(\alpha)/R(\alpha+\sqrt{\lambda})=e^{|b|\sqrt{\lambda}}\mathbb{E}e^{-\lambda H}$ increases from 1 to $R(\alpha)$ and is the transform of H divided by the transform $e^{-|b|\sqrt{\lambda}}$ of a one half stable law, exactly as for ζ in Lemma 16.7. Theorem 26.5

is unconditional for every basepoint $\alpha > 1$, and what it does not give is the representation at $\alpha = \frac{1}{2} + a$: there the poles of R at the spectrum s_j contribute positive Cauchy terms to the Thorin density and its zeros at $s_j - 1$ contribute negative ones, and because of Theorem 26.4(iv) the two series are each divergent; the trivial poles of R at the nonpositive integers, of constant multiplicity, supply the compensating logarithmic growth. Whether $R(\frac{1}{2}+a)/R(\frac{1}{2}+a+\sqrt{\lambda})$, with the pole at $s = 1$ removed, is a GGC transform for small $a > 0$ is Open Problem 26.1.

THEOREM 26.6 (random geodesic integers are Poisson Dirichlet; this book). *Let* $\mathfrak{n}_\sigma$ *be the random geodesic integer with* $\mathbb{P}(\mathfrak{n}_\sigma = \mathfrak{n}) = N(\mathfrak{n})^{-\sigma}/R(\sigma)$, $\sigma > 1$. *Its exponents* e_P *are independent geometric variables with* $\mathbb{P}(e_P \geq k) = N(P)^{-k\sigma}$, *and as* $\sigma \downarrow 1$: *the point process of rescaled sizes* $(\sigma-1)e_P\ell(P)$ *converges to the Poisson process with intensity* $x^{-1}e^{-x}dx$, *the Gamma process; the ranked normalised sizes* $e_P\ell(P)/\log N(\mathfrak{n}_\sigma)$ *converge to* PD(1); *the largest primitive geodesic factor has Dickman's law in the limit; and the number of distinct primitive factors is asymptotically normal with mean and variance* $\log\frac{1}{\sigma-1}$.

PROOF. The proof of Theorem 20.18, through Lemma 20.19, uses of the primes only two things: $\sum_{N(P)\leq x}\ell(P) \sim x$, which is the prime geodesic theorem by partial summation, and $\sum_P N(P)^{-2} < \infty$. Both hold for the geodesics, and the labels may be taken trivial. The Poisson Dirichlet limit, Dickman's law and the normal law for the number of factors then follow exactly as in Theorems 24.1, 24.2 and 24.4 with $\theta = 1$, the order of the pole of R at $s = 1$, which is simple because the zero of Z at $s = 1$ is simple ($\lambda_0 = 0$ has multiplicity one on a connected surface) and $Z(2) \neq 0$. □

The uniform analogue, a geodesic integer chosen uniformly among those of norm at most x, obeys the same limit laws with $\log x$ in place of $\log\frac{1}{\sigma-1}$, since the arguments of Chapter 24 use only the prime number theorem with a power saving in the error term, which the prime geodesic theorem supplies. What is missing is a Chebotarev theorem: the geodesics carry natural labels, the discriminant d modulo q, the type of the order, the sign of the norm of the fundamental unit, and the densities of these labels in the ordering by $N(P)$ would be the base measure of a Dirichlet process of geodesics. Some are known. Sarnak [481] showed that the reciprocal geodesics, those conjugate to their own inverse in $\mathrm{PSL}_2(\mathbb{Z})$, which correspond to the forms f equivalent to $-f$, are only about $\frac{3}{8}\sqrt{x}$ in number with norm below x, so they form a class of density zero and the Dirichlet process does not see them; the densities of the discriminant classes modulo q are Open Problem 26.2.

26.5 EIGENVALUE ONE QUARTER AND PARITY

A Maass form with $\lambda = \frac{1}{4}$ has $r = 0$ and gamma factor $\Gamma_{\mathbb{R}}(s+\epsilon)^2$. A two dimensional Artin representation ρ is even when $\det \rho(c) = 1$ for complex conjugation c, so that $\rho(c) = I$ or $\rho(c) = -I$, and its gamma factor is then $\Gamma_{\mathbb{R}}(s)^2$ or $\Gamma_{\mathbb{R}}(s+1)^2$ respectively; an odd representation, $\det \rho(c) = -1$, has $\Gamma_{\mathbb{R}}(s)\Gamma_{\mathbb{R}}(s+1) = \Gamma_{\mathbb{C}}(s)$, the gamma factor of a holomorphic form of weight one. So the Maass forms that can be Galois are those of eigenvalue exactly $\frac{1}{4}$, of parity ϵ with $\rho(c) = (-1)^\epsilon I$, and the case $\rho(c) = I$ is exactly the case in which the Galois heat trace of Definition 20.29 is computed with the Maass gamma factor $\Gamma_{\mathbb{R}}(s)^2$.

THEOREM 26.7 (Maass, Hecke; Langlands, Tunnell). *Let ρ be an irreducible two dimensional Artin representation that is even. If ρ is dihedral, tetrahedral or octahedral there is a Maass cusp form u_ρ of eigenvalue $\frac{1}{4}$ and parity ϵ, $\rho(c) = (-1)^\epsilon I$, on some $\Gamma_0(N)$ with character, with $L(s,\rho) = L(s,u_\rho)$; in the dihedral case u_ρ is one of Maass's original forms, induced from a ray class character of a real quadratic field. If ρ is icosahedral, the existence of u_ρ is open in general. Conversely it is conjectured that every Maass cusp form of eigenvalue $\frac{1}{4}$ whose Hecke eigenvalues are algebraic integers arises in this way.*

Theorem 20.30 now reads as follows. For a self dual ρ with $\rho(c) = I$ and root number $+1$, the heat trace $\mathcal{W}_\rho^+$ of Definition 20.29 computed with the Maass gamma factor $\Gamma_{\mathbb{R}}(s)^2$ is completely monotone if and only if $\Lambda(s,\rho)$ is entire and GRH holds; and if ρ is odd, so that it has a weight one form, the same function computed with the wrong parity differs by $\Delta(x) = \frac{1}{4}\mathbb{E}e^{-xC^2}$ with C hyperbolic secant distributed, and under GRH fails to be completely monotone. The difference comes from the gamma factors alone: at $s = \frac{1}{2} + it$ the reflection formula of Γ'/Γ gives $\operatorname{Re}\frac{d}{ds}\log\frac{\Gamma_{\mathbb{R}}(s+1)}{\Gamma_{\mathbb{R}}(s)} = \frac{\pi}{2}\operatorname{Re}\cot\left(\frac{\pi}{4} + \frac{i\pi t}{2}\right) = \frac{\pi}{2}\operatorname{sech}\pi t$, and $\operatorname{sech}\pi t$ is the density of C; under GRH the measure of the function computed with the wrong parity is $\frac{1}{2}\sum \delta_{\gamma^2}$ less a quarter of the law of C^2, which has a positive density on $(0,\infty)$, so it is not positive. The parity of a two dimensional Artin representation is therefore which of two gamma factors makes its heat trace a Laplace transform, and the Maass form of eigenvalue $\frac{1}{4}$, when it exists, is the object whose heat trace that is. The even icosahedral case is Open Problem 26.3: there the heat trace is defined from Frobenius traces, its complete monotonicity is equivalent to Artin's conjecture and GRH together, and the Maass form whose existence Artin's conjecture asserts would have eigenvalue exactly $\frac{1}{4}$ on $\Gamma_0(N)$ for the Artin conductor N; to our knowledge no Maass form of eigenvalue exactly $\frac{1}{4}$ that is not dihedral, tetrahedral or octahedral has been exhibited.

26.6 The Eichler Selberg trace formula

The holomorphic analogue of (26.5) is the formula for the trace of a Hecke operator on $S_k(\mathrm{SL}_2(\mathbb{Z}))$, which Chapter 9 uses in the form of Serre's vertical Sato Tate theorem and which is stated here. Let $H(N)$ be the Hurwitz class number, the number of $\mathrm{SL}_2(\mathbb{Z})$ classes of positive definite binary quadratic forms of discriminant $-N$ weighted by the reciprocal of the order of the automorphism group in $\mathrm{PSL}_2(\mathbb{Z})$, so that the forms equivalent to $a(x^2+y^2)$ and $a(x^2+xy+y^2)$ count $\frac12$ and $\frac13$, with $H(0)=-\frac{1}{12}$, and let $P_k(t,n)$ be the coefficient of X^{k-2} in $(1-tX+nX^2)^{-1}$, so that $P_k(t,n)=(\varrho^{k-1}-\bar\varrho^{\,k-1})/(\varrho-\bar\varrho)$ for the roots $\varrho,\bar\varrho$ of X^2-tX+n.

Theorem 26.8 (Eichler, Selberg [172, 496]; see Zagier [347, 592]). *For even $k\ge 4$ and $n\ge 1$,*

$$\operatorname{tr}\bigl(T_n \mid S_k(\mathrm{SL}_2(\mathbb{Z}))\bigr) = -\frac12 \sum_{t^2\le 4n} P_k(t,n)H(4n-t^2) - \frac12 \sum_{dd'=n} \min(d,d')^{k-1},$$

the first sum over all integers t with $t^2\le 4n$ and the second over the positive divisors.

The formula was checked against the Hecke matrices computed by PARI on S_k for every even k from 12 to 40 and every $n\le 30$, 450 cases, without a discrepancy; the first case is $\operatorname{tr} T_2|S_{12}=\tau(2)=-24$, and on S_{24} the two eigenvalues of T_2 sum to 1080.

Its probabilistic reading is the one Chapter 9 gave the Hecke operators. For $n=p^m$ write $a_p(f)=2p^{(k-1)/2}\cos\theta_p(f)$ for each of the $d_k=\dim S_k$ normalised eigenforms; then $a_{p^m}(f)=p^{m(k-1)/2}U_m(\cos\theta_p(f))$ with U_m the Chebyshev polynomial of the second kind, and Theorem 26.8 is a formula for the empirical moment

$$\frac{1}{d_k}\sum_f U_m(\cos\theta_p(f)) = \frac{\operatorname{tr} T_{p^m}}{d_k\, p^{m(k-1)/2}}.$$

The term $t^2=4n$, present only when m is even, contributes $\frac{(k-1)}{12}p^{m(k-2)/2}$ through $H(0)=-\frac{1}{12}$, and since $d_k\sim\frac{k}{12}$ this term alone gives the limit $p^{-m/2}$ of the moment as $k\to\infty$, which is $\int U_m(\cos\theta)\,d\mu_p$ for the Plancherel measure μ_p of the $(p+1)$ regular tree, the Kesten McKay law of Chapter 9: the identity term of the trace formula is the return probability of the random walk on the tree, and the class number terms measure the finite graph's departure from the tree. The remaining terms are $o(1)$ in k (Conrey, Duke and Farmer [95] at level one, Serre [508] in general), which is the vertical Sato Tate theorem, Theorem 9.4. For $p=2$ and all 3234 eigenforms of level one and weight at most 400 the normalised eigenvalues $a_2(f)/2^{(k-1)/2}$ have moments 1.485, 3.683, 10.61 against $\frac32$, $\frac{15}{4}$, $\frac{87}{8}$ for μ_2 and 1, 2, 5 for the Sato Tate law, and Kolmogorov distance 0.0055 to μ_2 against 0.089 to Sato Tate (Figure 26.6). The vertical and the horizontal limits differ because at fixed p

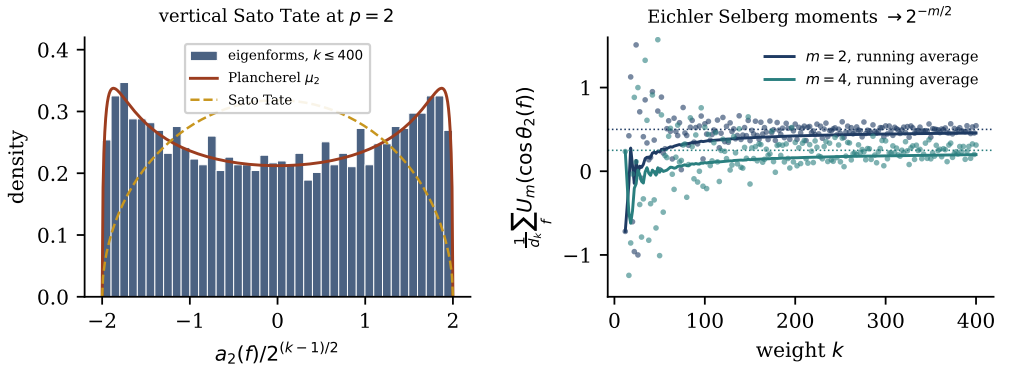


Figure 26.6: Left: Serre's vertical Sato Tate law at $p = 2$, the normalised Hecke eigenvalues $a_2(f)/2^{(k-1)/2}$ of the 3234 level one eigenforms of weight at most 400 against the Plancherel measure μ_2 of the 3 regular tree and the Sato Tate law. Right: the empirical moments $\frac{1}{d_k}\sum_f U_m(\cos\theta_2(f))$ for $m = 2$ and $m = 4$, computed from the Eichler Selberg formula for every even weight up to 400 (dots), and their running averages over k (lines), approaching $2^{-m/2}$.

the eigenvalues see the tree, while at fixed f and $p \to \infty$ the measures μ_p converge to the semicircle.

26.7 What this adds to the programme

The chapter adds the second half of the automorphic spectrum and finds the book's dictionary intact on it: heat trace for explicit formula, eigenvalues for zeros, geodesics for primes, complete monotonicity for the Riemann hypothesis, with Brownian motion on the surface as the probabilistic object that makes the trace formula a statement about return probabilities. It adds two unconditional theorems in the Thorin framework, Theorems 26.5 and 26.6, which are the sine squared lemma and the Poisson Dirichlet theorem for a Beurling system of primes, and one theorem, Theorem 26.4, that is the Galois heat trace theorem with the Riemann hypothesis replaced by Selberg's eigenvalue conjecture, and with the complete monotonicity it characterises proved for $\mathrm{SL}_2(\mathbb{Z})$ and, after the shift of Kim and Sarnak, for every congruence subgroup. It also adds the obstruction of Theorem 26.4(iv), which says why the zeros of an L function can be a Thorin measure and the eigenvalues of a surface cannot, and which separates the arithmetic case from the geometric one by a logarithm. Finally it places the even icosahedral problem where it belongs, among the Maass forms of eigenvalue $\frac{1}{4}$, and shows that the parity of a two dimensional Artin representation is a statement about which gamma factor, holomorphic or Maass, its Galois heat trace wants.

26.8 Notes

Maass introduced his forms in [378] and Selberg the trace formula and the zeta function in [496]; Hejhal's two volumes [252, 253] and Iwaniec [275] are the references for the noncompact case, and Sarnak's survey [478] for the eigenvalue conjecture. McKean's reading of the trace formula through Brownian motion is [389]; the heat kernel (26.2) is there and in Chavel [101]. The computation of the eigenvalues of $\mathrm{SL}_2(\mathbb{Z})$ is Hejhal's [254], with rigorous certification by Booker, Strömbergsson and Venkatesh [76]; the values quoted are from the LMFDB [371]. Weyl's law for $\mathrm{SL}_2(\mathbb{Z})$ is Selberg's; the lower order terms are in Hejhal. The Kim Sarnak bound is [315]. The prime geodesic theorem for $\mathrm{SL}_2(\mathbb{Z})$ and its reading through class numbers is Sarnak [475], reciprocal geodesics are [481], and the exponent $\frac{25}{36}$ is Soundararajan and Young [523]. The Eichler Selberg formula is in [172] and [496]; the form used is Zagier's, in his appendix to Lang [347] and in [592], and the vertical equidistribution is Serre's [508], with the level one case in Conrey, Duke and Farmer [95]. Theorems 26.4, 26.5 and 26.6 are new as formulated; their content is Selberg's, Thorin's and Kingman's, assembled through Chapters 16, 20 and 24.

26.9 Exercises

1. Show that the hyperbolic Brownian motion of Section 26.2, started at i, has $\mathbb{E}\log y_t = -t$ and that $\log y_t / t \to -1$ almost surely, so that on $\mathbb{H}$ the process converges to the boundary; explain why on X it does not.

2. Verify from (26.2) that $p_t(0) \sim (4\pi t)^{-1}$ as $t \to 0$, and that p_t integrates to 1 over $\mathbb{H}$.

3. Derive $\pi_\Gamma(x) \sim x/\log x$ from the trace formula with $h(r) = e^{-t(1/4+r^2)}$ as $t \to \infty$, using only $\lambda_0 = 0$ and $\lambda_1 > 0$.

4. Show that $R(s) = Z(s+1)/Z(s)$ has a simple pole at $s = 1$ with residue $\lim_{s\to 1}(s-1)R(s) = Z(2)/Z'(1)$, and compute $Z'(1)$ and $Z(2)$ for the modular surface from the geodesics of norm at most 10^6.

5. For the Hecke Maass form with $r_1 = 9.533695$, look up $\lambda(2)$ in the LMFDB, check the Ramanujan bound $|\lambda(2)| \le 2$, and write down the Thorin comb of Corollary 26.1.

6. Prove that the hyperbolic terms of (26.5) sum to $O(t^{-1/2}e^{-\ell_1^2/4t})$ as $t \to 0$, ℓ_1 the length of the shortest geodesic, so that the identity term alone gives Weyl's law.

7. Using Theorem 26.8, find all pairs (k, p) with $k \le 40$ and $p \le 30$ for which $\operatorname{tr} T_p | S_k = 0$.

8. Compute $\int U_m(\cos\theta)\,d\mu_p$ for all m and check that the even moments are $p^{-m/2}$.

26.10 Open problems

Open Problem 26.1 (the Thorin representation at the centre). Let $R(s) = Z(s+1)/Z(s)$ for a compact surface, and $R^*(s) = (s-1)R(s)$. Determine for which $a \in (0, \frac{1}{2})$ the function $R^*(\frac{1}{2}+a)/R^*(\frac{1}{2}+a+\sqrt{\lambda})$ is, after removal of a drift $e^{|b|\sqrt{\lambda}}$, the Laplace transform of a GGC, and compute its Thorin density numerically for the Bolza surface or for a Hurwitz surface of small genus, where the spectrum and geodesics are both available. The question is whether the logarithmic divergences of Theorem 26.4(iv) cancel into a positive density.

Open Problem 26.2 (a Chebotarev theorem for geodesics). Order the primitive geodesics of the modular surface by norm and colour each by the residue of its discriminant d modulo q, or by whether d is fundamental. Determine the densities of the colours, prove the Dirichlet process limit of Theorem 26.6 with that base measure, and identify the analogue of the $L(1,\chi)$ in the Mertens constants. The density of the fundamental discriminants among geodesics ordered by norm is not the density $6/\pi^2$ of fundamental discriminants among all discriminants, because the units weight the count.

Open Problem 26.3 (even icosahedral representations). For an even icosahedral Artin representation ρ of conductor N with $\rho(c) = I$ and trivial determinant, so that ρ is self dual, compute the heat trace $\mathcal{W}_\rho^+$ of Section 26.5 from Frobenius traces and test its complete monotonicity numerically, and search for a Maass form of eigenvalue $\frac{1}{4}$ on $\Gamma_0(N)$ with the Hecke eigenvalues $\operatorname{tr}\rho(\mathrm{Frob}_p)$ by Hejhal's algorithm. A single such form would be the first even icosahedral instance of reciprocity; the computation is within reach for the smallest conductors.

Open Problem 26.4 (cutoff for Brownian motion on $X(N)$). On the principal congruence covers $X(N)$ the area grows like N^3 while λ_1 stays bounded below by $\frac{975}{4096}$. Determine whether Brownian motion on $X(N)$ started at a fixed point exhibits cutoff in total variation at a time of order $\log N$, as random walks on expander graphs do, and how the cusps, whose number grows with N, affect the profile.

Computational Problem 26.1 (eigenvalues and geodesics at scale). Objective: compute the first 10^4 eigenvalues of $\mathrm{SL}_2(\mathbb{Z})$ by Hejhal's algorithm with certified error, and the primitive geodesics of norm at most 10^9 from the class numbers and units of the discriminants below $4\cdot 10^9$; verify the trace formula for the modular surface, with its elliptic, parabolic and Eisenstein terms, for the heat kernel at $t = 0.05$, 0.1 and 0.2 to the accuracy of the truncations; and compute the Thorin

density of Theorem 26.5 at $\alpha = 1.1$, where thousands of geodesics contribute. Starting point: `m1.py` and `m5.py`.

CHAPTER 27

EXTREME VALUES AND THE L FUNCTIONS OF ETA PRODUCTS

Euler's pentagonal number theorem expands $\prod_{n\geq 1}(1-q^n)$ as a lacunary series, and Jacobi's triple product does the same for theta functions. A probabilist reads these products differently. A product of numbers in $(0,1)$ is the probability that independent events all occur, and $1-e^{-\lambda y}$ is the probability that an exponential clock of rate λ has rung by time y. So $\prod_n(1-e^{-2\pi n y})$ is the probability that every clock in an infinite family has rung by time y: it is the distribution function of a maximum. This leads to a representation, new in this book, of the L function of Ramanujan's Δ, and of every multiplicative eta product, as the expectation of a regularised incomplete gamma function at an extreme value. For real $s>0$ it is a probability, so $L(E_{11a},1)$, an Artin L value, Catalan's constant and $\pi/4$ become probabilities that a gamma variable beats a maximum of exponentials. Modularity becomes self similarity of a law, and we obtain a Monte Carlo method. The representation does not locate the nontrivial zeros, and we explain why.

27.1 THE PRODUCT AS A DISTRIBUTION FUNCTION

Let $E_1, E_2, \ldots$ be independent standard exponential variables and set

$$M=\max_{n\geq 1}\frac{E_n}{2\pi n}, \qquad P(y)=\prod_{n\geq 1}\bigl(1-e^{-2\pi n y}\bigr) \quad (y>0).$$

The Dedekind eta function is $\eta(\tau)=q^{1/24}\prod_{n\geq 1}(1-q^n)$ with $q=e^{2\pi i\tau}$, so that $\eta(iy)=e^{-\pi y/12}P(y)$.

PROPOSITION 27.1 (this book). *The variable M is almost surely finite and positive, and $\mathbb{P}(M\leq y)=P(y)$ for $y>0$. The function P is continuous and strictly increasing, and*

$$1-P(y)\sim e^{-2\pi y} \quad (y\to\infty), \qquad P(y)\sim y^{-1/2}e^{-\pi/(12y)} \quad (y\to 0).$$

The density of M is $\frac{2\pi}{24}\bigl(1-E_2(iy)\bigr)P(y)$, where $E_2=1-24\sum_{N\geq 1}\sigma(N)q^N$ is the quasimodular Eisenstein series of weight two.

PROOF. Fix $y>0$. The events $A_n=\{E_n\leq 2\pi n y\}$ are independent with $\mathbb{P}(A_n)=1-e^{-2\pi n y}$, and $\{M\leq y\}=\bigcap_n A_n$. By continuity along the decreasing events

$\bigcap_{n\le N} A_n$, $\mathbb{P}(M \le y) = \lim_N \prod_{n\le N}(1-e^{-2\pi n y})$. Since $\sum_n e^{-2\pi n y} < \infty$ the product converges to a positive limit, which is $P(y)$, and $\prod(1-x_n) \ge 1 - \sum x_n$ gives

$$1 - \frac{e^{-2\pi y}}{1-e^{-2\pi y}} \le P(y) \le 1 - e^{-2\pi y}.$$

This yields the right tail and $M < \infty$ almost surely. Uniform convergence of $\sum_n \log(1-e^{-2\pi n y})$ on $[a,\infty)$ for each $a > 0$ gives continuity, and each factor is strictly increasing. Differentiating the logarithm termwise,

$$\frac{P'(y)}{P(y)} = \sum_{n\ge 1} \frac{2\pi n\, e^{-2\pi n y}}{1-e^{-2\pi n y}} = 2\pi \sum_{N\ge 1} \sigma(N) e^{-2\pi N y} = \frac{2\pi}{24}\big(1 - E_2(iy)\big).$$

For small y we use $\eta(-1/\tau) = \sqrt{\tau/i}\,\eta(\tau)$, which at $\tau = iy$ reads $\eta(i/y) = \sqrt{y}\,\eta(iy)$. Inserting $\eta(iy) = e^{-\pi y/12} P(y)$ on both sides gives

$$P(y) = y^{-1/2}\, e^{\pi y/12 - \pi/(12y)}\, P(1/y). \tag{27.1}$$

As $y \to 0$, $P(1/y) \to 1$, so $P(y) \sim y^{-1/2} e^{-\pi/(12y)}$; in particular $M > 0$ almost surely. □

The right tail is that of the first clock alone. The left tail decays like $e^{-\pi/(12y)}$, so $\mathbb{E}[M^{-a}] < \infty$ for every $a > 0$, and this is what will make our representations valid on the whole complex plane. The mean has a closed form, $\mathbb{E}[M] = 2/\sqrt{3} - 3/\pi = 0.1997709\ldots$ (Exercise 1).

27.1.1 Gumbel behaviour and the Rényi representation

Because $1 - P(y) \sim e^{-2\pi y}$, the law of M lies in the Gumbel domain of attraction [191, 216]: $P\big((z+\log m)/(2\pi)\big)^m \to e^{-e^{-z}}$ as $m \to \infty$, and at $z = 0$ the left side is 0.34475, 0.36235, 0.36733 for $m = 24, 100, 1000$, against $e^{-1} = 0.36788$. Rényi's representation [455] writes the maximum of m independent standard exponentials, in law, as $\sum_{j\le m} E'_j/j$ with E'_j independent standard exponentials, a generalized gamma convolution with Thorin measure $\sum_{j\le m}\delta_j$ (Chapter 13). Our M has *different* rates $2\pi n$, and it is not a sum of independent exponential variables at all (Exercise 6); in exchange, the arithmetic of the rates makes P modular.

27.2 Ramanujan's Delta as an extreme value law

Let $E_{j,n}$ ($j \le 24$, $n \ge 1$) be independent standard exponentials and

$$M^* = \max_{j\le 24,\, n\ge 1} \frac{E_{j,n}}{2\pi n}, \qquad F(y) = \mathbb{P}(M^* \le y) = P(y)^{24},$$

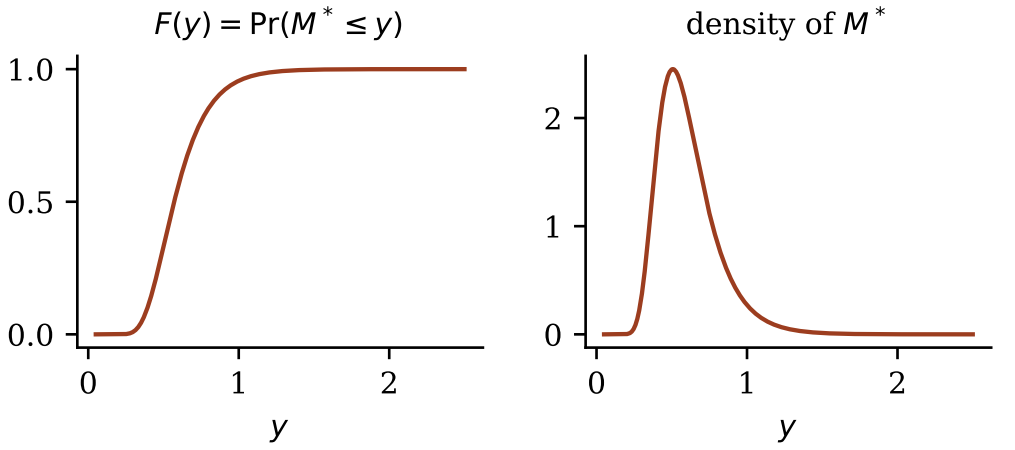


Figure 27.1: The distribution function $F(y) = \prod_n (1 - e^{-2\pi n y})^{24}$ of $M^* = \max_{j \le 24,\, n \ge 1} E_{j,n}/(2\pi n)$, with $E_{j,n}$ independent standard exponential variables (left), and its density (right). The function $e^{-2\pi y} F(y)$ is $\Delta(iy)$.

so that M^* is the maximum of 24 independent copies of M. Since $\Delta = \eta^{24} = q \prod_n (1 - q^n)^{24}$,

$$\Delta(iy) = e^{-2\pi y} F(y) \qquad (y > 0). \tag{27.2}$$

Figure 27.1 shows F and its density.

For complex s and $c > 0$ let $Q(s, c) = \Gamma(s, c)/\Gamma(s)$ be the regularised upper incomplete gamma function,

$$Q(s, c) = \frac{1}{\Gamma(s)} \int_c^\infty t^{s-1} e^{-t}\, dt.$$

For fixed $c > 0$, $\Gamma(\cdot, c)$ and $1/\Gamma$ are entire, so $Q(\cdot, c)$ is entire and vanishes at $s = 0, -1, -2, \ldots$. For real $s > 0$, $Q(s, c) = \mathbb{P}(G_s \ge c)$, where G_s has the gamma law of shape s and unit rate.

THEOREM 27.2 (this book). *For every complex s,*

$$L(s, \Delta) = \mathbb{E}\big[Q(s, 2\pi M^*)\big].$$

For real $s > 0$, equivalently, $L(s, \Delta) = \mathbb{P}(G_s \ge 2\pi M^*)$ with G_s independent of M^*.

PROOF. Recall from Chapter 2 that for $\operatorname{Re} s > 7$ the Dirichlet series converges absolutely and, by termwise integration,

$$\Lambda(s, \Delta) = (2\pi)^{-s} \Gamma(s) L(s, \Delta) = \int_0^\infty \Delta(iy) y^{s-1} dy.$$

Step 1. By (27.2), $0 < \Delta(iy) \le e^{-2\pi y}$, and modularity gives $\Delta(iy) = y^{-12}\Delta(i/y)$, which is at most $y^{-12}e^{-2\pi/y}$. Hence $\int_0^\infty \Delta(iy)y^{\sigma-1}dy < \infty$ for every real σ, locally uniformly in σ, so the Mellin integral defines an entire function that continues $\Lambda(s,\Delta)$, and $L(s,\Delta) = (2\pi)^s\Lambda(s,\Delta)/\Gamma(s)$ is entire, as Hecke showed [250].

Step 2. Fix s with $\sigma = \operatorname{Re} s$. For $m > 0$,

$$Q(s, 2\pi m) = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1}e^{-t}\, 1\{t > 2\pi m\}\, dt.$$

Take expectations and exchange the order. Fubini's theorem applies, because by Tonelli's theorem the absolute double integral is $\int_0^\infty t^{\sigma-1}e^{-t}F(t/2\pi)\,dt = (2\pi)^\sigma \int_0^\infty y^{\sigma-1}\Delta(iy)\,dy$, finite by Step 1; in particular $\mathbb{E}|Q(s, 2\pi M^*)| < \infty$. As M^* has no atoms, $t = 2\pi y$ and (27.2) give

$$\mathbb{E}[Q(s, 2\pi M^*)] = \frac{(2\pi)^s}{\Gamma(s)} \int_0^\infty y^{s-1}\Delta(iy)\,dy = \frac{(2\pi)^s\Lambda(s,\Delta)}{\Gamma(s)} = L(s,\Delta).$$

At the poles of Γ both sides vanish. For real $s > 0$, condition on M^*. □

COROLLARY 27.3. *For real $s > 0$, $0 < L(s,\Delta) < 1$.*

PROOF. G_s and $2\pi M^*$ are independent with positive densities on $(0,\infty)$, so the event $\{G_s \ge 2\pi M^*\}$ and its complement both have positive probability. □

The lower bound is positivity of the Mellin integral of $\Delta(iy)$. The next statement is where the probabilistic form helps: a coupling of gamma variables proves it.

COROLLARY 27.4 (this book). *On $s > 0$, $L(s,\Delta)$ is strictly increasing, and $L(s,\Delta) \uparrow 1$ as $s \to \infty$.*

PROOF. For $0 < s < t$ realise $G_t = G_s + G'$ with G' of shape $t - s$, independent of (G_s, M^*). Then $\{G_s \ge 2\pi M^*\} \subset \{G_t \ge 2\pi M^*\}$, and the difference $\{G_s < 2\pi M^* \le G_s + G'\}$ has positive probability. Since $G_s \to \infty$ in probability and $M^* < \infty$ almost surely, $\mathbb{P}(G_s \ge 2\pi M^*) \to 1$. □

Numerically, $L(1,\Delta) = 0.03744$, $L(3,\Delta) = 0.31524$, $L(6,\Delta) = 0.79212$, $L(10,\Delta) = 0.97981$, $L(12,\Delta) = 0.99454$; these values from PARI/GP [432] agree with quadrature of $\mathbb{E}[Q(s, 2\pi M^*)]$ to sixteen digits, also at complex points such as $s = 6 + 5i$. Figure 27.2 shows the curves for four eta products.

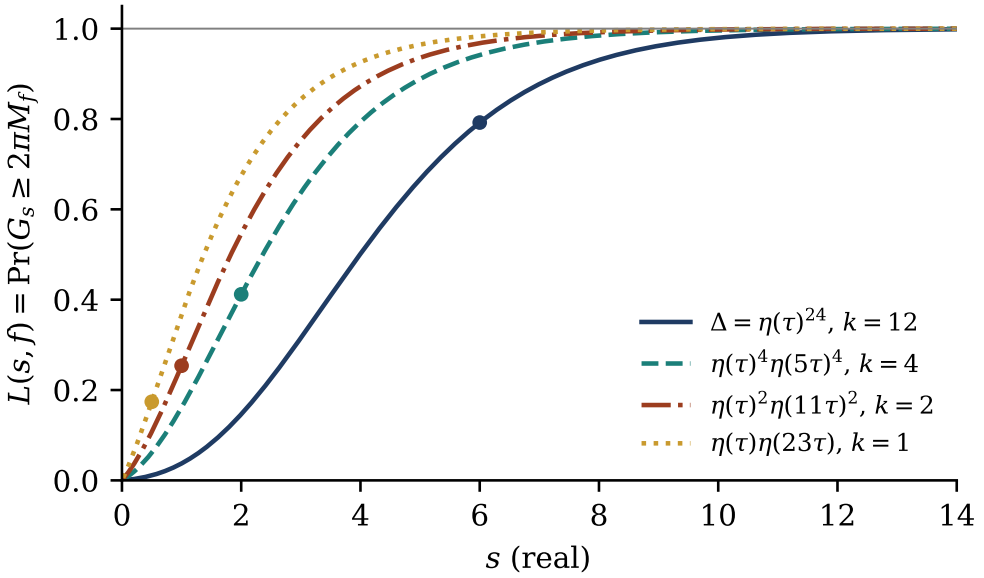


Figure 27.2: $L(s,f) = \mathbb{P}(G_s \geq 2\pi M_f)$ for real s, for the multiplicative eta products $\eta(\tau)^{24}$, $\eta(\tau)^4\eta(5\tau)^4$, $\eta(\tau)^2\eta(11\tau)^2$ and $\eta(\tau)\eta(23\tau)$, of weights $k = 12, 4, 2, 1$; G_s is a gamma variable of shape s, independent of the extreme value M_f of Theorem 27.7. Each curve increases from 0 to 1; dots mark the central values $L(k/2, f)$.

27.3 Modularity as self similarity of the law

At $\tau = iy$ the transformation $\Delta(-1/\tau) = \tau^{12}\Delta(\tau)$ reads $\Delta(i/y) = y^{12}\Delta(iy)$. In terms of F it is a statement about the law of M^*.

Proposition 27.5 (this book). *For all $y > 0$,*

$$F(y) = y^{-12}\, e^{2\pi(y-1/y)}\, F(1/y).$$

Consequently $F(y) \sim y^{-12}e^{-2\pi/y}$ as $y \to 0$, and $1 - F(y) \sim 24e^{-2\pi y}$ as $y \to \infty$.

Proof. Inserting (27.2) into $\Delta(i/y) = y^{12}\Delta(iy)$ gives

$$e^{-2\pi/y}F(1/y) = y^{12}e^{-2\pi y}F(y),$$

which rearranges to the identity (equivalently, raise (27.1) to the power 24). The left tail follows from $F(1/y) \to 1$, the right tail from $1 - P^{24} \sim 24(1 - P)$. □

We checked the identity at $y = 0.5, 0.8, 1.3$ to fifteen digits. So the law of M^* on $(0,1)$ is determined by its law on $(1,\infty)$, and modularity exchanges the two tails:

$\mathbb{P}(1/M^* \geq x) \sim x^{12}e^{-2\pi x}$ is a gamma tail of shape 13, while $1 - F(y) \sim 24e^{-2\pi y}$ is the exponential tail of the 24 slowest clocks, those with $n = 1$.

The functional equation $\Lambda(s, \Delta) = \Lambda(12 - s, \Delta)$ is an integrated form of this self similarity. A sharper reading recovers the Pólya kernel Φ_Δ of Chapter 13 as a conditional law. Put $V_s = G_s/2\pi$.

PROPOSITION 27.6 (this book). *Let $W = \log(G_6/2\pi)$ with G_6 independent of M^*. Conditionally on $\{G_6 \geq 2\pi M^*\}$, W has density Φ_Δ, and for every complex z*

$$\Lambda(6 + z, \Delta)/\Lambda(6, \Delta) = \mathbb{E}\big[e^{zW} \;\big|\; G_6 \geq 2\pi M^*\big].$$

PROOF. Let $g_6(y) = (2\pi)^6 y^5 e^{-2\pi y}/120$ be the density of V_6. For bounded measurable h,

$$\mathbb{E}\big[h(V_6)1\{M^* \leq V_6\}\big] = \int_0^\infty hFg_6\,dy = \frac{(2\pi)^6}{120}\int_0^\infty h(y)y^5\Delta(iy)\,dy.$$

Extend to $h(y) = y^z$ by dominated convergence, using the bounds in Step 1 of the proof of Theorem 27.2, and divide by the case $z = 0$. □

In Bayesian terms Φ_Δ is a posterior: a log gamma prior of shape 6 on W, likelihood $F(e^W)$, and by self similarity a posterior symmetric about 0, with variance 0.1123518 as in Chapter 13.

27.4 MULTIPLICATIVE ETA PRODUCTS

Let $r_m \geq 0$ be integers with $\sum_m m\,r_m = 24$, and put

$$f(\tau) = \prod_m \eta(m\tau)^{r_m} = q\prod_m\prod_{n\geq 1}(1 - q^{mn})^{r_m} = \sum_{n\geq 1} a_n q^n,$$

of weight $k = \frac{1}{2}\sum_m r_m$. Dummit, Kisilevsky and McKay [157] determined which have multiplicative coefficients; our search over all 1575 partitions of 24 (testing $a_{mn} = a_m a_n$ for coprime m, n with $mn \leq 300$) finds the same thirty: twenty eight of integral weight 1 to 12, and $\eta(8\tau)^3$, $\eta(24\tau)$ of weights $\frac{3}{2}$, $\frac{1}{2}$. Each is a holomorphic modular form for a congruence subgroup [327]. The theorem below needs only that $L(s, f) = \sum a_n n^{-s}$ converges absolutely in a right half plane, and its proof checks this directly for every eta product. Let

$$M_f = \max_m \max_{j\leq r_m} \max_{n\geq 1} \frac{E_{m,j,n}}{2\pi mn}, \qquad F_f(y) = \mathbb{P}(M_f \leq y) = \prod_m P(my)^{r_m},$$

the maximum of r_m independent copies of M/m for each m.

THEOREM 27.7 (this book). *Let $f = \prod_m \eta(m\tau)^{r_m}$ with $r_m \geq 0$ and $\sum_m m\, r_m = 24$. Then $f(iy) = e^{-2\pi y} F_f(y)$ for $y > 0$, $L(s,f)$ continues to an entire function, and for every complex s*

$$L(s,f) = \mathbb{E}\big[Q(s, 2\pi M_f)\big].$$

For real $s > 0$, $L(s,f) = \mathbb{P}(G_s \geq 2\pi M_f)$; in particular $0 < L(s,f) < 1$, and $L(s,f)$ increases strictly to 1 as $s \to \infty$.

PROOF. Since $\eta(imy) = e^{-\pi m y/12} P(my)$ and $\sum_m m r_m = 24$, we get $f(iy) = e^{-2\pi y} F_f(y)$. Next, $(\operatorname{Im}\tau)^6 |\Delta(\tau)|$ is invariant under $\mathrm{SL}_2(\mathbb{Z})$ and tends to 0 at the cusp, so it is bounded, and $|\eta(\tau)| \leq C(\operatorname{Im}\tau)^{-1/4}$. Hence $|f(x+iy)| \leq C_f y^{-k/2}$, and $a_n = e^{2\pi n y} \int_0^1 f(x+iy) e^{-2\pi i n x}\, dx$ at $y = 1/n$ gives $|a_n| \leq e^{2\pi} C_f n^{k/2}$. So $L(s,f)$ converges absolutely for $\operatorname{Re} s > k/2 + 1$, where termwise integration gives $(2\pi)^{-s}\Gamma(s) L(s,f) = \int_0^\infty f(iy) y^{s-1}\, dy$. By (27.1), $P(x) \leq e^{\pi/12} x^{-1/2} e^{-\pi/(12x)}$ for $x \leq 1$, so $F_f(y) \leq C y^{-k} e^{-c/y}$ for small y, with $c = \frac{\pi}{12} \sum_m r_m/m > 0$. These two facts replace the appeal to Chapter 2 and Step 1 in the proof of Theorem 27.2. Since F_f has a positive derivative on $(0,\infty)$, the rest of that proof and the proofs of Corollaries 27.3 and 27.4 apply word for word. □

Multiplicativity is not used in the proof. It is what gives the left side an Euler product, and so what makes the right side arithmetic.

27.4.1 An elliptic curve

The product $\eta(\tau)^2 \eta(11\tau)^2$ is the newform of weight 2 and level 11 attached by Eichler and Shimura [171, 514] to the elliptic curve $E = 11a$, $y^2 + y = x^3 - x^2 - 10x - 20$. Here M_f is the maximum of two copies of M and two copies of $M/11$, and since $Q(1,c) = e^{-c}$ the theorem at $s = 1$ reads

$$L(E,1) = \mathbb{E}\big[e^{-2\pi M_f}\big] = \mathbb{P}(E_0 \geq 2\pi M_f) = 0.2538419\ldots,$$

with E_0 a standard exponential independent of M_f. The value agrees with PARI/GP to sixteen digits, and $L(E,1)/\Omega_E = 0.2000000$ with $\Omega_E = 1.2692093$, the value $c_{11}/|E(\mathbb{Q})_{\mathrm{tors}}|^2 = 5/25$ (Tamagawa number 5 at 11, torsion of order 5) predicted by the Birch and Swinnerton Dyer formula [89] with trivial Tate Shafarevich group.

The probability is positive, so the analytic rank is 0, and hence the Mordell Weil rank is 0 by Kolyvagin [329] with Gross and Zagier [228]. But positivity is equally visible from the Mellin integral of $f(iy) > 0$, so this adds interpretation, not information; the same holds for all eight weight two multiplicative eta products, of levels $11, 14, 15, 20, 24, 27, 32, 36$.

27.4.2 Higher weight, weight one and theta series

For $\eta(\tau)^4\eta(5\tau)^4$, of weight 4 and level 5, $L(2,f) = 0.4118613$ is the probability that $G_2 \geq 2\pi M_f$; Table 27.1 gives more.

The weight one product $\eta(\tau)\eta(23\tau)$ is classical. It is half the difference of the theta series of $x^2+xy+6y^2$ and $2x^2+xy+3y^2$, and its coefficient at a prime $p \neq 23$ is $N_p - 1$, with N_p the number of roots of $x^3 - x - 1$ modulo p (checked for $p < 400$; Chapter 23 compares all prime powers below $2 \cdot 10^5$). By Hecke [251] its L function is the Artin L function $L(s,\rho) = \zeta_{\mathbb{Q}(\theta)}(s)/\zeta(s)$ of the two dimensional representation of S_3 of Chapter 3, with $\theta = 1.3247\ldots$ the real root of $x^3 - x - 1$. Theorem 27.7 says that this Artin L function is an expectation,

$$L(s,\rho) = \mathbb{E}\big[Q\big(s, 2\pi \max(M, M'/23)\big)\big],$$

with M, M' independent copies of M. At $s = 1$ the class number formula for $\mathbb{Q}(\theta)$ (signature $(1,1)$, class number 1, regulator $\log\theta$) gives

$$\mathbb{P}\big(E_0 \geq 2\pi \max(M, M'/23)\big) = \frac{2\pi \log\theta}{\sqrt{23}} = 0.3684093\ldots,$$

which we checked to sixteen digits. At the centre, $L(\frac{1}{2},\rho) = 0.1740363$.

The two forms of half integral weight are theta series of Dirichlet characters,

$$\eta(24\tau) = \sum_{n\geq 1} \chi_{12}(n) q^{n^2}, \qquad \eta(8\tau)^3 = \sum_{n\geq 1} \chi_{-4}(n)\, n\, q^{n^2},$$

with Dirichlet series $L(2s, \chi_{12})$ and $L(2s-1, \chi_{-4})$; Chapter 18 builds the Pólya kernels of $L(s,\chi_{12})$ and $L(s,\chi_{-4})$ from them. For independent copies M_1, M_2, M_3 of M,

$$\frac{\pi}{4} = \mathbb{P}\Big(E_0 \geq \frac{\pi}{4} \max_{j\leq 3} M_j\Big), \qquad \beta(2) = \mathbb{P}\Big(G_{3/2} \geq \frac{\pi}{4} \max_{j\leq 3} M_j\Big),$$

where $\beta(2) = L(2,\chi_{-4}) = 0.9159656$ is Catalan's constant, and $L(1,\chi_{12}) = \log(2+\sqrt{3})/\sqrt{3} = \mathbb{P}(G_{1/2} \geq \pi M/12) = 0.7603460$. All three were checked by quadrature to twelve digits.

Every entry of Table 27.1 lies in $(0,1)$, and each row increases with s, as Theorem 27.7 requires. Within each weight the values grow with the level: the clocks of $\eta(m\tau)$ ring at rates $2\pi mn$, so large m makes M_f small and easy to beat.

27.5 Zeros as a probabilistic statement

If ρ is a zero of $L(s,\Delta)$, Theorem 27.2 gives $\mathbb{E}[Q(\rho, 2\pi M^*)] = 0$. At the first zero $\rho = 6 + 9.2223794\,i$ our quadrature of the left side has modulus below 10^{-15}. At

f	N	k	$L(\frac{k}{2},f)$	$L(\frac{k+1}{2},f)$	$L(k,f)$
$\eta(\tau)^{24}$	1	12	0.7921228	0.8393455	0.9945437
$\eta(\tau)^8\eta(2\tau)^8$	2	8	0.6812925	0.7543068	0.9710803
$\eta(\tau)^6\eta(3\tau)^6$	3	6	0.5600387	0.6552302	0.9190337
$\eta(2\tau)^{12}$	4	6	0.7805377	0.8543050	0.9862573
$\eta(\tau)^4\eta(5\tau)^4$	5	4	0.4118613	0.5320174	0.7938049
$\eta(3\tau)^8$	9	4	0.7599657	0.8578161	0.9757577
$\eta(\tau)^3\eta(7\tau)^3$	7	3	0.3329818	0.4672118	0.6875637
$\eta(\tau)^2\eta(11\tau)^2$	11	2	0.2538419	0.4070039	0.5460480
$\eta(\tau)\eta(2\tau)\eta(7\tau)\eta(14\tau)$	14	2	0.3302237	0.5008415	0.6414713
$\eta(\tau)\eta(3\tau)\eta(5\tau)\eta(15\tau)$	15	2	0.3501508	0.5228611	0.6614752
$\eta(6\tau)^4$	36	2	0.7010911	0.8618161	0.9400130
$\eta(\tau)\eta(23\tau)$	23	1	0.1740363	0.3684093	0.3684093
$\eta(2\tau)\eta(22\tau)$	44	1	0.3157625	0.5772184	0.5772184
$\eta(12\tau)^2$	144	1	0.6120576	0.8706850	0.8706850

Table 27.1: $L(s,f)=\mathbb{P}(G_s\ge 2\pi M_f)$ at $s=\frac{k}{2},\frac{k+1}{2},k$ for fourteen multiplicative eta products f of level N and weight k (for $k=1$ the last two columns coincide), computed with PARI/GP (`lfunetaquo`) and confirmed by quadrature; the functional equation of each form was checked numerically.

the trivial zeros $s=0,-1,-2,\dots$, $Q(s,c)=0$ for *every* $c>0$, so the integrand itself vanishes. The nontrivial zeros arise by cancellation. For fixed $c>0$, $s\mapsto Q(s,c)$ also has complex zeros. For $c=2\pi y$, with y the 10%, 50% and 90% quantiles of M^*, the zeros of smallest positive imaginary part are near $5.293+6.081i$, $7.003+7.009i$ and $9.669+8.264i$, off the line $\operatorname{Re}s=6$; the zeros of $L(s,\Delta)$ appear only after averaging. The Riemann hypothesis for Δ takes the following form, which is a reformulation, not progress.

CONJECTURE 27.8 (Riemann hypothesis for Δ, extreme value form). *If* $0<\operatorname{Re}s<12$ *and* $\mathbb{E}[Q(s,2\pi M^*)]=0$, *then* $\operatorname{Re}s=6$.

Hecke's formula

$$\Lambda(s,\Delta)=\sum_{n\ge1}\tau(n)\big[\Gamma(s,2\pi n)(2\pi n)^{-s}+\Gamma(12-s,2\pi n)(2\pi n)^{s-12}\big],$$

obtained by splitting the Mellin integral at $y=1$ and applying modularity on $(0,1)$, is also an incomplete gamma expansion, but its terms carry the signs of $\tau(n)$; ours averages one kernel against a probability law. For Riemann's ξ the analogous probabilistic formula is $2\xi(s)=\mathbb{E}[Y^s]$ of Biane, Pitman and Yor [82], with Y a constant multiple of the maximum of a Brownian excursion (Chapter 13). Proposition 27.6 gives $\Lambda(s,\Delta)$ a probabilistic reading of the same kind, and on the reciprocal side $\xi(\frac12)/\xi(\frac12+\sqrt{s})$ is the Laplace transform of a generalized

gamma convolution if and only if the Riemann hypothesis holds (Polson [441]; Theorem 14.8).

Why does this not prove the Riemann hypothesis? First, positivity of a mixing law controls the transform on the real axis (Corollary 27.4), not the cancellation that creates zeros. Second, Theorem 27.7 holds for every eta product with $\sum m r_m = 24$, multiplicative or not, so it does not see the Euler product, and without an Euler product zeros can leave the line, as Davenport and Heilbronn showed [126]. Third, the known sufficient conditions for a Fourier transform to have only real zeros [444, 133, 417] rest on special structure of the kernel (an explicit form in the theorems of Pólya and de Bruijn, the Lee Yang property in Newman's), and nothing we have shown about M^* supplies such structure. What the representation does provide is an exact probabilistic formula and an unbiased estimator of L at complex points, though one whose variance grows exponentially with the height, as the next section shows.

27.6 Monte Carlo

Theorem 27.2 turns evaluation of $L(s,\Delta)$ into sampling. We truncate the maximum at $n \le N$ and call the result $M_N^* \le M^*$. Since $M_N^* \ne M^*$ requires some clock with $n > N$ to ring after M_N^*, the union bound gives

$$\mathbb{P}\big(M^* > y \ge M_N^*\big) \le 24 \sum_{n>N} e^{-2\pi n y} = \frac{24\, e^{-2\pi(N+1)y}}{1 - e^{-2\pi y}}.$$

For real $s > 0$ the bias is $\mathbb{P}(M_N^* \le V_s < M^*)$, at most the integral of the smaller of 1 and this bound against the density of V_s, which for $s = 6$ and $N = 400$ is 1.2×10^{-13}. The algorithm draws $E_{j,n}$ for $j \le 24$, $n \le N$, sets $m_r = \max_{j,n} E_{j,n}/(2\pi n)$, and returns the mean and standard error of $Q(s, 2\pi m_r)$ over R draws. With $N = 400$ and $R = 20000$ draws we obtained 0.7929 ± 0.0012 (one standard error) for $L(6,\Delta) = 0.7921$. The empirical 10%, 50%, 90% quantiles of M^* were 0.394, 0.569, 0.869, against exact values 0.3935, 0.5708, 0.8650. For real $s > 0$ the estimator is the conditional expectation of $1\{G_s \ge 2\pi M_N^*\}$ given M_N^*, so by the Rao Blackwell theorem its variance is smaller, in our runs by a factor of about five. For a fixed sample, $s \mapsto R^{-1} \sum_r Q(s, 2\pi m_r)$ is an entire empirical L function. On the critical line, however, the summands are large while L is not: for fixed c, $|Q(6+it, c)| \sim c^6 e^{-c} e^{\pi t/2} / (\sqrt{2\pi}\, t^{13/2})$ as $t \to \infty$, because $1/|\Gamma(6+it)|$ grows like $e^{\pi t/2}$, and the standard deviation of $Q(6+it, 2\pi M^*)$, computed by quadrature, is 0.18 at $t = 0$, 22 at the first zero $t = 9.22$ and 4.0×10^6 at $t = 20$. Locating zeros by simulation therefore requires a cancellation that grows exponentially with the height.

27.7 HITTING TIMES

If coupon (j,n) is collected at time $E_{j,n}/(2\pi n)$, the number of coupons still missing at time $t > 0$ is finite, with mean $24\sum_n e^{-2\pi nt}$. So M^* is the time at which a collection that starts with infinitely many missing coupons is completed: the hitting time of the empty set by the decreasing Markov process of missing coupons, which enters from an infinite configuration. For real $s > 0$, $L(s,\Delta)$ is the probability that the collection is completed before an independent gamma clock of shape s and rate 2π rings.

The laws that Biane, Pitman and Yor [82] build from theta functions are laws of Brownian functionals. Is M a Brownian functional? We do not know, but a related identity holds. Let $K = \sup_{t\le 1}|b_t|$ for a standard Brownian bridge b, so that $\mathbb{P}(K \le x) = \sum_{n\in\mathbb{Z}}(-1)^n e^{-2n^2x^2}$ is Kolmogorov's law [328].

PROPOSITION 27.9 (this book). *K^2 has the law of the maximum of independent exponential variables, one of rate $2j$ for each even $j \ge 2$ and two of rate $2j$ for each odd $j \ge 1$. With M_1, M_2, M_3 independent copies of M, and K independent of M_3,*

$$\max(M_1, M_2) \stackrel{d}{=} \max\bigl(K^2/\pi,\ M_3/2\bigr).$$

PROOF. Jacobi's triple product gives

$$\sum_{n\in\mathbb{Z}}(-1)^n q^{n^2} = \prod_{n\ge 1}(1-q^{2n})(1-q^{2n-1})^2.$$

With $q = e^{-2x^2}$ the left side is $\mathbb{P}(K^2 \le x^2)$ and each factor $1-q^j$ is the probability that an exponential variable of rate $2j$ is at most x^2; independence gives the first claim. The same identity in the form $\sum_n(-1)^n q^{n^2} = \eta(\tau)^2/\eta(2\tau)$, at $q = e^{-2\pi y}$, reads $P(y)^2 = \mathbb{P}(K^2 \le \pi y)\,P(2y)$, which is the second claim. □

We checked it at $y = 0.1, 0.2, 0.5$ to fifteen digits. It suggests, but does not establish, that M can be realised through passage or exit times of a diffusion. The simplest mechanism is excluded: the passage time of a one dimensional diffusion from an entrance or reflecting boundary to a fixed level is a sum of independent exponential variables [308], and M is not (Exercise 6).

OPEN PROBLEM 27.1. Find a natural Brownian or Bessel functional with the law of M, or of M^*, or show that none exists among first passage and last exit times of one dimensional diffusions.

27.8 NOTES

That $\prod_n(1-e^{-2\pi ny})$ is a distribution function is elementary, and such products pervade the probabilistic literature on theta functions, from Kolmogorov [328] to

Biane, Pitman and Yor [82]. What we have not seen before is the representation of L functions of eta products as expectations of incomplete gamma functions at an extreme value, with its consequences: special values as probabilities, monotonicity, modular self similarity, and the posterior reading of the Pólya kernel. These are the results we label as ours.

Martin [382] classified the multiplicative eta quotients; for quotients our argument fails, since $1/(1-e^{-2\pi n y})$ is not a probability. Köhler [327] is the reference for eta products and theta series, Gumbel [229] for extreme values, and Thorin [546] and Bondesson [73] for gamma convolutions. Hecke's formula underlies the evaluation of L functions in PARI/GP and the LMFDB [371].

27.9 Exercises

1. Prove $\mathbb{E}[M] = 2/\sqrt{3} - 3/\pi$. (Integrate the pentagonal series for $1 - P(y)$ termwise, pair k with $-k$, and evaluate $\int_0^1 x(1-x^2)/(1+x^3)\,dx = 2\pi/(3\sqrt{3}) - 1$.)

2. Using $\mathbb{P}(aG_s < E) = \mathbb{E}[e^{-aG_s}] = (1+a)^{-s}$ for a standard exponential E independent of G_s, and a union bound, prove for real $s > 1$ that $1 - L(s,f) \le \sum_m r_m \sum_{n \ge 1} (1+mn)^{-s}$. Evaluate the bound for Δ at $s = 12$ and compare with $1 - L(12,\Delta) = 0.0054563$.

3. Show that M^* has density $2\pi(1 - E_2(iy))F(y)$. Use the transformation law $E_2(-1/\tau) = \tau^2 E_2(\tau) + 6\tau/(\pi i)$ and Proposition 27.5 to find its behaviour as $y \to 0$.

4. Prove $\mathbb{E}[K^2] = \pi^2/12$ from Kolmogorov's series. Deduce from Proposition 27.9 that $\mathbb{E}[\max(M_1, M_2)] \le \pi/12 + \mathbb{E}[M]/2$, and check it numerically.

5. For real $s \in (0, 12)$ let W_s be $\log(G_s/2\pi)$ conditioned on $\{G_s \ge 2\pi M^*\}$. Show that $\mathbb{E}[e^{zW_s}] = \Lambda(s+z,\Delta)/\Lambda(s,\Delta)$, that the law of W_s is the exponential tilt of the law of W_6 by $e^{(s-6)u}$, and that $W_{12-s} \overset{d}{=} -W_s$.

6. Show that $\mathbb{E}[e^{-uM}] = \sum_{k \in \mathbb{Z}} (-1)^k u/(u + \pi k(3k-1))$ for $\operatorname{Re} u > 0$, and, using $\sum_{k \in \mathbb{Z}} (-1)^k/(k-a) = -\pi/\sin \pi a$, that with $w = \sqrt{1 - 12u/\pi}$

$$\mathbb{E}\left[e^{-uM}\right] = \frac{u}{w}\Big(\csc\frac{\pi(1-w)}{6} - \csc\frac{\pi(1+w)}{6}\Big).$$

   Deduce that the meromorphic continuation of $\mathbb{E}[e^{-uM}]$ vanishes at $u = -35\pi/12$, and conclude that M is not a sum of independent exponential variables. (The Laplace transform of $\sum_j E_j/\lambda_j$ is the reciprocal of the entire function $\prod_j (1 + u/\lambda_j)$.)

27.10 Open computational problems

Computational Problem 27.1 (Tables of the laws M_f). Objective: reference tables of the laws M_f for all thirty multiplicative eta products. Objects and scale: $F_f(y) = \prod_m P(my)^{r_m}$, densities, quantiles at levels $10^{-6}, \dots, 1 - 10^{-6}$, means, variances and the first twenty negative moments, to 50 digits. Route: the pentagonal series for $my \geq 1$ and (27.1) above, in PARI/GP or mpmath; Newton iteration for quantiles. Deliverable: tables and a common plot of all thirty laws. Why it matters: these laws are the arithmetic content of Theorem 27.7, and their tails encode weight and level through self similarity.

Computational Problem 27.2 (Variance reduction and zero location). Objective: make $\mathbb{E}[Q(s, 2\pi M_f)]$ a practical estimator of $L(s, f)$ on the critical line. For Δ at $s = 6 + it$ the plain estimator has standard deviation 0.18 at $t = 0$, 22 at the first zero and 4.0×10^6 at $t = 20$, growing like $e^{\pi t/2} t^{-13/2}$. Objects and scale: Δ on $s = 6 + it$ and $\eta(\tau)^2 \eta(11\tau)^2$ on $s = 1 + it$ for $0 \leq t \leq 30$, with up to 10^9 draws. Route: GPU sampling of truncated maxima; control variates from the slowest clocks, which have equal rates (the 24 clocks of rate 2π for Δ, the two of rate 2π for the level 11 form), so that their maximum has Rényi's representation, and from the exact values on the real axis; stratification on quantiles of M_f; randomised quasi Monte Carlo through the inverse distribution function. Deliverable: for each method the variance as a function of t and its exponential rate, and the largest height up to which the zeros of the sample average locate the zeros of $L(s, f)$ to within 0.01, with error bars. Why it matters: an estimator able to locate zeros would give a probabilistic certificate of their positions, independent of the approximate functional equation, and the growth rate measures the cancellation that any probabilistic route to the zeros must overcome.

Computational Problem 27.3 (Eigenforms that are not eta products). Objective: extend the representation to Hecke eigenforms f for which $g_f(y) = e^{2\pi y} f(iy)$ is not a distribution function; when the root number is -1, $f(iy)$ changes sign. Since g_f is smooth and of bounded variation, with $g_f(0+) = 0$ and $g_f(\infty) = 1$, its Jordan decomposition gives $g_f = aF_1 - bF_2$ with distribution functions F_1, F_2 on $(0, \infty)$, $a - b = 1$, and $a + b$ equal to the total variation V_f of g_f, the least possible value; the proof of Theorem 27.7 then gives $L(s, f) = a\mathbb{E}[Q(s, 2\pi X_1)] - b\mathbb{E}[Q(s, 2\pi X_2)]$ with X_i of law F_i. Objects and scale: all newforms of weight 2 with rational coefficients and level up to 500, and the level one eigenforms of weights 16 to 24. Route: q expansions from PARI/GP `mfinit` or the LMFDB, with the Fricke involution for small y; V_f by quadrature of $|g_f'|$; linear programming over mixtures of the laws $\prod_m P(my)^{r_m}$ and their rescalings, to approximate F_1 and F_2 by extreme value laws. Deliverable: for each form V_f and $b = (V_f - 1)/2$, against level, weight, root number and rank, and the error of the best extreme

value approximations. Why it matters: b measures the distance from the positive world of this chapter; it is positive whenever $L(\frac{k}{2}, f) = 0$, in particular for every elliptic curve of positive rank, since then $L(1, f) = 0$ by Kolyvagin's theorem [329].

Part V

Inference and Open Problems

CHAPTER 28

RIGHT HAAR MEASURE, TRANSFORMATION MODELS AND THE P* FORMULA

In 1933 Alfréd Haar proved that every separable locally compact group carries a measure invariant under translation [231]; von Neumann's uniqueness theorems and Weil's book of 1940 completed the theory in the generality used today. Statisticians met the same measure from the other side. Fisher's fiducial argument [190], Fraser's structural inference [200] and the invariant estimation theory of Stein [525] and Hora and Buehler [265] live on groups such as the affine group of the line, which are not unimodular: their left and right Haar measures differ, and the right one is the prior whose posterior reproduces the fiducial distribution and whose invariant credible sets have exact frequentist coverage. Barndorff Nielsen's p^* formula [37] uses the left one: in a transformation model $|\hat{\jmath}|^{1/2} d\hat{g}$ is a left Haar measure, which is why p^* is exact there.

This chapter puts the two traditions side by side, because the affine group of the line is, up to isomorphism, the triangular subgroup of $\mathrm{SL}_2(\mathbb{R})$ that acts simply transitively on the upper half plane. Sato Tate groups are compact, so left and right Haar measures agree and Haar measure is the law of Frobenius (Chapter 4); in Chapter 29 it enters the likelihood. The upper half plane, by contrast, is an orbit of a group that is not unimodular, and its hyperbolic area is the image of left Haar measure. Duke's theorem says that Heegner points equidistribute for this measure, a noncompact analogue of Sato Tate. Haar measures also normalise volumes, Tamagawa numbers and the period formulas behind central L values. The statistical theorems are classical, and we prove Theorems 28.11 and 28.14 in the generality of Definition 28.8; Proposition 28.16 on reference priors for conjugacy classes and the computations are new.

28.1 HAAR MEASURE ON LOCALLY COMPACT GROUPS

Left and right Haar measures can differ. We define the modular function that measures the difference, compute both measures for the affine group and the Borel subgroup of GL_2, and transport them to the upper half plane and the Bruhat Tits tree.

28.1.1 Existence, uniqueness and the modular function

Let G be a locally compact Hausdorff group. A *left Haar measure* is a nonzero Radon measure μ_l with $\mu_l(gE) = \mu_l(E)$ for all Borel E and $g \in G$; a right Haar measure satisfies $\mu_r(Eg) = \mu_r(E)$.

THEOREM 28.1 (Haar, von Neumann, Weil). *Every locally compact group has a left Haar measure, unique up to a positive constant.*

Haar [231] proved existence for second countable groups; von Neumann proved uniqueness for compact groups [556] and then for the groups of Haar's theorem [557]; and Weil [566] proved existence and uniqueness without countability assumptions. A modern proof is in [194]. If μ_l is a left Haar measure and $g \in G$, then $E \mapsto \mu_l(Eg)$ is again left invariant, so by uniqueness

$$\mu_l(Eg) = \Delta_G(g)\,\mu_l(E) \tag{28.1}$$

for a number $\Delta_G(g) > 0$ independent of E. This is our convention for the *modular function*; it is that of [194]. The map $\Delta_G : G \to \mathbb{R}_{>0}$ is a continuous homomorphism, and in differential form $d\mu_l(xg) = \Delta_G(g)\,d\mu_l(x)$.

LEMMA 28.2. *The measure $d\mu_r(x) = \Delta_G(x)^{-1} d\mu_l(x)$ is a right Haar measure, and every right Haar measure is a multiple of it. For a Lie group $\Delta_G(g) = |\det \mathrm{Ad}(g)|^{-1}$.*

PROOF. Substituting $x = yg$ and using (28.1),

$$\mu_r(Eg) = \int 1_{Eg}(x)\,\frac{d\mu_l(x)}{\Delta_G(x)} = \int 1_E(y)\,\frac{\Delta_G(g)\,d\mu_l(y)}{\Delta_G(y)\Delta_G(g)} = \mu_r(E).$$

Uniqueness follows from Theorem 28.1 applied to the opposite group. For the Lie group formula, let μ_l be given by a left invariant volume form ω, determined by its value ω_e on the Lie algebra. Right translation $R_g : x \mapsto xg$ commutes with left translations, so it pulls ω back to the left invariant form with value $\omega_e \circ \mathrm{Ad}(g^{-1}) = \det \mathrm{Ad}(g)^{-1}\,\omega_e$ at e, and $\mu_l(Eg) = \int_E |R_g^*\omega| = |\det \mathrm{Ad}(g)|^{-1}\mu_l(E)$. □

G is *unimodular* if $\Delta_G \equiv 1$, in which case left and right Haar measures coincide.

EXAMPLE 28.3 (Unimodular groups). (i) Compact groups: $\Delta_G(G)$ is a compact subgroup of $\mathbb{R}_{>0}$, hence $\{1\}$. So a compact group, in particular a Sato Tate group, has exactly one Haar probability measure, and the phrase "Haar measure" in Chapters 4 and 29 is unambiguous. (ii) $\mathrm{GL}_n(F)$ for $F = \mathbb{R}$ or $\mathbb{Q}_p$: on the matrix space $M_n(F)$ with additive Haar measure dx, left and right multiplication by g both scale dx by $|\det g|^n$, so $|\det x|^{-n}dx$ is a left and right Haar measure. (iii) $\mathrm{SL}_2(\mathbb{R})$ and $\mathrm{SL}_2(\mathbb{Q}_p)$ are perfect groups, equal to their commutator subgroups, and Δ_G is a homomorphism into an abelian group, so it is trivial. (iv) Connected nilpotent Lie groups: every g is $\exp X$ for some X in the Lie algebra, and $\mathrm{Ad}(g) = \exp(\mathrm{ad}\, X)$ is unipotent, so $\det \mathrm{Ad}(g) = 1$.

28.1.2 The affine group and the Borel subgroup

The simplest group that is not unimodular is the affine group of the line,

$$B = \left\{ (a,b) = \begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix} : a > 0,\ b \in \mathbb{R} \right\}, \qquad (a,b)(a',b') = (aa', ab' + b),$$

acting on $\mathbb{R}$ by $x \mapsto ax + b$.

PROPOSITION 28.4. *On B, $d\mu_l = a^{-2}\,da\,db$ and $d\mu_r = a^{-1}\,da\,db$, and $\Delta_B(a,b) = a^{-1}$.*

PROOF. Left translation by (α,β) is the map $(a,b) \mapsto (\alpha a, \alpha b + \beta)$, whose Jacobian matrix with respect to (a,b) is $\mathrm{diag}(\alpha,\alpha)$, of determinant α^2. Hence for the image point (a',b') we have $da'\,db'/a'^2 = \alpha^2\,da\,db/(\alpha^2 a^2) = da\,db/a^2$, so $a^{-2}da\,db$ is left invariant. Right translation by (α,β) is $(a,b) \mapsto (a\alpha, a\beta + b)$, with Jacobian matrix $\left(\begin{smallmatrix} \alpha & 0 \\ \beta & 1 \end{smallmatrix}\right)$ of determinant α; then $da'\,db'/a' = \alpha\,da\,db/(a\alpha) = da\,db/a$, so $a^{-1}da\,db$ is right invariant. Under the same right translation $da'\,db'/a'^2 = \alpha^{-1}da\,db/a^2$, so $\Delta_B(\alpha,\beta) = \alpha^{-1}$ by (28.1), in agreement with $d\mu_r = \Delta_B^{-1} d\mu_l$. □

For the Borel subgroup $B_2 = \left\{ \left(\begin{smallmatrix} a & b \\ 0 & d \end{smallmatrix}\right) \right\}$ of $\mathrm{GL}_2(F)$, $F = \mathbb{R}$ or $\mathbb{Q}_p$, the same Jacobian computation gives $d\mu_l = |a|^{-2}|d|^{-1}da\,db\,dd$, $d\mu_r = |a|^{-1}|d|^{-2}da\,db\,dd$ and $\Delta_{B_2} = |d/a|$ (Exercise 2). The inverse $|a/d|$ is the character usually written δ_B in representation theory, and the normalised induction that defines principal series representations twists by $\delta_B^{1/2}$ precisely to compensate for the failure of unimodularity.

28.1.3 Iwasawa decomposition

Let $G = \mathrm{SL}_2(\mathbb{R})$ with $K = \mathrm{SO}(2)$ and $B = B^+ = \{n(x)a(y)\}$, where $n(x) = \left(\begin{smallmatrix} 1 & x \\ 0 & 1 \end{smallmatrix}\right)$ and $a(y) = \mathrm{diag}(y^{1/2}, y^{-1/2})$, $y > 0$; or let $G = \mathrm{GL}_2(\mathbb{Q}_p)$ with $K = \mathrm{GL}_2(\mathbb{Z}_p)$ and B the upper triangular group. In both cases $G = BK$ (Iwasawa decomposition), and a Haar measure dg on the unimodular group G factors through left Haar measure on B:

$$\int_G f(g)\,dg = \int_B \int_K f(bk)\,d\mu_l(b)\,dk, \tag{28.2}$$

with dk the Haar probability measure on K [92, 194]. Formula (28.2) fixes the normalisation of dg; in the p adic case, normalising μ_l so that the compact group $B \cap K$ has measure one gives K measure one. The proof is the uniqueness argument again: $B \times K$ acts transitively on G by $g \mapsto bgk^{-1}$ with compact stabiliser $B \cap K$, so a measure on G invariant under this action is unique up to scale; Haar measure is one such measure because G is unimodular, and the image of $\mu_l \times dk$ under $(b,k) \mapsto bk$ is another. It is the left, not the right, Haar measure of B that appears.

28.1.4 The upper half plane

$\mathrm{SL}_2(\mathbb{R})$ acts on $\mathbb{H} = \{x+iy : y>0\}$ by Möbius transformations, the stabiliser of i is $\mathrm{SO}(2)$, and so $\mathbb{H} = \mathrm{SL}_2(\mathbb{R})/\mathrm{SO}(2)$. Since $n(x)a(y)\cdot z = yz+x$, the group B^+ is the affine group $(a,b) = (y,x)$ of Proposition 28.4, acting on $\mathbb{H}$ simply transitively, and the orbit map $\phi(a,b) = (a,b)\cdot i = b+ia$ is a bijection $B^+ \to \mathbb{H}$.

PROPOSITION 28.5. *The image under ϕ of left Haar measure $a^{-2}da\,db$ is the hyperbolic area $dx\,dy/y^2$; the image of right Haar measure is $dx\,dy/y$. Every B^+ invariant Radon measure on $\mathbb{H}$ is a multiple of $dx\,dy/y^2$, and this measure is invariant under all of $\mathrm{SL}_2(\mathbb{R})$. The measure $dx\,dy/y$ is not B^+ invariant.*

PROOF. In coordinates ϕ is $(a,b)\mapsto(x,y) = (b,a)$, so the images are as stated. Since $\phi(hg) = h\cdot\phi(g)$, a measure m on $\mathbb{H}$ is B^+ invariant exactly when $\phi_*^{-1}m$ is left invariant on B^+; by Theorem 28.1 these are the multiples of $a^{-2}da\,db$. For $g = \left(\begin{smallmatrix}\alpha & \beta\\ \gamma & \delta\end{smallmatrix}\right) \in \mathrm{SL}_2(\mathbb{R})$ the map $z\mapsto gz$ is holomorphic with derivative $(\gamma z+\delta)^{-2}$, so its real Jacobian is $|\gamma z+\delta|^{-4}$, while $\mathrm{Im}(gz) = y|\gamma z+\delta|^{-2}$; the two factors cancel in $dx\,dy/y^2$. Finally, for $c>0$ the measure $dx\,dy/y$ of cE is c times that of E. □

Thus hyperbolic area is the image of left Haar measure. The Iwasawa formula explains why it is invariant under the whole group although B^+ is not unimodular: on G/K with G unimodular and K compact the invariant measure is the image of dg, and by (28.2) this image is the image of $d\mu_l$ on B^+.

The standard fundamental domain of $\mathrm{SL}_2(\mathbb{Z})$ is $\mathcal{F} = \{|x|\le\tfrac12,\ x^2+y^2\ge 1\}$ (Figure 28.1), and its area is

$$\int_{-1/2}^{1/2}\int_{\sqrt{1-x^2}}^{\infty}\frac{dy}{y^2}\,dx = \int_{-1/2}^{1/2}\frac{dx}{\sqrt{1-x^2}} = 2\arcsin\tfrac12 = \frac{\pi}{3}. \tag{28.3}$$

Consequently $\frac{3}{\pi}\,dx\,dy/y^2$ is the invariant probability measure on $\mathrm{SL}_2(\mathbb{Z})\backslash\mathbb{H}$. The Petersson inner product $\langle f,g\rangle = \int_{\mathcal{F}} f\bar g\, y^k\,dx\,dy/y^2$ of weight k forms integrates against this measure, and Rankin and Selberg's unfolding, the source of $L(s,\Delta\times\Delta)$, uses the same invariance: with $\Gamma_\infty = \{\pm\left(\begin{smallmatrix}1 & m\\ 0 & 1\end{smallmatrix}\right) : m\in\mathbb{Z}\}$ and $E(z,s) = \sum_{\gamma\in\Gamma_\infty\backslash\mathrm{SL}_2(\mathbb{Z})}\mathrm{Im}(\gamma z)^s$,

$$\begin{aligned}\int_{\mathcal{F}}|\Delta(z)|^2y^{12}E(z,s)\,\frac{dx\,dy}{y^2} &= \int_0^\infty\int_0^1 |\Delta|^2y^{s+10}\,dx\,dy\\ &= \frac{\Gamma(s+11)}{(4\pi)^{s+11}}\sum_{n\ge1}\frac{\tau(n)^2}{n^{s+11}}\end{aligned}$$

[276]. The Mellin integral $\Lambda(s,\Delta) = \int_0^\infty \Delta(iy)y^s\,dy/y$ of Chapter 13 uses a third Haar measure, dy/y on the diagonal torus, which is abelian and therefore unimodular.

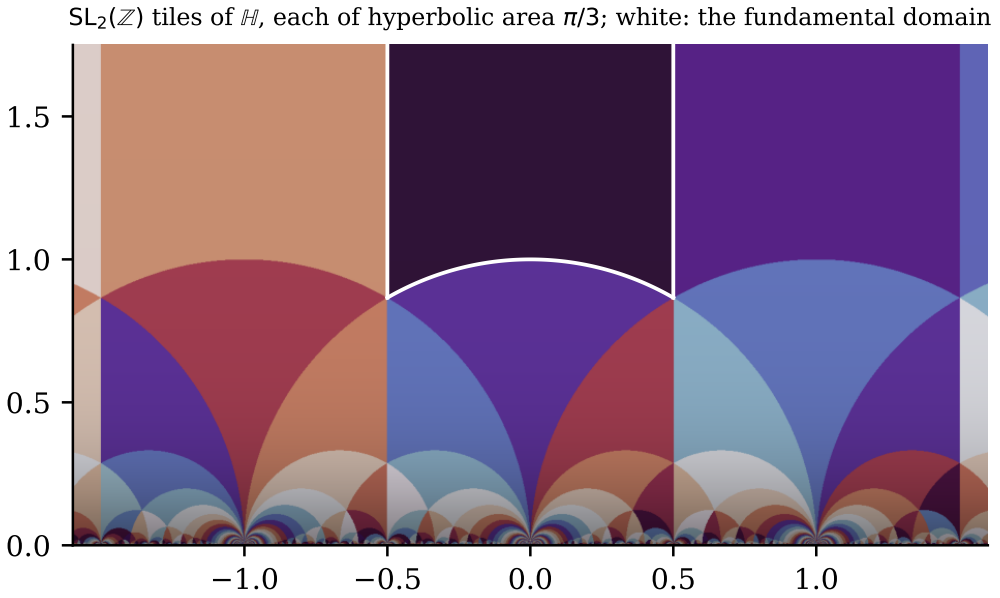


Figure 28.1: The tiles $\gamma\mathcal{F}$, $\gamma \in \mathrm{SL}_2(\mathbb{Z})$, of the upper half plane, coloured by reduction depth; the standard fundamental domain $\mathcal{F}$ is outlined in white. Each tile has hyperbolic area $\pi/3$ for the left Haar image $dx\,dy/y^2$.

28.1.5 The Bruhat Tits tree and two Haar shadows

For a prime p the vertex set of the Bruhat Tits tree $\mathcal{T}_p$ of Chapter 9 is

$$\mathrm{GL}_2(\mathbb{Q}_p)/\mathbb{Q}_p^{\times}\mathrm{GL}_2(\mathbb{Z}_p) = \mathrm{PGL}_2(\mathbb{Q}_p)/\mathrm{PGL}_2(\mathbb{Z}_p).$$

Normalise Haar measure on the unimodular group $\mathrm{PGL}_2(\mathbb{Q}_p)$ so that the open compact subgroup $\mathrm{PGL}_2(\mathbb{Z}_p)$ has measure one. Each vertex is a coset $g\mathrm{PGL}_2(\mathbb{Z}_p)$, of measure one, so the image of Haar measure on the vertex set is counting measure, the unique invariant measure on a discrete homogeneous space up to scale.

Two probability measures on $[-2,2]$ come from Haar measures in this picture. The Plancherel measure of $\mathrm{PGL}_2(\mathbb{Q}_p)$ restricted to the unramified tempered dual, for this normalisation of Haar measure, is the Kesten McKay law μ_p (Macdonald [379]; Chapter 9). The Sato Tate law μ_{ST} is the image of Haar measure on the compact group $\mathrm{SU}(2)$, the maximal compact subgroup of the dual group $\mathrm{SL}_2(\mathbb{C})$, under the map to conjugacy classes. One is a shadow of Haar measure on a noncompact p adic group, read through harmonic analysis; the other is a shadow of Haar measure on a compact real form of the complex dual group, read through Weyl's integration formula. They are related by the limit $\mu_p \Rightarrow \mu_{\mathrm{ST}}$ as $p \to \infty$

(Theorem 9.1), and arithmetically by Serre's vertical theorem (Theorem 9.4) against the horizontal Sato Tate theorem.

28.2 Equidistribution on a noncompact quotient

The quotient $\mathrm{SL}_2(\mathbb{Z})\backslash\mathbb{H}$ has finite volume but is not compact. Duke's theorem is the equidistribution theorem for its invariant probability measure $\frac{3}{\pi}\,dx\,dy/y^2$, the normalised image of left Haar measure.

28.2.1 Duke's theorem

Let $D < -4$ be a fundamental discriminant. A positive definite primitive binary quadratic form $Q = (a, b, c)$ with $b^2 - 4ac = D$ has a root $z_Q = (-b + i\sqrt{|D|})/(2a)$ in $\mathbb{H}$, and Q is reduced exactly when $z_Q \in \mathcal{F}$ (with the usual boundary conventions). The set Λ_D of these *Heegner points* (also called CM points) has $h(D)$ elements, the class number of $\mathbb{Q}(\sqrt{D})$ (Exercise 5).

Theorem 28.6 (Duke [167]). *As $D \to -\infty$ through fundamental discriminants, Λ_D becomes equidistributed in $\mathrm{SL}_2(\mathbb{Z})\backslash\mathbb{H}$ for $\frac{3}{\pi}\,dx\,dy/y^2$: for every Borel set $\Omega \subset \mathcal{F}$ whose boundary has measure zero,*

$$\frac{\#(\Lambda_D \cap \Omega)}{h(D)} \longrightarrow \frac{3}{\pi}\int_\Omega \frac{dx\,dy}{y^2}.$$

We describe the proof only at the level of its inputs. By the Weyl criterion on the noncompact quotient one must show that the normalised Weyl sums $h(D)^{-1}\sum_{z\in\Lambda_D}\varphi(z)$ tend to 0 for φ a Maass cusp form and for the continuous spectrum. For the Eisenstein series the Weyl sum is an explicit multiple of $\zeta(s)L(s,\chi_D)$, and a subconvex bound for Dirichlet L functions of Burgess type [93] gives a power saving. For cusp forms Duke related the Weyl sums, through a theta correspondence, to Fourier coefficients of Maass forms of weight one half, and he bounded these coefficients by extending Iwaniec's method for holomorphic forms of half integral weight [274]. Earlier, Linnik had proved the result under a congruence condition on D by an ergodic method [368].

The Langlands content is visible through Waldspurger's theorem [561]: the squares of the Fourier coefficients of a half integral weight form are proportional to the central values $L(\frac{1}{2}, f\otimes\chi_D)$ of the quadratic twists of its Shimura lift f, and the toric period formula [562] expresses $|\sum_{\Lambda_D}\varphi|^2$ through central values attached to φ and to the quadratic field. In this language equidistribution follows from, and is essentially equivalent to, a subconvex bound for these central values in the D aspect, and Iwaniec's estimate, extended by Duke to Maass forms, is such a bound. We do not attempt the exact constants, which depend on the Haar normalisations discussed in Section 28.6.

t	1	1.5	2	3	5	10
data	0.95193	0.63608	0.47524	0.31769	0.19104	0.09511
$3/(\pi t)$	0.95493	0.63662	0.47746	0.31831	0.19099	0.09549

Table 28.1: Heights of the 28251 Heegner points for the 60 fundamental discriminants in $[-1002999, -1002803]$ against the invariant tail $3/(\pi t)$; the row "data" is the empirical $\mathbb{P}(\operatorname{Im} z > t)$.

28.2.2 A computation

For the 60 fundamental discriminants D in $[-1002999, -1002803]$ we enumerated all reduced primitive forms (computation for this book). There are 28251 Heegner points, and for each D the count equals the class number $h(D)$ returned by PARI/GP. Above height $t \geq 1$ the fundamental domain is the strip $|x| \leq \frac{1}{2}$, so the invariant measure predicts

$$\mathbb{P}(\operatorname{Im} z > t) = \frac{3}{\pi} \int_t^\infty \frac{dy}{y^2} = \frac{3}{\pi t}, \qquad t \geq 1,$$

and, given $\operatorname{Im} z > 1$, a uniform law for $\operatorname{Re} z$ on $[-\frac{1}{2}, \frac{1}{2}]$. Table 28.1 and Figure 28.2 compare. The tail agrees with $3/(\pi t)$ to within $3 \cdot 10^{-3}$ at every height. In units of the binomial standard error $(P(1-P)/28251)^{1/2}$, $P = 3/(\pi t)$, which is only a yardstick (the points are not random, and z_Q and $-\bar{z}_Q$ have the same height), the deviation is 2.4 at $t = 1$ and below one at the other heights. Duke's proof gives a rate $|D|^{-\delta}$, and at $|D| \approx 10^6$ this is at least 0.03 for every $\delta \leq \frac{1}{4}$ (the Lindelöf value, Computational Problem 28.1), at least ten times the largest deviation. The table is therefore consistent with the theorem but does not test its rate. The horizontal coordinate above height one passes a Kolmogorov Smirnov test for uniformity with $p = 0.36$.

REMARK 28.7 (A noncompact Sato Tate theorem). Sato Tate says that the classes of Frobenius equidistribute for Haar measure on a compact group. Duke's theorem says that the CM points of discriminant D equidistribute for the invariant measure on $\mathrm{SL}_2(\mathbb{Z})\backslash\mathrm{SL}_2(\mathbb{R})/\mathrm{SO}(2)$, which on the Iwasawa coordinates is left Haar measure of B^+. In both, the limit law is a Haar measure and the proof needs an automorphic input, potential automorphy of symmetric powers [67] in one case and subconvexity for twisted central values in the other. The difference is the cusp: mass could escape to $y = \infty$, and the convergence of $\mathbb{P}(\operatorname{Im} z > t)$ to $3/(\pi t)$, which tends to 0 as $t \to \infty$, says that it does not.

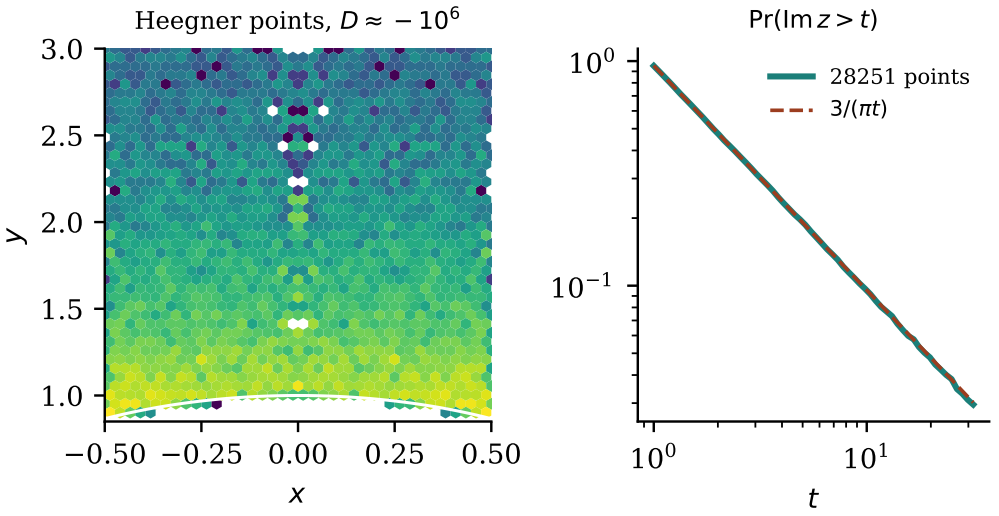


Figure 28.2: Left: density of the 28251 Heegner points of Table 28.1 in the fundamental domain, in hexagonal bins up to height 3; lighter cells hold more points, and the density decays like y^{-2}. Right: the empirical tail $\mathbb{P}(\operatorname{Im} z > t)$ against the left Haar prediction $3/(\pi t)$ on logarithmic axes.

28.3 Transformation models and the p* formula

In a transformation model the conditional law of the maximum likelihood estimator given an ancillary configuration is explicit, and Barndorff Nielsen's p^* formula reproduces it exactly; outside such models p^* is an approximation of high order.

28.3.1 Structural models

Following Fraser [200] and Barndorff Nielsen [37, 38], a *transformation model* is a family of laws generated from one law by a group. We work in the following setting.

Definition 28.8. Let G be a Lie group of dimension d acting freely on an open set $\mathcal{X} \subset \mathbb{R}^n$ by affine maps $x \mapsto A_g x + b_g$, and put $\chi(g) = |\det A_g|$, a continuous homomorphism $G \to \mathbb{R}_{>0}$. Let U have a density f on $\mathcal{X}$, twice continuously differentiable on the open set where it is positive. The *transformation model* is the family of laws of $X = gU$, $g \in G$, with densities

$$p(x; g) = f(g^{-1}x)\,\chi(g)^{-1}.$$

We assume that for every x in an open G invariant set $\mathcal{X}_0 \subset \mathcal{X}$ of full measure the likelihood $g \mapsto p(x; g)$ has a unique maximiser $\hat{g}(x)$, that $\hat{g}$ is smooth on

$\mathcal{X}_0$, and that the observed information $j(\hat g; x) = -\nabla^2 \ell(\hat g)$, $\ell(g) = \log p(x; g)$, is nonsingular in some (hence every) chart.

The location scale model $X_i = \mu + \sigma U_i$, $n \geq 2$, is the case $G = B$, $g = (\sigma, \mu)$, $\chi(g) = \sigma^n$, with $\mathcal{X} = \{x \in \mathbb{R}^n : \text{not all } x_i \text{ equal}\}$, on which B acts freely; the scale model $X_i = \theta U_i$ on $(0, \infty)^n$ is the case $G = \mathbb{R}_{>0}$.

LEMMA 28.9. *(i) $\hat g(hx) = h\hat g(x)$ for $h \in G$. (ii) The* configuration $a(x) = \hat g(x)^{-1} x$ *is a maximal invariant, so its law does not depend on g: it is ancillary. (iii) $x = \hat g(x)\, a(x)$, and $A = \{x \in \mathcal{X}_0 : \hat g(x) = e\}$ is a smooth submanifold of dimension $n - d$ such that $\Phi : G \times A \to \mathcal{X}_0$, $\Phi(h, a) = ha$, is a diffeomorphism.*

PROOF. (i) $p(hx; g) = f(g^{-1}hx)\chi(g)^{-1} = p(x; h^{-1}g)\chi(h)^{-1}$, so g maximises the left side if and only if $h^{-1}g$ maximises $p(x; \cdot)$, that is $g = h\hat g(x)$. (ii) $a(hx) = \hat g(x)^{-1}h^{-1}hx = a(x)$; and if $a(x) = a(y)$ then $y = \hat g(y)\hat g(x)^{-1}x$ lies on the orbit of x. Since $X = gU$, $a(X) = a(U)$ has the law of $a(U)$ for every g. (iii) By (i) the derivative of $\hat g$ along the orbit through x is onto, so $\hat g$ is a submersion and $A = \hat g^{-1}(e)$ is a submanifold; Φ is smooth with smooth inverse $x \mapsto (\hat g(x), \hat g(x)^{-1}x)$. □

LEMMA 28.10. *Under Φ, Lebesgue measure on $\mathcal{X}_0$ becomes $\chi(h)\, d\mu_l(h)\, \kappa(da)$ for a measure κ on A with a smooth positive density.*

PROOF. The pull back $\nu(F) = \lambda(\Phi(F))$ of Lebesgue measure λ under the diffeomorphism Φ has a smooth positive density, $\nu = q(h, a)\, d\mu_l(h)\, da$ with da a smooth volume on A. Since $\Phi(g'h, a) = g'\Phi(h, a)$ and $\lambda(g'E) = \chi(g')\lambda(E)$, the measure ν satisfies $\nu(L_{g'}F) = \chi(g')\nu(F)$ for $L_{g'}(h, a) = (g'h, a)$. As μ_l is left invariant this says $q(g'h, a) = \chi(g')q(h, a)$, hence $q(h, a) = \chi(h)q(e, a)$. Put $\kappa(da) = q(e, a)\, da$. □

*28.3.2 Exactness of p**

Barndorff Nielsen's formula approximates the conditional density of the maximum likelihood estimator given an ancillary a by

$$p^*(\hat g; g \mid a) = c(a)\, |j(\hat g; x)|^{1/2} \exp\{\ell(g) - \ell(\hat g)\}, \tag{28.4}$$

where x is reconstructed from $(\hat g, a)$, ℓ is the log likelihood, the density is taken with respect to Lebesgue measure in a chart of G, and $c(a)$ makes it integrate to one.

THEOREM 28.11 (Barndorff Nielsen [36, 37]). *In the transformation model of Definition 28.8, conditionally on the maximal invariant $a(X) = a$, the estimator $\hat g$ has density*

$$p(\hat g; g \mid a) = \frac{f(g^{-1}\hat g\, a)\, \chi(g^{-1}\hat g)}{c_0(a)} \qquad \text{with respect to } d\mu_l(\hat g), \tag{28.5}$$

$c_0(a) = \int_G f(ka)\chi(k)\,d\mu_l(k)$. *Moreover* $|j(\hat g;x)|^{1/2}d\hat g = |j(e;a)|^{1/2}\,d\mu_l(\hat g)$ *in any chart, with* μ_l *normalised to agree with chart Lebesgue measure at* e, *and* (28.4) *is exactly* (28.5).

For the location scale group the exact conditional density is due to Fisher [189]; the identification with p^*, and the point that $|\hat\jmath|^{1/2}d\hat g$ is left invariant, are Barndorff Nielsen's. The proof below is written for the generality of Definition 28.8.

PROOF. By Lemma 28.10 the joint density of $(\hat g, a) = \Phi^{-1}(X)$ with respect to $d\mu_l(h)\,\kappa(da)$ is $p(ha;g)\chi(h) = f(g^{-1}ha)\chi(g^{-1}h)$. Dividing by its integral over h gives (28.5), and $c_0(a) < \infty$ for κ almost every a by Fubini, since the joint density integrates to one.

For the information, fix $x = \hat g a$ and write $\ell_x(g) = \log p(x;g)$. From $p(\hat g a;g) = f(g^{-1}\hat g a)\chi(g)^{-1}$ we get $\ell_x(g) = \ell_a(\hat g^{-1}g) - \log\chi(\hat g)$. In a chart let ψ denote $g \mapsto \hat g^{-1}g$. Since e maximises ℓ_a (because $\hat g(a) = e$), its gradient vanishes there, so the chain rule gives no first order term and

$$j(\hat g;x) = D\psi(\hat g)^{\top}\, j(e;a)\, D\psi(\hat g), \qquad |j(\hat g;x)|^{1/2} = |\det D\psi(\hat g)|\,|j(e;a)|^{1/2}.$$

A left invariant measure has chart density $m(g) = m(e)|\det D(L_{g^{-1}})(g)|$, which is $|\det D\psi(\hat g)|$ at $g = \hat g$ when $m(e) = 1$. This proves the second claim. Finally

$$\exp\{\ell_x(g) - \ell_x(\hat g)\} = \frac{f(g^{-1}\hat g a)\chi(g)^{-1}}{f(a)\chi(\hat g)^{-1}} = \frac{f(g^{-1}\hat g a)\chi(g^{-1}\hat g)}{f(a)},$$

so $p^*\,d\hat g = c(a)|j(e;a)|^{1/2}f(a)^{-1}f(g^{-1}\hat g a)\chi(g^{-1}\hat g)\,d\mu_l(\hat g)$, which is (28.5) for $c(a) = f(a)/(c_0(a)|j(e;a)|^{1/2})$. □

EXAMPLE 28.12 (Location and scale). For $X_i = \mu + \sigma U_i$ with $\hat g = (\hat\sigma, \hat\mu)$, the configuration is $a = (a_i)$ with $a_i = (x_i - \hat\mu)/\hat\sigma$, $\chi(g) = \sigma^n$, and by Proposition 28.4 $d\mu_l = d\hat\mu\,d\hat\sigma/\hat\sigma^2$. With f now the density of each U_i, Theorem 28.11 gives the conditional density of $(\hat\mu, \hat\sigma)$ given a as proportional to $\hat\sigma^{n-2}\sigma^{-n}\prod_i f((\hat\mu - \mu + \hat\sigma a_i)/\sigma)$, Fisher's formula. Directly, $j(\hat g;x) = \hat\sigma^{-2}H(a)$ with $H(a)$ a function of the configuration alone, so $|\hat\jmath|^{1/2} = \hat\sigma^{-2}|H(a)|^{1/2}$ is the left Haar density.

EXAMPLE 28.13 (Exponential scale family). Let $X_1, \dots, X_n$ be independent with density $\theta^{-1}e^{-x/\theta}$ on $(0,\infty)$. Then $\ell(\theta) = -n\log\theta - n\bar x/\theta$, $\hat\theta = \bar x$, $\ell(\hat\theta) = -n\log\hat\theta - n$, and $\ell''(\theta) = n\theta^{-2} - 2n\bar x\theta^{-3}$, so $j(\hat\theta) = n/\hat\theta^2$. Hence

$$p^*(\hat\theta;\theta) = c\,\frac{n^{1/2}}{\hat\theta}\,\frac{\hat\theta^n e^n}{\theta^n}\,e^{-n\hat\theta/\theta} = c\,n^{1/2}e^n\,\frac{\hat\theta^{n-1}}{\theta^n}\,e^{-n\hat\theta/\theta}.$$

The exact law of $\bar X$ is Gamma with shape n and rate n/θ, with density

$$p(\hat\theta;\theta) = \frac{(n/\theta)^n\hat\theta^{n-1}e^{-n\hat\theta/\theta}}{\Gamma(n)},$$

the same function of $\hat\theta$, so after normalisation, $c = n^{n-1/2}e^{-n}/\Gamma(n)$, p^* is exact. (The configuration $x/\bar x$ is independent of $\bar x$ here, so conditioning is vacuous.) With the unnormalised constant $c = (2\pi)^{-1/2}$ the ratio p^*/p is $\Gamma(n)e^n/(\sqrt{2\pi}\,n^{n-1/2}) = 1 + \frac{1}{12n} + O(n^{-2})$ by Stirling's formula; numerically 1.0168, 1.0084, 1.0042 for $n = 5, 10, 20$.

28.3.3 Beyond group models

Outside transformation models p^* is an approximation. For a full exponential family and n independent observations, $\hat\theta$ is sufficient, no ancillary is needed, and p^* coincides with the saddlepoint approximation of Daniels [122] to the density of the sufficient statistic, transformed to $\hat\theta$. With $c = (2\pi)^{-d/2}$ its relative error is $O(n^{-1})$. After renormalisation the relative error is $O(n^{-3/2})$ on normal deviation regions, where $\hat\theta - \theta = O(n^{-1/2})$, and remains $O(n^{-1})$ for larger deviations [37, 38, 454]. In curved exponential families the order $O(n^{-3/2})$ holds conditionally on an ancillary that is exact or approximately ancillary to sufficiently high order [36, 38]. Laplace's method explains the shape: expanding $\ell(\theta) - \ell(\hat\theta)$ to second order around $\hat\theta$ makes (28.4) with $c = (2\pi)^{-d/2}$ approximately a normal density for $\hat\theta$ with mean θ and covariance $\hat\jmath^{-1}$, and $(2\pi)^{-d/2}|\hat\jmath|^{1/2}$ is its normalising constant. Example 28.13 shows two of these levels: the constant $(2\pi)^{-1/2}$ is off by Stirling's factor $1 + \frac{1}{12n} + O(n^{-2})$, while the renormalised formula is exact because the model is a group model.

28.4 Right Haar priors

We prove that the right Haar prior gives invariant credible sets exact frequentist coverage, compute the shortfall of the left Haar prior for normal errors, and check both by simulation.

28.4.1 Structural, fiducial and Bayes

The posterior of the right Haar prior coincides with Fraser's structural distribution and with Fisher's fiducial distribution, and its invariant credible sets are exact confidence sets for every sample size [200, 525, 265]; see also [45, 169]. We give a complete proof in the setting of Definition 28.8.

Theorem 28.14 (Fraser; Stein; Hora and Buehler). *Let the prior be $d\mu_r(g)$, and let $x \in \mathcal{X}_0$ have configuration a with $c_0(a) < \infty$, which holds for almost every x. Under the posterior, the pivot $k = g^{-1}\hat g(x)$ has law*

$$Q_a(dk) = c_0(a)^{-1} f(ka)\chi(k)\, d\mu_l(k),$$

and under the sampling law P_g, conditionally on $a(X) = a$, the pivot $g^{-1}\hat g(X)$ has the same law Q_a. Consequently, if $S_a \subset G$ satisfies $Q_a(S_a) = 1 - \alpha$ for each a, the set

$C(x) = \{g : g^{-1}\hat{g}(x) \in S_{a(x)}\}$ has posterior probability $1-\alpha$ and $P_g(g \in C(X)) = 1-\alpha$ for every $g \in G$ and every n.

PROOF. The posterior is proportional to $p(x;g)\,d\mu_r(g)$. Inversion carries μ_r to a left Haar measure, and by (28.1) right translation by $\hat{g}$ multiplies left Haar measure by a constant; so the image of μ_r under $g \mapsto k = g^{-1}\hat{g}$ is a constant multiple of μ_l. With $g = \hat{g}k^{-1}$ and $x = \hat{g}a$,

$$p(x;\hat{g}k^{-1}) = f(k\hat{g}^{-1}\hat{g}a)\,\chi(\hat{g}k^{-1})^{-1} = f(ka)\chi(k)\chi(\hat{g})^{-1},$$

and $\chi(\hat{g})$ is constant given x. So the posterior of k is Q_a, a proper law because $c_0(a) < \infty$. Under P_g, Theorem 28.11 gives the conditional density $f(g^{-1}\hat{g}a)\chi(g^{-1}\hat{g})/c_0(a)$ of $\hat{g}$ with respect to $d\mu_l(\hat{g})$, and $k = g^{-1}\hat{g}$ is a left translate of $\hat{g}$, so its density with respect to $d\mu_l(k)$ is $f(ka)\chi(k)/c_0(a)$. The coverage statement follows by conditioning on a and then averaging over it. □

The proof uses only the equivariance $\hat{g}(hx) = h\hat{g}(x)$ and the decompositions of Lemmas 28.9(iii) and 28.10, and these hold for any smooth equivariant map $\mathcal{X}_0 \to G$ in place of $\hat{g}$. So Theorem 28.14 needs neither a unique maximum likelihood estimator nor any smoothness of f: in the location scale model the pair $(s, \bar{x})$, with s the sample standard deviation, serves on all of $\mathcal{X}$.

COROLLARY 28.15. *In the location scale model with prior $d\mu\,d\sigma/\sigma$, the equal tailed $1-\alpha$ posterior intervals for μ and for σ have frequentist coverage exactly $1-\alpha$ for every $n \geq 2$ and every error density f.*

PROOF. Let $\hat{g} = (\hat{\sigma}, \hat{\mu})$ be $(s, \bar{x})$, which by the remark above is allowed for every f, and write $k = g^{-1}\hat{g} = (\rho, \tau)$ with $\rho = \hat{\sigma}/\sigma$ and $\tau = (\hat{\mu} - \mu)/\sigma$. Then $(\mu - \hat{\mu})/\hat{\sigma} = -\tau/\rho$ and $\sigma/\hat{\sigma} = 1/\rho$ are functions of k alone. The posterior quantiles of μ are $\hat{\mu} + \hat{\sigma}q_{\pm}(a)$ with $q_{\pm}(a)$ the quantiles of $-\tau/\rho$ under Q_a, so the event that the interval covers μ is $\{-\tau/\rho \in [q_-(a), q_+(a)]\}$, of the form in Theorem 28.14. The same argument applies to σ. □

28.4.2 Left Haar priors

Jeffreys' rule gives $d\mu\,d\sigma/\sigma^2$ for location and scale [282], the left Haar measure. Since $d\mu_l = \Delta_G\,d\mu_r$, the same computation shows that under the left Haar prior the pivot has law proportional to $\Delta_G(\hat{g}k^{-1})Q_a(dk) \propto \Delta_G(k)^{-1}Q_a(dk)$, a tilt of Q_a. For location and scale $\Delta_G(k)^{-1} = \rho$, so the tilt favours large $\hat{\sigma}/\sigma$, that is small σ: intervals for μ are too short, those for σ sit too low, and in the examples below coverage falls below nominal. For normal errors this is explicit (our calculation): the left Haar posterior of μ is a t law with n rather than $n-1$ degrees of freedom, centre $\bar{x}$ and scale $S^{1/2}/n$, $S = \sum(x_i - \bar{x})^2$, and that of S/σ^2 is χ^2_n rather than χ^2_{n-1}. Writing $t_{\nu,q}$ and $\chi^2_{\nu,q}$ for q quantiles and T_ν, χ^2_ν for random variables with

prior	Pólya variable of Δ μ	σ	normal μ	σ
right Haar $d\mu\, d\sigma/\sigma$	0.9005	0.898	0.909	0.908
left Haar $d\mu\, d\sigma/\sigma^2$	0.860	0.865	0.869	0.870

Table 28.2: Coverage of 90% equal tailed posterior intervals, $n = 5$; 4000 replications for the Pólya shape (standard error 0.0047), 1000 for the normal shape.

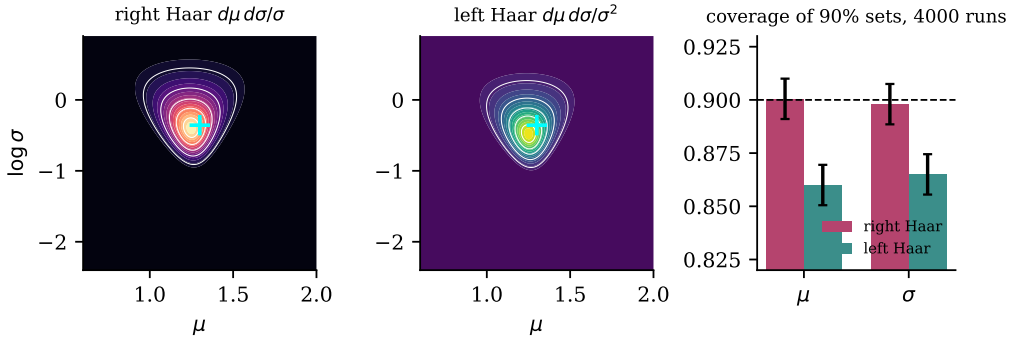


Figure 28.3: Left and centre: right and left Haar posterior densities of $(\mu, \log\sigma)$ for one data set of size 5 with Pólya errors; the cross marks the true value $(1.3, \log 0.7)$. Right: coverage of 90% equal tailed intervals for μ and σ over 4000 replications, with bars of two Monte Carlo standard errors; the dashed line is the nominal level.

these laws, the exact coverages of the 90% equal tailed intervals at $n = 5$ are $\mathbb{P}(|T_4| \le t_{5,0.95}\sqrt{4/5}) = 0.854$ for μ and $\mathbb{P}(\chi^2_{5,0.05} \le \chi^2_4 \le \chi^2_{5,0.95}) = 0.861$ for σ.

28.4.3 A computation with the Pólya variable of Δ

As error law we take the Pólya variable W of Chapter 13, with density $\Phi_\Delta(u) \propto \Delta(ie^u)e^{6u}$: symmetric, with doubly exponential tails much lighter than Gaussian ones. With $n = 5$, true $(\mu, \sigma) = (1.3, 0.7)$ and 4000 replications, we computed 90% equal tailed posterior intervals on a grid centred and scaled by each data set, so that the computation is itself equivariant (computation for this book). Table 28.2 and Figure 28.3 report the result. The right Haar coverages 0.9005 and 0.898 are within one Monte Carlo standard error (0.0047) of 0.9, as Corollary 28.15 predicts; the left Haar coverages 0.860 and 0.865 are 8.4 and 7.4 standard errors short. For normal errors (1000 replications, standard error about 0.01) the coverages are 0.909, 0.908 (right) and 0.869, 0.870 (left), against the exact values 0.9 and 0.854, 0.861.

28.4.4 Matching and paradoxes

Theorem 28.14 makes the right Haar prior an *exact* probability matching prior for μ and for σ, stronger than the first order matching of Welch and Peers [573]. Exactness needs only the free action. Optimality needs more: for an invariant loss the Bayes procedure of the right Haar prior is the best equivariant procedure [265, 169], and by the Hunt Stein theorem the best equivariant procedure is minimax when G is amenable, as the solvable group B is; for the general linear group acting on covariance matrices, which is not amenable, Stein showed that it need not be [170].

Dawid, Stone and Zidek [129] showed that improper priors can produce marginalisation paradoxes, in which the posterior of a parameter θ depends on the data only through a statistic z whose law depends only on θ, and yet no prior on θ reproduces it from the likelihood of z. In group models the paradox is avoided when the prior is right Haar measure on the group that leaves θ invariant, times any prior on θ. Exercise 6 works the classical case $\theta = \mu/\sigma$: here the relevant group is the scale group, the prior free of paradox is $d\mu\, d\sigma/\sigma^2$, and the right Haar prior $d\mu\, d\sigma/\sigma$ of the location scale group produces the paradox. "Right" is relative to the group that leaves the parameter of interest invariant; this is the starting point of Chang and Eaves.

28.5 Reference priors for orbits

When the parameter of interest is an orbit of a group acting on the parameter space, reference priors combine Haar measure on the group with a prior on the space of orbits. Conjugacy classes, and hence Satake parameters, are orbits of this kind.

28.5.1 Chang and Eaves

Let G act freely on a parameter space, and let the parameter of interest be the orbit, equivalently a maximal invariant θ. Chang and Eaves [100] showed that the prior combining right Haar measure on G with a limiting form of Jeffreys' prior for the maximal invariant coincides with Bernardo's reference prior [48, 46] when the orbit is the parameter of interest. Their prior can be computed without a product parametrisation, and their examples include the correlation matrix and the eigenvalues of a covariance matrix. For a one dimensional maximal invariant, Datta and J. K. Ghosh [123] compared reference, Chang Eaves and probability matching priors, judged them by marginalisation paradoxes and by frequentist matching, and concluded that the Chang Eaves and Berger Bernardo reference priors are the most attractive; Datta and M. Ghosh [124] studied which noninformative priors are invariant under reparametrisation.

28.5.2 Conjugacy classes as orbits

The Satake parameter of an unramified local representation is a conjugacy class in the dual group, an orbit of the adjoint action; when the representation is tempered it is a class in a compact group K, such as a Sato Tate group. We propose (this book) the following model for inference on such a class from noisy group elements. Let K be a compact connected Lie group with Haar probability measure dh, let $\varphi \geq 0$ be a smooth class function with $\int \varphi \, dh = 1$, and observe $Y_1, \dots, Y_n$ independent with density

$$p(y \mid M) = \varphi(M^{-1}y) \quad \text{with respect to } dh, \qquad M = \eta \, t \, \eta^{-1}, \tag{28.6}$$

that is $Y = M\varepsilon$ with ε of density φ. Examples are the heat kernel $\varphi_s = \sum_\rho \dim \rho \, e^{-s c_\rho} \chi_\rho$ (c_ρ the Casimir eigenvalue) and, for $K = \mathrm{SU}(2)$, the von Mises Fisher law $\varphi(\varepsilon) \propto \exp(\kappa \operatorname{Re} \operatorname{tr} \varepsilon)$. The parameter of interest is the class of M, coordinatised by t in a Weyl chamber T^+ of a maximal torus T; the nuisance parameter is $u = \eta T \in K/T$. Conjugation is not free on K, so the theorem of Chang and Eaves does not apply as stated. We instead compute the reference prior directly, with the two group algorithm of Berger and Bernardo [46], ordered as (class, conjugator): on a compact parameter space with information matrix $I = \left(\begin{smallmatrix} I_{11} & I_{12} \\ I_{21} & I_{22} \end{smallmatrix}\right)$,

$$\pi(u \mid t) \propto |I_{22}(t,u)|^{1/2}, \qquad \pi(t) \propto \exp\Big\{ \int \pi(u \mid t) \log |I_{11\cdot 2}(t,u)|^{1/2} \, du \Big\}, \tag{28.7}$$

with $I_{11\cdot 2} = I_{11} - I_{12} I_{22}^{-1} I_{21}$.

PROPOSITION 28.16 (this book). *Let K be compact, connected and simple, and suppose the Fisher information of* (28.6) *at $M = e$ is finite and nonsingular. Then:*

(i) The Fisher metric is $c\,n$ times a fixed bi invariant Riemannian metric on K, for a constant $c = c(\varphi) > 0$; Jeffreys' prior for M is Haar measure, and its image on classes has the Weyl density $|W|^{-1} \prod_\alpha |e^{i\alpha(t)} - 1|$ (product over all roots) with respect to the Haar probability measure dt of T: it is the Sato Tate measure of K.

(ii) The reference prior (28.7) *is the product of the K invariant probability measure on K/T and the uniform (Haar) measure dt on the Weyl chamber. Its image on classes is the image of Haar measure on the torus T.*

For $K = \mathrm{SU}(2)$ the reference prior for the angle $\theta \in [0, \pi]$ is $d\theta/\pi$, so $x = 2\cos\theta$ has the arcsine law, against the Sato Tate prior $\frac{2}{\pi} \sin^2\theta \, d\theta$. For $K = \mathrm{USp}(4)$ it is uniform on $0 \leq \theta_2 \leq \theta_1 \leq \pi$, against the density proportional to $(\cos\theta_1 - \cos\theta_2)^2 \sin^2\theta_1 \sin^2\theta_2$ of Proposition 29.4.

PROOF. (i) The group $K \times K$ acts on parameters and data by $(g_1, g_2) \cdot M = g_1 M g_2^{-1}$, $(g_1, g_2) \cdot y = g_1 y g_2^{-1}$, preserving dh, and $p(g_1 y g_2^{-1} \mid g_1 M g_2^{-1}) =$

$\varphi(g_2 M^{-1} y g_2^{-1}) = \varphi(M^{-1}y)$ because φ is a class function. So the Fisher metric is invariant under left and right translations. At e it is an $\mathrm{Ad}(K)$ invariant inner product on the Lie algebra $\mathfrak{k}$, and since K is simple Ad is irreducible, so by Schur's lemma the inner product is a multiple c of a fixed invariant one; $c > 0$ by the nonsingularity assumption, and independence multiplies it by n. The Riemannian volume of a bi invariant metric is Haar measure, and the image on classes is given by Weyl's integration formula.

(ii) On the regular set write $M = \eta t \eta^{-1}$ with t in the interior of the Weyl chamber; since the centraliser of a regular t is T, the map $(\eta T, t) \mapsto M$ is a diffeomorphism onto the regular set, whose complement has Haar measure zero. For $Y \in \mathfrak{t}$ the tangent vector $\frac{d}{ds}\eta t e^{sY}\eta^{-1}$ is the left translate by $\eta t \eta^{-1}$ of $\mathrm{Ad}(\eta)Y$; for $X \in \mathfrak{t}^{\perp}$, $\frac{d}{ds}\eta e^{sX} t e^{-sX}\eta^{-1}$ is the left translate of $\mathrm{Ad}(\eta)(\mathrm{Ad}(t^{-1})X - X)$. The bi invariant metric is Ad invariant, $\mathfrak{t}^{\perp}$ is $\mathrm{Ad}(t)$ stable, so torus and orbit directions are orthogonal: $I_{12} = 0$ and $I_{11\cdot 2} = I_{11} = cn \,\|\cdot\|^2_{\mathfrak{t}}$, constant in (t, u). On a root plane $\mathfrak{k}_\alpha \subset \mathfrak{t}^{\perp}$, $\mathrm{Ad}(t^{-1})$ is a rotation by the angle $\alpha(t)$, so $\|\mathrm{Ad}(t^{-1})X - X\| = |e^{i\alpha(t)} - 1|\,\|X\|$, and therefore $|I_{22}(t,u)|^{1/2} = (cn)^{\dim K/T/2}\prod_{\alpha>0}|e^{i\alpha(t)} - 1|^2\, v(u)$, with v the density of the invariant volume of K/T. In (28.7), normalising over $u \in K/T$ removes the factor depending on t, so $\pi(u \mid t)$ is the invariant probability on K/T; and $I_{11\cdot 2}$ is constant, so $\pi(t)$ is uniform. The image of dt on $T^+ = T/W$ is the image of Haar measure on T. For $\mathrm{SU}(2)$ the chamber is $\theta \in [0, \pi]$, and for $\mathrm{USp}(4)$, W is generated by sign changes and the swap, with chamber $0 \le \theta_2 \le \theta_1 \le \pi$. $\square$

Proposition 28.16 splits Weyl's formula into two parts. The Vandermonde factor $\prod_\alpha |e^{i\alpha} - 1|$ is the volume of the conjugacy orbit, nuisance information about the conjugator, and the reference prior removes it. The result agrees with the formal Chang Eaves recipe: Haar measure on the conjugating group times Jeffreys' prior for t with η known, which is uniform because I_{11} is constant. The two priors on classes of $\mathrm{SU}(2)$ are far apart: by Lemma 29.2,

$$\mathrm{KL}\Big(\frac{d\theta}{\pi}\Big\|\frac{2}{\pi}\sin^2\theta\, d\theta\Big) = \log 2, \qquad \mathrm{KL}\Big(\frac{2}{\pi}\sin^2\theta\, d\theta\Big\|\frac{d\theta}{\pi}\Big) = 1 - \log 2,$$

using $\int_0^\pi \log\sin\theta\, d\theta = -\pi\log 2$. The reference prior is the normalised continuous part of the law μ_{CM} of Chapter 4: the reference prior for a class of $\mathrm{SU}(2)$ is the Sato Tate law of the torus. Which prior to use depends on the question. For the class of a single Frobenius element of a curve without complex multiplication, the Sato Tate law is the correct prior. For the class of an unknown group element observed through isotropic noise, with no arithmetic information, the reference prior is uniform on the torus. Anisotropic noise, or an unknown noise level, breaks the block structure and is left as Computational Problem 28.2.

28.6 Haar measure behind L values

Haar measures normalise the volumes in special value formulas; we record three instances without proofs.

Volumes and zeta values. With $dx\,dy/y^2$, $\operatorname{vol}(\mathrm{SL}_2(\mathbb{Z})\backslash\mathbb{H}) = \pi/3 = 2\zeta(2)/\pi$ by (28.3). More generally, with Haar measure on $\mathrm{SL}_n(\mathbb{R})$ normalised through Lebesgue measure on matrices, $\operatorname{vol}(\mathrm{SL}_n(\mathbb{Z})\backslash\mathrm{SL}_n(\mathbb{R})) = \zeta(2)\zeta(3)\cdots\zeta(n)$; this goes back to Minkowski and was proved by Siegel [518], whose analytic theory of quadratic forms, including the average measure of [517], led to the Siegel Weil formula.

Tamagawa numbers. For a semisimple group H over $\mathbb{Q}$, an invariant differential form of top degree defines Haar measures on every $H(\mathbb{Q}_v)$; their product is a measure on $H(\mathbb{A})$ independent of the form by the product formula, and $\tau(H) = \operatorname{vol}(H(\mathbb{Q})\backslash H(\mathbb{A}))$. Weil proved $\tau(\mathrm{SL}_2) = 1$ [570]. The mechanism is visible in a sketch. Strong approximation gives

$$\mathrm{SL}_2(\mathbb{Q})\backslash\mathrm{SL}_2(\mathbb{A}) = \mathrm{SL}_2(\mathbb{Z})\backslash\Big(\mathrm{SL}_2(\mathbb{R}) \times \prod_p \mathrm{SL}_2(\mathbb{Z}_p)\Big),$$

so $\tau(\mathrm{SL}_2)$ is the product of the archimedean volume $\operatorname{vol}(\mathrm{SL}_2(\mathbb{Z})\backslash\mathrm{SL}_2(\mathbb{R})) = \zeta(2)$ and the volumes $\#\mathrm{SL}_2(\mathbb{F}_p)/p^3 = 1 - p^{-2}$ of the groups $\mathrm{SL}_2(\mathbb{Z}_p)$, all for the form measure; since $\prod_p (1 - p^{-2}) = \zeta(2)^{-1}$, the product is 1.

Period formulas. Waldspurger's formula [562] expresses the square of a toric period of a GL_2 automorphic form, for example a Weyl sum over Heegner points, as a central value $L(\frac{1}{2}, \pi_K)$ divided by $L(1, \pi, \mathrm{Ad})$, times local factors. Ichino and Ikeda [268] formulated the analogue for $\mathrm{SO}(n+1) \times \mathrm{SO}(n)$ periods, as a refinement of the Gross Prasad conjecture, and verified it in low rank cases. In both, the global constant is fixed only once Tamagawa measures are chosen on the groups and their subgroups, and each local factor is an integral of matrix coefficients against local Haar measure. The Duke computation of Section 28.2 is the case where all these normalisations are hidden in $\frac{3}{\pi}$.

28.7 Notes

The Iwasawa integration formula and the half plane are treated in [92]. Fisher's conditional argument for location and scale is [189] and his fiducial argument [190]; structural inference is Fraser's [200]. The p^* formula, its exactness for transformation models and its accuracy elsewhere are due to Barndorff Nielsen [36, 37], with the systematic treatment in [38] and the survey [454]; the saddlepoint method is Daniels' [122]. Stein [525] and Hora and Buehler [265] identified Bayes procedures under right Haar priors with best invariant procedures and fiducial distributions; Eaton's lectures [169] give the general theory of group invariance

in statistics and Berger [45] the decision theoretic setting; Eaton and George [170] trace Stein's work on invariance from the Hunt Stein theorem to his counterexamples for groups that are not amenable. Marginalisation paradoxes are from [129]; reference priors from [48, 46], and for orbits from Chang and Eaves [100], compared by Datta and J. K. Ghosh [123]; Datta and M. Ghosh [124] treat their invariance. Duke's theorem is [167], building on Iwaniec [274], with Waldspurger's formulas [561, 562] giving the L function interpretation; Linnik's ergodic method is in [368]. Proposition 28.5 and Theorems 28.11 and 28.14 are classical; only the proofs of the two theorems in the generality of Definition 28.8 are written for this book. Ours are Proposition 28.16, the exact normal coverages of Section 28.4 and the computations, all elementary consequences of the cited theory.

28.8 Exercises

1. Verify Proposition 28.4 through Lemma 28.2: compute $\mathrm{Ad}(a,b)$ on the Lie algebra of B in the basis $\left(\begin{smallmatrix}1&0\\0&0\end{smallmatrix}\right)$, $\left(\begin{smallmatrix}0&1\\0&0\end{smallmatrix}\right)$, and check $\Delta_B = |\det \mathrm{Ad}\,|^{-1} = a^{-1}$. Show that no nonzero Radon measure on $\mathbb{R}$ is invariant under all the maps $x \mapsto ax + b$, although B acts transitively on $\mathbb{R}$.

2. For the Borel subgroup of $\mathrm{GL}_2(F)$, $F = \mathbb{R}$ or $\mathbb{Q}_p$, prove that
$$d\mu_l = |a|^{-2}|d|^{-1}\,da\,db\,dd, \qquad d\mu_r = |a|^{-1}|d|^{-2}\,da\,db\,dd$$
and $\Delta_{B_2} = |d/a|$. Check that Δ_{B_2} is trivial on the unipotent radical and on the centre.

3. Show that for the normal location scale model p^* is exact without conditioning, by computing $j(\hat\mu,\hat\sigma) = \mathrm{diag}(n, 2n)/\hat\sigma^2$ and comparing (28.4) with the joint density of $\bar X$ and $\hat\sigma^2 = S/n$, using independence and $S/\sigma^2 \sim \chi^2_{n-1}$. Identify the constant c.

4. Compute the hyperbolic area of the fundamental domain of $\Gamma_0(2)$ by showing that it is the union of $[\mathrm{SL}_2(\mathbb{Z}) : \Gamma_0(2)] = 3$ translates of $\mathcal{F}$, and of $\Gamma_0(p)$ in general. Derive (28.3) also from the Gauss Bonnet formula for the triangle with angles $\pi/3$, $\pi/3$, 0.

5. Let $D < -4$ be a fundamental discriminant. Show that $Q \mapsto z_Q$ is injective, that $\mathrm{SL}_2(\mathbb{Z})$ equivalence of forms corresponds to the action on roots, and that each class of primitive positive definite forms of discriminant D contains exactly one reduced form; deduce that $\#\Lambda_D = h(D)$. Show that $\mathrm{Im}\, z_Q = \sqrt{|D|}/(2a)$, so that $\mathrm{Im}\, z_Q > t$ if and only if $a < \sqrt{|D|}/(2t)$, and use this to compute the tail counts of Table 28.1 from the list of reduced forms.

6. (Marginalisation paradox.) Let $X_1, \dots, X_n$ be independent $N(\mu, \sigma^2)$, $\theta = \mu/\sigma$, $T = \sum X_i^2$, $U = \sum X_i$, $z = U/\sqrt{T}$, and take the prior $\sigma^{-r}\,d\mu\,d\sigma$. Show

that the posterior of θ is proportional to

$$e^{-n\theta^2/2}\int_0^\infty v^{n+r-3}e^{-v^2/2+\theta z v}dv,$$

which depends on the data only through z, and that the law of z has density proportional to $(1-z^2/n)^{(n-3)/2}e^{-n\theta^2/2}\int_0^\infty v^{n-1}e^{-v^2/2+\theta z v}dv$. Conclude that a prior $\pi(\theta)$ reproduces the first posterior from the law of z if and only if $r=2$, and observe that $d\mu\,d\sigma/\sigma^2 = d\theta\,d\sigma/\sigma$ is Lebesgue measure in θ times the Haar measure $d\sigma/\sigma$ of the scale group $x\mapsto cx$, $c>0$, which leaves θ and z invariant.

28.9 Open computational problems

Computational Problem 28.1 (Duke rates at large discriminant). Objective: measure the rate in Theorem 28.6. Objects: Heegner points for all fundamental discriminants with $|D|\le 10^9$, and for samples of discriminants near 10^{10}, 10^{11} and 10^{12} (class numbers up to about 10^6). Route: enumerate reduced forms with PARI/GP `qfbred` and `qfbclassno` on a cluster, or generate the class group from prime forms with GPU batch reduction. Deliverable: the discrepancy of Λ_D against $\frac{3}{\pi}dx\,dy/y^2$ over hyperbolic disks and over the tail sets $\operatorname{Im} z > t$, and the Weyl sums of the first ten Maass cusp forms (from the LMFDB), with fitted exponents δ in $|D|^{-\delta}$. Why it matters: Waldspurger's formula turns the Weyl sums into central L values, so the fitted exponents test the subconvexity input and the Lindelöf prediction $\delta=\frac14$ for the cusp form sums.

Computational Problem 28.2 (Reference priors for anisotropic noise). Objective: extend Proposition 28.16 beyond bi invariant noise. Objects: $\mathrm{SU}(2)$ and $\mathrm{USp}(4)$ with matrix Fisher noise $\varphi(\varepsilon)\propto\exp(\operatorname{Re}\operatorname{tr}(F\varepsilon))$ for nonscalar F, and the heat kernel φ_s with the time s unknown. Route: Fisher information by Weyl integration and Gauss quadrature on the torus, then the Berger Bernardo algorithm with the class first, and the Chang Eaves recipe with right Haar measure on the stabiliser free part. Deliverable: the reference prior on classes as a density on the Weyl chamber, its Kullback Leibler distance to the Weyl density and to the torus Haar measure, and frequentist coverage of reference posterior sets for the class at $n=5,20,100$. Why it matters: it decides whether the Vandermonde factor of the Sato Tate measure survives in objective inference on Satake parameters when the noise is not symmetric.

Computational Problem 28.3 (p* and right Haar intervals for angular models). Objective: measure what the group structure buys for p^* and for Haar priors in models built from Sato Tate data. Objects: the location and scale models $x_i=\mu+\sigma\varepsilon_i$ with ε_i distributed by μ_{ST}, by the $\mathrm{USp}(4)$ trace law and by Φ_Δ;

the von Mises model on the circle with unknown rotation, a group model, and with unknown rotation and concentration, a full exponential family that is not a group model; n from 5 to 200. Route: exact conditional densities and constants $c(a)$ from Theorem 28.11 by quadrature over the group, exact and saddlepoint densities for the two parameter von Mises model, and 10^5 replications per setting in vectorised code. Deliverable: tables of the ratio $(2\pi)^{-d/2}/c(a)$ of unnormalised p^* to the exact conditional density, as a function of n and of the configuration; of the relative error of renormalised p^* in the two parameter von Mises model; and of the coverage of right and left Haar intervals. Why it matters: in the group models renormalised p^* and right Haar intervals are exact whatever the error law (Theorems 28.11 and 28.14), but the accuracy of the unnormalised formula rests on Laplace's method. The $\mathrm{USp}(4)$ trace density vanishes to fourth order at ± 4 and Φ_Δ is smooth, so those models are regular, while the semicircle vanishes like a square root at ± 2 and has infinite Fisher information for location; the tables show whether the $O(n^{-1})$ accuracy of the constant survives in this nonregular model, and how the exact group answers compare with the $O(n^{-3/2})$ approximation outside group models.

COMPUTATIONAL PROBLEM 28.4 (CM points and Hecke orbits on Shimura curves). Objective: test equidistribution for Haar measure on compact quotients of $\mathbb{H}$. Objects: the Shimura curves attached to the maximal orders of the quaternion algebras over $\mathbb{Q}$ of discriminant 6, 10 and 14; CM points of discriminant $|D| \leq 10^8$, and Hecke orbits $T_{p^k}x$ for the primes $p \leq 7$ not dividing the discriminant and $p^k \leq 10^7$. Route: Magma or Sage quaternion arithmetic for optimal embeddings, explicit fundamental domains (Voight's algorithms) for the hyperbolic coordinates. Deliverable: discrepancy against the normalised hyperbolic area, the decay rate in $|D|$ and in p^k, and for Hecke orbits a comparison with the rate $(k+1)p^{-k/2}$ that the Ramanujan conjecture for the Maass forms of the curve predicts (the random walk generated by T_p has the corresponding Kesten rate $(2\sqrt{p}/(p+1))^k$ of Chapter 9). Why it matters: there is no cusp here, so the noncompact effects of Table 28.1 disappear; Hecke orbits are images of the Bruhat Tits tree of $\mathrm{PGL}_2(\mathbb{Q}_p)$ in the curve, which joins Chapter 9 to the invariant measure of this chapter, and their decay rate is a numerical test of the Ramanujan conjecture for Maass forms, which is open.

CHAPTER 29

BAYESIAN IDENTIFICATION OF SATO TATE GROUPS

When Fité, Kedlaya, Rotger and Sutherland classified the Sato Tate groups of abelian surfaces they did two things at once: they proved that exactly 52 compact groups occur, up to conjugacy in $\mathrm{USp}(4)$, as Sato Tate groups of abelian surfaces over number fields, and they matched curves to groups by comparing empirical moments of the normalised L polynomial coefficients with the moments of each candidate [192]. The second activity is statistical inference, though it is rarely described that way. This chapter makes the inference explicit: the data are the normalised traces x_p, the parameter is a compact group G, and the model is the image of Haar measure on G under the trace. Bayes factors computed from this working model are consistent, with rates given by Kullback Leibler divergences, as a deterministic theorem about the actual traces whenever they are equidistributed for the true law, atoms included, and that law has finite divergence from a candidate (Theorem 29.1). We test the procedure on three genus 2 curves, one of which lies outside the candidate set, and use that failure to argue for the full 52 group likelihood.

29.1 THE INFERENCE PROBLEM

Let $A/\mathbb{Q}$ be an abelian variety of dimension g with good reduction outside N. For $p \nmid N$ the normalised Frobenius s_p is a conjugacy class of the Sato Tate group $G = \mathrm{ST}(A) \subset \mathrm{USp}(2g)$, and the general Sato Tate conjecture (Conjecture 3.11) asserts that (s_p) is equidistributed for the image μ_G of Haar measure. We observe $x_p = a_p/\sqrt{p} = \operatorname{tr} s_p \in [-2g, 2g]$. Given $x_{p_1}, \dots, x_{p_n}$, which G?

For elliptic curves ($\mathrm{SU}(2)$ or $N(\mathrm{U}(1))$) complex multiplication is cheap to detect directly, but the statistical answer is the template for higher genus, where the candidates are many.

29.2 EXCHANGEABILITY AS A WORKING MODEL

The traces are determined by the curve, so a likelihood for them is a working model. By de Finetti's theorem [134, 258] an infinite exchangeable sequence is a mixture of independent and identically distributed sequences, so to model (x_p) as exchangeable is to model it as an independent sample from an unknown law with a prior on that law. Here the prior is supported on the finitely many laws μ_G, and

the posterior over G is the posterior over the de Finetti measure. A frequentist reading of the same likelihood is a working model in the same sense.

The only property of independent sampling used by the first order asymptotics of Bayes factors, their exponential rates, is the strong law of large numbers for the log likelihood ratio. For arithmetic data it is replaced by equidistribution, $n^{-1}\sum_{k\le n}\varphi(x_{p_k})\to\int\varphi\,d\mu_G$ for continuous φ, which plays the role of the ergodic theorem. It does not supply independence, so finite sample error probabilities are calibrated only insofar as the traces fluctuate like random ones, as Chapter 5 suggests they do. Proposition 1.5 gives the limitation: equidistribution is blind to alterations on sets of density zero, and so is every limit in this chapter.

29.3 A GENERAL CONSISTENCY THEOREM

Fix a compact set $K\subset\mathbb{R}$, a finite set $D\subset K$, and the reference measure ν equal to Lebesgue measure on K plus counting measure on D. Candidate laws P_0,P_1 and the true law μ_0 have ν densities f_0,f_1,m, so their atoms lie in D. Write $h=\log(f_1/f_0)$, with values in $[-\infty,\infty]$, and for a clipping level $M>0$ write $h_M=\max(-M,\min(M,h))$. For a deterministic sequence (x_k) in K define the clipped log Bayes factor

$$\log B_n^M=\sum_{k\le n}h_M(x_k).$$

We say that (x_k) is *equidistributed for μ_0 relative to D* if $n^{-1}\#\{k\le n:x_k=a\}\to\mu_0(\{a\})$ for each $a\in D$ and $n^{-1}\sum_{k\le n,\,x_k\notin D}\delta_{x_k}$ converges weakly to the restriction of μ_0 to $K\setminus D$. When μ_0 has no atoms this follows from equidistribution by the portmanteau theorem.

THEOREM 29.1 (Consistency under equidistribution; this book). *Suppose (x_k) is equidistributed for μ_0 relative to D, that h is finite on D, that on $K\setminus D$ the function h agrees almost everywhere with a continuous map $\tilde h:K\to[-\infty,\infty]$, through which h is evaluated off D, and that $\mathrm{KL}(\mu_0\|P_0)<\infty$. Then for every M,*

$$\lim_{n\to\infty}\frac{1}{n}\log B_n^M=\int h_M\,d\mu_0,$$

and as $M\to\infty$ the right side tends to

$$\int h\,d\mu_0=\mathrm{KL}(\mu_0\|P_0)-\mathrm{KL}(\mu_0\|P_1)\in[-\infty,\infty).$$

In particular, if $\mu_0=P_0$ the double limit is $-\mathrm{KL}(P_0\|P_1)$.

PROOF. Divide by n and split the sum according to whether $x_k\in D$. The first part is

$$\sum_{a\in D}h_M(a)\,\frac{\#\{k\le n:x_k=a\}}{n}\longrightarrow\sum_{a\in D}h_M(a)\,\mu_0(\{a\}).$$

On the second part $h_M = \tilde{h}_M$, a bounded continuous function on K, so weak convergence gives the limit $\int_{K\setminus D} \tilde{h}_M \, d\mu_0 = \int_{K\setminus D} h_M \, d\mu_0$. Adding proves the first claim.

For the second, write $h_M = \min(h^+, M) - \min(h^-, M)$; by monotone convergence the integrals increase to those of $h^{\pm}$, and it suffices that $\int h^+ d\mu_0 < \infty$. Since $\mathrm{KL}(\mu_0 \| P_0) < \infty$, $f_0 > 0$ almost everywhere for μ_0, and on $\{m > 0\}$ we have $h = \log(f_1/m) + \log(m/f_0)$. Since $\log y \le y$,

$$\int \Big(\log \frac{f_1}{m}\Big)^+ d\mu_0 \le \int f_1 \, d\nu = 1, \qquad \int \Big(\log \frac{m}{f_0}\Big)^- d\mu_0 \le \int f_0 \, d\nu = 1,$$

so $\int |\log(m/f_0)| \, d\mu_0 \le \mathrm{KL}(\mu_0 \| P_0) + 2 < \infty$. Hence h^+ is integrable and $\int h \, d\mu_0 = -\mathrm{KL}(\mu_0 \| P_1) + \mathrm{KL}(\mu_0 \| P_0)$. □

When the truth is one of the two candidates the rate is a single divergence, up to sign; in general it is a difference of two, as in the case without complex multiplication below, where $\mu_0 = \mu_{\mathrm{ST}}$ is not the candidate M_{ST}. Clipping lets weak convergence act on h, which may be unbounded; the independent version in Appendix A rests on the strong law of large numbers and needs no clipping.

29.4 Elliptic curves: the constants

With $D = \{0\}$ the models are $M_{\mathrm{ST}} : x_p \sim (1-\varepsilon)\mu_{\mathrm{ST}} + \varepsilon\delta_0$ and $M_{\mathrm{CM}} : x_p \sim \mu_{\mathrm{CM}} = \frac{1}{2}\delta_0 + \frac{1}{2}$ arcsine, where the arcsine law on $[-2,2]$ has density $(\pi\sqrt{4-x^2})^{-1}$. The small atom ε keeps a single supersingular prime from being decisive; curves without complex multiplication have infinitely many of them [175]. The log likelihood ratio of M_{CM} to M_{ST} is

$$h(x) = -\log(1-\varepsilon) - \log(4-x^2) \quad (x \neq 0), \qquad h(0) = \log \frac{1}{2\varepsilon},$$

since $\frac{1}{2}(\pi\sqrt{4-x^2})^{-1}$ divided by $(1-\varepsilon)(2\pi)^{-1}\sqrt{4-x^2}$ is $((1-\varepsilon)(4-x^2))^{-1}$. Off D it extends continuously to $[-2,2]$ with values in $(-\infty, \infty]$.

Lemma 29.2. $\int_0^\pi \sin^2\theta \log\sin\theta \, d\theta = \frac{\pi}{4}(1 - 2\log 2)$. *Consequently*

$$\int \log(4-x^2) \, d\mu_{\mathrm{ST}} = 1, \qquad \int_{-2}^{2} \log(4-x^2) \, \frac{dx}{\pi\sqrt{4-x^2}} = 0.$$

Proof. On $(0,\pi)$, $\log\sin\theta = -\log 2 - \sum_{k\ge1} k^{-1}\cos 2k\theta$, with convergence in L^2. Multiply by $\sin^2\theta = \frac{1}{2}(1-\cos 2\theta)$ and integrate: $\int_0^\pi \cos 2k\theta \, d\theta = 0$ and $\int_0^\pi \cos 2\theta \cos 2k\theta \, d\theta = \frac{\pi}{2}$ if $k = 1$ and 0 otherwise. The constant term contributes $-\frac{\pi}{2}\log 2$ and the $k = 1$ term contributes $\frac{\pi}{4}$, which gives the first formula. With $x = 2\cos\theta$ we have $\log(4-x^2) = \log 4 + 2\log\sin\theta$, and μ_{ST} becomes $\frac{2}{\pi}\sin^2\theta \, d\theta$

on $[0,\pi]$, so $\int \log(4-x^2)\,d\mu_{\mathrm{ST}} = \log 4 + \frac{4}{\pi}\cdot\frac{\pi}{4}(1-2\log 2) = 1$. Under the arcsine law θ is uniform and the constant term alone survives: $\log 4 + \frac{2}{\pi}(-\pi\log 2) = 0$. □

COROLLARY 29.3 (Rates for elliptic curves; this book). *Let $E/\mathbb{Q}$ be an elliptic curve and $0<\varepsilon<\frac{1}{2}$. If E has no complex multiplication, $\lim_M \lim_n n^{-1}\log B_n^M = -1 - \log(1-\varepsilon)$. If E has complex multiplication, the limit is $\frac{1}{2}\log\frac{1}{2\varepsilon} - \frac{1}{2}\log(1-\varepsilon) > 0$.*

PROOF. Without complex multiplication (x_p) is μ_{ST} equidistributed by Theorem 4.9, hence equidistributed for μ_{ST} relative to $\{0\}$. Apply Theorem 29.1 with $\mu_0=\mu_{\mathrm{ST}}$, $P_0=M_{\mathrm{ST}}$ and $P_1=\mu_{\mathrm{CM}}$: here $\mathrm{KL}(\mu_{\mathrm{ST}}\|M_{\mathrm{ST}}) = -\log(1-\varepsilon)$, and $\mathrm{KL}(\mu_{\mathrm{ST}}\|\mu_{\mathrm{CM}}) = \int\log(4-x^2)\,d\mu_{\mathrm{ST}} = 1$ by Lemma 29.2. With complex multiplication, (x_p) is equidistributed for μ_{CM} (Section 4.6), and relative to $\{0\}$, because $a_p=0$ at the good primes inert in the CM field, which have density $\frac{1}{2}$, while $a_p\neq 0$ at the split primes of good reduction, which are ordinary. Now $\mu_0=P_1$, and by the arcsine half of Lemma 29.2 the limit is

$$\mathrm{KL}(\mu_{\mathrm{CM}}\|M_{\mathrm{ST}}) = \frac{1}{2}\log\frac{1/2}{\varepsilon} + \frac{1}{2}\bigl(-\log(1-\varepsilon) - 0\bigr).$$ □

With $\varepsilon=10^{-3}$ and $M=20$ the theoretical rates are -0.999 and 3.108 nats per prime. For $E_1: y^2=x^3+x+1$ (conductor 496, no complex multiplication) and $E_2: y^2=x^3-x$ (complex multiplication by $\mathbb{Z}[i]$) over the good primes below 40000 we observed -0.979 and 3.102; after only 50 primes $\log B = -15.9$ and $+160.4$ respectively (Figure 29.1). The slow start for E_1 comes from its supersingular primes 3, 17, 179 and 227 among the first 50, each of which adds $\log(1/(2\varepsilon)) = 6.21$ instead of about -1. The CM decision is faster because half the primes give $a_p=0$, which M_{ST} scores as probability ε.

29.5 SEQUENTIAL TESTING

Wald's sequential probability ratio test [559] of P_0 against P_1 stops when $\log B_n$ leaves the interval

$$\Bigl(\log\frac{\beta}{1-\alpha},\ \log\frac{1-\beta}{\alpha}\Bigr),$$

where α and β are the target probabilities of rejecting P_0 when it holds and of accepting it when P_1 holds. For independent observations Wald and Wolfowitz proved that no test whose error probabilities are at most those of the sequential probability ratio test has a smaller expected sample size under either hypothesis [563], and Wald's identity, ignoring overshoot, gives $\mathbb{E}_1[N] \approx \log(1/\alpha)/\mathrm{KL}(P_1\|P_0)$ for small α,β. By Theorem 29.1 the arithmetic walk $\log B_n$ drifts at the rate computed there, so the same count, with that rate in place of the divergence, describes the actual traces as $\alpha,\beta\to 0$; the error probabilities

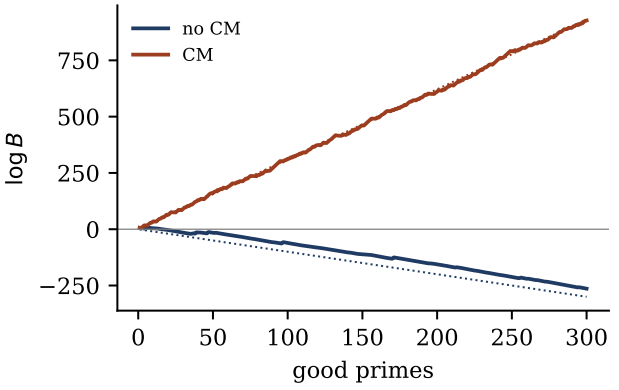


Figure 29.1: Log Bayes factor of M_{CM} against M_{ST} over the first 300 good primes for $E_1 : y^2 = x^3 + x + 1$ (no complex multiplication) and $E_2 : y^2 = x^3 - x$ (complex multiplication), with $\varepsilon = 10^{-3}$ and clipping at $M = 20$. Dotted lines: the limiting slopes -0.999 and 3.108 of Corollary 29.3.

themselves come only from the working model. With $\alpha = \beta = 10^{-6}$ and the rates of Corollary 29.3, the count is 13.8 primes without complex multiplication and 4.4 with it; on the actual traces the test stops at the 29th good prime for E_1, delayed by the supersingular primes 3 and 17, and at the 4th for E_2. In genus 2 the divergences of Table 29.1 are as small as 0.117, and the approximation asks for about 120 primes.

29.6 Moments versus likelihood

Moments are linear combinations of the Weyl averages

$$W_\rho(n) = \frac{1}{n} \sum_{k \le n} \chi_\rho(s_{p_k}),$$

with ρ running over the irreducible representations of $\mathrm{USp}(2g)$, and the vector $(\int \chi_\rho \, d\mu_G)_\rho$ is the Weyl fingerprint of G. If the log likelihood ratio has a uniformly convergent Peter Weyl expansion $h = \sum_\rho c_\rho \chi_\rho$, then

$$\frac{1}{n} \log B_n = \sum_\rho c_\rho \, W_\rho(n).$$

The log Bayes factor is a linear functional of the Weyl averages, so the vector of Weyl averages is sufficient in the limit: sequences with the same limiting fingerprint have the same Bayes factor rates for every pair of candidates. The likelihood chooses the weights c_ρ optimally (Neyman Pearson); matching a few

moments is valid but inefficient. In genus 2 the conjugacy class of s_p in $\mathrm{USp}(4)$ is determined by the normalised coefficients (a_1, a_2) of the L polynomial, and the trace a_1 alone loses information: the 55 candidate groups of [192], of which 52 occur, give only 37 distinct laws of a_1, while the laws of a_1 and of a_2 together distinguish all of them. No trace likelihood separates two groups with the same law of a_1.

29.7 Genus two: three connected candidates

We compare three connected groups through their trace laws on $[-4, 4]$. Write $s(x) = (2\pi)^{-1}\sqrt{4 - x^2}$ for the semicircle density.

Proposition 29.4 (Weyl integration on $\mathrm{USp}(4)$). *A class function φ on $\mathrm{USp}(4)$, written in terms of the eigenvalue angles $e^{\pm i\theta_1}, e^{\pm i\theta_2}$, satisfies, for the Haar probability measure dh,*

$$\int_{\mathrm{USp}(4)} \varphi \, dh = \frac{8}{\pi^2} \int_{[0,\pi]^2} \varphi \, (\cos\theta_1 - \cos\theta_2)^2 \sin^2\theta_1 \sin^2\theta_2 \, d\theta_1 \, d\theta_2.$$

So $x_i = 2\cos\theta_i$ have joint density $\frac{1}{2}(x_1 - x_2)^2 s(x_1)s(x_2)$, and $t = x_1 + x_2$ has density

$$f_{\mathrm{USp}(4)}(t) = \frac{1}{2} \int (2x - t)^2 s(x) s(t - x) \, dx, \qquad f_{\mathrm{USp}(4)}(0) = \frac{64}{15\pi^2},$$

with even moments $1, 3, 14$.

Proof. The first display is the case $g = 2$ of the density (6.1), a consequence of Theorem 6.1; its constant is $2^4/(2!\,\pi^2) = 8/\pi^2$. The change of variables uses $s(x)\,dx = \frac{2}{\pi}\sin^2\theta\,d\theta$ and $(x_1 - x_2)^2 = 4(\cos\theta_1 - \cos\theta_2)^2$. At $t = 0$,

$$f(0) = 2\int x^2 s(x)^2 \, dx = \frac{1}{2\pi^2} \int_{-2}^{2} (4x^2 - x^4) \, dx = \frac{64}{15\pi^2},$$

and, for independent semicircular x_1, x_2,

$$\mathbb{E}_{\mathrm{USp}(4)} t^2 = \tfrac{1}{2}\mathbb{E}\big[(x_1 + x_2)^2 (x_1 - x_2)^2\big] = \mathbb{E}x_1^4 - (\mathbb{E}x_1^2)^2 = 1;$$

the other moments are similar. □

The second candidate, $\mathrm{SU}(2) \times \mathrm{SU}(2)$ embedded block diagonally, has trace density $s * s$ (value $8/(3\pi^2)$ at 0, moments $2, 10, 70$). For a genus 2 curve C over $\mathbb{Q}$ it arises exactly when all endomorphisms of $\mathrm{Jac}(C)_{\overline{\mathbb{Q}}}$ are defined over $\mathbb{Q}$ and $\mathrm{End}(\mathrm{Jac}(C)_{\overline{\mathbb{Q}}}) \otimes \mathbb{R} \simeq \mathbb{R} \times \mathbb{R}$ [192], for instance when $\mathrm{Jac}(C)$ is isogenous over $\mathbb{Q}$ to a product of two elliptic curves without complex multiplication that are not isogenous over $\overline{\mathbb{Q}}$. The third, $\mathrm{SU}(2)$ embedded diagonally, has trace $2x$ with

x semicircular, density $(8\pi)^{-1}\sqrt{16-t^2}$ and moments $4, 32, 320$; it arises exactly when the same holds with $M_2(\mathbb{R})$ in place of $\mathbb{R} \times \mathbb{R}$, for instance when $\mathrm{Jac}(C)$ is isogenous over $\mathbb{Q}$ to the square of an elliptic curve without complex multiplication.

Table 29.1 gives the divergences, computed by quadrature of the convolution integrals and confirmed by Monte Carlo. Near ± 4 the $\mathrm{USp}(4)$ density vanishes like $(4-|t|)^4$, the diagonal density only like $(4-|t|)^{1/2}$, so the log likelihood ratio of the diagonal group against $\mathrm{USp}(4)$ is large where the diagonal law still puts appreciable mass, and $\mathrm{KL}(\mathrm{diag} \| \mathrm{USp}(4)) = 1.061$ is more than twice its reverse 0.456.

$P \setminus Q$	$\mathrm{USp}(4)$	$\mathrm{SU}(2) \times \mathrm{SU}(2)$	diagonal
$\mathrm{USp}(4)$	0	0.117	0.456
$\mathrm{SU}(2) \times \mathrm{SU}(2)$	0.168	0	0.161
$\mathrm{SU}(2)$ diagonal	1.061	0.262	0

Table 29.1: Kullback Leibler divergences $\mathrm{KL}(P \| Q)$ in nats between the trace laws of three connected Sato Tate groups in genus 2, with P indexing rows and Q columns. They are the per prime error exponents of Theorem 29.1.

29.8 Three curves

We counted points on three genus 2 curves at all odd primes below 10^5 that do not divide the discriminant of the defining polynomial (9589 or 9590 primes per curve), and computed posteriors with a uniform prior, likelihoods from the trace densities above and clipping at 20 (Figures 29.2 and 29.3).

$C_1 : y^2 = x^5 - x + 1$. The quintic has Galois group S_5, so by a theorem of Zarhin [593] the Jacobian has no endomorphisms over $\overline{\mathbb{Q}}$ beyond $\mathbb{Z}$, and its group is $\mathrm{USp}(4)$. The even moments $1.020, 3.166, 15.40$ are within about two standard errors of the $\mathrm{USp}(4)$ values $1, 3, 14$. The posterior probability of $\mathrm{USp}(4)$ is 0.426 after 10 primes, 0.984 after 50 and 0.99998 after 100. The per prime log Bayes factor of $\mathrm{SU}(2) \times \mathrm{SU}(2)$ against $\mathrm{USp}(4)$ is -0.111, against the theoretical -0.117, and that of the diagonal group is -0.447, against -0.456. For C_1 the theorem is conditional on the general Sato Tate conjecture.

$C_2 : y^2 = x^6 + 3x^4 + x^2 + 1$. The involutions $(x, y) \mapsto (-x, \pm y)$ split the Jacobian over $\mathbb{Q}$, up to isogeny, into $E : y^2 = x^3 - 2x + 2$ and $E' : y^2 = x^3 + x^2 + 3x + 1$, so $a_p(C_2) = a_p(E) + a_p(E')$. Their j invariants $-13824/19$ and $32768/19$ are not integers, so neither has complex multiplication. Curves without complex multiplication that are isogenous over $\overline{\mathbb{Q}}$ are, over $\mathbb{Q}$, isogenous up to a quadratic twist, which would force $|a_5(E)| = |a_5(E')|$; since $a_5(E) = 1$ and $a_5(E') = -3$, E and E' are not isogenous over $\overline{\mathbb{Q}}$. The group is therefore $\mathrm{SU}(2) \times \mathrm{SU}(2)$, and by the joint Sato Tate theorem (Theorem 4.11) the limits for C_2 hold unconditionally. The

moments $1.999, 9.91, 68.3$ match this group, its posterior is 0.998 after 50 primes, and the per prime rate against $\mathrm{USp}(4)$ is 0.167, against the theoretical 0.168.

$C_3 : y^2 = x^6 + x^3 + 1$. The moments $1.98, 15.8, 158.8$ fit none of the candidates. The reason is visible in the data: $a_p = 0$ for every prime $p \equiv 2 \bmod 3$ and for no prime $p \equiv 1 \bmod 3$ in our range, so the trace law has an atom of mass 0.501 at 0, and on the remaining primes the even moments $3.96, 31.7, 318$ are close to $4, 32, 320$. This suggests a disconnected group with identity component $\mathrm{SU}(2)$ diagonal; among the 55 groups of [192] exactly two, denoted E_2 and $J(E_1)$ there, have trace moments $2, 16, 160$, and only the law of a_2 separates them. The curve has extra automorphisms, $(x, y) \mapsto (\zeta_3 x, y)$ and $(x, y) \mapsto (1/x, y/x^3)$, and its group lies outside our candidate set.

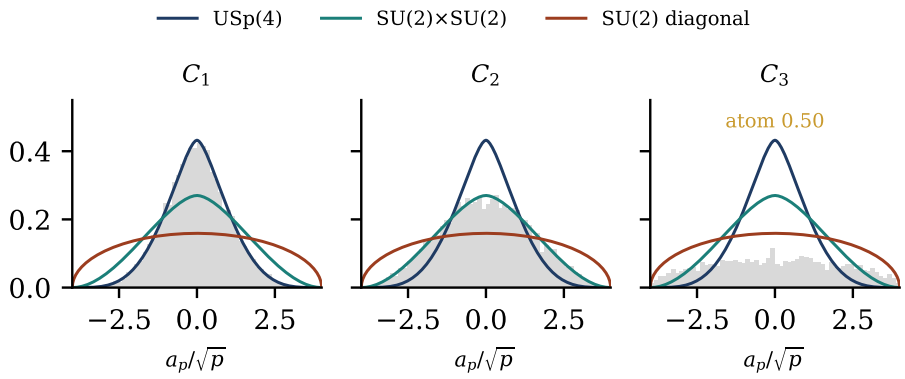


Figure 29.2: Histograms of $a_p/\sqrt{p}$ for C_1, C_2 and C_3 at the good primes $p < 10^5$, against the trace densities of $\mathrm{USp}(4)$, $\mathrm{SU}(2) \times \mathrm{SU}(2)$ and $\mathrm{SU}(2)$ diagonal. For C_3 the histogram shows the primes with $a_p \neq 0$, scaled to total mass 0.50; the atom at 0 carries the rest.

29.9 Misspecification

Berk [47] showed that when the true law μ_0 is outside a parametric model, the posterior concentrates, under regularity conditions, on the Kullback Leibler projection, the set of parameters minimising $\mathrm{KL}(\mu_0 \| P_\theta)$; see [322] for infinite dimensional models. Theorem 29.1 is the deterministic form for two candidates: the rate $\mathrm{KL}(\mu_0 \| P_0) - \mathrm{KL}(\mu_0 \| P_1)$ favours the candidate closer to μ_0.

For C_3 the true law $\mu_0 \approx \frac{1}{2}\delta_0 + \frac{1}{2}\,\mathrm{law}(2x)$ has an atom and every candidate is continuous, so every divergence $\mathrm{KL}(\mu_0 \| \mu_G)$ is infinite and the projection is undefined. A likelihood computed from densities scores the atom by the density value at 0, and with f_G the trace density of G the posterior then concentrates on the maximiser of $\frac{1}{2}\log f_G(0) + \frac{1}{2}\mathbb{E}\log f_G(2x)$, with x semicircular. The values $f_G(0)$

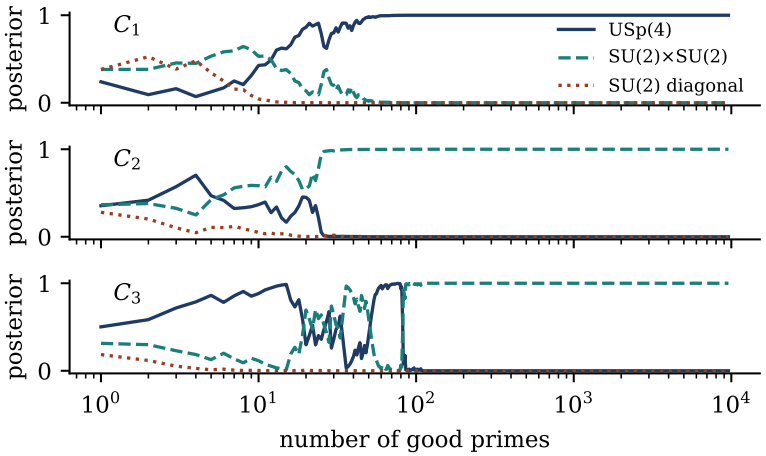


Figure 29.3: Posterior probabilities of $\mathrm{USp}(4)$, $\mathrm{SU}(2) \times \mathrm{SU}(2)$ and $\mathrm{SU}(2)$ diagonal for C_1, C_2 and C_3 against the number of good primes (logarithmic scale), uniform prior. For C_3, whose group is not a candidate, the posterior settles on $\mathrm{SU}(2) \times \mathrm{SU}(2)$ after about 100 primes.

are 0.432, 0.270 and 0.159 for $\mathrm{USp}(4)$, $\mathrm{SU}(2) \times \mathrm{SU}(2)$ and the diagonal group, so the atom favours the peaked $\mathrm{USp}(4)$ density, the continuous part favours the diagonal group, and the compromise is $\mathrm{SU}(2) \times \mathrm{SU}(2)$: predicted rates against $\mathrm{USp}(4)$ are 0.164 (product) and 0.031 (diagonal), observed 0.158 and 0.021, and the posterior on the product exceeds 0.99 after 100 primes. This is an artefact of how the atom is scored.

The honest reading for C_3 is *none of the candidates*: a posterior near one measures fit relative to the list. The moments expose the problem at once: for every candidate at least one of them is off by dozens of standard errors (for $\mathrm{SU}(2) \times \mathrm{SU}(2)$ the second moment fits, the fourth and sixth do not). The likelihood is far less sensitive: the average log likelihood under the winner is -1.80 per prime, against -1.75 expected if the traces followed the winning law, a gap of about seven standard errors after 9590 primes. The remedy is expansion to all 52 groups of [192], which forces the likelihood to carry discrete components: atoms at $a_1 = 0$ from disconnected groups with components on which the trace vanishes identically. With Lebesgue measure plus atoms as reference, as in Theorem 29.1, the atom alone separates C_3 from every connected candidate: each prime $p \equiv 2 \bmod 3$ adds $\log(1/(2\varepsilon))$ to the log Bayes factor if the connected candidates carry a small atom ε at 0, as in the elliptic case, and the clipping level M if they carry none.

29.10 Nonparametric alternatives and arithmetic priors

Distrusting the candidate list, one can put a nonparametric prior on the trace law. Dirichlet process draws [186] are discrete, so against a parametric μ_G a Pólya tree [356] or a Dirichlet process mixture is more natural; the Bayes factor of μ_G against such a prior is a test of fit of μ_G itself rather than a comparison within the list. Alternatively, a prior on moment sequences through the canonical moments of Dette and Studden [147], free coordinates in $(0,1)$ for the moment space of $[-4,4]$, tests whether the empirical sequence lies near some fingerprint. A roughness prior in the sense of Good and Gaskins, on densities of the eigenvalue angles that vanish on the walls of the alcove, has, formally, the Sato Tate law of a simply connected group as its mode, and for $\mathrm{SU}(2)$ it shrinks the empirical moments of the symmetric powers towards those of that law (Theorem 23.12). The Dirichlet process has already appeared in Chapter 24, not as a prior but as the limit law of the normalised Frobenius labelled factorisation of a random integer, with the Haar measure of Chebotarev's theorem as its base measure.

Arithmetic should inform the prior. The Sato Tate group of an abelian surface A is determined by its Galois endomorphism type, the real algebra $\mathrm{End}(A_{\overline{\mathbb{Q}}}) \otimes \mathbb{R}$ with its Galois action [192], which can be computed rigorously [107]; in particular, the component group of the Sato Tate group is the Galois group of the smallest field over which all endomorphisms are defined. For C_3 the automorphism $(x,y) \mapsto (\zeta_3 x, y)$ induces an endomorphism of the Jacobian that is defined over $\mathbb{Q}(\sqrt{-3})$ but not over $\mathbb{Q}$, so the group of C_3 is disconnected and none of the three connected candidates can be right, a fact available before any prime is examined. Once the endomorphism type is known, Frobenius data test the Sato Tate conjecture rather than estimate the group. Across a database, group frequencies among curves of bounded conductor give an empirical Bayes prior [464], dominated by $\mathrm{USp}(4)$.

29.11 Notes

The moment method in genus 2 goes back to Kedlaya and Sutherland [302]; the classification [192] and the survey [532] systematise it. Equidistribution is proved for many but not all of the 52 groups, and is open for $\mathrm{USp}(4)$ (Chapter 3).

The core of Theorem 29.1, that the average log likelihood ratio converges to a difference of divergences, is the argument of Wald's proof that maximum likelihood is consistent [560], with equidistribution in place of the strong law of large numbers; the divergence is that of Kullback and Leibler [335]. Theorem 29.1 and Corollary 29.3 are ours; they assume equidistribution and prove nothing new about Sato Tate.

29.12 Exercises

1. Prove $\int_0^{\pi} \log \sin\theta \, d\theta = -\pi \log 2$ without Fourier series, by the substitution $\theta \mapsto 2\theta$ and symmetry. Deduce that the arcsine mean of $\log(4 - x^2)$ is 0.
2. Show that $\mathrm{KL}(\mu_{\mathrm{ST}} \| \mu_{\mathrm{CM}}) = 1$ while $\mathrm{KL}(\mu_{\mathrm{CM}} \| \mu_{\mathrm{ST}}) = \infty$, and explain why the atom ε is needed. Show that both limits in Corollary 29.3 improve as ε decreases, and that the price is the jump $\log(1/(2\varepsilon))$ that each supersingular prime of a curve without complex multiplication adds to $\log B_n$. For which ε is the rate without complex multiplication at most -0.99, and what is the smallest rate with complex multiplication on that range?
3. Using Proposition 29.4, show that the $\mathrm{USp}(4)$ trace has fourth moment 3 and that $f_{\mathrm{USp}(4)}(t)$ vanishes like $(4 - |t|)^4$ as $|t| \to 4$. Explain the asymmetry in Table 29.1.
4. (Probabilistic.) From Wald's identity and optional stopping derive the approximation $\mathbb{E}_1[N] \approx \log(1/\alpha)/\mathrm{KL}(P_1 \| P_0)$. Evaluate it for $\mathrm{SU}(2) \times \mathrm{SU}(2)$ against $\mathrm{USp}(4)$ with $\alpha = 10^{-6}$.
5. For the limiting law of C_3 and each of the three connected candidates G, compute numerically
$$\tfrac{1}{2} \log f_G(0) + \tfrac{1}{2} \mathbb{E} \log f_G(2x), \qquad x \text{ semicircular},$$
and verify that $\mathrm{SU}(2) \times \mathrm{SU}(2)$ wins.
6. Show that altering (x_k) on a set of indices of density zero leaves every limit in Theorem 29.1 unchanged.

29.13 Open computational problems

Computational Problem 29.1 (The group of $y^2 = x^6 + x^3 + 1$). Objective: identify the Sato Tate group of C_3 among the 52 groups of [192] from Frobenius data; its trace moments point to E_2 and $J(E_1)$ in the notation of [192], which share the law of a_1. Data: the normalised coefficients (a_1, a_2) of the L polynomial at all good primes below 2^{30}. Route: `smalljac` or average polynomial time point counting [244]; moments of a_1 and a_2 up to order 12 and the frequency of $a_1 = 0$ by residue class mod 3, compared with the tables of [192] and checked against the endomorphism type [107]. Deliverable: the group, its posterior against the other group with the same law of a_1, and the number of primes needed to reach $1 - 10^{-9}$. Why it matters: it completes this chapter's misspecified example and measures what a_2 adds.

Computational Problem 29.2 (A likelihood library, a classifier and the error exponent matrix). Objective: build the joint densities of (a_1, a_2) for all 52 groups,

atoms and lower dimensional components included, as densities against a reference measure in the sense of Theorem 29.1; compute the 52×52 matrix of divergences $\mathrm{KL}(\mu_G \| \mu_{G'})$ for (a_1, a_2) and for a_1 alone (infinite where μ_G charges a set that $\mu_{G'}$ does not, and zero for a_1 alone when two groups share its law); and run Wald's test on every genus 2 curve in the LMFDB [371] with primes up to 10^7. Route: Weyl integration on identity components in Sage, Monte Carlo on cosets (10^8 draws per group), exact arithmetic on atoms. Deliverable: the library, the divergence matrix with its near ties, a confusion matrix against the LMFDB labels, and stopping times against $\log(1/\alpha)/\mathrm{KL}$. Why it matters: the divergences are the error exponents of every test, and the near ties are the hard instances. The library and the divergence matrix are the new objects here; the run to 10^7 is a pilot for Computational Problem 3.2, which goes to 2^{30}, and the library must reproduce the Weyl fingerprints of Computational Problem 4.2.

COMPUTATIONAL PROBLEM 29.3 (Genus three with an empirical Bayes prior). Objective: extend the classifier to the 410 groups of genus 3 [193], building on Sutherland's computations. Data: plane quartics and hyperelliptic curves of genus 3 with discriminant below 10^7, primes up to 2^{26}. Route: average polynomial time point counting [244] and a Dirichlet prior on group frequencies estimated across the database. Deliverable: frequencies, posteriors, and curves fitted by no group. Why it matters: with 410 candidates, shared trace laws and near ties should be far more common than in genus 2, and the experiment measures how much of the classification Frobenius data can resolve. It builds on the genus 3 sample of Computational Problem 3.2, adding a prior estimated across the database and a search for the curves that no group fits.

CHAPTER 30

OPEN PROBLEMS

In his 1900 address Hilbert remarked that a branch of science is alive as long as it offers an abundance of problems. Langlands' letter to Weil of 1967 and his 1970 paper [350] supplied such an abundance, and more than half a century of work has settled only part of it. This chapter collects twelve problems on which the probabilistic point of view of this book has something definite to say. We do not pretend that probability will settle the central conjectures: the limits proposition (Proposition 1.5) shows that no functional of the empirical law of Frobenius traces can detect automorphy. What probability does supply is a language for quantitative refinements: rates, fluctuations, large deviations, extreme values, hitting times and sequential tests. For each problem we give the statement, its status with attributions, what probability and our computations contribute, and what would be needed. Where we do not know the right answer, the problem is posed as a question. Chapter 31 collects computational versions of several of them. Chapter 17 poses two further problems, on the squarefree error under the Riemann hypothesis and on the signs of the Pólya kernels of imaginary quadratic fields (Open Problems 17.1 and 17.2), Chapter 18 four on Dirichlet's β function and the lines to the right of the centre (Open Problems 18.1 to 18.4), and Chapter 19 two on Pólya kernels and signed Thorin measures (Open Problems 19.1 and 19.2). Chapter 20 poses eight problems on reciprocity (Open Problems 20.1 to 20.8). Chapter 21 poses two on isogeny graphs and hitting times (Open Problems 21.1 and 21.2). Chapter 22 poses three, on the Hasse Weil bound, on patching and on Mazur's theorem (Open Problems 22.1 to 22.3). Chapter 23 poses two, on the stable recovery of Frobenius traces from a window of the transported density and on penalised estimators of Thorin densities (Open Problems 23.1 and 23.2). Chapter 24 poses three, on an upper bound for the rate of the Dirichlet process limit, on the signs of the Mertens deficits of fields with group S_n and on an arithmetic counterpart of Bondesson's product theorem (Open Problems 24.1 to 24.3). Chapter 25 poses three, on the discrepancy of Hecke angles, on the Euler Kronecker constants of Hecke characters and on an effective Gamma shape test (Open Problems 25.1 to 25.3), and Chapter 26 four, on the Thorin representation at the centre, on a Chebotarev theorem for geodesics, on the even icosahedral representations and on the cutoff of Brownian motion on $X(N)$ (Open Problems 26.1 to 26.4).

30.1 Spectral problems

Open Problem 30.1 (Ramanujan Petersson for Maass forms). Let ϕ be a Hecke Maass cusp form for $\mathrm{SL}_2(\mathbb{Z})$ or for a congruence subgroup, with Satake parameters $\{\alpha_p, \alpha_p^{-1}\}$ at an unramified prime p, real Hecke eigenvalue $\lambda_\phi(p) = \alpha_p + \alpha_p^{-1}$ and Laplace eigenvalue $\lambda = \frac{1}{4} + r^2$. Show that $|\alpha_p| = 1$ for every such p, equivalently $|\lambda_\phi(p)| \le 2$, and that r is real, which is Selberg's eigenvalue conjecture $\lambda \ge \frac{1}{4}$.

Status. For holomorphic forms the analogue is Deligne's theorem [140]; Maass forms have no motive and no geometric route is known. Selberg proved $\lambda \ge \frac{3}{16}$ and conjectured $\lambda \ge \frac{1}{4}$ [497]; for $\mathrm{SL}_2(\mathbb{Z})$ itself the eigenvalue part is classical, since the cuspidal spectrum starts far above $\frac{1}{4}$. The best general bounds, $|\alpha_p| \le p^{7/64}$ and $\lambda \ge \frac{975}{4096}$, are due to Kim and Sarnak [315], from the automorphy of Sym^4. Sarnak's density hypothesis [482], proved in many cases, says that exceptional parameters are rare in a quantitative sense.

What probability and computation contribute. By the trace formula the eigenvalues $\lambda_\phi(p)$, for ϕ on $\mathrm{SL}_2(\mathbb{Z})$ ordered by Laplace eigenvalue, equidistribute with respect to the Plancherel measure μ_p [476]: the spectral measure at the root o of simple random walk S on the $(p+1)$ regular tree, rescaled to $[-2, 2]$ (Chapter 9). Its moments $\int x^{2m}\, d\mu_p = (p+1)^{2m} p^{-m} \mathbb{P}(S_{2m} = o)$ grow like 4^m up to a polynomial factor. Hence if, in a family of V forms, the averages of $\lambda_\phi(p)^{2m}$ match these moments up to $m \approx c \log V$, Markov's inequality bounds $\#\{\phi : |\lambda_\phi(p)| \ge 2 + \delta\}$ by $V(1 + \delta/2)^{-2m}$ times a polynomial in m, that is, by $V^{1 - 2c \log(1+\delta/2)}$ up to such factors: the density hypothesis read as a large deviation bound for a random walk spectral measure. For the holomorphic analogue τ, the primes below 10^6 give $\max |\tau(p)/p^{11/2}| = 1.99920$.

What would be needed. Family averages cannot exclude a single exception. Individual bounds follow from automorphy of all $\mathrm{Sym}^n \phi$ by the tensor power trick [350]; Newton and Thorne proved this automorphy for holomorphic newforms [419, 420] through their Galois representations, which Maass forms in general lack.

Open Problem 30.2 (Lang and Trotter). Let $E/\mathbb{Q}$ be an elliptic curve of conductor N_E without complex multiplication, let $t \in \mathbb{Z}$, and let $\pi_{E,t}(x) = \#\{p \le x : \ p \nmid N_E,\ a_p(E) = t\}$. Show that

$$\pi_{E,t}(x) \ \sim \ C_{E,t}\, \pi_{1/2}(x), \qquad \pi_{1/2}(x) = \int_2^x \frac{du}{2\sqrt{u}\, \log u},$$

whenever the constant $C_{E,t}$ predicted by Lang and Trotter [353] is positive; when $C_{E,t} = 0$ a congruence obstruction on the image of Galois makes the set finite.

Status. Open for every pair (E,t) with $C_{E,t}>0$. Elkies proved that every elliptic curve over $\mathbb{Q}$ has infinitely many supersingular primes [175], and, with Kaneko and M. R. Murty, that $\pi_{E,0}(x)=O(x^{3/4})$ [176]; for every t, $\pi_{E,t}(x)=O(x(\log\log x)^2/(\log x)^2)$, as claimed by V. K. Murty [410] and proved by Thorner and Zaman [549]. Under the generalized Riemann hypothesis Zywina [597], improving Serre [506] and Murty, Murty and Saradha [411], proved $\pi_{E,0}(x)=O(x^{3/4}(\log x)^{-1/2})$ and $\pi_{E,t}(x)=O(x^{4/5}(\log x)^{-3/5})$. All are far above the predicted order $x^{1/2}/\log x$. On average over families the conjecture holds: Fouvry and Murty [199] for $t=0$, David and Pappalardi [128] for $t\neq 0$.

What probability and computation contribute. The conjecture is a local limit theorem with arithmetic corrections: the Sato Tate density at 0 is $1/\pi$, the lattice spacing of x_p is $1/\sqrt{p}$, and summing $1/(\pi\sqrt{p})$ over primes gives $(2/\pi)\pi_{1/2}(x)$, which a Haar measure computation on the image of Galois in $\mathrm{GL}_2(\hat{\mathbb{Z}})$ then corrects. For the curve $E_1\colon y^2=x^3+x+1$ of Chapter 5 the counts of supersingular primes are 10, 23, 79, 210 at $x=10^4,\dots,10^7$, with ratios to $\pi_{1/2}(x)$ of 0.66, 0.65, 0.89, 0.91.

What would be needed. Apart from the unconditional bound for $t=0$, which comes from singular moduli, the upper bounds above come from forms of the Chebotarev theorem in the division fields $\mathbb{Q}(E[\ell])$, whose error terms grow with the degree of the field even under the generalized Riemann hypothesis. An asymptotic requires control of a set of primes of density zero at the scale $x^{1/2}/\log x$, finer than any form of the Chebotarev theorem now available.

OPEN PROBLEM 30.3 (Optimal discrepancy and a functional limit theorem). Let $E/\mathbb{Q}$ be without complex multiplication, let $p_1<p_2<\cdots$ be its primes of good reduction, F_n the empirical distribution function of $x_{p_1},\dots,x_{p_n}$, F_{ST} that of μ_{ST}, and $D_n=\sup_x|F_n(x)-F_{\mathrm{ST}}(x)|$.

(a) Show that $D_n=O(n^{-1/2+\delta})$ for every $\delta>0$.

(b) (Question.) Let E be chosen uniformly among the curves without complex multiplication of conductor at most X, and let $n=\lfloor X^c\rfloor$ with $c>0$ fixed. For which c does $\sqrt{n}\,(F_n-F_{\mathrm{ST}})\circ F_{\mathrm{ST}}^{-1}$ converge in law to a Brownian bridge as $X\to\infty$, and $\sqrt{n}\,D_n$ to the Kolmogorov law?

Status. Akiyama and Tanigawa [11] conjectured (a) and showed that it implies the Riemann hypothesis for $L(s,E)$; Mazur [386] stated the extension to every $L(s,\mathrm{Sym}^m E)$, under analytic hypotheses that Newton and Thorne now supply (Chapter 5). Unconditional effective forms of the Sato Tate theorem give only rates of logarithmic type [548], and the Riemann hypothesis for every $L(s,\mathrm{Sym}^m E)$ gives $D_n\ll n^{-1/4}$ up to logarithms (V. K. Murty [409]; Rouse and Thorner [466] give an explicit form for squarefree level). For a single curve, with n random on a

logarithmic scale, the analogue of (b) has a negative answer under the Riemann hypothesis for $L(s, E)$, as explained below; the family version is, as far as we know, new.

What probability and computation contribute. For independent draws (a) holds almost surely and (b) is Donsker's theorem. For a single curve, under the Riemann hypothesis for $L(s, E)$, the explicit formula argument of Section 5.3 makes the first Chebyshev mode $n^{-1/2}\sum_{j\le n} x_{p_j}$ of order $(\log n)^{-1/2}$ in logarithmic mean square, whereas a Brownian bridge limit would make this continuous linear functional of the process standard normal; hence no randomisation of n on a logarithmic scale, such as n logarithmically uniform on $[N, N^2]$, produces a Brownian bridge for one curve; under the Riemann hypothesis for $L(s, \mathrm{Sym}^m E)$ every fixed mode $n^{-1/2}\sum_{j\le n} U_m(x_{p_j}/2)$ tends to 0 in the same way, and Conjecture 5.3 predicts that $\sqrt{n}\, D_n$ itself tends to 0 in logarithmic density. In a family one expects the primes up to p_n to behave independently while p_n is small compared with the conductors; a central limit theorem for the first mode is known for eigenforms of level one whose weight grows faster than every power of n [413]. The explicit formula takes over once p_n exceeds the conductor of $\mathrm{Sym}^m E$, about N_E^m, and its diagonal terms suggest that the mth mode has variance about $\min(1, m/c)$: a transition at $c = 1$, below which every mode has variance near 1 and above which the first mode is too small for a bridge. The data of Section 5.3 are consistent with so slow a decay: $\sqrt{n}\, D_n$ shows no drift for E_1, and across 200 curves a Kolmogorov Smirnov test does not separate its law from the Kolmogorov law.

What would be needed. For (a), square root cancellation in the mode sums $\sum_{j\le n} U_m(x_{p_j}/2)$ uniformly for m up to about $n^{1/2}$; the Riemann hypothesis for each $L(s, \mathrm{Sym}^m E)$ bounds the mth mode sum only by $m\sqrt{n}$ up to logarithms, because the number of zeros of bounded height grows linearly in m, and this is why it yields only $n^{-1/4}$. For (b), in the regime $c < 1$ a central limit theorem for the modes jointly as m grows, with tightness; in the regime $c > 1$ the variances of the low modes, which depend on the pair correlation of the low lying zeros of the symmetric power L functions in the family.

30.2 Inverse problems and symmetry

Open Problem 30.4 (An inverse problem for automorphy). (Question.) Is there a criterion, stated through quantitative properties of the twisted sums $\sum_{p\le x} x_p \chi(p) \log p$ at the square root scale (rates in the law of large numbers, mean squares on the logarithmic scale, large deviation rates), uniform over Dirichlet characters χ of growing conductor, that decides whether a sequence (x_p) in $[-2, 2]$ indexed by the primes is close to the normalised Hecke eigenvalues

$x_p(f) = a_p(f)/p^{(k-1)/2}$ of a holomorphic newform f of weight k with trivial nebentypus, in the sense that $\sum_{p\le x}(x_p - x_p(f))\chi(p)\log p = o(\sqrt{x})$ for every χ?

Status. The converse theorems of Hecke [250], Weil [571], Jacquet and Langlands [277] and Cogdell and Piatetski Shapiro [108] are inverse theorems of this type, but analytic: they start from the analytic continuation and functional equations of twisted L functions. Chapter 20 restates the hypotheses of Hecke's theorem as a symmetry of a Pólya kernel built from the coefficients (Proposition 20.27), and the holomorphy and the generalized Riemann hypothesis of a self dual Artin L function as the complete monotonicity of a heat trace computed from Frobenius traces, the conductor and the gamma factor (Theorem 20.30); these are reformulations, and Open Problems 20.1 and 20.5 pose related questions. Chapter 23 computes the Pólya kernel of an Artin L function by Bromwich inversion on the critical line, given that the L function is entire; that the result is modular is again the content of Weil's converse theorem (Proposition 23.8).

What probability and computation contribute. By Corollary 1.6 and the remark after it, no statistic that is unchanged when the x_p are altered on a set of primes of density zero, such as a limit of averages with weight $O(1/\pi(x))$ per prime, twisted or not, can decide automorphy. Rates at the square root scale cannot decide it either, which is why the question asks only for closeness: if S is an infinite set of primes with $\theta_S(x) = \sum_{p\in S,\, p\le x}\log p = o(\sqrt{x})$, such as the primes p_{2^j}, altering the $x_p(f)$ within $[-2,2]$ at every prime of S changes each twisted sum by at most $4\theta_S(x) = o(\sqrt{x})$ and leaves every such rate unchanged, while by Proposition 1.5 the altered sequence agrees at all but finitely many primes with no such newform. Pointwise bounds do not suffice: independent variables with the Sato Tate law satisfy, almost surely, $\sum_{p\le x} x_p\chi(p)\log p \ll \sqrt{x}\log(qx)$ for all characters χ of conductor $q \le x$, and are almost surely close to no newform, since the newforms are countable and for each of them the untwisted sum of the differences fluctuates on the scale $\sqrt{x\log x}$. Mean squares on the logarithmic scale separate them: under the generalized Riemann hypothesis, for a newform f the function $e^{-y/2}\sum_{p\le e^y} x_p(f)\chi(p)\log p$ is almost periodic in mean square, with frequencies the ordinates of the zeros of $L(s, f\otimes\chi)$, and so bounded in mean square, while for independent variables its mean square grows like y. Square root cancellation for every twist, even with a logarithmic loss, makes the twisted L functions holomorphic and nonvanishing in $\operatorname{Re} s > \frac{1}{2}$, but it does not give their functional equations.

What would be needed. A theorem converting such rates into the hypotheses of Weil's converse theorem, not for (x_p) itself, which may carry an alteration that the rates cannot see, but for a sequence reconstructed from the rates, for instance

from the frequencies above; we do not know whether such a statement can be true without the root numbers.

OPEN PROBLEM 30.5 (General Sato Tate groups and symmetry types). (a) For abelian varieties and motives over number fields, determine the Sato Tate group and prove equidistribution with respect to its Haar measure. (b) For natural families of L functions, prove the Katz and Sarnak predictions for low lying zeros for test functions of unrestricted Fourier support.

Status. For elliptic curves without complex multiplication over totally real fields, (a) is the theorem of Clozel, Harris, Shepherd Barron and Taylor, completed by Barnet Lamb, Geraghty, Harris and Taylor [103, 266, 540, 67]; over CM fields it was proved by Allen, Calegari, Caraiani and seven coauthors [16]. For abelian surfaces there are exactly 52 groups [192]; equidistribution is known for many with extra endomorphisms [284] but not for $\mathrm{USp}(4)$. For abelian threefolds Fité, Kedlaya and Sutherland announced a classification with 410 groups [193]. For (b), Katz and Sarnak [298, 299] prove the function field analogue; over number fields only restricted support is known [271], as for the n level correlations of a single L function [471].

What probability and computation contribute. Both parts concern pushforwards of Haar measure: in (a) to the conjugacy classes of the group, in (b) to the eigenvalue statistics of the symmetry group. In (b), restricted support means that the n level densities are known only against test functions whose Fourier transforms have bounded support, which falls short of determining them, much as finitely many moments fall short of determining a law. The Weyl averages of E_1 and of $E_2\colon y^2 = x^3 - x$ in Figure 4.2 of Chapter 4 match the generic and the CM law; how fast such agreement identifies the group is Open Problem 30.11.

What would be needed. For (a), already for a generic abelian surface, potential automorphy of the Galois representation composed with every irreducible representation of $\mathrm{USp}(4)$, not only the symmetric powers; for (b), a number field substitute for monodromy.

OPEN PROBLEM 30.6 (Functoriality beyond known cases). Prove functoriality for $\mathrm{GL}_m \times \mathrm{GL}_n \to \mathrm{GL}_{mn}$ in general, and for $\mathrm{Sym}^n : \mathrm{GL}_2 \to \mathrm{GL}_{n+1}$ for Maass forms and all n.

Status. Known cases include Sym^2 [212], $\mathrm{GL}_2 \times \mathrm{GL}_2$ [452], $\mathrm{GL}_2 \times \mathrm{GL}_3$ and Sym^3 [316], Sym^4 (Kim [315]), all symmetric powers for holomorphic newforms [419, 420], and the transfers from quasi split classical groups to GL_N [25]; see [24, 211].

What probability and computation contribute. Functoriality has consequences that Frobenius data can test. For an elliptic curve without complex multiplication, automorphy and cuspidality of every symmetric power give the Sato Tate law, and the even moments of x_p for E_1 over the good primes below 10^7 agree with the Catalan numbers $1, 2, 5, 14, 42, 132$ to a relative error below 6×10^{-4} (Chapter 5). For cuspidal π_1, π_2 on GL_2, cuspidality of $\pi_1 \boxtimes \pi_2$ on GL_4 gives

$$\sum_{p \le x} \frac{|\operatorname{tr} A_p(\pi_1)|^2 \, |\operatorname{tr} A_p(\pi_2)|^2}{p} \sim \log\log x,$$

which for two elliptic curves without complex multiplication that are not twists of each other is the moment identity $\iint x^2 y^2 \, d\mu_{\mathrm{ST}}(x) \, d\mu_{\mathrm{ST}}(y) = 1$.

What would be needed. The known cases came from the Langlands Shahidi method and converse theorems (Gelbart and Jacquet, Ramakrishnan, Kim and Shahidi, Kim), from the trace formula (Arthur) and from automorphy lifting (Newton and Thorne). By the limits proposition the checks above are functionals of empirical laws and cannot replace such methods.

30.3 Graphs

Open Problem 30.7 (Ramanujan graphs of all degrees). Give a deterministic algorithm that, for each $d \ge 3$ and all large n with nd even, constructs in time polynomial in n a nonbipartite d regular graph on n vertices whose nontrivial eigenvalues satisfy $|\lambda| \le 2\sqrt{d-1}$.

Status. The arithmetic constructions of Chapter 10 (Lubotzky, Phillips and Sarnak; Margulis; Morgenstern; Pizer) give special sequences of n only; the probabilistic results of that chapter give nearly Ramanujan graphs (Friedman, Bordenave), bipartite Ramanujan multigraphs of every degree and every even number of vertices (Marcus, Spielman and Srivastava) and, since about 69 percent of random d regular graphs are Ramanujan (Huang, McKenzie and Yau [263], a 2024 preprint), existence for all large n and a randomised polynomial time construction (sample a graph and test its spectrum). Deterministic polynomial time constructions reach $|\lambda| \le 2\sqrt{d-1} + \varepsilon$ for every $d \ge 3$ and $\varepsilon > 0$ (Mohanty, O'Donnell and Paredes [402]), on exactly n vertices for all large n with nd even (Alon [17]); the problem asks for $\varepsilon = 0$, which, as far as we know, no deterministic construction achieves for nonbipartite graphs of all large sizes.

What probability and computation contribute. The existence results beyond the arithmetic constructions are probabilistic: the trace method of Friedman and

Bordenave, the interlacing polynomials of Marcus, Spielman and Srivastava, and the edge universality around the Kesten McKay law of Huang, McKenzie and Yau.

What would be needed. Derandomisation of the random constructions, which Cohen [109] achieved for the bipartite construction of Marcus, Spielman and Srivastava, or an arithmetic source of graphs of every degree: the arithmetic constructions above come from groups acting on the tree of PGL_2 over a local field, whose degree is always a prime power plus one.

30.4 Thorin measures and extreme values

Open Problem 30.8 (Hilbert's eighth problem for GL_2 in Thorin form). Let $\Lambda(s,\Delta) = (2\pi)^{-s}\Gamma(s)L(s,\Delta)$, write its zeros as $6 \pm i\gamma_k$ with $\operatorname{Re}\gamma_k > 0$, and let $k_\Delta(x) = \sum_k e^{-\gamma_k^2 x}$ for $x > 0$. Prove by a probabilistic argument that k_Δ is completely monotone on $(0,\infty)$.

Status. Integrating termwise, $\sum_k (\gamma_k^2 + s)^{-1} = \int_0^\infty e^{-sx} k_\Delta(x)\,dx$ for $s > 0$, and by the Hadamard product the left side is $(d/ds)\log\Lambda(6+\sqrt{s},\Delta)$. By Bernstein's theorem and the equivalence with generalized gamma convolutions in Chapter 13 (Theorem 13.8) the problem is equivalent to the Riemann hypothesis for Δ, hence open. The parallel statement for ξ is Polson's reformulation of Hilbert's eighth problem [441], building on Biane, Pitman and Yor [82]; the theory of generalized gamma convolutions is due to Thorin [546] and Bondesson [73].

What probability and computation contribute. A probabilistic proof would exhibit the law with Laplace transform $\Lambda(6)/\Lambda(6+\sqrt{s})$, whose Lévy density is $x^{-1}k_\Delta(x)$, as a generalized gamma convolution built without the zeros, for instance from hitting or occupation times of a diffusion, as Bessel processes produce the laws of Biane, Pitman and Yor. Equivalently, the characteristic function of the Pólya variable W must have only real zeros, which is Pólya's question [444], with the de Bruijn Newman heat flow [133, 417, 465] as a natural deformation. The product over the 404 zeros up to height 400, with a tail correction, reproduces $\Lambda(6)/\Lambda(6+\sqrt{s})$ for $s \le 400$ to about four significant digits (relative error at most 1.9×10^{-4}), and $\operatorname{Var} W = 0.112352$ against 0.112348 from the zeros with the tail correction (Chapter 13). Chapter 19 computes a smoothed signed version of this Thorin measure by Stieltjes inversion to height 30.

What would be needed. A positive measure τ on $(0,\infty)$ with $k_\Delta(x) = \int e^{-tx}\,\tau(dt)$, the Thorin measure of the problem, constructed without reference to the zeros, for example from an occupation or hitting time law; such a

construction would prove the Riemann hypothesis for Δ. Positivity of k_Δ alone, which is infinite divisibility, would not suffice. Open Problem 20.4 asks whether a positive summation of the prime side of the explicit formula can play the part that Fejér's kernel plays for the circle (Proposition 20.9).

OPEN PROBLEM 30.9 (The extreme value representation). Let M^* be the maximum of 24 independent copies of $M = \max_{n\geq 1} E_n/(2\pi n)$, with E_n independent standard exponential variables, so that $L(s,\Delta) = \mathbb{E}\, Q(s, 2\pi M^*)$ with $Q(s,c) = \Gamma(s,c)/\Gamma(s)$ (Chapter 27).

(a) (Question.) Can properties of the law of M^*, such as the self similarity $F(y) = y^{-12}e^{2\pi(y-1/y)}F(1/y)$ of its distribution function given by modularity, be used to locate or count the zeros of $L(s,\Delta)$?

(b) (Question.) For which newforms f is there a random variable M_f, the maximum of countably many independent variables with distribution functions of the form $(1-e^{-cy})^\kappa$ with $c,\kappa > 0$, such that $f(iy) = e^{-2\pi y}\mathbb{P}(M_f \leq y)$ for all $y > 0$?

Status. The representation is proved in Chapter 27 for Δ and for every eta product $\prod_m \eta(m\tau)^{r_m}$ with $r_m \geq 0$ and $\sum_m m\, r_m = 24$, among them the multiplicative eta products of Dummit, Kisilevsky and McKay [157] (Theorems 27.2 and 27.7), so (b) has a positive answer for those of them that are newforms. Part (a) asks for a route to Conjecture 27.8. As far as we know, both questions are new.

What probability and computation contribute. By Corollaries 27.3 and 27.4, $L(s,\Delta)$ lies in $(0,1)$ and increases on $s > 0$. On the critical line the functional equation gives

$$\Lambda(6+it,\Delta) = 2\,\mathbb{E}\int_{\max(1,M^*)}^{\infty} e^{-2\pi y} y^5 \cos(t\log y)\, dy,$$

so (a) asks for the sign changes in t of an explicit expectation, the analogue of Hardy's function. By uniqueness of Laplace transforms, (b) holds exactly when the exponents κ_m in the formal product $q^{-1}f = \prod_{m\geq 1}(1-q^m)^{\kappa_m}$ are all nonnegative and the product converges for $0 < q < 1$; for nonnegative κ_m this convergence is equivalent to f having no zeros in the upper half plane. Chapter 27 checks the representation to 16 digits and by Monte Carlo.

What would be needed. For (a), a property of the specific law of M^* that fails for nearby laws, since expectations built from nearby laws need not have their zeros on a line. For (b), a description of the newforms with no zeros in the upper half plane, whose divisors are supported on the cusps, together with the signs of their exponents κ_m; eta quotients are the basic examples, Borcherds products

are a natural framework for the exponents, and Martin's classification of the multiplicative eta quotients [382] is the place to start.

30.5 Hitting times, inference and bias

Open Problem 30.10 (Kemeny constants of isogeny graphs at finite level). For primes $N \equiv 1 \bmod 12$ let K_N be the Kemeny constant of simple random walk on the supersingular 2 isogeny graph in characteristic N, which is 3 regular on $h = (N-1)/12$ vertices, so that $K_N/h \to 2$ (Theorem 10.6). (Question.) Does the empirical distribution of $K_N - 2h$ over the primes $N \le Y$ in this class converge as $Y \to \infty$, and is $K_N - 2h$ bounded?

Status. The first order law is Theorem 10.6 of Chapter 10, proved from Serre's vertical equidistribution [508] and the Brandt matrix identification [438]. The second order term has not been studied as far as we know; hitting times matter for isogeny based hash functions [98].

What probability and computation contribute. By Theorem A.4 and Remark 10.7,

$$K_N = \sum_{j \ge 2} \frac{1}{1 - \lambda_j/3} = \frac{3}{2} \sum_{m \ge 0} 2^{-m} \operatorname{tr}\big(T_{2^m} \mid S_2(\Gamma_0(N))\big),$$

where $\lambda_2, \dots, \lambda_h$ are the eigenvalues of T_2 on $S_2(\Gamma_0(N))$. The Eichler Selberg trace formula writes $\operatorname{tr} T_{2^m}$ as $(N+1)/12$ for even m, plus terms independent of N, minus half the sum of the Hurwitz class numbers $H(2^{m+2} - t^2)$ weighted by $1 + \left(\frac{D}{N}\right)$, $D = t^2 - 2^{m+2}$. The main terms add up to $(N+1)/6 = 2h + \frac{1}{3}$, so that, formally, $K_N - 2h = c_0 + \sum_D w_D \left(\frac{D}{N}\right)$ with explicit constants c_0 and $w_D \le 0$, the sum running over negative fundamental discriminants $D \ne -3, -4$. On this progression $\left(\frac{D}{N}\right)$ is the product of the signs $\varepsilon_q = \left(\frac{q}{N}\right)$ over the primes $q \ne 3$ that divide D to an odd power, and by Dirichlet's theorem, for each finite set of primes q, the signs ε_q are asymptotically independent and fair as N varies. This suggests that $K_N - 2h$ converges in law to the random multiplicative series obtained by taking the ε_q independent. Size estimates suggest that the weights are square summable, so that this series converges in mean square, but not absolutely summable, so that $K_N - 2h$ may be unbounded, with large negative values at levels where $\varepsilon_q = 1$ for many small primes q. Over the 207 levels $1000 < N < 8000$ computed for this book, $K_N - 2h$ lies between -17.7 and 6.9 with mean -7.0; its mean is -11.0 when $\left(\frac{-7}{N}\right) = 1$ and -3.3 when $\left(\frac{-7}{N}\right) = -1$ (Chapter 10).

What would be needed. Control of the trace formula with weights 2^{-m} up to m of order $2\log_2 N$, beyond which the Ramanujan bound controls the tail;

cancellation in the class number sums for discriminants larger than N, where the symbols $\left(\frac{D}{N}\right)$ are periodic in D rather than independent; and joint equidistribution of the signs ε_q for q up to a power of N, a large sieve statement.

OPEN PROBLEM 30.11 (Bayesian identification of Sato Tate groups). (a) Extend Theorem 29.1 to the 52 Sato Tate groups of abelian surfaces, whose laws of the normalised coefficients (a_1, a_2) can have atoms and components on lower dimensional sets, and to motives of higher rank. (b) Make it effective: bound the number of primes after which the log Bayes factor favours the true group against every other candidate, polynomially in the logarithm of the conductor and without assuming the Riemann hypothesis.

Status. Theorem 29.1 gives the double limit $\lim_{M\to\infty}\lim_{n\to\infty} n^{-1}\log B_n^M = \mathrm{KL}(\mu_0\|P_0) - \mathrm{KL}(\mu_0\|P_1)$ for the log Bayes factor $\log B_n^M$ of P_1 against P_0 clipped at level M, when P_0, P_1 and the limit law μ_0 of the data have densities on a compact set with respect to Lebesgue measure plus counting measure on a fixed finite set D and the data are equidistributed for μ_0 relative to D, under the further conditions stated there; Chapter 29 applies it to elliptic curves and to three candidates in genus 2. The classification [192] matches curves to groups by moments, not by sequential tests; several of the 52 groups share the law of a_1, so the pair (a_1, a_2) is needed. For independent observations Wald's test [559] is optimal for simple hypotheses [563], with error exponents given by the divergences. For elliptic curves, effective forms of the Sato Tate theorem [466, 548] yield bounds of the kind asked in (b), polynomial in $\log N_E$ under the Riemann hypothesis for the symmetric power L functions and much weaker unconditionally.

What probability and computation contribute. With $\mu_0 = P_0 = \mu_G$ for the true group G and $P_1 = \mu_{G'}$ for a candidate G', the double limit is $-d_{GG'}$ with $d_{GG'} = \mathrm{KL}(\mu_G\|\mu_{G'})$, so a sequential test with threshold $\log(1/\beta)$ stops after about $\log(1/\beta)/d_{GG'}$ primes. When μ_G charges a set that $\mu_{G'}$ does not, the divergence is infinite, and the small atoms and clipping of Chapter 29 make it large and finite; the near ties, which set the error exponents, come from groups that differ only on components of small mass. The 52×52 divergence matrix is part of Computational Problem 29.2.

What would be needed. For (a), reference measures adapted to the lower dimensional components, so that the clipping argument still applies; equidistribution for the true group, which is open for $\mathrm{USp}(4)$; and, in higher rank, the classification of the groups. For (b), unconditional effective equidistribution with power rates, which would also bear on Open Problem 30.3.

OPEN PROBLEM 30.12 (The rank bias). Let $E/\mathbb{Q}$ have conductor N_E and analytic rank r_E, let $S_E(x) = \sum_{p \le x,\, p \nmid N_E} a_p \log p/p$, and assume the Riemann hypothesis for $L(s,E)$. Then $S_E(x) - (\frac{1}{2} - r_E)\log x$, with $\log x$ uniform on $[0,Y]$ and $Y \to \infty$, converges in law to a distribution μ_E with mean C_E; let Z_E have the law μ_E shifted to mean zero. (Question.) As E runs through the curves of conductor at most X and analytic rank $r \in \{0,1\}$, does the law of $Z_E/\log N_E$ converge as $X \to \infty$, and is the limit the one predicted by the low lying zeros of the corresponding orthogonal family?

Status. Under the Riemann hypothesis for $L(s,E)$ the explicit formula and the method of Rubinstein and Sarnak [470] give the existence of μ_E, and identify it as the law of $C_E - \sum_{\gamma>0} 2\sin(\theta_\gamma)/\gamma$, with independent uniform phases θ_γ, when the positive ordinates γ of the noncentral zeros are linearly independent over $\mathbb{Q}$; Sarnak's letter [480] explains such biases through the explicit formula. The link between the rank and partial Euler products goes back to Goldfeld [219] and Kuo and Murty [336]; see Mazur [386]. The family question is, as far as we know, open.

What probability and computation contribute. The explicit formula gives $S_E(x) - (\frac{1}{2} - r_E)\log x = C_E - \sum_{\gamma>0} 2\sin(\gamma \log x)/\gamma + o(1)$ in logarithmic mean square: the prime squares produce the $\frac{1}{2}$, the central zero of order r_E the $-r_E$, and the other zeros an almost periodic oscillation (Chapter 5). Under linear independence Z_E has variance $V_E = 2\sum_{\gamma>0} \gamma^{-2}$, the same form as $\operatorname{Var} W$ in Open Problem 30.8. For the four curves of Chapter 5, the central Taylor coefficients computed in Chapter 14 give $V_E = 0.255$, 0.379, 0.814 and 1.452 for 11a1, 37a1, 389a1 and 5077a1, increasing as the lowest noncentral zero descends from 6.36 to 2.05. The fitted slopes 0.46, -0.59, -1.43, -2.51 against 0.5, -0.5, -1.5, -2.5 (Chapter 5) are consistent with oscillations of standard deviation $\sqrt{V_E}$, between 0.5 and 1.2, over the range $6.9 \le \log x \le 16.1$. Since the lowest zeros of a curve of large conductor lie at heights of order $1/\log N_E$, V_E is typically of order $(\log N_E)^2$, which explains the normalisation. If the normalised ordinates $t_j = \gamma_j \log N_E/(2\pi)$ follow the Katz Sarnak limit, the even orthogonal one for $r = 0$ and the odd one with its central zero removed for $r = 1$, then $Z_E/\log N_E$ should converge in law to $-\pi^{-1}\sum_j \sin(\theta_j)/t_j$, with (t_j) the points of that limit and θ_j independent and uniform. For $r = 0$ this limit has infinite variance, because the even orthogonal ensemble does not repel the lowest zero from the central point; for $r = 1$ the repulsion makes the variance finite. The slope test at high rank is Computational Problem 5.3.

What would be needed. Each μ_E needs the Riemann hypothesis for $L(s,E)$. For the family law, V_E is dominated by the lowest zeros, so one needs the distribution

of the lowest zero itself, which the restricted support results for the one level density [271] do not determine (Open Problem 30.5).

30.6 Notes

The conjectures behind Open Problems 30.4 and 30.6 go back to Langlands' paper [350]; Arthur [24] and Gelbart [211] are standard overviews. Sarnak's density hypothesis and vertical Plancherel law [476, 482] are the model for the probabilistic reading of Open Problem 30.1. Lang and Trotter [353] derived their constant from a probabilistic model of Frobenius in the image of Galois, the model behind Open Problem 30.2. Mazur's essay [386] is the source of the error term perspective behind Open Problems 30.3 and 30.12; in both, the explicit formula makes the arithmetic sums smaller and more structured than independent ones, the theme of the analysis of Chebyshev's bias by Rubinstein and Sarnak [470] and of Sarnak's letter [480]. The Thorin and extreme value problems are specific to this book.

30.7 Open computational problems

The full programme is in Chapter 31; the two problems below belong to this chapter.

Computational Problem 30.1 (Discrepancy at scale). Objective: test the transition at $c = 1$ suggested in Open Problem 30.3(b), and the slow decay predicted for single curves. It extends Computational Problem 5.1, which runs E_1 and a family to a fixed prime bound, by tying the number of primes to the conductor, $n = \lfloor X^c \rfloor$. Scale: for E_1 and 10 further curves without complex multiplication, all primes below 10^{10}; for the family, 10^3 curves with conductor in $[X, 2X]$ from the LMFDB [371] for $X = 10^3$ and 10^4, with $n = \lfloor X^c \rfloor$ for $c = \frac{1}{2}, 1, \frac{3}{2}, 2$, so at most 10^8 primes per curve. Route: average polynomial time point counting [243, 244], or PARI/GP `ellap` in parallel. Deliverables: for each X and c, the variances across the family of the modes $n^{-1/2} \sum_{j \le n} U_m(x_{p_j}/2)$ for $m \le 20$, compared with $\min(1, m/c)$, and the law of $\sqrt{n}\, D_n$ against the Kolmogorov law with a Kolmogorov Smirnov statistic; for the single curves, the paths of $\sqrt{n}\, D_n$ and of the modes for $m \le 200$ in $\log n$. Why it matters: the 200 curves of Chapter 5 have discriminants below 10^7 in absolute value and use the primes below 10^6, so c is of order 1 for them, and their standard deviation 0.219 against 0.2603 is consistent with a mild deficit there; varying c separates the two regimes, and a deficit for $c > 1$ would indicate that the explicit formula, not independence, governs the fluctuations.

Computational Problem 30.2 (Second order Kemeny constants). Objective: test the random series model of Open Problem 30.10, which Computational Problem 10.1 approaches through a regression on the symbols for small discriminants,

whereas here the full predicted series is tested. Data: the values of $K_N - 2h$ for the 19564 primes $N \equiv 1 \bmod 12$ below 10^6 (h up to 83330) produced in Computational Problem 10.1. Route: compute the weights w_D for $|D| \le 10^6$ from the trace formula, as $-\frac{3}{4}$ times the sum of $2^{-m}H(2^{m+2} - t^2)$ over the pairs (m, t) for which $t^2 - 2^{m+2}$ is D times a square, with PARI/GP `qfbhclassno`; compare $K_N - 2h$ with $c_0 + \sum_D w_D \left(\frac{D}{N}\right)$; and simulate the random multiplicative series with independent signs ε_q. Deliverables: the residuals of the prediction, the empirical law of $K_N - 2h$ against the simulated law (quantile plot and Kolmogorov Smirnov statistic), and the levels with the most extreme values together with their signs ε_q for $q \le 100$. Why it matters: agreement would support reducing Open Problem 30.10 to the joint equidistribution of Legendre symbols, and the extreme levels test whether $K_N - 2h$ is bounded.

CHAPTER 31

A COMPUTATIONAL PROGRAMME

Chapters 1 to 30 of this book pose computational problems that we have deliberately left open. They are not exercises. Each one asks for a computation at a scale beyond what we carried out for the book, and each was chosen because its answer would sharpen a theorem, test a conjecture, or calibrate a heuristic that appears in the text. This chapter collects them into a single programme, organises them into workstreams, and sets out the standards we propose for carrying them out. The programme is the agenda of the second author's computational work over the coming years, and we hope that readers will take up parts of it.

31.1 PRINCIPLES

The computations in this book are experiments on deterministic objects. A sequence of Frobenius traces is not a sample from a distribution; it is a fixed arithmetic sequence that, by theorem or by conjecture, behaves in the limit like one. This has consequences for how results should be reported.

1. Certified where possible. Point counts, Hecke matrices, isogeny graphs and L function zeros can be computed exactly or with rigorous error bounds. A claim such as "every nontrivial Ihara pole lies on the circle" should come with the arithmetic that certifies it, not only a floating point check.

2. Statistical where necessary. When a conjecture predicts a limiting law or constant (a Brownian bridge, a Kolmogorov distribution, a Lang and Trotter constant), the test statistic, the null law and the reference population must be stated in advance. Cross sectional comparisons across many curves, as in Chapter 5, are often more informative than a single long sequence, because they turn one deterministic path into many approximately independent ones.

3. Scaled with care. Many statistics converge slowly. The scaled discrepancy $\sqrt{n}\, D_n$ and the rank bias are predicted to approach their limits at logarithmic rates (Chapter 5), and the Lang and Trotter counts of that chapter, which grow like $\sqrt{x}/\log x$ and number at most a few hundred below 10^7, have a relative Poisson error of order $x^{-1/4}$ up to logarithmic factors. In both cases a factor of ten in the range of primes buys little. The programme favours

breadth (more curves, more levels, more forms) over depth whenever the conjectured limit is a law across families.

4. Reproducible. Every table should be regenerable from a script and a seed. Appendix B prints the core routines and lists the scripts and stored results behind the numbers of this book, together with an archive of the scripts in a layout with relative paths, but several of them still read files from the wrong folder, so making every table regenerable from a single command comes before scaling the computations up.

31.2 WORKSTREAMS

We group the problems into six workstreams. Problem N.M denotes Computational Problem N.M, the Mth computational problem of Chapter N (Problem 5.2 is the second computational problem of Chapter 5).

31.2.1 Workstream I: Frobenius data at scale

This workstream builds the basic data: traces of Frobenius for elliptic curves, genus two Jacobians and modular forms, at scales well beyond those of this book. The engine is a point counting pipeline (Problem 5.4), using baby step giant step and the average polynomial time methods of Harvey [243] and of Harvey and Sutherland [244] for elliptic and hyperelliptic curves, and GPU kernels for the embarrassingly parallel part. On this foundation sit the Sato Tate atlas (Problem 1.1), the Weyl averages and fingerprints (Problems 4.1 and 4.2), joint laws of pairs and twists (Problems 3.3 and 4.3), the Brownian bridge test (Problems 5.1 and 30.1), the Lang and Trotter constants (Problem 5.2), Koblitz's conjecture on the primes at which the number of points is prime, whose local densities are the Eisenstein shapes of Chapter 22 (Problem 22.2), the rank bias at high rank (Problem 5.3), the Chebotarev races (Problem 3.1), Gamma process tests of Chebotarev laws, with their Euler Kronecker drifts, and of joint Sato Tate laws (Problem 20.3), Gamma shape estimates for symmetric powers and GL_3 motives (Problem 20.4, a Dirichlet density counterpart of Problems 4.1 and 4.2), the three colourings of the prime factors of uniform random integers by Frobenius class, with the proportions of integers that avoid a class (Problem 24.1), the Mertens constants of the fields of $x^n - x - 1$, whose group is S_n (Problem 24.2), and tables of $\tau(p)$ (Problem 2.1). The impostor detection problem (Problem 1.2) is the statistical counterpart of Proposition 1.5.

31.2.2 Workstream II: families, symmetry and random matrices

The second workstream concerns families of L functions. Its problems are the one level density for quadratic twists of 11a (Problem 6.1), moments of quadratic

L functions against the Keating and Snaith predictions (Problem 6.2), nearest neighbour spacings for the zeros of $L(s, \Delta)$ at heights from 10^8 on (Problem 6.3, which continues Problem 12.3), Hecke eigenvalue tables for newforms of level up to 10^4 and even weight up to 12 (Problem 2.2), near coincidences of newforms (Problem 1.3), the congruences between newforms of level M and newforms of level dividing Mq, new at q, predicted by Ribet's level raising condition, read as vanishing discrepancy Thorin masses (Problem 22.3), and the use of the functional equation defect as a test statistic (Problem 2.3), extended to the twisted Pólya kernels of Weil's converse theorem and to Artin representations (Problem 20.2). The computational core is tables of Hecke eigenvalues and fast evaluation of L functions and their zeros across large families, for which the LMFDB [371] and the rigorous methods of Booker and Platt are the natural references.

31.2.3 Workstream III: buildings, graphs and hitting times

The third workstream extends Chapters 9 and 10 and the first section of Chapter 21. It asks for the rate of vertical convergence of Hecke spectra to the Plancherel measures (Problem 9.1), quotients of a regular tree by congruence and by random noncongruence subgroups of a Lubotzky Phillips Sarnak lattice (Problem 9.2), Plancherel statistics for GL_3 through Hecke operators on quotients of the building of $\mathrm{PGL}_3(\mathbb{Q}_2)$ by lattices from definite unitary groups in three variables (Problem 9.3), the second order term of the Kemeny constant (Problems 10.1 and 30.2), the laws of the critical groups of isogeny graphs at fixed degree against the Haar model of Munier and Shnidman and the Cohen Lenstra law (Problem 21.1), hitting times between supersingular curves, which govern the cost of naive path search in isogeny cryptography, for N up to 10^7 (Problem 10.2), other degrees and ordinary isogeny volcanoes (Problem 10.3), and Ihara zeta functions of random lifts (Problem 10.4). Here exact linear algebra on sparse graphs (isogeny graphs with $N \leq 10^5$, so up to about 8300 vertices, and building quotients with up to 10^5 vertices), together with Monte Carlo walks on graphs with up to 8×10^5 vertices, is the main tool.

31.2.4 Workstream IV: L functions as laws

The fourth workstream concerns the probabilistic representations of Part IV. It includes the value distribution of $\log L(\sigma + it, \Delta)$ against its two random Euler product models (Problem 11.1), large deviation rate functions (Problem 11.2), universality (Problem 11.3), the Thorin measure of Δ to height 10^5 (Problem 13.1), the de Bruijn Newman constant of Δ (Problem 13.2), complete monotonicity across weights and levels (Problem 13.3), the degenerate Thorin atom for curves of rank two (Problem 13.4), tables of the extreme value laws M_f for all multiplicative eta products (Problem 27.1), variance reduction and zero location by simulation

(Problem 27.2), and the extension to eigenforms that are not eta products (Problem 27.3). The Thorin framework of Chapters 14 and 15 adds its own problems: certified Thorin Hankel determinants computed from central derivatives (Problem 14.1), the Herglotz test (Problem 14.2), collision times of the Thorin atoms of Δ (Problem 14.3), a regularised Thorin computation for a Selberg zeta function (Problem 14.4), comb statistics against the zeros of $L(s, \Delta)$ (Problem 15.1), large deviations of passage times on finite isogeny graphs against the tree formula (Problem 15.2), local parabolic tests of the Ramanujan conjecture for Maass forms (Problem 15.3), and Thorin combs of curves of genus three and four (Problem 15.4). Chapter 16 adds four more: a certified Patie test with Bernstein gamma interpolation (Problem 16.1), transported Thorin densities at basepoints inside the critical strip (Problem 16.2), first passage tables and large deviations for Euler subordinators (Problem 16.3), and van Dantzig and Pólya frequency tests for the partner variables of L functions (Problem 16.4). Chapter 17 asks for the squarefree error beyond 10^{18} (Problem 17.1), the oscillations of the counts of squarefree ideals of $\mathbb{Q}(i)$ and $\mathbb{Q}(\sqrt{-3})$ (Problem 17.2), and the signs of the Pólya kernels of imaginary and of real quadratic fields (Problems 17.3 and 17.4). Chapter 18 asks whether, for the real characters of conductor up to 10^3, $\operatorname{Re} \Lambda'/\Lambda$ is smallest on the real axis along the lines to the right of the centre (Problem 18.1), carries its computations to the Hecke characters of the Gaussian field (Problem 18.2), asks for the response of its criteria to synthetic off line zeros (Problem 18.3), and adds $L(s, \chi_{-4})$ to Problem 16.2. Chapter 19 asks for the signs of Pólya kernels of degree up to four (Problem 19.1), zero counts by the argument principle in ball arithmetic (Problem 19.2) and the window leakage of signed Thorin densities (Problem 19.3). Chapter 20 adds the Galois heat trace of an icosahedral Artin representation with trivial determinant, hence self dual and even (Problem 20.1), where a certified violation of complete monotonicity would disprove Artin's conjecture or the generalized Riemann hypothesis for it, although finitely many such inequalities are not known to confirm them, even to a finite height; it needs the complete monotonicity pipeline of Problem 13.3 and the ball arithmetic of Problem 14.1. Chapter 21 adds the explosion cumulants of the same representation, the integrals of its heat trace against 1 and x, computed from its Frobenius classes, conductor and gamma factor (Problem 21.2): on the range of x that the primes control they are compared with the zeros, and in full with the power sums $\sum \gamma^{-2}$ and $\sum \gamma^{-4}$ that the route of Problem 14.1 computes from central derivatives. It also asks which real odd characters of conductor below 10^6 have a positive signed exit density, which excludes a real zero of their L function (Problem 21.3). Chapter 22 asks for the integrals of the heat trace against 1, x and x^2 for one elliptic curve from each isogeny class of conductor below 10^3, of either root number, with the constant $\frac{r}{2}$ of a central zero of order r removed; under GRH for $L(s, E)$, and when $r = 0$, they are the first three cumulants of an explosion time (Problem 22.1).

Chapter 23 asks for its inversions on elliptic curves of both root numbers, the Frobenius traces from Bohr means of the transported density, the Pólya kernel from Bromwich inversion and the Cauchy superpositions over the zeros (Problem 23.1); for the zeros recovered from the derivatives of the heat trace by the Post Widder formula, at the orders that the tests of Problem 20.1 compute (Problem 23.2); and for the laws of the Thorin variables of ξ, Δ and $L(s, \chi_{-4})$ at basepoints in and to the right of the critical strip, by Talbot inversion (Problem 23.3). Chapter 24 asks for the laws of the class shares of the zeta random integer near $\sigma = 1$, against the Beta limits of the Dirichlet process (Problem 24.3). These problems require high precision arithmetic (central derivatives to hundreds or thousands of digits in Problems 14.1 and 16.1) and, for Lévy densities near the origin, very many zeros; they are where the probabilistic reformulations of the Riemann hypothesis for L functions of low degree can be stress tested.

31.2.5 Workstream V: inference

The fifth workstream collects the problems of inference. Chapter 28 contributes Duke equidistribution rates for Heegner points (Problem 28.1), reference priors on conjugacy classes under anisotropic noise (Problem 28.2), the accuracy of p^* and the coverage of right Haar intervals in location scale and von Mises models (Problem 28.3), and CM points and Hecke orbits on Shimura curves (Problem 28.4).

The workstream also turns Chapter 29 into a working classifier. It asks for the Sato Tate group of $y^2 = x^6 + x^3 + 1$ (Problem 29.1), a likelihood library for all 52 groups of abelian surfaces with a sequential classifier and the matrix of error exponents (Problem 29.2), empirical Sato Tate groups in genus two and three (Problem 3.2), and an empirical Bayes prior over groups in genus three (Problem 29.3).

31.2.6 Workstream VI: exponential sums, random groups and log correlated fields

Three further chapters, two in Part II and one in Part IV, extend the programme. Chapter 7 asks for horizontal statistics of Kloosterman angles (Problem 7.1), the law of the maximum of Kloosterman paths against the random Fourier series of Kowalski and Sawin (Problem 7.2), sums of Kloosterman sums as a probe of the spectral gap (Problem 7.3), and Birch paths (Problem 7.4). Chapter 8 asks for Cohen Lenstra statistics for all $|D| \leq 10^9$ and in windows near 10^{10} and 10^{11}, with the secondary term separated (Problem 8.1), Selmer groups and ranks by height (Problem 8.2), Shafarevich Tate groups against the alternating cokernel model (Problem 8.3), and class groups of cubic and quartic fields (Problem 8.4). Chapter 12 asks for Selberg's central limit theorem and the hybrid constant for $L(s, \Delta)$ at heights up to 10^8 (Problem 12.1), the law of the maxima predicted by

Fyodorov, Hiary and Keating (Problem 12.2), pair correlation and spacings of the zeros of $L(s, \Delta)$ at larger height (Problem 12.3), and Harper's constant for random multiplicative functions (Problem 12.4).

31.3 The problems at a glance

Table 31.1 lists all problems with the scale we propose and a priority. Priority A marks problems whose answer would change a statement in this book; B marks problems that would extend the data substantially; C marks longer term or exploratory problems.

Table 31.1: Open computational problems.

No.	Problem	Scale	Pr.
1.1	Sato Tate atlas for elliptic curves	conductor $\leq 10^5$	B
1.2	Detecting impostors	$p < 10^8$	C
1.3	Near coincidences of newforms	level $\leq 10^4$	C
2.1	Ramanujan tau at scale	$p < 10^{11}$	B
2.2	Hecke eigenvalue tables	level $\leq 10^4$	B
2.3	Functional equations as a test statistic	level $\leq 10^3$	C
3.1	Chebotarev statistics and bias races	$p \leq 10^{10}$	B
3.2	Empirical Sato Tate groups in genus 2 and 3	LMFDB, $p \leq 2^{30}$	A
3.3	Joint Sato Tate independence	conductor $< 10^4$, $p < 10^9$	B
4.1	Weyl averages at scale	$p \leq 10^{10}$	B
4.2	Fingerprints of genus two Jacobians	all 52 groups	A
4.3	Joint laws of pairs and twists	10^3 pairs, $p \leq 10^9$	B
5.1	Brownian bridge test at scale	$p \leq 10^{10}$; 10^3 curves, $p \leq 10^9$	A
5.2	Lang Trotter constants	$p \leq 10^{12}$	A
5.3	Bias at high rank	rank ≥ 4, $p < 10^9$	A
5.4	A GPU point counting pipeline	$p \leq 10^{10}$	A
6.1	One level density, quadratic twists of 11a	conductor $11d^2 \leq 10^8$	B
6.2	Keating Snaith versus moments of quadratic L functions	$\lvert d \rvert \leq 10^9$	B
6.3	Spacings of zeros of $L(s, \Delta)$	heights 10^8 to 10^{20}	C
7.1	Horizontal statistics of Kloosterman angles	$p \leq 10^8$	B
7.2	Law of the maximum of Kloosterman paths	$p \leq 10^6$	B

No.	Problem	Scale	Pr.
7.3	Sums of Kloosterman sums and the spectral gap	$x \leq 10^7$	A
7.4	Birch paths	$p \leq 10^6$	C
8.1	Cohen Lenstra at scale with the secondary term	all $\lvert D\rvert \leq 10^9$, windows to 10^{11}	A
8.2	Selmer groups and ranks by height	$H \leq 10^{10}$	A
8.3	Shafarevich Tate groups against BKLPR	conductor $\leq 5 \cdot 10^5$	B
8.4	Class groups of cubic and quartic fields	cubic $\lvert D\rvert \leq 10^9$, quartic $\lvert D\rvert \leq 10^8$	C
9.1	Rate of vertical convergence	levels $\leq 10^5$	A
9.2	Noncongruence quotients	index $\leq 10^3$	C
9.3	Plancherel statistics for GL_3	10^5 vertices	C
10.1	Second order term of the Kemeny constant	$N \leq 10^6$	A
10.2	Hitting times on large supersingular graphs	$N \leq 10^7$	B
10.3	Other degrees and ordinary curves	$\ell \leq 7$	B
10.4	Ihara zeta for random lifts	degree $k \leq 100$, 10^4 lifts	C
11.1	Value distribution of $\log L(\sigma + it, \Delta)$	$t \leq 10^6$	B
11.2	Large deviation rate functions	$t \leq 10^7$	C
11.3	Testing universality	$t \leq 10^7$	C
12.1	Selberg's theorem and the hybrid constant at height 10^8	$T \leq 10^8$	A
12.2	The law of the maximum	heights 10^{10} to 10^{22}	A
12.3	Pair correlation and spacings for Δ to height 10^7	height 10^5; blocks at 10^6, 10^7	B
12.4	Harper's constant and critical chaos	$x \leq 10^{10}$	B
13.1	Thorin measure of Δ to height 10^5	2.8×10^5 zeros	A
13.2	de Bruijn Newman constant of Δ	$\lambda \leq \frac{1}{8}$, height 10^4	B
13.3	Monotonicity across weights and levels	weight $\leq$ 24, level ≤ 500	B
13.4	Rank two curves and the degenerate atom	50 curves, height 10^3	B
14.1	Certified Thorin Hankel determinants	LMFDB, size 20	A
14.2	The Herglotz test in the upper half plane	conductor $< 10^4$	B
14.3	Collision times of the Thorin atoms of Δ	10^5 zeros	B
14.4	Regularised Thorin computation for a Selberg zeta	10^4 eigenvalues	C

No.	Problem	Scale	Pr.
15.1	Comb statistics against the zeros of Δ	height 10^5	A
15.2	Large deviations on finite isogeny graphs	$N \le 10^5$	B
15.3	Kim Sarnak versus Ramanujan through local parabolas	LMFDB Maass forms	B
15.4	Thorin combs of curves of genus three and four	$q \le 13$	C
16.1	Certified Patie test and Bernstein gamma interpolation	order 100	A
16.2	Transported Thorin densities in the strip	$\frac{1}{2} < \alpha \le 1$	A
16.3	First passage tables and large deviations for Euler subordinators	$x \le 50$, $m \le 10^8$	B
16.4	Van Dantzig and Pólya frequency tests across LMFDB	conductor $< 10^4$	B
17.1	The squarefree error beyond 10^{18}	$10^{18} < x \le 10^{19}$	B
17.2	Oscillations of squarefree ideals	$\mathbb{Q}(i)$, $\mathbb{Q}(\sqrt{-3})$, norm $\le 10^{14}$	C
17.3	Pólya kernels of imaginary quadratic fields	$\lvert d\rvert \le 10^7$	B
17.4	Pólya kernels of real quadratic fields	$d \le 10^5$	C
18.1	Minimum on the real axis for real characters	conductor $\le 10^3$	A
18.2	Hecke characters of the Gaussian field	$k \le 5$, height 10^3	B
18.3	Synthetic off line zeros	χ_{-4}, height 10^3	C
19.1	Signs of Pólya kernels across degrees	degree ≤ 4, conductor $\le 10^3$	B
19.2	Zero counts by the argument principle in ball arithmetic	$T \le 200$	A
19.3	Window leakage of signed Thorin densities	$T \le 35$	C
20.1	Even icosahedral representations	one representation, $j \le 10^3$, $x \le 4$, $p \le 10^{10}$	C
20.2	Kernel symmetry at level N	conductor $\le 10^4$	C
20.3	Gamma process tests at scale	degree ≤ 10, $\lvert d_K\rvert < 10^6$; conductor $\le 10^4$	B
20.4	Gamma shapes for ℓ adic representations	$p \le X = 10^9$, $\sigma - 1 \ge 5/\log X$	C

No.	Problem	Scale	Pr.
21.1	Two prime tables	primes $N \equiv 1 \bmod 12$ below 10^5, $\ell \leq 13$	B
21.2	Explosion cumulants from Frobenius	the representation of Problem 20.1	C
21.3	Signed exit laws and real zeros	real odd χ, conductor $< 10^6$	C
22.1	Explosion cumulants of elliptic curves	conductor $< 10^3$, both root numbers, $p^m \leq 10^8$	C
22.2	Koblitz's conjecture and Eisenstein shapes	20 curves, $\ell \leq 10^{10}$	B
22.3	Level raising congruences	level ≤ 200, $3 \leq p \leq 7$, $q < 50$	C
23.1	Four faces of elliptic curves	conductor < 100, $n \leq 100$	B
23.2	Zeros from the heat trace by Post Widder	order $\leq 10^3$, first 50 zeros	C
23.3	Thorin laws in the strip by Talbot inversion	$\frac{1}{2} < \alpha \leq 2$, $t \leq 10^4$	C
24.1	Colourings at scale	S_3, S_4, C_3; $x \leq 10^{15}$	B
24.2	Mertens constants of symmetric fields	$x^n - x - 1$, $3 \leq n \leq 7$; $p \leq 10^{10}$	C
24.3	Laws of class shares near $\sigma = 1$	S_3, S_4; $\sigma - 1 \geq 10^{-5}$	B
25.1	Class number one fields	nine fields, $p < 10^9$	B
26.1	Eigenvalues and geodesics at scale	10^4 eigenvalues, $N(P) \leq 10^9$	B
27.1	Tables of the laws M_f	30 eta products	A
27.2	Variance reduction and zero location	$t \leq 30$, 10^9 draws	B
27.3	Eigenforms that are not eta products	weight 2, level ≤ 500	C
28.1	Duke rates at large discriminant	all $\lvert D\rvert \leq 10^9$, samples to 10^{12}	B
28.2	Reference priors for anisotropic noise	SU(2), USp(4)	B
28.3	p^* and right Haar intervals for angular models	$5 \leq n \leq 200$	C
28.4	CM points and Hecke orbits on Shimura curves	$\lvert D\rvert \leq 10^8$, $p^k \leq 10^7$	C
29.1	The group of $y^2 = x^6 + x^3 + 1$	$p < 2^{30}$	A
29.2	Likelihood library, classifier and error exponent matrix	52 groups	A
29.3	Genus three with an empirical Bayes prior	410 groups, $p \leq 2^{26}$	C

No.	Problem	Scale	Pr.
30.1	Discrepancy at scale	$p < 10^{10}$; 10^3 curves, 10^8 primes	A
30.2	Second order Kemeny constants	$N \leq 10^6$	A

31.4 A FIRST YEAR PLAN

If we had to choose, we would begin with the following sequence, in which later steps reuse the infrastructure of earlier ones.

1. The point counting pipeline (Problem 5.4), validated against the data of Chapter 5, and then used for the Brownian bridge test across a thousand curves (Problems 5.1 and 30.1). This tests, as far as numerics can, whether the fluctuations of Frobenius traces are those of independent draws.
2. The Lang and Trotter constants for $y^2 = x^3 + x + 1$ and two further Serre curves (Problem 5.2), compared with the formula from the Galois image. This is a sharp test of a conjecture that probability motivates but cannot yet prove.
3. The second order term of the Kemeny constant (Problems 10.1 and 30.2), computed from tables of class numbers through the Eichler Selberg trace formula, checked against exact spectra of supersingular isogeny graphs, and compared with a random multiplicative series model.
4. The Thorin measure of Δ to height 10^5 (Problem 13.1) and the tables of M_f (Problem 27.1), the two computations that put the new representations of Part IV on a firm numerical footing.
5. Certified Thorin Hankel determinants for every L function in the LMFDB of small conductor (Problem 14.1), computed from central derivatives in interval arithmetic; this is the cheapest systematic test of the Thorin form of the Riemann hypothesis, and it needs no zeros.
6. The 52 group likelihood library (Problem 29.2), beginning with the curve $y^2 = x^6 + x^3 + 1$ (Problem 29.1), whose atom at zero already shows that the library must include disconnected groups.

31.5 SOFTWARE AND DATA

Most computations in this book used PARI/GP through its Python interface, with NumPy, SciPy and mpmath for numerics and linear algebra. For the programme we expect to add Sage and Magma for modular forms and isogeny graphs, Sutherland’s smalljac and average polynomial time code for genus two and three, the

LMFDB as a source of curves and forms with precomputed invariants, and GPU code for point counting and Monte Carlo. Section B.7 lists the surviving scripts and result files.

31.6 Notes

The idea that computation is the experimental arm of number theory goes back at least to Gauss's tables of primes, and in the modern era to Birch and Swinnerton Dyer's computations on the EDSAC 2 computer in Cambridge that led to their conjecture. The LMFDB is the most systematic expression of this tradition for the Langlands programme. Our emphasis on statistical standards follows the view, implicit throughout this book, that once a conjecture predicts a law, its numerical test is a problem in statistics, and the usual disciplines of statistics (prespecified tests, reference populations, honest reporting of uncertainty) apply.

CHAPTER A

A PROBABILITY TOOLKIT

This appendix collects the probabilistic facts used in the book, with proofs where they are short and references where they are standard. Readers trained in number theory will find here what they need from probability; readers trained in probability may use it as a list of conventions. Throughout, a law means a Borel probability measure, $\mu_n \Rightarrow \mu$ denotes weak convergence of laws, and $X_n \Rightarrow X$ the same for random variables with those laws.

A.1 WEAK CONVERGENCE AND THE METHOD OF MOMENTS

A sequence of laws μ_n on a metric space S converges weakly to μ if $\int f\,d\mu_n \to \int f\,d\mu$ for every bounded continuous f. By the portmanteau theorem this is equivalent to $\mu_n(A) \to \mu(A)$ for every Borel set A whose boundary is μ null, and on $\mathbb{R}$ to convergence of distribution functions at continuity points of the limit [60]. A law μ on $\mathbb{R}$ is determined by its characteristic function $\varphi_\mu(t) = \int e^{itx}\,\mu(dx)$, and by Lévy's continuity theorem $\mu_n \Rightarrow \mu$ if and only if $\varphi_{\mu_n}(t) \to \varphi_\mu(t)$ for every $t \in \mathbb{R}$; more generally, if φ_{μ_n} converges pointwise to a function that is continuous at 0, that function is the characteristic function of a law μ and $\mu_n \Rightarrow \mu$ [62].

On a compact interval the method of moments is clean, because polynomials are dense in the continuous functions.

THEOREM A.1 (Moments determine compactly supported laws). *Let μ, ν be laws on a compact interval $I = [a, b]$ with $\int x^k\,d\mu = \int x^k\,d\nu$ for all $k \geq 0$. Then $\mu = \nu$. Moreover, if μ_n are laws on I and $\int x^k\,d\mu_n \to \int x^k\,d\mu$ for every k, then $\mu_n \Rightarrow \mu$.*

PROOF. Let $f \in C(I)$ and $\epsilon > 0$. By the Weierstrass approximation theorem there is a polynomial P with $\sup_I |f - P| < \epsilon$. Then

$$\left|\int f\,d\mu - \int f\,d\nu\right| \leq \left|\int (f-P)\,d\mu\right| + \left|\int P\,d\mu - \int P\,d\nu\right| + \left|\int (P-f)\,d\nu\right| < 2\epsilon,$$

because the middle term vanishes by linearity. Hence $\int f\,d\mu = \int f\,d\nu$ for all $f \in C(I)$. Continuous functions determine a finite Borel measure on a compact metric space: for a closed set F the functions $f_j(x) = \max(0, 1 - j\,d(x, F))$ decrease to the indicator of F, so by dominated convergence $\mu(F) = \nu(F)$, and closed sets form a

π system generating the Borel sets, so $\mu = \nu$ by Dynkin's theorem. For the second statement, the same three term estimate gives $\limsup_n |\int f\,d\mu_n - \int f\,d\mu| \le 2\epsilon$ for every $\epsilon > 0$. □

The theorem applies to the normalised traces $x_p \in [-2,2]$: if the empirical moments converge to those of μ_{ST} (the Catalan numbers in even degree, zero in odd degree) then the empirical laws converge weakly to μ_{ST}. On noncompact spaces moments need not determine laws. Stieltjes [528] showed this for the lognormal law: for $|a| \le 1$ the densities $(2\pi)^{-1/2}x^{-1}e^{-(\log x)^2/2}\big(1 + a\sin(2\pi\log x)\big)$ on $(0,\infty)$ all have the moments $e^{k^2/2}$ of the lognormal law. Carleman's condition $\sum_{k\ge1} m_{2k}^{-1/(2k)} = \infty$ on the even moments m_{2k} is a standard sufficient criterion for uniqueness [10], and if μ is determined by its moments and $\int x^k\,d\mu_n \to \int x^k\,d\mu$ for every k, then $\mu_n \Rightarrow \mu$ [62]. Since $\{x^k\}_{k\le m}$ and $\{U_k(x/2)\}_{k\le m}$ span the same space of polynomials, the theorem may equally be stated with Chebyshev moments, which is the form used for $\mathrm{SU}(2)$.

A.2 Equidistribution, Haar measure and Peter Weyl

A sequence (θ_n) in $\mathbb{R}/\mathbb{Z}$ is equidistributed if its empirical laws $N^{-1}\sum_{n\le N}\delta_{\theta_n}$ converge weakly to Lebesgue measure.

Theorem A.2 (Weyl [576]). *(θ_n) is equidistributed in $\mathbb{R}/\mathbb{Z}$ if and only if, for every integer $k \ne 0$, $N^{-1}\sum_{n\le N} e^{2\pi i k\theta_n} \to 0$ as $N \to \infty$.*

The criterion is the case $\mathrm{U}(1) \cong \mathbb{R}/\mathbb{Z}$ of Theorem 4.3, in the form for sequences indexed by the integers noted after its proof. Every compact Hausdorff group G carries a unique probability measure μ_G invariant under left translation, Haar measure; it is also invariant under right translation and inversion. For $\mathrm{SU}(2)$ the trace $x = 2\cos\theta$ of a Haar element has the semicircle law $\mu_{\mathrm{ST}}(dx) = \frac{1}{2\pi}\sqrt{4-x^2}\,dx$ on $[-2,2]$, by the Weyl integration formula, which writes the Haar integral of a class function with value $F(\theta)$ on the conjugacy class of eigenvalues $e^{\pm i\theta}$ as $\frac{2}{\pi}\int_0^\pi F(\theta)\sin^2\theta\,d\theta$.

Theorem A.3 (Peter and Weyl [436]). *The irreducible unitary representations ρ of a compact group G are finite dimensional, the characters χ_ρ of the pairwise inequivalent ones form an orthonormal basis of the class functions in $L^2(G,\mu_G)$, and their finite linear combinations are uniformly dense in the continuous class functions.*

Hence a sequence (g_n) in G has equidistributed conjugacy classes, meaning $N^{-1}\sum_{n\le N} f(g_n) \to \int f\,d\mu_G$ for every continuous class function f, if and only if $N^{-1}\sum_{n\le N}\chi_\rho(g_n) \to 0$ for every nontrivial irreducible ρ. For $\mathrm{SU}(2)$ the irreducible representations are the Sym^n, $n \ge 0$, of dimension $n+1$, with characters $U_n(x/2)$, where $U_n(\cos\theta) = \sin((n+1)\theta)/\sin\theta$ is the Chebyshev polynomial of the second

kind.

Equidistribution is measured by the Kolmogorov discrepancy. For a continuous distribution function F and the empirical distribution function F_n of $x_1, \dots, x_n$, put $D_n = \sup_t |F_n(t) - F(t)|$. If the x_i are independent with distribution function F, then $\sqrt{n}\, D_n \Rightarrow \mathcal{K} = \sup_{0\le u\le 1} |B(u)|$ for a Brownian bridge B, whichever continuous F is used (Kolmogorov [328]). The Kolmogorov distribution has $\mathbb{P}(\mathcal{K} \le x) = \sum_{k\in\mathbb{Z}} (-1)^k e^{-2k^2x^2}$ for $x > 0$, mean $\sqrt{\pi/2}\,\log 2 = 0.8687$, second moment $\pi^2/12$ and standard deviation 0.2603.

A.3 Tauberian theorems and infinitely divisible laws

The Wiener Ikehara theorem [580, 579]: let μ be a nonnegative measure on $[0,\infty)$ such that $F(s) = \int e^{-st}\,\mu(dt)$ converges for $\Re s > 1$. If $F(s) - c/(s-1)$ extends continuously to the closed half plane $\Re s \ge 1$ for some constant c, then $e^{-t}\mu([0,t]) \to c$ as $t \to \infty$. For a Dirichlet series $F(s) = \sum_n b_n n^{-s}$ with $b_n \ge 0$ (take $\mu = \sum_n b_n \delta_{\log n}$) the conclusion is $x^{-1}\sum_{n\le x} b_n \to c$, the form used in the proof of Theorem 4.8. The Hardy, Littlewood and Karamata theorem [292, 63] states that for a nonnegative measure μ on $[0,\infty)$ with Laplace transform $\hat\mu(\lambda) = \int e^{-\lambda x}\,\mu(dx)$ finite for $\lambda > 0$, a number $\rho \ge 0$ and a slowly varying function ℓ, $\hat\mu(\lambda) \sim \lambda^{-\rho}\ell(1/\lambda)$ as $\lambda \downarrow 0$ if and only if $\mu([0,x]) \sim x^\rho \ell(x)/\Gamma(1+\rho)$ as $x \to \infty$; the same holds with $\lambda \to \infty$ and $x \to 0$.

A law on $\mathbb{R}$ is infinitely divisible if for every $m \ge 1$ it is the m fold convolution of a law. Such laws are described by the Lévy Khintchine theorem [485] (Theorem 11.1), and for an Euler product F whose logarithm has nonnegative coefficients, $t \mapsto F(\sigma+it)/F(\sigma)$ is a compound Poisson characteristic function at every real σ where the series for $\log F$ converges absolutely, as in Theorems 11.2 and 11.4.

A.4 Random series

Kolmogorov's two series theorem: if Y_k are independent real random variables, $\sum_k \mathbb{E}Y_k$ converges and $\sum_k \operatorname{Var} Y_k < \infty$, then $\sum_k Y_k$ converges almost surely and in L^2. The three series theorem gives the exact criterion: for independent Y_k, $\sum_k Y_k$ converges almost surely if and only if for some (equivalently every) $c > 0$ the three series $\sum_k \mathbb{P}(|Y_k| > c)$, $\sum_k \mathbb{E}[Y_k 1_{\{|Y_k|\le c\}}]$ and $\sum_k \operatorname{Var}(Y_k 1_{\{|Y_k|\le c\}})$ converge [62]. For $X = \sum_k a_k \sin\theta_k$ with real a_k and independent phases θ_k uniform on $[0,2\pi)$, the terms have mean 0 and variance $a_k^2/2$, so the series converges almost surely and in L^2 when $\sum_k a_k^2 < \infty$, with $\operatorname{Var} X = \frac12 \sum_k a_k^2$ and characteristic function $\prod_k J_0(a_k t)$. Kronecker's theorem links deterministic and random: if $\gamma_1, \dots, \gamma_K$ are linearly independent over $\mathbb{Q}$, then $(\gamma_k y)_{k\le K}$ is equidistributed in $(\mathbb{R}/2\pi\mathbb{Z})^K$ as y ranges over $[0, Y]$, $Y \to \infty$, which means that the image of normalised Lebesgue measure

on $[0, Y]$ under $y \mapsto (\gamma_k y \bmod 2\pi)_{k \le K}$ converges weakly to Haar measure. This is the mechanism behind Bohr and Jessen [70], Rubinstein and Sarnak [470] and the rank bias of Section 5.4 and Open Problem 30.12.

A sequence of real random variables Y_n satisfies a large deviation principle with good rate function I if $I : \mathbb{R} \to [0, \infty]$ is lower semicontinuous with compact sublevel sets, $\limsup_n n^{-1} \log \mathbb{P}(Y_n \in C) \le -\inf_C I$ for every closed C, and $\liminf_n n^{-1} \log \mathbb{P}(Y_n \in O) \ge -\inf_O I$ for every open O. By the Gärtner Ellis theorem [143, Theorem 2.3.6], if $\Lambda(\theta) = \lim_n n^{-1} \log \mathbb{E} e^{n\theta Y_n}$ exists in $(-\infty, \infty]$ for every real θ, if 0 is an interior point of $\{\Lambda < \infty\}$, and if Λ is lower semicontinuous, differentiable on the interior of its domain and steep, meaning $|\Lambda'(\theta)| \to \infty$ as θ approaches a finite endpoint of the domain from inside, then Y_n satisfies the large deviation principle with the good rate function $\Lambda^*(y) = \sup_\theta (\theta y - \Lambda(\theta))$.

A.5 Generalized gamma convolutions and Pólya frequency functions

By Bernstein's theorem [49, 578] a function f on $(0, \infty)$ is completely monotone, $(-1)^n f^{(n)} \ge 0$ for all $n \ge 0$, if and only if it is the Laplace transform $f(x) = \int_{[0,\infty)} e^{-xt}\, \tau(dt)$ of a nonnegative measure τ, which is then unique. A law on $[0, \infty)$ is a generalized gamma convolution (GGC) if it is a weak limit of finite convolutions of gamma laws [546, 73]; equivalently

$$\mathbb{E} e^{-sX} = \exp\Big(-as - \int_{(0,\infty)} \log\Big(1 + \frac{s}{t}\Big)\, \tau(dt)\Big), \qquad s \ge 0,$$

with $a \ge 0$ and a positive measure τ on $(0, \infty)$, the Thorin measure, such that $\int \log(1 + 1/t)\, \tau(dt) < \infty$ (Definition 13.2). By Frullani's integral a GGC is infinitely divisible with drift a and Lévy density $x^{-1} \int e^{-tx}\, \tau(dt)$, and conversely every infinitely divisible law on $[0, \infty)$ of this form is a GGC; the sum $\sum_k E_k / c_k$ of independent standard exponentials, with $c_k > 0$ and $\sum_k c_k^{-1} < \infty$, has Thorin measure $\sum_k \delta_{c_k}$; and a GGC Laplace transform extends holomorphically to $\mathbb{C} \setminus (-\infty, 0]$ without zeros (Lemma 13.3), which drives the Thorin reformulation of Part IV.

A function f on $\mathbb{R}$ is a Pólya frequency function if $\det[f(x_i - y_j)]_{i,j \le m} \ge 0$ for every $m \ge 1$ and all $x_1 < \cdots < x_m$, $y_1 < \cdots < y_m$. Schoenberg's theorem [488, 293] (Theorem 13.1) characterises the integrable ones, not almost everywhere zero, by their bilateral Laplace transforms $1/\Psi$, with Ψ of Laguerre Pólya type; in its notation, $f / \int f$ is the density of $\delta + \sqrt{2\gamma}\, Z + \sum_k \delta_k (E_k - 1)$, with Z standard normal, the E_k standard exponential, and $Z, E_1, E_2, \ldots$ independent.

A.6 Extreme values

The maximum M_n of n independent random variables with distribution function F has distribution function F^n. By the Fisher Tippett Gnedenko theorem [191, 216,

358], if $(M_n - b_n)/a_n$ converges in law to a nondegenerate limit for some constants $a_n > 0$ and b_n, then the limit is, after a change of location and scale, a Gumbel law $\exp(-e^{-x})$, a Fréchet law $\exp(-x^{-\alpha})$ on $x > 0$, or a Weibull law $\exp(-(-x)^{\alpha})$ on $x < 0$, with $\alpha > 0$. By the Rényi representation [455] the maximum of n independent standard exponentials has the law of $\sum_{j \le n} E_j'/j$, with E_j' independent standard exponentials, a GGC with Thorin measure $\sum_{j \le n} \delta_j$. The infinite version used in the book is

$$\mathbb{P}\Big(\max_{n \ge 1} \frac{E_n}{cn} \le y\Big) = \prod_{n \ge 1} \big(1 - e^{-cny}\big), \qquad y > 0,$$

for $c > 0$; the maximum is almost surely finite because $\sum_n e^{-cny} < \infty$. With $c = 2\pi$ the product is $e^{\pi y/12}\eta(iy)$, the starting point of Chapter 27 (Proposition 27.1 and (27.2)).

A.7 Markov chains on graphs

Let P be the transition matrix of an irreducible Markov chain on a finite set V of size n with stationary law π, so $\pi(x) > 0$ for all x. The chain is reversible if $\pi(x)P(x,y) = \pi(y)P(y,x)$. Simple random walk on a connected d regular graph, whose adjacency matrix A is a symmetric matrix of nonnegative integers with all row sums d (loops and multiple edges are allowed), has $P = A/d$ and is reversible with uniform π.

Spectral decomposition. If P is reversible, then $D^{1/2}PD^{-1/2}$ with $D = \operatorname{diag}(\pi)$ is symmetric, so P has real eigenvalues $1 = \mu_1 > \mu_2 \ge \cdots \ge \mu_n \ge -1$ and eigenvectors $\phi_1 = 1, \phi_2, \ldots, \phi_n$ orthonormal in $L^2(\pi)$. Then

$$P^t(x,y) = \pi(y) \sum_{j=1}^{n} \mu_j^t \phi_j(x)\phi_j(y).$$

The strict inequality $\mu_2 < 1$ is irreducibility, and $\mu_n = -1$ if and only if the chain has period 2 (for a connected graph, if and only if it is bipartite). Averaged return probabilities $n^{-1} \operatorname{tr} P^t = n^{-1} \sum_j \mu_j^t$ are the moments of the spectral measure.

Hitting times. Let $\tau_y = \min\{t \ge 0 : X_t = y\}$ and $h(x,y) = \mathbb{E}_x \tau_y$. The function $h(\cdot, y)$ is the unique solution of the system $h(y,y) = 0$ and, for $x \ne y$, $h(x,y) = 1 + \sum_z P(x,z)h(z,y)$. With $\tau_y^+ = \min\{t \ge 1 : X_t = y\}$, Kac's formula gives the mean return time $\mathbb{E}_y \tau_y^+ = 1/\pi(y)$ [14].

Mixing and the spectral gap. The spectral decomposition also controls convergence to equilibrium. With $\mu_* = \max(\mu_2, |\mu_n|)$, the Cauchy Schwarz inequality gives

$$\Big|\frac{P^t(x,y)}{\pi(y)} - 1\Big| \le \mu_*^t \Big(\sum_{j\ge 2} \phi_j(x)^2\Big)^{1/2} \Big(\sum_{j\ge 2} \phi_j(y)^2\Big)^{1/2} \le \frac{\mu_*^t}{\sqrt{\pi(x)\pi(y)}},$$

since $\sum_j \phi_j(x)^2 = 1/\pi(x)$ by orthonormality. For a d regular graph this gives $|P^t(x,y) - 1/n| \le \mu_*^t$, which is empty for a bipartite graph, where $\mu_* = 1$. A connected Ramanujan graph that is not bipartite, so that with $q = d-1$ every adjacency eigenvalue λ with $|\lambda| \ne d$ has $|\lambda| \le 2\sqrt{q}$ and $-d$ is not an eigenvalue, has $\mu_* \le \rho_q = 2\sqrt{q}/(q+1)$, the spectral radius of simple random walk on the d regular tree [309]; hence $|P^t(x,y) - 1/n| \le \rho_q^t$.

Kemeny's constant. Kemeny and Snell [304] observed that the quantity $K = \sum_y \pi(y) h(x,y)$ does not depend on the starting state x. This is Kemeny's constant: the expected time to hit a target drawn from the stationary law. In our convention $h(y,y) = 0$; putting the mean return time $1/\pi(y)$ on the diagonal instead would add 1 to K. Let Π be the matrix with every row equal to π, and let $Z = \sum_{t\ge 0}(P^t - \Pi)$ be the fundamental matrix. By the spectral decomposition $P^t(x,y) - \pi(y) = \pi(y)\sum_{j\ge 2} \mu_j^t \phi_j(x)\phi_j(y)$, so if $\mu_n > -1$ the series converges and summing the geometric series gives

$$Z(x,y) = \pi(y) \sum_{j\ge 2} \frac{\phi_j(x)\phi_j(y)}{1-\mu_j}.$$

If $\mu_n = -1$ the chain is periodic, the series is understood in the Abel sense, $Z = \lim_{r\uparrow 1} \sum_t r^t (P^t - \Pi)$, and the displayed formula remains valid because $(1 - r\mu_j)^{-1} \to (1-\mu_j)^{-1}$ for every $j \ge 2$. In all cases we take the displayed formula as the definition of Z.

THEOREM A.4. *Let P be irreducible and reversible with eigenvalues $1 = \mu_1 > \mu_2 \ge \cdots \ge \mu_n \ge -1$. Then for all x, y*

$$h(x,y) = \frac{Z(y,y) - Z(x,y)}{\pi(y)},$$

and for every x

$$K = \sum_y \pi(y)\, h(x,y) = \sum_{j=2}^{n} \frac{1}{1-\mu_j}.$$

In particular, for simple random walk on a connected d regular graph with adjacency eigenvalues $d = \lambda_1 > \lambda_2 \ge \cdots \ge \lambda_n$, $h(x,y) = n\big(Z(y,y) - Z(x,y)\big)$ and $K = \sum_{j\ge 2}(1 - \lambda_j/d)^{-1}$.

PROOF. The matrix $(\sqrt{\pi(x)}\,\phi_j(x))_{x,j}$ is orthogonal, so its rows are orthonormal as well:

$$\sum_{j=1}^{n} \pi(y)\phi_j(x)\phi_j(y) = 1_{x=y}.$$

Since $\phi_1 = 1$, this gives

$$\sum_{j\geq 2} \pi(y)\phi_j(x)\phi_j(y) = 1_{x=y} - \pi(y).$$

Using $P\phi_j = \mu_j\phi_j$ and $\mu_j/(1-\mu_j) = 1/(1-\mu_j) - 1$ we find

$$(PZ)(x,y) = \pi(y)\sum_{j\geq 2} \frac{\mu_j\phi_j(x)\phi_j(y)}{1-\mu_j} = Z(x,y) - 1_{x=y} + \pi(y).$$

Fix y and define $g(x) = (Z(y,y) - Z(x,y))/\pi(y)$. Then $g(y) = 0$, and for $x \neq y$,

$$\begin{aligned}\sum_z P(x,z)g(z) &= \frac{Z(y,y) - (PZ)(x,y)}{\pi(y)} \\ &= \frac{Z(y,y) - Z(x,y) - \pi(y)}{\pi(y)} = g(x) - 1.\end{aligned}$$

So g solves the hitting time equations, and by uniqueness $g = h(\cdot, y)$. (Uniqueness: the difference u of two solutions vanishes at y and is harmonic off y. If $\max u > 0$, it is attained at some $x \neq y$, harmonicity makes every z with $P(x,z) > 0$ a maximum point as well, and irreducibility carries the maximum along a path to y, a contradiction; the same argument applies to $-u$.) Now

$$\sum_y \pi(y)h(x,y) = \sum_y Z(y,y) - \sum_y Z(x,y).$$

The second sum vanishes, because $\sum_y \pi(y)\phi_j(y) = \langle \phi_j, \phi_1\rangle_\pi = 0$ for $j \geq 2$. The first is the trace of Z, and $\sum_y \pi(y)\phi_j(y)^2 = 1$ by orthonormality, so

$$\operatorname{tr} Z = \sum_{j\geq 2} \frac{1}{1-\mu_j}\sum_y \pi(y)\phi_j(y)^2 = \sum_{j\geq 2}\frac{1}{1-\mu_j}.$$

This is independent of x, and the regular graph case follows with $\pi(y) = 1/n$ and $\mu_j = \lambda_j/d$. □

A.8 DIVERGENCE, BAYES FACTORS AND CONSISTENCY

The Kullback Leibler divergence [335] $\mathrm{KL}(\mu\|\nu) = \int \log(d\mu/d\nu)\,d\mu$, set equal to $+\infty$ unless $\mu \ll \nu$, is nonnegative, zero only for $\mu = \nu$, not symmetric, and jointly

convex in (μ, ν). For independent observations $X_1, X_2, \ldots$ from f_0, the Bayes factor of f_1 against f_0 [294] is $B_n = \prod_{i \le n} f_1(X_i)/f_0(X_i)$, and by the strong law of large numbers

$$\frac{1}{n} \log B_n \to -\mathrm{KL}(f_0 \| f_1) \qquad \text{almost surely,}$$

with limit $-\infty$ if the divergence is infinite. This is the principle behind the rates of Chapter 29, where equidistribution replaces the law of large numbers. Wald's sequential probability ratio test [559], whose stopping rule Chapter 29 states with the target probabilities α of rejecting f_0 when it holds and β of accepting it when f_1 holds, needs for small α and β, ignoring overshoot, about $\log(1/\beta)/\mathrm{KL}(f_0 \| f_1)$ observations under f_0 and $\log(1/\alpha)/\mathrm{KL}(f_1 \| f_0)$ under f_1, and is optimal in expected sample size among tests with no larger error probabilities [563]. Doob's theorem [163] gives posterior consistency at almost every parameter value under the prior, for a parametric model with a measurable and identifiable parametrisation, and the maximum likelihood estimator is consistent under Wald's conditions [560]. If the true density is one of finitely many candidate densities $f_1, \ldots, f_m$, say f_t, with $0 < \mathrm{KL}(f_t \| f_i)$ for $i \neq t$, then by the law of large numbers above the posterior probability of each wrong model i is $\exp(-n(\mathrm{KL}(f_t \| f_i) + o(1)))$ almost surely. If the truth f_0 lies outside the list, the posterior concentrates on the candidates that minimise $\mathrm{KL}(f_0 \| f_i)$ [47], and the exponential rates are differences of divergences.

A.9 Notation

The table lists the symbols that are used in more than one chapter or are central to the argument of a chapter, with the place where each is defined, while symbols that serve a single argument are defined where they occur.

Symbol	Meaning	Where defined
Number theory and L functions		
a_p	Trace of Frobenius at a good prime p, $a_p = p + 1 - \#E(\mathbb{F}_p)$ for an elliptic curve E; the Hecke eigenvalue $a(p)$ of a newform	Chapter 1; Section 3.2
x_p, θ_p	Normalised trace $x_p = a_p/p^{(k-1)/2} \in [-2, 2]$ ($a_p/\sqrt{p}$ in weight 2), $x_p = 2\cos\theta_p$ with $\theta_p \in [0, \pi]$, and $x_p = \mathrm{tr}\, A_p(\pi)$ in the unitary normalisation; θ_p is also the Frobenius class in $K^\natural$ and an independent Sato Tate angle in Chapters 11, 12, 15	Chapter 1; Theorem 1.4; Theorem 4.3
π	Cuspidal automorphic representation of $\mathrm{GL}_n(\mathbb{A}_\mathbb{Q})$ with unitary central character, contragredient $\tilde{\pi}$; π_f is the representation of a newform f	Chapter 2
$\pi(x)$	Number of primes $p \le x$	Chapter 1

Symbol	Meaning	Where defined
$Q(x)$, $Q_\alpha(x)$, $D(s)$	$Q(x)=\sum_{n\le x}\mu(n)^2$, the number of squarefree $n\le x$, and $D(s)=\sum_n \mu(n)^2 n^{-s}=\zeta(s)/\zeta(2s)$; $Q_\alpha(x)$ is the sum of $\mu^2(\mathfrak{a})N(\mathfrak{a})^{-\alpha}$ over the ideals $\mathfrak{a}$ of norm at most x of a number field; Chapter 17 writes $\Delta(x)=Q(x)-6x/\pi^2$, which is not Ramanujan's Δ	Chapter 17
$A_p(\pi)$, $\alpha_{j,p}$	Satake parameter of π at an unramified prime p, a semisimple class in $\mathrm{GL}_n(\mathbb{C})$ with eigenvalues $\alpha_{j,p}$	Chapter 2; Theorem 3.7
$L(s,\pi)$	Standard L function $\prod_p \det(I-A_p(\pi)p^{-s})^{-1}$ in the unitary (analytic) normalisation, critical line $\Re s=\frac12$; the default for π and for $\mathrm{Sym}^n f$	Chapter 1; Chapter 2
$L(s,f)$, $L(s,\Delta)$	Classical normalisation $\sum_n a(n)n^{-s}$ of a newform of weight k, centre $s=\frac{k}{2}$, so $L(s,\pi_f)=L(s+\frac{k-1}{2},f)$; for Δ it is $\sum_n \tau(n)n^{-s}$, centre $s=6$	Theorem 2.5; Chapter 13
$L(s,\Delta)$	Unitary normalisation $\sum_n \tau(n)n^{-11/2-s}$, centre $s=\frac12$; it is the classical $L(s+\frac{11}{2},\Delta)$	Chapter 11; Chapter 15
$L(s,E)$	L function of an elliptic curve $E/\mathbb{Q}$, $\prod_p(1-a_pp^{-s}+p^{1-2s})^{-1}$ at good p, in the classical normalisation with centre $s=1$	Section 2.4; Theorem 2.7
E_1, E_2	The curves $y^2=x^3+x+1$ (conductor 496, no complex multiplication) and $y^2=x^3-x$ (complex multiplication by $\mathbb{Z}[i]$, conductor 32); in Chapter 21, $E_1(v)=\int_v^\infty e^{-x}x^{-1}dx$ is the exponential integral	Chapter 1; Figure 1.1; Theorem 21.10
$\Lambda(s,\pi)$	Completed L function $\gamma(s,\pi)L(s,\pi)=\varepsilon(\pi)\Lambda(1-s,\tilde\pi)$ with $\gamma(s,\pi)=N_\pi^{s/2}\prod_j\Gamma_{\mathbb{R}}(s+\mu_j)$ and $\Gamma_{\mathbb{R}}(s)=\pi^{-s/2}\Gamma(\frac{s}{2})$, $\Gamma_{\mathbb{C}}(s)=\Gamma_{\mathbb{R}}(s)\Gamma_{\mathbb{R}}(s+1)=2(2\pi)^{-s}\Gamma(s)$; for an elliptic curve, $\Lambda(s,E)=N_E^{s/2}(2\pi)^{-s}\Gamma(s)L(s,E)$, centre 1 (Chapters 2 and 22)	Section 14.2; Chapter 2; Corollary 20.33
$\Lambda(s,\Delta)$	Completed function $(2\pi)^{-s}\Gamma(s)L(s,\Delta)=\Lambda(12-s,\Delta)$, centre 6 (classical); $(2\pi)^{-s-11/2}\Gamma(s+\frac{11}{2})L(s,\Delta)$, centre $\frac12$ (unitary; half of $\Lambda(s,\pi)$ of the row above, by Legendre duplication)	Theorem 2.5; Chapter 13; Section 15.2
$\beta(s)$, χ_{-4}, χ_{12}	Dirichlet's $\beta(s)=L(s,\chi_{-4})$, completed as $\Lambda(s)=(4/\pi)^{s/2}\Gamma(\frac{s+1}{2})\beta(s)$, and $\Lambda_{12}(s)=(12/\pi)^{s/2}\Gamma(\frac{s}{2})L(s,\chi_{12})$, with Pólya kernels built from η^3 and η	Section 18.1; Section 18.10
$\varepsilon(\pi)$, ε_f, $\varepsilon(\chi)$, $w(E)$	Root number, $\lvert\varepsilon(\pi)\rvert=1$; classically $\Lambda(s,f)=\varepsilon_f\Lambda(k-s,f)$, $\varepsilon_f=\pm i^k$; $\varepsilon(\chi)$ for a Dirichlet character, $w(E)=\varepsilon_f=\pm1$ for an elliptic curve E	Theorem 2.5; Section 14.2; Chapter 8

Symbol	Meaning	Where defined
$g(\chi)$, $g(m)$	Gauss sum $g(\chi) = \sum_{a \bmod N} \chi(a) e^{2\pi i a/N}$ of a Dirichlet character modulo N; for primitive χ with $\chi(-1) = (-1)^\delta$, $\varepsilon(\chi) = g(\chi)/(i^\delta\sqrt{N})$; quadratic Gauss sum $g(m) = \sum_{n \bmod m} e^{2\pi i n^2/m}$ for odd m	Chapter 2; Corollary 20.5
Δ, $\tau(n)$, q	$\Delta(\tau) = q\prod_{n\geq 1}(1-q^n)^{24} = \sum_n \tau(n)q^n$, the weight 12 cusp form (τ in the upper half plane, nome $q = e^{2\pi i\tau}$, Ramanujan function $\tau(n)$); $\tau(p)/p^{11/2} = 2\cos\theta_p$	Section 2.1; Example 2.2
$G_{\mathbb{Q}}$, Frob_p	Absolute Galois group $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, and the Frobenius class at an unramified prime p	Section 3.1
N	Conductor N_π or N_E, or level of a newform in $S_k(\Gamma_0(N))$	Section 2.4
$S_k(\Gamma_0(N))$, T_n	Cusp forms of weight k and level N, and the Hecke operators T_n, with T_p acting on a newform by $a(p)$	Proposition 2.4; Section 2.4
$\lambda_f(p)$	Unitary Hecke eigenvalue $a_f(p)/p^{(k-1)/2} = x_p \in [-2,2]$ of a newform f; $\lambda(p)$ for Δ and $\lambda_\phi(p)$ for a Maass form ϕ	Section 6.4; Open Problem 30.1
$\Lambda(n)$, $\Lambda_E(n)$, $\Lambda_\pi(n)$, $\Lambda_\rho(n)$	The von Mangoldt function; coefficients of $-L'/L$: $\Lambda_E(p^k) = p^{k/2}\,2\cos(k\theta_p)\log p$ (classical, centre 1) and $-\frac{L'}{L}(s,\pi) = \sum_n \Lambda_\pi(n)n^{-s}$ (unitary), and for an Artin representation ρ, $\Lambda_\rho(p^m) = \mathrm{tr}\,\rho(\mathrm{Frob}_p^m)\log p$ at unramified p	Chapter 4; Section 5.4; Chapter 16; Definition 20.29
$\pi_{E,t}$, $C_{E,t}$	Lang Trotter count $\pi_{E,t}(x) = \#\{p \leq x : p \nmid N,\ a_p = t\} \sim C_{E,t}\sqrt{x}/\log x$	Chapter 5; Open Problem 30.2
$S(x)$, r	$S(x) = \sum_{p\leq x,\, p\nmid N} a_p \log p/p \approx (\frac{1}{2} - r)\log x$ under the Riemann hypothesis for $L(s,E)$ (rank bias), where $r = \mathrm{ord}_{s=1} L(s,E)$ is the analytic rank; in Chapter 17, $S(x) = \sum_n \mu(n)^2 e^{-n/x}$	Section 5.4; Open Problem 30.12
γ_k, γ_{E}, γ_K	Ordinates of zeros: $\frac{1}{2} \pm i\gamma_k$ for $\Lambda(s,\pi)$ (unitary), $6 \pm i\gamma_k$ for $\Lambda(s,\Delta)$ (classical), all real exactly under the Riemann hypothesis; γ_{E} is Euler's constant; γ_K is Ihara's Euler Kronecker constant of a number field K	Equation (14.3); Theorem 13.7; Section 5.4; Theorem 20.26
κ, κ_K, c_K, κ_C	In Chapter 17, $\sum_{n\leq x}\mu(n)^2/n = \frac{6}{\pi^2}(\log x + \kappa) + O(x^{-1/2})$ with $\kappa = 24\log A - \gamma_{\mathrm{E}} - 2\log 2\pi$, A the Glaisher Kinkelin constant; over a number field K, $c_K = \mathrm{Res}_{s=1}\zeta_K(s)$ and the constant becomes $\kappa_K = \gamma_K - 2\zeta_K'(2)/\zeta_K(2)$; in Chapters 20 and 24, κ_C is the Euler Kronecker drift of a conjugacy class C; other chapters use κ as a local letter	Proposition 17.1; Theorem 17.4; Theorem 20.26

Symbol	Meaning	Where defined
$\delta(C)$, b_C, c_C, h_C, K_C	In Chapter 24, for a conjugacy class C of $G = \mathrm{Gal}(K/\mathbb{Q})$: its Haar measure $\delta(C) = \lvert C\rvert/\lvert G\rvert$ (written $\mu(C)$ in Chapter 20); the Mertens constants b_C and c_C, the limits of $\sum_{p\le t,\, p\in C} 1/p - \delta(C)\log\log t$ and of $\sum_{p\le t,\, p\in C} \log p/p - \delta(C)\log t$ over the unramified primes; $h_C(s) = P_C(s) - \delta(C)\log\frac{1}{s-1}$, as in the proof of Theorem 20.26; and $K_C = e^{-h_C(1)}/\Gamma(1-\delta(C))$, the constant of the proportion $K_C(\log x)^{-\delta(C)}$ of the integers $n \le x$ with no prime factor in C	Theorems 24.5, 24.7 and 24.8
Groups and Haar measure		
h, $h^\natural$, χ_ρ	Normalised Haar measure of a compact group K (written dh in Chapter 3), its image $h^\natural$ on the space $K^\natural$ of conjugacy classes, and the character $\chi_\rho = \mathrm{tr}\,\rho$	Chapter 4; Theorem 4.3
μ_G	Haar probability measure of a compact group G; for $G = \mathrm{ST}(A) \subset \mathrm{USp}(2g)$ also its image under the trace, a law on $[-2g, 2g]$	Theorem A.3; Chapter 29
$\bar\rho_{E,p}$, $G_{E,p}$, χ_p	Mod p Galois representation of an elliptic curve E, its image $G_{E,p} \subset \mathrm{GL}_2(\mathbb{F}_p)$ (written $G(\ell)$ in Chapter 5), and the mod p cyclotomic character $\chi_p = \det\bar\rho_{E,p}$	Section 22.1; Definition 22.10; Proposition 22.11
a_ρ, $\mathbb{P}_\sigma$, $d(\rho,\rho')$	Frobenius trace variable $a_\rho(p^m) = \mathrm{tr}\,\rho(\mathrm{Frob}_p^m)$ on the prime powers p^m, $p \notin S$, under $\mathbb{P}_\sigma(\{p^m\}) \propto p^{-m\sigma}/m$, and $a_\pi(p^m)$, the power sum of the Satake parameters of a cuspidal π; the Frobenius distance $d(\rho,\rho') = \lim_{\sigma\downarrow 1}(\mathbb{E}_\sigma\lvert a_\rho - a_{\rho'}\rvert^2)^{1/2}$	Section 20.3; Theorem 20.13
μ_{ST}	Sato Tate (semicircle) law $\frac{1}{2\pi}\sqrt{4-x^2}\,dx$ on $[-2,2]$: the law of $x = 2\cos\theta$ under Haar measure on $\mathrm{SU}(2)$	Chapter 1; Proposition 4.5
U_n, Sym^n	Chebyshev polynomial of the second kind, $U_n(\cos\theta) = \sin((n+1)\theta)/\sin\theta$; $\mathrm{tr}\,\mathrm{Sym}^n = U_n(x/2)$ is the character of the nth symmetric power of $\mathrm{SU}(2)$, orthonormal in $L^2(\mu_{\mathrm{ST}})$	Proposition 4.5; Theorem A.3
μ_{CM}	Complex multiplication law $\frac{1}{2}\delta_0 + \frac{1}{2}\,dx/(\pi\sqrt{4-x^2})$ on $[-2,2]$; $\int U_n(x/2)\,d\mu_{\mathrm{CM}}$ is 1 if $4 \mid n$ and 0 otherwise	Section 4.6; Proposition 4.10
$\mathrm{ST}(A)$	Sato Tate group of an abelian variety $A/\mathbb{Q}$ of dimension g, a compact subgroup of $\mathrm{USp}(2g)$; $\mathrm{SU}(2)$ or $N(\mathrm{U}(1))$ for an elliptic curve	Definition 3.10; Conjecture 3.11

Symbol	Meaning	Where defined
s_p, $W_\rho(n)$	Normalised Frobenius class of A at p in $G = \mathrm{ST}(A)$, $x_p = \operatorname{tr} s_p \in [-2g, 2g]$; Weyl average $W_\rho(n) = n^{-1}\sum_{k\le n}\chi_\rho(s_{p_k})$, with limit $\int \chi_\rho\, d\mu_G$ (the Weyl fingerprint); written $W_n(x)$ in Chapter 4 and W_m in Chapter 5, and unrelated to the Pólya variable W	Section 3.6; Chapter 29
Random matrices, families and random models		
$Z_N(\theta)$, Λ_N	Characteristic polynomial $\det(I - U_N e^{-i\theta})$ of a Haar unitary matrix U_N of size N, and its value $\Lambda_N = Z_N(0) = \det(I - U_N)$ at $\theta = 0$	Section 6.5; Chapter 12, Proposition 12.5
N	Size of a random matrix in $\mathrm{U}(N)$, $\mathrm{USp}(2N)$ or $\mathrm{SO}(2N)$, matched to arithmetic by $N \approx \log \frac{T}{2\pi}$ at height T and $N \approx \log c_f$ for a family	Section 6.1; Section 6.3
$W(G)$	One level density of the symmetry type $G \in \{\mathrm{U}, \mathrm{USp}, \mathrm{SO(even)}, \mathrm{SO(odd)}, \mathrm{O}\}$: $\mathbb{E}\sum_j \phi(N\theta_j/2\pi) \to \int \phi\, W(G)$	Section 6.3
$D(f;\phi)$, c_f	One level density statistic $\sum_{\gamma_f} \phi(\frac{\gamma_f}{2\pi}\log c_f)$ of the zeros $\frac{1}{2} + i\gamma_f$ of $L(s,f)$, with analytic conductor c_f	Section 6.3; Theorem 6.5
$\mathrm{Kl}(a,b;p)$, $K(t)$	Kloosterman sum $\sum_{x\in\mathbb{F}_p^\times} e((ax + b\bar{x})/p)$, with $\mathrm{Kl}(a;p)/\sqrt{p} = 2\cos\theta_{p,a} \in [-2,2]$; random Fourier series $K(t) = \sum_h c_h(t)\mathrm{ST}_h$ over independent ST_h with law μ_{ST}	Theorem 7.2; Definition 7.9
μ_{CL}, $\eta_n(p)$	Cohen Lenstra measure $\mu_{\mathrm{CL}}(G) = \eta_\infty(p)/\lvert\operatorname{Aut} G\rvert$ on finite abelian p groups, with $\eta_n(p) = \prod_{k=1}^n (1 - p^{-k})$	Definition 8.1
U_p, X_p	Independent uniform phases on the unit circle (the Steinhaus model) in the random Euler product $\prod_p (1 - U_p p^{-s})^{-1}$; written X_p in Chapter 11	Theorem 11.9; Proposition 12.7
Graphs and walks		
q, $\mathcal{T}_q$	Prime (or integer ≥ 2) q, and the Bruhat Tits tree of $\mathrm{PGL}_2(\mathbb{Q}_q)$: the $(q+1)$ regular tree whose vertices are homothety classes of lattices in $\mathbb{Q}_q^2$	Chapter 9; Theorem 9.1
A, A_n, X, B_n	On $\mathcal{T}_q$: adjacency operator $A = T_q$, sphere operators $A_n f(v) = \sum_{d(v,w)=n} f(w)$, $X = q^{-1/2}A$ and $B_n = q^{-n/2}A_n = U_n(X/2) - q^{-1}U_{n-2}(X/2)$ for $n \ge 2$	Theorem 9.1
μ_q	Kesten McKay law, the spectral measure of X at a vertex of $\mathcal{T}_q$, with $\mu_q \Rightarrow \mu_{\mathrm{ST}}$ as $q \to \infty$; written μ_ℓ, μ_p elsewhere	Theorem 9.1; Chapters 10 and 28

Symbol	Meaning	Where defined
$G_N(\ell)$, $B(\ell)$, N	Supersingular ℓ isogeny graph in characteristic N, a prime $N \equiv 1 \bmod 12$ (the level of $S_2(\Gamma_0(N))$), $(\ell+1)$ regular on $(N-1)/12$ vertices, with adjacency matrix the Brandt matrix $B(\ell)$; $G_N(q)$ with $\ell = q$ in Chapter 9; in Chapter 21, for any prime N, $P = B(\ell)/(\ell+1)$ is the isogeny walk, reversible for $\pi_i \propto 1/w_i$ with $w_i = \lvert \mathrm{Aut}(E_i)\rvert/2$	Theorem 10.4; Section 21.1
h	Number of vertices $(N-1)/12$ of $G_N(\ell)$, and for any prime N the number $g+1$ of supersingular curves, g the genus of $X_0(N)$ (Chapter 21); also the class number $h(D)$ of Chapters 8 and 28	Chapter 10; Theorem 10.6
P, π	Transition matrix and stationary law of a reversible Markov chain; $P = A/d$ and π uniform for a d regular graph with adjacency matrix A; in Chapters 23 and 24, P is a Dirichlet process (see Γ_μ)	Appendix A, Markov chains
μ_j, ϕ_j	Eigenvalues $1 = \mu_1 > \mu_2 \geq \cdots \geq \mu_n \geq -1$ of P, with eigenvectors ϕ_j orthonormal in $L^2(\pi)$; $\mu_j = \lambda_j/d$ for adjacency eigenvalues λ_j. In Chapter 15 they are p_j, and $\mu_j = 1 - r_j$ are Dirichlet rates	Appendix A, Markov chains; Chapter 15
τ_y, $h(x,y)$	Hitting time $\tau_y = \min\{t \geq 0 : X_t = y\}$ and mean hitting time $h(x,y) = \mathbb{E}_x \tau_y$	Appendix A, Markov chains; Chapter 10
K, K_N, Z	$K = \sum_y \pi(y) h(x,y) = \sum_{j \geq 2} (1-\mu_j)^{-1}$, the Kemeny constant; fundamental matrix $Z = \sum_{t \geq 0}(P^t - \Pi)$; K_N for $G_N(\ell)$ with $K_N/h \to \ell/(\ell-1)$; $K_N(\ell) = \sum_f (\ell+1)/(\ell+1-a_\ell(f))$ for the isogeny walk at any prime N	Theorem A.4; Theorem 10.6; Theorem 21.1
$H_N(\ell)$	Hecke GGC with Thorin measure $\sum_f \delta_{\ell+1-a_\ell(f)}$ over the normalised eigenforms of $S_2(\Gamma_0(N))$, a sum of independent exponential variables with mean $K_N(\ell)/(\ell+1)$	Definition 22.6; Theorem 22.7
$\tau(G)$, $J_0(N)$, C_N	Number of spanning trees of a graph G, the order of its critical group; the Jacobian of $X_0(N)$, and for prime N its cuspidal subgroup $C_N = J_0(N)(\mathbb{Q})_{\mathrm{tors}}$, cyclic of order the numerator of $(N-1)/12$	Theorem 21.2; Corollary 21.3
ρ, ρ_q, ρ_ℓ	Kesten rate $2\sqrt{q}/(q+1)$, the spectral radius of simple random walk on the $(q+1)$ regular tree; ρ_ℓ for $G_N(\ell)$	Theorem 9.1; Chapter 10
λ_*, μ_*	$\lambda_* = \max\{\lvert\lambda_j\rvert : \lvert\lambda_j\rvert \neq q+1\}$ for a $(q+1)$ regular graph; $\mu_* = \max(\mu_2, \lvert\mu_n\rvert) = \lambda_*/(q+1)$ if it is not bipartite	Appendix A, Markov chains; Proposition 10.3

Symbol	Meaning	Where defined
$\Lambda(\theta)$, $\Lambda^*(y)$	Limiting cumulant generating function $\Lambda(\theta) = \lim_n n^{-1} \log \mathbb{E}e^{n\theta Y_n}$ and rate function $\Lambda^*(y) = \sup_\theta(\theta y - \Lambda(\theta))$	Appendix A, Gärtner Ellis; Theorem 15.16
Infinitely divisible laws, Thorin measures and GGCs		
(a, β^2, ν)	Lévy Khintchine triple of an infinitely divisible law: drift a, Gaussian variance β^2 and Lévy measure ν on $\mathbb{R} \setminus \{0\}$; in Lemma 16.7 and Chapter 23, ν and ν_α are x times the Lévy measure of $\log f$ at the basepoint	Theorem 11.1; Lemma 16.7
τ, a	Thorin measure τ and drift $a \geq 0$ of a generalized gamma convolution (GGC), with $\mathbb{E}e^{-sX} = \exp(-as - \int \log(1 + s/t)\, \tau(dt))$; its densities U, U_ϵ, u_Γ and V_a (Chapter 16; $T_\alpha = \frac{1}{2} V_{\alpha - 1/2}$ in Chapter 18), the signed density u_ε (Chapter 19) and U_α (Chapter 22, the density U_{ar} of Chapter 16 for the Rankin Selberg product D of E; in Chapter 23, U_{ar} of any Euler product at the basepoint α, signed when its coefficients are) and local combs $\tau_{\theta,L}$ (Chapter 15); τ_Q, the Thorin measure of a Frobenius functional (Theorem 20.21); Tamagawa number $\tau(H)$	Definition 13.2; Equation (14.1)
k, k_π, $\mathcal{W}_\pi$, $\mathcal{W}_\rho$, $\mathcal{W}_E$	Lévy function: a GGC has Lévy density $k(x)/x$ with $k(x) = \int e^{-tx} \tau(dt)$, and $k_\pi(x) = \sum_k e^{-t_k x}$ for π, written $\mathcal{W}_\pi$ (the heat trace) in Chapter 16, where for self dual π it is formed from Ξ_π^{sd}; $\mathcal{W}_\rho$ is the Galois heat trace, defined from Frobenius traces, the conductor and the gamma factor, and $\mathcal{W}_E$ that of an elliptic curve (Chapter 22)	Equation (14.4); Theorem 13.8; Definition 20.29
t_k, S_n	Thorin atoms $t_k = \gamma_k^2$, where $\pm i\gamma_k$ are the zeros of Ξ_π, and power sums $S_n = \sum_k t_k^{-n}$	Equations (14.3) and (14.4)
ξ, Ξ, Φ	Riemann's $\xi(s) = \frac{1}{2} s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s)$ and $\Xi(t) = \xi(\frac{1}{2} + it) = \int \Phi(u) e^{itu}\, du$, with Pólya's kernel Φ; an unsubscripted Ξ is always Riemann's	Chapter 13; Equation (13.1)
Ξ_ζ, Ξ_π, Ξ_π^{sd}, F_π	$\Xi_\zeta(z) = \xi(\frac{1}{2} + z) = \Xi(-iz)$; $\Xi_\pi(z) = \Lambda(\frac{1}{2} + z, \pi)\Lambda(\frac{1}{2} + z, \tilde{\pi})/z^{2r}$, $r = \mathrm{ord}_{s=1/2}\, \Lambda(s, \pi)$; $\Xi_\pi^{\mathrm{sd}} = \Lambda(\frac{1}{2} + z, \pi)/z^r$ if $\pi \cong \tilde{\pi}$, so $\Xi_\pi = (\Xi_\pi^{\mathrm{sd}})^2$; $F_\pi(s) = \Xi_\pi(\sqrt{s})/\Xi_\pi(0) = \prod_k (1 + s/t_k)$; in Chapter 19 the clock $G_\pi(s) = \Xi_\pi^{\mathrm{sd}}(\sqrt{s})/\Xi_\pi^{\mathrm{sd}}(0)$ of a self dual π, so $G_\pi^2 = F_\pi$ ($G_\zeta = F_\zeta$)	Definition 14.5

Symbol	Meaning	Where defined
Φ_π, W_π	Pólya kernel with $\int \Phi_\pi(u)e^{zu}\,du = \Xi_\pi(z)/\Xi_\pi(0)$, and Pólya variable W_π with density $\Phi_\pi \geq 0$; W of Chapter 13 uses Ξ_Δ^{sd} and has density $\Delta(ie^u)e^{6u}/\Lambda(6,\Delta)$ (with Ξ_Δ it would be a sum of two copies); the kernels Φ_f and Φ_χ of Chapter 20 and Φ_E of Chapter 22, with Pólya variable W_E, are not normalised: $\int \Phi_f(u)e^{tu}\,du = \Lambda(\frac{k}{2}+t)$; nor are Φ_β and Φ_{12} (Chapters 18 and 19), Φ_K (Chapter 17) and the kernel $\Phi_\rho = 2\Phi_f$ of Chapter 23, computed by Bromwich inversion, with $\int \Phi_\rho(u)e^{tu}\,du = \Lambda(\frac{1}{2}+t,\rho)$, the factor 2 coming from $\Gamma_{\mathbb{C}}$	Lemma 14.12; Theorem 13.6; Equation (20.4); Proposition 23.8
X, Y, Z_π, T	The GGC $\sum_k E_k/t_k$ with drift 0 and Thorin measure $\sum_k \delta_{t_k}$, named X in Chapters 13 and 14, Y in Chapters 12 and 17, Z_π in Chapter 16 and T in Chapter 21 (T_E in Chapter 22, under GRH for $L(s,E)$), where it is the explosion time of a pure birth process with rates t_k; in Chapter 17, $T = W_\zeta/2$ is half the Pólya variable of ζ, with moment generating function M_T	Theorem 14.7; Theorem 16.6; Theorem 21.11
E_k, L_k	Independent standard exponential variables E_k and Laplace variables L_k (density $\frac{1}{2}e^{-\lvert x\rvert}$); E_k is also the Eisenstein series, E_2 quasimodular in Chapter 27	Theorem 13.8; Chapter 16; Section 2.1
V, V_π	Van Dantzig operator $VF(t) = 1/F(it)$ on characteristic functions, and the series $V_\pi = \sum_k L_k/\gamma_k$; in Chapter 18, $V = \mathrm{Var}\, X_\beta = 2S_1$ for the Pólya variable X_β of $L(s,\chi_{-4})$	Definition 16.1; Theorem 16.6
(X,H)	Wald couple: $H \geq 0$ and $\mathbb{E}e^{sX}\,\mathbb{E}e^{-s^2H} = 1$ for small $s \geq 0$; under the Riemann hypothesis (W_π, Z_π) is one	Definition 16.3
H, H_a, C	The GGC H of the sine squared lemma, $\mathbb{E}e^{-sH} = e^{b\sqrt{s}}f(\alpha)/f(\alpha+\sqrt{s})$ with $b = (f'/f)(\alpha)$, whose tail Chapter 23 computes; the GGC H_a with $\mathbb{E}e^{-sH_a} = \Xi_\pi(a)/\Xi_\pi(a+\sqrt{s})$ and Thorin density V_a, formed in Chapter 23 from Ξ_π^{sd}, where V_a is, in the variable $r = \sqrt{z}$, a superposition over the zeros of folded Cauchy laws, C being a standard Cauchy variable	Lemma 16.7; Theorem 16.8; Theorems 23.9 and 23.10
Θ_σ, P_σ	Parabolic Thorin condition: every zero of $\Lambda(s,\pi)$ has $\lvert\Re s - \frac{1}{2}\rvert \leq \sigma$, equivalently every atom lies in $-P_\sigma$; Θ_0 is the Riemann hypothesis, $\Theta_{1/2}$ is unconditional	Definition 14.15
$H_{\pi,\lambda}$, λ_π, Λ	Gaussian tilt $H_{\pi,\lambda}(t) = \int e^{\lambda u^2}\Phi_\pi(u)e^{itu}\,du$, and the de Bruijn Newman constant $\lambda_\pi \leq \frac{1}{8}$ in the book's units; the literature constant is $\Lambda = 4\lambda_0$ for ζ, with $\Lambda \leq \frac{1}{2}$	Theorem 14.17; Section 14.6; Chapter 13

Symbol	Meaning	Where defined
N_σ, e_p, $N_\sigma^{(\vartheta)}$	Zeta distributed random integer, $\mathbb{P}(N_\sigma = n) = n^{-\sigma}/\zeta(\sigma)$, with independent geometric exponents e_p (written N and G_p in Chapter 11); $\log N_\sigma$ is the zeta subordinator at time 1, with Lévy measure Π_σ; $N_\sigma^{(\vartheta)}$, whose logarithm is the zeta subordinator at time ϑ, has the law $d_\vartheta(n)n^{-\sigma}/\zeta(\sigma)^\vartheta$, with d_ϑ the coefficients of ζ^ϑ	Theorem 11.2; Proposition 16.10; Lemma 20.17; Theorem 24.4
Γ_μ, Ξ_σ, $\mathcal{M}_\sigma$, $\mathcal{P}_\sigma$	Gamma random measure with shape measure μ; the Frobenius labelled random measure of $(\sigma-1)\log N_\sigma$ (a subscripted Ξ that is not a completion) and the marked point process of its prime factors; normalised, $\mathcal{P}_\sigma = \Xi_\sigma/\Xi_\sigma(Y)$ (and μ when $\Xi_\sigma = 0$), which converges in law to the Dirichlet process P with base measure μ	Theorem 20.18; Theorem 24.3
$Q_{E,p}$, $c(E,p)$, c_p	Eisenstein discrepancy $Q_{E,p}(\ell) = 1\{p \nmid |E(\mathbb{F}_\ell)|\}$ and Eisenstein shape $c(E,p) = \mu_{G_{E,p}}\{g : \det(1-g) \neq 0\}$, the local density of Koblitz's conjecture and the shape of the Gamma limit of $\Xi_\sigma(Q_{E,p})$; $c_p = 1-(p^2-2)/((p^2-1)(p-1))$ when $G_{E,p} = \mathrm{GL}_2(\mathbb{F}_p)$	Definition 22.10; Proposition 22.11
PD(ϑ), V_k, W_k	Poisson Dirichlet law with parameter ϑ: the law of the ranked jumps of a standard Gamma subordinator on $[0,\vartheta]$ divided by their sum, written $V_1 \geq V_2 \geq \ldots$; for $\vartheta = 1$ the stick breaking sequence $W_k = (1-U_1)\cdots(1-U_{k-1})U_k$, with independent uniform U_i, is its size biased order (the GEM law); not the operator V, the densities V_a or the Pólya variable W	Section 24.1; Theorem 24.4
Extreme values and eta products		
$\eta(\tau)$	Dedekind eta function $q^{1/24}\prod_{n\geq1}(1-q^n)$, and eta products $f = \prod_m \eta(m\tau)^{r_m}$ with $\sum_m m\, r_m = 24$	Chapter 27; Theorem 27.7
M, $P(y)$	$M = \max_{n\geq1} E_n/(2\pi n)$ for standard exponentials E_n, so $\mathbb{P}(M \leq y) = P(y) = \prod_{n\geq1}(1-e^{-2\pi n y})$; M is also a clipping level (Chapter 29), $M(x;u_0,\ldots,u_k)$ is the Curry Schoenberg B spline with knots $u_0,\ldots,u_k$ (Chapter 23), and $M(t) = \sum_{d\leq t}\mu(d)$ is the Mertens function in Chapter 17	Proposition 27.1; Chapter 15; Section 23.7
M^*, M_f, F	Maximum M^* of 24 independent copies of M, with $F(y) = \mathbb{P}(M^* \leq y) = P(y)^{24}$ and $\Delta(iy) = e^{-2\pi y}F(y)$; M_f is the analogue for an eta product f	Theorem 27.2; Theorem 27.7

Symbol	Meaning	Where defined
$Q(s,c)$, G_s	Regularised incomplete gamma function $\Gamma(s,c)/\Gamma(s)$, and a gamma variable G_s of shape s independent of M^*: $L(s,\Delta) = \mathbb{E}\, Q(s, 2\pi M^*) = \mathbb{P}(G_s \geq 2\pi M^*)$ for $s > 0$ in the classical normalisation	Theorem 27.2
Statistics and inference		
D_n, F_n, $\mathcal{K}$	Kolmogorov discrepancy $D_n = \sup_t \lvert F_n(t) - F(t)\rvert$ of the empirical distribution F_n of $x_1, \dots, x_n$; $\sqrt{n}\, D_n \Rightarrow \mathcal{K}$ (Kolmogorov law) for independent draws (K in Chapters 5, 27)	Appendix A, equidistribution; Section 5.3
$\mathrm{KL}(\mu\|\nu)$	Kullback Leibler divergence $\int \log(d\mu/d\nu)\, d\mu$, equal to $+\infty$ unless $\mu \ll \nu$	Appendix A, Bayes factors
B_n, B_n^M, h, h_M	Bayes factor $B_n = \prod_{i \leq n} f_1(X_i)/f_0(X_i)$ of f_1 against f_0, not the sphere operators B_n; with the log likelihood ratio $h = \log(f_1/f_0)$ and its clipping $h_M = \max(-M, \min(M, h))$, $\log B_n^M = \sum_{k \leq n} h_M(x_k)$	Appendix A, Bayes factors; Theorem 29.1
M_{ST}, M_{CM}, ε	Models $M_{\mathrm{ST}} = (1-\varepsilon)\mu_{\mathrm{ST}} + \varepsilon\delta_0$ and $M_{\mathrm{CM}} = \mu_{\mathrm{CM}}$ for the traces x_p, with a small atom ε at 0	Corollary 29.3

Chapter B

Code for the Computations

This appendix prints the core routines behind the figures and tables of the book in condensed form, and then lists the scripts and result files that accompany the source. The computations use PARI/GP [432] through its Python interface cypari2, with NumPy, SciPy and mpmath. The release that produced the stored results was not recorded, except that `stage4c.json` names 2.15.4. The values in the comments come from a rerun of all listings in 2026 with PARI/GP 2.17.2 and cypari2 2.2.4, and they agree with the numbers quoted in the chapters; selected listings give the same values under PARI/GP 2.15.4, the series named in the reference. The listings are cumulative: later ones use the names defined by earlier ones, starting from these lines.

```
import math
import numpy as np
import mpmath as mp
import cypari2
pari = cypari2.Pari()
pari.allocatemem(2*10**9)       # a PARI stack of two gigabytes
```

B.1 Frobenius traces

Naive point counting by Legendre symbols, vectorised over residues, serves for primes up to about 10^5 and as an independent check on PARI's `ellap`, which we use for all primes up to 10^7.

```
def ap_naive(a, b, p):          # y^2 = x^3 + a x + b mod p
    x = np.arange(p, dtype=np.int64)
    f = (x*x % p*x + a*x + b) % p
    sq = np.zeros(p, np.int8); sq[(x*x) % p] = 1
    leg = np.where(f == 0, 0, np.where(sq[f] == 1, 1, -1))
    return -int(leg.sum())

def arr(expr):                  # a GP integer vector as a NumPy array
    t = str(pari(f"Vecsmall({expr})"))
    return np.array(t[t.index("[")+1:t.rindex("]")].split(","),
                    dtype=np.int64)

pari("E=ellinit([0,0,0,1,1]); P=primes([3,10^7]);"
     "V=vector(#P,i,ellap(E,P[i]));")
P, V = arr("P"), arr("V")
```

```
good = P != 31                  # 31 divides the discriminant -496
P, V = P[good], V[good]         # 664577 primes
x = V/np.sqrt(P)                # normalised traces in [-2, 2]
print([round(float(np.mean(x**k)), 4) for k in (2, 4, 6)])
# [1.0, 1.9994, 4.9976]: the Catalan numbers 1, 2, 5
assert all(ap_naive(1, 1, p) == v for p, v in zip(P[:500], V[:500]))
```

The even moments in the comment are those of Table 5.1. The curve $y^2 = x^3 - x$ is treated in the same way, with `[0,0,0,-1,0]` in place of `[0,0,0,1,1]` and no excluded prime.

For a genus two curve $y^2 = f(x)$ over $\mathbb{F}_p$, with p odd and not dividing the leading coefficient or the discriminant of f, the number of points is $\#C(\mathbb{F}_p) = p + \sum_{x \in \mathbb{F}_p} \left(\frac{f(x)}{p}\right) + n_\infty$, where n_∞ counts the points at infinity: $n_\infty = 1$ if $\deg f$ is odd, and for $\deg f$ even $n_\infty = 2$ or 0 according as the leading coefficient is a square modulo p or not. The trace of Frobenius is $p + 1 - \#C(\mathbb{F}_p)$.

```
def ap_genus2(c, p):            # y^2 = f(x), f = c[0] x^d + ... + c[d]
    x = np.arange(p, dtype=np.int64)
    v = np.zeros(p, dtype=np.int64)
    for a in c: v = (v*x + a) % p               # Horner: f(x) mod p
    sq = np.zeros(p, np.int8); sq[(x*x) % p] = 1
    chi = np.where(v == 0, 0, np.where(sq[v] == 1, 1, -1))
    inf = 1 if len(c) % 2 == 0 else (2 if sq[c[0] % p] else 0)
    return 1 - int(chi.sum()) - inf             # p + 1 - #C(F_p)

print(ap_genus2([1, 0, 0, 0, -1, 1], 101))      # -8: y^2 = x^5 - x + 1
```

For $y^2 = x^5 - x + 1$ over $\mathbb{F}_{101}$ the trace is -8, the value behind the characteristic polynomial $T^4 + 8T^3 - 16T^2 + 808T + 10201$ of Chapter 15. For the three curves of Chapter 29 the function reproduces the traces stored in `stage4c.json` (Section B.7) at all good primes below 10^5.

B.2 Supersingular isogeny graphs

The following PARI/GP function builds the ℓ isogeny graph on the supersingular j invariants over $\mathbb{F}_{p^2}$, the graph $G_N(\ell)$ of Chapter 10 with $N = p$. It takes the least $D \geq 3$ such that $-D$ is a fundamental discriminant and p is inert in $\mathbb{Q}(\sqrt{-D})$, so that every root modulo p of the Hilbert class polynomial of discriminant $-D$ is supersingular, and starts from one of them. It explores the neighbours of a vertex j through the roots of the classical modular polynomial $\Phi_\ell(j, Y)$. For supersingular j all $\ell + 1$ roots, counted with multiplicity, lie in $\mathbb{F}_{p^2}$, so every irreducible factor is linear, the entry A_{jr} of the result is the multiplicity of the root r, and every row of A sums to $\ell + 1$. The function is saved as `ss_graph.gp` in `code/work/` and in `hitting/`; that file differs from the listing in the names and declarations of some

variables, in the layout of a few statements and in a comment, and gives the same matrices for $\ell \in \{2,3,5,7\}$ and the primes $p \neq \ell$ with $7 \leq p \leq 400$, and for ten primes $p \equiv 1 \bmod 12$ between 1000 and 8000 with $\ell \in \{2,3\}$.

```
ssgraph(p,l)={
  my(T=ffinit(p,2,'a), g=ffgen(T,'a), D=3, fa, j0,
     seen=Map(), stack, edges=List(), n=1, A,
     P=polmodular(l));
  while(kronecker(-D,p)!=-1 || !isfundamental(-D), D++);
  fa=factor(subst(polclass(-D),'x,'y)*g^0);
  for(i=1,#fa[,1], if(poldegree(fa[i,1])==1,
     j0=-polcoef(fa[i,1],0)/polcoef(fa[i,1],1); break));
  mapput(seen,j0,1); stack=List([j0]);
  while(#stack>0, my(j=stack[#stack]); listpop(stack);
    fa=factor(subst(subst(P,'x,j),'y,'Y)*g^0);
    for(i=1,#fa[,1],
      my(r=-polcoef(fa[i,1],0)/polcoef(fa[i,1],1));
      if(!mapisdefined(seen,r),
         n++; mapput(seen,r,n); listput(stack,r));
      listput(edges,[mapget(seen,j),mapget(seen,r),fa[i,2]])));
  A=matrix(n,n);
  for(k=1,#edges, my(e=edges[k]); A[e[1],e[2]]+=e[3]);
  A
}
```

For $p \equiv 1 \bmod 12$ neither $j = 0$ nor $j = 1728$ is supersingular, so no vertex has an automorphism other than ± 1, and A is the symmetric Brandt matrix of Chapter 10, of size $(p-1)/12$. For other p the curves with $j = 0$ or $j = 1728$ can be supersingular and have extra automorphisms, the matrix is in general not symmetric, and the book does not use it. We ran the function for $\ell = 2, 3$ at the 207 primes $p \equiv 1 \bmod 12$ between 1000 and 8000: all 414 matrices are symmetric with these sizes and row sums $\ell + 1$, and their Kemeny constants and the largest absolute values of their nontrivial eigenvalues, divided by $2\sqrt{\ell}$, agree with `stage2.json`.

B.3 Kemeny constants and hitting times

The function `graph` turns the output of `ssgraph` into a NumPy matrix. The function `kemeny` is the spectral formula $K = \sum_{j \geq 2} (1 - \lambda_j/d)^{-1}$ of Theorem A.4. The function `hitting_times` solves the hitting time equations directly, one target at a time, in $O(n^4)$ operations, and `hitting_fast` computes the same matrix in $O(n^3)$ operations from the fundamental matrix, $h(x,y) = n\big(Z(y,y) - Z(x,y)\big)$ with $Z = (I - P + \Pi)^{-1} - \Pi$, where Π has every entry equal to $1/n$.

```
pari('read("ss_graph.gp")')
```

```
def graph(p, l):                 # adjacency matrix of the l graph at p
    pari(f"A=ssgraph({p},{l})"); n = int(pari("#A"))
    return np.array([[int(pari(f"A[{i},{j}]")) for j in range(1, n+1)]
                     for i in range(1, n+1)], dtype=float)

def kemeny(M):                   # M: adjacency of an (l+1) regular graph
    d = M.sum(1)[0]
    ev = np.sort(np.linalg.eigvalsh(M))
    return np.sum(1/(1 - ev[:-1]/d))

def hitting_times(M):            # H[u,v] = E_u tau_v
    n = len(M); P = M/M.sum(1)[0]; I = np.eye(n)
    H = np.zeros((n, n))
    for v in range(n):
        idx = [i for i in range(n) if i != v]
        H[idx, v] = np.linalg.solve((I - P)[np.ix_(idx, idx)],
                                    np.ones(n - 1))
    return H                     # H.mean() equals kemeny(M)

def hitting_fast(M):             # the same matrix, O(n^3) operations
    n = len(M); P = M/M.sum(1)[0]; Pi = np.ones((n, n))/n
    Z = np.linalg.inv(np.eye(n) - P + Pi) - Pi
    return n*(np.diag(Z)[None, :] - Z)

M = graph(1009, 2)               # 84 vertices, 3 regular
print(kemeny(M), hitting_times(M).mean(), hitting_fast(M).mean())
# all three are 156.766
```

Chapter 10 reports the direct computation at $N = 1009$, 2017 and 4129.

B.4 The extreme value representation

Let M^* be the maximum of 24 independent copies of $\max_{n \geq 1} E_n/(2\pi n)$, with E_n standard exponential variables, as in Appendix A and Chapter 27. For real $s > 0$, $L(s, \Delta) = \mathbb{E}\, Q(s, 2\pi M^*)$, where $Q(s, c) = \Gamma(s, c)/\Gamma(s)$ is the regularised upper incomplete gamma function. The simulation truncates the maximum at $n \leq 400$, which biases the estimate of $L(6, \Delta)$ by at most 1.2×10^{-13} (Monte Carlo section of Chapter 27).

```
from scipy.special import gammaincc        # regularised Q(s,c)
rng = np.random.default_rng(3); n = np.arange(1, 401)
Ms = np.concatenate([
    np.max(rng.exponential(size=(1000, 24, 400))
           / (2*np.pi*n), axis=(1, 2)) for _ in range(20)])
q = gammaincc(6, 2*np.pi*Ms)               # Q(6, 2 pi M*), 20000 draws
print(q.mean(), q.std()/np.sqrt(len(q)))   # 0.7929 and 0.0012

# exact value by quadrature: L(s,Delta) is the integral of
# Delta(iy) (2 pi y)^s dy/(y Gamma(s)) over y > 0
```

```
def EQ(s):
    D = lambda y: mp.mpf(float(pari(f"real(eta(I*{y},1)^24)")))
    g = lambda y: D(y)*(2*mp.pi)**s*y**(s-1)/mp.gamma(s)
    return mp.quad(g, [0.01, 0.1, 0.3, 1, 3, 12])
print([mp.nstr(EQ(s), 7) for s in (3, 6, 10)])
# ['0.3152416', '0.7921228', '0.9798091']
```

The values in the last comment are $L(3,\Delta)$, $L(6,\Delta)$ and $L(10,\Delta)$, and the Monte Carlo value 0.7929 is within one standard error of $L(6,\Delta)$. In the quadrature, PARI's `eta(z,1)` is the Dedekind eta function itself, whereas `eta(z)` omits the factor $e^{i\pi z/12}$, and $\Delta(iy) = \eta(iy)^{24}$. The integrand is negligible outside $[0.01, 12]$ for the moderate s used here.

B.5 Zeros of $L(s,\Delta)$ and the GGC product

With $\Lambda(s) = (2\pi)^{-s}\Gamma(s)L(s,\Delta)$ and zeros $6 \pm i\gamma_k$, the Hadamard product gives $\Lambda(6)/\Lambda(6+\sqrt{s}) = \prod_k (1 + s/\gamma_k^2)^{-1}$ (Chapter 13). The listing multiplies over the zeros with $0 < \gamma \le 400$ and corrects for the others by the tail $\sum_{\gamma > T} \gamma^{-2} \approx (\log(T/2\pi) + 1)/(\pi T)$ with $T = 400$, which follows from the zero density $\pi^{-1}\log(t/2\pi)$. Here `lfunlambda` is PARI's completed L function, and only the quotient of its values matters.

```
pari("mf=mfinit([1,12]); L=lfunmf(mf,mfDelta());")
z = [float(t) for t in pari("lfunzeros(L,400)") if t > 0]
lam = lambda s: float(pari(f"lfunlambda(L,{s})"))
T = 400; tail = (math.log(T/(2*math.pi)) + 1)/(math.pi*T)
def ggc(s):                    # Lambda(6)/Lambda(6+sqrt s)
    return math.exp(-sum(math.log1p(s/t**2) for t in z)
                    - s*tail)

print(len(z), ggc(100), lam(6)/lam(16))
# 404 0.0069741 0.0069732: corrected product and exact quotient
```

Without the tail the product is too large by the factor $e^{s\cdot\mathrm{tail}}$, which is 1.51 at $s = 100$ and 5.16 at $s = 400$. With it, the relative error is below 2×10^{-4} on a grid of step 0.05 in $(0, 400]$, the worst value being 1.9×10^{-4} near $s = 227$.

B.6 Bayes factors for Sato Tate groups

The function returns the running sum over the primes of the log likelihood ratios of the model M_{CM} against M_{ST} of Chapter 29, each term clipped to $[-M, M]$. The argument `zero` marks the primes with $a_p = 0$, and `eps` is the mass ε that M_{ST} puts on 0.

```
def log_bayes(x, zero, eps=1e-3, M=20):    # CM against SU(2)
```

```
    s = np.sqrt(np.clip(4 - x**2, 1e-12, None))
    l_st = np.where(zero, np.log(eps),
                    np.log(1 - eps) + np.log(s/(2*np.pi)))
    l_cm = np.where(zero, np.log(.5), np.log(.5/(np.pi*s)))
    return np.cumsum(np.clip(l_cm - l_st, -M, M))

m = P < 40000                  # good primes below 40000, curve E_1
b = log_bayes(x[m], V[m] == 0)
print(b[49], b[-1]/len(b))     # -15.9 and -0.979 per prime
```

Here `x`, `P` and `V` are the arrays of the first listing, for $E_1 : y^2 = x^3 + x + 1$. The two numbers are the log Bayes factor after 50 good primes and the average per prime over the good primes below 40000. For $E_2 : y^2 = x^3 - x$ the same computation gives 160.4 and 3.102. The limiting slopes are -0.999 and 3.108 (Chapter 29).

B.7 Scripts and results

The stage scripts behind the numbers of the book, and the JSON files with their results, sit in the folder that holds the source, next to `book/`; those of Chapters 19, 20, 21, 22, 23, 24, 25 and 26 are in the subfolders `scholze/`, `recip/`, `hitting/`, `flt/`, `laplace/code/`, `pd/code/`, `gl1/code/` and `maass/code/`. The folder `code/` holds copies of the stage scripts other than `stage4c.py`, and of the scripts of Chapters 20 and 21, in a layout with relative paths, described below, together with scripts kept only there: `compute.py`, `etacheck.py`, the scripts of five figures and pilot computations for some of the open problems. Table B.1 lists the scripts with the chapters that use their results, those kept only in `code/` last.

Table B.1: Computation scripts and the chapters that use them.

Script	Computes	Chapters
`stage1.py`	Traces of Frobenius to 10^7 for $y^2 = x^3 + x + 1$, $y^2 = x^3 - x$ and the curves 11a1, 37a1, 389a1 and 5077a1, and $\tau(p)$ to 10^6; Sato Tate discrepancies, also for 200 random curves to 10^6; rank bias; Lang Trotter counts	1, 2, 4, 5
`stage2.py`, `stage2b.py`	Spectra and Kemeny constants of the supersingular 2 and 3 isogeny graphs at the 207 primes $p \equiv 1 \bmod 12$ between 1000 and 8000; hitting time matrices at $p =$ 1009, 2017 and 4129	9, 10
`stage3.py`	Pólya kernel of Δ and its 404 zeros up to height 400, the GGC product, the extreme value representation	13, 27
`stage4.py`, `stage4b.py`	Traces of three genus two curves to 10^5; Bayes factors for Sato Tate groups of genus two (superseded by `stage4c.py`)	29

Script	Computes	Chapters
`stage4c.py`	Traces of the same curves, their trace densities by quadrature, the Kullback Leibler table, posteriors and rates of the chapter	29
`stage5.py`	Thorin measures of eleven L functions, Hankel determinants, passage times on trees and isogeny graphs, the Weil example	14, 15
`stage6.py`, `stage6b.py`	The van Dantzig test for Δ, its heat trace from zeros and from primes, zeros recovered from primes by a sliding bump, the $2\sin^2$ transport check	16
`stage7.py`, `stage7b.py`	The Patie class survey of L functions, simulated first passage times for the zeta subordinator, high precision Patie tests for ξ, Δ, $L(s, \chi_{-4})$ and the curve 11a1	16
`stage8.py`	Heegner points, Duke's theorem, coverage of right and left Haar priors	28
`stage9a.py`, `stage9a2.py`	Selberg's central limit theorem, the Fyodorov Hiary Keating maxima, spacings and pair correlation for Δ	12
`stage9b.py`	Maxima of random fields, branching random walk, Harper's model	12
`stage9c.py`	Cohen Lenstra and Poonen Rains statistics	8
`stage9d.py`	Katz and Kowalski Sawin statistics for Kloosterman sums	7
`scholze/signed_thorin.py`, `scan_p2p3.py`, `offline_test.py`, `certify.py`	The signed Thorin density u_ε of ζ, $L(s, \chi_{12})$ and $L(s, \Delta)$ by Stieltjes inversion and their zeros on the critical line by sign changes; its comparison with the Lorentzian smoothing of these zeros; a toy function with one pair of zeros moved off the line; winding numbers of Λ around rectangles	19
`recip/data.py`, `laws.py`, `second.py`, `heat.py`, `recip_compute.py`, `recip_law.py`, `recip_gauss.py`	Frobenius classes for $x^3 - x - 1$ and $x^3 + x + 1$, the symbols $(\frac{7}{p})$ and $(\frac{p}{7})$, and traces of 11a1 and 37a1 for the primes below 10^7; Gamma process Laplace transforms, zero probabilities and Euler Kronecker drifts; the discrete Gaussian modulo c and heat flow reciprocity; Pólya kernel symmetry, the Galois heat trace and Gamma shapes; Frobenius distances, joint laws and conditional means for the primes below 10^6; Gauss sum reciprocity and its Brownian limit	20
`hitting/graphs.py`, `iso_run.py`, `snf_run.py`, `sandpile.py`, `test37.py`, `hitrates.py`, `other.py`, `explosion.py`, `expl_frob.py`, `exitcheck.py`	Isogeny walks for $\ell = 2, 3, 5, 7$ at 58 prime levels against T_ℓ from modular symbols (Kemeny constants, spanning trees, Kirchhoff indices); exact Kemeny constants and tree counts at the primes $N \equiv 1 \bmod 12$ below 700; Smith normal forms at ten levels; critical groups at the 48 primes $N \equiv 1 \bmod 12$ with $37 \le N < 1300$ and the comparison at $N = 37$; Chebotarev hitting rates for $x^3 - x - 1$; mean explosion times from central derivatives, from zeros, and from Frobenius classes with the conductor and the gamma factor; the signed exit law	21

Script	Computes	Chapters
`flt/data.py`, `flt.py`, `thorin_ec.py`, `flt_eis.py`, `frey_like.py`	The primes below 10^7 and the traces of 11a1 and 37a1 (the same script as `recip/data.py`); Wilton's congruence for the primes below $3 \cdot 10^5$, the newforms of level 77 against 11a1 modulo 3, dimensions of $S_2(\Gamma_0(N))$ and counts of Hurwitz quaternions; Pólya kernels of 11a1 and 37a1, and the heat trace of 11a1 from the explicit formula against its zeros, with its integral; Eisenstein shapes, Haar counts in $\mathrm{GL}_2(\mathbb{F}_p)$, shares and Laplace transforms for 37a1 and 11a1; the shapes of the Frey type curve of $3 + 5^3 = 2^7$	22
`laplace/code/switch23.py`, `switch23b.py`, `zeta_switch.py`, `zeta_tail.py`, `splines/data.py`, `checks.py`, `gg_mple.py`	Frobenius traces of the representation of $x^3 - x - 1$ against the q expansion of $\eta(z)\eta(23z)$ at the prime powers below $2 \cdot 10^5$, and their transported densities at $\alpha = \frac{3}{2}$; Gaussian means recovering $\Lambda_\rho(n)$, $n \leq 64$, from the q expansion, and Bromwich inversion of $\Lambda(s, \rho)$, computed as ζ_K/ζ, on the critical line; the Thorin densities of Ξ_ζ against Cauchy superpositions over the first 1000 zeros of ζ, and Talbot and Post Widder inversion of the Thorin law of ζ at $\alpha = 2$; its density on $[0.0005, 400]$ with its mass, mean and tail constant; the primes below 10^7 and the traces of 11a1 and 37a1 (the same script as `recip/data.py`); Weyl denominators, and the Hecke GGC $H_{67}(2)$ as a Gamma variable times a B spline variable; Good and Gaskins estimates for 11a1 and $y^2 = x^3 - x$	23
`pd/code/data.py`, `pd1.py`, `pd2.py`, `op_pd.py`, `op_pd2.py`	The primes below 10^7 with the Frobenius classes of $x^3 - x - 1$ (the same script as `recip/data.py`); 400000 uniform random integers below 10^7, with their largest prime factors, numbers of prime factors and three colourings of the prime factors by Frobenius class; Mertens constants per class, the predicted colourings, the Dirichlet mean check and the Laplace transform of a Frobenius functional at $\sigma = 1.01$, from the Euler product over the primes below 10^7 and the prime number theorem beyond; the constants b_C from $L(1, \chi)$, the probabilities that a class is empty, the share means and the Rankin Selberg colouring; the constants c_C from $\frac{L'}{L}(1, \chi)$, the constants K_C and the proportions of the sampled integers with no prime factor in a class	24
`gl1/code/g1.py`, `g3.py`, `g4.py`, `g5.py`, `sym2test.py`	The Gaussian primes below 10^7 with their primary generators, the traces of the curve 32a2, the discrepancy and Weyl sums of the Hecke angles and the moments of complex multiplication; Rankin Selberg Laplace transforms of 32a2 over $\mathbb{Q}$ and over $\mathbb{Q}(i)$; Mertens constants modulo 4, atoms at zero, Landau's constant and a sieve to 10^8; the zeros of $L(s, \chi_4)$ to height 3000 with the product check; a check that the symmetric square of 32a2 in PARI has a trivial Euler factor at 2	25

Script	Computes	Chapters
`maass/code/m1.py` to `m7.py`, `lmfdb_r.py`	The prime geodesics of the modular surface of norm at most 10^6, from the class numbers and units of the discriminants below $4 \cdot 10^6$, with their counting function against li; 64 paths of Brownian motion on the modular surface to time $2 \cdot 10^4$; the Eichler Selberg trace formula against Hecke matrices for weights up to 40 and $n \leq 30$; Serre's vertical Sato Tate law at $p = 2$ for the eigenforms of level one and weight up to 400, with the Eichler Selberg moments of orders 2 and 4; the Thorin density of $Z(s+1)/Z(s)$ at $\alpha = 2$ with the sine squared check; the error term of the prime geodesic theorem and its periodogram; the first 100 spectral parameters of level one, copied from the LMFDB	26
Kept only in `code/book_build/` and `code/book_figscripts/`		
`compute.py`	Traces of $y^2 = x^3 + x + 1$ and $y^2 = x^3 - x$ below 4×10^4 by Legendre symbols, supersingular primes and the fraction of $a_p = 0$, the Bayes factors of Section B.6	1, 5, 29
`etacheck.py`	The extreme value representation against PARI at real points for four eta products	27
`ch03_chebotarev.py`	Frobenius classes of $x^3 - x - 1$ for the primes below 10^7, from which `ch03_chebfig.py` draws the figure of the chapter	3
`ch06_figs.py`	The one level densities of the symmetry types; spacings of 400 Haar unitary matrices of size 200; a trace check with 4000 matrices of size 50, of which the chapter prints a rerun with another seed	6
`ch11_lvalues.py`	$L(s, f)$ on the real axis for four eta products	27
`ch12_posterior.py`	Trace densities of the genus two groups by quadrature, posterior paths, the Kullback Leibler table and the rates of the three curves, from the traces written by `stage4b.py`	29

Each stage script writes the JSON file of its own stage, and random draws use seeded NumPy generators. The later scripts `stage2b.py`, `stage4b.py`, `stage6b.py` and `stage9a2.py` overwrite the JSON files of stages 2, 4, 6 and 9a, and the table describes these later runs, not the earlier and smaller `stage2.py` and `stage4.py`. The Kullback Leibler divergences, posteriors and rates in `stage4.json` come from histogram densities that alias; `stage4c.py` evaluates the densities by quadrature, reads no file and reproduces the genus two numbers of Chapter 29. The file `stage6.json` lacks the heat trace computed by `stage6.py`, a selection of which is in Table 16.1. The file `stage8.json` has 4000 replications and a block of normal error checks, under key names that `stage8.py` does not write. That script runs 2000 replications on grids too narrow for the Pólya error law and gives coverages near 0.65 for σ; the normal block is computed by `code/scratch/dbg8.py`, and the coverage block of `stage8.py`, run with the wider grids of `dbg8.py` and 4000 replica-

tions, agrees with Table 28.2 within Monte Carlo error. The file `stage9a_var.json` is read by `stage9a2.py` and written by no saved script.

The figures are drawn by `figs.py`, `figs2.py`, `fig3.py`, `fig3b.py`, `fig9.py`, `fig_parabola10b.py`, `stage5fig.py` and `stage7fig.py` from the JSON files and the arrays written by the stage scripts; by the five figure scripts kept only in `code/`, of which `ch12_posterior.py` also draws `genus2` in the layout of the book (`figs2.py` draws another version); for Chapter 18, by `beta/fig_beta.py`; and, for Chapters 20 to 26, by the scripts of `recip/`, `hitting/`, `flt/`, `laplace/code/`, `pd/code/`, `gl1/code/` and `maass/code/`. Four figures, `rankbias`, `langtrotter`, `sf_atoms` and `sf_thorin_strip`, are written by no saved script (for the first two `figs.py` draws plots of the same data, saved as `/tmp/rb.pdf` and `/tmp/lt.pdf`, which differ from the figures of the book). The scripts next to `book/` do not run as they stand: stages 1 to 4 (except `stage4c.py`) and the figure scripts hardcode the path `/home/claude/langlands/` of the sandbox where they ran, and several scripts read arrays written by earlier scripts, of which only the JSON results were kept, not the arrays or `stage6_coeffs.json`. Their copies in `code/book_build/` replace that path by `../work/` and are run from their own folder, with the files shared between folders in `code/work/`, which also holds `ss_graph.gp`. In the cases we tested (`stage3.py`, the trace arrays of `stage4b.py`, `compute.py`, `etacheck.py` and the computations of the figure scripts of Chapters 3, 6, 27 and 29) they reproduce the numbers of the chapters, but not every script runs as it stands: several figure scripts, `stage5.py` and `stage8.py` read from their own folder files that other scripts write to `code/work/`, or the reverse, a few paths into `/tmp` remain, and `stage4c.py` is not in the archive.

Chapters 17 and 18 came without scripts; their text says how their numbers were computed. The script `beta/fig_beta.py`, written for this book, computes and draws the two figures of Chapter 18 (mpmath at 20 digits, matplotlib). The four scripts of Chapter 19, in `scholze/`, use mpmath, read no files and run as they stand; all but `offline_test.py` take the function, the height and the grid sizes as arguments and print their results as JSON. The script `signed_thorin.py` computes the signed density u_ε on a grid from Λ'/Λ, and the zeros on the critical line, for the three functions at 20 digits; its outputs are kept as `zeta_out.json`, `chi12_out.json` and `delta_out.json`. The script `scan_p2p3.py` compares u_ε with the Lorentzian smoothing of the located zeros, for ζ at $\varepsilon = 1$ and $\frac{1}{2}$ and for $L(s, \chi_{12})$ at $\varepsilon = 1$, in the files `s_zeta_e1.json`, `s_zeta_e05.json` and `s_chi12_e1.json`, and `offline_test.py` prints the minimum of u_ε at $\varepsilon = 2$ and 20 for a toy function built from the first five zeros of ζ, with the third left on the line or moved off it by 0.3. The script `certify.py` counts zeros by unwrapping the phase of Λ at sampled points of a rectangle (400 per side, fixed in the script) and compares the count with the sign changes of Λ on the critical line; its flag `certified` records only that the two agree. It works at 12 digits and with no interval arithmetic or bound on the

phase steps, so its counts are numerical, not certified; its output was not kept. The grid sizes of the kept outputs were not recorded with them.

The seven computational scripts of Chapter 20, in `recip/`, use relative paths and run as they stand, `data.py` first, since `laws.py` and `second.py` read the array file `data.npz` that it writes, which is not kept. The nine figures of the chapter are drawn by `recip_figs.py`, `recip_gauss_fig.py`, `recip_law_fig.py`, `figs.py` and `fig_second.py` of that folder; the last two write `gamma_process.pdf`, `zero_heat.pdf` and `second_order.pdf`, which the book prints under the names `recip_gammaproc.pdf`, `recip_zeroheat.pdf` and `recip_drift.pdf`.

The ten computational scripts of Chapter 21, in `hitting/`, use PARI/GP through cypari2, except `hitrates.py` and `exitcheck.py`. Six scripts hard-code paths under `/home/claude/langlands/`, the sandbox where they ran: `graphs.py`, `sandpile.py`, `test37.py` and `fig_graph.py` read `ss_graph.gp` there, and `hitrates.py` and `expl_frob.py` read `newrecip/data.npz`, the array file written by `recip/data.py`, which `other.py` looks for in `../newrecip/`; `iso_run.py` and `snf_run.py` read `iso.gp`. Both GP files are kept in `hitting/`. Of the four figures of the chapter, `figs.py` draws `hit_isogeny`, `hit_rates` and `hit_exit`, and `fig_graph.py`, which also needs networkx, draws `hit_graph`. Copies of the scripts of Chapters 20 and 21 are in `code/ch17_reciprocity/`, `code/ch17_gamma_process/` and `code/ch18_hitting/` (the folder names follow an earlier numbering of the chapters), where `data.npz` and `ss_graph.gp` are read from `code/work/`.

The five computational scripts of Chapter 22, in `flt/`, use relative paths and run as they stand, `data.py` first, since `thorin_ec.py`, `flt_eis.py` and `thorin_fig.py` read the array file `data.npz` that it writes, which is not kept; all of them except `flt_eis.py` use PARI/GP through cypari2. Of the five figures of the chapter, the script `figs.py`, which also uses cypari2, draws `flt_labels` and `flt_congruence`; the script `thorin_fig.py`, which also reads the file `thorin_ec.json`, draws the figures `flt_kernels` and `flt_thorin` and writes the file `thorin_fig.json`; and the script `fig_eis.py` draws `flt_eisenstein` from the files `flt_eis.json` and `frey_like.json`.

The computational scripts of Chapter 23, in `laplace/code/` and its subfolder `splines/`, use relative paths and run as they stand, `splines/data.py` (the same script as `recip/data.py`) first, since `gg_mple.py` reads the array file `data.npz` that it writes, which is not kept. The scripts `switch23.py` and `switch23b.py` read no file and use PARI/GP through cypari2. The script `zeta_switch.py` uses mpmath and reads the first 1000 zeros of ζ from `zeros1000.json`, which it computes and writes if the file is missing; `checks.py` uses sympy and cypari2, and `gg_mple.py` SciPy and cypari2. Each writes a JSON file named after it, the first two `switch23_a.json` and `switch23_b.json`, and rerun, they reproduce the stored files to rounding, the Good and Gaskins coefficients to the tolerance of the opti-

miser. The tail data of Section 23.5, stored in `zeta_switch.json` as sent, are written by no saved script; `zeta_tail.py`, written for this book, recomputes them with mpmath, obtains the mass and the mean by inverting their own transforms instead of by trapezoid sums, and writes `zeta_tail.json`. Of the three figures of the chapter, `figs.py` draws `lap_faces` and `lap_zeta` from the files of `switch23.py`, `switch23b.py`, `zeta_switch.py` and `zeta_tail.py`, and `fig_gg.py` in `splines/`, which also uses cypari2 and SciPy, draws `lap_spline` from those of `checks.py` and `gg_mple.py`; both write into a folder `figs/` below their own. Both were edited to draw the figures in the style of the book, `fig_gg.py` now computing the B spline formula exactly rather than from the samples, and the versions as sent are kept as `figs.original.py` and `splines/fig_gg.original.py`.

The computational scripts of Chapter 24, in `pd/code/`, use relative paths. The script `data.py`, the same as `recip/data.py`, runs first, since `pd1.py`, `pd2.py`, `op_pd.py` and `op_pd2.py` read the array file `data.npz` that it writes, which is not kept; `op_pd2.py` also reads the file `pd_samples.npz` written by `pd1.py`, which is not kept either. The scripts `data.py`, `op_pd.py` and `op_pd2.py` use PARI/GP through cypari2, `op_pd.py` also mpmath and SciPy, and `pd1.py` and `pd2.py` SciPy. The four that read `data.npz` each write a JSON file named after them, and rerun, they reproduce the stored files to rounding. A few printed numbers come instead from checks made for this book: the share means of Section 24.5 and their deficits, from a finer quadrature of the integrals of `op_pd.py`, whose trapezoid sums are about 0.07 percent too high; the bound 0.066 of Section 24.1, the supremum over all v, where `pd1.json` has the maximum 0.056 over a grid; the value 0.2096 there, which is exact, where `pd2.json` has the Monte Carlo value 0.2095; and the sums over the primes below 10^9 in Section 24.5 and the deficits of the other fields in Open Problem 24.2. The two figures of the chapter, `pd_integers` and `pd_dirichlet`, are drawn by `figs.py` from `pd1.json`, `pd2.json` and `pd_samples.npz`, into a folder `figs/` below its own; it was edited to draw them in the style of the book, and the version as sent is kept as `figs.original.py`.

The scripts of Chapters 25 and 26, in `gl1/code/` and `maass/code/`, use relative paths, and all but `g4.py`, `m2.py`, `m5.py`, `m7.py`, the list `lmfdb_r.py` and the two figure scripts use PARI/GP through cypari2. They have not been rerun for this book: the JSON files kept with them are those of the runs behind the chapters. In `gl1/code/`, `g1.py` runs first, with the bound 10^7 as its argument, since `g4.py` and `figs.py` read the array file `gl1_data.npz` that it writes; `figs.py` also reads `chi4_zeros.npy`, written by `g5.py`, and neither array is kept. The file `twosquares.json`, counts of the sums of two squares that the chapter does not quote, is written by no saved script. In `maass/code/`, `m1.py` runs first, with the bound 10^6 as its argument, since `m5.py`, `m7.py` and `figs.py` read the geodesics that it writes to `geodesics.npz`; `figs.py` also reads `bm_traj.npz` from `m2.py`, `vertical_p2.npy` from `m4.py` and `m7.npz` from `m7.py`, of which only the

last is kept, writes `zeta_terms.json`, and imports the spectral parameters of `lmfdb_r.py`, a list copied from the LMFDB. Both figure scripts save the figures into `../book/figs/`, a path of the layout in which they ran.

The routines above and the stored numbers are the starting point for the programme of Chapter 31.

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